OpenText™ Service Manager

Use

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1. Use

This section provides information on how to use Service Manager.

Service Desk (Codeless)

Service Desk (Hybrid)

Streamlined Interaction(Codeless)

Incident Management (Codeless)

Incident Management (Hybrid)

Problem Management

Change Management (Codeless)

Change Management (Hybrid)

Request Fulfillment (Codeless)

Request Management (Hybrid)

Configuration Management

Knowledge Management

Service Catalogs

Service Level Management

Smart Analytics

Service Manager Reports

Service Manager Collaboration

Service Manager Mobile Applications

Survey

1.1. Service Desk (Codeless)

You can use Service Manager Service Desk to restore normal service to your users as quickly as possible by resolving interactions on first contact or by immediately escalating interactions for proper handling. Service Desk helps you manage all user interactions, whether they are reported by using Self-Service website or directly to your service desk.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Service Desk User Roles

Key concepts

Describes the most important things that you need to know about how the Service Desk module functions.

Service Desk overview

What's a Service Desk interaction?

Out-of-box Service Desk categories

Working with Service Desk interaction records

Related records in Service Desk

User workflows

Describes the workflows used by Service Manager and the sub tasks that end users need to perform to execute the workflows.

Service Desk workflows and user tasks

Additional resources

Processes and Best Practices

1.1.1. Service Desk overview

Service Manager Service Desk enables you to restore normal service to your users as quickly as possible by resolving user requests on first contact or by immediately escalating them for proper handling. Service Desk helps you manage all user interactions, whether reported by using a Self-Service website or directly to your service desk. Service Desk enables you to achieve the following results:

- Control the relationship between Service Desk and other Service Manager applications
- Require Service Desk interactions to follow a set process
- Log and prioritize all calls to the service desk
- Categorize types of interactions and keep track of their resolution
- Provide first-line investigation and problem diagnosis, and quickly escalate user requests that can't be resolved within agreed-on time limits to fulfillment processes
- Request a service, provide information, or track previous requests without the assistance of a Service Desk Agent
- Communicate with customers and users to keep them informed of progress, impending changes, agreed-on outages, and other notifications.

Related topics

User Interaction Management

1.1.1.1. User Interaction Management

The Service Manager Service Desk application enables a Service Desk Agent to document and track calls. Service Desk provides one-click access to other Service Manager applications to automatically enter information received during the call.

Information that's obtained during a call can be used to open an incident record if a reported issue requires further action. Information can also be added to a record in another Service Manager application, such as Change Management.

User Interaction Management provides users with a single point of contact (the Service Desk) for all their IT Service related questions, requests, and issues, and provides resolutions to user requests within the agreed service level targets.

User Interaction Management includes all direct interactions between a user and the Service Desk by phone or by email. User Interaction Management also includes all user activities that occur by use of the self-service web portal (for example, searching the Knowledge base, checking for status updates, or logging an interaction).

Related topics

Service Desk overview
What's a Service Desk interaction?
Working with Service Desk interaction records

1.1.1.2. What's a Service Desk interaction?

Each time a user contacts the Service Desk, the Service Desk Agent uses Service Manager to create an interaction record. The Service Desk Agent records the user name, the name of the component about which the user is calling, and a description of the user request. After the Service Desk Agent collects this information, the Service Desk Agent performs the actions that are required to resolve the user request.

- If the service request can be resolved without escalating it to an incident, the Service Desk Agent can close the interaction record.
- If the service request can't be resolved without escalating it to an incident, the Service Desk Agent searches for existing incidents that affect the same component or one of the parent assets of that component.
    - If an existing incident is found, the Service Desk Agent can associate the current interaction with the existing incident record.
    - If an existing incident record is not found, the Service Desk Agent can register a new incident based on the Service Desk interaction. Service Manager copies information from the interaction record into the newly created incident record.

For example, if a user can't print to a network printer, the user contacts the Service Desk for assistance. The Service Desk Agent populates an interaction record with the relevant information. Because the issue can't be resolved immediately, the Service Desk Agent opens an incident, and the incident is assigned to a technician. The technician discovers that the printer network connection is broken. The technician fixes the connection and closes the incident. The Service Desk Agent contacts the user and instructs the user to attempt printing to the network printer. If the user can successfully print, the Service Desk Agent can close the interaction. If the user still can't print, the Service Desk Agent may reopen the related incident record.

Service Desk interaction list

The Service Desk interaction list provides general information about interaction records.

Column name	Description
Status	The status of the interaction record.
Description	A brief description of the reported interaction.
Interaction ID	The interaction record number.
Impact	The level of impact this issue has on the company and its end users. The impact can be classified as one of the following values: • 1–Enterprise • 2–Site/Department • 3–Multiple Users • 4–User

Column name	Description
Urgency	The importance of the service request to the recipient. The urgency can be classified as one of the following values. The default value is 4–Low. • 1–Critical (Severe business disruption) The business unit or subunit is unable to operate; a critical system component has failed or is severely impaired. • 2–High (Major business disruption) A critical user or user group is unable to operate, or the business unit experiences a significant reduction in system performance. • 3–Average (Minor business disruption) A single user is unable to operate with no available work around. • 4–Low (Minor disruption) A single user or user group is experiencing incidents, but a workaround is available.
Priority	The priority value is calculated by using the initial impact and urgency values.

What's a category?

Service Manager uses the category to which a record is assigned to determine what work needs to be performed in order to close an interaction. For example, a service

interruption logged in the incident category could have a different workflow than a service catalog request category interaction. The steps to manage and resolve the incident category interaction may differ from the steps for a service catalog request category interaction.

The category of the interaction determines which Process Designer workflow the interaction uses, and the workflow determines which forms and required fields are displayed. In addition, the category of the interaction determines which Service Manager process is invoked when you escalate an interaction. Categories of the complaint or compliment type are handled entirely as other types of interactions; however, you can't escalate these interactions.

Note

Your organization may have customized many workflows to handle the interaction, and may also have applied different service agreements to those workflows.

Out-of-box Service Desk categories

The following table lists the out-of-box Service Desk categories. Depending on the category that you select, Service Manager may require you to further categorize the interaction record by selecting an Area and Subarea.

Category	Use
Complaint	Interactions that report less than satisfactory service or performance.
Compliment	Interactions that report exemplary service or performance.
Incident	Interactions that are opened specifically to start the incident management process.
Problem	Interactions that are opened specifically to start the problem management process.
Request for Administration	Interactions that are opened specifically to request assistance from a system administrator, such as a password resets or rights to access a specific area.
Request for Change	Interactions that are opened specifically to start the change process.
Request for Information	Call or incidents that request information only.
Service catalog	Interactions that are requests for items from the service catalog.

Categories can be created and revised by the System Administrator. Therefore, additional categories may be available to you.

The Notify By field specifies how to contact the person or customer who initiated an interaction. If the interaction record is associated with another record, the information in the Notify By field affects how the interaction record is closed. For example, if you associate an incident record to an interaction record, the related interaction record closes according to the information in the Notify By field when the incident record is closed.

Notify By setting	Description
None	The interaction record closes.
Email	Service Manager sends an email to the contact who is listed in the interaction record to inform the contact that the related record is closed. Service Manager then closes the interaction record.
Telephone	Notification by telephone adds a required action to the interaction record. The required action explains why the customer needs to be called back. This setting also prevents the interaction record from being closed until all required actions are complete.

Note

1. An administrator can add page and fax notifications.
2. In Service Manager Codeless, you can configure HTML Rulesets from the workflow to send notifications as needed.

Affected CI field in Service Desk

The information that's available to populate the Affected CI field depends on the service recipient.

For example, if the recipient has individual or departmental subscriptions, a list of CIs associated with those subscriptions opens. If the recipient doesn't have any subscriptions and is a self-service user, no list is generated. However, if a recipient doesn't have any subscriptions and is not a self-service user, the following behavior occurs:

- If an interaction recipient is specified, a list of the CIs assigned to that recipient opens.
- If a location is specified, a list of the CIs assigned to that location opens.

- If a company is specified in a multi-company system, a list of the CIs assigned to that company opens.
- If a device type is specified, a list of the CIs of that type opens.

Note

The only way to obtain a list of all CIs is to click the Affected CI field before you enter any recipient information.

Required actions

The Required Actions function enables you to track the steps that are necessary to resolve an interaction record. You can view any required actions for an interaction record by opening the Required Actions section of the interaction form. The Required Actions section is available only for interaction records that have required actions. You can add a required action to existing interaction records that have related incident records only. You can't close an interaction record that's associated to incomplete required actions.

For example, if you use the Full Help Desk model to close an interaction and the Notify By field is set to "telephone" a required action is automatically created and must be closed before the interaction can be closed. Therefore, the Service Desk Agent is required to call the customer and verify the issue is fixed, close the required action, and then close the interaction.

1.1.1.3. Working with Service Desk interaction records

The Service Desk functions that are available to you depending on your security role. You may be able to access all Service Desk functions, or you may be limited by the capability words that are assigned to you in your operator record.

Depending on your security role, the following Service Desk functions may be available to you.

Function	Desccription
Create New Interaction	From a blank Service Desk interaction form, you can log information about a service request, check for related interaction records, and add service request information to other Service Manager records, such as incident records. From this form, you can: • Escalate the interaction to an incident, a problem, a change, or a request. • Log a complaint or compliment. • Request an item from the Service Catalog.
Interaction Queue	View the Service Desk interaction queue. You can also view other queues, such as the queues for incidents, problems, requests, changes, and any associated tasks lists.
Search Interactions	Specify filtering information to locate a specific Service Desk interaction record or set of records.

Hover-over forms for interaction records

Service Desk interaction record forms include hover-over forms. A hover-over form is displayed when you move the mouse over a field that supports hover-over forms. Hover-over forms are only displayed when the field contains data. The data that's displayed in hover-over forms is read-only.

Service Desk interaction record forms contain the following hover-over fields.

Field	Subform fields
Contact	Full Name, Telephone, Email
Service Recipient	Full Name, Telephone, Email

Related records in Service Desk

Service Manager lists all related records for an interaction record in the Related Records section. All information is read-only, but you can click a record ID to view the related record. The related records are listed in alphabetic order by the record file name: Change (cm3r), Request (request), Incident (probsummary). Within each module, the records are sorted by ID number, in ascending order.

The Service Desk Related Records section contains the following fields.

Field	Description
ID	The related record number in the database.
Type	Identifies the related record as related to a change, request, or incident.

Activities section

The Activities section allows users to enter new updates for a record or to view journal updates and historic activities for a record.

Journal Updates

Field	Description
New Update Type	Specifies or categorizes the activity update (for example, communication with the customer)
Visible to Customer?	Makes the update visible to customers so they can view related interactions via self service **Note** This option can be bypassed by setting the **Bypass Visibility Option** flag to **true** in the activityactions table. When this flag is set to **true**, automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to **false**, all activities are visible to customers if the **Customer Visible** option is selected.

Field	Description
New Update	Used to enter notes to explain and describe updates made for the record. If Journaling is enabled, the text entered here is displayed in Journal Updates. If Activities are enabled, the text entered in this field is displayed as an activity record for the selected activity type. **Note** The System Administrator is responsible for enabling Journaling and Activities.
Journal Updates	Displays text entered in the **New Update** field together with the operator's name and a timestamp for the update. This field displays information when Journaling is enabled.

Activity Logs

Select an activity type from the **Activity Type** drop-down list or select a user from the **User** drop-down list to filter the activity logs.

The activity log lists activities for the current record in order of occurrence, with the newest at the beginning of the list. Each activity log displays the following information:

- Date/Time
- Operator
- Description
- Type
- Visible to Customer icon (if you have selected the **Visible to Customer** checkbox)

Service Catalog requests

You can track your Service Catalog requests on the Service Desk update interaction form. When you submit a request by using the Service Catalog, you can view the request record in Service Desk. An interaction record for a Service Catalog request includes two additional sections, Approvals and Catalog Items.

Approvals section

The Approvals section contains the following subsections.

Subsections	Description
Current Approvals	Displays a table that lists current catalog requests and the approval details
Approval Log	Displays a log of the approval actions which have been performed regarding the catalog requests

Catalog Items section

The Catalog Items section contains the list of items for the catalog request and the details of the each item in the request. This section also includes the **Edit Cart** button, which links you to the Service Catalog so that you can change your catalog request.

Related topics

Service Desk workflows and user tasks
What's a Service Desk interaction?
Associating a Service Desk interaction with another application record

1.1.1.4. Associating a service desk interaction with another application record

You can associate a Service Desk interaction record with an existing request or request for change. Before you associate an interaction record with another record, you must know the ID number of the other record. For example, if the ID number of an existing quote is Q1001, you need this number to create the association.

Related topics

Service Desk overview
What's a Service Desk interaction?
Working with Service Desk interaction records

1.1.1.5. Automatic updates of Service Desk records

When a Service Desk interaction record is related to a change, incident, problem or request record, the status of the interaction record and certain activities in the activity log may be updated by the related change, incident, problem or request record.

Activity logs are copied from the related record to an interaction only when the activity is customer visible. You can identify where an activity originated by the following designations in the activity log of the Service Desk interaction:

Origins of activity logs

Originating module:	Logged as
Incident	Incident change
Service Request	Request change
Change	RFC change
Problem	Problem change

The following tables describe the when the status updates occur.

Note

The arrow indicates the direction of the change in status. During the initial instantiation of the related record, the Service Desk interaction record triggers the corresponding related change, incident, problem or request record and that record is set to the corresponding initial status. Thereafter, the direction is reversed. When you modify the related record, the Service Desk interaction is modified. For example, when you set a related incident to "Resolved," the status of the interaction also changes to "Resolved."

Service Desk/Incident Management

Interaction	Direction	Incident
Dispatched	→	Categorize (Initial status)
Callback or Closed	←	Closed
Resolved	←	Resolved

Service Desk/Request Fulfillment

Interaction	Direction	Request
Dispatched	→	Open (Initial status)
Callback or Closed	←	Closed
Resolved	←	Fulfilled

Service Desk/Problem Management

Interaction	Direction	Problem
Dispatched	→	Categorize (Initial status)
Callback or Closed	←	Closed

Service Desk/Change Management

Interaction	Direction	Change (phase based)
Dispatched	→	Registration and Categorization (initial phase)
Resolved	←	Post Implementation Review
Callback or Closed	←	Closure

1.1.1.6. Priority, impact, and urgency

Priority is how an individual service desk interaction, change request, incident, or problem fits into the ongoing sequence of tasks required to close the interaction, change request, incident, or problem. It also indicates how soon the work should begin. Determining the priority of a single service desk interaction, change request, incident, or problem depends on how many other defects need attention, the risk of delay, and the resources available to fix it.

Impact is the potential business vulnerability. There is no global value; it is subjective and each business must set and modify its own impact value list.

Urgency is a value that reflects how soon the defect must be resolved to avoid business consequences. It identifies how soon you must react to avert or reduce the impact of the defect on customers.

Assigning values to impact and urgency is subjective. Priority is a Service Manager calculation based on the values you specify for impact and urgency. As you can experience assigning impact and urgency values, you will refine your decision criteria. A service desk interaction, change request, incident, or problem that's isolated can have a low impact initially, but a high urgency because of the potential for damage if the defect becomes widespread. For example, a new computer virus is a problem that can escalate quickly.

Related topics

Change Management overview
Incident Management overview
Problem Management overview
Service Desk overview

1.1.2. Service Desk user roles

The following table describes the responsibilities of the Service Desk user roles.

Service Desk user roles

Role	Responsibilities
Service Desk User	• Report all IT-related requests to the service desk or use the self-service web pages • Validate solutions and answers that are provided by the IT department to a registered service request
Service Desk Agent	• Open interactions based on contact with the user • Match user interaction to incidents, problems, known errors, or knowledge documents • Solve and close interactions • Provide status updates to users on request • Register incidents based on user interactions and assign them to the correct support group • Register requests for change, based on user interactions • Register service requests, based on a user interactions • Validate solutions provided by support groups • Report and verify solution to users • Monitor the Service Level Agreement (SLA) targets of all registered incidents and escalate the incidents if required • Communicate service outages to all users

Related topics

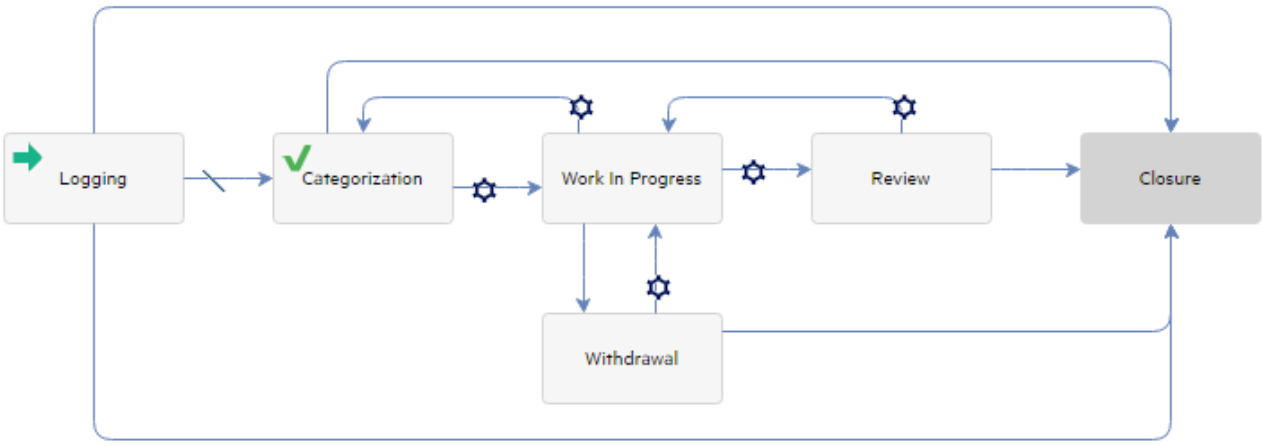
User roles
Add a user role record
Delete a user role record
Search for a user role record

1.1.3. Service Desk workflows and user tasks

Every user contact with the Service Desk is logged in an interaction record. User Interaction Management is the process for handling all interactions with the service desk that are received from self-service webpages or directly from service desk personnel. This process streamlines service desk activities, thereby decreasing the workload for second line support teams.

Service Desk includes the following out-of-box workflows:

Service Desk

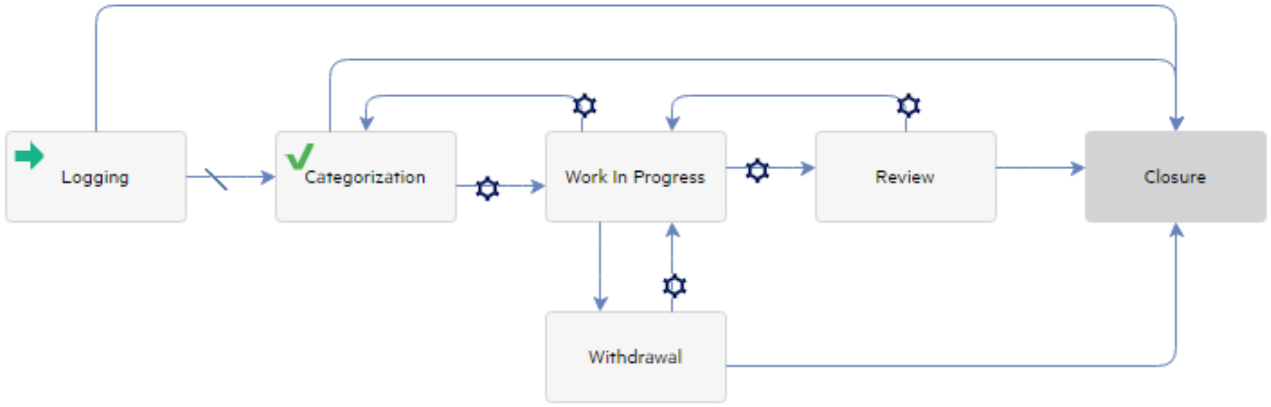


The Service Desk workflow is a sequence of connected steps in the life cycle of a support or service request. In the workflow, a Service Desk request goes through several phases to complete the life cycle.

The Service Desk workflow consists of the following phases and tasks.

Workflow phase	Tasks	User role
Logging	• Request a service catalog item or bundle • Create a self-service request • Close a self-service request • Resubmit a self-service request • Update a self-service request • View a self-service request	Service Desk User
Categorization	• Create a new Service Desk interaction • Update an interaction record • Cancel a new Service Desk interaction	Service Desk Agent
Work in progress		Service Desk Agent
Review		Service Desk Agent
Closure	• Close an existing Service Desk interaction	Service Desk Agent
Withdrawal		Service Desk Agent

Service Catalog



When a user submits a Service Catalog request through the ESS portal, a Service Desk interaction is created and follows the Service Catalog workflow.

The Service Desk workflow consists of the following phases and tasks.

Workflow phase	Tasks	User role
Logging	• Request a service catalog item or bundle • Create a self-service request • Close a self-service request • Resubmit a self-service request • Update a self-service request • View a self-service request	Service Desk User
Categorization	• Update an interaction record • Cancel a new Service Desk interaction	Service Desk Agent
Work in progress		Service Desk Agent
Review		Service Desk Agent
Closure	• Close an existing Service Desk interaction	Service Desk Agent
Withdrawal		Service Desk Agent

Related topics

Service Desk overview
User Interaction Management
Automatic updates of Service Desk records

1.1.3.1. Access Interaction Management reports

User Roles: Service Desk Agent

The Reporting tool in Service Manager provides a number of out-of-box reports on the interaction data in your system. You can view these reports through a dashboard named Service Desk Overview (Global). You can also create your own dashboards to display other reports of your interest.

To access Interaction Management reports, follow these steps:

1. Click **Service Desk** > **Service Desk Overview**.

   By default, the **Service Desk Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. If needed, click the **New Dashboard** button to add your own dashboards. For details, see Create a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default dashboard.

1.1.3.2. Request a Service Catalog item or bundle

Applies to User roles: User

You can easily order products and services from Service Catalog without needing to contact the service desk. Items in the Service Catalog can include an item picture, description, attachments, and purchasing options. Some items in the Service Catalog are for information only, such as instructions on how to order items from an outside service. Information-only items only have an item picture, description, and attachments; they don't include purchasing options.

> **Note**
>
> You may also order one or more products and related service tasks by choosing a bundle.

To request an item or bundle from the Service Catalog, follow these steps:

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.
3. Click the item you want to request to view the Order Information for that item.

   > **Note**
   >
   > Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.

4. In the Requested for field, select the Individual who will receive the good or service.
5. Select the quantity and any other item options, and then click **Add to Cart** to add the item you want to your cart.
6. From the Order from Catalog home page, click **View Cart/Checkout**.
7. Click **Submit Request**.
8. Type any required information justifying the request.
9. Click **Submit**.

Related topics

Service Catalogs
Out-of-box Service Catalog
Shopping cart
Service Catalog interface
Service Catalog items and bundles
Service Catalog connectors
Approve a request

1.1.3.3. Create a self-service request

Applies to User roles: User

If you can't find an existing solution to your incident or problem in the knowledge base, you can create a self-service request for a Service Desk Agent to resolve.

To create a self-service request, follow these steps:

1. Click **Submit a Request**.
2. In the Submit a Request window, verify the **Service Recipient** and **Primary Contact** fields.
3. Select your notification preferences in the **Notify By** field.
4. Select the **Urgency** of your issue.
5. Type a title and a description for your issue.
6. Click **Add Files**, and then follow these steps to add any relevant attachments:
   - Click **Browse** to locate your attachment.
   - Select the file to add.
   - Click **OK**.

   Note

   Service Desk agents or other operators can access your submitted attachments directly from the fulfillment records that are linked to your request.

7. Click **Submit**.

   Note

   After you submit your self-service request, Service Manager automatically creates the corresponding fulfillment record in the system so that your self-service request can be handled appropriately.

Related topics

Close a self-service request
View a self-service request
Update a self-service request
Resubmit a self-service request

1.1.3.4. Create a self-service request

Applies to User roles: User

You can close a self-service request if you are no longer experiencing the problem or if you have another reason for closing the request. If you need to resubmit the request, you can do so from your list of closed self-service requests.

To close a self-service request, follow these steps:

1. Click **View Open Requests**. Service Desk displays a table of your open requests and the status of each request.
2. Select the Service Desk interaction record that you want to close.
3. Click **Update**, type an update description if necessary, and then click **Close Request**.
4. Select a reason to close the request, and then click **Submit**. The Service Desk interaction record is closed.

Note

When you close your self-service request, an activity update is added to the corresponding back-end fulfillment records to reflect that you have closed the self-service request.

Related topics

Self-service access
Searching records
Create a self-service request
View a self-service request
Update a self-service request
Resubmit a self-service request

1.1.3.5. Resubmit a self-service request

Applies to User roles: User

If you disagree with a proposed solution to a closed request, you can resubmit the request and provide a reason why the resolution is unsatisfactory. When you resubmit a closed interaction record, Service Manager assigns you a new interaction identification number to track the request.

To resubmit a self-service request, follow these steps:

1. Click **View Closed Requests**.
2. Select the Service Desk interaction record that you want to resubmit.
3. Click **Resubmit**. Service Desk creates a new interaction record with a different identification number. However, the new record displays the same field information as the original Service Desk interaction record.
4. Type the reason why you are resubmitting the request.
5. Click **Save & Exit**.

Related topics

Self-service access
Searching records
Create a self-service request
View a self-service request
Update a self-service request
Close a self-service request

1.1.3.6. Update a self-service request

Applies to User roles: User

You can update any of your open self-service requests with additional information to assist the Service Desk Agent.

To update a self-service request, follow these steps:

1. Click **View Open Requests** to display a table of your open requests and the status of each request.
2. Select the Service Desk interaction record that you want to update.
3. Click **Update**, and then enter any new request details, additional descriptions, or add relevant attachments as needed.
4. Click **Save & Exit**.

Note

If the Service Desk interaction record is linked to a request , an incident, or a change record, your updates are automatically copied to the **Activities** section in the linked record.

Related topics

Self-service access
Searching records
Create a self-service request
View a self-service request
Resubmit a self-service request
Close a self-service request

1.1.3.7. View a self-service request

Applies to User roles: User

You can view your service requests to view the status of any outstanding interactions or to view a list of all your past requests.

To view a self-service request, follow these steps:

1. Click **View Open Requests** or **View Closed Requests** to display a table of requests and the status of each request.

   The status of a Service Desk interaction is based on the status of its fulfillment record:

   - Once the fulfillment record is created and linked to the interaction, the interaction status moves to “Dispatched” automatically.
   - Once the fulfillment record is in the “Pending Customer” status, the interaction status moves to “Pending Customer” automatically.
   - Once the fulfillment record is abandoned or closed, the interaction status moves to “Closed” automatically.
   - If the fulfillment record is an incident, when the incident moves to the "Resolved" status, the interaction status also moves to “Resolved” automatically.

2. Select the Service Desk interaction record that you want to view.

3. Click **Back** to return to the list of service requests.

Related topics

Self-service access
Searching records
Create a self-service request
Update a self-service request
Resubmit a self-service request
Close a self-service request

1.1.3.8. Create a new Service Desk interaction

User role: Service Desk Agent

You can create a Service Desk interaction to collect information about a service or support request from a user such as a complaint, an incident, a request for change, a service catalog request, and so on. Categorize the service request, and then solve or escalate the request.

To register a new Service Desk interaction:

1. Click **Service Desk** > **Create New Interaction**.

   Service Manager prompts you to choose a category, after which the new interaction form is displayed. However, if a default category is defined in **Service Desk** > **Administration** > **Settings**, you're not prompted to choose a category, and the the new interaction form is displayed with the category field pre-populated.

   Tip

   To change the category of an interaction, select **More** > **Change Category**.

2. Specify the **Contact** for the interaction.

   Note

   If you use the Fill function to replace the information in a field that already has data, clear the field before you click **Fill**. If the field contains data, the Fill function can't operate.

3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. In the **Affected Service** field, select the affected service. (Optional in Logging phase)
6. In the **Affected CI** field, select the affected CI. (Optional in Logging phase)

> **Note**
>
> The **Default Impact** and **Priority** values of the affected CI are automatically populated (if those values are populated in the CI record) to the **Impact** and **Urgency** fields of the interaction record. You can manually change these auto-populated values if needed.

7. Select Subcategory and Area. (Optional in Logging phase)
8. Type a title and description for the interaction.
9. Determine if there is a known solution or workaround for the user's issue:
    1. Search available knowledgebases.

    > **Note**
    >
    > When you use the **Search Knowledge** option from the **More** menu in the Interaction form, Service Manager uses the interaction description text as the search query.

    2. If you find an existing solution or workaround in the knowledgebases, use the solution and then close the interaction.
10. If you are still unable to solve the issue, select the Impact and Urgency of the request, and then click **Escalate**.

    Service Manager creates both an interaction and an incident (or a problem, an RFC, and so on) and associates the interaction to the new record.

    When an interaction is escalated to other ITIL processes (such as Incident, Problem, RFC, and so on), the status of interaction will be changed to "Dispatched" automatically. Additionally, the Process Targets that are created to measure the process time from an interaction's Open status to Dispatched status will stop measuring when the interaction is escalated and has a Dispatched status.

1.1.3.9. Update an interaction record

Applies to User roles: Service Desk Agent

You can update an interaction record with additional information to assist the applicable assignment group.

To update an interaction record, follow these steps:

1. Locate the Service Desk interaction record by using one of the following methods:
   - Select a record from your interaction queue.
   - Use search or advanced search to find one or more records.
2. Update any fields that require revision.

   Note

   If you are modifying a field that uses the Fill function, clear any existing data before using the function.
3. Click **Save** to update the Service Desk interaction record.

Related topics

Searching records

1.1.3.10. Cancel a new Service Desk interaction

Applies to User roles: Service Desk Agent

You can cancel a new Service Desk interaction without saving it. For example, if the user no longer experiences the problem, you can cancel the interaction.

To cancel a new Service Desk interaction, follow these steps:

1. Use search or advanced search to find one or more records.
2. While in the New Interaction form, click **Cancel**.
3. Click **Yes** to cancel the interaction without saving the changes.

Related topics

Service Desk overview
Searching records

1.1.3.11. Close a first-time resolved Service Desk interaction

Applies to User roles: Service Desk Agent

You can close a Service Desk interaction on first intake if you are able to resolve the interaction for the user in the Logging phase.

To close a first time Service Desk interaction:

1. From the System Navigator, click **Service Desk** > **Create New Interaction** or search for an existing interaction.
2. Specify the information in the mandatory fields.
3. Type a solution for the interaction.
4. Click the **Close Interaction** button in the toolbar.

   Service Manager displays the **Close Interaction** page.
5. Select a **Completion Code** for the interaction from the drop-down list.
6. (Optional) Type any **Completion Comments**.
7. Click **Finish**.

   The status of the interaction changes to **Closed**. The interaction is removed from the Interaction To Do list.

Related topics

Service Desk overview

1.1.3.12. Close an existing Service Desk interaction

Applies to User roles: Service Desk Agent

You can close an existing Service Desk interaction record if the user accepts the proposed solution. If you or the user disagree with the proposed solution, don't close the interaction. Instead, resubmit the related incident for further investigation. Depending on your company policy, you may also need to document the interaction's solution in a knowledge base for future use.

To close an existing Service Desk interaction record, follow these steps:

1. Click **Service Desk** > **Interaction Queue**. Service Manager displays the list of open interactions.
2. Locate the Service Desk interaction record that you want to close.
3. Use the contact for this interaction to determine which user to notify of the solution.
4. If the user accepts the solution, select a closure code, and then click **Close Interaction**. If the user rejects the solution, reopen the related incident or create a new incident, and then link the incident to the unsolved interaction.

Related topics

Service Desk overview
Searching records

1.1.3.13. Using mass update with Service Desk interaction record lists

The mass update feature enables you to update the value of one or more fields in a list of records. The system provides a template form that displays the current field values and enables you to type or select a new value for the displayed fields. In compliance with the data policy of your organization (as recorded in the Data Policy form), the Mass Update template form doesn't display read-only fields in the records. The mass update option is available from both interaction and incident record lists.

When you perform a mass update operation, make sure that the value you type or select for a particular field is the value you want in all the records that you selected. When you update incident records, you must also update the incident activity data.

System Administrators can determine if a field is displayed in the Mass Update template form by directly editing the Data Policy record for the form. To remove a field from the Mass Update form, change the Usage Type column for the field to System. If a System Administrator wants to grant Template Mass Update permissions to other operators, the System Administrator must update the operator profile to include the Template Mass Update and Complex Mass Update capabilities.

Update multiple interaction records

Applies to User roles: Service Desk Manager

You can use the mass update feature to update one or more values in multiple interaction records. Remember that the value you enter for a particular field becomes the value for all of the records you selected when you perform a mass update.

To update multiple records, follow these steps:

1. Locate the records by using one of the following methods:
    - Go to the Interaction Queue to display a list of records.
    - Use search or advanced search to find one or more records.
2. Click **More** or the **More Actions** icon, and then select **Mass Update**.
3. Double-click the field that you want to update.
4. Type the value for the field or use the Find feature to display a list of potential values for the field.
5. Click **Next**, and then repeat steps 3 and 4 for additional fields as needed.
6. Update the Activity Action field for the updates you made in the Mass Update template, and then click **Execute**.

To use complex update to update multiple records:

1. Locate the records by using one of the following methods:
    - Go to the Interaction Queue to display a list of records.
    - Use search or advanced search to find one or more records.
2. Select the records that you want to update.
3. Click **Mass Update**, and then click **Complex Update**.
4. Type the instructions (in RAD syntax) to be executed one time at the beginning of the mass add or mass update operation.
5. Type the instructions (in RAD syntax) for action on each record.
6. Click **Execute**.

1.1.3.14. Set a reminder for an interaction record

Applies to User roles: Service Desk Agent

You can set a reminder to notify you if an existing interaction record meets certain conditions by a specified time. For example, you may want to be reminded if an interaction is still open or is not updated by a given date.

To set a reminder, follow these steps:

1. Locate an existing interaction.
2. Click **More** or the **More Actions** menu, and then select **Set Reminder**.
3. Choose one of the following options to specify when the reminder occurs:
    - Click **Remind At** to set a reminder for a particular date and time.

      Note

      You can click **Fill** to display a calendar tool to select the date and time of the reminder.

    - Click **Remind In** to set the reminder to occur after a particular time interval. If you choose this option, you must specify the time interval in hours, minutes, and seconds (use the 00:00:00 format), and then select a shift from the list.
4. In the **Remind If** field, select from the following options any conditions that you want the interaction record to meet before a notification occurs:
    - Always (default)
    - ... has not been updated
    - ... is still open
    - ... is still assigned to me
5. In the Pop-up Message area of the form, select from the following options the type of notification that you want to receive:
    - Pop-up
    - Email
    - SM Mail (Service Manager Internal Mail)

Note

If you select the Pop-up notification method, type the reminder message format in the **Message** field. If you select the Email or SM Mail notification methods, type the title you want the notification to display in the **Title** field, and then select **Message Type**.

6. Click **OK**.

1.1.3.15. Service Catalog support

When a user (the service recipient) submits a Service Catalog request through the ESS portal, a Service Desk interaction is created in the "Service Catalog" category to help you fulfill the Service Catalog request for the service recipient.

A Service Desk agent can also submit a Service Catalog request on behalf of a user requesting the service. The request triggers Service Desk interaction.

Order from Service Catalog

Applies to User roles: Service Desk Agent

You can submit a Service Catalog request on behalf of a user requesting the service (the service recipient). The request triggers Service Desk interaction.

To register a Service Catalog request on behalf of a service recipient:

1. Click **Service Desk** > **Create New Interaction**, and then select a category if no default category is configured.

   The Interaction Details screen is displayed.

   Note

   If the interaction is initially registered in the "service catalog" category and you don't follow the steps in this section to order from catalog, a message will pop up to remind you that you can order from catalog:

   The shopping cart is empty now. You may order enterprise products and services from service catalog via "Order from Catalog" in the more options list.

2. Click the cursor into the **Service Recipient** field.
3. Click the **Fill Field** button immediately to the right of the **Service Recipient** field.

   The Search Contact Information screen is displayed.

4. Click the cursor into the **Contact Name** field and click **Search**.
5. Click the name of the person whose request you will fill.

   That person's name appears as the service recipient.

6. Using the **More** drop-down list, select **Order From Catalog**.

> **Note**
>
> The **Order From Catalog** option is only available for interactions in "Logging" or "Categorization" phase. For interactions in other phases that need order from catalog, if the interaction is not escalated and is not a service catalog request or support request, you can change the interaction category to "service catalog" or change the interaction status to "Categorize", then the interaction will move to the "Categorization" phase, where the **Order From Catalog** option is available.

7. Add items to your cart by doing one of the following:
   - Search for the desired cart items:
     The items for which you search are filtered against the service recipient's capability words. Items that match the search criteria don't display in search results unless the service recipient has the required capability words.
   - Select from **Most Popular Requests**:
     The Most Popular Requests list displays only the cart items that are allowed by the service recipient's capability words.
   - Browse the service catalog:
     The service catalog displays only the categories, items, and bundles that are allowed by the service recipient's capability words.
8. After you add the desired items to your cart, click **View Cart**.
9. If needed, click **Remove** to remove an item or click **View Catalog** to add more items to your cart, and then click **View Cart** again.

> **Note**
>
> Be sure to click **View Cart** each time before you click **Return to Request** in the next step; otherwise Service Manager will ask if you wish to cancel. If you select **Yes**, your requested items will be canceled.

10. Click **Return to Request**.

11. Fill in the blank fields on the Interaction Details screen. Fields with asterisks require an entry.
12. Click **Save**.

    The catalog items you requested appear in the Catalog Items section of the interaction record.

1.1.3.16. Callback mechanism

In the Full Service Desk Model, if you set the **Notify By** field to "Telephone" in an interaction and link the interaction to an incident, the interaction will automatically move to the **Review** phase and the **Callback** status once the related incident is closed and a Required Action record is associated with the interaction to track the customer callback activity prior to interaction closure.

When the interaction is in the **Callback** status, the system will not close the interaction automatically. To close the interaction, go to the **Required Action** tab, click for action details, and select "The customer agrees that the incident is fixed" after you call and confirm with the customer.

Tip

To enable the Full Service Desk Model, go to **Service Desk** > **Administration** > **Environment**, and then select **Full Service Desk Model** in the **Service Desk Record Relationship Models** section.

1.1.3.17. Example: Service Desk Workflow

The following example demonstrates the minimum needed steps to resolve the out-of-box Service Desk workflow:

Phase	User Actions	Options
Logging	1. Specify the **Contact** for the interaction. 2. In the **Service Recipient** field, select the service recipient. 3. In the **Notify By** field, select the user's preferred notification method. 4. In the **Affected Service** field, select the affected service. 5. (Optional) In the **Affected CI** field, select the affected CI. 6. Enter a title in the **Title** field. 7. Provide a description of the interaction in the **Description** field. 8. Set the **Impact** and **Urgency** fields. 9. Click **Save** to save the interaction. Or, depending on the category, you may move on to close the interaction or escalate the interaction further.	This is the phase that's initiated when an end user register a new interaction (**Service Desk** > **Create New Interaction**). From the Logging phase, you can also move to any of the following: • Closure

Phase	User Actions	Options
Categorization	1. Click the **Fill** button to specify the **Subcategory**. 2. Click the **Fill** button to specify the **Area**. 3. Set the **Status** field to **Assign**. 4. Click the **Fill** button for **Assignment Group**, and then select an appropriate assignment group. 5. Click the **Fill** button for **Assignee**, and then select an appropriate assignee. 6. Set the **Status** field to **Dispatched** or **In Progress**. 7. Click **Save**.	From the Categorization phase, you can also move to any of the following: • Move to the Work In Progress phase. • Jump directly to the Closure phase.
Work In Progress	1. Enter the **Solution**. 2. Set the **Status** field to **Resolved**. 3. Click **Save**.	From the Work In Progress phase, you can also move to any of the following: • Move to the Withdrawal phase. • Return to the Categorization phase.

Phase	User Actions	Options
Review	1. Review the data entered for the interaction.	From the Review phase, you can also move to any of the following: • Return to the Work In Progress phase.
Withdrawal	1. From the Work In Progress phase, click the **Withdraw** button on the toolbar. Service Manger displays the **Withdraw Interaction** page. 2. Enter the **Reason for Withdrawal** and then click **Finish**. The interaction moves to the Withdrawal phase and the status changes to Withdraw Requested.	From the Withdrawal phase, you can also move to any of the following: • Return to the Work In Progress phase.

Phase	User Actions	Options
Closure	1. Click the **Close Interaction** button. 2. Select the appropriate **Completion Code**. 3. (Optional) Enter the **Completion Comments**. 4. Click **Finish**.	

1.1.4. Service Desk administration

Service Desk administration consists of the following:

- Service Desk Environment: Service Desk application contains an environment record for Interactions. This record contains options that defines the functionality of the Service Desk application for all Service Desk users.
- Settings: You can configure whether escalation is allowed for Interactions, and you can set the default category for new Interactions.

Configure the Service Desk settings

Applies to User roles: System Administrator

To configure Service Desk settings:

1. Click **Service Desk** > **Administration** > **Settings** in the System Navigator.
2. Configure the following settings for the Service Desk application:

Setting	Description
Allow Escalation	Whether or not you can escalate the Interactions. In the out-of-box setting, escalation is allowed. **Note** You can only escalate an Interaction when the **Allow Escalation** option is selected both in the **Settings** menu here and in the category definition for the specific category assigned to the Interaction.
Enable "Create New Interaction" Menu Entry	This setting determines whether the Interactions will be created to follow the legacy Interaction workflows (select the checkbox) or the new streamlined workflows introduced as of SM 9.41 (clear the checkbox). For more information, see Enable streamlined Interaction.

Setting	Description
Default Category	The default category when you create a new Interaction. The out-of-box value for this parameter is None. The default category is used when you click **Create New Interaction**: If no default category is defined, you are prompted to select a category before the Interaction detail screen appears. If the default category is defined, the Interaction detail screen is displayed directly and the **Category** field is populated with the default category. **Note** The default category doesn't apply to the ESS requests. For support requests submitted through ESS, Interactions are always created in the "incident" category.

Setting	Description
Email Notification Template for the "Visible to Customer" Option	The default email notification template when you enter new updates for a record (such as a Request) that is related to an Interaction with the **Visible to Customer** checkbox selected in this record. The out-of-box value for this field is **RM Visible to Customer notification**. After you select the **Visible to Customer** checkbox on the **Activities** tab in a record that is related to an Interaction and then enter new updates, Service Manager sends an email notification to the Service Recipient of the Interaction using the specified template. If no default template is defined, Service Manager no longer sends any email notifications when you enter new updates for a record (such as a Request) that is related to an Interaction with the **Visible to Customer** checkbox selected in this record.
Chat Assignment End Lead Time (hh:mm:ss)	

3. Click **Save**.

Configure the Service Desk environment

Applies to User roles: System Administrator

To configure a Service Desk environment record:

1. Click **Service Desk** > **Administration** > **Environment**.

2. Configure the settings for your Service Desk environment.

   You can specify the Service Desk Record Relationship Model as follows:

   - **Full Service Desk Model**: In this model, the state of a Service Desk Interaction record changes when each related record closes. The closed state of an Interaction record depends on the notification value chosen for the "Notify By" field in the Interaction record. For example, if the notification value is "Telephone," the Interaction record has a required action before it is closed. This action describes why the customer must be contacted. It also prevents the Interaction record from closing until all required actions are complete. In this case, the Interaction record goes into the Callback state before it is closed. For more information, see Full Service Desk model.
   - **All Records close Independently**: In this model, all Service Desk Interaction records close independently. The state of related records doesn't affect closing an Interaction record, and closing the Interaction record doesn't affect related records.
   - **Close Interactions when Related Record closes**: In this model, when the last related record closes, the Service Desk Interaction record closes.
   - **Cannot close Related Record until Interactions are closed**: In this model, records related to a Service Desk Interaction can't close until the Interaction record is closed.
   - **Cannot close Interactions until Related Records are closed**: In this model, a Service Desk Interaction record can't close until all related Incident Management records, Change Management records, and Request Management records are closed.

   You can use the following table for more information on these fields:

Check box	Description
Delay Assigning Interaction ID?	If selected (set to true), prevents Service Manager from assigning an identifier to the Interaction immediately. When this option is active, there is no unique identifier to tie an attachment within the file when you first open it. Therefore, you can't save attachments when you open the ticket. You can add attachments when saving an update after the record has a unique identifier. **Note** This option doesn't work under the following circumstances: ◦ When the system automatically creates an Interaction from a chat request in background. ◦ When a Service Desk agent manually creates an additional Interaction for a chat request.
User Operator Full Name?	If selected (set to true), uses the operator's Full Name instead of the login ID when time stamping reports (on open, update, and so on) for records in the application environment

Check box	Description
Return to Blank Interaction?	If selected (set to true), returns the user to a blank (new) Interaction screen after the creation of an Incident record. If cleared (set to false), the user will return to the Interaction they created the Incident from.
Post resolution to Interactions?	If selected (set to true), posts the resolution of a closed ticket to the related Interaction report.
Copy Associations of Cloned Interactions?	Copies the associations of a cloned Interaction.

You can use the following table to specify information about the Post Back links:

Post Back Link	Description
IM Post back Link:	The link record used to post information from a related Incident to the Interaction when the Incident is closed.
ChM Post back Link:	The link record used to post information from a related Incident to the Interaction when the Change is closed
RF Post back Link:	The link record used to post information from a related Incident to the Interaction when the request is closed.
PbMPost back Link:	The link record used to post information from a related Incident to the Interaction when the Problem is closed.
Copy/Open Link:	The record from which to copy the data. When you copy a record, data from the record specified here is copied to the new record based on the link record.

The following four fields specif information about the Service Desk Manager, Group, Phone Number and Email. These fields are used in notifications and for assigning the Interaction to a default group.

Tip

You can use the field help to view the description for each setting. To view the help on field:

- Web client: Select a field, and then press **F1**.
- Windows client: Select a field, and then press **Ctrl+H**.

3. Click **Save**.

Add a Service Desk assignment group

Applies to User Roles: System Administrator

Service Desk assignment groups specify the Service Desk agents in the group so that when someone creates a new Interaction it can be assigned to the applicable group. Typically, assignment groups are organized by location and expertise. To facilitate Interaction response, new assignment groups can be added to the out-of-box assignment groups for Service Desk.

To add a Service Desk assignment group:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.
   1. Enter the name of the manager of the assignment group.
   2. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   3. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide response when you create and populate a new assignment group.

   (Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.
9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.
10. Follow these steps to configure the Supported Services field:

    1. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    2. Select **Business Service** in the CI Type field, and then click **Search**.
    3. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    4. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.
11. Click **Save** and **OK**.

1.1.5. Service Desk configuration

Service Desk configuration enables you to configure alert, interaction categories, workflow, solution matching, and so on.

Create an interaction category

Applies to User roles: System Administrator

If you are a Service Manager Administrator, you may want to create new interaction categories. You may want to create a category when you require a different workflow that specifies unique forms and business rules.

> **Note**
>
> If you do create a new category, you must also add corresponding subcategories and areas.

Service Manager provides a number of out-of-box category records that you can use or modify. For more information, see Out-of-box Service Desk categories.

To create a new category record:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New**.
4. Type the name of the category.
5. Specify the **Apply To** option to indicate the modules that can share this category.

Note

The **Apply To** flag indicates whether the category is shared across different modules. For example, if the **Apply To** is set to "Interaction/Incident", this category will be created in both the interaction category table and the incident category table. Typical shared categories in the out-of-box system include:

- complaint
- incident
- problem
- request for information
- request for administration

For an Interaction Category, the following rules apply:

- Once a value is specified in the **Apply To** field and the record is saved, the value can't be modified.
- If **Apply To** is set to "Interaction" only, the **Allow Escalation** flag becomes read-only.

  The following table describes the Allow Escalation settings for the out-of-box categories that only apply to Interaction.

Category	Allow Escalation	Notes
▪ compliment ▪ service catalog	Disabled (Read-only)	This is because "Apply To Interaction" means the category is not shared among other modules, and interactions of this category don't need to trigger a record in another module. A compliment can be closed immediately or worked through the phases as a normal interaction. A service catalog interaction can be resolved by ordering items from the catalog.
▪ complaint ▪ incident ▪ problem ▪ request for administration ▪ request for change ▪ Request for information	Enabled (Editable)	Although the category is not shared among other modules, an interaction of one of these categories should be allowed to trigger a change request or a service request.

The **Apply To** options that are used in Interaction Categories include:

- Interaction
- Interaction/Incident
- Interaction/Incident/Problem
- Interaction/Problem

6. Specify whether the category is active and whether escalation is allowed.

> **Note**
>
> If the **Active** checkbox is unchecked, you can't use this category when creating a new record.

7. Type the category description.
8. Select a workflow for the category.
9. Click **Save** to view the **Workflow** tab and other associated tabs.
10. Click **Subcategories** tab to create subcategories for the interaction category.
11. Click **Service Desk Approval** tab to configure approval settings.

> **Note**
>
> The settings in the **Service Desk Approval** tab are only for service catalog approval. That is, in the out-of-box system, they are only valid for the service catalog category.

12. Click **Save**.

> **Note**
>
> The Interaction category name is read-only after the category is created.

Add a new interaction subcategory

Applies to User Roles: System Administrator

Subcategories are a way to refine a Service Desk category into smaller groups. For example, the complaint category can have service delivery and support subcategories.

To add a subcategory:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New** to create a new category or click **Search** to select an existing category.
4. In the **Subcategories** section, click the **Link New Subcategories** button.
5. Type a subcategory name.
6. Type a description for the subcategory.
7. Click **Save** to add the new subcategory.

Add a new interaction area

Applies to User roles: System Administrator

Areas are a way to further refine a Service Desk subcategory. For example, the support subcategory in the complaint category can have availability, functionality, and performance areas.

To add an area:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New** to create a new category or click **Search** to select an existing category.
4. In the **Subcategories** section, click the **Link New Subcategories** button to create a new subcategory or select an existing subcategory from the list.
5. In the **Areas** section, click the **Link New Areas** button.
6. Type a name for the area.
7. Type a description for the area.
8. Click **Save** to add the new area.

Service Desk solution matching

When you escalate an interaction record to an incident, a list of incidents or problems appears based on the pre-defined solution matching configuration. You can then select the matching records and link them to the interaction. Alternatively, you can specify the ID of an existing incident or problem, and then link it to the interaction. The linked records are shown in the **Related Records** section of the interaction.

Note

When you escalate from interaction to incident, instead of selecting the assignment group through a wizard, the assignment group of the escalated incident is populated according to the following rules:

- If an interaction is escalated to an incident of the incident category, the **Config admin group** field defined in the affected service or affected CI of the interaction is used.
- If an interaction is escalated to an incident of other categories, the Service Desk group in the Service Desk environment is used.

You can also configure an auto-assignment rule depending on your organization's requirements.

In an out-of-box system, the following configurations are provided for Service Desk interaction solution matching:

- Find an incident with a matching configuration item
- Find an incident with a matching service
- Find an incident with a matching title
- Find a problem with a matching configuration item
- Find a problem with a matching service
- Find a problem with a matching title

To modify the out-of-box solution matching configurations, navigate to **Service Desk** > **Configuration** > **Solution Matching**.

1.1.6. Security

The topics in this section explain the Service Desk security roles, security areas, and rights.

Setting rights to interactions

Applies to User roles: Service Desk Manager

The Service Desk Manager can set rights to interactions using the role-based security. The Service Desk and Service Desk Configuration areas contain default security rights and settings. The rights are copied to new roles created for these areas. However, the settings are inherited only if there are no settings specified on the roles.

Note

Whenever the roles in an operator record are updated, the operator must log out and then log in for the changes to take effect.

Service Desk security areas

Applies to User roles: System Administrator

The security areas for Service Desk are Service Desk and Service Desk Configuration. These areas contain the default security rights and settings for Interactions in the Service Desk module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

These security areas are used to set permissions to operators to provide access to particular area of Service Desk. For example, for the Service Desk Analyst security role, set rights as **View** for the Service Desk area. Then, an operator with this security role can only view the Service Desk items under Service Desk module. The following table lists the areas and the relevant Service Desk menu items the operators can access.

Area	System Navigator menu items for this area
Service Desk	This area contains the default security rights and settings for Service Desk requests in the Service Desk module. For example, Create New Interaction, Interaction Queue, and Search Interactions.

Area	System Navigator menu items for this area
Service Desk Configuration	This area contains the default security rights and settings for Service Desk administration and configuration, which are dedicatedly used by Service Desk module. For example, Settings and Interaction Categories. **Note** When you set the security rights for a security role in the Service Desk Configuration area: • The **View** right is to view the settings defined in the **Administration** menu and the **Configuration**menu. • The **Update** right is to update the values of existing settings defined in the **Administration** menu and the **Configuration** menu. • The **New** and **Delete** rights are to create and delete a setting in the **Configuration** menu, such as category. • The **Admin** right is to add, edit, or delete the settings in the **Administration** > **Settings** menu.

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Service Desk and Service Desk Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Service Desk	TRUE	FALSE	Never	Never	FALSE	FALSE
Service Desk Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Service Desk and Service Desk Configuration areas.

Service Desk security roles and settings

The out-of-box security roles for the Service Desk module include the following:

- service tech
- service desk agent
- service desk analyst
- service desk manager
- self service
- approver
- service desk process owner

Tip

If you have upgraded from a previous version of Service Desk, which is not Process Designer based, see Service Desk security profiles mappings for more information about the mappings between the legacy Service Desk profiles and the current Process Designer security roles.

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Service Desk area specified in the corresponding security roles. See the table below for the out-of-box security rights of the new security roles in the Service Desk area and the Service Desk Configuration area. This table only lists the new security roles that have different settings with the default rights.

Service Desk area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
service desk analyst	TRUE	TRUE	Always	Always	FALSE	TRUE	FALSE
service desk manager	TRUE	TRUE	Always	Always	TRUE	TRUE	FALSE
service desk process owner	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

Service Desk Configuration area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
service desk manager	TRUE	TRUE	Always	Always	FALSE	FALSE	FALSE
service desk process owner	TRUE	FALSE	Never	Never	FALSE	FALSE	FALSE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

For information about the out-of-box role rights in the Common Configuration area, see Out-of-box role rights in the Common Configuration area.

1.2. Service Desk (Hybrid)

You can use Service Manager Service Desk to restore normal service to your users as quickly as possible by resolving interactions on first contact or by immediately escalating interactions for proper handling. Service Desk helps you manage all user interactions, whether they are reported by using Self-Service website or directly to your service desk.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Service Desk User

Service Desk Agent

Key concepts

Describes the most important things that you need to know about how the Service Desk module functions.

Service Desk overview

What's a Service Desk interaction?

Service Desk out-of-box categories

Related records in Service Desk

Working with Service Desk interaction records

User workflows

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users need to perform to execute the workflows.

Self-Service user interactions (SO0.1)

User interaction handling (SO0.2)

Interaction closure (SO0.4)

1.2.1. Service Desk overview

Service Manager Service Desk enables you to restore normal service to your users as quickly as possible by resolving user requests on first contact or by immediately escalating them for proper handling. Service Desk helps you manage all user interactions, whether reported by using a Self-Service website or directly to your service desk. Service Desk enables you to achieve the following results:

- Control the relationship between Service Desk and other Service Manager applications
- Require Service Desk interactions to follow a set process
- Log and prioritize all calls to the service desk
- Categorize types of interactions and keep track of their resolution
- Provide first-line investigation and problem diagnosis, and quickly escalate incidents and service requests that can't be resolved within agreed-on time limits
- Request a service, provide information, or track previous requests without the assistance of a Service Desk Agent
- Increase accessibility through a single point of contact and information
- Communicate with customers and users to keep them informed of progress, impending changes, agreed-on outages, and other notifications.

Related topics

User Interaction Management

1.2.1.1. User Interaction Management

The Service Manager Service Desk application enables a Service Desk Agent to document and track calls. Service Desk provides one-click access to other Service Manager applications to automatically enter information received during the call.

Information that's obtained during a call can be used to open an incident record if a reported issue requires further action. Information can also be added to a record in another Service Manager application, such as Change Management.

User Interaction Management provides users with a single point of contact (the Service Desk) for all their IT Service related questions, requests, and issues, and provides resolutions to user requests within the agreed service level targets.

User Interaction Management includes all direct interactions between a user and the Service Desk by phone or by email. User Interaction Management also includes all user activities that occur by use of the self-service web portal (for example, searching the Knowledge base, checking for status updates, or logging an interaction).

Related topics

Service Desk overview
What's a Service Desk interaction?
Working with Service Desk interaction records
Reopen an incident

1.2.1.2. What's a Service Desk interaction?

Each time a user contacts the Service Desk, the Service Desk Agent uses Service Manager to create an interaction record. The Service Desk Agent records the user name, the name of the component about which the user is calling, and a description of the user request. After the Service Desk Agent collects this information, the Service Desk Agent performs the actions that are required to resolve the user request.

- If the service request can be resolved without escalating it to an incident, the Service Desk Agent can close the interaction record.
- If the service request can't be resolved without escalating it to an incident, the Service Desk Agent searches for existing incidents that affect the same component or one of the parent assets of that component.
    - If an existing incident is found, the Service Desk Agent can associate the current interaction with the existing incident record.
    - If an existing incident record is not found, the Service Desk Agent can register a new incident based on the Service Desk interaction. Service Manager copies information from the interaction record into the newly created incident record.

For example, if a user can't print to a network printer, the user contacts the Service Desk for assistance. The Service Desk Agent populates an interaction record with the relevant information. Because the issue can't be resolved immediately, the Service Desk Agent opens an incident, and the incident is assigned to a technician. The technician discovers that the printer network connection is broken. The technician fixes the connection and closes the incident. The Service Desk Agent contacts the user and instructs the user to attempt printing to the network printer. If the user can successfully print, the Service Desk Agent can close the interaction. If the user still can't print, the Service Desk Agent may reopen the related incident record.

Service Desk interaction list

The Service Desk interaction list provides general information about interaction records.

Column name	Description
Status	The status of the interaction record.
Description	A brief description of the reported interaction.
Interaction ID	The interaction record number.
Impact	The level of impact this issue has on the company and its end users. The impact can be classified as one of the following values: • 1-Enterprise • 2-Site/Department • 3-Multiple Users • 4-User

Column name	Description
Urgency	The importance of the service request to the recipient. The urgency can be classified as one of the following values. The default value is 4-Low. • 1-Critical (Severe business disruption) The business unit or subunit is unable to operate; a critical system component has failed or is severely impaired. • 2-High (Major business disruption) A critical user or user group is unable to operate, or the business unit experiences a significant reduction in system performance. • 3-Average (Minor business disruption) A single user is unable to operate with no available workaround. • 4-Low (Minor disruption) A single user or user group is experiencing incidents, but a workaround is available.
Priority	The priority value is calculated by using the initial impact and urgency values.

What's a category?

Service Manager uses the category to which a record is assigned to determine what work needs to be performed in order to close an interaction or incident record. For

example, an interaction record that describes an email incident requires different work than an interaction record regarding a search for equipment for a new employee.

The Service Desk and Incident Management applications each provide the same list of categories from which to choose. When an interaction is escalated to an incident, the category in the new incident record is the same as the category assigned to the associated interaction record. The category you assign to a record determines the following information:

- Who is responsible for resolving a call report or incident record (the default assignment group)
- Which information you need to open the interaction record or incident record
- Which solutions to search for in the Knowledge base
- The applicable Service Level Agreements
- The default severity of the record
- The default priority of the record
- How quickly a record must be resolved
- The time interval for escalating a record to a higher alert stage
- Who must be notified as the record is escalated

The processing logic for each category is the same. However, Service Manager may use different record forms for each category. The sections displayed on the forms depend on the assigned category because each category has different information requirements. For example, the information that's required to fix a software error is different from the information that's required to solve an equipment failure.

Out-of-box Service Desk categories

The following table lists the out-of-box Service Desk categories. Depending on the category that you select, Service Manager may require you to further categorize the interaction record by selecting an Area and Subarea.

Category	Use
Complaint	Interactions in which the user reports less than satisfactory service or performance. This is a Hybrid mode category, that's created by using your existing Classic mode category when you upgrade to Service Manager Hybrid.
Complaint codeless	Interactions in which the user reports less than satisfactory service or performance.
Compliment	Interactions that report exemplary service or performance.
Incident	Interactions that are opened specifically to start an incident management process. This is a Hybrid mode category, that's created by using your existing Classic mode category when you upgrade to Service Manager Hybrid.
Incident codeless	Interactions that are opened specifically to start an incident management process.
Problem Codeless	Interactions that are opened specifically to start the problem management process.
Request for Administration	Interactions that are opened specifically to request assistance from a system administrator, such as a password resets or rights to access a specific area.

Category	Use
Request for Change	Interactions that are opened specifically to start a change process. This is a Hybrid mode category, that's created by using your existing Classic mode category when you upgrade to Service Manager Hybrid.
Request for Change codeless	Interactions that are opened specifically to start a change process.
Request for Information	Call or incidents that request information only. This is a Hybrid mode category, that's created by using your existing Classic mode category when you upgrade to Service Manager Hybrid.
Request for Information codeless	Call or incidents that request information only.
Service catalog codeless	Interactions that are requests for items from the service catalog.
Service request	Interactions that are opened specifically to start the request management process.

Categories can be created and revised by the System Administrator. Therefore, additional categories may be available to you.

Notify By field

The Notify By field specifies how to contact the person or customer who initiated an interaction. If the interaction record is associated with another record, the information in the Notify By field affects how the interaction record is closed. For example, if you associate an incident record to an interaction record, the related interaction record closes according to the information in the Notify By field when the incident record is closed.

Notify By setting	Description
None	The interaction record closes.
Email	Service Manager sends an email to the contact who is listed in the interaction record to inform the contact that the related record is closed. Service Manager then closes the interaction record.
Telephone	Notification by telephone adds a required action to the interaction record. The required action explains why the customer needs to be called back. This setting also prevents the interaction record from being closed until all required actions are complete.

Note

There is no dependency between the Notify By field and the Notification Engine. You must configure any notifications to the person in the Notify By field manually.

Affected CI field in Service Desk

The information that's available to populate the Affected CI field depends on the service recipient.

For example, if the recipient has individual or departmental subscriptions, a list of CIs associated with those subscriptions opens. If the recipient doesn't have any subscriptions and is a self-service user, no list is generated. However, if a recipient doesn't have any subscriptions and is not a self-service user, the following behavior occurs:

- If an interaction recipient is specified, a list of the CIs assigned to that recipient opens.
- If a location is specified, a list of the CIs assigned to that location opens.
- If a company is specified in a multi-company system, a list of the CIs assigned to that company opens.

- If a device type is specified, a list of the CIs of that type opens.

Note

The only way to obtain a list of all CIs is to click the Affected CI field before you enter any recipient information.

Required actions

The Required Actions function enables you to track the steps that are necessary to resolve an interaction record. You can view any required actions for an interaction record by opening the Required Actions section of the interaction form. The Required Actions section is available only for interaction records that have required actions. You can add a required action to existing interaction records that have related incident records only. You can't close an interaction record that's associated to incomplete required actions.

For example, if you use the Full Help Desk model to close an interaction and the Notify By field is set to "telephone" a required action is automatically created and must be closed before the interaction can be closed. Therefore, the Service Desk Agent is required to call the customer and verify the issue is fixed, close the required action, and then close the interaction.

1.2.1.3. Working with Service Desk interaction records

The Service Desk functions that are available to you depend on your user profile. You may be able to access all Service Desk functions, or you may be limited by the capability words that are assigned to you in your operator record.

Depending on your user profile, the following Service Desk functions may be available to you.

Function	Description
Create New Interaction	From a blank Service Desk interaction form, you can log information about a service request, check for related interaction records, and add service request information to other Service Manager records, such as incident records. From this form, you can escalate the request to an incident record.
Interaction Queue	View the Service Desk interaction queue. You can also view other queues, such as the incident queue, problem queue, Change Management change and task lists, and quotes and order lists for all users.
Search Interactions	Specify filtering information to locate a specific Service Desk interaction record.
Security Files	Add new users, edit the profile of an existing user, maintain macros, views, and probable cause records. Requires administrative rights.

Hover-over forms for interaction records

Service Desk interaction record forms include hover-over forms. A hover-over form is displayed when you move the mouse over a field that supports hover-over forms. Hover-over forms are only displayed when the field contains data. The data that's displayed in hover-over forms is read-only.

Service Desk interaction record forms contain the following hover-over fields.

Field	Subform fields
Contact	Full Name, Telephone, Email
Service Recipient	Full Name, Telephone, Email
Affected CI	Critical CI, Pending Change

Related records in Service Desk

Service Manager lists all related records for an interaction record in the Related Records section. All information is read-only, but you can click a record ID to view the related record. The related records are listed in alphabetic order by the record file name: Change (cm3r), Interaction (incident), Known Error (knownerror), Request (ocmq), Incident (probsummary). Within each module, the records are sorted by ID number, in ascending order.

The Service Desk Related Records section contains the following fields.

Field	Description
ID	The related record number in the database.
Type	Identifies the related record as related to a change, interaction, known error, request, or incident.

Activities section

The Activities section allows users to enter new updates for a record or to view journal updates and historic activities for a record.

Journal Updates

Field	Description
New Update Type	Specifies or categorizes the activity update (for example, communication with the customer)
Visible to Customer?	Makes the update visible to customers so they can view related interactions via self service **Note** This option can be bypassed by setting the **Bypass Visibility Option** flag to “true” in the activityactions table. When this flag is set to “true,” automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to “false,” all activities are visible to customers if the **Customer Visible** option is selected.

Field	Description
New Update	Used to enter notes to explain and describe updates made for the record. If Journaling is enabled, the text entered here is displayed in Journal Updates. If Activities are enabled, the text entered in this field is displayed as an activity record for the selected activity type. **Note** The System Administrator is responsible for enabling Journaling and Activities.
Journal Updates	Displays text entered in the **New Update** field together with the operator's name and a timestamp for the update. This field displays information when Journaling is enabled.

Activity Logs

Select an activity type from the **Activity Type** drop-down list or select a user from the **User** drop-down list to filter the activity logs.

The activity log lists activities for the current record in order of occurrence, with the newest at the beginning of the list. Each activity log displays the following information:

- Date/Time
- Operator
- Description
- Type
- Visible to Customer icon (if you have selected the **Visible to Customer** checkbox)

Service Catalog requests

You can track your Service Catalog requests on the Service Desk update interaction form. When you submit a request by using the Service Catalog, you can view the request record in Service Desk. An interaction record for a Service Catalog request includes two additional sections, Approvals and Catalog Items.

Approvals section

The Approvals section contains the following subsections.

Subsections	Description
Current Approvals	Displays a table that lists current catalog requests and the approval details
Approval Log	Displays a log of the approval actions which have been performed regarding the catalog requests

Catalog Items section

The Catalog Items section contains the list of items for the catalog request and the details of the each item in the request. This section also includes the **Edit Cart** button, which links you to the Service Catalog so that you can change your catalog request.

Related topics

Service Desk overview
What's a Service Desk interaction?
Associating a Service Desk interaction with another application record
Reopen an incident

1.2.1.4. Associating a service desk interaction with another application record

You can associate a Service Desk interaction record with an existing request or request for change. Before you associate an interaction record with another record, you must know the ID number of the other record. For example, if the ID number of an existing quote is Q1001, you need this number to create the association.

Related topics

Service Desk overview
What's a Service Desk interaction?
Working with Service Desk interaction records

Reopen an incident

1.2.1.5. Priority, impact, and urgency

Priority is how an individual service desk interaction, change request, incident, or problem fits into the ongoing sequence of tasks required to close the interaction, change request, incident, or problem. It also indicates how soon the work should begin. Determining the priority of a single service desk interaction, change request, incident, or problem depends on how many other defects need attention, the risk of delay, and the resources available to fix it.

Impact is the potential business vulnerability. There is no global value; it is subjective and each business must set and modify its own impact value list.

Urgency is a value that reflects how soon the defect must be resolved to avoid business consequences. It identifies how soon you must react to avert or reduce the impact of the defect on customers.

Assigning values to impact and urgency is subjective. Priority is a Service Manager calculation based on the values you specify for impact and urgency. As you can experience assigning impact and urgency values, you will refine your decision criteria. A service desk interaction, change request, incident, or problem that's isolated can have a low impact initially, but a high urgency because of the potential for damage if the defect becomes widespread. For example, a new computer virus is a problem that can escalate quickly.

Related topics

Change Management overview
Incident Management overview
Service Desk overview

1.2.2. Service Desk user roles

The Service Desk module has the following user roles:

Service Desk User

The Service Desk User user role has the following responsibilities:

- Report all IT-related requests to the service desk or use the self-service web pages
- Validate solutions and answers that are provided by the IT department to a registered service request

To execute this role, the following tasks are available to the Service Desk User:

Note

For an overview of how these tasks fit in the incident management workflow, see Service Desk workflows and user tasks

ITIL Workflows	User Tasks
Self-Service user interactions (SO0.1)	• Request a service catalog item or bundle • Create a self-service request • Close a self-service request • Resubmit a self-service request • Update a self-service request • View a self-service request
User interaction handling (SO0.2)	• None
Interaction closure (SO0.4)	• None

Service Desk Agent

The Service Desk Agent user role has the following responsibilities:

- Open interactions based on contact with the user
- Match user interaction to incidents, problems, known errors, or knowledge documents
- Solve and close interactions
- Provide status updates to users on request
- Register incidents based on user interactions and assign them to the correct support group
- Register requests for change, based on user interactions
- Register service requests, based on a user interactions
- Validate solutions provided by support groups
- Report and verify solution to users
- Monitor the Service Level Agreement (SLA) targets of all registered incidents and escalate the incidents if required
- Communicate service outages to all users

To execute this role, the following tasks are available to the Service Desk Agent:

Note

For an overview of how these tasks fit in the incident management workflow, see Service Desk workflows and user tasks

ITIL Workflows	User Tasks
Self-Service user interactions (SO0.1)	• None
User interaction handling (SO0.2)	• Register a complaint • Register a request for change • Register a service catalog request • Register an incident • Register a request for information • Update an interaction record • Cancel a new Service Desk interaction • Reopen an incident
Interaction closure (SO0.4)	• Close a first time Service Desk interaction • Close an existing Service Desk interaction

Related topics

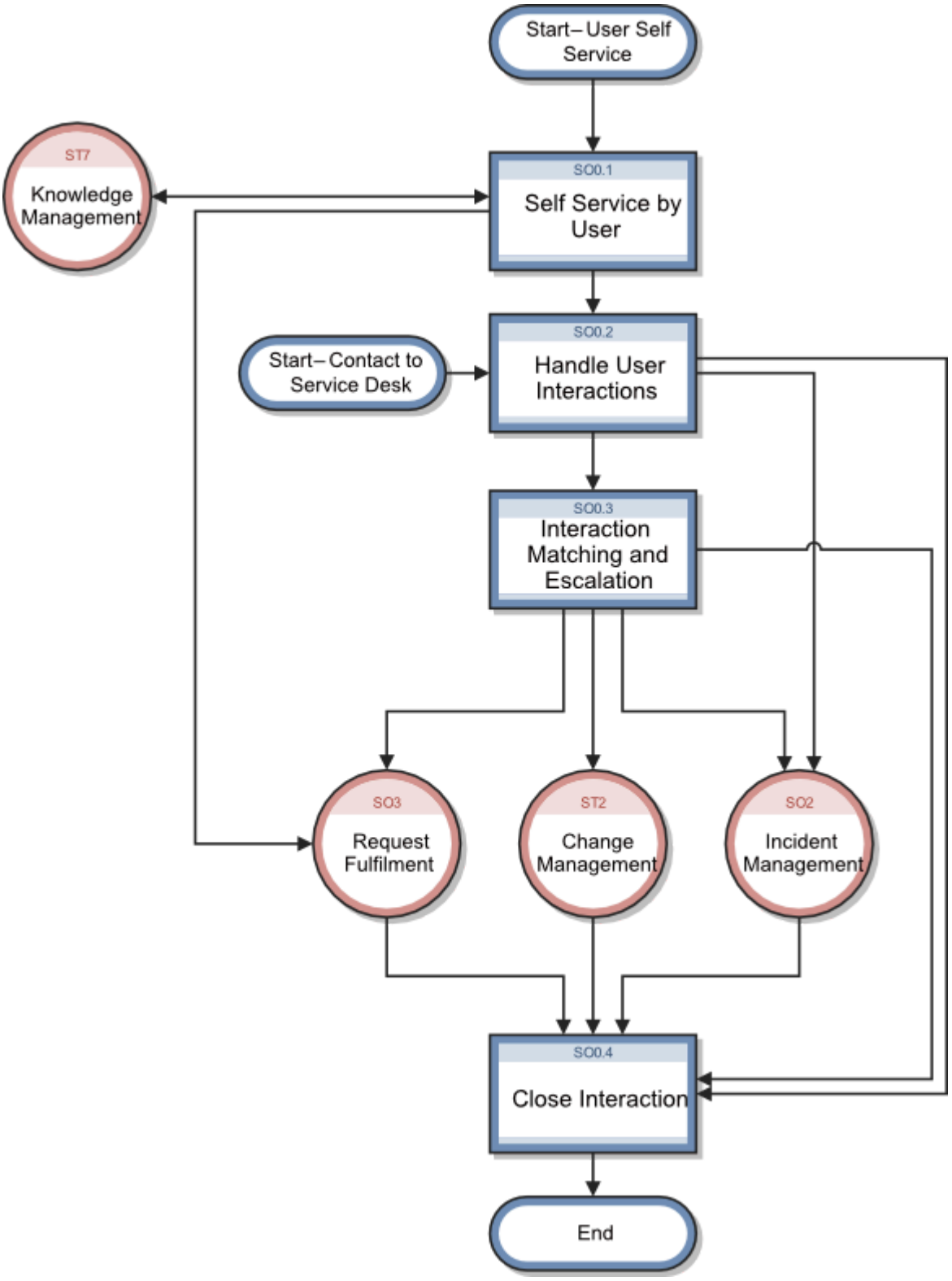
User roles
Add a user role record
Delete a user role record
Search for a user role record

1.2.3. Service Desk workflows and user tasks

Every user contact with the Service Desk is logged in an interaction record. User Interaction Management is the process for handling all interactions with the service desk that are received from self-service webpages or directly by service desk personnel.

This process streamlines service desk activities, thereby decreasing the workload for second line support teams.

User Interaction Management workflow



The User Interaction Management process consists of the following workflows and tasks:

ITIL Workflow	Tasks
Self-Service user interactions (SO0.1)	• Request a service catalog item or bundle • Create a self-service request • Close a self-service request • Resubmit a self-service request • Update a self-service request • View a self-service request
User interaction handling (SO0.2)	• Register a complaint • Register a request for change • Register a service catalog request • Register an incident • Register a request for information • Update an interaction record • Cancel a new Service Desk interaction • Reopen an incident
Interaction matching and escalation (SO0.3)	
Interaction closure (SO0.4)	• Close a first time Service Desk interaction • Close an existing Service Desk interaction

To view more workflow diagrams and other information about this process, refer to Service Manager *Processes and Best Practices*.

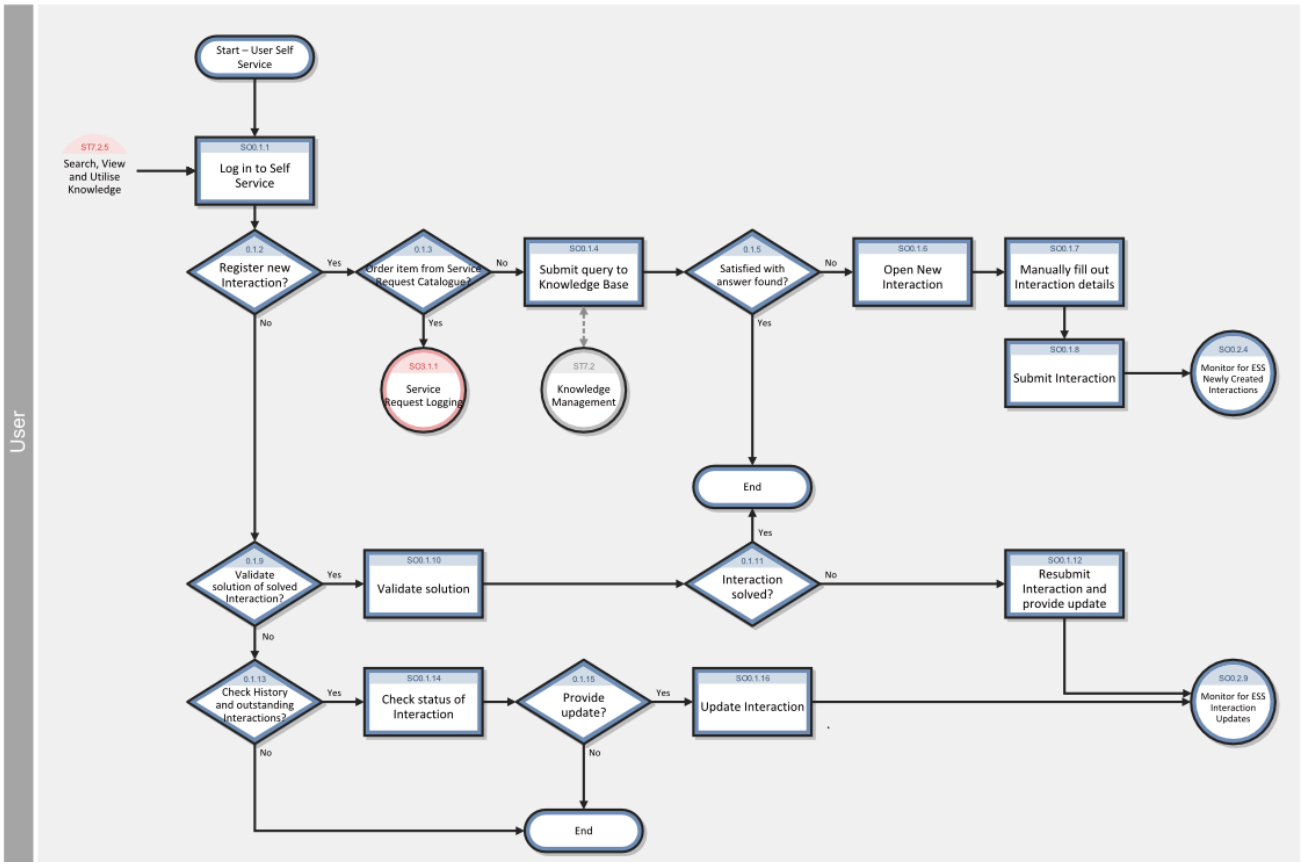
Related topics

Service Desk overview
User Interaction Management

1.2.3.1. Self-Service user interactions (SO0.1)

By using the self-service web environment, users can perform the following activities without contacting the Service Desk:

- Search the Knowledge Base to find an answer to a question or issue
- Monitor the status of previously reported interactions
- Log new interactions
- Order items from the service catalog



1.2.3.1.1. Request an item or bundle

Applies to User roles: User

You can easily order products and services from Service Catalog without needing to contact the service desk. Items in the Service Catalog can include an item picture, description, attachments, and purchasing options. Some items in the Service Catalog are for information only, such as instructions on how to order items from an outside service. Information-only items only have an item picture, description, and attachments; they don't include purchasing options.

> **Note**
>
> You may also order one or more products and related service tasks by choosing a bundle.

To request an item or bundle from the Service Catalog, follow these steps:

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.
3. Click the item you want to request to view the Order Information for that item.

   > **Note**
   >
   > Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.

4. In the Requested for field, select the Individual who will receive the good or service.
5. Select the quantity and any other item options, and then click **Add to Cart** to add the item you want to your cart.
6. From the Order from Catalog home page, click **View Cart/Checkout**.
7. Click **Submit Request**.
8. Type any required information justifying the request.
9. Click **Submit**.

Related topics

Out-of-box Service Catalog
Shopping cart
Service Catalog interface
Service Catalog items and bundles
Service Catalog connectors
Approve a request
Self-service access

1.2.3.1.2. Create a self-service request

Applies to User roles: User

If you can't find an existing solution to your incident or problem in the knowledge base, you can create a self-service request for a Service Desk Agent to resolve.

To create a self-service request, follow these steps:

1. Click **Submit a Request**.
2. In the Submit a Request window, verify the **Service Recipient** and **Primary Contact** fields.
3. Select your notification preferences in the **Notify By** field.
4. Select the **Urgency** of your issue.
5. Type a title and a description for your issue.
6. Click **Add Files**, and then follow these steps to add any relevant attachments:
    - Click **Browse** to locate your attachment.
    - Select the file to add.
    - Click **OK**.
7. Click **Submit**.

Related topics

Close a self-service request
View a self-service request
Update a self-service request
Resubmit a self-service request

1.2.3.1.3. Close a self-service request

Applies to User roles: User

You can close a self-service request if you are no longer experiencing the problem or if you have another reason for closing the request. If you need to resubmit the request, you can do so from your list of closed self-service requests.

To close a self-service request, follow these steps:

1. Click **View Open Requests**. Service Desk displays a table of your open requests and the status of each request.
2. Select the Service Desk interaction record that you want to close.
3. Click **Update**, type an update description if necessary, and then click **Close Request**.
4. Select a reason to close the request, and then click **Submit**. The Service Desk interaction record is closed.

Related topics

Self-service access
Searching records
Create a self-service request
View a self-service request
Update a self-service request
Resubmit a self-service request

1.2.3.1.4. Resubmit a self-service request

Applies to User roles: User

If you disagree with a proposed solution to a closed request, you can resubmit the request and provide a reason why the resolution is unsatisfactory. When you resubmit a closed interaction record, Service Manager assigns you a new interaction identification number to track the request. Service Manager automatically links the newly-created request to your closed interaction record and sends the new request to a Service Desk Agent for further diagnosis.

To resubmit a self-service request, follow these steps:

1. Click **View Closed Requests**.
2. Select the Service Desk interaction record that you want to resubmit.
3. Click **Resubmit**. Service Desk creates a new interaction record with a different identification number. However, the new record displays the same field information as the original Service Desk interaction record.
4. Type the reason why you are resubmitting the request.
5. Click **Save & Exit**.

Related topics

Self-service access
Searching records
Create a self-service request
View a self-service request
Update a self-service request
Close a self-service request

1.2.3.1.5. Update a self-service request

Applies to User roles: User

You can update any of your open self-service requests with additional information to assist the Service Desk Agent.

To update a self-service request, follow these steps:

1. Click **View Open Requests** to display a table of your open requests and the status of each request.
2. Select the Service Desk interaction record that you want to update.
3. Click **Update**, and then enter any new request details, additional descriptions, or add relevant attachments as needed.
4. Click **Save & Exit**.

Related topics

Self-service access
Searching records
Create a self-service request
View a self-service request
Resubmit a self-service request
Close a self-service request

1.2.3.1.6. View a self-service request

Applies to User roles: User

You can view your service requests to view the status of any outstanding interactions or to view a list of all your past requests.

To view a self-service request, follow these steps:

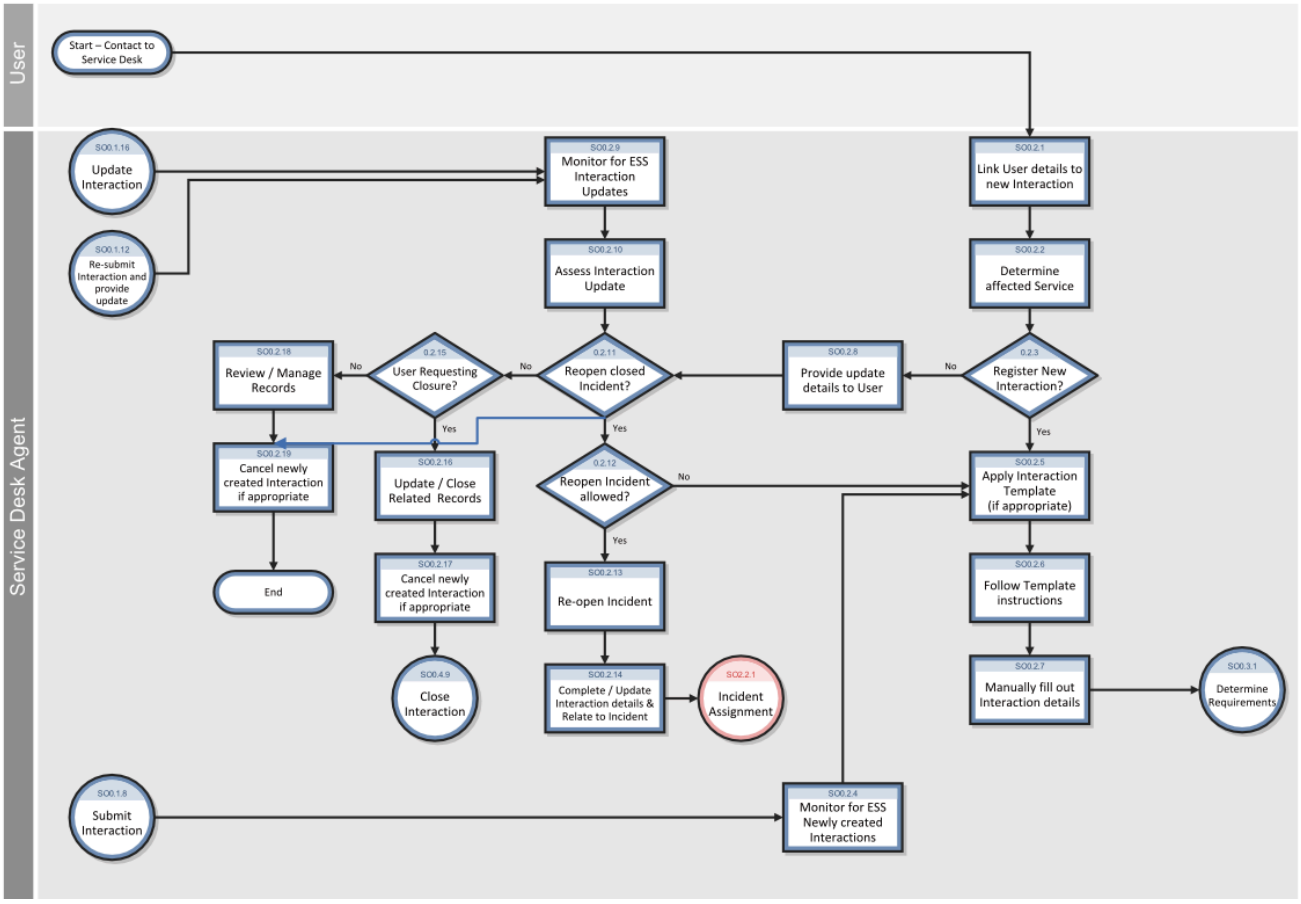
1. Click **View Open Requests** or **View Closed Requests** to display a table of requests and the status of each request.
2. Select the Service Desk interaction record that you want to view.
3. Click **Back** to return to the list of service requests.

Related topics

Self-service access
Searching records
Create a self-service request
Update a self-service request
Resubmit a self-service request
Close a self-service request

1.2.3.2. User interaction handling (SO0.2)

The Service Desk is responsible for handling all user interactions that are received by the self-service web portal, email or phone. The Service Desk attempts to resolve an interaction on first contact when the user makes first contact with the Service Desk.



Related topics

Service Desk overview
User Interaction Management
Service Desk workflows and user tasks
Register a complaint
Register a request for change
Register a service catalog request
Register an incident
Register a request for information
Update an interaction record
Cancel a new Service Desk interaction
Reopen an incident

1.2.3.2.1. Register a complaint

Applies to User roles: Service Desk Agent

If a user contacts you regarding a service complaint or a concern with the performance of a configuration item (CI), you can create a Service Desk interaction to collect information about the user's service request. Categorize the service request as a complaint, and then solve or escalate the request.

To register a complaint, follow these steps:

1. Click **Service Desk** > **Create New Interaction**.

   Service Manager prompts you to choose a category, after which the new interaction form is displayed. However, if a default category is defined in **Service Desk** > **Administration** > **Settings**, you're not prompted to choose a category, and the new interaction form is displayed with the category field pre-populated.

2. Select the contact for the interaction. The associated fields fill automatically.

   Note

   If you use the Fill function to replace the information in a field that already has data, clear the field before you click **Fill**. If the field contains data, the Fill function can't operate.

3. In the **Notify By** field, select the user's preferred notification method.
4. In the **Service Recipient** field, select the service recipient. The associated field fills automatically.
5. In the **Affected Service** field, select the affected service.
6. In the **Affected CI** field, select the affected CI.

Note

- The **Default Impact** and **Priority** values of the affected CI are automatically populated to the **Impact** and **Urgency** fields of the interaction record. You can manually change these auto-populated values if needed.
- If you specify the **Impact** and **Urgency** values first and then specify an affected CI with an empty **Default Impact** or **Priority** value, the **Impact** or **Urgency** value that you selected for the interaction record is cleared.

7. Type a title and description for the interaction.
8. Select the area and subarea.
9. Determine if there is a known solution or workaround for the user's issue. To do this, follow these steps:
    1. Search the available knowledge bases.

        Note

        When you use the Search Knowledge option from the New Interaction form, Service Manager uses the interaction description text as the search query.

    2. If you find an existing solution or workaround in the knowledge bases, close the interaction by using the solution.
    3. If you are unable to solve the user's issue, search for a problem or known error.
    4. If you find a workaround in the problem or known error records, close the interaction by using the workaround.
10. If you are still unable to solve the issue, select the impact and urgency of the request, and then click **Escalate**. Service Manager creates both an interaction and an incident, and then associates the interaction to the incident.

Related topics

What's a Service Desk interaction?
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
Searching records

Register an incident
Register a request for change
Register a request for information
Register a service catalog request
Update an interaction record
Cancel a new Service Desk interaction
Search for a solution

1.2.3.2.2. Register a request for change

Applies to User roles: Service Desk Agent

If a user contacts you to request a change to a service or configuration item (CI), you can create a Service Desk interaction record to collect information about the service request. Categorize the service request as a request for change, and then solve or escalate the request.

To register a request for change, follow these steps:

1. Click **Service Desk** > **Create New Interaction**.

   Service Manager prompts you to choose a category, after which the new interaction form is displayed. However, if a default category is defined in **Service Desk** > **Administration** > **Settings**, you're not prompted to choose a category, and the new interaction form is displayed with the category field pre-populated.

2. Select the contact for the interaction. The associated fields fill automatically.

   Note

   If you use the Fill function to replace the information in a field that already contains data, clear the field before you click Fill. If the field contains data, the Fill function doesn't work.

3. Select the user's preferred notification method.
4. Select the service recipient. The associated field fills automatically.
5. Select the affected service.
6. Select the affected CI.

Note

- The **Default Impact** and **Priority** values of the affected CI are automatically populated to the **Impact** and **Urgency** fields of the interaction record. You can manually change these automatically populated values if needed.
- If you specify the **Impact** and **Urgency** values first and then specify an affected CI with an empty **Default Impact** or **Priority** value, the **Impact** or **Urgency** value that you selected for the interaction record is cleared.

7. Type a title and description for the interaction.
8. Select the subarea.

Note

The system provides the area (service portfolio) because in an out-of-box system, this is the only area available.

9. If you are able to fulfill the user's request, close the interaction and document your solution.
10. If you are unable to fulfill the user's request, select the impact and urgency of the request, and then click **Escalate**.
11. The Escalation wizard starts. Follow the onscreen instructions. Service Manager creates both an interaction and a change request, and then associates the interaction to the change request. Service Manager displays the New Interaction form when the wizard finishes.

Related topics

What's a Service Desk interaction?
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
Searching records
Register a complaint
Register a request for information
Register an incident
Register a service catalog request
Update an interaction record

Cancel a new Service Desk interaction

Search for a solution

1.2.3.2.3. Register a service catalog request

Applies to User roles: Service Desk Agent

You can submit a service catalog request on behalf of a user who requests the service (the service recipient). The request triggers the creation of a Service Desk interaction record.

To register a service catalog request on behalf of a service recipient, follow these steps:

1. Click **Service Desk** > **Create New Interaction**.

   Service Manager prompts you to choose a category, after which the new interaction form is displayed. However, if a default category is defined in **Service Desk** > **Administration** > **Settings**, you're not prompted to choose a category, and the new interaction form is displayed with the category field pre-populated.

2. Move the cursor to the **Service Recipient** field.
3. Click the **Fill Field** button immediately to the right of the **Service Recipient** field to display the Search Contact Information screen.
4. Move the cursor to the **Contact Name** field, and then click **Search**.
5. Click the name of the person whose request you will fill. That person's name appears as the service recipient.
6. Select the affected service.
7. Select the affected CI.

   Note

   - The **Default Impact** and **Priority** values of the affected CI are automatically populated to the **Impact** and **Urgency** fields of the interaction record. You can manually change these automatically populated values if needed.
   - If you specify the **Impact** and **Urgency** values first and then specify an affected CI with an empty **Default Impact** or **Priority** value, the **Impact** or **Urgency** value that you selected for the interaction record is cleared.

8. In the **More** drop-down list, select **Order from Catalog**.

> **Note**
>
> Once you click **Order from Catalog**, you can't change the service recipient. If you must change the service recipient, cancel the current interaction and create a new one.

9. Use one of the following methods to add items to your cart:
   - **Search for the desired cart items**

     The items for which you search are filtered against the service recipient's capability words. Items that match the search criteria don't display in search results unless the service recipient has the required capability words.
   - **Select from Most Popular Requests**

     The Most Popular Requests list displays only the cart items that are allowed by the service recipient's capability words.
   - **Browse the service catalog**

     The service catalog displays only the categories, items, and bundles that are allowed by the service recipient's capability words.
10. After you add the desired items to your cart, click **View Cart**.
11. If needed, click **Remove** to remove an item or click **View Catalog** to add more items to your cart, and then click **View Cart** again.

> **Note**
>
> Be sure to click **View Cart** each time before you click **Return to Request** in the next step. If you don't do this, Service Manager asks if you wish to cancel. If you select **Yes**, your requested items are canceled.

12. Click **Return to Request**.
13. Fill in the blank fields on the Interaction Details screen, and then click **Save**.

The catalog items you requested appear in the Catalog Items section of the interaction record.

Related topics

What's a Service Desk interaction?
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
More Actions menu
Searching records
Register a complaint
Register a request for information
Register an incident
Register a request for change
Update an interaction record
Cancel a new Service Desk interaction
Search for a solution

1.2.3.2.4. Register an incident

Applies to User roles: Service Desk Agent

If a user contacts you to report an incident, you can create a Service Desk interaction record to collect information about the service request. Categorize the service request as an incident, and then solve or escalate the request.

To register an incident, follow these steps:

1. Click **Service Desk** > **Create New Interaction**.

   Service Manager prompts you to choose a category, after which the new interaction form is displayed. However, if a default category is defined in **Service Desk** > **Administration** > **Settings**, you're not prompted to choose a category, and the the new interaction form is displayed with the category field pre-populated.

2. Select the contact for the interaction. The associated fields fill automatically.

   Note

   If you use the Fill function to replace the information in a field that already contains data, clear the field before you click Fill. If the field contains data, the Fill function doesn't work.

3. Select the user's preferred notification method.
4. Select the service recipient. The associated field fills automatically.
5. Select the affected service.
6. Selected the affected CI.

Note

- The **Default Impact** and **Priority** values of the affected CI are automatically populated to the **Impact** and **Urgency** fields of the interaction record. You can manually change these automatically populated values if needed.
- If you specify the **Impact** and **Urgency** values first and then specify an affected CI with an empty **Default Impact** or **Priority** value, the **Impact** or **Urgency** value that you selected for the interaction record is cleared.

7. Type a title and description for the interaction.
8. Select the Area and Subarea.
9. Follow these steps to determine if there is a known solution or workaround for the user's issue:
    1. Search the available knowledge bases.

        Note

        When you use the Search Knowledge option from the New Interaction form, Service Manager uses the interaction description text as the search query.

    2. If you find an existing solution or workaround in the knowledge bases, close the interaction by using the solution.
    3. If you are unable to solve the user's issue, search for a problem or known error.
    4. If you find a workaround in the problem or known error records, close the interaction by using the workaround.
10. If you are still unable to solve the user's issue, select the impact and urgency of the request, and then click **Escalate**.
11. The Escalation wizard opens. Follow the onscreen instructions. Service Manager creates both an interaction record and an incident record, and then associates the interaction to the incident.

Related topics

What's a Service Desk interaction?
Working with Service Desk interaction records

Associating a Service Desk interaction with another application record
Searching records

Register a complaint
Register a request for change
Register a request for information
Register a service catalog request
Reopen an incident
Update an interaction record
Cancel a new Service Desk interaction
Search for a solution

1.2.3.2.5. Register a request for information

Applies to User roles: Service Desk Agent

If a user contacts you to request information, you can create a Service Desk interaction record to collect information about the service request. Categorize the service request as a request for information, and then solve or escalate the request.

To register a request for information, follow these steps:

1. Click **Service Desk** > **Create New Interaction**.

   Service Manager prompts you to choose a category, after which the new interaction form is displayed. However, if a default category is defined in **Service Desk** > **Administration** > **Settings**, you're not prompted to choose a category, and the new interaction form is displayed with the category field pre-populated.

2. Select the contact for the interaction. The associated fields fill automatically.

   Note

   If you use the Fill function to replace the information in a field that already contains data, clear the field before you click Fill. If the field contains data, the Fill function doesn't work.

3. Select the user's preferred notification method.
4. Select the service recipient. The associated field fills automatically.
5. Select the affected service.
6. Select the affected CI.

Note

- The **Default Impact** and **Priority** values of the affected CI are automatically populated to the **Impact** and **Urgency** fields of the interaction record. You can manually change these automatically populated values if needed.
- If you specify the **Impact** and **Urgency** values first and then specify an affected CI with an empty **Default Impact** or **Priority** value, the **Impact** or **Urgency** value that you selected for the interaction record is cleared.

7. Type a title and description for the interaction.
8. Select the area. Service Manager selects the subarea field automatically.
9. Determine if there is a known solution or workaround for the user's issue. To do this, follow these steps:
    1. Search the available knowledge bases.

    Note

    The system provides the area (service portfolio) because in an out-of-box system, this is the only area available.

    2. If you find an existing solution or workaround in the knowledge bases, close the interaction by using the solution.
    3. If you are unable to solve the user's issue, search for a problem or known error.
    4. If you find a workaround in the problem or known error records, close the interaction by using the workaround.
10. If you are still unable to solve the issue, select the impact and urgency of the request, and then click **Escalate**. Service Manager creates both an interaction and an incident and associates the interaction to the incident.

Related topics

What's a Service Desk interaction?
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
Searching records
Register a complaint

Register a request for change
Register an incident
Register a service catalog request
Update an interaction record
Cancel a new Service Desk interaction
Search for a solution

1.2.3.2.6. Update an interaction record

Applies to User roles: Service Desk Agent

You can update an interaction record with additional information to assist the applicable assignment group.

To update an interaction record, follow these steps:

1. Locate the Service Desk interaction record by using one of the following methods:
   - Select a record from your interaction queue.
   - Use search or advanced search to find one or more records.
2. Update any fields that require revision.

   Note

   If you are modifying a field that uses the Fill function, clear any existing data before using the function.
3. Click **Save** to update the Service Desk interaction record.

Related topics

Searching records

1.2.3.2.7. Cancel a new Service Desk interaction

Applies to User roles: Service Desk Agent

You can cancel a new Service Desk interaction without saving it. For example, if the user no longer experiences the problem, you can cancel the interaction.

To cancel a new Service Desk interaction, follow these steps:

1. Use search or advanced search to find one or more records.
2. While in the New Interaction form, click **Cancel**.
3. Click **Yes** to cancel the interaction without saving the changes.

Related topics

Service Desk overview
Interaction closure (SO0.4)
Searching records

1.2.3.2.8. Reopen an incident

Applies to User roles: Service Desk Agent

If a user who is not satisfied with the solution provided for a previously registered interaction contacts you, you can reopen the incident record to address the user's concerns.

Note

You can't reopen an incident record if more than two weeks have passed since you notified the user of the solution. In this situation, create a new interaction record.

To reopen an incident, follow these steps:

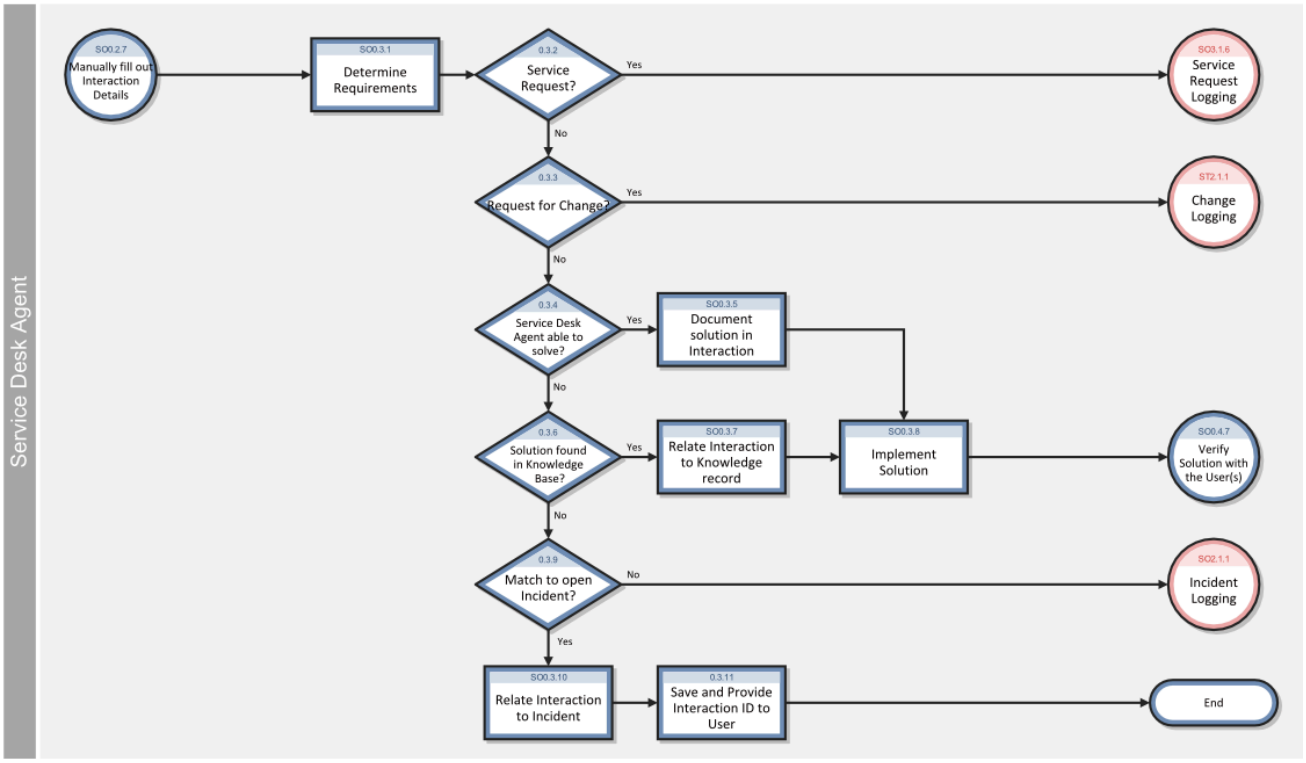
1. Click **Service Desk** > **Search Interactions**.
2. Use search or advanced search to find one or more records.
3. Double-click the interaction record to open it.
4. Open the Related Records section, locate the incident, and then double-click the incident record ID to open it.
5. Click **Reopen**, and then click **Save**.

Related topics

What's a Service Desk interaction?
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
More Actions menu
Searching records
Register an incident
Register a complaint
Register a request for change
Register a request for information
Register a service catalog request

1.2.3.3. Interaction matching and escalation

When an interaction is received, the Service Desk Agent determines whether the interaction is a service request or a request for change, and logs the request if appropriate. If the Service Desk Agent is not able to resolve the issue, the incident can either be related to an existing incident or logged as a new incident.

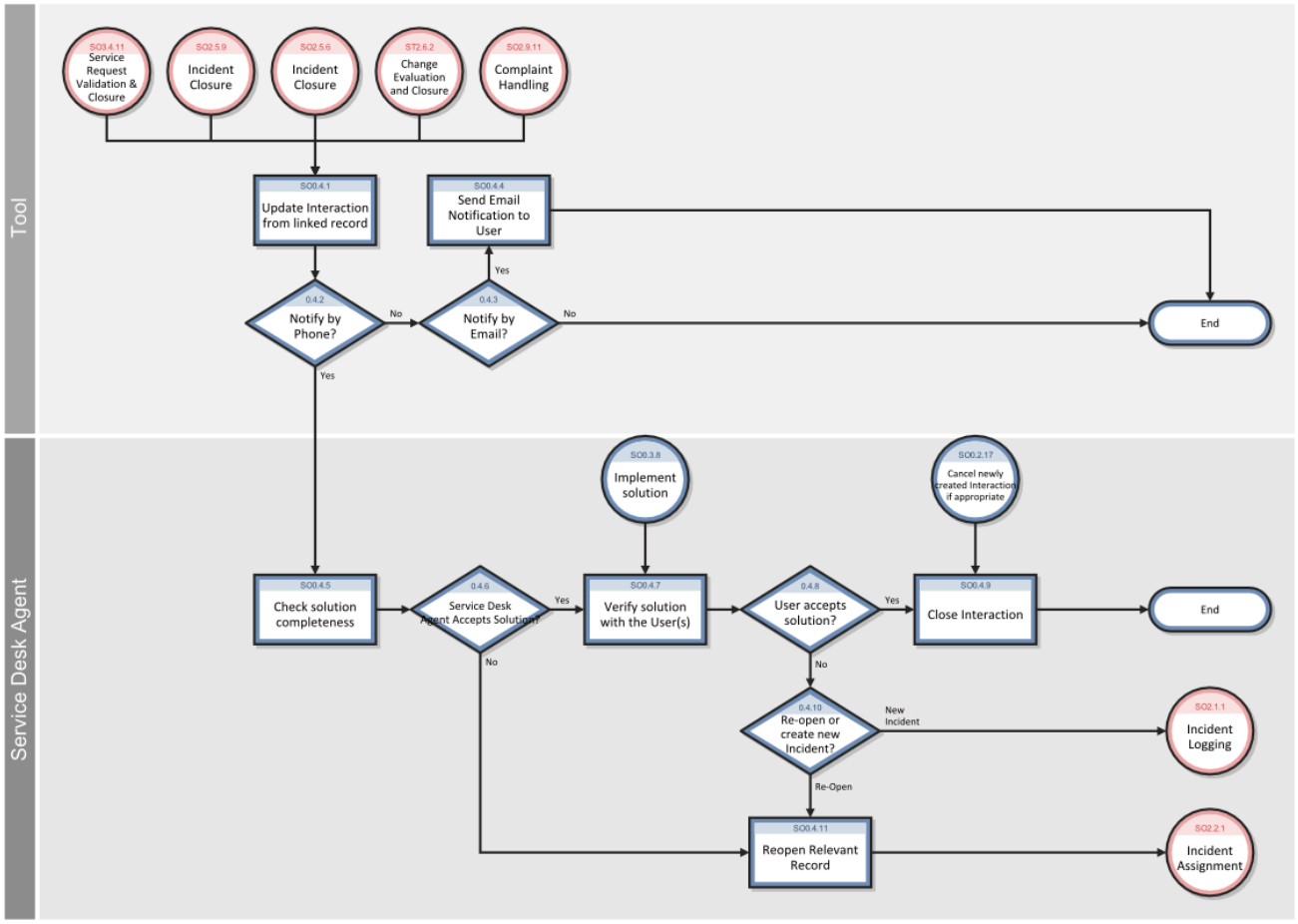


Related topics

Service Desk overview
User Interaction Management
Service Desk workflows and user tasks

1.2.3.4. Interaction closure

The interaction closure process occurs when an interaction is resolved by the Service Desk without escalation or solved when a related incident, change, or request is resolved. Based on user preferences, the Service Desk communicates the solution to the user by phone or email.



Close a first time resolved Service Desk interaction

Applies to User roles: Service Desk Agent

You can close a Service Desk interaction record without escalating it to an incident if you are able to resolve the issue immediately. Depending on your company policies, you may also need to document the solution in a knowledge base for future use.

To close a first time Service Desk interaction, follow these steps:

1. Use search or advanced search to find one or more records.
2. In the "Interaction Details" section of the Service Desk interaction form, select the closure code and type a solution for the interaction.
3. Click **Close Interaction**.

Close an existing Service Desk interaction

Applies to User roles: Service Desk Agent

You can close an existing Service Desk interaction record if the user accepts the proposed solution. If you or the user disagree with the proposed solution, don't close the interaction. Instead, resubmit the related incident for further investigation. Depending on your company policy, you may also need to document the interaction's solution in a knowledge base for future use.

To close an existing Service Desk interaction record, follow these steps:

1. Click **Service Desk** > **Interaction Queue**. Service Manager displays the list of open interactions.
2. Locate the Service Desk interaction record that you want to close.
3. Use the contact for this interaction to determine which user to notify of the solution.
4. If the user accepts the solution, select a closure code, and then click **Close Interaction**. If the user rejects the solution, reopen the related incident or create a new incident, and then link the incident to the unsolved interaction.

Related topics

Service Desk overview
User Interaction Management
Service Desk workflows and user tasks

1.2.3.5. Miscellaneous tasks

You can also perform the following tasks in the Service Desk module.

Using mass update with Service Desk interaction record lists

The mass update feature enables you to update the value of one or more fields in a list of records. The system provides a template form that displays the current field values and enables you to type or select a new value for the displayed fields. In compliance with the data policy of your organization (as recorded in the Data Policy form), the Mass Update template form doesn't display read-only fields in the records. The mass update option is available from both interaction and incident record lists.

When you perform a mass update operation, make sure that the value you type or select for a particular field is the value you want in all the records that you selected. When you update incident records, you must also update the incident activity data.

System Administrators can determine if a field is displayed in the Mass Update template form by directly editing the Data Policy record for the form. To remove a field from the Mass Update form, change the Usage Type column for the field to System. If a System Administrator wants to grant Template Mass Update permissions to other operators, the System Administrator must update the operator profile to include the Template Mass Update and Complex Mass Update capabilities.

Update multiple interaction records

Applies to User roles: Service Desk Manager

You can use the mass update feature to update one or more values in multiple interaction records. Remember that the value you enter for a particular field becomes the value for all of the records you selected when you perform a mass update.

To update multiple records, follow these steps:

1. Locate the records by using one of the following methods:
    - Go to the Interaction Queue to display a list of records.
    - Use search or advanced search to find one or more records.
2. Click **More** or the **More Actions** icon, and then select **Mass Update**.
3. Double-click the field that you want to update.

4. Type the value for the field or use the Find feature to display a list of potential values for the field.
5. Click **Next**, and then repeat steps 3 and 4 for additional fields as needed.
6. Update the Activity Action field for the updates you made in the Mass Update template, and then click **Execute**.

To use complex update to update multiple records:

1. Locate the records by using one of the following methods:
   - Go to the Interaction Queue to display a list of records.
   - Use search or advanced search to find one or more records.
2. Select the records that you want to update.
3. Click **Mass Update**, and then click **Complex Update**.
4. Type the instructions (in RAD syntax) to be executed one time at the beginning of the mass add or mass update operation.
5. Type the instructions (in RAD syntax) for action on each record.
6. Click **Execute**.

Duplicate an existing interaction record

Applies to User roles: Service Desk Agent

You can duplicate an existing interaction record to copy information from an existing record into a new record. For example, you duplicate an existing interaction record because a user is experiencing a repeat of a past problem or issue.

To duplicate an existing interaction record, follow these steps:

1. Click **Service Desk** > **Search Interactions**.
2. Use search or advanced search to find one or more records.
3. Double-click the record to open it.
4. Click **More** or the **More Actions** icon, and then and select **Duplicate Interaction**. Service Manager assigns the next available ID number to the duplicated interaction.
5. Add or change information in the interaction record as needed.

Note

You must clear any existing data in a field to use the Fill and Find functions to select a new value.

6. Click **Save & Exit**. If the record that you duplicated has any related records, a message that informs you the record has related records and asks if you want to copy the associations is displayed.
    - Click **Yes** to copy associated records and return to the interaction queue.
    - Click **No** if you don't want to copy the associated records to the new interaction record. The Escalation Details wizard starts, and you can relate the new record to an existing incident, an existing known error, or a new incident.

Add a required action

Applies to User roles: Service Desk Agent

A required action is an action that a service recipient or your service organization requires you to perform before an interaction can be closed. For example, the service recipient may want to be contacted by a certain date. When you make this a required action, nobody can close this interaction until someone contacts the service recipient.

By default, Service Manager defines one required action: When the Notify By field for an interaction is set to "telephone" the Service desk Agent must complete and close the required action before the interaction can be closed.

When you add a required action to an interaction record, the system updates the status of the record to Open-Callback, which means that the specified required action must be completed before the interaction and any associated incidents can be closed. The Service Desk Agent specifies what the required action is when they add the required action information to the interaction record. Required Actions lists a description of the action required for the interaction and a link to any associated incidents.

To add a required action, follow these steps:

1. Click **Service Desk** > **Search Interactions**.
2. Use search or advanced search to find one or more records.
3. Double-click the record to open it.
4. Click **More** or the **More Actions** icon, and then select **Add Required Action**.
5. Type the action details and a reason for the required action.
6. Select the incident number to associate with this interaction.

> **Note**
>
> You can click **Fill** to display a list of incidents.

7. Click **OK**.

> **Note**
>
> You can't close the interaction until someone performs the required action.

Set a reminder for an interaction record

Applies to User roles: Service Desk Agent

You can set a reminder to notify you if an existing interaction record meets certain conditions by a specified time. For example, you may want to be reminded if an interaction is still open or is not updated by a given date.

To set a reminder, follow these steps:

1. Locate an existing interaction.
2. Click **More** or the **More Actions** menu, and then select **Set Reminder**.
3. Choose one of the following options to specify when the reminder occurs:
   - Click **Remind At** to set a reminder for a particular date and time.

     > **Note**
     >
     > You can click **Fill** to display a calendar tool to select the date and time of the reminder.

   - Click **Remind In** to set the reminder to occur after a particular time interval. If you choose this option, you must specify the time interval in hours, minutes, and seconds (use the 00:00:00 format), and then select a shift from the list.
4. In the **Remind If** field, select from the following options any conditions that you want the interaction record to meet before a notification occurs:
   - Always (default)
   - ... has not been updated

- ... is still open
- ... is still assigned to me

5. In the Pop-up Message area of the form, select from the following options the type of notification that you want to receive:
   - Pop-up
   - Email
   - SM Mail (Service Manager Internal Mail)

   > **Note**
   >
   > If you select the Pop-up notification method, type the reminder message format in the **Message** field. If you select the Email or SM Mail notification methods, type the title you want the notification to display in the **Title** field, and then select **Message Type**.

6. Click **OK**.

Contents

- 1 Using mass update with Service Desk interaction record lists
  - 1.1 Update multiple interaction records
- 2 Duplicate an existing interaction record
- 3 Add a required action
- 4 Set a reminder for an interaction record

1.2.4. Service Desk administration

Service Desk administration consists of the following:

- Service Desk Environment: Service Desk application contains an environment record for interactions. This record contains options that defines the functionality of the Service Desk application for all Service Desk users.
- Settings: You can configure whether escalation is allowed for interactions, and you can set the default category for new interactions.

Configure the Service Desk settings

Applies to User roles: System Administrator

To configure Service Desk settings:

1. Click **Service Desk** > **Administration** > **Settings** in the System Navigator.
2. Configure the following settings for the Service Desk application:

Setting	Description
Allow Escalation	Whether or not you can escalate the interactions. In the out-of-box setting, escalation is allowed. **Note** You can only escalate an interaction when the **Allow Escalation** option is selected both in the **Settings** menu here and in the category definition for the specific category assigned to the interaction.
Enable "Create New Interaction" Menu Entry	This setting determines whether the interactions will be created to follow the legacy interaction workflows (select the checkbox) or the new streamlined workflows introduced as of SM 9.41 (clear the checkbox).

Setting	Description
Default Category	The default category when you create a new interaction. The out-of-box value for this parameter is None. The default category is used when you click **Create New Interaction**: If no default category is defined, you are prompted to select a category before the interaction detail screen appears. If the default category is defined, the interaction detail screen is displayed directly and the **Category** field is populated with the default category. **Note** The default category doesn't apply to the ESS requests. For support requests submitted through ESS, interactions are always created in the "incident" category.

Setting	Description
Email Notification Template for the "Visible to Customer" Option	The default email notification template when you enter new updates for a record (such as a Change or a Request) that is related to an Interaction with the **Visible to Customer** checkbox selected in this record. The out-of-box value for this field is **RM Visible to Customer notification**. After you select the **Visible to Customer** checkbox on the **Activities** tab in a record that is related to an Interaction and then enter new updates, Service Manager sends an email notification to the contact of the Interaction using the specified template. If no default template is defined, Service Manager no longer sends any email notifications when you enter new updates for a record (such as a Change or a Request) that is related to an Interaction with the **Visible to Customer** checkbox selected in this record.
Chat Assignment End Lead Time (hh:mm:ss)	

3. Click **Save**.

Configure the Service Desk environment

Applies to User roles: System Administrator

To configure a Service Desk environment record:

1. Click **Service Desk** > **Administration** > **Environment**.

2. Configure the settings for your Service Desk environment.

   You can specify the Service Desk Record Relationship Model as follows:

   - **Full Service Desk Model**: In this model, the state of a Service Desk interaction record changes when each related record closes. The closed state of an interaction record depends on the notification value chosen for the "Notify By" field in the interaction record. For example, if the notification value is "Telephone," the interaction record has a required action before it is closed. This action describes why the customer must be contacted. It also prevents the interaction record from closing until all required actions are complete. In this case, the interaction record goes into the Open-Callback state before it is closed. For more information, see Full Service Desk model.
   - **All Records close Independently**: In this model, all Service Desk interaction records close independently. The state of related records doesn't affect closing an interaction record, and closing the interaction record doesn't affect related records.
   - **Close Interactions when Related Record closes**: In this model, when the last related record closes, the Service Desk interaction record closes.
   - **Cannot close Related Record until Interactions are closed**: In this model, records related to a Service Desk interaction can't close until the interaction record is closed.
   - **Cannot close Interactions until Related Records are closed**: In this model, a Service Desk interaction record can't close until all related Incident Management records, Change Management records, and Request Management records are closed.

   You can use the following table for more information on these fields:

Check box	Description
Delay Assigning Interaction ID?	If selected (set to true), prevents Service Manager from assigning an identifier to the Interaction immediately. When this option is active, there is no unique identifier to tie an attachment within the file when you first open it. Therefore, you can't save attachments when you open the ticket. You can add attachments when saving an update after the record has a unique identifier. **Note** This option doesn't work under the following circumstances: ◦ When the system automatically creates an Interaction from a chat request in background. ◦ When a Service Desk agent manually creates an additional Interaction for a chat request.
User Operator Full Name?	If selected (set to true), uses the operator's Full Name instead of the login ID when time stamping reports (on open, update, and so on) for records in the application environment

Check box	Description
Return to Blank Interaction?	If selected (set to true), returns the user to a blank (new) Interaction screen after the creation of an Incident record. If cleared (set to false), the user will return to the Interaction they created the Incident from.
Post resolution to Interactions?	If selected (set to true), posts the resolution of a closed ticket to the related Interaction report.
Copy Associations of Cloned Interactions?	Copies the associations of a cloned interaction.

You can use the following table to specify information about the Post Back links:

Post Back Link	Description
IM Post back Link:	The link record used to post information from a related Incident to the Interaction when the Incident is closed.
ChM Post back Link:	The link record used to post information from a related Incident to the Interaction when the Change is closed
RF Post back Link:	The link record used to post information from a related Incident to the Interaction when the request is closed.
PbMPost back Link:	The link record used to post information from a related Incident to the Interaction when the Problem is closed.
Copy/Open Link:	The record from which to copy the data. When you copy a record, data from the record specified here is copied to the new record based on the link record.

The following four fields specif information about the Service Desk Manager, Group, Phone Number and Email. These fields are used in notifications and for assigning the interaction to a default group.

Tip

You can use the field help to view the description for each setting. To view the help on field:

- Web client: Select a field, and then press **F1**.
- Windows client: Select a field, and then press **Ctrl+H**.

3. Click **Save**.

Add a Service Desk assignment group

Applies to User Roles: System Administrator

Service Desk assignment groups specify the Service Desk agents in the group so that when someone creates a new interaction it can be assigned to the applicable group. Typically, assignment groups are organized by location and expertise. To facilitate interaction response, new assignment groups can be added to the out-of-box assignment groups for Service Desk.

To add a Service Desk assignment group:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.
   1. Enter the name of the manager of the assignment group.
   2. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   3. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide response when you create and populate a new assignment group.

   (Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.
9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.
10. Follow these steps to configure the Supported Services field:

    1. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    2. Select **Business Service** in the CI Type field, and then click **Search**.
    3. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    4. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.
11. Click **Save** and **OK**.

1.2.5. Service Desk configuration

Service Desk configuration enables you to configure alert, interaction categories, workflow, solution matching, and so on.

Create an interaction category

Applies to User roles: System Administrator

If you are a Service Manager Administrator, you may want to create new interaction categories. You may want to create a category when you require a different worfklow that specifies unique forms and business rules.

> **Note**
>
> If you do create a new category, you must also add corresponding subcategories and areas.

Service Manager provides a number of out-of-box category records that you can use or modify. For more information, see Out-of-box Service Desk categories.

To create new category record:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New**.
4. Type the name of the category.
5. Specify the **Apply To** option to indicate the modules that can share this category.

Note

The **Apply To** flag indicates whether the category is shared across different modules. For example, if the **Apply To** is set to "Interaction/Incident", this category will be created in both the interaction category table and the incident category table. Typical shared categories in the out-of-box system include:

- complaint
- incident
- problem
- request for information
- request for administration

For an Interaction Category, the following rules apply:

- Once a value is specified in the **Apply To** field and the record is saved, the value can't be modified.
- If **Apply To** is set to "Interaction" only, the **Allow Escalation** flag becomes read-only.

  The following table describes the Allow Escalation settings for the out-of-box categories that only apply to Interaction.

Category	Allow Escalation	Notes
▪ compliment ▪ service catalog	Disabled (Read-only)	This is because "Apply To Interaction" means the category is not shared among other modules, and interactions of this category don't need to trigger a record in another module. A compliment can be closed immediately or worked through the phases as a normal interaction. A service catalog interaction can be resolved by ordering items from the catalog.
▪ complaint ▪ incident ▪ problem ▪ request for administration ▪ request for change ▪ Request for information	Enabled (Editable)	Although the category is not shared among other modules, an interaction of one of these categories should be allowed to trigger a change request or a service request.

The **Apply To** options that are used in Interaction Categories include:

- Interaction
- Interaction/Incident
- Interaction/Incident/Problem
- Interaction/Problem

6. Specify whether the category is active and whether escalation is allowed.

> **Note**
>
> If the **Active** checkbox is unchecked, you can't use this category when creating a new record.

7. Type the category description.
8. Select a workflow for the category.
9. Click **Save** to view the **Workflow** tab and other associated tabs.
10. Click **Subcategories** tab to create subcategories for the interaction category.
11. Click **Service Desk Approval** tab to configure approval settings.

> **Note**
>
> The settings in the **Service Desk Approval** tab are only for service catalog approval. That is, in the out-of-box system, they are only valid for the service catalog category.

12. Click **Save**.

> **Note**
>
> The Interaction category name is read-only after the category is created.

Add a new interaction subcategory

Applies to User Roles: System Administrator

Subcategories are a way to refine a Service Desk category into smaller groups. For example, the complaint category can have service delivery and support subcategories.

To add a subcategory:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New** to create a new category or click **Search** to select an existing category.
4. In the **Subcategories** section, click the **Link New Subcategories** button.
5. Type a subcategory name.
6. Type a description for the subcategory.
7. Click **Save** to add the new subcategory.

Add a new interaction area

Applies to User roles: System Administrator

Areas are a way to further refine a Service Desk subcategory. For example, the support subcategory in the complaint category can have availability, functionality, and performance areas.

To add an area:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New** to create a new category or click **Search** to select an existing category.
4. In the **Subcategories** section, click the **Link New Subcategories** button to create a new subcategory or select an existing subcategory from the list.
5. In the **Areas** section, click the **Link New Areas** button.
6. Type a name for the area.
7. Type a description for the area.
8. Click **Save** to add the new area.

Service Desk solution matching

When you escalate an interaction record to incident, a list of incidents or problems appears based on the pre-defined solution matching configuration. You can then select the matched records and link them to the interaction. Alternatively, you can specify the ID of the existing incident or problem, and then link it to the interaction. The linked records are shown in the **Related Records** section of the interaction.

Note

When you escalate from interaction to incident, instead of selecting the assignment group through a wizard, the assignment group of the escalated incident is populated according to the following rules:

1. If an interaction is escalated to an incident of the incident category, the **Config admin group** field defined in affected service or affected CI of the interaction is used.
2. If an interaction is escalated to an incident of other categories, the Service Desk group in the Service Desk environment is used.

You can also configure an auto-assignment rule depending on your organization's requirements.

In an out-of-box system, the following configurations are provided for Service Desk interaction solution matching:

- Find an incident with a matching configuration item
- Find an incident with a matching service
- Find an incident with a matching title
- Find a problem with a matching configuration item
- Find a problem with a matching service
- Find a problem with a matching title

To modify the out-of-box solution matching configurations, navigate to **Service Desk** > **Configuration** > **Solution Matching**.

1.2.6. Security

The topics in this section explain the Service Desk security roles, security areas, and rights.

Setting rights to interactions

Applies to User roles: Service Desk Manager

The Service Desk Manager can set rights to interactions using the role-based security. The Service Desk and Service Desk Configuration areas contain default security rights and settings. The rights are copied to new roles created for these areas. However, the settings are inherited only if there are no settings specified on the roles.

Note

Whenever the roles in an operator record are updated, the operator must log out and then log in for the changes to take effect.

Service Desk security areas

Applies to User roles: System Administrator

The security areas for Service Desk are Service Desk and Service Desk Configuration. These areas contain the default security rights and settings for Interactions in the Service Desk module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

These security areas are used to set permissions to operators to provide access to particular area of Service Desk. For example, for the Service Desk Analyst security role, set rights as **View** for the Service Desk area. Then, an operator with this security role can only view the Service Desk items under Service Desk module. The following table lists the areas and the relevant Service Desk menu items the operators can access.

Area
Service Desk
Service Desk Configuration

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Service Desk and Service Desk Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Service Desk	TRUE	FALSE	Never	Never	FALSE	FALSE
Service Desk Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Service Desk and Service Desk Configuration areas.

Service Desk security roles and settings

The out-of-box security roles for the Service Desk module include the following:

- service tech
- service desk agent
- service desk analyst
- service desk manager
- self service
- approver
- service desk process owner

Tip

If you have upgraded from a previous version of Service Desk, which is not Process Designer based, see Service Desk security profiles mappings for more information about the mappings between the legacy Service Desk profiles and the current Process Designer security roles.

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Service Desk area specified in the corresponding security roles. See the table below for the out-of-box security rights of the new security roles in the Service Desk area and the Service Desk Configuration area. This table only lists the new security roles that have different settings with the default rights.

Area Name

Role Name

View

New

Update

Delete/Close

Modify Template

Expert

Admin rowspan='4'|Service Desk

service desk analyst

TRUE

TRUE

Always

Always

FALSE

TRUE

FALSE

service desk manager

TRUE

TRUE

Always

Always

TRUE

TRUE

FALSE

service desk process owner

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

system administrator

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE rowspan='3'|Service Desk Configuration

service desk manager

TRUE

TRUE

Always

Always

FALSE

FALSE

FALSE

service desk process owner

TRUE

FALSE

Never

Never

FALSE

FALSE

FALSE

system administrator

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

For information about the out-of-box role rights in the Common Configuration area, see Out-of-box role rights in the Common Configuration area.

1.3. Streamlined Interaction (Codeless)

You can use Service Manager Service Desk to log user requests and trigger appropriate fulfillment processes for proper handling. Service Desk helps you manage all user interactions, whether they are reported by using Self-Service website or directly to your service desk.

Note

The topics in this section are based on the Streamlined Interaction solution, which is introduced as of Service Manager 9.41. The Streamlined Interaction feature is optional and is disabled by default. You need to manually enable it before you read the topics in this section. To enable the Streamlined Interaction solution, see Enable streamlined interaction.

If you want to read the help topics for the out-of-box Service Desk module (Codeless mode), see the help content in Service Desk help hub.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Service Desk User Roles

Key concepts

Describes the most important things that you need to know about how the Service Desk module functions.

Service Desk overview

What's a Service Desk interaction?

Out-of-box Service Desk categories

Working with Service Desk interaction records

Working with Service Desk interaction records

User workflows

Describes the workflows used by Service Manager and the sub tasks that end users need to perform to execute the workflows.

Service Desk workflows and user tasks

Additional resources

Processes and Best Practices

1.3.1. Service Desk overview

Service Manager Service Desk enables you to restore normal service to your users as quickly as possible by resolving user requests on first contact or by immediately escalating them for proper handling. Service Desk helps you manage all user interactions, whether reported by using a Self-Service website or directly to your service desk. Service Desk enables you to achieve the following results:

- Control the relationship between Service Desk and other Service Manager applications
- Require Service Desk interactions to follow a set process
- Log and prioritize all calls to the service desk
- Categorize types of interactions and keep track of their resolution
- Provide first-line investigation and problem diagnosis, and quickly escalate user requests that can't be resolved within agreed-on time limits to fulfillment processes
- Request a service, provide information, or track previous requests without the assistance of a Service Desk Agent
- Communicate with customers and users to keep them informed of progress, impending changes, agreed-on outages, and other notifications.

Related topics

User Interaction Management
Processes and Best Practices

1.3.1.1. Streamlined Interaction introduction

To increase IT agents' efficiency and to enhance user experience, Service Manager introduces a new, streamlined interaction solution as of version 9.41. This streamlined interaction solution provides the following features and benefits:

- A simplified, intuitive form for interaction logging that requires fewer fields to fill and fewer clicks

  Note

  The interaction records still exist at the portal level only to exchange updates with the IT customers.

- A streamlined workflow that requires fewer steps in interaction handling
- A much easier way to manage recall and to measure process KPIs with all information at the same place
- Status synchronization between interaction and fulfillment records:
  - End users have full visibility of fulfillment status
  - IT agents can focus on fulfillment records
- Visibility on the attachment of interaction in the fulfillment record
- Consolidated menu entry for interaction creation, which is based on whether Smart Analytics and the streamlined interaction solution are enabled or not

Note

The streamlined interaction solution is disabled by default. This enables you to continue to use your existing interaction workflows without any issues or additional configuration after you upgrade to the current version of Service Manager.

To use the new streamlined interaction solution, simply manually enable it after you upgrade to Service Manager Codeless or Service Manager Hybrid (version 9.41 or later). For information about how to enable the streamlined interaction solution, see Enable streamlined interaction.

Once you enable the streamlined interaction solution, it replaces the previous interaction solution .

Related topics

Service Desk overview
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
Create a streamlined interaction

1.3.1.2. User Interaction Management

The Service Manager Service Desk application enables a Service Desk Agent to document and track calls. Service Desk provides one-click access to other Service Manager applications to automatically enter information received during the call.

Information that's obtained during a call can be used to open an incident record if a reported issue requires further action. Information can also be added to a record in another Service Manager application, such as Change Management.

User Interaction Management provides users with a single point of contact (the Service Desk) for all their IT Service related questions, requests, and issues, and provides resolutions to user requests within the agreed service level targets.

User Interaction Management includes all direct interactions between a user and the Service Desk by phone or by email. User Interaction Management also includes all user activities that occur by use of the self-service web portal (for example, searching the Knowledge base, checking for status updates, or logging an interaction).

Related topics

Service Desk overview
What's a Service Desk interaction?
Working with Service Desk interaction records

1.3.1.3. What's a Service Desk interaction?

Each time a user contacts the Service Desk, the Service Desk Agent uses Service Manager to create an interaction record. The Service Desk Agent records the user name, the name of the component about which the user is calling, and a description of the user request. After the Service Desk Agent collects this information, the Service Desk Agent performs the actions that are required to resolve the user request.

- If the user request is a complaint or compliment, the Service Desk manager can handle the interaction and close it.
- If the user request can be resolved by the Service Desk agent on first contact, the Service Desk agent will use the interaction to trigger an incident or a service request, and then close the incident or service request directly with the solution provided. Then, the interaction will be closed automatically.
- If the user request can't be resolved without escalation, the Service Desk agent can register a new record in a fulfillment process (such as the incident, service request, problem, or change fulfillment process) based on the Service Desk interaction. Service Desk copies information from the interaction record into the newly created fulfillment record.

For example, consider a user who can't print to a network printer:

- The user contacts the service desk for assistance.
- The Service Desk agent populates an interaction record with the relevant information.
- The Service Desk agent opens an incident from the interaction, and the incident is assigned to a technician.
- The technician discovers that the printer network connection is broken.
- The technician fixes the connection and closes the incident. The interaction is automatically closed accordingly.
- The Service Desk agent contacts the user and instructs the user to attempt printing to the network printer.
- If the user can successfully print, the Service Desk agent can send out survey based on the closed interaction. If the user still can't print, the Service Desk agent may register a new interaction and then trigger the interaction to a new incident.
- If the user wishes to report a related or new issue, the Service Desk agent opens a new interaction detailing the new issue that the user needs to report.

What's a category?

Service Manager uses the category to which a record is assigned to determine what work needs to be performed in order to close an interaction. For example, a service interruption logged in the incident category could have a different workflow than a service catalog request category interaction. The steps to manage and resolve the incident category interaction may differ from the steps for a service catalog request category interaction.

The category of the interaction determines which Process Designer workflow the interaction uses, and the workflow determines which forms and required fields are displayed. In addition, the category of the interaction determines which Service Manager process is invoked when you escalate an interaction. Categories of the complaint or compliment type are handled entirely as other types of interactions; however, you can't escalate these interactions.

Note

Your organization may have customized many workflows to handle interactions, and may also have applied different service agreements to those workflows.

Out-of-box Service Desk categories

The following table lists the out-of-box Service Desk categories. Depending on the category that you select, Service Manager may require you to further categorize the interaction record by selecting a Subcategory and Area.

Category	Description
complaint	Interactions in which the user reports less than satisfactory service or performance.
compliment	Interactions in which the user reports exemplary service or performance.
incident	Interactions that are opened specifically to start the incident management process.
problem	Interactions that are opened specifically to start the problem management process.
request for administration	Interactions that are opened specifically to request assistance from a system administrator, such as a password resets or rights to access a specific area.
request for change	Interactions that are opened specifically to start the change process.
request for information	Interactions that are requests for information only.
service catalog	Interactions that are requests for items from the service catalog.
service request	Interactions that are opened specifically to start the request fulfillment process.
support catalog	Interactions that are requests for support from the support catalog.

Categories can be created and revised by the System Administrator. Therefore, additional categories may be available to you.

Notify By field

The Notify By field specifies how to contact the person or customer who initiated an interaction. If the interaction record is associated with another record, the information in the Notify By field affects how the interaction record is closed. For example, if you associate an incident record to an interaction record, the related interaction record closes according to the information in the Notify By field when the incident record is closed.

Notify By setting	Description
None	The interaction record closes.
E-mail	Service Manager sends an email to the contact who is listed in the interaction record to inform the contact that the related record is closed. Service Manager then closes the interaction record.
Telephone	The contact who is listed in the interaction record is called back to be informed that the related record is resolved.

Note

In Service Manager Codeless, you can configure HTML Rulesets from the workflow to send notifications as needed.

Related topics

Service Desk overview
Working with Service Desk interaction records
Associating a Service Desk interaction with another application record
Create a streamlined interaction

1.3.1.4. Working with Service Desk interaction records

The Service Desk functions that are available to you depend on your user profile. You may be able to access all Service Desk functions, or you may be limited by the capability words that are assigned to you in your operator record.

Depending on your security role, the following Service Desk functions may be available to you.

Function	Description
Create Streamlined Interaction	From a blank Service Desk interaction form, you can log information about a service request, and add service request information to other Service Manager records, such as incident records. From this form, you can: • Trigger the interaction to an incident, a problem, a change, or a request. • Log a complaint or compliment. • Request an item from the Service Catalog or Support Catalog.
Interaction Queue	View the Service Desk interaction queue. You can also view other queues, such as the queues for incidents, problems, requests, changes, and any associated tasks lists.
Search Interactions	Specify filtering information to locate a specific Service Desk interaction record or set of records.

Hover-over forms for interaction records

Service Desk interaction record forms include hover-over forms. A hover-over form is displayed when you move the mouse over a field that supports hover-over forms. Hover-over forms are only displayed when the field contains data. The data that's displayed in hover-over forms is read-only.

Service Desk interaction record forms contain the following hover-over fields.

Field	Subform fields
Contact	Full Name, Telephone, Email
Service Recipient	Full Name, Telephone, Email

Related records in Service Desk

Service Manager lists all related records for an interaction record in the **Related Records** section. All information is read-only, but you can click a record ID to view the related record. The related records are listed in alphabetic order by the record file name: Change (cm3r), Request (request), Incident (probsummary). Within each module, the records are sorted by ID number, in ascending order.

The **Related Records** section in Service Desk contains the following fields.

Field	Description
ID	The number of the related record in the database.
Type	• Trigger To: The interaction is escalated to a different fulfillment process, such as Incident Management, Change Management, Request Fulfillment, and so on. • Fulfillment: When a service catalog or support catalog request (interaction) is submitted and approved, the fulfillment records are created accordingly.
Phase	The current phase of the related record.
Status	The current status of the related record.
Title	The title of the related record.

Activities section

The Activities section allows users to enter new updates for a record or to view journal updates and historic activities for a record.

Note

The Activities section is only available in the interactions that are categorized as complaints or compliments.

Journal Updates

Field	Description
New Update Type	Specifies or categorizes the activity update (for example, communication with the customer)
Visible to Customer?	Makes the update visible to customers so they can view related interactions via self service **Note** This option can be bypassed by setting the **Bypass Visibility Option** flag to **true** in the activityactions table. When this flag is set to **true**, automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to **false**, all activities are visible to customers if the **Customer Visible** option is selected.

Field	Description
New Update	Used to enter notes to explain and describe updates made for the record. If Journaling is enabled, the text entered here is displayed in Journal Updates. If Activities are enabled, the text entered in this field is displayed as an activity record for the selected activity type. **Note** The System Administrator is responsible for enabling Journaling and Activities.
Journal Updates	Displays text entered in the **New Update** field together with the operator's name and a timestamp for the update. This field displays information when Journaling is enabled.

Activity Logs

Select an activity type from the **Activity Type** drop-down list or select a user from the **User** drop-down list to filter the activity logs.

The activity log lists activities for the current record in order of occurrence, with the newest at the beginning of the list. Each activity log displays the following information:

- Date/Time
- Operator
- Description
- Type
- Visible to Customer icon (if you have selected the **Visible to Customer** checkbox)

Service Catalog requests

You can track your Service Catalog requests on the Service Desk update interaction form. When you submit a request by using the Service Catalog, you can view the request record in Service Desk. An interaction record for a Service Catalog request includes two additional sections, Approvals and Catalog Items.

Approvals section

The Approvals section contains the following subsections.

Subsections	Description
Current Approvals	Displays a table that lists current catalog requests and the approval details
Approval Log	Displays a log of the approval actions which have been performed regarding the catalog requests

Catalog Items section

The Catalog Items section contains the list of items for the catalog request and the details of the each item in the request. This section also includes the **Edit Cart** button, which links you to the Service Catalog so that you can change your catalog request.

Related topics

Service Desk workflows and user tasks
What's a Service Desk interaction?
Associating a Service Desk interaction with another application record

1.3.1.5. Triggering a fulfillment process from a streamlined Service Desk interaction

You can trigger a fulfillment process when you create a Service Desk interaction record by selecting a category. When the fulfillment process is triggered, a fulfillment record is created and linked to the interaction.

Note

The interactions that are in the "complaint" or "compliment" categories are handled in the Service Desk module without creating a new fulfillment record.

The following table describes the categories and their corresponding processes in the out-of-box system.

Interaction category	Fulfillment process to be triggered
incident	Incident Management
request for information	Request Fulfillment
problem	Problem Management
service request	Request Fulfillment
request for administration	Request Fulfillment
request for change	Change Management
service catalog	Service Catalog (Request a service catalog item or bundle) After you order an item or bundle, based on the connector definition of the ordered item or bundle, one or more fulfillment records are created and linked to the interaction.
support catalog	Support Catalog (Request a non-cart item) After you order a support catalog item, based on the connector definition of the ordered item, a fulfillment record is created and linked to the interaction.
complaint	Complaint process (the "Streamlined Complaint or Compliment" workflow) in Service Desk.
compliment	Compliment process (the "Streamlined Complaint or Compliment" workflow) in Service Desk

Interaction status

If a Service Desk interaction is linked to a fulfillment record such as an incident, the system automatically updates the interaction status based on the status of its fulfillment record:

- Once the fulfillment record is created and linked to the interaction, the interaction status moves to “Dispatched” automatically.
- Once the fulfillment record is in the “Pending Customer” status, the interaction status moves to “Pending Customer” automatically.
- Once the fulfillment record is abandoned or closed, the interaction status moves to “Closed” automatically.
- If the fulfillment record is an incident, when the incident moves to the "Resolved" status, the interaction status also moves to “Resolved” automatically. The interaction solution is also updated automatically according to the solution that was updated in the incident record.

If you want to configure the out-of-status mapping rules, see Configure status mapping between streamlined interactions and fulfillment records.

Related topics

Service Desk overview
What's a Service Desk interaction?
Service Desk workflows and user tasks

1.3.2. Service Desk user roles

The following table describes the responsibilities of the Service Desk user roles.

Service Desk user roles

Role	Responsibilities
User	• Report all IT-related requests to the service desk or use the self-service web pages • Validate solutions and answers that are provided by the IT department to a registered service request
Service Desk Agent	• Open interactions based on contact with the user • Solve and close interactions • Provide status updates to users on request • Register incidents based on user interactions and assign them to the correct support groups • Register requests for change, based on user interactions • Register service requests, based on user interactions • Register problems, based on user interactions • Validate solutions provided by support groups • Report and verify solutions to users • Monitor the Service Level Agreement (SLA) targets of all registered interactions • Communicate service outages to all users

Role	Responsibilities
Service Desk Manager	• Appoint people to the required roles • Manage resources assigned to the service desk • Manage service desk activities • Attend CAB meetings • Report any issue that could significantly impact the business to senior managers • Take overall responsibility for incident and service request handling on the service desk • Monitor and report on service desk performance • Make improvements to the service desk

Related topics

User roles
Add a user role record
Delete a user role record
Search for a user role record

1.3.3. Service Desk workflows and user tasks

Every user contact with the Service Desk is logged in an interaction record. User Interaction Management is the process for handling all interactions with the service desk that are received from self-service webpages or directly from service desk personnel. This process streamlines service desk activities, thereby decreasing the workload for second line support teams.

In an out-of-box system, depending on the interaction categories, different fulfillment processes are triggered to help fulfill the user requests. Only the interactions that are in the "complaint" or "compliment" categories are handled in the Service Desk module. The following table describes the categories and their corresponding processes in the out-of-box system.

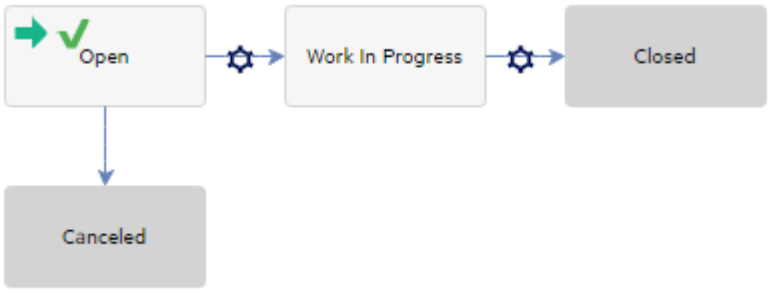
Interaction category	Workflow	Fulfillment process to be triggered
incident	Streamlined Service Desk	Incident Management
request for information	Streamlined Service Desk	Request Fulfillment
problem	Streamlined Service Desk	Problem Management
service request	Streamlined Service Desk	Request Fulfillment
request for administration	Streamlined Service Desk	Request Fulfillment
request for change	Streamlined Service Desk	Change Management
service catalog	Streamlined Service Catalog	Service Catalog (Request a service catalog item or bundle) After you order an item or bundle, based on the connector definition of the ordered item or bundle, one or more fulfillment records are created and linked to the interaction.

Interaction category	Workflow	Fulfillment process to be triggered
support catalog	Streamlined Service Desk	Support Catalog (Request a non-cart item) After you order a support catalog item, based on the connector definition of the ordered item, a fulfillment record is created and linked to the interaction.
complaint	Streamlined Complaint or Compliment	Complaint process (the "Streamlined Complaint or Compliment" workflow) in Service Desk.
compliment	Streamlined Complaint or Compliment	Compliment process (the "Streamlined Complaint or Compliment" workflow) in Service Desk

For the categories that are associated with the "Streamlined Service Desk" or "Streamlined Service Catalog" workflow, when the fulfillment process is triggered, a fulfillment record is created and linked to the original interaction. When the fulfillment record is closed, the corresponding interaction record is also closed automatically.

For the complaint and compliment categories, the "Streamlined Complaint or Compliment" workflow is provided so that the interactions in these categories can be handled appropriately in the Service Desk module without triggering a fulfillment process.

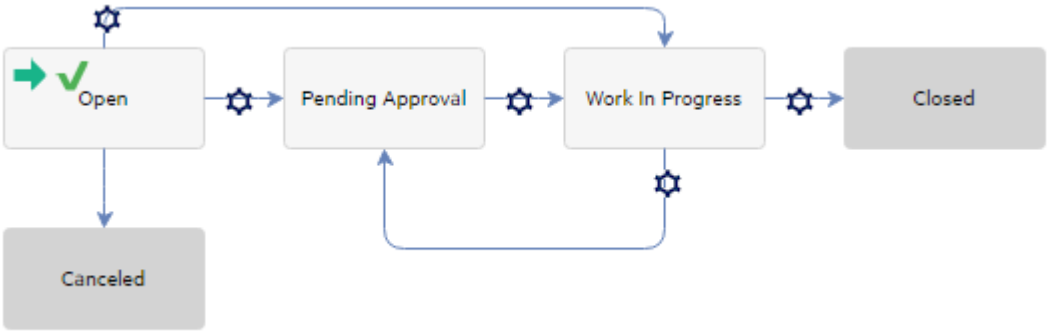
- Streamlined Service Desk



The Streamlined Service Desk workflow handles interactions in the following categories and triggers corresponding fulfillment processes:

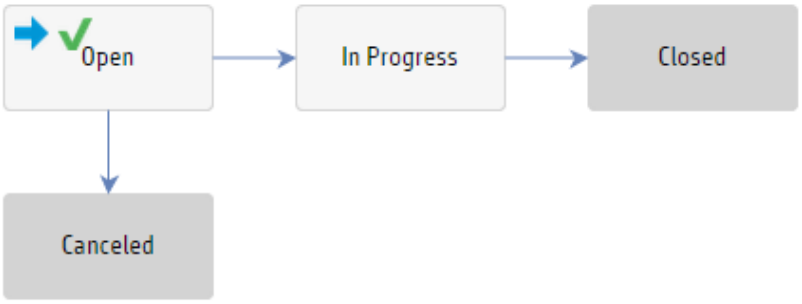
- incident
- request for information
- problem
- service request
- request for administration
- request for change
- support catalog

- Streamlined Service Catalog



The Streamlined Service Catalog workflow handles service catalog requests. After you order a service catalog item, based on the connector definition of the ordered item, one or more fulfillment records are created and linked to the interaction.

- Streamlined Complaint or Compliment



The Streamlined Complaint or Compliment workflow handles user complaints and compliments.

To view more workflow diagrams and other information about the Service Desk process, refer to *Process Designer Best Practices* that's linked to in the related topics.

Related topics

Service Desk overview
User Interaction Management
Processes and Best Practices

1.3.3.1. Self-Service user interactions

By using the self-service web environment, users can perform the following activities without contacting the Service Desk:

- Search the Knowledge Base to find an answer to a question or issue
- Monitor the status of previously reported interactions
- Log new interactions
- Order items from the service catalog

1.3.3.1.1. Request a Service Catalog item or bundle

Applies to User roles: User

You can easily order products and services from Service Catalog without needing to contact the service desk. Items in the Service Catalog can include an item picture, description, attachments, and purchasing options. Some items in the Service Catalog are for information only, such as instructions on how to order items from an outside service. Information-only items only have an item picture, description, and attachments; they don't include purchasing options.

> **Note**
>
> You may also order one or more products and related service tasks by choosing a bundle.

To request an item or bundle from the Service Catalog, follow these steps:

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.
3. Click the item you want to request to view the Order Information for that item.

   > **Note**
   >
   > Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.

4. In the Requested for field, select the Individual who will receive the good or service.
5. Select the quantity and any other item options, and then click **Add to Cart** to add the item you want to your cart.
6. From the Order from Catalog home page, click **View Cart/Checkout**.
7. Click **Submit Request**.
8. Type any required information justifying the request.
9. Click **Submit**.

Related topics

Service Catalogs
Out-of-box Service Catalog
Shopping cart
Service Catalog interface
Service Catalog items and bundles
Service Catalog connectors
Approve a request

1.3.3.1.2. Create a self-service request

Applies to User roles: User

If you can't find an existing solution to your incident or problem in the knowledge base, you can create a self-service request for a Service Desk Agent to resolve.

To create a self-service request, follow these steps:

1. Click **Submit a Request**.
2. In the Submit a Request window, verify the **Service Recipient** and **Primary Contact** fields.
3. Select your notification preferences in the **Notify By** field.
4. Select the **Urgency** of your issue.
5. Type a title and a description for your issue.
6. Click **Add Files**, and then follow these steps to add any relevant attachments:
   - Click **Browse** to locate your attachment.
   - Select the file to add.
   - Click **OK**.

   Note

   Service Desk agents or other operators can access your submitted attachments directly from the fulfillment records that are linked to your request.

7. Click **Submit**.

   Note

   After you submit your self-service request, Service Manager automatically creates the corresponding fulfillment record in the system so that your self-service request can be handled appropriately.

Related topics

Searching records
Close a self-service request
Resubmit a self-service request

Update a self-service request
View a self-service request

1.3.3.1.3. Close a self-service request

Applies to User roles: User

You can close a self-service request if you are no longer experiencing the problem or if you have another reason for closing the request. If you need to resubmit the request, you can do so from your list of closed self-service requests.

To close a self-service request, follow these steps:

1. Click **View Open Requests**. Service Desk displays a table of your open requests and the status of each request.
2. Select the Service Desk interaction record that you want to close.
3. Click **Update**, type an update description if necessary, and then click **Close Request**.
4. Select a reason to close the request, and then click **Submit**. The Service Desk interaction record is closed.

> **Note**
>
> When you close your self-service request, an activity update is added to the corresponding back-end fulfillment records to reflect that you have closed the self-service request.

Related topics

Self-service access
Searching records
Create a self-service request
Resubmit a self-service request
Update a self-service request
View a self-service request

1.3.3.1.4. Resubmit a self-service request

Applies to User roles: User

If you disagree with a proposed solution to a closed request, you can resubmit the request and provide a reason why the resolution is unsatisfactory. When you resubmit a closed interaction record, Service Manager assigns you a new interaction identification number to track the request.

To resubmit a self-service request, follow these steps:

1. Click **View Closed Requests**.
2. Select the Service Desk interaction record that you want to resubmit.
3. Click **Resubmit**. Service Desk creates a new interaction record with a different identification number. However, the new record displays the same field information as the original Service Desk interaction record.
4. Type the reason why you are resubmitting the request.
5. Click **Save & Exit**.

Related topics

Self-service access
Searching records
Create a self-service request
Close a self-service request
Update a self-service request
View a self-service request

1.3.3.1.5. Update a self-service request

Applies to User roles: User

You can update any of your open self-service requests with additional information to assist the Service Desk Agent.

To update a self-service request, follow these steps:

1. Click **View Open Requests** to display a table of your open requests and the status of each request.
2. Select the Service Desk interaction record that you want to update.
3. Click **Update**, and then enter any new request details, additional descriptions, or add relevant attachments as needed.
4. Click **Save & Exit**.

Note

If the Service Desk interaction record is linked to a request , an incident, or a change record, your updates are automatically copied to the **Activities** section in the linked record.

Related topics

Self-service access
Searching records
Create a self-service request
Close a self-service request
Resubmit a self-service request
View a self-service request

1.3.3.1.6. View a self-service request

Applies to User roles: User

You can view your service requests to view the status of any outstanding interactions or to view a list of all your past requests.

To view a self-service request, follow these steps:

1. Click **View Open Requests** or **View Closed Requests** to display a table of requests and the status of each request.

   The status of a Service Desk interaction is based on the status of its fulfillment record:

   - Once the fulfillment record is created and linked to the interaction, the interaction status moves to “Dispatched” automatically.
   - Once the fulfillment record is in the “Pending Customer” status, the interaction status moves to “Pending Customer” automatically.
   - Once the fulfillment record is abandoned or closed, the interaction status moves to “Closed” automatically.
   - If the fulfillment record is an incident, when the incident moves to the "Resolved" status, the interaction status also moves to “Resolved” automatically.

2. Select the Service Desk interaction record that you want to view.

3. Click **Back** to return to the list of service requests.

Related topics

Self-service access
Searching records
Create a self-service request
Close a self-service request
Resubmit a self-service request
Update a self-service request

1.3.3.2. Create a streamlined interaction

User role: Service Desk Agent

You can create an interaction to collect information about a service or support request from a user. After you categorize the user request (such as an incident, a request for change, a service catalog request, a complaint, and so on), you can proceed with the process that associated with the category to solve the user request.

To register a new Service Desk interaction, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. In the **Contact** field, select the contact person for the interaction.

   > **Note**
   >
   > If you use the Fill function to replace the information in a field that already has data, clear the field before you click **Fill**. If the field contains data, the Fill function can't operate.

3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. Type a description for the interaction.
6. In the **Category** field, select the category for the interaction.
7. (Optional) In the **Subcategory** and **Area** fields, select the subcategory and area for the interaction.
8. In the **Affected Service** field, select the affected service.
9. Click the **Continue** button. After you click **Continue**, an interaction record is created and the fulfillment process associated with the selected category is initiated. Then, you can follow the workflow of the triggered process to handle the user request.

   The following table describes the categories and their corresponding processes in the out-of-box system.

Interaction category	Fulfillment process to be triggered
incident	Incident Management
request for information	Request Fulfillment
problem	Problem Management
service request	Request Fulfillment
request for administration	Request Fulfillment
request for change	Change Management
service catalog	Service Catalog (Request a service catalog item or bundle) After you order an item or bundle, based on the connector definition of the ordered item or bundle, one or more fulfillment records are created and linked to the interaction.
support catalog	Support Catalog (Request a non-cart item) After you order a support catalog item, based on the connector definition of the ordered item, a fulfillment record is created and linked to the interaction.

Interaction category	Fulfillment process to be triggered
complaint	Complaint process in Service Desk
compliment	Compliment process in Service Desk

Tip

If the fulfillment record has not been created, you can click **Cancel** or **Back** to go back to the interaction logging form. Then, you can change the interaction category to trigger another fulfillment process.

10. Enter the information as necessary in the fulfillment record, and then click **Save**.

1.3.3.3. Change the category of a standalone interaction

Applies to User roles: Service Desk Agent

If an interaction is a standalone interaction without any related fulfillment records, you can change the category of the interaction, and then trigger the corresponding fulfillment process. For example, you may need to trigger a different fulfillment process to resume your work on a user request.

Note

The **Change Category** option is only available for the standalone interaction without any related fulfillment records. If the interaction is already in the Dispatched status and linked to a fulfillment record, you can't change the category of the interaction.

To change the category of an open interaction, follow these steps:

1. Click **Service Desk** > **Interaction Queue**. Service Manager displays the list of interactions.
2. Locate and open a Service Desk interaction record that's in the Open status and is not a service catalog or support catalog request.
3. On the interaction detail form, click **More** > **Change Category**.
4. Select a category, and then click **Yes** for confirmation.
5. Click the **Continue** button. The corresponding fulfillment process is triggered.
6. Follow the triggered fulfillment process to fulfill the user request.

Related topics

Service Desk overview
Service Desk workflows and user tasks
Create a streamlined interaction
Searching records

1.3.3.4. Cancel a new Service Desk interaction

Applies to User roles: Service Desk Agent

You can cancel an interaction (not a service catalog or support catalog request) that's in the Open status. For example, if the user no longer experiences the problem, you can cancel the interaction without triggering a fulfillment process.

> **Note**
>
> The **Cancel Interaction** button is only available for the interaction that's still in the Open status and is not a service catalog or support catalog request. If the interaction is in the Dispatched status, you can't cancel the interactions.

To cancel a Service Desk interaction, follow these steps:

1. Click **Service Desk** > **Interaction Queue**. Service Manager displays the list of interactions.
2. Locate and open a Service Desk interaction record that's in the Open status and is not a service catalog or support catalog request.
3. On the interaction detail form, click the **Cancel Interaction** button.
4. click **Yes** to confirm that you want to cancel this interaction.

   The interaction is now in the Canceled status.

Related topics

Service Desk overview
Searching records

1.3.3.5. Close a first-time resolved Service Desk interaction

Applies to User roles: Service Desk Agent

If a Service Desk interaction is triggered to an incident or a request record, you can resolve the user request on the first intake in the Logging phase of the triggered incident or request record. After the related incident or request is resolved, the Service Desk interaction is also closed automatically.

To close a first time Service Desk interaction that's triggered to an incident or a request record, follow these steps:

1. From the System Navigator, click **Service Desk** > **Create Streamlined Interaction**.
2. Select a category that's associated with the Incident Management or Request Fulfillment process.

   In an out-of-box system, the incident category is associated the Incident Management process while the service request, request for information, and request for administration are associated with the Request Fulfillment process.
3. Specify the information in other mandatory fields.
4. Click **Continue**. The interaction is created and the New Incident or New Request form is displayed.
5. On the New Incident or New Request form, specify the information in the mandatory fields.
6. Type a solution for the incident or request.
7. Click the **Close Directly** button on the toolbar.

   Service Manager displays the **Close Incident** or **Close Request** page.
8. Specify the information in the mandatory fields such as **Closure Code**.
9. Click **Finish**.

   The status of the incident or request changes to **Closed**. Meanwhile, the original interaction is also closed automatically along with the incident or request.

Related topics

Service Desk overview
Searching records

1.3.3.6. Handle a complaint or compliment

Applies to User roles: Service Desk Manager

If a user contacts you regarding a service complaint or compliment, you can create a Service Desk interaction to collect information about the user's service request. Categorize the service request as a complaint or compliment, and then solve the request.

To register a complaint or compliment, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. Specify the **Contact** for the interaction.

   > **Note**
   >
   > If you use the Fill function to replace the information in a field that already has data, clear the field before you click **Fill**. If the field contains data, the Fill function can't operate.

3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. In the **Description** field, type a description for the interaction.
6. In the **Category** field, select "complaint" or "compliment" as the category.
7. (Optional) In the **Subcategory** and **Area** fields, select the subcategory and area for the interaction.
8. (Optional) In the **Affected Service** field, select the affected service.
9. Click the **Continue** button. An interaction record is created and the interaction detail form is displayed.
10. On the **Assignment** tab, specify the assignment group and the assignee.
11. Click **Save**.

To close a complaint or compliment, follow these steps:

1. Locate and open the complaint or compliment interaction record that's in the "Open" status.
2. Make sure that the **Assignment Group** and the **Assignee** fields are specified, and the click **Fulfill**. The interaction status changes to "In Progress."
3. In the **Closure Code** field, specify a closure code.

4. In the **Solution** field, type the solution information.
5. Click **Close**. The complaint or compliment interaction record is closed.

Related topics

Service Desk overview
Searching records
Create a streamlined interaction

1.3.3.7. Access Service Desk views

Applies to User Roles: Service Desk Agent

Streamlined Service Desk views contained in the Favorites and Dashboards navigation pane enable you to easily and quickly access specific types of records.

To view the streamlined Service Desk views, click **Favorites and Dashboards** > **Streamlined Service Desk**. Service Manager provides the following default streamlined Service Desk views:

- Interactions with initial process target – Breached
- Interactions with initial process target – Breach within 1 hour
- Interactions with initial process target – Breach within 4 hours
- Open Complaints and Compliments Assigned to Me

Related topics

Views and favorites

1.3.3.8. Service Catalog support

When a user (the service recipient) submits a Service Catalog or Support Catalog request through the ESS portal, a Service Desk interaction is created in the “service catalog” or specified category to help the Service Desk agent to fulfill the request for the service recipient.

A Service Desk agent can also submit a Service Catalog request (**Service Catalog** > **Order from Catalog**) or a Support Catalog request (**Service Catalog** > **Non-cart Catalog Requests**) on behalf of a user who requests the service through an email or phone call. The Service Catalog request triggers a Service Desk interaction in the "service catalog" category while the Support Catalog request triggers an interaction in the specified category.

Order from Service Catalog

Applies to User Roles: Service Desk Agent

You can submit a Service Catalog or Support Catalog request on behalf of a user requesting the service (the service recipient). The request triggers Service Desk interaction in the "service catalog" or specified category.

Register a Service Catalog request

To register a Service Catalog request on behalf of a service recipient, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. In the **Contact** field, select the contact person for the interaction.
3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. Type a description for the interaction.
6. In the **Category** field, select "service catalog" as the category for the interaction.
7. Click **Continue**. The form for selecting service catalog items is displayed.

> **Note**
>
> You can also search for and open an existing interaction that's in the "Open" status and the "service catalog" category, and then click the **Continue** button.

8. Add items to your cart by doing one of the following:
   - Search for the desired items:
     The items for which you search are filtered against the service recipient's capability words. Items that match the search criteria don't display in search results unless the service recipient has the required capability words.
   - Select from **Most Popular Requests**:
     The Most Popular Requests list displays only the cart items that are allowed by the service recipient's capability words.
   - Browse the service catalog:
     The service catalog displays only the categories, items, and bundles that are allowed by the service recipient's capability words.
9. After you add the desired items to your cart, click **View Cart/Checkout**.
10. If needed, click **Remove** to remove an item or click **Return to Catalog** to add more items to your cart, and then click **View Cart/Checkout** again.
11. Click **Resubmit Request**.

    The catalog items you requested appear in the **Catalog Items** section of the interaction record. Once this Service Catalog request is approved, fulfillment records are created and linked to the interaction based on the connector definition of the ordered items.

Register a Support Catalog request

To register a Support Catalog request on behalf of a service recipient, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. In the **Contact** field, select the contact person for the interaction.
3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.

5. Type a description for the interaction.
6. In the **Category** field, select "support catalog" as the category for the interaction.
7. Click **Continue**. The form for selecting support catalog items is displayed.

   > **Note**
   >
   > You can also search for and open an existing interaction that's in the "Open" status and the "support catalog" category, and then click the **Continue** button.

8. Select your support catalog item by doing one of the following:

   - Search for the desired items:
     The items for which you search are filtered against the service recipient's capability words. Items that match the search criteria don't display in search results unless the service recipient has the required capability words.
   - Select from **Most Popular Requests**:
     The Most Popular Requests list displays only the cart items that are allowed by the service recipient's capability words.
   - Browse the support catalog:
     The support catalog displays only the categories and items that are allowed by the service recipient's capability words.

9. After you select the desired item, specify the required information, and then click **Submit**.

   The support catalog item you requested appears in the **Catalog Items** section of the interaction record. In addition, based on the connector definition of the ordered item, a fulfillment record is created and linked to the interaction.

1.3.3.9. Using mass update with Service Desk interaction record lists

The mass update feature enables you to update the value of one or more fields in a list of records. The system provides a template form that displays the current field values and enables you to type or select a new value for the displayed fields. In compliance with the data policy of your organization (as recorded in the Data Policy form), the Mass Update template form doesn't display read-only fields in the records. The mass update option is available from both interaction and incident record lists.

When you perform a mass update operation, make sure that the value you type or select for a particular field is the value you want in all the records that you selected. When you update incident records, you must also update the incident activity data.

System Administrators can determine if a field is displayed in the Mass Update template form by directly editing the Data Policy record for the form. To remove a field from the Mass Update form, change the Usage Type column for the field to System. If a System Administrator wants to grant Template Mass Update permissions to other operators, the System Administrator must update the operator profile to include the Template Mass Update and Complex Mass Update capabilities.

Update multiple interaction records

Applies to User roles: Service Desk Manager

You can use the mass update feature to update one or more values in multiple interaction records. Remember that the value you enter for a particular field becomes the value for all of the records you selected when you perform a mass update.

To update multiple records, follow these steps:

1. Locate the records by using one of the following methods:
    - Go to the Interaction Queue to display a list of records.
    - Use search or advanced search to find one or more records.
2. Click **More** or the **More Actions** icon, and then select **Mass Update**.
3. Double-click the field that you want to update.
4. Type the value for the field or use the Find feature to display a list of potential values for the field.
5. Click **Next**, and then repeat steps 3 and 4 for additional fields as needed.

6. Update the Activity Action field for the updates you made in the Mass Update template, and then click **Execute**.

To use complex update to update multiple records:

1. Locate the records by using one of the following methods:
   - Go to the Interaction Queue to display a list of records.
   - Use search or advanced search to find one or more records.
2. Select the records that you want to update.
3. Click **Mass Update**, and then click **Complex Update**.
4. Type the instructions (in RAD syntax) to be executed one time at the beginning of the mass add or mass update operation.
5. Type the instructions (in RAD syntax) for action on each record.
6. Click **Execute**.

Contents

- 1 Update multiple interaction records

1.3.3.10. Access Service Desk reports

User Roles: Service Desk Agent

The Reporting tool in Service Manager provides a number of out-of-box reports on the interaction data in your system. You can view these reports through a dashboard named Service Desk Overview (Global). You can also create your own dashboards to display other reports of your interest.

To access Interaction Management reports, follow these steps:

1. Click **Service Desk** > **Service Desk Overview**.

   By default, the **Service Desk Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. If needed, click the **Create New Dashboard** button to add your own dashboards. For details, see Create a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default dashboard.

1.3.4. Service Desk administration

Service Desk administration consists of the following:

- Service Desk Environment: Service Desk application contains an environment record for interactions. This record contains options that defines the functionality of the Service Desk application for all Service Desk users.
- Settings: You can enable the "Create New Interaction" menu entry, and you can set the default category for new interactions.

1.3.4.1. Enable streamlined interaction

Applies to User roles: System Administrator

To enable the streamlined interaction solution, follow these steps:

1. Click **Service Desk** > **Administration** > **Settings** in the System Navigator, and then clear the **Enable “Create New Interaction” Menu Entry** option.
2. Manually change the workflows that are associated to interaction categories to the streamlined interaction workflows according to the following table:

Interaction category	Associated workflow in SM Codeless (Out-of-box)	Workflow for streamlined interaction	Fulfillment process to be triggered from a streamlined interaction
incident	ServiceDesk	Streamlined Service Desk	Incident Management
request for information	ServiceDesk	Streamlined Service Desk	Request Fulfillment
problem	ServiceDesk	Streamlined Service Desk	Problem Management
service request	ServiceDesk	Streamlined Service Desk	Request Fulfillment
request for administration	ServiceDesk	Streamlined Service Desk	Request Fulfillment
request for change	ServiceDesk	Streamlined Service Desk	Change Management

Interaction category	Associated workflow in SM Codeless (Out-of-box)	Workflow for streamlined interaction	Fulfillment process to be triggered from a streamlined interaction
support catalog	N/A	Streamlined Service Desk	Support Catalog (Request a non-cart item) After you order a support catalog item, based on the connector definition of the ordered item, a fulfillment record is created and linked to the interaction.

Interaction category	Associated workflow in SM Codeless (Out-of-box)	Workflow for streamlined interaction	Fulfillment process to be triggered from a streamlined interaction
service catalog	Service Catalog	Streamlined Service Catalog	Service Catalog (Request a service catalog item or bundle) After you order an item or bundle, based on the connector definition of the ordered item or bundle, one or more fulfillment records are created and linked to the interaction.
complaint	ServiceDesk	Streamlined Complaint or Compliment	Complaint process (the "Streamlined Complaint or Compliment" workflow) in Service Desk.

Interaction category	Associated workflow in SM Codeless (Out-of-box)	Workflow for streamlined interaction	Fulfillment process to be triggered from a streamlined interaction
compliment	ServiceDesk	Streamlined Complaint or Compliment	Compliment process (the "Streamlined Complaint or Compliment" workflow) in Service Desk

3. Manually activate the "support catalog" interaction category (that is, set the **Active** setting to TRUE).
4. (Optional) Enable the following out-of-box views for Streamlined Interaction by changing the Audience and Ownership from the default value ("falcon") to the appropriate operators in your organization:

   Folder:

   - Streamlined Service Desk

   Views:

   - Interactions with initial Process Target – Breached;
   - Interactions with initial Process Target – Breach within 1 hour;
   - Interactions with initial Process Target – Breach within 4 hours;
   - Open Complaints and Compliments Assigned to Me
5. Log out and log back on.

   Now, the streamlined interaction is enabled and you can create streamlined interactions from **Service Desk** > **Create Streamlined Interaction**.

Note

If Smart Analytics is enabled, the menu entry for streamlined interaction is still **Create Smart Interaction** after you enable streamlined interaction, but the underlined interaction workflows are now supported by streamlined interaction. For more information, see Consolidated menu entry for interaction creation.

Related topics

Service Desk administration

1.3.4.2. Configure the Service Desk settings

Applies to User roles: System Administrator

To configure Service Desk settings, follow these steps:

1. Click **Service Desk** > **Administration** > **Settings** in the System Navigator.
2. Configure the following settings for the Service Desk application:

Setting	Description
Enable "Create New Interaction" Menu Entry	This setting determines whether the interactions will be created to follow the legacy interaction workflows (select the checkbox) or the new streamlined workflows introduced as of SM 9.41 (clear the checkbox). The streamlined interaction solution is disabled by default so that if you want to run your previous interaction processes (workflows) as in 9.40 Codeless, you can continue to use them without any issues or additional configuration when you upgrade to Service Manager 9.41 or later. To enable the streamlined interaction solution, see Enable streamlined interaction. **Note** The menu entry for creating new interactions also depends on whether the SM Smart Analytics feature is enabled or not. For more information, see *Consolidated menu entry for interaction creation* below.

Setting	Description
Default Category	The default category when you create a new interaction. The out-of-box value for this parameter is None. The default category is used when you click **Create New Interaction**: ◦ If no default category is defined, you are prompted to select a category before the interaction detail screen appears. ◦ If the default category is defined, the interaction detail screen is displayed directly and the **Category** field is populated with the default category. **Note** The default category doesn't apply to the ESS requests. For support requests submitted through ESS, interactions are always created in the "incident" category or the category suggested by SM Smart Analytics.

If you use streamlined interaction and later you decide to select the **Enable "Create New Interaction" Menu Entry** checkbox to go back to the legacy interaction workflow, you need to go into the Interaction category records and change the workflow back to the non-streamlined workflows. The following table is for your reference:

Interaction category	Associated non-streamlined workflow in SM Codeless	Associated workflow in streamlined interaction solution
complaint	ServiceDesk	Streamlined Complaint or Compliment
compliment	ServiceDesk	Streamlined Complaint or Compliment
incident	ServiceDesk	Streamlined Service Desk
problem	ServiceDesk	Streamlined Service Desk
request for administration	ServiceDesk	Streamlined Service Desk
request for change	ServiceDesk	Streamlined Service Desk
request for information	ServiceDesk	Streamlined Service Desk
service request	ServiceDesk	Streamlined Service Desk
support catalog	N/A	Streamlined Service Desk

Interaction category	Associated non-streamlined workflow in SM Codeless	Associated workflow in streamlined interaction solution
service catalog	Service Catalog	Streamlined Service Catalog

3. Click **Save**.

Consolidated menu entry for interaction creation

Depending on whether Smart Analytics is enabled and whether the new streamlined interaction solution is used, the menu entries for creating new interactions are different. To learn how the menu entries are consolidated, see the following table:

Smart Analytics	Enable "Create New Interaction" Menu Entry	Consolidated menu entry for interaction creation
Enabled	Checked (to use the original interaction solution as in SM 9.40)	Two menu entries are available: • **Create New Interaction** • **Create Smart Interaction**
Enabled	Unchecked (to use the new streamlined interaction solution)	**Create Smart Interaction**
Disabled	Checked	Create New Interaction
Disabled	Unchecked	Create Streamlined Interaction

Note

For operators with the "SysAdmin" capability word, all the three menu entries are visible regardless the system setting.

Related topics

Service Desk administration

1.3.4.3. Configure the Service Desk environment

Applies to User roles: System Administrator

To configure a Service Desk environment record, follow these steps:

1. Click **Service Desk** > **Administration** > **Environment**.
2. Configure the settings for your Service Desk environment.

   You can use the following table for more information on the options that you can set by selecting or clearing the checkbox:

Check box	Description
Delay Assigning Interaction ID?	If selected (set to true), prevents Service Manager from assigning an identifier to the Interaction immediately. When this option is active, there is no unique identifier to tie an attachment within the file when you first open it. Therefore, you can't save attachments when you open the ticket. You can add attachments when saving an update after the record has a unique identifier. **Note** This option doesn't work under the following circumstances: ◦ When the system automatically creates an Interaction from a chat request in background. ◦ When a Service Desk agent manually creates an additional Interaction for a chat request.
User Operator Full Name?	If selected (set to true), uses the operator's Full Name instead of the login ID when time stamping reports (on open, update, and so on) for records in the application environment

You can use the following table to specify information about the Post Back links:

Post Back Link	Description
IM Post back Link:	The link record used to post information from a related incident to the interaction when the incident is closed.
ChM Post back Link:	The link record used to post information from a related change to the Interaction when the change is closed
RM Post back Link:	The link record used to post information from a related request to the Interaction when the request is closed.
PbM Post back Link:	The link record used to post information from a related problem to the Interaction when the problem is closed.
Copy/Open Link:	The record from which to copy the data. When you copy a record, data from the record specified here is copied to the new record based on the link record.

You can specify information about the Service Desk Manager Email, Service Desk Group, Service Desk Phone Number, and Service Desk Email. These fields are used in notifications and for assigning the interaction to a default group.

Tip

You can use the field help to view the description for each setting. To view the help on field:

- Web client: Select a field, and then press **F1**.
- Windows client: Select a field, and then press **Ctrl+H**.

3. Click **Save**.

Note

If the legacy interaction workflows (not the new streamlined interaction workflows introduced as of SM 9.41) are used, you need to manually enable the following settings so that they can take effect:

Setting	Description
Service Desk Record Relationship Models	• **Full Service Desk Model**: In this model, the state of a Service Desk interaction record changes when each related record closes. The closed state of an interaction record depends on the notification value chosen for the "Notify By" field in the interaction record. For example, if the notification value is "Telephone," the interaction record has a required action before it is closed. This action describes why the customer must be contacted. It also prevents the interaction record from closing until all required actions are complete. In this case, the interaction record goes into the Callback state before it is closed. For more information, see Full Service Desk model. • **All Records close Independently**: In this model, all Service Desk interaction records close independently. The state of related records doesn't affect closing an interaction record, and closing the interaction record doesn't affect related records. • **Close Interactions when Related Record closes**: In this model, when the last related record closes, the

Setting	Description
	Service Desk interaction record closes. • **Cannot close Related Record until Interactions are closed**: In this model, records related to a Service Desk interaction can't close until the interaction record is closed. • **Cannot close Interactions until Related Records are closed**: In this model, a Service Desk interaction record can't close until all related Incident Management records, Change Management records, and Request Management records are closed.
Return to Blank Interaction?	If selected (set to true), returns the user to a blank (new) Interaction screen after the creation of an Incident record. If cleared (set to false), the user will return to the Interaction they created the Incident from.
Post resolution to Interactions?	If selected (set to true), posts the resolution of a closed ticket to the related Interaction report.
Copy Associations of Cloned Interactions?	Copies the associations of a cloned interaction.

Related topics

Service Desk administration

1.3.4.4. Configure status mapping between streamlined interactions and fulfillment records

Applies to User roles: System Administrator

In the out-of-box system, the status of a streamlined interaction is dependent on the status of its related fulfillment record. For the out-of-box status mapping rules, see Interaction status. However, you may want to define your own status mapping to meet your business needs. To do this, you can open the status mapping script (StatusSyncServiceConfig), and then modify the configuration item (STATUS_MAPS_STREAMLINED) that defines status mapping between streamlined interactions and fulfillment records.

Note

In the StatusSyncServiceConfig script, you can see two configuration items (STATUS_MAPS_ESCALATION and STATUS_MAPS_STREAMLINED):

- STATUS_MAPS_STREAMLINED is the configuration item that defines the status mapping between the streamlined interactions and the fulfillment records, which have a "Trigger To" or "Fulfillment" relationship.
- STATUS_MAPS_ESCALATION is the configuration item that defines the status mapping between the non-streamlined interaction and the fulfillment records, which have a "Escalation'" relationship.

To configure status mapping between streamlined interactions and fulfillment records, follow these steps:

1. Click **Tailoring** > **Script Library** in the System Navigator.
2. In the **Name** field, type StatusSyncServiceConfig, and then click **Search**. The detailed script is displayed.
3. Search for and locate the STATUS_MAPS_STREAMLINED configuration item.
4. Modify the codes in STATUS_MAPS_STREAMLINED as needed. The following example defines the status mapping between streamlined interactions and incidents (in a "Trigger To" or "Fulfillment" relationship). See the comments inline for instructions.

```
//The file name of incidents
probsummary: {
//The status field of incidents
statusField: "problem.status",
statusMap: {
"Pending Customer": {
status: "Pending Customer",
priority: 0
},
"Resolved": {
 //When the incident is Resolved, the interaction status
changes to “Resolved” as well.
status: "Resolved",
//The status mapping priority.
priority: 0,
syncFields: [
//Copy the incident resolution to the interaction.
{
source: "resolution",
target: "resolution",
override: true
}]
}
},
//When the request is not in the status mapping above, use
the default status.
defaultStatus: {
status: "Dispatched",
priority: 1
},
//When the request is in the ignored status, the system
doesn't perform status synchronization.
ignoreStatus: ["Closed"]
}
```

> **Note**
>
> When an interaction has more than one fulfillment records that need status synchronization, the interaction may be mapped to different statuses. In this case, the status mapping with the highest priority will be used. For the priority number,‘0’ is the highest priority.

5. Click **Save**.
6. From **Miscellaneous** > **System Status**, stop and then restart the "scheduledAction" process.

   The updated status mapping between streamlined interactions and fulfillment records now takes effect.

If you want to modify the STATUS_MAPS_ESCALATION configuration item, you can refer to the following example that defines the status mapping between the non-streamlined interactions and requests, which are in the "Escalation" relationship.

```
//the file name of requests
request: {
//The status field of requests
statusField: "status",
statusMap: {
//When the request is fulfilled, the interaction is changed to
the “Resolved” status.
"Fulfilled": "Resolved",
}
}
```

Related topics

Service Desk administration

1.3.4.5. Add a Service Desk assignment group

Applies to User Roles: System Administrator

Service Desk assignment groups specify the Service Desk agents in the group so that when someone creates a new interaction it can be assigned to the applicable group. Typically, assignment groups are organized by location and expertise. To facilitate interaction response, new assignment groups can be added to the out-of-box assignment groups for Service Desk.

To add a Service Desk assignment group:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.
   1. Enter the name of the manager of the assignment group.
   2. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   3. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide response when you create and populate a new assignment group.

   (Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in

this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.
9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.
10. Follow these steps to configure the Supported Services field:
    1. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    2. Select **Business Service** in the CI Type field, and then click **Search**.
    3. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    4. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.
11. Click **Save** and **OK**.

Related topics

Service Desk administration

1.3.5. Service Desk configuration

Service Desk configuration enables you to configure alert, interaction categories, workflow, solution matching, and so on.

Create an interaction category

Applies to User roles: System Administrator

If you are a Service Manager administrator, you may want to create new interaction categories. You may want to create a category when you require a different worfklow that specifies unique forms and business rules.

Note

If you do create a new category, you must also add corresponding subcategories and areas.

Service Manager provides a number of out-of-box category records that you can use or modify. For more information, see Out-of-box Service Desk categories.

Note

The **Allow Escalation** option is not applicable to the streamlined interactions.

To create new interaction category record, follow these steps:

1. Click **Service Desk** > **Configuration** > **Interaction Categories**.
2. Click **New**.
3. Type the name of the category.
4. Specify the **Apply To** option to indicate the modules that can share this category.

Note

The **Apply To** flag indicates whether the category is shared across different modules. For example, if the **Apply To** is set to "Interaction/Incident", this category will be created in both the interaction category table and the incident category table. Typical shared categories in the out-of-box system include:

- incident
- problem
- request for information
- support catalog

For an Interaction Category, the following rule applies:

- Once a value is specified in the **Apply To** field and the record is saved, the value can't be modified.

The **Apply To** options that are used in Interaction Categories include:

- Interaction
- Interaction/Incident
- Interaction/Incident/Problem
- Interaction/Problem

5. Specify whether the category is active.

Note

If the **Active** checkbox is unchecked, you can't use this category when creating a new record.

6. Type the category description.
7. Select a workflow for the category.
8. Specify the fulfillment process that's associated with this interaction category. When an interaction record is created in this category, the specified process will be initiated.

Note

- The following table lists the processes in the out-of-box system and the recommended workflows that you need to specify in step 8.

Fulfillment process	Workflow
Incident Management	Streamlined Service Desk
Problem Management	Streamlined Service Desk
Request Fulfillment	Streamlined Service Desk
Change Management	Streamlined Service Desk
Service Desk	Streamlined Complaint or Compliment

- When the fulfillment process is reconfigured for a category, the configuration will take effect after you re-login Service Manager.

9. Click **Save** to view the **Workflow** tab and other associated tabs.
10. Click **Subcategories** tab to create subcategories for the interaction category.
11. Click **Service Desk Approval** tab to configure approval settings.

Note

The settings in the **Service Desk Approval** tab are only for service catalog approval. That is, in the out-of-box system, they are only valid for the service catalog category.

12. Click **Save**.

> **Note**
>
> The Interaction category name is read-only after the category is created.

Add a new interaction subcategory

Applies to User roles: System Administrator

Subcategories are a way to refine a Service Desk category into smaller groups. For example, the complaint category can have service delivery and support subcategories.

To add a subcategory:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New** to create a new category or click **Search** to select an existing category.
4. In the **Subcategories** section, click the **Link New Subcategories** button.
5. Type a subcategory name.
6. Type a description for the subcategory.
7. Click **Save** to add the new subcategory.

Add a new interaction area

Applies to User roles: System Administrator

Areas are a way to further refine a Service Desk subcategory. For example, the support subcategory in the complaint category can have availability, functionality, and performance areas.

To add an area:

1. Click **Service Desk**.
2. Click **Configuration** > **Interaction Categories**.
3. Click **New** to create a new category or click **Search** to select an existing category.
4. In the **Subcategories** section, click the **Link New Subcategories** button to create a new subcategory or select an existing subcategory from the list.
5. In the **Areas** section, click the **Link New Areas** button.
6. Type a name for the area.

7. Type a description for the area.
8. Click **Save** to add the new area.

1.3.6. Security

The topics in this section explain the Service Desk security roles, security areas, and rights.

Setting rights to interactions

Applies to User roles: Service Desk Manager

The Service Desk Manager can set rights to interactions using the role-based security. The Service Desk and Service Desk Configuration areas contain default security rights and settings. The rights are copied to new roles created for these areas. However, the settings are inherited only if there are no settings specified on the roles.

Note

Whenever the roles in an operator record are updated, the operator must log out and then log in for the changes to take effect.

Service Desk security areas

Applies to User roles: System Administrator

The security areas for Service Desk are Service Desk and Service Desk Configuration. These areas contain the default security rights and settings for Interactions in the Service Desk module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

These security areas are used to set permissions to operators to provide access to particular area of Service Desk. For example, for the Service Desk Analyst security role, set rights as **View** for the Service Desk area. Then, an operator with this security role can only view the Service Desk items under Service Desk module. The following table lists the areas and the relevant Service Desk menu items the operators can access.

Area	System Navigator menu items for this area
Service Desk	This area contains the default security rights and settings for Service Desk requests in the Service Desk module. For example, Create New Interaction, Interaction Queue, and Search Interactions.

Area	System Navigator menu items for this area
Service Desk Configuration	This area contains the default security rights and settings for Service Desk administration and configuration, which are dedicatedly used by Service Desk module. For example, Settings and Interaction Categories. **Note** When you set the security rights for a security role in the Service Desk Configuration area: • The **View** right is to view the settings defined in the **Administration** menu and the **Configuration**menu. • The **Update** right is to update the values of existing settings defined in the **Administration** menu and the **Configuration** menu. • The **New** and **Delete** rights are to create and delete a setting in the **Configuration** menu, such as category. • The **Admin** right is to add, edit, or delete the settings in the **Administration** > **Settings** menu.

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Service Desk and Service Desk Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Service Desk	TRUE	FALSE	Never	Never	FALSE	FALSE
Service Desk Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Service Desk and Service Desk Configuration areas.

Service Desk security roles and settings

The out-of-box security roles for the Service Desk module include the following:

- service tech
- service desk agent
- service desk analyst
- service desk manager
- self service
- approver
- service desk process owner

Mapping between previous security profiles and current PD security roles

The following table lists the mapping relationship between previous Service Desk security profiles and current PD security roles in the Service Desk module.

Security Profile	Security Role/Area
DEFAULT	DEFAULT/Service Desk
approver	approver/Service Desk
helpdesk tech	helpdesk tech/Service Desk
initiator	initiator/Service Desk
self service	self service/Service Desk
service desk agent	service desk agent/Service Desk
service desk manager	service desk manager/Service Desk
sysadmin	sysadmin/Service Desk
N/A	service desk analyst/service desk process owner/Service Desk

Field mapping between security profiles and PD security rights/settings

The following table lists the mapping of fields in legacy Service Desk security profiles and Process Designer security roles.

Security profile settings	Process Designer security rights and settings
New	New
Close	Delete/Close
Update	Update
View	View
Can notify	View
Alternate views	View
Advanced Search	View
Template Mass Update	Expert
Complex Mass Update	Expert
Approval groups	Approval Groups
Allowed Statuses	Allowed Statuses
Can Create Personal Views	Can Create Personal Views
Can Create System Views	Can Create System views
Lock on Display	Lock on Display
Modify Templates	Modify Templates
Delegate Approvals	Can Delegate Approvals
Initial view	Initial View
List Format	List Format
Manage Format	Manage Format
ESS initial format	ESS initial format

Security profile settings	Process Designer security rights and settings
ESS edit format	ESS edit format
ESS search format	ESS search format
ESS list format	ESS list format
Default Template	Default Template
Append query	Append query
New Thread: View → Search	New Thread: View → Search
New Thread: Search → List	New Thread: Search → List
New Thread: List → Edit	New Thread: List → Edit
New Thread: View → Edit	New Thread: View → Edit

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Service Desk area specified in the corresponding security roles. See the table below for the out-of-box security rights of the new security roles in the Service Desk area and the Service Desk Configuration area. This table only lists the new security roles that have different settings with the default rights.

Service Desk area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
service desk analyst	TRUE	TRUE	Always	Always	FALSE	TRUE	FALSE
service desk manager	TRUE	TRUE	Always	Always	TRUE	TRUE	FALSE
service desk process owner	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

Service Desk Configuration area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
service desk manager	TRUE	TRUE	Always	Always	FALSE	FALSE	FALSE
service desk process owner	TRUE	FALSE	Never	Never	FALSE	FALSE	FALSE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

For information about the out-of-box role rights in the Common Configuration area, see Out-of-box role rights in the Common Configuration area.

1.4. Incident Management (Codeless)

Service Manager Incident Management can automate the reporting and tracking of a single incident or of a group of incidents, and help you to achieve service performance that meets Service Level Agreement (SLA), Operation Level Agreement (OLA), and Underpinning Contract (UC) targets.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Operator

Incident Analyst

Incident Coordinator

Incident Manager

Service Desk Agent

Service Desk Manager

Key concepts

Describes the most important things that you need to know about how the Incident Management module functions.

Incident Management overview

What's an incident?

Incident Details section

Assignment groups

Incident Management views

User workflows

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Incident Logging

Incident Categorization

Incident Investigation

Incident Recovery

Incident Review

Incident Closure

Administrator tasks

Describes the Administrator tasks in the Incident Management module.

Configure the Incident Management environment

Configure Incident Management settings

Add an Incident Management assignment group

Update multiple incident records

Additional resources

Processes and Best Practices

1.4.1. Incident Management overview

You can use Service Manager Incident Management to automate reporting and tracking of a single incident or a group of incidents. Incident Management restores normal service operation as quickly as possible and minimizes the adverse impact on business operations, thus ensuring that the best possible levels of service quality and availability are maintained. Incident Management helps you achieve service performance that meets Service Level Agreement (SLA), Operation Level Agreement (OLA), and Underpinning Contract (UC) targets. Incident Management enables you to achieve the following results:

- Require Incident records to follow a set process
- Define the users who are responsible for an type incident, and automatically notify them incident opens or escalates
- Issue alerts or escalate an incident to properly meet the agreed-upon terms of the service contract
- Post device outages to the SLA application
- Plan outages based on services that could be affected, based on the CIs specified in an incident or change

Related topics

Incident Management administrator tasks
Incident Management contract management records
Incident Management and service level agreements
Processes and Best Practices

1.4.1.1. What's an incident?

Incidents include any event which disrupts, or which could disrupt, a service. This includes events that are communicated directly by Users, either through the Service Desk or through an automated interface between Event Management and Incident Management tools.

Incidents can also be reported and logged by support staff, who may notify the Service Desk if they notice an issue. Not all events are logged as incidents. Many classes of events aren't related to disruptions at all, but are indicators of normal operation or are simply informational.

Related topics

Incident Management overview

1.4.1.2. Affected services option for Incidents and Changes

If you want to see a list of services that are potentially affected by an outage related to the CIs specified in an Incident or Change record, click the **View Affected Services** command on the **More** menu. When you open an Incident record, you can examine detailed information for any of the CIs in the list to determine the potential impact of an outage relating to critical, dependent services. The **View Affected Services** command also enables you to plan appropriately when you open a Change record. You can determine how a planned outage affects these services.

When you click **View Affected Services** on the **More** menu, the system processes the outage spreads relating to the CIs in the Incident or Change record by browsing the configuration relationship records in the `cirelationship` table. This process is independent of any outage dependency information specified in the relationship records. No matter how many hierarchy levels a CI is above or below a service, the system browses all levels from the CI in the record to build a list of CIs. After this list is built, the system searches for any outage dependency information specified in the relationship records for those CIs in the list and removes any CIs that don't meet the conditions for outage dependencies. This final list of CIs then opens and you can select a CI for detailed information from the Configuration Management record.

Note

- The list of CIs generated does *not* include the CIs specified in the primary Incident or Change record.
- When you click **View Affected Services** on the **More** menu, the system displays neither the Retired and Consumed services nor the services that have dependencies on retired CIs.

Affected Item field in Incident Management

The information available to populate the **Affected Item** field on the CIs and Services tab depends on conditions related to the service recipient. For example, if the recipient has individual or department subscriptions, a list of CIs associated with those subscriptions opens. If a recipient doesn't have any subscriptions, the following behavior occurs:

- If there is an incident recipient, a list of the CIs assigned to that recipient opens.
- If there is a location, a list of the CIs assigned to that location opens.

- If the system has several companies and there is a company specified, a list of the CIs assigned to that company opens.
- If there is a device type specified, a list of the CIs of that type opens.

Note

If you want to obtain a list of all CIs, don't enter any recipient information before you click the **Affected Item** field.

Related topics

Outages

1.4.1.3. Creating an incident

Service Manager allows you to create an incident via several ways.

You can directly create a new incident in the In the Incident Management module by clicking **Incident Management** > **Create New Incident**.

You can also create an incident by escalating an interaction. For example, if a user can't print to the network printer, the user calls the Service Desk for help. The Service Desk Agent creates a service desk interaction record to capture the service request. If the agent discovers that the issue can't be resolved during the phone call, the agent escalates the interaction into an incident.

The Incident Coordinator reviews the information and verifies that it is assigned to the correct group. An Incident Analyst resolves the issue or escalates it to initiate a change. When the issue is resolved, the Service Desk Agent informs the user that the record is closed. All phases of this incident are tracked, from opening the service desk interaction record, escalating the incident, resolving the incident, and closing the record.

Activities section

The Activities section allows users to enter new updates for a record or to view journal updates and historic activities for a record.

Journal Update

Field	Description
New Update Type	Specifies or categorizes the activity update (for example, communication with the customer)
Visible to Customer?	Makes the update visible to customers so they can view related interactions via self service **Note** This option can be bypassed by setting the **Bypass Visibility Option** flag to **true** in the activityactions table. When this flag is set to **true**, automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to **false**, all activities are visible to customers if the **Customer Visible** option is selected.

Field	Description
New Update	Used to enter notes to explain and describe updates made for the record. If Journaling is enabled, the text entered here is displayed in Journal Updates. If Activities are enabled, the text entered in this field is displayed as an activity record for the selected activity type. **Note** The System Administrator is responsible for enabling Journaling and Activities.
Journal Updates	Displays text entered in the **New Update** field together with the operator's name and a timestamp for the update. This field displays information when Journaling is enabled.

Activity Logs

Select an activity type from the **Activity Type** drop-down list or select a user from the **User** drop-down list to filter the activity logs.

The activity log lists activities for the current record in order of occurrence, with the newest at the beginning of the list. Each activity log displays the following information:

- Date/Time
- Operator
- Description
- Type
- Visible to Customer icon (if you have selected the **Visible to Customer** checkbox)

Incident Details section

Use the following information as a guide when you add data to an incident.

Field	Description
Title	A short description summarizing the incident.
Description	A description of the incident in more detail.
Incident ID	The system-generated unique ID for this incident.

Field	Description
Status	Displays the status of the incident. These statuses are available in the out-of-box: **Categorize** means the incident is being categorized. **Assign** meas the incident is assigned. **Work In Progress** means work is underway to resolve this incident. **Pending customer** means awaiting action from the customer who reported the incident. **Pending Evidence** means that you need evidence from the customer or vendor. **Pending Vendor/Supplier** means awaiting action from a vendor/supplier. **Pending Parent Incident** menas tThe incident has been linked to a parent incident and is waiting for the resolution of the parent incident. This status is automatically set when an incident is linked to a parent incident. **Pending Other** means that you need something from an outside source other than customer or vendor. **Resolved** means there is a resolution, but it has not yet been verified by the customer. **Suspended** means the customer has agreed to suspend the incident for a period of time; the incident doesn't appear in your Inbox during this period.
	The current phase of this incident.

Field	Description
Primary Affected Service	The service affected by the registered issue. Only services the service recipient has a subscription for can be selected.
Affected CI	The configuration item (CI) the incident is registered for. Using Fill shows the CIs that are part the affected service. Other CIs can be entered manually.
CI is operational (no outage)	If selected (set to true), indicates the item is currently operational and there is no outage.
Outage Start Time	A date/time stamp for when the service outage started.
Outage End Time	A date/time stamp for when the service outage ended.
Requested By	The name of the operator who opens the incident.
Contact Person	The contact person for this incident.
Location	The service recipient's location
Major Incident	When this option is selected, the incident requires the Major Incident Review process.
Escalated	Select this option to mark the incident for escalation. When this option is selected, the **Escalation Team** section appears.
Incident Manager	An incident manager need be specified when the **Major Incident** or **Escalated** option is selected.

Field	Description
Parent Incident	This option indicates if the incident is a parent incident. You can link one or multiple child incidents to a parent incident if they are related in a certain way.
Link to Parent Incident	The parent incident of the current incident. This field is unavailable when the **Parent Incident** option is selected.
Category	The out-of-box system has four incident categories: Incident, Complaint, Request for Administration, and Request for Information. When you escalate an interaction into an incident, the new incident inherits the category from the interaction.
Subcategory	The subcategory of the incident.
Area	Depending on the selection of a category, a different list appears describing the area of concern for the incident.
Assignment Group	The support group assigned to work on this incident.
Assignee	The name of the person assigned to work on this incident. This person is a member of the assigned support group.
Impact	The impact the incident has on the business. The impact and urgency are used to calculate the priority.

Field	Description
Urgency	The urgency indicates how pressing the incident is for the organization.
Priority	The order in which to address this incident in comparison to others.
Source	The source of the incident.
Expected Solution Time	The expected time when the incident is resolved.

Note: When you view an existing incident, shaded fields aren't available for more input. Incident Management populates these fields from information stored in associated records. You can update the associated record to increase the amount of information in these read-only fields.

Related Records in Incident Management

Service Manager lists all related records for the incident on the Related Records section. All information is read-only, but a user can click a record ID to view the related record. The related records are listed in alphabetic order by the module file name: Change (cm3r), Interaction (incidents), Incident (probsummary), Request (request), Problem (rootcause). Within each module the records are sorted by ID in ascending order.

The Related Records section contains the following fields.

Field	Description
ID	The related record number in the database.
Type	The link type between the related record and the incident.
Phase	The current phase of the related record.
Status	The current status of the related record.
Title	The title of the related record.

SLT section

Service Manager populates the information in the SLT section with one or more Service Level Agreement records associated with the asset. If no agreement exists, the section is blank. All the information is read-only.

The **Next Expiration** field displays the next immediate deadline from all matching SLTs. The SLT section contains the following subsections that list the details of the specific SLTs.

SLT type	Description
Process Targets	Lists all of the matching SLTs from all of the associated SLAs.
Uptime Targets	Lists the percentage of time that all SLTs are available.
Max Duration Targets	Lists the acceptable amount of time for an outage for all SLTs.
Upcoming Alerts	Lists the alerts for all of the SLTs.

Note: When you view an existing record, shaded fields aren't available for more input. Incident Management populates these fields from information stored in associated

records. You can update the associated record to increase the amount of information in these read-only fields.

Hover-over forms for incident records

In Incident Management, incident forms for open, update, and close include hover-over forms. A hover-over form opens for a field that supports hover-over forms when a user moves the mouse over the field. The hover-over form only opens when the field contains data. The data displayed on hover-over forms is read only.

The incident record forms (open, update, close) contain the following hover-over fields.

Field	Subform fields
Contact	Full Name, Telephone, Email
Affected CI	Critical CI, Pending Change

Related topics

Incident Management overview

1.4.1.4. Posting outages

The Incident Management application can be used to post outage information about devices in the system into the SLA application when you open or close incidents. The system posts outages both manually and automatically.

When posting outages manually, Incident Management displays the Post Which Outages? form for the selected device when you open and close and incident. Use this form to specify the exact start and stop time of the outage instead of the time the outage was reported. Click **OK** to continue opening or closing the incident.

The current time is the default value in manual mode. You may accept the default or type the exact time.

Related topics

Incident Management overview
Priority, impact, and urgency
Creating an incident

1.4.1.5. Incident Management contract management records

Contract Management integrates service contract information and tracking into the service desk. You can enable the application or set automatic labor and parts calculation features for Contract Management in the Contract Management configuration record.

Related topics

Incident Management overview
Incident Management administrator tasks
Incident Management and service level agreements

1.4.1.6. Incident Management and service level agreements

Incident Management supports the selection of more than one applicable SLA for an incident. When you create an incident, you can choose a Customer SLA for the contact, one or more applicable Service SLAs for the contact's subscriptions to a service, or no SLAs at all. Service SLAs only apply if the incident references a Business Service, the contact has a subscription to the service, and the subscription references an SLA. The following describes the system's process for adding SLAs to an incident.

- If one SLA is associated with the incident based on the contact, the Customer SLA is added to the incident.
- If the contact has an Individual Subscription for the CI, the Service SLA from that subscription is added to the incident.
- If the contact has a Department Subscription for the CI, the Service SLA from that subscription is added to the incident.
- If the contact has neither, then no Service SLA is added to the incident

The SLAs should contain all Service Level Targets (SLTs) that define the business rules for all response and availability metrics. You can choose as many SLTs as necessary to describe your response or availability commitment. If necessary, you can add more SLTs that meet your criteria.

When you view the new record, the **SLT** section lists the SLTs that apply to the incident.

Open the related topics to find the definitions for Customer SLA and Service SLA.

Related topics

Incident Management overview
Incident Management administrator tasks
Incident Management contract management records

1.4.1.7. Parent and child incidents

The parent-child relationship allows you to link multiple incidents that are related in a certain way. For example, a network outage results in multiple similar incidents from different users. You can link these incidents under a parent incident. In this way, the activities that occur to the parent incident are automatically recorded under the **Activities** section in all child incidents and related Service Desk interactions (if there are any). In addition, you can also configure if the child incidents are automatically closed when the parent incident is closed.

Related topics

Incident Management overview
Set a parent incident
Set a child incident

1.4.1.8. Major incident

If the impact or priority of an incident increases, the incident may become a major incident. For a major incident, you may need problem management activities to identify the root cause, and take appropriate actions and plans to prevent future major incidents from occurring.

After you mark an incident as a major incident, the incident has a **Major Incident Review** section and the **Review Details** field in this section is mandatory in the Review phase. Service Manager also notifies the specified Incident Manager about the major incident.

Related topics

Mark an incident as a major incident

1.4.1.9. Incident access control

The Incident Management module utilizes security roles and security rights to authorize users in creating, modifying, and closing records. These settings allows you to grant a user with proper rights. For example, a user can create and close records, but can't modify records that aren't assigned to this user.

Service Manager also provides a security control to segment data between multiple customers. The Mandantan controls identify which customer data a particular user or group can access or update.

Related topics

Controlling user access and security
Application profiles
Application Setup Application Setup
Capability word model

1.4.1.10. Incident Updates: Incident Diagnosis Details

Service Manager provides an audit trail capability that identifies each step in the diagnosis and resolution of an Incident record. Each time the record is updated, Service Manager creates a separate historical activity record for that specific Incident record, including which operators took what actions at what times. An authorized user may leaf through each historical activity record of an Incident record to develop a comprehensive understanding of the history and resolution of the Incident record.

The Incident record also includes a field to capture the resolution of the record.

Incident logging: manual

Service Manager allows the following approaches to manually create an Incident record:

- Authorized Service Desk users can create an Incident record directly within the Service Manager Incident Management module.
- Service Manager also includes a web-based, self-service interface that allows end users to create service requests through an easy-to-complete form. Once the form is complete, Service Manager routes the request through the Service Desk module to the appropriate Service Desk technician, who can then escalate the request to an Incident record. Users can track, update, or close their requests through this interface.
- An Incident record can be created through the web services and event services interfaces. These interfaces are used by existing integrations to network and systems management (NSM) tools (both the company's and third party's) and could be used in customer-created integrations to home-grown tools.

Incident logging: templates and related incidents

You can quickly create a new Incident record by copying an existing Incident record, or by applying templates. The Service Manager template functionality allows authorized users to create templates for common tasks in the Service Desk, Incident, and Change Management modules. Templates include predefined information for some or all fields of a record. A user can create an Incident record and apply a template, which can automatically fill in some or all fields of the record to save the user's effort.

1.4.1.11. Alerts and escalation

There are several ways of providing alerts and escalations in Service Manager:

1. Notifications: Service Manager provides a broad and deep solution to deliver notifications to incident stakeholders whenever the incident is opened, updated or closed. Administrators have complete control of the following areas in the notification engine:
    1. The notification vehicle (email, page, etc) and format
    2. The conditions on which the notification is sent
    3. The receiver of the notification
2. Alerts: The administrator can configure alerts in the incident workflow and SLM targets. When these targets are met, specified users receive alerts and notifications.
3. Escalation: The **Escalated** option in an incident record allows you to mark an incident for escalation. Once you select this option, you must specify an appropriate incident manager. After you save the incident, the incident manager and the incident assignee will then receive a notification about this escalated incident.
4. Service Level monitoring: Alerts and escalations can be triggered through evaluation of service levels as defined in the Service Level Management module. This approach is used to set Service Level Targets and Service Level Agreements with associated actions should service levels approach breach conditions. In this way, escalation is accomplished proactively to avoid breach rather than react to breach.

Incident escalation: Functional escalation

Service Manager includes pre-defined workflows reflecting the best practices based on ITIL 2011 guidance. These best practices come from the company's professional services organization and are developed through working with some of the primary partner companies.

By default, Incidents are assigned to the service desk for acceptance as an Incident and assignment to the appropriate group.

Based on customer process requirements, assignment can be automated. Service Manager can be configured to automatically define record assignment based on values in any of the Incident record fields.

For example, escalations can be defined in the category record associated with an incident record or in SLAs that are applied to the record. Escalations include

notifications and alerts (as the record passes through escalation levels). Service Manager can send alerts to different users, managers, and IT staff as the record is escalated.

1.4.1.12. Field-Level Controls

Service Manager provides two basic methods for setting field level controls including making the field mandatory. The system provides two methods to indicate that a field is mandatory in the UI.

If the field only needs to be mandatory on a screen-by-screen basis, you can set the field as mandatory via a RuleSet validation. This control can be visually indicated via a property in forms designer.

If the field needs to be mandatory for all access points throughout the system, you can set a workflow-level or table-level Rulset validations by using the supplied configuration tools. In this case, the visual indication is automatic.

Visually on the screen, the system identifies a mandatory field with a red asterisk.

1.4.1.13. Incident record data model

Incident Management enables you to track different types of information related to an incident, such as categorization, prioritization, description, contact information, and so on.

Incident data model: Contact details

Service Manager stores related contact information, including the operator who opened the incident and the contact person for the reported issue, with the preferred method of contact (email, phone, or other).

Incident data model: Incident source

When the source of the incident is a person, the Interaction record linked to the Incident record shows the basic information of the source. For example, the following information:

- Whether the Incident is opened through the self-service interface
- The Incident trigger
- The location of the issue

When the source of the incident is an event trigger, the Title and Description fields of the Incident record contain that information, which references the source event displayed in the External ID field of the Event Browser in Operations Manager.

Incident data model: Categorization

Service Manager categorizes or classifies Incidents according to the following information:

- Category
- Subcategory
- Area

In an out-of-box Service Manager system, Service Manager automatically applies the Incident category to the new Incident that you create. If the Incident Management module has other customized categories, you need to select a category when you create a new Incident. Based on the selected category, Service Manager presents a filtered set of choices for Area and Subarea.

The Incident record also captures the ID of the failed CI or service, including the failing component.

Incident data model: Priority

The Incident record form includes fields to capture impact, urgency, and priority. Service Manager automatically derives the priority code based on the impact and urgency codes, but this mechanism can be manually overridden.

Service Manager automatically assigns urgency based on the Category of a record. However, system administrators can define additional rules for assignment based on any parameter, such as the CI that's the subject of the record, the caller who is the subject of the record, an SLA associated with the record, and so on.

An Incident record has the following impact and urgency codes.

Impact codes:

- 1 - Enterprise
- 2 - Site/Dept
- 3 - Multiple Users
- 4 - User

Urgency and priority codes:

- 1 - Critical
- 2 - High
- 3 - Average
- 4 - Low

Incident data model: Priority calculation

Service desk users complete the **Impact** and **Urgency** fields and Service Manager uses these fields to calculate the priority of the Incident. The **Priority** field is read-only to protect this calculated value.

Incident data model: Assignment

The Incident record stores the assignee group and an individual assignee within the group, if known. The Incident record also captures the name of the vendor/supplier to which the record has been assigned, if appropriate.

Incident data model: Symptoms

An Incident record has a field for the brief summary (the Title) of the issue and a free text field for the details of the problem or symptoms of the fault, which can be used as the search information for the built-in Knowledge Management capability.

Incident data model: Status

Service Manager tracks the status of an Incident in the **Status** field. Service Manager provides the following out-of-box status codes, other codes can be easily added to meet customer requirements.

- **Categorize**
- **Assign**
- **Work In Progress**
- **Pending customer**
- **Pending Evidence**
- **Pending Vendor/Supplier**
- **Pending Other**
- **Resolved**
- **Suspended**

Incident data model: Resolution

Help desk operators are prompted to enter record resolution information when they close records, including a Closure Code (this is a mandatory field). Service Manager provides out-of-box Closure Codes, which can be tailored to meet the customer's requirements. Resolutions can be promoted as candidates for the knowledgebase. Knowledgebase entries can also be used as the resolution for the Incident record.

Incident data model: Closure

Service Manager allows users to enter a Closure Code when a record is closed. The Closure Code can be compared to the category recorded at open time to report on trends. In addition, Service Manager automatically captures the time and date of Incident closure for reporting.

1.4.1.14. Incident and Service Request separation

Service Requests can be handled within the Service Desk module, supported by the Service Catalog module. Service Requests are tracked as Service Desk Interactions and can be fulfilled using Incident Management, Change Management, Request Management, or another fulfillment engine specified by in the Service Catalog configuration. Service Requests that don't originate from the Service Catalog and instead come directly to the Service Desk can be routed to the appropriate fulfillment mechanism by the Service Desk Analyst or they can be completed and closed at the Service Desk.

Incident integration: Configuration Management system

Service Manager is fully integrated with the UCMDB CMS solution. This is a federated solution that integrates with the company-provided discovery tools as well as those provided by other vendors.

Incident integration: Change Management

Out-of-box, the Service Manager Incident Management module is fully integrated with the Change Management module. Users can open Change request records from Incident records. Service Manager can pass relevant information from the Incident record to the Change request record. The two records are linked automatically.

Authorized users can access the Change Management module, create Changes associated with an Incident record, search for Change requests, view the Calendar, link an Incident record to an open Change request, and so on.

Incident integration: Incident matching with RFCs

Service Manager allows linkage between Incidents and Changes, and provides tools for users to establish the linkage. Users can link an existing Change or a newly created Change to an Incident. Service Manager can use business rules to manage the relationships. For example, Service Manager can use business rules to set all the related Incidents to “Resolved” when the related Change is resolved. Related users can then check if these Incidents can be closed.

Incident integration: Knowledge Management

The Service Manager Knowledge Management module allows end users and service-desk personnel to get speedy and accurate answers to their questions directly from any Service Manager application screen. End users can access the Knowledge Management module through the self-service interface to search for potential problem resolutions before contacting the service desk.

Knowledge can come from any Service Manager data source—Incidents, Known Errors and so on—or any external data source. Knowledge can also include Hot News.

However, the Knowledge Management module is more than just the technical components needed to store answers. Because knowledge articles are created via the Knowledge-Centered Support (KCS) process, users can be confident in the accuracy and "just-in-time" solution quality of the answers. The KCS process is an industry best practice for knowledge management for which Service Manager has received external certification.

Service Manager Knowledge Management is a fully integrated support-oriented knowledge management solution that supports KCS standards and guidelines. The Knowledge Management module provides a natural language search engine and rich-text authoring tools that enable users to search, update, and author knowledge articles. Knowledge Management integrates with Interaction, Incident, and Problem management modules so that users can search and use knowledge from existing incidents or problems while attempting to resolve a new incident or problem. Users can also use this integration with Interactions, Incidents, and Problems to create new knowledge. The rich-text editor allows users to include various types of images and documents as attachments that can be linked to other documents or included as part of an existing document.

As part of the company's Closed Loop Incident Process (CLIP) solution, the integration between Service Manager Knowledge Management module and Operations Orchestration (OO) allows automatic execution of run-books related to change management tasks in the context of Service Manager Knowledge Documents. This allows for better and higher levels of automation.

For example, a Service Desk user diagnoses an Incident (or an end user who does self-help diagnosis) and searches for a Knowledge Management document with matching symptoms to the issue. When the user finds such a document, the user can execute a script from a link on the Knowledge Management page itself. The Knowledge Management link triggers a web service action that tells the OO server to run a specific pre-defined script on a specific server. The OO server then sends appropriate commands to the specified server to execute the `runbook` commands. When the target server receives the `runbook` commands, the server executes

operating system commands or database commands as pre-defined to accomplish the specified action. This action could be a workaround or a fix, applied to the environment selected by the Service Desk (or end) user.

Incident integration: Problem Management

Service Manager provides separate modules for creating and managing Incident and Problem records, including Known Error records. However, because these modules are fully integrated and share data seamlessly, Incident, Problem, and Known Error records can be linked. Once linked, the relationships between the records are maintained. Users can create Problem records from Incident records (and vice versa), and Service Manager automatically copies all relevant information from the Incident record to the Problem record to minimize retyping. These records are automatically linked. Users can also manually link records.

Known Error records are created as part of the Problem Management process. The Service Manager Problem Management workflow contains two sub-processes:

- Problem Control. This process analyzes the problem, finds the root cause, and allows users to gather all data needed to initiate a new problem review. Problem Control includes two phases:
  - Identification and Classification. This phase includes creating the Problem record. Classification identifies relationships, urgency, assesses the impact on the customer’s business service, determines priority, and assigns the problem to a specialist or support group.
  - Investigation and Diagnosis. This phase examines the symptoms of the problem to identify the root cause.
- Error Control. This process manages correcting the root cause identified by Problem Control. The objective of Error Control is to effect permanent changes to Configuration Items (CIs) with known errors. Phase 1 of Error Control is Error assessment. Phase 2 of Error Control is closing the known error record.

While Service Manager will not automatically close all Incident records linked to a Problem or Known Error record, Service Manager facilitates closing linked records by listing all linked Incident records that need to be closed. Service Manager can be tailored to automatically close these records.

However, we don't believe that this supports best practices. Service Manager Problem Management is designed to address infrastructure errors, not immediate issues. Therefore, the technician should manage the closure of the Incident separately to help ensure the customer has received the necessary support.

Incident integration: Incident matching with Problem

Incident matching and association is a foundational function in Service Manager. When a user creates an Incident record, Service Manager prompts the user with a list of similar Incident and Problem records. Service Manager can automatically check for potentially similar records based on similar known errors, root causes, incident records, incident record duplicates on a device, or incident record duplicates on parents.

The user can then determine if the new Incident record is identical or similar to an existing Incident record (and can then link the records), or can be resolved based on the workaround information in an existing Problem record.

Incident integration: Request Fulfillment

Out-of-box, the Service Manager Incident Management module is fully integrated with the Request Fulfillment module. Users can open Request records from Incident records, and Service Manager can pass relevant information from the Incident record to the Request record. The two records will be linked automatically.

Service Manager also includes a web-based, self-service interface that allows end users to create service requests through an easy-to-complete form. Once the form is complete, Service Manager routes the request through the Service Desk module to the appropriate Service Desk technician, who can then escalate the request to an Incident record. Users can track, update, or close their requests through this interface.

Incident integration: Service Level Management

Service Level Agreements (SLA), Operational Level Agreements (OLA), and Underpinning Contracts (UC) are created in the Service Manager Service Level Management module. Each SLA has multiple Service Level Targets (SLT), which can define required response or resolution timeframes associated with Incident records, or required uptime for CIs or services. When an Incident record is created, Service Manager applies the applicable SLA (based on the business unit for which the Incident is created, the CI that's the subject of the record, or other parameters) and populates response and resolution time SLT, as well as availability SLTs, in the Incident record.

This information can be used to prioritize record resolution, and Service Manager displays upcoming SLT deadlines, escalations, and alerts in the Incident record.

Related topics

What's a Service Desk interaction?

1.4.2. Incident Management user roles

There are specific roles associated with Incident Management. Service Manager uses role-based permissions to enable you to complete a task that's appropriate to your role. System Administrators manage and assign these permissions.

The Incident Management module has the following user roles:

Role	Responsibilities
Operator	• Register incident based on an event and assign to the correct support group.
Incident Analyst	• Review and accept or reject assigned incidents. • Investigate and diagnose the incident. • Document incident resolution or workaround in the Service Management application. • Implement incident resolution. • Verify that the incident is resolved and close the incident.
Incident Coordinator	• Review and accept or reject incidents assigned to the support group. • Handle incidents escalated by an Incident Analyst of the support group. • Monitor Operational Level Agreements (OLA) and Underpinning Contracts (UC) targets of the support group.
Incident Manager	• Handle incidents escalated by the Incident Coordinator or by the Service Desk Agent. • Determine and execute the appropriate escalation actions. • Request an Emergency Change if required.

Role	Responsibilities
Service Desk Agent	• Open interactions based on contact with the user • Match user interaction to incidents, problems, known errors, or knowledge documents • Solve and close interactions • Provide status updates to users on request • Register incidents based on user interactions and assign them to the correct support group • Register requests for change, based on user interactions • Register service requests, based on a user interactions • Validate solutions provided by support groups • Report and verify solution to users • Monitor the Service Level Agreement (SLA) targets of all registered incidents and escalate the incidents if required • Communicate service outages to all users
Service Desk Manager	• Handle incidents that are categorized as Complaints.

Related topics

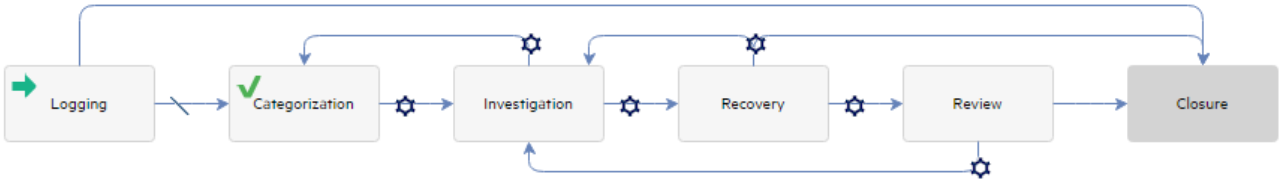
User roles
Add a user role record
Delete a user role record
Search for a user role record

1.4.3. Incident Management workflows and user tasks

Incident Management enables you to categorize and track various types of incidents (such as service unavailability or performance issues and hardware or software failures), and ensures that incidents are resolved within agreed-on service level targets.

The incident workflow is a sequence of connected steps in the life cycle of an incident. In the workflow, an incident goes through several phases to complete the life cycle.

Incident Management workflow and its phases

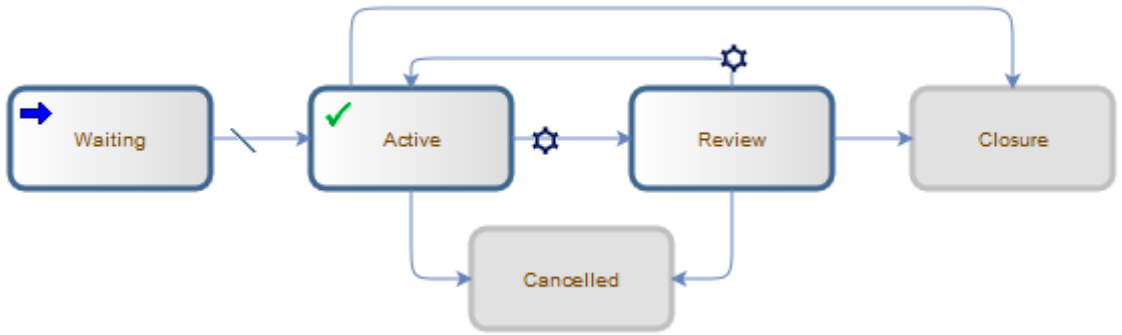


The following tasks are associated with the Incident Management workflow:

Workflow Phase	User Tasks
Logging	• Create a new incident from a user interaction • Create New Incidents from Monitoring System Notifications • Review and Update Incident Information • Open an incident
Categorization	• Reassign an Incident • Assign an Incident
Investigation	• Change Incident Status to Pending User Information • Change Incident Status to Pending Vendor/Supplier Investigation • Document an Existing Solution or Workaround in an Incident • Escalate an incident • Reassign an incident for additional support
Recovery	• Test the Incident Resolution • Change Incident Resolution • Reassign Incident Resolution

Workflow Phase	User Tasks
Review	• Reject an Incident Resolution • Reject an Incident Resolution with an Associated Change or Service Request • Close an Incident • Close an Incident with an Associated Interaction or Event • Handle complaints
Closure	None

Incident task workflow



Workflow phases can consist of one or multiple tasks to be completed to proceed to the next phase. You can use Incident Management tasks to categorize, investigate, resolve, or review an incident.

The following tasks are associated with incident tasks:

Incident Task Phase	User Tasks
Waiting	Create an incident task
Active	None
Review	• Cancel an opened task for an incident record • Close an opened task for an incident record
Closure	None
Cancelled	None

Note

Incident Task can be created by clicking the **Link New Task** button at any Incident phases except the Logging and Closure phases. All the tasks must be closed before Incident closure.

You need to perform some minimum steps to resolve an out-of- box incident. The following example demonstrates these steps:

Phase:	User Actions:	Options
Logging	1. Enter a title in the **Title** field. 2. Provide a description of the problem in the **Description** field. 3. Click the **Fill** button and then select an appropriate value to specify the **Primary Affected Service** field. 4. If necessary, set the **Impact** and **Urgency** fields. 5. Click **Save**. 6. If you want to link an existing incident record, select the incident in the solution candidates and then click **Link Selected Record**. Otherwise, click **Continue** to create a new incident.	• This is the phase that's initiated when an end user Opens a New Problem (**Incident Management > Create New Incident**)

Phase:	User Actions:	Options

Categorization	1. Click the **Fill** button to specify the **Subcategory**.
	    1. Select an appropriate subcategory.
	    2. Select an appropriate area.
	2. Click the **Fill** button for the **Assignment Group**, and then select an appropriate assignment group.
	3. Click the **Fill** button for the **Assignee**, and then select an appropriate assign person.
	4. Set the Status field to **Work In Progress**.
	5. Click **Save**.	

Investigation	1. Enter the **Solution** field in the **Proposed Solution** tab. 2. Click **Save**.	From the Investigation Phase, you can also move to any of the following: • Return to the Categorization Phase.

Phase:	User Actions:	Options
Recovery	1. Set the Status field to **Resolved**. 2. Click **Save**.	From the Recovery Phase, you can also move to any of the following: • Return to the Investigation Phase. • Jump directly to the Closure Phase.
Review	1. Review the data entered for the Incident.	From the Review Phase, you can also move to any of the following: • Return to the Investigation Phase.
Closure	1. Click **Close** Button. 2. Select the appropriate **Closure Code**. 3. Click **Finish**.	

Related topics

Incident Management overview
Processes and Best Practices

1.4.3.1. Access Incident Management reports

Applies to User roles: Incident Manager, Incident Coordinator

The Reporting tool in Service Manager provides a number of out-of-box reports on the incident data in your system. You can view these reports through a dashboard named Incident Overview (Global). You can also create your own dashboards to display other reports of your interest.

To access Incident Management reports, follow these steps:

1. Click **Incident Management** > **Incident Overview**.

   By default, the **Incident Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. Add your own dashboards as needed. For details, see Update a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default one.

1.4.3.2. Create a new incident from a user interaction

Part of Workflow(s): Incident Management: Logging

Applies to User Roles: Service Desk Agent

You can create an incident from a user interaction. You can also update an incident from a user interaction that has already been triggered to an incident.

Create an incident from a user interaction

To create an incident from a user interaction, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. Specify the **Contact** for the interaction.
3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. Type a description for the interaction.
6. In the **Category** field, select the **incident** or **request for information** category.
7. (Optional) In the **Subcategory** and **Area** fields, select the subcategory and area for the interaction.
8. In the **Affected Service** field, select the affected service.
9. Click the **Continue** button. The New Incident form is displayed, with fields populated with the values from the interaction.
10. Proceed with opening a new incident by updating fields such as **Primary Affected Service**, and then click **Save**.
11. Search possible solutions in the form that appears. If matching solutions are found, select the corresponding record, and then click **Link Selected Record**.

Note

You can also perform this step after an incident is created. To do this, open the incident and then select **Solution Matching** from the **More** menu in the Detail List toolbar.

12. Complete the incident form with any other relevant information.
13. Click **Save & Exit**.

Update the new incident

To update the incident from the related user interaction, follow these steps:

1. Click **Service Desk** > **Interaction Queue**. Service Manager opens the Interaction queue.
2. Click the interaction in the queue to display the details of the record.
   Those items with a status of **Dispatched** are user interactions that have been triggered to Incident Management (or another module).
3. On the record details page, click the **Related Records** tab, and then click the related incident that you want to update.
4. Proceed with updating the incident. After closing the incident form, you are returned to the user interaction list.

Related topics

Searching records
Create New Incidents from Monitoring System Notifications
Review and Update Incident Information

1.4.3.3. Create new incidents from monitoring system notifications

Part of Workflow(s): Incident Management: Logging

Applies to User Roles: Operator

You can create an incident when notified by a monitoring system of an event in the Information and Communication Technology (ICT) infrastructure.

To create an incident from a monitoring system event notification:

1. Click **Incident Management** > **Create New Incident**.
2. Select the incident category.

   > **Note**
   >
   > If a default category is specified in the Incident Settings, Service Manager directly assigns the incident to the default category and skips this step. For more information, see Configure Incident Management settings.

3. Click **Apply Template**.
4. Select the appropriate template. The template automatically fills in some fields with predefined values.
5. Type a title and description for the incident.
6. In Incident Detail, verify or complete the following required entries. Use the Fill button when available to display a list of values for the field.
   - **Category**
   - **Subcategory**
   - **Area**
   - **Impact**
   - **Urgency**
7. Click **Fill** to select an Assignment Group.
8. Click **Fill** or **Find** to select the applicable Service affected by the incident.
9. Complete the form with any other relevant information.
10. Click **Save & Exit**.

Incident Management opens a new incident, assigns a unique number with the prefix IM, and places the incident in the queue for the Incident Coordinator.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Create a new incident from a user interaction
Review and Update Incident Information
Apply a template to complete an incident
Searching records

1.4.3.4. Review and update incident information

Part of Workflow(s): Incident Management: Logging

Applies to User Roles: Service Desk Agent

To update incident information:

1. Click **Incident Management > Incident Queue**.
2. Use search or advanced search to find one or more records.
3. Double-click the target record.
4. Review the information in the record to verify that it is complete and correct.
5. For incident records that are complete:
    1. Click **Fill** to select the applicable Assignment Group for the incident.
    2. Follow the procedures defined by your company to provide the interaction number and the SLA target to the customer.
        - To see a list of the related interactions,open the **Related Records** section.
        - To see information about the SLA, Open the **SLT** section.
6. For incident records that are incomplete or incorrect, gather the required information and update the form. The required information includes:
    - The title and description of the incident
    - The service provided by the affected item (for example, email)
    - The Incident details: Category, Area, Subarea, Impact, Urgency, Priority
7. Click **Fill** to select the applicable Assignment Group for the incident, if it's not already assigned to a group.
8. Use internally-defined procedures to provide the interaction number and the SLA target to the customer. The incident is now assigned to the assignment group you selected.
9. Click **Save & Exit** to return the incident queue.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Searching records
Create a new incident from a user interaction
Create New Incidents from Monitoring System Notifications

1.4.3.5. Assign an incident

Part of Workflow(s): Incident Management: Categorization

Applies to User Roles: Incident Coordinator

The Incident Coordinator reviews Open status incidents and assigns the record to an Incident Analyst for further investigation and diagnosis.

The process of assigning an incident consists of two steps. First, review an incident to check whether your group can resolve it. Next, assign the incident to an Incident Analyst.

To review and assign an incident:

1. Click **Incident Management** > **Incident Queue**.
2. Select the **Incidents by Assignment Group** view.
3. Review the incident records in the list to decide the ones for further investigation of your group.

   Review each incident record and look at the contents of the Affected CI and Description fields. Check the candidate for the incident assignment.
4. Open the **Categorization and Assignment** section.
5. Click **Fill** to select an assignment group from the Assignment Group field and then click **Fill** to select an assignee from the Assignee field.
6. Click **Save & Exit**.

Related topics

Searching records
Reassign an Incident

1.4.3.6. Reassign an incident

Part of Workflow(s): Incident Management: Categorization

Applies to User Roles: Incident Analyst, Incident Coordinator

At times you will need to reassign incident records when an Incident Analyst is unavailable.

To reassign an incident:

1. Click **Incident Management** > **Incident Queue** > **By Assignment Group**.
2. Select a list of Incident Analyst assignments for a particular analyst.
    1. Click the list in the **View Records for which Assignment Group?** field, and then select an assignment group.
    2. Click **OK**.
3. To complete the reassignment of an incident:
    1. Select the incident record to be reassigned.
    2. Clear the assignee in the **Assignee** field, and then click **Fill** to select an operator from the list of names to reassign the incident to that person.
    3. Click **Save & Exit**.

Related topics

Searching records
Assign an Incident

1.4.3.7. Change incident status pending user information

Part of Workflow(s): Incident Management: Investigation

Applies to User Roles: Incident Analyst

If you need additional information from the user, contact the user directly. While waiting for information from the user, change the incident status to pending user information status.

Note

You can configure the **Suspend processing for these states** option in the corresponding Service Level Target (SLT) of the Incident so that the SLT stops time interval tracking for a pending incident. Otherwise, the SLT continues to track the incident as an open incident.

In a pending status, the assignment group remains the same.

1. Click **Incident Management** > **Incident Queue**.
2. Select the **Open incidents assigned to my group** view, which is sorted on the service level agreement (SLA) target date. The view is sorted by target date with the oldest date listed first.
3. Select an open incident, and then look at the category in the Category field. The category is inherited from the Service Desk interaction process. Valid entries are complaint, incident, or request for information. The Area and Subarea fields are also populated according to the selected category.
4. Click the list in the **Status** field and change the status to an applicable pending status.
5. When the user delivers the requested information, change the status to accepted or work in progress. Click the list in the **Status** field and update the status.
6. Click **Save & Exit**.

Related topics

Searching records
Change Incident Status to Pending Vendor/Supplier Investigation
Document an Existing Solution or Workaround in an Incident

1.4.3.8. Change incident status pending vendor/supplier investigation

When you find that a vendor needs to investigate and diagnose the cause of an incident, update the incident's status to indicate that the investigation is pending vendor feedback.

Part of Workflow(s): Incident Management: Investigation

Applies to User Roles: Incident Analyst

To update an incident record status to pending vendor information:

1. Click **Incident Management** > **Incident Queue**.
2. Select the **Open incidents assigned to my group** view. Based on the service level agreement (SLA) target date, Service Manager sorts the view by target date with the oldest date listed first.
3. Select an open incident, and then look at the category in the **Category** field. Inheriting from the Service Desk interaction process, valid entries in the category are **complaint**, **incident**, or **request for information**. The Area and Subarea fields are also populated according to the selected category.
4. Click the list in the **Status** field and change the status to **Pending Vendor/Supplier**.

   Note: When an incident record is at a status of pending vendor, the service level agreement clock continues to track the incident as an open incident. The assignment group remains the same.
5. Open the **Activities** section.
6. Click **Fill** in the Vendor/Supplier field to generate a list of vendors/suppliers and choose one.

   Note: The Vendor/Supplier field is only valid when the incident Status is Pending Vendor/Supplier.
7. Enter the reference number (obtained from the vendor/supplier) in the **Vendor/Supplier Record** field.

   Note: When the vendor/supplier provides the requested information, you can change the status to work in progress.

8. Click **Save & Exit**.

Related topics

Searching records
Change Incident Status to Pending User Information
Document an Existing Solution or Workaround in an Incident

1.4.3.9. Document an existing solution or workaround in an incident

Part of Workflow(s): Incident Management: Investigation

Applies to User Roles: Incident Analyst

When you find an existing solution that resolves an open incident, you can verify the effectiveness of the solution and then use the solution to resolve the incident.

After you verify that the solution works, perform the following steps to attach the solution to the incident record:

1. From the knowledgebase document or record that contains the solution, do the following:
    1. Click **Use Solution** to add the solution to the Solution field of the incident record..
    2. View the published document, and then select it as a solution. The system reopens the record with the Solution field updated.

       Note: For information about attaching a file, see the related topics.
2. Update the incident detail:
    1. Open the Activities section.
    2. Select the applicable update type in **New Update Type**.
    3. Add other necessary information and notes in **New Update**.
    4. Check **Visible to Customer** to make the information available to customers.

       Note

       This option can be bypassed by setting the **Bypass Visibility Option** flag to “true” in the activityactions table. When this flag is set to “true,” automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to “false,” all activities are visible to customers if the **Customer Visible** option is selected.
3. Click **Save & Exit**.

Relates topics

Search for a solution
Change Incident Status to Pending User Information
Change Incident Status to Pending Vendor/Supplier Investigation
Searching records

1.4.3.10. Test the incident resolution

Part of Workflow(s): Incident Management: Recovery

Applies to User Roles: Incident Analyst

One of the tasks of the Incident Analyst is to review and test the resolution described in the Solution field of an incident record. When the test is successful, the incident is assigned for incident closure. When the resolution test fails, the Incident Analyst returns the incident to incident investigation and resolution or assigns it to incident escalation.

To update tested incident resolutions:

1. Click **Incident Management > Incident Queue** or view your To Do LIst.
2. View and identify any incidents whose resolution (Solution field) can be tested.
3. Test the resolution described in a test environment that mimics the production environment.
4. When the resolution fails:
    1. If the incident requires an escalation, update the **Assignment Group** and **Assignee** fields with the Incident Coordinator responsible for the Incident Escalation process.
    2. If the incident doesn't require an escalation, update the **Assignment Group** and **Assignee** fields with the applicable Incident Analyst responsible for the investigation and diagnosis process or incident escalation.
    3. In both cases, in the Activities section, select a **New Update Type** and then in **New Update** type an explanation of the test failure.
5. When resolution testing is successful:
    1. Update the **Assignment Group** and **Assignee** fields with the applicable Incident Analyst responsible for the incident closure process.
    2. In the Activities section, select the **New Update Type** and then in **New Update** type a description of the test results.
6. Click **Save & Exit**.

Related topics

Change Incident Resolution
Reassign Incident Resolution
Searching records

1.4.3.11. Change incident resolution

Part of Workflow(s): Incident Management: Recovery

Applies to User Roles: Incident Analyst

One of the tasks of the Incident Analyst is to review and identify incidents received from the incident investigation and diagnosis process that need to be escalated to the Incident Coordinator because the implementation of the resolution requires a change.

To initiate a change for an incident:

1. Click **Incident Management > Incident Queue** or view your To Do list.
2. View and identify any incidents whose resolution (shown in the Solution field) can only be accomplished with a change.
3. Update the **Assignment Group** and **Assignee** fields with the applicable group and Incident Coordinator.
4. In the Activities section, select the **New Update Type** and then in **New Update** type a description of the change required to implement the resolution.
5. Click **Save & Exit**.

Related topics

Searching records
Test the Incident Resolution
Reassign Incident Resolution

1.4.3.12. Reassign incident resolution

Part of Workflow(s): Incident Management: Recovery

Applies to User Roles: Incident Analyst

One of the tasks of the Incident Analyst is to review and identify incidents received from the incident investigation and diagnosis process that need to be reassigned to another group, because the Incident Analyst doesn't have the necessary permissions to implement the resolution.

To reassign an incident:

1. Click **Incident Management > Incident Queue** or view your To Do list.
2. View and identify any incidents with a resolution (shown in the Solution field) that can't be accomplished because you don't have the applicable permissions.
3. Update the **Assignment Group** and **Assignee** fields with the applicable group and operator.
4. In the **Activities** section, do the following:
    1. From the **New Update Type** field list, select an update date.
    2. In the **New Update** field, type a description of the permissions required to implement the resolution.
5. Click **Save & Exit**.

Related topics

Searching records
Test the Incident Resolution
Change Incident Resolution

1.4.3.13. Reject an incident resolution

Part of Workflow(s): Incident Management: Review

Applies to User Roles: Incident Analyst

If the solution assigned to an incident is found to not resolve the incident completely, the incident must be returned to the incident investigation and diagnosis process for additional analysis. For example, a solution may appear to solve the problem, but when the customer uses the solution, the problem persists. Incidents that have associated service requests or changes require additional processing.

To reassign an incident for additional investigation and diagnosis:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident for review.
3. Review the incident resolution description in the Solution field to verify that it is correct and complete. Verify that all information in the detail record is complete.
4. When the solution doesn't resolve the incident and the incident doesn't include an associated service request or change:
    1. Update the **Status** field to Work In Progress.
    2. Update the **Assignment Group** and **Assignee** fields with the applicable group and Incident Analyst.
    3. In the Activities section, do the following:
        1. From the **New Update type** field list, select an update type.
        2. In the **New Update** field, type an explanation that describes the problem with the proposed solution.
    4. Click **Save & Exit** to return to your To Do queue.
5. If the incident includes an associated service request or change, the incident requires additional processing.

Related topics

Searching records
Reject an Incident Resolution with an Associated Change or Service Request
Close an Incident
Close an Incident with an Associated Interaction or Event

1.4.3.14. Reject an incident resolution with an associated change or service request

Part of Workflow(s): Incident Management: Review

Applies to User Roles: Incident Analyst

If the solution provided for an incident is found to not resolve the incident completely, then the Incident Analyst must review the incident for necessary changes or requests. When incident closure rejects the incident solution and the incident has an associated change, incident closure updates the status of the incident to pending change and requests that Change Management reopen the change. When incident closure rejects the incident solution and the incident has an associated service request, incident closure updates the status of the incident to pending change and requests that the Service Desk reopen the request.

To reassign a rejected incident solution with an associated change or service request:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident for review.
3. Review the incident resolution description in the Solution field to verify that it is correct and complete.
4. When the solution doesn't resolve the incident and the incident includes an associated service request:
    - Update the **Status** field to Pending Change.
    - Update the **Assignment Group** and **Assignee** fields with the applicable Service Desk group and Service Desk Agent.
    - In the **Activities** section, do the following:
        - From the **New Update Type** field list, select an update type.
        - In the **New Update** field, type an explanation that describes the problem with the proposed solution and that also mentions any associated requests that must be reopened.
    - Click **Save & Exit** to return to your To Do queue.
5. When the solution doesn't resolve the incident, and the incident includes an associated change:
    - Update the **Status** field to Pending Change.
    - Update the **Assignment Group** and **Assignee** fields with the applicable Change Management group and Change Coordinator.
    - In the **Activities** section, do the following:
        - From the **New Update Type** field list, select an update type.

        - In the **New Update** field, type an explanation that describes the problem with the proposed solution and that also mentions any associated changes that must be reopened.
    - Click **Save & Exit** to return to your To Do queue.

Related topics

Searching records
Reject an Incident Resolution
Close an Incident
Close an Incident with an Associated Interaction or Event

1.4.3.15. Escalate an incident

Part of Workflow(s): Incident Management: Investigation

Applies to User Roles: Incident Coordinator

If the Incident Coordinator determines that the incident can be solved by escalating an incident, the Incident Coordinator proceeds as follows:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** queue.
2. Click the open incident record.
3. Click the **Related Records** tab.
4. Select an appropriate link type from the **Link Type** drop-down list box.
5. Click **Link New Record**.
6. Select an appropriate category on the page that appears.
7. Follow the instructions to continue creating a new record.

Related topics

Reassign an Incident for Additional Support
Searching records

1.4.3.16. Reassign an incident for additional support

Part of Workflow(s): Incident Management: Investigation

Applies to User Roles: Incident Manager

After an Incident Manager is notified that an incident is not going to be resolved on time and that the service level agreement (SLA) is breached or in danger of being breached, the Incident Manager reassigns the incident for additional support.

The Incident Manager reviews the SLA target times and determines that the incident requires reassignment to another level of support. The Incident Manager reassigns the incident to another support group.

To reassign an incident for additional supports:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** list.
2. Click the open incident.
3. Clear the assignment group in the **Assignment Group** field, and then click **Fill** to select the new assignment group.
4. Click **Save & Exit**.

Note: You can contact the Service Desk to request that the Service Desk send an information bulletin to affected users and stakeholders.

Related topics

Searching records
Escalate an Incident

1.4.3.17. Monitor interaction queue for service level agreement breaches

Part of Workflow(s):

Service Level Agreement Monitoring

Applies to User Roles:

Service Desk Agent

The Service Desk Agent monitors the interaction queues for interactions with associated incidents that have not been resolved in the time specified by the Service Level Agreement (SLA). When this occurs, the Service Desk Agent begins the escalation procedure for the incident by assigning the incident to the applicable Incident Manager and assignment group based on the services affected by the incident.

To reassign breached incidents for additional analysis and diagnosis:

1. Click **Service Desk > Interaction Queue** or view your To Do queue.
2. In the View list, select Monitor SLA Interaction - Breached.
3. From the list, select an interaction record with a linked incident (Open - Linked) record.
4. To view the related incident, click **More** or the More Actions icon and select **Related > Incidents > View**.
5. Double-click the **Incident ID**.
6. Update the **Assignment Group** and **Assignee** fields based on the information in the Service and Affected CI fields.
7. In the Activities section, do the following:
    1. In the **New Update Type** field list, select an update type.
    2. In the **New Update** field. type an explanation to describe the reason for the escalation and any relevant details.
8. Click **Save & Exit** to return to the Interaction queue.

1.4.3.18. Monitor interaction queue for potential service level agreement breaches

Part of Workflow(s): Service Level Agreement Monitoring

Applies to User roles: Service Desk Agent

The Service Desk Agent monitors the interaction queues for interactions with associated incidents that are within a few hours of breaching as specified by the Service Level Agreement (SLA) for the type of incident. For those items that are within one hour of breaching, the Service Desk Agent coordinates with the Incident Coordinator who escalates the incident associated with the escalation and communicate the expected resolution to the applicable users and user groups. For those items that are within four hours or one day of breaching, the Service Desk Agent monitors the applicable queues until the interactions are resolved or escalated.

To monitor the interaction queues for potential breached interactions:

1. Click **Service Desk > Interaction Queue** or view your To Do queue.
2. In the View list, select the **Monitor SLA Interaction - Breach within 1 hour** queue.
3. To view the incident details for an interaction in the queue:
    1. View the related incident.
    2. Click **More** or the More Actions icon and select **Related > Incidents > View**.
    3. Double-click the **Incident ID**.
4. Contact the Incident Coordinator assigned to the related incident, and either begin the Incident Escalation process or contact the affected users and groups with the expected resolution details.
5. For any interactions in the other Monitor SLA Interaction queues, continue to monitor the queues until the items are resolved or require escalation.

1.4.3.19. Handle Complaints

Part of Workflow(s): Complaint Handling

Applies to User Roles: Service Desk Manager

The Service Desk Agent assigns an incident containing a user complaint to the Service Desk Group. When the incident is saved, it is automatically assigned to the Service Desk Manager. The Service Desk Manager reviews and accepts the incident to research and resolve the complaint.

To manage an incident based on a user complaint:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** queue.
2. Click the open incident record and review the contents of the User complaint.

   Note: Investigate the complaint by reviewing the relevant information and talking to the people involved.

3. When you have finished searching for an answer or solution to satisfy the user, select the Activities section to add appropriate notes.
4. Enter the resolution in the **Solution** field and verify that the information is correct and complete.
5. Select **Resolved** from the Status field.
6. Click **Close** to close the incident.
7. Select an appropriate code from the **Close Code** field in the form that appears.
8. Click **Finish**. The status changes to Closed.

Related topics

Incident Management overview
Service Desk overview
Associating a Service Desk interaction with another application record
More Actions menu
Searching records

1.4.3.20. Open an incident

Applies to User Roles: Configuration Auditor, Incident Coordinator, Incident Manager

You can open incident records as part of many Service Management processes, including interaction management, event management, configuration verification and audit, and incident management.

Note: For information on opening or creating incident records in the interaction management process and event management process, see the related topics.

To open an incident record:

1. Click **Incident Management** > **Create New Incident**.
2. Select the incident category.

   > **Note**
   >
   > If a default category is specified in the Incident Settings, Service Manager directly assigns the incident to the default category and skips this step. For more information, see Configure Incident Management settings.

3. Type an appropriate title in the **Title** field and then give appropriate description in the **Description** field.
4. Click **Fill** to select an Assignment Group.
5. Click **Fill** to select the applicable **Affected Service**.
6. Click **Fill** to select the **Affected CI**.

   > **Note**
   >
   > The **Default Impact** and **Priority** values of the affected CI are automatically populated to the **Impact** and **Urgency** fields of the incident record. You can manually change these auto-populated values if needed.

7. Click **Save**.
8. Search possible solutions in the form that appears. If matching solutions are found, select the corresponding record, and then click **Link Selected Record**.

> **Note**
>
> You can also perform this step after an incident is created. To do this, open the incident and then select **Solution Matching** from the **More** menu in the Detail List toolbar.

9. Complete the incident form with any other relevant information.
10. Click **Save & Exit**.

> **Note**
>
> You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Incident Management overview
Create New Incidents from Monitoring System Notifications
Create a new incident from a user interaction
Access Incident Management views
Update an incident
Relate a record to an incident record
Resolve an incident
Close an Incident
Searching records

1.4.3.21. View a list of services potentially affected by an outage

Applies to User Roles: Incident Coordinator, Incident Manager

The View Affected Services menu option enables you to view a list of services that are potentially affected by an outage relating to CIs that are specified in an incident or change record. By viewing the affected services list, you can determine the potential impact of an outage related to critical, dependent services when you open an incident. You can also use the affected services list to plan when opening a change.

Note

The list of CIs generated doesn't include the CIs specified in the primary incident or change record.

To view a list of affected services:

1. Perform one of the following actions.
   - Open an existing incident or change record. Use search or advanced search to find one or more records.
   - Proceed as if you are going to add a new incident or change record.
2. If you are in the process of adding a new record, fill in the **Affected CI**.
3. Click **More** or the More actions icon and select **View Affected Services**. The CI Identifier record opens.
4. Open the **Affected Services** section. A list of affected services opens or a message, stating that there are no services affected by the outage.
   Note: For change records, a list of all affected business services is returned in all cases, whether the record has one primary CI or multiple CIs.
5. Select a CI from the list for detailed information from the Configuration Management record.
6. When you finish viewing the affected services, click **Cancel**.

Related topics

Outages
More Actions menu
Searching records
Enable automatic outage posting
Disable automatic outage posting

Create a new incident from a user interaction
Create New Incidents from Monitoring System Notifications
Create a new change record

1.4.3.22. Apply a template to complete an incident

Applies to User Roles: Incident Analyst, Incident Coordinator, Incident Manager

Templates enable users to quickly complete an incident record by automatically populating fields with the applicable information.

Note: Make sure that the user profile is enabled to use templates. For additional information on enabling a profile to use templates, see the related topics.

To apply a template to complete an incident:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** queue.
2. Select an incident record.
3. Click **Apply Template**. The Select Incident Template wizard opens.
4. Double-click the template that you want to apply to the selected incident. You are returned to the selected incident record and the fields are automatically filled in with the values set in the template.
5. Modify other fields, as needed.
6. Click **Save & Exit**.

Note: Users authorized to use this template can also modify the template to meet their needs.

Related topics

Application profiles
User roles
Searching records

1.4.3.23. Access Incident Management views

Applies to User Roles: Incident Analyst, Incident Coordinator, Incident Manager

Incident Management views contained in the Favorites and Dashboards navigation pane enable you to easily and quickly access specific types of records. When the database changes, the dashboard contents change to reflect current activity.

To view your favorites, click **Favorites and Dashboards** > **Incident Management**. Service Manager provides the following default Incident Management views:

- (1) Monitor OLA-UC Incident - Breached
- (2) Monitor OLA-UC Incident - Breach within 1 hour
- (3) Monitor OLA-UC Incident - Breach within 4 hours
- (4) Monitor OLA-UC Incident - Breach within 1 day
- All Open Incidents
- Incidents by Assignment Group
- Open Incidents Assigned to Me
- Open Incidents I Own

Note: The Monitor OLA-UC breached views contain incident records that measure the performance of individual support groups and applicable vendors. The records that are close to breaching an agreement or that already have breached an agreement can be seen in these views.

Related topics

Views and favorites

1.4.3.24. Relate a record to an incident record

Applies to User Roles: Incident Analyst, Incident Coordinator, Incident Manager

As part of incident processing, you can relate or associate an existing incident, change, request, or problem to an incident. Use related records to associate an incident with any applicable incidents, changes, requests, or problems so that status changes or updates that you make will also be made to associated records.

To relate a record to an incident record:

1. Click **Incident Management** > **Incident Queue** or view your To Do queue.
2. Select a target record.
3. Click the **Related Records** tab.
4. Select an appropriate type from the **Link Type** drop-down list box.
5. Click **Link Existing Record**.
6. When the Associating Records form opens, type the ID of the record.

   You can also use **Search** in the Associating Records dialog box to locate the applicable ID number.
7. Click **OK**.

Related topics

Incident Management overview
More Actions menu
Searching records
Access Incident Management views
Open an incident
Update an incident
Resolve an incident
Close an Incident

1.4.3.25. Update an incident

Applies to User roles: Incident Analyst, Incident Coordinator, Incident Manager

Sometimes you need to update an existing incident record for reasons, such as reassigning the incident to another group or changing the priority of the incident record. When you update the incident record, you can also include helpful information in the Update field of the Activities section.

To update an incident:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Select a record to display it.
4. Make any necessary changes. For example, if you want to reassign the incident record to another group:
    - Clear the assignment group in the **Assignment Group** field, and then click **Fill**. A list of assignment groups is displayed.
    - Select the new assignment group from the list.
5. Open the Activities section to do the following:
    - Add notes in the **Update** field with the reason why the incident has been updated or to add other helpful information.
    - If necessary, click the **New Update Type** list to categorize the activity update.
6. When you are finished with your updates, click **Save & Exit**.

Note: When you view an existing incident, shaded fields are read-only fields. Incident Management populates these fields from information stored in associated records. You can update the associated record to increase the amount of information in this incident record.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Incident Management overview
Searching records
Access Incident Management views
Open an incident
Relate a record to an incident record
Resolve an incident
Close an Incident

1.4.3.26. Resolve an incident

Applies to User Roles: Incident Analyst, Incident Coordinator, Incident Manager

When you find an existing solution that resolves an incident, you can attach the solution to the incident record and close the record. You can also save the resolution as a candidate for the knowledgebase.

To resolve an incident:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Double-click a record to display it in the Incident form.
4. Click the **Activities** tab and then do the following:
    1. In the **Update Type** field list, select an update type.
    2. In the **Update** field, type a note to explain how you resolved the incident.
5. Click the **Proposed Solution** tab and then type the solution in the **Solution** field.

    If the incident is a potential problems candidate, select the **Problem Candidate** checkbox.
6. Select **Resolved** from the **Status** drop-down list box.
7. Click **Save**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Incident Management overview
Searching records
Access Incident Management views
Open an incident

Update an incident
Relate a record to an incident record
Close an Incident

1.4.3.27. Close an incident

Part of Workflow(s): Incident Management: Review

Applies to User roles: Incident Analyst

Incidents that have been resolved satisfactorily and have no associated events, interactions, or changes are closed by specifying a closure code and updating the status to closed. Incidents that have associated events, interactions, or changes require additional processing.

To close an incident:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident that's ready to be closed.
3. Click **Close**. The Close Incident form is displayed.
4. Set the following fields:

   Subcategory and **Area**: You can change the **Subcategory** and **Area** fields, and the change will be saved to the incident when you click **Finish**.

   Closure Code: Select the code that best describes the closure.

   Problem Candidate: Select this checkbox if you want to mark this incident as a problem candidate.

   Completion Comments: Enter your comment for the closure.
5. Click **Finish**. The status changes to Closed.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Searching records
Close an Incident with an Associated Interaction or Event
Relate a record to an incident record
Reject an Incident Resolution
Reject an Incident Resolution with an Associated Change or Service Request

1.4.3.28. Close a first-time resolved incident

Applies to User roles: Service Desk Agent, Incident Coordinator

When you create a new incident, you can close the incident directly if you are able to resolve the user request on the first intake in the Logging phase of the incident. If the incident is triggered from an interaction, the associated interaction record is closed automatically.

To close a first-time resolved incident, follow these steps:

1. Open the incident record that's in the Logging phase.
2. In the **Solution** filed, type a solution for the incident.
3. Click the **Resolve Directly** button on the toolbar.

   Service Manager displays the **Close Incident** page.
4. Specify information for the **Subcategory** and **Closure Code** fields.
5. Specify information for other optional fields such as **Completion Comments**.
6. Click **Finish**. The status of the incident changes to Closed.

Related topics

Searching records
Close an Incident

1.4.3.29. Close an incident with an associated interaction or event

Part of Workflow(s): Incident Management: Review

Applies to User Roles: Incident Analyst

Incidents that are resolved satisfactorily that have associated events or interactions are closed by specifying a closure code and updating the status to closed. Incidents that have associated changes require additional processing.

To close an incident with an associated interaction or event:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident that's ready to be closed.
3. Review the incident resolution description in the Solution field to verify that it is correct. Make sure that all information in the detail record is complete.
4. If this incident has an associated interaction, update the **Assignment Group** and **Assignee** fields with the Service Desk group and Service Desk Agent responsible for the Close Interaction process.
5. If this incident has an associated event, update the **Assignment Group** and **Assignee** fields with the Group and Operator responsible for Event Management.
6. Click **Close**. The Close Incident form is displayed.
7. Use Fill to select an applicable code for the **Closure Code** field.
8. Enter your comments in the **Completion Comments** field.
9. Click **Finish**. The status changes to Closed.

Related topics

Searching records
Close an Incident
Reject an Incident Resolution
Reject an Incident Resolution with an Associated Change or Service Request

1.4.3.30. Close multiple incident records

Applies to User Roles:

System Administrator

You can close multiple incident records or a set of incident records using mass close. For example, you may want to close multiple incident records that have been rejected.

To close multiple incident records:

1. Click **Incident Management > Search Incidents**.
2. Use search or advanced search to find one or more records. Select the View list of incident records you want to close. For example, select Rejected Incidents to close multiple incident records that have been rejected.
3. Using the scroll bar and the **ctrl** key, click the individual records that you want to include in the mass close. Each record you select remains highlighted.
4. Click **Mass Close**. The Mass Close Incidents form opens.
5. Provide resolution information in the following fields:
    - **Fix Type**: select permanent or temporary.
    - If the service delivery units were unable to fix the issue, select **SDU unable to fix**.
    - Click **Fill** in the **Resolution Code** field to generate a record list of possible resolutions and choose one.
    - Click **Fill** in the **Resolution Analysis Code** field and choose one.
    - Click **Next**.
    - Type an explanation in the **Solution** field.
6. Click **Finish**. The mass close process starts, and then displays each incident number as it is closed. A message also states that related interactions will be processed normally.

1.4.3.31. Create an incident task

Applies to User roles: Incident Coordinator

After an incident is created, you can create the tasks for categorizing, investigating, resolving, and reviewing the incident.

To create an incident task:

1. Select an incident that's not closed.
2. Click the **Link New Task** button in the **Tasks** tab of this incident.
3. Select a category for the new task.

   Service Manager opens a task information form.
4. Complete the form with all required information.
5. Click **Save** or **Save & Exit**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Cancel an opened task for an incident record
Close an opened task for an incident record

1.4.3.32. Cancel an opened task for an Incident record

Applies to User roles: Incident Coordinator, Incident Manager, Incident Analyst

If an incident record has an opened task, you can cancel the open task by using the Cancel Task feature.

To cancel an opened task for an incident record:

1. Click **Incident Management** > **Search Incidents**, enter your search criteria, and then click **Search**.
2. Select an incident and then click the **Tasks** tab.

   A task list appears if this change contains opened tasks.
3. Double-click the task that you want to cancel.
4. Click **Cancel Task** from the **More** drop-down list.
5. Enter the Task Outcome and then click Finish.

The opened task is now cancelled:

- The status of the opened task is set to Cancelled.
- The automatic transition moves the workflow phase to Cancelled.
- The Completion Code is set to Cancelled.

Related topics

Create an incident task
Close an opened task for an incident record

1.4.3.33. Close an opened task for an incident record

Applies to User roles: Incident Coordinator, Incident Manager, Incident Analyst

If an incident record has opened tasks, you can close them by using the Close Task feature.

Note

You can't close a high priority (priority is High or Critical) incident task if this incident task is not in the Review phase.

To close an opened task for an incident record:

1. Click Incident **Management** > **Search Incidents**, enter your search criteria, and then click **Search**.
2. Select an incident and then select the **Tasks** tab.

   A task list appears if this change contains opened tasks.
3. Double-click the task that you want to close.
4. Provide necessary information for task completion.
5. Click **Close Task**.
6. Enter the Completion Code and Task Outcome, and then click **Finish**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

The opened task is now closed:

- The status of the opened task is set to Closed.

- The task is moved to the Closure phase.
- The Completion Code is set to the value you provided.

Related topics

Create an incident task
Cancel an opened task for an incident record

1.4.3.34. Create other types of record from an incident

You can create other types of record from an incident record, and the created record is automatically linked to the incident record.

To do this, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Open the incident from which you want to create another type of record.
4. In the **Related Records** section, select one of the following options from the **Link Type** drop-down list. These options defines the type of the new record and the relation between the new record and the incident.
    - Caused By Changes
    - Caused Changes
    - Caused By Incidents
    - Caused Incidents
    - Related Requests
    - Related Known Errors
    - Related Problems
5. Click **Link New Record**.
6. Follow the on-screen instructions to complete the creation of the new record.

After you create the new record, the new record is automatically linked with the incident.

1.4.3.35. Set a parent incident

Before you can create a parent-child relationship between incidents, you must set an incident as the parent incident.

To do this, follow these steps:

1. Open the incident record that you want to use as the parent incident.
2. In incident details, select the **Parent Incident** checkbox.
3. Click **Save**.

The **Child Incidents** section then appears under **Related Records**.

If you clear the **Parent incident** checkbox, the **Child Incidents** section then disappears. When you save the change, the parent incident ID is removed from the **Link to Parent Incident** field in all child incidents. That is, the parent-child relationships between these incidents are dismissed. In addition, if the **Notify incident owner if Parent Incident flag was unchecked** option is selected in incident settings, a notification is sent to all assignees of the child incidents.

Related topics

Parent and child incidents
Incident Management overview
Set a child incident
Unlink a child incident

1.4.3.36. Set a child incident

Follow these steps to set and link a child incident to a parent incident:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Open the incident that you want to make as the child.
4. Specify the parent incident in the **Link to Parent Incident** field. To do this, you can use one of the following methods:

   > **Note**
   >
   > The incident that you specify in this field must be a parent incident. That is, the **Parent Incident** option of this incident is selected.

   - Enter the ID of the parent incident. By this method, you can specify any incident that's marked as a parent.
   - Click the **Fill Field Link to Parent Incident** button. A list of parent incident candidates then appears. Select a parent incident and then click **Link Selected Record**.

     By this method, you can only specify a parent incident that matches one of the following criteria, which you can use to filter existing parent incidents.

     - An incident with a matching title
     - An incident with a matching related service or configuration item
     - An incident with a matching service or configuration item
5. Set the **Subcategory**, **Area** and **Assignment Group** fields.
6. Click **Save**.

After you link a child incident to a parent incident, the child incident appears in the **Child Incidents** section in the parent incident.

Related topics

Parent and child incidents
Incident Management overview
Set a parent incident
Unlink a child incident

1.4.3.37. Unlink a child incident

To unlink a child incident from the parent incident, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Open the parent incident.
4. In the **Child Incidents** section, select the child incident that you want to unlink, and then click **Unlink Selected Record**.

   > **Note**
   >
   > You can also select and unlink multiple child incidents, or simply click **Unlink All** to unlink all child incidents.

5. Click **Save**.

Related topics

Parent and child incidents
Incident Management overview
Set a parent incident
Set a child incident

1.4.3.38. Mark an incident as a major incident

Applies to User roles: Incident Analyst

To mark an incident as a major incident, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Double-click a record to display it in the Incident form.
4. Select the **Major Incident** checkbox and then specify an incident manager.

   A new **Major Incident Review** section then appears in incident details.
5. Click **Save**.

The specified incident manager receives an email about the creation of the major incident.

1.4.3.39. Mark an incident for escalation

Applies to User roles: Incident Coordinator

To mark an incident for escalation, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records.
3. Double-click a record to display it in the Incident form.

1. Select the **Escalated** checkbox and then specify an incident manager.
2. Specify an escalation team in the **Escalation Team** section.
3. Click **Save**.

The incident assignee and the incident manager then receive an email about the incident escalation.

1.4.4. Incident Management administrator tasks

A System Administrator can access the Operator record for the following information types and functions:

- User and contact information
- Application profile privileges
- Mandanten utility

Configure Incident Management settings

Applies to User roles: System Administrator

To configure Incident settings:

1. Click **Incident Management** > **Administration** > **Settings** in the System Navigator.
2. Click **Incident** and then configure the following settings for the Incident application:

Setting	Description
Notify incident owner if Parent Incident flag was unchecked	When the Parent Incident flag is unchecked, a notification is sent to all the assignees of the child incidents that were related to this parent incident
Default Category	The default category when you register a new Incident. The out-of-box value for this parameter is "none".
Parent to Child Incident Default Status	When any of the following scenarios occurs, the status of the child incident changes to the default status that's specified in this setting. ◦ The parent-child relationship is dismissed from the parent incident ◦ The parent incident is closed and the **Parent Incident Relationship Model** is **Parent Incident and Child Incidents close independently**
Parent Incident Relationship Model:	Whether to close a child incidents when its parent incident is closed

Setting	Description
Post back link to Child	The link file that posts the changes of certain fields in the parent incident to the corresponding fields in child incidents. This function works only when you select **Close Child Incidents when Parent Incident closed** in **Parent Incident Relationship Model**. You can customize the fields that are posted back to the child incidents. For more information about links, see Links.

3. Optionally, click the **Incident Task** and then configure the following settings for an incident task:

Setting	Description
Default Category	The default category when you register a new incident task. The out-of-box value for this parameter is "none".

4. Click **Save** and then click **OK**.

Configure the Incident Management environment

Applies to User roles: System Administrator

Incident Management provides an environment record in which you define the operational environment for the application by allowing or disallowing specific operator actions. Administrators can modify the environment parameters to match the operational requirements of their organization.

To configure the Incident Management environment:

1. Click **Incident Management** > **Administration** > **Environment** in the System Navigator.
2. Select or clear the parameters for your Incident Management environment.

   Tip

   You can use the field help to view the description for each setting. To view the help on field:

   - Web client: Select a field, and then press **F1**.
   - Windows client: Select a field, and then press **Ctrl+H**.

3. Click **Save**.
4. Click **OK**.

Create a template to complete incident records

Applies to User Roles:

System Administrator

Templates enable users to quickly complete incident records by automatically populating various fields with necessary information. You can create as many templates as you need while authorizing select user roles for each template. The users authorized to use a particular template will then be able to apply that template when completing an incident.

Note

Make sure that the user profile is enabled to use templates.

To create a template, follow these steps:

1. Click **Tailoring** > **Templates**, and then click **New**.
2. Type a name in the **Template name** field.
3. Select Incident in the **Table name** field, and then click **Next**.
4. Click **Fill** for each user role that you are authorizing to use this template.
5. Modify the fields as required to meet the needs of your template by double-clicking in the field to assign a value to each field you select.
6. Click **Add** to exit the template wizard.
7. Click **Save**.

8. Click **OK**. The roles authorized for this template can now use or modify this template.

Alternatively, you can create a template from an existing record. To do this, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Enter an Incident number in the **Incident ID** field, and then click **Search**.
3. Click the incident record.
4. Click **More** or the More Actions icon and select **Create Template from Record**.
5. Modify the name in the **Template name** field.
6. Click **Fill** for each user role that you are authorizing to use this template.
7. Modify the fields as required to meet the needs of your template by double-clicking in the field to assign a value to each field you select.
8. Click **Add** to exit the template wizard.
9. Click **Save**.
10. Click **OK**.

Assignment groups

An assignment group is a list of users who are responsible for an Incident record. Incident Management notifies the group when an Incident record opens or escalates. Assignment groups make the routing and escalation of Incident records easier.

For example, the Service Desk Agent receives a service request to fix a disabled workstation. The agent creates an Incident record, and then assigns the incident to the Hardware assignment group. The hardware technician determines that the hard drive must be replaced. Because the drive must be purchased, the technician assigns the Incident record to a Request Fulfillment assignment group for acquisition. If the hard drive purchase is delayed, and the delay impacts the closure of the Incident record in a reasonable amount of time, Incident Management automatically escalates the Incident record and assigns it to the Request Fulfillment manager.

Service Manager administrators create Assignment groups as part of the Ongoing Maintenance process.

Add an Incident Management assignment group

Applies to User Roles: System Administrator

Incident Management assignment groups specify the Incident Coordinator and Operators in the group so that when someone creates a new incident it can be assigned to the applicable group. Typically, assignment groups are organized by

location and expertise. To facilitate incident response, new assignment groups can be added to the out-of-box assignment groups for Incident Management.

To add an Incident Management assignment group:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.
   1. Enter the name of the manager of the assignment group.
   2. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   3. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide response when you create and populate a new assignment group.

   (Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.
7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

(Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.
9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.
10. Follow these steps to configure the Supported Services field:
    1. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    2. Select **Business Service** in the CI Type field, and then click **Search**.
    3. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    4. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.
11. Click **Save** and **OK**.

Using mass update with Incident Management record lists

Mass update enables you to select multiple records from a list of records and then to update the value in a field or several fields in the selected records. The system provides a template form that displays the fields for the selected records and allows a user to change the value of any of the displayed fields. The Mass Update template form doesn't display all fields in the records. For example, fields marked read only in the data policy don't display. You can also do a Complex Update on the selected records. A Complex Update uses RAD expressions containing variables and concatenated fields to populate another field in the records.

When doing a mass update, you should remember that the value you enter for a particular field becomes the value for that field for all of the selected records. Whenever you update incident records, you must also update the incident activity data.

Mass update is available for incident records and incident queue records. The ADMIN and SYSADMIN profiles in the out-of-box system provide the Template Mass Update

and Complex Mass Update capability

A System Administrator can edit the datadict table and probsummary file so that a field doesn't appear in the list of fields displayed by the Mass Update wizard. On the Data Policy form, change the Usage Type column for the field to System.

Update multiple incident records

Applies to User Roles: System Administrator

You can use mass update to update a value in one or more fields for multiple incident records. For example, if you want to update multiple high priority incident records, select the high priority view of the Incident queue. The value you enter for a particular field becomes the value for that field for all the records you selected for mass update.

You can use complex update to enter expressions as instructions for actions to be performed on multiple incident records.

To update multiple incident records by using **Mass Update**:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records. Select the View list of incident records that you want to update. For example, select High Priority Incidents to update multiple high priority incident records.
3. Use the arrow keys to move to a row and then use the space bar to select individual records. A range of records can be selected using **Shift**+Spacebar. You can also use the scroll bar and the **ctrl** key to select individual records.
4. Click **Mass Update**.
5. Update the fields to be included in the mass update:
    1. Double-click the field that you want to update, and then type the value for the field in the box or use the Fill feature to display a list of potential values for the field.
    2. Click **Next**.
    3. Continue updating the fields that you want included in the mass update.
    4. Click **Next** after each update.
    5. Click the **Activity Update** field to document what's included in the mass update. When the mass update is complete, these notes display in the Journal Update of the Activities section for each record.
6. Click **Execute**. Mass update starts, and then displays each incident number as it is updated.
7. If an incident record can't be included in the mass update, you can choose to:
    - **Retry**: Retry the mass update.

- **Skip**: Skip the incident record to exclude it from the mass update and continue with the next record.
- **End**: End the mass update, and retry later.

To update multiple incident records, using **Complex Update**:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records. Select the View list of incident records you want to update. For example, select High Priority Incidents to update multiple high priority incident records.
3. Click **Mass Update**.
4. Click **Complex Update**.
5. Follow the instructions in the complex update form and type the expressions in the **Instructions to be executed ONCE at the beginning of mass add/update** field and the **Instructions for action on EACH RECORD** field.
6. Click **Execute**.
7. If an incident record can't be included in the complex update, you can choose to:
   - **Bypass**: Bypass the invalid message and include the incident record in the complex update.
   - **Skip**: Skip the incident record to exclude it from the complex update and continue with the next record.
   - **End**: End the complex update, and retry later.

Incident Management downtime records

You can access Incident Management downtime records by using the Incident Management Security Administration Utility to show the availability of a selected device. Availability measures the ability of a device or component to provide service within a measured time frame. There are three downtime measurements.

Type of availability	Description
Explicit unavailability	The downtime experienced by the failing device.
Implicit unavailability	The downtime experienced because of the failure of a parent or controlling device.
Perceived unavailability	The explicit or implicit downtime during normal business hours. The total reflects only working hours.

Create a downtime record

Applies to User Roles: System Administrator

As part of monitoring the availability of devices in the system, a System Administrator creates downtime records for a specific device.

To create a downtime record:

1. Click **System Administration** > **Base Configuration** > **Monitoring** > **Downtime**.
2. Type the applicable data in the **Logical Name**, **Location**, **Contact Name**, **Type**, and **Table Name** fields.
3. Click **Add**.
4. Click **OK**.

Reset downtime

Applies to User Roles: System Administrator

To reset downtime:

1. Click **System Administration** > **Base Configuration** > **Monitoring** > **Downtime**.
2. Click **More** > **Reset**.
3. Click **Schedule** to set a time for the reset to occur.
4. Click **OK**.

Create a note

Applies to User Roles: System Administrator

Any note that an Administrator or Operator user creates is saved with the associated Service Manager records. You can add more notes each time you select the Notes option. Each set of notes has a time stamp and the name of the operator who created the note set.

You must have notes enabled by your Service Manager System Administrator to use this feature. To enable notes for an Incident Management operator, select notes on the Incident Management Security Profile form. For the out-of-box system, only the System Administrator has notes enabled.

To create a note:

1. Do one of the following to locate a record:
    1. Select an incident record from a queue.
    2. Use search or advanced search to find one or more records.
2. Click **More** or the More Actions icon and select **Notes**.
3. Type the information in the **Add New Note** box.
4. Click **Save**.
5. Click **Back**.
6. Click **Save & Exit**.

1.4.5. Incident configuration

Incident configuration enables you to configure alert, Incident categories, workflow, solution matching, and so on.

Create an incident category

Applies to User roles: System Administrator

If you are a Service Manager Administrator, you may want to create an incident category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides out-of-box category records that you can use or modify.

> **Note**
>
> When a category is set as the default category in settings, don't delete it to avoid unpredictable issues.

To create new category record:

1. Click **Incident Management**.
2. Click **Configuration** > **Incident Categories**.
3. Click **New**.
4. Type the name of the incident category.
5. Clear the **Active** checkbox if you don't want the new category to appear in the category list.

   > **Note**
   >
   > If you clear the **Active** checkbox, this category can't be used to create new incidents.

6. Type the category description.
7. Select a workflow for the category.
8. Click **Save**.

9. **Optional**: Click the **Link New Subcategories** button to create a new subcategory for the incident category.

Note

The **Apply To** flag indicates whether the category is shared across different modules.

For example, if the **Apply To** is set to "Interaction/Incident", then this category will be created in both the interaction category table and the incident category table.

Typical shared categories in the out-of-box configuration include Complaint, Request for Information, Incident, and Request for Administration.

For the **Apply To** flag in Incident Category:

When creating an incident category, the **Apply To** option is predefined as "Incident" only and can't be modified. However, an incident category can also be created when creating an interaction category with the **Apply To** option set as "Interaction/Incident" or "Interaction/Incident/Problem".

When searching for an incident category, all the **Apply To** values that include Incident can be used.

The following list is used in Incident Category:

1 - Incident

3 - Interaction/Incident

4 - Interaction/Incident/Problem

Note

Incident category name is read-only after the category is created.

Create an incident task category

Applies to User roles: System Administrator

If you are a Service Manager Administrator, you may want to create an incident task category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides out-of-box category records that you can use or modify.

To create a new incident task category record:

1. Click **Incident Management**.
2. Double-click **Configuration** > **Incident Task Categories**.
3. Click **New**.
4. Type the name of the incident task category.
5. Type a description of the problem task category.
6. Clear the **Active** checkbox if you don't want the new category to appear in the category list.

   Note

   If you clear the **Active** checkbox, this category can't be used to create new incident tasks.

7. Select a workflow for the category. The **Workflow** tab is displayed.
8. Click **Save**.

Note

Incident task category name is read-only after the category is created.

Add a new subcategory to an incident category

User Roles: System Administrator and Implementer

You can add a new subcategory directly for an incident category.You can view a list of subcategories and their record details associated with the current category.

To add a new subcategory for an incident category:

1. Click **Incident Management** > **Configuration** in the System Navigator.
2. Click **Incident Categories** and then click **Search** in the form that appears.
3. Select the Incident Category for which you want to add a subcategory.
4. Click the **Subcategories** tab in the Incident Category Definition page, and then click the **Link New Subcategories** button.

5. Type a subcategory name.
6. Type a description for the subcategory.
7. Click **Save** to add the new subcategory.
8. **Optional**: Click the **Link New Areas** button to create areas for the incident subcategories. When you have finished, click **Save**.

Add a new area for an incident subcategory

Applies to User roles: System Administrator and Implementer

You can add a new area directly for an incident subcategory.You can view a list of areas and their record details associated with the current subcategory.

To add a new area for an incident subcategory:

1. Click **Incident Management** > **Configuration** in the System Navigator.
2. Click **Incident Categories** > **Search**.
3. Select the Incident Category for which you want to add an area.
4. Under the **Subcategories** section in the Incident Category Definition page, select the subcategory for which you want to add an area.
5. Type an area name.
6. Type a description for the area.
7. Click **Save** to add the new area.

Incident solution matching

When you create an incident, a list of potentially related incidents appears based on the pre-defined solution matching configuration. You can then select the matched records and update them directly. You can also run solution matching for an existing incident that's not closed. To do this, select the incident and then click **More** > **Solution Matching** from the Detail List toolbar.

In an out-of-box system, the following options are provided for Incident solution matching:

- Find an incident with a matching service or configuration item
- Find an incident with a matching related service or configuration item
- Find an incident with a matching title
- Find a known error with a matching service or configuration item
- Find a known error with a matching title
- Find a problem with a matching service or configuration item
- Find a problem with a matching title

To configure Incident solution matching options, navigate to **Incident Management** > **Configuration** > **Solution Matching**.

Note

If you create an incident from Service Desk escalation or link to a new incident from the Related Records section, the above incident solution matching will be ignored.

Incident management downtime record

You can access Incident Management downtime records by using the Incident Management Security Administration Utility to show the availability of a selected device. Availability measures the ability of a device or component to provide service within a measured time frame. There are three downtime measurements.

Type of availability	Description
Explicit unavailability	The downtime experienced by the failing device.
Implicit unavailability	The downtime experienced because of the failure of a parent or controlling device.
Perceived unavailability	The explicit or implicit downtime during normal business hours. The total reflects only working hours.

Add a downtime record

Applies to User roles: System Administrator

As part of monitoring the availability of devices in the system, a System Administrator adds downtime records for a specific device.

To add a downtime record:

1. Do one of the following:

- Click **Incident Management** > **Configuration** > **Downtime Records**.
- Click **System Administration** > **Base System Configuration** > **Monitoring** > **Downtime**.

2. Type the applicable data in the **Logical Name**, **Location**, **Contact Name**, **Type**, and **Table Name** fields.
3. Click **Add**.
4. Click **OK**.

Contents

- 1 Create an incident category
- 2 Create an incident task category
- 3 Add a new subcategory to an incident category
- 4 Add a new area for an incident subcategory
- 5 Incident solution matching
- 6 Incident management downtime record
  - 6.1 Add a downtime record

1.4.6. Security

The following sections explain the Incident Management security roles, security areas, and rights.

Incident security areas

The security areas for Incident are Incident, Incident Tasks, and Incident Management Configuration. These areas contain the default security rights and settings for the Incident Management module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

These security areas are used to set permissions to operators to provide access to particular area of Incident. The following table lists the areas and the relevant Incident menu items the operators can access.

Area	System Navigator menu items for this area
Incident	This area contains the default security rights and settings for Incident. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.
Incident Task	This area contains the default security rights and settings for Incident Tasks. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.

Area	System Navigator menu items for this area
Incident Management Configuration	This area contains the default security rights and settings for Incident Management administration and configuration.. For example, Settings, Incident Categories, Incident Task Categories, and Solution Matching. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.

Area	System Navigator menu items for this area

	Note
	When you set the security rights for a security role in the Incident Management Configuration area:
	• The **View** right is to view the settings defined in the **Administration** menu and the **Configuration** menu.
	• The **Update** right is to update the values of existing settings defined in the **Administration** menu and the **Configuration** menu.
	• The **New** and **Delete** rights are to create and delete a setting in the **Configuration** menu, such as category.
	• The **Admin** right is to add, edit, or delete the settings in the following menus:
	  ◦ **Administration** > **Settings**.
	  ◦ **Administration** > **Configuration** > **Solution Matching**

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Incident, Incident Tasks, and Incident Management Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Incident	TRUE	FALSE	Never	Never	FALSE	FALSE
Incident Tasks	TRUE	FALSE	Never	Never	FALSE	FALSE
Incident Management Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Incident, Incident Tasks, and Incident Management Configuration areas.

Incident security roles and settings

The out-of-box security roles for the Incident module include the following:

- Incident process owner
- Incident coordinator
- Incident analyst
- Incident manager
- Incident task assignee
- Configuration auditor
- Operator

Tip

If you have upgraded from a previous version of Incident Management, which is not Process Designer based, see Incident Management security profiles mappings for more information about the mappings between the legacy Incident Management profiles and the current Process Designer security roles.

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Incident area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Incident, Incident Tasks, and Incident Management Configuration areas. These tables only list the new security roles that have different settings with the default rights.

Area Name: Incident

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
incident process owner	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

Area Name: Incident Tasks

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
configuration auditor	TRUE	TRUE	Never	Never	FALSE	FALSE	FALSE
incident analyst	TRUE	TRUE	Always	Always	FALSE	TRUE	FALSE
incident coordinator	TRUE	TRUE	Always	Always	FALSE	TRUE	FALSE
incident manager	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
incident process owner	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
incident task assignee	TRUE	FALSE	When assigned to workgroup	When assigned to workgroup	FALSE	FALSE	FALSE
initiator	TRUE	TRUE	Never	Never	FALSE	FALSE	FALSE
operator	TRUE	TRUE	When assigned	When assigned	FALSE	FALSE	FALSE

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
problem manager	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
SD agent/ manager	TRUE	FALSE	When assigned	Never	FALSE	FALSE	FALSE
service tech	TRUE	TRUE	Always	Always	FALSE	FALSE	FALSE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

Area Name: Incident Management Configuration

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
incident manager	TRUE	TRUE	Always	Always	FALSE	FALSE	FALSE
incident process owner	TRUE	FALSE	Never	Never	FALSE	FALSE	FALSE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

For information about the out-of-box role rights in the Common Configuration area, see Out-of-box role rights in the Common Configuration area.

1.5. Incident Management (Hybrid)

Service Manager Incident Management can automate the reporting and tracking of a single incident or of a group of incidents, and help you to achieve service performance that meets Service Level Agreement (SLA), Operation Level Agreement (OLA), and Underpinning Contract (UC) targets.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Operator

Incident Analyst

Incident Coordinator

Incident Manager

Service Desk Agent

Service Desk Manager

Key concepts

Describes the most important things that you need to know about how the Incident Management module functions.

Incident Management overview

What's an incident?

Alerts and escalation

Incident Details

Categories

Assignment groups

Incident Management views

User workflows

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Incident Logging (SO2.1)

Incident Assignment (SO2.2)

Incident Investigation and Diagnosis (SO2.3)

Incident Resolution and Recovery (SO2.4)

Incident Closure (SO2.5)

Incident Escalation (SO2.6)

Service Level Agreement Monitoring (SO2.7)

OLA and UC Monitoring (SO2.8)

Complaint Handling (SO2.9)

Administrator tasks

Describes the Administrator tasks in the Incident Management module.

Add an Incident Management assignment group

Create a template to complete incident records

Update multiple incident records

1.5.1. Incident Management overview

You can use Service Manager Incident Management to automate reporting and tracking of a single incident or a group of incidents. Incident Management restores normal service operation as quickly as possible and minimizes the adverse impact on business operations, thus ensuring that the best possible levels of service quality and availability are maintained. Incident Management helps you achieve service performance that meets Service Level Agreement (SLA), Operation Level Agreement (OLA), and Underpinning Contract (UC) targets. Incident Management enables you to achieve the following results:

- Categorize types of incidents and keep track of their resolution
- Require Incident records to follow a set process
- Define the users who are responsible for an type incident, and automatically notify them incident opens or escalates
- Issue alerts or escalate an incident to properly meet the agreed-upon terms of the service contract
- Post device outages to the SLA application
- Plan outages based on services that could be affected, based on the CIs specified in an incident or change
- Quickly detect and resolve incidents, resulting in lower downtime and higher service availability

Related topics

Incident Management administrator tasks
Incident Management summary link records
Incident Management contract management records
Incident Management and service level agreements

1.5.1.1. What's an incident?

Incidents include any event which disrupts, or which could disrupt, a service. This includes events that are communicated directly by Users, either through the Service Desk or through an automated interface between Event Management and Incident Management tools.

Incidents can also be reported and logged by support staff, who may notify the Service Desk if they notice an issue. Not all events are logged as incidents. Many classes of events aren't related to disruptions at all, but are indicators of normal operation or are simply informational.

Related topics

Incident Management overview

1.5.1.2. Affected services option for Incidents and Changes

If you want to see a list of services that are potentially affected by an outage related to the CIs specified in an Incident or Change record, click the **View Affected Services** command on the **More** menu. When you open an Incident record, you can examine detailed information for any of the CIs in the list to determine the potential impact of an outage relating to critical, dependent services. The **View Affected Services** command also enables you to plan appropriately when you open a Change record. You can determine how a planned outage affects these services.

When you click **View Affected Services** on the **More** menu, the system processes the outage spreads relating to the CIs in the Incident or Change record by browsing the configuration relationship records in the `cirelationship` table. This process is independent of any outage dependency information specified in the relationship records. No matter how many hierarchy levels a CI is above or below a service, the system browses all levels from the CI in the record to build a list of CIs. After this list is built, the system searches for any outage dependency information specified in the relationship records for those CIs in the list and removes any CIs that don't meet the conditions for outage dependencies. This final list of CIs then opens and you can select a CI for detailed information from the Configuration Management record.

Note

The list of CIs generated does *not* include the CIs specified in the primary Incident or Change record.

Affected Item field in Incident Management

The information available to populate the **Affected Item** field on the CIs and Services tab depends on conditions related to the service recipient. For example, if the recipient has individual or department subscriptions, a list of CIs associated with those subscriptions opens. If a recipient doesn't have any subscriptions, the following behavior occurs:

- If there is an incident recipient, a list of the CIs assigned to that recipient opens.
- If there is a location, a list of the CIs assigned to that location opens.
- If the system has several companies and there is a company specified, a list of the CIs assigned to that company opens.
- If there is a device type specified, a list of the CIs of that type opens.

Note

If you want to obtain a list of all CIs, don't enter any recipient information before you click the **Affected Item** field.

Related topics

Outages

1.5.1.3. Submitting an incident

Service Manager allows you to create an incident via several ways.

You can directly create a new incident in the In the Incident Management module by clicking **Incident Management** > **Create New Incident**.

You can also create an incident by escalating an interaction. For example, if a user can't print to the network printer, the user calls the Service Desk for help. The Service Desk Agent creates a service desk interaction record to capture the service request. If the agent discovers that the issue can't be resolved during the phone call, the agent escalates the interaction into an incident.

The Incident Coordinator reviews the information and verifies that it is assigned to the correct group. An Incident Analyst resolves the issue or escalates it to initiate a change. When the issue is resolved, the Service Desk Agent informs the user that the record is closed. All phases of this incident are tracked, from opening the service desk interaction record, escalating the incident, resolving the incident, and closing the record.

Activities section

The Activities section allows users to enter new updates for a record or to view journal updates and historic activities for a record.

New update

Field	Description
New Update Type	Specifies or categorizes the activity update (for example, communication with the customer)
Visible to Customer?	Makes the update visible to customers so they can view related interactions via self service **Note** This option can be bypassed by setting the **Bypass Visibility Option** flag to “true” in the activityactions table. When this flag is set to “true,” automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to “false,” all activities are visible to customers if the **Customer Visible** option is selected.

Field	Description
New Update	Used to enter notes to explain and describe updates made for the record. If Journaling is enabled, the text entered here is displayed in Journal Updates. If Activities are enabled, the text entered in this field is displayed as an activity record for the selected activity type. **Note** The System Administrator is responsible for enabling Journaling and Activities.

Journal updates

Journal updates displays text entered in the **New Update** field together with a timestamp for the update. This field displays information when Journaling is enabled.

Activity type

To filter the list by the type of activity, select an activity type, and then click **Filter**. Service Manager opens a new record list that displays the records of that activity type.

The activities list displays activities for the current record. The activities are listed in order of occurrence, with the most recent activity displayed first. The following information is displayed for each activity:

- Date/Time
- Type
- Operator
- Description

Incident Details section

Use the following information as a guide when you add data to an incident.

Field	Description
Incident ID	The system-generated unique ID for this incident.
Status	Displays the status of the incident. These statuses are available out-of-box: **Pending other** You need something from an outside source other than customer or vendor. **Work in progress** means work is underway to resolve this incident. **Reject** means the record is rejected. No resolution at this time. **Resolved** There is a resolution, but it has not yet been verified by the customer. **Pending Vendor/Supplier** means awaiting action from a vendor/supplier. **Pending customer** means awaiting action from the customer who reported the incident. **Closed** means the record is resolved. **Open** The incident has been opened but it's not currently being worked on. **Accepted** You accept responsibility for the record. **Suspended** means the record is suspended.
Contact	The contact person for this incident.
Location	The service recipient's location

Field	Description
Affected Service	The service affected by the registered issue. Only services the service recipient has a subscription for can be selected.
Affected CI	The configuration item (CI) the incident is registered for. Using Fill shows the CIs that are part the affected service. Other CIs can be entered manually.
CI is operational (no outage)	If selected (set to true), indicates the item is currently operational and there is no outage.
Outage Start	A date/time stamp for when the service outage started.
Outage End	A date/time stamp for when the service outage ended.
Service Contract	The contract covering the affected equipment.
SLA Target Date	The date and time when the next Service Level Objective (SLO) breach will occur.
Title	A short description summarizing the incident.
Description	A description of the incident in more detail.
Assignment Group	The support group assigned to work on this incident.

Field	Description
Assignee	The name of the person assigned to work on this incident. This person is a member of the assigned support group.
Vendor/Supplier	The name of the vendor/supplier the incident has been assigned to.
Vendor/Supplier Record	This is the number the vendor uses to identify the record.
Category	The category is inherited from the interaction. Valid entries are Incident or Complaint in the out-of-box system
Area	Depending on the selection of a category, a different list appears describing the area of concern for the incident.
Subarea	The third level of categorizing an incident.
Impact	The impact the incident has on the business. The impact and urgency are used to calculate the priority.
Urgency	The urgency indicates how pressing the incident is for the organization.
Priority	The order in which to address this incident in comparison to others.
Closure Code	A set of pre-defined codes that describe how the incident has been closed.
Solution	A description of the solution used to close the incident.

Note: When you view an existing incident, shaded fields aren't available for more input. Incident Management populates these fields from information stored in associated records. You can update the associated record to increase the amount of information in these read-only fields.

Related Records in Incident Management

Service Manager lists all related records for the incident on the Related Records section. All information is read-only, but a user can click a record ID to view the related record. The related records are listed in alphabetic order by the module file name: Change (cm3r), Interaction (incidents), Request (ocmq or request), Incident (probsummary), Problem (rootcause). Within each module the records are sorted by ID in ascending order.

The Related Records section contains the following fields.

Field	Description
ID	The related record number in the database.
Type	Identifies the related record as a Change, Interaction, Request, Incident, Problem.

SLT section

Service Manager populates the information in the SLT section with one or more Service Level Agreement records associated with the asset. If no agreement exists, the section is blank. All the information is read-only.

The **Next Expiration** field displays the next immediate deadline from all matching SLTs. The SLT section contains the following subsections that list the details of the specific SLTs.

SLT type	Description
Process Targets	Lists all of the matching SLTs from all of the associated SLAs.
Uptime Targets	Lists the percentage of time that all SLTs are available.
Max Duration Targets	Lists the acceptable amount of time for an outage for all SLTs.
Upcoming Alerts	Lists the alerts for all of the SLTs.

Note: When you view an existing record, shaded fields aren't available for more input. Incident Management populates these fields from information stored in associated records. You can update the associated record to increase the amount of information in these read-only fields.

Hover-over forms for incident records

In Incident Management, incident forms for open, update, and close include hover-over forms. A hover-over form opens for a field that supports hover-over forms when a user moves the mouse over the field. The hover-over form only opens when the field contains data. The data displayed on hover-over forms is read only.

The incident record forms (open, update, close) contain the following hover-over fields.

Field	Subform fields
Contact	Full Name, Telephone, Email
Affected CI	Critical CI, Pending Change

Related topics

Incident Management overview

1.5.1.4. Alerts and escalation

Incident Management can escalate an Incident record automatically when no one updates the record within a specified time. An alert process triggers the escalation and sends a message that an Incident record is not resolved. Escalation is the process of advancing through the alert stages. Service Manager has the following alert stages:

- Alert Stage 1
- Alert Stage 2
- Alert Stage 3
- DEADLINE ALERT

Each category record specifies a time interval for each alert stage. An unresolved Incident record reaches the next alert stage if no one updates the record after the specified time. When someone updates an Incident record that's in Alert Stage 1, 2, or 3, the alert status resets and the next stage becomes Alert Stage 1. The DEADLINE ALERT stage is always scheduled for a fixed interval after the opening of the Incident record. When an Incident record reaches the DEADLINE ALERT stage, it remains at this stage and the update to the record can't reset the alert stage until the Incident record is resolved.

The following list describes different users who might be notified when an Incident record escalates:

- All contacts in the current contact list for the Assignment Group (except the manager)
- The user who opened the Incident record
- The current primary contact in the owner group
- The current manager of the Assignment Group
- The current manager of the owner group
- The service level manager and client manager of the company and department specified in the Incident record

Alerts and calendars

The clocks that manage alerts don't need to run on a 24-hour schedule. By default, all alert clocks run on a 24-hour, seven-days-a-week schedule. However, you can override the 24-hour schedule with a different calendar for a Deadline Alert group that's available fewer hours a day, or fewer days a week. Then the alert clocks for that group run only during the duty hours defined by the associated calendar. For

example, if your employees work from 9 am to 5 pm, you can set the alert clocks to run only during these hours.

Reassignment alert groups

Each time a user reassigns an incident, Incident Management updates that incident. When an update occurs, there is no escalation triggered, even if there is no progress to resolve the incident. Therefore, it is possible to reassign an incident without resolving it. Incident Management enables you to set a threshold limit on the number of times an incident can be reassigned in the category record. When this threshold is reached, Incident Management notifies the Reassignment Alert group.

When an incident is reassigned, the assignment profile context changes. If an incident is reassigned to an assignment group on Alert Stage 2, the reassignment on Alert Stage 3 depends on the assignment profile definition of the group that received the incident in the previous escalation.

Related topics

Incident Management overview
Incident Management administrator tasks
Incident Management summary link records
Incident Management contract management records
Incident Management and service level agreements

1.5.1.5. Categories

Categories within Request Management combine items that would normally be listed independently. Without categories, the request process would be more time-consuming. Categories reduce or eliminate the need to search for items and services.

When you create a primary category for the catalog, you begin a process which contains ever-increasing levels of specification.

Request categories

Service Manager has three out-of-box request categories from which you can choose to order products and services, and accomplish an employee office move. When you open a request, you see the following out-of-box request category selections:

- Customer Procurement Requests
- Human Resources
- Employee Office Move Process

After you select a request category, you see the primary catalog categories.

Primary catalog categories

Primary catalog categories provide a bundled approach to ordering from the catalog that combines related categories and line items. For example, you can add labor to more than one primary category so that it's not overlooked and doesn't create delays if it is requested too late.

Primary catalog categories and line items are displayed directly after the quote categories. For example, when you select quote category Customer Procurement Requests, Service Manager displays the primary Catalog.

The primary catalog categories combine related items and services. For example, select Computers & Related for the following item types and services:

- Computer Accessories
- Desktop Computers
- Handheld and PDAs
- Hardware Requests
- Request Hardware Upgrades
- Installation Services
- Individual Monitors
- Notebook Computers

If you want a computer monitor only, click Individual Monitors.

The summary of New Employee Setup includes furniture, computer equipment, installation services, and everything needed for a new employee. You can add items to, or take them away from, your order as your needs change.

Line item categories

Line item categories are smaller and more numerous than primary catalog item categories. Line item categories are the building blocks of primary categories; however, they are still categories, built of parts. The parts are individual line items, such as an DEFAULT mouse or an ABC 17" monitor, which you can find in a line item category called Computers & Related along with many other individual line items associated with a computer setup. Line items include both specific parts and services.

You can restrict line item categories to certain quote and order groups the same way as primary catalog categories. On the first page of the catalog, select from the primary catalog category. The catalog displays line item categories in subsequent catalog pages.

The line item category is the next step down in specification from the primary category. Line items are more numerous and specific to the individual groups of products a company holds in inventory than are the primary categories. Line items include both specific parts and services.

Selecting a line item category from the catalog presents a list of actual line items, or parts, available for that category.

Phantom line items

Phantoms are place holders, or pseudo parts, used to organize a collection of multiple items. They are used to provide flexibility in defining component relationships within the catalog, and are usually not physical items. A phantom can be considered an umbrella type of level for several parts selections of the same type. For example, beneath the New Employee Setup primary category the New Accounts & Access line item provides a selectable option. This item itself doesn't display on the quote or order when it is selected; the catalog displays each of the associated parts as individual line items, such as internet access, email account, and network ID.

A phantom part is a parent of the specific items within it, and you can select specific items (or parts) through the phantom part.

Phantoms are assigned part numbers, but they themselves usually aren't ordered or placed on a quote.

There are two ways to designate a listing as a phantom.

Designate a model file as a phantom

Applies to User roles: Request Administrator, System Administrator

To designate a model file as a phantom:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Model**.
2. Add optional search criteria, and then click **Search**.

   A list of model records opens.
3. Select the target model record to update.
4. Select the **Reorder** tab.
5. In the **Reorder Type** field, select **Phantom**.
6. In the Catalog tab, do the following:
    1. Select the **Part Conditions** subtab.
    2. Clear the **Copy to LI?** field.
7. Click **Save**.

1.5.1.6. Posting outages

The Incident Management application can be used to post outage information about devices in the system into the SLA application when you open or close incidents. The system posts outages both manually and automatically.

When posting outages manually, Incident Management displays the Post Which Outages? form for the selected device when you open and close and incident. Use this form to specify the exact start and stop time of the outage instead of the time the outage was reported. Click **OK** to continue opening or closing the incident.

The current time is the default value in manual mode. You may accept the default or type the exact time.

Related topics

Incident Management overview
Priority, impact, and urgency
Creating an incident
Cause codes and probable cause

1.5.1.7. Cause codes and probable cause

The Cause Code field is an optional field that enables you to define the probable cause of an incident. If you use a standard series of cause codes, you can easily track Incident records that have a common cause. Probable cause records defined by the Service Manager Administrator generate cause codes to simplify the reporting and tracking of Incident records. The cause code links the probable cause record to the Incident record and enables you to categorize and assign Incident records.

Related topics

Incident Management overview
Priority, impact, and urgency
Creating an incident
Posting outages

1.5.1.8. Incident Management summary link records

Summary links associate fields in one file or form with other records and other files. For example, there is a special link record, `build.problem.summary2`, used to copy information from the problem record in the rootcause table to the incident record in the probsummary table whenever the appropriate problem record is updated. This link record uses the $pfile variable in place of the normal $File variable when building expressions. It processes expressions after copying the fields rather than before as in normal link processing.

The Fill and Find functions also use summary links.

Related topics

Incident Management administrator tasks
Incident Management contract management records
Incident Management and service level agreements

1.5.1.9. Incident Management contract management records

Contract Management integrates service contract information and tracking into the service desk. You can enable the application or set automatic labor and parts calculation features for Contract Management in the Contract Management configuration record.

Related topics

Incident Management overview
Incident Management administrator tasks
Incident Management summary link records
Incident Management and service level agreements

1.5.1.10. Incident Management and service level agreements

Incident Management supports the selection of more than one applicable SLA for an incident. When you create an incident, you can choose a Customer SLA for the contact, one or more applicable Service SLAs for the contact's subscriptions to a service, or no SLAs at all. Service SLAs only apply if the incident references a Business Service, the contact has a subscription to the service, and the subscription references an SLA. The following describes the system's process for adding SLAs to an incident.

- If one SLA is associated with the incident based on the contact, the Customer SLA is added to the incident.
- If the contact has an Individual Subscription for the CI, the Service SLA from that subscription is added to the incident.
- If the contact has a Department Subscription for the CI, the Service SLA from that subscription is added to the incident.
- If the contact has neither, then no Service SLA is added to the incident

The SLAs should contain all Service Level Targets (SLTs) that define the business rules for all response and availability metrics. You can choose as many SLTs as necessary to describe your response or availability commitment. If necessary, you can add more SLTs that meet your criteria.

When you view the new record, the **SLT** section lists the SLTs that apply to the incident.

Open the related topics to find the definitions for Customer SLA and Service SLA.

Related topics

Incident Management overview
Incident Management administrator tasks
Incident Management summary link records
Incident Management contract management records
Incident Management and service level agreements

1.5.1.11. Incident Management macro list editor

Macros are actions that are driven by predefined conditions, which execute when Incident Management saves a record in the database. Macro actions are associated with files and reflect certain states in the records of those files.

You can create, edit, and delete macros by using the Incident Management Security Administration Utility.

Related topics

Incident Management overview
Incident Management administrator tasks
Categories
Views and favorites

1.5.1.12. Incident Audit Trail

For all records in Service Manager, a date time stamp and a user stamp are updated in the record when it is created or saved.

Service Manager provides an audit trail capability that identifies each step taken in the resolution of an Incident record. Each time the record is updated (whether manually or automatically), a separate historical activity is created for the Incident record. An authorized user may review each historical activity to develop a comprehensive understanding of the history of the record and its resolution, including which operators took what actions at what times. These updates are displayed in an aggregated journal of all updates.

Service Manager also provides an auditing feature that records modifications to fields within the Service Manager database. Field modifications are detected by comparing the fields in the original version of a record to the updated version. When modifications are detected, an Audit Log entry is recorded for each changed field. The Audit Log entry shows the name of the modified field, the old and new versions of the data, the current date and time, and the current operator's user ID.

Related topics

Audit processing

1.5.1.13. Incident Updates: Incident Diagnosis Details

Service Manager provides an audit trail capability that identifies each step in the diagnosis and resolution of an Incident record. Each time the record is updated, Service Manager creates a separate historical activity record for that specific Incident record, including which operators took what actions at what times. An authorized user may leaf through each historical activity record of an Incident record to develop a comprehensive understanding of the history and resolution of the Incident record.

The Incident record also includes a field to capture the resolution of the record.

Incident logging: manual

Service Manager allows the following approaches to manually create an Incident record:

- Authorized Service Desk users can create an Incident record directly within the Service Manager Incident Management module.
- Service Manager also includes a web-based, self-service interface that allows end users to create service requests through an easy-to-complete form. Once the form is complete, Service Manager routes the request through the Service Desk module to the appropriate Service Desk technician, who can then escalate the request to an Incident record. Users can track, update, or close their requests through this interface.
- An Incident record can be created through the web services and event services interfaces. These interfaces are used by existing integrations to network and systems management (NSM) tools (both the company's and third party's) and could be used in customer-created integrations to home-grown tools.

Incident logging: templates and related incidents

You can quickly create a new Incident record by copying an existing Incident record, or by applying templates. The Service Manager template functionality allows authorized users to create templates for common tasks in the Service Desk, Incident, and Change Management modules. Templates include predefined information for some or all fields of a record. A user can create an Incident record and apply a template, which can automatically fill in some or all fields of the record to save the user's effort.

1.5.1.14. Alerts and Escalation

There are several ways of providing alerts and escalations in Service Manager:

1. Notifications: Service Manager provides a broad and deep solution to deliver notifications to incident stakeholders whenever the incident is opened, updated or closed. Administrators have complete control of the following areas in the notification engine:
    1. The notification vehicle (email, page, etc) and format
    2. The conditions on which the notification is sent
    3. The receiver of the notification
2. Alerts: The Incident Management module includes the ability to set criteria for the generation of alerts. For example, the administrator can set rules for three stages of alert and a deadline condition. When these targets are met, specified users receive alerts and notifications through the Service Manager tool, email, pager, or fax etc.
3. Escalations: Business rules can be used to trigger automated escalations of records under various conditions. For example, the following conditions:
    - The status of a record remains unchanged for a long time
    - A record has been reassigned too often between assignment groups

    In addition, to document the violation of these rules the administrator can configure Service Manager to act in certain ways should these rules be breached. For example, reassign the record to the assignee's manager if a record remains in the Open state for too long.
4. Service Level monitoring: Alerts and escalations can be triggered through evaluation of service levels as defined in the Service Level Management module. This approach is used to set Service Level Targets and Service Level Agreements with associated actions should service levels approach breach conditions. In this way, escalation is accomplished proactively to avoid breach rather than react to breach.

1.5.1.15. Field-Level Controls

Service Manager provides two basic methods for setting field level controls including making the field mandatory. The system provides two methods to indicate that a field is mandatory in the UI.

If the field only needs to be mandatory on a screen-by-screen basis, you can set the field as mandatory via a RuleSet validation. This control can be visually indicated via a property in forms designer.

If the field needs to be mandatory for all access points throughout the system, you can set a workflow-level or table-level Rulset validations by using the supplied configuration tools. In this case, the visual indication is automatic.

Visually on the screen, the system identifies a mandatory field with a red asterisk.

1.5.1.16. Incident record data model

Incident Management enables you to track different types of information related to an incident, such as categorization, prioritization, description, contact information, and so on.

Incident data model: Contact details

Service Manager stores related contact information, including the operator who opened the incident and the contact person for the reported issue, with the preferred method of contact (email, phone, or other).

Incident data model: Incident source

When the source of the incident is a person, the Interaction record linked to the Incident record shows the basic information of the source. For example, the following information:

- Whether the Incident is opened through the self-service interface
- The Incident trigger
- The location of the issue

When the source of the incident is an event trigger, the Title and Description fields of the Incident record contain that information, which reference the source event displayed in the External ID field of the Event Browser in Operations Manager.

Incident data model: Categorization

Service Manager categorizes or classifies Incidents according to the following information:

- Category
- Subcategory
- Area

In an out-of-box Service Manager system, Service Manager automatically applies the Incident category to the new Incident that you create. If the Incident Management module has other customized categories, you need to select a category when you create a new Incident. Based on the selected category, Service Manager presents a filtered set of choices for Area and Subarea.

The Incident record also captures the ID of the failed CI or service, including the failing component.

Incident data model: Priority

The Incident record form includes fields to capture impact, urgency, and priority. Service Manager automatically derives the priority code based on the impact and urgency codes, but this mechanism can be manually overridden.

Service Manager automatically assigns urgency based on the Category of a record. However, system administrators can define additional rules for assignment based on any parameter, such as the CI that's the subject of the record, the caller who is the subject of the record, an SLA associated with the record, and so on.

An Incident record has the following impact and urgency codes.

Impact codes:

- 1 - Enterprise
- 2 - Site/Dept
- 3 - Multiple Users
- 4 - User

Urgency and priority codes:

- 1 - Critical
- 2 - High
- 3 - Average
- 4 - Low

Incident data model: Priority calculation

Service desk users complete the **Impact** and **Urgency** fields and Service Manager uses these fields to calculate the priority of the Incident. The **Priority** field is read-only to protect this calculated value.

Incident data model: Assignment

The Incident record stores the assignee group and an individual assignee within the group, if known. The Incident record also captures the name of the vendor/supplier to which the record has been assigned, if appropriate.

Incident data model: Symptoms

An Incident record has a field for the brief summary (the Title) of the issue and a free text field for the details of the problem or symptoms of the fault, which can be used as the search information for the built-in Knowledge Management capability.

Incident data model: Status

Service Manager tracks the status of an Incident in the **Status** field. Service Manager provides the following out-of-box status codes, other codes can be easily added to meet customer requirements.

- **Categorize**
- **Assign**
- **Work In Progress**
- **Pending customer**
- **Pending Evidence**
- **Pending Vendor/Supplier**
- **Pending Other**
- **Resolved**
- **Suspended**

Incident data model: Resolution

Help desk operators are prompted to enter record resolution information when they close records, including a Closure Code (this is a mandatory field). Service Manager provides out-of-box Closure Codes, which can be tailored to meet the customer's requirements. Resolutions can be promoted as candidates for the knowledgebase. Knowledgebase entries can also be used as the resolution for the Incident record.

Incident data model: Closure

Service Manager allows users to enter a Closure Code when a record is closed. The Closure Code can be compared to the category recorded at open time to report on trends. In addition, Service Manager automatically captures the time and date of Incident closure for reporting.

1.5.1.17. Incident and Service Request separation

Service Requests can be handled within the Service Desk module, supported by the Service Catalog module. Service Requests are tracked as Service Desk Interactions and can be fulfilled using Incident Management, Change Management, Request Management, or another fulfillment engine specified by in the Service Catalog configuration. Service Requests that don't originate from the Service Catalog and instead come directly to the Service Desk can be routed to the appropriate fulfillment mechanism by the Service Desk Analyst or they can be completed and closed at the Service Desk.

Incident integration: Configuration Management system

Service Manager is fully integrated with the UCMDB CMS solution. This is a federated solution that integrates with the company-provided discovery tools as well as those provided by other vendors.

Incident integration: Change Management

Out-of-box, the Service Manager Incident Management module is fully integrated with the Change Management module. Users can open Change request records from Incident records. Service Manager can pass relevant information from the Incident record to the Change request record. The two records are linked automatically.

Authorized users can access the Change Management module, create Changes associated with an Incident record, search for Change requests, view the Calendar, link an Incident record to an open Change request, and so on.

Incident integration: Incident matching with RFCs

Service Manager allows linkage between Incidents and Changes, and provides tools for users to establish the linkage. Users can link an existing Change or a newly created Change to an Incident. Service Manager can use business rules to manage the relationships. For example, Service Manager can use business rules to set all the related Incidents to “Resolved” when the related Change is resolved. Related users can then check if these Incidents can be closed.

Incident integration: Knowledge Management

The Service Manager Knowledge Management module allows end users and service-desk personnel to get speedy and accurate answers to their questions directly from any Service Manager application screen. End users can access the Knowledge Management module through the self-service interface to search for potential problem resolutions before contacting the service desk.

Knowledge can come from any Service Manager data source—Incidents, Known Errors and so on—or any external data source. Knowledge can also include Hot News.

However, the Knowledge Management module is more than just the technical components needed to store answers. Because knowledge articles are created via the Knowledge-Centered Support (KCS) process, users can be confident in the accuracy and "just-in-time" solution quality of the answers. The KCS process is an industry best practice for knowledge management for which Service Manager has received external certification.

Service Manager Knowledge Management is a fully integrated support-oriented knowledge management solution that supports KCS standards and guidelines. The Knowledge Management module provides a natural language search engine and rich-text authoring tools that enable users to search, update, and author knowledge articles. Knowledge Management integrates with Interaction, Incident, and Problem management modules so that users can search and use knowledge from existing incidents or problems while attempting to resolve a new incident or problem. Users can also use this integration with Interactions, Incidents, and Problems to create new knowledge. The rich-text editor allows users to include various types of images and documents as attachments that can be linked to other documents or included as part of an existing document.

As part of the company's Closed Loop Incident Process (CLIP) solution, the integration between Service Manager Knowledge Management module and Operations Orchestration (OO) allows automatic execution of run-books related to change management tasks in the context of Service Manager Knowledge Documents. This allows for better and higher levels of automation.

For example, a Service Desk user diagnoses an Incident (or an end user who does self-help diagnosis) and searches for a Knowledge Management document with matching symptoms to the issue. When the user finds such a document, the user can execute a script from a link on the Knowledge Management page itself. The Knowledge Management link triggers a web service action that tells the OO server to run a specific pre-defined script on a specific server. The OO server then sends appropriate commands to the specified server to execute the `runbook` commands. When the target server receives the `runbook` commands, the server executes

operating system commands or database commands as pre-defined to accomplish the specified action. This action could be a workaround or a fix, applied to the environment selected by the Service Desk (or end) user.

Incident integration: Problem Management

Service Manager provides separate modules for creating and managing Incident and Problem records, including Known Error records. However, because these modules are fully integrated and share data seamlessly, Incident, Problem, and Known Error records can be linked. Once linked, the relationships between the records are maintained. Users can create Problem records from Incident records (and vice versa), and Service Manager automatically copies all relevant information from the Incident record to the Problem record to minimize retyping. These records are automatically linked. Users can also manually link records.

Known Error records are created as part of the Problem Management process. The Service Manager Problem Management workflow contains two sub-processes:

- Problem Control. This process analyzes the problem, finds the root cause, and allows users to gather all data needed to initiate a new problem review. Problem Control includes two phases:
  - Identification and Classification. This phase includes creating the Problem record. Classification identifies relationships, urgency, assesses the impact on the customer’s business service, determines priority, and assigns the problem to a specialist or support group.
  - Investigation and Diagnosis. This phase examines the symptoms of the problem to identify the root cause.
- Error Control. This process manages correcting the root cause identified by Problem Control. The objective of Error Control is to effect permanent changes to Configuration Items (CIs) with known errors. Phase 1 of Error Control is Error assessment. Phase 2 of Error Control is closing the known error record.

While Service Manager will not automatically close all Incident records linked to a Problem or Known Error record, Service Manager facilitates closing linked records by listing all linked Incident records that need to be closed. Service Manager can be tailored to automatically close these records.

However, we don't believe that this supports best practices. Service Manager Problem Management is designed to address infrastructure errors, not immediate issues. Therefore, the technician should manage the closure of the Incident separately to help ensure the customer has received the necessary support.

Incident integration: Incident matching with Problem

Incident matching and association is a foundational function in Service Manager. When a user creates an Incident record, Service Manager prompts the user with a list of similar Incident and Problem records. Service Manager can automatically check for potentially similar records based on similar known errors, root causes, incident records, incident record duplicates on a device, or incident record duplicates on parents.

The user can then determine if the new Incident record is identical or similar to an existing Incident record (and can then link the records), or can be resolved based on the workaround information in an existing Problem record.

Incident integration: Request Management

Out-of-box, the Service Manager Incident Management module is fully integrated with the Request Management module. Users can open Request records from Incident records, and Service Manager can pass relevant information from the Incident record to the Request record. The two records will be linked automatically.

Service Manager also includes a web-based, self-service interface that allows end users to create service requests through an easy-to-complete form. Once the form is complete, Service Manager routes the request through the Service Desk module to the appropriate Service Desk technician, who can then escalate the request to an Incident record. Users can track, update, or close their requests through this interface.

Incident integration: Service Level Management

Service Level Agreements (SLA), Operational Level Agreements (OLA), and Underpinning Contracts (UC) are created in the Service Manager Service Level Management module. Each SLA has multiple Service Level Targets (SLT), which can define required response or resolution timeframes associated with Incident records, or required uptime for CIs or services. When an Incident record is created, Service Manager applies the applicable SLA (based on the business unit for which the Incident is created, the CI that's the subject of the record, or other parameters) and populates response and resolution time SLT, as well as availability SLTs, in the Incident record.

This information can be used to prioritize record resolution, and Service Manager displays upcoming SLT deadlines, escalations, and alerts in the Incident record.

Related topics

What's a Service Desk interaction?
Register an incident

1.5.1.18. User Satisfaction

The Service Manager survey integration solution allows users to integrate Service Manager applications with the third-party survey solution MarketTools CustomerSat. This solution uses Service Manager extracted data and survey results to produce powerful reports by using the CustomerSat reporting engine.

The Service Manager survey integration solution enables users to survey their customers immediately after their helpdesk experience to achieve the following goals:

- Understand key drivers of satisfaction
- Identify areas for improvement
- Quickly identify and take action to address critical issues
- Hold each employee accountable for the service experience by using feedback metrics to rate agents
- Identify coaching opportunities

The Service Manager survey integration has the following abilities:

- Send out surveys based on Service Desk interactions
- Identify survey recipients based on department, activity with system, time, and other product and demographic options
- Uses Service Manager extracted data and survey results to produce powerful reports and dashboards by using the CustomerSat reporting engine

Related Reference

SM CustomerSat (PDF)

1.5.2. Incident Management user roles

The Incident Management module has the following user roles:

Operator

The Operator user role has the following responsibilities:

- Register incident based on an event and assign to the correct support group.

To execute this role, the Operator has following tasks available:

Note

For an overview of how these tasks fit in the Incident Management workflow, see Incident Management workflows and user tasks.

ITIL Workflows	User Tasks
Incident logging and categorization (SO2.1)	• Create New Incidents from Monitoring System Notifications

Incident Analyst

The Incident Analyst user role has the following responsibilities:

- Review and accept or reject assigned incidents.
- Investigate and diagnose the incident.
- Document incident resolution or workaround in the Service Management application.
- Implement incident resolution.
- Verify that the incident is resolved and close the incident.

To execute this role, the Incident Analyst can perform the following tasks.

Note

For an overview of how these tasks fit in the Incident Management workflow, see Incident Management workflows and user tasks.

ITIL Workflows	User Tasks
Incident Assignment (SO2.2)	• Reassign an Incident
Incident Investigation and Diagnosis (SO2.3)	• Change Incident Status to Pending User Information • Change Incident Status to Pending Vendor Investigation • Document an Existing Solution or Workaround in an Incident • Reject an Incident
Incident Resolution and Recovery (SO2.4)	• Test the Incident Resolution • Change Incident Resolution • Reassign Incident Resolution
Incident Closure (SO2.5)	• Close an Incident • Close an Incident with an Associated Interaction or Event • Reject an Incident Resolution • Reject an Incident Resolution with an Associated Change or Service Request

Incident Coordinator

The Incident Coordinator user role has the following responsibilities:

- Review and accept or reject incidents assigned to the support group.
- Handle incidents escalated by an Incident Analyst of the support group.
- Monitor Operational Level Agreements (OLA) and Underpinning Contracts (UC) targets of the support group.

To execute this role, the Incident Coordinator can perform the following tasks.

Note

For an overview of how these tasks fit in the Incident Management workflow, see Incident Management workflows and user tasks.

ITIL Workflows	User Tasks
Incident Assignment (SO2.2)	• Assign an Incident • Reassign an Incident
Incident Escalation (SO2.6)	• Escalate an Incident with a Service Request • Open an Emergency Change for an Incident Escalation
OLA and UC Monitoring (SO2.8)	• Monitor OLA and UC Performance

Incident Manager

The Incident Manager user role has the following responsibilities:

- Handle incidents escalated by the Incident Coordinator or by the Service Desk Agent.
- Determine and execute the appropriate escalation actions.
- Request an Emergency Change if required.

To execute this role, the Incident Manager has following tasks available:

Note

For an overview of how these tasks fit in the Incident Management workflow, see Incident Management workflows and user tasks.

ITIL Workflows	User Tasks
Incident Escalation (SO2.6)	• Reassign an Incident for Additional Support

Service Desk Agent

The Service Desk Agent user role has the following responsibilities:

- Open interactions based on contact with the user
- Match user interaction to incidents, problems, known errors, or knowledge documents
- Solve and close interactions
- Provide status updates to users on request
- Register incidents based on user interactions and assign them to the correct support group
- Register requests for change, based on user interactions
- Register service requests, based on a user interactions
- Validate solutions provided by support groups
- Report and verify solution to users
- Monitor the Service Level Agreement (SLA) targets of all registered incidents and escalate the incidents if required
- Communicate service outages to all users

To execute this role, the Service Desk Agent has following tasks available:

Note

For an overview of how these tasks fit in the Incident Management workflow, see Incident Management workflows and user tasks.

ITIL Workflows	User Tasks
Incident logging and categorization (SO2.1)	• Create a new incident from a user interaction • Review and Update Incident Information
Service Level Agreement Monitoring (SO2.7)	• Monitor Interaction Queue for Potential Service Level Agreement Breaches • Monitor Interaction Queue for Service Level Agreement Breaches

Service Desk Manager

The Service Desk Manager user role has the following responsibilities:

- Handle incidents that are categorized as Complaints.

To execute this role, the Service Desk Manager has following tasks available:

Note

For an overview of how these tasks fit in the Incident Management workflow, see Incident Management workflows and user tasks.

ITIL Workflows	User Tasks
Complaint Handling (SO2.9)	• Handle Complaints

Related topics

User roles

Add a user role record

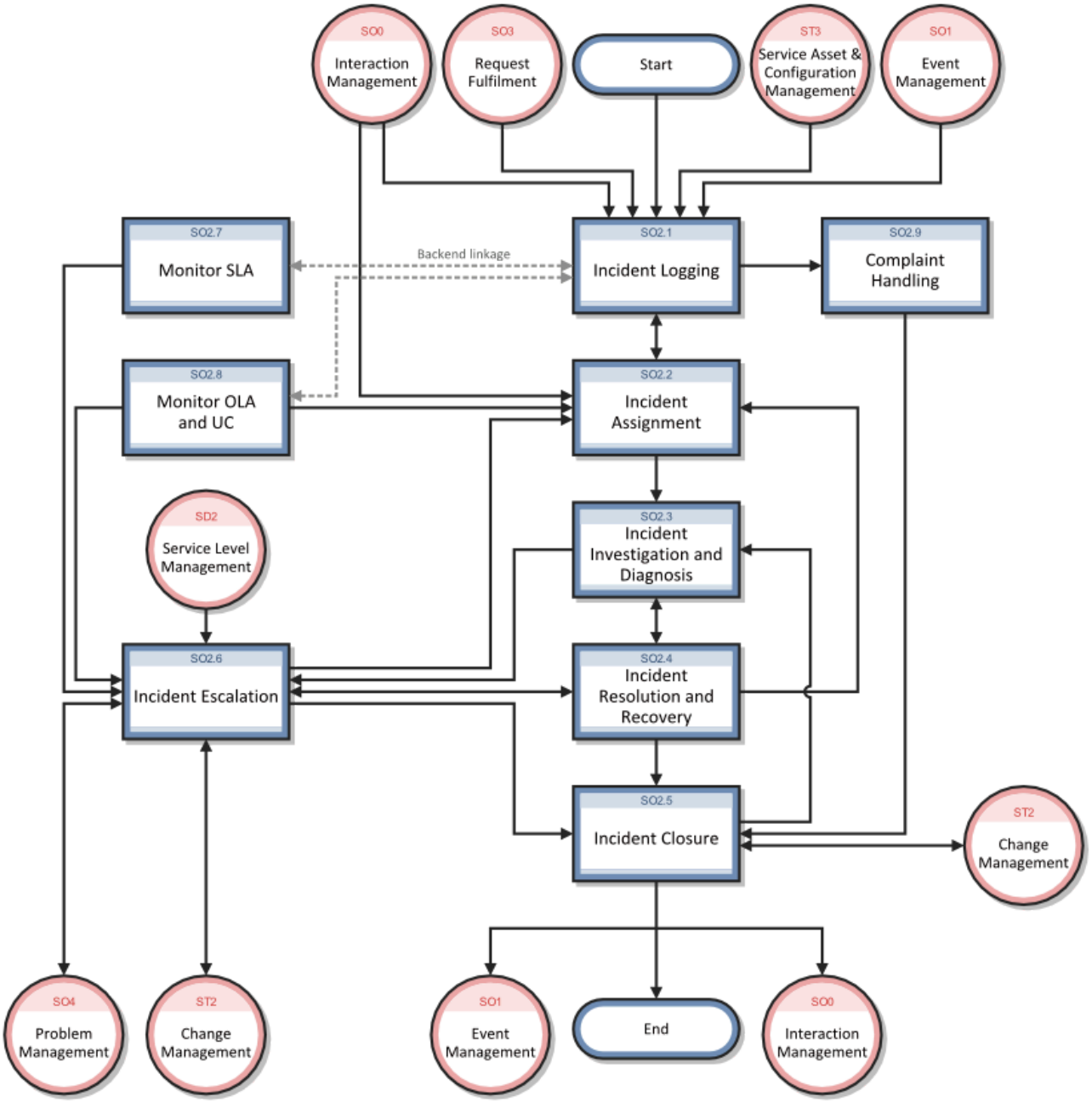
Delete a user role record

Search for a user role record

1.5.3. Incident Management workflows and user tasks

Incident Management enables you to categorize and track various types of incidents (such as service unavailability or performance issues and hardware or software failures), and ensures that incidents are resolved within agreed-on service level targets.

Incident Management workflow



ITIL Workflows	User Tasks
Incident logging and categorization (SO2.1)	• Create a new incident from a user interaction • Create New Incidents from Monitoring System Notifications • Review and Update Incident Information
Incident Assignment (SO2.2)	• Reassign an Incident • Assign an Incident
Incident Investigation and Diagnosis (SO2.3)	• Change Incident Status to Pending User Information • Change Incident Status to Pending Vendor Investigation • Document an Existing Solution or Workaround in an Incident • Reject an Incident
Incident Resolution and Recovery (SO2.4)	• Test the Incident Resolution • Change Incident Resolution • Reassign Incident Resolution
Incident Closure (SO2.5)	• Close an Incident • Close an Incident with an Associated Interaction or Event • Reject an Incident Resolution • Reject an Incident Resolution with an Associated Change or Service Request

ITIL Workflows	User Tasks
Incident Escalation (SO2.6)	• Escalate an Incident with a Service Request • Open an Emergency Change for an Incident Escalation • Reassign an Incident for Additional Support
Service Level Agreement Monitoring (SO2.7)	• Monitor Interaction Queue for Potential Service Level Agreement Breaches • Monitor Interaction Queue for Service Level Agreement Breaches
OLA and UC Monitoring (SO2.8)	• Monitor OLA and UC Performance
Complaint Handling (SO2.9)	• Handle Complaints

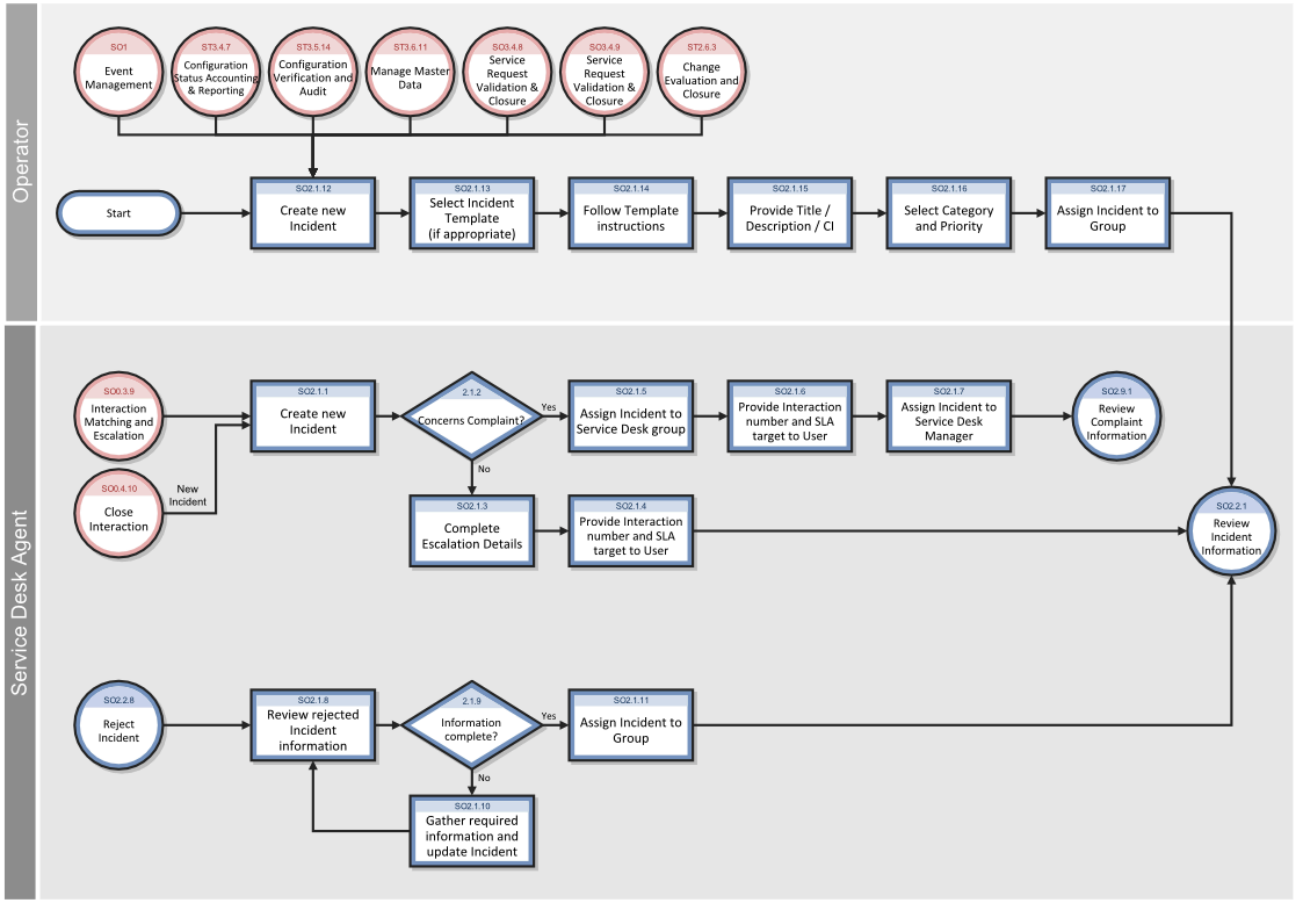
Related topics

Incident Management overview

Incident Management workflows and user tasks

1.5.3.1. Incident logging and categorization (SO2.1)

Incidents are initiated and logged as part of the Interaction Management process or as part of the Event Management process, depending on the source and nature of the incident. All relevant information relating to incidents must be logged so that a full historical record is maintained. By maintaining accurate and complete incident records, future assigned support group personnel are better able to resolve recorded incidents.



Related topics

Incident Management overview
Create New Incidents from Monitoring System Notifications
Create a new incident from a user interaction
Review and Update Incident Information
Incident Management workflows and user tasks

1.5.3.1.1. Create a new incident from a user interaction

Part of Workflow(s): Incident logging and categorization (SO2.1)

Applies to User Roles: Service Desk Agent

You can create an incident from a user interaction because it can't be solved immediately. You can also update an incident from a user interaction that has already been escalated.

To create an incident from a user interaction:

1. Click **Service Desk** > **Create New Interaction**. Service Manager opens a blank New Interaction form and assigns the issue a unique interaction ID.
2. Select the contact for the interaction. The associated fields fill automatically.
3. Select the user's preferred notification method.
4. Select the service recipient. The associated field fills automatically.
5. Select the service or affected configuration item (CI).
6. Type a title and description for the interaction.
7. Type one or more free-form tags (keywords or terms) for the interaction. The tags you enter are searchable and can be used as keywords to highlight important information. For more information, see Working with tags.
8. Type some description for the interaction.
9. When you determine that you are unable to solve the user's issue, select the **Impact** and **Urgency** of the interaction, and then click **Escalate**. The Escalation Wizard opens with a list of previously-reported incidents that contain the same issue.
10. Browse the list of incidents to find a record that matches the issue being reported. If you select a record, do the following.
    1. Click **link** to link the incident record to the user interaction record. A message displays, stating that the interaction has been associated with the incident.
    2. Click **OK**.
    3. Click **Back** to go back to the interaction record. A message displays stating the interaction has been added.
    4. Click **Save & Exit**.
11. If no incidents match the issue being reported in the user interaction record, click **Create New Incident**, fill in **Location** and **Assignment**, and then click **Next**. A list of potentially-related problems is opened. You can do one of the following:

- Link to an existing problem.
    1. Click **link** to link the problem record to the new incident record. A message displays, stating the new incident record has been associated with the existing problem.
        - Click **OK**.
        - Click **Continue**. You are returned to the interaction record.
        - Click **Save & Exit**.
- If no problems relate to the new incident, click **Continue** to proceed with opening a new incident. After the new incident is opened, you are returned to the user interaction.
    - Click **Save & Exit**.

To update the new incident:

1. Click **Service Desk** > **Interaction Queue**. Service Manager opens the Interaction queue.
2. Double-click the interaction in the queue to display the details of the record.

Note: Those items with a status of **Open – Linked** are user interactions that have been escalated to Incident Management.

1. For interactions that are complaints or concerns:
    1. Click **More** or the More Actions icon and select **Related > Incidents > View** to view and select the related incident.
    2. Select Service Desk in Assignment Group.
    3. Select the name of the Service Desk Manager for Assignee.
    4. Using internally-defined procedures, provide the interaction number and the SLA target to the customer. The incident is now assigned to the Service Desk Manager.
    5. Click **Save & Exit**.
    6. Click **Back** to return to the next interaction in the queue.
2. For interactions that aren't complaints or concerns:
    1. Click **More** or the More Actions icon and select **Related > Incidents > View** to view and select the related incident.
    2. Click **Fill** to select the applicable Assignment Group for the incident.
    3. Use internally defined procedures to provide the interaction number and the SLA target to the customer.
    4. Click **Save & Exit**.
    5. Click **Back** to return to the next interaction in the queue.

Related topics

Create New Incidents from Monitoring System Notifications
Review and Update Incident Information
Register an incident

Incident logging and categorization (SO2.1)

1.5.3.1.2. Create new incidents from monitoring system notifications

Part of Workflow(s): Incident logging and categorization (SO2.1)

Applies to User Roles: Operator

You can create an incident when notified by a monitoring system of an event in the Information and Communication Technology (ICT) infrastructure.

To create an incident from a monitoring system event notification:

1. Click **Incident Management** > **Create New Incident**.
2. Click **Apply Template**.
3. Select the appropriate template. The template automatically fills in some fields with predefined values.
4. Type a title and description for the incident.
5. In Incident Detail, verify or complete the following required entries. Use the Fill button when available to display a list of values for the field.
    - **Category**
    - **Area**
    - **Subarea**
    - **Impact**
    - **Urgency**
    - **Priority**
6. Click **Fill** to select an Assignment Group.
7. Click **Fill** or **Find** to select the applicable Service affected by the incident.
8. Complete the form with any other relevant information.
9. Click **Save & Exit**.

Incident Management opens a new incident, assigns a unique number with the prefix IM, and places the incident in the queue for the Incident Coordinator.

Related topics

Create a new incident from a user interaction
Review and Update Incident Information
Searching records
Incident logging and categorization (SO2.1)

1.5.3.1.3. Review and update incident information

Part of Workflow(s): Incident logging and categorization (SO2.1)

Applies to User Roles: Service Desk Agent

When a Service Desk Agent receives an incident that has been rejected by an assignment group because of incomplete information in the record, the agent gathers the missing information and updates the incident record. The agent then reassigns the incident to the applicable group and informs the user, if necessary.

To update incident information:

1. Click **Incident Management > Incident Queue**.
2. Use search or advanced search to find one or more records.
3. Double-click the target record.
4. Review the information in the record to verify that it is complete and correct.
5. For incident records that are complete:
    1. Click **Fill** to select the applicable Assignment Group for the incident.
    2. Follow the procedures defined by your company to provide the interaction number and the SLA target to the customer.
        - To see a list of the related interactions, click **More** or the More Actions icon and select **Related Records** > **Interactions**.
        - To see information about the SLA, Open the **SLA** section.
6. For incident records that are incomplete or incorrect, gather the required information and update the form. The required information includes:
    - The title and description of the incident
    - The service provided by the affected item (for example, email)
    - The Incident details: Category, Area, Subarea, Impact, Urgency, Priority
7. Click **Fill** to select the applicable Assignment Group for the incident, if it's not already assigned to a group.
8. Use internally-defined procedures to provide the interaction number and the SLA target to the customer. The incident is now assigned to the assignment group you selected.
9. Click **Save & Exit** to return the incident queue.

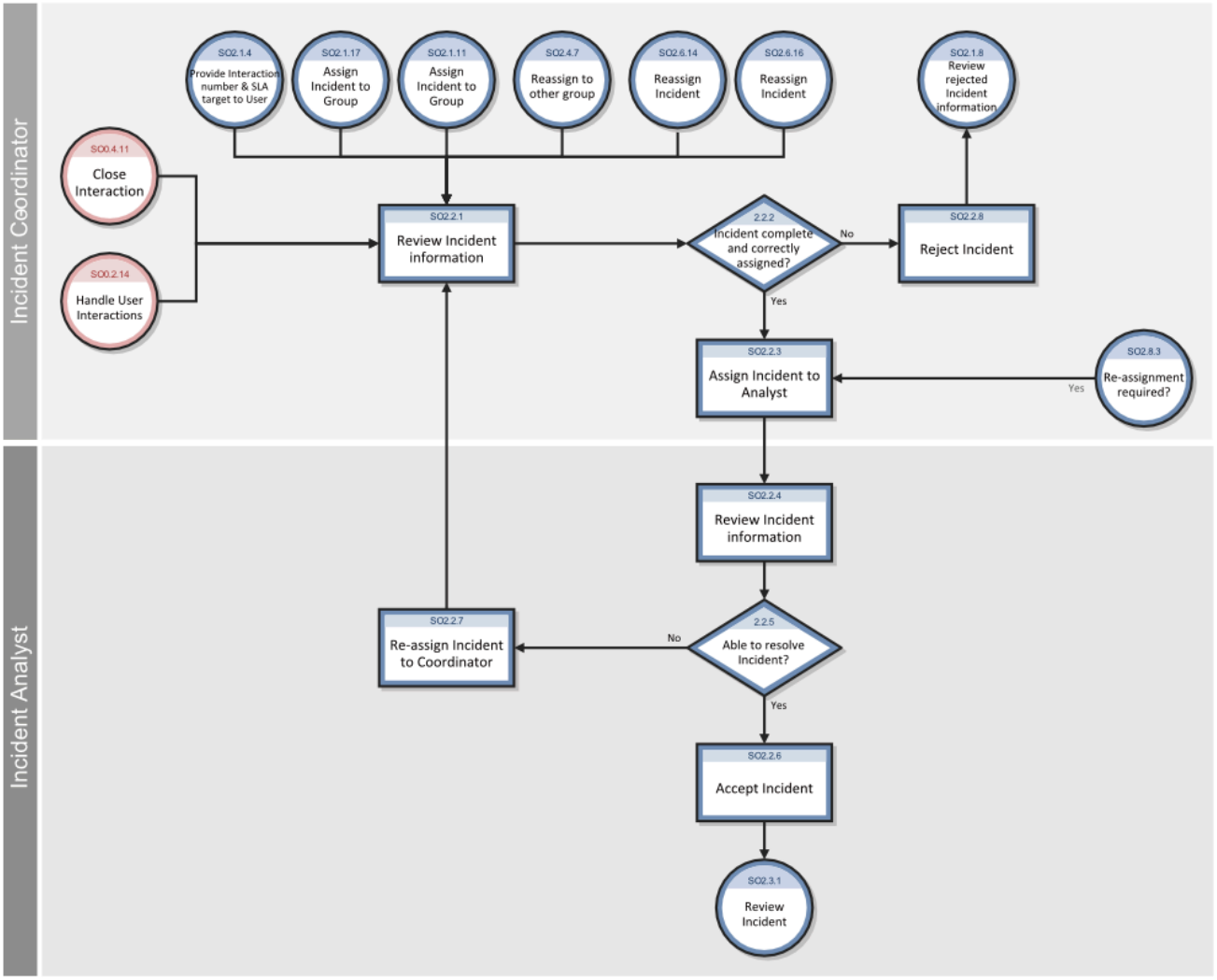
Related topics

Create a new incident from a user interaction
Create New Incidents from Monitoring System Notifications

Incident logging and categorization (SO2.1)

1.5.3.2. Incident Assignment (SO2.2)

Incident records are logged from an interaction by a Service Desk Agent or from an event by an Operator. The Incident Coordinator monitors the incident queue, reviews open status incidents, and determines from the information provided whether to accept or reject incident records. When an incident record is accepted, it is assigned to an Incident Analyst for further investigation and diagnosis.



Related topics

Incident Management overview
Assign an Incident
Reassign an Incident
Incident Management workflows and user tasks

1.5.3.2.1. Assign an incident

Part of Workflow(s): Incident Assignment (SO2.2)

Applies to User Roles: Incident Coordinator

The Incident Coordinator reviews open status incidents and determines from the information provided whether to accept or reject incident records. After an incident record is accepted, the Incident Coordinator assigns the record to an Incident Analyst for further investigation and diagnosis.

Assigning an incident is a two-step process. First, review an incident to determine if it can be resolved by your group. Next, assign the incident to an Incident Analyst, or reject the incident and return it to the Service Desk.

To review and assign an incident:

1. Click **Incident Management** > **Incident Queue**.
2. Select the **Incidents by Assignment Group** view.
3. Review the incident records in the list to determine which ones can be assigned to your group for further investigation.
    1. Review each incident record and look at the contents of the Affected CI and Description fields.
    2. Determine whether to accept or reject an incident record.
4. To accept an incident record, do the following:
    1. Click the list in the **Status** field and change the status to **Accepted**.
    2. Clear the assignee in the **Assignee** field, and then click the **Fill** button. A list of names opens.
    3. Select a name from the list to assign the incident to an Incident Analyst.
    4. Click **Save & Exit**.
5. To reject an incident and reassign it to the Service Desk, do the following:
    1. Click the list in the **Status** field and change the status to **Rejected**.
    2. Click **Save**. A notes box opens and asks you to enter the reason for rejection.
    3. Enter the reason for rejecting the incident.
    4. Click **Save**.
    5. Click **Save & Exit** to exit.

Related topics

Reassign an Incident
Incident Assignment (SO2.2)

1.5.3.2.2. Reassign an incident

Part of Workflow(s): Incident Assignment (SO2.2)

Applies to User Roles: Incident Analyst, Incident Coordinator

At times you will need to reassign incident records when an Incident Analyst is unavailable.

To reassign an incident:

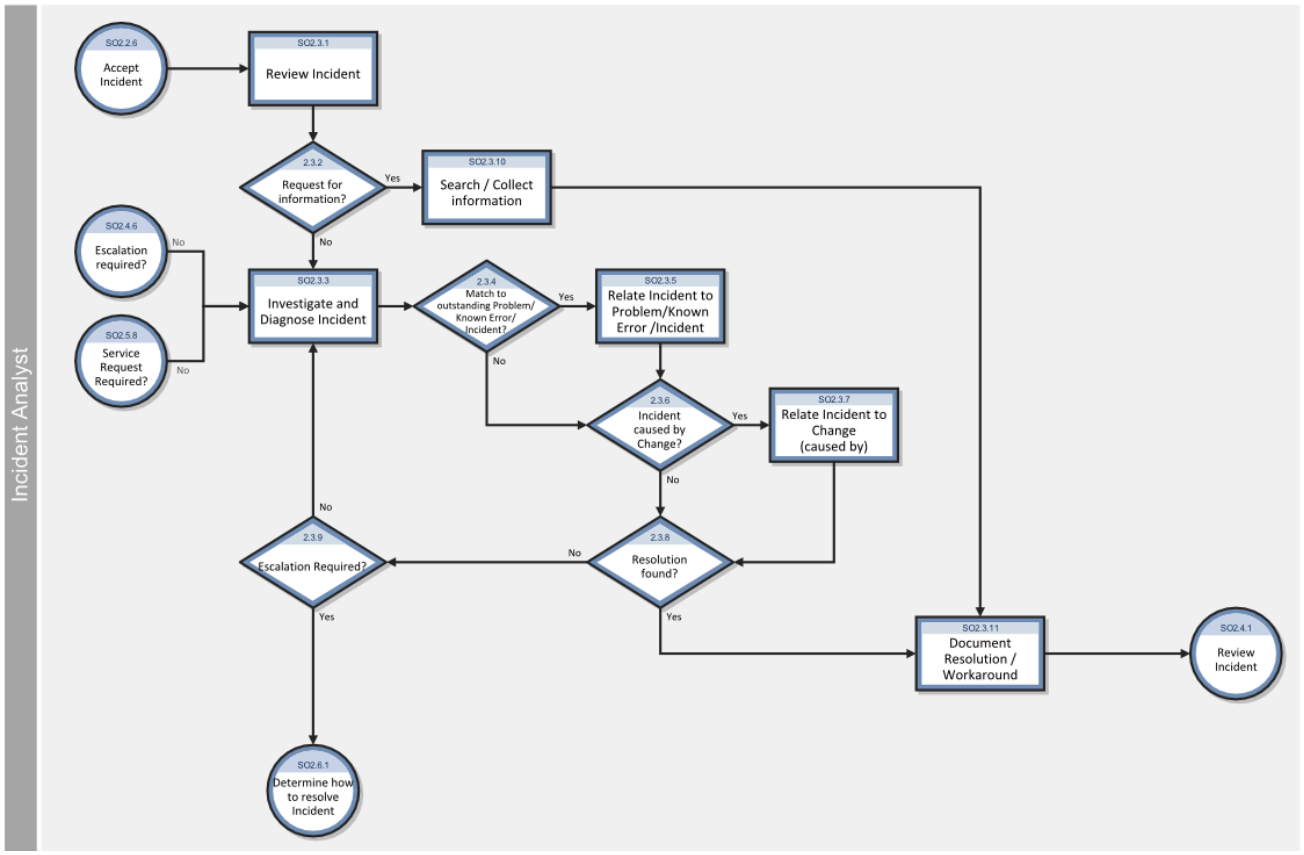
1. Click **Incident Management** > **Incident Queue** > **By Assignment Group**.
2. Select a list of Incident Analyst assignments for a particular analyst.
    1. Click the list in the **View Records for which Assignment Group?** field, and then select an assignment group.
    2. Click **OK**.
3. To complete the reassignment of an incident:
    1. Select the incident record to be reassigned.
    2. Clear the assignee in the **Assignee** field, and then click **Fill** to select an operator from the list of names to reassign the incident to that person.
    3. Click **Save & Exit**.

Related topics

Assign an Incident
Incident Assignment (SO2.2)

1.5.3.3. Incident Investigation and Diagnosis (SO2.3)

Each support group involved with handling incidents must perform investigation and diagnosis tasks to determine the cause and solution to the incident. All actions performed by support group personnel are documented in the incident record, so that a complete historical record of all activities is maintained at all times.



Related topics

Incident Management overview
Change Incident Status to Pending User Information
Change Incident Status to Pending Vendor Investigation
Document an Existing Solution or Workaround in an Incident
Reject an Incident
Incident Management workflows and user tasks

1.5.3.3.1. Change incident status pending user information

Part of Workflow(s): Incident Investigation and Diagnosis (SO2.3)

Applies to User Roles: Incident Analyst

If you need additional information from the user, contact the user directly. While waiting for information from the user, change the incident status to pending user information status.

1. Click **Incident Management** > **Incident Queue**.
2. Select the **Open Incidents Assigned To Me** view, which is based on the service level agreement (SLA) target date. The view is sorted by target date with the oldest date listed first.
3. Select an open incident, and then look at the category in the Category field. The category is inherited from the Service Desk interaction process. Valid entries are complaint, incident, or request for information. The Area and Subarea fields are also populated according to the selected category.
4. Click the list in the **Status** field and change the status to an applicable pending status.

   Note: When an incident record is set to a pending status, the service level agreement clock continues to track the incident as an open incident. The assignment group remains the same.
5. When the user delivers the requested information, change the status to accepted or work in progress. Click the list in the **Status** field and update the status.
6. Click **Save & Exit**.

Related topics

Change Incident Status to Pending Vendor Investigation
Document an Existing Solution or Workaround in an Incident
Reject an Incident
Incident Investigation and Diagnosis (SO2.3)

1.5.3.3.2. Change incident status pending vendor investigation

When you determine that a vendor needs to investigate and diagnose the cause of an incident, you are required to update the status of the incident record to indicate that the investigation is pending vendor feedback.

Part of Workflow(s): Incident Investigation and Diagnosis (SO2.3)

Applies to User Roles: Incident Analyst

To update an incident record status to pending vendor information:

1. Click **Incident Management** > **Incident Queue**.
2. Select the **Open Incidents Assigned To Me** view, which is based on the service level agreement (SLA) target date. The view is sorted by target date with the oldest date listed first.
3. Select an open incident, and then look at the category in the Category field. The category is inherited from the Service Desk interaction process. Valid entries are complaint, incident, or request for information. The Area and Subarea fields are also populated according to the selected category.
4. Click the list in the **Status** field and change the status to **Pending vendor**.
    - **Note**: When an incident record is at a status of pending vendor, the service level agreement clock continues to track the incident as an open incident. The assignment group remains the same.
5. Click **Fill** in the **Vendor** field to generate a list of vendors and choose one.
    - **Note**: The Vendor field is only valid when the incident Status is Pending Vendor.
6. Enter the reference number (obtained from the vendor) in the **Reference Number** field.
    - **Note**: When the vendor provides the requested information, you can change the status to accepted or work in progress.
7. Click **Save & Exit**.

Related topics

Change Incident Status to Pending User Information
Document an Existing Solution or Workaround in an Incident
Reject an Incident
Incident Investigation and Diagnosis (SO2.3)

1.5.3.3.3. Document an existing solution or workaround in an incident

Part of Workflow(s): Incident Investigation and Diagnosis (SO2.3)

Applies to User Roles: Incident Analys

When you find an existing solution that resolves an open incident, you can verify the effectiveness of the solution and then use the solution to resolve the incident.

After you verify that the solution works, perform the following steps to attach the solution to the incident record:

1. From the knowledgebase document or record that contains the solution, do the following:
    1. Click **Use Solution** to add the solution to the Solution field of the incident record..
    2. View the published document, and then select it as a solution. The system reopens the record with the Solution field updated.

    Note: For information about attaching a file, see the related topics.
2. Update the incident detail:
    1. Open the Activities section.
    2. Select the applicable update type in **New Update Type**.
    3. Add other necessary information and notes in **New Update**.
    4. Check **Visible to Customer** to make the information available to customers.

    > **Note**
    >
    > This option can be bypassed by setting the **Bypass Visibility Option** flag to “true” in the activityactions table. When this flag is set to “true,” automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to “false,” all activities are visible to customers if the **Customer Visible** option is selected.
3. Click **Save & Exit**.

Related topics

Search for a solution
Change Incident Status to Pending User Information
Change Incident Status to Pending Vendor Investigation
Reject an Incident
Incident Investigation and Diagnosis (SO2.3)

1.5.3.3.4. Reject an incident

Part of Workflow(s): Incident Investigation and Diagnosis (SO2.3)

Applies to User Roles: Incident Analyst

If you determine that you can't perform work on an incident, then you must change the incident status to rejected before closing or exiting the incident record.

To reject an incident:

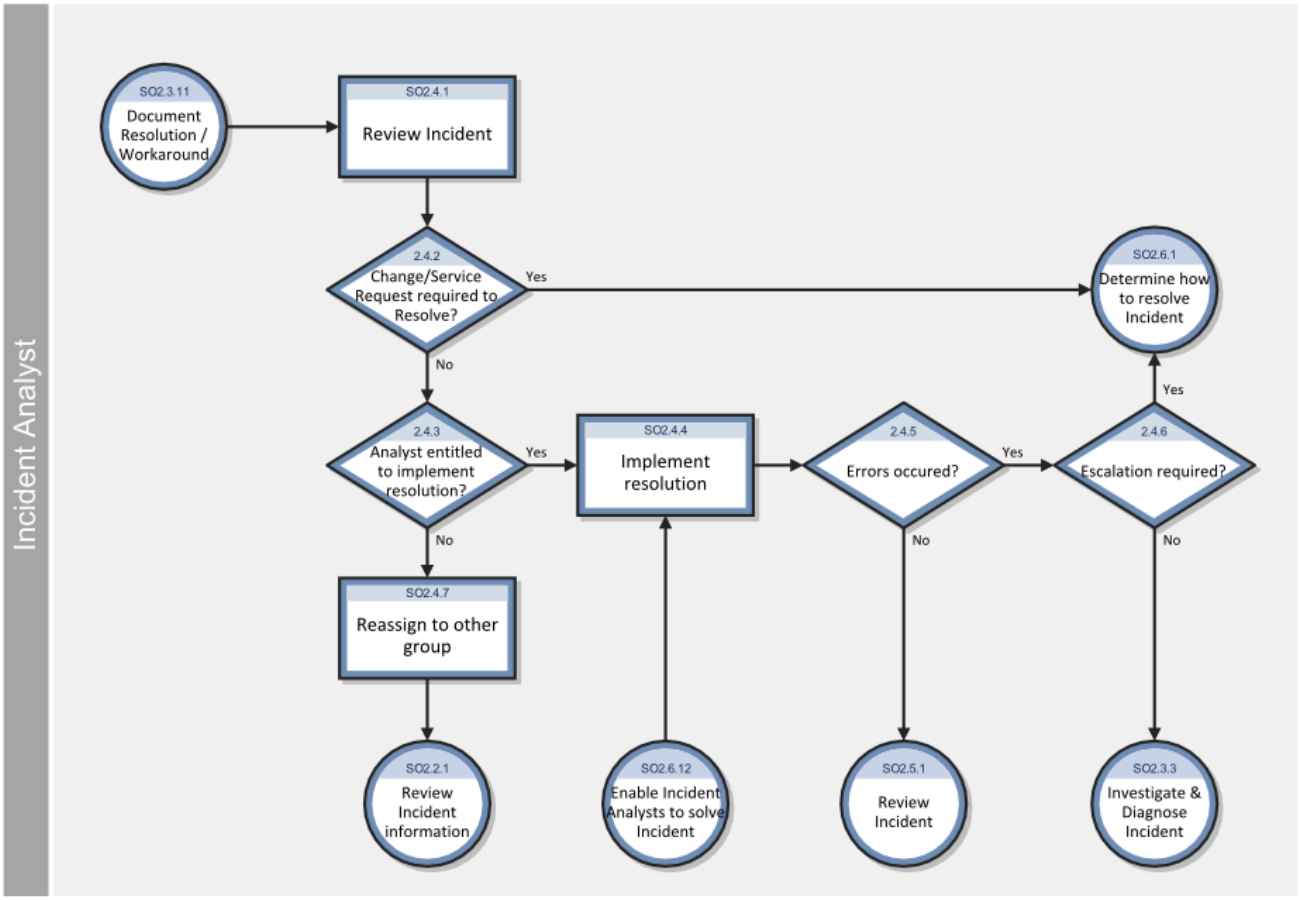
1. In the Status field, change the status to **Rejected**.
2. Clear the **Assignee** field.
3. Click **Fill** in the **Assignee** field to generate a list of Operator names in the assignment group, and then choose the name of the coordinator to whom this incident will be reassigned.
4. In the **Activities** section, do the following:
    1. From the **New Update Type** field list, select an update type to categorize the activity update.
    2. In the **New Update** field, type a description of the update.
5. Click **Save**. You are prompted for the reason for the rejection.
6. Enter the reason for the rejection, and then click **Save**. The system indicates the incident has been updated.
7. Click **Save & Exit**.

Related topics

Change Incident Status to Pending User Information
Change Incident Status to Pending Vendor Investigation
Document an Existing Solution or Workaround in an Incident
Incident Investigation and Diagnosis (SO2.3)

1.5.3.4. Incident Resolution and Recovery (SO2.4)

As part of the incident resolution and recovery process, the Incident Analyst identifies and evaluates potential resolutions before those resolutions are applied and escalates incidents as necessary. The Incident Analyst may escalate incidents to the Incident Coordinator, including those incidents that require a change. If the Incident Analyst doesn't have the required level of permissions to implement a change, the Incident Analyst reassigns the incident to another group that can implement the resolution.



Related topics

Incident Management overview
Change Incident Resolution
Reassign Incident Resolution
Test the Incident Resolution
Incident Management workflows and user tasks

1.5.3.4.1. Test the incident resolution

Part of Workflow(s): Incident Resolution and Recovery (SO2.4)

Applies to User Roles: Incident Analyst

One of the tasks of the Incident Analyst is to review and test the resolution described in the Solution field of an incident record. When the test is successful, the incident is assigned for incident closure. When the resolution test fails, the Incident Analyst returns the incident to incident investigation and resolution or assigns it to incident escalation.

To update tested incident resolutions:

1. Click **Incident Management > Incident Queue** or view your To Do LIst.
2. View and identify any incidents whose resolution (Solution field) can be tested.
3. Test the resolution described in a test environment that mimics the production environment.
4. When the resolution fails:
    1. If the incident requires an escalation, update the **Assignment Group** and **Assignee** fields with the Incident Coordinator responsible for the Incident Escalation process.
    2. If the incident doesn't require an escalation, update the **Assignment Group** and **Assignee** fields with the applicable Incident Analyst responsible for the investigation and diagnosis process or incident escalation.
    3. In both cases, in the Activities section, select a **New Update Type** and then in **New Update** type an explanation of the test failure.
5. When resolution testing is successful:
    1. Update the **Assignment Group** and **Assignee** fields with the applicable Incident Analyst responsible for the incident closure process.
    2. In the Activities section, select the **New Update Type** and then in **New Update** type a description of the test results.
6. Click **Save & Exit**.

Related topics

Change Incident Resolution
Reassign Incident Resolution
Incident Resolution and Recovery (SO2.4)

1.5.3.4.2. Change incident resolution

Part of Workflow(s): Incident Resolution and Recovery (SO2.4)

Applies to User Roles: Incident Analyst

One of the tasks of the Incident Analyst is to review and identify incidents received from the incident investigation and diagnosis process that need to be escalated to the Incident Coordinator because the implementation of the resolution requires a change.

To initiate a change for an incident:

1. Click **Incident Management > Incident Queue** or view your To Do list.
2. View and identify any incidents whose resolution (shown in the Solution field) can only be accomplished with a change.
3. Update the **Assignment Group** and **Assignee** fields with the applicable group and Incident Coordinator.
4. In the Activities section, select the **New Update Type** and then in **New Update** type a description of the change required to implement the resolution.
5. Click **Save & Exit**.

Related topics

Searching records
Test the Incident Resolution
Reassign Incident Resolution
Incident Resolution and Recovery (SO2.4)

1.5.3.4.3. Reassign incident resolution

Part of Workflow(s): Incident Resolution and Recovery (SO2.4)

Applies to User Roles: Incident Analyst

One of the tasks of the Incident Analyst is to review and identify incidents received from the incident investigation and diagnosis process that need to be reassigned to another group, because the Incident Analyst doesn't have the necessary permissions to implement the resolution.

To reassign an incident:

1. Click **Incident Management > Incident Queue** or view your To Do list.
2. View and identify any incidents with a resolution (shown in the Solution field) that can't be accomplished because you don't have the applicable permissions.
3. Update the **Assignment Group** and **Assignee** fields with the applicable group and operator.
4. In the **Activities** section, do the following:
    1. From the **New Update Type** field list, select an update date.
    2. In the **New Update** field, type a description of the permissions required to implement the resolution.
5. Click **Save & Exit**.

Related topics

Test the Incident Resolution
Change Incident Resolution
Incident Resolution and Recovery (SO2.4)

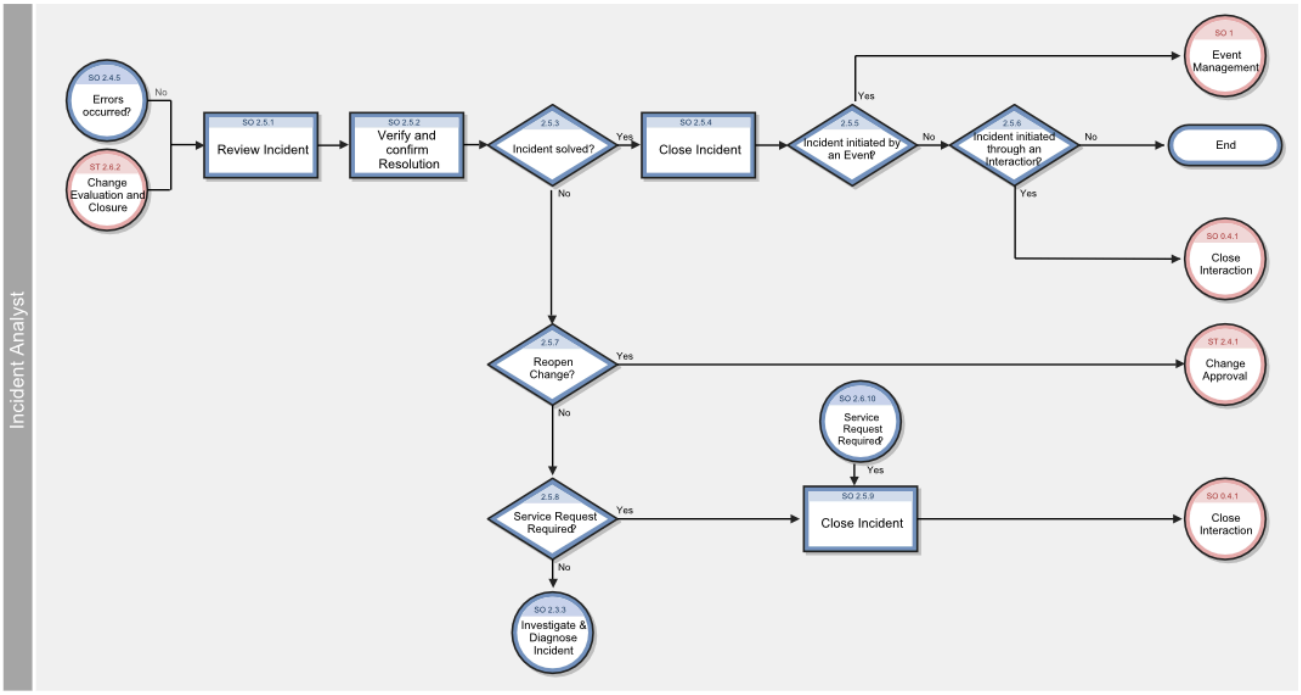
1.5.3.5. Incident Closure (SO2.5)

The Incident Closure process includes many steps to verify the success of implemented solutions and to verify that incident records are accurate and complete.

Service Manager is configured out-of-box to use a one-step Incident Closure process. Therefore, incident personnel can close the incident directly after resolving it. The Service Desk takes care of notifying the end user and closing the interaction that initiated the incident.

Legacy Service Manager customers who did not activate Service Desk and used a two-step incident close will find that this is no longer necessary, because the Service Desk application is now included. However, you can still choose to close incidents in two steps after you finish some configuration.

To close an incident in two steps, administrators need to configure the system to enable a two-step close process for incidents. In the two-step close process, a technician starts the close process when work on the incident is complete. The record is resolved, but not closed until an operator contacts the user and confirms the resolution. The operator then finishes closing the incident and the incident is inactivated.



Related topics

Incident Management overview
Close an Incident

Close an Incident with an Associated Interaction or Event
Reject an Incident Resolution
Reopen an incident
Incident Management workflows and user tasks

1.5.3.5.1. Close an incident

Part of Workflow(s): Incident Closure (SO2.5)

Applies to User Roles: Incident Analyst

Incidents that have been resolved satisfactorily and have no associated events, interactions, or changes are closed by specifying a closure code and updating the status to closed. Incidents that have associated events, interactions, or changes require additional processing.

To close an incident:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident that's ready to be closed.
3. Click **Fill** in the **Closure Code** field to select an applicable code.
4. Enter the resolution in the **Solution** field and verify that the information is correct and complete.
5. Click **Close Incident**. The status changes to Closed.

1. Click **Save & Exit**.
2. In the Confirm Outage Information form, update or accept the specified outage start and end times and click **OK**. You are returned to your To Do queue. The status is updated to Closed.

Related topics

Close an Incident with an Associated Interaction or Event
Reject an Incident Resolution
Reject an Incident Resolution with an Associated Change or Service Request
Incident Closure (SO2.5)

1.5.3.5.2. Close an incident with an associated interaction or event

Part of Workflow(s): Incident Closure (SO2.5)

Applies to User Roles: Incident Analyst

Incidents that are resolved satisfactorily that have associated events or interactions are closed by specifying a closure code and updating the status to closed. Incidents that have associated changes require additional processing.

To close an incident with an associated interaction or event:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident that's ready to be closed.
3. Review the incident resolution description in the Solution field to verify that it is correct. Make sure that all information in the detail record is complete.
4. Update the **Status** field to closed.
5. Use Fill to select an applicable code for the **Closure Code** field.
6. If this incident has an associated interaction, update the **Assignment Group** and **Assignee** fields with the Service Desk group and Service Desk Agent responsible for the Close Interaction process.
7. If this incident has an associated event, update the **Assignment Group** and **Assignee** fields with the Group and Operator responsible for Event Management.
8. Click **Close Incident**.
9. Click **Save & Exit**.
10. In the Confirm Outage Information form, update or accept the specified outage start and end times. Click **OK**. You are returned to your To Do queue.

Related topics

Close an Incident
Reject an Incident Resolution
Reject an Incident Resolution with an Associated Change or Service Request
Incident Closure (SO2.5)

1.5.3.5.3. Reject an incident resolution

Part of Workflow(s): Incident Closure (SO2.5)

Applies to User Roles: Incident Analyst

If the solution assigned to an incident is found to not resolve the incident completely, the incident must be returned to the incident investigation and diagnosis process for additional analysis. For example, a solution may appear to solve the problem, but when the customer uses the solution, the problem persists. Incidents that have associated service requests or changes require additional processing.

To reassign an incident for additional investigation and diagnosis:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident for review.
3. Review the incident resolution description in the Solution field to verify that it is correct and complete. Verify that all information in the detail record is complete.
4. When the solution doesn't resolve the incident and the incident doesn't include an associated service request or change:
    1. Update the **Status** field to Work In Progress.
    2. Update the **Assignment Group** and **Assignee** fields with the applicable group and Incident Analyst.
    3. In the Activities section, do the following:
        1. From the **New Update type** field list, select an update type.
        2. In the **New Update** field, type an explanation that describes the problem with the proposed solution.
    4. Click **Save & Exit** to return to your To Do queue.
5. If the incident includes an associated service request or change, the incident requires additional processing.

Related topics

Reject an Incident Resolution with an Associated Change or Service Request
Close an Incident
Close an Incident with an Associated Interaction or Event
User interaction handling (SO0.2)

1.5.3.5.4. Reject an incident resolution with an associated change or service request

Part of Workflow(s): Incident Closure (SO2.5)

Applies to User Roles: Incident Analyst

If the solution provided for an incident is found to not resolve the incident completely, then the Incident Analyst must review the incident for necessary changes or requests. When incident closure rejects the incident solution and the incident has an associated change, incident closure updates the status of the incident to pending change and requests that Change Management reopen the change. When incident closure rejects the incident solution and the incident has an associated service request, incident closure updates the status of the incident to pending change and requests that the Service Desk reopen the request.

To reassign a rejected incident solution with an associated change or service request:

1. Click **Incident Management > Incident Queue** or view your To Do queue.
2. Select an incident for review.
3. Review the incident resolution description in the Solution field to verify that it is correct and complete.
4. When the solution doesn't resolve the incident and the incident includes an associated service request:
    - Update the **Status** field to Pending Change.
    - Update the **Assignment Group** and **Assignee** fields with the applicable Service Desk group and Service Desk Agent.
    - In the **Activities** section, do the following:
        - From the **New Update Type** field list, select an update type.
        - In the **New Update** field, type an explanation that describes the problem with the proposed solution and that also mentions any associated requests that must be reopened.
    - Click **Save & Exit** to return to your To Do queue.
5. When the solution doesn't resolve the incident, and the incident includes an associated change:
    - Update the **Status** field to Pending Change.
    - Update the **Assignment Group** and **Assignee** fields with the applicable Change Management group and Change Coordinator.
    - In the **Activities** section, do the following:

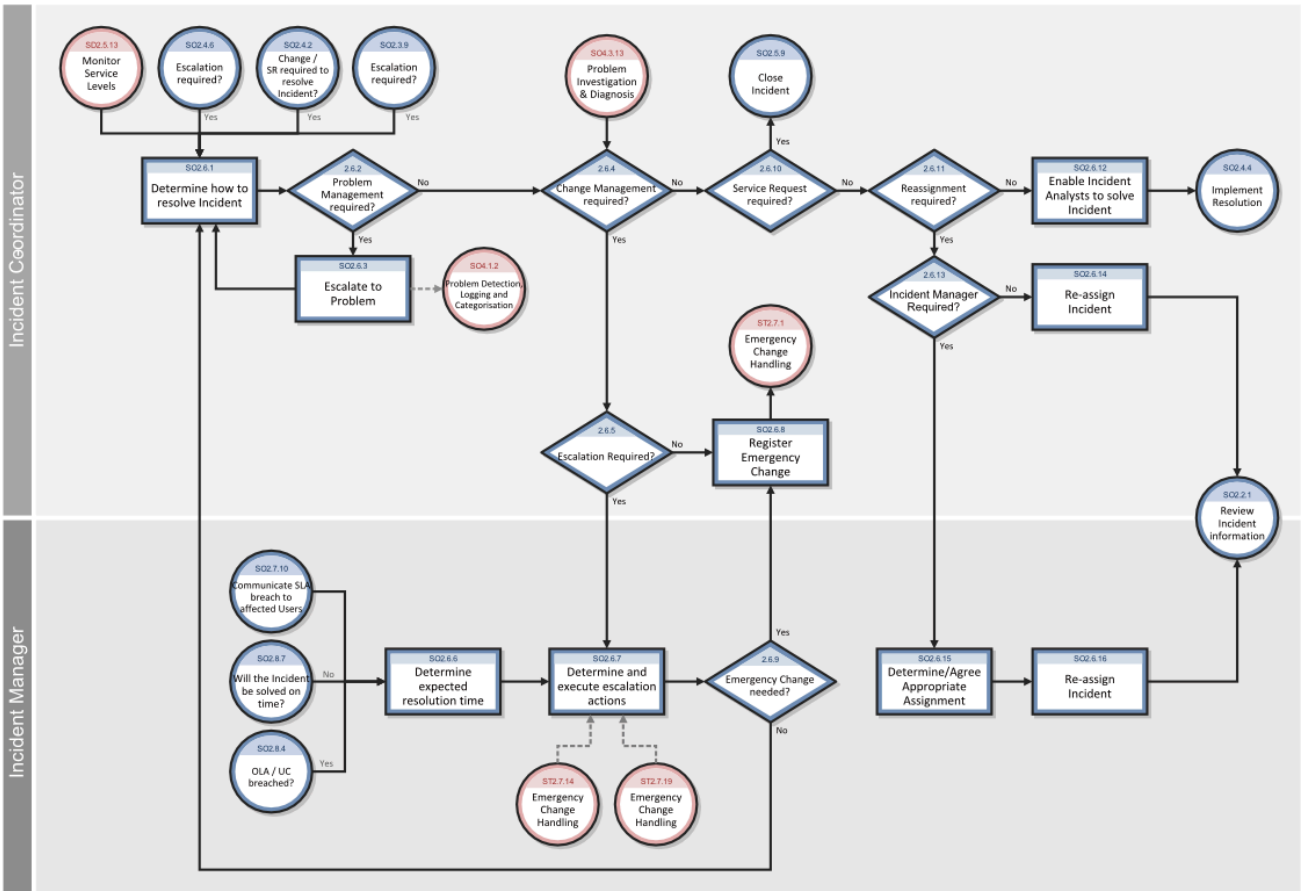
        - From the **New Update Type** field list, select an update type.
        - In the **New Update** field, type an explanation that describes the problem with the proposed solution and that also mentions any associated changes that must be reopened.
    - Click **Save & Exit** to return to your To Do queue.

Related topics

Reject an Incident Resolution
Close an Incident
Close an Incident with an Associated Interaction or Event
Incident Closure (SO2.5)

1.5.3.6. Incident Escalation (SO2.6)

When an Incident Analyst is unable to solve an assigned incident within the target time, the analyst escalates the incident to the Incident Coordinator. The Incident Coordinator determines how the incident can best be resolved by consulting the Incident Analyst and, if needed, other Incident Analysts. If an incident is severe (for example, designated as Priority 1), the appropriate IT managers must be notified so that they can anticipate and prepare for an escalation.



Related topics

Incident Management overview
Escalate an Incident with a Service Request
Open an Emergency Change for an Incident Escalation
Reassign an Incident for Additional Support
Incident Management workflows and user tasks

1.5.3.6.1. Escalate an incident with a service request

Part of Workflow(s): Incident Escalation (SO2.6)

Applies to User Roles: Incident Coordinator

If the Incident Coordinator determines that the incident can be solved by initiating a service request, the Incident Coordinator proceeds as follows:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** queue.
2. Click the open incident record.
3. Open a service request. Click **More** or the More Actions icon and select **Related** > **Quotes** > **Open**. The Request Category form opens.
4. Select one of the categories for this request.
    - Customer Procurement Requests
    - Human Resources
    - Employee Office Move Process
5. If you selected **Customer Procurement Requests**, perform the following steps:
    1. Select from the **Master Catalog**.
    2. Select the category of the item you are requesting.
    3. Select the item(s) you wish to request. A list of the item(s) you selected opens and shows the total cost.
    4. Click **Add Items** to add more items to your request. A summary list of your selections opens and shows total cost.
    5. Click **Submit Request**.
    6. Fill in the required fields (such as requested delivery date and reason for request) in the **New Request** form.
6. If you selected **Human Resources**, perform the following steps:
    1. Select a category. You can choose to select defaults and also to modify the quantity.
    2. Click **Add Items** to add more items to your request. A summary list of your selections opens and shows total cost.
    3. Click **Submit Request**.
    4. Fill in the required fields (such as start date, department, and manager) in the human resources form.
7. If you selected **Employee Office Move Process**, perform the following steps:
    1. Select from the available list of options. You can choose to select defaults and also to modify the quantity. A summary list of your selections opens and shows total cost.

2. Click **Add Items** to add more items to your request.
3. Click **Submit Request**.
4. Fill in the required fields (such as first name, last name, and effective date) in the employee office move request form.

8. Click **Save**. Service Manager creates a request record. The items you requested are listed in the Line Items section, and the incident record is listed in the Related Records section.
9. Click **Save & Exit**. You are returned to the incident record. The new request appear in the Related Records section of the incident record.

Related topics

More Actions menu
Open an Emergency Change for an Incident Escalation
Reassign an Incident for Additional Support
Incident Escalation (SO2.6)

1.5.3.6.2. Open an emergency change for an incident escalation

Part of Workflow(s): Incident Escalation (SO2.6)

Applies to User Roles: Incident Coordinator

When an Incident Analyst is not able to solve an assigned incident on time, the analyst escalates the incident to the Incident Coordinator. The Incident Coordinator determines how the incident can best be resolved by consulting the Incident Analyst and, if needed, other Incident Analysts.

If the Incident Coordinator determines that the incident can only be solved by implementing an emergency change, the Incident Coordinator proceeds by opening an emergency change.

To open an emergency change for an incident escalation:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** list.
2. Click an existing incident record.
3. Open an emergency change from the incident record More Actions menu.
    1. Click **More** or the More Actions icon and select **Related > Changes > Open**.
    2. Select **Emergency Change** to mark this as an emergency change.
    3. In the **Urgency** field list, update the urgency code choosing **1-Critical**.
    4. Enter the name of the person who initiated this change in the **Initiated by** field.
    5. Click **Fill** in the **Assignment Group** field to generate a list of assignment groups, and then choose the applicable assignment group. The Change Coordinator field is populated with the name of the Change Coordinator for that assignment group.
    6. Update all other required fields.
4. Click **Save**. The emergency change procedure starts. You can notify the Change Coordinator that the request has occurred and that the emergency change procedure has started.
5. Return to the related incident record.
6. Click the list in the **Status** field of the incident record to update the status to **Pending Change**.
7. In the Activities section, do the following:
    1. Select the **New Update Type**.

2. In the **New Update** field, type a description of the update, including information about the new emergency change record.

8. Click **Save & Exit**.

Related topics

More Actions menu
Escalate an Incident with a Service Request
Reassign an Incident for Additional Support
Incident Escalation (SO2.6)

1.5.3.6.3. Reassign an incident for additional support

Part of Workflow(s): Incident Escalation (SO2.6)

Applies to User Roles: Incident Manager

After an Incident Manager is notified that an incident is not going to be resolved on time and that the service level agreement (SLA) is breached or in danger of being breached, the Incident Manager reassigns the incident for additional support.

The Incident Manager reviews the SLA target times and determines that the incident requires reassignment to another level of support. The Incident Manager reassigns the incident to another support group.

To reassign an incident for additional supports:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** list.
2. Click the open incident.
3. Clear the assignment group in the **Assignment Group** field, and then click **Fill** to select the new assignment group.
4. Click **Save & Exit**.

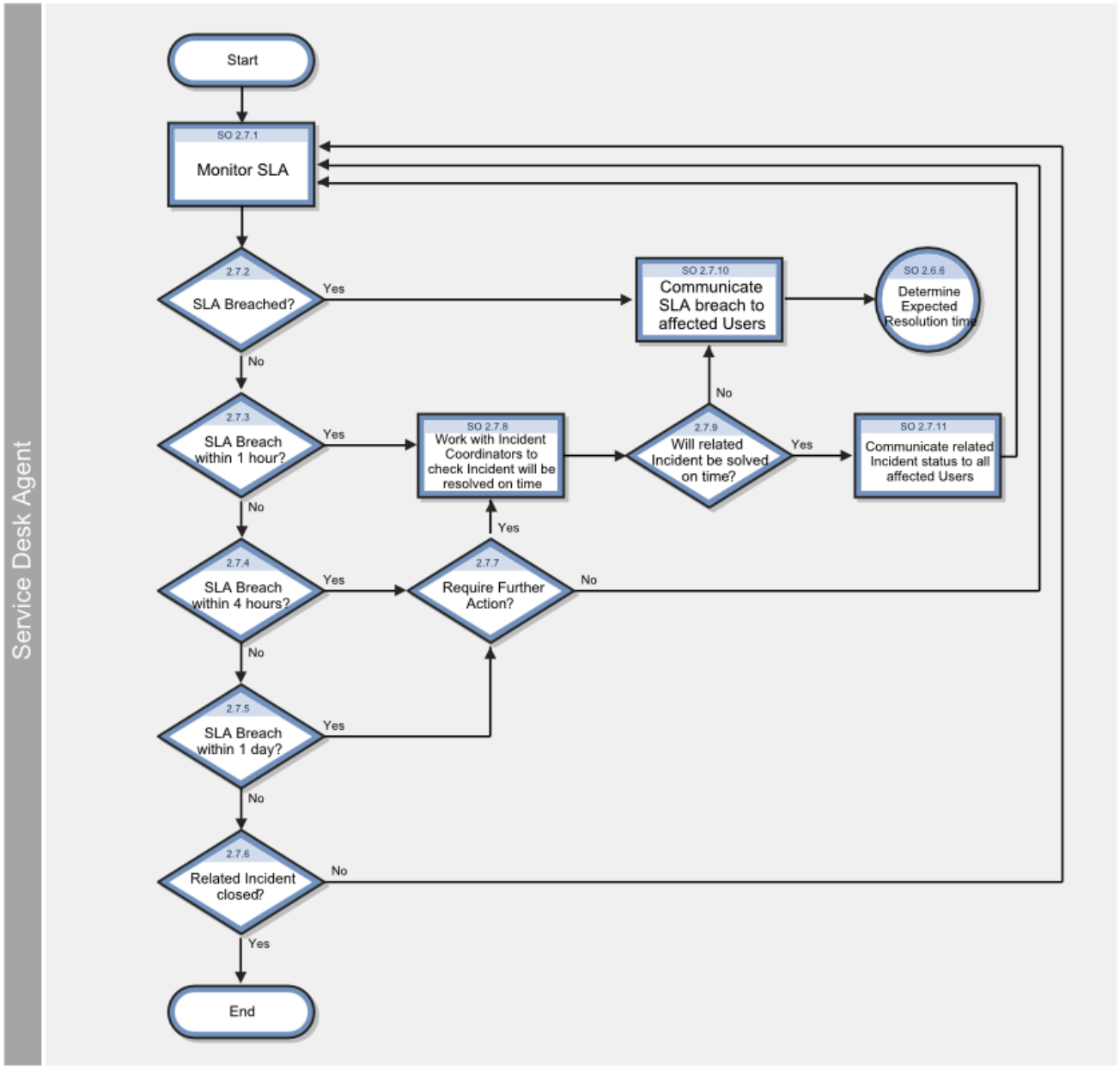
Note: You can contact the Service Desk to request that the Service Desk send an information bulletin to affected users and stakeholders.

Related topics

Escalate an Incident with a Service Request
Open an Emergency Change for an Incident Escalation
Incident Escalation (SO2.6)

1.5.3.7. Service Level Agreement Monitoring (SO2.7)

Service level agreements (SLAs) contain standards for incident resolution performance. This process describes the activities to monitor all interactions related to incidents from initialization to resolution. SLA monitoring also determines whether time targets for incident resolution are met, and indicates whether escalation is required to meet the target resolution date according to the associated SLA. SLA monitoring is an ongoing process performed by the Service Desk.



1.5.3.7.1. Monitor interaction queue for service level agreement breaches

Part of Workflow(s): Service Level Agreement Monitoring (SO2.7)

Applies to User Roles: Service Desk Agent

The Service Desk Agent monitors the interaction queues for interactions with associated incidents that have not been resolved in the time specified by the Service Level Agreement (SLA). When this occurs, the Service Desk Agent begins the escalation procedure for the incident by assigning the incident to the applicable Incident Manager and assignment group based on the services affected by the incident.

To reassign breached incidents for additional analysis and diagnosis:

1. Click **Service Desk > Interaction Queue** or view your To Do queue.
2. In the View list, select Monitor SLA Interaction - Breached.
3. From the list, select an interaction record with a linked incident (Open - Linked) record.
4. To view the related incident, click **More** or the More Actions icon and select **Related > Incidents > View**.
5. Double-click the **Incident ID**.
6. Update the **Assignment Group** and **Assignee** fields based on the information in the Service and Affected CI fields.
7. In the Activities section, do the following:
    1. In the **New Update Type** field list, select an update type.
    2. In the **New Update** field. type an explanation to describe the reason for the escalation and any relevant details.
8. Click **Save & Exit** to return to the Interaction queue.

1.5.3.7.2. Monitor interaction queue for potential service level agreement breaches

Part of Workflow(s): Service Level Agreement Monitoring (SO2.7)

Applies to User Roles: Service Desk Agent

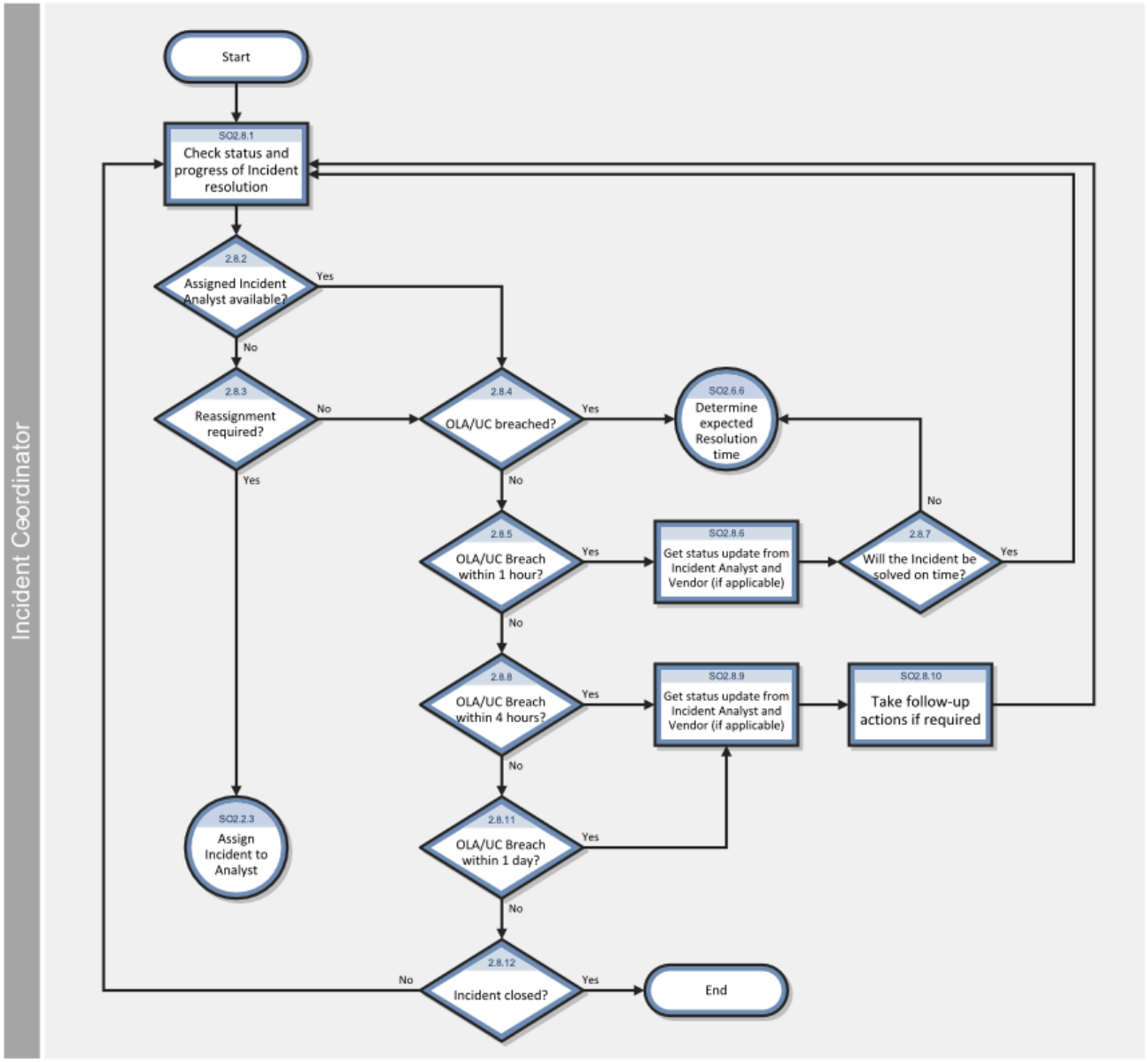
The Service Desk Agent monitors the interaction queues for interactions with associated incidents that are within a few hours of breaching as specified by the Service Level Agreement (SLA) for the type of incident. For those items that are within one hour of breaching, the Service Desk Agent coordinates with the Incident Coordinator who escalates the incident associated with the escalation and communicate the expected resolution to the applicable users and user groups. For those items that are within four hours or one day of breaching, the Service Desk Agent monitors the applicable queues until the interactions are resolved or escalated.

To monitor the interaction queues for potential breached interactions:

1. Click **Service Desk > Interaction Queue** or view your To Do queue.
2. In the View list, select the **Monitor SLA Interaction - Breach within 1 hour** queue.
3. To view the incident details for an interaction in the queue:
    1. View the related incident.
    2. Click **More** or the More Actions icon and select **Related > Incidents > View**.
    3. Double-click the **Incident ID**.
4. Contact the Incident Coordinator assigned to the related incident, and either begin the Incident Escalation process or contact the affected users and groups with the expected resolution details.
5. For any interactions in the other Monitor SLA Interaction queues, continue to monitor the queues until the items are resolved or require escalation.

1.5.3.8. OLA and UC Monitoring (SO2.8)

One measure of the successful resolution of incidents is the performance of the individual support groups and applicable vendors. The performance of support groups is measured by targets set up within Operation Level Agreements (OLAs). The performance of vendors is measured by targets set up in the Underpinning Contracts (UCs).



1.5.3.8.1. Monitor OLA and UC performance

Part of Workflow(s): OLA and UC Monitoring (SO2.8)

Applies to User Roles: Incident Coordinator

Service guarantees exist between IT and other departments in the organization to track hardware or software asset availability and response performance. For example, the IT department might guarantee that a department server will be available 98% of the time and that 99% of the time IT will respond to an outage involving that device within one hour. You can use Service Level Agreements (SLAs) to monitor the performance of Operational Level Agreements (OLAs) in support groups and Underpinning Contracts (UCs) with applicable vendors.

The Incident Coordinator monitors all incidents assigned to the support group and applicable vendors. The Incident Coordinator can reassign an incident to another group or escalate an incident to the Incident Manager if the target time is exceeded or close to being exceeded.

To monitor OLA and UC performance:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** queue.
2. Click the list in the **Queue** field and select Incident.
3. Select an OLA-UC breach view to see which incidents need follow-up or reassignment. Talk to the Incident Analyst or vendor working on the incident to receive a status update and discuss possible resolutions. The OLA-UC breach views include:
    - Breach within 1 hour
    - Breach within 4 hours
    - Breach within 1 day

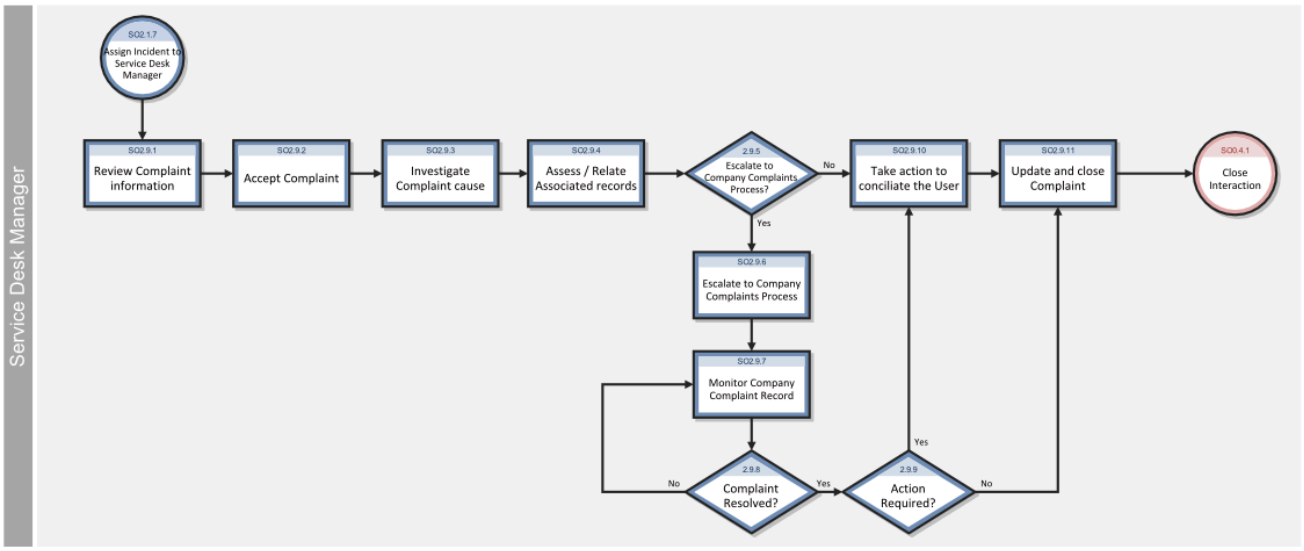
Note: You can add these views to your Favorites in the navigation tree for easy access.

1. When you locate an incident that has not been resolved on time, the **SLA Breached** field in the SLA section is checked.
2. Based on your review of the incident and discussion with the Incident Analyst or vendor, you determine how to proceed with resolving the incident. For example:
    - Reassign to a second- or third-line support group with more knowledge.
    - Escalate the incident. Reassign the incident to the Incident Manager, and then contact the Incident Manager so the escalation process can begin.

3. Open the **Activities** section, and then provide the applicable entries for the **New Update Type** and **New Update** fields to add current status information and actions required to resolve the incident.
4. Click **Save & Exit**.

1.5.3.9. Complaint Handling (SO2.9)

Complaint handling is the process by which the Service Desk Manager handles complaints. The complaint category is typically used to indicate less than satisfactory service received by a user in the support or service delivery categories.



Related topics

Incident Management overview
Alerts and escalation
Service Desk overview
Associating a Service Desk interaction with another application record
Handle Complaints
Incident Management workflows and user tasks

1.5.3.9.1. Handle complaints

Part of Workflow(s): Complaint Handling (SO2.9)

Applies to User Roles: Service Desk Manager

The Service Desk Agent assigns an incident containing a user complaint to the Service Desk Group. When the incident is saved, it is automatically assigned to the Service Desk Manager. The Service Desk Manager reviews and accepts the incident to research and resolve the complaint.

To manage an incident based on a user complaint:

1. Click **Incident Management** > **Incident Queue** or view your **To Do** queue.
2. Click the open incident record and review the contents of the User complaint.
3. Click the list in the **Status** field and select **Accepted**.

Note: Investigate the complaint by reviewing the relevant information and talking to the people involved.

1. When you have finished searching for an answer or solution to satisfy the user, select the Activities section to add notes in the **Update** field.
2. Click **Fill** in the **Closure Code** field to select an applicable code.
3. Enter the resolution in the **Solution** field and verify that the information is correct and complete.
4. Click **Close Incident**.
5. Click **Save & Exit**. The status changes to Closed.

Related topics

Incident Management overview
Alerts and escalation
Service Desk overview
Associating a Service Desk interaction with another application record
More Actions menu
Complaint Handling (SO2.9)

1.5.3.10. Miscellaneous tasks

You can also perform the following tasks in the Service Desk module.

Using mass update with Service Desk interaction record lists

The mass update feature enables you to update the value of one or more fields in a list of records. The system provides a template form that displays the current field values and enables you to type or select a new value for the displayed fields. In compliance with the data policy of your organization (as recorded in the Data Policy form), the Mass Update template form doesn't display read-only fields in the records. The mass update option is available from both interaction and incident record lists.

When you perform a mass update operation, make sure that the value you type or select for a particular field is the value you want in all the records that you selected. When you update incident records, you must also update the incident activity data.

System Administrators can determine if a field is displayed in the Mass Update template form by directly editing the Data Policy record for the form. To remove a field from the Mass Update form, change the Usage Type column for the field to System. If a System Administrator wants to grant Template Mass Update permissions to other operators, the System Administrator must update the operator profile to include the Template Mass Update and Complex Mass Update capabilities.

Update multiple interaction records

Applies to User roles: Service Desk Manager

You can use the mass update feature to update one or more values in multiple interaction records. Remember that the value you enter for a particular field becomes the value for all of the records you selected when you perform a mass update.

To update multiple records, follow these steps:

1. Locate the records by using one of the following methods:
    - Go to the Interaction Queue to display a list of records.
    - Use search or advanced search to find one or more records.
2. Click **More** or the **More Actions** icon, and then select **Mass Update**.
3. Double-click the field that you want to update.

4. Type the value for the field or use the Find feature to display a list of potential values for the field.
5. Click **Next**, and then repeat steps 3 and 4 for additional fields as needed.
6. Update the Activity Action field for the updates you made in the Mass Update template, and then click **Execute**.

To use complex update to update multiple records:

1. Locate the records by using one of the following methods:
   - Go to the Interaction Queue to display a list of records.
   - Use search or advanced search to find one or more records.
2. Select the records that you want to update.
3. Click **Mass Update**, and then click **Complex Update**.
4. Type the instructions (in RAD syntax) to be executed one time at the beginning of the mass add or mass update operation.
5. Type the instructions (in RAD syntax) for action on each record.
6. Click **Execute**.

Duplicate an existing interaction record

Applies to User roles: Service Desk Agent

You can duplicate an existing interaction record to copy information from an existing record into a new record. For example, you duplicate an existing interaction record because a user is experiencing a repeat of a past problem or issue.

To duplicate an existing interaction record, follow these steps:

1. Click **Service Desk** > **Search Interactions**.
2. Use search or advanced search to find one or more records.
3. Double-click the record to open it.
4. Click **More** or the **More Actions** icon, and then and select **Duplicate Interaction**. Service Manager assigns the next available ID number to the duplicated interaction.
5. Add or change information in the interaction record as needed.

Note

You must clear any existing data in a field to use the Fill and Find functions to select a new value.

6. Click **Save & Exit**. If the record that you duplicated has any related records, a message that informs you the record has related records and asks if you want to copy the associations is displayed.
    - Click **Yes** to copy associated records and return to the interaction queue.
    - Click **No** if you don't want to copy the associated records to the new interaction record. The Escalation Details wizard starts, and you can relate the new record to an existing incident, an existing known error, or a new incident.

Add a required action

Applies to User roles: Service Desk Agent

A required action is an action that a service recipient or your service organization requires you to perform before an interaction can be closed. For example, the service recipient may want to be contacted by a certain date. When you make this a required action, nobody can close this interaction until someone contacts the service recipient.

By default, Service Manager defines one required action: When the Notify By field for an interaction is set to "telephone" the Service desk Agent must complete and close the required action before the interaction can be closed.

When you add a required action to an interaction record, the system updates the status of the record to Open-Callback, which means that the specified required action must be completed before the interaction and any associated incidents can be closed. The Service Desk Agent specifies what the required action is when they add the required action information to the interaction record. Required Actions lists a description of the action required for the interaction and a link to any associated incidents.

To add a required action, follow these steps:

1. Click **Service Desk** > **Search Interactions**.
2. Use search or advanced search to find one or more records.
3. Double-click the record to open it.
4. Click **More** or the **More Actions** icon, and then select **Add Required Action**.
5. Type the action details and a reason for the required action.
6. Select the incident number to associate with this interaction.

> **Note**
>
> You can click **Fill** to display a list of incidents.

7. Click **OK**.

> **Note**
>
> You can't close the interaction until someone performs the required action.

Set a reminder for an interaction record

Applies to User roles: Service Desk Agent

You can set a reminder to notify you if an existing interaction record meets certain conditions by a specified time. For example, you may want to be reminded if an interaction is still open or is not updated by a given date.

To set a reminder, follow these steps:

1. Locate an existing interaction.
2. Click **More** or the **More Actions** menu, and then select **Set Reminder**.
3. Choose one of the following options to specify when the reminder occurs:
   - Click **Remind At** to set a reminder for a particular date and time.

     > **Note**
     >
     > You can click **Fill** to display a calendar tool to select the date and time of the reminder.

   - Click **Remind In** to set the reminder to occur after a particular time interval. If you choose this option, you must specify the time interval in hours, minutes, and seconds (use the 00:00:00 format), and then select a shift from the list.
4. In the **Remind If** field, select from the following options any conditions that you want the interaction record to meet before a notification occurs:
   - Always (default)
   - ... has not been updated

- ... is still open
- ... is still assigned to me

5. In the Pop-up Message area of the form, select from the following options the type of notification that you want to receive:
   - Pop-up
   - Email
   - SM Mail (Service Manager Internal Mail)

   Note

   If you select the Pop-up notification method, type the reminder message format in the **Message** field. If you select the Email or SM Mail notification methods, type the title you want the notification to display in the **Title** field, and then select **Message Type**.

6. Click **OK**.

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  - 1.1 Update multiple interaction records
- 2 Duplicate an existing interaction record
- 3 Add a required action
- 4 Set a reminder for an interaction record

1.5.4. Incident Management administrator tasks

A System Administrator can access the Operator record for the following information types and functions:

- User and contact information
- Application profile privileges
- Mandanten utility

Configure Incident Management settings

Applies to User roles: System Administrator

To configure Incident settings:

1. Click **Incident Management** > **Administration** > **Settings** in the System Navigator.
2. Click **Incident** and then configure the following settings for the Incident application:

Setting	Description
Notify incident owner if Parent Incident flag was unchecked	When the Parent Incident flag is unchecked, a notification is sent to all the assignees of the child incidents that were related to this parent incident
Default Category	The default category when you register a new Incident. The out-of-box value for this parameter is "none".
Parent to Child Incident Default Status	When any of the following scenarios occurs, the status of the child incident changes to the default status that's specified in this setting. ◦ The parent-child relationship is dismissed from the parent incident ◦ The parent incident is closed and the **Parent Incident Relationship Model** is **Parent Incident and Child Incidents close independently**
Parent Incident Relationship Model:	Whether to close a child incidents when its parent incident is closed

Setting	Description
Post back link to Child	The link file that posts the changes of certain fields in the parent incident to the corresponding fields in child incidents. This function works only when you select **Close Child Incidents when Parent Incident closed** in **Parent Incident Relationship Model**. You can customize the fields that are posted back to the child incidents. For more information about links, see Links.

3. Optionally, click the **Incident Task** and then configure the following settings for an incident task:

Setting	Description
Default Category	The default category when you register a new incident task. The out-of-box value for this parameter is "none".

4. Click **Save** and then click **OK**.

Configure the Incident Management environment

Applies to User roles: System Administrator

Incident Management provides an environment record in which you define the operational environment for the application by allowing or disallowing specific operator actions. Administrators can modify the environment parameters to match the operational requirements of their organization.

To configure the Incident Management environment:

1. Click **Incident Management** > **Administration** > **Environment** in the System Navigator.
2. Select or clear the parameters for your Incident Management environment.

   Tip

   You can use the field help to view the description for each setting. To view the help on field:

   - Web client: Select a field, and then press **F1**.
   - Windows client: Select a field, and then press **Ctrl+H**.

3. Click **Save**.
4. Click **OK**.

Create a template to complete incident records

Applies to User Roles: System Administrator

Templates enable users to quickly complete incident records by automatically populating various fields with necessary information. You can create as many templates as you need while authorizing select user roles for each template. The users authorized to use a particular template will then be able to apply that template when completing an incident.

Note

Make sure that the user profile is enabled to use templates.

To create a template, follow these steps:

1. Click **Tailoring** > **Templates**, and then click **New**.
2. Type a name in the **Template name** field.
3. Select Incident in the **Table name** field, and then click **Next**.
4. Click **Fill** for each user role that you are authorizing to use this template.
5. Modify the fields as required to meet the needs of your template by double-clicking in the field to assign a value to each field you select.
6. Click **Add** to exit the template wizard.
7. Click **Save**.
8. Click **OK**. The roles authorized for this template can now use or modify this template.

Alternatively, you can create a template from an existing record. To do this, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Enter an Incident number in the **Incident ID** field, and then click **Search**.
3. Click the incident record.
4. Click **More** or the More Actions icon and select **Create Template from Record**.
5. Modify the name in the **Template name** field.
6. Click **Fill** for each user role that you are authorizing to use this template.
7. Modify the fields as required to meet the needs of your template by double-clicking in the field to assign a value to each field you select.
8. Click **Add** to exit the template wizard.
9. Click **Save**.
10. Click **OK**.

Assignment groups

An assignment group is a list of users who are responsible for an Incident record. Incident Management notifies the group when an Incident record opens or escalates. Assignment groups make the routing and escalation of Incident records easier.

For example, the Service Desk Agent receives a service request to fix a disabled workstation. The agent creates an Incident record, and then assigns the incident to the Hardware assignment group. The hardware technician determines that the hard drive must be replaced. Because the drive must be purchased, the technician assigns the Incident record to a Request Fulfillment assignment group for acquisition. If the hard drive purchase is delayed, and the delay impacts the closure of the Incident record in a reasonable amount of time, Incident Management automatically escalates the Incident record and assigns it to the Request Fulfillment manager.

Service Manager administrators create Assignment groups as part of the Ongoing Maintenance process.

Add an Incident Management assignment group

Applies to User Roles: System Administrator

Incident Management assignment groups specify the Incident Coordinator and Operators in the group so that when someone creates a new incident it can be assigned to the applicable group. Typically, assignment groups are organized by location and expertise. To facilitate incident response, new assignment groups can be added to the out-of-box assignment groups for Incident Management.

To add an Incident Management assignment group:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   > **Tip**
   >
   > To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.
   1. Enter the name of the manager of the assignment group.
   2. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   3. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide response when you create and populate a new assignment group.

   (Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.
7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.
9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.
10. Follow these steps to configure the Supported Services field:
    1. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    2. Select **Business Service** in the CI Type field, and then click **Search**.
    3. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    4. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.
11. Click **Save** and **OK**.

Using mass update with Incident Management record lists

Mass update enables you to select multiple records from a list of records and then to update the value in a field or several fields in the selected records. The system provides a template form that displays the fields for the selected records and allows a user to change the value of any of the displayed fields. The Mass Update template form doesn't display all fields in the records. For example, fields marked read only in the data policy don't display. You can also do a Complex Update on the selected records. A Complex Update uses RAD expressions containing variables and concatenated fields to populate another field in the records.

When doing a mass update, you should remember that the value you enter for a particular field becomes the value for that field for all of the selected records. Whenever you update incident records, you must also update the incident activity data.

Mass update is available for incident records and incident queue records. The ADMIN and SYSADMIN profiles in the out-of-box system provide the Template Mass Update and Complex Mass Update capability

A System Administrator can edit the datadict table and probsummary file so that a field doesn't appear in the list of fields displayed by the Mass Update wizard. On the Data Policy form, change the Usage Type column for the field to System.

Update multiple incident records

Applies to User Roles: System Administrator

You can use mass update to update a value in one or more fields for multiple incident records. For example, if you want to update multiple high priority incident records, select the high priority view of the Incident queue. The value you enter for a particular field becomes the value for that field for all the records you selected for mass update.

You can use complex update to enter expressions as instructions for actions to be performed on multiple incident records.

To update multiple incident records by using **Mass Update**:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records. Select the View list of incident records that you want to update. For example, select High Priority Incidents to update multiple high priority incident records.
3. Use the arrow keys to move to a row and then use the space bar to select individual records. A range of records can be selected using **Shift**+Spacebar. You can also use the scroll bar and the **ctrl** key to select individual records.
4. Click **Mass Update**.
5. Update the fields to be included in the mass update:
    1. Double-click the field that you want to update, and then type the value for the field in the box or use the Fill feature to display a list of potential values for the field.
    2. Click **Next**.
    3. Continue updating the fields that you want included in the mass update.
    4. Click **Next** after each update.
    5. Click the **Activity Update** field to document what's included in the mass update. When the mass update is complete, these notes display in the Journal Update of the Activities section for each record.
6. Click **Execute**. Mass update starts, and then displays each incident number as it is updated.
7. If an incident record can't be included in the mass update, you can choose to:
    - **Retry**: Retry the mass update.
    - **Skip**: Skip the incident record to exclude it from the mass update and continue with the next record.
    - **End**: End the mass update, and retry later.

To update multiple incident records, using **Complex Update**:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records. Select the View list of incident records you want to update. For example, select High Priority

Incidents to update multiple high priority incident records.
3. Click **Mass Update**.
4. Click **Complex Update**.
5. Follow the instructions in the complex update form and type the expressions in the **Instructions to be executed ONCE at the beginning of mass add/update** field and the **Instructions for action on EACH RECORD** field.
6. Click **Execute**.
7. If an incident record can't be included in the complex update, you can choose to:
    - **Bypass**: Bypass the invalid message and include the incident record in the complex update.
    - **Skip**: Skip the incident record to exclude it from the complex update and continue with the next record.
    - **End**: End the complex update, and retry later.

Incident Management downtime records

You can access Incident Management downtime records by using the Incident Management Security Administration Utility to show the availability of a selected device. Availability measures the ability of a device or component to provide service within a measured time frame. There are three downtime measurements.

Type of availability	Description
Explicit unavailability	The downtime experienced by the failing device.
Implicit unavailability	The downtime experienced because of the failure of a parent or controlling device.
Perceived unavailability	The explicit or implicit downtime during normal business hours. The total reflects only working hours.

Create a downtime record

Applies to User Roles: System Administrator

As part of monitoring the availability of devices in the system, a System Administrator creates downtime records for a specific device.

To create a downtime record:

1. Click **System Administration** > **Base Configuration** > **Monitoring** > **Downtime**.
2. Type the applicable data in the **Logical Name**, **Location**, **Contact Name**, **Type**, and **Table Name** fields.
3. Click **Add**.
4. Click **OK**.

Reset downtime

Applies to User Roles: System Administrator

To reset downtime:

1. Click **System Administration** > **Base Configuration** > **Monitoring** > **Downtime**.
2. Click **More** > **Reset**.
3. Click **Schedule** to set a time for the reset to occur.
4. Click **OK**.

Create a note

Applies to User Roles: System Administrator

Any note that an Administrator or Operator user creates is saved with the associated Service Manager records. You can add more notes each time you select the Notes option. Each set of notes has a time stamp and the name of the operator who created the note set.

You must have notes enabled by your Service Manager System Administrator to use this feature. To enable notes for an Incident Management operator, select notes on the Incident Management Security Profile form. For the out-of-box system, only the System Administrator has notes enabled.

To create a note:

1. Do one of the following to locate a record:
    1. Select an incident record from a queue.
    2. Use search or advanced search to find one or more records.
2. Click **More** or the More Actions icon and select **Notes**.
3. Type the information in the **Add New Note** box.
4. Click **Save**.
5. Click **Back**.
6. Click **Save & Exit**.

1.5.5. Incident configuration

Incident configuration enables you to configure alert, Incident categories, workflow, solution matching, and so on.

Create an incident category

Applies to User roles: System Administrator

If you are a Service Manager Administrator, you may want to create an incident category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides out-of-box category records that you can use or modify.

> **Note**
>
> When a category is set as the default category in settings, don't delete it to avoid unpredictable issues.

To create new category record:

1. Click **Incident Management**.
2. Click **Configuration** > **Incident Categories**.
3. Click **New**.
4. Type the name of the incident category.
5. Clear the **Active** checkbox if you don't want the new category to appear in the category list.

   > **Note**
   >
   > If you clear the **Active** checkbox, this category can't be used to create new incidents.

6. Type the category description.
7. Select a workflow for the category.
8. Click **Save**.

9. **Optional**: Click the **Link New Subcategories** button to create a new subcategory for the incident category.

Note

The **Apply To** flag indicates whether the category is shared across different modules.

For example, if the **Apply To** is set to "Interaction/Incident", then this category will be created in both the interaction category table and the incident category table.

Typical shared categories in the out-of-box configuration include Complaint, Request for Information, Incident, and Request for Administration.

For the **Apply To** flag in Incident Category:

When creating an incident category, the **Apply To** option is predefined as "Incident" only and can't be modified. However, an incident category can also be created when creating an interaction category with the **Apply To** option set as "Interaction/Incident" or "Interaction/Incident/Problem".

When searching for an incident category, all the **Apply To** values that include Incident can be used.

The following list is used in Incident Category:

1 - Incident

3 - Interaction/Incident

4 - Interaction/Incident/Problem

Note

Incident category name is read-only after the category is created.

Create an incident task category

Applies to User roles: System Administrator

If you are a Service Manager Administrator, you may want to create an incident task category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides out-of-box category records that you can use or modify.

To create a new incident task category record:

1. Click **Incident Management**.
2. Double-click **Configuration** > **Incident Task Categories**.
3. Click **New**.
4. Type the name of the incident task category.
5. Type a description of the problem task category.
6. Clear the **Active** checkbox if you don't want the new category to appear in the category list.

   Note

   If you clear the **Active** checkbox, this category can't be used to create new incident tasks.

7. Select a workflow for the category. The **Workflow** tab is displayed.
8. Click **Save**.

Note

Incident task category name is read-only after the category is created.

Add a new subcategory to an incident category

User Roles: System Administrator and Implementer

You can add a new subcategory directly for an incident category.You can view a list of subcategories and their record details associated with the current category.

To add a new subcategory for an incident category:

1. Click **Incident Management** > **Configuration** in the System Navigator.
2. Click **Incident Categories** and then click **Search** in the form that appears.
3. Select the Incident Category for which you want to add a subcategory.
4. Click the **Subcategories** tab in the Incident Category Definition page, and then click the **Link New Subcategories** button.

5. Type a subcategory name.
6. Type a description for the subcategory.
7. Click **Save** to add the new subcategory.
8. **Optional**: Click the **Link New Areas** button to create areas for the incident subcategories. When you have finished, click **Save**.

Add a new area for an incident subcategory

Applies to User roles: System Administrator and Implementer

You can add a new area directly for an incident subcategory.You can view a list of areas and their record details associated with the current subcategory.

To add a new area for an incident subcategory:

1. Click **Incident Management** > **Configuration** in the System Navigator.
2. Click **Incident Categories** > **Search**.
3. Select the Incident Category for which you want to add an area.
4. Under the **Subcategories** section in the Incident Category Definition page, select the subcategory for which you want to add an area.
5. Type an area name.
6. Type a description for the area.
7. Click **Save** to add the new area.

Incident solution matching

When you create an incident, a list of potentially related incidents appears based on the pre-defined solution matching configuration. You can then select the matched records and update them directly. You can also run solution matching for an existing incident that's not closed. To do this, select the incident and then click **More** > **Solution Matching** from the Detail List toolbar.

In an out-of-box system, the following options are provided for Incident solution matching:

- Find an incident with a matching service or configuration item
- Find an incident with a matching related service or configuration item
- Find an incident with a matching title
- Find a known error with a matching service or configuration item
- Find a known error with a matching title
- Find a problem with a matching service or configuration item
- Find a problem with a matching title

To configure Incident solution matching options, navigate to **Incident Management** > **Configuration** > **Solution Matching**.

Note

If you create an incident from Service Desk escalation or link to a new incident from the Related Records section, the above incident solution matching will be ignored.

Incident management downtime record

You can access Incident Management downtime records by using the Incident Management Security Administration Utility to show the availability of a selected device. Availability measures the ability of a device or component to provide service within a measured time frame. There are three downtime measurements.

Type of availability	Description
Explicit unavailability	The downtime experienced by the failing device.
Implicit unavailability	The downtime experienced because of the failure of a parent or controlling device.
Perceived unavailability	The explicit or implicit downtime during normal business hours. The total reflects only working hours.

Add a downtime record

Applies to User roles: System Administrator

As part of monitoring the availability of devices in the system, a System Administrator adds downtime records for a specific device.

To add a downtime record:

1. Do one of the following:

- Click **Incident Management** > **Configuration** > **Downtime Records**.
- Click **System Administration** > **Base System Configuration** > **Monitoring** > **Downtime**.

2. Type the applicable data in the **Logical Name**, **Location**, **Contact Name**, **Type**, and **Table Name** fields.
3. Click **Add**.
4. Click **OK**.

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- 5 Incident solution matching
- 6 Incident management downtime record
  - 6.1 Add a downtime record

1.5.6. Security

The following sections explain the Incident Management security roles, security areas, and rights.

Incident security areas

The security areas for Incident are Incident, Incident Tasks, and Incident Management Configuration. These areas contain the default security rights and settings for the Incident Management module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

These security areas are used to set permissions to operators to provide access to particular area of Incident. The following table lists the areas and the relevant Incident menu items the operators can access.

Area
Incident
Incident Task
Incident Management Configuration

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Incident, Incident Tasks, and Incident Management Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Incident	TRUE	FALSE	Never	Never	FALSE	FALSE
Incident Tasks	TRUE	FALSE	Never	Never	FALSE	FALSE
Incident Management Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Incident, Incident Tasks, and Incident Management Configuration areas.

Incident security roles and settings

The out-of-box security roles for the Incident module include the following:

- Incident process owner
- Incident coordinator
- Incident analyst
- Incident manager
- Incident task assignee
- Configuration auditor
- Operator

Tip

If you have upgraded from a previous version of Incident Management, which is not Process Designer based, see Incident Management security profiles mappings for more information about the mappings between the legacy Incident Management profiles and the current Process Designer security roles.

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Incident area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Incident, Incident Tasks, and Incident Management Configuration areas. This table only lists the new security roles that have different settings with the default rights.

Area Name

Role Name

View

New

Update

Delete/Close

Modify Template

Expert

Admin rowspan='2'|Incident

incident process owner

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

system administrator

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE rowspan='12'|Incident Tasks

configuration auditor

TRUE

TRUE

Never

Never

FALSE

FALSE

FALSE

incident analyst

TRUE

TRUE

Always

Always

FALSE

TRUE

FALSE

incident coordinator

TRUE

TRUE

Always

Always

FALSE

TRUE

FALSE

incident manager

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

incident process owner

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

incident task assignee

TRUE

FALSE

When assigned to workgroup

When assigned to workgroup

FALSE

FALSE

FALSE

initiator

TRUE

TRUE

Never

Never

FALSE

FALSE

FALSE

operator

TRUE

TRUE

When assigned

When assigned

FALSE

FALSE

FALSE

problem manager

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

SD agent/manager

TRUE

FALSE

When assigned

Never

FALSE

FALSE

FALSE

service tech

TRUE

TRUE

Always

Always

FALSE

FALSE

FALSE

system administrator

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE rowspan='3'|Incident Management Configuration

incident manager

TRUE

TRUE

Always

Always

FALSE

FALSE

FALSE

incident process owner

TRUE

FALSE

Never

Never

FALSE

FALSE

FALSE

system administrator

TRUE

TRUE

Always

Always

TRUE

TRUE

TRUE

For information about the out-of-box role rights in the Common Configuration area, see Out-of-box role rights in the Common Configuration area.

1.6. Problem Management

Service Manager Problem Management can help you to identify the underlying reasons for one or more incidents, implement workarounds, identify known errors, and provide permanent solutions that minimize the effects of incidents caused by errors in the IT infrastructure.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Problem Management user roles

Key concepts

The most important things you need to know about how the Problem Management module functions and interacts with other modules.

Problem Management overview

Proactive and reactive Problem Management

Problem Control overview

Error Control overview

Problem Management searches

Incident Management relationship

Change Management relationship

User workflows

Describes the ITIL worflows as implemented by Service Manager and the sub-tasks that end users need to perform to execute the workflows.

Problem Management workflows and user tasks

Administrator tasks

Describes the individual steps an Administrator needs to take to configure the Problem Management workflows, assign problem categories, and Problem phases/error phases.

Configure the Problem Management settings

Configure the Problem Management environment

Additional resources

Processes and Best Practices

1.6.1. Problem Management overview

You can use Service Manager Problem Management to identify the underlying reasons for one or more incidents, implement workarounds, identify known errors, and provide permanent solutions that minimize the effects of incidents caused by errors in the IT infrastructure and to prevent problem recurrence. Over the long term, this leads to a reduced volume of incidents as well as saved time and money. Problem Management enables you to achieve the following results:

- Identify errors in IT infrastructure, record them, track their history, find resolutions for them, and prevent recurrence
- Create automatic alerts and notifications that indicate when a problem, task, or known error opens or the owner or status changes
- Escalate problems automatically if not resolved in a timely manner
- Record resolutions and make them easily available to affected user groups
- Find opportunities for improvements and make the necessary tools easily accessible
- React to issues related to incidents
- Proactively resolve issues before incidents occur
- Reduce incidents against your service assets and configuration items

Related topics

Problem Management implementation
Proactive and reactive Problem Management
Incident Management relationship
Change Management relationship
Priority, impact, and urgency
Problem Control overview
Error Control overview

1.6.1.1. Problem Management implementation

You should implement Problem Management to run concurrently with incident Management because it relies heavily on data gathered through the incident management process. The quality of proactive Problem Management depends on monitoring service successfully and the quality of the data recorded. Issues and incidents must be identified, recorded, classified, investigated, and diagnosed by knowledgeable users. To be effective, Problem Management requires the following input:

- Well recorded incidents and issues
- Reactive analysis of incidents and issues
- Proactive analysis of the IT infrastructure
- Input from involved personnel, such as support staff, developers, vendors, trainers
- Discovered solutions

It often works well to begin with reactive analysis of incidents and issues, then continue with proactive analysis of the IT infrastructure after gathering data. Proactive Problem Management relies heavily on established service monitoring and data gathering.

Related topics

Problem Management overview
Proactive and reactive Problem Management
Incident Management relationship
Change Management relationship
Priority, impact, and urgency

1.6.1.2. Proactive and reactive Problem Management

Service Manager Problem Management provides a lifecycle approach to managing problems that's integrated with other modules of Service Manager, including Incident Management, Change Management, Release Management, and Knowledge Management. It consists of reactive and proactive approaches to managing problems:

- Reactive Problem Management, which is generally executed as part of Service Operation based on incident history.
- Proactive Problem Management, which is initiated in Service Operation, but generally driven as part of Continual Service Improvement.

Problem Management is the process responsible for managing the lifecycle of all problems. Incident Management and Problem Management are separate processes although they are closely linked. Incident Management focuses on the restoration of service to users, whereas Problem Management focuses on identifying and removing the causes of incidents.

Problem Management includes the activities required to diagnose the root cause of incidents and to determine the resolution to related problems. It is also responsible for ensuring that the resolution is implemented through the appropriate control procedures, especially Change Management and Release Management.

Problem Management also maintains information about problems and the appropriate workarounds and resolutions, which helps the organization to reduce the number and impact of incidents over time. In this respect, Problem Management has a strong interface with Knowledge Management, and tools such as the Known Error Database are used for both.

Problem Management provides improved service quality and reliability. The number of recurring incidents should decrease as you increase the number of permanent solutions. As your Problem Management system matures, the amount of elapsed time in the find-to-fix ratio should decrease.

Related topics

Problem Management overview
Problem Management implementation
Incident Management relationship
Change Management relationship
Priority, impact, and urgency

1.6.1.3. Using mass update with record lists in Problem Management

Mass update enables you to select multiple records from a list of records and then to update the value in a field or several fields in the selected records. The system provides a template form that displays the fields for the selected records and allows a user to change the value of any of the displayed fields. The Mass Update template form doesn't display all fields in the records. For example, fields marked as read only in the data policy don't display. Mass update is available for problem records, problem task records, and known error list records. It is also available for problem queues, problem task queues, and known error queues.

When doing a mass update, you should remember that the value you enter for a particular field becomes the value for that field for all of the records you selected.

A system administrator can edit the appropriate datadict table so that a field doesn't appear in the list of fields displayed by the Mass Update template form. On the Data Policy form, change the Usage Type column for the field to System.

1.6.1.4. Incident Management relationship

Problem Management and incident Management are related. Incident Management must provide effective incident classification and tracking to provide good data for Problem Management analysis.

The Knowledge Base that Problem Management builds and maintains is a solution repository for new incidents. Matching incidents to problems and known errors is the first step in spotting trends. Subsequently, trend analysis helps you remove errors before they affect a large segment of users.

Interoperability

Service Manager has a built-in Interoperability feature that automatically populates related problem and incident records with updated information.

Interoperability enables two or more components to exchange information within Service Manager. Service Manager schedules transparent updates to a table or application to reflect changes in another table or application.

The automatic process begins when you update a record that has links to another record. For example, if a problem or known error record has associated incident and change records, an update to the change record should trigger automatic updates to the problem and incident records.

When the ioevents.schedule RAD application runs, it uses copies of the original record and the changed record, then selects the ioevents record with a matching table name. If the condition in the actions array is true, the application creates a schedule record with a list of actions to occur. When the ioevents.process RAD application runs, it processes the schedule record and the associated ioaction record with the update information.

These record updates are transparent in normal processing. Background schedulers make the changes without affecting the current client/server session performance. There is no intervention required by the user to ensure all updates post correctly.

The Interoperability feature uses these tables and applications.

Component	Description	Table field	Content
ioevents table	Contains schedule records to trigger events listed in the ioactions table.	name	Action name.
		description	A brief description.
		javascript	The name of the JavaScript that runs when the action occurs.
ioactions table	Contains action or event records processed by the ioevents.schedule RAD routine. Each action record contains a JavaScript to complete the action.	filename	The table name that the event references.
		actions	An array of structures with information about interoperability actions and conditions.
		name	A field in the actions array of structures that contains the name of the interoperability action.
		condition	A field in the actions array of structures containing the condition that triggers the action.

Component	Description	Table field	Content
ioevents.schedule RAD routine	Processes the ioevents table at regular intervals to locate records with pending actions. If the event condition is true, ioevents.process processes the actions record in the ioactions table.		
ioevents.process RAD routine	Processes the actions records in the ioactions table.		
Interoperability Helpers(ScriptLibrary record)	Contains JavaScript functions to create schedule records manually.		

Creating schedule records manually

There is a ScriptLibrary record named interoperabilityHelpers. This record contains JavaScript functions that enable you to create schedule records manually. When Service Manager calls these functions, they generate the same records that ioevents.schedule creates automatically. Administrators may find it necessary to schedule updates at another time if a record to be processed is locked and processing is delayed. You can use the scheduleAction(record,action) JavaScript function to create these schedule records manually.

Advanced Interoperability tailoring

Application developers can add scripts for new actions or tailor out-of-box scripts to run differently. In all custom applications, ensure that you refer to the ioaction table

record as vars.$L_file.

Contents

- 1 Interoperability
  - 1.1 Creating schedule records manually
  - 1.2 Advanced Interoperability tailoring

1.6.1.5. Change Management relationship

Problem Management and Change Management are also related. Change Management is the tool that tracks and implements a Request for Change (RFC), which permanently changes the infrastructure and prevents future incidents.

When the RFC is complete, the Problem Management process reviews the change before the known error closes.

Related topics

Problem Management overview
Problem Management implementation
Proactive and reactive Problem Management
Incident Management relationship
Priority, impact, and urgency

1.6.1.6. Problem Management users

Each user must have an operator record that contains personal information. User information includes name, address, phone numbers, login name, and password for Service Manager. Service Manager operator records also store capability words for each user. Without an operator record, a user can't log onto Service Manager.

Related topics

Alerts
Views and favorites
Problem Management assignment groups
Problem Management Macro List Editor
Problem Management tables
Problem Management link records
Problem Management and Service Level Management
Add an operator record with the User Quick Add utility
Update an operator record

1.6.1.7. Problem Management phases

Problem Management phases are the activities in the life cycle of a problem. The out-of-box Information Technology Infrastructure Library (ITIL) workflow category matches the best practice recommendations of ITIL. The following table shows the ITIL process and the out-of-box Service Manager process.

ITIL activity	**ITIL workflow**	**Service Manager phases**
Problem Control	Problem Identification	Logging
	Problem Classification	Categorization
	Problem Investigation	Investigation
	Problem Resolution	Resolution, Review
Error Control	Error Identification	Logging
	Error Assessment	
	Error Resolution	
	Close Error	Closure

1.6.1.7.1. Problem Control overview

The Problem Control objective is to return the user to full functionality as quickly as possible. Problem Control identifies the cause of the problem and delivers a solution or a workaround back to the service desk managing the incident.

Related topics

Problem Control activities
Tracking and monitoring problems

1.6.1.7.1.1. Problem Control activities

Service Manager has an action for each Problem Control activity defined in ITIL.

Problem Control activity

ITIL workflow	Service Manager phases	Possible Service Manager actions
Problem Identification	Logging	Click **Problem Management** > **Create New Problem** or Click **Problem Management** > **Search Problems**
Problem Classification	Categorization	Click **Problem Management** > **Search Problems**
Problem Investigation	Investigation	Click **Problem Management** > **Search Problems** You can: Select a problem record that's in the Investigation phase. Open the **Tasks** tab and click **Link New Task** to create a problem task record.
Problem Resolution	Resolution, Review	Click **Problem Management** > **Search Problems**

Related topics

Problem Control overview
Tracking and monitoring problems
Problem Management phases

1.6.1.7.1.2. Tracking and monitoring problems

Service Manager can track individual problems and their associated activities. A complex problem can generate several tasks, potential workarounds, and can require updates to existing incidents. Problem Management tracks every aspect of a problem during these problem Control activities and can produce charts and dashboards that visually describe the status of the problem.

Service Manager applications also have an integrated alert and notification system that keeps the primary reporting resources informed about the problem. You can specify when Service Manager should send these notifications and who should receive them.

Related topics

Problem Control overview
Problem Control activities
Problem Management phases

1.6.1.7.1.3. Phase 1: Logging

Input to the problem identification activity comes from two sources, incident Management and proactive Problem Management. When an incident doesn't match existing problems or known errors, incident Management passes the information to problem Control. Opening a new problem record stores the relevant details in the rootcause table.

Creating a problem

An administrator or user creates a problem record when one or more conditions occur:

- When you analyze incident Management data, there are recurring incidents caused by a single error.
- A problem or known error record doesn't exist for the identified error.
- There are incidents that don't have problem or known error record matches.
- Independent infrastructure analysis identifies a potential problem that's not yet related to a reported incident.
- A critical incident requires an immediate solution.

The problem record captures all critical data to ensure that you can track the error to a final resolution.

Identify the problem

If you analyze incident data, you may find more than one incident that describes the same error, or incidents that don't match identified problems or known errors. For example, three separate service desk interactions in a single day report a network outage. The circumstances are similar and the affected Configuration Items are the same. The three incidents generate a problem record that describes the outage.

Ongoing analysis of the infrastructure might also identify a problem that's likely to cause errors in the future. For example, a technician finds a computer virus on the mail server, or the network administrator learns that there is a powerful new virus propagated by e-mail attachments.

When you identify the problem, there is a permanent record that contains information about the affected Configuration Item (CI) and related CIs, as well as the primary and secondary assignment groups who own the problem.

Contents

- 1 Creating a problem
  - 1.1 Identify the problem

1.6.1.7.1.4. Phase 2: Categorization

After you create the problem record, the problem classification process begins. Classification identifies relationships, urgency, assesses the impact on the customer's business service, determines priority, and assigns the problem to a specialist or support group. Problem classification is an important activity because it identifies the relationships that the problem has with other services provided, and assesses the amount of effort required to research the problem and recover from it.

Categorize the problem

Problem Management has four categorization levels that refine the problem description. Statistical analysis of problems enables you to spot trends and identify areas that need further analysis. The categories are:

- Category
- Area
- Subarea
- Problem Type

For example, there are 10 out-of-box categories for Configuration Items. If the Category is `problem`, the Area is `failure`, and the Subarea is `job failed`, there are seven different problem Types that you can choose. This refinement provides data for trend analysis.

Classify the problem

Classification determines how much effort is necessary to restore the Configuration Items to service and return the user to full functionality. Classification is a subjective assessment based on impact, urgency, and priority. Each has a numeric assignment.

Classification	Description	Possible values
Impact	The potential business vulnerability. There is no global value; each business must set its own impact criteria.	1 – Enterprise 2 – Site/Dept 3 – Multiple Users 4 – User
Urgency	How soon the problem must be resolved to avoid business consequences. It identifies the amount of time that you have to avert or reduce the impact of the problem on customers.	1 – Critical 2 – High 3 – Average 4 – Low
Priority	How an individual problem fits into the ongoing sequence of problem resolution. Indicates when a problem can be addressed. Determining the priority of a single problem depends on how many problems need attention, the risk of delay, and the resources to fix the problem. Summarizes the assessment of urgency plus impact on customers.	1 – Critical 2 – High 3 – Average 4 – Low

Problem Management calculates priority to be the average of Impact plus Urgency. Fractional results round to the lower number. For example, if the Impact is 4 and the Urgency is 3, the Priority is 3.

Problem record information

Problem records are usually linked to incident records and contain some of the same information. As the problem moves through successive phases, you should add information about the solution, workaround, and link to the known error record. There are two ways to create a problem record:

- Open a related problem record from an existing incident record. The required information from the incident record automatically populates the problem record.
- Open a problem record directly. You must provide all required information.

The problem record includes the following information.

Field	Required?	Description
Title	Yes	Title of the problem.
Description	Yes	Summary of the problem.
Affected Service	Yes	The service that's affected by the problem.
Status	No	Categorize, Assigned, Work in Progress, or Closed.
Problem ID	Yes	System-generated problem record number with a PM prefix.
Phase	No	The current workflow phase of the problem.
Target Resolution Date	No	Date promised to the customer. All subordinate tasks to resolve the problem must have due dates that don't exceed this date.
Primary CI	No	Click **Fill** to select the primary hardware or software resource affected.

Field	Required?	Description
Affected CI Count	No	A system-generated count of the number of CIs affected by the outage. The count doesn't include the Primary CI. Affected CI count is based on the number of items entered in the Assessment section. It is calculated based on what is in the Assessment section in the Affected CIs table. The information in this field is read-only.
Category	Yes	This field is prepopulated with the value "problem".
Area	Yes	This field is prepopulated with data from an escalated incident. Service Manager displays different lists of areas depending on the category you selected.
Subarea	Yes	This field is prepopulated with data from an escalated incident. Service Manager displays different lists of subareas depending on the area you selected.

Field	Required?	Description
Problem Source	No	The source of the problem. Click the drop-down list to choose one of these values: 1 – User 2 – Event 3 – Group 4 – Incident
Contact Information	No	If the problem is created from an interaction, this field is populated with the value in the corresponding field in the interaction record.
Related incident Count	No	Problem Management calculates the number of related incidents. The information in this field is read-only.

Field	Required?	Description
Impact	Yes	A numeric value that measures the potential business vulnerability of the customer. It may reflect the effect on related agreements or expected service levels. Click the drop-down list to choose one of these values: 1 – Enterprise 2 – Site/Dept 3 – Multiple Users 4 – User
Urgency	No	A numeric value that measures how soon the problem must be resolved to avoid business consequences. Click the drop-down list to choose one of these values: 1 – Critical 2 – High 3 – Average 4 – Low

Field	Required?	Description
Priority	No	How soon the problem needs resolution, based on impact and urgency. The values are: 1 – Critical 2 – High 3 – Average 4 – Low The information in this field is read-only. Problem Management calculates Priority based on the Impact and Urgency values that you choose.
Root Cause Description	Yes	Detailed description of the problem.
Assignment Group	Yes	The group tasked with resolving the problem.
Problem Coordinator	Yes	Click **Fill** to select a problem coordinator.
Major Problem	No	When checked, indicates that the problem is a major problem, and a **Problem Manager** field appears. In addition, the **Major Problem Review** tab appears in the problem record, and the Review phase becomes mandatory.

Field	Required?	Description
Attachments section	No	Click **Add File...** to attach a document to the problem.
Related Records section	No	Service Manager automatically populates the table entry with related record information.
Workflow section	No	Graphic representation of the Problem Management workflow for the current record.

Note

To open a problem record, your operator record security profile must include the Open privilege. The SysAdmin capability word includes this privilege.

Searching for a Configuration Item

You can specify a Configuration Item (CI) in the problem record. Problem Management transfers the primary CI information in the problem record to the known error record. If the CI that caused the issue is an associated CI, you must make it the primary CI in the problem record, or edit the CI information in the known error record.

When you search for a CI from a problem form, There are three checkboxes that add criteria to the query: CI Down, Critical CI, and Pending change. Service Manager logical fields (checkboxes) can have four values: true, false, null, and unknown. If you don't select a checkbox, it has a null value and any query ignores it. If you select the checkbox, it has a value of true. If you clear the checkbox, it has a value of false.

When you search for a CI, ensure that you consider these possible checkbox values when you construct your query.

Problems with multiple Configuration Items

If a problem affects more than one Configuration Item (CI), you must repeat the steps in the Investigation phase for each CI. Problem Management enables you to open a separate task for each CI and assign an owner to complete the Investigation activities for the CI. As you open each task, Problem Management automatically links the individual task record to the main problem record.

Contents

- 1 Categorize the problem
- 2 Classify the problem
- 3 Problem record information
- 4 Searching for a Configuration Item
- 5 Problems with multiple Configuration Items

1.6.1.7.1.5. Advancing a problem to the next phase

When you complete any phase within Problem Control (or Error Control), you can move the problem to the next logical phase within the category. If you follow the out-of-box workflow, you can close a problem only when it is in the last phase of problem Control. If you have a customized problem Control and Error Control progression, the best practice is to complete each phase before you advance to the next one.

To advance a problem to the next phase, you must change its status and complete all of the fields that are required for that status.

When you complete the last phase in Problem Control, you must open a Known Error record to continue advancing through Error Control.

1.6.1.7.1.6. Phase 3: Investigation

When the specialist or support group formally accepts responsibility for the problem, the problem investigation process begins. Problem investigation examines the symptoms of the problem to identify the root cause. A correct analysis of the root cause enables you to identify the solution correctly during the Error Control process. It may be necessary to offer a workaround to restore full or partial service to the customer while you wait for a permanent solution.

Documenting the root cause

The root cause is the reason why the problem exists. Identifying and documenting the root cause leads to a workaround and known error status. Identifying the root cause also formally stores this information in the database for future reference. The following table shows a sample problem and different root causes that affect the workaround and final solution.

Problem	Root Cause
Service Manager upgrade fails	User error. The user did not uninstall the previous version first.
Service Manager upgrade fails	Training required. The user did not know that the client version must be the same as the server version.
Service Manager upgrade fails	Configuration Item error. The operating system is unsupported.

Contents

- 1 Documenting the root cause

1.6.1.7.1.7. Phase 4: Resolution

Problem resolution and closure is the final activity in Problem Control. The objective is to test the root cause candidate. Then you can reclassify the problem as a known error. You might also open a Request for Change (RFC) during this activity.

Due dates

Problem Management uses date fields to set expectations for problem resolution.

Type of record	Date field name	Comments
Problem	Expected Resolution Date	You must specify a future date.
Task	Due Date	This date must occur on or before the expected resolution date for the problem unless the problem resolution date has already expired. The task owner can't edit this date and time after initial entry.
	Expected Resolution Date	Read-only. This field has a value only if the parent problem record has an expected resolution date specified before you create the task. You can't adjust the expected resolution date of the problem to fit the task.
Known error	Expected Resolution Date	This time and date aren't dependent on the expected resolution date of the problem record.

Notes:

1. The default time (00:00:00) is midnight of the date specified. You may want to edit it to fall within daytime business hours.
2. Problem Management allows you to open a task for a problem record with an expired resolution date when necessary. The notification system continues to notify you that the resolution date of the problem with a status of past due is expired. You must close the new task before you can close the problem record with the expired resolution date.

Contents

- 1 Due dates

1.6.1.7.2. Error Control overview

The Error Control objective is to effect permanent changes to Configuration Items (CIs) with known errors. Making permanent changes reduces the number of current and prospective incidents. Integrated Service Manager Error Control and Change Management processes enable you to track the entire workflow and update all affected records automatically.

Error Control activities

Service Manager has an appropriate action for each Error Control activity.

ITIL activity	ITIL workflow	Service Managerphases	Possible Service Manager actions
Error Control	Error Identification	Logging	Click **Problem Management** > **Search Known Errors**. For a known error record that's in the Logging and phase, you can: • Elaborate the description of the underlying cause and possible workaround (if identified). • Define a temporary fix or permanent solution for the known error. Different solution alternatives can be evaluated until a definitive solution can be proposed to the Problem Manager.
	Error Assessment		

ITIL activity	ITIL workflow	Service Managerphases	Possible Service Manager actions
	Error Resolution		• Create a Request for Change record. To do so, select the known error record, click **More** or the More Actions icon and choose **Open Related Change**.
	Close Error	Closure	Click **Problem Management** > **Search Known Errors**. You can select a known error record in the Closure phase to record the resolution and close the known error.

Tracking and monitoring errors

The team that identified the root cause is not necessarily the team who will locate and apply the resolution. The first stage of Error Control is important for a smooth hand off of the error. The team that tracks the error keeps sources on the primary reporting matrix updated as necessary.

The steps in tracking and monitoring errors are:

- Obtain the error status.

- Reassess the error status.
- Update primary reporting matrix about the error status.
- Track any relevant data and ensure that the problem, known error, incident, and change records are all updated.

Service Manager posts updates to affected records automatically.

Phase 1: Logging

This phase is critical because it is where technical experts study the error to understand it and develop a solution.

Although a known error is the usual result of completing the last phase of Problem Control, an internal organization, such as Development or Customer Support can also identify known errors. For example, Development might notice a known error when performing current or new development activities such as upgrades, enhancements, or technology acquisitions. In this case, Development would Create a new problem record and verify the error during the Error assessment phase.

A permanent solution may require a change to the affected Configuration Item (CI). If this is the case, the error assessment team can open a Request for Change (RFC). Service Manager enables you to open an RFC from the known error record. It also updates all affected records automatically.

Known errors

Known errors are related records that link to the parent problem record, or to change records that also relate to the parent problem record. Problem records reside in the rootcause table; known error records reside in the knownerror and rootcause tables.

Creating a known error record has certain restrictions:

- To open a known error from a problem record, a Problem Management administrator must enable the option in *both* the phase record and the user profile record.
- You can't open a task record from a known error record.
- The problem record must be in the Investigation phase if you follow the out-of-box ITIL workflow.
- If you open a known error record from a problem record, Problem Management copies the Description field from the problem record.

Error identification

The first phase of Error Control captures all information about the known error. A known error is the usual result of completing the last phase of problem Control.

An internal organization, such as Development or Customer Support may identify known errors, but the primary record must be the problem record, which becomes the parent of the known error record. For example, Development might notice a known error when performing current or new development activities such as upgrades, enhancements, or technology acquisitions. The first step is to open the problem record, then step through the problem Control phases before opening an associated known error record. The known error is verified during the Error identification phase.

When a known error evolves from problem Control phases, the known error has an identified root cause and a workaround, or an explanation of why root cause and workaround information doesn't apply.

Change requests

Error control often initiates a Request for Change (RFC) to implement a solution. The solution may require a technical change, revised process, training, or other organizational change. When you create an RFC, it becomes the responsibility of change Management. The priority of the RFC depends on the impact of the problem on business activities.

As long as Known error resolution is pending, the status of the RFC must be reported back to Problem Management and recorded in the known error record. The known error can't close until the RFC closes.

Phase 2: Closure

Closing the record is the last Error Control activity and the last Problem Management activity. Do not close the record until you ensure that the error is resolved, and the Problem Management process functioned correctly.

You can't close a known error record if there are any related Request for Change (RFC) records open.

Error resolution

The input to Error resolution is from the Error assessment phase, which produces a solution. This phase applies the solution to the known error through a Request for Change (RFC), submitted to change Management. The Change Management process applies the solution to the Configuration Item (CI) to eliminate the known error.

You must verify that the Request for Change (RFC) is complete, removes the error in the infrastructure, and doesn't introduce a new error before you can close the known error record.

When does a known error close?

Closing the known error record depends on these conditions:

- All related Request for Change (RFC) records must be closed.
- The Known Error Details section must have information about a Solution, Root Cause, and Workaround before you can close the known error record.
- The record must be in the Error Closure phase.

Contents

- 1 Error Control activities
- 2 Tracking and monitoring errors
- 3 Phase 1: Logging
  - 3.1 Known errors
    - 3.1.1 Error identification
    - 3.1.2 Change requests
- 4 Phase 2: Closure
  - 4.1 Error resolution
  - 4.2 When does a known error close?

1.6.1.7.3. Using Problem Management

Users and administrators must know how to complete certain tasks that are associated with Problem Management. For example, you need to be familiar with the Knowledge Base, be able to find problem and known error records, and create reminders. For more information about working in Problem Management, see the related topics.

1.6.1.7.4. Problem Management searches

Problem Management enables you to search for problem records and display them in a list. You can also save these queries as views. The search form fields enable you to narrow the search parameters.

There are three types of searches available:

- Use a **Basic Search** to look for records using key search criteria.
- Use an **Advanced Search** to find records opened, updated, or closed within a certain time frame.
- Use an **IR Search** to locate specific text within records.

1.6.1.8. Problem Management assignment groups

When you Create a new problem, the assignment group that you specify is a list of users who are responsible for a problem.

Administrators define the available assignment groups for problems, tasks, and known errors. It is important to make the same assignment groups available for problems, tasks, and errors. If an assignment group is available for a problem, but not for a task or known error, Service Manager generates an error message when you create a new task or known error record. The user must choose an available assignment group to continue.

To define Problem Management assignment groups, add a list of available groups to the Problem Management Security Role. If you don't define Problem Management assignment groups, Problem Management makes all assignment groups available when you create problem, task, or known error records.

Related topics

Alerts
Views and favorites
Problem Management users
Problem Management Macro List Editor
Problem Management tables
Problem Management link records
Problem Management and Service Level Management

1.6.1.9. Problem Management Macro List Editor

Macros are actions, driven by predefined conditions, that execute when Problem Management saves a record in the database. Macro actions are associated with files and reflect certain states in the records of those files.

You can create, edit, and delete macros from the Tailoring tools.

1.6.1.10. Problem Management link records

Problem Management uses the following link records to link data:

- PMscrelate.problem.task
- rootcause
- rootcausetask
- screlate.rootcause
- screlate.rootcausetask

Related topics

Alerts
Views and favorites
Problem Management users
Problem Management assignment groups
Problem Management Macro List Editor
Problem Management and Service Level Management

1.6.1.11. Problem Management tables

Problem Management uses the following tables.

Record type	Table	Record prefix
Problem Management record	rootcause	PM
Problem Management task record	rootcausetask	PM (Task records append a second sequential number to the parent task number)
Problem Control category record	rootcausecat	none
Problem Control phase record	rootcausephase	none
Problem Control task category record	rootcausetaskcat	none
Known error record	rootcause	KE
Error Control category record	rootcause	none
Error Control phase record	rootcause	none
Incident record	probsummary	IM
Request for Change (RFC) record	cm3r	C

Click **System Definition** > **Tables** > *anytablename* to see associated fields, keys, links, forms, triggers, records.

Related topics

Alerts
Views and favorites
Problem Management users
Problem Management assignment groups
Problem Management Macro List Editor
Problem Management and Service Level Management

1.6.1.12. Problem Management and Service Level Management

Problem Management supports the selection of more than one applicable SLA for a problem record. When you open a problem, you can choose a Customer SLA for the contact, a Customer SLA for the contact and one or more applicable Service SLAs for the contact's subscriptions to a service, or no SLAs at all. Service SLAs only apply if the problem references a Business Service, the contact has a subscription to the service, and the subscription references an SLA. The following describes the system's process for adding SLAs to a problem record.

- If one SLA is associated with the problem based on the contact, the Customer SLA is added to the problem.
- If the contact has an Individual Subscription for the CI, the Service SLA from that subscription is added to the problem.
- If the contact has a Department Subscription for the CI, the Service SLA from that subscription is added to the problem.
- If the contact has neither, then no Service SLA is added to the problem.

The SLAs should contain all Service Level Targets (SLTs) that define the business rules for all process and service metrics. You can choose as many SLTs as necessary to describe your process and service commitment. If necessary, you can add more SLTs that meet your criteria.

When you view the new record, the SLA section lists the Process Time Objectives that apply to the problem.

See the related topics to view the definitions for Customer SLA and Service SLA.

Related topics

Alerts
Views and favorites
Problem Management users
Problem Management assignment groups
Problem Management Macro List Editor
Problem Management tables
Problem Management link records

1.6.1.13. Escalation and Notification

Service Manager can escalate Problems based on thresholds defined in the Problem Management module or thresholds associated with SLAs, which can be applied to Problem records. Alerting, escalation, reporting, and breach tracking are part of the integration between the Problem Management and Service Level Management modules.

1.6.1.14. Historical problem records

You can search for and access historical Problem and Known Error records through the Problem Management module during the investigation process.

1.6.1.15. Incident trending for problem identification

Service Managerfacilitates incident trending for pro-active problem identification. Problem Managers can use charts and dashboards to view historical interactions or incident data.

The charts and dashboards allow sorting by certain incident characteristics which demonstrate relatedness (such as incident rates, related CI, category/area/subarea, and affected service). Therefore, the analyst is able to study the incidents types that occur on certain IT infrastructure or services, to measure both the frequency of those incidents over time and the rate of change (increase/decrease), to pinpoint which CIs/Services experience certain types of issues more frequently, etc.

With this information, Problem Managers can analyze the patterns and trends to draw hypotheses for the incidents and targets. More targeted queries as a result of initial analysis can support Total Quality Control (TQC) methodology that helps progress the investigation from identifying symptoms to uncovering root causes to prevent future incidents.

For example, reporting on historical incident data and determining an increasing trend of downtime incidents related to a particular CI might indicate the need for maintenance or replacement of that CI. An increasing trend of degraded-response-time incidents against a given service might be traced back to a capacity inadequacy on an infrastructure component of that service.

In addition, key information such as Related Incident Counts is associated with a Problem Record. This information can be consolidated in reports showing time-line related trends.

1.6.1.16. Integration

Service ManagerProblem Management is fully integrated with other Service Managermodules such as Change Management, Incident Management, Configuration Management, and Knowledge Management.

Integration: Change Management

Service ManagerProblem Management and Change Management modules are fully integrated, so Known Error records can be linked to Change requests. When a Change request is created from a Known Error record, the link is established automatically. Links between the existing records can also be created manually. Links are referenced in each linked record.

Integration: CMDB

Each Problem Record includes links to all related/affected CIs. The user can drill down into any CI record to view its CMDB definition (as appropriate for the user based on user roles and security restrictions). Important information such as CI relationship and impact analysis is greatly aided by this integration.

Integration: Configuration Management

Each Problem record references the primary CI that's the subject of the Problem record, as well as other affected CI records.

Authorized users can access CI records from the Problem record to gather additional information on the subject CIs. Authorized users can also directly access the Configuration Management module (the Service Manager CMDB) to gather Problem Management related information through direct record access or queries.

Integration: Incident Management

Service Managerprovides separate modules for creating and tracking Incident records and Problem records, but you can link the records from these modules. You can open Problem records from Incident records (and vice versa), and Service Manager automatically copies all relevant information from the source record to the target record and links the records. You can also link the records manually, or link multiple records from one module to multiple records from the other module. Links between records are referenced in linked record.

Integration: Knowledge Management

Problem information can be promoted to Service Manager Knowledge Management module; when you search for potential resolutions in Knowledge Management, you can find applicable workarounds from Problem records.

Service Manager Knowledge Management module is a fully integrated solution for knowledge-management support. It supports Knowledge-Centered Support (KCS) standards and guidelines by providing a natural language search engine and rich-text authoring tools that enable you to search, update, and author knowledge articles.

Knowledge Management integrates with Interaction, Incident, and Problem records so that you are able to search for and use knowledge from the existing Incidents or Problems while attempting to resolve a new Incident or Problem, or to create new knowledge. You can also use the rich-text editor to include image files and document files of various types as attachments that can be linked to other documents or included as part of an existing document.

Contents

- 1 Integration: Change Management
- 2 Integration: CMDB
- 3 Integration: Configuration Management
- 4 Integration: Incident Management
- 5 Integration: Knowledge Management

1.6.1.17. Known Error

There are two main phases to the Problem Management process: Problem Control and Error Control. In the first phase, the Problem Manager or technician assigned to the process studies infrastructure trends and analyzes services and CIs to determine possible failure points. In the second phase, the analyst documents the findings related to an identified problem, and initiates a corrective course of action that may include: documenting a workaround; recording a Known Error condition pending future action; or creating a Request for Change. The “recording of a Known Error condition pending future action” could be the result of a defect that requires resolution from a third party such as a software patch or firmware upgrade.

Problems and Known Errors can be opened against multiple environments, including development, testing, and production environments, by using Category/Area/Sub-Area and/or CI names that identify the environment following the same general process.

Known Errors are visible in linked Incident records. Similar Known Errors can be searched for during Incident creation. Known Errors are also visible in related Change records. Known Errors are indexed by the search engine and presented in knowledge search results.

Known Error found in the development environment

Known Error records can be opened directly to show the known errors found in the development environment. This is a useful activity as Known Error records are automatically used to support other processes including Incident and Knowledge Management. Automatic searches and matching of problem descriptions in an Incident creation or knowledge search can highlight Known Error records for the user.

Contents

- 1 Known Error found in the development environment

1.6.1.18. Problem record data model

Problem Management enables you to track different types of information related to a problem, such as categorization, prioritization, description, contact information, and so on.

1.6.1.18.1. Assignment

Service Manager Problem Management routes and assigns records to predefined users and user groups based on Category of the problem, or the skill set, location, and availability (these routing rules require configuration). Problem records can be assigned to individuals or groups. The status or Problem records can also be viewed through service queues and the reporting applications.

Related topics

Assign a Problem Management task
Schedule a problem
Reassign a problem for additional support

1.6.1.18.2. Categorization

Each problem record includes tiered categorization that identifies the problem and can associate it with a specific service, as well as with a category/area/sub-area.

In addition, each Problem is associated with one or more affected CIs that are linked via their own relationship tracking to one or more affected Services. The user can visualize the list of impacted CIs and Services as they relate to the Problem.

1.6.1.18.3. Closure

Problem closure activities trigger the prompting of a closure code that identifies the type of activity category associated with the ability to close the Problem Record.

The possible values of Cause Code and Resolution are:

- Automatically Closed
- Not Reproducible
- Out of Scope
- Request Rejected
- Solved by Change/Service Request
- Solved by User Instruction
- Solved by Workaround
- Unable to solve
- Withdrawn by User

1.6.1.18.4. Contact information

The contact for the problem, the Problem Coordinator, is a field in the Problem record with a link to the Contact record. The Contact record for that Problem Coordinator contains all his/her contact information, including name and preferred method of contact.

Click on the magnifying glass icon next to the **Problem Coordinator** field to execute a Find action, which links to the Contact record. The path of the linkage is from Problem-Coordinator to the Operator table, which then links to the Contact table.

Related topics

Types of links
Find functionality
Operator records
Synchronization between contact and operator records
Add an operator record
Create a contact record from an operator record

1.6.1.18.5. Problem description

Each Problem record includes a field for providing detailed descriptive information recording the symptoms and other relevant information.

Related topics

Problem record data model

1.6.1.18.6. Problem lifecycle

The Service Manager Problem Management module implements ITIL processes for problem management. The module helps ensure that standardized methods and procedures are used for all problem investigation activities. Separate workflows are provided for the Problem and Known Error processes. By its out-of-box workflows, Problem Management enables personnel and management to:

- Create Problem and Known Error records, assigning a unique number against which all activities can be tracked and reported
- Within a Problem and Known Error record, identify the tasks required to determine the root cause of the Problem or resolve the Known Error
- Follow, track, and query Problem and Known Error records through their entire lifecycle. Workflow phases identify the logical sequence of the repeatable steps within the problem management lifecycle. There is also a visualization of the current, previous, and potential future phases of the Problem or Known Error.

1.6.1.18.7. Problem resolution

In addition to storing analysis activities in the Activity list, the Problem Record also stores specific resolution and recommendation instructions in their own fields which can be used to populate useful, related Knowledge Management database entries. The date/time stamps of the resolution activities are also stored.

Related topics

Update a problem

1.6.1.18.8. Problem status

Each Problem Record includes a Status field that's populated with an appropriate pre-defined list of lifecycle states.

Related topics

Update a problem
Close a problem
Defer a problem
Create a new problem not based on an incident

1.6.1.18.9. Source of problem records

Each Problem record stores the information about the initial originator/creator of the information. A problem will often be created from one or more related incidents. These can be seen in the related incidents section of the problem form.

When the source of the related incident(s) to a problem is a person, then basic information-such as whether the Incident was opened through the self-service interface, the details of the submitter, the Incident trigger, and location of the issue-is shown in the Interaction record linked to the Incident record.

When the source of the related incident(s) to a problem is an event trigger, then the incident record contains that information in the Title and Description fields. These fields reference the source event displayed in the External ID field of the Event Browser in Operations Manager.

Related topics

What's a Service Desk interaction?
Create a new incident from a user interaction
Create a self-service request
View related OMi event details from an incident
Relate a record to an incident record

1.6.1.19. Problem prioritization

Problem records include fields to capture impact and urgency codes. Urgency can be automatically or manually populated based on the Category of the Problem record. Impact is populated manually or could be related to other information in the problem record such as the related service or CIs. Service Manageruses these values to automatically derive the priority of the record.

Problem prioritization based on associated records

Using the related records function in Service Manager, operators may increase the severity of a Problem. Service Manager can also be configured to increase the severity or impact rating of a Problem record based on the number of linked Incident records or the number of the affected end users .

Contents

- 1 Problem prioritization based on associated records

1.6.1.20. Problem records creation

Service Manager provides various mechanisms for creating Problem records:

- Problem records can be created manually from a blank Problem screen.
- Problem records can be created manually from an incident to support a reactive problem activity.
- Problem records can be created automatically based on a configured event criteria.
- When Incident records are opened, automatic matching can also be triggered.

Problem record identifier

Service Manager tracks problems in individual Problem Record, and each is assigned with its own unique identifier. The default is to automatically assign a unique number using a format such as PM<XXX>. This helps the user not only identify the record uniquely, but also know that the record comes from Problem Management.

1.6.1.21. Problem workarounds

Problem solutions and workarounds can be communicated to the service desk staff in two ways.

- System administrators can configure a help desk operator's interface to automatically check for potentially similar records based on similar Known Errors, root causes, Incident records, Incident record duplicates on a device, or Incident record duplicates on parents.
- When a user creates an Incident record, Service Manager prompts the user with a list of similar Problem records, which the user can access for workaround information. In addition, Problem information can be promoted to the Service Manager Knowledge Management module; when users search for potential resolutions in Knowledge Management, applicable workarounds from Problem records are presented.

1.6.1.22. Logs of problem record updates

Each problem record includes the date/time/activity logging that describes any action taken on the record and the lifecycle activity that it is related to. The log is maintained automatically, and the user is prompted to enter descriptive information when taking an action so that the details of the action can be properly logged.

Problem diagnostic information

Each Problem record includes an activity audit trail that stores the date, time, and text for each root cause analysis activity. This information is shown within an “Activities” section in the Problem record.

Related topics

Update a problem
Create a new problem from an incident
Create a new problem not based on an incident

1.6.2. Problem Management user roles

The following table describes the responsibilities of the Problem Management user roles.

Problem Management user roles

Role	Responsibilities
Problem Manager	• Communicate with stakeholders if required • Inform the Change Manager if required • Defer problems if needed • Decide on investigation of problems • Register Request for Changes or Service Requests to solve problems • Validate proposed solutions to problems • Validate the outcome of closed changes and close problem • Validate that a problem is solved • Conduct problem review and document lessons learned • Close problem and inform stakeholders • Monitor the problem resolution progress and perform the required action
Problem Coordinator	• Periodically perform analysis to see if new problems need to be registered • Register problems • Categorize and prioritize problems • Assign work to the Problem Analysts • Schedule the problem resolution • Coordinate root cause analysis and diagnosis

Role	Responsibilities
Problem Analyst	• Investigate and diagnose assigned problems for workarounds and/or root causes • Review and accept or reject assigned errors problems or problem tasks • Investigate and diagnose assigned problems and propose solutions and workarounds • Identify known errors • Implement corrective actions

Related topics

User roles
Add a user role record
Delete a user role record
Search for a user role record

1.6.3. Problem Management workflows and user tasks

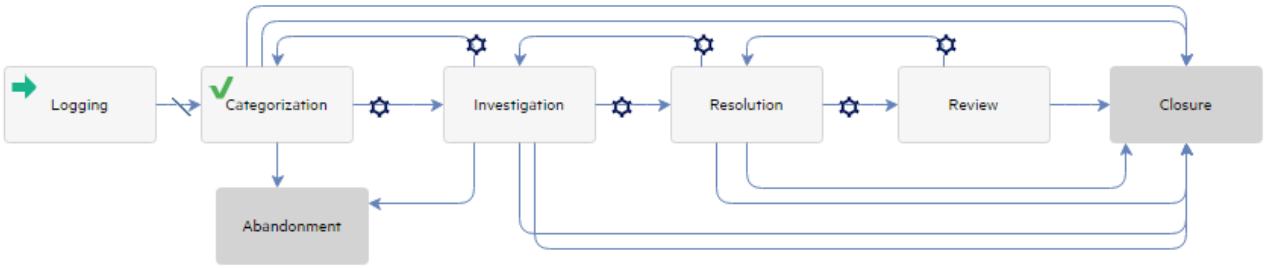
The Problem Management process includes the activities required to identify and classify problems, to diagnose the root cause of incidents, and to determine the resolution to related problems. It is responsible for ensuring that the resolution is implemented through the appropriate control processes, such as Change Management.

Problem Management includes the following out-of-box workflows:

Problem Management

The Problem Management workflow is a sequence of connected steps in the life cycle of a support or service request. In the workflow, a Problem record goes through several phases to complete the life cycle.

The Problem Management workflow consists of the following phases and tasks.



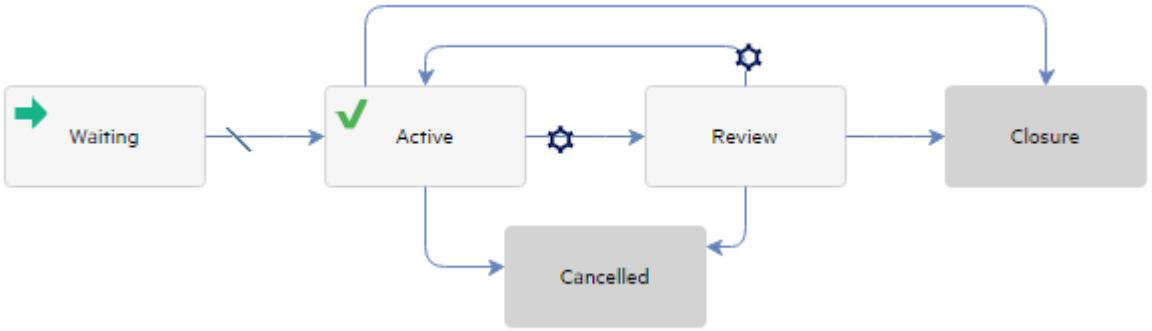
Workflow phase	Tasks
Logging	• Find problem candidates • Review incidents • Create a new problem from an incident • Create a new problem not based on an incident
Categorization	• Associate a problem with an existing incident • Update a problem • Defer a problem • Evaluate a problem • Close a problem that will not be fixed • Access the knowledgebase • Review deferred problems
Investigation	• Update a problem • Schedule a problem • Close a problem that will not be fixed • Review deferred problems • Access the knowledgebase • Document the workaround • Reassign a problem for additional support • Schedule a problem • Set a reminder for a problem record • View records related to a problem
Resolution	• Update a problem • Document the root cause

Workflow phase	Tasks
Review	• Close a problem
Closure	• None
Abandonment	• None

Problem Task

The Problem Task workflow is a sequence of connected steps in the life cycle of a support or service request. In the workflow, a Problem task goes through several phases to complete the life cycle.

The Problem Task workflow consists of the following phases and tasks.



Workflow phase	Tasks
Waiting	• Open a Problem Management task
Active	• Assign a Problem Management task • Investigate and diagnose a Problem Management task • Test the workaround
Review	• Close a problem task
Closure	• None
Cancelled	• None

Known Error

The Known Error workflow is a sequence of connected steps in the life cycle of a support or service request. In the workflow, a Known Error goes through a single phase before closure.

The Known Error workflow consists of the following phases and user tasks.

Workflow phase	Tasks
Logging	• Open a known error • Update a known error • Close a known error • Assign a known error record to schedule a fix • Associate a change request with a known error record
Closure	• None

To see more workflow diagrams and more information about these workflows, refer to *Service Manager Processes and Best Practices* linked to in the related topics.

Related topics

Problem Management overview
Processes and Best Practices

1.6.3.1. Problem Management views

Part of Workflow(s): All

The Problem Management views contained in the Favorites and Dashboards navigation pane allow you to easily and quickly access specific types of records, including known errors, problems, and problem tasks.

Note

Service Manager provides default Problem Management views. Views that are available to a specific role or roles aren'ted below.

Available views for known errors

Click **Favorites and Dashboards** > **Problem Management** > **Known Error**.

The following views are available to the Problem Analyst:

- All Open Known Errors
- Open Known Errors Assigned to Me
- Open Known Errors Assigned to My Group

The following views are available to the Problem Coordinator:

- All Open Known Errors
- Open Known Errors Assigned to My Group

The following views are available to the Problem Manager:

- All Known Errors
- All Closed Known Errors
- All Open Known Errors
- Known Errors Not Assigned in Known error investigation Phase
- Known Errors Where Solution Time Is Not Met

Available views for problems

Click **Favorites and Dashboards** > **Problem Management** > **Problem**.

The following views are available to the Problem Analyst:

- All Open Problems
- High Priority Problems

- Open Problems Assigned to My Group
- Problem Tickets Owned by Me

The following views are available to the Problem Coordinator:

- All Open Problems
- High Priority Problems
- Open Problems Assigned to Me
- Open Problems Assigned to My Group
- Problem Tickets Owned by Me

The following views are available to the Problem Manager:

- All Open Problems
- High Priority Problems
- Open Problems Assigned to Me
- Problems in Problem Prioritization and Planning Phase
- Problems Not Assigned in Problem Investigation and Diagnosis Phase
- Problem Tickets Owned by Me

Available views for problem tasks

Click **Favorites and Dashboards** > **Problem Management** > **Problem Tasks**.

The following views are available to the Problem Analyst:

- All Open Problem Tasks Assigned to Me

Related topics

Views and favorites

1.6.3.2. Find problem candidates

Part of Workflow(s): Problem Management: Logging

Applies to User roles: Problem Coordinator

There are times when you may need to create a problem before any incidents are reported. Typically, such problems arise from periodic reviews of incident trend analysis or from vendor/supplier published issues. You can also obtain lists of specific incidents to review from your stakeholders, from external reporting systems, or by searching for incidents flagged as problem candidates. You should set up periodic reviews for all areas mentioned below, and any other areas that might be problematic.

To find problem candidates, follow these steps:

1. Review known error information published by your suppliers to identify potential problems.
2. Review events for all systems to identify potential functional problems.
3. Review performance, capacity, and availability for all systems to identify potential performance and capacity issues.
4. Review data provided by security management to identify potential security problems.
5. Search for priority 1 incidents in the database. To do this, follow these steps:
    1. Open **Problem Management** > **Search Problems**.
    2. In the **Look For** list, select **Incident**.
    3. In the **Priority field** list, select **1 - Critical**.
    4. Use search or advanced search to find one or more records.Service Manager displays incidents with a priority of 1.
6. Search for problem candidates. To do this, follow these steps:
    1. Open **Problem Management** > **Search Problems**.
    2. In the **Look For** list, select **Problem**.
    3. In the **View** list, select **Open Problem Management Candidates**.
    4. Use search or advanced search to find one or more records.Service Manager displays incidents flagged as problem candidates.
7. Search for problem candidates in the incidents queue. To do this, follow these steps:
    1. Open **Problem Management** > **Search Problems**.
    2. In the **Look For** list, select **Incident**.
    3. In the **View list**, select **All Open Incidents**.
    4. Use search or advanced search to find one or more records.

5. Refine your search by adding the following kinds of incidents, or other candidates, as required.

   - Closed incidents
   - Candidates resolved through a workaround or temporary fix
   - Candidates not linked to problems or known errors
   - Suspected problems (as identified by stakeholders)

   Service Manager displays incidents that meet the search criteria.

Related topics

Review incidents
Create a new problem from an incident
Create a new problem not based on an incident
Associate a problem with an existing incident
Associate a problem with an existing problem

1.6.3.3. Review incidents

Part of Workflow(s): Problem Management: Logging

Applies to User roles: Problem Coordinator

You should periodically review incidents (whether they have been closed or are currently open) to determine if they are the result of a new problem or if they match an existing problem. Your analysis of incident data may reveal similar or reoccurring incidents, which may indicate a problem. By default, Service Manager requires that all closed incidents not resolved through a permanent fix must be related to a problem. You can match such incidents to existing problems or create a new problem record for them.

To review incidents, follow these steps:

1. Open the incident that you want to review.
2. Review the details of the incident.
3. Determine whether the incident is already related to an existing problem or known error.
4. Search for problems and known errors similar to this incident (such as same classification, same CI, or same product or model).
5. If the incident is similar to an existing problem, associate the incident to that problem. You can relate a configuration item (CI) group to the problem or create a new CI group to define the set of CIs that have this problem.

   Note

   The Incident Management staff may have already linked some incidents to existing problems.

   Service Manager automatically updates the incident count of a problem record when you associate an incident to it.
6. If you can't find a matching problem or known error, create a new problem record.

Related topics

Find problem candidates
Create a new problem from an incident
Create a new problem not based on an incident

Associate a problem with an existing incident
Update a problem

1.6.3.4. Create a new problem from an incident

Part of Workflow(s): Problem Management: Logging

Applies to User roles: Problem Coordinator

When you create a problem record from an incident, Service Manager automatically associates the incident record to the problem record. Every time someone associates another incident to this problem, Service Manager updates the incident count for the problem. The more related incidents a problem has, the greater the impact the problem has on an organization and the higher priority it should have.

Note

By default, problem record identification numbers have a prefix of PM.

To create a new problem record based on an existing incident record, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records. Specify search criteria to find the incidents that you want to make new problems for.
3. Select an incident from the record list.
4. In the **Related Records** tab, select **Related Problems** from the **Link Type** drop-down list, and then click **Link New Record**.
5. Fill out the remainder of the problem record as needed.
6. Click **Save**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

1.6.3.5. Create a new problem not based on an incident

Part of Workflow(s): Problem Management: Logging

Applies to User roles: Problem Coordinator

There are times when you may need to create a problem before any incidents have been reported. Typically, such problems arise from periodic reviews of incident trend analysis or from vendor/supplier published issues. You can address these problems as part of an organization's preventive or scheduled maintenance. Should any incidents arise from these problems, you can associate them with the problem after creating it.

Note

By default, problem record identification numbers have a prefix of PM.

To create a new problem record not based on an existing incident record, follow these steps:

1. Click **Problem Management** > **Create New Problem**.
2. In the **Title** field, type a title that accurately describes the problem.
3. In the **Description** field, type the following details:
    - The problem
    - The symptoms, including any error messages
    - The impact from a business perspective
    - The frequency of reoccurrence
    - The conditions under which any service disruption occurs
4. Fill in the following required fields, and any other required or relevant fields.

Note

Problems created from Incidents or other records may contain some prepopulated fields.

- Select the **Affected Service**.
- Select the **Status**.
- Select the **Impact** and **Urgency**.

> **Note**
>
> Service Manager selects the default priority based on the impact and urgency values.

5. Click **Save**.

> **Note**
>
> You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

1.6.3.6. Associate a problem with an existing incident

Part of Workflow(s): Problem Management: Categorization

Applies to User roles: Problem Coordinator

Any time that you discover that an outstanding problem causes an incident, you should link the incident to the problem record. It's important to link incidents to problems so that you can decide the impact of the problem in your organization. The more related incidents a problem has, the greater the impact of the problem. Monitoring the number of related incidents will also help you identify problems that require additional work to resolve.

To associate a problem with an existing incident, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select an existing problem from the record list.
4. In the **Related Records** tab, select **Related Incidents** from the **Link Type** drop-down list, and then click **Link Existing Record**.
5. Specify the incident number or click **Search** to view the incident search form.
6. Add optional filtering criteria, and then click **Search** again to display an incident record list.
7. Double-click the incident record you want to associate with the problem record. The system displays the incident record ID.
   - Click **OK** to link the incident record to the problem record.
   - Click **Cancel** to stop the search and exit.
   - Click **Search** to view the incident record search form again. Click **Search** again to display the incident record list a second time.
8. Click **OK**. Problem Management populates the Related Records section with information about the related incident.

Related topics

Searching records
Find problem candidates

Review incidents
Create a new problem from an incident
Create a new problem not based on an incident
Update a problem

1.6.3.7. Associate a problem with an existing problem

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Analyst, Problem Coordinator, Problem Manager

If you find that an existing problem record caused a problem, you should link the problem with an existing problem record. It's important to link the problem, so you can analyze the impact of the problem on your organization.

To associate a problem with an existing problem record, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select an existing problem from the record list.
4. In the **Related Records** tab, select **Related Problems** from the **Link Type** drop-down list, and then click **Link Existing Record**.
5. Specify the problem number or click **Search** to view the incident search form.
6. Add optional filtering criteria, and then click **Search** again to display a problem record list.
7. Double-click the problem record you want to associate with the current problem. Service Manager displays the problem record ID.
   - Click **OK** to link the problem record to the current problem.
   - Click **Cancel** to stop the search and exit.
   - Click **Search** to view the problem record search form again. Click **Search** again to display the problem record list again.
8. Click **OK**. Problem Management populates the Related Records section with information about the related problem.

Related topics

Associate a problem with an existing incident
Create a new problem from a problem record

1.6.3.8. View records related to a problem

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Analyst

Problem Coordinator

Problem Manager

Part of the review process for monitoring problems includes reviewing related records. When you review records related to a problem, you can see how many incidents and other record types are affected by the reported problem. As the Problem Manager reviews and monitors problems and known errors, this is an opportunity to see if any trends are developing, and to resolve any issues quickly.

To view a related record, follow these steps:

1. Click **Problem Management** > **Search problems**.
2. Use search or advanced search to find one or more records.
3. Select a problem from the record list.
4. In the **Related Records** tab, click on the record number of the related record that you want to view. The related record opens.
5. Click **Cancel** to return to the problem record.
6. Click **Save & Exit**.

1.6.3.9. Create a new problem from a problem record

Part of Workflow(s): Problem Management: All

Applies to User roles: Problem Coordinator, Problem Manager

There are times when you may need to open a new problem from an existing problem record. Whether a periodic review of incident trend analysis or as part of an organization's preventive or scheduled maintenance, you can open a related problem record that will be associated with the existing problem record.

To open a new problem from an existing problem record, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select a target problem record.
4. In the **Related Records** tab, select **Related Problems** from the **Link Type** drop-down list, and then click **Link New Record**. A New Problem form opens.
5. Add the required information to create the new problem record.
6. Click **Save & Exit**. You are returned to the existing problem record.
7. In the **Activities** tab of the original problem record, document the current activities. In the **New Update Type** field, select the applicable update type.
8. In the **New Update** field, type notes to explain the new open problem and any other current activity information.
9. Click **Save & Exit**.

Related topics

Associate a problem with an existing incident
Associate a problem with an existing problem

1.6.3.10. Create a new problem from a user interaction

Part of Workflow(s): Problem Management: Logging

Applies to User Roles: Service Desk Agent

You can create a problem from a user interaction. You can also update a problem from a user interaction that has already been triggered to a problem.

Create a problem from a user interaction

To create a problem from a user interaction, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. Specify the **Contact** for the interaction.
3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. Type a description for the interaction.
6. In the **Category** field, select the **problem** category.
7. (Optional) In the **Subcategory** and **Area** fields, select the subcategory and area for the interaction.
8. In the **Affected Service** field, select the affected service.
9. Click the **Continue** button. The New Problem form is displayed, with fields populated with the values from the interaction.
10. Proceed with opening a new problem by updating fields such as **Affected Service**, and then click **Save**.
11. Search possible solutions in the form that appears. If matching solutions are found, select the corresponding record, and then click **Link Selected Record**. If you don't want to link a problem to the new problem or if no matching problems are found, click **Continue**.
12. Complete the problem form with any other relevant information.
13. Click **Save & Exit**.

Update the new problem

To update the new problem from the related user interaction, follow these steps:

1. Click **Service Desk** > **Interaction Queue**. Service Manager opens the Interaction queue.

2. Click the interaction in the queue to display the details of the record.
   Those items with a status of **Dispatched** are user interactions that have been triggered to Problem Management (or another module).
3. On the record details page, click the **Related Records** tab, and then click the related problem that you want to update.
4. Proceed with updating the problem. After closing the problem form, you are returned to the user interaction list.

Contents

- 1 Create a problem from a user interaction
- 2 Update the new problem

1.6.3.11. Update a problem

Part of Workflow(s): Problem Management: Categorization

Problem Management: Investigation

Problem Management: Resolution

Problem Management: Review

Applies to User roles: Problem Analyst

Problem Coordinator

Problem Manager

You may need to update problem records many times. Typically, you add relevant information to the Activities tab to describe the work you performed to resolve the problem. The Activities tab provides you with information about the work other operators have performed to resolve the problem. You can use the Activity tab to view the history of other activities, such as when the record opened or when the status of the record changed.

You should record all the investigation activities in the known error to track the effort associated with the problem, task, or known error. After the current activities for a phase are complete, you advance the known error record to the next phase.

To update a problem, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select an existing problem from the record list.
4. Click **Refresh** to synchronize the data.
5. To document test results provided by the Problem Analyst, follow these steps:
    1. In the **Investigation and Resolution** tab, enter a detailed update in **Suggested Workaround** field. Include the necessary actions to implement the solution and any possible risks.
    2. Enter the following information to document the estimated costs and resources:
        - The estimated number of days to apply the fix
        - The estimated costs

3. In the **Affected Configuration Items** tab, click the **Fill** icon to select any affected CIs.
4. If the solution needs to be implemented through the Change Management process, open a change request.

6. In the **Activities** tab, select the applicable update type in the **New Update Type** drop-down list.
7. In the **New Update** field, type the description of the update.
8. Consider whether the priority of resolving this problem has changed. If you need to update the priority of this record, select the **Categorization** tab, and then follow these steps:
   - Update the **Impact**, if necessary.
   - Update the **Urgency**, if necessary.
9. Consider whether the problem needs further investigation. If so, update the **Status** field to **Deferred**.
10. Click **Save & Exit**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Note

- If you update the workaround before the Problem Analyst is finished diagnosing and testing, make sure you notify the Problem Analyst.
- If any outstanding incidents matched the problem, notify the Incident Analyst.

1.6.3.12. Evaluate a problem

Part of Workflow(s): Problem Management: Categorization

Problem Management: Investigation

Applies to User roles: Problem Analyst

Problem Coordinator

The goal of evaluating a problem is to determine what priority resolving the problem has in your organization. A problem's priority is a combination of the problem's impact, urgency, severity, frequency, and risk. For example, the frequency of reoccurring incidents may influence the urgency to resolve the problem. Furthermore, you may need to perform a risk assessment to determine the problem's impact on the business, such as whether the problem affects service availability or customer satisfaction. Due to resource constraints, it is important to focus on those problems that have the highest impact on the business.

To evaluate a problem, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Discuss the problem with stakeholders (for example, during a meeting) and answer the following questions:
    - What's the priority of resolving this problem?
        - Update the **Impact**, if necessary.
        - Update the **Urgency**, if necessary.
    - Is this problem correctly documented? If the problem is incorrectly documented, update the problem documentation.
    - Does the problem need further investigation?
        - If the problem is still relevant, update the Status field to **Deferred** and schedule it for investigation at a later date.
        - If the problem is no longer relevant, close the problem record.
4. In the Activities tab, select an update type in the **New Update Type** field.
5. In the **New Update** field, type a description of your update.
6. Click **Save & Exit**.
7. If you have determined that you can close the problem record, follow these steps:

- Make sure all known errors are closed. For more information, see the related topics.
- Complete the necessary actions until the problem record progresses to the Review phase.
- Click **Close** in the taskbar, confirm that the subcategory and area are correct, and then select an applicable closure code.
- Click **Finish** to close the problem.

1.6.3.13. Schedule a problem

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Coordinator

Problem Manager

When you schedule a problem, set the target dates based on the priority and impact on affected services and whether there is a workaround or fix available. Assign or reassign the problem to a group with the skills necessary to find a resolution to the problem.

To schedule a problem:

1. Click **Problem Management** > **Search problems**.
2. Use search or advanced search to find one or more records.
3. Select a record from the list.
4. In the **Investigation and Resolution** tab, enter values in the **Expected Root Cause Identified Date** and **Expected Problem Resolution Date** fields.

   Note

   The Investigation and Resolution tab is only available once the problem moves to the Investigation phase.

5. In the Activities section, click the **New Update Type** field list and select the applicable update type.
6. In the **New Update** field, type notes to include the current activity update.
7. Click **Save**or **Save & Exit**.
8. Notify the Problem Coordinator and the stakeholders about the plans and resources assigned to this problem.

1.6.3.14. Close a problem that will not be fixed

Part of Workflow(s): Problem Management: Categorization

Problem Management: Investigation

Applies to User roles: Problem Coordinator

Problem Manager

If, after investigation, you determine that an open problem is not considered to be a problem by the stakeholders, or that the problem is invalid or a duplicate, you can close it without fixing it.

To close a problem that will not be fixed, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Double-click the problem record that you want to close.
4. In the Activities tab, select the applicable type from the **New Update Type** field.
5. In the **New Update** field, enter the reason that this problem will not be fixed.
6. Click **Close** in the taskbar, confirm that the subcategory and area are correct, and then select an applicable closure code.
7. Click **Finish** to close the problem.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

1.6.3.15. Defer a problem

Part of Workflow(s):

Problem Management: Categorization

Problem Management: Investigation

Applies to User roles:

Problem Coordinator

Problem Manager

If you want to defer a problem, you must specify the reason for deferral. For example, if after investigation you decide to delay scheduling a problem for resolution, you can defer it. Or, if the Problem Analyst was unable to successfully test and validate a fix and you want to continue investigating a solution, you can defer the problem. You can later finalize how to handle the problem when you review deferred problems.

To defer a problem, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Double-click the problem record that you want to defer.
4. In the Status drop-down list, select **Deferred**.
5. In the Activities section, select the applicable update type from the **New Update Type** field list.
6. In the **New Update** field, type the reason for the deferral.
7. Click **Save & Exit**.

Note

Make sure you update the status of the problem during your regularly-scheduled review of deferred problems.

1.6.3.16. Review deferred problems

Part of Workflow(s): Problem Management: Categorization

Problem Management: Investigation

Applies to User roles: Problem Manager

You should periodically review problems that are deferred to determine whether additional actions are required. As you evaluate these problems, you can determine if they should be deferred again, reassigned for further investigation, or closed if the problem is no longer relevant.

To review deferred problems:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records. You can refine your search by selecting **Deferred** in the Status field.
3. Double-click the problem record you want to review.
4. If necessary, discuss the problem with stakeholders to determine the current answers to the following questions:
    - What's the priority of resolving this problem?
        - Update the **Impact**, if necessary.
        - Update the **Urgency**, if necessary.
    - Is this problem correctly documented? If the problem is incorrectly documented, update the problem documentation.
    - Does the problem need further investigation?
        - If the problem is still relevant, the problem retains a **Deferred** status, and you schedule it for investigation at a later date.
        - If the problem is no longer relevant, you close the problem record.
5. In the **Activities** tab, click the **New Update Type** drop-down list, and then select the update type.
6. In the **New Update** field, type notes to update information about the current activity.
7. Click **Save & Exit**.
8. If you determine that you can close the problem record, follow these steps:
    1. Click **Close** in the taskbar, confirm that the subcategory and area are correct, and then select an applicable closure code.
    2. Click **Finish** to close the problem.

1.6.3.17. Reassign a problem for additional support

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Manager

After the Problem Manager has reviewed the outstanding problems and known error records and determined that the action needed is to escalate the problem for further investigation, the Problem Manager reassigns the problem record to another level of support.

To reassign a problem record, follow these steps:

1. Click **Problem Management** > **Search Problems**
2. Use search or advanced search to find one or more records. If you searched for a specific problem ID, the problem record you specified opens. Otherwise a list of problem records opens.
3. Double-click the problem record that you want to reassign.
4. Update the **Status** to **Work in progress**.
5. Clear the assignment group in the **Assignment Group** field, and then click **Fill** to select a new assignment group.
6. Clear the assignee in the **Assignee** field, and then click **Fill** to select an assignee from the new assignment group.
7. In the **Activities** tab, click the **New Update Type** field list and select the applicable update type.
8. In the **New Update** field, type notes to include your current activity update.
9. Click **Save & Exit**.

Related topics

Using Problem Management
Problem Management searches
More Actions menu
Close a known error
Close a problem
Review deferred problems
Schedule a problem
Set a reminder for a problem record
Update a problem
View records related to a problem

1.6.3.18. Set a reminder for a problem record

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Analyst

Problem Coordinator

Problem Manager

You can set a reminder to notify you if an existing problem record meets certain conditions by a specified time. For example, you may want to be reminded if a problem is still open or is not updated by a given date.

When you set the reminder, you can choose when and how you want to be notified. You will find this particularly helpful for persistent problems that need your immediate attention.

To set a reminder, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select a problem from the list of records.
4. Click **More** or the More Actions icon and select **Set Reminder**.
5. In the **Set Reminder** area, select one of the following reminder options:
    - **Remind At**:

      Set a reminder for a particular day and time. Click **Fill** to display a calendar to choose a date and time for the reminder.
    - **Remind In**:

      Set the reminder to occur after a particular time interval. If you choose this option, you must specify the time interval in hours, minutes, and seconds using the 00:00:00 format. You must also select a shift to specify the work schedule that this time interval is **Based On**.
6. Select one of the following conditions in the **Remind if** drop-down list:

- **Always** (the default value)
- **Problem has not been updated**
- **Problem is still open**
- **Problem is still assigned to me**

7. In the **Pop-up Message** area, select one of the following types of notification:
   - **Pop-up**
   - **Email**
   - **SM Mail**
8. If you selected **Pop-up**, type the message that you want to appear in the reminder in the **Message** field.
9. If you selected **Email** or **SM Mail**, type a title for the mail in the **Title** field., and then select the type of mail message to send in the **Message Type** area.
10. Click **OK**.

1.6.3.19. Document the root cause

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Coordinator, Problem Manager

When a root cause has been identified for a problem record, you can document the root cause in the problem record and add other necessary and helpful information.

To document the root cause of a problem, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select the record to be updated.
4. Select the **Investigation and Resolution** tab.
5. In the **Root Cause** field, type the root cause information.
6. In the **Expected Root Cause Identified Date** field, select a target identification date.
7. In the **Activities** tab, click the **New Update Type** field list and select the applicable update type.
8. In the **New Update** field, type notes to include your current activity update.
9. Click **Save & Exit**.

Related topics

Phase 2: Categorization
Assign a Problem Management task
Close a problem task
Document the workaround
Investigate and diagnose a Problem Management task
Open a known error
Test the workaround

1.6.3.20. Document the workaround

Part of Workflow(s): Problem Management: Investigation

Applies to User roles: Problem Analyst, Problem Coordinator, Problem Manager

A workaround is a temporary solution to a problem. It might be a strategy that diverts your work flow around the affected configuration item or changes the service temporarily until the problem is resolved. For example, the workaround for a planned maintenance outage on printer A is to create a temporary connection to printer B. The additional number of users for printer B will increase the workload beyond normal levels, but the users for printer A can continue working until it returns to full service.

When the workaround is documented by the assigned Problem Analyst or Problem Coordinator, it also becomes the workaround in the known error record. Once the workaround is validated, you should inform the affected stakeholders.

When a workaround description is entered and saved in a problem record, Problem Management automatically records the workaround information in the related incident, if the incident is still open.

To document a workaround, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select the record to update.
4. In the **Investigation and Resolution** tab, type the workaround in the **Suggested Workaround** field.
5. In the **Activities** tab, select the applicable update type in the **New Update Type** field.
6. In the **New Update** field, type notes to include your current activity update.
7. Click **Save & Exit**.

Related topics

Assign a Problem Management task
Close a problem task
Document the root cause
Investigate and diagnose a Problem Management task
Open a known error
Test the workaround

1.6.3.21. Access the knowledgebase

Part of Workflow(s): Problem Management: Categorization, Problem Management: Investigation

Applies to User roles: Problem Analyst, Problem Coordinator, Problem Manager

The central knowledgebase is the default database for any knowledge search in Service Manager. You can search the knowledgebase by using the standard Service Manager knowledge application, or you can choose other knowledgebases to search.

To access the Service Manager knowledgebase, follow these steps:

1. Click **Problem Management** > **Search Knowledgebase**.
2. Use search or advanced search to find one or more records. For example, type a word or phrase in the **Search for** field.

Note

You can search for the following libraries:

- Knowledge Library
- Known Errors
- Problems
- Incidents
- Interactions

1.6.3.22. Open a Problem Management task

Part of Workflow(s): Waiting

Applies to User roles: Problem Coordinator

Problem Manager

When a problem is prioritized and the investigation and resolution activities are planned (such as deadlines for the root cause analysis, solution investigation, and resolution target dates), you can create the tasks for investigating, diagnosing, and testing a workaround.

To open (or create) a Problem Management task, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Select the target record.
4. In the **Tasks** tab, click **Link New Tasks**and then choose a task category.
5. Enter values in the **Title**, **Description**, **Urgency**, and **Task** fields.
6. Click **Save** to move the task to the Active phase.

1.6.3.23. Investigate and diagnose a Problem Management task

Part of Workflow(s): Active

Applies to User roles: Problem Analyst

A Problem Coordinator will assign a problem task to the Problem Analyst who has the required skills and resources to investigate the problem. The Problem Analyst is then expected to take the information provided in the task to investigate and duplicate the problem, and then find a root cause and determine a workaround for it.

To work with a current problem task, follow these steps:

1. Click **Problem Management** > **Problem Task Queue** or view your To Do list.
2. Select the target record. You can see the description of the problem, due date, category, urgency, and other details provided in the task.
3. In the **Status** field, update the status to **Work In Progress**.
4. You can begin your investigation, trying to reproduce the problem so that it is clear to you what went wrong. You can also search incident records to see if any might match the problem. For information on associating a problem record with an incident record(s), see the related topics.
5. If you can't reproduce the problem, document your activities in the **Activities** tab. To do this, follow these steps:
    1. In the **New Update Type** drop-down list, select the applicable update code.
    2. In the **New Update** field, add notes to document what you did while trying to investigate, diagnose, and duplicate the problem. Include your time spent performing the investigation.
    3. Notify the Problem Coordinator so the coordinator can determine what other resources and skills are required to diagnose the problem.
    4. Click **Close Task**.
6. If you find the root cause and determine a workaround, document your workaround in the **Activities** tab. To do this, follow these steps:
    1. In the **New Update Type** field list, select the applicable update code.
    2. In the **New Update** field, add notes to document what you did to duplicate the problem and type a description of the temporary workaround to fix the problem. Include your time spent during the investigation.
7. Click **Save & Exit**.

8. Test your workaround, and then document the results in the problem task.

1.6.3.24. Close a problem task

Part of Workflow(s): Active Review

Applies to User roles: Problem Analyst, Problem Coordinator, Problem Manager

To close a problem task, follow these steps:

1. Click **Problem Management** > **Search Problem Tasks**.
2. Use search or advanced search to find one or more records.
3. Double-click the problem task that you want to close.
4. In the **Activities** tab, select the applicable update type from the **New Update Type** list.
5. In the **New Update**field, add any activity notes.
6. Click **Close** in the taskbar.
7. In the in the **Closure Code** drop-down list, select a closure code.
8. In the **Task Outcome** field, enter an outcome.
9. Click **Finish** to close the problem task.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Problem Management overview
Assign a Problem Management task
Document the root cause
Document the workaround
Investigate and diagnose a Problem Management task
Open a known error
Test the workaround

1.6.3.25. Test the workaround

Part of Workflow(s): Active

Applies to User roles: Problem Analyst

One of the tasks of the Problem Analyst is to review and test the workaround identified and documented in the problem task. These tests help to validate the suitability for resolving related incidents. When the test is successful, the Problem Coordinator documents the workaround in the problem record and creates a known error record.

To update tested workarounds, follow these steps:

1. Click **Problem Management** > **Problem Task Queue** or view your To Do list.
2. View and identify any problem tasks whose workaround can be tested.
3. Test the workaround described in the **Activities** tab, using a test environment that mimics the production environment.
4. If the workaround fails, follow these steps:
    1. Document your findings in the **Activities** tab.
    2. Select the applicable update code from the **New Update Type** field list.
    3. Add notes to the **New Update** field with the results of your testing. If applicable, document the reasons a root cause is not found. Include your time spent during the investigation.

Note

Notify the Problem Coordinator of the test results who then escalates the problem to the Problem Manager. The Problem Manager determines other resources and skills required to diagnose the problem.

5. If the workaround is successful and has been validated, follow these steps:
    1. Document your findings in the **Activities** tab.
    2. Select the applicable update type from **New Update Type** field list.
    3. In the **New Update** field, type a description of the test results.

Note

Notify the Problem Coordinator of the test results. The Problem Coordinator validates the test results and makes additional determinations, such as whether the root cause is related to an outstanding known error, and then creates a new known error.

6. Click **Close Task**.

1.6.3.26. Close a problem

Part of Workflow(s): Resolution

Review

Applies to User roles: Problem Analyst

Problem Manager

The Problem Analyst verifies that all related known errors are closed or resolved before closing the problem record. If you try to close a problem that has open related known errors, the following message is displayed:

The Problem has open associated Known Errors. Force those Known Errors closed?

If you answer "yes," the open known errors are closed, and the solution to the problem is cascaded to all those known errors (but not to known errors that were already closed). If you answer "no," only the problem is closed, and the known errors remain unchanged.

The Problem Analyst then consults with the Problem Manager to determine if a problem review and problem report are required. If a review is required, the Problem Manager documents the results of the review as part of the problem report. As part of the problem closure and review process, the phase of the problem record is updated to Problem Closure . When the review is complete, the problem is closed with the applicable closure code.

To close a problem, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.
3. Double-click the problem record that you want to close.
4. In the Activities tab, select the applicable type from the **New Update Type** field.
5. In the **New Update** field, enter the reason that this problem will not be fixed.
6. Click **Close** in the taskbar, confirm that the subcategory and area are correct, and then select an applicable closure code.
7. Click **Finish** to close the problem.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

1.6.3.27. Assign a Problem Management task

Part of Workflow(s): Active

Applies to User roles: Problem Coordinator

The problem investigation and diagnosis process in Problem Management is aimed at identifying the root cause of a problem. After the Problem Coordinator determines the required skills and resources needed to investigate a problem, the Problem Coordinator creates a task and assigns it to a Problem Analyst who meets the requirements. If there are multiple configuration items (CIs) involved in the problem, the Problem Coordinator should create a separate task for each CI. This allows the Problem Coordinator to assign different specialists or external resources to investigate the problem.

To assign a Problem Management task, follow these steps:

1. Click **Problem Management** > **Search Problem Tasks**.
2. Use search or advanced search to find one or more records.
3. Select the target record.
4. Add the following information to the task:
    - **Due Date**
    - **Assignment Group**
    - **Assignee**
5. Click **Save & Exit**. Service Manager updates task records with related record information as updates occur.

1.6.3.28. Open a known error

Part of Workflow(s): Known Error: Logging

Applies to User roles: Problem Coordinator

Problem Manager

A known error is an issue for which a root cause (the underlying cause of a problem, or one or more incidents) has been diagnosed and a solution or workaround has been determined. When the Problem Coordinator validates a workaround that has been successfully tested by the Problem Analyst, the Problem Coordinator determines whether the root cause is related to an outstanding known error. You can open a known error only for problems in the Problem and Investigation phase and if there is a primary configuration item specified in the problem record.

If there is an existing known error, the Problem Coordinator relates this problem to the outstanding known error.

If there is not an existing known error, the Problem Coordinator opens a new known error record, and then moves the phase of the problem record to the Problem Resolution phase.

Note

Known error records can be opened by the assigned Problem Coordinator from a problem record, or directly (that is, unassociated with an existing problem record). When opened from a problem record, Problem Management automatically populates the new record with the relevant information from the problem record.

To open a known error record from a problem record, follow these steps:

1. Click **Problem Management** > **Search Problems**.
2. Use search or advanced search to find one or more records.

> **Note**
>
> Known error records can be only be opened by the assigned Problem Coordinator, so if you intend to open a known error you should include the assigned Problem Coordinator as part of your search criteria.

3. Select a record.
4. In the **Related Records** tab, select **Related Known Errors** from the **Link Type** drop-down list, and then click **Link New Record**. A New Problem form opens.
5. Complete any required fields on the form.
6. Click **Save & Exit**. You are returned to the problem record form.
7. Click **Save & Exit**.

To open a known error record directly, follow these steps:

1. Click **Problem Management** > **Create New Known Error**.
2. In the **Title** field, type a title that accurately describes the problem.
3. In the **Description** field, type the details of the known error.
4. Fill in the following required fields, and any other required or relevant fields.
    - Select the **Affected Service**.
    - Select the **Impact** and **Urgency**.

      > **Note**
      >
      > Service Manager selects the default priority based on the impact and urgency values.

    - In the **Root Cause** field, type a description of the root cause of the known error.
    - In the **Workaround** field, type a description of the workaround of the known error.
5. Click **Save & Exit**.

1.6.3.29. Update a known error

Part of Workflow(s): Known Error: Logging

Applies to User roles: Problem Coordinator

Problem Manager

You should always update a known error record with the outcome of the known error investigation. If the solution is validated, the Problem Coordinator updates the known error with the information about the solution, including the cost and resources needed to implement the solution and any possible risks. You can view ==Related topics== to see what work others have done to determine the known error, a workaround, and the time spent working on the investigation.

You should record all the investigation activities in the known error to track the effort associated with the problem, task, or known error. After the current activities for a phase are complete, you advance the known error record to the next phase.

To update a known error, follow these steps:

1. Click **Problem Management** > **Search Known Errors**.
2. Use search or advanced search to find one or more records.
3. Select an existing known error from the record list.
4. Type detailed information in the **Solution** field. You will find this information in the known error task where the Problem Analyst documented the details. Make sure that you include cost, resource information, and any possible risks.
5. In the **Activities** tab, select the applicable update type from the **New Update Type** field list.
6. Type additional updates in the **New Update** field.
7. Click **Save & Exit**.

1.6.3.30. Assign a known error record to schedule a fix

Part of Workflow(s): Known Error: Logging

Applies to User roles: Problem Manager

When you have validated a solution to a known error and determined that a Problem Analyst can resolve the required approach, follow these steps:

1. Click **Problem Management** > **Search Known Errors**.
2. Use search or advanced search to find one or more records.
3. Select the known error record that you want to assign from the record list.
4. Click the Fill icon in the **Assignment Group** field to select the assignment group. The Problem Coordinator for the group is automatically displayed in the **Assignee** field.

   If you need to reassign this known error record to a different coordinator, click **Fill** to generate a record list of Problem Coordinator names, and then choose one.
5. Note the resolution information from the problem management meeting.

   • On the **Activities** tab, click **New Update Type**, and then select **Problem meeting**.
   • Type notes in **New Update** to add information about the proposed fix and requested fix date. The Problem Coordinator uses this information to schedule the fix for a Problem Analyst to resolve.
6. Click **Save & Exit**. The system notifies the Problem Coordinator of this updated record.

1.6.3.31. Associate a change request with a known error record

Part of Workflow(s): Known Error: Logging

Applies to User roles: Problem Manager

When you have validated a solution to a known error and determined that the required approach can be resolved with a change request, follow these steps:

1. Click **Problem Management** > **Search Known Errors**.
2. Use search or advanced search to find one or more records.
3. Select a known error record in the Logging phase from the record list.
4. In the **Related Records** tab, select **Caused Change** or **Solved by Changes** from the **Link Type** drop-down list, and then click **Link Existing Record**.
5. Specify the change number or click **Search** to view the change search form.
6. Add optional filtering criteria, and then click **Search** again to display a change record list.
7. Double-click the change record you want to associate with the current known error. The change record ID is displayed.
    - Click **OK** to link the chage record to the current problem.
    - Click **Cancel** to stop the search and exit.
    - Click **Search** to view the change record search form again. Click **Search** again to redisplay the change record list.
8. Click **OK**. Problem Management populates the Related Records section with information about the related change.

1.6.3.32. Close a known error

Part of Workflow(s): Known Error: Logging

Applies to User roles: Problem Coordinator

Problem Analyst

Problem Manager

After you verify that a known error meets the closure conditions, you can close the known error. For example, when the Problem Coordinator receives notification from the Change Management process that a solution is successfully implemented and verifies that the known error is resolved, the known error record can be closed.

To update and close a known error, follow these steps:

1. Click **Problem Management** > **Search Known Errors**.
2. Use search or advanced search to find one or more records.
3. Double-click the known error record that you want to update and close.
4. In the **Activities** tab, select the applicable update type in the **New Update Type** field.
5. In the **New Update** field, type current activity update notes.
6. Click **Save**.
7. Click **Close** in the taskbar, confirm that the subcategory and area are correct, and then select an applicable closure code.

1.6.3.33. Access Problem Management reports

Applies to User roles: Problem Manager, Problem Coordinator

The Reporting tool in Service Manager provides a number of out-of-box reports on the problem data in your system. You can view these reports through a dashboard named Problem Overview (Global). You can also create your own dashboards to display other reports of your interest.

To access Problem Management reports, follow these steps:

1. Click **Problem Management** > **Problem Overview**.

   By default, the **Problem Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. Add your own dashboards as needed. For details, see Update a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default one.

1.6.3.34. Example: Problem workflow

The following example demonstrates the minimum needed steps to resolve the out-of-box Problem workflow:

Phase	User Actions	Options
Logging	1. Enter a title in the **Title** field. 2. Provide a description of the problem in the **Description** field. 3. Click the **Fill** button for **Affected Service**, and then select an appropriate value(s) that describes the **Affected Service**. 4. If necessary, set the **Impact** and **Urgency** fields. 5. Click **Save**. Service Manager displays a solution matching screen.	This is the phase that's initiated when an end user Opens a New Problem (**Problem Management** > **Open a New Problem**). **Note** The out-of-box default category is **-None-**. You can change the default category in **Problem Management** > **Administration** > **Settings**

Phase	User Actions	Options

Categorization	1. Click the **Fill** button to specify the **Subcategory**.
	    1. Select an appropriate area.
	    2. Select an appropriate subcategory.
	2. Set the Status field to **Assign**.
	3. Click the **Fill** button for the **Assignment Group**, and then select an appropriate assignment group.
	4. Click the **Fill** button for the **Assignee**, and then select an appropriate assignee.
	5. Set the Status field to **Work In Progress**.
	6. Click **Save**.
		From the Categorization Phase, you can also move to any of the following:
		• Jump directly to the Closure Phase.
		• Abandon the Problem.

Phase	User Actions	Options
Investigation	1. Set the **Expected Resolution Date**. 2. Set the **Root Cause Identified Date**. 3. Enter the **Root Cause**. 4. Click **Save**.	From the Investigation Phase, you can also move to any of the following: • Jump directly to the Closure Phase. • Abandon the Problem. • Return to the Categorization Phase.
Resolution	1. Set the **Solution Identified Date**. 2. Enter the **Solution**. 3. Set that Status to **Resolved**. 4. Click **Save**.	From the Resolution Phase, you can also move to any of the following: • Jump directly to the Closure Phase. • Return to the Investigation Phase.
Review	1. Review the data entered for the Problem.	From the Review Phase, you can also move to any of the following: • Return to the Resolution Phase.

Phase	User Actions	Options
Closure	1. Click the **Close** button. 2. Select the appropriate **Closure Code**. 3. Enter the **Closure Comments**. 4. Click **Finish**.	

Note

Notes:

- In all the phases except the Logging, Abandonment, and Closure phases, you can create or link a known error from the **Related Records** tab of the problem record.
- In all the phases except the Logging, Abandonment, and Closure phases, you can create a problem task by clicking the **Link New Task** button. All tasks must be closed before you can close a Problem.

1.6.4. Problem Management Administration

Problem Management administration consists of the following:

- Problem Management Environment: Problem Management application contains an environment record for problems. This record contains options that defines the functionality of the Problem Management application for all Problem Management users.
- Settings: You can set the default category for new problems and problem tasks.

Configure the Problem Management settings

Applies to User roles: System Administrator

To configure the Problem Management settings, follow these steps:

1. Click **Problem Management** > **Administration** > **Settings**.
2. In the topmost pane, select **Problem** or **Problem Task**.
3. Configure the **Default Category** setting for the Problem or Problem Task. This setting defines the default category that's used when you register a new Problem or Problem Task. The out-of-box value for this setting is **None**.
4. Click **Save**.

Configure the Problem Management environment

Applies to User roles: System Administrator

System Administrators can modify the Problem Management environment parameters to match the operational requirements of their organization.

To configure a Problem Management environment, follow these steps:

1. Click **Problem Management** > **Administration** > **Environment**.
2. Select or clear the parameters and populate the required fields as appropriate to the Problem Management environment.
3. Configure the settings for your Problem environment.

Tip

You can use the field help to view the description for each setting.
To view the help on field:

- Web client: Select a field, and then press **F1**.
- Windows client: Select a field, and then press **Ctrl+H**.

4. Click **Save**.

Contents

- 1 Configure the Problem Management settings
- 2 Configure the Problem Management environment

1.6.5. Problem Management Configuration

Problem Management configuration enables you to configure alert, problem categories, workflow, solution matching, and so on.

Create a Problem Category

User Roles: System Administrator

If you are a Service Manager Administrator, you may want to create a problem category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides out-of-box category records that you can use or modify.

Note

When a category is set as the default category in settings, don't delete it to avoid unpredictable issues.

To create a new problem category record, follow these steps:

1. Click **Problem Management**.
2. Click **Configuration** > **Problem Categories**.
3. In the menu of the **Problem Category** pane, click **New**.
4. Type the name of the problem category.
5. Clear the **Active** checkbox if you don't want the new category to appear in the category list. If the **Active** checkbox is cleared, the new category is unavailable when you create a new record.
6. Type a description of the problem category.
7. Select a workflow for the category.
   The **Workflow** tab is displayed.
8. Click **Save**.
   The **Subcategories** tab is displayed.
9. **Optional**: Click the **Subcategories** tab to create subcategories for the problem category. When you have finished, click **Save**.

Note

The **Apply To** flag indicates whether the category is shared across different modules.

For example, if the **Apply To** is set to "Interaction/Incident", then this category will be created in both the interaction category table and the incident category table.

Typical shared categories in OOB include Complaint, Request for Information, Incident, and Request for Administration.

For the **Apply To** flag in Problem Category:

When creating a problem category, the **Apply To** option is predefined as "Problem" only and can't be modified. However, an problem category can also be created when creating an interaction category with the **Apply To** option set as "Interaction/Incident/Problem".

When searching for a problem category, all the **Apply To** values that include Problem can be used.

The following list is used in Problem Category:

2 - Problem

4 - Interaction/Incident/Problem

Note

Problem category name is read-only after the category is created.

Create a Problem Task Category

User Roles: System Administrator

If you are a Service Manager Administrator, you may want to create a problem task category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides out-of-box category records that you can use or modify.

To create a new problem task category record, follow these steps:

1. Click **Problem Management**.
2. Click **Configuration** > **Problem Task Categories**.
3. In the menu of the **Problem Task Category** pane, click **New**.

4. Type the name of the problem task category.
5. Clear the **Active** checkbox if you don't want the new category to appear in the category list. If the **Active** checkbox is cleared, the new category is unavailable when you create a new record.
6. Type a description of the problem task category.
7. Select a workflow for the category.
   The **Workflow** tab is displayed.
8. Click **Save**.

Note

To avoid unpredictable issues, it is highly recommended that you don't change the problem task category name after it is created.

Add a new subcategory to a problem category

User Roles: System Administrator

You can add a new subcategory to a problem category directly. Additionally, you can view a list of the subcategories that are associated with the current category, together with the record details for each subcategory.

To add a new subcategory for a problem category, follow these steps:

1. Click **Problem Management** > **Configuration** in the system navigator.
2. Click **Problem Categories** > **Search**.
3. Select the problem category to which you want to add a subcategory.
4. Under the **Subcategories** tab, click the **Link New Subcategories** button.
5. Type a name for the new subcategory.
6. Type a description for the subcategory.
7. Click **Save** to add the new subcategory.
8. **Optional**: Click the **Link New Areas** button to create problem areas for the subcategory.
9. Click **Save**.

Add a new problem area to a subcategory

User Roles: System Administrator

You can add a new problem area to a subcategory directly. Additionally, you can view a list of the areas that are associated with the current subcategory, together with the record details for each area.

To add a new problem area to a subcategory, follow these steps:

1. Click **Problem Management** > **Configuration** in the system navigator.
2. Click **Problem Categories** > **Search**.
3. Select the problem category with which the subcategory is associated.
4. Under the **Subcategories** tab, select the subcategory to which you want to add a new area.
5. Click the**Link New Areas** button.
6. Type a name for the new area.
7. Type a description for the area.
8. Click **Save** to add the new area.

Problem Management solution matching

When you create a new problem, Service Manager enables you to link that problem to an existing problem. Service Manager includes the following out-of-box queries, with which you can search for existing problems:

- Find a problem or known error with a matching configuration item
- Find a problem or known error with a matching service
- Find a problem or known error with a matching title

To modify an out-of-box Problem management solution matching query, follow these steps:

1. Click **Problem Management** > **Configuration** > **Solution Matching**.
2. In the topmost pane, select the query that you want to modify.
3. Modify the **Name**, **Match**, and **With** fields as appropriate to your needs, and then click **Save**.

Contents

- 1 Create a Problem Category
- 2 Create a Problem Task Category
- 3 Add a new subcategory to a problem category
- 4 Add a new problem area to a subcategory
- 5 Problem Management solution matching

1.6.6. Security

The topics in this section explain the Problem Management security roles, security areas, and rights.

Problem Management Security Areas

User Role: System Administrator

The security areas for Problem Management are Problem, Problem Task, and Problem Management Configuration. These areas contain the default security rights and settings for Interactions in the Problem Management module. The security rights settings will be inherited by the new roles created in an area when no settings are specified in the security role.

Area	System navigator menu items for this area
Problem	This area contains the default security rights and settings for Problem. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.
Problem Tasks	This area contains the default security rights and settings for Problem Tasks. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.

Area	System navigator menu items for this area
Problem Management Configuration	This area contains the default security rights and settings for Problem Management Configuration. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role. **Note** When you set the security rights for a security role in the Problem Management Configuration area: • The **View** right is to view the settings defined in the **Administration** menu and the **Configuration** menu. • The **Update** right is to update the values of existing settings defined in the **Administration** menu and the **Configuration** menu. • The **New** and **Delete** rights are to create and delete a setting in the **Configuration** menu, such as category. • The **Admin** right is to add, edit, or delete the settings in the **Administration** > **Settings** menu.

Area	System navigator menu items for this area

Default rights

When you create new security roles, they inherit the default rights that are defined in the security area. The following table describes the out-of-box rights that are defined in the Problem, Problem Tasks, and Problem Management Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Problem	TRUE	FALSE	Never	Never	FALSE	FALSE
Problem Tasks	TRUE	FALSE	Never	Never	FALSE	FALSE
Problem Management Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Problem, Problem Tasks, and Problem Management Configuration areas.

Problem Management security roles and settings

The out-of-box security roles for Problem Management module include the following:

- Problem process owner
- Problem coordinator
- Problem analyst
- Problem task assignee
- Reviewer

Tip

If you have upgraded from a previous version of Problem Management, which is not Process Designer based, see Problem Management security profiles mappings for more information about the mappings between the legacy Problem Management profiles and the current Process Designer security roles.

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Problem area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Problem, Problem Tasks, and Problem Management Configuration areas. This table only lists the new security roles that have different settings with the default rights.

Problem area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
problem process owner	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

Problem Tasks area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
problem process owner	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
problem task assignee	TRUE	FALSE	When assigned to workgroup	When assigned to workgroup	FALSE	FALSE	FALSE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

Problem Management Configuration area

Role Name	View	New	Update	Delete /Close	Modify Template	Expert	Admin
problem manager	TRUE	TRUE	Always	Always	FALSE	FALSE	FALSE
problem process owner	TRUE	FALSE	Never	Never	FALSE	FALSE	FALSE
system administrator	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE

1.7. Change Management (Codeless)

Service Manager Change Management tracks changes to service assets and configuration items in your infrastructure.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Change Requester

Change Approver

Change Coordinator

Change Owner

Change Manager

Change Advisory Board (CAB)

E-CAB

Key concepts

Describes the most important things that you need to know about how the Change Management module functions.

Change Management overview

What are changes?

What's an approval?

Approval delegation

Priority, impact, and urgency

Release Management

Workflows and user tasks

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Change Proposal (ST 2.0)

Standard Change (ST 2.2)

Normal Change (ST 2.3)

Emergency Change (ST 2.4)

Administrator tasks

Describes the Administrator tasks in the Change Management module.

Change Management security and access control

Managing approvals in Change Management

Configuring Change Management categories

Configuring change models

Configuring Change Management workflows

Managing alerts

Additional resources

Processes and Best Practices

1.7.1. Change Management overview

All changes to the Service Portfolio or service catalog are implemented through Change Management and the changes that are managed by the Service Transition life cycle stage are defined and agreed. Standardized methods and procedures are used for efficient and prompt handling of all changes, in order to minimize the impact of change-related incidents on business continuity, service quality and re-work.

To be effective, Change Management requires the following input:

- Policy and strategies for change and release
- Request for change
- Change proposal
- Plans (change, transition, release, deployment, test, evaluation, and rendition)
- Current change schedule and projected service outage (PSO)
- Current assets or configuration items
- As-planned configuration baseline
- Test results, test report, and evaluation report

All updates to changes and releases are recorded against service assets and/or configuration items in the Configuration Management System.

Related topics

What are changes?
Priority, impact, and urgency

1.7.1.1. What are changes?

Changes are the records submitted that initiate a change in an IT service. Each change has a life cycle of approvals, alerts, tasks (if required), phases, and closure. When you submit a change request, Change Management assigns a unique change record number.

Change identifiers

Service Manager automatically creates a unique ID for each RFC and Change Task when the record is opened. This field is mandatory and read-only. The format of the ID is configurable to meet customer requirements.

Time and date stamp

Service Manager time and date stamps all events (manual or automated) associated with a Change record, from the time the record is opened until it is closed. These fields are mandatory and read-only.

Priority, impact, and urgency

Priority is how an individual service desk interaction, change request, incident, or problem fits into the ongoing sequence of tasks required to close the interaction, change request, incident, or problem. It also indicates how soon the work should begin. Determining the priority of a single service desk interaction, change request, incident, or problem depends on how many other defects need attention, the risk of delay, and the resources available to fix it.

Impact is the potential business vulnerability. There is no global value; it is subjective and each business must set and modify its own impact value list.

Urgency is a value that reflects how soon the defect must be resolved to avoid business consequences. It identifies how soon you must react to avert or reduce the impact of the defect on customers.

Assigning values to impact and urgency is subjective. Priority is a Service Manager calculation based on the values you specify for impact and urgency. As you can experience assigning impact and urgency values, you will refine your decision criteria. A service desk interaction, change request, incident, or problem that's isolated can have a low impact initially, but a high urgency because of the potential for damage if the defect becomes widespread. For example, a new computer virus is a problem that can escalate quickly.

Related topics

What's an approval?
Managing alerts

1.7.1.2. Change prioritization

The Change form includes fields to capture the impact and urgency of the RFC. These fields can be set as part of the Change Model using the Templates feature. The fields can also be populated based on the Service or CI that's identified for the Change. Service Manager automatically calculates priority based on impact and urgency. The lists of values used for impact, urgency, and priority along with the calculation used to determine priority can be modified by the customer.

Impact codes include:

- 1 – Enterprise
- 2 – Site/Department
- 3 – Multiple Users
- 4 – User

Urgency and Priority codes include:

- 1 - Critical
- 2 - High
- 3 - Average
- 4 – Low

Closure phase

The Change workflows provided in Service Manager contain a Closure phase that indicates that the Change is closed. In the workflow viewer this phase is with a solid gray background so it is visually evident that Changes in this phase aren't active.

Closure codes are captured in the Post Implementation Review phase and can be configured by the customer.

Post implementation review

Service Manager Change workflows contain a Post Implementation Review phase that supports notifying the person responsible for the review and capturing the details of the review when it is performed.

Remediation plans

Service Manager provides an out-of-box field that can be used to record the remediation procedures for a specific RFC. The information in the Remediation Plan

field could reference attached documents or external sources for remediation plan documentation.

The remediation plan review is part of the Deployment CAB approval process of the Normal Change workflow.

Contents

- 1 Closure phase
- 2 Post implementation review
- 3 Remediation plans

1.7.1.3. Change scheduling

Service Manager provides an embedded Change Calendar (also referred to as Service Manager Calendar) for a Forward Schedule of Change. This calendar is based on a Calendar widget and is accessible to anyone with view rights for Changes. Users with only view rights will have a read-only view and users with update rights have read/write access.

This calendar displays time period records and associated business records in a graphic and intuitive user interface. It enables users in different locations and departments of your organization to optimize their task planning accordingly.

Service Manager Calendar has a huge benefit from being a part of Service Manager. Its time period rules engine more effectively integrates time period policy into the change process, with notifications of time period violations and the capability to require additional approvals if time periods rules are violated. It is easier to initially configure and maintain.

Projected service outage

Service Manager has fields to capture the Scheduled Downtime Start and Scheduled Downtime End, as well as the Actual Downtime Start and Actual Downtime End that's the result of the Change.

This information is visible in the Configuration Item or Business Service fields to show the Projected Service Outages for that CI or Service.

Related topics

Calendar Administration
Notifying Service Subscribers of a Planned Outage

1.7.1.4. Assessment capability

Service Manager offers the following assessment capabilities:

- Manual assessment of the impact of the change by the Submitter, Coordinator, or Assignee.
- Collision detection: One or more RFCs for the same CI are proposed at the same time.
- Resource limitations: when two or more RFCs are proposed using the same resources to alert the CAB to potential resource constraints.
- Risk impact calculated using business rules and CMS relationship information at the CI level.
- History of previous similar changes and how successful they were.

Related topics

Assess the risk and impact of a change

1.7.1.5. Change notification and escalation

Change Management leverages Process Designer workflows, but not direct HTML Mail rulesets, in contrast to all other Process Designer based modules. Instead, Change Management uses calls to the notification engine. As such, there are numerous out-of-box notifications available. These notifications can be customized easily to provide broad and deep control and oversight of the notifications and escalation behaviors throughout the workflow.

The Change Management workflows deliver alert options via the alert engine. This solution enables the definition of alerts including:

- When the alert should be evaluated
- What work schedule is appropriate for determining the alert evaluation
- What conditions of the ChM record are indicative that the alert should trigger
- Who should be notified when the alert has triggered
- What actions should take place

1.7.1.6. Change Management Audit trail

For all records in Service Manager, a date time stamp and a user stamp are updated in the record when it is created or saved.

Service Manager provides an audit trail capability that identifies each step taken in the resolution of a change or task. Each time they are updated (whether manually or automatically), a separate historical activity is created for the record. An authorized user may review each historical activity to develop a comprehensive understanding of the history of the record and its resolution, including which operators took what actions at what times.

Service Manager also provides an auditing feature that records modifications to fields within the database. Field modifications are detected by comparing the fields in the original version of a record to the updated version. When modifications are detected, an Audit Log entry is recorded for each changed field showing the name of the modified field, the old and new version of the data, the current date/time, and the current operator's user ID.

Related topics

Audit processing
Invoke auditing from Format Control
Processes and Best Practices

1.7.1.7. Change archival

Closure of processed changes and tasks is a standard part of the workflow. The closure action not only supports saving of the record but separate validations, rules, and possibly even separate forms to ensure business success. Closing a change doesn't remove it from the database.

The most typical approach for archiving records is to leverage the chosen RDBMS tools from Oracle or MS SQL Server to perform their database administration for purging and archiving.

Service Manager provides an archive and purge tool that can be configured by table and criteria for archiving. This is a facility only the administrator would use. This capability allows the ability to archive either manually or on a scheduled basis.

One new capability in this area is the ability to take advantage of the archive tool suite. These tools allow data to be archived out of Service Manager into a system accessible file system. The production or alternative instance of Service Manager can then be used to access these data records.

Related topics

Revision control
Archive records

1.7.1.8. What are notifications?

Messages are generated by Service Manager events, such as opening or closing a change or task. Administrators can edit these messages, add new messages, change the conditions that trigger the messages, and select who will receive the messages.

The Notification Engine normalizes notification across the applications, removing the need for each module to define its own notification process (like cm messages).

The notification engine can be called by PD rule type "Send Notifications", which can be attached to PD workflows at both the phase and workflow levels. Alternatively, you can use Process Designer Send HTML Mail rules to call HTML notifications from the Process Designer workflow.

Change access control

Change Management utilizes the new Process Designer security controls to permit authorized users to create, modify, and close changes and change tasks. The system provides Update authorization options for **Never**, **Always**, **When Assigned** and **When assigned to workgroup**. For example, one user may be able to create and close changes or tasks but only modify changes assigned to them.

The system provides the construct of a user role which allows administration of many users with a single role definition.

Access to functionality within each module is governed by Service Manager security utilities, which define access based on role and user.

Service Manager provides a security control to segment data between multiple customers. The Mandantan controls identify which customer data a particular user or group can access, update, or both.

Contents

- 1 Change access control

1.7.1.9. Change records and CIs

Service Manager Change Management provides the ability to link Configuration Items (CIs) to Change records. Change Tasks can be created based on the list of CIs in the Change record.

The Change record will also include a list of Affected Services that are based on the topology information for the linked CIs.

Change Management Integration: Release Management

Service Manager Change Management includes a Release Management category that provides the ability to schedule and authorize coordinated release activities that span multiple RFCs. The Release record can contain links to one or more Change records. This allows several RFCs to be assessed, scheduled, approved, and deployed as a group. The execution of the changes happens in the individual Change and Change Task records that are linked to the Release. The execution of the individual Changes can be supported by the integration to Operations Orchestration (a run a run book automation workflow tool) that automates steps of software deployment through Server Automation and Client Automation depending on the target being deployed to.

Change Management Integration: Incident Management

Service Manager allows the creation of links from a Change record to an Incident record. RFCs can be created from an Incident or an Incident can be related to an existing RFC. The Incident can also be indirectly related to a Change record through the Problem Management module.

Change Management Integration: Problem Management

Service Manager provides the ability to create Changes from Known Errors and to link Changes to existing Known Errors as recommended in the best practice documentation. Service Manager can also associate Problems and Changes through configuration of the system.

The related records tab shows established relationships between Problems and Known Errors, which are stored as separate records in Service Manager, and Changes.

Change Management Integration: Service Desk

Email can be generated when an RFC is created, updated or closed and sent via standard email delivery tools to individuals or groups. Emails can be generated and sent automatically or manually.

Service Desk agents can be given access to the forward schedule of changes and their security roles used to control their access rights. You may want to assign them read-only access to the forward schedule of changes to avoid the entire Service Desk team from consuming change licenses.

If an Incident record has one or more related Change records, the Service Desk agent can click on the link to see the related RFC(s).

Additionally, Service Desk agents (or anyone using Service Manager) can use charts and views to get visibility into Change Management information, such as a list of Changes scheduled for the next week.

Users can easily create these charts and lists on the fly, and save them to their favorites, or they can be used to create dashboards for groups of users.

Authorized users can drill down into the details of records that are behind the charts and see individual records and their details.

Change Management Integration: uCMDB

Service Manager has an out-of-box integration with the CMS solution (UCMDB).

The CMS strategy is to leverage web services and use contributing CMDB and other data repositories through a common view of business services provided by the UCMDB. As such, the integration between UCMDB and Service Manager is using web services.

UCMDB and the discovery tools provide detailed CI and relationship information built using an innovative discovery capability. Essentially, by listening on the network and identifying applications and hardware “talking” through standard protocols, CMS uses a spiral discovery approach, interrogating applications to gain information about levels 2 through 7. This discovery capability understands how applications and hardware are related and shows this in graphical form.

The information presented in the CMS is representative of the actual state and can be viewed within Service Manager in the “Actual State” tab in a CI record.

This is an alternative view of the same data. Service Manager also provides a deliberately simpler visualization of the CIs and their relationships designed to provide

the Service Desk user a summary of the information held in the CMS.

Indicators are used on the CI icons to visually show the user there are open records for the CI.

Change Management assessment capabilities are integrated with UCMDB to provide impact analysis information based on the topological view of the CI data.

Also, UCMDB provides the ability to model the potential impact of changes on the fly without physically making the change.

1.7.1.10. Role-based authorization

Change Management uses a role-based authorization system. Different members of the CAB can have different roles and therefore different rights in the Change Management module. In addition, members of a Change Group can be identified as either Approvers or Members of the group.

Service Manager also provides the ability to temporarily assign approval capabilities to another user as long as that user also has the appropriate authorization. This is called approval delegation and customers often use this for when they are going on vacation or will be out of the office for a period of time.

1.7.1.10.1. What's an approval?

An approval occurs when a user (approver) or a group of users (approval group) agrees to accept the risk, cost, and responsibility to implement a change request or complete a task. The System Administrator defines membership in an approval group in each user's operator record according to the User Role. Change requests and tasks contain approval requirements. Approvals give Change Management the ability to stop work and to begin certain work activities.

As an approver, you can also be part of a change message group that consists of reviewers and approvers. If you are an approver for a change message group, your task is to accept or deny the changes your group must approve.

When you select the **Approve All Selected** option, all the records in the list that are awaiting your approval are immediately approved.

Note

Change Management doesn't prompt you to verify that this is the action you want to take.

Related topics

What's an approval sequence?

1.7.1.10.2. What's an approval sequence?

An approval sequence is the order in which approval requirements are activated. The lowest sequence numbers are available for approval activity. When approval occurs, the next highest number is activated. Some approvals must be sequential. In this case, approval sequence numbers reflect the required order. If an approval group shares the same sequence number, the group members can approve in any order.

1.7.1.10.3. What's an approval type?

An approval type is a type of approval required for an approval definition record in the **ApprovalDef** table. The following table describes the approval types in the out-of-box system.

Type	Description
All must approve	All members of the group must approve the request. **Note** This is the default value when no approval type is specified for an approval definition.
One must approve	Only one member must approve the request.
Quorum	A majority of the approval group must approve the request.
All must approve - immediate denial	All approvers must approve the request. The first denial updates the status of the current approval request to Deny. Meanwhile, the first denial also updates the status of other pending approval requests to Denied By System and sets the value in the Approved by field to System. Other approvers don't need to take any action.

Type	Description
One must approve by sequence approval level	When this option is selected, the following behavior applies: • When a sequence is defined with a group of approvers, if one approver approves the request, the entire sequence is approved and the request is passed to the next sequence for approval. Only when all sequences are approved, the request is approved. • Only when all approvers for a sequence have denied the request, the entire sequence is denied. If one sequence is denied, the request is denied.

Related topics

What's an approval?
Approval status

1.7.1.10.4. Approval options in Change Management

Change Management enables you to approve, deny, or retract changes or tasks.

Approval option	Description
Approve	The approver accepts the need for the change or task, and approves commitment of the resources required to fulfill the request. When all approvals are complete, work begins. When you choose this option, the record shifts to browse mode, and the retract option is available. If you're not a member of a group with approval rights to this change request, Change Management generates an error message.
Deny	The approver is unwilling to commit the required resources, or doesn't consider the change or task to be essential. No further approvals are possible until the denial is retracted. An administrative procedure should be set up to handle a denial. If you select deny, a dialog box opens with a prompt to specify the reason for your action. Type an explanation and click **OK**.
Retract	The approver accepts the need for the change, but is unwilling to commit the resources or perhaps there are technical incidents at the present time. Retract removes a previous approval or denial and resets the change request to pending approved status, which requires a new approval cycle. If you select retract, a dialog box opens with a prompt to specify the reason for your action. Type an explanation and click **OK**.

Note

Rights for the override option are granted to users with **Admin** or **Expert** security rights for the Change area.

Related topics

What's an approval?
What's an approval sequence?
Configuring approvals in Change Management

1.7.1.10.5. Defining approvers

The Change security roles include a setting that defines whether users with that Role are allowed to approve Changes. The Change Tasks security roles provide the same setting for approval of Change Tasks.

Within the workflow for Change Management, Service Manager allows for approvals to be assigned to either individuals or Assignment Groups. Users who are approvers for the Assignment Group and who have the “Can Approve” right will be able to approve that individual change (or task).

As a change approver, an operator must meet all of the following conditions:

- The operator is assigned a security role that has the **Can Approve** right for the Change security area (**System Administration** > **Security** > **Roles**).
- The operator is defined as an approver of an assignment group (**System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**).

However, if the operator is an approval delegate (that is, a change approver delegates an approval to the operator), the operator only needs the **Can Approve** right and doesn't need to be in an assignment group.

1.7.1.10.6. Approval options

Changes may only be rejected by users who are assigned to approve the Change and who have the “Can Approve” change security right. The Approval Type set in the Approval Definition record will control whether a single denial of approval will result in the change being rejected (All must approve – immediate denial) or if denials by some members of the Assignment Group can still allow the change to be approved.

Approvers have the options to approve, deny, or retract change records. For more information, see Approval options in Change Management.

1.7.1.10.7. Approval status

In any workflow phase with an approval, the Approval Status field will indicate if the Change has been approved, denied, or is still pending the approval process. A Change will not be allowed to progress past any phase with approvals until all the required approvals have been given. In change records, the Approvals section reflects only all approvals that must be performed in the current phase of the change workflow, not all approvals that are required in the entire change workflow.

Alerts and escalations can be built into the change process to prompt approvers as the RFC approval phase runs out of time, and basic checks such as not allowing an approval after the proposed planned start date and time are also possible.

Approval can also be sequenced to allow for certain approvals to be given (perhaps the technical team's approval that the proposed RFC is technically feasible) before sending approval requests to managers for a business approval.

What's approval status?

Approval status describes the condition of the change or task in the current phase and in the Change Management workflow. Each change request phase is marked with one of the following status options:

- Approved
- Denied
- Pending

Individual approval requirements within a phase are marked with one of the following status options:

- Approved
- Denied
- Pending
- Future

Pending approvals are awaiting action. Future approvals will be acted on following action on the pending approvals.

When a pending phase is approved, its status becomes **approved**. The next set of future approvals becomes **pending**, and subsequent approvals remain in the **future** status.

Related topics

What's an approval?

1.7.1.10.8. Approval delegation

Approval delegation is an optional feature that enables users with approval rights to temporarily delegate their approval authority to another qualified operator. Operators with the **Can Delegate Approvals** or **Delegate Approvals** option enabled in their application security roles can delegate some or all their approvals by using the Approval Delegation wizard.

Using the **Approval Delegation** wizard, an operator can grant another qualified operator the right to temporarily view and act on items in the operator's approval queue. The wizard offers the following delegation options:

- Delegate all approvals to another qualified operator
- Delegate approvals from a particular application to another qualified operator
    - Delegate approvals directly assigned to you as an operator
    - Delegate approvals assigned to you as a member of an approval group
- Delegate approvals from a specified start date to a specified end date

The **Approval Delegation** wizard enables an operator to create any number of approval delegation combinations, including delegating the same approvals to multiple operators at the same time. A delegator can also update an existing approval delegation to change the delegation start and end dates and the delegate's name.

Note

Service Manager tracks all changes to approval delegations using the standard field auditing capability.

When delegates log on to Service Manager, they see both their own and any delegated approvals in their approval list. For security reasons, delegates always retain their original application profiles and operator records. Service Manager determines what temporary rights delegates have when they view or act on an approval.

For more information about the tasks that relate to approval delegation, refer to the following topics:

What happens when I delegate approval authority?
What happens when I receive delegated approval authority?
Copy an approval delegation

Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

1.7.1.11. Release Management

Release Management is a category of Change Management that ensures that the Configuration Management Database (CMDB) is kept up to date, that changes are appropriately managed, and that all new software and hardware is stored in the Definitive Software Library (DSL) and Definitive Hardware Store (DHS). After one or more changes are developed, tested, and packaged into releases for deployment, Release Management is responsible for introducing these changes and managing their release. Release Management also contributes to the efficient introduction of changes by combining them into one release and deploying them together.

Release Management relies on the Change Management and Configuration Management applications.

Goal

The purpose of Release Management is to ensure that all changes are deployed successfully in the least disruptive manner. Release Management is responsible for the following functions:

- Driving the release strategy, which is the over arching design, plan, and approach for deployment of a change into production in collaboration with the Change Advisory Board (CAB)
- Determining the readiness of each release based on release criteria (such as quality of release, release package and production environment readiness, training and support plans, rollout and backout plans, and risk management plan)

Benefits

Release Management offers the following benefits for users:

- Provides a packaged release for all changes deployed into production and only deploys changes approved by change management
- Provides two different types of releases: hardware and software. Once the release type is determined, the application provides the appropriate tasks
- Provides the ability to manage changes of groups such as Configuration Item (CI) groups and business services
- Provides the ability to terminate a release at any time except in the Training phase or once it has been installed or verified
- Provides the ability to back-out the change
- Provides the availability of optional training phase

Related topics

Release Management phases
Terminating a release change
Create a new change with a Release Management category

1.7.1.11.1. Release Management phases

The Release Management category includes twelve phases, which follows the recommended Information Technology Infrastructure Library (ITIL) process by providing you with a set of approvals and tasks that you can expand.

Phase	Description
Assess Release	Determine the importance or value of a release.
Plan and Design Authorization	Approve the plan and design phase of the release.
Release Plan and Design	Plan releases in line with requirements resulting from approved changes. Analyze all the affected Configuration Items (CIs) and CI Groups, review what software or hardware is needed to accomplish the plan, and determine the cost. This phase requires the plan and design approval before going to the next step.
Build and Test Authorization	Approve the build and test phase of the release.
Release Build and Test	Build effective release packages for the deployment of one or many changes into production. Test release mechanisms to ensure minimum disruption to the production environment.
Release Training	Identify and fulfill training requirements for various users. This phase is optional and is activated when you select high Impact Assessment for the release change.
Distribution Authorization	Approve the distribution and roll out of the release.

Phase	Description
Release Distribution	The distribution or roll out of either a hardware or software release into the IT environment.
Release Remediation	A plan detailing how a specific release can be undone after being applied, if deemed necessary.
Release Verification	Review preparation for the release to ensure maximum successful deployments.
Abandoned	Abandon a release in any phase prior the distribution phase.
Closure	After the verification phase the release is closed since the release cycle is completed.

Release Management: Assess Release phase

The first step in the release process is to determine the importance and value of a release. If the assessment determines that a release is indeed necessary, the next step is to plan and design the release.

Release Management: Plan and Design Authorization phase

In this phase the Release Manager works with the relevant stakeholders, such as service owners or project managers to approve the Plan and Design phase of the release. If unapproved the release is abandoned.

Release Management: Release Plan and Design phase

The third step in the release process is the creation of a plan identifying the activities and the resources required to successfully deploy a release into the production environment. The following is a summary of the release planning activities.

- Identifying scope and content of a change, and the release requirements for successful deployment
- Performing a risk assessment for the release and obtaining the appropriate approvals
- Prioritizing, planning, and scheduling release activities
- Establishing a suitable team for the release if required
- Coordinating with experts and interested parties to determine the required resources and strategy for the release
- Documenting and tracking all release planning activities

Release Management: Build and Test Authorization phase

The Release Manager reviews the Planning and Design activities and discusses the results with all relevant stakeholders to decide whether to approve the activities required to build and test the release. In case the release is rejected the Release Manager either initializes a rework of the plan and design steps or is abandoning the release.

Release Management: Release Build and test phase

After the release team agrees on a release plan, members of the release team identify and develop the processes, tools, and technologies required to deploy the release into production. The following is a summary of the release building activities.

- Selects a suitable release mechanism for the change that's a strategic fit, is repeatable, and is consistent
- Designs and builds a release package for the change that allows it to be successfully deployed
- Tests the release package to ensure that it delivers the change effectively in line with requirements
- Ensures the release package is updated to the Configuration Management Database (CMDB) as a pending release package and that any new script is added to the Definitive Software Library (DSL)

Release Management has two out-of-box tasks created for the Build and Test phase:

- Identify the affected systems (setting up the affected Configuration Items (CIs) – hardware and software
- Create release (software development, software push, hardware install, or other similar tasks) – software only

Release Management: Release Training phase

For some releases, it is necessary to train users, Service Desk staff, technical support teams, and operations staff in the skills and abilities they require to perform their job roles. Training may take the form of classroom training courses, workshops, or seminars, but it could also be provided by on-demand computer-based training (CBT) courses and simulations.

The Release Manager identifies and describes the type of training to be provided, when the training is to be conducted, how it is to be administered, and the reasons for providing it. A detailed estimate of cost may also be required.

This phase is optional if the impact is low. When you select **1-Enterprise** or **2-Site/Dept** for the impact during the Assess phase, the training phase is mandatory.

Release Management: Distribution Authorization phase

All stakeholders of the release review the Build and Test results and approving / abandoning / rejecting the planned activities. If the Build and Test results are approved, the release manager updates the implementation schedule and authorizes the release distribution to commence. In case the release is rejected the Release Manager either initializes a rework of the build and test / training activities or is abandoning the release.

Release Management: Release Distribution phase

After acceptance training and testing is complete, the next step is to distribution and roll out the release. The process depends on the type and nature of the release and on the selected release mechanism. In all cases, however, the Release Manager should be provided with status reports and, where appropriate, tools and technologies that enable tracking and monitoring of deployment progress. As changes are made to IT components during deployment, corresponding changes are made to the configuration items and relationships modeling them in the CMDB.

The following is a summary of the activities in the Distribution and Rollout phase:

- Selects a suitable deployment group for the release
- Ensures users are adequately trained in a timely manner for the maximum benefit from the release
- Deploys the release into the production environment
- Reviews the deployed release, taking into account feedback and comments from all parties involved

- Passes the feedback to the change manager for the change review process

Release Management has one or more out-of-box Update System tasks created for the Distribution and Rollout phase.

Release Management: Release Remediation phase

If the release fails to meet expectations or if serious problems are encountered during deployment, Problem Management may be required to help identify and diagnose the root cause of the problem. If a suitable fix or workaround can be found, it should be documented and a Request for Change (RFC) created to deploy the fix or workaround into the production environment. If no fix or workaround is found, it may be appropriate to use the Remediation procedures to remove the release from that environment. If the release needs to be backed out, this should also be recorded, including any information that supports this decision.

The Remediation option is available during the Distribution and Rollout phase or the Verification phase.

Release Management: Release Verification phase

After the release is distributed, the Release Manager confirms that the release is working correctly before proceeding with further deployments. The Verification phase ensures that the deployment of the release is successful and meets all required criteria.

When the release distribution phase is complete, the release process should ensure that any results and information about any workarounds or Request for Changes (RFCs) raised in support of the release are recorded. If the release was remediated, this should also be recorded, including any information that supports this decision.

Release Management: Abandoned phase

You can abandon a release in any phase before the distribution phase was approved. Prior distribution the release is still in planning and design steps hence not widely used by an audience. In these phases the release might be abandoned due to various reasons. When the release is abandoned, the release process should ensure that the reasons for this decision are properly documented. Once abandoned, the release is closed. It’s not foreseen to reopen an abandoned release.

Release Management: Closure

After the verification phase the release is closed since the release cycle is completed. When you reopen a closed release, you return to the verification phase since this is

the last completed phase in the cycle.

Related topics

Release Management
Terminating a release change
Create a new change with a Release Management category

1.7.1.11.2. Terminating a release change

You can terminate a release in any phase except in the Training phase or after it has been installed or verified. If you reopen a change that was closed, you are returned to the assess phase to determine whether or not a release is still required. This behavior is different than if a release is closed. In that case, the release cycle is complete and therefore closed. When you reopen a closed release, you return to the verification phase since this is the last completed phase in the cycle.

Related topics

Release Management
Release Management phases
Create a new change with a Release Management category

1.7.1.12. Change Management and Service Level Agreements

Change Management enables you to select one or more Service Level Agreements (SLA) to relate to a change record. When you open a change, you can choose one Customer SLA for the contact, the same Customer SLA or a different one for the contact and one or more applicable Service SLAs for the contact's subscriptions to a service, or no SLAs at all. Service SLAs only apply if the change references a Business Service, the contact has a subscription to the service, and the subscription references an SLA. The following list describes the system's process for relating SLAs to a change record.

- If one SLA is associated with the change based on the contact, the Customer SLA is added to the change.
- If the contact has an Individual Subscription for the Configuration Item (CI), the Service SLA from that subscription is added to the change.
- If the contact has a Department Subscription for the CI, the Service SLA from that subscription is added to the change.
- If the contact has neither, then no Service SLA is added to the change.

The SLAs should contain all Service Level Targets (SLTs) that define the business rules for all response and availability metrics. You can choose as many SLTs as necessary to describe your response or availability commitment. If necessary, you can add more SLTs that meet your criteria.

When you view the new record, the SLT section lists the SLTs related to the change.

Related topics

Change Management overview

1.7.2. Change Management user roles

The out-of-box Change Management user role descriptions use these security roles, application profiles, and capability words. In order to use Change Management, you must assign the appropriate user role descriptions to your personnel. If necessary, you can modify these user role descriptions or add new ones.

User role	Security roles	Contract Profile	Configuration Profile	Capability Words
E-CAB	• DEFAULT • emergency group	DEFAULT	DEFAULT	• partial.key • change request • change task • problem management • expedite change • service desk • user.favorites • incident management • EditContacts • data administrator • public.favorites • data administrator • OCMQ • OCMO • OCML

User role	Security roles	Contract Profile	Configuration Profile	Capability Words
Change Analyst	• DEFAULT • change analyst change • change analyst tasks		DEFAULT	• partial.key • change request • change task • problem management • expedite change • service desk • user.favorites • incident management

User role	Security roles	Contract Profile	Configuration Profile	Capability Words
Change Approver	• DEFAULT • change approver		DEFAULT	• partial.key • change request • change task • problem management • expedite change • service desk • user.favorites • incident management

User role	Security roles	Contract Profile	Configuration Profile	Capability Words
Change Coordinator	• DEFAULT • change coordinator change • change coordinator tasks		DEFAULT	• partial.key • change request • change task • problem management • expedite change • service desk • user.favorites • incident management

User role	Security roles	Contract Profile	Configuration Profile	Capability Words
Change Manager	• DEFAULT • change manager		DEFAULT	• partial.key • change request • change task • problem management • expedite change • service desk • user.favorites • incident management • ChMAdmin • OCMQ • OCML

1.7.3. Change Management workflows and user tasks

Change Management controls the process to request, manage, approve, and control changes that modify your organization's infrastructure. This managed infrastructure includes assets such as, network environments, facilities, telephony, and resources. For user requests for products and services, refer to Request Fulfillment overview.

Change Management automates the approval process and eliminates the need for memos, E-mail, and phone calls, based on the following workflows.

Note

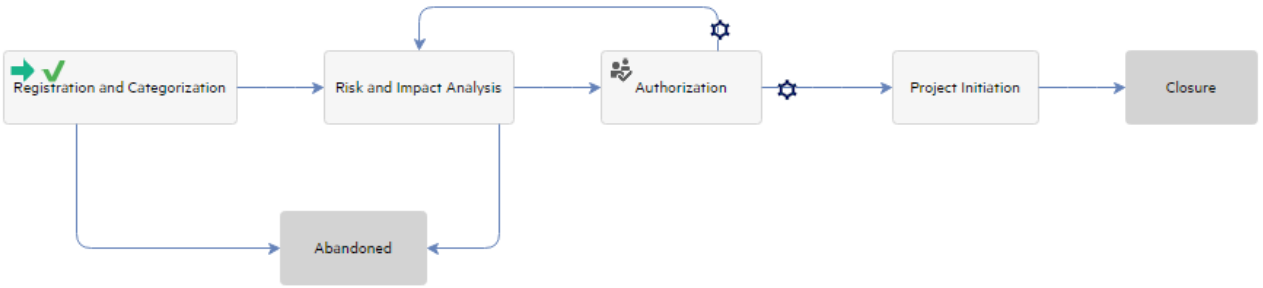
The Standard, Normal, and Emergency change workflows have an implementation phase, which is named Execution, Deployment, or Implementation, respectively. The names are by design and selected to match the types of activities for the flows as specified in ITIL. The implementation phase in each workflow is named differently because the activities are different.

Change Proposal (ST 2.0)

Change proposals are submitted to change management before chartering new or changed services in order to ensure that potential conflicts for resources or other issues are identified. Authorization of the change proposal doesn't authorize implementation of the change but simply allows the service to be chartered so that the service design activities can commence.

A change proposal is used to communicate a high-level description of the change. This change proposal is normally created by the service portfolio management process and is passed to change management for authorization. In some organizations, change proposals may be created by a programme management office or by individual projects.

The following figure depicts the Change Proposal workflow in Process Designer.



The following table lists the user tasks during each phase of the workflow.

Phase	Tasks
Registration and Categorization	• Create a new change record • Update a change request • Return a change request to the requester • Assign a change owner for a change request
Risk and Impact Analysis	• Assess the risk and impact of a change
Authorization	• Approve a change request
Project Initiation	• Create a change request from a change proposal
Closure	• Review and close a change
Abandoned	• Abandon a change request

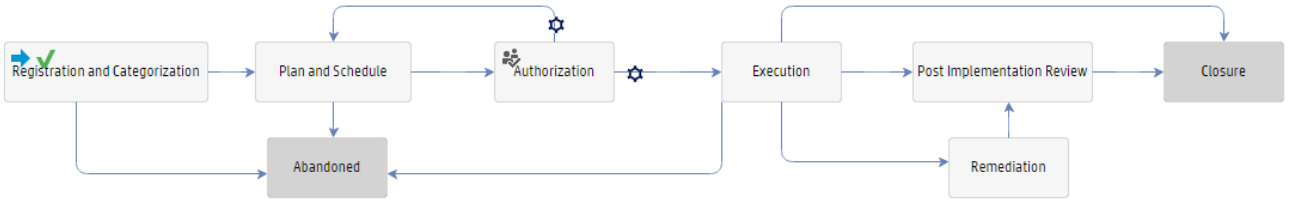
Standard Change (ST 2.2)

A Standard Change is a pre-authorized change that follows a standard procedure, for example, a password reset or the provision of standard equipment to a new employee. Standard Change uses the Change Model configured in the system which can pre-populate the information in the change record on registering the Change.

For an activity to be accepted as a Standard Change, the following requirements must be met:

- The documented tasks must be commonly known and proven
- Authority will be given in advance based on predetermined criteria
- The chain of events can be initiated by a functional Service Desk
- Budgetary approval will typically be predetermined or within the control of the Change Requester

The following figure depicts the Standard Change workflow in Process Designer.



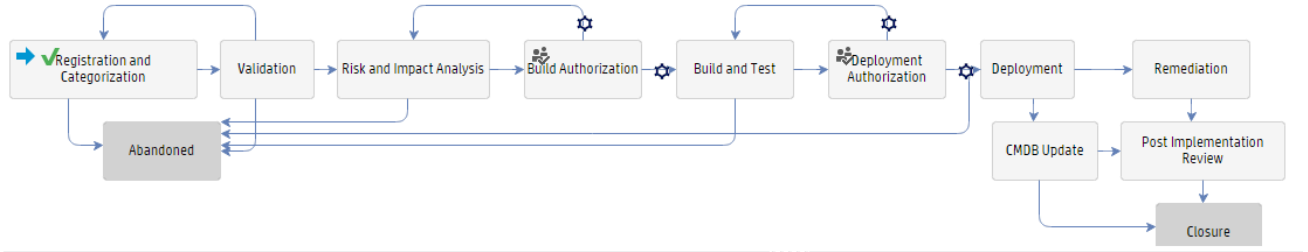
The following table lists the user tasks during each phase of the workflow.

Phase	Tasks
Registration and Categorization	• Create a new change record • Update a change request • Return a change request to the requester • Assign a change owner for a change request
Plan and Schedule	• Prioritize a change request • Plan and schedule a Standard change
Authorization	• Approve a change request
Execution	• Implement a Standard change • Cancel open tasks for a change record • Close a change task
Abandoned	• Abandon a change request
Remediation	• Review and close a change
Post Implementation Review	• Review and close a change
Closure	• Review and close a change

Normal Change (ST 2.3)

A Normal Change is a change that's categorized, prioritized, planned and that follows all approvals before deployment. The Normal Change activities describe the steps necessary to process a Normal Change by coordinating work effort. Normal Change can be further categorized as Major and Minor. Both Major and Minor changes go through the same workflow except that, for a Normal Major RFC the TCAB Approval phase is a manual transition & the change has to be approved manually by TCAB members. In a Normal Minor RFC the TCAB Approval phase is auto approved, this phase is optional in Minor RFC but it is still configured in Out of Box. The Change must be appropriately reviewed, approved, and executed successfully through the Normal Change process at least once prior to acceptance.

The following figure depicts the Normal Change workflow in Process Designer.



The following table lists the user tasks during each phase of the workflow.

Phase	Tasks
Registration and Categorization	• Create a new change record • Update a change request • Return a change request to the requester • Assign a change owner for a change request
Abandoned	• Abandon a change request
Validation	• Validate a Normal change
Risk and Impact Analysis	• Assess the risk and impact of a change
Build Authorization	• Approve a change request
Build and Test	• Build and test a change
Deployment Authorization	• Approve a change request
Deployment	• Implement a Normal change • Cancel open tasks for a change record • Close a change task
Remediation	• Review and close a change

Phase	Tasks
CMDB Update	• Update the CMDB for associated configuration items
Post Implementation Review	• Review and close a change
Closure	• Review and close a change

Emergency Change (ST 2.4)

An Emergency Change can also be initiated in the Incident Management process. They should be used only to repair an IT service error that's negatively impacting the business at a high level of severity. Changes that are intended to make an immediately required business improvement are handled as normal changes, although they may be assigned a high priority based on the urgency of the required business improvement.

The emergency change process follows the normal change process, except for the following:

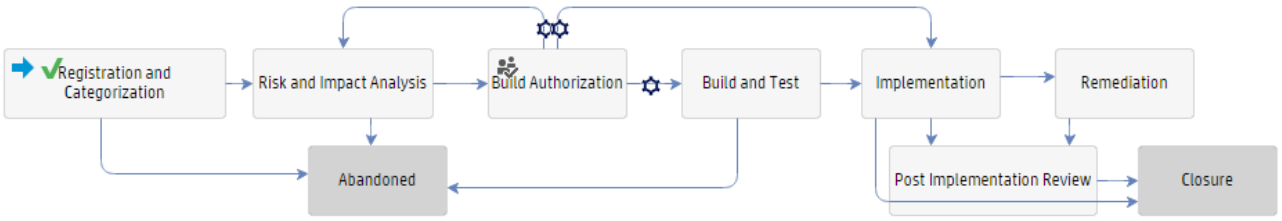
- Approval is given by the Emergency Change Approval Board (E-CAB) instead of waiting for a regular CAB meeting.
- Testing may be reduced, or in extreme cases eliminated, if doing so is considered necessary to deliver the change immediately.
- Updating of the change request and configuration data may be deferred, till normal working hours.

If the E-CAB decides to handle an emergency change as a normal change, the emergency change is re-categorized and implemented using the Normal Change process.

The following user roles are involved in Emergency Change Handling:

- Change Manager
- E-CAB
- Change Owner

The following figure depicts the Emergency Change workflow in Process Designer.



The following table lists the user tasks during each phase of the workflow.

Phase	Tasks
Registration and Categorization	• Create a new change record • Update a change request • Return a change request to the requester • Assign a change owner for a change request
Risk and Impact Analysis	• Assess the risk and impact of a change
Build Authorization	• Approve a change request
Build and Test	• Build and test a change • Use Task Planner to plan change tasks
Abandoned	• Abandon a change request
Implementation	• Implement an Emergency change • Cancel open tasks for a change record • Close a change task
Remediation	• Review and close a change
Post Implementation Review	• Review and close a change

Phase	Tasks
Closure	• Review and close a change

Related topics

Change Management user roles

1.7.3.1. Create a new change record

User Roles: Problem Manager, Change Coordinator, Release Manager

Sometimes you will need to create a new change request, which is not automatically created from an existing record. Depending on how your administrator has configured the Change Management global settings, you open a new change either by selecting a change model or by selecting a change category. By default, you open a new change by selecting a change model.

To create a new change request, follow these steps:

1. Click **Change Management** > **Create New Change** in the System Navigator.

   Or

   From the To Do Queue, select **Change** in the **Queue** field, and click **New**.

   Note

   If your home page is not the To Do Queue and you have no access to Change Management menu items, click **Miscellaneous** > **My Preferences** to change your home page to **To Do Queue**, and then log off and log back in.

2. If a list of change models is displayed (which are grouped by change category and subcategory), drill down the change categories to select a change model for the new change request. If a list of change categories is displayed, select a change category for the new change request.

   Tip

   If you are creating a new change based on change models, you can use the search function on the form to quickly locate the change model that you want to use.

   Service Manager displays a new form, automatically assigns a change ID, and populates the following fields with values based on your previous selections:

   - Phase
   - Category
   - Change Requester

The **Change Model** and **Subcategory** fields are empty if you selected a change category in this step. If you like, you can click **More** or the More Actions menu and select **Apply Change Model** to select a change model. The **Category** and **Subcategory** fields are automatically updated based on the change model you apply.

3. Update the **Change Requester** field if you are creating the change request on behalf of someone else.
4. Complete the following mandatory fields:
    - Title
    - Requested End Date
    - Reason for Change
    - Service
    - Impact
    - Urgency
    - Description
    - Effect of not Implementing
5. Optionally, in the **Scope** field, enter a description of the change scope.
6. Select an Assignment Group, and a Change Coordinator.
7. If you need to upload an attachment, click the **Attachments** tab, and then click **Add Files** to upload one or more attachments.

    Note

    To remove an uploaded attachment, select the attachment file, and click **Remove**.

8. Complete the form with other required information and any additional information, and click **Save**.
9. Service Manager automatically calculates the Priority value based on the Impact and Urgency values you selected. A notification of the change is sent to the selected change initiator.

    Service Manager automatically populates the **Priority** field with a value that's calculated based on the Impact and Urgency values you selected.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Change Management workflows and user tasks

1.7.3.2. Create a change request from a change proposal

User Roles: Change Coordinator

When a change proposal enters the Project Initiation phase, you need to create a change request based on the change proposal.

Tip

In Knowledge Management, the Reference document type has the following two document subtypes for Change Proposals:

- **Business Case**: provides a business justification for a change proposed by a formal Change Proposal
- **Project Information**: provides information about the project proposed by a formal Change Proposal

To create a change request from a change proposal, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change proposal you want to update.
3. Click the **Related Records** tab.
4. In the **Link Type** field, select **Related Changes**.
5. Click **Link New Record**.
6. Create a new change request based on the change proposal.

   Once the new change request is created, the change request and change proposal are associated with each other on their **Related Records** tabs.

Related topics

Change Management workflows and user tasks

1.7.3.3. Create a new change with a Release Management category

Applies to User roles: Change Manager, Change Coordinator

Change Management provides an out-of-box change category named Release Management, which is based on the out-of-box Release Management workflow. You create a new change either from a change model (by default) or from a change category depending on how your system administrator has set up Change Management.

Before you can create a new change with a Release Management category, your Change Management must meet either of the following prerequisites:

- Change Management is configured to open new changes from change models, and has a Change Model defined for the Release Management category; the Release Management category has one or more subcategories defined.
- Change Management is configured to open new changes from change categories and has already a Release Management change category defined.

To open a new change with a Release Management category, follow these steps:

1. Click **Change Management** > **Create New Change**.
2. Perform one of the following steps depending on your Change Management settings:
    - Select a change model that has the Release Management category and a subcategory defined.
    - Select **Release Management** from the list of change categories.
3. Select the release type (hardware or software) from the **Release Type** list.
4. Fill in the all required fields and other additional fields as needed.
5. Click **Save**.
6. Click **Next Phase**.

    The Assess phase is closed and the change advances to the Plan and Design phase for the out-of-box workflow.

1.7.3.4. Apply a change model to an existing change record

User Roles: Change Manager, Change Coordinator, Change Requester

Change Management provides an **Apply Change Model** option that you can use to apply a change model to an existing change record, which may or may not have already a change model specified.

After you apply a new change model to a change record, the following behaviors apply to the change record:

- The change automatically enters the first phase of the workflow associated with the new change model.
- The field values defined in the change model will overwrite the original values in the change record.
- All tasks in the waiting phase are automatically canceled because no one has worked on these tasks yet; the tasks after the waiting phase must be manually canceled before applying the change model, so that you are aware that applying the new change model will disrupt your original change process as you might have already started working on some tasks.

To apply a change model to a change record, follow these steps:

1. In the System Navigator, click **Change Management** > **Search Change**, and then click **Search**. Or, click **Change Management** > **Change Queue**.
2. Open the change record to which you want to apply a new change model.
3. Click **More** > **Apply Change Model**.

   Note

   In the out-of-the-box setting, this option is available only for change records that use the Project Proposal, Emergency Change, Normal Change, or Standard Change workflow and are in a phase before Authorization. If the change record has open tasks, Service Manager prompts you to cancel all the open tasks manually before applying a change model.

4. Select the desired change model from the list.

5. Click **Yes** to apply the change model.

   Now you have applied the selected change model to the change record. The change phase moves to the first phase of its workflow.

Related topics

Configuring change models

1.7.3.5. Update a change request

Applies to User roles: Problem Manager, Change Coordinator, Release Manager

You may get notifications (for example, by telephone or email) when you need to update a Change. For example, you may have assigned the Change to the wrong Change Coordinator or the Change Manager may disagree with the impact assessment or risk categorization and request an update.

To update a change request, follow these steps:

1. Click **Change Management** > **Search Changes** from the System Navigator.
2. Locate the change request by performing a search.
3. On the **Activities** tab, select an appropriate type in the **New Update Type** field, and then type a description of your update in the **New Update** field
4. Modify the applicable fields and then click **Save**.
5. Click **More** or the **More Actions** icon and then select **Change Phase**. Double-click the applicable phase, and click **Yes** to confirm the phase change.
6. Click **Save**.

Note

You can add parts and labor costs tracking to an Incident, Problem, Change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Or, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Searching records
Create a new change record

1.7.3.6. Update multiple change requests

Applies to User roles: Change Manager

You can update more than one change request at a time by using the Mass Update template form or the Complex Update function.

To update multiple change requests by using the Mass Update template form, follow these steps:

1. From your To Do queue, select **Change** in the **Queue** list and then select the desired view.

   Alternatively, click **Change Management** > **Search Changes** to use the Search to generate a record list. Use search or advanced search to find one or more records.

2. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
3. From the **List** menu, select **Mass Update**.
4. Double-click the field you want to update.
5. Type the value for the field in the text box or use the Fill feature to display a list of potential values for the field.
6. Click **Next**.
7. Continue updating fields and click **Execute** when you complete your updates.

To update multiple change records by using the Complex Update function, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and and then select the desired view.
2. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
3. From the **List** menu, select **Mass Update**.
4. Click **Complex Update**.
5. Type the RAD code in the **Complex Update** form.
6. Click **Execute**.

1.7.3.7. Update multiple task records

Applies to User roles: Change Manager

You can update more than one task at a time by using either the Mass Update template form or by using the Complex Update function, which enables you to enter RAD codes to further customize your update.

To update multiple task records by using Mass Update, follow these steps:

1. Click **Change Management** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
4. Click **Mass Update**. The **Mass Update** wizard opens.
5. Double-click the field you want to update.
6. Type the value for the field in the text box or use the Find feature to display a list of potential values for the field.
7. Click **Next**.
8. Continue updating fields and click **Execute** when you complete your updates.

To update multiple tasks by using Complex Update, follow these steps:

1. Click **Change Management** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
4. Click **Mass Update**.
5. Click **Complex Update**.
6. Type the RAD code in the **Complex Update** form.
7. Click **Execute**.

1.7.3.8. Return a change request to the requester

Applies to User roles: Change Coordinator

If the information in the change form is incomplete, you may contact the requester to get the missing information and then enter it in the **Updates** section of the change record. If the change is assigned to the wrong support group, you should return it to the requester by changing the phase back to Change Registration and Categorization and then contacting the requester to have the requester correct the change record. The requester then places the request in the next phase of the change workflow.

To return a change request to the requester, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list, and then select **My Open Changes** from the **View** list.
2. Click a change request from the record list.
3. Open the **Updates** section and select the type of update in the **New Update Type** field.
4. Type the reason for returning the change in the **New Update** field.
5. Click **More** or the **More Actions** icon and then select **Change Phase**.
6. Double-click the **Registration and Categorization** phase.

   The change is returned to this phase.
7. Contact the requester (by telephone or email) to let the requester know that the change was returned to the Registration and Categorization phase and that the requester must correct it.

Related topics

More Actions menu
Create a new change record

1.7.3.9. Abandon a change request

User Roles: Change Coordinator, Change Owner

You may need to abandon a change request that's in the Registration and Categorization, Validation, Risk and Impact Analysis, or Build and Test phase. For example, you may need to abandon a standard change that should have initially been handled using the request fulfillment process or because the change is invalid. The reason for the abandonment could be that the change request is not logical or feasible, is not in line with company standards and policies, or it has already been raised and rejected in an earlier phase.

Note

You can abandon a change request only when its change workflow has an **Abandoned** phase defined.

To abandon an existing change request, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to abandon.
3. Click **More** or the More Actions menu.
4. Select **Abandon**.
5. Confirm that the currently selected subcategory is correct. If not, select the correct one.
6. Select a closure code, and enter your closure comments.
7. Click **Finish**.

   The phase of the change record changes to **Abandoned**, and the closure code and comments you entered appear on the **Closure** tab.

Related topics

Create a new change record

1.7.3.10. Assign a change owner for a change request

User Roles: Change Coordinator

When you receive a change request that's in the Registration and Categorization phase, you need to assign a change owner for it.

To assign a change owner for a change request, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to update.
3. In the **Change Owner** field, specify a person who will be the change request owner.
4. Click **Save**.
5. Click a button that reflects the name of the transition associated with the Registration and Categorization phase. For example, for a Change Proposal, click the **Request Risk and Impact Analysis** button.

   The change request moves to the next phase in the change workflow.

Related topics

Create a new change record

1.7.3.11. Assess the risk and impact of a change

User Roles: Change Manager, Change Coordinator

When a change moves to the Risk and Impact Analysis phase, you need to assess its risk and impact.

> **Note**
>
> Only Change Proposals, Normal and Emergency changes have a Risk and Impact Analysis phase in their workflows.

To assess the risk and impact of a change, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to update.
3. Depending on the type of Change, follow the steps below to move the Change to the next phase.

If this is a Change Proposal or an Emergency change, do the following:

1. Update the **Impact** field as needed.
2. Complete the **Financial Impact** field if needed.
3. In the **Risk Assessment** field, select a value from the drop-down list.

   This field becomes read-only if you have selected the **Enable Risk Assessment** checkbox. You can either click the **Calculate Risk** button next to this field or click **More** > **Calculate Risk** to calculate the risk. See Calculate risks for more information.
4. (For an Emergency change only) Deselect the **Build and Test Required** checkbox if you want to the change to bypass the Build and Test phase of the out-of-the-box workflow.

   > **Note**
   >
   Service Manager selects this checkbox by default for Emergency changes.

5. Complete the mandatory fields: **Scheduled Implementation Start**, **Scheduled Implementation End**, **Implementation Plan**, and **Remediation Plan**.
6. (For an Emergency change only) Click the **Change Details** tab, and deselect the **Review Required** checkbox if you don't want to force the change to go through the Post Implementation Review phase of the out-of-the-box workflow.

   > **Note**
   >
   > This checkbox is unavailable for a Change Proposal. Service Manager selects this checkbox by default for an Emergency change. For more information about this setting, see Mark a change as Review Required.

7. Click **Request Authorization**.

If this is a Normal change, do the following:

1. Update the **Risk Assessment** field as needed.
2. Update the **Impact** field as needed.
3. Complete the **Financial Impact** field if needed.
4. Deselect the **Review Required** checkbox if you don't want to force the change to go through the Post Implementation Review phase.

   > **Note**
   >
   > Service Manager selects this checkbox automatically for a Normal change. For more information about this setting, see Mark a change as Review Required.

5. Complete the **Remediation Plan** field.
6. Click **Request Authorization**.

Related topics

Mark a change as Review Required
Plan and schedule a Standard change

1.7.3.11.1. Mark a change as review required

User Roles: Change Manager, Change Coordinator

When a Normal or Emergency change moves to the **Risk and Impact Analysis** phase or a Standard change moves to the **Plan and Schedule** phase, you can mark the change as review required by selecting a **Review Required** checkbox.

- When this option is selected, the change must go through the Post Implementation Review phase of the out-of-box workflow. Additionally, when the change record enters the Post Implementation Review phase, the system automatically sends a notification to the Change Coordinator and Change Manager.
- When this option is not selected, the change can bypass this review phase.

Note

This checkbox appears in the **Risk and Impact Analysis** phase or the **Plan and Schedule** phase, and becomes read-only after this phase.

The following table describes how the **Review Required** settings affect the change workflows.

Workflow	Review Required is selected	Review Required is not selected
Normal Change	This is the default setting. A **Close** button is unavailable in the CMDB Update phase.	A **Close** button is available in the CMDB Update phase, which allows the change to directly advance to the Closure phase.
Emergency Change	This is the default setting. A **Close** button is unavailable in the Implementation phase.	A **Close** button is available in the Implementation phase, which allows the change to directly advance to the Closure phase.
Standard Change	A **Close** button is unavailable in the Execution phase.	This is the default setting. A **Close** button is available in the Execution phase, which allows the change to directly advance to the Closure phase.

To mark a change as Review Required, follow these steps:

1. If it is a Normal or Emergency change, you do so as part of the risk and impact analysis task. For details, see Assess the risk and impact of a change.
2. If it is a Standard change, you do so as part of the planning and scheduling task. For details, see Plan and schedule a Standard change.

Related topics

Assess the risk and impact of a change
Plan and schedule a Standard change

1.7.3.12. Prioritize a change request

User Roles: Change Owner

When a change requester creates a change request, a change owner is specified. Once the change request is reviewed for completeness and accuracy, the change owner then needs to prioritize the change request.

To prioritize a change request, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to prioritize.
3. Update the **Impact** and **Urgency** values of the change request.

   The change request's **Priority** value is automatically changed according to the Impact and Urgency values you selected.
4. Click **Save**.

Related topics

Create a new change record
Assess the risk and impact of a change

1.7.3.13. Plan and schedule a Standard change

User Roles: Change Owner

Once a Standard change is prioritized, the change owner needs to plan the resources required and also review revises and update the schedule for the Standard change as needed.

To plan and schedule a change, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the Standard change record you want to update.
3. Click **Request Plan and Schedule**. The change moves to the Plan and Schedule phase.
4. Complete the following fields:
    - Scheduled Implementation Start
    - Scheduled Implementation End
    - Implementation Plan
    - Remediation Plan
5. On the **Change Details** tab, perform these steps:
    1. Select the **Review Required** checkbox if you want to force the change to go through the Post Implementation Review phase of the out-of-box workflow. By default, this checkbox is not selected.
    2. Complete other required fields as needed.
6. Click **Save**.
7. If the scheduled change conflicts with an existing Freeze or Maintenance period that's defined in Time Period Management, a warning occurs. If you want, you can adjust the schedule and plans:
    1. Click **OK** to close the warning.
    2. Go to **Time Period Management** > **Search Time Periods**.
    3. Search for time periods whose category is **Local Maintenance Window** or **Regional Freeze Period**.

4. Adjust the schedule and plans of the Standard change according to your search results.
5. Click **Save**.

8. If the change has affected configuration items, add them in the **Affected Configuration Items** field.
9. If the affected configuration items have a scheduled down time, select the **Configuration Item(s) Down** checkbox, and complete the **Scheduled Downtime Start** and **Scheduled Downtime End** fields.
10. Click **Save**.
11. Set the expected change states of the attributes of the affected configuration items. To do so, click **More** or the More Actions menu and select **Set Expected States**, or click the **Associated CIs** tab and then click **Add Attributes/Relationships**.

    A wizard opens. Follow the instructions described in Set the expected state of a configuration item.
12. Click **Request Authorization**.

    The Standard change moves to the Authorization phase.

Related topics

Approve a change request

1.7.3.14. Use Task Planner to plan change tasks

User Roles: Change Coordinator, Change Manager

By using Task Planner from a change record, you can plan change tasks for the current or future phases in the ongoing change.

> **Note**
>
> You are recommended to use Task Planner from a change record to plan tasks for it. Using the **Change Management** > **Create New Task** option or the **Create New Task** option from the **More** menu of a change record is not recommended.

To plan change tasks in the task planner from a change record, follow these steps:

1. Do one of the following in the System Navigator:
    1. Click **Change Management** > **Change Queue**
    2. Click **Change Management** > **Search Changes**, enter your search criteria, and then click **Search**.
2. Find the change request you want to create tasks from, and then double-click the record to open it.
3. Click the **Tasks** tab, and then click the **Edit** button.

    The task planner graphical interface supports panning. You can click **Zoom in** or **Zoom out** in the toolbar or use the mouse wheel to zoom in or zoom out the task planner graphical interface.
4. Do the following:
    1. Click **Add Task** on the toolbar to add a task to the task planner.
    2. Type a task name in the **Title** field.
    3. Select a task category from the drop-down list.
    4. Optionally, select a task template from the drop-down list.
    5. From the **Open In Phase** drop-down list, select a change phase in which the task will be created.
    6. Optionally, from the **Close By Phase** drop-down list, select a change phase by the end of which the task must be closed.

> **Note**
>
> The Close By Phase must be no earlier than the Open In Phase. If left blank, the Close By Phase defaults to the Open In Phase. If the Open In Phase in the change task you just planned is same as the current phase of the change record, the change task is created with a Task Number assigned. Otherwise, the task will not be created until the change reaches the Open In Phase defined for this task.

7. In the **Task Condition** field, specify a condition under which the task is to be created by using the Condition Editor. For details, see Using the Condition Editor.

> **Note**
>
> You can configure the **Mark as required/Set properties as read-only** option only when planning a task in a change model. If you select this option, you can't modify or delete the task from a change record of this change model, which is indicated by a red dot at the upper left corner of the task box in task planner (see Add a task to a change model). When you plan a task in a change record, this setting is not configurable and not used.

5. Click **Add Task** to add another task.

   Or

   To create a sequential task, hover the mouse on the new task added in the graphical interface until you see a hand symbol, and then drag the mouse pointer to add a new task.

6. If needed, resize the task planner graphical interface by expanding the details form or resizing the task planner window.

7. If needed, specify an existing task as dependent on another task by adding a connector between two sequential tasks.

> **Tip**
>
> You can reposition the workflow by moving the outline boxes of the tasks or by clicking **Auto Layout** to allow task planner to automatically arrange the tasks and dependencies.

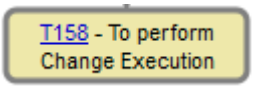
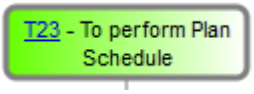
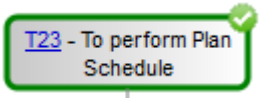
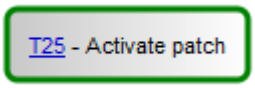
8. Click **Save** to save the planned tasks.

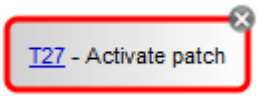
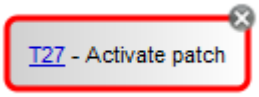
   Once a task is created, you can't delete this task or modify its properties in task planner. You can click the task number in graphical interface to modify its details.

   If a task is not marked as required in the change model and has not been created, you can click **Delete** to remove this planned task, or you can modify its properties. However, you can't move the Open In Phase of a planned task to a phase prior to the current phase of the change.

   If you see a red dot on the top left corner of a task, it means that this task can't be modified or deleted from the change record, because the **Mark as required/Set properties as read-only** option is selected for this task in the change model.

   The following table describes various status values of a task and their graphic styles in task planner.

Task Status	Graphic Style	Remarks
0-Planned		
0-Planned (opened)		This style appears when a task is created but still has a Planned status because it is dependent on another task to complete.
1-Ready		
2-Assigned		
3-In Progress		
4-Blocked		
20-Completed		
21-Completed with problem		

Task Status	Graphic Style	Remarks
30-Canceled		
31-Withdrawn		
32-Failed		

Related topics

Create a new change record

1.7.3.15. Cancel open tasks for a change record

User Roles: Change Coordinator, Change Manager, Change Analyst

If a change record has more than one open task in the current change phase and you want to cancel some or all of them, you can cancel them one by one using the **Cancel Task** option or cancel some or all of them at once by using the **Mass Cancel** feature.

To cancel an open task of a change record, follow these steps:

1. Click **Change Management** > **Search Changes**, enter your search criteria, and then click **Search**.
2. Locate the change record by performing a search.
3. Click the **Tasks** tab of the change record, and then click the open task.

   The task record opens.
4. Click **More**, and then select **Cancel Task**.
5. Select a Closure Code and enter your Closure Comments.
6. Click **Finish**.

   The task phase and status are changed to **Canceled**.

To cancel multiple open tasks for a change record, follow these steps:

1. Click **Change Management** > **Search Changes**, enter your search criteria, and then click **Search**.
2. Select a change, and then click **More** > **View Opened Tasks**.

   A list of tasks appears. These tasks were created either before or in the current change phase and aren't yet closed.
3. Select the tasks that you want to cancel, and then click **Mass Cancel** from the task bar.
4. Click **Yes**.

   The Closure Code is automatically set to **6 – Canceled**, and the Closure Comments field is populated with text "**Mass Canceled.**"
5. Update the Closure Comments field if needed, and then click **Finish**.

The status of each selected open task is set to **Canceled**, and the automatic transition in the task workflow moves the task phase to **Canceled**.

Related topics

Use Task Planner to plan change tasks

1.7.3.16. Validate a Normal change

User Roles: Change Coordinator

Once a Normal change request is created, the change coordinator needs to validate the change request. If the validation fails, the change request is returned to the Registration and Categorization phase; if the validation passes, the change request is moved to the next phase.

To validate a Normal change request, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the Normal change record you want to validate.
3. If the change request is still in Registration and Categorization phase, click **Request Validation**. The change request advances to the Validation phase.
4. Review the change request.
5. If the change request is invalid, do the following:
    - Click **Validation Rejected**. The change record returns to the **Registration and Categorization** phase.
    - Contact the change requester (by email or phone) and let the requester know that the change was returned to the Registration and Categorization phase and that the requester must correct it.
6. If the change request is valid, click **Validation Accepted**. The change record moves to the Risk and Impact Analysis phase.

Related topics

Use Task Planner to plan change tasks
Assess the risk and impact of a change

1.7.3.17. Relate a change to another record

Applies to User roles: Service Desk Agent, Problem Manager, Change Coordinator

You can relate a change record to another existing or new Incident, Problem, Known Error, Request, Service Catalog item or Change record. When a record is related to a change, the system tracks the history of this activity. This includes the changes that are related or dropped and the history of phase changes and transitions.

In the Related Records section, you can view the current related records, associate an existing record with the change record, or create a new record that's automatically related to the change being viewed. When you associate a change record with another record, note the ID number of that record so you can verify it in the **Related Records** section.

To relate a change record to an existing record, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the change record that you want to relate to another existing record.
3. On the **Related Records** tab, select one of the following link types:
    - **Caused By Changes**: indicates that the current change is caused by the change it will be linked to.
    - **Related Changes**: indicates that the current change is related to the change it will be linked to.
    - **Caused By Incidents**: indicates the current change is caused by the incident it will be linked to.
    - **Related Incidents**: indicates the current change is related to the incident it will be linked to.
    - **Related Requests**: indicates the current change is related to the request it will be linked to.
    - **Caused Known Errors**: indicates the current change will cause the known error it will be linked to.
    - **Caused Problems**: indicates the current change will cause the problem it will be linked to.
    - **Solved Known Errors**: indicates the current change will solve the known error it will be linked to.
    - **Solved Problems**: indicates the current change will solve the problem it will be linked to.
    - **Related Service Catalog Items**: indicates that the current change is related to the Service Catalog item it will be linked to.

4. To relate the change record to an existing record, click **Link Existing Record**. A form is opened, allowing you to link the record in either way as follows:
    - Type the record ID in the text box and then click **OK**.
    - Click the **Search** link. In the search form that's displayed, specify your search criteria and then click **Search**. From the search results, select the record ID that you want. The selected record ID appears in the text box. Click **OK**.

    Service Manager automatically adds the selected record to the Related Records list.
5. To relate the change record to a new record, click **Link New Record**. Follow the new record creation process based on the selected link type to create a new record. The new record is automatically linked to the current change record once created.
6. Click **Save**.

Related topics

Unrelate a change to another record

1.7.3.18. Unrelate a change to another record

Applies to User roles: Service Desk Agent, Problem Manager, Change Coordinator

You can relate a change record with another existing or new Incident, Problem, Known Error, Request, or Change record. If the situation changes you can remove the relationship between the change record and the other record.

To unrelate a change record to an existing record, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the change record that you want to unrelate to another existing record.
3. On the **Related Records** tab, select a linked record.
4. Click **Unlink Record**.
5. Click **Yes** to confirm the deletion.

Related topics

Relate a change to another record

1.7.3.19. Approve a change request

User roles: Change Approver

Once a change has advanced to any of the following phases, you need to review and then approve the change:

- A Change Proposal: the Authorization phase
- A Normal change: the Build Authorization or the Deployment Authorization phase
- An Emergency change: the Build Authorization phase
- A Standard change: the Authorization phase

To approve a change, follow these steps:

1. Sign in to Service Manager as the approver of the change. For example, Change.Approver.
2. Click **Approval Inbox**. Changes awaiting your approval are displayed.
3. Select the change you want to approve, and click **View**.
4. Review the change schedule, plans, and other information.
5. When you have finished your review, click **Cancel**.
6. Select the change you want to process, and approve or deny it according to your review result:

   To approve the change, click **Approve**. The change automatically moves to the next phase.

   To reject the change, click **Deny**, enter a rejection reason, add additional comments, and then click **OK**. The change returns to the previous phase.

Note

The Rejection Reason can be one of the following values:

Financial
Insufficient Information
Requester Not Authorized
Technical
Resource Availability
Not Aligned with Strategy
Incomplete Impact Assessments

Related topics

Approval options in Change Management
Update an active approval delegation
Delegate approvals to another operator
Disable an active approval delegation
Copy an approval delegation

1.7.3.20. Build and test a change

Applies to User roles: Change Coordinator, Change Manager

Once a Normal or an Emergency change has entered the Build and Test phase, you need to have a Build and Test plan in place, and then complete the tasks based on the plan. When the tasks are completed, you need to update the change record with the Build and Test information.

To build and test a Normal or an Emergency change, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the Normal change you want to update by performing a search.
3. Check that all change tasks that are associated with the Build and Test phase are completed.
4. Go to the **Tasks** tab, and close any open tasks. For details, see Close a change task.

   Caution

   You can't close the change if it has any open tasks.

5. Continue with the following steps depending on the type of Change.

If this is a Normal change, do the following:

1. Complete the following fields:
   - Scheduled Implementation Start
   - Scheduled Implementation End
   - Implementation Plan
   - Remediation Plan
   - Build and Test Plan
   - Build and Test Result
2. If the test result is unacceptable, click **More** or the More Actions menu and select **Abandon** to abandon the change.
3. If the result is acceptable, click **Request Authorization** to advance to the Deployment Authorization phase.

4. If Service Manager reports that the scheduled implementation period conflicts with any Time Period freeze window, make adjustments to the schedule.

   The Normal change is now awaiting approval.

If this is an Emergency change, do the following:

1. Complete the following fields:
   - Build and Test Plan
   - Build and Test Result
2. If the test result is unacceptable, click **More** or the More Actions menu and select **Abandon** to abandon the change.
3. If the result is acceptable, click **Request Implementation** to advance to the Implementation phase.

The next step is to implement the change.

Related topics

Implement a Normal change
Implement a Standard change
Implement an Emergency change
Close a change task

1.7.3.21. Implement a Standard change

Applies to User roles: Change Coordinator, Change Manager

Once a Standard change is approved, it enters the Execution phase. When the change is implemented and all change tasks are closed, you need to update the change record with actual implementation information.

To implement a Standard change, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the Standard change by performing a search.
3. Check that all change tasks that are associated with the Execution phase are completed.
4. Go to the **Tasks** tab, and close any open tasks. For details, see Close a change task.

   Caution

   You will not be able to close the change if it has any open tasks.

5. Complete the following fields:
   - Actual Implementation Start
   - Actual Implementation End
   - Implementation Comments
6. Complete or update other additional fields if needed.
7. Click **Save** to save the change.

   Next, you need to review and close the change.

Related topics

Change Management workflows and user tasks
Close a change task

1.7.3.22. Implement an Emergency change

Applies to User roles: Change Coordinator, Change Manager

Once an Emergency change has entered the Implementation phase, you need to implement the change. Once the change is implemented and all change tasks are closed, you should update the change record with actual implementation information.

To implement an Emergency change, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the Emergency change by performing a search.
3. Check that all change tasks that are associated with the Implementation phase are completed.
4. Go to the **Tasks** tab, and close any open tasks. For details, see Close a change task.

   Caution

   You will not be able to close the change if it has any open tasks.

5. Complete the following fields:
   - Actual Implementation Start
   - Actual Implementation End
   - Implementation Comments
6. Complete or update other additional fields if needed.
7. Click **Save** to save the change.

   Next, you need to review and close the change.

Related topics

Change Management workflows and user tasks
Close a change task

1.7.3.23. Implement a Normal change

Applies to User roles: Change Coordinator, Change Manager

Once a Normal change has entered the Deployment phase, you need to implement the change. Once the change is implemented and all change tasks are closed, you should update the change record with actual implementation information.

To implement a Normal change, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the Normal change by performing a search.
3. Check that all change tasks that are associated with the Deployment phase are completed.
4. Go to the **Tasks** tab, and close any open tasks. For details, see Close a change task.

   Caution

   You will not be able to close the change if it has any open tasks.

5. Complete the following fields:
   - Actual Implementation Start
   - Actual Implementation End
   - Implementation Comments
6. Complete or update other additional fields if needed.
7. Click **Save** to save the change.

Next, you can click **Request CMDB Update** or **Remediation** as needed to move the change to the **CMDB Update** or **Remediation** phase, and continue to review and close the change. For details, see Review and close a change.

Related topics

Change Management workflows and user tasks
Close a change task

1.7.3.24. Update the CMDB for associated configuration items

Applies to User roles: Change Coordinator, Change Manager

The Normal change workflow contains an optional CMDB Update phase. If a Normal change affects some configuration items (CIs), after the deployment of the change, you need to update the Configuration Management Database (CMDB) for the affected CIs.

This CMDB Update process works as follows:

1. When the change is in a phase prior to Deployment (for example, the Build and Test phase), you specify the proposed changes for the affected CIs of the change.
    1. Click **Change Management** > **Search Changes**.
    2. Locate the Normal change by performing a search.
    3. In the **Affected Configuration Item** field on the **Change Details** tab, specify the affected CIs of the change if you have not already done so.
    4. On the **Associated CIs** tab, click **Add Attributes/Relationships**.

        A wizard opens, asking you to select which CIs you want to update with this change.
    5. Select one CI from the list.

        A list of modification options is displayed. By default, the **Set Attribute Name and Values** option is selected.
    6. Select an option. For example, select the default option: **Set Attribute Name and Values**.

        Note

        The subsequent steps are based on the default option. The steps for other options are similar.
    7. Specify a new value for an attribute of the selected CI.
        1. Click **Add New Field**.
        2. In the **Field in** field, select an attribute.
        3. In the **Value** box that appears, enter a new value for the attribute.

4. Click **Next**.

   The attribute name and its new value appears in the attribute list. To remove an attribute from the list, click the attribute row and click **Remove Field**.

8. Repeat the previous step to specify new values for more attributes, and then click **Finish**.

   You are returned to the Associated CIs tab, where the selected CIs and attributes are listed, along with the old and new values of the attributes. Each of the listed item has a Status of **Proposed**.

2. Let the change go through the Deployment Authorization phase and advance to the Deployment phase.

   On the Associated CIs tab, each planned attribute/relationship moves from the **Proposed** status to **Planned**.

3. When the change has been implemented, do the following in the change record:
   1. On the Associated CIs tab, click each attribute or relationship item and manually change their status from **Planned** to **Implemented**.
   2. Click **Save Changes**.

4. Click **Request CMDB Update** from the task bar in the change record.

   The change record moves to the CMDB Update phase.

5. Let the change go through the rest of the phases in the change workflow until it has reached the phase immediately before Closure.
6. On the **Associated CIs** tab of the change record, do the following:

   1. Click each attribute or relationship item, and change their status from **Implemented** to **Validated**.

      Note

      If the attribute or relationship modification is not physically completed, change the status from **Implemented** to **Validation Failed** or **Canceled**. Be aware that you can't close the change unless each of the attribute or relationship modifications is in either the Validated or the Canceled status.

   2. Click **Save Changes**.

7. Complete any required fields in the change, and then do the following to close the change:
    1. Click **Close**, select a closure code and enter your closure comments.
    2. Click **Finish**.

       The change is closed. Additionally, on the Associated CIs tab, the attribute or relationship items are changed to the **Processed** status.
8. Verify that the affected CIs of the change are automatically updated with the proposed changes.
    1. Go to **Configuration Management** > **Resources** > **Search CIs**.
    2. Locate each affected CI of the change by performing a search.
    3. If you selected **Set Attribute Name and Values** previously, open the **CI Changes** section. The attribute value change should be listed in the Historic Attribute Changes subsection with a status of **processed**, and the change record is listed as a Linked Record. Locate the CI attribute field, which has changed to the new value.
    4. If you selected **Set Relationship** previously, open the **Relationship Changes** section. The Historic Relationship Changes subsection should display a relationship that has been added for the change, and the change record is listed as a Linked Record. Go to the **Relationships** section of the change record, where you should see the new relationship that was created for this change.

Now, you have completed CMDB Update for the change. Next, you need to review and close the change. For details, see Review and close a change.

Related topics

Change Management workflows and user tasks
Review and close a change

1.7.3.25. Close a change task

User roles: Change Manager, Change Coordinator, Change Analyst

After a change task is complete, the person listed in the Assignee field of the task record can close the task.

To close a task, follow these steps:

1. Click **Change Management** > **Tasks** > **Task Queue**.

   The **Change Task** queue opens.

2. Choose the **All Open Change Tasks** or **Open Change Tasks Assigned to Me** task queue from the **View** list:

   Change Management displays the selected task queue.

3. Click the task record you want to close to view its details.
4. Change the task status to **Completed** or **Completed with Problems**.
5. Enter the **Actual Start** and **Actual End** dates.
6. Click **Close**.
7. Select the appropriate **Closure Code**.
8. Add any relevant comments in the **Closure Comments** field.
9. Click **Finish**. The change task closes.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Another way is, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Change Management workflows and user tasks
Use Task Planner to plan change tasks
Cancel open tasks for a change record

1.7.3.26. Review and close a change

Applies to User roles: Change Coordinator, Change Manager

The process of reviewing and closing a change varies with the change workflow: Change Proposal, Standard, Emergency, or Normal.

Change Proposal

Once a change request has been created from a change proposal, you can review and close the change proposal.

To review and close a change proposal, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the change proposal you want to close.
3. Click the **Related Records** tab, and open the linked change request to verify that it contains correct and sufficient information.
4. Return to the change proposal, and click **Close**.
5. Enter a closure code and closure comments.
6. Click **Finish**.

Standard, Emergency, or Normal

Once a Standard, Normal, or an Emergency change is implemented, you need to review the change implementation, and then select one of the options as described in the following table.

Caution

You can't reopen a Change Proposal, Standard, Emergency, or Normal change request once it is closed.

No.	Option	Description	Notes
1	Directly close the change	Select this option if the change is successful and can be closed immediately.	Standard and Emergency changes only
2	Perform a Post-Implementation Review (PIR) and then close the change	Select this option if you need to perform a PIR before closing the change record. The purpose of a PIR is to confirm the following items before closing the change: • The change meets its objectives. • The change requester and stakeholders are satisfied with the results • Unanticipated effects have been avoided • Lessons learned are incorporated in future changes	Standard and Emergency changes only

No.	Option	Description	Notes
3	Update the Configuration Management Database (CMDB) and then close the change	Select this option if the change is successful and you need to update the CMDB for the affected CIs before closing the change record.	Normal changes only
4	Update the CMDB and then perform a PIR and close the change	Select this option if you need to update the CMDB for the affected CIs and need to perform a PIR before closing the change record.	Normal changes only
5	Conduct remediation and then perform a PIR and close the change	Select this option when remediation is needed before performing a PIR and closing the change.	Standard, Emergency, and Normal changes

To review and close a Standard, Normal, or an Emergency change, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Locate the change you want to close.
3. Review the change implementation results, and proceed to one of the following options.
4. Option 1: Directly close the change.
    1. Click **Close**.
    2. If the subcategory being displayed is incorrect, select the correct one.
    3. Select a closure code, and add closure comments.
    4. Click **Finish**.

5. Option 2: Request a PIR before closing the change.
    1. Click **Request PIR**. The change moves to the Post Implementation Review phase.
    2. Perform a post-implementation review.
    3. In the **Review Results** field, enter your review results.
    4. Click **Close**.
    5. If the subcategory being displayed is incorrect, select the correct one.
    6. Select a closure code, and add closure comments.
    7. Click **Finish**.
6. Option 3: Update the CMDB and then close the change.
    1. Update the CMDB for associated CIs of the change. For details, see Update the CMDB for associated configuration items.

        Caution

        Make sure that all CI attribute/relationship changes (if any) are in **Validated** or **Canceled** status. If any of them are still in **Proposed** status, you will not be able to close the change later.

    2. Click **Close** to close the change.
7. Option 4: Update the CMDB and then perform a PIR and close the change.
    1. Update the CMDB for associated CIs of the change. For details, see Update the CMDB for associated configuration items.
    2. Continue as described in option 2.
8. Option 5: Conduct remediation before performing a PIR and then closing the change.
    1. Click **Remediation**.
    2. In the **Remediation Method** field, select **Full** or **Partial**.
    3. In the **Remediation Comments** field, enter your comments.
    4. Continue as described in option 2.

Related topics

Change Management workflows and user tasks

1.7.3.27. Reopen a change request or task

Applies to User roles: Change Coordinator

Sometimes you need to reopen a closed change request or a change task, to move it back to the previous phase.

Caution

You can't reopen a Project Proposal, Standard, Emergency, or Normal change request.

To reopen a change request or task, follow these steps:

1. Click **Change Management** > **Search Changes** or **Change Management**> **Search Tasks**.
2. Select **closed** from the **Status** list.
3. Use search or advanced search to find one or more records.
4. Click a record to view its detail.
5. Click the **Reopen** button.
6. Edit the record as necessary.
7. Click **Save & Exit**.

   The change request or change task changes to the reopened state.

Related topics

Close a change task
Review and close a change

1.7.3.28. View the alert log of a change

User Roles: Change Coordinator

To view the alert log of a change record, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to view.
3. Click **More** or the More Actions menu, and select **View Alert Log**.

Related topics

Create a new change record

1.7.3.29. Change the phase of a change

User Roles: Change Coordinator

You can use the **Change Phase** option to bypass some phases of a change workflow and move the change record to a specific phase.

To change the phase of a change record, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to view.
3. Click **More** or the More Actions menu, and select **Change Phase**.
4. Select the destination phase.

   The change records directly moves to the selected phase.

Related topics

Change Management workflows and user tasks
Change the category of a change

1.7.3.30. Change the category of a change

User Roles: Change Coordinator

You can use the Change Phase option to bypass some phases of a change workflow and move the change record to a specific phase.

To change the phase of a change record, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to view.
3. Click **More** or the More Actions menu, and select **Change Category**.
4. Select a change category.

   The selected change category is applied to the change record.

Related topics

Change Management workflows and user tasks
Change the phase of a change

1.7.3.31. Print a change request

Applies to User roles: Change Manager, Change Coordinator

To print a change request, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Select a record from the record list.
4. Click the **Print Page** icon at the upper right of the record detail pane.
5. Right-click the page and save the page as an HTML file.
6. Open the HTML file in your web browser and print it from the browser to a PDF file or a hard copy.

Related topics

More Actions menu
Searching records

1.7.3.32. Print a change request list

Applies to User roles: Change Manager, Change Coordinator

To print a change request list, follow these steps:

1. Click **Change Management** > **Changes** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Click the **Print Page** icon at the upper right of the record list pane. The list of changes opens in another page.
4. Right-click the page and save the page as an HTML file.
5. Open the HTML file in your web browser and print it from the browser to a PDF file or a hard copy.

Related topics

More Actions menu
Searching records

1.7.3.33. Set a reminder for a change request

Applies to User Roles: Change Manager, Change Coordinator, Change Analyst, Change Approver

From an existing change request, you can schedule a reminder to be sent to you at a specified time.

To set a reminder, follow these steps:

1. Click **Change Management** >**Search Changes**.
2. Use search or advanced search to find one or more records.
3. Click the record to view its detail.
4. Click **More** or the **More Actions** icon and then select **Set Reminder**.
5. Select one of the following to set the time when the reminder occurs:
    - Click **Remind At** to set a reminder for a particular day and time. Click **Fill** to display a calendar where you can select the date and time for the reminder.
    - Click **Remind In** to set the reminder to occur at a particular interval. If you choose this option, specify the time interval (in 00:00:00 format), and select a shift from the **Based On** list.
6. In the **Remind if** field, select an option from the list:
    - Always (the default)
    - Change is still open
    - Change is still assigned to me
7. In the **Pop-up Message** area of the form, select the type of notification you want to receive.
8. If you select **Pop-up**, type the reminder message in the **Message** field.
9. If you select **Email** or **SM Mail**, Service Manager displays a **Title** field where you can enter a title for the email message and two Message Type option buttons.
10. Select the message type option in the Message Type frame.
    - Click **Send Change Record** to include a copy of the change record in the email.
    - Click **Fixed Text** and type the message you want to appear in the email.
11. Click **OK**.

Related topics

Update a change request

Relate a change to another record

1.7.3.34. View a list of services potentially affected by an outage

Applies to User roles: Incident Coordinator, Incident Manager, Change Coordinator, Change Manager

The **View Affected Services** menu option enables you to view a list of services that are potentially affected by an outage relating to CIs that are specified in an incident or change record. By viewing the affected services list, you can determine the potential impact of an outage related to critical, dependent services when you open an incident or a change. You can also use the affected services list to plan when opening a change.

> **Note**
>
> The list of CIs generated doesn't include the CIs specified in the primary incident or change record. Also, if no outage is scheduled, potentially affected services will not be calculated.

To view a list of affected services, follow these steps:

1. Perform one of the following actions.
   - Open an existing incident or change record. Use search or advanced search to find one or more records.
   - Proceed as if you are going to add a new incident or change record.
2. If you are in the process of adding a new record, fill in the **Affected CI** field.
3. Click **More** or the More actions icon and select **View Affected Services**. The CI Identifier record opens.
4. Open the **Affected Services** section. A list of affected services opens or a message, stating that there are no services affected by the outage.

   > **Note**
   >
   > For change records, a list of all affected business services is returned in all cases, whether the record has one primary CI or multiple CIs.

5. Select a CI from the list for detailed information from the Configuration Management record.

6. When you finish viewing the affected services, click **Cancel**.

Related topics

Outages
More Actions menu

1.7.3.35. Send a notification from a task or change request

Applies to User roles: Change Coordinator

To send a notification from a task or change request, follow these steps:

1. Do one of the following to generate a record list:
    - Click **Change Management** > **Search Changes**.
    - Click **Change Management** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Click a record to view its detail.
4. Open the **More Actions** menu on the change or task form.
5. Click **Notify**.
6. Add one or more recipients to the **To** text box, or type an assignment group in the **Group** text box.

    Tip: If you have set a valid group email for members in the Assignment Groups definition, the system sends emails to this group email address.
7. Select one of the following notification methods:
    - **Mail**: Uses the internal Service Manager Mail application (not Email).
    - **Email**: Uses the Service Manager Email application.

      If you select **Email**, you can specify the names of the recipients in the **CC** and **BCC** fields as needed. In addition, you can click **Attachment** > **Add Files** to add attachments to the email message as needed.
8. Modify the subject and the message in the notification as needed.
9. Click **Send**.

Related topics

More Actions menu
Searching records

1.7.3.36. Manually calculate Time Period conflicts for a change

User Roles: Change Manager, Change Coordinator

> **Note**
>
> This feature applies only when Service Manager Calendar and the Time Period Management functionality are used.

When planning a change in the out-of-box system, if you enter a **Scheduled Implementation Start** and a **Scheduled Implementation End** value and then the change record is saved, Service Manager automatically calculates possible conflicts of the time range you enter with existing Time Period freeze and maintenance windows. If Service Manager has detected any conflicts, it displays a warning and turns on the **Time Period Conflict** flag in the change record.

> **Note**
>
> By default, the **Disable the Pop-up Window to warn time period conflicts** option is not selected in the Time Period global settings, which means Time Period conflict warning messages are displayed in a pop-up window by default; when this option is selected, such warnings are displayed only in the message history box.

Automatic Time Period conflict calculation

Time Period conflict calculation is defined in the Calendar Mapping record for the Change object. To view this record, follow these steps:

1. Navigate to **System Administration** > **Ongoing Maintenance** > **Calendar Administration** > **Calendar Mappings**.
2. In the Object field, select **Change**.
3. Click **Search**. The Calendar Mapping record for Change opens.
4. Scroll down to the Time Period Mapping section, and view the following settings:
    - Field for Start Date
    - Field for End Date

- Time Period Mapping Query
- Query for Time Period Conflict Calculation

For a description of each setting, see Display records from an application module in the calendar.

Manual Time Period conflict calculation

Automatic time period conflict calculation is triggered only when changes are saved to the fields in a change record that are mapped to **Field for Start Date** and **Field for End Date** in the Time Period Mapping for Change. There are times when Time Period conflicts need to be manually calculated. For example, when there are updates made to Time Period records or to the Time Period Mapping definition for the Change object, or when you have changed the Category of a change record while the Category field is being referenced in the **Query for Time Period Conflict Calculation** expression, you need to manually trigger Time Period conflict calculation.

To manually calculate Time Period conflicts for a change record, follow these steps:

1. Click **Change Management** > **Search Changes** in the System Navigator.
2. Locate the change record you want to view.
3. Click **More** or the More Actions menu, and select **Calculate Time Period Conflicts**.

   Note

   This option is available only for a change that's not closed and is eligible for Time Period conflict calculation based on the settings in the Calendar Mapping for the Change object.

   Service Manager recalculates Time Period conflicts for the change record and performs the following actions:

   - Displays the following message when the recalculation is complete: `Recalculation of time period conflicts complete`.
   - Clears or turns on the **Time Period Conflict** flag in the change record according to the recalculation result.

Related topics

Calculate time period conflicts

1.7.3.37. Access Change Management reports

User Roles: Change Manager; Change Coordinator

The Reporting tool in Service Manager provides a number of out-of-box reports on the change data in your system. You can view these reports through the out-of-the-box dashboards named **CAB/eCAB Dashboard (Global)**, **Change KPI (Global)**, **Change Overview (Global)**, and **Change Risk Analysis (Global)**. You can also create your own dashboards to display other reports of your interest.

To access Change Management reports, follow these steps:

1. Click **Change Management** > **Change Overview**.

   By default, the **Change Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. If needed, click the **New Dashboard** button to add your own dashboards. For details, see Create a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default dashboard.

1.7.3.38. Assess and calculate risks

In this release, Service Manager provides the out-of-box (OOB) risk analysis function. After you enable the **Risk Assessment** function, click the **Calculate Risk** button or click **More** > **Calculate Risk** to calculate the risk involved of each Change based on the risk factors that you defined in Service Manager.

A **Risk Factor** is defined as an element to help calculate the risk of a change request. Each risk factor contains a score that contributes to the final risk result. A risk factor has two natures:

- **User Based**: This type is basically a question that has to be manually answered by the change implementer. The risk score is dependent on the answer.
- **System based**: This type triggers an action in Service Manager (such as a query or a JavaScript) to calculate some results without manual intervention.

You need to create and manage your required risk factors, which can be reused in risk groups later on.

A **Risk Group** is an element that bundles several risk factors together. A Risk Group is evaluated against which changes the risk factors run. You can define the mapping between the risk factor scores and the final risk result in a risk group.

After the calculation is completed, the system opens a **Risk Result Report** and displays the latest version of the risk result. This report provides an overall summary of the risk analysis for the selected change request, including the creator of the report, total score, risk assessment result, all related risk factors and the corresponding answers, collision status, and individual score for each risk factor in this report. This information enables you to pinpoint the most significant factors contributing towards the risk level of the selected change request. See the following screenshot as an example:

Risk Result Report - C10071

Created By: falcon

Risk Result Version: Version - 2 - 07/14/19 20:56:02 (Latest Version)

Total Score	**154**
Risk Assessment	**5 - Very High Risk**

Risk Score Definition	Risk Assess Definition
0 - 10	0 - No Risk
11 - 30	1 - Low Risk
31 - 60	2 - Some Risk
61 - 90	3 - Moderate Risk
91 - 140	4 - High Risk
141 - 999	**5 - Very High Risk**

Risk Factor	Answer	Collision	Score
How well is the implementation (incl. implementation plan) automated?	Partiallly automated	-	10
How and to what extent is the change tested?	Manually tested	-	20
What is the complexity of the change?	Medium complexity	-	10
Is support from other departments required for the implementation of the change?	-	-	0
Is one of the payment streams impacted by the change?	Yes	-	30
Has this change been tested in an environment that simulates Production and all critical/high defects closed?	Yes	-	1
Is this a change that has failed previously without root cause being identified and remediated ?	Yes, a similar change failed before	-	50
What is the impacted user base of the Affected Service?	0 to 50	No	1
Is this a change that impacts the availability of a SOX application ?	-	No	1
How many CIs are affected by this change?	Less than 10 CIs affected	No	10
How many other Business Services are impacted?	No other service impacted	No	1
Is there a conflict with a freeze/maintenance window?	Maintenance window conflict	Yes	20
Is there a conflict with other changes for the same implementation time and services?	No conflict or 1 conflict.	No	0
Total Score			**154**

If there are any collisions, the system documents this details in the Change record, which is also visualized in the Change Calendar.

1.7.3.38.1. Create a risk factor

User Roles: Change Administrator, System Administrator

A risk factor is an element that helps to calculate the risk of a change request. Each risk factor contains a score which contributes to the final risk result. You need to create and manage your required risk factors, which can be reused in risk groups later on.

Follow these steps to add a risk factor:

1. From the system navigator, click **Change Management** > **Configuration** > **Risk Factor**.
2. Click **New**.
3. Select either **User Based** or **System Based** as the Assessment Type.
    - **User Based**: This type is basically a question which has to be manually answered by the change implementor. The risk score is dependent on the answer.
    - **System Based**: This type triggers an action in Service Manager (such as a query or a JavaScript) to calculate some results without manual intervention.
4. Follow these steps if you select **User Based**:
    1. Type a question in the **Question** field.
    2. Select the **Mandatory** checkbox to make the current question obligatory. Otherwise, the change implementer can ignore the question.
    3. Select an answer type to be displayed on the questionnaire.

        If you select **Drop-down List**, **Check Box**, or **Radio Buttons**, you must provide multiple answers and the related scores as well. Each answer must have its own unique risk score.

        Tip

        - A change implementer can provide multiple selections for the **Check Box** answer type during a risk calculation process.
        - The score for the **Comment** answer type is 0.
5. Follow these steps if you select **System Based**:

1. Type a question in the **Question** field.
2. Specify a default score for the question.
3. Fill in the table in the **Score** section as necessary.

Field	Description
Score	Specify a score for the linked condition.
Condition	Specify a condition to flexibly build the automated risk evaluation. You can run RAD expressions to perform complex calculations by using JavaScript. For example, `nullsub(tpconflict in $L.file,0)=1`. `$L.file` is the change record that's related to risk assessment.
Label	Type an answer for the question, which will be displayed in the risk result report.
Collision	Select whether to mark the condition as a collision or not. As soon as a condition of type collision is valid, the system sets a flag for the change record in the Service Manager Calendar.

4. Select a table from the **Detail Table** drop-down list, and then type a query in the **Detail Query** field. You can configure these two settings to associate a risk factor with the related records in Service Manager. Later, you can view a list of these records by clicking the risk factor's link in a Risk Result Report.
6. Click **Save & Exit**.

Edit a risk factor

Follow these steps to edit a risk factor:

1. From the system navigator, click **Change Management** > **Configuration** > **Risk Factor**.
2. Click **Search** to view a list of existing risk factor records.
3. Select a risk factor record and then update it as necessary. Note that this risk factor will be updated in all linked risk groups as well.
4. Click **Save** and **OK**.

Localize a risk factor

Follow these steps to localize a risk factor:

1. Configure the language in Service Manager.

   Some languages may already be enabled on the Service Manager side. Others may not be enabled. To verify whether your target language is enabled, or to enable your target language, follow these steps:

   1. From the Service Manager client, expand the left navigation and click **Tailoring** > **Database Manager**.
   2. From the **Table** field, type or select **language**.
   3. Click the **Search** icon.

      Service Manager returns a list of Service Manager language files.

   4. Double-click the desired language file.
   5. From the **Language Identification** dialog, click the **Search** icon.

      A list of available languages appears. If the language you want has an active flag set to true, the language is enabled, and your task is complete. If the flag is false, proceed to the next step.

   6. Select the language that you want to enable.
   7. Check the empty **Active for logins** checkbox.
   8. Click **Save** to update the language record.
   9. Click **OK**.
   10. Repeat step 1 through step 9 to add more languages.
   11. Click **Back** to exit this task.
   12. Restart the Service Manager client to verify your changes.

2. Configure the user profile.

   This step is to associate a default language with the individual user in their Service Manager user profile. For users in a multi-lingual environment, one

might prefer to search the catalog in French, while another user would be more comfortable with Spanish.

Note

If the user doesn't exist in the Service Manager system, you can follow these steps to navigate to the Operator Record dialog, but you must create the user before you complete the language association task.

1. In the **System Navigator** pane, click **System Administration** > **Ongoing Maintenance** > **Operators** to edit the user's operator record.
2. Click **Search**.
3. Select the user from the **Login Name** or **Full Name** column.
4. Click the **Login Profile** tab.
5. Click the **Language** drop-down list to select the default language for that user.
6. Click **Save**.
7. Click **OK**.
8. Repeat step 2 through step 8 if you have more than one operator record to configure.
9. Click **Back** to exit this task.

3. Select a risk factor record.
4. From the detail list toolbar of the selected record, click **More** > **Localize this record**. Service Manager displays the Create Localized Data wizard.
5. Choose the language from the drop-down list, and then click **Next**.
6. Replace the English question and answers with equivalent values in the language you selected in the previous step.
7. Click **Finish**.

1.7.3.38.2. Create a risk group

User Roles: Change Administrator, System Administrator

A Risk Group is an element that bundles several risk factors together. A Risk Group is evaluated against which changes the risk factors are executed. You can define the mapping between the risk factor scores and the final risk result in a risk group.

Follow these steps to add a risk group:

1. From the system navigator, click **Change Management** > **Configuration** > **Risk Group**.
2. Click **New**.
3. In the **Owner** field, select an owner who has the ownership for the risk group record.
4. In the **Title** field, enter a unique name for the risk group.
5. In the **Description** field, enter descriptive text for the risk group.
6. Click **Edit** under the **Usage Condition** field to open the Condition Editor. You can build a condition or a set of conditions to link a risk group to a change type, against which the assessment is executed. The target table is cm3r.

   For example, you can specify that the risk group only applies to a Change record whose category is Release Management. The **Condition** field is populated with the settings you specified. In this case, the system displays `(Category in CurrentRecord = "Release Management")` in the **Usage Condition** field.
7. Specify scores and then select the corresponding risk level in the **Score/Risk mapping** table as necessary.
8. Add one or more risk factors to the risk group as necessary.

   Tip

   The **Total Scores (min/max)** field displays the minimum and maximum scores derived from the risk factors. When you add or remove the risk factors from the risk group, the system recalculates the minimum and maximum total scores.

   - To add an existing risk factor:

     1. Click **Link Existing Factor**.

2. In the search form that's displayed, enter your search criteria and then click **Search**.
3. Select the checkboxes of the risk factors that you want to link, and then click **Select**. The selected risk factors are added to the list.

- To create a new risk factor and link it to the risk group, click **Add Factor**. The system opens the **Risk Group Tailoring - Add a Risk Factor** page. See Create a risk factor for detailed instructions.
- To remove a risk factor from the group, select the checkboxes before the risk factor, and then click **Remove Factor**.

9. Click **Save**.
10. (Optional) Click **Preview** to get a preview of the questionnaire. Note that the system displays the User Based risk factors only.
11. Click **Activate**. The risk group moves to the Active phase.

    After you create a new risk group and save it, the risk group is in the Draft phase and therefore not available for risk calculation. If the risk group is ready for use, you must activate it to make it available to users.

> **Note**
>
> You can activate multiple risk groups. If there are multiple risk groups available for risk calculation, the system automatically selects the first risk group in sequential order.

Edit a risk group

Risk groups are managed by a workflow of three phases: Draft, Active, and Inactive. You can edit a risk group only when it is in the Draft phase. If a risk group is in the Active or Inactive phase, you have to move it back to the Draft phase before you can edit it.

Follow these steps to edit a risk group:

1. From the system navigator, click **Change Management** > **Configuration** > **Risk Group**.
2. Click **Search** to view a list of existing risk factor records.
3. Select a risk group record and then update it as necessary.

   - If the risk group record is in the Inactive phase, click **Revise** to move it to the Draft phase.

- If the risk group record is in the Active phase, click **Inactivate** and then click **Revise** to move it to the Draft phase.
- If the risk group record is in the Draft phase, edit it directly.

4. Click **Save**.

1.7.3.38.3. Calculate risks

User Roles: Change Manager, Change Coordinator

You can calculate the risk involved in the implementation of each change based on the risk factors that you defined in Service Manager.

Follow these steps to calculate risks:

1. Enable the risk assessment function.

   To do so, click **Change Management** > **Administration** > **Settings**. Select the **Enable Risk Assessment** checkbox, and then click **Save** and **OK**.

   The system displays a **Calculate Risk** button next to the **Risk Assessment** field. In addition, the **More** > **Calculate Risk** option is also available.

   Note

   - The **Calculate Risk** button is only available in the first phase when the **Risk Assessment** field is mandatory for a Change. This may vary for different Change categories. For example:
     - Normal Change: Validation phase
     - Standard Change: Plan and Schedule phase
     - Change Proposal and Emergency Change: Risk and Impact Analysis phase
     - Release Management: Assess Release phase
   - The **More** > **Calculate Risk** option is available when the status of a Change is Open. This option is unavailable in the Abandoned, Cancelled, and Closed phases.
   - The calculate risk function is unavailable for Unplanned Change and Subscription Change.

2. Create a series of risk factors. See Create a risk factor for detailed instructions.
3. Create one or more risk groups. See Create a risk group for detailed instructions.
4. Open a Change Proposal or an Emergency Change, and then click the **Change Details** tab.

5. Click the **Calculate Risk** button next to the **Risk Assessment** field or click **More** > **Calculate Risk**. The system automatically assigns an active risk group that matches the usage condition as you defined to perform the risk calculation.

   If the risk group consists of user-based risk factors, the system opens the **User Based Risk Factor** page. Provide answers to the questionnaire, and then click **Calculate**. Note that the questions with asterisks are mandatory.

   If the risk group consists of system-based risk factors only, the system automatically evaluates the risk.

6. After the calculation completes, the system opens the **Risk Result Report** page and displays the latest version of the risk result by default. This report provides an overall summary of the risk analysis for the selected change request, including the creator of the report, total score, risk assessment result, all related risk factors and the corresponding answers, collision status, and individual score for each risk factor in this report. You can click the link of a risk factor to view a list of conflicted records. This information enables you to pinpoint the most significant factors contributing towards the risk level of the selected change request.

   If there are any collisions, the system sets a flag in the Service Manager Calendar for this Change record. If you have configured the Detailed Table and Detailed Query settings for a risk factor, you can hover the mouse over the **Risk** or **Collision** column for the Change record, and then click the **View Risk Result** link to access the Risk Result Report.

   If you have calculated the risks several times before, select a historical risk result report from the **Risk Result Version** list to view the details of a previous version.

7. Click **Back** to exit the report. Later, you can click **More** > **Show Risk Result** to view the risk result again.

1.7.3.38.4. Tailor the customized forms for risk calculation

User Roles: System Administrator, Change Administrator

In this release, Service Manager provides the risk assessment and calculation function by default. However, to use the risk calculation function on your customized formats or Change workflows, you must complete the following tasks.

- Task 1: Add the **Calculate Risk** button to your format
- Task 2: Set the Risk Assessment field to read-only

Add the Calculate Risk button to your format

This task describes how to add the Calculate Risk button to your customized forms.

Follow these steps:

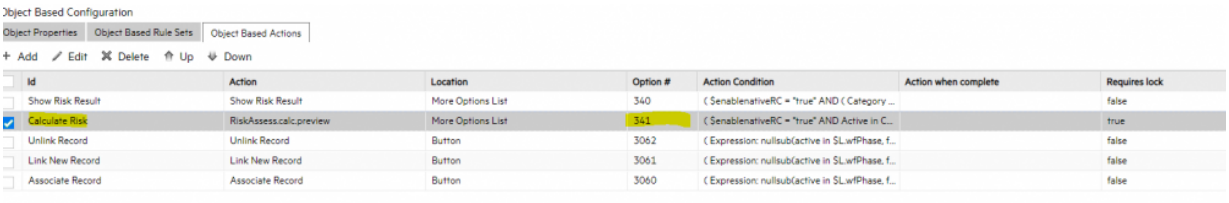
1. Sign in to Service Manager as a system administrator.
2. Type `fd` in the Service Manager command line, and then press Enter.
3. In the Form field, type the format name to which you need to add the Calculate Risk button. For example, `chm.subform.detail.reg`.
4. Click **Search**.
5. Click **Design**.
6. Place a button next to the Risk Assessment field.
7. Specify the format properties as follows:

   Visible Condition: `[$enablenativeRC]=true & <your customized conditions>`

   Image File: Image filename for the button.

   Enabled: Select this checkbox.

   ButtonID: The option ID (Option #) of Calculate Risk in the Change Workflow Actions. As illustrated in the following sample screenshot, the default Option ID is 341.



8. Click **OK**.

Set the Risk Assessment field to read-only

This task describes how to set the risk assessment field to read-only. To use the risk assessment and calculation function, you must set this field to read-only. Later, when working with this function, the system automatically fills the **Risk Assessment** combo box after you calculate the risks. You no longer need to select a value in this combo box manually.

Follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Type `fd` in the Service Manager command line, and then press Enter.
3. In the Form field, type the format name you'd like to update the read-only property of the Risk Assessment field.
4. Click **Search**.
5. Click **Design**.
6. Select the Risk Assessment combo box.
7. In the **Read-Only Condition** property field, type `[$enablenativeRC]=true`.
8. Click **OK**.

Note

If you have tailored the Change workflow, you can modify the **Show Risk Result** and **Calculate Risk** action conditions as required in Object Based Action.

1.7.4. Change Management administration

Change Management administration provides the capability to control the Configuration Management application functionality, manage change groups, and create change request based on change model, category, or subcategory.

Change Management administration consists of the following:

- Change Environment: Change Management contains an environment record for changes. This record contains an option that defines the functionality of the Change Management application for all Change Management users.
- Groups: This enables you to group operators to be a member or approver of change groups.
- Settings: You can create a change request based on the change model or category by selecting the appropriate option in the Change Settings page. You can also enable sending email notifications through Event Messages to Change management.
- Task Environment: Change Management contains an environment record for tasks. This record contains an option that defines the functionality of Change Management application for all Change Management users.

View the Change Management background processor

Applies to User roles: System Administrator

The default system start-up record (startup) lists all background agents, or processes, that start up each time you start Service Manager. Most specify 60-second wake-up intervals. This list includes despooler, report, alert (for Incident Management), change, availability, agent, marquee, lister, linker, and event. To add any other start-up records to the system startup record, type the new start-up records information at the bottom of the agent array on the system start-up record. The system startup default record processes records that are in the schedule file. The appropriate background agent picks up the related schedule record for processing.

To view the Change Management background processor, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Startup Information**.

2. In the Type field, type change.startup. This is the default start-up record for the Change Management background processor.
3. Use search or advanced search to find one or more records.

Configure Change Management assignment groups

Applies to User roles: System Administrator

Change Management assignment groups enable you to group operators who can be members or approvers of the same group. You can create change assignment groups to contain members and approvers who operate on a common area of responsibility.

Service Manager sends a notification to members of the corresponding groups whenever a new change request is created or an existing change is modified. Approvers of a group can accept or deny the approval requirement of a change if at least one of their security roles has the **Can Approve** right for the Change area.

To add a change group, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, enter the name of the manager of the assignment group. Select the work schedule of the assignment group from the **Calendar Name** drop-down list. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide response when you create and populate a new assignment group.

(Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.

9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.

10. Follow these steps to configure the Supported Services field:

    a. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.

    b. Select **Business Service** in the CI Type field, and then click **Search**.

    c. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.

    d. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.

11. Click **Save** and **OK**.

Configure global settings for Change Management

Applies to User roles: System Administrator

Change settings are global settings that are applied to the entire Change Management module.

To configure change settings, follow these steps:

1. Click **Change Management** > **Administration** > **Settings** in the System Navigator.

   Ignore the **Enable Legacy Event Messages** checkbox. The Process Designer change workflows don't use legacy event messaging. Instead, notifications are called from RuleSets.

2. To specify the default link file for mapping change model fields to change records, enter the name of the link file in the **Default Open/Apply Link** text box.

   Note

   You can view and reconfigure the field mappings defined in the link file. For details, see Configuring change models.

3. To enable risk assessment, select the **Enable Risk Assessment** checkbox.

   Note

   Service Manager provides the risk analysis function by default. However, to use the risk calculation function on your customized formats or Change workflows, you must perform additional manual configurations. For details, see Tailor the customized forms for risk calculation.

4. To enable change model filtering based on a script function you defined, enter the script function name in the **Change Model Restriction Script Function** text box. For example, ChangeModelRestriction.getRestrictionSql.

   Note

   The specified script function is used to filter change models that you can see when creating a new change or applying a change model to a change, according to the conditions defined in the script.

   To define the query conditions in the script, click **Tailoring** > **Script Library** in the System Navigator, search for the script or create a new script, and then define the script function for change model filtering.

5. To enable users to create new change requests based on change categories, select **Category** under **Select On Open**.

6. To enable users to create new change requests based on change models, select **Change Model** under **Select On Open**.

   Note

   Be default, **Change Model** is selected.

7. To configure the change model hierarchy, type up to four column names to group the change model in the **Group Change Model by** text box. For example, to group the change model by category and template, type category, changeTemplate.

   Note

   • The change model hierarchy configuration takes effect only when **Change Model** is selected under **Select on Open**.
   • Use the column names specified in the `changeModel` table.
   • You can configure a maximum of four levels in the change model hierarchy by specifying the columns in a comma-separated list.

8. Click **Save**.

   Note

   You can create or edit custom settings by selecting **Add New Setting** or **Edit Settings** from the **More** menu.

Enable Change Manager to prompt reason for relating and unrelating a change

Applies to User roles: System Administrator

The screlconfig record controls whether or not a text box appears in which the operator can type a reason for relating or unrelating a change record to another change record in Change Management. The Relation Configuration Record form for Change Management (the cm3r table) contains two checkboxes: **Prompt Reason for Relating** and **Prompt Reason for Unrelating**. When these checkboxes are selected, the system prompts the operator with a text box in which the operator can provide a reason for relating or unrelating a change to another change.

To configure the screlconfig record for Change Management history, follow these steps:

1. Type db in the command line field.
2. Type screlconfig in the **Form** field on the **Database Manager** page.
3. Use search or advanced search to find one or more records.
4. Click **Search** again to display a specific list of records with Relation Configuration Records.
5. Select **cm3r** from the record list to display the **Relation Configuration Record** form in Change Management.
6. Make the changes to the **Prompt Reason for Unrelating** and **Prompt Reason for Relating** checkboxes.
7. Click **Save** and **OK**.

Configuring Change Management workflows

A change workflow is a sequence of connected steps in the life cycle of a change record. In the workflow, a change record goes through several phases to complete the life cycle. A change workflow phase may consist of one or multiple tasks to be completed before the change record can proceed to the next phase.

Similarly, a change task workflow is a sequence of connected steps in the life cycle of a change task.

Change Management uses the following out-of-box workflows that are ITIL aligned.

Change workflows

- Project Proposal
- Standard Change
- Normal Change
- Emergency Change

Task workflows

- Generic Task

The change and task workflows are associated with change categories and change task categories. The workflows are applied when you create change and change task records based on the categories.

Access Change Management workflows

User Role: System Administrator

Change Management workflows (including change workflows, change task workflows, and change model workflows) are accessible from the **Change Workflows** menu item.

To access Change Management workflows, follow these steps:

1. Click **Change Management** > **Configuration** > **Change Workflows**.

   All Change Management workflows are displayed in the list.

2. Click **Change Management** on the taskbar, and select the **Change**, **Change Management Configuration**, or **Change Tasks** filter to display a specific type of workflow.

   > **Tip**
   >
   > You cannot modify the workflows marked as **Proprietary**. Instead, you can create your own workflows by copying them and making necessary modifications.

3. To view a workflow, double-click it from the list.

Create a Change Management workflow

User Roles: System Administrator

Each change category or change task category must have a workflow associated with it. Before you can create a change category or change task category, you must create a workflow for it. You can create a new workflow from scratch, or copy an existing workflow and make necessary modifications.

> **Tip**
>
> Change Management has the following out-of-box workflows, which can't be modified but can be used as a template when creating your own workflows: Change Proposal, Emergency Change, Normal Change, Standard Change, and Generic Task.

To create a change workflow or change task workflow from scratch, follow these steps:

1. Create a basic workflow.

    a. Navigate to **Change Management** > **Configuration** > **Change Workflows**.

    b. Click **New**. The **Workflow properties** window opens.

    c. Type a name and description for the workflow.

    d. Select the table to associate with the workflow. For example, select **cm3r** for Change, or **cm3t** for Change Task.

    e. Click **OK**, and then click **Save**. The new workflow appears in the workflow list if you click **Change Workflows**.

    f. Log off and log back in.

2. Add phases to the workflow. For details, see Add a phase.
3. Create transitions between the phases. For details, see Create a default workflow transition, Create a manual workflow transition, and Create an automatic workflow transition.
4. Click **Save**.

To create a change workflow or change task workflow from an existing one, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Copy Existing Workflow**.
2. Select a change or change task workflow from the list.

    Tip

    You cannot modify the workflows marked as **Proprietary**. Instead, you can create your own workflows based on them.

3. Follow the instructions provided by the workflow cloning wizard to make a copy of the selected workflow. For details, see Copy an existing workflow.
4. Make necessary modifications as you do when creating a new workflow from scratch.

1.7.5. Change Management configuration

Change Management configuration provides the capability to configure the following items:

- Change categories and subcategories
- Task categories
- Change models
- Change workflows
- Change approvals and alerts
- Change messages

1.7.5.1. Configuring Change Management categories

Change Management categorizes changes and change tasks by change category and task category.

Each change category has a change workflow associated with it and can have subcategories. Each task category has a task workflow associated with it and available change phases defined for it.

To access change categories, click **Change Management** > **Configuration** > **Change Categories,** and then click **Search**.

Each task category contains a number of change phases and has a workflow associated with it. To access task categories, click **Change Management** > **Configuration** > **Task Categories,** and then click **Search**.

Service Manager provides a number of out-of-box change categories and task categories that you can use or customize.

Create a change category or task category

User Roles: System Administrator

You can create new category records either by copying and modifying an existing record or by creating a new one. Service Manager provides out-of-box category records that you can use or modify.

To create new category record, follow these steps:

1. Navigate to **Change Management**> **Configuration** > **Change Categories** or **Task Categories**.
2. Click **New**.
3. Type the name of the category.
4. Type a category description.
5. In the **Availability** field, type true.

> **Note**
>
> If you type false, the category will not be available to select from the category list when you open or update a change.

6. Select a workflow for the category.
7. Select a template for the category if required.
8. Optionally, select the **Assign number before commit?** checkbox so that Service Manager assigns an identification number .
9. Click **Save** to view the **Workflow** tab and other associated tabs.
10. For change category, click the **Subcategories** tab to create subcategories for the change category.
11. For task category, click the **Available Phases** tab and select one or more change phases from the drop-down list to associate the task category with the change phases. The newly created task category definition will be applied to the change workflow phases.

    If you leave **Available Phases** blank for a task category, you can use this task category in all change phases.

    Note

    The list of available phases is retrieved from all of the change workflows.

12. Click **Save**.

    Note

    To delete the category, click **Delete**.

Add a new subcategory

User Roles: System Administrator

Subcategories are a way to divide a Change Category into smaller groups. For example, the **Hardware** category can have the **Install Hardware** and **Move Hardware** subcategories.

To add a subcategory, follow these steps:

1. Click **Change Management** > **Configuration** in the System Navigator.
2. Click **Subcategories**.
3. Type a category name in the Category text box.

Or

Click the **Fill Field Category** icon to select a change category.

4. Type the subcategory name in the **Subcategory** text box.
5. Type the subcategory description.
6. Click the **Fill Field** icon in the **Company** text box to associate a company record with the subcategory.
7. Click **Save** to add the new subcategory of a change category.

Add a new subcategory from a change category

User Roles: System Administrator and Implementer

You can add a new subcategory directly from a change category. This reduces the number of access points required for you to have in order to create a subcategory. You can view a list of subcategories and their record details associated with the current category.

1. Click **Change Management** > **Configuration** in the System Navigator.
2. Click **Change Categories** > **Search**.
3. Select the Change Category for which you want to add a subcategory.
4. In the Change Category Definition page, click the **Subcategories** tab.
5. Click **Add subcategory**.
6. Type a subcategory name.
7. Type a description for the subcategory.
8. Click the **Fill** icon for the **Company** text box to associate one or more company records with the subcategory.
9. Click **Save** to add the new subcategory.

   Service Manager creates the new subcategory and adds it to the change category.

Related topics

Change Management configuration

1.7.5.2. Configuring change models

A change model is a record that's used to pre-define the contents of a specific type of Request for Change (RFC), including the information used to populate the RFC and tasks needed to complete the change. When you open a change request using a change model, all information is added automatically to the change without any action needed by the operator.

When creating a new change model, you can define the tasks to be completed and the dependencies between multiple tasks. When you create a new change, a list of change model templates appears. In the change model template, change models are grouped by category and subcategory. You can apply a change model to a change using the change model template.

Create a new change model

User Roles: Change Coordinator, Change Manager

To create a new change model, follow these steps:

Template 1

Field	Value
Description	Install Windows 11 on new employee's laptop
Brief Description	Install Windows 11
Assignment Group	(empty)

Template 2

Field	Value
Description	(empty)
Brief Description	Install Windows 10
Assignment Group	Hardware

Merge Result

Field	Value
Description	Install Windows 11 on new employee's laptop
Brief Description	Install Windows 10
Assignment Group	Hardware

Click **Change Management** > **Configuration** > **Change Models** in the System Navigator.

1. Click **New**.
2. Type the change model ID.
3. Type the change model name.
4. Type the change description.
5. Click the **Fill Field Category** icon in the **Category** text box to select a change category.

   Note

   After the change model is created, you can't change the category specified in the change model. If you want to change the category, you can create a new change model.

6. Click the **Fill Field Subcategory** icon in the **Subcategory** text box to select a change subcategory.
7. Select a change template from the drop-down list.

   If needed, you can specify more than one template. When users open a new change from this model, Service Manager automatically applies all specified templates in the order listed by merging their field values. The following table lists two example templates and their merge result.
8. Select a wizard from the drop-down list.

> **Note**
>
> The wizard helps you to accomplish certain tasks when it executes. When you open a change model from a change, the wizard associated with the change model executes. For example, you can run a wizard to prompt the user for a device type when adding a device record.

9. Complete the form with all of the required information.
10. Click **Save.**

Add a task to a change model

User Roles: Change Manager

Using Task Planner, you can add, edit, or delete tasks, which are part of change model record. In the graphical interface, you can specify tasks as dependent on other tasks by adding a connector between two tasks. Tasks that don't have dependencies may be open in parallel but are still governed by the **Open In Phase** and **Close By Phase** attributes, as well as the task conditions.

Change task attributes

When adding a task to a model, specify the following attributes:

- Title
- Task Category
- Task Template
- Open In Phase – Species the starting change phase in which the task will be activated for the change. If the task is dependent on another task, it will have a Planned status.
- Close By Phase – Specifies the ending change phase by which the task must be closed. If there are still open tasks for a phase, the change will not be allowed to go to the next phase. If left blank, the ending phase will match the starting phase.
- Mark as required/Set properties as read-only – Specifies whether or not this task can be re-planned or removed when you plan tasks using Task Planner from a change record.
- Task Condition – Defines the conditions for task creation.
    - The task condition is evaluated at each update of the change record when the change is in the Open In Phase defined for this task.

- If you leave this field blank, the condition will evaluate to true.
- When the condition evaluates to true after a change update, the task is created and linked with the change record.
- Once a task has been created and linked with the change, the condition for this task will not be re-evaluated anymore.
- When the condition is true in the first instance but false after the change is updated, the system doesn't automatically close the task and you need to close it manually.
- The condition of a planned task is shown as read-only in Task Planner from the change record.
- If a task is dependent on a conditional task that has not been created because its condition evaluates to false, this task will still be created (with a Ready status) when the change reaches the Open In Phase defined for this task.

As these tasks are created for a Change, task information is automatically applied to the new task record.

For task dependencies to function, each task status must follow a convention to use a numeric value for the task status value. The global list “Change Task Status” contains the status values listed in the following table.

Value List	Display List
0	Planned
1	Ready
2	Assigned
20	Completed
21	Completed with Problems
3	In Progress
30	Cancelled
31	Withdrawn
32	Failed
4	Blocked

Tasks that don't have any dependencies are automatically opened with the“1=Ready” status. Tasks that are dependent on other tasks are opened with the “0-Planned” status. Once all tasks that another task is dependent on have reached a status of 20 or greater, the dependent task will have its status changed to “1-Ready”.

To see a working example, view the **Generic Task** workflow. It utilizes format “chm.task”, and the statuses above. The initial “Waiting” phase will transition to “Active” when the task status > 0 which means it's not planned.

Using the behavior of the tasks statuses set by Process Designer, you can create your own task flow or modify a copy of the **Generic Task** workflow. If you are using task statuses that aren't using the numeric values, task dependencies will not function.

You can move a task from one phase to another by setting appropriate conditions. The following condition is an example to move a task from the “Waiting” phase to “Active”.

```
(Status in CurrentRecord> 0 AND Variable $L.tableAccess.update =
true)
```

You can set multiple dependencies for a task. If a task is set with dependencies, the task can start only after all the dependent tasks are completed. The following table outlines the various statuses of a child task which is “dependent” on the parent task when you open a change.

Parent Task	Child Task
Ready	Planned
Completed	Ready
Completed with Problems	Ready
Canceled	Ready
Withdrawn	Ready
Failed	Ready
Blocked	Ready

To add a task to a change model, follow these steps:

1. Click **Change Management** > **Configuration** > **Change Models** in the System Navigator.
2. Click **Search** to select a change model.

   Or

   Click **New** to create a new change model record.

   Note

   Select a category to display the task editor in the Task Information section.

3. In the Task Information section, click the **Edit** button, and then click **Add Task**. A new task is added in Task Planner.
4. Select the new task, click the **Task-Properties** tab, and perform these steps:
   1. Type a task description in the **Title** text box.
   2. Select a task category from the drop-down list.

3. Optionally, select a task template from the drop-down list.
4. Select a change phase to create the task from the **Open In Phase** list.
5. Optionally, select a change phase to close the task from the **Close By Phase** list.

> **Note**
>
> The ending phase must be no earlier than the starting phase. If left blank, the ending phase will be same as the starting phase.

6. Optionally, select the **Mark as required/Set properties as read-only** checkbox if you don't want this task to be modified or deleted when you plan tasks in Task Planner from a change record.

> **Note**
>
> After you select the checkbox, a red dot appears on the top left corner of the task.

7. Optionally, add conditions to create the task.

> **Note**
>
> “CurrentRecord” in the condition editor refers to the Change record.

> **Caution**
>
> You can't add a new task until you define mandatory details of the previous task in the Task Planner window.

5. Click **Add Task** to add another task.

   Or

   Hover the mouse on the new task added in the graphical interface until you see a hand symbol, and then drag the mouse pointer to add a new task.

6. If needed, click **Delete** in the toolbar to delete a task.
7. (Optional) After you finish adding tasks, click **Auto Layout** to allow task planner to automatically arrange the tasks and dependencies.

8. Click **OK** to save the change model tasks.

Note

- You can edit a task by selecting the task in the graphical interface and modify its properties.
- The Task Planner graphical interface supports panning.
- You can click **Zoom in** or **Zoom out** in the toolbar or use the mouse wheel to zoom in or zoom out the Task Planner graphical interface.
- You can reposition the workflow by moving the outline box within the graphical interface.
- You can resize the Task Planner graphical interface by expanding the details form or resizing the Task Planner window.

Map change model fields into change records

User Roles: System Administrator and Implementer

By configuring a predefined link file, you can customize the fields to be copied from a change model to a change record. You can modify a global link file used as the default for all change models, and you can further specify a link file for an individual change model. The link file for a specific change model takes precedence over the global link file.

To configure a default change model field mapping for all change models, follow these steps:

1. Click **Change Management** > **Administration** > **Settings** in the System Navigator.

   The Change Setting screen appears. The default link file for all the change models is specified in the **Default Open/Apply Link** text box.

2. Click the **Find Related Information** icon on the right of the **Default Open/Apply Link** text box to open the link file.
3. Click the line containing the mapping information in the table, and then click **More** > **Select Line**.
4. Modify the field mapping in the **Source Field (Fill To/Post From)** column and the **Target Field (Fill From/Post To)** column.

> **Note**
>
> Enter the field names of the `changeModel` table in the **Source Field (Fill To/Post From)** column, and enter the field names of the `cm3r` table in the **Target Field (Fill From/Post To)** column. The source and target fields in each mapping row should have the same data type.

5. Click **Save**.

> **Note**
>
> If you want to use a different link file, enter the name of your link file in the **Default Open/Apply Link** text box.

To configure a field mapping for one specific change model:

1. Click **Tailoring** > **Tailoring Tools** > **Links** in the System Navigator.
2. In the **Name** field, type the name of the form or file. If desired, type a descriptive name or phase in the **Description** field.
3. Click **New** to create the new link record.
4. Click a line in the table, and then click **More** > **Select Line**.
5. Type descriptive relationship information in the appropriate text boxes.
6. Configure a mapping using the **Source Field (Fill To/Post From)** column and the **Target Field (Fill From/Post To)** column.

   > **Note**
   >
   > Enter the field names of the `changeModel` table in the **Source Field (Fill To/Post From)** column, and enter the field names of the `cm3r` table in the **Target Field (Fill From/Post To)** column. The source and target fields in each mapping row should have the same data type.

7. Click **Save** to add the new link record.
8. Click **Change Management** > **Configuration** > **Change Models** in the System Navigator.
9. Click **Search** to select a change model.

   Or

   Click **New** to create a new change model record.

10. In the **Open/Apply Link** field, click the Fill icon to select the link file you created.
11. Click **Save**.

Use change models in the Service Catalog connector

User Roles: System Administrator

When adding a new Service Catalog item using the “Open a Change” connector, you can select a change model to open a change.

To use a change model when creating a service catalog item using the “Open a Change” connector:

1. Click **Service Catalog** > **Administration** > **Manage Items** in the System Navigator.
2. Click **Add New Service Item**.

   The New Service Item Wizard appears.
3. Enter the information as needed for the new Service Catalog item, and then click **Next**.
4. Select **Open a Change** as the connector type and specify the parent category that this new item belongs to.
5. Click **Next**.
6. Select change category, change subcategory, and change model, and then click **Next**.
7. Specify all the fields to define the initial state of the change request this item creates.

Note

All these fields are mandatory for creating the change requests. They must be specified either here or in the selected change model. If you leave any of these fields empty here, make sure this field has been defined in the selected change model. Otherwise, the change will not be created automatically after a request is approved.

The values specified here take precedence over those defined in the selected change model (or in the change templates used by the change model).

8. Click **Finish**.

After completing these steps, a Service Catalog item is created with the “Open a Change” connector. When a user orders this Service Catalog item and the request gets approved by the approver, a change record using the selected change model will be created. When the change record has gone through the change workflow and is closed, the interaction record is automatically closed.

Enable or disable the Apply Change Model option for a change workflow phase

User Roles: System Administrator, Change Manager

In the out-of-box setting, the **Apply Change Model** option is available only for change records that use the Project Proposal, Emergency Change, Normal Change, or Standard Change workflow and are in a phase before Authorization.

You can enable or disable this option for a phase of a change workflow by configuring the **Chm Apply Change Model** action in the workflow phase.

To enable the “Apply Change Model” option for a change workflow phase, follow these steps:

Note

If you want to edit a **Proprietary** workflow or use it as a model, create a copy of this **Proprietary** workflow.

1. Click **Change Management** > **Configuration** > **Change Workflows** from the System Navigator. All change workflows are displayed.
2. Open the change workflow that you want to configure.
3. Select a phase in the workflow, and then click the **Actions** tab.
4. Click **Add** to add an action with the following settings:
   - **Id**: type Apply Change Model
   - **Action**: select **Chm Apply Change Model**
   - **Location**: select **More Options List**
   - **Action Condition**: Select **RAD Expression**, and then enter $L.tableAccess.update=true
   - **Requires lock**: select the checkbox (true)
5. Click **Save**.
6. To disable this option for this phase, select the **Apply Change Model** action, and click **Delete**.

Clone a change model

User Roles: System Administrator, Change Manager

To clone a change model with task information, follow these steps:

Note

If you want to edit a **Proprietary** workflow or use it as a model, create a copy of this **Proprietary** workflow.

1. Click **Change Management** > **Configuration** > **Change Models** from the System Navigator. All change models are displayed.
2. Click **Search**, and then select a change model.
3. Click **Clone**. The system opens the **Clone Model** page.
4. Specify a new unique ID and a name for the cloned change model record.
5. Select the **Copy task information from another model** checkbox, and then click the **Fill** icon to select another change model from which the task information will be copied.

   Note that if the change model you selected has a different task workflow than the current one, the copied task workflow may not work correctly if you use this cloned model.
6. Click **OK**.

Related topics

Change Management configuration

1.7.5.3. Configuring approvals in Change Management

Change Management requires approvals to be complete before the change request can move to the next logical phase. In some cases, approvals can be defined based on associated Service Level Agreements (SLA), so that managers and sponsors associated with the SLA will be listed as approvers for some phases.

After the approvals are complete, you can move the request to any desired phase by setting a condition in the condition editor of a transition. You can set the transition condition to move the change request to another phase or remain in the current phase by setting the statuses to “Denied” and “Pending” respectively. For example, you can set the following conditions for the transitions in the “ECAB Approval” phase to move the change request to different phases.

Condition	Action Performed
Set the following condition in the transition from the “ECAB Approval” phase to the “Build and Test” phase `Approval Status in CurrentRecord= approved AND Build Test Required in CurrentRecord= true`	Change request moves from the “ECAB Approval” phase to the next phase, “Build and Test”.
Set the following condition in the transition from “ECAB Approval” to “Implementation” `Approval Status in CurrentRecord= approved AND Build Test Required in CurrentRecord= false`	Change request moves from the “ ECAB Approval” phase to “Implementation”.
Set the following condition in the transition from “ECAB Approval” to “Risk and Impact Analysis”. `Approval Status in CurrentRecord= denied`	Change request moves from the “ECAB Approval” phase to the previous phase, “Risk and Impact Analysis”.

A change request may require other approvals:

- Change requests with an impact of 3 (Major Impact) also require approval by an IT director.
- Change requests with an impact other than 1 (Minor Impact) must be approved by a member of the Change Advisory Board (CAB).

Related topics

What's an approval?
Global variables available for approval delegation
Approval options in Change Management

1.7.5.4. Managing alerts

Alerts trigger a series of checkpoints in a change or task to ensure that the required work activities occur within the specified time frames. Alerts serve as reminders to keep a change or task on schedule.

Alerts are timed or delayed conditions that are true or false. Requests for change proceed in phases, according to a predefined schedule. Alerts monitor the progress of these phases. An alert condition becomes true when circumstances require an automated response. Change Management treats it as an event, and sends the predefined notifications. For example, the late notice alert notifies a designated management group that a request for change is overdue for approval, and updates the alert status to include late notice.

Users can define any number of standard or customized alerts for any phase, control who is notified for each alert, and control the naming convention used for the alert itself. Alerts support other functions within the system.

Function	Description
Alert messaging	The event manager generates messages to designated recipients as a result of an alert, which updates the original request.
Batch scheduling	Change Management schedules all alerts associated with a phase at once when the phase occurs.

What are the alert controls?

The Phase record enables you to set alert controls.

Alert control	Description
Reset	Sets the status of all current Alert records associated with the current request to inactive. Marks the last action field as reset. Schedules a calculate alert record to recalculate the alerts and restart the alerts process.
Recalc	Retrieves each Alert associated with the request for change and performs the following processing: • If the current alert status is active, the alert condition is reevaluated, the alert is updated to reflect the correct status, and processing ends. • If the current status is not active, the Schedule Condition field is reevaluated. If Schedule Condition= true, Change Management updates the following fields: ◦ Status is set to scheduled. ◦ Last Action is set to recalc. ◦ Action Time is set to current date/time. ◦ Schedule Condition is reevaluated. If true, Alert Time is recalculated and Status updated to scheduled. If false, Status is set to not required.

Alert processing files

There are two primary files used in alert processing:

- The Alert Definition (AlertDef) file defines the alerts used by all phases

AlertDef defines the basic alert information for each named alert and all general alert definitions. AlertDef is a static file.

- The Current Alerts (Alert) file tracks the alerts created for each phase

  Alert is an active file.

Access an alert definition record

Applies to User roles: System Administrator

Alerts trigger a series of checkpoints in a change or task to ensure that the required work activities occur within the specified time frames. Alerts serve as reminders to keep a change or task on schedule. You can access alert definition records in order to modify, add, or delete alert definitions.

To access an Alert Definition record, follow these steps:

1. Click **Change Management** > **Configuration** > **Alerts**.
2. Use search or advanced search to find one or more records.
3. Click a record to view it in the **Alert Definition** form.

Create an Alert Definition record

Applies to User roles: System Administrator

To create alert definition records for use by the Change Management, Problem Management, and Request Fulfillment applications, follow these steps:

1. Click **Change Management** > **Configuration** > **Alerts**.
2. Problem Management: Click **Problem Management** > **Configuration** > **Alert Definitions**.
3. Request Fulfillment: Click **Request Fulfillment** > **Configuration** > **Alert Definitions**.
4. Type the alert name in the **Alert Name** field.
5. Type the alert description in the **Description** field.
6. Complete the remaining fields in the alert definition record form. If necessary, press **Ctrl+H** to view help for each field.
   - **Scheduling** tab: describes the calculations that trigger when the alert occurs.
   - **Update** tab: describes Format Control considerations and notification messages to be sent.
   - **Work Schedule** tab: specifies the work schedule to which the alert belongs.

- **Time Zone** tab: enables you to perform the following actions:
  - Define a specific time zone.
  - Define a RAD expression to set the time zone alert.
  - Obtain the time zone from a table lookup.

7. Click **Add**.

1.7.5.5. Configuring Change Management notifications

In the out-of-box system, Change Management calls the notification engine to send notifications. You can find the out-of-box Change Management notifications from **System Administration** > **Tailoring** > **Notifications** > **Notifications**. When creating custom notifications for Change Management, you can also use Process Designer rulesets to call the Send HTML Email rule type to send notifications.

Access a notification definition record

Applies to User roles: System Administrator

Notifications are messages generated by Service Manager events, such as opening or closing a change or task. You can edit these messages, add or delete messages, change the conditions that trigger the messages, and select which people receive the messages.

When configuring Change Management notifications, you can use either the notification engine or Process Designer rulesets to call the Send HTML EMail rule type. The following steps apply when you use the notification engine.

To access a Notification Definition record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Use search or advanced search to find one or more records.
3. Click a record in the record list to view it in the **Notification Definition** form.

View a message class record

Applies to User roles: System Administrator

Service Manager contains several default message classes that enable a user to define additional messages to display in Change Management.

This task applies only when Process Designer based Change Management calls the notification engine to send notifications. It doesn't apply if Change Management uses Send HTML Email rulesets to send notifications.

To view a Message Class record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Click **Log Message Class** (or any other message class type).

3. Use search or advanced search to find one or more records.
4. Click a record to view its detail in the related **Message Class File** form.

Add a notification definition record

Applies to User roles: System Administrator

The Notification Engine is primarily responsible for sending messages that are generated by Service Manager events, such as opening or closing a quote or order. System Administrators can edit these messages, add new messages, change the conditions under which the messages will be sent out, and select who will receive the messages.

To add a notification definition record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**. The **Notification Definition** form opens.
2. Type or select the following information.

Field	Description
Name	Type the name of the notification record.
Condition	Type true to enable the notification in all conditions. Type an expression that's true or false to specify a condition when Service Manager should send the notification. Type false to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.
Message tab	Use this tab to define the message type, delivery method, and conditions under which Service Manager should send the notification.
Msg Class	Type the message class for the notification. This value must match a message class definition in the message record.
Msg No	Type the message number for the notification. This value must match a message number definition in the message record.

Field	Description
Arguments	Type any RAD expressions to display information from the record or event that triggered the notification. Service Manager displays this information in the text of the notification. You can enter an array of arguments by enclosing all arguments in brackets and separating each argument with a comma. For example: {argument1,argument2,argument3} **Note** Arguments typically take the following form: `<field name> in $L.file` where `<field name>` is the name of a field in the record that triggered the notification.
Condition	Type true to enable the notification in all conditions. Type an expression that can be true or false to define a condition when Service Manager should send the notification. Type false to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.

Field	Description
Format	Type the format control record used to display the triggering record as part of the notification.

Field	Description
Notify Method	Type or select the delivery method to be used to send the notification. This value must match a delivery method defined in the msgtype table. **Note** ◦ The notification applications listed below must be defined in RAD code and exist in Service Manager to work. ◦ The message type must be enabled (true) by selecting "Active?" in the msgtype table. See the following list for valid message types. ◦ connect: Send a message to an external device. ◦ email: Send an email. ◦ log: Create a log entry for a message. ◦ mail: Send internal mail. ◦ msg: Send an internal message to a user's current Service Manager session (for example, "Incident IM1011 has been assigned to your monitor list since...").

Field	Description
	Note If the user is not currently connected to a Service Manager session, the message is not sent. ◦ print: Print message.
Recipients	Type one of the following to identify who should receive the notification: ◦ A contact name ◦ An operator name ◦ Array of approval group names ◦ A RAD expression that gathers a contact or operator name from some field in the triggering record
Group File	Type the name of the group receiving the notification. You can specify the On Call group to send the notification to for those operators who meet the On Call criteria.

Field	Description
Group Area	Type the name of the group area receiving the notification. You can use this field only with ocmgroups and cm3groups group tables. Valid choices include: ◦ ocmgroups -All -Line Items -Orders -Quotes ◦ cm3groups -All -Changes -Tasks
Subgroup	Type the name of the subgroup area receiving the notification. You can use this field only with ocmgroups and cm3groups group tables. Valid choices include: ◦ All ◦ Approvers ◦ Members
Email/Mail Subject Line tab	Use this tab to type any custom information that should appear in e-mail notifications.

Field	Description
Msg Class	Type the message class for the subject of the notification. This value must match a message class definition in the message record.
Msg No	Type the message number to use as the subject of the notification. This value must match a message number definition in the message record.

Field	Description

Arguments	Type any RAD expressions used to display information from the record or event that triggered the notification. Service Manager displays this information in the subject of the notification. You can enter an array of arguments by enclosing all arguments in brackets and separating each argument with a comma. For example:
	`{argument1,argument2,argument3}`
	Note
	Arguments typically take the following form:
	`<field name> in $L.file`
	where `<field name>` is the name of a field in the record that triggered the notification.

3. Click **Add**.

1.7.6. Security

Service Manager controls access to Change Management functions based on security roles, security areas, and rights.

Change Management security areas

User Role: System Administrator

The security areas in Change Management are Changes, Change Tasks, and Change Management Configuration. These areas contain the default security rights and settings for Change Requests in the Change Management module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

The change areas are used to set permissions to operators to provide access to particular area of Change Management. For example, for the Change Coordinator security role, set change area as Change Management Configuration and rights as **View** and **Admin**. Then, an operator with this security role can view and configure only the Change Management Configuration items under Change Management.

The following table lists the areas and the relevant Change Management menu items the operators can access.

Security area	System Navigator menu items for this area
Change	This area contains the default security rights and settings for change requests in the Change Management module. For example, **Create New Change**, **Change Queue**, **Task Queue**, and **Search Changes**.
Change Tasks	This area contains the default security rights and settings for change tasks in the Change Management module. For example, **Create New Tasks**, and **Search Tasks**.
Change Management Configuration	This area contains the default security rights and settings for Change Management configuration. For example, **Alerts**, **Approvals**, **Change Categories**, **Change Models**, **Change Workflows**, **Messages**, **Subcategories**, and **Task Categories**.

The default rights defined in areas are inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Change, Change Task and Change Management Configuration areas.

Security area	View	New	Update	Delete/ Close	Expert	Admin
Change	TRUE	FALSE	Never	Never	FALSE	FALSE
Change Tasks	TRUE	FALSE	Never	Never	FALSE	FALSE
Change Management Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Change Management security roles and settings

User Role: System Administrator

The out-of-box security roles for Change Management include the following:

- change advisory board (CAB)
- change analyst change
- change analyst tasks
- change approver
- change coordinator
- change coordinator change
- change coordinator tasks
- change domain expert (CDE)
- change implementer
- change manager
- change owner
- change requestor

Tip

If you have upgraded from a previous version of Change Management, which is not Process Designer based, see Change Management security profiles mappings for more information about the mappings between the legacy Change Management profiles and the current Process Designer security roles.

Rights and settings

To view the rights and settings that are defined for out-of-box Change Management security roles, follow these steps:

1. Navigate to **System Administration** > **Security** > **Roles**.
2. In the **Name** field, select a Change Management security role.
3. Click **Search**.
4. Double-click a security area to open the Rights form.

   This form lists the rights and settings of the security role for the selected security area.
5. View the **Rights** and **Settings** sections in the form.

   The following table describes possible rights for Change Management security areas.

Security role rights	Description
New	Can create new records in this area
View	Can view records in this area.
Expert	Has Expert rights in this area
Admin	Has Admin rights in this area
Modify Template	Can modify template records in this area
Update	Can update records in this area
Delete/Close	Can delete or close records in this area
Allowed Categories	Specifies the categories in this area that users with this role can select
Allowed Statuses	Specifies the status values in this area that users with this role can select

The following table describes the settings available for Change Management security areas.

Setting	Description
Allow Inefficient Query	When this option is true, users with this role can run incomplete (inefficient) queries in this area, but receive a warning message. The Skip Inefficient Query Warning option overrides this setting. If not selected, users with this role can't run inefficient queries in this area.
Skip Inefficient Query Warning	Turns off inefficient query warnings for this area. This option overrides the **Allow Inefficient Query** option.
Can Approve	Users with this role can approve records in this area.
Can Delegate Approvals	Users with this role can delegate their own approvals in this area to another user.
Reopen	Users with this role can reopen records in this area.
Change Manage Format	The name of the form to display as the default change queue form. If left blank, **sc.manage.chm** is used.
Default Category	Unused
Default Task Category	Unused
Initial Change View	Unused. Service Manager automatically saves the Change view that users with the role opened last time.

Setting	Description
Initial Task View	Unused. Service Manager automatically opens the Change Task view that users with this role opened last time.
Initial Format	The initial format that's used when users with this role search change records. If left blank, **cm3r.search** is used.
List Format	Also known as QBE Format. This is the QBE form to use when displaying records to users with this role. If left blank, **cm3r.qbe** is used.
Task Manage Format	The name of the form to display as the default task queue form. If left blank, **sc.manage.cmt** is used.
Append Query	This field stores an expression to append to all queries in this area run by users with this role. This expression restricts the records that the user with this security role can see, by appending (adding on) this query to everything the user does in this module. You must use the field name from the dbdict in the expression. For example, you can enter priority.code="3" so that users with this security role can only see Priority 3 records.

6. View the Security Folders section.

   This section specifies security folders that are accessible to this Role for records of this area. For more information, see Add folder permissions to a security role.

The out-of-box security folders available in Service Manager are **DEFAULT** and **advantage**. You can also create security folders to meet your business needs. By default, all security folders are assigned to a new security role created. Once a role is created and rights are configured, you can modify the security rights for a role within an area.

Configuring Change Management security

User Role: System Administrator

Change Management Administration enables you to perform the following Change Management security configurations:

Set rights to changes and change tasks

The Change Administrator can set rights to changes and tasks using the role-based security. The Changes, Change Tasks, and Change Management Configuration areas contain default security rights and settings. The rights are copied to new roles created for these areas. However, the settings are inherited only if there are no settings specified on the roles.

Note

Whenever the roles in an operator record are updated, the operator must sign out and then sign in for the changes to take effect.

Add security roles and settings

User Role: System Administrator

To create a security role and assign rights and settings to it, follow these steps:

1. Click **System Administration** > **Security** > **Roles**.
2. Click **New**. The security role form is displayed.
3. Type a security role name.
4. Type a security role description.
5. Click **Save**.
6. Select a security area. The security rights and settings form is displayed.

   Under **Rights**, select the rights to be assigned to the security role.

For example, set **Expert** rights for the security role. The **Expert** security right enables the operator to view alert logs, opened tasks, affected services, and clocks of change requests. It also enables the operator to set reminders, send notifications, create hot news, and associate change requests to incidents, interactions, requests, and known errors.

7. Under **Settings**, add required settings.
8. Under **Folders**, add folder permissions to the security role.

   The out-of-box security folders available in Service Manager are DEFAULT and advantage. You can also create security folders to meet your business needs. By default, all security folders are assigned to a new security role created. Once a role is created and rights are configured, you can modify the security rights for a role within an area.

Modify an operator record to enable Change Management access

Applies to User roles: System Administrator

To modify an operator record to enable Change Management access, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Use search or advanced search to find one or more records.
3. Click the operator record that you want to view.
4. Click the **General** tab.
5. Add the desired Change Management security roles to the **Security Roles** table.
6. Click **Save**.
7. Click **OK**.

Enable approval delegation

Approval delegation is an optional feature that enables users with approval rights to temporarily delegate their approval authority to another qualified operator. For more information, see Approval delegation.

To enable an operator to delegate approval authority, a System Administrator must make changes to a specific security role of the operator.

The following example illustrates how to enable approval delegation for the Change Management application.

To edit the security role to which you want to grant approval delegation authority, follow these steps:

1. Sign in to Service Manager with a System Administrator account.
2. Click **System Administration** > **Security** > **Roles**.

   Service Manager displays a list of security roles.
3. (Optional) In the **Name** field, type the name of the security role to which you want to grant approvals, for example, **change approver**.
4. Click **Search**. The security role or a list of security roles opens.
5. Locate the security role you want to edit, and select a Change Management security area, **Change** or **Change Tasks**.
6. Select **Can Delegate Approvals** under **Settings**.
7. Click **Save**.

Note

It is a best practice to enable the **Delegate Approvals** or **Can Delegate Approvals** option only for operators who can also view and approve objects in the application.

Now, the operator can delegate approvals for the selected security area. For information about how to delegate approvals to another operator, see Delegate approvals to another operator.

Related topics

Process Designer security model
Add folder permissions to a security role

1.8. Change Management (Hybrid)

Service Manager Change Management tracks changes to service assets and configuration items in your infrastructure.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Change Analyst

Change Coordinator

Change Manager

Change Approver

Release Packaging and Build Manager

E-CAB

Key concepts

Describes the most important things that you need to know about how the Change Management module functions.

Change Management overview

What are changes?

Default change categories

What's an approval?

Approval delegation

Priority, impact, and urgency

Release Management

Workflows and user tasks

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Change logging (ST 2.1)

Change review (ST 2.2)

Change assessment and planning (ST 2.3)

Change approval (ST 2.4)

Coordinate change implementation (ST 2.5)

Change evaluation and closure (ST 2.6)

Emergency change handling (ST 2.7)

Change Management tasks

Administrator tasks

Describes the Administrator tasks in the Change Management module.

Managing approvals in Change Management

Managing Change Management messages

Managing events

Managing alerts

Additional resources

Processes and Best Practices

1.8.1. Change Management overview

All changes to the Service Portfolio or service catalog are implemented through Change Management and the changes that are managed by the Service Transition life cycle stage are defined and agreed. Standardized methods and procedures are used for efficient and prompt handling of all changes, in order to minimize the impact of change-related incidents on business continuity, service quality and re-work.

To be effective, Change Management requires the following input:

- Policy and strategies for change and release
- Request for change
- Change proposal
- Plans (change, transition, release, deployment, test, evaluation, and rendition)
- Current change schedule and projected service outage (PSO)
- Current assets or configuration items
- As-planned configuration baseline
- Test results, test report, and evaluation report

All updates to changes and releases are recorded against service assets and/or configuration items in the Configuration Management System.

Related topics

What are changes?
Priority, impact, and urgency
Hover-over forms for change records

1.8.1.1. What are changes?

Changes are the records submitted that initiate a change in an IT service. Each change has a life cycle of approvals, alerts, tasks (if required), phases, and closure. When you submit a change request, Change Management assigns a unique change record number.

Before you process a change, an administrator must define user profiles and groups for your Service Manager installation.

Change identifiers

Service Manager automatically creates a unique ID for each RFC and Change Task when the record is opened. This field is mandatory and read-only. The format of the ID is configurable to meet customer requirements.

Time and date stamp

Service Manager time and date stamps all events (manual or automated) associated with a Change record, from the time the record is opened until it is closed. These fields are mandatory and read-only.

Priority, impact, and urgency

Priority is how an individual service desk interaction, change request, incident, or problem fits into the ongoing sequence of tasks required to close the interaction, change request, incident, or problem. It also indicates how soon the work should begin. Determining the priority of a single service desk interaction, change request, incident, or problem depends on how many other defects need attention, the risk of delay, and the resources available to fix them.

Impact is the potential business vulnerability. There is no global value; it is subjective and each business must set and modify its own impact value list.

Urgency is a value that reflects how soon the defect must be resolved to avoid business consequences. It identifies how soon you must react to avert or reduce the impact of the defect on customers.

Assigning values to impact and urgency is subjective. Priority is a Service Manager calculation based on the values you specify for impact and urgency. As you can experience assigning impact and urgency values, you will refine your decision criteria. A service desk interaction, change request, incident, or problem that's isolated can have a low impact initially, but a high urgency because of the potential

for damage if the defect becomes widespread. For example, a new computer virus is a problem that can escalate quickly.

Related topics

Managing events
What's an approval?
Managing alerts

1.8.1.2. When should I use Change Management instead of Request Management?

- Change Management handles any change to your business that modifies or disrupts the current state of that environment. Usually these modifications or disruptions affect multiple users or business units.
- Request Management handles common user requests for products and services. These requests usually affect only the person making the request, or a subordinate group of employees.

Related topics

Change Management overview
Request Management overview

1.8.1.3. Change categorization

Service Manager supports both major and minor changes using the Normal Change category and workflow. The subcategory then further differentiates the Change between minor and major. Significant can be added as a subcategory.

1.8.1.4. Hover-over forms for change records

Change Management change forms for open, update, and close include hover-over forms. A hover-over form opens for a field that supports hover-over forms when the user moves the mouse over the field. The hover-over form only displays when the field contains data. The data displayed on hover-over forms is read only.

The change record forms (open, update, close) contain the following hover-over fields.

Field	Subform fields
Initiated by	Full Name, Telephone, Email

1.8.1.5. Managing approvals in Change Management

Approvals are based on associated Service Level Agreements (SLA). Managers and sponsors associated with the SLA must be listed as approvers for each phase. Change Management requires approvals to be complete before the change request can move to the next logical phase.

A change request may require other approvals, depending on the category listed in the categorization field:

- Change requests with an impact of 3 (Major Impact) also require approval by an IT director.
- Change requests with an impact other than 1 (Minor Impact) must be approved by a member of the Change Advisory Board (CAB).

Related topics

Change Management overview
What's an approval?
What's an approval sequence?
What's approval status?
Approval options in Change Management
Approve, deny, or retract a change
Approve multiple change requests

1.8.1.5.1. What's an approval?

An approval occurs when a user (approver) or a group of users (approval group) agrees to accept the risk, cost, and responsibility to implement a change request or complete a task. The System Administrator defines membership in an approval group in each user's operator record according to the User Role. Change requests and tasks contain approval requirements. Approvals give change management the ability to stop work and to begin certain work activities.

As an approver, you can also be part of a change message group that consists of reviewers and approvers. If you are an approver for a change message group, your task is to accept or deny the changes your group must approve.

When you select the **Mass Approve** option, all the records in the list that are awaiting your approval are immediately approved.

Note

Change Management doesn't prompt you to verify that this is the action you want to take.

Related topics

What's an approval sequence?
What's approval status?
Approvals and security profiles in Change Management
Approval options in Change Management

1.8.1.5.2. What are approvers?

As an approver, your user profile must contain the approval type for the approvals you will make, and you must be a member of the approval group associated with each approval type that's listed in your profile. For example, the out-of-box Change Management Coordinator profile assigns the approval type of Assessment. If you are a member of the Assessment approval group, you can approve any change request that requires that approval type.

However, if your user profile changes and you no longer have the Assessment approval type, you can't approve change requests that require the Assessment approval type. Even if you remain a member of the approver list for the Assessment approval type, you can't approve unless your individual profile specifies this type.

Related topics

Change Management overview
What are changes?
Managing approvals in Change Management
Managing alerts

1.8.1.5.3. What's approval status?

Approval status describes the condition of the change or task in the current phase and in the Change Management workflow. Each change request phase is marked with one of the following status options:

- Approved
- Denied
- Pending

Individual approval requirements within a phase are marked with one of the following status options:

- Approved
- Denied
- Pending
- Future

Pending approvals are awaiting action. Future approvals will be acted on following action on the pending approvals.

When a pending phase is approved, its status becomes **approved**. The next set of future approvals becomes **pending**, and subsequent approvals remain in the **future** status.

Related topics

What's an approval?
What's an approval sequence?
Approvals and security profiles in Change Management
Approval options in Change Management

1.8.1.5.4. What's an approval sequence?

An approval sequence is the order in which approval requirements are activated. The lowest sequence numbers are available for approval activity. When approval occurs, the next highest number is activated. Some approvals must be sequential. In this case, approval sequence numbers reflect the required order. If an approval group shares the same sequence number, the group members can approve in any order.

Related topics

What's an approval?
What's approval status?
Approval options in Change Management

1.8.1.5.5. What's an approval type?

An approval type is a type of approval required for an approval definition record in the **ApprovalDef** table. The following table describes the approval types in the out-of-box system.

Type	Description
All must approve	All members of the group must approve the request. **Note** This is the default value when no approval type is specified for an approval definition.
One must approve	Only one member must approve the request.
Quorum	A majority of the approval group must approve the request.
All must approve - immediate denial	All approvers must approve the request. The first denial updates the status of the current approval request to Deny. Meanwhile, the first denial also updates the status of other pending approval requests to Denied By System and sets the value in the Approved by field to System. Other approvers don't need to take any action.

Type	Description
One must approve by sequence approval level	When this option is selected, the following behavior applies: • When a sequence is defined with a group of approvers, if one approver approves the request, the entire sequence is approved and the request is passed to the next sequence for approval. Only when all sequences are approved, the request is approved. • Only when all approvers for a sequence have denied the request, the entire sequence is denied. If one sequence is denied, the request is denied.

Related topics

What's an approval?
What's approval status?
Approvals and security profiles in Change Management
Approval options in Change Management

1.8.1.5.6. Approval options in Change Management

Change Management enables you to approve, deny, or retract changes or tasks.

Approval option	Description
Approve	The approver accepts the need for the change or task, and approves commitment of the resources required to fulfill the request. When all approvals are complete, work begins. When you choose this option, the record shifts to browse mode, and the retract option is available. If you're not a member of a group with approval rights to this change request, Change Management generates an error message.
Deny	The approver is unwilling to commit the required resources, or doesn't consider the change or task to be essential. No further approvals are possible until the denial is retracted. An administrative procedure should be set up to handle a denial. If you select deny, a dialog box opens with a prompt to specify the reason for your action. Type an explanation and click **OK**.
Retract	The approver accepts the need for the change, but is unwilling to commit the resources or perhaps there are technical incidents at the present time. Retract removes a previous approval or denial and resets the change request to pending approved status, which requires a new approval cycle. If you select retract, a dialog box opens with a prompt to specify the reason for your action. Type an explanation and click **OK**.

Note

Rights for the override option are granted to users with **Admin** or **Expert** security rights for the Change area.

Related topics

What's an approval sequence?
Configuring approvals in Change Management
Approve, deny, or retract a change

1.8.1.5.7. Approval delegation

Approval delegation is an optional feature that enables users with approval rights to temporarily delegate their approval authority to another qualified operator. Operators with the **Delegate Approvals** or **Can Delegate Approvals** option enabled in their application profiles can delegate some or all of their approvals by using the Approval Delegation wizard.

Using the **Approval Delegation** wizard, an operator can grant another qualified operator the right to temporarily view and act on items in that operator's approval queue. The wizard offers the following delegation options:

- Delegate all approvals to another qualified operator
- Delegate approvals from a particular application to another qualified operator
  - Delegate approvals directly assigned to you as an operator
  - Delegate approvals assigned to you as a member of an approval group
- Delegate approvals from a specified start date to a specified end date

The **Approval Delegation** wizard enables an operator to create any number of approval delegation combinations, including delegating the same approvals to multiple operators at the same time. Delegators can also update an existing approval delegation to change the delegation start and end dates, as well as change the delegate's name.

Note

Service Manager tracks all changes to approval delegations using the standard field auditing capability.

When delegates log on to Service Manager, they see both their own and any delegated approvals in their approval list. For security reasons, delegates always retain their original application profiles and operator records. Service Manager determines what temporary rights delegates have when they view or act on an approval.

For more information about the tasks that relate to approval delegation, refer to the following topics:

What happens when I delegate approval authority?
Temporary rights of an approval delegate
What happens when I receive delegated approval authority?
Copy an approval delegation

Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

1.8.1.5.8. Approvals and security profiles in Change Management

If you are an authorized approver for any approval group, the group names, such as CHGCOMM or CUS, appear in your security profile. You may approve all changes and tasks assigned to your approval groups.

Related topics

What's an approval?
What's an approval sequence?
What's approval status?
Approval options in Change Management

1.8.1.5.9. Approval options in Change Management

Change Management enables you to approve, deny, or retract changes or tasks.

Approval option	Description
Approve	The approver accepts the need for the change or task, and approves commitment of the resources required to fulfill the request. When all approvals are complete, work begins. When you choose this option, the record shifts to browse mode, and the retract option is available. If you're not a member of a group with approval rights to this change request, Change Management generates an error message.
Deny	The approver is unwilling to commit the required resources, or doesn't consider the change or task to be essential. No further approvals are possible until the denial is retracted. An administrative procedure should be set up to handle a denial. If you select deny, a dialog box opens with a prompt to specify the reason for your action. Type an explanation and click **OK**.
Retract	The approver accepts the need for the change, but is unwilling to commit the resources or perhaps there are technical incidents at the present time. Retract removes a previous approval or denial and resets the change request to pending approved status, which requires a new approval cycle. If you select retract, a dialog box opens with a prompt to specify the reason for your action. Type an explanation and click **OK**.

Note

Rights for the override option are granted to users with **Admin** or **Expert** security rights for the Change area.

Related topics

What's an approval sequence?
Configuring approvals in Change Management
Approve, deny, or retract a change

1.8.1.5.10. Approval phase for KM Ddocument category

There is a sample change phase "KM Approval" for the "KM Document" change category in the out-of-box system. This approval phase can be tailored for any Knowledge Management or administrator approval group, notification, or alert.

Note

The Approve Internal and Approval External buttons on a Knowledge Management change request relate to the KCS status for a document only when it is in final form and may be viewed by an internal or external audience. The Approve Internal and Approval External status of a document is not related to Change Management approval categories or phases.

Knowledge Management uses the standard RAD application, cm.close, functionality for improved integration with Change Management so that approvals, alerts, and notifications are generated as appropriate for the change request.

KM Approval is present in the system as a sample phase only in Change Management. It's not part of the Knowledge Management document workflow unless the system is configured to make KM Approval a phase in Change Management.

1.8.1.6. Message Group Definition record

Service Manager Change Management uses message groups to identify the members of a work group, also known as a Message group. Message groups include two types:

- Members receive work messages sent to the group, such as event and alert messages. For example, notifying members about the progress of requests and tasks that their group is responsible for managing. Members automatically act as reviewers.
- Approvers approve requests for the group. Approvers usually receive notifications when a request is awaiting their approval.

Approvers aren't automatically reviewers. If an approver must receive all group notifications, whether related to work activities or to approvals, be sure to add them to both the Members list and to the Approvers list.

The Message Group Definition record stores the login IDs of all members and approvers who receive notifications and messages during a change project.

1.8.1.7. Default change categories

Change Management has the following default change categories.

Category	Description
CI Group	Manages Configuration Items Group changes.
Default	
Hardware	Manages hardware changes.
KM Document	Manages a Knowledge document.
Maintenance	Manages maintenance-related changes.
Network	Manages network-related changes.
Release Management	Manages the releases of hardware and software.
Software	Manages software-related changes.
Subscription	Manages changes to business service subscriptions.
Unplanned Change	Manages exceptions identified by a discovery process in the actual state of configuration items. For example, a computer CI might have exceptions in the system's IP address or amount of RAM installed. **Note** The Discovery Event Manager uses this category to identify exceptions to the expected CI state.

1.8.1.8. Managing Change Management messages

Change Management sends messages in response to an event. They are directed to specific operators listed in the event record and contain values from the specific fields in quote, order, and line item records that cause the initial event.

Change Management message processing includes the following actions:

- Check for Field Name and Operators in the cm3rmessages record or ends processing if neither Field Name nor Operators exist
- Record the generic message (from the cm3rmessages record Append Text field) in the msglog
- Send the standard message and the mail message to the operators defined in the Operators field of the cm3rmessages record.
- Send a message to the operators defined in those fields referenced by the Field Name field of the cm3rmessages record

  Change Management assumes the content of these fields is a group. If this group name is found in the cm3rgroups file, then either the Members or Approvers (depending on the Member List field) of that group are added to a working list. If the group name doesn't exist, the system searches the operator file. If an operator record is found, it is added to the working list.
- Check the working list for operators
- Send the message

What are message classes?

Service Manager provides several default message classes that you can use to define additional messages to display in Change Management.

How does Change Management process messages?

The following must occur for Change Management to process a message.

1. The record must exist.
2. The message flag in the phase definition record (cm3rcatphase/cm3tcatphase) must be true.

3. The operators and field names must be valid message groups, contacts, or operators.
4. The event must either be an alert definition (AlertDef) or a message (cm3messages). The event syntax generally used to check for an event is as follows:

   ```
   if (condition=true) then ($cm3messages file.$events.pntr in
   $cm3messages =“event name”;$cm3messages
   file.$events.pntr+=1)
   ```

   where:

   - The variable `$cm3messages` is an array of character strings that tracks the events occurring during a particular phase of processing.
   - The variable `$cm3messages.events.pntr` is a pointer to the next array element that records an event name.
5. When you add an event to the array, it is important to increment the pointer by 1. If you omit this step, the event previously recorded is overwritten.
6. Define the event that's scheduled if the condition is true in the cm3rmessages file.

Add a Message Class record

Applies to User Roles: System Administrator

Service Manager provides several default message classes that you can use to define additional messages to display in Change Management.

To add a Message Class record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Log Message Class** (or any other message class type). A blank **Message Class File** form opens.
2. Type the message class name and description.
3. Click **Add**.

Display the list of message classes

Applies to User Roles: System Administrator

Service Manager provides several default message classes that a user can use to define additional messages to display in Change Management.

To display the list of message classes, follow these steps:

1. Click **Tailoring** > **Notifications**.
2. Click one of the following message types. They represent the possible action types for different message classes.
    - **External Mail Message Class**
    - **Internal Mail Message Class**
    - **Log Message Class**
    - **On Screen Message Class**
    - **Print Message Class**

View a message class record

Applies to User Roles: System Administrator

Service Manager contains several default message classes that enable a user to define additional messages to display in Change Management.

To view a Message Class record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Click **Log Message Class** (or any other message class type).
3. Use search or advanced search to find one or more records.
4. Click a record to view its detail in the related **Message Class File** form.

1.8.1.9. Managing events

Events are system occurrences triggered by new or updated change requests, such as creating or approving a change. Events can trigger special processing, such as alerts and messages. When an alert condition becomes true, Change Management treats it as an event, and sends predefined notifications. You can create other customized routines to run as part of event processing.

The Events file contains the names and definitions of all valid Change Management events.

Change Management includes several default events, such as cm3r approved and cm3r closed. You can add others, according to your business requirements. For example, there may be activities or conditions to be checked or unique events that require user notification.

What controls events?

Event controls are options for processing messages and events. The change or task Phase definition record contains control information that defines when Change Management processes events for the named phase.

The cm3rmessages file

You must define all event names in the cm3rmessages file, or else no event processing can occur. Events are processed when the phase of the change or task is defined in the Phases field of the cm3rmessages file.

The ocmevents record

If the Phase field is NULL on the ocmevents record, the event is processed for all phases.

Checklist: Adding new events

You can perform the following software management tasks to add new events:

1. Activate the Change Management background processor (cm3r)
2. Activate the Change Management (cm3r) alerts schedule record
3. Set the environment record
4. Define the Messages/Events option for the phase
5. Set any RAD or Format Control definition to track a custom event

6. Define the operator groups in the cm3rmessages record in the cm3rgroups file
7. Define the appropriate operator records
8. Define the event in the cm3rmessages file
9. Define the Format Name in the format file
10. Validate the Message Notification controls

Contents

- 1 What controls events?
  - 1.1 The cm3rmessages file
  - 1.2 The ocmevents record
- 2 Checklist: Adding new events

1.8.1.10. Managing alerts

Alerts trigger a series of checkpoints in a change or task to ensure that the required work activities occur within the specified time frames. Alerts serve as reminders to keep a change or task on schedule.

Alerts are timed or delayed conditions that are true or false. Requests for change proceed in phases, according to a predefined schedule. Alerts monitor the progress of these phases. An alert condition becomes true when circumstances require an automated response. Change Management treats it as an event, and sends the predefined notifications. For example, the late notice alert notifies a designated management group that a request for change is overdue for approval, and updates the alert status to include late notice.

Users can define any number of standard or customized alerts for any phase, control who is notified for each alert, and control the naming convention used for the alert itself. Alerts support other functions within the system.

Function	Description
Alert messaging	The event manager generates messages to designated recipients as a result of an alert, which updates the original request.
Batch scheduling	Change Management schedules all alerts associated with a phase at once when the phase occurs.

What are the alert controls?

The Phase record enables you to set alert controls.

Alert control	Description
Reset	Sets the status of all current Alert records associated with the current request to inactive. Marks the last action field as reset. Schedules a calculate alert record to recalculate the alerts and restart the alerts process.
Recalc	Retrieves each Alert associated with the request for change and performs the following processing: • If the current alert status is active, the alert condition is reevaluated, the alert is updated to reflect the correct status, and processing ends. • If the current status is not active, the Schedule Condition field is reevaluated. If Schedule Condition= true, Change Management updates the following fields: ◦ Status is set to scheduled. ◦ Last Action is set to recalc. ◦ Action Time is set to current date/time. ◦ Schedule Condition is reevaluated. If true, Alert Time is recalculated and Status updated to scheduled. If false, Status is set to not required.

Alert processing files

There are two primary files used in alert processing:

- The Alert Definition (AlertDef) file defines the alerts used by all phases

AlertDef defines the basic alert information for each named alert and all general alert definitions. AlertDef is a static file.

- The Current Alerts (Alert) file tracks the alerts created for each phase

  Alert is an active file.

Access an alert definition record

Applies to User roles: System Administrator

Alerts trigger a series of checkpoints in a change or task to ensure that the required work activities occur within the specified time frames. Alerts serve as reminders to keep a change or task on schedule. You can access alert definition records in order to modify, add, or delete alert definitions.

To access an Alert Definition record, follow these steps:

1. Click **Change Management** > **Configuration** > **Alerts**.
2. Use search or advanced search to find one or more records.
3. Click a record to view it in the **Alert Definition** form.

Create an Alert Definition record

Applies to User roles: System Administrator

To create alert definition records for use by the Change Management, Problem Management, and Request Fulfillment applications, follow these steps:

1. Click **Change Management** > **Configuration** > **Alerts**.
2. Problem Management: Click **Problem Management** > **Configuration** > **Alert Definitions**.
3. Request Fulfillment: Click **Request Fulfillment** > **Configuration** > **Alert Definitions**.
4. Type the alert name in the **Alert Name** field.
5. Type the alert description in the **Description** field.
6. Complete the remaining fields in the alert definition record form. If necessary, press **Ctrl+H** to view help for each field.
   - **Scheduling** tab: describes the calculations that trigger when the alert occurs.
   - **Update** tab: describes Format Control considerations and notification messages to be sent.
   - **Work Schedule** tab: specifies the work schedule to which the alert belongs.

- **Time Zone** tab: enables you to perform the following actions:
  - Define a specific time zone.
  - Define a RAD expression to set the time zone alert.
  - Obtain the time zone from a table lookup.

7. Click **Add**.

1.8.1.11. Change workflows

The Service Manager Change Management module implements ITIL processes for change management. The module ensures that standardized methods and procedures are used for efficient and prompt handling of all changes to an organization's infrastructure, from the time the need for the Change is realized until the time the Change is fully executed. Workflows for Normal, Standard, and Emergency changes are provided and additional workflows can be added by the customer if required by their processes. Each of these workflows enables personnel and management to:

- Create a request for change (RFC) record, assigning a unique number against which all activity can be tracked and reported
- Within a change record, identify the tasks (work processes) required to accomplish the change
- Follow, track, and query changes through the entire implementation cycle

  Workflow phases identify the logical sequence of repeatable steps within the change implementation cycle. There is also a visualization of the current phase, previous, and potential future phases of the RFC.

Response-time objectives are available for both changes and change tasks. These objectives set goals for the amount of time required to move the record from one phase to another and have associated business rules, alerts, escalations, and notifications.

Change workflow types

Specific workflows for Normal, Standard, and Emergency changes are provided and additional workflows can be added by the customer if required by their processes.

Contents

- 1 Change workflow types

1.8.1.12. Assessment capability

Service Manager offers the following assessment capabilities:

- Manual assessment of the impact of the change by the Submitter, Coordinator, or Assignee.
- Collision detection: One or more RFCs for the same CI are proposed at the same time.
- Resource limitations: when two or more RFCs are proposed using the same resources to alert the CAB to potential resource constraints.
- Risk impact calculated using business rules and CMS relationship information at the CI level.
- History of previous similar changes and how successful they were.

Related topics

Assess change impact and risk

1.8.1.13. Change prioritization

The Change form includes fields to capture the impact and urgency of the RFC. These fields can be set as part of the Change Model using the Templates feature. The fields can also be populated based on the Service or CI that's identified for the Change. Service Manager automatically calculates priority based on impact and urgency. The lists of values used for impact, urgency, and priority along with the calculation used to determine priority can be modified by the customer.

Impact codes include:

- 1 – Enterprise
- 2 – Site/Department
- 3 – Multiple Users
- 4 – User

Urgency and Priority codes include:

- 1 - Critical
- 2 - High
- 3 - Average
- 4 – Low

Closure phase

The Change workflows provided in Service Manager contain a Closure phase that indicates that the Change is closed. In the workflow viewer this phase is with a solid gray background so it is visually evident that Changes in this phase aren't active.

Closure codes are captured in the Post Implementation Review phase and can be configured by the customer.

Post implementation review

Service Manager Change workflows contain a Post Implementation Review phase that supports notifying the person responsible for the review and capturing the details of the review when it is performed.

Back-out plans

Service Manager provides an out-of-box field that can be used to record the back-out procedures for a specific RFC. The information in the Back Out Plan field could

reference attached documents or external sources for back out documentation.

The back-out plan review is part of the Deployment CAB approval process of the Normal Change workflow.

Contents

- 1 Closure phase
- 2 Post implementation review
- 3 Back-out plans

1.8.1.14. Change scheduling

Service Manager provides an embedded Change Calendar (also referred to as Service Manager Calendar) for a Forward Schedule of Change. This calendar is based on a Calendar widget and is accessible to anyone with view rights for Changes. Users with only view rights will have a read-only view and users with update rights have read/write access.

This calendar can display time period records and associated business records in a graphic and intuitive user interface. It enables users in different locations and departments of your organization to optimize their task planning accordingly.

Service Manager Calendar has a huge benefit from being a part of Service Manager. Its time period rules engine more effectively integrates time period policy into the change process, with notifications of time period violations and the capability to require additional approvals if time periods rules are violated. It is easier to initially configure and maintain.

Projected service outage

Service Manager has fields to capture the Scheduled Downtime Start and Scheduled Downtime End, as well as the Actual Downtime Start and Actual Downtime End that's the result of the Change.

This information is visible in the Configuration Item or Business Service fields to show the Projected Service Outages for that CI or Service.

Related topics

Calendar Administration
Notifying Service Subscribers of a Planned Outage

1.8.1.15. Change records and CIs

Service Manager Change Management provides the ability to link Configuration Items (CIs) to Change records. Change Tasks can be created based on the list of CIs in the Change record.

The Change record will also include a list of Affected Services that are based on the topology information for the linked CIs.

Change Management Integration: Release Management

Service Manager Change Management includes a Release Management category that provides the ability to schedule and authorize coordinated release activities that span multiple RFCs. The Release record can contain links to one or more Change records. This allows several RFCs to be assessed, scheduled, approved, and deployed as a group. The execution of the changes happens in the individual Change and Change Task records that are linked to the Release. The execution of the individual Changes can be supported by the integration to Operations Orchestration (a run a run book automation workflow tool) that automates steps of software deployment through Server Automation and Client Automation depending on the target being deployed to.

Change Management Integration: Incident Management

Service Manager allows the creation of links from a Change record to an Incident record. RFCs can be created from an Incident or an Incident can be related to an existing RFC. The Incident can also be indirectly related to a Change record through the Problem Management module.

Change Management Integration: Problem Management

Service Manager provides the ability to create Changes from Known Errors and to link Changes to existing Known Errors as recommended in the best practice documentation. Service Manager can also associate Problems and Changes through configuration of the system.

The related records tab shows established relationships between Problems and Known Errors, which are stored as separate records in Service Manager, and

Changes.

Change Management Integration: Service Desk

Email can be generated when an RFC is created, updated or closed and sent via standard email delivery tools to individuals or groups. Emails can be generated and sent automatically or manually.

Service Desk agents can be given access to the forward schedule of changes and their profiles used to control their access rights. You may want to assign them read-only access to the forward schedule of changes to avoid the entire Service Desk team from consuming change licenses.

If an Incident record has one or more related Change records, the Service Desk agent can click on the link to see the related RFC(s).

Additionally, Service Desk agents (or anyone using Service Manager) can use charts and views to get visibility into Change Management information, such as a list of Changes scheduled for the next week.

Users can easily create these charts and lists on the fly, and save them to their favorites, or they can be used to create dashboards for groups of users.

Authorized users can drill down into the details of records that are behind the charts and see individual records and their details.

Change Management Integration: uCMDB

Service Manager has an out-of-box integration with the CMS solution (UCMDB).

The CMS strategy is to leverage web services and use contributing CMDB and other data repositories through a common view of business services provided by the UCMDB. As such, the integration between UCMDB and Service Manager is using web services.

UCMDB and the discovery tools provide detailed CI and relationship information built using an innovative discovery capability. Essentially, by listening on the network and identifying applications and hardware “talking” through standard protocols, CMS uses a spiral discovery approach, interrogating applications to gain information about levels 2 through 7. This discovery capability understands how applications and hardware are related and shows this in graphical form.

The information presented in the CMS is representative of the actual state and can be viewed within Service Manager in the “Actual State” tab in a CI record.

This is an alternative view of the same data. Service Manager also provides a deliberately simpler visualization of the CIs and their relationships designed to provide the Service Desk user a summary of the information held in the CMS.

Indicators are used on the CI icons to visually show the user there are open records for the CI.

Change Management assessment capabilities are integrated with UCMDB to provide impact analysis information based on the topological view of the CI data.

Also, UCMDB provides the ability to model the potential impact of changes on the fly without physically making the change.

1.8.1.16. Change notification and escalation

Change Management leverages the new Process Designer workflows and rules. As such, there are numerous out-of-box notifications available. These rules can be customized easily to provide broad and deep control and oversight of the notifications and escalation behaviors throughout the workflow.

The Change Management workflows deliver alert options via the alert engine. This solution enables the definition of alerts including:

- When the alert should be evaluated
- What work schedule is appropriate for determining the alert evaluation
- What conditions of the ChM record are indicative that the alert should trigger
- Who should be notified when the alert has triggered
- What actions should take place

1.8.1.17. Change Management Audit trail

For all records in Service Manager, a date time stamp and a user stamp are updated in the record when it is created or saved.

Service Manager provides an audit trail capability that identifies each step taken in the resolution of a change or task. Each time they are updated (whether manually or automatically), a separate historical activity is created for the record. An authorized user may review each historical activity to develop a comprehensive understanding of the history of the record and its resolution, including which operators took what actions at what times.

Service Manager also provides an auditing feature that records modifications to fields within the database. Field modifications are detected by comparing the fields in the original version of a record to the updated version. When modifications are detected, an Audit Log entry is recorded for each changed field showing the name of the modified field, the old and new version of the data, the current date/time, and the current operator’s user ID.

1.8.1.18. What are notifications?

Messages are generated by Service Manager events, such as opening or closing a change or task. Administrators can edit these messages, add new messages, change the conditions that trigger the messages, and select who will receive the messages.

The Notification Engine normalizes notification across the applications, removing the need for each module to define its own notification process (like cm messages).

The Notification Engine is called from either the document engine Processes, or, in a few cases, directly from RAD for some modules.

1.8.1.19. Change archival

Closure of processed changes and tasks is a standard part of the workflow. The closure action not only supports saving of the record but separate validations, rules, and possibly even separate forms to ensure business success. Closing a change doesn't remove it from the database.

The most typical approach for archiving records is to leverage the chosen RDBMS tools from Oracle or MS SQL Server to perform their database administration for purging and archiving.

Service Manager provides an archive and purge tool that can be configured by table and criteria for archiving. This is a facility only the administrator would use. This capability allows the ability to archive either manually or on a scheduled basis.

One new capability in this area is the ability to take advantage of the archive tool suite. These tools allow data to be archived out of Service Manager into a system accessible file system. The production or alternative instance of Service Manager can then be used to access these data records.

1.8.1.20. Change Access Control

Change Management utilizes the new Process Designer security controls to permit authorized users to create, modify, and close changes and change tasks. The system provides Update authorization options for **Never**, **Always**, **When Assigned** and **When assigned to workgroup**. For example, one user may be able to create and close changes or tasks but only modify changes assigned to them.

The system provides the construct of a user role which allows administration of many users with a single role definition.

Access to functionality within each module is governed by Service Manager security utilities, which define access based on role and user.

Service Manager provides a security control to segment data between multiple customers. The Mandantan controls identify which customer data a particular user or group can access, update, or both.

1.8.1.21. Release Management

Release Management is a category of Change Management that ensures that the Configuration Management Database (CMDB) is kept up to date, that changes are appropriately managed, and that all new software and hardware is stored in the Definitive Software Library (DSL) and Definitive Hardware Store (DHS). After one or more changes are developed, tested, and packaged into releases for deployment, Release Management is responsible for introducing these changes and managing their release. Release Management also contributes to the efficient introduction of changes by combining them into one release and deploying them together.

Release Management relies on the Change Management and Configuration Management applications.

Goal

The purpose of Release Management is to ensure that all changes are deployed successfully in the least disruptive manner. Release Management is responsible for the following functions:

- Driving the release strategy, which is the over arching design, plan, and approach for deployment of a change into production in collaboration with the Change Advisory Board (CAB)
- Determining the readiness of each release based on release criteria (such as quality of release, release package and production environment readiness, training and support plans, rollout and backout plans, and risk management plan)

Benefits

Release Management offers the following benefits for users:

- Provides a packaged release for all changes deployed into production and only deploys changes approved by change management
- Provides two different types of releases: hardware and software. Once the release type is determined, the application provides the appropriate tasks
- Provides the ability to manage changes of groups such as Configuration Item (CI) groups and business services
- Provides the ability to terminate a release at any time except in the Training phase or once it has been installed or verified
- Provides the ability to back-out the change
- Provides the availability of optional training phase

Related topics

Release Management phases
Terminating a release change

1.8.1.21.1. Release Management phases

The Release Management category includes seven phases, which follows the recommended Information Technology Infrastructure Library (ITIL) process by providing you with a set of approvals and tasks that you can expand.

Phase	Description
Assess	Determine the importance or value of a release.
Plan and design	Plan releases in line with requirements resulting from approved changes. Analyze all the affected Configuration Items (CIs) and CI Groups, review what software or hardware is needed to accomplish the plan, and determine the cost. This phase requires the plan and design approval before going to the next step.
Build and test	Build effective release packages for the deployment of one or many changes into production. Test release mechanisms to ensure minimum disruption to the production environment.
Training	Identify and fulfill training requirements for various users. This phase is optional and is activated when you select high Impact Assessment for the release change.
Distribution and rollout	The introduction of either a hardware or software release into the IT environment.
Back out	A plan detailing how a specific release can be undone after being applied, if deemed necessary.
Verification	Review preparation for the release to ensure maximum successful deployments.

Release Management: Assess phase

The first step in the release process is to determine the importance and value of a release. If the assessment determines that a release is indeed necessary, the next step is to plan and design the release.

Release Management has no out-of-box tasks or approvals created for the Assess phase.

Release Management: Plan and design phase

The second step in the release process is the creation of a plan identifying the activities and the resources required to successfully deploy a release into the production environment. The following is a summary of the release planning activities.

- Identifying scope and content of a change, and the release requirements for successful deployment
- Performing a risk assessment for the release and obtaining the appropriate approvals
- Prioritizing, planning, and scheduling release activities
- Establishing a suitable team for the release if required
- Coordinating with experts and interested parties to determine the required resources and strategy for the release
- Documenting and tracking all release planning activities

Release Management has no out-of-box tasks for the Plan and Design phase. However, it contains the plan and design approval type. This approval must be met in order to advance to the next phase (Build and Test).

Release Management: Build and test phase

After the release team agrees on a release plan, members of the release team identify and develop the processes, tools, and technologies required to deploy the release into production. The following is a summary of the release-building activities.

- Selects a suitable release mechanism for the change that is a strategic fit, is repeatable, and is consistent
- Designs and builds a release package for the change that allows it to be successfully deployed
- Tests the release package to ensure that it delivers the change effectively in line with requirements
- Ensures the release package is updated to the Configuration Management Database (CMDB) as a pending release package and that any new script is

added to the Definitive Software Library (DSL)

Release Management has two out-of-box tasks created for the Build and Test phase:

- Identify the affected systems (setting up the affected Configuration Items (CIs) – hardware and software
- Create release (software development, software push, hardware install, or other similar tasks) – software only

Tip

To view these tasks, select the **Release Build and Test** phase and then click **Additional Phase Information**. Select the **Auto Open Tasks** tab and there are two tasks defined for the **Build and Test** and **Risk and Impact Analysis** categories.

Any changes to the Build and Test phase will affect both Release Management and Normal workflows. This phase of Release Management contains the build and test approval type. This approval must be met in order to advance to the next phase (Training or Distribution and Rollout).

Note

If the test fails, you return to the plan and design phase to reevaluate the scope and content of the change.

Release Management: Training phase

For some releases, it is necessary to train users, Service Desk staff, technical support teams, and operations staff in the skills and abilities they require to perform their job roles. Training may take the form of classroom training courses, workshops, or seminars, but it could also be provided by on-demand computer-based training (CBT) courses and simulations.

The Release Manager identifies and describes the type of training to be provided, when the training is to be conducted, how it is to be administered, and the reasons for providing it. A detailed estimate of cost may also be required.

This phase is optional and is activated when you select high Impact Assessment for the release change during the Assess phase.

The Training phase is the only phase in the Release Management process that does not include the option to either terminate or back out of the release.

Release Management: Distribution and rollout phase

After acceptance training and testing is complete, the next step is to deploy the release into the production environment. The process depends on the type and nature of the release and on the selected release mechanism. In all cases, however, the Release Manager should be provided with status reports and, where appropriate, tools and technologies that enable tracking and monitoring of deployment progress. As changes are made to IT components during deployment, corresponding changes are made to the configuration items and relationships modeling them in the CMDB.

The following is a summary of the activities in the Distribution and Rollout phase:

- Selects a suitable deployment group for the release
- Ensures users are adequately trained in a timely manner for the maximum benefit from the release
- Deploys the release into the production environment
- Reviews the deployed release, taking into account feedback and comments from all parties involved
- Passes the feedback to the change manager for the change review process

Release Management has one out-of-box Update System task created for the Distribution and Rollout phase.

This phase of Release Management contains the Release Distribution and Rollout approval type. This approval must be met in order to advance to the next phase (Verification).

Release Management: Backout phase

If the release fails to meet expectations or if serious problems are encountered during deployment, Problem Management may be required to help identify and diagnose the root cause of the problem. If a suitable fix or workaround can be found, it should be documented and a Request for Change (RFC) created to deploy the fix or workaround into the production environment. If no fix or workaround is found, it may be appropriate to use the backout procedures to remove the release from that environment. If the release needs to be backed out, this should also be recorded, including any information that supports this decision.

The Backout option is available during the Distribution and Rollout phase or the Verification phase.

Release Management: Verification phase

After the release is deployed, the Release Manager confirms that the release is working correctly before proceeding with further deployments. The Verification phase ensures that the deployment of the release is successful and meets all required criteria.

When the release deployment phase is complete, the release process should ensure that any results and information about any workarounds or Request for Changes (RFCs) raised in support of the release are recorded. If the release needs to be backed out, this should also be recorded, including any information that supports this decision.

This phase of Release Management contains the verification approval type.

Related topics

Release Management
Terminating a release change

1.8.1.21.2. Terminating a release change

You can terminate a release in any phase except in the Training phase or after it has been installed or verified. If you reopen a change that was closed, you are returned to the assess phase to determine whether or not a release is still required. This behavior is different than if a release is closed. In that case, the release cycle is complete and therefore closed. When you reopen a closed release, you return to the verification phase since this is the last completed phase in the cycle.

Related topics

Release Management
Release Management phases

1.8.2. Change Management user roles

The Change Management module has the following user roles:

Change Analyst

The Change Analyst user role has the following responsibilities:

- May be involved in the change assessment and planning phase to deliver input to the Change Coordinator when assessing the change impact
- Verifies that tasks are correctly assigned and rejects tasks if needed
- Builds, tests, and implement changes based on the change plan
- Executes the backup plan if required

To execute this role, the Change Analyst has following tasks available:

Note

For an overview of how these tasks fit in the Change Management workflow, see Change Management workflows and user tasks.

ITIL workflow	Tasks
Change logging (ST 2.1)	• None
Change review (ST 2.2)	• None
Change assessment and planning (ST 2.3)	• None
Change approval (ST 2.4)	• None
Coordinate change implementation (ST 2.5)	• Validate a change task • Reject a change task • Build and test a change • Implement a change and perform a production test • Execute a fall-back scenario
Change evaluation and closure (ST 2.6)	• None
Emergency change handling (ST 2.7)	• None
Change Management tasks	• Close a change task • Set a reminder for a change request

Change Coordinator

The Change Coordinator user role has the following responsibilities:

- Register changes
- Review the changes received and validate the change after filtering

- Complete registration of changes by applying the correct change model and providing the change details
- Coordinate the assessment phase of the change, and create change planning based upon the assessment information
- After approval, coordinate the change implementation according to the planning created earlier
- Assign tasks to the Change Analysts and perform action based on the outcome of the change implementation
- After implementation, evaluate the change handling and close the change

To execute this role, the Change Coordinator has following tasks available:

Note

For an overview of how these tasks fit in the Change Management workflow, see Change Management workflows and user tasks.

ITIL workflow	Tasks
Change logging (ST 2.1)	• Create a new change request • Update a change request
Change review (ST 2.2)	• Review a new change request • Relate a new change to a change • Delete a related change to a change • Return a change request to the requester • Reject a change request
Change assessment and planning (ST 2.3)	• Assess change impact and risk • Plan and schedule a change
Change approval (ST 2.4)	• None
Coordinate change implementation (ST 2.5)	• Schedule a change implementation • Process a successful change implementation • Process an unsuccessful change implementation
Change evaluation and closure (ST 2.6)	• None
Emergency change handling (ST 2.7)	• None

ITIL workflow	Tasks
Change Management tasks	• Create a change task • Update a change task • Close a change task • Reopen a change request or task • Print a change request • Print a change request list • Associate a change with another change • Set a reminder for a change request • Send a notification from a task or change request

>

The Change Manager user role has the following responsibilities:

- Review all changes after the assessment and planning phase and address the right Change Approvers
- Organize Change Advisory Board meeting if necessary
- Update the change after approval is given
- Periodically review changes in a Post Implementation Review and determine and execute follow-up actions
- Coordinate all activities in case the emergency change handling procedure is triggered

To execute this role, the Change Manager has following tasks available:

Note

For an overview of how these tasks fit in the Change Management workflow, see Change Management workflows and user tasks.

ITIL workflow	Tasks
Change logging (ST 2.1)	• None
Change review (ST 2.2)	• Relate a new change to a change • Delete a related change to a change • Reject a change request
Change assessment and planning (ST 2.3)	• None
Change approval (ST 2.4)	• Prepare a change for approval • Process an approved, denied, or retracted change
Coordinate change implementation (ST 2.5)	• None
Change evaluation and closure (ST 2.6)	• Evaluate change handling • Close a change • Generate a list of closed changes for review
Emergency change handling (ST 2.7)	• Log an emergency change • Approve an emergency change • Recategorize an emergency change

ITIL workflow	Tasks
Change Management tasks	• Update a change task • Close a change task • Print a change request • Print a change request list • Set a reminder for a change request • Update multiple change requests • Update multiple task records

Change Approver

The Change Approver user role has the following responsibilities:

Approve or deny Change when requested. This can be either electronically by using the service management tool or by using a Change Advisory Board (CAB) or Emergency-Change Advisory Board (E-CAB) meeting.

To execute this role, the Change Approver has following tasks available:

Note

For an overview of how these tasks fit in the Change Management workflow, see Change Management workflows and user tasks.

ITIL workflow	Tasks
Change logging (ST 2.1)	• None
Change review (ST 2.2)	• None
Change assessment and planning (ST 2.3)	• None
Change approval (ST 2.4)	• Approve, deny, or retract a change • Approve multiple change requests
Coordinate change implementation (ST 2.5)	• None
Change evaluation and closure (ST 2.6)	• None
Emergency change handling (ST 2.7)	• None
Change Management tasks	• Set a reminder for a change request

Release Packaging and Build Manager

The Release Packaging and Build Manager user role has the following responsibilities:

Change Analyst who transfers the new release from development to test environment or from test to production environment.This role can't be fulfilled by the Change Analyst who has built the new release.

To execute this role, the Release Packaging and Build Manager has following tasks available:

Note

For an overview of how these tasks fit in the Change Management workflow, see Change Management workflows and user tasks.

ITIL workflow	Tasks
Change logging (ST 2.1)	• None
Change review (ST 2.2)	• None
Change assessment and planning (ST 2.3)	• None
Change approval (ST 2.4)	• None
Coordinate change implementation (ST 2.5)	• None
Change evaluation and closure (ST 2.6)	• None
Emergency change handling (ST 2.7)	• None
Change Management tasks	• None

E-CAB

The E-CAB user role has the following responsibilities:

Selection of Change Approvers who need to provide approval in case of an Emergency Change.

To execute this role, the E-CAB has following tasks available:

Note

For an overview of how these tasks fit in the Change Management workflow, see Change Management workflows and user tasks.

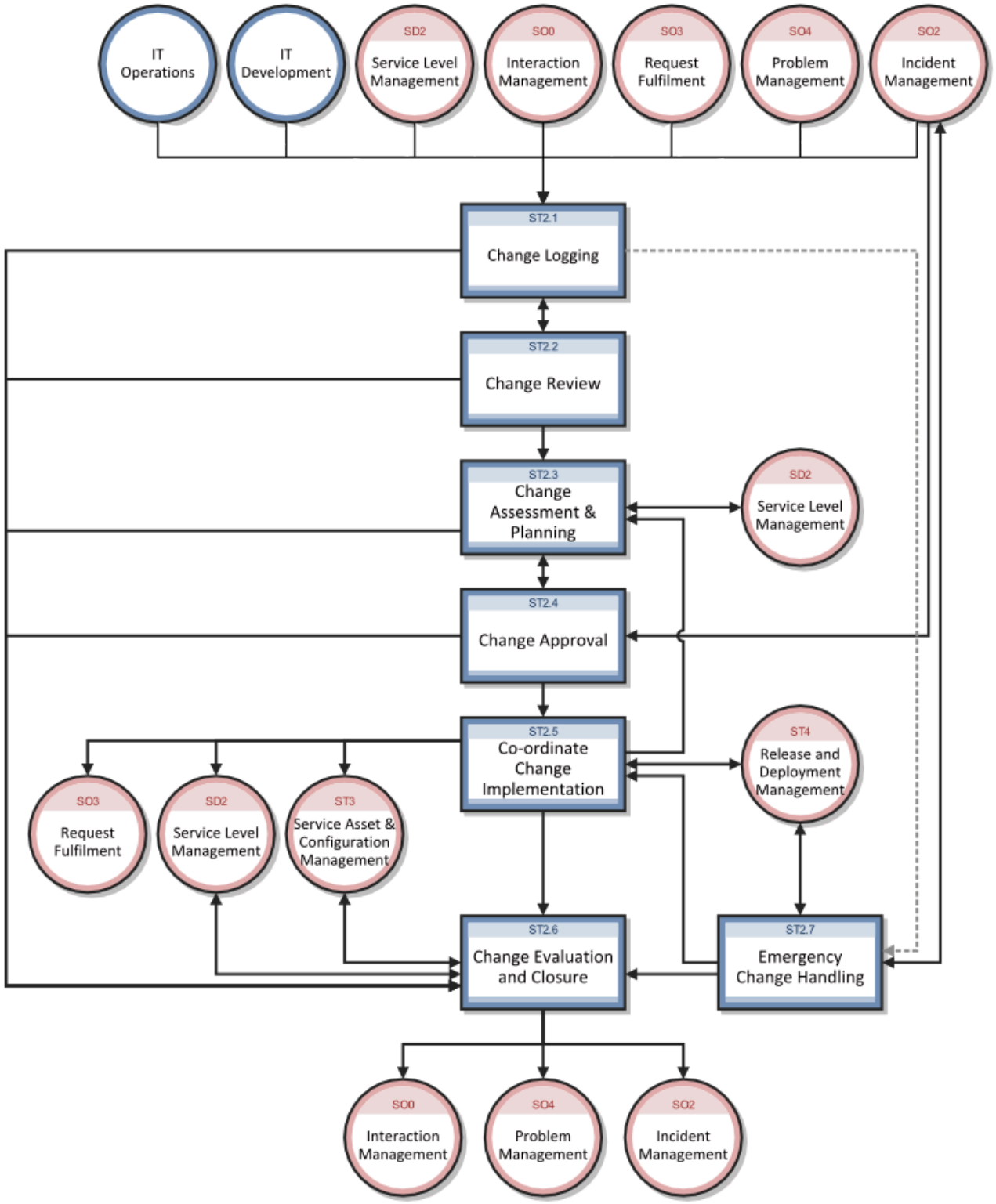
ITIL workflow	Tasks
Change logging (ST 2.1)	• None
Change review (ST 2.2)	• None
Change assessment and planning (ST 2.3)	• None
Change approval (ST 2.4)	• Approve, deny, or retract a change • Approve multiple change requests
Coordinate change implementation (ST 2.5)	• None
Change evaluation and closure (ST 2.6)	• None
Emergency change handling (ST 2.7)	• None
Change Management tasks	• Set a reminder for a change request

1.8.3. Change Management workflows and user tasks

The Change Management process includes the activities necessary to control changes to service assets and configuration items across the entire service life cycle. It provides standard methods and procedures to use when implementing all changes.

The Change Management process consists of the following workflows and user tasks.

Change Management workflow


IT Operations
IT Development
SD2
Service Level Management
SO0
Interaction Management
SO3
Request Fulfilment
SO4
Problem Management
SO2
Incident Management
ST2.1
Change Logging
ST2.2
Change Review
ST2.3
Change Assessment & Planning
SD2
Service Level Management
ST2.4
Change Approval
ST2.5
Co-ordinate Change Implementation
ST4
Release and Deployment Management
SO3
Request Fulfilment
SD2
Service Level Management
ST3
Service Asset & Configuration Management
ST2.6
Change Evaluation and Closure
ST2.7
Emergency Change Handling
SO0
Interaction Management
SO4
Problem Management
SO2
Incident Management

ITIL workflow	Tasks
Change logging (ST 2.1)	• Create a new change request • Update a change request
Change review (ST 2.2)	• Relate a new change to a change • Delete a related change to a change • Review a new change request • Return a change request to the requester • Reject a change request
Change assessment and planning (ST 2.3)	• Assess change impact and risk • Plan and schedule a change
Change approval (ST 2.4)	• Approve, deny, or retract a change • Approve multiple change requests • Prepare a change for approval • Process an approved, denied, or retracted change

ITIL workflow	Tasks
Coordinate change implementation (ST 2.5)	• Validate a change task • Reject a change task • Build and test a change • Implement a change and perform a production test • Execute a fall-back scenario • Schedule a change implementation • Process a successful change implementation • Process an unsuccessful change implementation
Change evaluation and closure (ST 2.6)	• Evaluate change handling • Close a change • Generate a list of closed changes for review
Emergency change handling (ST 2.7)	• Log an emergency change • Approve an emergency change • Recategorize an emergency change

ITIL workflow	Tasks
Change Management tasks	• Create a change task • Update a change task • Set a reminder for a change request • Close a change task • Reopen a change request or task • Print a change request • Print a change request list • Associate a change with another change • Send a notification from a task or change request • Update multiple change requests • Update multiple task records

To see the workflow diagrams and other information about this process, refer to *Service Manager Processes and Best Practices*.

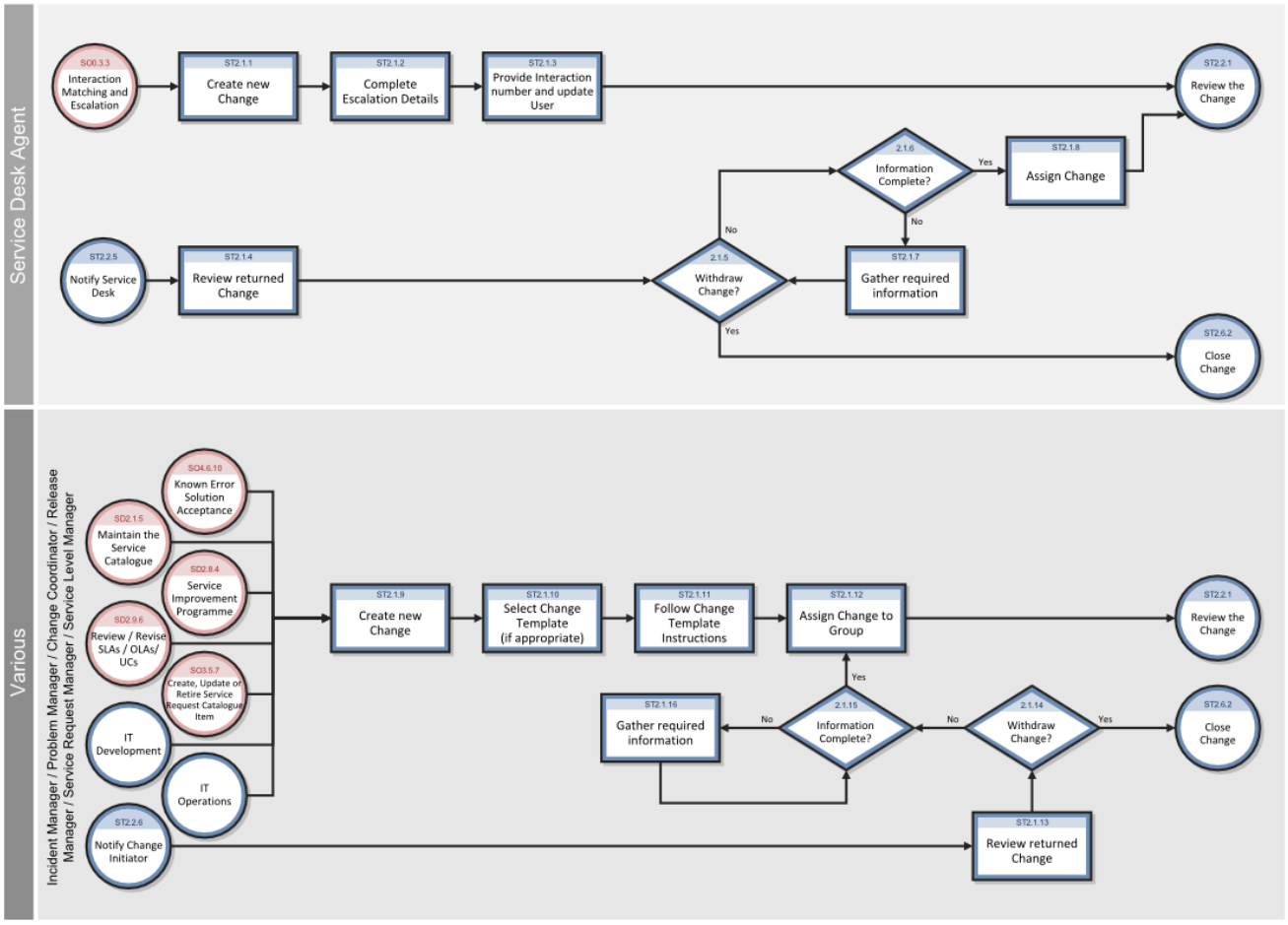
Related topics

Change Management overview

1.8.3.1. Change logging (ST 2.1)

An individual or organizational group that requires a change can initiate a Request for Change (RFC). Change requests can be initiated as part of a variety of management processes, including User Interaction Management, Incident Management, Problem Management, and Release Management. Each RFC must be registered in an identifiable way. Service Manager provides change templates that standardize and speed up the change logging process.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Create a new change request
Create change request from user interaction
Update a change request
Withdraw a change request
Change Management workflows and user tasks

1.8.3.1.1. Create change request from user interaction

Applies to User roles: Service Desk Agent

As a Service Desk Agent, you can escalate an interaction to a change record that's linked to the interaction.

To create a change request from a user interaction, follow these steps:

1. Click **Service Desk** > **Interaction Queue** or select **Interaction** from your To Do Queue list.
2. From the record list, double-click the open-idle Interaction that you want to escalate to a change.
3. Click **Fill** for the **Category** field.
4. Double-click **Request for Change** and select the subarea.
5. Click **Escalate**. The **Escalation Details** wizard opens.
6. Click the list fields to select the Location, Assignment, and Requested End Date for the change.

   Note

   A Change Coordinator is automatically assigned based on your selections.

7. Click **Finish**. A notification message with the change number displays above the interaction detail fields and the Service Desk application notifies the original end user of the actions taken.
8. Click **Save & Exit**.

Related topics

Create a new change request
Update a change request
Withdraw a change request

1.8.3.1.2. Create a new change request

Part of Workflow(s): Change logging (ST 2.1)

Applies to User roles: Change Coordinator, Release Manager

Sometimes you will need to create a new change request that's not based on an existing interaction or known error.

To create a new change request, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Click **New**. Service Manager displays a list of categories specific to your user profile.
3. Click a category for the new change request. Service Manager displays a new form and populates the following fields based on your previous selections:
   - Change ID
   - Phase
   - Status
   - Approval Status
4. Complete the form with all of the required information and any additional information.
5. If you are logged in as a Problem Manager and want to associate an existing known error with this change, use the **More** button or the **More Actions** icon to associate a known error record with this change.
6. Click **Save**. A message displays the change number and a notification of the change is sent to the selected change initiator.
7. Click **More** or the **More Actions** icon and then select **Change Phase**.
8. Select the applicable phase. For example, Change Review. The phase status changes to the phase you selected.

Related topics

Create change request from user interaction
Update a change request
Withdraw a change request

1.8.3.1.3. Update a change request

Part of Workflow(s): Change logging (ST 2.1)

Applies to User roles: Service Desk Agent, Change Coordinator, Release Manager

You may get notifications (for example, by telephone or email) that a Change you created has returned to the Change Logging or Change Assessment and Planning phase. In this case, you need to update this Change. For example, you may have assigned the Change to the wrong Change Coordinator or the Change Manager may disagree with the impact assessment or risk categorization and request an update.

To update a change request, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Select the change request to update from the record list.
3. Click the **Activities** tab. Use the applicable entries on the **Activity Logs** tab to view the details of the updates needed for the change record.
4. Modify the applicable fields and then click **Save**.
5. Click **More** or the **More Actions** icon and then select Change Phase. Double-click the applicable phase.
6. Click **Save & Exit**.

Related topics

Create a new change request
Create change request from user interaction
Withdraw a change request

1.8.3.1.4. Withdraw a change request

Applies to User roles: Service Desk Agent, Service Desk Manager

In some cases, the Change Coordinator will return a change request, which the Service Desk Agent or Service Desk Manager will need to withdraw.

To withdraw a change request, follow these steps:

1. From the To Do queue, select **Change** from the **Queue** list.
2. Expand **Phase: Change Logging**, and then click a change request to display its detail.
3. Click **Withdraw**. A notification appears, confirming the change request's closed status.

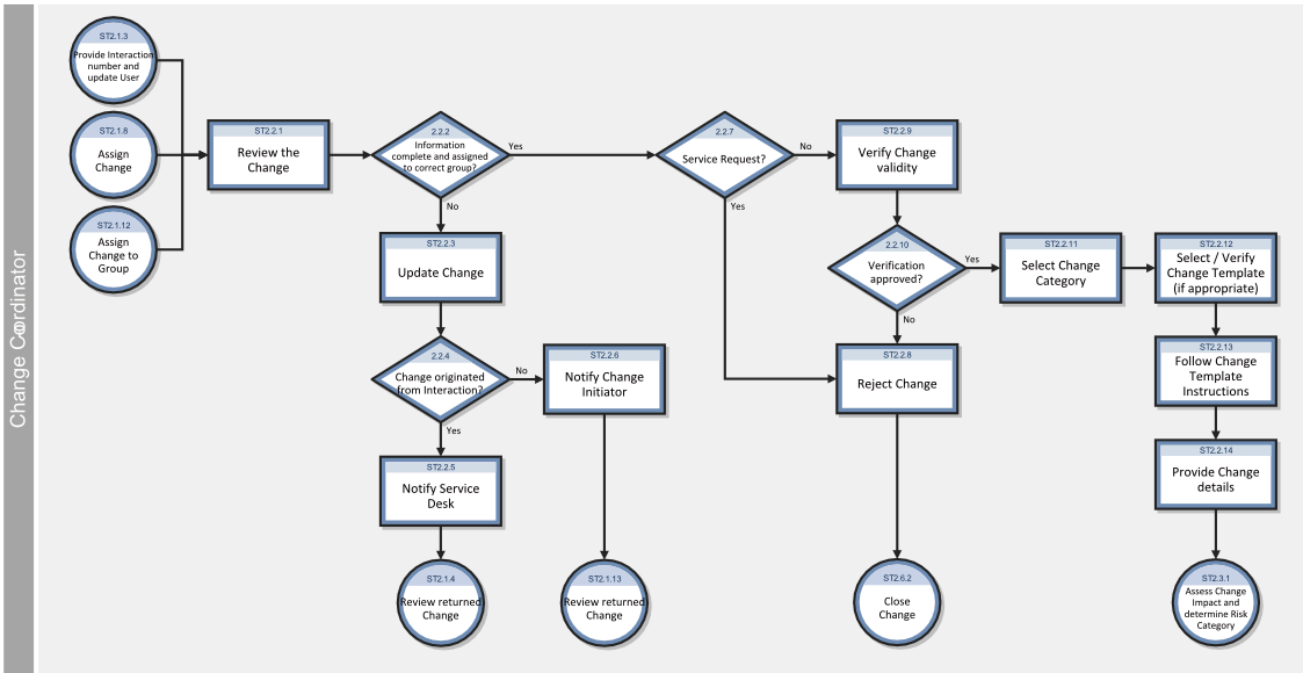
Related topics

Create a new change request
Create change request from user interaction
Update a change request

1.8.3.2. Change review (ST 2.2)

After a change request is logged, the Change Coordinator verifies that the request is logical, feasible, necessary, and complete. If more information is needed, the Change Coordinator will request that the initiator update the request. The Change Coordinator also checks to see if the change has already been previously submitted and rejected. If a requested change doesn't meet the requirements, the Change Coordinator rejects the change and communicates the reason for the rejection to the change initiator.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Review a new change request
Return a change request to the requester
Reject a change request
Change Management workflows and user tasks

1.8.3.2.1. Review a new change request

Part of Workflow(s): Change review (ST 2.2)

Applies to User roles: Change Coordinator

After a user creates a new change request or a Service Desk Agent escalates an interaction to a change, the change request moves to the Change Review phase. You can then review the information to verify that all the information you need is provided and that the change is assigned to the correct support group.

To review a new change request, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and **My Open Changes** from the **View** list.
2. Expand **Phase: Change Review** from the record list, and then click the change request to view its detail.
3. Check to see if the change request is a standard request to be handled by the request fulfillment process. If it is, select **Reject** from the **More Actions** menu to reject the change.
4. Enter the reason for the rejection and click **Finish** to move to the Evaluation and Change Closure phase.
5. Review the information in the form to determine if it is complete and verify that the change is assigned to the correct support group. If more information is needed or the support group is incorrect, send the request back to the requester or to the Service Desk.
6. If the request status in the **Category** field is **Default**, select **Change Category** from the **More Actions** menu and select the appropriate category. Fill in any other required fields.
7. Click **Close Phase**. The phase status changes from Change Review to Change Assessment & Planning.

Related topics

Implement a change and perform a production test
Reject a change request
Return a change request to the requester

1.8.3.2.2. Relate a new change to a change

Part of Workflow(s): Change review (ST 2.2)

Applies to User roles: Change Manager, Change Coordinator

To relate a new change to an existing change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Select the change from the record list.
3. Click **More** or the **More Actions** icon and then select **Related** > **Changes** > **Open**.
4. Complete all of the required information for the change record form.
5. Click **Save & Exit**.

You can view the Change in the Related Records section to verify the newly related change record.

Related topics

Associating a change with another record
Associate a change with another change

1.8.3.2.3. Delete a related change to a change

Part of Workflow(s): Change review (ST 2.2)

Applies to User roles: Change Manager, Change Coordinator

If a change is related to another change, the relationship between the two can be deleted so that the two changes are no longer related to each other. Note that deleting a related change doesn't delete either change record. Both change records continue to exist, but they are no longer related to each other.

To delete a related change to a change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Select the change from the record list.
3. From the **More Actions** menu, select **Related** > **Changes** > **View**.
4. Select the change record you want to delete (unrelate) from the list of related changes.

   To select a record in the web client, click a non-link section of the record.

   To select a record in the Windows client, click the record.
5. Click **Delete**.
6. Click **OK** to confirm that you want to delete this record from the list of related change records.
7. Type a reason for unrelating the changes in the text box.

   Note

   You may not be required to provide a reason for unrelating the changes in some installations.
8. Click **OK**.

You can view the changes in the **Related Records** section to verify that the change records are no longer related.

Related topics

Associating a change with another record
Relate a new change to a change

1.8.3.2.4. Return change request to requester

Part of Workflow(s): Change review (ST 2.2)

Applies to User roles: Change Coordinator

If the information in the change form is incomplete, you may contact the requester to get the missing information and then enter it in the Activities section of the change record. If the change is assigned to the wrong support group, you should return it to the requester by changing the phase back to Change Logging and then contacting the requester to have the requester correct the change record. The requester then places the request in the Change Review phase.

To return a change request to the requester, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list, and then select **My Open Changes** from the **View** list.
2. Click a change request from the record list.
3. Open the **Activities** section and select the type of update in the **New Update Type** field.
4. Type the reason for returning the change in the **New Update** field.
5. Click **More** or the **More Actions** icon and then select **Change Phase**.
6. Double-click **Change Logging**.

   The phase status changes from Change Review to Change Logging.
7. Contact the requester (by telephone or email) to let the requester know that the change was returned to the Change Logging phase and that the requester must correct it.

Related topics

Implement a change and perform a production test
Reject a change request
Review a new change request

1.8.3.2.5. Reject a change request

Part of Workflow(s): Change review (ST 2.2)

Applies to User roles: Change Manager, Change Coordinator

You may need to reject a change request during the change review. For example, you may need to reject a standard change that should have initially been handled using the request fulfillment process or because the change is invalid. The reason for the rejection could be because it's not logical or feasible, is not in line with company standards and policies, or it has already been raised and rejected in an earlier phase. You may also reject a change during the Change Assessment and Planning phase if change assessors don't support approval.

To reject a change request, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and **My Open Changes** from the **View** list.
2. Click a change from the record list and select **Reject** from the **More Actions** menu.
3. In the **Rejection Details** wizard, type the reason for the rejection and click **Finish**.

   The change is now in the Change Evaluation and Closure phase.

Related topics

Implement a change and perform a production test
Review a new change request

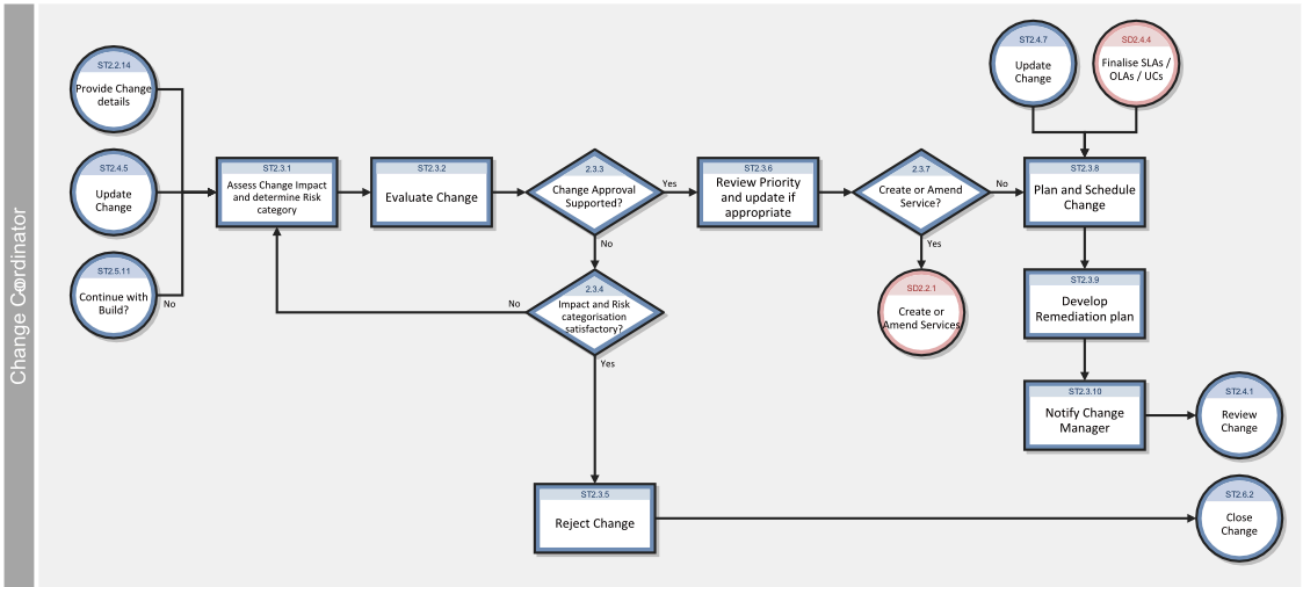
1.8.3.3. Change assessment and planning (ST 2.3)

For all normal changes, the Change Coordinator assesses the need for a change based on answers to the following questions:

- Who is the requestor that initiated the need for the change?
- What's the reason for the change?
- What's the result required from the change?
- What are the risks involved in the change?
- What resources are required to deliver the change?
- Who is responsible for the build, test, and implementation of the change?
- What's the relationship between this change and other changes?

Based on the answers to these questions, the change is categorized, prioritized, and planned, and then a remediation plan is developed.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Assess change impact and risk
Plan and schedule a change
Change Management workflows and user tasks

1.8.3.3.1. Assess change impact and risk

Part of Workflow(s): Change assessment and planning (ST 2.3)

Applies to User roles: Change Coordinator

After a change is reviewed and its validity is verified, it moves to the Change Assessment and Planning phase. You can then start the impact and resource assessment.

To register your assessment, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and then select **My Open Changes** from the **View** list.
2. Expand **Phase: Change Assessment Planning** from the record list, and then click the change request.
3. Select or revise the values in the **Impact**, **Urgency**, and **Risk Assessment** fields in the **Change Details** section. You can also use the **Description** field for additional notes.

   Be sure to consider the following impacts in your assessment:

   - Impact of change on business operations
   - Impact on infrastructure and customer service
   - Impact on other services that run on the same infrastructure
   - Impact on infrastructures in the organization that aren't Information Technology (IT) related
   - Impact of not implementing the change
   - IT, business, and other resources required to implement the change
   - Current change schedule (CS) and projected service outage (PSO)
   - Additional ongoing resources required if the change is implemented
   - Impact on the continuity plan, capacity plan, security plan, regression test scripts, data and test environment, and Service Operations practices

   Note

   After you add values to the Impact and Urgency fields, the Priority field is automatically calculated based on those values.

4. Contact the change assessors for their evaluation of the change, and then update the change with their comments.

The result of the evaluation is:

- Temporary approval
- Readjust the impact and risk categorization
- Reject the change and update the description with an explanation for the rejection

  Click **More** or the **More Actions** icon and then select **Reject** to forward the change to the Evaluation and Closure phase. In the **Rejection Details** wizard, type the reason for the rejection and then click **Finish**.

5. Determine whether or not the change needs to be implemented using Release and Deployment Management due to a functional change to the service. If so, select the **Release Management** checkbox in the **Change Details** section.
6. Click **Save & Exit**.
7. After completing these steps, plan and schedule the change if necessary.

Related topics

Reject a change request
Plan and schedule a change
More Actions menu
Change assessment and planning (ST 2.3)

1.8.3.3.2. Plan and schedule a change

Part of Workflow(s): Change assessment and planning (ST 2.3)

Applies to User roles: Change Coordinator

After assessing the change impact and risk and registering the results of your assessment, you can plan and schedule the change.

To plan and schedule a change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and **My Open Changes** from the **View** list.
2. Expand **Phase: Change Assessment Planning** from the record list and click the change request.
3. In the **Change Details** section, enter the **Planned Start** and **Planned End** dates.
4. Type the details of the plan changes in the **Plan** field.
5. Click **Close Phase**. The phase status changes from Change Assessment & Planning to Change Approval.

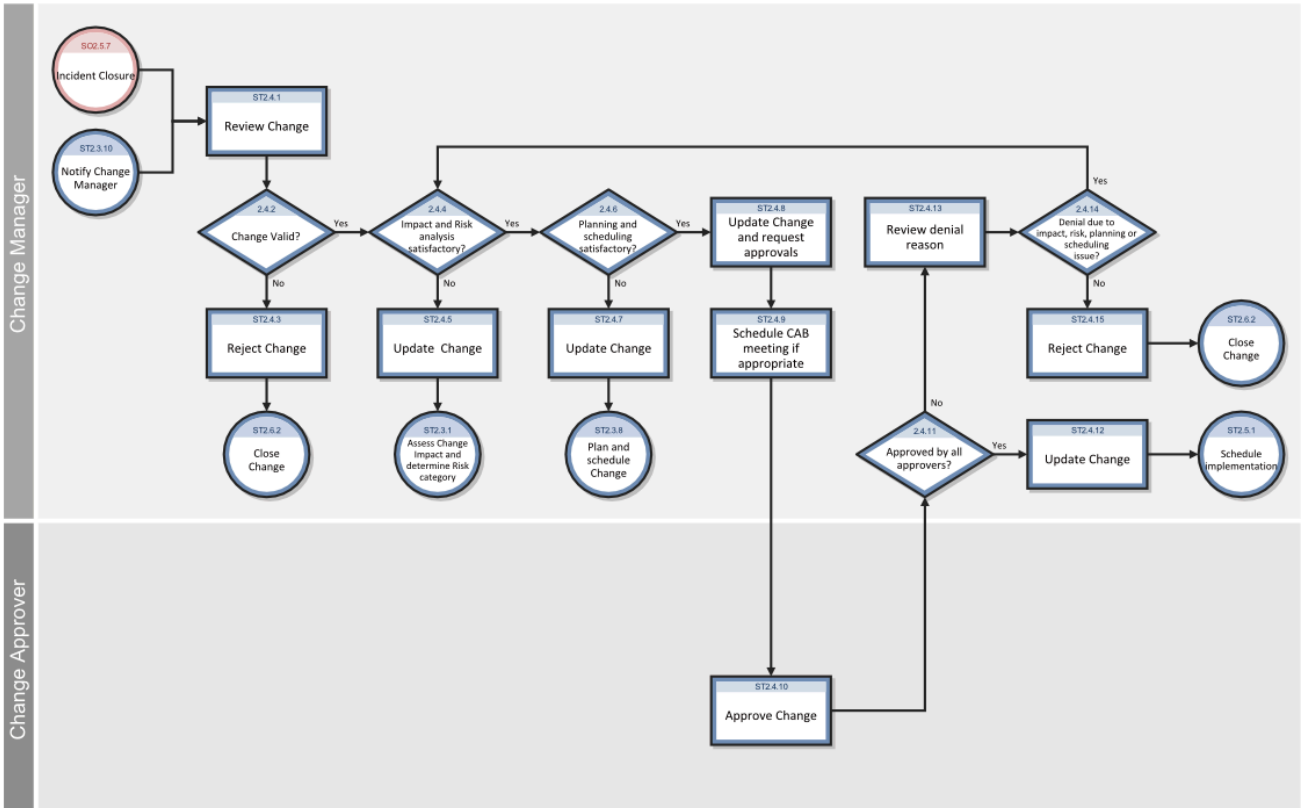
Related topics

Change assessment and planning (ST 2.3)
Assess change impact and risk
Reject a change request

1.8.3.4. Change approval (ST 2.4)

Every change requires a formal authorization from a change authority, which may be a role, person, or group of people. The levels of authorization for a particular type of change are judged by the type, size, or risk of the change.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Prepare a change for approval
Approve, deny, or retract a change
Process an approved, denied, or retracted change
Approve multiple change requests
Change Management workflows and user tasks

1.8.3.4.1. Prepare a change for approval

Part of Workflow(s): Change approval (ST 2.4)

Applies to User roles: Change Manager

After a change is planned and scheduled by the Change Coordinator, it moves to the Prepare for Change Approval phase. During this phase, the Change Manager reviews the change for completeness and validity and then rejects or updates the change or prepares the change for approval.

To prepare a change for approval, follow these steps:

1. From your To Do Queue, select **Change** from the **Queue** list and **All Open Changes** from the **View** list.
2. Expand **Phase: Prepare for Change Approval** from the record list, and click a change request to view its detail.
3. Review the change to verify that it is valid. Reasons a change could be considered invalid are if the information is incomplete, it's not compliant with change and company policies, or it has already been raised and rejected in an earlier phase.
4. If the change is invalid, select **Reject** from the **More Actions** menu on the details toolbar to reject the change. In the **Rejection Details** wizard, type the reason for the rejection and then click **Finish**.
5. If the change is valid, assess the change impact and risk categorization.
6. If you disagree with the impact assessment and risk categorization:
    1. Update the description with the reason you disagree.
    2. From the **More Actions** menu on the details toolbar, click **Change Phase** > **Change Assessment & Planning**. Service Manager returns the change to the Change Assessment and Planning phase.
7. If you agree with the impact assessment and risk categorization, review the change schedule and plan.
8. If you disagree with the change schedule and plan:
    1. Update the description with the reason you disagree.
    2. From the **More Actions** menu on the details toolbar, click **Change Phase** > **Change Assessment & Planning**. Service Manager returns the change to the Change Assessment and Planning phase.
9. If you agree with the schedule and change plan, determine the level of authorization based on the type, size, or risk of the change, and then select the change approvers.

10. Determine whether a Change Advisory Board (CAB) meeting must be scheduled to discuss the change approval or if the change can be authorized by using email or using the automatic notification system.
11. Click **Close Phase**. Service Manager moves the change to the Change Approval phase.

Related topics

Assess change impact and risk
Reject a change request

1.8.3.4.2. Approve, deny, or retract a change

Part of Workflow(s): Change approval (ST 2.4)

Applies to User roles: Change Approver

You can approve or deny a change that's pending approval only if you are a member of the necessary approval group and you are assigned the appropriate Change Management user profile. You can also retract a change that has been previously approved or denied, if you are unwilling to commit resources or know of technical incidents that affect the request.

To process a change approval request, follow these steps:

1. From your To Do Queue, select **Change** from the **Queue** list and then select **Awaiting My Approval** from the **View** list.
2. Click to open the change request you want to approve.
3. Open the **Approvals** section to determine which approvals are required. The Current Approvals form lists the approval type and approval status.
4. Double-click **Approval** in the **Approval Type** field to learn which approval group is assigned this approval type. When a change request requires more than one approval, the approvals must occur in the order listed in the Sequence column.
5. Review the change information and determine how you will process the approval request.
6. To approve a change, click **Approve**.

   Service Manager changes the **Approval Status** to **approved**, and the Change Manager updates the change and passes it to the Change Coordinator for implementation.
7. To deny a change, click **Deny**. Type the reason you are denying the change, and then click **OK**.

   Service Manager changes the **Approval Status** to **denied**, and no further approvals are possible until the denial is retracted.
8. To retract a change, click **Retract**. Type the reason you are retracting the change, and then click **OK**.

   Service Manager changes the **Approval Status** to **pending**, and the change request requires a new approval cycle to progress.

Note

Select **Retract** to remove a previous approval or denial and reset the change request to **pending.**

Related topics

Change Management overview
Managing approvals in Change Management
Approval options in Change Management
More Actions menu
Approve multiple change requests

1.8.3.4.3. Process an approved, denied, or retracted change

Part of Workflow(s): Change approval (ST 2.4)

Applies to User roles: Change Manager

After a change is approved, denied, or retracted by the Change Approver, you can move it to the applicable phase. If the change is approved, update the change and move it to the Implementation procedure. If the change is denied or retracted, return the change to the Change Assessment and Planning procedure for updates or reject the change.

To process an approved, denied, or retracted change, follow these steps:

1. From your To Do Queue, select **Change** from the **Queue** list and **All Open Changes** from the **View** list.
2. Select a change request to view its detail.
3. Review the **Approval Status** field.
4. If the change was approved by all approvers:
    1. In the **Activities** section, select an update type from the **New Update Type** list and then type the approval information in the **New Update** field.
    2. Click **Close Phase**. Service Manager closes the current phase, and it may move the change to the next phase, depending on the phase configuration and work flow.
5. If the change was denied or retracted by all approvers:
    1. In the **Activities** section, select an update type from the **New Update Type** list and then type the denial or retraction information in the **New Update** field.
    2. If the change was denied or retracted because of impact, risk planning, or scheduling issues, click **More** or the **More Actions** icon and then select **Change Phase**. Change the phase to the phase used for assessment and planning.
    3. If the change was denied for a reason other than impact, risk planning, or scheduling, click **More** or the **More Actions** icon and then select **Reject** to reject the change request.
    4. Open the **Activities** section to provide a reason for the rejection.

The change is now closed.

Related topics

Change Management overview
Approve multiple change requests
Reject a change request

1.8.3.4.4. Approve multiple change requests

Part of Workflow(s): Change approval (ST 2.4)

Applies to User roles: Change Approver

You can approve multiple change requests by using mass approval. When you select the **Mass Approve** option, all the requests on the list that are waiting for your approval are immediately approved.

To approve multiple change records, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Click **Mass Approve** in the **More Actions** menu to approve all records in the list that are pending your approval.

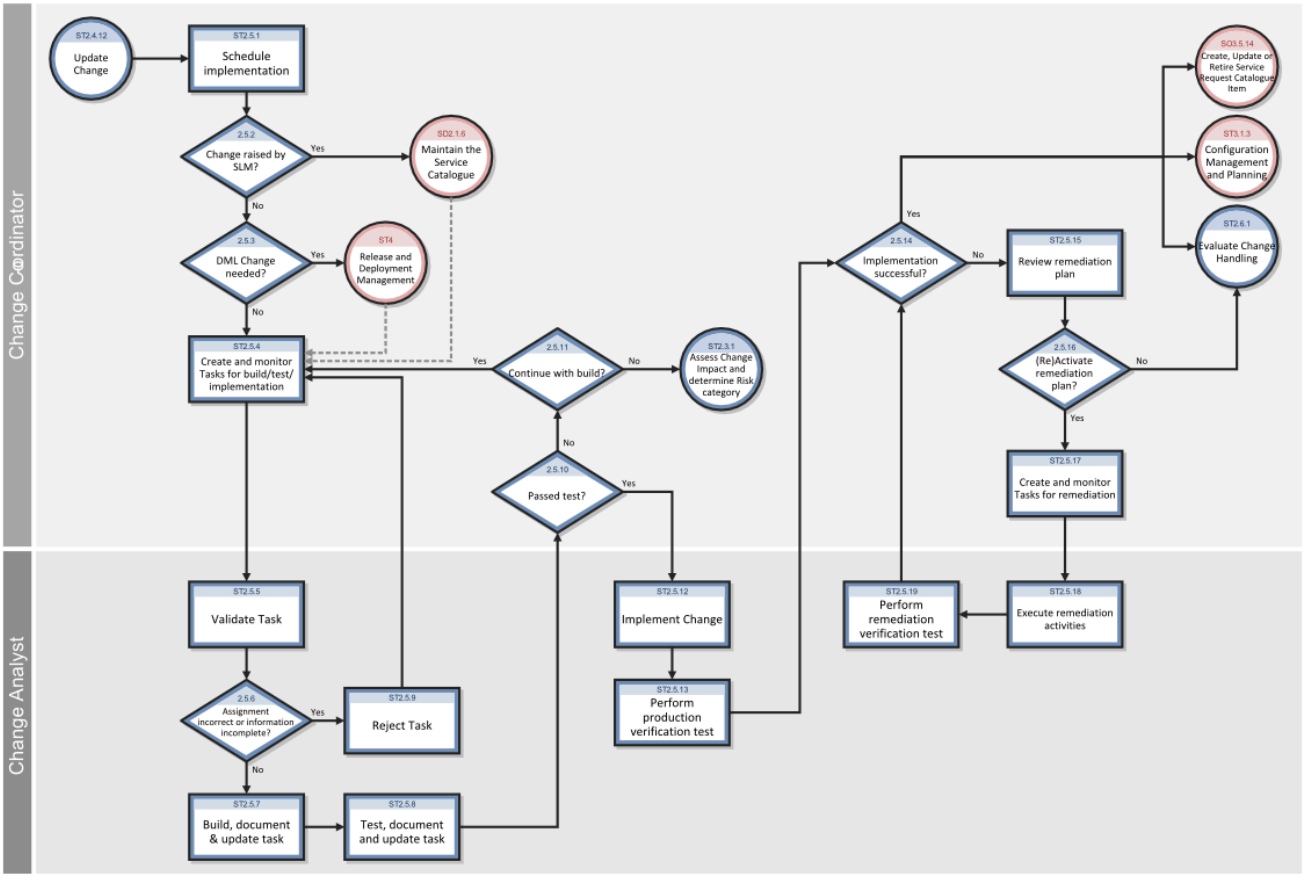
Related topics

Change Management overview

1.8.3.5. Coordinate change implementation (ST 2.5)

Authorized change requests should be passed to the relevant technical groups for building, testing, and implementing the change. The Change Coordinator schedules tasks for the build, test, and implementation phases and assigns those tasks to the responsible Change Analysts. Change Management is responsible for ensuring that changes are implemented as scheduled. The actual implementation of authorized changes is performed by Change Analysts in the specialist groups.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Schedule a change implementation
Create a change task
Validate a change task
Reject a change task
Build and test a change
Implement a change and perform a production test

Process a successful change implementation
Process an unsuccessful change implementation
Execute a fall-back scenario
Change Management workflows and user tasks

1.8.3.5.1. Schedule a change implementation

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Coordinator

When a change is approved, it moves to the Coordinate Change Implementation phase and you can schedule it for implementation.

To schedule a change implementation, follow these steps:

1. Click **Change Management** > **Change Queue**.
2. Expand **Phase: Change Implementation** in the record list.
3. Find the change request you want to schedule. Use search or advanced search to find one or more records.
4. Double-click to open a record.
5. In the **Change Details** section, review the **Planned Start** and **Planned End** dates established in the Change Assessment and Planning phase and update the dates if necessary.
6. Determine whether the change needs to be implemented using Release and Deployment Management, due to a change in the Definitive Media Library (DML). If so, click the **Release Management** checkbox. Service Manager adds the change to the Release Management queue. After Release Management completes the release package, the change is returned to the Change Management process.
7. After you schedule the change implementation, you can create tasks for building, testing, and implementing the change.

Related topics

Create a change task
Searching records
Coordinate change implementation (ST 2.5)

1.8.3.5.2. Create a change task

Applies to User Roles: Change Coordinator

After a change is scheduled, you can create the tasks for building, testing, and implementing the change.

To create a change task, follow these steps:

1. Click **Change Management** > **Change Queue**.
2. Expand **Phase: Change Implementation** in the record list.
3. Use search or advanced search to find one or more records.
4. Find the change request you want to create tasks from, and click the record to open it.
5. Click **More** or the **More Actions** icon and then select **Create New Task**.
6. Select a category for the new task.

   Service Manager opens a task information form. Service Manager automatically copies the following information from the change request:

   - Category
   - Status
   - Parent Change
   - Description
   - Impact
   - Urgency
   - Priority
   - Planned Start
   - Planned End

   Note

   Click **Apply Template** if you want to use a predefined task template for the new task.

7. Complete the information in the **Change Task Details** section.
8. Assign the task to the designated Assignment Group and scheduled Change Analyst.
9. Click **Save & Exit**.
10. Repeat the above steps for each new task that you want to open. Service Manager automatically relates the task to the change.

Related topics

Close a change task
Searching records
Coordinate change implementation (ST 2.5)

1.8.3.5.3. Validate a change task

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Analyst

When a change task is assigned to you or your group, verify that the information is complete and that the task is correctly assigned before implementing the task.

To validate a change task, follow these steps:

1. Click **Change Management** > **Task Queue**.
2. Select **Change Task** from the **Queue** list, and then select **Open Change Tasks Assigned to My Group** from the **View** list.
3. Find the change implementation task you want to validate. Use search or advanced search to find one or more records.
4. Click a record to view its detail.
5. Review the task to verify that it is correctly assigned and that the information needed to implement it is complete.
6. If the task information and assignment are correct, build and test the change.
7. If the task information or assignment is incorrect, reject the task.

Related topics

Reject a change task
Build and test a change
Searching records
Coordinate change implementation (ST 2.5)

1.8.3.5.4. Reject a change task

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Analyst

You can reject a change task during the Coordinate Change implementation phase if the task information is incomplete or the task is incorrectly assigned.

To reject a change task, follow these steps:

1. From your To Do queue, select **Change Task** from the **Queue** list and **Open Change Tasks Assigned to My Group** from the **View** list.
2. Click a task from the record list and select **Reject** from the **More Actions** menu.
3. In the **Rejection Details** wizard, type the reason for the rejection, and then click **Finish**. Service Manager returns the task to the Change Coordinator to be corrected or reassigned.

Related topics

Validate a change task
Coordinate change implementation (ST 2.5)

1.8.3.5.5. Build and test a change

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Analyst

After you verify that the change implementation tasks contain complete information and are correctly assigned, you can build and test the scheduled change. All hardware changes, software changes, and new releases must be tested before implementing them in a production environment.

> **Note**
>
> It is important to document all infrastructure changes and test results.

To build and test a change, follow these steps:

1. Click **Change Management** > **Task Queue**.
2. Click **Change Task** in the **Queue** list and **Open Change Tasks Assigned to My Group** in the **View** list.
3. Find the change implementation tasks for building and testing the change, and double-click the records to open the selected tasks. If you don't see the change implementation tasks in the record list, you can click **Search**. Use search or advanced search to find one or more records.
4. Perform the activities needed to complete the change implementation tasks for building and testing the change.
5. Update the change task record with the following information:
   - Assignment Group
   - Affected CI
   - Actual Start date
   - Actual End date
6. Document the infrastructure changes or test results. You can use the Work Notes section to document this information or attach the information as a separate file by using the Attachments section.
7. After the change implementation tasks for building and testing the change are complete, close the change tasks:
   1. Click **Close Task**.
   2. Select the Closure Code and type a closure comment.
   3. Click **Save & Exit**. Service Manager sets the task state to Closed.
8. Notify the scheduled Change Coordinator that change implementation tasks for building and testing the change are complete. The Change Coordinator will then

verify that the change meets the test criteria and is authorized to be implemented in the production environment.

Related topics

Searching records
Coordinate change implementation (ST 2.5)

1.8.3.5.6. Implement a change and perform a production test

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Analyst

After the Change Coordinator verifies that a change meets the test criteria and is authorized to be implemented in the production environment, you can implement the change and perform a production test as scheduled.

Note

It is important to document all infrastructure changes and test results.

To implement a change and perform a production test, follow these steps:

1. Click **Change Management** > **Task Queue**.
2. Click **Change Task** in the **Queue** list, and then click **Open Change Tasks Assigned to My Group** in the **View** list.
3. Find the change implementation tasks for implementing the change and performing a production test. Use search or advanced search to find one or more records.
4. Double-click to open the records.
5. Perform the activities required to complete the change implementation tasks for implementing the change and performing a production test.
6. Update the change task record with the following information:
   - Assignee
   - Affected CI
   - Actual Start
   - Actual End
7. Document the infrastructure changes or test results. You can use the **Work Notes** section to document this information or attach the information as a separate file in the **Attachments** section.
8. After the change implementation tasks for implementing the change and performing a production test are complete, close the tasks:
   1. Click **Close Task**.
   2. Select the closure code and type a closure comment.
   3. Click **Save & Exit**. Service Manager sets the task state to Closed.

9. Notify the scheduled Change Coordinator that the change implementation tasks for implementing the change and performing a production test are complete. The Change Coordinator will then verify that the implementation is successful.

Related topics

Searching records
Coordinate change implementation (ST 2.5)

1.8.3.5.7. Process a successful change implementation

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Coordinator

After you verify that a change is successfully implemented in the production environment, you can trigger the configuration management process to update the Configuration Management System (CMS) data model for the affected configuration item (CI).

To process a successful change implementation, follow these steps:

1. Click **Change Management** > **Task Queue**.
2. Select **Change Task** from the **Queue** list and **Open Change Tasks Assigned to My Group** from the **View** list.
3. Find the change implementation tasks for implementing the change and performing a production test. Use search or advanced search to find one or more records.
4. Double-click the records to open them.
5. Verify that the change tasks were completed successfully.
    - If the change implementation was successful, go on to the next step.
    - If the change implementation caused a severe disruption in the production environment, activate the remediation plan by creating change implementation tasks to execute a fall back scenario. Assign the tasks to the responsible Change Analyst.
6. Update the Change form description with the implementation remarks.
7. Create a change implementation task requesting an update to the CMS data model, and assign it to the responsible Configuration Manager.

   When all tasks are complete and the CI changes are registered in the CMS, Service Manager automatically moves the phase from Change Implementation to Change Evaluation and Closure.

Related topics

Validate a change task
Process an unsuccessful change implementation

Create a change task
Coordinate change implementation (ST 2.5)

1.8.3.5.8. Process an unsuccessful change implementation

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Coordinator

If a change causes a severe disruption in the production environment, you can activate the remediation plan developed in the Change Assessment and Planning phase to reverse the change and return the environment to the prechange state.

To process an unsuccessful change implementation, follow these steps:

1. Click **Change Management** > **Task Queue**.
2. Select **Change Task** from the **Queue** list and **Open Change Tasks Assigned to My Group** from the **View** list.
3. Find the change implementation tasks for implementing the change and performing a production test. Use search or advanced search to find one or more records.
4. Click a task record to view its detail.
5. Verify that the change tasks were not completed successfully.
    - If the change implementation was unsuccessful, go on to the next step.
    - If the change implementation was successful, trigger the configuration management process to update the Configuration Management System (CMS) for the affected configuration item (CI).
6. Update the Change form description with details about the unsuccessful implementation.
7. Activate the remediation plan by creating change tasks to execute the fall back scenario, and assign the tasks to the responsible Change Analyst.
8. When all tasks are complete, Service Manager automatically moves the phase from Change Implementation to Change Evaluation and Closure.

Related topics

Process a successful change implementation
Create a change task
Reject a change task
Coordinate change implementation (ST 2.5)

1.8.3.5.9. Execute a fall-back scenario

Part of Workflow(s): Coordinate change implementation (ST 2.5)

Applies to User roles: Change Analyst

If a change causes a severe disruption in the production environment, the Change Coordinator activates the remediation plan by creating tasks to execute a fall-back scenario. You will then perform the activities required to return the environment to the prechange state.

To execute a fall-back scenario, follow these steps:

1. Click **Change Management** > **Task Queue**.
2. Click **Change Task** in the **Queue** list and **Open Change Tasks Assigned to My Group** in the **View** list.
3. Find the change implementation tasks for executing the fall-back scenario. Use search or advanced search to find one or more records.
4. Double-click the records to open them.
5. Perform the activities required to complete the fall-back scenario change tasks.
6. Update the change task form with the following information:
    - Assignee
    - Affected CI
    - Actual Start
    - Actual End
7. Document the results. You can use the **Work Notes** section to document this information or attach the information as a separate file in the **Attachments** section.
8. After the fall-back scenario change tasks are complete, close the tasks:
    1. Click **Close Task**.
    2. Select the closure code and type a closure comment.
    3. Click **Save & Exit**. Service Manager sets the task state to Closed.

When all change tasks are complete, Service Manager automatically moves the phase from Change Implementation to Change Evaluation and Closure.

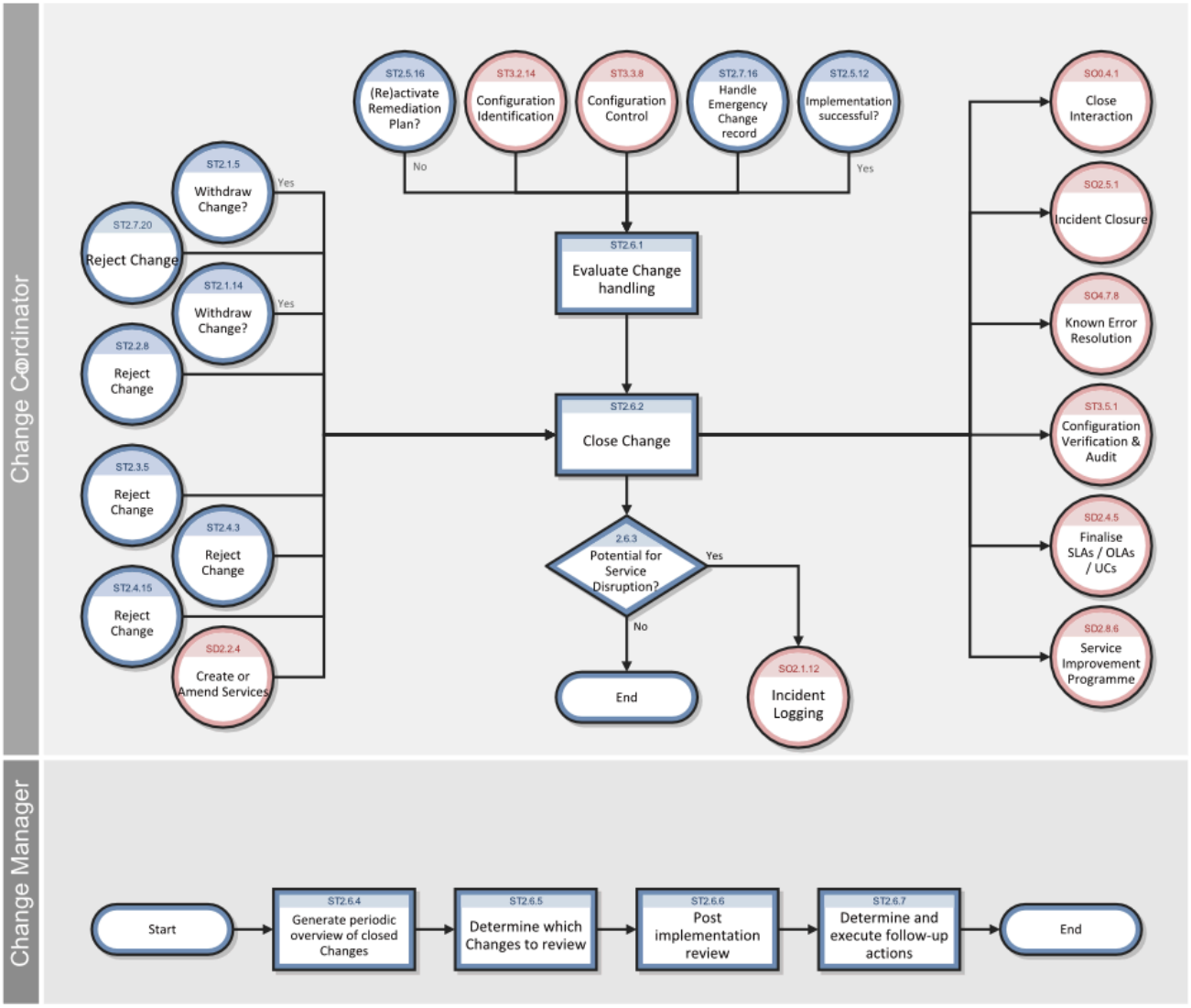
Related topics

Validate a change task
Reject a change request
Coordinate change implementation (ST 2.5)

1.8.3.6. Change evaluation and closure (ST 2.6)

After a change is completed, the results must be reported for evaluation to those responsible for managing changes, and then presented for stakeholder agreement. This process includes the closing of related user interactions, incidents, and known errors.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Evaluate change handling
Close a change

Generate a list of closed changes for review

Change Management workflows and user tasks

1.8.3.6.1. Evaluate change handling

Part of Workflow(s): Change evaluation and closure (ST 2.6)

Applies to User roles: Change Manager

After a change implementation is complete, the change advances to the Change Evaluation and Closure phase. You must verify that the change was handled correctly and that the administration of the change is complete. After evaluating the change handling and determining that the information in the record is complete, you can close the change.

To evaluate change handling, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Select a record to view its details.
4. Open the **Related Records** section and then open each related record to review and verify that each record is correct.
5. Open the **Activities** section to enter any relevant implementation and process remarks.
6. In the **Change Details** section, select the applicable closure code from the list and type a comment in the **Closure Comments** field.
7. Make any other necessary updates in the record.
8. Click **Save & Exit**.

Related topics

Close a change
Generate a list of closed changes for review

1.8.3.6.2. Close a change

Part of Workflow(s): Change evaluation and closure (ST 2.6)

Applies to User roles: Change Manager

After you evaluate the change handling, update it by summarizing any implementation or process remarks, and determine that the administration of the change is complete, you can close the change.

To close a change, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Select the record you want to close.
4. Click **Close Change**.
5. Complete all required inputs. You must include the time format even if the task was completed in whole day increments. For example, if a task was completed in exactly three days, type 3 00:00:00 ddd hh:mm:ss.
6. Select the **Closure Code** and type any comments in the **Closing Comments** field.
7. Click **Save & Exit**.

When the change is closed, all related interactions are set to Open-Callback status if the Notify By method is telephone. If the Notify By method is an automatically-generated email or no notification is required, and there are no related records, the status is set to Closed.

Related topics

Evaluate change handling
Generate a list of closed changes for review

1.8.3.6.3. Generate a list of closed changes for review

Part of Workflow(s): Change evaluation and closure (ST 2.6)

Applies to User roles: Change Manager

You should periodically generate a list of changes that have been closed since the last review. You can then review the list of closed changes to determine which closed changes should be reviewed during a Post-Implementation Review.

To generate a list of closed changes, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records. You can select the date of the last review from the **Closed After** list, and then click **Search**. A list of changes closed since the date you entered is displayed.

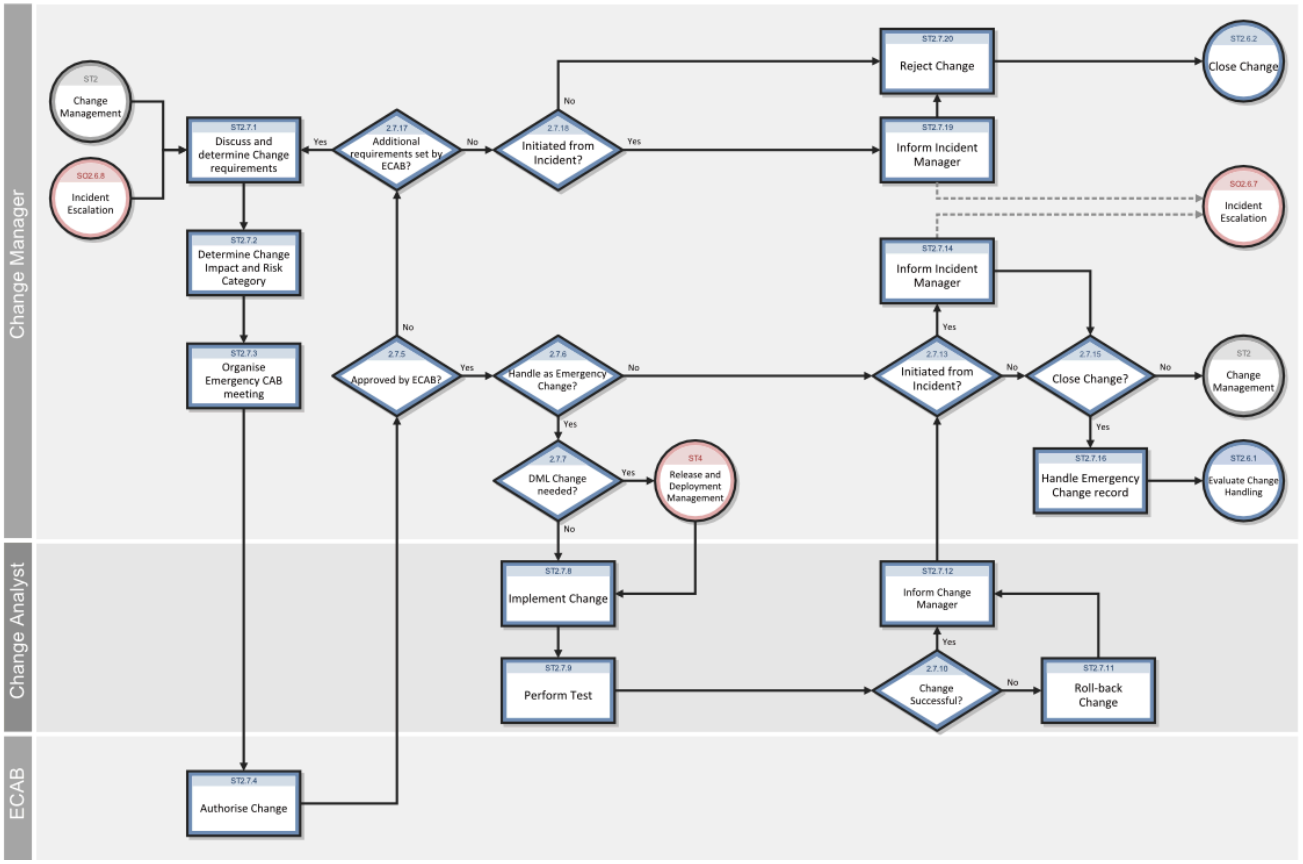
Related topics

Evaluate change handling
Close a change
Change evaluation and closure (ST 2.6)

1.8.3.7. Emergency change handling (ST 2.7)

Emergency changes can only be initiated from within the Incident Management process. They should be used only to repair an IT service error that's negatively impacting the business at a high level of severity. Changes that are intended to make an immediately required business improvement are handled as normal changes, although they may be assigned a high priority based on the urgency of the required business improvement.

Details for this process are displayed in the following flowchart.



Related topics

Change Management overview
Log an emergency change
Approve an emergency change
Recategorize an emergency change
Change Management workflows and user tasks

1.8.3.7.1. Log an emergency change

Part of Workflow(s): Emergency change handling (ST 2.7)

Applies to User roles: Change Manager

If an IT incident is negatively impacting the business at a high severity level, the Incident Coordinator may escalate it and create an emergency change. The Incident Manager will notify you of the new change and inform you that the emergency change procedure must be started.

To log an emergency change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Click **Search**, and then select the **Emergency Change** checkbox.
3. Click **Search** to display a list of emergency changes.
4. Select a change request record to view its details.
5. Discuss the requirements with the Incident Manager, and then select or revise the values in the **Impact** and **Risk Assessment** fields based on your assessment.
6. Discuss the implementation with the Change Analyst, and then complete the **Planned Start** date, **Planned End** date, and **Affected CI(s)**.
7. Click **Close** to forward the change to the Change Approval phase.

Related topics

Reject a change request
Plan and schedule a change
Change assessment and planning (ST 2.3)

1.8.3.7.2. Approve an emergency change

Part of Workflow(s): Emergency change handling (ST 2.7)

Applies to User roles: Change Manager

Emergency changes must be approved by the Emergency Change Approval Board (E-CAB). The E-CAB consists of members authorized to make decisions about high-impact emergency changes. The E-CAB decides whether to approve the change, and if it should be handled as an emergency change. The Change Manager registers the results of E-CAB decisions.

To approve an emergency change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Click **Search**, and then select the **Emergency Change** checkbox.
3. Use search or advanced search to find one or more records.
4. Click the emergency change request to view the details.
5. Contact the E-CAB to obtain the results of their meeting. Their decision affects how you will proceed.
    - If the E-CAB approves the change and authorizes that it be handled as an emergency change, continue with step 6 to continue the approval process for the emergency change.
    - If the E-CAB approves the change but doesn't authorize it to be handled as an emergency change, re-categorize it as a normal change.
    - If the E-CAB doesn't approve the change because additional requirements are needed, go back to the Incident Manager to obtain additional information.
    - If the E-CAB rejected the change, click **More** or the **More Actions** icon and then select **Reject** to reject the change. In the **Rejection Details** wizard, type the reason for the rejection and then click **Finish**.
6. Click **More** or the **More Actions** icon and then select **Approval** > **Approve**.
7. Click **Save**.
8. Click **Close** to forward the emergency change to the implementation phase.

Related topics

Reject a change request
Plan and schedule a change
Change assessment and planning (ST 2.3)

1.8.3.7.3. Recategorize an emergency change

Part of Workflow(s): Emergency change handling (ST 2.7)

Applies to User roles: Change Manager

The Emergency Change Approval Board (E-CAB) may decide that a change originally logged as an emergency change is not that critical anymore, and should in fact be treated using the normal change procedure. The Change Manager registers the results of their decision and recategorizes the emergency change.

To recategorize an emergency change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and **Emergency Changes** from the **View** list.
2. Use search or advanced search to find one or more records.
3. Click a emergency change request to view the details.
4. Uncheck the **Emergency Change** checkbox.
5. Change the value in the **Urgency** field to an appropriate value.
6. Update the change with the decisions made in the E-CAB meeting.
7. Check the **Assignment**, **Affected CI**, and other fields to see whether any changes are necessary.
8. Schedule the change by entering dates in the **Planned Start** field and the **Planned End** field.
9. Click **Save**.
10. Click **Close Phase** to close the current phase.

Related topics

Change assessment and planning (ST 2.3)
Reject a change request
Plan and schedule a change

1.8.3.8. Change Management tasks

Approved change requests generate tasks which must be completed to complete the change request. These tasks typically include a description of the task, the urgency and priority of the task, task scheduling information, and the task assignment.

Change Management tasks include:

- Opening, assigning, and associating a task with a change
- Searching for a task
- Managing task categories, environments, and phases
- Using the task queue

Related topics

Change Management overview
Create a new change request
Review a new change request
Set a reminder for a change request
Create a change task
Close a change task
Implement a change and perform a production test

1.8.3.8.1. Access Change Management reports

User Roles: Change Manager; Change Coordinator

The Reporting tool in Service Manager provides a number of out-of-box reports on the change data in your system. You can view these reports through the out-of-the-box dashboards named **Change Overview (Global)**, **Change KPI (Global)** and **CAB/eCAB Dashboard (Global)**. You can also create your own dashboards to display other reports of your interest.

To access Change Management reports, follow these steps:

1. Click **Change Management** > **Change Overview**.

   By default, the **Change Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. If needed, click the **New Dashboard** button to add your own dashboards. For details, see Create a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default dashboard.

1.8.3.8.2. Create a change task

Applies to User Roles: Change Coordinator

After a change is scheduled, you can create the tasks for building, testing, and implementing the change.

To create a change task, follow these steps:

1. Click **Change Management** > **Change Queue**.
2. Expand **Phase: Change Implementation** in the record list.
3. Use search or advanced search to find one or more records.
4. Find the change request you want to create tasks from, and click the record to open it.
5. Click **More** or the **More Actions** icon and then select **Create New Task**.
6. Select a category for the new task.

   Service Manager opens a task information form. Service Manager automatically copies the following information from the change request:

   - Category
   - Status
   - Parent Change
   - Description
   - Impact
   - Urgency
   - Priority
   - Planned Start
   - Planned End

   Note

   Click **Apply Template** if you want to use a predefined task template for the new task.

7. Complete the information in the **Change Task Details** section.
8. Assign the task to the designated Assignment Group and scheduled Change Analyst.
9. Click **Save & Exit**.
10. Repeat the above steps for each new task that you want to open. Service Manager automatically relates the task to the change.

Related topics

Close a change task
Searching records
Coordinate change implementation (ST 2.5)

1.8.3.8.3. Update a change task

Applies to User roles: Change Manager, Change Coordinator

To update a change task, follow these steps:

1. Click **Change Management** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Click a record to view its detail.
4. Modify the record.
5. Click **Save & Exit**.

Related topics

Change Management tasks
Create a change task
Close a change task
Reopen a change request or task

1.8.3.8.4. Close a change task

User roles: Change Manager, Change Coordinator, Change Analyst

After a task is complete, the person listed in the Assignee field of the task record can close the task.

To close a task, follow these steps:

1. Click **Change Management** > **Task Queue**. The **Change Task** queue opens.
2. Choose one of the following task queues from the **View** list:
   - **All Open Change Tasks**
   - **Open Change Tasks Assigned to Me**
3. Change Management displays the selected task queue. Click the task record you want to close to view its detail.
4. Click **Close Task**.
5. Enter the **Actual Start** and **Actual End** dates.
6. Select the appropriate **Closure Code**.
7. Add any relevant comments in the **Closure Comments** field.
8. Click **OK**.

Related topics

Create a change task

1.8.3.8.5. Reopen a change task

Applies to User roles: Change Coordinator

To reopen a change request or task, follow these steps:

1. Click **Change Management** > **Search Changes** or **Tasks** > **Search Tasks**.
2. Select **closed** from the **Status** list.
3. Use search or advanced search to find one or more records.
4. Click a record to view its detail.
5. Click **Reopen**.
6. Edit the record as necessary.
7. Click **Save & Exit**.

Related topics

Close a change
Close a change task

1.8.3.8.6. Print a change request

Applies to User roles: Change Manager, Change Coordinator

To print a change request, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Select a record from the record list.
4. Click **More** and select **Print** or click the **Print Page** icon.

Related topics

Review a new change request
Update a change request

1.8.3.8.7. Print a change request list

Applies to User roles: Change Manager, Change Coordinator

To print a change request list, follow these steps:

1. Click **Change Management** > **Search Changes**.
2. Use search or advanced search to find one or more records.
3. Select multiple records you want to print.
4. Click the **Print Page** icon and select **List View**.

Related topics

Review a new change request
Update a change request

1.8.3.8.8. Associating a change with another record

You can associate a change record with an existing Service Desk interaction, incident, known error, quote, or another change record. If the situation changes you can remove the relationship between the change record and the other record.

When a record is related to a change, the system tracks the history of this activity. This includes the changes that are related or dropped and the history of phase changes and transitions.

In the Related Records section, you can view the current related records, associate an existing record with the change record, or open a new change that's automatically related to the change being viewed.

When you associate a change record with another record, note the ID number of that record so you can verify it in the Related Records section.

Related topics

Relate a new change to a change
Delete a related change to a change

1.8.3.8.9. Associate a change with another change

Applies to User roles: Service Desk Agent, Change Coordinator

To associate a change with another existing change, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list.
2. Select a change from the record list.
3. Open the **Related Records** section.
4. Click **More** or the **More Actions** icon and then select **Related** > **Changes** > **Associate**.
5. Type the change record number or click **Search** to find the change record.
6. Click **OK**.

You can verify that the related change record is associated with this change in the **Related Records** section.

Related topics

Associating a change with another record
Approval phase for KM Document category
Relate a new change to a change

1.8.3.8.10. Set a reminder for a change request

Applies to User roles: Change Manager, Change Coordinator, Change Analyst, Change Approver

From an existing change request, you can schedule a reminder to be sent to you at a specified time.

To set a reminder, follow these steps:

1. Click **Change Management** >**Search Changes**.
2. Use search or advanced search to find one or more records.
3. Click the record to view its detail.
4. Click **More** or the **More Actions** icon and then select **Set Reminder**.
5. Select one of the following to set the time when the reminder occurs:
    - Click **Remind At** to set a reminder for a particular day and time. Click **Fill** to display a calendar where you can select the date and time for the reminder.
    - Click **Remind In** to set the reminder to occur at a particular interval. If you choose this option, specify the time interval (in 00:00:00 format), and select a shift from the **Based On** list.
6. In the **Remind if** field, select an option from the list:
    - Always (the default)
    - Change is still open
    - Change is still assigned to me
7. In the **Pop-up Message** area of the form, select the type of notification you want to receive.
8. If you select **Pop-up**, type the reminder message in the **Message** field.
9. If you select **Email** or **SM Mail**, Service Manager displays a **Title** field where you can enter a title for the email message and two Message Type option buttons.
10. Select the message type option in the Message Type frame.
    - Click **Send Change Record** to include a copy of the change record in the email.
    - Click **Fixed Text** and type the message you want to appear in the email.
11. Click **OK**.

1.8.3.8.11. View a list of services potentially affected by an outages

Applies to User Roles: Incident Coordinator, Incident Manager

The View Affected Services menu option enables you to view a list of services that are potentially affected by an outage relating to CIs that are specified in an incident or change record. By viewing the affected services list, you can determine the potential impact of an outage related to critical, dependent services when you open an incident. You can also use the affected services list for planning when opening a change.

Note

The list of CIs generated doesn't include the CIs specified in the primary incident or change record.

To view a list of affected services:

1. Perform one of the following actions.
   - Open an existing incident or change record. Use search or advanced search to find one or more records.
   - Proceed as if you are going to add a new incident or change record.
2. If you are in the process of adding a new record, fill in the **Affected CI**.
3. Click **More** or the More actions icon and select **View Affected Services**. The CI Identifier record opens.
4. Open the **Affected Services** section. A list of affected services opens or a message, stating that there are no services affected by the outage.

   Note

   For change records, a list of all affected business services is returned in all cases, whether the record has one primary CI or multiple CIs.

5. Select a CI from the list for detailed information from the Configuration Management record.
6. When you finish viewing the affected services, click **Cancel**.

Related topics

Create a new incident from a user interaction
Create New Incidents from Monitoring System Notifications
Create a new change request

1.8.3.8.12. Using Mass Update with Change Management record lists

Mass update enables you to select multiple records from a list of records and update them all at one time. You can update the value in a single field or in several fields in the selected records. The system provides a template form that displays the fields for the selected records and enables you to change the value of any of the fields displayed by the form. The Mass Update template form doesn't display all fields in the records. For example, fields marked as read-only in the data policy don't display. You can also do a complex update on the records you selected. A complex update uses rad expressions containing variables and concatenated fields to populate another field in the selected records.

When you perform a Mass Update, the value you enter for a particular field becomes the value for that field for all of the records you selected.

Mass Update is available for change request records, task records, and change queues. The ADMIN and SYSADMIN profiles in the out-of-box system provide the Template Mass Update and Complex Mass Update capability.

A System Administrator can edit the datadict table cm3r or cm3t so that a field doesn't appear in the list of fields displayed by the Mass Update template form. On the Data Policy form, change the Usage Type column for the field to System.

Update multiple change requests

Applies to User roles: Change Manager

You can update more than one change request at a time by using the Mass Update template form or the Complex Update function.

To update multiple change requests by using the Mass Update template form, follow these steps:

1. From your To Do queue, select **Change** in the **Queue** list and then select the desired view.

   Alternatively, click **Change Management** > **Search Changes** to use the Search to generate a record list. Use search or advanced search to find one or more records.

2. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
3. From the **List** menu, select **Mass Update**.
4. Double-click the field you want to update.
5. Type the value for the field in the text box or use the Fill feature to display a list of potential values for the field.
6. Click **Next**.
7. Continue updating fields and click **Execute** when you complete your updates.

To update multiple change records by using the Complex Update function, follow these steps:

1. From your To Do queue, select **Change** from the **Queue** list and and then select the desired view.
2. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
3. From the **List** menu, select **Mass Update**.
4. Click **Complex Update**.
5. Type the RAD code in the **Complex Update** form.
6. Click **Execute**.

Update multiple task records

Applies to User roles: Change Manager

You can update more than one task at a time by using either the Mass Update template form or by using the Complex Update function, which enables you to enter RAD codes to further customize your update.

To update multiple task records by using Mass Update, follow these steps:

1. Click **Change Management** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
4. Click **Mass Update**. The **Mass Update** wizard opens.
5. Double-click the field you want to update.
6. Type the value for the field in the text box or use the Find feature to display a list of potential values for the field.
7. Click **Next**.
8. Continue updating fields and click **Execute** when you complete your updates.

To update multiple tasks by using Complex Update, follow these steps:

1. Click **Change Management**> **Tasks** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Holding down the **Ctrl** key as you click on each record to select multiple records in the list that you want to update.
4. Click **Mass Update**.
5. Click **Complex Update**.
6. Type the RAD code in the **Complex Update** form.
7. Click **Execute**.

Contents

- 1 Update multiple change requests
- 2 Update multiple task records

1.8.3.8.13. Change Management and Service Level Agreements

Change Management enables you to select one or more Service Level Agreements (SLA) to relate to a change record. When you open a change, you can choose one Customer SLA for the contact, the same Customer SLA or a different one for the contact and one or more applicable Service SLAs for the contact's subscriptions to a service, or no SLAs at all. Service SLAs only apply if the change references a Business Service, the contact has a subscription to the service, and the subscription references an SLA. The following list describes the system's process for relating SLAs to a change record.

- If one SLA is associated with the change based on the contact, the Customer SLA is added to the change.
- If the contact has an Individual Subscription for the Configuration Item (CI), the Service SLA from that subscription is added to the change.
- If the contact has a Department Subscription for the CI, the Service SLA from that subscription is added to the change.
- If the contact has neither, then no Service SLA is added to the change.

The SLAs should contain all Service Level Targets (SLTs) that define the business rules for all response and availability metrics. You can choose as many SLTs as necessary to describe your response or availability commitment. If necessary, you can add more SLTs that meet your criteria.

When you view the new record, the SLT section lists the SLTs related to the change.

Related topics

Change Management overview

1.8.4. Change Management administration

Change Management administration provides the capability to control the Configuration Management application functionality, manage change groups, and create change request based on change model, category, or subcategory.

Change Management administration consists of the following:

- Change Environment: Change Management contains an environment record for changes. This record contains an option that defines the functionality of the Change Management application for all Change Management users.
- Groups: This enables you to group operators to be a member or approver of change groups.
- Settings: You can create a change request based on the change model or category by selecting the appropriate option in the Change Settings page. You can also enable sending email notifications through Event Messages to Change management.
- Task Environment: Change Management contains an environment record for tasks. This record contains an option that defines the functionality of Change Management application for all Change Management users.

View the Change Management background processor

Applies to User roles: System Administrator

The default system start-up record (startup) lists all background agents, or processes, that start up each time you start Service Manager. Most specify 60-second wake-up intervals. This list includes despooler, report, alert (for Incident Management), change, availability, agent, marquee, lister, linker, and event. To add any other start-up records to the system startup record, type the new start-up records information at the bottom of the agent array on the system start-up record. The system startup default record processes records that are in the schedule file. The appropriate background agent picks up the related schedule record for processing.

To view the Change Management background processor, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **System** > **Startup Information**.

2. In the Type field, type change.startup. This is the default start-up record for the Change Management background processor.
3. Use search or advanced search to find one or more records.

Configure Change Management assignment groups

Applies to User roles: System Administrator

Change Management assignment groups enable you to group operators who can be members or approvers of the same group. You can create change assignment groups to contain members and approvers who operate on a common area of responsibility.

Service Manager sends a notification to members of the corresponding groups whenever a new change request is created or an existing change is modified. Approvers of a group can accept or deny the approval requirement of a change if at least one of their security roles has the **Can Approve** right for the Change area.

To add a change group, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.
   1. Enter the name of the manager of the assignment group.
   2. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   3. Select the time zone.
5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many

users should provide response when you create and populate a new assignment group.

(Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.
9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.
10. Follow these steps to configure the Supported Services field:

    1. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    2. Select **Business Service** in the CI Type field, and then click **Search**.
    3. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    4. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.

11. Click **Save** and **OK**.

Configure global settings for Change Management

Applies to User roles: System Administrator

Change settings are global settings that are applied to the entire Change Management module.

To configure change settings, follow these steps:

1. Click **Change Management** > **Administration** > **Settings** in the System Navigator.

   Ignore the **Enable Legacy Event Messages** checkbox. The Process Designer change workflows don't use legacy event messaging. Instead, notifications are called from RuleSets.

2. To specify the default link file for mapping change model fields to change records, enter the name of the link file in the **Default Open/Apply Link** text box.

   > **Note**
   >
   > You can view and reconfigure the field mappings defined in the link file. For details, see Configuring change models.

3. To enable change model filtering based on a script function you defined, enter the script function name in the **Change Model Restriction Script Function** text box. For example, ChangeModelRestriction.getRestrictionSql.

   > **Note**
   >
   > The specified script function is used to filter change models that you can see when creating a new change or applying a change model to a change, according to the conditions defined in the script.
   >
   > To define the query conditions in the script, click **Tailoring** > **Script Library** in the System Navigator, search for the script or create a new script, and then define the script function for change model filtering.

4. To enable users to create new change requests based on change categories, select **Category** under **Select On Open**.

5. To enable users to create new change requests based on change models, select **Change Model** under **Select On Open**.

   > **Note**
   >
   > Be default, **Change Model** is selected.

6. To configure the change model hierarchy, type up to four column names to group the change model in the **Group Change Model by** text box. For example, to group the change model by category and template, type category, changeTemplate.

   Note

   - The change model hierarchy configuration takes effect only when **Change Model** is selected under **Select on Open**.
   - Use the column names specified in the `changeModel` table.
   - You can configure a maximum of four levels in the change model hierarchy by specifying the columns in a comma-separated list.

8. Click **Save**.

   Note

   You can create or edit custom settings by selecting **Add New Setting** or **Edit Settings** from the **More** menu.

Enable Change Manager to prompt reason for relating and unrelating a change

Applies to User roles: System Administrator

The screlconfig record controls whether or not a text box appears in which the operator can type a reason for relating or unrelating a change record to another change record in Change Management. The Relation Configuration Record form for Change Management (the cm3r table) contains two checkboxes: **Prompt Reason for Relating** and **Prompt Reason for Unrelating**. When these checkboxes are selected, the system prompts the operator with a text box in which the operator can provide a reason for relating or unrelating a change to another change.

To configure the screlconfig record for Change Management history, follow these steps:

1. Type db in the command line field.
2. Type screlconfig in the **Form** field on the **Database Manager** page.
3. Use search or advanced search to find one or more records.

4. Click **Search** again to display a specific list of records with Relation Configuration Records.
5. Select **cm3r** from the record list to display the **Relation Configuration Record** form in Change Management.
6. Make the changes to the **Prompt Reason for Unrelating** and **Prompt Reason for Relating** checkboxes.
7. Click **Save** and **OK**.

Configuring Change Management workflows

A change workflow is a sequence of connected steps in the life cycle of a change record. In the workflow, a change record goes through several phases to complete the life cycle. A change workflow phase may consist of one or multiple tasks to be completed before the change record can proceed to the next phase.

Similarly, a change task workflow is a sequence of connected steps in the life cycle of a change task.

Change Management uses the following out-of-box workflows that are ITIL aligned.

Change workflows

- Project Proposal
- Standard Change
- Normal Change
- Emergency Change

Task workflows

- Generic Task

The change and task workflows are associated with change categories and change task categories. The workflows are applied when you create change and change task records based on the categories.

Access Change Management workflows

User Role: System Administrator

Change Management workflows (including change workflows, change task workflows, and change model workflows) are accessible from the **Change Workflows** menu item.

To access Change Management workflows, follow these steps:

1. Click **Change Management** > **Configuration** > **Change Workflows**.

All Change Management workflows are displayed in the list.

2. Click **Change Management** on the task bar, and select one of the following filters to display a specific type of workflows:
   - Change
   - Change Management Configuration
   - Change Tasks

   > **Tip**
   >
   > You cannot modify the workflows marked with **Proprietary**. You can create your own workflows by copying them and making necessary modifications.

3. To view a workflow, double-click it from the list.

Create a Change Management workflow

User Roles: System Administrator

Each change category or change task category must have a workflow associated with it. Before you can create a change category or change task category, you must create a workflow for it. You can create a new workflow from scratch, or copy an existing workflow and make necessary modifications.

> **Tip**
>
> Change Management has the following out-of-box workflows, which can't be modified but can be used as a template when creating your own workflows: Change Proposal, Emergency Change, Normal Change, Standard Change, and Generic Task.

To create a change workflow or change task workflow from scratch, follow these steps:

1. Create a basic workflow.
   1. Navigate to **Change Management** > **Configuration** > **Change Workflows**.
   2. Click **New**. The **Workflow properties** window opens.
   3. Type a name and description for the workflow.
   4. Select the table to associate with the workflow. For example, select **cm3r** for Change, or **cm3t** for Change Task.

5. Click **OK**, and then click **Save**.

   The new workflow appears in the workflow list if you click **Change Workflows**.

6. Log off and log back in.

2. Add phases to the workflow. For details, see Add a phase.
3. Create transitions between the phases. For details, see Create a default workflow transition, Create a manual workflow transition, and Create an automatic workflow transition.
4. Click **Save**.

To create a change workflow or change task workflow from an existing one, follow these steps:

1. Click **Tailoring** > **Process Designer** > **Copy Existing Workflow**.
2. Select a change or change task workflow from the list.

   Tip

   You cannot modify the workflows marked with **Proprietary**. You can create your own workflows based on them.

3. Follow the instructions provided by the workflow cloning wizard to make a copy of the selected workflow. For details, see Copy an existing workflow.
4. Make necessary modifications as you do when creating a new workflow from scratch.

1.8.5. Change Management configuration

Change Management configuration provides the capability to configure the following items:

- change categories and subcategories
- task categories
- change models
- change workflows
- change approvals and alerts
- change messages

1.8.5.1. Configuring Change Management categories

Change Management categorizes changes and change tasks by change category and task category.

Each change category has a change workflow associated with it and can have subcategories. Each task category has a task workflow associated with it and available change phases defined for it.

To access change categories, click **Change Management** > **Configuration** > **Change Categories,** and then click **Search**.

Each task category contains a number of change phases and has a workflow associated with it. To access task categories, click **Change Management** > **Configuration** > **Task Categories,** and then click **Search**.

Service Manager provides a number of out-of-box change categories and task categories that you can use or customize.

Create a change category or task category

User Roles: System Administrator

You can create new category records either by copying and modifying an existing record or by creating a new one. Service Manager provides out-of-box category records that you can use or modify.

To create new category record, follow these steps:

1. Navigate to **Change Management**> **Configuration** > **Change Categories** or **Task Categories**.
2. Click **New**.
3. Type the name of the category.
4. Type a category description.
5. In the **Availability** field, type true.

> **Note**
>
> If you type false, the category will not be available to select from the category list when you open or update a change.

6. Select a workflow for the category.
7. Select a template for the category if required.
8. Optionally, select the **Assign number before commit?** checkbox so that Service Manager assigns an identification number .
9. Click **Save** to view the **Workflow** tab and other associated tabs.
10. For change category, click the **Subcategories** tab to create subcategories for the change category.
11. For task category, click the **Available Phases** tab and select one or more change phases from the drop-down list to associate the task category with the change phases. The newly created task category definition will be applied to the change workflow phases.

    If you leave **Available Phases** blank for a task category, you can use this task category in all change phases.

    Note

    The list of available phases is retrieved from all of the change workflows.

12. Click **Save**.

    Note

    To delete the category, click **Delete**.

Add a new subcategory

User Roles: System Administrator

Subcategories are a way to divide a Change Category into smaller groups. For example, the **Hardware** category can have the **Install Hardware** and **Move Hardware** subcategories.

To add a subcategory, follow these steps:

1. Click **Change Management** > **Configuration** in the System Navigator.
2. Click **Subcategories**.
3. Type a category name in the Category text box.

Or

Click the **Fill Field Category** icon to select a change category.

4. Type the subcategory name in the **Subcategory** text box.
5. Type the subcategory description.
6. Click the **Fill Field** icon in the **Company** text box to associate a company record with the subcategory.
7. Click **Save** to add the new subcategory of a change category.

Add a new subcategory from a change category

User Roles: System Administrator and Implementer

You can add a new subcategory directly from a change category. This reduces the number of access points required for you to have in order to create a subcategory. You can view a list of subcategories and their record details associated with the current category.

1. Click **Change Management** > **Configuration** in the System Navigator.
2. Click **Change Categories** > **Search**.
3. Select the Change Category for which you want to add a subcategory.
4. In the Change Category Definition page, click the **Subcategories** tab.
5. Click **Add subcategory**.
6. Type a subcategory name.
7. Type a description for the subcategory.
8. Click the **Fill** icon for the **Company** text box to associate one ore more company records with the subcategory.
9. Click **Save** to add the new subcategory.

   Service Manager creates the new subcategory and adds it to the change category.

Related topics

Change Management configuration

1.8.5.2. Configuring change models

A change model is a record that's used to pre-define the contents of a specific type of Request for Change (RFC), including the information used to populate the RFC and tasks needed to complete the change. When you open a change request using a change model, all information is added automatically to the change without any action needed by the operator.

When creating a new change model, you can define the tasks to be completed and the dependencies between multiple tasks. When you create a new change, a list of change model templates appears. In the change model template, change models are grouped by category and subcategory. You can apply a change model to a change using the change model template.

Create a new change model

User Roles: Change Coordinator, Change Manager

To create a new change model, follow these steps:

1. Click **Change Management** > **Configuration** > **Change Models** in the System Navigator.
2. Click **New**.
3. Type the change model ID.
4. Type the change model name.
5. Type the change description.
6. Click the **Fill Field Category** icon in the **Category** text box to select a change category.

   Note

   After the change model is created, you can't change the category specified in the change model. If you want to change the category, you can create a new change model.

7. Click the **Fill Field Subcategory** icon in the **Subcategory** text box to select a change subcategory.
8. Select a change template from the drop-down list.

   If needed, you can specify more than one template. When users open a new change from this model, Service Manager automatically applies all specified

templates in the order listed by merging their field values. The following table lists two example templates and their merge result.

Template 1

Field	Value
Description	Install Windows 11 on new employee's laptop
Brief Description	Install Windows 11
Assignment Group	(empty)

Template 2

Field	Value
Description	(empty)
Brief Description	Install Windows 10
Assignment Group	Hardware

Merge Result

Field	Value
Description	Install Windows 11 on new employee's laptop
Brief Description	Install Windows 10
Assignment Group	Hardware

9. Select a wizard from the drop-down list.

> **Note**
>
> The wizard helps you to accomplish certain tasks when it executes. When you open a change model from a change, the wizard associated with the change model executes. For example, you can run a wizard to prompt the user for a device type when adding a device record.

10. Complete the form with all of the required information.
11. Click **Save**.

Add a task to a change model

User Roles: Change Manager

Using Task Planner, you can add, edit, or delete tasks, which are part of change model record. In the graphical interface, you can specify tasks as dependent on other tasks by adding a connector between two tasks. Tasks that don't have dependencies may be open in parallel but are still governed by the **Open In Phase** and **Close By Phase** attributes, as well as the task conditions.

Change task attributes

When adding a task to a model, specify the following attributes:

- Title
- Task Category
- Task Template
- Open In Phase – Species the starting change phase in which the task will be activated for the change. If the task is dependent on another task, it will have a Planned status.
- Close By Phase – Specifies the ending change phase by which the task must be closed. If there are still open tasks for a phase, the change will not be allowed to go to the next phase. If left blank, the ending phase will match the starting phase.
- Mark as required/Set properties as read-only – Specifies whether or not this task can be re-planned or removed when you plan tasks using Task Planner from a change record.
- Task Condition – Defines the conditions for task creation.
    - The task condition is evaluated at each update of the change record when the change is in the Open In Phase defined for this task.
    - If you leave this field blank, the condition will evaluate to true.

- When the condition evaluates to true after a change update, the task is created and linked with the change record.
- Once a task has been created and linked with the change, the condition for this task will not be re-evaluated anymore.
- When the condition is true in the first instance but false after the change is updated, the system doesn't automatically close the task and you need to close it manually.
- The condition of a planned task is shown as read-only in Task Planner from the change record.
- If a task is dependent on a conditional task that has not been created because its condition evaluates to false, this task will still be created (with a Ready status) when the change reaches the Open In Phase defined for this task.

As these tasks are created for a Change, task information is automatically applied to the new task record.

For task dependencies to function, each task status must follow a convention to use a numeric value for the task status value. The global list “Change Task Status” contains the status values listed in the following table.

Value List	Display List
0	Planned
1	Ready
2	Assigned
20	Completed
21	Completed with Problems
3	In Progress
30	Cancelled
31	Withdrawn
32	Failed
4	Blocked

Tasks that don't have any dependencies are automatically opened with the“1=Ready” status. Tasks that are dependent on other tasks are opened with the “0-Planned” status. Once all tasks that another task is dependent on have reached a status of 20 or greater, the dependent task will have its status changed to “1-Ready”.

To see a working example, view the **Generic Task** workflow. It utilizes format “chm.task”, and the statuses above. The initial “Waiting” phase will transition to “Active” when the task status > 0 which means it's not planned.

Using the behavior of the tasks statuses set by Process Designer, you can create your own task flow or modify a copy of the **Generic Task** workflow. If you are using task statuses that aren't using the numeric values, task dependencies will not function.

You can move a task from one phase to another by setting appropriate conditions. The following condition is an example to move a task from the “Waiting” phase to “Active”.

```
(Status in CurrentRecord> 0 AND Variable $L.tableAccess.update =
true)
```

You can set multiple dependencies for a task. If a task is set with dependencies, the task can start only after all the dependent tasks are completed. The following table outlines the various statuses of a child task which is “dependent” on the parent task when you open a change.

Parent Task	Child Task
Ready	Planned
Completed	Ready
Completed with Problems	Ready
Canceled	Ready
Withdrawn	Ready
Failed	Ready
Blocked	Ready

To add a task to a change model, follow these steps:

1. Click **Change Management** > **Configuration** > **Change Models** in the System Navigator.
2. Click **Search** to select a change model.

   Or

   Click **New** to create a new change model record.

   > **Note**
   >
   > Select a category to display the task editor in the Task Information section.

3. In the Task Information section, click the **Edit** button, and then click **Add Task**. A new task is added in Task Planner.
4. Select the new task, click the **Task-Properties** tab, and perform these steps:
   1. Type a task description in the **Title** text box.
   2. Select a task category from the drop-down list.
   3. Optionally, select a task template from the drop-down list.
   4. Select a change phase to create the task from the **Open In Phase** list.
   5. Optionally, select a change phase to close the task from the **Close By Phase** list.

      > **Note**
      >
      > The ending phase must be no earlier than the starting phase. If left blank, the ending phase will be same as the starting phase.

   6. Optionally, select the **Mark as required/Set properties as read-only** checkbox if you don't want this task to be modified or deleted when you plan tasks in Task Planner from a change record.

      > **Note**
      >
      > After you select the checkbox, a red dot appears on the top left corner of the task.

   7. Optionally, add conditions to create the task.

Note

"CurrentRecord" in the condition editor refers to the Change record.

Caution

You can't add a new task until you define mandatory details of the previous task in the Task Planner window.

5. Click **Add Task** to add another task.

   Or

   Hover the mouse on the new task added in the graphical interface until you see a hand symbol, and then drag the mouse pointer to add a new task.

6. If needed, click **Delete** in the toolbar to delete a task.
7. (Optional) After you finish adding tasks, click **Auto Layout** to allow task planner to automatically arrange the tasks and dependencies.
8. Click **OK** to save the change model tasks.

Note

- You can edit a task by selecting the task in the graphical interface and modify its properties.
- The Task Planner graphical interface supports panning.
- You can click **Zoom in** or **Zoom out** in the toolbar or use the mouse wheel to zoom in or zoom out the Task Planner graphical interface.
- You can reposition the workflow by moving the outline box within the graphical interface.
- You can resize the Task Planner graphical interface by expanding the details form or resizing the Task Planner window.

Map change model fields into change records

User Roles: System Administrator and Implementer

By configuring a predefined link file, you can customize the fields to be copied from a change model to a change record. You can modify a global link file used as the default for all change models, and you can further specify a link file for an individual

change model. The link file for a specific change model takes precedence over the global link file.

To configure a default change model field mapping for all change models, follow these steps:

1. Click **Change Management** > **Administration** > **Settings** in the System Navigator.

   The Change Setting screen appears. The default link file for all the change models is specified in the **Default Open/Apply Link** text box.

2. Click the **Find Related Information** icon on the right of the **Default Open/Apply Link** text box to open the link file.
3. Click the line containing the mapping information in the table, and then click **More** > **Select Line**.
4. Modify the field mapping in the **Source Field (Fill To/Post From)** column and the **Target Field (Fill From/Post To)** column.

   Note

   Enter the field names of the `changeModel` table in the **Source Field (Fill To/Post From)** column, and enter the field names of the `cm3r` table in the **Target Field (Fill From/Post To)** column. The source and target fields in each mapping row should have the same data type.

5. Click **Save**.

Note

If you want to use a different link file, enter the name of your link file in the **Default Open/Apply Link** text box.

To configure a field mapping for one specific change model:

1. Click **Tailoring** > **Tailoring Tools** > **Links** in the System Navigator.
2. In the **Name** field, type the name of the form or file. If desired, type a descriptive name or phase in the **Description** field.
3. Click **New** to create the new link record.
4. Click a line in the table, and then click **More** > **Select Line**.
5. Type descriptive relationship information in the appropriate text boxes.

6. Configure a mapping using the **Source Field (Fill To/Post From)** column and the **Target Field (Fill From/Post To)** column.

   > **Note**
   >
   > Enter the field names of the `changeModel` table in the **Source Field (Fill To/Post From)** column, and enter the field names of the `cm3r` table in the **Target Field (Fill From/Post To)** column. The source and target fields in each mapping row should have the same data type.

7. Click **Save** to add the new link record.
8. Click **Change Management** > **Configuration** > **Change Models** in the System Navigator.
9. Click **Search** to select a change model.

   Or

   Click **New** to create a new change model record.

10. In the **Open/Apply Link** field, click the Fill icon to select the link file you created.
11. Click **Save**.

Use change models in the Service Catalog connector

User Roles: System Administrator

When adding a new Service Catalog item using the “Open a Change” connector, you can select a change model to open a change.

To use a change model when creating a service catalog item using the “Open a Change” connector:

1. Click **Service Catalog** > **Administration** > **Manage Items** in the System Navigator.
2. Click **Add New Service Item**.

   The New Service Item Wizard appears.

3. Enter the information as needed for the new Service Catalog item, and then click **Next**.
4. Select **Open a Change** as the connector type and specify the parent category that this new item belongs to.

5. Click **Next**.
6. Select change category, change subcategory, and change model, and then click **Next**.
7. Specify all the fields to define the initial state of the change request this item creates.

   Note

   All these fields are mandatory for creating the change requests. They must be specified either here or in the selected change model. If you leave any of these fields empty here, make sure this field has been defined in the selected change model. Otherwise, the change will not be created automatically after a request is approved.

   The values specified here take precedence over those defined in the selected change model (or in the change templates used by the change model).

8. Click **Finish**.

After completing these steps, a Service Catalog item is created with the “Open a Change” connector. When a user orders this Service Catalog item and the request gets approved by the approver, a change record using the selected change model will be created. When the change record has gone through the change workflow and is closed, the interaction record is automatically closed.

Enable or disable the Apply Change Model option for a change workflow phase

User Roles: System Administrator, Change Manager

In the out-of-box setting, the **Apply Change Model** option is available only for change records that use the Project Proposal, Emergency Change, Normal Change, or Standard Change workflow and are in a phase before Authorization.

You can enable or disable this option for a phase of a change workflow by configuring the **Chm Apply Change Model** action in the workflow phase.

To enable the “Apply Change Model” option for a change workflow phase, follow these steps:

Note

If you want to edit a **Proprietary** workflow or use it as a model, create a copy of this **Proprietary** workflow.

1. Click **Change Management** > **Configuration** > **Change Workflows** from the System Navigator. All change workflows are displayed.
2. Open the change workflow that you want to configure.
3. Select a phase in the workflow, and then click the **Actions** tab.
4. Click **Add** to add an action with the following settings:
    - **Id**: type Apply Change Model
    - **Action**: select **Chm Apply Change Model**
    - **Location**: select **More Options List**
    - **Action Condition**: Select **RAD Expression**, and then enter $L.tableAccess.update=true
    - **Requires lock**: select the checkbox (true)
5. Click **Save**.
6. To disable this option for this phase, select the **Apply Change Model** action, and click **Delete**.

Clone a change model

User Roles: System Administrator, Change Manager

To clone a change model with task information, follow these steps:

Note

If you want to edit a **Proprietary** workflow or use it as a model, create a copy of this **Proprietary** workflow.

1. Click **Change Management** > **Configuration** > **Change Models** from the System Navigator. All change models are displayed.
2. Click **Search**, and then select a change model.
3. Click **Clone**. The system opens the **Clone Model** page.
4. Specify a new unique ID and a name for the cloned change model record.
5. Select the **Copy task information from another model** checkbox, and then click the **Fill** icon to select another change model from which the task information will be copied.

Note that if the change model you selected has a different task workflow than the current one, the copied task workflow may not work correctly if you use this cloned model.

6. Click **OK**.

Related topics

Change Management configuration

1.8.5.3. Configuring approvals in Change Management

Change Management requires approvals to be complete before the change request can move to the next logical phase. In some cases, approvals can be defined based on associated Service Level Agreements (SLA), so that managers and sponsors associated with the SLA will be listed as approvers for some phases.

After the approvals are complete, you can move the request to any desired phase by setting a condition in the condition editor of a transition. You can set the transition condition to move the change request to another phase or remain in the current phase by setting the statuses to “Denied” and “Pending” respectively. For example, you can set the following conditions for the transitions in the “ECAB Approval” phase to move the change request to different phases.

Condition	Action Performed
Set the following condition in the transition from the “ECAB Approval” phase to the “Build and Test” phase `Approval Status in CurrentRecord= approved AND Build Test Required in CurrentRecord= true`	Change request moves from the “ECAB Approval” phase to the next phase, “Build and Test”.
Set the following condition in the transition from “ECAB Approval” to “Implementation” `Approval Status in CurrentRecord= approved AND Build Test Required in CurrentRecord= false`	Change request moves from the “ ECAB Approval” phase to “Implementation”.
Set the following condition in the transition from “ECAB Approval” to “Risk and Impact Analysis”. `Approval Status in CurrentRecord= denied`	Change request moves from the “ECAB Approval” phase to the previous phase, “Risk and Impact Analysis”.

A change request may require other approvals:

- Change requests with an impact of 3 (Major Impact) also require approval by an IT director.
- Change requests with an impact other than 1 (Minor Impact) must be approved by a member of the Change Advisory Board (CAB).

Related topics

What's an approval?
Approval options in Change Management

1.8.5.4. Managing alerts

Alerts trigger a series of checkpoints in a change or task to ensure that the required work activities occur within the specified time frames. Alerts serve as reminders to keep a change or task on schedule.

Alerts are timed or delayed conditions that are true or false. Requests for change proceed in phases, according to a predefined schedule. Alerts monitor the progress of these phases. An alert condition becomes true when circumstances require an automated response. Change Management treats it as an event, and sends the predefined notifications. For example, the late notice alert notifies a designated management group that a request for change is overdue for approval, and updates the alert status to include late notice.

Users can define any number of standard or customized alerts for any phase, control who is notified for each alert, and control the naming convention used for the alert itself. Alerts support other functions within the system.

Function	Description
Alert messaging	The event manager generates messages to designated recipients as a result of an alert, which updates the original request.
Batch scheduling	Change Management schedules all alerts associated with a phase at once when the phase occurs.

What are the alert controls?

The Phase record enables you to set alert controls.

Alert control	Description
Reset	Sets the status of all current Alert records associated with the current request to inactive. Marks the last action field as reset. Schedules a calculate alert record to recalculate the alerts and restart the alerts process.
Recalc	Retrieves each Alert associated with the request for change and performs the following processing: • If the current alert status is active, the alert condition is reevaluated, the alert is updated to reflect the correct status, and processing ends. • If the current status is not active, the Schedule Condition field is reevaluated. If Schedule Condition= true, Change Management updates the following fields: ◦ Status is set to scheduled. ◦ Last Action is set to recalc. ◦ Action Time is set to current date/time. ◦ Schedule Condition is reevaluated. If true, Alert Time is recalculated and Status updated to scheduled. If false, Status is set to not required.

Alert processing files

There are two primary files used in alert processing:

- The Alert Definition (AlertDef) file defines the alerts used by all phases

AlertDef defines the basic alert information for each named alert and all general alert definitions. AlertDef is a static file.

- The Current Alerts (Alert) file tracks the alerts created for each phase

  Alert is an active file.

Access an alert definition record

Applies to User roles: System Administrator

Alerts trigger a series of checkpoints in a change or task to ensure that the required work activities occur within the specified time frames. Alerts serve as reminders to keep a change or task on schedule. You can access alert definition records in order to modify, add, or delete alert definitions.

To access an Alert Definition record, follow these steps:

1. Click **Change Management** > **Configuration** > **Alerts**.
2. Use search or advanced search to find one or more records.
3. Click a record to view it in the **Alert Definition** form.

Create an Alert Definition record

Applies to User roles: System Administrator

To create alert definition records for use by the Change Management, Problem Management, and Request Fulfillment applications, follow these steps:

1. Click **Change Management** > **Configuration** > **Alerts**.
2. Problem Management: Click **Problem Management** > **Configuration** > **Alert Definitions**.
3. Request Fulfillment: Click **Request Fulfillment** > **Configuration** > **Alert Definitions**.
4. Type the alert name in the **Alert Name** field.
5. Type the alert description in the **Description** field.
6. Complete the remaining fields in the alert definition record form. If necessary, press **Ctrl+H** to view help for each field.
   - **Scheduling** tab: describes the calculations that trigger when the alert occurs.
   - **Update** tab: describes Format Control considerations and notification messages to be sent.
   - **Work Schedule** tab: specifies the work schedule to which the alert belongs.

- **Time Zone** tab: enables you to perform the following actions:
  - Define a specific time zone.
  - Define a RAD expression to set the time zone alert.
  - Obtain the time zone from a table lookup.

7. Click **Add**.

1.8.5.5. Configuring Change Management notifications

In the out-of-box system, Change Management calls the notification engine to send notifications. You can find the out-of-box Change Management notifications from **System Administration** > **Tailoring** > **Notifications** > **Notifications**. When creating custom notifications for Change Management, you can also use Process Designer rulesets to call the Send HTML Email rule type to send notifications.

Access a notification definition record

Applies to User roles: System Administrator

Notifications are messages generated by Service Manager events, such as opening or closing a change or task. You can edit these messages, add or delete messages, change the conditions that trigger the messages, and select which people receive the messages.

To access a Notification Definition record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Use search or advanced search to find one or more records.
3. Click a record in the record list to view it in the **Notification Definition** form.

View a message class record

Applies to User roles: System Administrator

Service Manager contains several default message classes that enable a user to define additional messages to display in Change Management.

To view a Message Class record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Click **Log Message Class** (or any other message class type).
3. Use search or advanced search to find one or more records.
4. Click a record to view its detail in the related **Message Class File** form.

Add a notification definition record

Applies to User roles: System Administrator

The Notification Engine is primarily responsible for sending messages that are generated by Service Manager events, such as opening or closing a quote or order. System Administrators can edit these messages, add new messages, change the conditions under which the messages will be sent out, and select who will receive the messages.

To add a notification definition record, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**. The **Notification Definition** form opens.
2. Type or select the following information.

Field	Description
Name	Type the name of the notification record.
Condition	Type true to enable the notification in all conditions. Type an expression that's true or false to specify a condition when Service Manager should send the notification. Type false to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.
Message tab	Use this tab to define the message type, delivery method, and conditions under which Service Manager should send the notification.
Msg Class	Type the message class for the notification. This value must match a message class definition in the message record.
Msg No	Type the message number for the notification. This value must match a message number definition in the message record.

Field	Description
Arguments	Type any RAD expressions to display information from the record or event that triggered the notification. Service Manager displays this information in the text of the notification. You can enter an array of arguments by enclosing all arguments in brackets and separating each argument with a comma. For example: {argument1,argument2,argument3} **Note** Arguments typically take the following form: `<field name> in $L.file` where `<field name>` is the name of a field in the record that triggered the notification.
Condition	Type true to enable the notification in all conditions. Type an expression that can be true or false to define a condition when Service Manager should send the notification. Type false to disable the notification in all conditions. By default, Service Manager treats a blank entry as true.

Field	Description
Format	Type the format control record used to display the triggering record as part of the notification.
Notify Method	Type or select the delivery method to be used to send the notification. This value must match a delivery method defined in the msgtype table. **Note** - The notification applications listed below must be defined in RAD code and exist in Service Manager to work. - The message type must be enabled (true) by selecting "Active?" in the msgtype table. See the following table for valid message types.

Message type	Desctiption
connect	Send a message to an external device.
email	Send email.
log	Create a log entry for a message.

Field	Description
mail	Send internal mail.
msg	Send an internal message to a user's current Service Manager session (for example, "Incident IM1011 has been assigned to your monitor list since..."). **Note** If the user is not currently connected to a Service Manager session, the message is not sent.
print	Print message.
Recipients	Type one of the following to identify who should receive the notification: ◦ A contact name ◦ An operator name ◦ Array of approval group names ◦ A RAD expression that gathers a contact or operator name from some field in the triggering record
Group File	Type the name of the group receiving the notification. You can specify the On Call group to send the notification to for those operators who meet the On Call criteria.

Field	Description
Group Area	Type the name of the group area receiving the notification. You can use this field only with ocmgroups and cm3groups group tables. Valid choices include: ◦ ocmgroups -All -Line Items -Orders -Quotes ◦ cm3groups -All -Changes -Tasks
Subgroup	Type the name of the subgroup area receiving the notification. You can use this field only with ocmgroups and cm3groups group tables. Valid choices include: ◦ All ◦ Approvers ◦ Members
Email/Mail Subject Line tab	Use this tab to type any custom information that should appear in e-mail notifications.

Field	Description
Msg Class	Type the message class for the subject of the notification. This value must match a message class definition in the message record.
Msg No	Type the message number to use as the subject of the notification. This value must match a message number definition in the message record.

Field	Description

Arguments	Type any RAD expressions used to display information from the record or event that triggered the notification. Service Manager displays this information in the subject of the notification. You can enter an array of arguments by enclosing all arguments in brackets and separating each argument with a comma. For example:
	`{argument1,argument2,argument3}`
	Note
	Arguments typically take the following form:
	`<field name> in $L.file`
	where `<field name>` is the name of a field in the record that triggered the notification.

3. Click **Add**.

Send a notification from a task or change request

Applies to User roles: Change Coordinator

To send a notification from a task or change request, follow these steps:

1. Do one of the following to generate a record list:
    - Click **Change Management** > **Search Changes**.
    - Click **Change Management** > **Search Tasks**.
2. Use search or advanced search to find one or more records.
3. Click a record to view its detail.
4. Open the **More Actions** menu on the change or task form.
5. Click **Notify**.
6. Add one or more recipients to the **To** text box, or type an assignment group in the **Group** text box.
7. Select one of the following notification methods:
    - **Mail**: Uses the internal Service Manager Mail application (not Email).
    - **Email**: Uses the Service Manager Email application.

      If you select **Email**, you can specify the names of the recipients in the **CC** and **BCC** fields as needed. In addition, you can click **Attachment** > **Add Files** to add attachments to the email message as needed.
8. Modify the subject and the message in the notification as needed.
9. Click **Send**.

Contents

- 1 Access a notification definition record
- 2 View a message class record
- 3 Add a notification definition record
- 4 Send a notification from a task or change request

1.8.6. Security

Service Manager controls access to Change Management functions based on security roles, security areas, and rights.

Change Management security areas

User Role: System Administrator

The security areas in Change Management are Changes, Change Tasks, and Change Management Configuration. These areas contain the default security rights and settings for Change Requests in the Change Management module. The security right settings will be inherited by the new roles created in an area when no settings are specified in the security role.

The change areas are used to set permissions to operators to provide access to particular area of Change Management. For example, for the Change Coordinator security role, set change area as Change Management Configuration and rights as **View** and **Admin**. Then, an operator with this security role can view and configure only the Change Management Configuration items under Change Management.

The following table lists the areas and the relevant Change Management menu items the operators can access.

Security area	System Navigator menu items for this area
Change	This area contains the default security rights and settings for change requests in the Change Management module. For example, **Create New Change**, **Change Queue**, **Task Queue**, and **Search Changes**.
Change Tasks	This area contains the default security rights and settings for change tasks in the Change Management module. For example, **Create New Tasks**, and **Search Tasks**.
Change Management Configuration	This area contains the default security rights and settings for Change Management configuration. For example, **Alerts**, **Approvals**, **Change Categories**, **Change Models**, **Change Workflows**, **Messages**, **Subcategories**, and **Task Categories**.

The default rights defined in areas are inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Change, Change Task and Change Management Configuration areas.

Security area	View	New	Update	Delete/ Close	Expert	Admin
Change	TRUE	FALSE	Never	Never	FALSE	FALSE
Change Tasks	TRUE	FALSE	Never	Never	FALSE	FALSE
Change Management Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Change Management security roles and settings

User Role: System Administrator

The out-of-box security roles for Change Management include the following:

- change advisory board (CAB)
- change analyst change
- change analyst tasks
- change approver
- change coordinator
- change coordinator change
- change coordinator tasks
- change domain expert (CDE)
- change implementer
- change manager
- change owner
- change requestor

Tip

If you have upgraded from a previous version of Change Management, which is not Process Designer based, see Change Management security profiles mappings for more information about the mappings between the legacy Change Management profiles and the current Process Designer security roles.

Rights and settings

To view the rights and settings that are defined for out-of-box Change Management security roles, follow these steps:

1. Navigate to **System Administration** > **Security** > **Roles**.
2. In the **Name** field, select a Change Management security role.
3. Click **Search**.
4. Double-click a security area to open the Rights form.

   This form lists the rights and settings of the security role for the selected security area.
5. View the **Rights** and **Settings** sections in the form.

   The following table describes possible rights for Change Management security areas.

Security role rights	Description
New	Can create new records in this area
View	Can view records in this area.
Expert	Has Expert rights in this area
Admin	Has Admin rights in this area
Modify Template	Can modify template records in this area
Update	Can update records in this area
Delete/Close	Can delete or close records in this area
Allowed Categories	Specifies the categories in this area that users with this role can select
Allowed Statuses	Specifies the status values in this area that users with this role can select

The following is a list of settings available for Change Management security areas.

- **Allow Inefficient Query**: When this option is true, users with this role can run incomplete (inefficient) queries in this area, but receive a warning message. The Skip Inefficient Query Warning option overrides this setting. If not selected, users with this role can't run inefficient queries in this area.
- **Skip Inefficient Query Warning**: Turns off inefficient query warnings for this area. This option overrides the **Allow Inefficient Query** option.
- **Can Approve**: Users with this role can approve records in this area.
- **Can Delegate Approvals**: Users with this role can delegate their own approvals in this area to another user.
- **Reopen**: Users with this role can reopen records in this area.

- **Change Manage Format**: The name of the form to display as the default change queue form. If left blank, **sc.manage.chm** is used.
- **Default Category**: Unused
- **Default Task Category**: Unused
- **Initial Change View**: Unused. Service Manager automatically saves the Change view that users with the role opened last time.
- **Initial Task View**: Unused. Service Manager automatically opens the Change Task view that users with this role opened last time.
- **Initial Format**: The initial format that's used when users with this role search change records. If left blank, **cm3r.search** is used.
- **List Format**: Also known as QBE Format. This is the QBE form to use when displaying records to users with this role. If left blank, **cm3r.qbe** is used.
- **Task Manage Format**: The name of the form to display as the default task queue form. If left blank, **sc.manage.cmt** is used.
- **Append Query**: This field stores an expression to append to all queries in this area run by users with this role. This expression restricts the records that the user with this security role can see, by appending (adding on) this query to everything the user does in this module. You must use the field name from the dbdict in the expression. For example, you can enter priority.code="3" so that users with this security role can only see Priority 3 records.

6. View the Security Folders section.

   This section specifies security folders that are accessible to this Role for records of this area. For more information, see Add folder permissions to a security role.

   The out-of-box security folders available in Service Manager are **DEFAULT** and **advantage**. You can also create security folders to meet your business needs. By default, all security folders are assigned to a new security role created. Once a role is created and rights are configured, you can modify the security rights for a role within an area.

Configuring Change Management security

User Role: System Administrator

Change Management Administration enables you to perform the following Change Management security configurations:

Set rights to changes and change tasks

The Change Administrator can set rights to changes and tasks using the role-based security. The Changes, Change Tasks, and Change Management Configuration areas contain default security rights and settings. The rights are copied to new roles created for these areas. However, the settings are inherited only if there are no settings specified on the roles.

Note

Whenever the roles in an operator record are updated, the operator must sign out and then sign in for the changes to take effect.

Add security roles and settings

User Role: System Administrator

To create a security role and assign rights and settings to it, follow these steps:

1. Click **System Administration** > **Security** > **Roles**.
2. Click **New**.

   The security role form is displayed.
3. Type a security role name.
4. Type a security role description.
5. Click **Save**.
6. Select a security area.

   The security rights and settings form is displayed.

   1. Under **Rights**, select the rights to be assigned to the security role.

      For example, set **Expert** rights for the security role. The **Expert** security right enables the operator to view alert logs, opened tasks, affected services, and clocks of change requests. It also enables the operator to set reminders, send notifications, create hot news, and associate change requests to incidents, interactions, requests, and know errors.
7. Under **Settings**, add required settings.
8. Under **Folders**, add folder permissions to the security role.

   The out-of-box security folders available in Service Manager are DEFAULT and advantage. You can also create security folders to meet your business needs. By default, all security folders are assigned to a new security role created. Once a

role is created and rights are configured, you can modify the security rights for a role within an area.

Modify an operator record to enable Change Management access

Applies to User roles: System Administrator

To modify an operator record to enable Change Management access, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Use search or advanced search to find one or more records.
3. Click the operator record that you want to view.
4. Click the **General** tab.
5. Add the desired Change Management security roles to the **Security Roles** table.
6. Click **Save**.
7. Click **OK**.

Enable approval delegation

Approval delegation is an optional feature that enables users with approval rights to temporarily delegate their approval authority to another qualified operator. For more information, see Approval delegation.

To enable an operator to delegate approval authority, a System Administrator must make changes to a specific security role of the operator.

The following example illustrates how to enable approval delegation for the Change Management application.

To edit the security role to which you want to grant approval delegation authority, follow these steps:

1. Sign in to Service Manager with a System Administrator account.
2. Click **System Administration** > **Security** > **Roles**.

   Service Manager displays a list of security roles.
3. Optionally, in the **Name** field, type the name of the security role to which you want to grant approvals, for example, **change approver**.
4. Click **Search**. The security role or a list of security roles opens.

5. Locate the security role you want to edit, and select a Change Management security area, **Change** or **Change Tasks**.
6. Select **Can Delegate Approvals** under **Settings**.
7. Click **Save**.

Note

It is a best practice to enable the **Delegate Approvals** or **Can Delegate Approvals** option only for operators who can also view and approve objects in the application.

Now, the operator can delegate approvals for the selected security area. For information about how to delegate approvals to another operator, see Delegate approvals to another operator.

Related topics

Process Designer security model
Add folder permissions to a security role

1.9. Request Fulfillment (Codeless)

Service Manager Request Fulfillment manages user requests for products and services. Requests can affect only the person making the request, or a subordinate group of employees. Examples include password resets, individual PC upgrades, and new employee setup.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Request Fulfillment user roles

Key concepts

Describes the most important things that you need to know about how the Request Fulfillment module functions.

Request Fulfillment overview

Key elements of Request Fulfillment

Workflows and user tasks

Describes the workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Request Fulfillment workflows and user tasks

Request Task workflows and user tasks

Administrator tasks

Describes the Administrator tasks in the Request Fulfillment module.

Configure the Request Fulfillment settings

Additional resources

Processes and Best Practices

1.9.1. Request Fulfillment overview

The Service Manager Request Fulfillment is an application used to manage user requests for products and services.

The Request Fulfillment application enables business staff to improve their productivity or the quality of business services and products. It can also help reduce the cost of providing services and reduce people effort involved in requesting and receiving access to services. Moreover, the use of Request Fulfillment application can increase the control level of an organization's services and the number of fulfilled requests.

Request Fulfillment includes the following key features:

- Request model, which defines the prerequisites, required authorizations, and sequenced or parallel standard tasks to fulfill the service request
- A detailed, customizable catalog of products
- Scheduling of service requests and tasks
- Automated request fulfillment
- Order and stock management
- Interaction with other Service Manager applications, such as Service Catalog, Configuration Management, Service Desk, Incident Management, Change Management, and Service Level Management. Service Catalog (Front Office) and Request Fulfillment catalog (Back Office product catalog) work together to provide an end-to-end process of service requests, from the end user portal (Service Catalog), to the automated dispatching and control of every tasks for the delivery of the service.
- Integration with other products, including:
  - Providing a common web service interface so that other products are able to access service requests and tasks
  - Integration with Asset Manager for request fulfillment billing

For Request Fulfillment processes and best practices, see the corresponding sections in Processes and Best Practices .

Related topics

Request Fulfillment security roles

Request Fulfillment administration

Request Fulfillment configuration

Request Fulfillment workflows and user tasks

Differences between Request Fulfillment and Change Management

Processes and Best Practices

1.9.1.1. Key elements of Request Fulfillment

Request Fulfillment includes the following key elements.

Request Model

A repeatable model of handling a particular category of service request. A request model defines specific agreed tasks that need to be followed to fulfill the service request of this category. Request models can be very simple, with no requirement for authorization (for example, password reset), or can be complex with many sequential or parallel tasks that require authorization (for example, provision of an existing IT service).

Request models are the catalog of service requests from the fulfillment perspective. Usually, the request models are referenced in the connector of a Front Office catalog item (svcCatalog, connector "Open New Request").

Request models define the delivery workflow, through the list of their tasks, controlled by the task planner. Each task defined in a request model has a task category, which defines the role and behavior of the task (labor, product purchase, CMDB synchronization, and so on).

Product Catalog

In a request model, tasks from the Purchase category are used to order a product from a vendor, and to create the corresponding CIs. The product catalog defines which products can be purchased. The product definition is used to process the purchase task, and to become the model of the corresponding CIs.

The product catalog is a predefined catalog of products (hardware, software, or called parts). The product catalog defines the models of independent items that may be requested or ordered.

The product catalog supports general information of parts, including part number, description, cost, and manufacturer. The inventory of the specific item is also available in the product catalog portal. The receiving details of the purchased items can be pre-defined in the product catalog.

Catalog items are represented as records in the `productCatalog` table.

Vendors/suppliers

Vendors/suppliers are internal or external providers of parts. Vendors/suppliers have a many-to-many relationship with catalog items, and may or may not directly interact with Service Manager.

Vendors/suppliers are represented as records in the `vendor` table. The terms under which a specific vendor/suppliers will provide a specific catalog item are stored in the `modelvendor` table.

Requests

A request is a high-level record that defines the basic request information such as requester, required dates, coordinator, and description. Request records are the "tickets" that trace the workflow of a request from the user perspective, data entry and request task addition, through authorizations, fulfillment, and follow-up.

Request records are stored in the `request` table.

Request Tasks

A request task is a low level record that defines one of the steps to fulfill the service request. Request tasks can be created automatically according to the request model definition or manually by the authorized users. After creation, request tasks are assigned to internal workgroups or external vendors/suppliers.

Request tasks are stored in the `requestTask` table.

Orders

Order records are a type of request record that traces the workflow of an actual order of a part item or several part items from the ordering and receiving perspective. Orders are created manually by authorized users or by automated background processes when the reorder rules defined for stock management are satisfied.

Order records are stored in the `request` table, with a particular "Order" category.

Authorization processing

The authorization process automates and formalizes the technical and business evaluation by the appropriate levels of management of requests. Authorizations control accepts risk, cost, and responsibility of a request and its tasks. Authorizations create "chains" of groups who may be required to approve requests before they can advance within their lifecycle. Authorizations can have conditions attached, such as total cost, lead time requirements, and impact.

Authorizations can be defined either in request model or in the dedicated Authorization phase of the request fulfillment workflow.

1.9.1.2. Differences between Request Fulfillment and Change Management

Request Fulfillment and Change Management are separate processes, but they are closely related. Request Fulfillment handles common user requests for products and services. These requests usually affect only the person making the request, or a subordinate group of employees. Change Management handles any change to your business that modifies or disrupts the current state of that environment. Usually these modifications or disruptions affect multiple users or business units.

- Request Fulfillment
  - Handles common requests for products and services
  - Affects a small or limited number of users
  - Scope is limited
- Change Management
  - Manages changes (implementations) that modify a business environment
  - Affects many users
  - Scope is often large, including large groups or multiple business units

Related topics

Request Fulfillment overview
Request Fulfillment security roles
Request Fulfillment administration
Request Fulfillment configuration
Request Fulfillment workflows and user tasks
Processes and Best Practices

1.9.1.3. Request Fulfillment and service level agreements

Process target in Service Level Management is supported by requests and request tasks. The following are the out-of-box process targets:

- Request with global lead time defined

  The duration is defined in the global.lead.time field of the request record.

- Request Task with planned lead time defined

  The duration is defined in the planned.lead.time field of the request task record.

- Other out-of-box process targets without lead time defined, such as:
  - Platinum - Request without global lead time defined (duration is 12:00:00)
  - Platinum - Request Task without planned lead time defined (duration is 12:00:00)
  - Gold - Request without global lead time defined (duration is 1 00:00:00)
  - Gold - Request Task without planned lead time defined (duration is 1 00:00:00)
  - Silver - Request without global lead time defined (duration is 3 00:00:00)
  - Silver - Request Task without planned lead time defined (duration is 3 00:00:00)
  - Bronze - Request without global lead time defined (duration is 5 00:00:00)
  - Bronze - Request Task without planned lead time defined (duration is 5 00:00:00)

Related topics

Request Fulfillment overview
Request Fulfillment security roles
Request Fulfillment administration
Request Fulfillment configuration
Request Fulfillment workflows and user tasks
Processes and Best Practices

1.9.1.4. Request Charges

Service Manager can track labor costs and parts costs in service request fulfillment processes.

Meanwhile Service Manager can pass cost and billing information for goods and services to an external financial tool. In out-of-box, integration is created using Connect-It between Service Manager and Asset Manager.

Costs of Request Fulfillment

Each request displays the fulfillment cost (including incremental ongoing costs associated with service options).

Request billing

By integrating Service Manager with Asset Manager, you can view the request billing report in Asset Manager.

You can find the "Cost of SM Service Request" widgets in the Homepage widgets library of Asset Manager. If you want to show these widgets in anyone’s homepage, you need to manually add them. For more instructions on adding widgets to homepage, see the "Customizing Web Clients > Homepage of the Web Client" section in the Asset Manager Tailoring guide.

All credit type expense line records will be assigned to the **IT Finance** billing center, which you can find in the out-of-box database of Asset Manager. If you don't want to use the **IT Finance** billing center in your production environment, you can change the Connect-It scenarios to allocate a new billing center.

Related topics

Labor costs
Parts costs
Service Manager integration methods and tools

1.9.1.5. Request Integration

Integration: Service Desk or Incident Management

In Service Manager, the service requests are stored as Service Desk interactions of a specific type. Service Manager Service Catalog can easily integrate with the Incident Management module through open fulfillment connector architecture.

Integration: Change Management

Change Management is a standard fulfillment engine for service requests.

Service Manager will copy shared information from the service request to the change request record and automatically link the records. Links between these records, once created, can't be broken by users. Links to other records are listed in each linked record, and users can click on links to access other linked records.

This link allows Service Manager to correctly propagate the status changes between the change request and the service request.

Integration: Release and Deployment Management

All Service Manager modules are fully integrated. Release and Deployment Management records can be quickly opened from a service request, if required. Service Manager will copy shared information from the service request to the release request record (created in the Change Management module) and automatically link the records. Release request records can also be opened automatically from service request records based on business rules defined to meet your needs. Links between these records, once created, can't be broken by users. Links to other records are listed in each linked record, and users can click on links to access other linked records.

Integration: Configuration Management System

One option in a catalog item is to create a subscription to a service (or to any type of configuration item).

When the service request is created for this type of catalog item, the service request is linked to the CIs through the catalog items it references.

Integration: Service Catalog

In Service Manager, the creation of a Service Catalog request can starts with the Service Catalog by using Self-Service web or reported directly to service desk via

phone or email. One or more entries in the catalog are linked to the Service Catalog request through a cart item (request line).

Integration: Third-Party

Bi-directional integration with third-party solutions can be created using Connect-It or the Web Services API in Service Manager.

1.9.2. Request Fulfillment security roles

There are specific roles associated with Request Fulfillment. Service Manager uses role-based permissions to enable you to complete a task that's appropriate to your role. System Administrators manage and assign these permissions.

The following table describes the responsibilities of the Request Fulfillment security roles.

Request Fulfillment security roles

Role	Responsibilities
Request Process Owner	• Designs request fulfillment models and workflows • Ensures process technicians have the required knowledge and the required technical and business understanding to deliver the process, and understand their role in the process • Communicates process information or changes as appropriate to ensure awareness • Reviews opportunities for process enhancements and for improving the efficiency and effectiveness of the process • Works with other process owners to ensure there is an integrated approach to the design and implementation of request fulfillment, incident management, event management, access management, and problem management
User	• Uses Self Service or the Service Desk to log appropriate Service Requests
Requester	• Initiates a Service Request through the Request Fulfillment portal on behalf of users

Role	Responsibilities
Request Coordinator	• Provides a single point of contact and end-to-end responsibility to ensure submitted Service Requests have been processed • Provides initial triage of Service Requests to determine which IT resources should be engaged to fulfill them • Assigns a Service Request to the correct group • Communicates Service Requests to other IT resources that will be involved in fulfilling them • Escalates Service Requests in line with established Service Level Targets
Request Analyst	• Carries out one or more activities in the fulfillment of Service Requests • Updates the records to show that activities have been carried out correctly • Responsible for the provisioning of Service Requests within the agreed SLA
Request Approver	• Reviews Service Request details • Confirms Service Request details are correct • Approves/Rejects Service Requests

Role	Responsibilities
Request Manager	• Plans and manages support for request fulfillment tools and processes • Handles staff, customer and management concerns, requests, issues and enquiries • Involved in Service Request escalation • Ensures request fulfillment activities operate in line with service level targets
Stock Manager	• Responsible for managing stock and defining reorder rules for different stockrooms

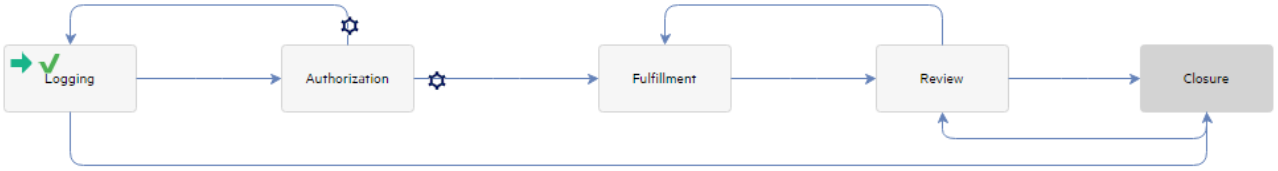
1.9.3. Request Fulfillment workflows and user tasks

The request workflow is a sequence of connected steps in the life cycle of a request. In the workflow, a request goes through several phases to complete the life cycle. Workflow phases can consist of one or multiple tasks that must be completed to proceed to the next phase.

The following are the out-of-box workflows for the Request Fulfillment module.

Request

The Request workflow manages the generic request fulfillment process.



The Request workflow consists of the following phases and tasks.

Workflow phase	Tasks
Logging	• Create a new request record • Create a new request from a user interaction • Create a request by copying an existing request record • Apply a request template • Create a request template • Apply a request model to an existing request record • Update a request record • Update multiple request records • Link a request to another record • Set a reminder for a request • Send a notification from a request • View audit log from a request • Access the alert log when viewing a request • Close a request directly in the Logging phase
Authorization	• Approve or deny a request • Approve multiple request records • Update a request record • Escalate a request • Update multiple request records • Set a reminder for a request • Send a notification from a request • View audit log from a request • Access the alert log when viewing a request • Create a request template

Workflow phase	Tasks
Fulfillment	• Suspend a request • Update a request record • Create a request task • Escalate a request • Update multiple request records • Link a request to another record • Set a reminder for a request • Send a notification from a request • View audit log from a request • Access the alert log when viewing a request • View or edit the cost of a request • Create a request template
Review	• Update a request record • Update multiple request records • Escalate a request • Link a request to another record • Set a reminder for a request • Send a notification from a request • View audit log from a request • Access the alert log when viewing a request • View or edit the cost of a request • Create a request template
Closure	• Close a request record • Reopen a request record • Send a notification from a request • View audit log from a request • Access the alert log when viewing a request • Create a request template

Order

The Order workflow manages purchase and stock management in the Request Fulfillment application.

The Order workflow consists of the following phases and tasks.

Workflow phase	Tasks
Order	• Create a new request record • Create a new request from a user interaction • Create a request by copying an existing request record • Apply a request template • Update a request record • Update multiple request records • Set a reminder for a request • Send a notification from a request • View audit log from a request • Access the alert log when viewing a request • View or edit the cost of a request • Create a request template
Closure	• Close a request record • View audit log from a request • Send a notification from a request • Access the alert log when viewing a request • Create a request template

For examples that demonstrate the minimum needed steps to complete the out-of-box request workflows, see Examples: request workflows.

To view more workflow diagrams and other information about this process, such as related ITIL processes, refer to *Process Designer Best Practices* that's linked to in the related topics.

Related topics

Process Designer
Examples: request task workflows
Create a workflow
Processes and Best Practices

1.9.3.1. Examples: Request Workflow

The following examples demonstrates the minimum needed steps to complete the out-of-box request workflows.

Request workflow

Phase	User Actions	Options
Logging	1. Enter a title in the **Title** field. 2. Provide a description of the request in the **Description** field. 3. Click the **Fill** button for **Requestor**, and then select the requester. **Note** The assign ment group is pre-filled by the out-of-box assign ment rule automat ically. 4. Select the **Impact** for this request. 5. Select the **Urgency** for this request. 6. Specify other fields such as **Affected CI** and **Affected Service** according to your business needs.	This is the first phase when you create a new request (**Request Fulfillment** > **Create New Request**) based on a category or request model that uses the "Request" workflow. You create a request by selecting a category or a request model, depending on the "Select on Open" setting in **Request Fulfillment** > **Administration** > **Settings**. **Tip** If you are creating a new request based on request models, you can use the search function on the form to quickly locate the request model that you want to use.

Phase	User Actions	Options
	7. Click **Save**. 8. Select an expected date to finish the request on behalf of the requester if the request is not created from a service catalog request. **Note** If the user has selected the delivery target when ordering the service item or bundle, the expected finish date is automatically filled by the system.	

Phase	User Actions	Options
Authorization	1. Click the **Submit for Approval** button.	By default, if a request is denied and is not updated after seven days, the request will be automatically closed with the "Denied request fulfillment" closure code. **Tip** System administrators can edit the "rm.close.denied.request" rule set to define the days to automatically close the denied requests.

Phase	User Actions	Options
	Note In the out-of-box system, no approval definition is defined in the Authorization phase, so after you click **Submit for Approval**, the request will directly go to the Fulfillment phase. System administrators can add approval definitio	

Phase	User Actions	Options
	ns in the Authorization phase based on business needs. If approvals are defined in the Authorization phase, the request will be in pending approval status and the request form can't be edited until the request is approved or denied.	

Phase	User Actions	Options
	When the request is approved, it automatically moves to the Fulfillment phase; when request is denied, it automatically moves back to the Logging phase.	

Phase	User Actions	Options
Fulfillment	1. Update the request record as necessary based on your fulfillment actions and progress. 2. If request tasks are defined in the Fulfillment phase, complete the request tasks as required.	**Note** In the Fulfillment phase, you can suspend the request. For more information, see Suspend a request.
Review	1. Click the **Review** button after all the request tasks defined in the Fulfillment phase are completed. 2. Review and update the information in the request.	From the Review phase, you can also return to the Fulfillment phase by clicking the **Back to Fulfill** button.
Closure	1. Click the **Close** button. 2. Enter or change the subcategory for the request. 3. Select the appropriate closure code. 4. Enter the closure comments. 5. Click **Finish**.	From the Closure phase, if you have the "Reopen" right in the "Request" area, you can click the **Reopen** button to reopen the closed request. After you click **Reopen**, the request is moved back to the Review phase.

Note

You can create a request task by using the task planner in the **Tasks** section. All tasks must be closed before you can close a request. For more information about how to create a request task, see Create a request task.

Order workflow

Phase	User Actions	Options
Order	1. Enter a title in the **Title** field. 2. Provide a description of the request in the **Description** field. 3. Click the **Fill** button for **Requestor** to select the requester. **Note** The assign ment group is pre-filled by the out-of-box assign ment rule automat ically. 4. Select a value in the **Impact** field. 5. Select a value in the **Urgency** field. 6. Select an expected date to finish the request. 7. If necessary, set other fields for this	This is the first phase when you create a new request (**Request Fulfillment** > **Create New Request**) based on a category or request model that uses the "Order" workflow. You can create a request by selecting a category or a request model, depending on the "Select on Open" setting in **Request Fulfillment** > **Administration** > **Settings**. **Tip** If you are creating a new request based on request models, you can use the search function on the form to quickly locate the request model that you want to use.

Phase	User Actions	Options
	request, and then click **Save**. 8. Select an expected date to finish the request on behalf of the requester if the request is not created from a service catalog request, and then click **Save**. **Note** If the user has selected the delivery target when ordering the service item or bundle, the expected finish date is automatically filled by the system.	**Note** If you use the "Order" request model, a purchase task is automatically created as defined in the task plan.

Phase	User Actions	Options
	9. Finish the purchase task associated with the request.	
Closure	1. Click the **Close** button. 2. Enter or change the subcategory for the request. 3. Select the appropriate closure code. 4. Enter the closure comments. 5. Click **Finish**.	

Tip

To check the activity log for a request record, go to the **Activities** section of the record. The **Activities** section enables you to enter new updates or view journal updates and logged activities (such as "Open" and "Update") for a record.

1.9.3.2. Create a new request record

Security Roles: Requester, Request Coordinator

After a service catalog request is approved, the fulfillment records (change or request) are created according to connector definitions linked to the ordered items or bundles. If the connector defined in the ordered items or bundles is "Open New Request," a request record will be created automatically and linked to the Service Desk interaction record.

However, you may need to create a new request record, which is not automatically created through service catalog or from an existing record such as an interaction. Depending on how your administrator has configured the global settings (the "Select on Open" setting in **Request Fulfillment** > **Administration** > **Settings**), you open a new request either by selecting a request model or by selecting a request category. By default, you open a new request record by selecting a request model.

To create a new request record directly from the Request Fulfillment module, follow these steps:

1. Do one of the following:
    - Click **Request Fulfillment** > **Create New Request** in the System Navigator.
    - From the To Do Queue, select **Request** in the **Queue** field, and click **New**.

    Note

    If your home page is not the To Do Queue, click **Miscellaneous** > **My Preferences** to change your home page to **To Do Queue**, and then log off and log back in.

2. If a list of request models (which are grouped by request category and subcategory) is displayed, drill down the request categories to select a request model for the new request record. If a list of request categories is displayed, select a request category for the new request record.

    Tip

    If you are creating a new request based on request models, you can use the search function on the form to quickly locate the request model that you want to use.

Service Manager displays a new form, automatically assigns a request ID, and populates the fields that are defined in the request model or based on the request category that you selected.

3. If you are creating the request for someone other than yourself, change **Requested For** to the recipient.
4. Complete the following mandatory fields:
   - Title
   - Description
   - Requestor
   - Assignment Group
   - Impact
   - Urgency
5. Complete the form with any required or additional information such as **Affected CI** and **Affected Service**, and then click **Save**.

   A new request record is now created. Service Manager automatically populates the **Priority** field with a value that's calculated based on the Impact and Urgency values you selected.
6. Select an expected date to finish the request on behalf of the requester if the request is not created from a user interaction.

   Note

   If the user has selected the delivery target when ordering the service, the expected finish date will be automatically filled by the system.
7. Click **Save**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Request Fulfillment workflows and user tasks

1.9.3.3. Create a new request from a user interaction

Security Role: Service Desk Agent

As a Service Desk Agent, you can trigger an interaction that's in the “request for administration,” "request for information," or "service request" category to a request.

To a new request from a user interaction, follow these steps:

1. Click **Service Desk** > **Create Streamlined Interaction**. The New Interaction form is displayed.
2. Specify the **Contact** for the interaction.

   > **Note**
   >
   > If you use the Fill function to replace the information in a field that already has data, clear the field before you click **Fill**. If the field contains data, the Fill function can't operate.

3. In the **Service Recipient** field, select the service recipient.
4. In the **Notify By** field, select the user's preferred notification method.
5. Type a description for the interaction.
6. In the **Category** field, select request for administration, request for information, or service request.
7. In the **Affected Service** field, select the affected service.
8. Click the **Continue** button. The New Request form is displayed.
9. Enter necessary information to create the new request.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.4. Create a request by copying an existing record

Security Roles: Requestor, Request Coordinator

You can create a new request by copying an existing request record. The new request will start from the first phase in request fulfillment workflow. The task plan defined in the original request record is also copied to the new request record.

This "Copy Record" action is available in all request phases. Users who have the “New” right in the “Request” area can run this action.

To create a request by copying an existing record, follow these steps:

1. Click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Open the request record from which you want to create a new request.
3. Click **More**, and then click **Copy record**. A new request form opens with the information copied from the previous request record.
4. Modify the new request record as needed, and then click **Save**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.5. Update a request record

Security Roles: Request Coordinator, Request Analyst

After a request record is created, you may need to update that record to add or correct some information, change the status of the request, move a request to the next phase, or send some updates to the customer.

To update a request record, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate the request record that you want to update, and then double-click the record to open it.
3. Modify the applicable fields such as status in the request form.
4. On the **Activities** tab, do the following:
    1. In the **New Update Type** field, select an appropriate type.
    2. In the **New Update** field, type a description of your update.
    3. Select the **Visible to Customer** checkbox if you want to make this update visible to the end user.

    Note

    This option can be bypassed by setting the **Bypass Visibility Option** flag to “true” in the activityactions table. When this flag is set to “true,” automatic activities are invisible to customers, whether the **Customer Visible** option is selected or not. When the flag is set to “false,” all activities are visible to customers if the **Customer Visible** option is selected.
5. Click **Save**. The request record is now updated.
6. If needed, click the appropriate button to move the request record the next or previous phase.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.6. Suspend a request

Security Roles: Request Coordinator, Request Analyst

If you temporarily can't process the request, you can suspend the request when it is in the Fulfillment phase. You need to specify when to reactivate the request and the justification for request suspension.

When reactivated time is reached, the request is automatically unsuspended and the status is changed back to “In Progress”. If you manually change the status back to “In Progress” before the reactivated time is reached, you need to provide a justification in the unsuspension wizard.

To suspend a request, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate the request record that you want to suspend, and then double-click the record to open it.

   Note

   You can only suspend requests that are in the Fulfillment phase.

3. In the **Status** field, select **Suspended**, and then click **Save**.

   The Suspend Request wizard is displayed.
4. Select the reactivated time.
5. Enter the justification for suspending this request.
6. Click **Finish**. The request is now suspended.

Related topics

Request Fulfillment workflows and user tasks

1.9.3.7. Approve or deny a request

Security Roles: Request Approver

You can approve or deny a request that's pending approval only if you are a member of the necessary approval group and you are assigned the appropriate Request Fulfillment security right.

These actions are available in all phases except “Closure”. Users who have the “Can Approve” right in the “Request” area are authorized to run these actions.

To approve or deny a request in the approval inbox, follow these steps:

1. Sign in to Service Manager as the request approver.
2. Click **Approval Inbox**.

   Requests awaiting your approval are displayed.
3. Select the request that you want to approve, and click **View**.
4. Review the details of the request record.
5. After you finish your review, click **Cancel**.
6. Select the request that you want to process, and then approve or deny it according to your review result:
   - To approve the request, click **Approve**. The request moves to the next phase.
   - To reject the request, click **Deny**, enter a rejection reason and additional comments, and then click **OK**. The request returns to the previous phase.

From the request record, you can also approve or deny the request. In addition, you can retract a previous approval or denial action from the request record. To do so, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate the request record on which you want to perform the approval actions, and then double-click the record to open it.
3. Open the **Approvals** section, and then go to the **Current Approvals** subsection. The Current Approvals form lists the approval type and approval status.

4. Click an approval type to learn which approval group is assigned this approval type. When a request requires more than one approval, the approvals must occur in the order listed in the **Sequence** column.
5. Review the request information and determine how you will process the approval request.
6. To approve a request, click **Approve**.

   Service Manager changes the approval status to **approved**.
7. To deny a request, click **Deny**. Type the reason you are denying the request, and then click **OK**. Service Manager changes the approval status to **denied**, and no further approvals are possible until the denial is retracted.
8. To retract a previous approval or denial action, click **Retract**. Type the reason you are retracting the request, and then click **OK**. Service Manager changes the approval status to **pending**, and the request requires a new approval cycle to progress.

Related topics

Request Fulfillment workflows and user tasks

1.9.3.8. Escalate a request

Security Roles: Request Coordinator, Request Analyst

After the Logging phase, you can perform hierarchical escalation in a request record. After the request is flagged as "Escalated," the system notifies the specified escalation manager.

To escalate a request, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate the request record that you want to escalate, and then double-click the record to open it.
3. Select the **Escalated** checkbox. The **Request Manager** field is displayed.

   Note

   The **Escalated** checkbox is unavailable in the Logging phase.

4. Fill in the **Request Manager** field.
5. Click **Save**.

The request is now escalated and the request manager will receive a notification.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.9. Create a request template

Security Role: Request Manager

Templates enable users to quickly complete request records by automatically populating various fields with necessary information. You can create as many templates as you need while authorizing select user roles for each template. The users authorized to use a particular template will then be able to apply that template when completing a request.

You can create a request template either from the system navigator menu (**Tailoring** > **Templates**) or directly from an existing request record.

To create a template from the system navigator menu, follow these steps:

1. Click **Tailoring** > **Templates**.
2. Click **New**.
3. Type a name in the **Template name** field.
4. Select **Request** in the **Table name** field, and then click **Next**.
5. Click **Fill** for each role that you are authorizing to use this template.
6. Modify the fields as required to meet the needs of your template by clicking the field to assign a value.

   Note

   In Data Policy, you must not set the Usage Type of the fields that are used in the template to either **Deprecated** or **System**. For more information, see Usage Type column in Data Policy.

7. Click **Add**. The roles that are authorized for this template can now use or modify this template.

To create a template from an existing request record, follow these steps:

1. Click **Request Fulfillment** > **Search Requests**, and then search for the record that you want to create a template from.
2. From the request record, click **More** or the More Actions icon, and then select **Create Template from Record**.

   To use this action, you must have the “Modify Template” right in the “Request” area.

3. Edit the name in the **Template name** field.
4. Modify the fields as required to meet the needs of your template by clicking the field to assign a value.

   Note

   In Data Policy, you must not set the Usage Type of the fields that are used in the template to either **Deprecated** or **System**. For more information, see Usage Type column in Data Policy.

5. Click **Add**. The roles that are authorized for this template can now use or modify this template.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record
Apply a request template

1.9.3.10. Apply a request template

Security Roles: Requestor

Templates enable you to quickly complete a request record by automatically populating fields with the applicable information. You can apply a template when completing a request if you are authorized to use that particular template.

Note

Make sure that the “Supports Templates” option is enabled in the request object.

To apply a template to a request, follow these steps:

1. Click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Open the request record to which you want to apply a template.
3. Click **Apply Template**. The Select Request Template wizard opens.

   Note

   The **Apply Template** action is available in all request phases except "Closure". To run this action, you must have the “Update” or "New" right in the “Request” area. However, if you just have the “New” right, you can only run this action when in the “Logging” phase.

4. Click the template that you want to apply to the selected request. You are returned to the selected request record and the fields are automatically filled in with the values set in the template.

   Tip

   Users authorized to use this template can also modify the template to meet their needs.

5. Modify other fields, as needed.
6. Click **Save & Exit**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record
Create a request template

1.9.3.11. Apply a request model to an existing request record

Security Roles: Request Coordinator, Request Manager

A request model is designed to handle a particular category of service request. Specific agreed tasks are defined in the request model to fulfill the service request of this category.

You can apply a request model to an existing request record that doesn't have a request model applied or you can apply a request model to replace the previous request model.

In the out-of-box setting, the **Apply Request Model** option is only available for the request records in the "Logging" phase. Users who have the “Update” right in the “Request” area and the “Expert” right in the “Request Tasks” area are authorized to do this action.

After you apply a new request model to a request record:

- The field values defined in the request model will overwrite the values in the request record.
- All the existing tasks in the "Waiting" phase are automatically cancelled as these tasks have not been worked on. However, you must manually cancel the existing tasks after the "Waiting" phase before you can apply a new request model.

Note

Apply request model is unsupported in the Order workflow.

To apply a request model to a request record, follow these steps:

1. Do one of the following in the System Navigator:
    1. Click **Request Fulfillment** > **Search Request** , enter your search criteria, and then click **Search**.
    2. Click **Request Fulfillment** > **Request Queue**.
2. Open the request record that you want to apply a new request model to.

3. Click **More** > **Apply Request Model**.

   > **Note**
   >
   > In the out-of-box setting, the **Apply Request Model** option is only available for the request records in the "Logging" phase.
   >
   > If the request record has active tasks, you will be prompted to cancel all the active tasks manually before applying a request model.

4. Select the request model that you want to use from the list.
5. Click **Yes** to apply the request model.

   The selected request model is applied to the request record.

1.9.3.12. Update multiple request records

Security Role: Request Coordinator

You can use the mass update feature to update one or more values in multiple request records. When you perform a mass update, the value that you enter for a particular field becomes the value for that field in all the records that you selected.

In addition, you can use complex update to enter expressions as instructions to perform actions on multiple request records.

To update multiple request records by using **Mass Update**, follow these steps:

1. Locate the records by using one of the following methods:
   - Click **Request Fulfillment** > **Request Queue** to display a list of records.
   - Click **Request Fulfillment** > **Search Requests**, and then search for the records.
2. Select the records that you want to update.
3. Click **Mass Update**.
4. Click the field that you want to update.
5. Type the value for the field or use the Find feature to display a list of potential values for the field, and then click **Next**.
6. Repeat steps 4 and 5 for additional fields as needed.
7. Click **Execute**.

To update multiple request records by using **Complex Update**, follow these steps:

1. Locate the records by using one of the following methods:
   - Click **Request Fulfillment** > **Request Queue** to display a list of records.
   - Click **Request Fulfillment** > **Search Requests**, and then search for the records.
2. Select the records that you want to update.
3. Click **Mass Update**.
4. Click **Complex Update**.
5. Follow the instructions in the complex update form and type the expressions.
6. Click **Execute**.

Related topics

Request Fulfillment workflows and user tasks
Searching records

Create a new request record
Approve multiple request records

1.9.3.13. Approve multiple request records

Security Role: **Request Approver**

You can approve multiple requests by using mass approval.

When you select the **Mass Approve** option, all the requests on the list that are waiting for your approval are immediately approved.

This action is available for users who have the **Expert** right in the **Request** area.

To approve multiple request records, follow these steps:

1. Locate the records by using one of the following methods:
   - Click **Request Fulfillment** > **Request Queue** to display a list of records.
   - Click **Request Fulfillment** > **Search Requests**, and then search for the records.
2. Select the records that you want to approve.
3. Click **Mass Approve**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record
Update multiple request records

1.9.3.14. Link a request to another record

Security Role: Request Coordinator

In the **Related Records** section of a request record, you can view the current related records, associate an existing incident, change, or request record with the current request record, or open a new incident, change, or request record that's automatically related to the current request record.

> **Note**
>
> This is unsupported in the records that use the Order workflow.

To link a request to another record, follow these steps:

1. Do one of the following in the System Navigator:
    1. Click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
    2. Click **Request Fulfillment** > **Request Queue**.
2. Open the request record that you want to link to another record, and then go to the **Related Records** section.
3. If you want to link the request to an existing record, follow these steps:
    1. Select one of the following link types:
        - **Caused By Changes**
        - **Caused Changes**
        - **Related Incidents**
        - **Related Requests**
    2. Click the **Link Existing Record** button.
    3. Specify the record number or click **Search** to search for the existing record.
    4. Click **OK**.

    The system populates the **Related Records** section with information about the related record.

> **Tip**
>
> You can also link a change or incident record to existing request records from the **Related Records** section in the change or incident record.

4. If you want to link the request to a new record that you are about to create, follow these steps:
    1. Select one of the following link types:
        - **Caused Changes**
        - **Caused By Changes**
        - **Related Incidents**
        - **Related Requests**
    2. Click the **Link New Record** button.
    3. Enter the required information to create the new record.

        > **Note**
        >
        > Some fields are automatically populated based on the request record. For example, if you are creating a new request, the system copies the values in the **Title**, **Description**, **Impact**, and **Urgency** fields from the exiting request to the new request.

    4. Click **Save and Exit**.

    Request Fulfillment populates the **Related Records** section with information about the related record.

> **Tip**
>
> Similarly, from the **Related Records** section of a change or incident record, you can create a new request and associate it automatically with the change or incident record.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.15. Set a reminder for a request

Security Roles: Requestor, Request Coordinator

You can set a reminder to notify you if an existing request record meets certain conditions by a specified time. For example, you may want to be reminded if a request is still open or is not updated by a given date.

To set a reminder for a request, follow these steps:

1. Click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Open the request record for which you want to set a reminder.
3. Click **More** or the **More Actions** icon, and then select **Set Reminder**.
4. Do one of the following to set the time when the reminder occurs:
    - Click **Remind At** to set a reminder for a particular day and time.

      Tip

      Click **Fill** to display a calendar where you can select the date and time for the reminder.

    - Click **Remind In** to set the reminder to occur at a particular interval. If you choose this option, specify the time interval (in 00:00:00 format), and select a shift from the **Based On** list.
5. In the **Remind if** field, select a condition for the reminder to occur:
    - **Always.** (the default value)
    - Request has not been updated.
    - **Request is still open.**
    - Request is still assigned to me.
6. In the **Pop-up Message** area of the form, select the type of notification you want to receive.
7. If you select **Pop-up**, type the reminder message in the **Message** field that you want to appear in the reminder.
8. If you select **Email** or **SM Mail**, follow these steps:

1. Type a title for the mail in the **Title** field.
2. Select the type of mail message s in the **Message Type** area:
   - Click **Send Request Record** to include a copy of the request record in the email.
   - Click **Fixed Text**, and then type the message you want to appear in the email.

9. Click **OK**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.16. Send a notification from a request

Security Roles: Requestor, Request Coordinator

You can send a notification to one or more recipients in a particular request phase.

This "Notify" action is available in all request phases. Users who have the “View” right in the “Request” area can run this action.

To send a notification from a request record, follow these steps:

1. Click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Open the request record from which you want to send a notification.
3. Click **More** or the **More Actions** icon, and then click **Notify**.
4. Select the date and time to send this notification.
5. Add one or more recipients to the **To** text box, or type an assignment group in the **Group** text box.

   Tip: If you have set a valid group email for members in the Assignment Groups definition, the system sends emails to this group email address.
6. Select one of the following notification methods:
   - **Mail**: Uses the internal Service Manager Mail application (not email).
   - **Email**: Uses the Service Manager email application.

     If you select **Email**, you can specify the names of the recipients in the **CC** and **BCC** fields as needed. In addition, you can click **Attachment** > **Add Files** to add attachments to the email message as needed.
7. Modify the subject and the message in the notification as needed.
8. Click **Send**.

Related topics

Request Fulfillment workflows and user tasks

Searching records

Create a new request record

1.9.3.17. View audit log from a request

Security Role: Request Coordinator

After enabling, an audit application can monitor field value changes in a request or request task record.

A global setting **Audit Records** to enable or disable audit application is available in **Request Fulfillment** > **Administration** > **Settings**. When the setting value is `true`, the audit application will run when a request or request task gets created, updated, or closed. Otherwise, Service Manager will skip the audit application.

The View Audit Log action enables users to view the audit log of the current request record. This View Audit Log action is available in all request phases. The user who has the `Expert` right in the `Request` area can to run this action.

The out-of-the-box audit specification for the request table includes the following audit fields.

- Phase
- Status
- Approval Status
- Request Model
- Requested For
- Requester
- Assignment Group
- Assignee
- Request Coordinator

To view the audit log of a request, follow these steps:

1. Click **Request Fulfillment** > **Search Requests**, enter your search criteria, and then click **Search**.
2. Open your desired request record.
3. Click **More**, and then select **View Audit Log**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.18. Access the alert log when viewing a request

Security Role: Request Coordinator

The alert log is a file that lists currently scheduled and active alerts.

To access the alert log when viewing a request, click **More** from the record that you are viewing, and then click **View Alert Log**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.19. Create a scheduled maintenance task from a request record

Security Role: System Administrator

You can create a scheduled maintenance task from the current request record.

To a scheduled maintenance task from a request record, follow these steps:

1. Create a new request, or locate an existing request record in Request Fulfillment.
2. Click **More** or the More Actions icon from the request record, and then click **Generate Schedule Maintenance**.

   By default, the **Create One RM Request** option is selected on the **Effect** tab and several field values are copied to the maintenance task as the request details.
3. Edit the scheduled maintenance task as needed.
4. Click **Save**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.20. View or edit the cost of a request

Security Roles: Request Coordinator, Request Analyst

From the **Cost** tab in a request record, you can find the total cost and the currency information for the request. The total cost includes the labor cost for handling the request and the sum of costs in all request tasks. You can also add the labor cost for handling the request.

To view or edit the cost of a request, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate the request record, and then double-click the record to open it.
3. Click the **Cost** tab. The total cost of the request and the currency are displayed in corresponding fields.

   Note

   The total cost also includes the costs in individual request tasks. When an operator works on a request task, the operator can also view or edit the cost in that request task.

   In the request record, the currency of the cost reflects the display currency of the operator defined in the **Requested For** field.

4. To add the labor cost for the request, fill in the date, the name of the technician, and the hours that the technician has spent on this request.
5. Click **Save** to save your updates.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.21. Close a request directly in the Logging phase

Applies to User Roles: Service Desk Agent, Request Coordinator

When the streamlined interaction solution is enabled, you can close a request directly if you are able to resolve the user request on the first intake in the Logging phase of the request record. If the request is triggered from an Interaction, the associated interaction record is closed automatically.

To close a request in the Logging phase, follow these steps:

1. Open a request record that's in the Logging phase.
2. Go to the **Solution** tab, and then type a solution for the request.
3. Click the **Close Directly** button on the toolbar. Service Manager displays the **Close Request** page.
4. Enter or change the subcategory for the request.
5. Select the appropriate closure code.
6. Type the closure comments.
7. Click **Finish**. The request record is now successfully closed.

> **Note**
>
> If the request is triggered from an Interaction, the associated interaction record is also closed automatically.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Reopen a request record
Close a request record

1.9.3.22. Close a request record

Security Role: Request Coordinator

After request fulfillment and review, you can close the request record.

> **Note**
>
> If a service desk agent moves an interaction to the Withdrawal phase, the corresponding linked fulfillment request (if any) will be automatically moved to the Closure phase with the “Withdrawal requested by customer” closure code. In addition, all the request tasks in the request record will be cancelled automatically.

To close a request, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate and open the request record that you want to close.
3. Click the **Close** button.
4. Enter or change the subcategory for the request.
5. Select the appropriate closure code.
6. Enter the closure comments.
7. Click **Finish**. The request record is now successfully closed.

> **Note**
>
> You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Reopen a request record

1.9.3.23. Reopen a request record

Security Role: Request Coordinator

From the Closure phase in the Request workflow, if you have the "Reopen" right in the "Request" area, you can click the **Reopen** button to reopen the closed request. After you click **Reopen**, the request is moved back to the Review phase.

To reopen a request, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate and open the request record in the Closure phase that you want to reopen.
3. Click **Reopen**. The request is now moved back to the Review phase. When the request is re-opened, the original tasks will not be re-activated.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.3.24. Access Request Fulfillment reports

Security Roles: Request Manager; Request Coordinator

The Reporting tool in Service Manager provides a number of out-of-box reports on the request data in your system. You can view these reports through the dashboards named **Request Overview (PD) (Global)** and **Request KPI (Global)**. You can also create your own dashboards to display other reports of your interest.

To access Request Fulfillment reports, follow these steps:

1. Click **Request Fulfillment** > **Request Overview**.

   By default, the **Request Overview (PD) (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. If needed, click the **New Dashboard** button to add your own dashboards. For details, see My dashboards.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default dashboard.

1.9.4. Request Task workflows and user tasks

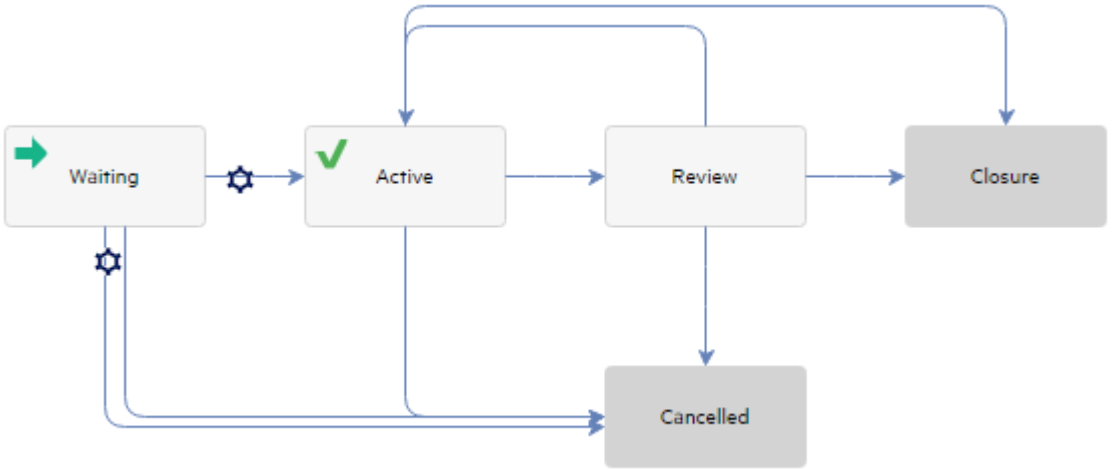
The request task workflow is a sequence of connected steps in the life cycle of a request task. In the workflow, a request task goes through several phases to complete the life cycle.

The following are the out-of-box workflows for request tasks:

Generic Request Task

The Generic Request Task workflow manages the process for generic request tasks without CMDB update, such as manual labor for internal vendors/suppliers.

The Generic Request Task workflow consists of the following phases and tasks.



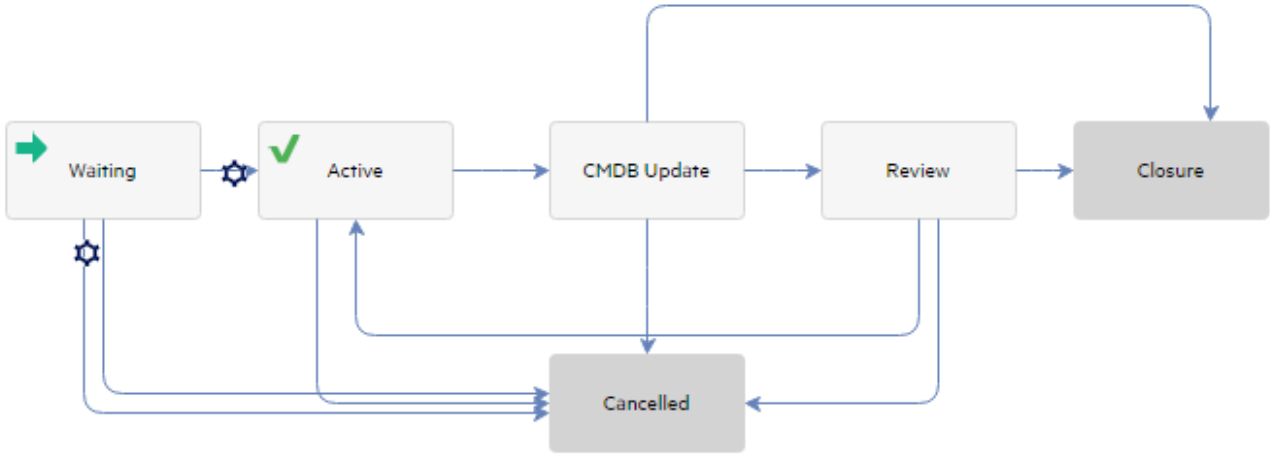
Workflow phase	Tasks
Waiting	• Create a request task • Create a request task template • Send a notification from a request task
Active	• Apply a request task template • Create a request task template • Create a change from a request task • Update multiple request tasks • Cancel multiple request tasks • Send a notification from a request task • Set a reminder for a request task • Access the alert log when viewing a request task • View audit log from a request task
Review	• Update multiple request tasks • Cancel multiple request tasks • Send a notification from a request task • Set a reminder for a request task • Access the alert log when viewing a request task • View audit log from a request task • Create a request task template

Workflow phase	Tasks
Closure	• Close a request task • Send a notification from a request task • Access the alert log when viewing a request task • View audit log from a request task • Create a request task template
Cancelled	• Send a notification from a request task • Access the alert log when viewing a request task • View audit log from a request task • Create a request task template

Request Task with CMDB Update

The Request Task with CMDB Update workflow manages the process for request tasks that involve CMDB updates such as installation, uninstallation, purchase, and reservation services.

The Request Task with CMDB Update workflow consists of the following phases and tasks.



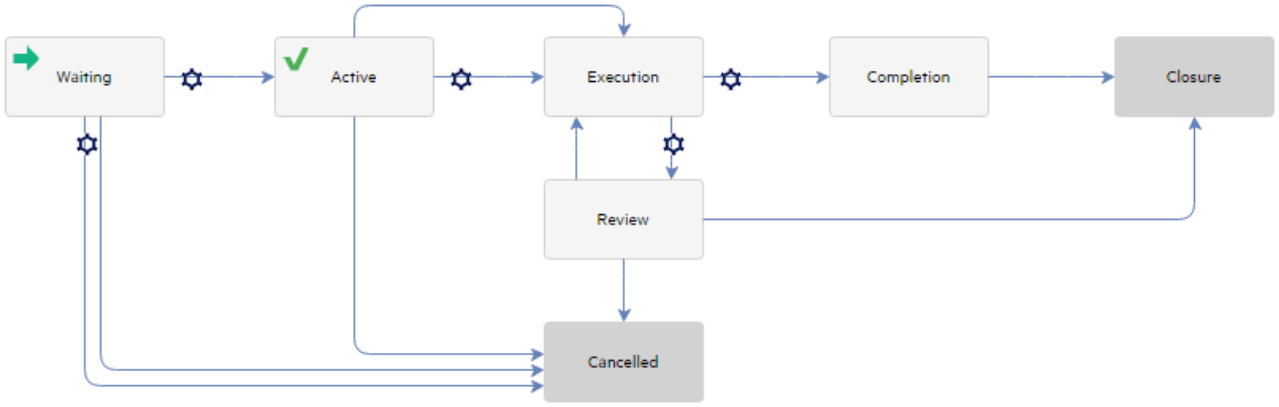
Workflow phase	Tasks
Waiting	• Create a request task • Create a request task template • Send a notification from a request task
Active	• Apply a request task template • Create a request task template • Create a change from a request task • Update multiple request tasks • Cancel multiple request tasks • Send a notification from a request task • Set a reminder for a request task • View audit log from a request task • Access the alert log when viewing a request task
CMDB Update	• Send a notification from a request task • Set a reminder for a request task • Access the alert log when viewing a request task • View audit log from a request task • Create a request task template

Workflow phase	Tasks
Review	• Send a notification from a request task • Set a reminder for a request task • View audit log from a request task • Access the alert log when viewing a request task • Update multiple request tasks • Create a request task template
Closure	• Close a request task • Send a notification from a request task • View audit log from a request task • Access the alert log when viewing a request task • Create a request task template
Cancelled	• Send a notification from a request task • View audit log from a request task • Access the alert log when viewing a request task • Create a request task template

Automated Task

The Automated Task workflow manages the Operations Orchestration (OO) flow execution process. For more information on the Automated Task integrated with the OO flow, see Automated fulfillment of request tasks.

The Automated Task workflow consists of the following phases and tasks.


Waiting
Active
Execution
Completion
Closure
Review
Cancelled

Workflow phase	Tasks
Waiting	• Create a request task • Create a request task template • Send a notification from a request task
Active	• Apply a request task template • Create a request task template • Update multiple request tasks • Cancel multiple request tasks • Send a notification from a request task • Set a reminder for a request task • View audit log from a request task • Access the alert log when viewing a request task
Execution	• Send a notification from a request task • Set a reminder for a request task • View audit log from a request task • Access the alert log when viewing a request task • Create a request task template
Review	• Send a notification from a request task • Set a reminder for a request task • Update multiple request tasks • View audit log from a request task • Access the alert log when viewing a request task • Create a request task template

Workflow phase	Tasks
Completion	• Send a notification from a request task • Set a reminder for a request task • View audit log from a request task • Access the alert log when viewing a request task • Create a request task template
Closure	• Close a request task • Send a notification from a request task • Access the alert log when viewing a request task • Create a request task template
Cancelled	• Send a notification from a request task • View audit log from a request task • Access the alert log when viewing a request task • Create a request task template

For examples that demonstrate the minimum needed steps to complete the out-of-box request task workflows, see Examples: request task workflows.

Related topics

Process Designer
Examples: request workflows
Create a workflow
Processes and Best Practices

1.9.4.1. Examples: Request Task Workflow

The following examples demonstrates the minimum needed steps to complete the following out-of-box request task workflows:

Generic Request Task workflow

Phase	User Actions	Options
Waiting	To move the task to the Active phase, complete the tasks that this task is dependent on and meet the conditions that are set for this task.	If the task condition is evaluated as “false”, the system moves this task to the Cancelled phase.

Phase	User Actions	Options
Active	1. Enter information for the required fields. 2. Complete the actual task as described in this request task record.	From the Active phase, you can move the task to the following phases: • Cancelled: Click **More** > **Cancel Task**. • Review: Click the **Review** button for high priority tasks (1-Critical or 2-High). **Note** High priority tasks (1-Critical or 2-High) must be reviewed before closure. • Closure: Click the **Close Task** button for low priority tasks (3-Average or 4-Low).

Phase	User Actions	Options
Review	1. Click the **Review** button. 2. Review the data entered for the request task.	From the Review phase, you can move the task to the following phases: • Fulfillment: Click the **Back to Fulfill** button. • Cancelled: Click **More** > **Cancel Task**. • Closure: Click the **Close Task** button.
Cancelled	To move the task to the Cancelled phase, click **More** > **Cancel Task**.	
Closure	1. Click the **Close Task** button. 2. Select the appropriate **Closure Code**. 3. Enter the **Closure Comments**. 4. Select the date on which the task is finished. 5. Click **Finish**.	

Request Task with CMDB Update workflow

Phase	User Actions	Options
Waiting	To move the task to the Active phase, complete the tasks that this task is dependent on and meet the conditions that are set for this task.	If the task condition is evaluated as “false”, the system moves this task to the Cancelled phase.
Active	1. Change the status to In Progress. 2. Update the information for the required fields if needed. For the "Purchase" task, follow steps: 1. Enter the purchase information in the **Purchase** tab, and then click **Save**. 2. Click the **Receive** button, and then review and update the CI details in the receiving form.	From the Active phase, you can move the task to the Cancelled phases by clicking **More** > **Cancel Task**.

Phase	User Actions	Options
CMDB Update	1. Click the **Update CMDB** button. 2. Review and update the CI information. 3. On the **CI Update History** tab, do one of the following: ◦ For purchase tasks, click **Submit to CMDB**. ◦ For other tasks, click **Update CMDB** 4. Click **Yes** to confirm the CMDB update. **Note** For more information, see Updating CMDB	From the CMDB Update phase, you can move the task to the following phases: • Cancelled: Click **More** > **Cancel Task**. • Review: Click the **Review** button for high priority tasks (1-Critical or 2-High). **Note** High priority tasks (1-Critical or 2-High) must be reviewed before closure. • Closure: Click the **Close Task** button for low priority tasks (3-Average or 4-Low).

Phase	User Actions	Options
Review	1. Click the **Review** button. 2. Review the data entered for the request task.	From the Review phase, you can move the task to the following phases: • Fulfillment: Click the **Back to Fulfill** button. • Cancelled: Click **More** > **Cancel Task**. • Closure: Click the **Close Task** button.
Cancelled	To move the task to the Cancelled phase, click **More** > **Cancel Task**.	
Closure	1. Click the **Close Task** button. 2. Select the appropriate **Closure Code**. 3. Enter the **Closure Comments**. 4. Select the date on which the task is finished. 5. Click **Finish**.	

Tip

To check the activity log for a request task record, go to the **Activities** section of the record. The **Activities** section enables you to enter new updates or view journal updates and logged activities such as "Open" and "Update" for a record.

1.9.4.2. Create a request task

Security Roles: Request Coordinator, Request Analyst

After a request is created, you can create tasks to fulfill the request.

To create a request task, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Queue** or **Search Requests**.
2. Locate and open the request record for which you want to create a task.
3. From the **Tasks** section, click the **Edit** button to open Task Planner.
4. Click **Add Task**.
5. Type a title for the new task.
6. Select a category for the new task.
7. Select a phase to define when the new task will be opened for fulfillment.
8. Select a phase to define by which the new task must be closed.
9. Complete the form with other information if needed, and then click **Save**.

Note

You can add parts and labor costs tracking to an incident, problem, change, or request or to any associated task of a record. To do this, navigate to the **Cost** tab, specify the currency, and then specify the date, part number, and quantity for any parts used. Alternatively, specify the date, technician name (used to derive the rate from the operator record), and hours worked for any labor. Service Manager will automatically calculate and roll up costs from any sub tasks into the **Total cost** field on the **Costs** tab.

For detailed instructions on using Task Planner, see Task Planner.

Related topics

Task Planner
Request Task workflows and user tasks
Searching records
Plan a task with Task Planner

1.9.4.3. Create a request task template

Security Role: Request Manager

Templates enable users to quickly complete request task records by automatically populating various fields with necessary information. You can create as many templates as you need while authorizing select user roles for each template. The users authorized to use a particular template will then be able to apply that template when completing a request task.

You can create a request task template either from the system navigator menu (**Tailoring** > **Templates**) or directly from an existing request task record.

To create a request task template from the system navigator menu, follow these steps:

1. Click **Tailoring** > **Templates**.
2. Click **New**.
3. Type a name in the **Template name** field.
4. Select **Request Task** in the **Table name** field, and then click **Next**.
5. Click **Fill** for each role that you are authorizing to use this template.
6. Modify the fields as required to meet the needs of your template by clicking the field to assign a value.
7. Click **Add**. The roles that are authorized for this template can now use or modify this template.

To create a template from an existing request task record, follow these steps:

1. Click **Request Fulfillment** > **Search Request Tasks**, and then search for the record that you want to create a template from.
2. From the request task record, click **More** or the More Actions icon, and then select **Create Template from Record**.

   Note

   To use this action, you must have the “Modify Template” right in the “Request Tasks” area.

3. Edit the name in the **Template name** field.

4. Modify the fields as required to meet the needs of your template by clicking the field to assign a value.
5. Click **Add**. The roles that are authorized for this template can now use or modify this template.

1.9.4.4. Apply a request task template

Security Role: Request Analyst

Templates enable you to quickly complete a request task record by automatically populating fields with the applicable information. You can apply a template when completing a request task if you are authorized to use that particular template.

> **Note**
>
> Make sure that the “Supports Templates” option is enabled in the request task object.

To apply a template to a request task, follow these steps:

1. Click **Request Fulfillment** > **Search Request Tasks**, enter your search criteria, and then click **Search**.
2. Select the request task record to which you want to apply a template.
3. From the request task record, click **Apply Template**. The Select Request Task Template wizard opens.

   > **Note**
   >
   > The **Apply Template** action is available in the "Active" and "Review" phases. To run this action, you must have the “Update” right in the “Request Tasks” area.

4. Click the template that you want to apply to the selected request task. You are returned to the selected request task record and the fields are automatically filled in with the values set in the template.

   > **Tip**
   >
   > Users authorized to use this template can also modify the template to meet their needs.

5. Modify other fields, as needed.
6. Click **Save & Exit**.

1.9.4.5. Create a change from a request task

Security Roles: Request Coordinator, Request Analyst

You can create a change record directly from a request task, so that the request task can be fulfilled by this change.

This action is only available in the “Active” phase. Users who have the “New” right in the “Change” area are authorized for this action.

To create a change from an existing request task, follow these steps:

1. Click **Request Fulfillment** > **Search Request Tasks**, enter your search criteria, and then click **Search**.
2. Open the request task record that you want to create a change from.
3. From the request task record, click **More** or the More Actions icon, and then select **Create Change**.
4. Select a change model or a category for the change.
5. Edit the values or type necessary information in the change record.

   Note

   The values in following fields are copied from the request task: title, description, impact, company, and folder.

6. Click **Save**

   A new change record is created and linked to the request task.

1.9.4.6. Update multiple request tasks

Security Roles: Request Analyst

You can use the mass update feature to update one or more values in multiple request tasks. When you perform a mass update, the value that you enter for a particular field becomes the value for that field in all the records that you selected.

In addition, you can use complex update to enter expressions as instructions to perform actions on multiple request tasks.

To update multiple request tasks by using **Mass Update**, follow these steps:

1. Locate the records by using one of the following methods:
    - Click **Request Fulfillment** > **Request Task Queue** to display a list of records.
    - Click **Request Fulfillment** > **Search Request Tasks**, and then search for the records.
2. Select the records that you want to update.
3. Click **Mass Update**.
4. Click the field that you want to update.
5. Type the value for the field or use the Find feature to display a list of potential values for the field, and then click **Next**.
6. Repeat steps 4 and 5 for additional fields as needed.
7. Click **Execute**.

To update multiple request tasks by using **Complex Update**, follow these steps:

1. Locate the records by using one of the following methods:
    - Click **Request Fulfillment** > **Request Task Queue** to display a list of records.
    - Click **Request Fulfillment** > **Search Request Tasks**, and then search for the records.
2. Select the records that you want to update.
3. Click **Mass Update**.
4. Click **Complex Update**.
5. Follow the instructions in the complex update form and type the expressions.
6. Click **Execute**.

1.9.4.7. Cancel multiple request tasks

Security Roles: Request Analyst

You can use the mass cancel feature to cancel one or more values in multiple request tasks.

Users with the “Expert” right in the “Request Tasks” area are authorized to run this action.

To cancel multiple request tasks by using **Mass Cancel**, follow these steps:

1. Locate the records by using one of the following methods:
    - Click **Request Fulfillment** > **Request Task Queue** to display a list of records.
    - Click **Request Fulfillment** > **Search Request Tasks**, and then search for the records.
2. Select the records that you want to cancel.
3. Click **Mass Cancel**.
4. Click **Yes**.
5. Enter your closure comments, and then click **Finish**.

1.9.4.8. Set a reminder for a request task

Security Roles: Request Coordinator, Request Analyst

You can set a reminder to notify you if an existing request task record meets certain conditions by a specified time. For example, you may want to be reminded if a request task is still open or is not updated by a given date.

To set a reminder for a request task, follow these steps:

1. Click **Request Fulfillment** > **Search Request Tasks**, enter your search criteria, and then click **Search**.
2. Open the request task record to which you want to set a reminder.
3. Click **More**, and then select **Set Reminder**.
4. Do one of the following to set the time when the reminder occurs:
    - Click **Remind At** to set a reminder for a particular day and time.

      Tip

      Click **Fill** to display a calendar where you can select the date and time for the reminder.

    - Click **Remind In** to set the reminder to occur at a particular interval. If you choose this option, specify the time interval (in 00:00:00 format), and select a shift from the **Based On** list.
5. In the **Remind if** field, select a condition for the reminder to occur:
    - **Always.** (the default value)
    - Request task has not been updated.
    - **Request task is still open.**
    - Request task is still assigned to me.
6. In the **Pop-up Message** area of the form, select the type of notification you want to receive.
7. If you select **Pop-up**, type the reminder message in the **Message** field that you want to appear in the reminder.

8. If you select **Email** or **SM Mail**, follow these steps:
    1. Type a title for the mail in the **Title** field.
    2. Select the type of mail message s in the **Message Type** area:
        - Click **Send Request Task Record** to include a copy of the request task record in the email.
        - Click **Fixed Text**, and then type the message you want to appear in the email.
9. Click **OK**.

1.9.4.9. Send a notification from a request task

Security Roles: Request Coordinator, Request Analyst

You can send a notification to one or more recipients in a particular request task phase.

This "Notify" action is available in all request task phases. Users who have the “View” right in the “Request Tasks” area can run this action.

To send a notification from a request task record, follow these steps:

1. Click **Request Fulfillment** > **Search Request Tasks**, enter your search criteria, and then click **Search**.
2. Open the request task record from which you want to send a notification.
3. Click **More**, and then click **Notify**.
4. Select the date and time to send this notification.
5. Add one or more recipients to the **To** text box, or type an assignment group in the **Group** text box.

   Tip: If you have set a valid group email for members in the Assignment Groups definition, the system sends emails to this group email address.
6. Select one of the following notification methods:
   - **Mail**: Uses the internal Service Manager Mail application (not email).
   - **Email**: Uses the Service Manager email application.

     If you select **Email**, you can specify the names of the recipients in the **CC** and **BCC** fields as needed. In addition, you can click **Attachment** > **Add Files** to add attachments to the email message as needed.
7. Modify the subject and the message in the notification as needed.
8. Click **Send**.

1.9.4.10. View audit log from a request task

Security Role: Request Analyst

After enabling, an audit application can monitor field value changes in a request or request task record.

A global setting **Audit Records** to enable or disable audit application is available in **Request Fulfillment** > **Administration** > **Settings**. When the setting value is `true`, the audit application will run when a request or request task gets created, updated, or closed. Otherwise, Service Manager will skip the audit application.

The View Audit Log action enables users to view the audit log of the current request task record. This View Audit Log action is available in all request task phases. The user who has the `Expert` right in the `Request Tasks` area can run this action.

The out-of-the-box audit specification for the request task table includes the following audit fields.

- Phase
- Status
- Assignment Group
- Assignee
- Part No
- Planned Lead Time
- Actual Lead Time

To view the audit log of a request task, follow these steps:

1. Click **Request Fulfillment** > **Search Request Tasks**, enter your search criteria, and then click **Search**.
2. Open your desired request task record.
3. Click **More**, and then select **View Audit Log**.

Related topics

Request Fulfillment workflows and user tasks
Searching records
Create a new request record

1.9.4.11. Access the alert log when viewing a request task

Security Roles: Request Analyst

The alert log is a file that lists currently scheduled and active alerts.

To access the alert log when viewing a request task, click **More** from the record that you are viewing, and then click **View Alert Log**.

1.9.4.12. Close a request task

Security Roles: Request Analyst

After you fulfill and review a request task, you can close the request task record.

To close a request task, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Request Task Queue** or **Search Request Tasks**.
2. Locate and open the request task that you want to close.
3. Click the **Close** button.
4. In the **Closure Code** drop-down list, select an appropriate closure code.
5. In the **Closure Comments** field, enter the closure comments.
6. Select the date on which the task is finished.
7. Click **Finish**. The request task record is now successfully closed.

Related topics

Request Task workflows and user tasks
Examples: request task workflows
Searching records
Create a request task
Cancel multiple request tasks

1.9.4.13. Task scheduling

- Expected Finish Date

  Expected Finish Date in request records reflects the expected delivery date from the end user perspective.

  If the delivery target is defined when ordering service catalog items or bundles, the value of Expected Finish Date in the request record will be calculated as the request creation time (submit.date) plus the delivery target.

  If the delivery target is not specified when ordering the service catalog items or bundles, or the request is an off-catalog request, the IT operator needs to manually fill in the Expected Finish Date on behalf of the requester.

- Delivery Date

  The Delivery Date reflects the planned delivery date from the fulfillment perspective.

  The Delivery Date in requests is set based on the request creation time plus the Global Lead Time defined in request model. Once the request record is created, the Delivery Date can't be changed.

- Global Lead Time

  In request records, the Global Lead Time is copied from the request model and is read-only in request records.

  In request models, you can specify the Global Lead Time. The reasonable scope for the value of Global Lead Time is greater than the longest task Planned Lead Time (considering task dependency) and less than sum of the planned lead time of all the tasks. If your specified value is not in this range, an error message is displayed and you can't save the request model.

  If the value of Global Lead Time in request model is blank, once the request model is saved, the Global Lead Time will be auto-calculated from task Planned Lead Time based on task plan.

- Planned Lead Time

  In request tasks, the planned lead time is populated according to the setting in the request model and it is read-only in request tasks.

In request models, the value of the planned lead time can be set manually when planning tasks.

- Planned Start Date and Planned End Date

  If the Global Lead Time is not zero, the Planned Start Date and Planned End Date of a request task are calculated by starting from the request “Delivery Date” and counting upstream from the last task to the first one according to the task plan. Task dependency and planned lead time will be considered. If the Global Lead Time is zero, they will be calculated based on the Expected Finish Date.

  If you adjust the task plan in a request, such as adding or deleting tasks, the Delivery Date and Expected Finish Date of the request aren't changed but the Planned Start Date and Planned End Date for all the tasks are re-calculated based on the Delivery Date (or Expected Finish Date) and the updated task plan.

- Contractual Lead Time

  Contractual Lead Time is defined in request task records. By default, Contractual Lead Time equals to Planned Lead Time. If the Contractual Lead Time is adjusted by authorized operator, the Planned Start Date and Planned End Date will be re-calculated accordingly.

  If the changed Contractual Lead Time is longer than Planned Lead Time, the request fulfillment SLA might be affected as Global Lead Time might be enlarged due to the change. In this case, a warning message is displayed to operator.

- Actual Start Date

  The Actual Start Date in the request task record is auto-populated when the request task is activated.

- Actual End Date

  The Actual End Date in the request task record is manually provided by the task assignee when the request task is completed.

- Calendar Name

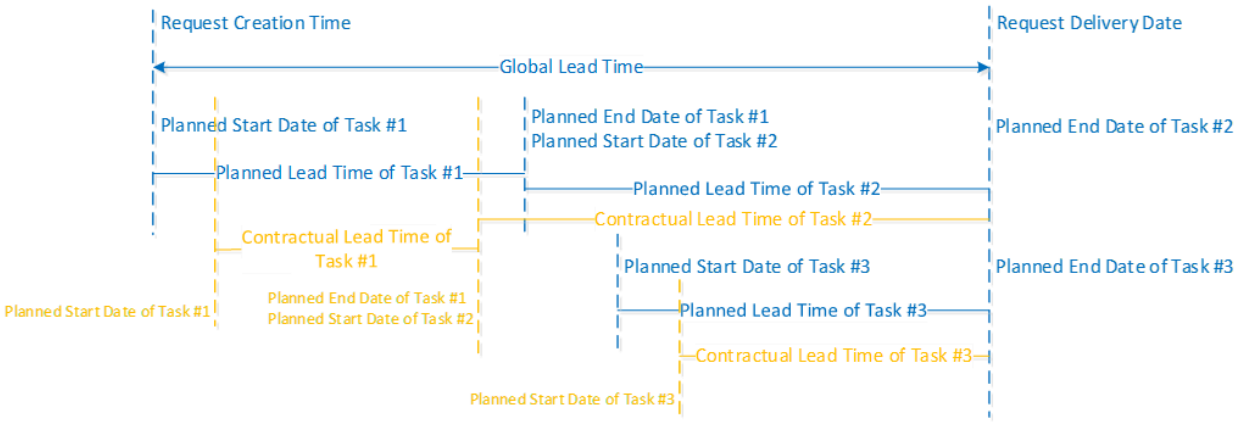
  Calendar Name enables you to select the work schedule that's used for request task scheduling. For example, if you select "Mon to Fri from 7am to 3pm", the period between 3pm and 7am, Saturday, and Sunday are excluded from calculations such as the Planned End Date.

  The Calendar Name field is configured in request models.

Note

Administrators can create custom calendars in the caldutyhours table.

The following figure illustrates the task scheduling process in the request fulfillment module.



Related topics

Request Task workflows and user tasks
Create a request task
Create a request model

1.9.4.14. Updating CMDB

In the out-of-box system, a "CMDB Update" phase is defined in the "Request Task with CMDB Update" request task workflow. This workflow is used by several out-of-box request task categories. When a request task in these task categories moves to the CMDB Update phase, users can update the CI information in a dedicated CMDB update form corresponding to the specific task category. After the CMDB update, the CI status in CMDB will be updated accordingly.

The following table describe the task categories that require CMDB update, the associated forms with the CI queries, and the corresponding CI status after CMDB update.

Task Category	Request Category	Associated form	CIs for CMDB update	CI Status after CMDB update
Purchase	Generic Request	rm.requestTask.CI.Creation.form	Lists the received CIs to be created	Reserved
Reservation	Generic Request	rm.requestTask.CI.Reservation.form	Pick up existing CIs from CMDB: “istatus=’In Stock’ and owner=department name of requested.for in parent request”	Reserved
Installation	Generic Request	rm.requestTask.CI.Installation.form	Either use the CI list reserved by prior task, or pick up the existing CIs from CMDB: “istatus=’Reserved’ and owner=the department name of requested.for in parent request”	In Use

Task Category	Request Category	Associated form	CIs for CMDB update	CI Status after CMDB update
Uninstallation	Generic Request	rm.requestTask.CI.Uninstallation.form	Pick up existing CIs from CMDB: “istatus=’In Use’ and owner=the department name of requested.for in parent request”	Retired
Purchase	Order	rm.requestTask.CI.Creation.form	Lists the received CIs to be created	In Stock

Note

If the number of CIs to be updated is no more than 50, the CMDB update will be executed in foreground. To avoid potential performance issue, if the number of CIs is greater than 50, a background schedule record will be generated to update CMDB.

Related topics

Request Task workflows and user tasks
Create a request task
Create a request model

1.9.5. Request Fulfillment administration

Request Fulfillment administration consists of the following:

- Environment: This environment record contains options that defines the functionality of the Request Fulfillment application for all Request Fulfillment users. For example, you can set the formats of request number and request task number.
- Settings: You can configure the general settings for the Request Fulfillment module. For example, you can configure to create a new request either by selecting a category or by selecting a request model.

Configure the Request Fulfillment environment

Security Roles: System Administrator

To configure Request Fulfillment environment settings, follow these steps:

1. Click **Request Fulfillment** > **Administration** > **Environment** in the System Navigator.
2. Configure the following environment settings as needed.

Setting	Description
Delay Assigning Request Number?	Defines whether or not the system delays assigning the request number until you save the record.
Skip Inefficient Query Warning?	Normally, it is desirable to warn an operator when they have entered an inefficient (partially or non-keyed) query. If you want to bypass the warning, select this checkbox.
Use Operator Full Name?	Defines whether or not the operator full name is used when time stamping reports (on open, update, and so on) for records in the application environment.
Request Number Controls	Defines the format of request number. See Number controls.
Request Task Number Controls	Defines the format of the request task number. See Number controls.
Unique Number Length	Defines the length of the unique number. **Note** This setting only takes effect on the **Parent No/Unique No (PPPNNN)** option in Request Task Number Controls.

Setting	Description
Separator	Defines the separator that's used in Request Number Controls and Request Task Number Controls.

3. Click **Save**.

Number controls

Number controls determine how your requests or request tasks will be numbered.

Option	Description
Yr/Unique No (YYNNN)	The current year (system-calculated), followed by a unique sequential number.
Yr-Mo/Unique No (YYMMNNN)	The current year and current month (system-calculated), followed by a unique sequential number.
Julian/Unique No (YYDDDNNN)	The current Julian date (system-calculated), followed by a unique sequential number.
Unique Number Only	A unique number without prefixes or suffixes.
Parent No/Unique No (PPPNNN)	The parent request number, followed by a unique sequential number determined. This option is only available for request task number control.

>

Security Roles: System Administrator

To configure Request Fulfillment settings, follow these steps:

1. Click **Request Fulfillment** > **Administration** > **Settings** in the System Navigator.
2. Configure the following settings as needed.

Setting	Description
Audit Records	The changes of field values in requests or request tasks can be monitored by the audit application. You can use this setting to enable or disable the audit application for requests and request tasks.
Select on Open	You can choose to open a new off-catalog request either based on category or based on request model. Default: Request Model

Setting	Description
Group Request Model By	To configure the request model hierarchy, type up to four column names to group the request models in the **Group Request Model By** text box. For example, to group request models by category and template, type category, request.template. **Note** - The request model hierarchy configuration takes effect only when **Request Model** is under **Select on Open**. - Use the column names specified in the `requestModel` table. - You can configure at most four levels in the request model hierarchy by specifying the column names in a comma separate list.

3. Click **Save**.

Notifications in Request Fulfillment

Request Fulfillment includes the following out-of-box notifications, which are sent in response to certain events.

Request Notifications

Request Task Notifications

1.9.6. Request Fulfillment configuration

Request Fulfillment configuration enables you to configure alerts, approvals, assignment groups, product catalog, request categories, request task categories, request models, workflows, and so on.

Create a request model

Create a request category

Add a subcategory to a request category

Create a request task category

Create a product catalog category

Create a product catalog item

Stock management

Add a Request Fulfillment assignment group

Add a Request Fulfillment approval definition

Clone a request model

1.9.6.1. Create a request model

Security Roles: Request Process Owner

A request model defines the prerequisites, required authorizations, and standard activities to fulfill a service request. When you create a request based on a request model, all information in the request model is added automatically to the request. In addition, you can plan fulfillment tasks with sequencing and dependencies by using the task planner in request model.

Service Manager provides the following out-of-box request models:

- New Employee On-boarding
- Employee Off-boarding
- Password Reset
- Software Package
- Order Hardware
- Application Access
- Sharepoint Access
- Request for Information
- Order

Different request models contain different pre-defined tasks for operators to fulfill the corresponding requests for end users.

To create a new request model, follow these steps:

Template 1

Field	Value
Description	Install Windows 11 on new employee's laptop
Brief Description	Install Windows 11
Assignment Group	(empty)

Template 2

Field	Value
Description	(empty)
Brief Description	Install Windows 10
Assignment Group	Hardware

Merge Result

Field	Value
Description	Install Windows 11 on new employee's laptop
Brief Description	Install Windows 10
Assignment Group	Hardware

1. From the System Navigator, click **Request Fulfillment** > **Configuration** > **Request Models**.
2. Click **New**.

Tip

Alternatively, you can create a new request model based on an existing request model. To do this, open an existing request model, and then click **More** > **Copy Request Model**. The request and task plan information in the original request model is copied to the new request model for your further modification. In addition, you can change the category in the new request model as needed.

3. Enter the request model ID.
4. Enter the request model name.
5. (Optional) Enter the request model description.
6. Click the **Fill Field Category** icon in the **Category** text box to select a request category.

> **Note**
>
> After the request model is created, you can't change the category specified in the request model. If you want to change the category, you can create a new request model.

7. Click the **Fill Field SubCategory** icon in the **SubCategory** text box to select a request subcategory.
8. (Optional) Clear the **Active** checkbox if you want to disable this new request model so that it's not available for selection when users create new request records.
9. (Optional) Select request templates from the **Request Template** drop-down list.

   If needed, you can specify more than one template. When users open a new request with this model, Service Manager automatically applies all specified templates in the order listed by merging their field values. The following table lists two example templates and their merged result.
10. (Optional) Select allowed roles from the **Allowed Roles** drop-down list.
11. (Optional) In the **Approvals** section, select the required approvals and set the request phases that the approvals apply to.

> **Note**
>
> Besides request models, you can also define Approvals on the workflow phases. If you have Approvals defined in both places, they both take effect.

12. (Optional) Add other information if needed, such as cost, department, and currency.
13. (Optional) Use task planner to add tasks to the request model. For more information, see Task Planner.
14. (Optional) Specify the global lead time. For more information, see Request task scheduling.
15. Click **Save**.

Related topics

Request Fulfillment configuration
Create a request category

1.9.6.2. Create a request category

Security Role: Request Manager

If you are a Service Manager Administrator, you may want to create a request category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides the following out-of-box category records that you can use or modify:

- Generic Request
- Order

To create new category record, follow these steps:

1. Click **Request Fulfillment** > **Configuration** > **Request Categories**.
2. Click **New**.
3. Enter the name of the request category.
4. Clear the **Active** checkbox if you don't want the new category to appear in the category list.

   Note

   By default, the **Active** checkbox is selected. If you clear the **Active** checkbox, this category can't be used to create new requests.

5. Enter the category description.
6. Select a workflow for the category.
7. Click **Save**.

   The new request category is created and the subcategories tab is displayed.

8. If you want to create a new subcategory for this request category, go to the **Subcategories** tab, and then click the **Link New Subcategories** button to create a new subcategory.

Note

To avoid breaking existing request records, the system doesn't allow you to delete the existing request categories or edit their names.

Related topics

Request Fulfillment configuration
Add a subcategory to a request category

1.9.6.3. Add a subcategory to a request category

Security Role: Request Manager

Request subcategory specifies the second level of classifying a Service Request. You can add a new subcategory to a request category directly. Additionally, you can view a list of the subcategories that are associated with the current category, together with the record details for each subcategory.

In the out-of-box system, the following subcategories are available for the “Generic Request” category:

- Employee Off-boarding
- Employee On-boarding
- Hardware
- Others
- Request for Administration
- Request for Access
- Request for Information
- Software

The following subcategory is available for the “Order” category :

- Order

To add a new subcategory for a request category, follow these steps:

1. Click **Request Fulfillment** > **Configuration** in the system navigator.
2. Click **Request Categories** > **Search**.
3. Select the request category to which you want to add a subcategory.
4. Under the **Subcategories** tab, click the **Link New Subcategories** button.
5. Type a name for the new subcategory.
6. Type a description for the subcategory.
7. Clear the **Active** checkbox if you don't want the new subcategory to appear in the subcategory list. If you clear the **Active** checkbox, this subcategory can't be used in request records.
8. Click **Save** to add the new subcategory.

Related topics

Request Fulfillment configuration
Create a request category

1.9.6.4. Create a request task category

Security Role: Request Manager

If you are a Service Manager administrator, you may want to create a request task category. To do this, you can modify an existing category record, or you can create a new category record. Service Manager provides the following out-of-box category records that you can use or modify:

- Purchase
- Reservation
- Labor
- Installation
- Uninstallation
- Automation

To create a new request task category record, follow these steps:

1. Click **Request Fulfillment**.
2. Click **Configuration** > **Request Task Categories**.
3. Click **New**.
4. Type the name of the request task category.
5. Clear the **Active** checkbox if you don't want the new category to appear in the category list. If the **Active** checkbox is cleared, the new category is unavailable when you plan a new task in task planner.
6. Type a description of the request task category.
7. Select a workflow for the category.

   The **Workflow** tab is displayed.
8. Click **Save**.

Note

To avoid breaking existing request task records, the system doesn't allow you to delete the existing request task categories or edit their names.

Related topics

Request Fulfillment configuration
Create a request category

1.9.6.5. Create a product catalog category

Security Role: Request Manager

You can define product catalog categories so that users can find their desired product catalog items easily. Product catalog items are organized in product catalog categories.

To create a product catalog, follow these steps:

1. Click **Request Fulfillment** > **Configuration** > **Product Catalog Categories**.
2. Click **New**.
3. In the **Name** field, type a name for the product catalog category.
4. Clear the **Active** checkbox if you don't want the new category to appear in the category list.

   Note

   If you clear the **Active** checkbox, this category can't be used to create new product catalogs.

5. (Optional) In the **Description** field, type a description for the product catalog category.
6. Click **Save**.

Related topics

Request Fulfillment configuration
Create a product catalog item

1.9.6.6. Create a product catalog item

Security Role: Request Manager

A product catalog item is the actual product or service in the Request Fulfillment process.

To create a product catalog item, follow these steps:

1. Click **Request Fulfillment** > **Configuration** > **Product Catalog**.
2. Click **New**.
3. In the **General** tab, type the information in the following fields:
    - **Part No.**: The unique identifier of the particular component.
    - **Brief Description**: A brief description of the product catalog item.
    - **Manufacturer**: The manufacturer of this component.
    - **Model**: The model name of the component, which is usually defined by the manufacturer.
    - **Product Catalog Category**: The category that this product catalog item belongs to.
4. (Optional) Add information in other fields in the **General** tab if needed. For example, you can configure the following fields for CI updates.
    - **Config. File**: Identifies the file name where corresponding CIs will be created. If it is left blank, the "device" file is used by default.
    - **CI Type**: Identifies the CI type of the corresponding CIs for the specific product catalog item.
5. (Optional) Add applicable information in the following tabs:
    - **Vendors**: Stores specific vendor/supplier information for a product.
    - **Receiving Information**: Defines how to generate record in Request Fulfillment receiving log.
        - **Receiving Format**:

          Specifies the form that's presented during receiving process. If it is left blank, the default value "ocmlrec.receive.g" is used.
        - **Asset Tag # Name**:

When the product catalog item is received, a CI is automatically created. The logical.name of the CI is constructed by consulting the record (the record name is defined in Asset Tag # Name) in the “number” table.

- **Field Name/Field Description/Required?/Default Value/Data Type**:

  Defines additional CI attributes that will be assigned to the CIs during CI creation.

- **Software**: Stores software information, such as licensing and installation information.
- **Current Quantities**: Displays quantities of items in different statuses for each stockroom.

  Tip

  To retrieve the latest amount of product catalog items in different statues for each stockroom, click **More** > **Take Inventory**.

- **CI Information**: Lists all the CIs that correspond to this product catalog item in cmdb.

6. Click **Save**.

Related topics

Request Fulfillment configuration
Create a product catalog category

1.9.6.7. Stock management

As a sub-module in Request Fulfillment, stock management manages the stock information, including stock rooms, reorder rules, model vendors, and inventory configuration.

To access stock management, from the System Navigator, click **Request Fulfillment** > **Stock Management**. Then, you can access the following menu items for stock management.

- **Stock Rooms**

  A stock room stores inventory information and generic information such as supporting locations and shipping address.

- **Reorder Rules**

  You can create reorder rules by specifying reorder point and reorder amount for particular product catalog item in each stock room.

- **Model Vendors**

  You can manage vendor/supplier information for particular product catalog item.

- **Inventory Configuration**

  The configuration is referenced by the “Take Inventory” feature which calculates the amount of installed, reserved, available, and unavailable product catalog items in each stock room.

Create a reorder rule

Security Roles: Stock Manager

You can define reorder rules for particular product catalog items to facilitate or automate the reordering process.

To create a reoder rule, follow these steps:

1. In the System Navigator, click **Request Fulfillment** > **Stock Management** > **Reorder Rules**.
2. Click **New**. The Reorder Rule form is displayed.
3. Create the reorder rule according to the following table:

Setting	Description
Part No.	The part number of the product catalog item for which you want to create a reorder rule.
Min. Ord. Amount	The minimum order quantity for the product catalog item. If an operator requests less than this amount, the system will raise the requested amount to this amount.
Max. Ord. Amount	The maximum order quantity for the product catalog item. If an operator requests more than this amount, the system will lower the requested amount down to this amount.
Lot Size (Ord.)	The order amount is a multiple of this number. If not, the system will adjust it accordingly.
Unit/Measure	The standard unit of measure for the product catalog item. The valid values are defined in the “unitofmeasure” table.

Setting	Description
Send email to Stock Manager/Generate order	◦ If **Send email to Stock Manager** is selected, a notification will be sent to the stock manager of the specific stockroom to inform the stock manager to create the order manually. If the stock manager for the specific stockroom is not specified, the default stock manager will be notified. ◦ If **Generate order** is selected, an order record with a purchase task is automatically generated based on the "Order" request model and the order assignee is the stock manager of the specific stockroom. If stock manager of the specific stockroom is not specified, the default stock manager will be assigned.
Default Stock Manager	When the stock manager of the specific stockroom is not specified, the default stock manager will be notified or assigned.
Stockroom/Reorder Point/Reorder Amount	You can specify the reorder point and reorder amount for each stockroom.

4. Click **Save**. The reorder rule is created.

Related topics

Request Fulfillment configuration

1.9.6.8. Add a Request Fulfillment assignment group

Applies to Security Roles: System Administrator

A Request Fulfillment assignment group is a list of users who are responsible for a request record. When a request or request task record opens or escalates, Request Fulfillment notifies the applicable group, which specifies the Request Coordinator, members, and approvers. Typically, assignment groups are organized by location and expertise. Assignment groups make the routing and escalation of request and request task records easier.

To add a Request Fulfillment assignment group,follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Groups** > **Assignment Groups**.

   Tip

   To get a list of existing assignment groups for your reference, click **Search**.

2. Type the name of the new assignment group in the **Assignment Group** field.
3. Click **Add**.
4. On the **Group** tab, complete the fields as described below.

   a. Enter the name of the manager of the assignment group.
   b. Select the work schedule of the assignment group from the **Calendar Name** drop-down list.
   c. Select the time zone.

5. Select the **Help Desk Group** checkbox so that the current assignment group supports End User Chat.
6. On the **Members** tab, insert the cursor in a blank line and use the Fill button to select the operators you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should provide responses when you create and populate a new assignment group.

(Optional) In the **Member Group Email** field, type a valid group email of members in this assignment group. Make sure that all members are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member of the assignment group.

7. On the **Approvers** tab, insert the cursor in a blank line and use the Fill button to select the approvers you want to assign to the group. There is no limit to the number of operators in an assignment group. However, consider how many users should approve when you create and populate a new assignment group.

   (Optional) In the **Approver Group Email** field, type a valid group email of approvers in this assignment group. Make sure that all approvers are included in this group email. When sending notifications, the system sends emails to this group email address only. Otherwise, the system separately sends the emails to each member in the assignment group.

8. On the **Assignments** tab, fill in the fields in Supported Departments, Supported Locations, and Supported Languages, respectively.

9. (Optional) Fill in the fields in Supported Companies if you have enabled the multi-tenant feature in your Service Manager system.

10. Follow these steps to configure the Supported Services field:

    a. Click **Configuration Management** > **Resources** > **Search CIs** to open the Search Configuration Item Records form.
    b. Select **Business Service** in the CI Type field, and then click **Search**.
    c. Select a configuration item, and then append an assignment group in the Support Groups field. For example, ASSET MANAGEMENT.
    d. Click **Save**. The selected configuration item is displayed in the Supported Services field of the ASSET MANAGEMENT assignment group.

11. Click **Save** and **OK**.

Related topics

Request Fulfillment configuration

1.9.6.9. Add a Request Fulfillment approval definition

Applies to Security Roles: System Administrator

The Approval Definition is a static file that defines the approval groups and the approval conditions.

An approval definition record defines the basics of an approval requirement. The approval condition for each requirement can be based on any field in the source record, such as cost and category.

To add a Request Fulfillment approval definition,follow these steps:

1. Click **Request Management** > **Configuration** > **Approval Definitions**.
2. In the **Name** field, type a unique name for this new record.
3. Add information on the form to set up the new approval definition record. For example, type an **Approval Description**, set the **Approval Type**, define approval condition, and add approval groups or operators.
4. Click **Add** to add the new record when you finish defining the approval definition.

Caution

When you are adding a new record from an existing record, make sure you don't click **Save**. Doing so will replace the existing record with the new record you are attempting to add.

Related topics

Request Fulfillment configuration

1.9.6.10. Clone a request model

Security Roles: Request Process Owner

To clone a request model with task information, follow these steps:

> **Note**
>
> If you want to edit a **Proprietary** workflow or use it as a model, create a copy of the this workflow.

1. Click **Request Fulfillment** > **Configuration** > **Request Models** from the System Navigator. All request models are displayed.
2. Click **Search**, and then select a request model.
3. Click **Clone**. The system opens the **Clone Model** page.
4. Specify a new unique ID and a name for the cloned request model record.
5. Select the **Copy task information from another model** checkbox, and then click the **Fill** icon to select another request model from which the task information will be copied.

   Note that if the request model you selected has a different task workflow than the current one, the copied task workflow may not work correctly if you use this cloned model.
6. Click **OK**.

Related topic

Request Fulfillment configuration

1.9.7. Security

The topics in this section explain the Request Fulfillment security roles, security areas, and rights.

Request Management security areas

Request Management security roles and settings

Out-of-box role rights in the Common Configuration area

Security folder and mandanten

Related topics

Request Fulfillment administration

1.9.7.1. Request Management security areas

The security areas for Request Fulfillment are Request, Request Tasks, and Request Management Configuration. These areas contain the default security rights and settings for requests and request tasks in the Request Fulfillment module. The security rights and settings will be inherited by the new roles created in an area when no settings are specified in the security role.

Area	System navigator menu items for this area
Request	This area contains the default security rights and settings for requests. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the role.
Request Tasks	This area contains the default security rights and settings for request tasks. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the role.

Area	System navigator menu items for this area
Request Management Configuration	This area contains the default security rights and settings for Request Fulfillment configuration. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the role. **Note** When you set the security rights for a security role in the Request Management Configuration area: • The **View** right is to view the settings defined in the **Configuration** menu. • The **Update** right is to update the values of existing settings defined in the **Configuration** menu. • The **New** and **Delete** rights are to create and delete a setting in the **Configuration** menu, such as category. • The **Admin** right enables operators to perform the administrative activities listed under the **Administration** menu of Request Fulfillment.

Area	System navigator menu items for this area

Default rights

When you create new security roles, they inherit the default rights that are defined in the security area. The following table describes the out-of-box rights that are defined in the Request, Request Tasks, and Request Management Configuration areas.

Area Name	View	New	Update	Delete/Close	Modify Template
Request	True	False	Never	Never	False
Request Tasks	True	False	Never	Never	False
Request Management Configuration	No right	No right	No right	No right	No right

Default settings

The default settings defined in areas will be inherited only if no value is specified for the settings on the role. In an out-of-box system, none of the default settings is checked or set in the Request, Request Tasks, and Request Management Configuration areas.

The following table describes the settings for the Request Fulfillment module.

Security Area	Setting (label)	Description
Request	Manager Group	Used by ApprovalDef: (mgr)/(MGR)
Request Request Tasks	Can Alternate Form	If selected (set to true), enables members of this role to select an alternate form for viewing the currently displayed record.
Request	Can Approve	If selected (set to true), enables members of this role to approve/deny requests/request tasks or retract previous approval/denial.
Request	Can Delegate Approvals	If selected (set to true), enables members of this role to delegate approvals.
Request Request Tasks	Skip Inefficient Query Warning	If selected (set to true), bypasses the inefficient query warning for members of this role; Normally, it is desirable to warn an operator when they have entered an inefficient (partially or non-keyed) query.
Request Request Tasks	Append Query	A query for Service Manager to append to all queries run by members of this role.

Security Area	Setting (label)	Description
Request Request Tasks	Initial Format	The name of the form to display when a member of this role first enters a given application.
Request Request Tasks	QBE Format	The QBE form to use when displaying records for members of this role.
Request	Request Initial View	The default view to display when the members of this role open the request queue.
Request	Request Queue Format	The default queue form to display when the members of this role open the request queue.
Request Tasks	Request Task Initial View	The default view to display when the members of this role open the request task queue.
Request Tasks	Request Task Queue Format	The default queue form to display when the members of this role open the request task queue.
Request	Edit Task Planner	If selected (set to true), enables members of this role to edit task planner in the request records.

Related topics

Security

Request Fulfillment security roles and settings

1.9.7.2. Request Management security roles and settings

The out-of-box security roles for Request Fulfillment module include the following:

- Requestor
- Request Approver
- Request Coordinator
- Request Analyst
- Request Manager
- Request Process Owner
- Stock Manager

Out-of-box role rights

The following table describes the out-of-box security rights for the security roles on the areas in Request Fulfillment module.

Request area

Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	True	True	Never	Never	False	False	False	Blank	Blank
Request Approver	True	False	Never	Never	False	False	False	Blank	Blank
Request Coordinator	True	True	Always	Always	False	True	False	Blank	Blank
Request Analyst	True	False	When assigned to workgroup	Never	False	False	False	Blank	Blank
Request Manager	True	True	Always	Always	True	True	True	Blank	Blank

Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Request Process Owner	True	True	Always	Always	False	False	False	Blank	Blank
Stock Manager	False	False	Never	Never	False	False	False	Blank	Blank

Request Tasks area

Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	True	True	Never	Never	False	False	False	Blank	Blank
Request Approver	True	False	Never	Never	False	False	False	Blank	Blank
Request Coordinator	True	True	Always	Always	False	True	False	Blank	Blank
Request Analyst	True	False	When assigned	When assigned	False	False	False	Blank	Blank
Request Manager	True	True	Always	Always	True	True	True	Blank	Blank
Request Process Owner	True	True	Always	Always	False	False	False	Blank	Blank

Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Stock Manager	False	False	Never	Never	False	False	False	Blank	Blank

Request Management Configuration area

Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	True	False	Never	Never	False	False	False	N/A	N/A
Request Approver	False	False	Never	Never	False	False	False	N/A	N/A
Request Coordinator	True	False	Never	Never	False	False	False	N/A	N/A
Request Analyst	False	False	Never	Never	False	False	False	N/A	N/A
Request Manager	True	True	Always	Always	True	True	True	N/A	N/A
Request Process Owner	True	True	Always	Always	False	False	False	N/A	N/A

Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Stock Manager	True	True	Always	Always	False	False	False	N/A	N/A

For information about the out-of-box role rights in the Common Configuration area, see Out-of-box role rights in the Common Configuration area.

The following table describes the out-of-box security rights for the Request Fulfillment security roles on the areas in other modules.

Note

The security roles in Process Designer Request Fulfillment have no rights on the areas that aren't listed in the following table, including:

- Security
- Knowledge Administration
- Change Management Configuration
- Service Desk Configuration
- Incident Management Configuration
- Areas for Problem Management

Change area & Incident area

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	True	False	Never	Never	False	False	False	Blank	Blank
Request Approver	True	False	Never	Never	False	False	False	Blank	Blank
Request Coordinator	True	True	Never	Never	False	False	False	Blank	Blank
Request Analyst	True	True	Never	Never	False	False	False	Blank	Blank
Request Manager	True	False	Never	Never	False	False	False	Blank	Blank
Request Process Owner	True	False	Never	Never	False	False	False	Blank	Blank

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Stock Manager	False	False	Never	Never	False	False	False	Blank	Blank

Change Tasks area & Incident Tasks area

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	True	False	Never	Never	False	False	False	Blank	Blank
Request Approver	True	False	Never	Never	False	False	False	Blank	Blank
Request Coordinator	True	False	Never	Never	False	False	False	Blank	Blank
Request Analyst	True	False	Never	Never	False	False	False	Blank	Blank
Request Manager	True	False	Never	Never	False	False	False	Blank	Blank
Request Process Owner	True	False	Never	Never	False	False	False	Blank	Blank

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Stock Manager	False	False	Never	Never	False	False	False	Blank	Blank

Service Desk area

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	True	False	Never	Never	False	False	False	Blank	Blank
Request Approver	True	False	Never	Never	False	False	False	Blank	Blank
Request Coordinator	True	False	Never	Never	False	False	False	Blank	Blank
Request Analyst	True	False	Never	Never	False	False	False	Blank	Blank
Request Manager	True	False	Never	Never	False	False	False	Blank	Blank
Request Process Owner	True	False	Never	Never	False	False	False	Blank	Blank

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Stock Manager	False	False	Never	Never	False	False	False	Blank	Blank

Tailoring area

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Requestor	False	False	Never	Never	False	False	False	N/A	N/A
Request Approver	False	False	Never	Never	False	False	False	N/A	N/A
Request Coordinator	False	False	Never	Never	False	False	False	N/A	N/A
Request Analyst	False	False	Never	Never	False	False	False	N/A	N/A
Request Manager	True	True	Always	Always	False	False	False	N/A	N/A
Request Process Owner	True	True	Always	Always	False	False	False	N/A	N/A

Role Name	View	New	Update	Delete/Close (delete)	Modify Template	Expert	Admin	Allowed Statuses	Allowed Categories
Stock Manager	False	False	Never	Never	False	False	False	N/A	N/A

Related topics

Security
Request Fulfillment security areas

1.9.7.3. Out-of-box role rights in the Common Configuration area

The Common Configuration area is used to set the permissions that control access to common configurations, including Alert Definitions, Approval Definitions, Assignment Groups, Service Desk/Problem Solution Matching, and Environment.

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Common Configuration area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Common Configuration area. This table only lists the new security roles that have different settings with the default rights.

Area	Role	View	New	Update	Delete/Close	Modify Template	Expert	Admin
Common Configuration	change analyst change	True	True	Always	Always	False	False	False
	change analyst tasks	True	True	Always	Always	False	False	False
	change approver	True	True	Always	Always	False	False	False
	change coordinator change	True	True	Always	Always	False	False	False
	change coordinator tasks	True	True	Always	Always	False	False	False
	change manager	True	True	Always	Always	False	False	False

Area	Role	View	New	Update	Delete/Close	Modify Template	Expert	Admin
	configuration auditor	True	True	Always	Always	False	False	False
	emergency group	True	True	Always	Always	False	False	False
	incident analyst	True	True	Always	Always	False	False	False
	incident coordinator	True	True	Always	Always	False	False	False
	incident manager	True	True	Always	Always	False	False	False
	initiator	True	True	Always	Always	False	False	False

Area	Role	View	New	Update	Delete/Close	Modify Template	Expert	Admin
	operator	True	True	Always	Always	False	False	False
	problem analyst	True	True	Always	Always	False	False	False
	problem coordinator	True	True	Always	Always	False	False	False
	problem manager	True	True	Always	Always	False	False	False
	reviewer	True	True	Always	Always	False	False	False
	SD agent/manager	True	True	Always	Always	False	False	False
	service desk agent	True	True	Always	Always	False	False	False

Area	Role	View	New	Update	Delete/Close	Modify Template	Expert	Admin
	service desk manager	True	True	Always	Always	False	False	False
	system administrator	True	True	Always	Always	True	True	True
	Requestor	True	False	Never	Never	False	False	False
	Request Approver	False	False	Never	Never	False	False	False
	Request Coordinator	True	False	Never	Never	False	False	False
	Request Analyst	False	False	Never	Never	False	False	False
	Request Manager	True	True	Always	Always	False	False	False

Area	Role	View	New	Update	Delete/Close	Modify Template	Expert	Admin
	Request Process Owner	True	True	Always	Always	False	False	False
	Stock Manager	False	False	Never	Never	False	False	False

1.9.7.4. Security folder and mandanten

The following tables in Request Fulfillment module support security folder and mandanten mechanism in the out-of-box system.

- request
- requestTask
- rmCategory
- rmTaskCat
- requestModel
- productCatalog
- prodCatalogCategory

Related topics

Security
Request Fulfillment security areas

1.10. Request Management (Hybrid)

Service Manager Request Management manages user requests for products and services. Requests can affect only the person making the request, or a subordinate group of employees. Examples include password resets, individual PC upgrades, and new employee setup.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Request Management user roles

Key concepts

Describes the most important things that you need to know about how the Request Management module functions.

Request Management overview

Pre-implementation planning

Approvals in Request Management

Quotes

Orders

Categories

Phases

User workflows

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Service Request Logging (SO 3.1)

Service Request Approval (SO 3.2)

Service Request Provisioning (SO 3.3)

Service Request Validation and Closure (SO 3.4)

Create, Update or Retire Service Request Catalog Item (SO 3.5)

Service Request Monitoring (SO 3.6)

Service Request Escalation (SO 3.7)

Administrator tasks

Describes the Administrator tasks in the Request Management module.

Add a profile

Edit environment records

Access approval definitions

Access approval logs

Reset the number control application

Access the alert log file

Edit group records

1.10.1. Request Management user roles

The Request Management module has the following user roles.

Requestor

The Requestor user role has the following responsibilities:

- Initiate a request through Request Management and search and view request records.

To execute this role, the Requestor has following tasks available:

ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote
Service Request Approval (SO 3.2)	
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	
Create, Update or Retire Service Request Catalog Item (SO 3.5)	
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Request Coordinator

The Request Coordinator user role has the following responsibilities:

- Coordinate requests. May also be responsible for receiving things into inventory.

To execute this role, the Request Coordinator has following tasks available:

ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote • Associate a request with a related record • Open a related record
Service Request Approval (SO 3.2)	• Access approvals from the search screens • View approval type group options
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	• Close an order
Create, Update or Retire Service Request Catalog Item (SO 3.5)	
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Request Reviewer

The Request Reviewer user role has the following responsibilities:

- Evaluate a Request Management quote for potential errors. A reviewer may not necessarily have approval rights.

To execute this role, the Request Reviewer has following tasks available:

ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote • Associate a request with a related record
Service Request Approval (SO 3.2)	• Access approvals from the search screens • View approval type group options
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	
Create, Update or Retire Service Request Catalog Item (SO 3.5)	
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Request Approver

The Request Approver user role has the following responsibilities:

- Approve requests before they can be processed and filled.

To execute this role, the Request Approver has following tasks available:

ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote • Associate a request with a related record • Open a related record
Service Request Approval (SO 3.2)	• Access approvals from the search screens • View approval type group options
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	
Create, Update or Retire Service Request Catalog Item (SO 3.5)	
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Request Helpdesk Tech

The Request Helpdesk Tech user role has the following responsibilities:

- Provide helpdesk assistance for opening new item requests.
- Process service requests.
- Oversee the request process through its cycle back to the customer.

To execute this role, the Request Helpdesk Tech has following tasks available:

ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote • Associate a request with a related record
Service Request Approval (SO 3.2)	
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	
Create, Update or Retire Service Request Catalog Item (SO 3.5)	
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Request Procurement Manager

The Request Procurement Manager user role has the following responsibilities:

- Process requests through the system.
- May be required to update inventory data when requested items arrive.

To execute this role, theRequest Procurement Manager has following tasks available:

ITIL workflow	Tasks
Service Request Logging (SO 3.1)	
Service Request Approval (SO 3.2)	
Service Request Provisioning (SO 3.3)	• Close an order
Service Request Validation and Closure (SO 3.4)	
Create, Update or Retire Service Request Catalog Item (SO 3.5)	
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Request Administrator

The Request Administrator user role has the following responsibilities:

- Administer Request Management applications.

To execute this role, the Request Administrator has following tasks available:

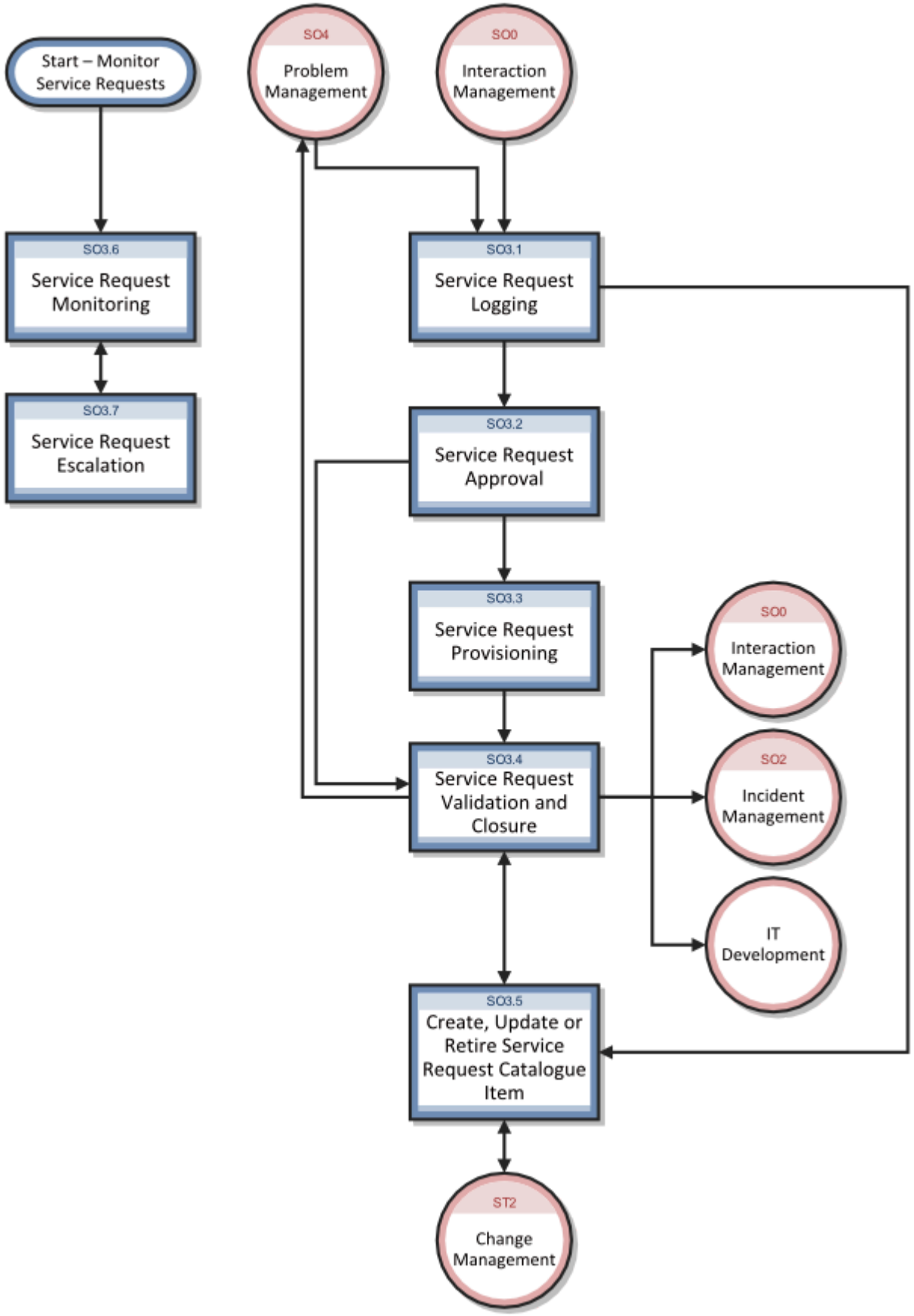
ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote • Associate a request with a related record • Open a related record
Service Request Approval (SO 3.2)	• Access approvals from the search screens
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	• Close an order
Create, Update or Retire Service Request Catalog Item (SO 3.5)	• Add a master category • Add a line item category • Access a line item view • Designate a model file as a phantom
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Related topics

User roles
Add a user role record
Delete a user role record
Search for a user role record

1.10.2. Request Management workflows and user tasks

The Request Management process includes the activities required to select items from the menu and submit a service request, to give financial and business approvals, to provision, and to fulfil service requests. It is responsible for ensuring that a IT support is offered for self-help practices and requests can be effectively fulfilled after needed approvals.



The Request Management process consists of the following workflows and tasks:

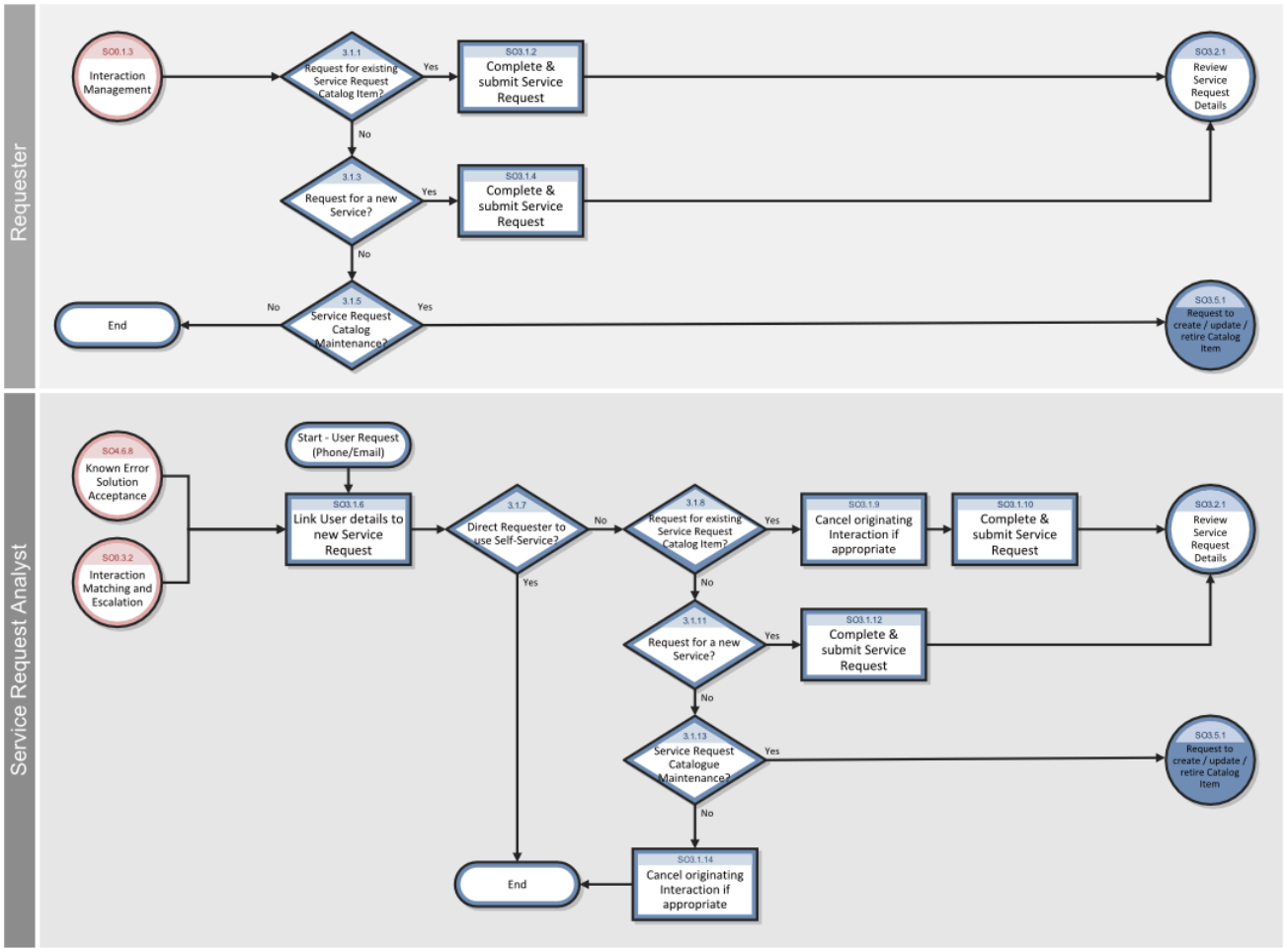
ITIL workflow	Tasks
Service Request Logging (SO 3.1)	• Open a quote • Associate a request with a related record • Open a related record
Service Request Approval (SO 3.2)	• Access approvals from the search screens • View approval type group options
Service Request Provisioning (SO 3.3)	
Service Request Validation and Closure (SO 3.4)	• Close an order
Create, Update or Retire Service Request Catalog Item (SO 3.5)	• Add a master category • Add a line item category • Access a line item view
Service Request Monitoring (SO 3.6)	
Service Request Escalation (SO 3.7)	

Related topics

Processes and Best Practices

1.10.2.1. Service Request Logging

The Service Request Logging process starts when a Requester uses Self Service Ticketing or the Service Desk to log appropriate Service Requests. A Service Request submitted by the Requester can be a request for existing Service Request Catalog Item, a request for a new service, or an amendment to the Service Request Catalog. The Service Request Analyst needs to link User Details to the new Service Request, analyze the request, and then decide what to do next. As a result of the Service Request Logging process, a Service Request will be submitted. An originating interaction could be cancelled if needed.



For more information on this process, refer to *Service Manager Processes and Best Practices*.

Open a quote

Applies to User roles: Request Help Desk Tech, Requestor, Request Coordinator, Request Approver, Request Administrator, System Administrator

To open a service request:

1. Click **Request Management** > **Quotes** > **Create New Quote**.
2. Select one of the categories for this request.
    - Customer Procurement Requests
    - Human Resources
    - Employee Office Move Process
3. If you selected **Customer Procurement Requests**, perform the following steps:
    1. Select from the **Master Catalog**.
    2. Select the category of the item you are requesting.
    3. Select the item(s) you wish to request. A list of the item(s) you selected opens and shows the total cost.
    4. Click **Add Items** to add more items to your request.
    5. Click **Submit Request**.
    6. Fill in the required fields (such as requested delivery date and reason for request) in the **New Request** form.
4. If you selected **Human Resources**, perform the following steps:
    1. Select a category. You can choose to select defaults and also to modify the quantity.
    2. Click **Add Items** to add more items to your request.
    3. Click **Submit Request**.
    4. Fill in the required fields (such as start date, department, and manager) in the human resources form.
5. If you selected **Employee Office Move Process**, perform the following steps:
    1. Select from the available list of options. You can choose to select defaults and also to modify the quantity. A summary list of your selections opens and shows total cost.
    2. Click **Add Items** to add more items to your request.
    3. Click **Submit Request**.
    4. Fill in the required fields (such as first name, last name, and effective date) in the employee office move request form.
6. Click **Save**. Service Manager creates the new quote.

Associate a request with a related record

Applies to User roles: Request Coordinator, Request Reviewer, Request Approver, Request Helpdesk Tech, System Administrator

You can associate a Request Management quote with an existing service desk interaction, incident, or change request. Before associating a quote with another record, make note of the record number. For example, an incident record begins with the prefix IM. The record number would look like IM1001.

In the following example, associate a Request Management quote with an Incident Management incident record. The procedure is the same for associating other types of records.

1. Click **Request Management** > **Quotes** > **Search Quotes** or **Quote Queue** and then click **Search** to open an existing Request Management quote.
2. Click **More** or the More Actions icon, and then select **Related** > **Incidents** > **Associate**.
3. Type the incident number or click **Search** to view the incident search form.
4. Add optional filtering criteria, and then click **Search** again to display an incident record list.
5. Select the incident record you want to associate with the quote record. The incident record number is displayed.
   - Click **OK** to link the incident record to the quote record.
   - Click **Search** to view the incident record search form again. Click **Search** again to redisplay the incident record list.
   - Click **Cancel** to stop the search and exit.
6. Click **Save & Exit**.

   Request Management populates the Related Records section with information about the related incident.

Note

Record numbers such as incidents (IM1001), changes (C10), and requests (Q1001) are case-sensitive.

Open a related record

Applies to User roles: Request Administrator, System Administrator

You can open a Change Management change, Incident Management incident record, or Service Desk interaction from Request Management. The new record is related to an existing quote.

The following example opens an Incident Management incident record. The procedure is the same for opening other types of records.

To open a related record:

1. Click **Request Management** > **Quotes** > **Quote Queue** or **Search Quotes** to access an existing Request Management quote.

2. Add optional search criteria, and then click **Search**.

   A list of quote records opens.

3. Select a target record.
4. Click **More** or the More Actions icon, and then select **Related** > **Incidents** > **Open**.

   Service Manager creates a new incident record and automatically fills in the Incident ID and Title of the new record.

5. Fill in the required fields and the remainder of the incident record, as needed.
6. Click **Save & Exit**.

   Note

   Depending on the information you provided in the new incident record, a list of potentially related incident records described below may or may not open. If the list doesn't appear, you will directly go to step 10.

7. A list of potentially related incident records opens. Browse through the related incident records list to see if you can associate the quote to an existing incident record.
8. If an existing incident record is related to the quote:
   1. Click to open this incident, and then click **Edit**.
   2. Click **More** or the More Actions icon, and select **Related** > **Quotes** > **Associate**, enter the current quote number and click **OK**. The quote is associated to the existing incident record.
   3. Click **Save & Exit**, and then **Back** twice. You are returned to the new incident record.
   4. Click **Cancel** to cancel opening the new incident. You are returned to the quote record, and the existing incident record displays in the Related Records section of the quote.
9. If none of the existing incident records is related to the quote, go to step 10.
10. Click **Create New Incident**.

> **Note**
>
> Depending on the information you provided in the new incident record, a list of potentially related problem records described below may or may not open. If the list doesn't appear, step 11 is skipped and the new incident record is opened and associated to the quote.

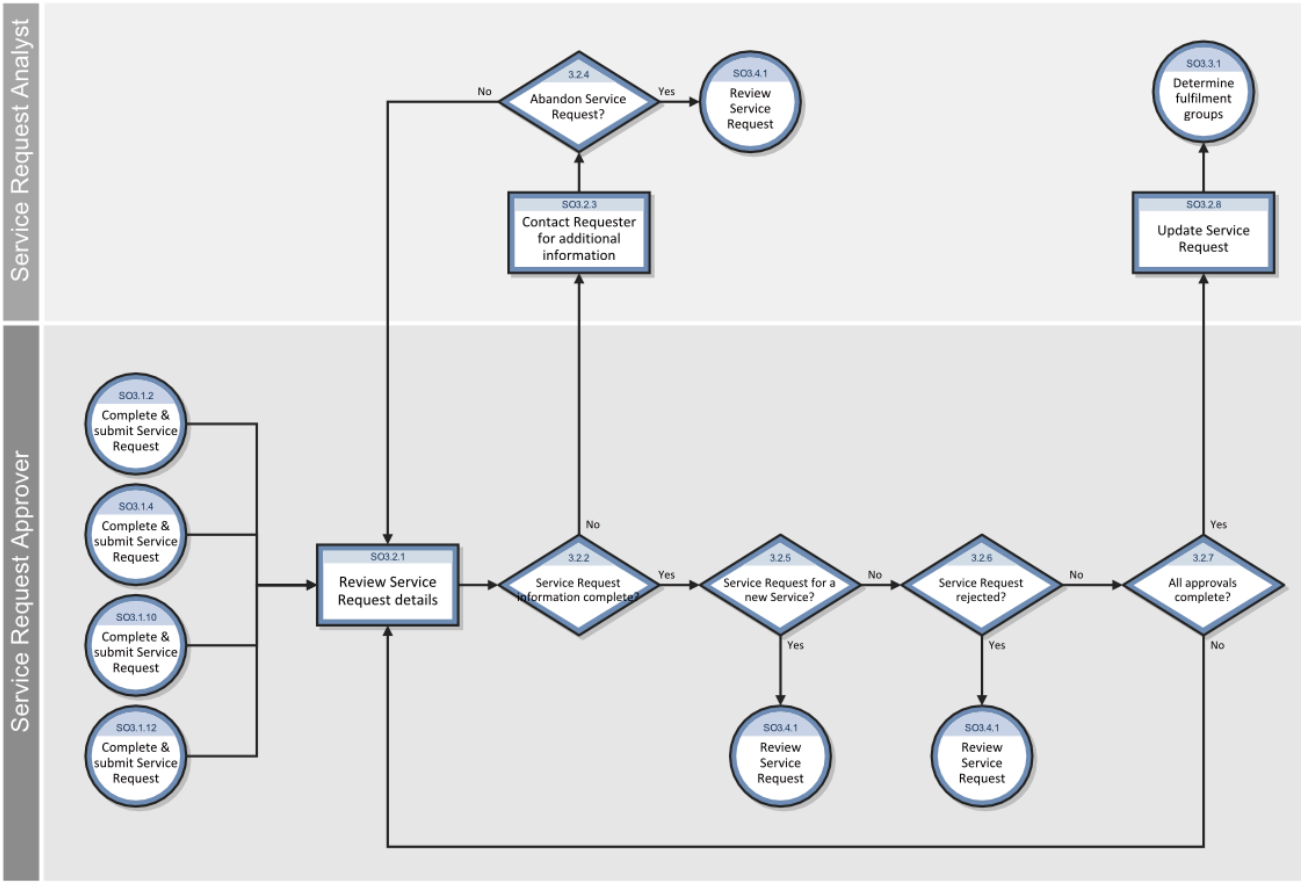
11. A list of potentially related problem records opens. Browse through the related problem records list to see if you can associate the new incident to an existing problem record. Do one of the following:
    - If none of the problem records can be related, click **Continue** to continue with opening the new incident record. The new incident is opened, and the quote is associated to the new record.
    - If a problem record can be related, click the problem record, and then click **link** to link the problem record and continue to browse the other records; or click **linkext** to link the problem record and exit. The new incident is opened with the problem record linked to it, and the quote is associated to the new incident record.

Related topics

Request Management overview
Request Management workflows and user tasks
Processes and Best Practices

1.10.2.2. Service Request Approval

A service request initiated by the Requester has the request and user information automatically included in the service request. After a service request is logged, the Service Request Approver reviews the Service Request details. If more information is needed, the Service Request Approver will contact the Requester to complete the information, and then approve or reject the request. Once all approvals have been received, the Service Request Analyst updates the Service Request and make sure all Service Request information is up-to-date.



For more information on this process, refer to *Service Manager Processes and Best Practices*.

Access approvals from the search screens

Applies to User roles: Request Approver, Coordinator, Request Administrator, System Administrator

Approvals can either be approved, retracted, or denied. Pending approvals await one of these actions.

To access approvals from a search screen:

1. Click one of the following:
   - **Request Management** > **Quotes** > **Search Quotes**
   - **Request Management** > **Orders** > **Search Orders**
2. Add optional search criteria, and then click **Search**.

   A record list of quotes or orders opens.

   Note

   If you have quotes authority in only one group, the list of current requests for that group opens.

   If you have approval authority for multiple groups, the Approval Group Select dialog box appears. Select the groups that you want to search.

   Request Management places a check mark beside the flagged group boxes and searches for requests currently awaiting approvals for that group. If all groups are checked, all groups are queried.
3. Select the target quote or order record.
4. Open the **Approvals** section, and go to the Current Approvals subsection.

   For the approvals in this quote or order, you see the approval type, approval status, and number of approved.
5. In the **Current Approvals subsection**, click an approval type.
6. Click **More** or the More Actions icon, and choose one of the following:

   - **Approve**
   - **Deny**
   - **Retract**

   The descriptions of these approval actions are as follows:

Action	Description
Approve	Gives authority to proceed with all items listed in the request.
Deny	Denies authority to proceed with all items listed in the request.
Retract	Removes previous approval actions.

7. If you made edits to approvals, click **Save** to save your changes.

View approval type group options

Applies to User Roles: All roles except Self-Service User

To view Approval Type group options:

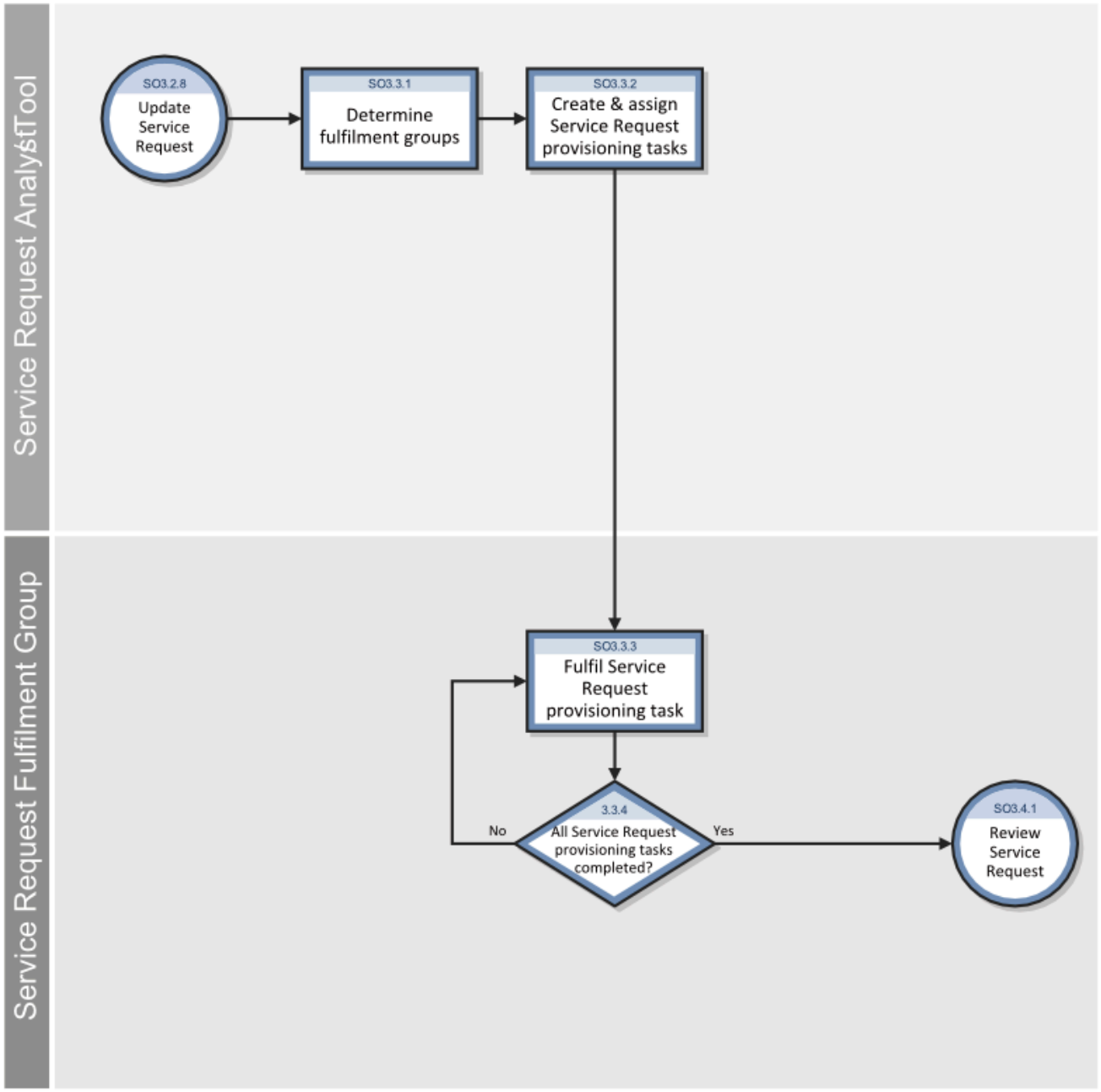
1. Click **Request Management** > **Quotes** > **Search Quotes** or **Quote Queue** and then click **Search** to open an existing quote record.
2. Open the **Approvals** section.
3. In the Current Approvals subsection, double-click an Approval Type group whose group options you want to view. The Approval form opens.
4. Do one of the following:
   - If the approval type has a status of pending, go to the Currently Pending Approvals area, and click a group listed in the Group/Operator Name field.
   - If the approval type has a status of approved or denied, go to the Completed Approval Actions area, and click a group listed in the Approval Group/Oper field.
5. Click **More** or the More Actions icon, and then select **Override** to view the options.

Related topics

Request Management overview
Request Management workflows and user tasks
Processes and Best Practices

1.10.2.3. Service Request Provisioning

In the Service Request Provisioning process, the Service Request Analyst identifies which Service Request Group/s is best able to fulfil the Service Request. This step can also be performed by Service Manager. The tool can automatically assign records to the appropriate group based on the record's categorization. After that, Service Request Provisioning tasks are created for the group to fulfil.



For more information on this process, refer to *Service Manager Processes and Best Practices*.

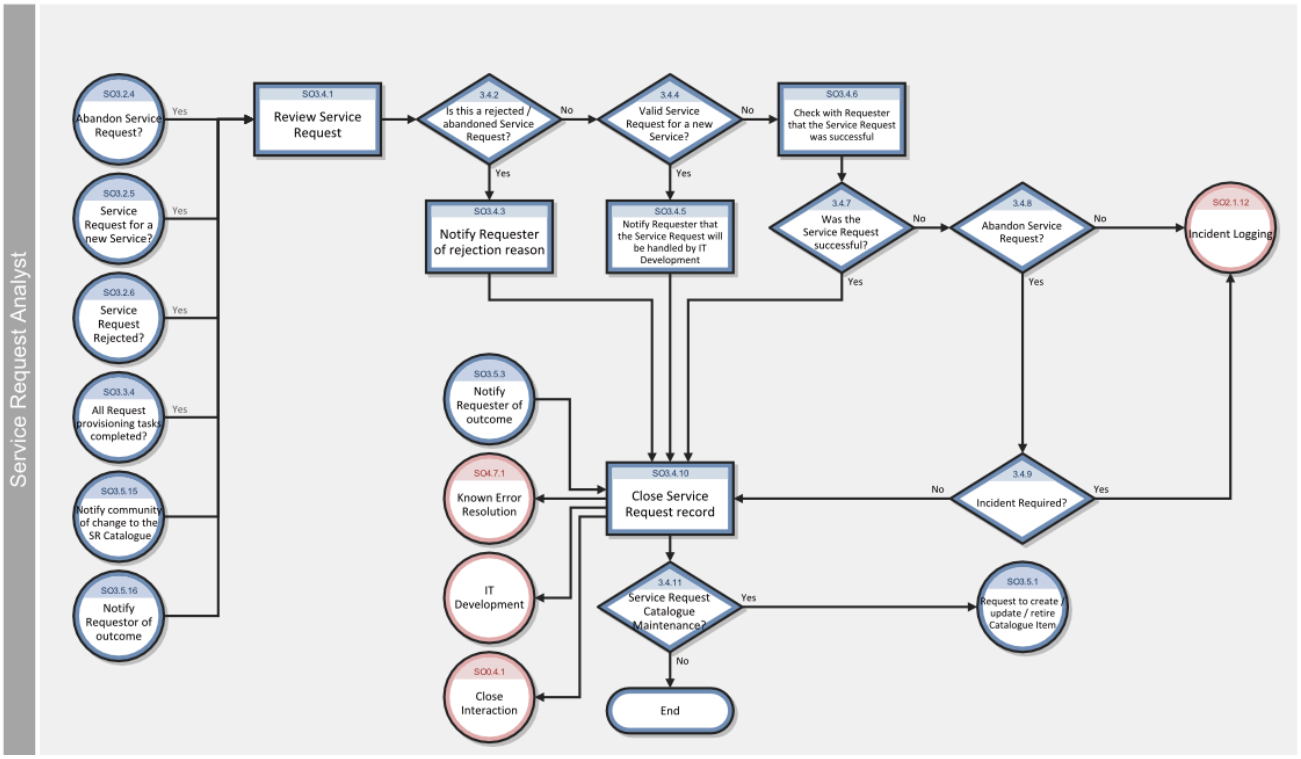
Related topics

Request Management overview
Request Management workflows and user tasks
Processes and Best Practices

1.10.2.4. Service Request Validation and Closure

After a Service Request is approved and fulfilled, the Service Request Analyst starts to review, validate, and then close the request. A Service Request can be closed when the Service Request Analyst finishes one of the following tasks:

- Notify the Requester the rejection reason, if the Service Request is abandoned and rejected.
- Notify the Requester that the Service Request will be handled by IT development after validating the Service Request for a new service.
- Check with the Service Requester that the Service Request is successfully fulfilled.
- Log a incident for the Requester if the Service Request is not successful.



For more information on this process, refer to *Service Manager Processes and Best Practices*.

Close an order

Service Manager allows you to close an order using the Close Order option even when it has open line items. However, closing an order with open line items will cancel the open line items. Normally, you close an order by closing all the line items of the order after confirming that all the parts have been received or the tasks have been

completed. Service Manager automatically closes the order when all the line items are closed.

To close an order:

1. Click **Request Management** > **Orders** > **Search Orders**.
2. Click **Search** to generate a record list.
3. Click a record to view it.
4. Click **More** or the More Actions icon, and select **Line Items**. If the order has more than one line item, a list of line items opens in the upper screen, and an Order Line Item Summary opens in the lower screen. If the order has only one line item, the Order Line Item Summary opens in the whole screen.
5. Check that all line items of the order have been received or their tasks are completed, and then close each open line item.
    - If the line item is a part for which the **Track Receiving?** option is enabled, do the following:
        1. Click **Receive** to begin the receiving process.

            Note

            The Receive button is available only for line items for which the **Track Receiving?** option is enabled. To see if this option is enabled for a part, go to **Request Management** > **Maintenance** > **Supporting Files** > **Model**, open the Model record of the part, and then select the **Reorder** tab.

        2. Verify the receiving information, and then click **OK**.
        3. Verify and adjust the quantities, and click **Post**.

            Note

            The line item quantities are posted to update the **Current Quantities** tab of the Model record of the part.

    - If the **Track Receiving?** option is disabled for the part, do the following:
        1. Change the Quantity Received to be equal to the Original Quantity.
        2. Click **Close Line Item**. You can only close a line item that has no outstanding balance.

- If the line item is a service for which the work is completed, click **Close Line Item** to close it.

> **Note**
>
> Scripts can be used during the close process for the operator to provide further comments about the work performed. The name of the Close Script is displayed on the line item's phase definition. When you click **OK**, the Request Management order summary form opens. A different receiving format can be designated on the line item category phase definition for the part selected from the catalog.

6. Click **Save & Exit**. You are returned to the order record.

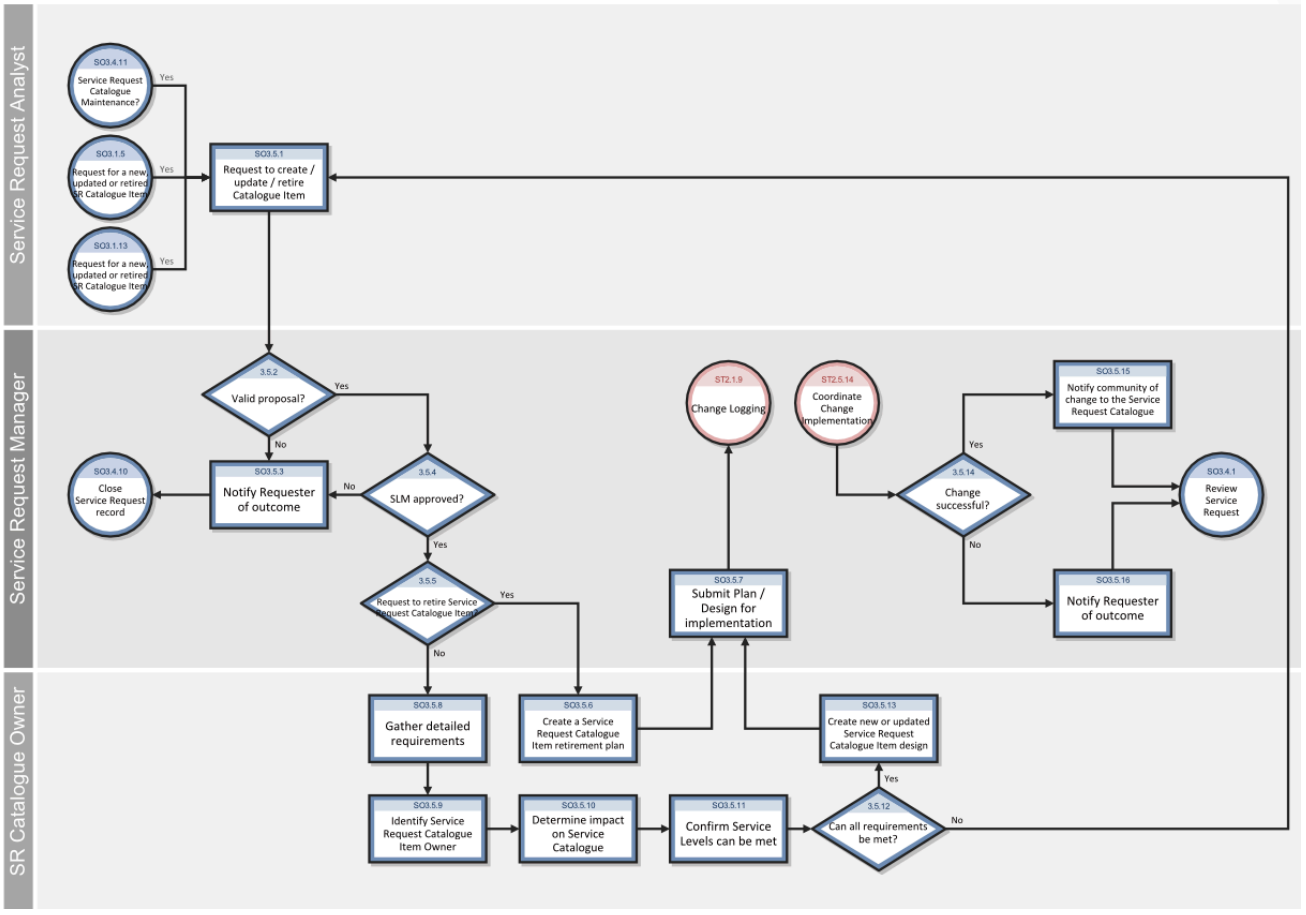
   Service Manager changes the order status to closed.

Related topics

Request Management overview
Request Management workflows and user tasks
Processes and Best Practices

1.10.2.5. Create, Update or Retire Service Request Catalog Item

The Service Request Analyst requests to update the Service Request Catalog when the Service Request Catalog Maintenance is needed. Service Request Catalog Owner is responsible for creating a Service Request Catalog Item retirement plan or a updated Service Request Catalog design after making sure all requirements can be met. Once the plan or design is submitted for implementation, it will be managed as part of the Change Management process. The Requester who initiated the Requester and appropriate stakeholders will be notified the change implementation results.



For more information on this process, refer to Processes and Best Practices.

Add a master category

User roles: Request Coordinator, Request Administrator, System Administrator

Categories within Request Management combine items that would normally be listed independently. Categories reduce or eliminate the need to search for items and services.

To add a master category:

1. Click **Request Management** > **Maintenance** > **Administration** > **Master Categories**.
2. Type the required unique **Name** for this master category, and then type a short **Description** of this catalog item (up to 30 characters). Record lists display this description and it should be meaningful for operators.
3. The Availability field determines if the operator can select this category while adding a line item. If the operator needs to have this control, type **true**. Otherwise, the default setting is false when the field is left blank.

   You can use a condition to create a more specific availability control.

   > **Note**
   >
   > The variable *$lo.ucapex* can be used to search for operator or group capability words and determine whether this master category will be available to particular operators. It can be used in the Availability field as part of a condition, such as the following.
   >
   > `index("Capability words",$lo.ucapex)>0`

4. The Display Categories field determines if the operator can see the list of line item categories under this master category. If the operator needs to have this control, type **true**. Otherwise, the default setting is false when the field is left blank. If the field is left blank (false), the line item category record list is not displayed; instead, all parts (or line items) with a line item category matching any of this master category's line item categories are displayed.
5. The Sequence field determines the sequence to sort the list of parts for this category.

   > **Note**
   >
   > You can modify the sort order within the Sequence field by selecting the category key in the model file.

6. Quote categories determine which quote categories access this master category and display this line item in the catalog. Type the name of this category in the **Quote Categories** field. If left blank, this category will be available for all operator quote categories.

> **Note**
>
> You can return to the record later to add additional quote categories if you don't know the names of the categories at this time.

7. Determine which order categories access this master category and display this line item in the catalog. Type the name of this category in the **Order Categories** field. If left blank (NULL), this category will be available for all operator order categories. The out-of-box system contains five order categories.

> **Note**
>
> You can return to the record later to add additional order categories, if you don't know the names of the categories at this time.

8. Determine which line item categories are to be available in this master category. Type the name of this category in the **Line Item Categories** field. These are the smaller groupings of products and services, which organize the available pool of resources. Currently there are just over 80 default line item categories. If you leave this field blank, all the line item categories will be available to this master category.
9. Click **Add**.

   The new master category is added.

Add a line item category

User roles: Request Coordinator, Request Administrator, System Administrator

Line item categories are the building blocks of master categories. The parts of line item categories are individual line items, such as a computer mouse or monitor. Line items include both specific parts and services.

To add a line item category:

1. Click **Request Management** > **Maintenance** > **Administration** > **Line Item Categories** or **Create Line Item Category**.
2. Type the **Name** and **Description** of the category.
3. In the Availability field, type **true** to make the new category visible. If you leave this field blank or type false, then the category is unavailable.

4. In the QBE format field, type the name of the record list format that you want to use if it is different from the default *ocml.qbe* form.
5. The **List Bitmap** field enables you to add images of your inventory items to the forms. Images are stored in the Service Manager repository and require a GUI client.
6. The Assign Number Before Commit? checkbox assigns a number to the line item before it opens a confirmation screen (provided that display option is active).
7. Click the first field in the Quote Categories box, and then click **Fill** to generate a list of categories.
8. Select a quote category from the list. The system adds it to the Quote Categories box.
9. Click the first field in the Order Categories field, and then click **Fill** to generate a list of categories.
10. Select an order category from the list. The system adds it to the Order Categories box.

    The Line Item Phase field displays the phase name for the category. Format Control automatically matches this to the existing category name.
11. Click **Add**.

Important

This item will not function properly until a phase definition record of the same name is committed to the database.

Note

When you save the new line item category, you need to add an associated phase definition record.

Access a line item view

Applies to User roles: Request Coordinator, Request Administrator, System Administrator

The Line Item view form displays the ordering controls that were set in the Catalog for the part and copied over to the line item during the request process (see Catalog Operations).

1. Click **Request Management** > **Line Items** > **Line Item Queue**.

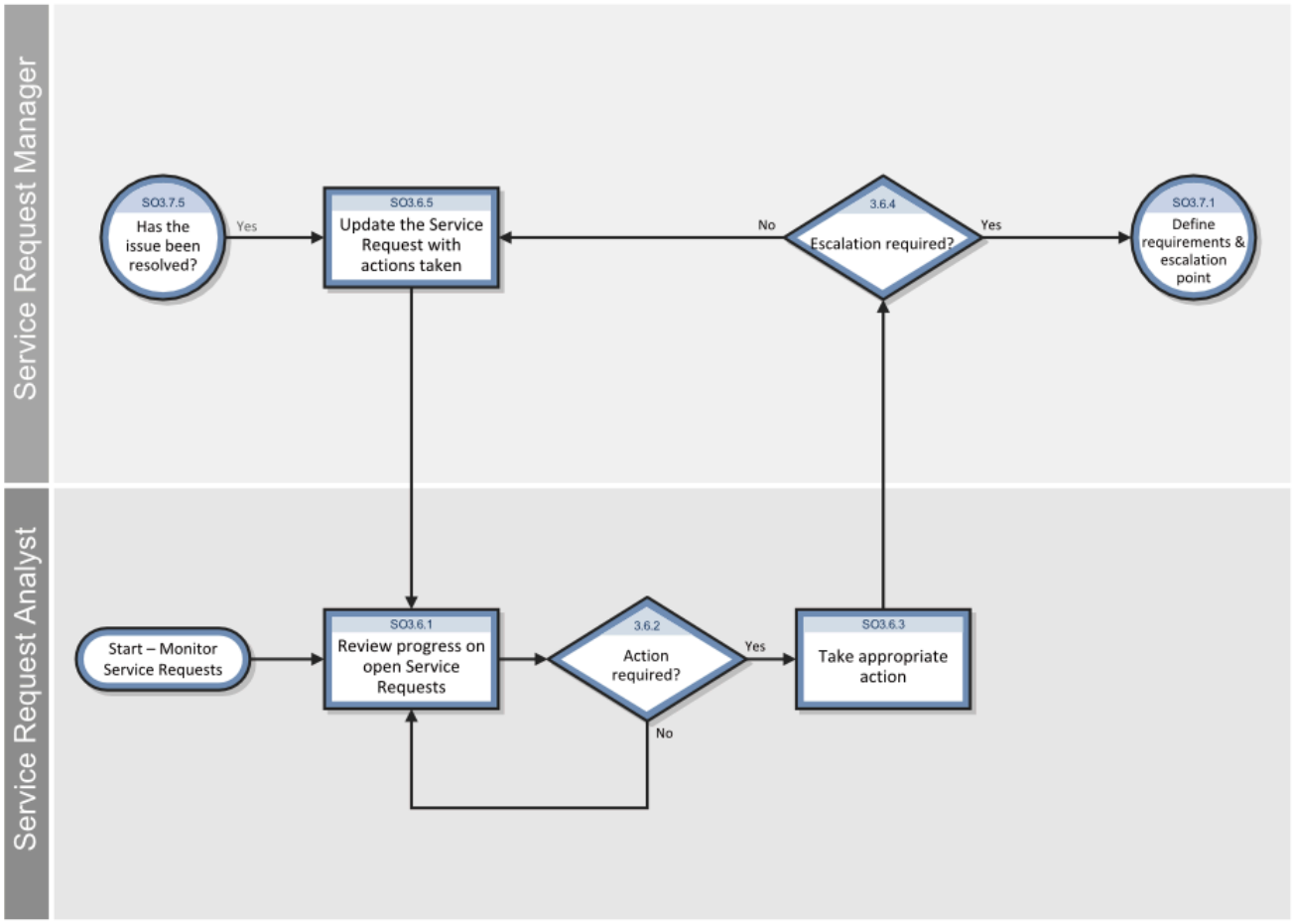
2. Use search or advanced search to find one or more records.
3. Select a target record to view.
4. To see different views of the data, click **More** or the More Actions icon, and then select **Alternate Forms** to select an alternate line item view.

Related topics

Request Management overview
Request Management workflows and user tasks
Processes and Best Practices

1.10.2.6. Service Request Monitoring

The Service Request Monitoring process describes the activities to monitor all open Service Requests from initialization to resolution. Service Request Monitoring also determines whether action or escalation is required to meet the target resolution according to the associated SLA. For example, action is needed if requests are greater than 50% of SLA expired. Service Request Monitoring is an ongoing process performed by the Service Request Analyst and the Service Request Manager.



For more information on this process, refer to *Service Manager Processes and Best Practices*.

Access Request Management reports

Security Roles: Request Manager; Request Coordinator

The Reporting tool in Service Manager provides a number of out-of-box reports on the request data in your system. You can view these reports through the dashboards named **Request Overview (Global)**' ' and **Request KPI (Global)**. You can also create your own dashboards to display other reports of your interest.

To access Request Fulfillment reports, follow these steps:

1. Click **Request Management** > **Request Overview**.

   By default, the **Request Overview (Global)** dashboard is displayed.

2. View the reports on the dashboard. For descriptions of these reports, see Report descriptions and usage.

3. If needed, click the **New Dashboard** button to add your own dashboards. For details, see My dashboards.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

   Tip

   You can click **Export** to export the reports on a dashboard to PDF format.

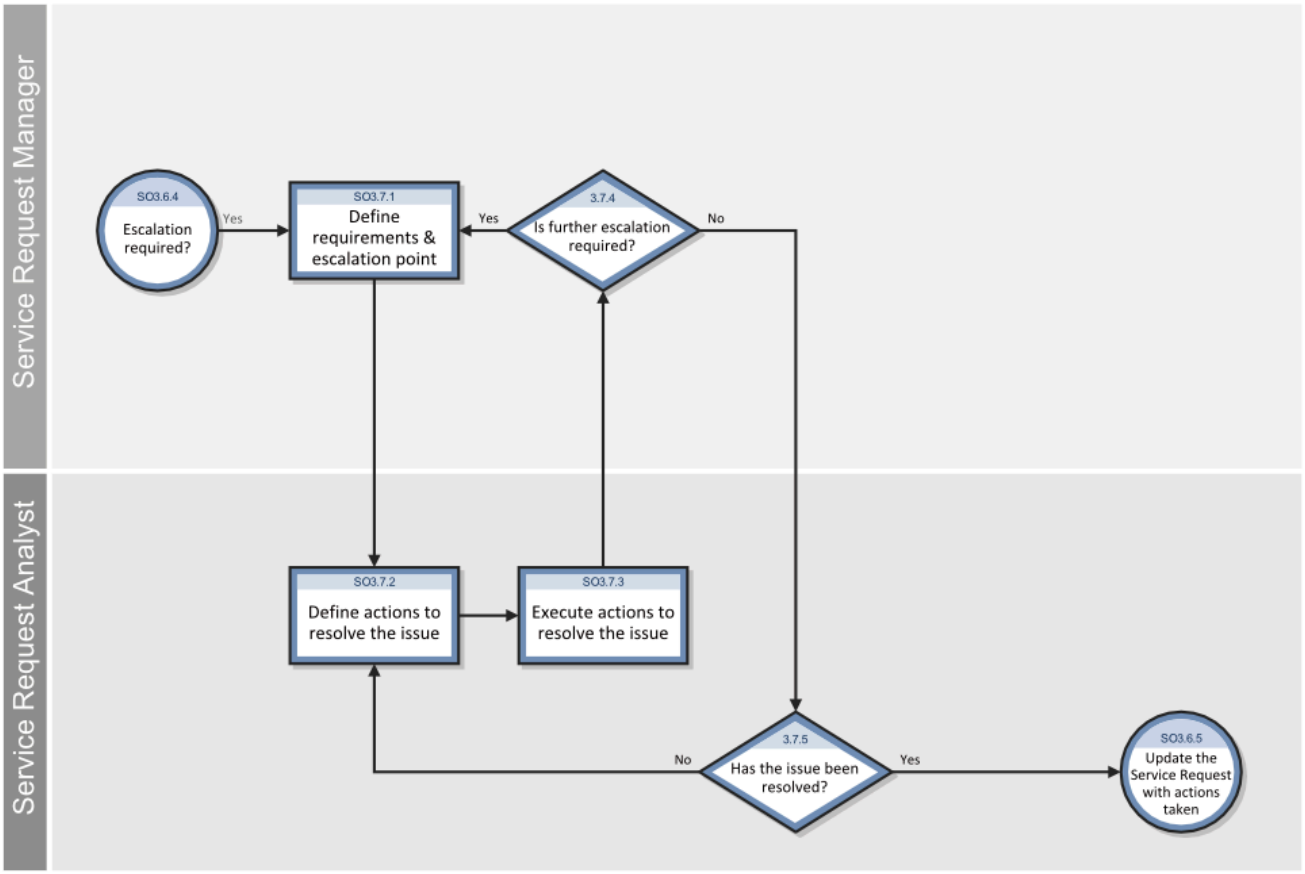
4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default dashboard.

Related topics

Request Management overview
Request Management workflows and user tasks
Processes and Best Practices

1.10.2.7. Service Request Escalation

When a Service Request Analyst reports to the Service Request Manager the action taken to resolve the issue with the Service Request, the manager determines whether escalation is needed. The Service Request Escalation process starts from the requirements and escalation point defined by the Service Request Manager, and the analyst takes care of defining actions that should be taken and action execution until the issue is resolved.



For more information on this process, refer to *Service Manager Processes and Best Practices*.

Related topics

Request Management overview
Request Management workflows and user tasks

1.10.3. Request Management administration

You can access Request Management administration from the Request Management application. You can configure the Request Management environment, add users, assign profiles, and ensure that individual users have rights that are appropriate for their roles

Related topics

Request Management overview
Request Management User Profile
Environment records
Categories
Posting
Posting

1.10.3.1. Security and access

Security in the Request Management process is essential. Most operators only need access to limited areas and items for creating quotes and ordering. Only certain operators should be allowed to approve quotes and orders, mark shipments as received, and respond to operators' requests for items. You can implement rules such as these with Request Management.

Request Management uses several files containing records on individual operators to secure the database and efficiently use quotes, orders, and line items.

Note: If there is a conflict in the settings for a particular operator, the User Profile record settings override the environment settings in Request Management. The category and phase record settings for the functional areas override the User Profile record settings. Ultimately, restrictions established in the category and phase definition records determine operator access privileges.

Related topics

Environment records
Setting up approvals

1.10.3.2. Request Management User Profile

Administrators assign user access privileges to enable Service Desk operators and others to use integrated Request Management functionality. The administrator often assigns these privileges by creating a user role and assigning that role to a group of users.

A typical user role would be Service Desk Agent. This role has indirect Request Management access. The typical privileges assigned to a user role would be requestor.

A Request Management user also has a profile that specifies security rights and approval groups.

The user profile controls the Request Management function access for users. The system administrator can assign the standard profile to a user, or establish user access with a unique profile.

The system administrator can specify user access for each of the three functional areas of Request Management or use one profile for access to all. If an operator has two profiles (such as one for the Quote area and another for All areas), the specific area profile takes precedence and overrides the All areas profile.

Generally, the standard user has minimal direct access to Request Management. This user has limited data views and limited options within the application. It is best to begin with a more restricted system and expand the options over time, rather than provide an open system and rescind permissions and accessibility to the product options at a later time.

Add a profile

Applies to User roles: System Administrator

Request Management profiles enable administrators to selectively assign access and security to users through the interaction of user roles and profiles. You can add Request Management profiles to the out-of-box profiles.

Note

If you're not sure what information is available in the fields on the tabs you select, you can click **Fill** to generate a record list of possible options, and then choose one.

To add a new profile:

1. Click **System Administration** > **Ongoing Maintenance** > **Profiles** > **Request Management Profiles**.
2. In the **Profile Name** field, type the name of the new profile.
3. Select the applicable rights, forms, and request options for the profile, and then click **Add**.
4. If you don't complete all required fields, a series of screens opens. Continue to modify the record to create your new operator and group profile. After you complete each screen, click **Save**.
5. Click **OK**.
6. To apply this and other operator profile record group updates to the applicable group definition records, click **More** or the More Actions icon and select **Rebuild Group**.
7. Click **Save** to save your changes.

To use an existing profile to add a new profile:

1. Click **System Administration** > **Ongoing Maintenance** > **Profiles** > **Request Management Profiles**.
2. Use search or advanced search to find one or more records.
3. Select a profile from the list that most closely matches the profile you want to add.
4. In the **Profile Name** field, type the name of the new profile.
5. Select the applicable rights, forms, and request options for the profile.
6. Continue to modify the record to create your new operator and group profile.
7. Click **Add**.

   Caution

   Make sure that you don't click **Save** because doing so will replace the existing profile with the new profile you are attempting to add.

8. Click **OK**.

9. To apply this and other operator profile record group updates to the applicable group definition records, right-click in the form (for Windows clients only) or click **More** to select **Rebuild Group**.
10. Click **Save** to save your changes.

Edit a profile

Applies to User roles: System Administrator

As the System Administrator, you can edit Request Management profiles. You can edit settings, such as the basic options, alert options, and approval options.

Note

You must set profile privileges separately for each record type.

To update a Request Management profile record:

1. Click **System Administration** > **Ongoing Maintenance** > **Profiles** > **Request Management Profiles**.
2. Add optional search criteria, and then click **Search**.

   A list of profile records opens.
3. Select a record from the list.
4. Review the restrictions for that profile and determine the exact restrictions or permissions you wish to enable for the user. Add or change the applicable information on the form.
5. When you are finished with your updates, click **Save**.
6. Click **OK**.

Related topics

Request Management overview
Add folder permissions to a security role

1.10.3.3. Environment records

Within Request Management, environment records define the overall operating environment and establish features for each functional area: quotes, line items, and orders. The environment records contain information specific to each operator, including the following:

- Access rights
- Numbering format and length
- Enabled/disabled alerts and events

Environment records are required to access each functional area within Request Management.

Edit environment records

The Request Management Environment record defines options that enable or disable Request Management application functionality for all Request Management users. The Request Management Environment record has default settings. However, you can configure these settings to meet your business needs.

To configure the Request Management environment settings for line items, orders, or quotes, do the following.

1. As a System Administrator, click the following to open application environment records:
    - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Line Item Environment**
    - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Order Environment**
    - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Quote Environment**

    Or click the following from Request Management:

    - **Request Management** > **Maintenance** > **Administration** > **Line Item Environment**
    - **Request Management** > **Maintenance** > **Administration** > **Order Environment**

    - **Request Management** > **Maintenance** > **Administration** > **Quote Environment**
2. Select new options or clear default options. Your changes redefine the Request Management environment for all users.
3. Determine the numbering format to establish for quotes, line items, or orders.
4. Decide on a separator character between your main quote or order number and the included line items or parts, such as Q1234-001 and O2345-100.
5. Determine the length of the unique number to be added (this will determine how often you may need to reset your numbers).
6. Stockroom option: When a line item has been completely received, select whether it should be marked as "In Stock" or closed.
7. When you have finished making your changes, click **Save** and **OK**.

Number controls

Each of the first three following options retrieves the unique sequential number according to the *ocmq*, *ocmo*, and *ocml* records in the number file. The number.reset application can be used to reset this unique number at the beginning of each year, or more often if necessary. These number controls determine how your quotes will be numbered.

Option	Description
Yr/Unique No (YYNNN)	The current year (system-calculated), followed by a unique sequential number.
Yr-Mo/Unique No (YYMMNNN)	The current year and current month (system-calculated), followed by a unique sequential number.
Julian/Unique No (YYDDDNNN)	The current Julian date (system-calculated), followed by a unique sequential number.
Unique Number Only	A unique number without prefixes or suffixes, retrieved from the number file (ocmq, ocml or ocmo classes). Separator causes a separator character to appear between your main quote or order number and the included line items or parts, such as, Q1234-001. Unique Number Length specifies the length of the unique number to be added (determines how often you may need to reset your numbers).
Parent No/Unique No (PPPNNN)	The Parent Quote or Order number, followed by a sequential number determined by the total.line.items field in the parent item. This option is only displayed in the line item environment record (*environment.ocml*).

Quote options

These options determine the overall behavior of quotes and how they will be processed.

Option	Description
Allow Access without Operator Profile	If true, operators without user profiles can access the area using the area's default profile.
Schedule Alerts?	Enables alerts to be turned on (true) or off (false) for the entire area, such as alerts added to the current alerts file.
Schedule Events?	Enables event processing to be turned on (true) or off (false) for the entire area, such as events added to the schedule file, or not. The default setting is false.
Operator ID/Full Name	Controls if the Operator's Login ID or the Full Name is recorded in the application's records. If true, the Operator ID is recorded; if false, the Full Name is recorded. The default is false. This information is recorded in the variable *ocm.ufname*.

Contents

- 1 Edit environment records
- 2 Number controls
- 3 Quote options

1.10.3.4. Number control reset application

Resetting the number file at a particular point may be required due to several scenarios, including the following:

- Heavy activity.
- Auditing requirements.
- Dates used as part of the numbering convention.

Number resetting is often performed on a monthly or annual basis, depending on your numbering scheme for that functional area.

The *number.reset* application is used to force the number file to be reset at a particular time. The application resets the number record to zero (0).

Important

Quote and Order line item numbers are stored in the same out-of-box Line Items file (*ocml*), distinguished by Q for quotes and O for orders. When you reset numbering, you need to take into account whether future records will conflict with current record numbers. The number must be a unique value, as defined in the database dictionary file for each of the *ocm* files.

A schedule record must be defined to run in any Service Manager background processor. This record should be set to execute the *number.reset* application.

Reset the number control application

Applies to User Roles: System Administrator

Resetting the number file at a particular point may be required due to heavy activity, audits, or dates used as part of the numbering convention.

Important

When you reset numbering, you need to take into account whether future records will conflict with current record numbers. The number must be a unique value, as defined in the database dictionary file for each of the `ocm` files.

To reset the number control application for the line item environment, order environment, or quote environment:

1. Click one of the following to open an application environment record:
   - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Line Item Environment**
   - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Order Environment**
   - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Quote Environment**
2. In the Number Controls section, make the necessary updates to reset the numbering in the application environment.
3. Click **Save**.
4. Click **OK**.

Contents

- 1 Reset the number control application

1.10.3.5. Model view

The model form is an alternate view of the model file. It displays the main part specifications and order fields, tied in with the current inventory, and provides control over when to reorder an item.

Add information to the model view

Applies to User roles: Request Administrator, System Administrator

The model form is an alternate view of the model file. It displays the main part specifications and order fields tied to the current inventory.

To add information to the model file:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Model**.

   A blank Request Management model form opens. Add information by creating a new record, or editing an existing record.

2. To edit the existing record:
   1. Use search or advanced search to find one or more records.
   2. Select a target record to edit.
   3. Edit the record.
   4. Click **Save** to save changes to an existing record.
3. To add a new record:
   1. Type the name of the **Model**.
   2. Add applicable information in the fields in the following tabs:

      Note

      If you're not sure what information is available in the fields on the tabs you select, see field help. To do this, place your cursor in the field, and then press Ctrl+H or select **Help on field** from the Help menu.

      - **General tab**: Stores information, such as the name of the file to post data to and the cost assigned to control the order of items that are presented for selection from the catalog when opening a line item.
      - **Current Quantities tab**: Stores stockroom information pertaining to quantities of items. For example, for serialized items, the number of

units in the device file listed as installed and for non-serialized items, the number of units that have been delivered over the life of the item.
- **Reorder tab**: Stores reorder information. For example, when an operator requests less than the amount entered in the Min Ord Amount field, then the amount the operator orders is adjusted upward to that amount.
- **Vendors tab**: Stores specific vendor/supplier information for a product.
- **Catalog tab**: Stores all the information about the inventories parts and items.
- **Software tab**: Stores software information, such as licensing and installation information.
- **Picture tab**: Insert an image into the model record. In a Windows client, right-click on the blank box,select **Add**, and then browse your directory to locate the image. In a Web client, click **Specify Image**, type the full path of the file in the **File location** text field or browse to the file, and then click **OK**.

3. Click **Add** to add a new record.
4. Click **OK**.

To use an existing model record to add a new model record:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Model**.

   A blank Request Management model form opens.

2. Click **Search** to open a record list.
3. Select a record from the list to add a new model record.
4. Clear the **Model** field and type a new model name.
5. Add or change information on the form, in each of the fields in all the tabs, to select the applicable information for the new model record.
6. Click **Add** to add the new record.

   Caution

   Make sure that you don't click **Save** because doing so will replace the existing model with the new model you are attempting to add.

7. Click **OK**.

Related topics

Catalog view

1.10.3.6. Catalog view

The *ocmco* (the co stands for component) or *ocmo.detail* form is a way to view the catalog. It displays the runtime processing control options for each part and the relationship of one part to another. Each item has its own unique component definitions.

Related topics

Model view
Catalog operations

1.10.3.7. Catalog operations

Creating the catalog is the first step in the request process and is one of the most important parts of the Request Management setup process.

The catalog includes three primary files: the *model*, *modelvendor*, and *vendor* files. The model table is a list of items and bundled items available for request, the vendor file is a list of vendors, and the modelvendor table connects the two.

An administrator with rights to create catalog items configures the system, and most operators have rights to access existing catalogs only. Operators don't have editorial control; they can't add items to the catalog that aren't already defined.

Model table

The *model* table is a detailed bill of materials for a part. This includes the hardware, software and services that comprise the item, as well as controls on the part and its components. The table contains information regarding what operators can do with the part, how components of this part are selected, how quote line items turn into orders, and whether to order or to consume from stock. Both the *model.g* form and the *ocmco.g* form can be used to display data contained in the model table.

The catalog defines an item’s components and their order generation dependencies. Controlling the order generation dependencies includes the purchase requisition, service, and work orders necessary to obtain the item.

The *Type: Category* field controls the grouping of similar Catalog items under a common name.

The model file defines the components an operator may order, using the Unique key of Part Number to identify items. It also contains all associated component relationships within the catalog.

The model record contains:

- The rules for processing a component when it is a line item as part of a quote or order
- References to component parts included with a parent part
- The names of approval definitions that a component must have when it is added to a quote or an order
- If a part has components, the rules for selecting the dependent components
- The status of quantities of a component: in use, available, on order

The information in the model record is copied to each line item in a quote.

You can view the model file in two different screens:

- *ocmco* (the catalog view)
- *model* (the model view)

Modelvendor table

The `modelvendor` table combines certain information from the `model` and `vendor` tables, such as the vendor's cost, lead time, time zone, and payment schedules.

Vendor table

The vendor table is shared throughout Service Manager as the source of manufacturer and retail vendor/supplier information. It also holds data on internal and external service providers. A vendor record for a particular manufacturer must exist before items from the manufacturer can be admitted to the Configuration Management and Request Management catalogs.

The vendor table provides contact and services information for a particular vendor/suppler.

Related topics

Pre-implementation planning

1.10.3.8. Macros

There are no specific macros for Request Management. The standard macro options are available through the macro definitions (*macro.definitions.g*) form. You can set these up to create actions from within Request Management. The standard options include the following:

- Fax 1 Assignment Group
- Fax 1 Person
- Fax Many People
- Mail 1 Assignment Group
- Mail 1 Person
- Mail Many People
- Page 1 Person
- Page Many People
- Page an Assignment Group

Applies to User roles: System Administrator

Macro conditions are expressions written in Service Manager RAD syntax that evaluate at run time. If the expression evaluates to true, the macro executes. If the expression doesn't evaluate to true, the macro can't proceed.

To access the macro lister and define a macro to use in Request Management:

1. Click **Tailoring** > **Tailoring Tools** > **Macros**.
2. Click **Add**.

   The macro editor opens. You can define a macro to use in Request Management.

3. In the **Macro Name** field, type a unique name for this new macro.
4. In the **Applies When** field, select a pre-defined event from the list of options when this macro should execute. For example, when a change request is saved.
5. In the **Macro Condition** field, define when this macro should execute. For example, to enter an expression for the condition when a service request is opened you can type `requested in $L.new="open"`.
6. Click **Set Parameters** to establish the parameters for this macro.

   Note: The available fields in this form vary depending on the value in the Macro Type field in the edit form for your new macro.

7. Provide additional information where needed. For example, send page to a specific phone or contact and construct a message.
8. Click **OK**.

   You are returned to the macro list form.
9. Click **Search** from the macro list to refresh the list of macros, and then click **OK**.

Related topics

Macro conditions
Definitions for macro forms

1.10.3.9. Posting

Posting is the process of copying data from a source record to a target record. The purpose of posting is to update similar fields in other records without having to open those records to modify each field.

A link record defined to support the posting process copies data from orders (source records) to quotes (target records). The Comment fields must start with the word *POST*.

Posting to a new Configuration Item (CI) record is done automatically for models that have a configuration file defined. Further posting of actions can occur, but requires system tailoring.

If the background posting process can't lock records, the posting is rescheduled for a certain time in the future (for a maximum number of times). If the risk of two users accessing and modifying a record simultaneously is unacceptable, consider alternatives for posting to records that need to be locked.

The process of posting is covered in the Format Control section of the Tailoring Help. In Request Management, the master Format Control records (*ocmq, ocmo, or ocml*) execute first; then the category Format Control (name=phase definition's default view). These Format Control records can call the *post.fc* application.

See the Tailoring Help for more information about posting and creating link records.

Post data using a link record

Applies to User Roles: System Administrator

One of the advantages of a relational database is the elimination of redundant information. This is accomplished by storing information about a particular subject in one place, or file, with links to other subjects. Links are a combination of data and link definitions, sets of conditions containing relationships for linked information.

The posting process requires you to create a standard Service Manager link record. The link record establishes the connection between similar fields in the source and target files.

To post data using a link record:

1. Click **Tailoring** > **Tailoring Tools** > **Links**.

   A blank Link File form opens.

2. Type the **Name** of the new link record you wish to add.
3. Click **New**.

   You receive a message that states the link record has been added.
4. To finish adding the record, type the following values in the first line of the new link record. Place your cursor in each of the following fields to enter the values for each.

Field	Value
Source Field Name	requested.by
Target File Name	contacts
Target Field Name	contact.name
Comments	POST

   Important: You must enter POST in upper case letters in the Comments field for the posting process to work.
5. Click **Save** to update the link record.
6. Click **OK**.

Related topics

Format Control
Link maintenance

1.10.3.10. Alerts, events, and messages

Events are system occurrences, triggered by user activity or certain conditions, that require additional user action. Examples of events in Request Management include open, update, close, and approval (of quotes).

Events in Request Management send messages to designated parties (operators or groups) within the system. For example, messages indicate that someone opened a quote and start the approval process by notifying an approver.

If a time limit set for accomplishing an approval is not met, an alert runs. An alert is an optional time-delayed event, which causes another event to send out a message.

As soon as a request is approved, that action constitutes an event. The event sends a message that indicates the state of the request.

Events, alerts, and messages build a chain of communication, notifying users of pending quotes and orders and the status of requested items throughout their life cycles.

Note: Setting up alerts, events, messages, and approvals requires the operation of the *ocm* background scheduler. The scheduler must be included in the system start-up schedule record. See Background processing and Generating orders for help in setting up these processes.

Related topics

Events and messages
Alerts

1.10.3.10.1. Alerts

Quotes, orders, and line items progress in phases according to a predefined schedule. Alerts monitor the progress of these phases and take action when approvals are overdue. For example, the late notice alert notifies a designated management group that a quote line item(s) is overdue for approval, and updates the alert status to include late notice.

The user can define any number of standard or customized alerts for any phase; control who is notified for each alert; and control the naming convention used for the alert itself.

Alerts support several functions within the system.

Function	Description
Alert Messaging	Alerts trigger events. The event manager generates messages to certain designated recipients as a result of an alert. The messages update the original request.
Batch Scheduling	All alerts associated with a phase are scheduled at once when the phase opens.

Alert processing

There are two primary files used in alert processing.

File Name	Description
Alert Definition (AlertDef)	Used by all phases to define the basic alert information for each named alert and and all general alert definitions (static file)
Current Alerts (Alerts)	Tracks the alerts created for each phase (active alert file)

Alert definitions

Quotes, orders, and line items progress in phases according to a predefined schedule. Alerts monitor the progress of these phases and take action when approvals are overdue. The Alert Definition is a static file that defines the basic alert information for each named alert and all general alert definitions.

View an alert definition record

Applies to User roles: System Administrator

To view alert definition records for use in the Change Management, Problem Management, Request Management, or Service Level Management applications, follow these steps:

1. Select one of the following paths:
    - Change Management: Click **Change Management** > **Maintenance** > **Alerts**.
    - Problem Management: Click **Problem Management** > **Administration** > **Alert Definitions**.
    - Request Management: Click **Request Management** > **Maintenance** > **Supporting Files** > **Alert Definitions**.
    - Service Level Management: Click **Service Level Management** > **Configuration** > **Alert Definitions**.
2. Type or select optional search criteria.
3. Click **Search**.
4. Click any record in the list to view the alert details.
5. You can add to or change information in the form.

Tip

If necessary, press **Ctrl+H** in the Windows client or press **F1** in the web client for each field.

6. If you make changes, click **Save**.

Alert log file

Quotes, orders, and line items progress in phases according to a predefined schedule. Alerts monitor the progress of these phases and take action when

approvals are overdue. The Alert log is a file that lists currently scheduled and active alerts.

Access the alert log file

Applies to User Roles: System Administrator

Quotes, orders, and line items progress in phases according to a predefined schedule. Alerts monitor the progress of these phases and take action when approvals are overdue.

To access the alert log file:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Alert Logs**. A blank search alert log form opens.
2. Add optional search criteria, and then click **Search**.

   A list of all current alert log records opens.
3. Click a record in the list to access the record.

Access the alert log when viewing a quote, line item, or order record

Applies to User Roles: Request Coordinator, Request Administrator, System Administrator

The alert log is a file that lists currently scheduled and active alerts.

To access the alert log when viewing a quote, line item, or order record:

1. Click one of the following:
   - **Request Management** > **Quotes** > **Quote Queue**
   - **Request Management** > **Line Items** > **Line Item Queue**
   - **Request Management** > **Orders** > **Order Queue**
2. Click **Search** in the request queue form.
3. Add optional search criteria, and then click **Search** to generate a list of all current records.
4. Select a record in the list that you want to view.
5. Click **More** or the More Actions icon, and then choose **Logs** > **Alert Log** to view the currently scheduled and active alerts.

Access phase definition alert controls

User roles: Request Coordinator, Request Administrator, System Administrator

You can access phase definition alert controls in Request Management for quotes, line items, or orders.

To access phase definition alert controls for quotes, line items, or orders:

1. Click one of the following:
   - **Request Management** > **Quotes** > **Quote Phases**
   - **Request Management** > **Line Items** > **Line Item Phases**
   - **Request Management** > **Orders** > **Order Phases**
2. Add optional search criteria, and then click **Search**.

   A list of phase records opens.
3. Click a phase name to open it in the form.
4. Click the **Alerts** tab. The phase record specifies certain alert controls. Alert controls on the phase record form include the following:

Control	Description
Reset	Sets the status of all current alert records associated with the current quote or order to inactive and marks the last action field as reset. Then it schedules a calculate alert record to recalculate the item's alerts and restart the alerts process.
Reeval	Retrieves each Alert associated with the quote or order and performs the following processing. If current alert status is active, Service Manager reevaluates the alert condition and updates the alert to reflect the correct status. If the alert status is not active, Service Manager reevaluates the alert condition. If this condition is true, Service Manager: 1. Sets the status to Scheduled. 2. Sets the last action to Recalc. 3. Sets the action time to the current date/time. 4. Reevaluates the Schedule Condition. If this condition is true, Service Manager recalculates the Alert Time and updates the status to Scheduled. If this condition is false, Service Manager sets the status to Not Required.

Related topics

Events and messages
Environment records
Alerts, events, and messages

Notifications and alerts
Pre-implementation planning

1.10.3.10.2. Events

Events are system occurrences triggered by the creation or update of quotes, orders, and line items. The following are some examples.

- Opening a quote
- Approving a quote
- Changing the dollar amount of an order

You can use events to trigger special processing, such as alerts and messages. When these specific activities occur, Request Management sends mail messages to users, as part of the default processing. You can run other customized routines for particular events.

Several default events are included with Request Management. Others may be added, according to your business needs (for example, activities or conditions that need to be checked or unique events warranting notification).

When an Alert Condition evaluates to true, Request Management treats it as an event, and notifications are sent for this alert condition.

Related topics

Alerts, events, and messages
Request Management event controls
Logical actions

1.10.3.10.2.1. Request Management event services

Increasingly, enterprise-wide network management tools depend on automation to detect activity on the network and to execute the appropriate procedures. These network incidents are often called alarms or alerts; Service Manager refers to them as events.

Service Manager includes Event Services enhancements in Request Management to support external event processing.

1.10.3.10.2.2. Request Management event controls

The option to schedule events is on the quote, order, and line item environment records (*environment.ocmq*, *.ocmo*, or *.ocml*). For information on editing the environment records, see the related topics.

The Schedule Events? field in the Quote Options structure sets when events are processed for categories and phases.

- If this checkbox is unchecked (false), events aren't scheduled for processing or recorded in the message log, *msglog* file.
- If this checkbox is checked (true), and Messages/Events? in the phase record (*ocmoptions*) is false, the event is scheduled and recorded in the *msglog*, but no other processing occurs.

The option for processing messages and events is located on the quote, order, and line item Phase definition (*ocmoptions*) record. See Phases for details on how to access and create phase definitions.

The Messages/Events field in the Controls tab sets the controls that define when events are processed for the particular phases named.

All event names must be defined in the *ocmevents* file, or no event processing can occur. See Events for more details.

Events are processed when the phase of the quote, order, or line item is defined in the Phases field of the *ocmevents* file. If the Phase field is empty on the *ocmevents* record, the event is processed for all phases.

The *Exec Cond* option in the *ocmevents* record controls, at the event level, when an event is processed.

1.10.3.10.2.3. Request Management events table

Enterprise-wide network management tools depend on automation to detect activity on the network and to execute the necessary procedures. Service Manager refers to these network incidents as events. The Request Management (*ocmevents*) file contains the names and definitions of all valid Request Management events.

1.10.3.10.2.4. Defining additional events

A new event can be defined and called by Format Control calculations, in the case that a specific condition must be checked for the event to occur at specialized times.

The syntax generally used to check for an event is as follows.

```
(x=o, q, or l): if (condition=true) then ($ocmx.events.pntr in
$ocmx.events=“event name”;$ocmx.events.pntr+=1)
```

- The variable *$ocmx.events* (where x = the area q, o, or l) is an array of character strings used to track the events that occur during a particular phase of processing.
- The variable *$ocmx.events.pntr* is a pointer to the next array element that can be used to record an event name.

Important: Once an event has been added to the array, it is important to increment the pointer by 1 (one). If this doesn't happen, the event previously recorded will be overwritten.

The event that's scheduled if the condition is true must be defined in the *ocmevents* file.

1.10.3.10.2.5. Process external events from SCAuto

If you have an integration, you can use the method described in the *ServiceCenter Automate (SCAuto) Software Development Kit (SDK)* to send events via email. However, as defined below, the event can be configured as an open, approval, denial, or any of the other accepted events in the system. Use the method described in the above guide for manipulating the data and placing it in the eventin table.

Service Manager applications have been set up to support incoming events for Request Management. The supported events are as follows.

Event	Description
rmqin	Supports Open, Update, and Close-phase of Quotes.
rmlin	Supports Open, Update, and Close-phase of Line Items.
rmoin	Supports Open, Update, and Close-phase of Orders.

With these three events, it is expected that the event will include the following initial data elements.

`number^operator^action`

Data Element	Description
Number	The item to work on.
Operator	Role to work as.
Action	What to do (open/update/close).

This event is also supported, with different initial data elements.

Event	Description
approval	Supports Approval actions on Quotes (Approve, Deny, and Retract).

The following initial data elements are included in the approval event.

`area^number^group^action^date^operator^comments`

Data Element	Description
Area	Quote (ocmq), order (ocmo), or line item (ocml).
Number	The item to work on.
Group	The Approval group.
Action	What to do (approve/deny/retract).
Date	Date of the transaction.
Operator	Role to work as.
Comments	Extra information, such as for billing or shipping.

Related topics

Events

1.10.3.10.2.6. Events and messages

The system can send messages to operators when events occur, such as the following:

- Opening a quote
- Approving a request
- Opening a line item
- Closing a phase

Event records are defined in the *ocmevents* file. These records indicate what message is sent when the event occurs. See Alerts, events, and messages for more information on events and messages.

Related topics

Approvals in Request Management
Setting up approvals
Alerts, events, and messages
Messages

1.10.3.10.2.7. Add new events

Applies to User Roles: System Administrator

Events in Request Management send messages to designated parties (operators or groups) within the system. For example, messages indicate that someone opened a quote and start the approval process by notifying an approver.

To add a new event:

A. Set the environment record to schedule events:

1. Click the following to edit the line item, order, and quote environment records.
    - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Line Item Environment**.
    - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Order Environment**.
    - **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Request Management Quote Environment**.

      The Request Management application environment form opens.
2. For each environment record, check the **Schedule Events?** field to set to true.
3. Click **Save**.
4. Click **OK**.

B. Define the Messages/Events option for the phase.

1. Click the following to edit the phase for line items, orders, and quotes.
    - **Request Management** > **Line Items** > **Line Item Phases**.
    - **Request Management** > **Orders** > **Order Phases**.
    - **Request Management** > **Quotes** > **Quote Phases**.
2. Click **Search** to open a record list.
3. Edit the messages and events field.
    - For line item phases: Select the Definition tab of the Controls section, and then type **true** in the **Msg/Events** field.
    - For order phases: Select the Definition tab of the Controls section, and then type **true** in the **Msg/Events** field.
    - For quote phases: Select the Controls, and then type **true** in the **Messages/Events** field.
4. Click **Save**.
5. Click **OK**.

C. Add the new event.

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Events**.

   A blank Request Management events form opens.

2. Type the name of the **Event**.
3. In the **Area** field, select from Quotes, Orders, Line Items, or All to designate the area for the event.
4. Type a **Description** of the event.
5. Continue to define the new event by completing the fields on the form. For example, type the designated operators or groups to send messages to.
6. Click **Add** to add a new record.

To use an existing event record to create a new event:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Events**.

   A blank Request Management events form opens.

2. Click **Search** to open a record list.
3. Select a record from the list to create a new event.
4. Clear the **Event** field and type a new event name.
5. Add or change information on the form to select the applicable information for the new event record.

6. Click **Add** to add the new record.

   Caution: Make sure that you don't click Save because doing so will replace the existing event with the new event you are attempting to add.

7. Click **OK**.

Related topics

Phases
Format Control
Request Management event controls
Defining additional events

1.10.3.10.2.8. Create a background event record

To create a background event record:

1. Click **Request Management** > **Maintenance** > **Administration** > **Backend Order Destinations**.
2. Start typing the appropriate values, or click **Search** in the toolbar to open a list of existing backend event records.
3. Click the name of an event on which you would like to base your new backend event.
4. Type a new backend event Name.
5. Modify the record to create your new backend event configuration record.
6. Click **Add** to add it to the *ocmbackend* file.

1.10.3.10.2.9. Event names and definitions

Request Management includes several predefined system events. You can define additional events as needed. The following partial record list shows some of the most commonly used predefined system events, and to which Request Management areas they apply.

Commonly Used Predefined System Events

The following table describes predefined system events.

Events	Area
alerts recalc	o (orders)
alerts recalc	q (quotes)
alerts reset	o
alerts reset	q
approval	o
approval	q
approval reeval	o
approval reeval	q
approval reset	o
approval reset	q
approved	o
approved	q
close category change	o
close - category change	q
close - partial received	l (line items)

Default List of System Event Names and Definitions

The following is the default list of system event names and definitions. (Some apply to certain areas, and others apply to all areas.)

Event Name	Description
alerts recalc	The quote or order’s alerts have been recalculated.
alerts reset	The quote or order’s alerts have been reset.
approval	An operator has approved one of the approval requirements of an item.
approval reeval	The approval requirements of a quote/order have been reevaluated.
approval reset	The approval requirements of a quote/order have been reset.
approved	All approval requirements for an item have been approved.
close - category change	The category of a quote/order has been manually changed.
close - partially received	A line item has been closed, but not all of the ordered quantities have been received.
close - phase change	The phase of an item has been closed due to a manual phase change.
close -- received	A line item has been fully received; or, all of the line items of an order have been fully received.
close final	The final phase of an item is complete. The item is closed.
close phase	An intermediary phase of an item is complete, but the item is still open.

Event Name	Description
copy&open	An item (quote, order, or line item) has been copied to a new item.
disapproval	One of an item's approval requirements has been denied.
disapproved	The quote/order is denied based on the Approval Status rules.
drop avail	A line item has been manually dropped from the available to order status.
due date notice	This contains sample system data.
gen order delayed	The background generate orders process has been delayed due to a posting process that has not yet completed. The generate orders schedule record has been updated to run at its next normally scheduled time.
late notice	This contains sample system data.
mark avail	A line item has been made available for order processing.
open	An item has been opened.
open next	The next phase of an item has been opened.
open next - category change	A new phase has been started due to a manual category change.
open next - phase change	The next phase has been started due to a manual phase change.

Event Name	Description
pending vendor - select	A line item is pending the selection/assignment of a vendor and can't be processed until the vendor is assigned.
post bg - error	The line item that was targeted for posting could not be found.
post bg - excessive resched	A background posting schedule record has been rescheduled more than three times.
post bg - process error	During background posting of line item data, an unknown processing error occurred.
post bg - quote not found	During background posting, the parent quote could not be found; no data was posted.
post bg - rescheduled	The parent quote/order could not be locked.
posting not complete	When quantity > 0, and Received + BO'd + Cancel + Returned = 0 (the fields on ocml.receive.post format)
reeval	For approvals or alerts.
reopen	A closed item has been reopened.
reset	For approvals and alerts.

Event Name	Description
shortage	The OCM Order Create background processor attempted to process a quote line item; there is insufficient available inventory to satisfy the requirement of the line item. There are no open orders for this Part.
sap order pricing	SAP associates and verifies prices for individual items to be ordered.
sap purchase req	SAP processed Service Manager request and queries for items available to order.
sap quote pricing	SAP associates prices with items that are displayed on the Service Manager quote.
unapproval	An operator has retracted one of the approval requirements of an item.
update	An item has been updated.
waiting	The OCM Order Create Background processor attempted to process a quote line item; there is insufficient available inventory, but there is already an outstanding order for the Part. Waiting on receipt of item.

Related topics

Request Management event services
Pre-implementation planning

1.10.3.10.2.10. Example event string start sequences

To approve a quote (such as number Q1234), acting as SYSTEM ADMINISTRATOR, within the security group management, send an *Approval* event with this string.

```
ocmq^Q1234^^approve^02/20/01 06:00^System.Admin^ok
```

To update a line item (such as L78), acting as a service desk agent, send an *rmlin* event with this string.

```
L78^Servicedesk.Agent^update^.....
```

To move an order (such as O89) to its next phase, acting as request administrator Rachel Boudreau, send an *rmoin* event with this string.

```
O89^Rachel.Boudreau^close^.....
```

Related topics

Pre-implementation planning
Add new events

1.10.3.10.2.11. External event services

The External Event Services enhancements include the following events.

Area	Action	Description
Quote	Approve	Approve a quote
Quote	Deny	Deny a quote
Quote	Retract	Retract a quote
Quote	Open	Create a new quote
Quote	Update	Update an existing quote
Quote	Close	Close current phase and go to next if exists
Order	Approve	Approve an order
Order	Deny	Deny an order
Order	Retract	Retract an order
Order	Open	Create a new order
Order	Update	Update an existing order
Order	Close	Close current phase and go to next if exists
Line Item	Open	Create a new line item
Line Item	Update	Update an existing line item
Line Item	Close	Close current phase and go to next, if exists

Related topics

Processing external events from SCAuto
Pre-implementation planning
Line item events
Order events
Quote events
Add new events

1.10.3.10.2.12. Line item events

The following tables describe line item events.

Map Name	Sequence	Position	File Name	Field
rmoappr	1	1	ocmo	number
rmoappr	1	2	ocmo	$L.void.operator
rmoappr	1	3	ocmo	$L.void.action
rmoappr	1	4	ocmo	$L.void.approval.action
rmoappr	1	5	ocmo	$L.void.approval.group
rmoappr	1	6	ocmo	open
rmoappr	1	7	ocmo	page
rmoappr	1	8	ocmo	category
rmoappr	1	9	ocmo	current.phase
rmoappr	1	10	ocmo	phase.num
rmoappr	1	11	ocmo	phase.start.date
rmoappr	1	12	ocmo	orig.operator
rmoappr	1	13	ocmo	last.operator
rmoappr	1	14	ocmo	impact
rmoappr	1	15	ocmo	total.line.items
Map Name	Sequence	Position	File Name	Field
rmoappr	1	16	ocmo	vendor

Map Name	Sequence	Position	File Name	Field
rmoappr	1	17	ocmo	alert
rmoappr	1	18	ocmo	alert.names
rmoappr	1	19	ocmo	approval.status
rmoappr	1	20	ocmo	pending.groups
rmoappr	1	21	ocmo	status
rmoappr	1	22	ocmo	priority
rmoappr	1	23	ocmo	reason
rmoappr	1	24	ocmo	business.area
rmoappr	1	25	ocmo	submit.date
rmoappr	1	26	ocmo	update.date
rmoappr	1	27	ocmo	close.date
rmoappr	1	28	ocmo	coordinator
rmoappr	1	29	ocmo	coord.dept
rmoappr	1	30	ocmo	total
Map Name	Sequence	Position	File Name	Field
rmoappr	1	31	ocmo	description
rmoappr	1	32	ocmo	justification
rmoappr	1	33	ocmo	comments
rmoappr	1	34	ocmo	invoice.no

Map Name	Sequence	Position	File Name	Field
rmoappr	1	35	ocmo	vendor.contact
rmoappr	1	36	ocmo	shipping.terms
rmoappr	1	37	ocmo	po.number
rmoappr	1	38	ocmo	completion.code
rmoappr	1	39	ocmo	completion.notes
rmoappr	1	40	ocmo	net.total
rmoappr	1	41	ocmo	tax.code
rmoappr	1	42	ocmo	tax.rate
rmoappr	1	43	ocmo	tax.amount
rmoappr	1	44	ocmo	ship.to.code
rmoappr	1	45	ocmo	vendor.zip
Map Name	Sequence	Position	File Name	Field
rmoappr	1	46	ocmo	bill.to.code
rmoappr	1	47	ocmo	shipping.carrier
rmoappr	1	48	ocmo	freight.on.board
rmoappr	1	49	ocmo	payment.terms

Map Name	Sequence	Position	File Name	Field
rmoappr	1	50	ocmo	freight.charges
rmoappr	1	51	ocmo	vendor.contract.no
rmoappr	1	52	ocmo	discount.rate
rmoappr	1	53	ocmo	discount.amt
rmoappr	1	54	ocmo	ship.to.ext
rmoappr	1	55	ocmo	bill.to.ext
rmoappr	1	56	ocmo	phase.desc
rmoappr	1	57	ocmo	project.id
rmoappr	1	58	ocmo	target.date
rmoappr	1	59	ocmo	shipping.terms
rmoappr	1	60	ocmo	orig.manager.group
rmoappr	1	61	ocmo	parts.summary
rmoappr	1	62	ocmo	foreign.id

Map Name	Sequence	Position	File Name	Field
rml	1	1	ocml	number
rml	1	2	ocml	$L.void.operator
rml	1	3	ocml	$L.void.action
rml	1	4	ocml	page
rml	1	5	ocml	phase.num
rml	1	6	ocml	phase.desc
rml	1	7	ocml	category
rml	1	8	ocml	current.phase
rml	1	9	ocml	status
rml	1	10	ocml	parent.quote
rml	1	11	ocml	parent.order
rml	1	12	ocml	parent.line.item
rml	1	13	ocml	gen.ord.li
rml	1	14	ocml	project.id
rml	1	15	ocml	alert
Map Name	Sequence	Position	File Name	Field
rml	1	16	ocml	alert.names
rml	1	17	ocml	orig.operator

Map Name	Sequence	Position	File Name	Field
rml	1	18	ocml	last.operator
rml	1	19	ocml	trans.type
rml	1	20	ocml	open
rml	1	21	ocml	avail.to.order
rml	1	22	ocml	business.area
rml	1	23	ocml	receiving.priority
rml	1	24	ocml	track.receiving
rml	1	25	ocml	consolidate
rml	1	26	ocml	combine
rml	1	27	ocml	deffered.select
rml	1	28	ocml	generate.order
rml	1	29	ocml	serialized
rml	1	30	ocml	reorder.type
Map Name	Sequence	Position	File Name	Field
rml	1	31	ocml	submit.date
rml	1	32	ocml	phase.start.date
rml	1	33	ocml	update.date

Map Name	Sequence	Position	File Name	Field
rml	1	34	ocml	close.date
rml	1	35	ocml	ordered.date
rml	1	36	ocml	received.date
rml	1	37	ocml	returned.date
rml	1	38	ocml	backorder.date
rml	1	39	ocml	cancelled.date
rml	1	40	ocml	requested.date
rml	1	41	ocml	projected.date
rml	1	42	ocml	target.date
rml	1	43	ocml	actual.date
rml	1	44	ocml	requested.lead.time
rml	1	45	ocml	target.lead.time
Map Name	Sequence	Position	File Name	Field
rml	1	46	ocml	normal.lead.time
rml	1	47	ocml	actual.lead.time
rml	1	48	ocml	duty.table

Map Name	Sequence	Position	File Name	Field
rml	1	49	ocml	coordinator
rml	1	50	ocml	coord.dept
rml	1	51	ocml	assigned.to
rml	1	52	ocml	assigned.dept
rml	1	53	ocml	quantity
rml	1	54	ocml	quantity.returned
rml	1	55	ocml	quantity.received
rml	1	56	ocml	quantity.bo
rml	1	57	ocml	quantity.cancelled
rml	1	58	ocml	quantity.balance
rml	1	59	ocml	from.stock
rml	1	60	ocml	description
Map Name	Sequence	Position	File Name	Field
rml	1	61	ocml	comments
rml	1	62	ocml	ship.to.code
rml	1	63	ocml	ship.to.ext
rml	1	64	ocml	bill.to.code
rml	1	65	ocml	bill.to.ext

Map Name	Sequence	Position	File Name	Field
rml	1	66	ocml	serial.no.
rml	1	67	ocml	shipping.terms
rml	1	68	ocml	completion.code
rml	1	69	ocml	completion.notes
rml	1	70	ocml	part.no
rml	1	71	ocml	part.desc
rml	1	72	ocml	manufacturer
rml	1	73	ocml	model
rml	1	74	ocml	model.ext
rml	1	75	ocml	unit.of.measure
Map Name	Sequence	Position	File Name	Field
rml	1	76	ocml	level
rml	1	77	ocml	sequence
rml	1	78	ocml	neg.level
rml	1	79	ocml	neg.seq
rml	1	80	ocml	unit.cost
rml	1	81	ocml	labor.hours
rml	1	82	ocml	labor.cost

Map Name	Sequence	Position	File Name	Field
rml	1	83	ocml	labor.overtime.hours
rml	1	84	ocml	labor.overtime.cost
rml	1	85	ocml	tax.code
rml	1	86	ocml	tax.rate
rml	1	87	ocml	tax.amount
rml	1	88	ocml	modelv.no
rml	1	89	ocml	vendor
rml	1	90	ocml	vendor.part.no
Map Name	Sequence	Position	File Name	Field
rml	1	91	ocml	vendor.contract.no
rml	1	92	ocml	vendor.contact
rml	1	93	ocml	vendor.dt
rml	1	94	ocml	payment.freq
rml	1	95	ocml	payment.terms
rml	1	96	ocml	no.of.payments
rml	1	97	ocml	payment.amount

Map Name	Sequence	Position	File Name	Field
rml	1	98	ocml	discount
rml	1	99	ocml	discount.amt
rml	1	100	ocml	total
rml	1	101	ocml	warehouse
rml	1	102	ocml	logical.name
rml	1	103	ocml	rma.no
rml	1	104	ocml	asset.no
rml	1	105	ocml	current.location
Map Name	Sequence	Position	File Name	Field
rml	1	106	ocml	warranty
rml	1	107	ocml	request.number
rml	1	108	ocml	cm3.access
rml	1	109	ocml	target.completion
rml	1	110	ocml	target.order
rml	1	111	ocml	actual.order.lead
rml	1	112	ocml	foreign.id
rml	1	113	ocml	contact.name
rml	1	114	ocml	location

Map Name	Sequence	Position	File Name	Field
rml	1	115	ocml	license.number

Related topics

Pre-implementation planning
External event services
Order events
Quote events
Link records
Example event string start sequences
Add new events

1.10.3.10.2.13. Quote events

The following table describes quote events.

Map Name	Sequence	Position	File Name	Field
rmq	1	1	ocmq	number
rmq	1	2	ocmq	$L.void.operator
rmq	1	3	ocmq	$L.void.action
rmq	1	4	ocmq	open
rmq	1	5	ocmq	page
rmq	1	6	ocmq	category
rmq	1	7	ocmq	current.phase
rmq	1	8	ocmq	phase.num
rmq	1	9	ocmq	phase.start.date
rmq	1	10	ocmq	orig.operator
rmq	1	11	ocmq	last.operator
rmq	1	12	ocmq	impact
rmq	1	13	ocmq	total.line.items
rmq	1	14	ocmq	alert
rmq	1	15	ocmq	alert.names
Map Name	Sequence	Position	File Name	Field
rmq	1	16	ocmq	approval.status

Map Name	Sequence	Position	File Name	Field
rmq	1	17	ocmq	pending.groups
rmq	1	18	ocmq	status
rmq	1	19	ocmq	priority
rmq	1	20	ocmq	reason
rmq	1	21	ocmq	business.area
rmq	1	22	ocmq	submit.date
rmq	1	23	ocmq	update.date
rmq	1	24	ocmq	close.date
rmq	1	25	ocmq	cancelled.date
rmq	1	26	ocmq	requestor.name
rmq	1	27	ocmq	requestor.dept
rmq	1	28	ocmq	coordinator
rmq	1	29	ocmq	coord.dept
rmq	1	30	ocmq	assigned.to
Map Name	Sequence	Position	File Name	Field
rmq	1	31	ocmq	assigned.dept
rmq	1	32	ocmq	description

Map Name	Sequence	Position	File Name	Field
rmq	1	33	ocmq	justification
rmq	1	34	ocmq	comments
rmq	1	35	ocmq	ship.to.code
rmq	1	36	ocmq	bill.to.code
rmq	1	37	ocmq	shipping.terms
rmq	1	38	ocmq	total.cost
rmq	1	39	ocmq	completion.code
rmq	1	40	ocmq	completion.notes
rmq	1	41	ocmq	li.mark.flag
rmq	1	42	ocmq	bill.to.ext
rmq	1	43	ocmq	ship.to.ext
rmq	1	44	ocmq	phase.desc
rmq	1	45	ocmq	project.id
Map Name	Sequence	Position	File Name	Field
rmq	1	46	ocmq	orig.manager.group
rmq	1	47	ocmq	requested.date
rmq	1	48	ocmq	admin.lead.time

Map Name	Sequence	Position	File Name	Field
rmq	1	49	ocmq	duty.table
rmq	1	50	ocmq	parts.summary
rmq	1	51	ocmq	foreign.id
rmq	1	52	ocmq	requestor.phone
rmq	1	53	ocmq	manager
rmq	1	54	ocmq	logical.name

Related topics

Pre-implementation planning
External event services
Line item events
Order events
Example event string start sequences

1.10.3.10.2.14. Order events

The following tables describe order events.

Map Name	Sequence	Position	File Name	Field
rmqappr	1	1	ocmq	number
rmqappr	1	2	ocmq	$L.void.operator
rmqappr	1	3	ocmq	$L.void.action
rmqappr	1	4	ocmq	$L.void.approval.action
rmqappr	1	5	ocmq	$L.void.approval.group
rmqappr	1	6	ocmq	open
rmqappr	1	7	ocmq	page
rmqappr	1	8	ocmq	category
rmqappr	1	9	ocmq	current.phase
rmqappr	1	10	ocmq	phase.num
rmqappr	1	11	ocmq	phase.start.date
rmqappr	1	12	ocmq	orig.operator
rmqappr	1	13	ocmq	last.operator
rmqappr	1	14	ocmq	impact
rmqappr	1	15	ocmq	total.line.items
Map Name	Sequence	Position	File Name	Field
rmqappr	1	16	ocmq	alert

Map Name	Sequence	Position	File Name	Field
rmqappr	1	17	ocmq	alert.names
rmqappr	1	18	ocmq	approval.status
rmqappr	1	19	ocmq	pending.groups
rmqappr	1	20	ocmq	status
rmqappr	1	21	ocmq	priority
rmqappr	1	22	ocmq	reason
rmqappr	1	23	ocmq	business.area
rmqappr	1	24	ocmq	submit.date
rmqappr	1	25	ocmq	update.date
rmqappr	1	26	ocmq	close.date
rmqappr	1	27	ocmq	cancelled.date
rmqappr	1	28	ocmq	requestor.name
rmqappr	1	29	ocmq	requestor.dept
rmqappr	1	30	ocmq	coordinator
Map Name	Sequence	Position	File Name	Field
rmqappr	1	31	ocmq	coord.dept
rmqappr	1	32	ocmq	assigned.to

Map Name	Sequence	Position	File Name	Field
rmqappr	1	33	ocmq	assigned.dept
rmqappr	1	34	ocmq	description
rmqappr	1	35	ocmq	justification
rmqappr	1	36	ocmq	comments
rmqappr	1	37	ocmq	ship.to.code
rmqappr	1	38	ocmq	bill.to.code
rmqappr	1	39	ocmq	shipping.terms
rmqappr	1	40	ocmq	total.cost
rmqappr	1	41	ocmq	completion.code
rmqappr	1	42	ocmq	completion.notes
rmqappr	1	43	ocmq	li.mark.flag
rmqappr	1	44	ocmq	bill.to.ext
rmqappr	1	45	ocmq	ship.to.ext
Map Name	Sequence	Position	File Name	Field
rmqappr	1	46	ocmq	phase.desc
rmqappr	1	47	ocmq	project.id
rmqappr	1	48	ocmq	orig.manager.group

Map Name	Sequence	Position	File Name	Field
rmqappr	1	49	ocmq	requested.date
rmqappr	1	50	ocmq	admin.lead.time
rmqappr	1	51	ocmq	duty.table
rmqappr	1	52	ocmq	parts.summary
rmqappr	1	53	ocmq	foreign.id
rmqappr	1	54	ocmq	requestor.phone
rmqappr	1	55	ocmq	manager
rmqappr	1	56	ocmq	logical.name

Map Name	Sequence	Position	File Name	Field
rmo	1	1	ocmo	number
rmo	1	2	ocmo	$L.void.operator
rmo	1	3	ocmo	$L.void.action
rmo	1	4	ocmo	open
rmo	1	5	ocmo	page
rmo	1	6	ocmo	category
rmo	1	7	ocmo	current.phase
rmo	1	8	ocmo	phase.num
rmo	1	9	ocmo	phase.start.date
rmo	1	10	ocmo	orig.operator
rmo	1	11	ocmo	last.operator
rmo	1	12	ocmo	impact
rmo	1	13	ocmo	total.line.items
rmo	1	14	ocmo	vendor
rmo	1	15	ocmo	alert
Map Name	Sequence	Position	File Name	Field
rmo	1	16	ocmo	alert.names

Map Name	Sequence	Position	File Name	Field
rmo	1	17	ocmo	approval.status
rmo	1	18	ocmo	pending.groups
rmo	1	19	ocmo	status
rmo	1	20	ocmo	priority
rmo	1	21	ocmo	reason
rmo	1	22	ocmo	business.area
rmo	1	23	ocmo	submit.date
rmo	1	24	ocmo	update.date
rmo	1	25	ocmo	close.date
rmo	1	26	ocmo	coordinator
rmo	1	27	ocmo	coord.dept
rmo	1	28	ocmo	total
rmo	1	29	ocmo	description
rmo	1	30	ocmo	justification
Map Name	Sequence	Position	File Name	Field
rmo	1	31	ocmo	comments
rmo	1	32	ocmo	invoice.no
rmo	1	33	ocmo	vendor.contact

Map Name	Sequence	Position	File Name	Field
rmo	1	34	ocmo	shipping.terms
rmo	1	35	ocmo	po.number
rmo	1	36	ocmo	completion.code
rmo	1	37	ocmo	completion.notes
rmo	1	38	ocmo	net.total
rmo	1	39	ocmo	tax.code
rmo	1	40	ocmo	tax.rate
rmo	1	41	ocmo	tax.amount
rmo	1	42	ocmo	ship.to.code
rmo	1	43	ocmo	vendor.zip
rmo	1	44	ocmo	bill.to.code
rmo	1	45	ocmo	shipping.carrier
Map Name	Sequence	Position	File Name	Field
rmo	1	46	ocmo	freight.on.board
rmo	1	47	ocmo	payment.terms
rmo	1	48	ocmo	freight.charges

Map Name	Sequence	Position	File Name	Field
rmo	1	49	ocmo	vendor.contract.no
rmo	1	50	ocmo	discount.rate
rmo	1	51	ocmo	discount.amt
rmo	1	52	ocmo	ship.to.ext
rmo	1	53	ocmo	bill.to.ext
rmo	1	54	ocmo	phase.desc
rmo	1	55	ocmo	project.id
rmo	1	56	ocmo	target.date
rmo	1	57	ocmo	shipping.terms
rmo	1	58	ocmo	orig.manager.group
rmo	1	59	ocmo	parts.summary
rmo	1	60	ocmo	foreign.id

Related topics

Pre-implementation planning
External event services
Line item events
Quote events
Example event string start sequences

1.10.3.10.2.15. Logical actions

There are twelve logical actions that run through four actual event classes.

Event Class	Description
rmlin	Used for open/update/close of all line items
rmoin	Used for open/update/close of orders
rmqin	Used for open/update/close of quotes
approval	Used to approve/deny/retract quotes and orders.

Related topics

Pre-implementation planning

1.10.3.10.2.16. Record modification classes

These three events (rmlin, rmqin, rmoin) are used to send record modification actions into Request Management for line items, quotes, and orders. The structure of their event strings is as follows.

Sequence	Description
1	Number of the object to be acted upon. If in rmlin, this will be a line item number. If in rmqin, this will be a quote number. If in rmoin, this will be an order number. In the open process, this field is discarded.
2	Operator ID who is executing this action. The action takes place within this operator's security context.
3	Action Taken, indicating what type of modification action to take. One of open, update, close.
4..*n*	Mapped data fields, containing changes to the record. Processed along with the approval.

Related topics

Pre-implementation planning

1.10.3.10.3. Messages

Service Manager sends messages in response to an event. They can be directed to specific operators listed in the event record, and contain values from certain fields in quote, order, and line item records that caused the initial event.

Request Management message processing involves the following:

- The background processor looking at the *ocmoptions* record for the phase or category that generated the event. If the record doesn't exist, processing ends.
- Evaluation of the Messages/Events option in the *ocmoptions* record. If false, processing ends.
- Checking for field name and operators in the *ocmevents* record. If none, processing ends (unless this event is delayed by *gen order*).
- Recording in the *msglog* the generic message (from the *ocmevents* record's Append Text field).
- Sending the standard message and the mail message to the operators defined in the Operators field of the *ocmevents* record.
- Sending a message to the operators defined in those fields referenced by the Field Name field of the *ocmevents* record.

The content of these fields is first assumed to be a group. If this group name is found in the *ocmgroups* file, then either the members or approvers (depending on the Member List field) of that group are added to a working list.

If this group name doesn't exist, the system searches the operator file; and if an operator record is found, it is added to the working list.

Message classes

Service Manager has several default message classes where a user can define additional messages to display in Request Management.

The Notifications menu contains the various message class action types:

- On-Screen (*msg*) – Send a message to the user's screen.
- Print (*print*) – Send a copy of the message to the receiver's default printer.
- Log (*log*) – Send a copy of the message to the msglog file.
- TSO (*TSO*) – Send a copy of the message to the receiver's TSO ID.
- External E-Mail (*email*) – Send a copy of the message to the receiver's e-mail address as specified in the operator or contacts record.
- Internal E-Mail (*email*) – Send a copy of the message to the receiver's internal Service Manager mailbox.

Add a message class record

Applies to User Roles: System Administrator

Service Manager has several default message classes where a user can define additional messages to display in Request Management.

To add a message class record:

1. Click **Tailoring** > **Notifications** > **Message Types**.
2. Click **Search**.
3. Select a message type from the record list on which to base your new record.
4. Clear the message **Type**, and then enter a type for the new message record.
5. Add or change information on the form to set up the new message record.
6. Check the **Active?** box if this is an active message.
7. Click **Add** to confirm the new record.

   Caution: Make sure you don't click Save because doing so will replace the existing message record with the new record you are attempting to add.

Notification engine

The Notification Engine is primarily responsible for sending messages that are generated by Service Manager events, such as opening or closing a quote or order. Administrators can edit these messages, add new messages, change the conditions under which the messages will be sent out, as well as select who will receive the messages.

The notification file works with the message file to define notifications for common system events. Administrators can modify the notification arguments that trigger the notification, as well as define who receives the notification.

Open the notifications file

Applies to User Roles: System Administrator

You can configure Service Manager to send notifications whenever specific system events occur. By default, Service Manager supports notifications for every type of status change event.

To open the notifications file:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Add optional search criteria, and then click **Search**.

The list of notification records opens.

3. Select a target record to view.

Note

For information on adding a notification definition record, adding a notification delivery method, or creating a custom notification, see the related topics.

Related topics

Routing orders to external purchase systems
Alerts, events, and messages
Events and messages

1.10.4. Request Management overview

Request Management is the Service Manager application used to manage user requests for products and services. Requests affect only the person making the request, or a subordinate group of employees. Examples include password resets, individual pc upgrades, and new employee setup. For changes to the infrastructure, refer to Change Management.

Request Management includes the following key features:

- Automated quote, manager approval, and order processing tracking for products and services.
- Detailed, customizable catalog of products and services, including bundled and sequenced parts and services.
- Scheduling and integration of service requests and work orders with purchase requests.
- Combination of multiple quotes into single or multiple orders, based on vendor/supplier.
- Provision for external vendors/suppliers and internal work groups.
- Integration with other Service Manager applications, such as Configuration Management and Change Management.
- Sequential and Conditional on-line quote entry and approvals.
- Automated mail notification and alerts for normal and exceptional events.
- Customer control, consolidation of acquisitions, and life cycle management.
- Quote—Order—Receiving—Posting process.

When should you use Change Management instead of Request Management? Here are some guidelines.

- Change Management handles any change to your business that modifies or disrupts the current state of that environment. Usually these modifications or disruptions affect multiple users or business units.
- Request Management handles common user requests for products and services. These requests usually affect only the person making the request, or a subordinate group of employees.

Related topics

Change Management overview
Request Management User Profile

1.10.4.1. Pre-implementation planning

Pre-implementation planning includes the following activities:

- Creating the catalog.
- Setting up the approval process.
- Setting up phases.
- Creating the order process.
- Setting up notifications and alerts.
- Considering how quotes should be handled when closed.
- Considering the activities for closing and receiving order line items.
- Establishing the level of user access to Request Management for users.
- Determining the implementation plan for configuration items (CIs).
- Determining what to report, and when.
- Selecting audit fields and who will maintain the audit records.
- Validating the implementation of Request Management to your office's expectations.

The related topics provide a synopsis of planning considerations for the implementation of a Request Management system. These need to be considered a foundation tool for initial planning, used in conjunction with technical consulting services.

Related topics

The catalog
Approvals in Request Management
Phases
User data entry
Order process
Notifications and alerts
Closing a quote
Closing and receiving order line items
Request Management User Profile
Configuration Items
Reporting
Auditing
Process validation
Structure of the catalog
Vendors/Suppliers

1.10.4.1.1. The catalog

The Request Management application is based on the concept of a catalog. A catalog, a list of parts and services, is a file that contains all of the items that a requester might want to order. This includes request categories, primary categories, and line items such as a new phone, move/add/change of a PC, and a new logon ID.

The Catalog contains all the items that users can request, including request category packages, such as Employee Office Move or whether they directly request to add something to their order at a later time.

Catalog line items require detailed definitions, such as specific model numbers, prices, and delivery lead times. Other items that can be tracked include descriptions, vendors/suppliers, and approval requirements. The Request Management system is only as detailed as the Catalog entries. If it is necessary for items like cable drops, labor related items, and training to be part of a quote, they must be defined in the Catalog.

A feature of the Catalog is the ability to group items in frequently ordered bundles. For example, a common desktop workstation is identified as a 3 GHz Pentium with a 120GB hard drive, 23″ monitor, 52X CD-ROM, and 2GB RAM. The requester could then just order a Standard Workstation rather than the individual components. You can also give the requester the option of changing components or ordering an individual component. This is true of services as well. A New Employee Office request may consist of a new phone connection, a LAN drop, a standard workstation, and a telephone.

Related topics

Catalog operations
Catalog view
Categories
Structure of the catalog

1.10.4.1.2. Structure of the catalog

An entry in the catalog (such as High End PC) may be subdivided into further component entries. These entries are also items in the catalog. When an order is placed, the top-level item (considered a Parent item, and possibly a place-holder phantom item) is ordered. The requester doesn't need to know any more about the line item than that. If the catalog item has been identified as being a bundle of lower level components, or Children items (also in the catalog), those items will automatically be broken out on the request. Each component then becomes a separate line item with its own costs and delivery dates.

The following table describes an example of a catalog entry.

Relationship	Part No	Catalog Name
Parent	1	High End PC
Child	8	Pentium 3 GHz CPU, 1 GB RAM
Child	12	21” Monitor
Child	25	Keyboard
Child	87	Mouse

When the customer requests a High End PC, all the underlying components are ordered and processed as line item entries.

A customer may also order the 21” Monitor as a separate line item. Each entry is a catalog item in its own right.

Related topics

Vendors/Suppliers
Approvals in Request Management

1.10.4.1.3. Primary catalog

The catalog is the most frequently encountered component of Request Management. It includes lists of items that are available for request, as well as bundled Request Categories. Categories include Customer Procurement Requests, Human Resources, and Employee Office Move, along with their components and lists of item types.

The primary catalog enables the user to select a predefined set of hardware, software, and services that constitute an item. This item is available for order through the catalog and includes all associated components. For example, the New Employee Setup item within the Human Resources category includes office furniture, CPU with monitor and mouse, and network connectivity. The user orders one item, New Employee Setup, and orders for all its components are sent out. Each component is then tracked as the order is processed, the product is received, and the product is installed or delivered to the user's site.

Many features of the Request Management application are available to the System Administrator only.

Related topics

Catalog operations
Categories

1.10.4.1.4. Primary catalog sections

Primary catalog sections are similar to those of any mail order catalog. Desks, chairs, new employee setup, and computer equipment are primary catalog sections. Each category divides into a subcategory. For example, relationship parts for a parent item in the catalog, such as High End PC, would include child parts Pentium 3 GHz CPU, 1 GB RAM; 21 inch monitor; keyboard; and mouse. Within each primary catalog section, the groups are further subdivided into line item categories.

Related topics

Approvals in Request Management
Structure of the catalog
Vendors/Suppliers

1.10.4.1.5. Phases

Phases determine when and how quotes, line items, and orders are processed. Phases control the activity allowed during each administrative step in the business process flow.

With a quote category, a phase indicates where in the process the quote currently stands, and who can modify the quote. A typical implementation has one quote category with two to three phases.

Phase	Definition
Initial phase	For setting up the quote with approvals.
Ordering phase	For order generation.
QA phase	For verifying successful fulfillment.

For example, where a phase requires that the operator who submitted the quote verify the satisfaction, you can set up an operator approval phase. You restrict the close or approval control of the quote to a request coordinator who must contact the originating operator to verify that the operator approves of the quote.

Not all operators have access to quotes and orders at all stages of the process. Typically, an implementation sets up one phase per order category, providing specific instructions for that category. Each line item category has only one phase, which defines the system behavior when that line is selected and processed.

Typically, Request Management is configured for three phases: User entry, coordinator, and ordering.

User entry

During this phase, the average user can access the quote and change information on it. The following is an example of information that the average user can change.

- Create the quote.
- Request items.
- Request a delivery date.

Coordinator

During this phase, the coordinator can replace the user's request for generic items with specific items and add detailed information to both the line item and the quote. The coordinator is responsible for gathering enough information to bill for the items and deliver them properly.

Ordering

During this phase, the quote becomes an order and the actual order is generated. Some considerations for phase definitions include the following.

- Are these phase names acceptable?
- What requirements are there before closing a phase and opening the next?
- Should users who don't have update capability in a phase have browse capability at that phase?
- What's the administrative lead time at the Coordinator phase?
- How long does it take for the coordinator(s) to review a user's request and process it into an order?

Add a quote phase

1. Click **Request Management** > **Quotes** > **Quote Phases**. A blank phase definition form opens.
2. Click **Search** to generate a record list.
3. Click the name of a phase definition on which you want to base your new quote phase. The phase definition form opens, containing the definition data of the selected quote phase.
4. Clear the name from the **Name** field, and then type a unique name for the new quote phase.
5. Modify the new phase record by providing your information in each of the following tabs. The information should be relevant to the system behavior and level of control you want an operator to have on a quote in this phase.
    - Definition
    - Controls - When updating or adding a new phase definition record, you can provide both the button label and the balloon help text in the Close Description field. The format is `Button label;Balloon help`.
    - Alerts
    - Approvals
    - Model/Line Items
    - Scripts/Views
6. Click **Add** to add the new phase to the system.

    A list of events opens.

Caution: Make sure that you don't click Save because doing so will replace the existing phase with the new phase you are attempting to add.

7. To restrict the new quote phase to certain events, select only those events.
    1. Double-click an event and review the event form to make sure you want to add the phase to this event.
    2. To go back to the list of events and continue to browse through the events, click **Back**.
    3. To add the phase to this event, click **OK**.
    4. Continue steps a. through c. to select the applicable events for this new quote phase.
    5. When you are done, click **Back**.
    6. Click **OK**.
8. To add the new quote phase to all events, do the following:
    1. Click **Add to All**.
    2. Click **OK**.

Add a line item phase

1. Click **Request Management** > **Line Items** > **Line Item Phases**. A blank line item phase definition form opens.
2. Click **Search** to generate a record list.
3. Click the name of a phase definition on which you want to base your new line item phase. The phase definition form opens, containing the definition data of the selected line item phase.
4. Clear the name from the **Name** field, and then type a unique name for the new line item phase.
5. In the **Parent Area** field, select Quotes or Orders as a valid parent area for a line item under this phase. When this field is left blank, all line items under this phase use this phase definition.
6. The **Modify Dates** field in the Model/Line Items tab enables you to change the ordering dates of a line item. To have the ability to change the dates, type **true**. If you leave this field blank or if it is set to false, you can't modify the dates.
7. Modify the rest of this new phase record with your information in the following tabs:
    - Definition
    - Alerts
    - Approvals
    - Model/Line Items
    - Scripts/Views
    - Reports

8. Click **Add** to add the new phase to the system.

   A list of reports opens.

   > **Caution**
   >
   > Make sure that you don't click Save because doing so will replace the existing phase with the new phase you are attempting to add.

9. Double-click the report name to add the phase to the report.

   A list of events opens.

10. To restrict the new line item phase to certain events, select only those events.
    1. Double-click an event and review the event form to make sure you want to add the phase to this event.
    2. To go back to the list of events and continue to browse through the events, click **Back**.
    3. To add the phase to this event, click **OK**.
    4. Continue steps a. through c. to select the applicable events for this new line item phase.
    5. When you are done, click **Back**.
    6. Click **OK**.
11. To add the new line item phase to all events, do the following:
    1. Click **Add to All**.
    2. Click **OK**.

Add an order phase

To add an order phase:

1. Click **Request Management** > **Orders** > **Order Phases**. A blank order phase definition form opens.
2. Click **Search** to generate a record list.
3. Click the name of a phase definition on which you want to base your new order phase. The phase definition form opens, containing the definition data of the selected order phase.
4. Clear the name from the **Name** field, and then type a unique name for the new order phase.
5. Modify the new phase record with your information and be sure to update the fields in the following tabs:
   - Definition
   - Alerts

- Approvals
- Model/Line Items
- Scripts/Views
- Reports

6. Click **Add** to add the new phase to the system.

   A list of events opens.

   Caution: Make sure that you don't click Save because doing so will replace the existing phase with the new phase you are attempting to add.

7. To restrict the new order phase to certain events, select only those events.
   1. Double-click an event and review the event form to make sure you want to add the phase to this event.
   2. To go back to the list of events and continue to browse through the events, click **Back**.
   3. To add the phase to this event, click **OK**.
   4. Continue steps a. through c. to select the applicable events for this new order phase.
   5. When you are done, click **Back**.
   6. Click **OK**.
8. To add the new order phase to all events, do the following:
   1. Click **Add to All**.
   2. Click **OK**.

1.10.4.1.6. User data entry

Depending on your implementation of Request Management, the requester may be required to enter data during the User Entry phase. For example, the requester enters contact information, location, and type of request on a description form.

Coordinator data entry fields

After the requester completes the description form, the request is routed to a coordinator who contacts the requester for specific information. Consider the following.

- What information do you need from the requester for each catalog item you have previously identified?
- What fields must be validated in a certain format, and which must be validated to exist in a supplementary table?

Some data items are defined as coming from another table. A common example of this is contact information.

During the process of opening a quote, the coordinator may be required to supply the first name, last name, department, and phone number of the individual requesting an item.

- Should this information be checked against the contacts table?
- What should be done if the individual is not yet in the contacts table?

Related topics

Pre-implementation planning

1.10.4.1.7. Order process

Things to consider when you create the order process:

- How often will orders be generated?
- Will orders be combined to minimize paperwork given to vendors/suppliers?
- What are the basic criteria to combine line items into batches and create new orders?
- How will Request Management interrelate with any external purchasing system already in use?

Related topics

Pre-implementation planning

1.10.4.1.8. Notifications and alerts

There are things to consider when you set up notifications and alerts:

- Which events generate notification and alerts?
- For each event, who is to be notified?
- For each event, what type of notification is to be used, such as, pager, e-mail, or fax?
- Which alerts must be defined?

Category	Event	Notify who?	Type of notification
Pager	Pending approval > 24 hrs.	IT Supervisor	e-mail
Pager	Not closed > 2 weeks	IT Supervisor	e-mail

Related topics

Alerts

1.10.4.1.9. Closing a quote

Consider the following questions when you close a quote:

- How will customer follow-up be handled?
- Will the operators performing follow-ups close the quotes?
- What information must be collected at completion time?

Related topics

Pre-implementation planning

1.10.4.1.10. Closing and receiving order line items

Consider the following questions when closing and receiving order line items:

- Who will enter receipt information into Request Management when parts arrive at the receiving dock? If you use Get.It resources, you don't need to receive items into Request Management.
- How will inventory be updated by this information?
- Will scripts be used when technicians close order line items for work/service/labor after their task is completed?

Although request processing through Request Management can be considered the front-end component to interface with an external purchasing system, Request Management does receive against the orders within the system.

If you use an external purchasing system, we recommend that you record the other system's Purchase Order Numbers (usually found on packing slips) on the order in Request Management:

- Create a field on the quote and/or order line item formats associated with the *ocml* file.
- Add the field to the dbDict.

You can now track the actual order generated through the external purchasing system and associate it with the original quote in Service Manager.

- Order line items for services are closed after the work is completed.
- Order line items for parts marked in the Catalog for receipt are received. These parts can be serialized (have a unique serial number) or non-serialized.

After it receives order line items that are combined from several different quote line items, Service Manager displays a distribution screen to post the receipt information back to the quote line items that generated the order.

You can configure the system to automatically close the original quote line item when you complete the receiving and posting process for an order line item.

Related topics

Closing and receiving

1.10.4.1.11. Configuration Items

Consider the following questions during configuration item (CI) planning:

- Are there plans for posting information about new CIs into the CI tables?
- Is there a schedule in place for implementation of the plan?
- What's the life cycle of the CI?
- What determines a change in the life cycle of the CI?
- Who determines where old CIs go?
- What's the CI trickle process?

Related topics

Pre-implementation planning

1.10.4.1.12. Reporting

Reporting planning.

- Are there specific reporting requirements?
- Will the reports be scheduled or on-demand reports? Will Request Management meet the reporting requirements defined in the customer’s design document?

Related topics

Pre-implementation planning

1.10.4.1.13. Auditing

Consider the following questions during audit planning:

- Which fields will be audited?
- Who will maintain the audit records?
- What are the reasons behind selecting to audit fields as opposed to tracking entire records via paging?

1.10.4.1.14. Process validation

You may need to validate the implementation of Request Management to your office's expectations. Changes may be required in both the tool and the process. Determine what you want to accomplish and how you can implement Request Management to meet those requirements.

- Does your office's process make sense in relation to the Request Management tool?
- Does Request Management make sense in relation to your office's best practices and process?

Related topics

Pre-implementation planning

1.10.4.2. Approvals in Request Management

The approval process automates and formalizes the evaluation of quotes and orders by the appropriate management entities. This process channels the risk, cost, and responsibility associated with a request to the proper levels.

An approval requirement is assigned when an item requires a decision-maker's evaluation. These approvals, though encompassing the line items and part numbers listed on the quote or order, operate only at the quotes and orders level. The status of individual line items may change (for example, from requested to ordered), but it is only the quote or order itself that's approved or denied.

Those operators authorized to issue approvals within each area and category of requests and orders are listed in the group definition records.

Approvals give controlling authorities (Approval groups) the responsibility of costs and risks associated with a request, the ability to stop work, and to control when certain work activities can proceed. They determine:

- Which categories of catalog items require approvals
- Which catalog items require approvals
- What are the Approval groups
- Who is in each group

Related topics

Security and access
Request Management overview
Approval classes
Approval actions
Approval components
Using approvals
Approval classes

1.10.4.2.1. Approval classes

Approval classes are used to send approval actions into Request Management. The structure of their event strings is as follows.

Sequence	Description
1	File name of the object to be acted upon. ocmq = quotes ocmo = orders
2	Number of the object to be acted upon.
3	Group for which the operator has rights to approve orders. If left blank, operator has rights to all groups.
4	Action taken, indicating whether approve, retract, or deny.
5	Date and time approval was sent. Mapped data fields containing changes to the record. Processed along with the approval.
6	Identification of the operator who is executing this approval. The approval happens in this operator's security context.

Related topics

Approvals in Request Management
Pre-implementation planning
Record modification classes

1.10.4.2.2. Approval sequence

Approvals can either be approved, retracted, or denied, and pending approvals await one of these actions. Approval groups for a request are placed in sequence, in the order in which their approval is required. If groups have the same sequence number, their approvals can be made independently of each other.

When a pending approval is approved, its status becomes approved and the quote or order moves on to the next phase.

Related topics

Approvals in Request Management
Approval components

1.10.4.2.3. Approval components

The Approval Status of a quote or order remains pending until appropriate approval actions have been taken. The following topics are the components of the approval process:

- Approval Status
- Approval Files and Features
- Approval Actions

Related topics

Approvals in Request Management
Gen by items
Approval status
Approval files and features
Approval actions

1.10.4.2.4. Approval status

The approval status of a quote or order is the current state of its required approvals. A record may be waiting for its items to be ordered or received, but no approvals are required for the status to change. A unique status is given to each line item within the quote or order, as well as the quote or order as a whole. Some phases require approvals, as a phase change most often involves the intervention of a decision-maker, indicating the request or order needs to be moved on to another department or area's responsibility.

The approval status of a quote or order reflects the current condition of all subordinate approval requirements associated with the parent quote or order. Request Management includes the following default approval statuses.

Approval Status	Description
Approved	All of the approval requirements have been approved.
Denied	The approval requirements have been denied.
Pending	Active and waiting for an approval action.

Related topics

Approvals in Request Management
Approval components
Approval files and features
Approval actions

1.10.4.2.5. Approval files and features

Two primary files contain approval information: the Approval Definitions and the Approval Log.

File Name	Description
Approval Definition (ApprovalDef)	Defines the approvals used by all phases (static file).
Approval Log (ApprovalLog)	Displays current approvals. • An unlimited number of approval requirements can be defined for each item. • Approvals may have conditions attached, such as Total Cost, Lead Time requirements, and Impact.

Related topics

Approvals in Request Management
Approval components
Approval status
Approval actions

1.10.4.2.6. Approval actions

Approval groups have the option to approve, deny or retract when they face an approval requirement.

Approve

Gives authority to proceed with the item or scheduled work, or to accept completed work. There are four predefined approval types.

- **All must approve** — All Groups/Operators defined in the Approval Definition must issue an approval before the status of the record is set to "approved." If only one or some (but not all) of the Groups/Operators issues an approval, then the status of the record is set to "pending."

  Example: You have three Groups/Operators in an Approval Definition and only one Group/Operator has approved the record. The Approval Status is updated to pending. The Approval table shows one currently pending approval, one future approval, and one completed approval action.

- **One must approve** — The record is approved with one approval from any Group/Operator of the approving group.
- **Quorum** — The record is approved as soon as a majority of the approving group indicate approval.
- **All must approve - immediate denial** — All Groups/Operators must approve the record. The first denial causes the status to change to Deny. All approvers don't need to register their approval action. Otherwise, the record is denied when all Groups/Operators of the approving group issue a denial.

Deny

Removes authority to proceed with an item or scheduled work, and reject completed work.

Retract

Removes a previously approved or denied action. This resets the approval status to pending.

Related topics

Approvals in Request Management
Approval components

Approval files and features
Approval status
Using approvals

1.10.4.2.7. Using approvals

Service Manager user profiles determine whether you see an Approve Requests button in the home menu after you log on. Only users with approval authority have access to the Approve Requests button. If you have access, click **Approve Requests** to begin.

To access approvals from the quote and order search screens, an operator's profile must meet the following requirements.

- The Approvals parameter for the current area (quote or order) must evaluate to true.
- The Approvals Group field must contain at least one group for the current Area.

Note

If a user doesn't have access rights to a particular area or to approval records, the status bar indicates this.

Related topics

Approvals in Request Management
Approval type group options
Approval definitions
Security and access

1.10.4.2.8. Group records

A group in Request Management represents a collection of operators who share a common set of responsibilities. Group definitions specify two divisions within the group: members (formerly reviewers), and approvers. Approvers have approval authority and receive approval information associated with this group. They are a subset of the member group; it is possible for all members to be approvers.

One of the primary purposes for groups is to ensure dissemination of system messages to relevant parties. Member groups or Approval groups may be listed in the *ocmevents* record. If these groups aren't also listed in the group record, they will not receive messages. The individuals of a group have the same authority as identified in their profile, but they don't receive messages.

An operator’s authority depends on the operator profile; whether an operator receives messages depends on membership in the group record.

1.10.4.2.9. Approval type group options

Approval Type groups have options to approve, deny, or retract approvals for currently pending or future approvals.

Approval Group Options	Description
Deny One | Retract One	Ability to approve, deny, or retract a current or future approval for this Approval Type group
Deny Current	Ability to approve or deny current approvals for this Approval Type group
Deny All | Retract All	Ability to approve all, deny all, or retract all approvals for this Approval Type group

Related topics

Approvals in Request Management
Using approvals
Approval definitions

1.10.4.2.10. Approval definitions

The Approval Definition is a static file that defines the approval groups and the approval conditions associated with a particular approval group, as well as their valid functional area.

An Approval Definition record defines the basics of an approval requirement. The approval condition for each requirement can be based on any field in the source record, such as cost, category, assignments, or priority.

Access approval definitions

Applies to User roles: Request Administrator, System Administrator

An approval definition record defines the basics of an approval requirement. The approval condition for each requirement can be based on any field in the source record, such as cost and category.

To access approval definitions:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Approval Definitions**.
2. Add optional search criteria, and then click **Search**.

   A record list of current approval definition records opens.
3. Click a record to open it.
   - Click **Next** or **Previous** to scroll through the list of records and see the specifics for each record.
   - Click **Back** to go back to the search screen.
   - Click **Delete** to delete a record.
   - Click **Save** if you make any changes.
4. To add a new record, do the following:
   1. In the **Name** field, clear the name of the current record and type a unique new name for this new record.
   2. Add or change information on the form to set up the new approval definition record. For example, type an **Approval Description** and set the **Approval Type**.
   3. Click **Add** to add the new record.

Caution

When you are adding a new record from an existing record, make sure you don't click Save. Doing so will replace the existing record with the new record you are attempting to add.

Related topics

Approvals in Request Management
Using approvals
Approval type group options

1.10.4.2.11. Approval log

The Approval Log lists all the approval actions performed for an approval request.

When you open a quote, order, or line item, the system searches *ApprovalDef* to determine that item's approval requirements. It searches for those definitions that match what appears in the Phase definition (*ocmoptions*) for the item, or in the Catalog definition (*ocmco*).

When the Approval log runs, it includes all currently approved items in the list.

Access approval logs

Applies to User roles: Request Administrator, System Administrator

The approval log lists all the approval actions performed for an approval request. When you open a quote, order, or line item, the system searches the approval definitions that match what appears in the phase definition for the item, or in the catalog definition.

To access approval logs:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Approval Logs**.

   An empty approval log form opens.

2. Add optional search criteria, and then click **Search** to generate a record list of current approved records.
3. Click a record to open it.
   - Click **Next** or **Previous** to scroll through the list of records and see the Approval Name for each record.
   - Click **Back** to go back to the search screen.
   - Click **Delete** to delete a record.

Related topics

Approvals in Request Management
Using approvals
Approval groups

1.10.4.2.12. Approval groups

The Approvers array of the Group Definition form displays the list of operators who issue approvals for that group. In addition, they can be designated to receive messages about system events involving quotes, orders, or line items.

Membership in this group is specified by the Members array values. Those operators specified in this array can also receive messages when quotes, orders, and line items are entered into the system.

To add a name to an array, update the operator record to use a request profile that specifies the group as an approval group.

What's an approval group?

An approval group consists of a list of reviewers and approvers. If you have approval rights, you may be the only approver, or represent a group whose approval is necessary for the request to move forward in the workflow. If a request requires approval by a named group, you must be a member of that group for your approval to be valid.

Examples of approval groups are listed below.

- Facilities
- Helpdesk
- LAN Support
- M/F Support
- Onsite Support
- WAN Support
- Training
- Telecoms
- Service Desk
- Procurement

Add group records

Applies to User roles: Request Administrator, System Administrator

To add group records:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Groups**.

   A blank group definition form opens.

2. Type the **Name** of the new group definition.
3. In the **Area** field, select from Quotes, Orders, Line Items, or All to designate the area for the group definition.
4. Type a **Description** of the group definition.
5. Continue to define the new group definition by completing the fields on the form. For example, set the **Work Schedule** and the **Time Zone**.
6. Click **Add**.
7. Click **More** or the More Actions menu to select **Rebuild Group** to update the Members and Approvers lists and include any profile record Message group additions or changes made in the profile record.
8. Click **Save** to save your changes.

To use an existing group record to create a new group definition:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Groups**.

   A blank group definition form opens.

2. Click **Search** to open a list of group records.
3. Select a record from the list to create your new group record.
4. Change the **Name** of the group definition.
5. Add or change information on the form to select the applicable information for the new group definition record.
6. Click **Add** to add the new record.

   Caution

   Make sure that you don't click **Save** because doing so will replace the existing group definition with the new group definition you are attempting to add.

7. Click **OK**.
8. Click **More** or the More Actions menu to select **Rebuild Group** to update the Members and Approvers lists and include any profile record Message group additions or changes made in the profile record.
9. Click **Save** to save your changes.

Edit group records

Applies to User roles: Request Administrator, System Administrator

To edit group records:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Groups**.
2. Add optional search criteria, and then click **Search**.

   A list of group records opens.
3. Select a record from the list to update.
4. Review the membership and approval lists for the message group definition. All membership additions or deletions are made through the operator profile record group changes.

   Add or change information on the form to select the applicable information for the group definition record.
5. Click **Save**.
6. Click **More** or the More Actions icon and select **Rebuild Group** to update the Members and Approvers lists and include any profile record message group additions or changes made in the profile record.
7. Click **Save** to save your changes.
8. Click **OK**.

Related topics

Approvals in Request Management
Events and messages

1.10.4.2.13. Setting up approvals

Setting up approval authorities and requirements involves several components: Security and Access, Phase definitions, and Catalog features.

In order for operators to have approval authority in Request Management, their operator records must include capability words in the operator record that indicate Request Management access, such as *SysAdmin*, *OCMAdmin*, *OCMO*. Configuring this is explained in Security and Access.

Within Request Management, the user must be included in the membership of a designated approval group, in the *ocmgroups* file. See Approval groups for more information.

The user's authority within Request Management must be defined in the designated Request Management profile, the *ocmprofile* record specified in the operator record. This includes:

- A list of the approval groups for which the user can issue approvals.
- Approval capabilities defined on the various checkbox tabs of the *ocmprofile* form. See Security and Access for configuration information.
- The approval groups for which this user has approval authority must be included in the definition of the approval that's required for the request or order.
- Approval requirements are defined on the Phase definition control (*ocmoptions*) or catalog part record control.
- Approval Definition (*ApprovalDef*) records must be defined according to the Phase definition approval group requirements. An approval definition record must exist for any group defined as the approval group of a particular Phase definition. See Approval definitions for more information.

Related topics

Security and access
Approval groups
Phases
Approval definitions

1.10.4.2.14. Gen by items

Gen by items are those items which cause an approval to be required. These items include the following.

- Individual line items
- Part numbers
- Quotes
- Orders

Although record types listed as possible *Gen by items* can generate individual approval requirements, the approval requirements for the line items and part numbers are consolidated to the parent quote or order. The line items and part numbers themselves aren't approved; the parent quote or order is approved.

For example, if you order a New Employee Setup package but you order the executive PC configuration instead of the standard configuration, the executive PC configuration line item is a Gen by item and causes an approval requirement.

Related topics

Approvals in Request Management
Approval components
Approval status
Approval files and features
Approval actions

1.10.4.2.15. Approval delegation

Approval delegation is an optional feature that enables users with approval rights to temporarily delegate their approval authority to another qualified operator. Operators with the **Delegate Approvals** or **Can Delegate Approvals** option enabled in their application profiles can delegate some or all of their approvals by using the Approval Delegation wizard.

Using the **Approval Delegation** wizard, an operator can grant another qualified operator the right to temporarily view and act on items in that operator's approval queue. The wizard offers the following delegation options:

- Delegate all approvals to another qualified operator
- Delegate approvals from a particular application to another qualified operator
  - Delegate approvals directly assigned to you as an operator
  - Delegate approvals assigned to you as a member of an approval group
- Delegate approvals from a specified start date to a specified end date

The **Approval Delegation** wizard enables an operator to create any number of approval delegation combinations, including delegating the same approvals to multiple operators at the same time. Delegators can also update an existing approval delegation to change the delegation start and end dates, as well as change the delegate's name.

Note

Service Manager tracks all changes to approval delegations using the standard field auditing capability.

When delegates log on to Service Manager, they see both their own and any delegated approvals in their approval list. For security reasons, delegates always retain their original application profiles and operator records. Service Manager determines what temporary rights delegates have when they view or act on an approval.

For more information about the tasks that relate to approval delegation, refer to the following topics:

What happens when I delegate approval authority?
Temporary rights of an approval delegate
What happens when I receive delegated approval authority?
Copy an approval delegation

Delegate approvals to another operator
Disable an active approval delegation
Update an active approval delegation
Global variables available for approval delegation
Views available for approval delegation

1.10.4.3. Categories

Categories within Request Management combine items that would normally be listed independently. Without categories, the request process would be more time-consuming. Categories reduce or eliminate the need to search for items and services.

When you create a primary category for the catalog, you begin a process which contains ever-increasing levels of specification.

Request categories

Service Manager has three out-of-box request categories from which you can choose to order products and services, and accomplish an employee office move. When you open a request, you see the following out-of-box request category selections:

- Customer Procurement Requests
- Human Resources
- Employee Office Move Process

After you select a request category, you see the primary catalog categories.

Primary catalog categories

Primary catalog categories provide a bundled approach to ordering from the catalog that combines related categories and line items. For example, you can add labor to more than one primary category so that it's not overlooked and doesn't create delays if it is requested too late.

Primary catalog categories and line items are displayed directly after the quote categories. For example, when you select quote category Customer Procurement Requests, Service Manager displays the primary Catalog.

The primary catalog categories combine related items and services. For example, select Computers & Related for the following item types and services:

- Computer Accessories
- Desktop Computers
- Handheld and PDAs
- Hardware Requests
- Request Hardware Upgrades
- Installation Services
- Individual Monitors
- Notebook Computers

If you want a computer monitor only, click Individual Monitors.

The summary of New Employee Setup includes furniture, computer equipment, installation services, and everything needed for a new employee. You can add items to, or take them away from, your order as your needs change.

Line item categories

Line item categories are smaller and more numerous than primary catalog item categories. Line item categories are the building blocks of primary categories; however, they are still categories, built of parts. The parts are individual line items, such as an DEFAULT mouse or an ABC 17" monitor, which you can find in a line item category called Computers & Related along with many other individual line items associated with a computer setup. Line items include both specific parts and services.

You can restrict line item categories to certain quote and order groups the same way as primary catalog categories. On the first page of the catalog, select from the primary catalog category. The catalog displays line item categories in subsequent catalog pages.

The line item category is the next step down in specification from the primary category. Line items are more numerous and specific to the individual groups of products a company holds in inventory than are the primary categories. Line items include both specific parts and services.

Selecting a line item category from the catalog presents a list of actual line items, or parts, available for that category.

Phantom line items

Phantoms are place holders, or pseudo parts, used to organize a collection of multiple items. They are used to provide flexibility in defining component relationships within the catalog, and are usually not physical items. A phantom can be considered an umbrella type of level for several parts selections of the same type. For example, beneath the New Employee Setup primary category the New Accounts & Access line item provides a selectable option. This item itself doesn't display on the quote or order when it is selected; the catalog displays each of the associated parts as individual line items, such as internet access, email account, and network ID.

A phantom part is a parent of the specific items within it, and you can select specific items (or parts) through the phantom part.

Phantoms are assigned part numbers, but they themselves usually aren't ordered or placed on a quote.

There are two ways to designate a listing as a phantom.

Designate a model file as a phantom

Applies to User roles: Request Administrator, System Administrator

To designate a model file as a phantom:

1. Click **Request Management** > **Maintenance** > **Supporting Files** > **Model**.
2. Add optional search criteria, and then click **Search**.

   A list of model records opens.
3. Select the target model record to update.
4. Select the **Reorder** tab.
5. In the **Reorder Type** field, select **Phantom**.
6. In the Catalog tab, do the following:
    1. Select the **Part Conditions** subtab.
    2. Clear the **Copy to LI?** field.
7. Click **Save**.

1.10.4.4. Phases

Phases determine when and how quotes, line items, and orders are processed. Phases control the activity allowed during each administrative step in the business process flow.

With a quote category, a phase indicates where in the process the quote currently stands, and who can modify the quote. A typical implementation has one quote category with two to three phases.

Phase	Definition
Initial phase	For setting up the quote with approvals.
Ordering phase	For order generation.
QA phase	For verifying successful fulfillment.

For example, where a phase requires that the operator who submitted the quote verify the satisfaction, you can set up an operator approval phase. You restrict the close or approval control of the quote to a request coordinator who must contact the originating operator to verify that the operator approves of the quote.

Not all operators have access to quotes and orders at all stages of the process. Typically, an implementation sets up one phase per order category, providing specific instructions for that category. Each line item category has only one phase, which defines the system behavior when that line is selected and processed.

Typically, Request Management is configured for three phases: User entry, coordinator, and ordering.

User entry

During this phase, the average user can access the quote and change information on it. The following is an example of information that the average user can change.

- Create the quote.
- Request items.
- Request a delivery date.

Coordinator

During this phase, the coordinator can replace the user's request for generic items with specific items and add detailed information to both the line item and the quote. The coordinator is responsible for gathering enough information to bill for the items and deliver them properly.

Ordering

During this phase, the quote becomes an order and the actual order is generated. Some considerations for phase definitions include the following.

- Are these phase names acceptable?
- What requirements are there before closing a phase and opening the next?
- Should users who don't have update capability in a phase have browse capability at that phase?
- What's the administrative lead time at the Coordinator phase?
- How long does it take for the coordinator(s) to review a user's request and process it into an order?

Add a quote phase

1. Click **Request Management** > **Quotes** > **Quote Phases**. A blank phase definition form opens.
2. Click **Search** to generate a record list.
3. Click the name of a phase definition on which you want to base your new quote phase. The phase definition form opens, containing the definition data of the selected quote phase.
4. Clear the name from the **Name** field, and then type a unique name for the new quote phase.
5. Modify the new phase record by providing your information in each of the following tabs. The information should be relevant to the system behavior and level of control you want an operator to have on a quote in this phase.
    - Definition
    - Controls - When updating or adding a new phase definition record, you can provide both the button label and the balloon help text in the Close Description field. The format is `Button label;Balloon help`.
    - Alerts
    - Approvals
    - Model/Line Items
    - Scripts/Views
6. Click **Add** to add the new phase to the system.

    A list of events opens.

Caution: Make sure that you don't click Save because doing so will replace the existing phase with the new phase you are attempting to add.

7. To restrict the new quote phase to certain events, select only those events.
    1. Double-click an event and review the event form to make sure you want to add the phase to this event.
    2. To go back to the list of events and continue to browse through the events, click **Back**.
    3. To add the phase to this event, click **OK**.
    4. Continue steps a. through c. to select the applicable events for this new quote phase.
    5. When you are done, click **Back**.
    6. Click **OK**.
8. To add the new quote phase to all events, do the following:
    1. Click **Add to All**.
    2. Click **OK**.

Add a line item phase

1. Click **Request Management** > **Line Items** > **Line Item Phases**. A blank line item phase definition form opens.
2. Click **Search** to generate a record list.
3. Click the name of a phase definition on which you want to base your new line item phase. The phase definition form opens, containing the definition data of the selected line item phase.
4. Clear the name from the **Name** field, and then type a unique name for the new line item phase.
5. In the **Parent Area** field, select Quotes or Orders as a valid parent area for a line item under this phase. When this field is left blank, all line items under this phase use this phase definition.
6. The **Modify Dates** field in the Model/Line Items tab enables you to change the ordering dates of a line item. To have the ability to change the dates, type **true**. If you leave this field blank or if it is set to false, you can't modify the dates.
7. Modify the rest of this new phase record with your information in the following tabs:
    - Definition
    - Alerts
    - Approvals
    - Model/Line Items
    - Scripts/Views
    - Reports

8. Click **Add** to add the new phase to the system.

   A list of reports opens.

   > **Caution**
   >
   > Make sure that you don't click Save because doing so will replace the existing phase with the new phase you are attempting to add.

9. Double-click the report name to add the phase to the report.

   A list of events opens.

10. To restrict the new line item phase to certain events, select only those events.
    1. Double-click an event and review the event form to make sure you want to add the phase to this event.
    2. To go back to the list of events and continue to browse through the events, click **Back**.
    3. To add the phase to this event, click **OK**.
    4. Continue steps a. through c. to select the applicable events for this new line item phase.
    5. When you are done, click **Back**.
    6. Click **OK**.
11. To add the new line item phase to all events, do the following:
    1. Click **Add to All**.
    2. Click **OK**.

Add an order phase

To add an order phase:

1. Click **Request Management** > **Orders** > **Order Phases**. A blank order phase definition form opens.
2. Click **Search** to generate a record list.
3. Click the name of a phase definition on which you want to base your new order phase. The phase definition form opens, containing the definition data of the selected order phase.
4. Clear the name from the **Name** field, and then type a unique name for the new order phase.
5. Modify the new phase record with your information and be sure to update the fields in the following tabs:
   - Definition
   - Alerts

- Approvals
- Model/Line Items
- Scripts/Views
- Reports

6. Click **Add** to add the new phase to the system.

   A list of events opens.

   Caution: Make sure that you don't click Save because doing so will replace the existing phase with the new phase you are attempting to add.

7. To restrict the new order phase to certain events, select only those events.
   1. Double-click an event and review the event form to make sure you want to add the phase to this event.
   2. To go back to the list of events and continue to browse through the events, click **Back**.
   3. To add the phase to this event, click **OK**.
   4. Continue steps a. through c. to select the applicable events for this new order phase.
   5. When you are done, click **Back**.
   6. Click **OK**.
8. To add the new order phase to all events, do the following:
   1. Click **Add to All**.
   2. Click **OK**.

1.10.4.5. Quotes

A Request Management quote defines basic request information such as requester, required dates, coordinator, and description. A quote doesn't contain detailed parts information. One quote can contain many line items, collectively identifying the goods and services a user is requesting. Once approved, one quote can generate one or more related orders.

Related topics

Phases

1.10.4.6. Orders

An order is a formal request that a vendor/supplier perform a service or deliver a product to a particular destination. The Ordering function of Request Management is particularly useful when vendors/suppliers are included in the business process. Vendors/suppliers are both internal and external providers of parts and services.

The best way to access order information is directly from line item records, rather than order records themselves. The line items provide more detailed information about items being ordered than the order header fields. More useful query fields (such as Vendor Name or Assignee) are displayed through line items. Here vendors/suppliers can access information about their work or procurement obligations.

Orders offer these benefits:

- The ability to track vendors/suppliers
- Preparation by service organizations (such as internal vendors/suppliers) for upcoming work orders
- Simultaneous updating or closing order line items without interfering with others' access to the quotes
- The use of Purchase Orders

Related topics

Routing orders to external purchase systems
Prepare for background ordering requirement
Phases

1.10.4.6.1. Order generation process

Orders generate in the background. The Request Management order generation process controls several features:

- Order frequency
- Treatment of on-hand inventory
- Combining line items into a common order line item

Locks

Request Management uses locks which restrict access to records to ensure that multiple users aren't applying updates to the same record at the same time. Locking is implemented at the control item level when a quote or order is viewed. When the quote or order is locked, all associated items are locked as well, including line items. For this reason, processing, such as the movement between and through phases, can't occur while the user is modifying the control item records.

The system's background order processing attempts to lock the parent quotes of line items being processed or ordered. Therefore, certain implementation issues need to be considered. These include the following:

- Scheduling of background order processing
- Establishing the use of the Inactivity Timer (for parameters to disable inactive users)

See the Server Security Help for details about the Inactivity Timer.

Background processing

The background processor (*ocm*) manages Request Management ordering and event processing. This processor's interval is set by default to check every 60 seconds for *ocm* schedule records to process.

There is one *ocm* class schedule record for the ordering process.

- OCM Create Order
  - Also called Background Order Demand
  - (for batch -- demand order processing) Executes the *rmo.create.order* application.

Related topics

Phases

1.10.4.6.2. Batch ordering

Batch ordering is controlled by the demand schedule records. There can be more than one of these demand schedule records in the schedule file at one time, to be processed at different intervals and executing different queries.

These schedule records determine when and how often to generate orders. They also define what field values cause a break to a new order, as quotes are processed. The options here include values found in the following.

- vendor
- vendor.contract.no
- trans.type
- bill.to.code
- ship.to.code
- shipping.terms
- tax.rate
- payment.terms

Determining what causes a break to a new order line item is also contained in the demand schedule records. The options include values from the following.

- payment.freq
- no.of.payments
- part.no
- unit.cost
- unit.of.measure
- discount

Related topics

Processing control fields
Orders
Order process
Order generation process

1.10.4.6.3. Processing control fields

Field	Description
Line Item Query	The query overrides the default query run against the *ocml* file. For example: `avail.to.order=true and reorder.type="b"` and `open=true and quantity.balance>0 and target.order<=tod()`
Order Category	When you specify an order category while you create an order, your order category overrides the default order category. The default order category is that associated with the line item as defined in the modelvendor record.

Order breaks and line item breaks

While tailoring an implementation's order process, the Order Breaks and Line Item Breaks array fields on the Demand Schedule record provide a list of field names in a sequence that must coincide with a key in the *ocml* database dictionary record. The system checks for field name differences as each record is processed, and this determines when to complete the current order or line item and start a new one (for example, when to "break").

Important: In the base system's *ocml* database dictionary record, the key contains five fields. Do *not* modify this key.

- avail.to.order
- reorder.type
- open
- quantity.balance
- target.order

OCM order create for Line Item Schedule record

- Also called Background Order Demand.

- For batch -- demand order processing. Runs the *ocmo.create.order* application.
- Uses high breaks for order breaks (for example, creates new order record).
- Uses low breaks for line item breaks (for example, creates a new line item entry within the current order).
- Will automatically use the "backend" field as the first high break. This field is used when certain orders are sent to an external system.

Related topics

Batch ordering
Prepare for background ordering requirement

1.10.4.6.4. Routing orders to external purchase systems

The processing for orders is the same as for quotes. For example, you can define approvals, alerts, associated events and messages for an order.

You have several methods available, specifically designed for orders, to interface with an external system.

- You can send email notifications.
- In Format Control, you can create an *eventout* record.
- If batch processing is used, you can print the order to a specific printer.

  Note: Printing the immediate ordering type is unsupported.
- Alternatively, you can perform a file unload process for recently created orders, and move the file to an external procurement system with receipt information sent to Service Manager.

Related topics

Order process
Pre-implementation planning

1.10.4.6.5. Prepare for background ordering requirement

You can configure Request Management to send orders to an external system automatically. When the background ordering of events and line items requirements is in place, those items are ready to be ordered (*avail-to-order* items). Orders created with an external back end defined are routed to the external system, according to the order's *ocmbackend* definition.

Define Line items as ready to order on the *ocml.view.default* form.

Related topics

Pre-implementation planning

1.10.4.6.6. Closing and receiving

After the quote is officially entered into the system, approvals are made, and the associated orders are generated, the order process moves into a receiving mode and closes the individual line items.

Line items, which make up orders, are traced back to the initiating quote which can be closed after all items (parts and services) are received and closed. Several automatic and manual procedures track and associate related processes.

Related topics

Closing and receiving order line items
Receiving log

1.10.4.6.7. Closing and receiving order line items

Consider the following questions when closing and receiving order line items:

- Who will enter receipt information into Request Management when parts arrive at the receiving dock? If you use Get.It resources, you don't need to receive items into Request Management.
- How will inventory be updated by this information?
- Will scripts be used when technicians close order line items for work/service/labor after their task is completed?

Although request processing through Request Management can be considered the front-end component to interface with an external purchasing system, Request Management does receive against the orders within the system.

If you use an external purchasing system, we recommend that you record the other system's Purchase Order Numbers (usually found on packing slips) on the order in Request Management:

- Create a field on the quote and/or order line item formats associated with the *ocml* file.
- Add the field to the dbDict.

You can now track the actual order generated through the external purchasing system and associate it with the original quote in Service Manager.

- Order line items for services are closed after the work is completed.
- Order line items for parts marked in the Catalog for receipt are received. These parts can be serialized (have a unique serial number) or non-serialized.

After it receives order line items that are combined from several different quote line items, Service Manager displays a distribution screen to post the receipt information back to the quote line items that generated the order.

You can configure the system to automatically close the original quote line item when you complete the receiving and posting process for an order line item.

Related topics

Closing and receiving

1.10.4.6.8. Receipt verification

The receiving process:

- Log the receipt of serialized equipment. The Catalog indicates whether an item is to be received and whether it is serialized.
- Serialized items must:
  - Be received.
  - Have a quantity of 1.
  - Have a Serial Number.
  - Post data to the model record.
  - Distribute (post) the received quantities to the original quote line items.

Related topics

Receiving log
Closing and receiving order line items

1.10.4.6.9. Confirmation

You can configure your business rules and user profiles to display a confirmation screen when you click **OK** to confirm receiving and distribution.

Users must provide confirmation for the process to complete, terminate, or reset. Depending on the process flow, receiving personnel may be able to provide this information.

The out-of-box system assigns the receipt to the first available line items.

- If there are any dependent line items ready for ordering, the system marks them accordingly.
- If there are no further line items to be ordered, the order goes to the next phase.
- If there is no next phase, the order is closed.

Related topics

Request Management User Profile
Receipt verification

1.10.4.6.10. Receiving log

The receiving log tracks the receipt of serialized and non-serialized equipment. The system populates this file after the receiving process.

A benefit of the receiving log (*ocmlrec*) record is that it provides the serial number of the particular Configuration Item (CI) or device for posting to the Configuration Management application.

Related topics

Closing and receiving
Format control and link records

1.10.4.6.11. Link records

You can copy data from different data records to a line item via a standard link record. Request Management checks for the following link records for the indicated process.

Note: The following are the Request Management link records which should be modified according to any renaming of Request Management files or forms.

Link Name	From	To	When Used
ocml.copy.quote	ocmq	ocml	When opening a line item from a quote
ocml.copy.order	ocmo	ocml	When opening a line item from an order
ocml.copy.model	model	ocml	When opening a line item from a catalog part
ocml.copy.modelvendor	modelvendor	ocml	When opening a line item from a catalog part and the user selected a vendor/supplier
ocml.receive.copy	ocml	ocmlrec	When receiving process copies data from the line item to the receiving log

If found, Request Management copies data from the source record to the line item based on the data fields defined at the detail level of the first element in the applicable link record.

Do not use these link records for any other purposes. Request Management always uses the first element of the link record for copying data from the source record to the line item. It doesn't examine any other link elements.

For more information on how to use link records and the roles played by link records, see the related topics.

Related topics

Links
Types of links

1.10.4.6.12. Format control and link records

The Format Control record for the *ocmlrec* form updates the receiving log. You can tailor it to implement the posting process.

The link record *ocml.receive.copy* copies data from the line item to the receiving log. You can modify this link record (and the *ocmlrec* database dictionary record) to track additional data fields.

Related topics

Receiving log

1.10.4.7. Vendors

The vendors/suppliers may be either external suppliers or internal departments. We recommend that you fill in the first 7 items. You may have as many vendors/suppliers as you need.

- Vendor/Supplier Information
- Catalog Description
- Vendor/Supplier Name
- Vendor/Supplier Part No.
- Transaction Type (purchase, lease, etc.)
- Cost (in $)
- Labor Hours
- Labor Cost/Hour
- Lead Time
- Overtime Cost/Hour
- Payment Frequency
- No. Of Payments
- Billing Code
- Shipping Terms
- Vendor/Supplier Contract No.
- Priority
- Type of Order (purchase, lease, return, rental, work)
- Warranty
- Discounts

1.10.4.8. Related records

You can open a Change Management change or an Incident Management incident record from Request Management. The new record is related to an existing request.

When in a service request record, the More Actions menu enables you to work with incidents, changes, and Service Desk interactions. It also enables you to:

- Create an association between existing records.
- View a list of associated records.
- Create new records.

1.10.4.9. Printing lists and detail records in Request Management

The Request Management interface allows users to print lists and detail records for quotes, orders, and line items.

Use Search to display the list of records or a detail record. Follow the instructions below to print in the windows client or the web client:

In the Windows client:

1. Click **File** > **Print** or click the **Print** icon.
2. Select one of the following options:
    - **List**: Print the list of records.

      You can specify the **Start** and **Count** fields to print a number of consecutive records in the list.
    - **Detail**: Print a single record.
3. Select the printer and then click **Print**.

In the web client:

1. Do one of the following actions:
    - Print the list of records: Click the **Print** icon on the Record List toolbar or press Ctrl+Alt+P.
    - Print a single record: Click the **Print** icon on the Record Detail toolbar or press Alt+P.
2. Perform one of the following actions in the newly-opened browser window:
    - Right-click the page and then click **Print**.
    - Press Ctrl+P.
3. Select the printer and then click **Print**.

Related topics

Request Management overview
Orders

1.10.4.10. Business Rules and Workflows

Business rules and workflows can be defined for each type of request, and can include routing, required approvals, notifications and alerts, fulfillment tasks, escalations, assignment, and so on.

For example, the Service Catalog module lets the catalog manager define how the service request needs to be routed for fulfillment.

Request processing is automated using other appropriate Service Manager modules. For example, the Service Manager Request Management module may be used to automate a “new employee setup” request, because this type of request has goods and services components that require defined ordering and sequencing processes that can be facilitated by the Request Management module.

Other types of service requests can be fulfilled in the Service Desk, Change Management, or other modules.

A catalog administrator defines the modules required to fulfill each service catalog item. Fulfillment can include creating sequenced or parallel tasks to fulfill the request, routing tasks to appropriate groups or staff, gathering required approvals, generating notifications and alerts, measuring response time based on applied SLAs, and so on.

The service catalog uses an open connector architecture, where new fulfillment connectors can be added to integrate with third-party or legacy tools.

Service Catalog Request Status

The requester is able to use the same web application to monitor the Service Catalog request status.

The requester can simply look at the list of open Service Catalog requests and follow the progression of the Service Catalog request through its lifecycle. The status is displayed in the list.

It is possible to gain more insight into what's happening during those phases. The requester has the ability to monitor the pending approvals, or to view the approvals that have already been granted.

Thus they know who might be holding up their Service Catalog request during the approval phase.

When the Service Catalog request is approved, the user has, in some cases, the ability to follow the progression of the Service Catalog request through fulfillment in

greater details. From the request details screen, the user can click on the View Related Records link. For example, for a Service Catalog request that's being fulfilled through a change request, it is possible to follow the lifecycle of the change request.

Service Catalog Request Progress

Service Manager time and date stamps each Service Catalog request, and provides an audit trail capability that identifies each step taken in the resolution of Service Catalog request throughout the Service Catalog request lifecycle. Each time the Service Catalog request is updated (whether manually or automatically), a separate historical activity is created for the Service Catalog request. An authorized user may leaf through each historical activity to develop a comprehensive understanding of the history of the record and its resolution, including which operators took what actions at what times.

Service Manager also tracks the Service Catalog request against defined service levels. Once the Service Catalog request is created, Service Manager automatically assigns appropriate service level targets, based on the request type, the requester, the category of the request, or other defined parameters. Progress against service level targets is captured in the request record; imminent or actual breaches of service levels can trigger alerts and escalations.

Service Catalog Request Cancellation

End users can submit Service Catalog requests through the Service Manager Service Catalog module, view the status of their Service Catalog requests at any time through an easy-to-use web interface, modify existing Service Catalog requests (for upgrades or downgrades, for example) or cancel Service Catalog requests that are no longer necessary.

Request Fulfillment

Once a service request is created in the Service Catalog module, Service Manager routes the request to the appropriate fulfillment module (Request Management, Change, and so on), and the fulfillment record is automatically routed to the appropriate group or staff member for completion. Routing is based on business rules defined to meet your requirements.

Alerts can be triggered when services are added to a fulfillment group’s queue.

Request Models

The Service Manager Request Management module can accommodate requests that have complex sequential and parallel tasks; multiple, tiered approval requirements;

complex routing requirements; and so on.

Related topics

Monitor Interaction Queue for Service Level Agreement Breaches
Monitor Interaction Queue for Potential Service Level Agreement Breaches

1.10.4.11. Financial Approval

The Service Catalog module supports routing end-user Service Catalog requests for required financial approvals.

The approval rules are defined in the Service Catalog. Some approval rules are defined globally and apply to all Service Catalog requests that match the required conditions (approvals can trigger based on the total price of the Service Catalog request). Other approval rules are directly attached to catalog items, and provide additional control in the service provisioning.

The requesters are able to review the progress of the approval process at any time by looking at their open Service Catalog requests in the web interface.

Moreover, approval routing can also be performed in other Service Manager modules used to fulfill requests. For example, the Service Manager Request Management module includes capabilities to route a request for multiple sequential or parallel approvals based on the type of request, the dollar value of the request, and so on.

Related topics

Service Catalog approvals

Approvals in Request Management

1.10.4.12. Self-Help Requests

Employees shop for service items using an easy-to-use web interface. Following is an example:

1. After selecting New Request, the Catalog is displayed to the User showing the top-level categories.
2. When the user selects a category, for example, "Personal Productivity Services", the list of categories or items group under Pers.Prod.Svx is displayed.
3. Selecting an item from the Personal Productivity Services list, then the next level of the category is displayed. For example, the user might select "Hardware bundles".
4. In this example, hardware bundles include items that can be ordered. The user can select an item to order. If there are specific questions that must be answered, they are prompted for the required information.
5. The user selects “Add to Cart” to save the item.
6. A shopping cart can be viewed, canceled, saved for ordering at a later time, modified, or saved as a template for placing future orders for the same set of items.
7. Once the user is ready to order the items in the cart, he/she submits the order, at which Service Manager asks for information about the request recipient and delivery information.
8. Service Manager prompts the user to enter the request recipient and delivery information.
9. Once the Service Catalog request is submitted, Service Manager displays the submitted Service Catalog request with the assigned ID.

Related topics

Self-service access
Request a service catalog item or bundle

1.10.4.13. Record Content

Each Service Catalog request record has several fields to document the details of the Service Catalog request. The Service Catalog request record captures the request initiator, and the service recipient, if different from the request initiator. The Service Catalog request record displays the fields to record the services requested. Each Service Catalog request record has a field to capture the request closure category. Activities performed to fulfill the request are captured in the Activities section and are date and time stamped with the name of the person performing the activities. Service Manager can support additional fields or even additional tables if necessary to meet customer needs.

Service Catalog Request Status

Each Service Catalog request record has a status field that indicates the current status of the Service Catalog request. Service Manager populates this field automatically based on work performed to date on the Service Catalog request. Status codes, as well as customer-defined status codes, are supported.

The requester is also able to check the current Service Catalog request status in the Service Catalog web interface.

Category of Support Request and Service Catalog Request

For support requests, the category determines the workflow used to fulfill the support requests. Combined with the area and subarea, it also is used for to report results and to determine the knowledgebase assignment for the event.

Depending on user’s selection on support request category, the Escalate Interaction wizard opens one of the following wizards:

- Escalate Interaction - Complaint wizard: creates a new incident record in the background, and assigns it to the service desk manager.
- Escalate Interaction - Incident wizard: requests further information, including location and assignment, and create an incident record.
- Escalate Interaction - RFI wizard: creates a new incident record in the background with the default category Request for Information (RFI). The RFI incident record is assigned to the service desk assignment group.
- Escalate Interaction - RFC wizard: creates a new change request in the background, in the review phase, with the category “default.”

For Service Catalog request created in the Service Catalog module, the Service Catalog request is routed to the Service Manager module responsible for fulfilling the Service Catalog request. The request fulfillment record is automatically created, categorized and classified based on business rules built into the request fulfillment process. Each customer can create business rules to meet individual requirements.

The module responsible for fulfilling the request is defined in the catalog item record.

In addition, a category corresponding to the fulfillment module is attached to the catalog item. This helps ensure not only that the Service Catalog request will be sent to the appropriate fulfillment engine, but also that it will be properly triaged by the fulfillment teams.

Priority, Impact, and Urgency identification

Priority, impact, and urgency indicators can be assigned automatically or manually to fulfill Service Catalog requests.

Impact is based on the scope of users impacted – the wider the users impacted the higher the impact. Urgency is based on the user input. Service Manager automatically calculates the value for Priority based on the values specified for impact and urgency.

When the Service Catalog request is sent for fulfillment, however, the corresponding fulfillment modules can reevaluate the impact according to its own rules (for example, a change request, incident, or request record).

Request Closure Category

Each request fulfillment record has a field to capture the closure category.

Related topics

Service Manager Processes and Best Practice

1.10.4.14. Request Charges

Service Manager can track labor costs and parts costs in service request fulfillment processes.

Meanwhile Service Manager can pass cost and billing information for goods and services to an external financial tool. In out-of-box, integration is created using Connect-It between Service Manager and Asset Manager.

Costs of Request Fulfillment

Each request displays the fulfillment cost (including incremental ongoing costs associated with service options).

Request billing

By integrating Service Manager with Asset Manager, you can view the request billing report in Asset Manager.

You can find the "Cost of SM Service Request" widgets in the Homepage widgets library of Asset Manager. If you want to show these widgets in anyone’s homepage, you need to manually add them. For more instructions on adding widgets to homepage, see the "Customizing Web Clients > Homepage of the Web Client" section in the Asset Manager Tailoring guide.

All credit type expense line records will be assigned to the **IT Finance** billing center, which you can find in the out-of-box database of Asset Manager. If you don't want to use the **IT Finance** billing center in your production environment, you can change the Connect-It scenarios to allocate a new billing center.

1.10.4.15. Request Escalation: Functional Escalation

When a Service Catalog request is approved, Service Manager automatically assigns the Service Catalog request to an individual or support group based on pre-defined parameters, such as the items ordered.

System administrators can define escalations and assignment rules to meet your requirements. For example, when fulfilled by the Request Management module, the Service Catalog request triggers the creation of a quote record. The category of the quote record dictates assignment to an individual or a service group. The role of the requester can dictate the service levels assigned to the request for the response and resolution time frame.

Service desk technicians can also manually escalate support requests.

1.10.4.16. Request Escalation: Hierarchic Escalation

Service Manager provides alerts and escalations, including the following:

- Escalations. Business rules can be used to trigger automated hierarchical escalation of a request if the status of the request remains unchanged for too long, the request is “bounced” too often between assignment groups, or if the request meets other predefined parameters or thresholds. Users can also manually escalate a request. The administrator can configure Service Manager to act in certain ways should these rules be breached. For example, if a request remains in open state for too long, Service Manager can be configured to escalate the record to the assigned operator’s manager.
- Alerts. Some modules have the ability to set criteria for the generation of alerts.
- For more sophisticated, cross-module alerts and escalations, the Service Manager Service Level Management module is used to set Service Level Targets and Service Level Agreements with associated actions for record SLA breaches.

Escalation thresholds can be defined for Service Catalog requests as well as the request fulfillment records. Escalations can also be associated with SLAs defined for Service Catalog requests or request fulfillment records.

1.10.4.17. Request integration

Integration: Service Desk or Incident Management

In Service Manager, the service requests are stored as Service Desk interactions of a specific type. Service Manager Service Catalog can easily integrate with the Incident Management module through open fulfillment connector architecture.

Integration: Change Management

Change Management is a standard fulfillment engine for service requests.

Service Manager will copy shared information from the service request to the change request record and automatically link the records. Links between these records, once created, can't be broken by users. Links to other records are listed in each linked record, and users can click on links to access other linked records.

This link allows Service Manager to correctly propagate the status changes between the change request and the service request.

Integration: Release and Deployment Management

All Service Manager modules are fully integrated. Release and Deployment Management records can be quickly opened from a service request, if required. Service Manager will copy shared information from the service request to the release request record (created in the Change Management module) and automatically link the records. Release request records can also be opened automatically from service request records based on business rules defined to meet your needs. Links between these records, once created, can't be broken by users. Links to other records are listed in each linked record, and users can click on links to access other linked records.

Integration: Configuration Management System

One option in a catalog item is to create a subscription to a service (or to any type of configuration item).

When the service request is created for this type of catalog item, the service request is linked to the CIs through the catalog items it references.

Integration: Service Catalog

In Service Manager, the creation of a Service Catalog request can starts with the Service Catalog by using Self-Service web or reported directly to service desk via

phone or email. One or more entries in the catalog are linked to the Service Catalog request through a cart item (request line).

Integration: Third-Party

Bi-directional integration with third-party solutions can be created using Connect-It or the Web Services API in Service Manager.

1.10.4.18. Request Management Predefined Reports

Request Management Predefined Reports

Service Manager includes two out-of-box options for generating operational reports using key performance indicators: Service Manager Reports and Database Reports. These tools allow users to generate predefined and ad hoc reports to display any information. The following describes these tools.

Service Manager Reports

The Service Manager Reports module provides reports and dashboards to enable faster analysis and improved time to resolution. These reports organize data into various chart formats, and the dashboards display one or more reports to provide global information about critical activities or metrics. These reports display relationships among categories of data. For example, one report might display the number of incidents per customer, while another displays the number of incidents by priority. Viewing these reports together as a dashboard enables you to make better business decisions, such as assigning resources to close incidents. Report Managers can share a report or dashboard. The SM Reports module aims to provide a lightweight reporting feature for active operational data, and the reports are therefore designed to retrieve, represent and visualize up to 100,000 active records out of millions. To define analytical reports against the entire data set, you need to use a third-party business intelligence tool.

Database Reports

Service Manager provides built-in report views in the database for operational reporting or regularly scheduled reporting requirements. You can either review the sample report views to determine which tables, fields, and filters are used for the development of the report or create your own management reports using these details. After your customization, you need to use a third-party reporting tool (such as Microsoft Power BI) to access the Service Manager database and then generate the reports as required.

Satisfaction Survey

Service Manager provides a native survey tool, which enables you to manually send a survey from a request record or to set a schedule survey by applying the internal out-of box survey connector.

Service Manager also provides other survey integration solutions. For more information, see Survey introduction.

1.10.4.19. Service Request Record

Service Catalog requests are created and approved in the Service Manager Service Catalog module.

While IT Technicians are able to create Service Catalog requests using the Service Manager Windows or Web clients, business users can do so through a simple, easy-to-use web client.

At this point, a Service Catalog request record is created, and a unique number is displayed back to the requester.

In Service Manager, Service Catalog requests are stored as Service Desk interactions but processed in other Service Manager modules, such as Change or Request Fulfillment.

Service Catalog Description

The catalog manager can also:

- Define Service Options for a Service Catalog item (for example, specify the color or size of a selected item). The options selected can be configured to trigger a cost adjustment for the item ordered.
- Expose optional service levels.
- Limit user's access to Service Catalog items based on their assigned role.

Finally, the management user interface also allows catalog managers to duplicate existing Service Catalog items, thereby facilitating the creation of new Service Catalog items.

After a Service Catalog item is defined, its details can be exposed to end users browsing or requesting services on the corporate intranet. For example, an e-mail Service Catalog item may list available options, costs, and service levels.

Service Manager includes dozens of predefined Service Catalog items and bundles that span major categories. All the services can include Service Descriptions so the user making the request has clear information on what each service truly includes.

The Service Catalog is organized hierarchically. You can use the categories, subcategories, and items that are provided to structure the catalog. You also have the flexibility to modify, copy, and remove any of these to fit your needs. Any Service Manager business service owner and the service catalog manager in particular, can create or maintain a catalog using a simplified Service Manager interface and wizards to guide them in adding or changing service offerings. User can easily get products

and services information from Service Catalog without needing to contact the service desk. Items in the Service Catalog can include an item picture, description, attachments, and purchasing options.

Types of catalog entries within categories and subcategories can include the following:

- A request for fulfillment activities without a related purchase of goods — for example: office move
- A purchase request with no related required fulfillment activities — for example: business cards
- A request for physical goods and associated fulfillment activities — for example: memory upgrade
- A bundle of multiple items, including multiple goods and fulfillment activities — for example: new employee setup
- Project or Request for Services — for example: application-hosting service

Request Matching

Service Manager can alert request fulfillment staff if an incoming request appears to be a duplicate of an existing request (same requester, same request type, same request timeframe, and so on).

Service Manager provides a smart indicator function, which provides information on related issues and gives users the ability to drill down to this data quickly to troubleshoot the issue. Smart indicators can be configured to show other interactions and incidents open and closed for the same user, incidents associated with the same service or incidents associated with the same configuration item (CI). Smart Indicators can also be configured to show known errors for a service, providing the ability to immediately communicate to the caller information related with the service.

As an example, after a user has submitted a service request, the request will be created first as an interaction. During this process, the smart indicator can be used to browse through other interactions while looking for associated service request with the same service recipient.

Meanwhile Service Manager provides other matching options. For example: In Incident Management Security Profile, customer can define how they want to check similar or duplicate Incidents. Then later if a user contacts service desk agent to report an Incident, agent can create a Service Desk interaction to collect information about the service request. Categorize the service request as an Incident, and then solve or escalate the request. When the service request results in an Incident system can automatically identify potentially matching Incidents.

Related topics

Request a service catalog item or bundle

1.10.4.20. Request Access Control

User profiles control authorized users to create, modify, and close records. Request Fulfillment users are controlled by the Service Desk module self-service profiles for the requesting behaviors.

Also, service catalog configuration rights are controlled by capability words for accessing and edit control of Service Catalog records.

In this model, there is an “operator” control record for each user of the system. This record contains a column holding the specific “capability words” which allow or restrict functionality. The “service catalog” capability word allows or restricts request of catalog items. The “svcCatDeptRequester” and “svcCatEmployeeRequester” allow or restrict users from requesting items from the catalog for others. The “svcCatAdmin” capability word allows or restricts edit of the Service Catalog items.

Fulfillment activities are controlled by the appropriate module where fulfillment will occur. The system provides options for Never, Always, and When Assigned. For example, one user may be able to create and close records but only modify records assigned to them. Access to functionality within each module is governed by Service Manager security utilities, which define access based on role and user.

Service Manager also provides a security control to segment data between multiple customers. “Mandantan” controls identify which customer data a particular user or group can access, update, or both.

Access to Services

The Service Catalog only displays the items that the logged-in user is allowed to order, whether the user browses or searches the catalog.

Access to a service catalog item is defined in the catalog item itself. In the Access tab in the catalog item details, the catalog administrator can list the roles that are allowed to order the item. A business user will be able to see a catalog item in the catalog and request it only if this user has been granted one of the roles listed.

In addition, access to a catalog item can be restricted using access filters. It is then possible to restrict a catalog item to a specific department, or a specific business unit. The access filters are also defined as part of the catalog item definition.

1.11. Configuration Management

Service Manager Configuration Management uses Process Designer to identify, define, and track organizational assets by creating and managing records for those items.

User roles

Describes the ITIL user roles as implemented by Service Manager.

Configuration Management user roles

Key concepts

Describes the most important things that you need to know about how the Configuration Management module functions.

Configuration Management overview

Configuration Management tables

Configuration item maintenance history

Searching Configuration Management

Generating a large number of incident records

Configuration Management security options

User workflow

Describes the workflow as implemented by Service Manager and the sub-tasks that end users must perform in order to execute the workflow.

Configuration item workflow

Administrator tasks

Describes the Administrator tasks in the Configuration Management module.

View a service level agreement for a configuration item

Configure the Configuration Management environment

Contract Management overview

Additional resources

Processes and Best Practices

1.11.1. Configuration Management overview

Service Manager Configuration Management identifies, defines, and tracks organizational assets by creating and managing records for those items. Other Service Manager applications can then access these records from a central repository. For example, when you create an incident record, Incident Management can access the hardware component details from the database and populate the new record with that information. This significantly reduces the time spent to resolve the incident, as well as alerts you to other potential incidents due to component relationships and dependencies defined in the database. Configuration Management enables you to achieve the following results:

- Accommodate changes to and reuse of standards and best practices
- Significantly reduce incident resolution time by using a central repository for critical infrastructure data that can be accessed by other applications
- Map configuration grouping and business relationships
- Meet business and customer control objectives and requirements
- Authorize changes and releases or resolve incidents and problems faster
- Minimize the number of quality and compliance issues caused by improper configuration of services and assets
- Optimize the use of service assets, IT configurations, capabilities, and resources

By using Configuration Management to manage IT assets, users and administrators or auditors can perform the following tasks:

User (Access based on User role)

- Add, edit, and save asset (CI) records to the Configuration Management database, such as software, hardware, office electronics, telecommunication devices, and furnishings
- Manage CIs using predefined views to find your workload
- View and modify software installation information
- View the maintenance schedule for a CI
- View and modify service level agreement (SLA) information
- Add CIs to a contract and manage existing contracts

Administrator or Auditor

- Identify, control, record, report, audit and verify service assets and Configuration Items, including versions, baselines, constituent components, and their attributes and relationships.
- Account for, manage, and protect the integrity of service assets and Configuration Items throughout the service life cycle by ensuring that only authorized components are used and only authorized changes are made.
- Ensure the integrity of the assets and configurations required to control services and IT infrastructure by establishing and maintaining an accurate and complete Configuration Management System (CMS).

1.11.1.1. Configuration Management tables

The information displayed on the Configuration Item form is the result of the device, attribute, and join files working together.

File or record	Definition
Asset record	A record that stores details about a piece of equipment or other component.
Device type file	Each asset in the Configuration Management database has a Configuration Item (CI) type. The CI type (the **devtype** file) contains one record for each CI defined. When you define a new CI type, the **devtype** file adds a new record.
Device file	Contains all asset records. Each record contains the unique name of the asset and the pointers to other files. For example, serial.no and vendor/supplier are common fields for both the server and modem device types.
Attribute file	Contains data specific to an asset, depending on its device type. If an attribute file exists for a device type, each asset record for that device type has a corresponding attribute record in the device type's attribute file. For example, a storage device type has an attribute file named storage and contains fields that are specific to a storage asset, such as storage.type. A software license device type has an attribute file named softwarelicense that contains fields that are specific to a software license asset, such as product.pool.

File or record	Definition
Join file	A record that displays information from the device file and attribute file for the specified device type. If you make changes to the data, Configuration Management writes the information into the corresponding records in the device file and attribute file. For example, a server device type has a join file named deviceserver. It contains fields that are a combination of the device file and the relevant attribute file, such as serial.no and vendor/supplier from the device file and printer.queues from the server file.

Configuration item relationships

Service Manager tracks upstream and downstream relationships between configuration items (CIs). A relationship between CIs means that there is some dependency between CIs. If an upstream CI has an interruption of service, Service Manager assumes that all CIs with a downstream relationship to the affected CI also have an interruption of service. For example, if a network router has an interruption of service, then all servers and PCs that connect to that router also have an interruption of service.

Any given CI typically has one upstream relationship and one or more downstream relationships. CIs can have logical or physical relationships based on the logical name of the configuration item. CI relationships are independent of baseline, actual, or managed states.

Configuration item descendant relationships

To improve the performance when selecting the CI for an interruption of service, Service Manager stores the relationships between the affected CI and all CIs with the descendant downstream relationships. Refer to Initialize CI descendant relationships to generate the descendant relationships.

When you delete a CI or add, update, or delete CI relationships, the **configuration** background process recalculates the descendant relationships.

Note

Start the **configuration** process from the sm.cfg file as a separate Service Manager process when your system has a large number of CIs and relations.

Device table

The device table is the central repository for Configuration Item (CI) information. The table contains a record for each CI (hardware or software). These records provide information to populate forms used by other Service Manager applications, such as Incident Management.

When you view asset records, the form tabs and field data vary, depending on the asset selected. Only information common to all network components is stored in the device table. Asset records are identified by the logical.name field.

Characteristics of the device table:

- Only one device table.
- One logical record in the device table for every tracked asset or CI.
- logical.name field is the unique identifier in the dbdict for the device table.

Devtype table

The devtype table contains a record for each type of CI that you are tracking.

Depending on the CI component type, each logical component in the device table may have a corresponding record in a separate attribute table. Attribute tables define all common and unique attributes that appear when you select a CI type.

Cigroup table

The cigroup table consists of all information used to define items in a CI group and a list of items included in the group. There is one record for each CI group defined.

Characteristics of the cigroup table:

- Contains the group name.
- The logical.name field is the unique identifier in the dbdict for the cigroup table.
- For list-based groups, contains a list of members.
- Baseline groups and Query-based groups are dynamic and therefore don't contain members.

- Specifies a group type, such as a query-based group, a list-based group, or a baseline group.
- If the group is a baseline group, it contains the baseline version and state. This doesn't apply to list-based or query-based groups.

Note

When setting thresholds for baseline group versioning, make sure that you set an appropriate number of versioning levels to grow with your environment. If you later need more version levels than specified at setup, you can increase the number of levels without recreating entire groups and relationships.

Attribute tables

Most hardware or software Configuration item (CI) types have a set of descriptive attributes. Generally, each CI type has attributes that are organized into logical attribute records in an Attribute table. There are exceptions. Not all CI types have attribute tables, and, although the CI type records are different, a class of CI types can have the same attributes. The attribute record links to the CI record using the same identifier in the logical.name field. All other information in the attribute record and the CI record is unique.

Characteristics of attribute tables:

- There is one attribute table for every CI type.
- Attribute tables have the same name as their CI type.
- The logical.name field is the unique identifier in the dbdict for the attribute table.

For example, the following is a list of the out-of-box attribute tables for standard network components:

- bridge
- circuit
- controller
- CPU
- FEP
- modem
- multiplexer
- path
- peripheral
- port

- switch
- workstation

Join tables

A join table is a temporary logical table that has the following properties and functions:

- Resides only in memory
- Contains no records
- Is not written to the database
- Points to the data in the device table and related attribute table.
- Does have a dbdict

The join table temporarily joins the fields in the device table record with the fields in the attribute table. Join tables are never populated. They only serve as pointers to fields that are extracted from the device table and the attribute tables and they display the joined data in a form. When you make changes to the join table by making changes to the data in a form, Service Manager writes the changes to the original device and attribute tables.

As a rule, the join table name has*device* as a prefix and the attribute table name as the suffix. For example, the join table name for a PC component is devicepc. The form that displays the joined data is named device.attribute, for example, `device.pc`.

Join tables are named according to the following conventions:

- Join table name is a combination of the device table name and its attribute table name (device.attribute.name).
- Logical.name field is the unique identifier in the dbdict for the device and attribute tables.

Creating subtables from an array of structures

Service Manager enables a dbdict administrator to manage data more effectively by creating subtables of unique and non-unique attributes within an array of structures. You can use this feature to achieve the following benefits:

- Improve mapping to external SQL database tables.
- Implement a more cost-effective solution for managing attribute information.
- Simplify queries.

The dbdict administrator can identify two subtable names for each array of structures in the dbdict. One table contains the names of unique attributes. The second table

names non-unique attributes. A pop-up utility dialog box enables you to identify which attributes are unique.

The subtable feature helps you create queries that can return detailed information. This type of available detail can improve business and management decisions. You can create subtables for an array of structures in any dbdict. Service Manager provided out-of-box subtables already created for all arrays of structures in the inventory dbdicts.

Configuration Management forms

Configuration Management uses three types of forms.

Form	Description
Device	The forms device and device.g display query results for fields common to all CI types. For example, a PC record may appear using device.g.
Attribute	There is one form for each attribute file. The form displays detailed attribute data. The form name matches the attribute file name. For example, if the attribute file name is circuit, the form name is circuit.
Join	There is one form for each join file. The form enables you to view, populate, and update fields in both device and attribute files. Normal functions are available to add, delete, and update the records. Special functions are available to access parents and children. For example, if the join file name is devicecircuit, the form name is device.circuit.

Configuration item types and subtypes

Service Manager includes the following out-of-box Configuration Item (CI) types and subtypes.

Type	Subtype
Application	Anti-Virus / Security Back-up Business Development Tools Entertainment Graphics Internet/Web Networking Operating System Reference Other
Business Service	Application Service Business Service Infrastructure Service
CI Group	Ad hoc Baseline
Computer	Desktop Dumb Terminal Host Laptop Logical Partition MAC Mainframe Server Terminal Server Tower Unix VAX Windows

Type	Subtype
Display Device	Monitor Projector
Example	Template of common device fields that you can use to create a new device type.
Furnishings	Armoire Artwork Bookcase Chair Computer Desk Desk Collection File Cabinet Meeting Table
Hand Held Devices	Blackberry Device Cell Phone GPS Device Pager PDA
Mainframe	Controller FEP Host CPU LPAR NCP

Type	Subtype
Network Components	ATM Switch Concentrator Firewall Gateway Hub LB Modem Network Component Network Interface Card RAS Router Switch Router Switch
Office Electronics	Calculator Camera Copy Machine Fax Machine Multifunction Net Printer Paper Shredder Printer Speaker Television Typewriter UPS VCR Word Processor

Type	Subtype
Software License	DBMS License Development Tool License Enterprise Management License Operating System License Outlook Productivity Tools License Project Management License Utility Software License
Storage	CD Burner CDRW Direct Attached Storage (DAS) HDD Network Attached Storage (NAS) Storage Area Network (SAN) ZIP
Telecommunications	Desk Phone Flush Wall Mount Headsets and Accessories NBX Paging Solution PBX Surface Mount

1.11.1.2. Configuration item maintenance history

When a maintenance task runs, the Maintenance History function tracks the number of incident records, change requests, or request management quotes the maintenance task generated. If you set the maintenance task to create a single incident, change, or quote, the number recorded is one (1). For maintenance tasks that generate a collection of incidents, changes, or quotes, the number recorded may be different every time the task runs.

For example, on May 1, two servers are running in City A, but by June 1, an additional server is added. Therefore, a task that generates a change for every server in City A would generate two (2) changes in May, and three (3) changes in June. The maintenance history function keeps track of these numbers.

1.11.1.3. Searching Configuration Management

You can search for a specific configuration item record or a set of configuration item records. For example, if you type the beginning letters of the configuration item in the Display Name field, such as sap, and click **Search**, a list of configuration items starting with sap is displayed. The first record in the list is displayed in the configuration item details form. If you leave all fields blank, click **Search** to perform a true query and return a list of all configuration item records.

What's a joined query?

In Configuration Management, joined queries enable you to search a specific device type by using the Search form for that device type. Join tables are used by Service Manager to make searches more efficient and useful. When you search for a specific device type, the system creates a join table that includes those items that have the attributes you searched for based on the search criteria you provide.

Contents

- 1 What's a joined query?

1.11.1.4. Generating a large number of incident records

The Scheduled Maintenance advanced query function provides an anti-spam feature. By default, the system generates a maximum of 50 incident records, no matter how many records the advanced query returns. This prevents users from accidentally creating a Scheduled Maintenance task that opens an incident record for every asset in Configuration Management. However, if you want to generate a larger number of incident records, and you have administrative rights, you can increase that number.

Related topics

Configuration Management overview
Contract Management overview
Searching Configuration Management

1.11.1.5. Service Portfolio Creation

The Service Manager Configuration Management module provides service asset and configuration management and can be used to manage service portfolio records. The Configuration Management process manages service assets to support other IT Service Management processes. Services are represented as CIs in Service Manager, which enables their access to all related support and delivery processes. Out-of-box lifecycle processes include:

- Defining a new service
- Discovering, defining, and managing CI relationships
- Adding service levels (service-oriented SLA)
- Publishing a service to the Service Manager Service Catalog
- Signing up customers and managing their subscriptions
- Fulfilling subscription requests through Service Manager Change Management
- Monitoring and supporting through Incident, Change, and Problem Management
- Monitoring SLAs and optimizing service delivery

Services are provided by IT to satisfy a range of business needs. Services can be delivered to individuals, departments, or an entire enterprise as defined by service subscriptions. Sample services can include email, a billing system, workstation backup or office automation. The service instance is the deployment of a service, modeled as a CI in the CMDB, and it can have related CIs, customers, incidents, and changes.

Once the service is deployed, the Service Catalog lists and describes all of the services that IT offers, providing customers with a view to browse and make requests. Catalog items can range from single user workstations to department or enterprise application support.

Users can create new CI record instances in various ways, including:

- Creation of a new blank record
- Cloning of an existing record

A wizard is provided to create a new CI.

Service Portfolios

The following fields capture Service Portfolio details.

19 fields described in ITIL	Corresponding Field in Service Manager
Service name	Display Name/CI Identifier
Service description	Description
Service status	Status
Service classification and criticality	Secrity/Sox/Export Control classification/Critical CI
Applications used	Software/Applications
Data and/or data schema used	In UCMDB, service CI can be mapped to its database (like Oracle) down to its schema/tablespace.
Business processes supported	In PPM, see screenshot below
Business owners	Owner
Business users	Subscribers
IT owners	CI Owner
Service warranty level, SLA and SLR references	In Service Catalog module
Supporting services	Relationships
Supporting resources	Supporting Groups
Dependent services	Relationships
Supporting OLAs, contracts and agreements	Service Contract
Service costs	In Asset Manager
Service charges (if applicable)	In Asset Manager and Application Portfolio Management

19 fields described in ITIL	Corresponding Field in Service Manager
Service revenue (if applicable)	Captured in Application Portfolio Management
Service metrics	Metrics

Attachments

Each service record can contain multiple attachments, including business case documents.

Tailoring Fields of Records

Customers can add fields to the service record and tailor them into any of the service views for update, review, and management.

Service Status

Service Manager supports two separate states for a Business Service – the Status field, which is common for all CIs, and the Service Status field that's only for Service type CIs.

Service Development

The service under development can be recorded using service status fields.

Third-Party Services

Each service record lists the vendor/supplier that provides the service, if applicable. Third-party information can be recorded in contract and vendor records that are associated with the service.

Service Portfolio Lifecycle

Service Manager provides a complete set of integrated lifecycle processes that support recording, updating, monitoring and tracking business and IT services from inception through pipeline and catalog to retirement. A service is defined as anything provided by IT to satisfy a range of business needs. Services may be delivered to individuals, departments, or an entire enterprise. Sample services may include: enterprise systems such as email, line of business applications such as a billing system, and fulfillment of individual employee requests such as a personal workstation backup service or an office automation bundle.

The lifecycle activities and processes supported in Service Manager include:

- Modeling service instances using Configuration Management
- Building and visualizing relationships to IT systems and CIs
- Publishing service definitions in the Service Catalog
- Defining supported Service Levels for each definition in the Service Catalog
- Self-service requesting of services for individuals and departments
- Fulfilling service requests with Change or Request Management
- Tracking the full lifecycle of subscriptions to services, from request, to fulfillment, to cancellation
- Monitoring and supporting the services through Incident, Change, and Problem Management
- Measuring and analyzing service delivery against metrics and goals
- Optimizing service delivery and support

An important aspect of the lifecycle is that it represents a continuous flow. Within the cycle, new services are defined while others may be approaching retirement. New customers are added as users of services, while continued analysis and optimization makes improvements in the delivery and support received by these customers.

CI Records And Lifecycles

The Service Manager Configuration Management module provides service asset and configuration management. The Configuration Management process manages service assets to support other IT Service Management processes. Services are represented as CIs in Service Manager, which enables their access to all related support and delivery processes.

The Service Manager Configuration Management module captures information on CIs and CI attributes in CI records. Service Manager uses the concept of Device Types to categorize CIs, and to define what information can be collected for each CI.

Each CI record has a status field. This field can be used to track the lifecycle status of a CI, including status codes such as: “In Use,” “In Stock,” “Retired/Consumed,” “Awaiting Receipt,” “Return for Maintenance,” “Return to Supplier,” “Missing,” “Planned/On order,” “Received,” “Reserved,” “Installed.” Status codes can be created to meet the customer’s needs, and Service Manager tracks a complete history of changes to the status field. By filtering on the specific value user can get a view/list of retired services.

Lifecycle transitions

Each service includes lifecycle status that marks its transition into Released, Operation, and Retirement stages.

Transition into these stages can be coupled with Change workflows that drive related activities. In addition, a Service Catalog view can display those services currently available for public consumption.

Lifecycle States

Each service record includes lifecycle states relating to the process of defining, analyzing, approving, chartering, and so on the service. Each state transition is typically matched to a Change workflow that drives the lifecycle tasks for the service.

Pipeline Services

The Service Manager Configuration Management module provides service asset and configuration management. The Configuration Management process manages service assets to support other IT Service Management processes. Services are represented as CIs in Service Manager, which enables their access to all related support and delivery processes.

The Service Manager Configuration Management module captures information on CIs and CI attributes in CI records. Service Manager uses the concept of Device Types to categorize CIs, and to define what information can be collected for each CI.

Each CI record has a status field. All CI's are tracked through their lifecycle using the Status field, and the Environment field with values such as "Development", "Test", "Production", "Failover." By filtering on the specific value user can get a view/list of services in the pipeline.

Related topics

Service life cycle management

Create a new configuration item (CI)

1.11.1.6. Service Inventory

List views of services in the Catalog/Portfolio can be configured and viewed by different user roles.

Service Portfolio Presentation

Service Manager displays service portfolio items, including planned, deployed, and retired services. Each service has associated costs, and demand for services can be recorded through Service Catalog requests.

The Service Catalog documents services at a general level so that they can be requested by business users. Catalog items can be linked to services managed in the Configuration Management database. This is how, when a request is fulfilled, a new subscription can be added to the service CI.

You can perform business planning in the Project and Portfolio Management tool suite, and the use reporting tools to compare actual versus forecasted consumption and costs.

1.11.1.7. Service Asset & Configuration Management (SACM) Reports

SACM Summary Reports

Name	Description
Configuration Item Relationships	The Configuration Item Relationships report breaks down the configuration items (CI) and their downstream relationships by type and then by subtype. If a CI has more than one downstream relationship, the report displays the total count. Users can double-click the count to view the full list of downstream relationships.
Configuration Item Summary	The Configuration Item Summary report breaks down the configuration items by type and then by status. This report contains a pie chart displaying all CIs by type. Users can double-click any piece of the chart to display a detailed breakdown of the CI type by its status. This report uses grouping to provide a view of total counts by type and status.

Name	Description
Percentage of Configuration Items Related to Other Configuration Items	The Percentage of Configuration Items Related to Other Configuration Items report provides an overview of the number of CIs within the CMDB for which the relationship with other CIs has been defined compared to the total number of CIs within the Configuration Management System (CMS). It only shows the relationship between CIs. This report contains a graphical representation of the breakdown of all CIs by relationship with other CIs using a pie chart. Users can double-click the “CI related to other CIs” piece of the pie to display a detailed breakdown of that CI type by its subtypes. This report organizes the information using groups and sub-groups to help users quickly view percentages and total counts by type and subtype of CIs.

SACM Compliance Reports

Name	Description
Actual policy statuses	Displays policy status information on all CIs for which there are defined policies
All authorized changes	Displays all authorized changes that occurred between two points in time
All detected changes	Displays all changes that were detected between two points in time
Authorized configuration comparison	Compares authorized changes in configuration between two points in time
Authorized planned changes	Displays all planned changes that were authorized between two points in time
Authorized unplanned changes	Displays all unplanned changes that were authorized between two points in time
Changes pending authorization	Displays all changes that are pending authorization
Configuration comparison	Compares configurations between two points in time
Detailed policy drift	Compares the policy status on all CIs for which there are defined policies and displays detailed policy information about the breached CIs between two points in time
In-breach CIs	Displays policy status information for all CIs whose policies are in breach

CMS Reports

Name	Description
Application Breakdown Report	This report displays a breakdown of applications and their versions. Major applications are Running Software CITs that have been assigned the MAJOR_APP class qualifier.
Asset Report	This report displays a list of all the CIs in a specified view, their attribute values, as well as their child CIs in the CMDB.
Changed Application Report	This report displays a bar graph showing the number of detected changes in applications within a defined period.
Changed Views Report	This report displays the number of changes that occurred in a specific view within a defined period.
CI Change Report	The CI Change report enables you to view and analyze the actual changes made to a specific CI and its children.
CMDB Utilization Report	This report displays general CMDB statistics. For example, the statistics can include the number of Impact rules created, how many CIs are contained in the CMDB, or how many snapshots have been taken.
Compare Archives Report	This report enables you to compare two archived reports or an archived report to the current version of the report.

Name	Description
Compare CIs Report	This report enables you to compare the hierarchies of two compound CIs. A compound CI is a CI that's linked to another CI by a relationship with the CONTAINER qualifier. You can display the compared CIs and their child CIs that appear in a selected view, or in the CMDB.
Compare Snapshots Report	This report enables you to compare two snapshots of a specific view taken at different times. It enables you to visualize the differences between the views by comparing the statuses of the view at the times the snapshots were taken.
Configuration Manager Policy Report	This report displays Configuration Manager policy compliance level data.
Database Breakdown Report	This report displays the breakdown of database types and versions. You can view all databases deployed in the system or the databases deployed at a specific location.
Delete Candidates Report	This report displays the CITs and relationships that were deleted within a defined period, and the ones that are about to be deleted.

Name	Description
Dependency Report	This report enables you to generate a list of all the relationships of the type Dependency that connect between servers in the selected view, or between the servers in the CMDB and the servers within the selected view. The report includes relationships that: • Exist between databases and clients • Exist between clients and servers • Are created as a result of an MQ connection
Discovery Errors Report	This report enables you to uncover specific problems that Data Flow Management encounters during a run, for example, incorrect credentials.
Generic Breakdown Report	This report displays the breakdown of CI types by the following: • The CITs grouped under the selected CIT in the CIT tree • A CIT attribute
Gold Master Report	This report compares one or more CIs to another CI that has been designated as a Gold Master, and specifies which of the compared CIs are configured differently from the Gold Master CI.
Impact Analysis Report	This report displays a list of CIs that are impacted in the system as a result of simulated changes.

Name	Description
Network Device Breakdown Report	This report displays a breakdown of network devices.
Node OS Breakdown Report	This report displays a breakdown of operating systems. You can view all deployed operating systems or you can view data for operating systems at a certain location or running on a certain subnet.
Number of Changes Report	This report displays the number of detected changes in a CIT (additions, deletions, or updates) within a defined time period.
Topology Report	This report displays the topology of a view in report format.

Contents

- 1 SACM Summary Reports
- 2 SACM Compliance Reports
- 3 CMS Reports

1.11.1.8. Data Validation

Integration with discovery tools supports monitoring the infrastructure that's part of the service. Any updates/changes can be validated against planned change records to help ensure no unplanned or unexpected variations are introduced into the environment that differ from what's expected and approved in the service record.

Related topics

Discovery Event Manager

1.11.1.9. Configuration Management Access Control

Service Portfolio Management leverages Configuration Management, which is controlled by security roles. These roles allow authorized users to create, modify, and close configuration items (CI). The system provides options for Never, Always, and When Assigned. For example, one user may be able to create and close CIs but only modify items assigned to them.

Access to functionality within each module is governed by Service Manager security utilities, which define access based on role and user.

Service Manager also provides a security control to segment data between multiple customers. “Mandanten” controls identify which customer data a particular user or group can access, update, or both.

1.11.1.10. Service Portfolio Auditing

Service Manager doesn't audit the Service Portfolio items but has a utility by which the system can create a copy of the catalog item record every time it is updated. In this way, all changes to the items are fully documented.

Service Manager date/time and user stamps each record when the record is created or saved.

Service Manager also tracks changes to CI attributes in the Configuration Management module.

Service Manager provides an audit trail capability that identifies the updates to the CIs/services. Each time the CI record is updated (whether manually or automatically), updates document the attributes that were updated.

Service Manager also provides an auditing feature that records modifications to fields within the Service Manager database. Field modifications are detected by comparing the fields in the original version of a record to the updated version. When modifications are detected, an Audit Log entry is recorded for each changed field showing the name of the modified field, the old and new version of the data, the current date/time, and the current operator’s user ID.

Related topics

Audit processing
Invoke auditing from Format Control

1.11.1.11. Service Portfolio Notification and Escalation

Throughout the Service Manager suite, several ways of providing alerts and escalations are provided. The Service Portfolio items and configuration management module workflows control the requesting behavior notification and alerts while the appropriate fulfillment module controls notification and escalation for fulfillment.

Throughout the Service Manager suite, there are several ways of defining and providing alerts and escalations. The Service Catalog items and Configuration Management module workflows control notification and alerts associated with requests, while the appropriate fulfillment module controls notification and escalation for fulfillment.

1. Notifications: Service Manager provides a broad and deep solution to deliver notifications to stakeholders whenever the request or fulfillment is opened, updated or closed. The notification engine allows administrators complete control over
    1. What notification vehicle (for example, email) and format to deliver
    2. On what conditions to send the notification
    3. To whom the notification should be sent
2. Alerts: The Service Desk module includes the ability to set criteria for the generation of alerts. For example the administrator can set rules for three stages of alert and a deadline condition. When these targets are met, specified users receive alerts and notifications through the Service Manager tool, email or other method. Fulfillment records in Request Management or Change Management have even more robust notification and alert capabilities.
3. Escalations. Business rules can be used to trigger automated escalations of records under various conditions. Examples include: should the status remain unchanged for too long; the record is “reassigned” too often between assignment groups; or for other rules as defined. In addition to documenting the violation of these rules, the administrator can configure Service Manager to act in certain ways should these rules be breached. For example, if a record remains in open state for too long, reassign it to the assigned user’s manager. Fulfillment records in request management or change management have even more robust notification and alert capabilities.

Out-of-box Service Manager provides various notifications during the lifecycle of services. Service Manager also provides notification controls are activated for CIs or for Service Catalog items. The alert and escalation capabilities aren't enabled out-of-box. However, these capabilities can be added quickly and easily via simple configuration settings.

1.11.1.12. Service Portfolio Archiving

Standard Service Manager workflows include deactivation of Service Catalog or CI records. These activities don't actively remove records from the system.

The most typical approach for archiving records is to leverage the chosen RDMS tools from Oracle or MS SQL to perform database administration for purging and archiving.

Service Manager provides an archive and purge tool that can be configured by table and criteria for archiving. This is a facility only the administrator would use. This capability allows the ability to archive either manually or on a scheduled basis.

One new capability in this area is the ability to take advantage of the company's archive tool suite. These tools allow data to be archived out of Service Manager into a system-accessible file system. The production or alternative instance of Service Manager can then be used to access these data records. See the link below for additional details.

Related topics

Revision control
Archive records

1.11.1.13. Integration: Service Catalog And Knowledge Management Modules

The service portfolio is captured in the Configuration Management module and can also be displayed to end users through the Service Catalog module. Service Manager also has a Knowledge Management module that can be used to capture information specific to services.

Related topics

Service Catalogs

1.11.1.14. Configuration Item Relationship Types

Relationship type	Description
Aggregation	The "Aggregation" relationship describes a whole-part structured relationship between two elements that does NOT include any impact or dependency.
ClientServer	This Relationship represents a Client-Server dependency between, for example, two RunningSoftware instances.
Composition	The "Composition" relationship represents a strictly structured relationship, which means that the composite (part) is part of the lifecycle of the whole and that someone would percieve the "whole" as a different thing when a part is taken away.
Connection	The "Connection" relationship type represents a (potentially) undirected relationship between two elements that doesn't represent any kind of structure or dependency. Both participating elements have an independent lifecycle. There is no status propagation or health impact between connected elements.
Containment	The "Containment" relationship describes the case where containee is contained in a container. The key aspect of this relationship is the notion of exclusivity. This means that container is the sole user of the contained object and other objects aren't allowed to use the containee.

Relationship type	Description
Dependency	The "Dependency" relationship describes the logical dependency of one element on another. A dependency is a semantic relationship where a change to the influent or independent element may affect the semantics of the dependent modeling element.
ExecutionEnvironment	The "ExecutionEnvironment" is a kind of dependency relation where an instance of the end1 class runs in an instance of the end2 class and therefore depends on the end2 instance. Example: "A J2eeApplication isRunBy a J2eeServer" and a "J2eeServer runs a J2eeApplication"
ManagedRelationship	PSEUDO-RELATIONSHIP. DO NOT USE! This relationship type is provided solely to establish the attribute inheritance down to the core relationship types (Dependency, Composition, etc). Class relationships should not be based off of this type.
Membership	The "Membership" relationship describes a non-structural relationship between two elements (called member and group). Examples: A NodeGroup contains Nodes. NodeGroup A may contain all nodes of DNS domain "*.deu.hp.com". Another NodeGroup B may contain all nodes with Windows operating system. A specific node may be member of both groups.

Relationship type	Description
Ownership	The "Ownership" relationship is a connection that expresses that an instance of the end1 class owns an instance of the end2 class. Ownership is distinguished from Responsibility, in that ownership means 'accountability'. E.g. A person owns a project (is accountable for driving the project) or a person owns the incident/problem (is accountable for its resolution).
Realization	An instance of the end1 class realizes an instance of the end2 class. Realization is has a stronger connotation than just "dependency" from which is a specialization. Often the user sees the two entities as not being different on the first glance.Example: A filesystem is an exported filesystem
Responsibility	The "Responsibility" relationship is a connection that expresses that an instance of the end1 class isResponsibleFor an instance of the end2 class. Responsibility is distinguished from Ownership in that it is NOT about "accountability", but rather "assignment". E.g. A person may 'own' a project, or a person may be 'responsible' for a project.

Relationship type	Description
TimeBoundConnection	Assets are connected to Contracts: a Contract may mention and involve zero or more Assets. An Asset may be mentioned in zero or more Contracts. This RelationshipType permits attributes such as contract start and end date to be attached to the relationship, which is necessary because the contract may involve different assets at different times.
Usage	The "Usage" relationship is a kind of "Dependency" where one element uses (and thus depends on) another element.

1.11.1.15. Service Levels and Subscriptions

Each service record has associations to one or more Service Levels offered to potential customers. In addition, the services are linked to actual customers and their contracted SLAs. SLAs can be associated to services, and performance against defined SLAs is displayed graphically in Service Manager.

Moreover, service subscriptions track the relationships between IT customers and the services they use. A service subscriber, either an individual user or an entire department, can request subscriptions to various services listed in the Service Catalog. A subscriber's list of subscriptions may reference access to shared services and individually assigned CIs. Subscriptions can include SLAs, history, custom options, and pending change requests.

Service Manager offers two modes of support for requesting service subscriptions: requests by using the Service Catalog and direct requests by using Change Management fulfillment.

Each subscription can be associated with one Service SLA. The subscription table stores the SLA ID for the Service SLA associated with the subscription. Service Manager supports multiple Service SLAs when an interaction, change request, incident, or problem record is created.

A power user can search for and view the Service SLA title when viewing a subscription record. A self-service user can view the Service SLA in a read-only find field. When a self-service user clicks the Find button, a detail window opens that contains the Service SLA ID, title, and description.

Service Subscriptions and Business Data

Service subscriptions track the relationships between business users and the services they use. A service subscriber, either an individual user or an entire department, can request subscriptions to various services listed in the Service Catalog. A subscriber's list of subscriptions may reference access to shared services and individually assigned CIs. Subscriptions can include SLAs, history, custom options, and pending change requests.

This function is also available in Project and Portfolio Management (PPM). PPM Application Portfolio Management (APM) module maps business processes to applications.

In UCMDB, the CI and CI relationships are governed by the Universal Data Model (UDM). From that viewpoint, a user can see three CI types: Business Process,

Service, and Business Service – all extensions of CI type Business Element.

The CI type BusinessProcess is defined as follows in UDM:

“A Business Process is a collection of inter (user-triggers or automatic), which solve a particular issue to define the functional needs of the business service being provided. A process can be hierarchically decomposed into business activities. Usage: Some of the business processes are provided to the external customers, others are internal processes that are essential for the effective management of the business. Example: Order processing, new employee on-boarding, etc.”

The UDM allows business process CIs to be connected “downward” to service CIs and “upward” to business service CIs.

The logic is supported by the definition of CI type “service” (below) in UDM which can represent an IT service:

“A Service is an abstract class which may represent an IT or Business Service, including Infrastructure Services which are delivered by IT in support of a Business Service.”

Service Subscriptions and Technical Data

Each service is linked to the applications, hardware, and other CIs (including other services) that comprise the service. OLAs, contracts, and other agreements can be created in the Service Level Management module and linked to the service.

Moreover each subscription can be associated with one service SLA. The subscription table stores the SLA ID for the service SLA associated with the subscription. Service Manager supports multiple service SLAs when an interaction, change request, incident, or problem record is created.

A power user can search for and view the service SLA title when viewing a subscription record. A self-service user can view the service SLA in a read-only find field. When a self-service user clicks the Find button, a detail window opens that contains the service SLA ID, title, and description.

1.11.2. Configuration Management user roles

The following table describes the responsibilities of the Configuration Management user roles.

User role	Responsibilities
Configuration Manager	• Manage the Configuration Management Plan and Policies
Configuration Administrator	• Based on successful implemented change, assess current Configuration Items (CIs) and adjust CI details if required. • Create new CIs if allowed and register all required information. • Label new CIs and relate CIs to contracts. • Periodically (or on request) generate status reports and distribute to stakeholders.
Configuration Auditor	• Review and validate CMS updates and create exception reports if needed. • Conduct configuration audits and perform appropriate actions if an unregistered component is detected or if a component is missing.
CMS/Tools Administrator	• Configure the data model, policies, and CI types in the service management tool

Related topics

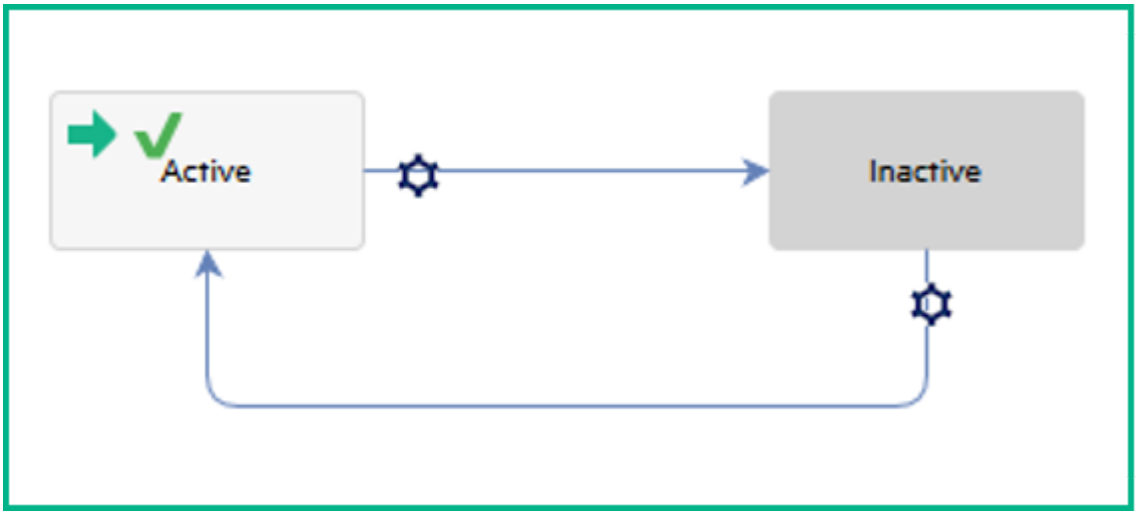
User roles
Add a user role record

Delete a user role record
Search for a user role record

1.11.3. Configuration Management workflow and user tasks

The Configuration Management process ensures that selected components of a complete IT service, system, or product (the configuration item) are identified, baselined, and maintained and those changes to them are controlled. It provides a configuration model of the services, assets, and infrastructure by recording the relationships between service assets and configuration items (CIs). It also ensures that releases into controlled environments and operational use are completed on the basis of formal approvals. It provides a configuration model of the services, assets, and infrastructure by recording the relationships between service assets and CIs.

The Configuration Management module bases on Process Designer. By default, the module uses the Configuration Item workflow.



Although you can create multiple Configuration Management workflows, Configuration Management supports the use of only one Process Designer workflow at the same time. To change a workflow that's currently used to another one, you must access the Object Definition of the `device` file, select **In Object** in the Workflow Location field, and then select another workflow in the Workflow field.

The record's status decide the workflow phase of a record that follows the Configuration Item workflow.

Status	Phase
Available	Active
In Use	Active
In Stock	Active
Awaiting Receipt	Active
Planned/On Order	Active
Received	Active
Reserved	Active
Installed	Active
Missing	Inactive
Return for Maintenance	Inactive
Return to Supplier	Inactive
Retired/Consumed	Inactive

The following table illustrates the tasks you can perform in each Configuration Item workflow phase.

Workflows phase	Common tasks
Active	• Evaluate the need for a Configuration Management System (CMS) data model update • Update Configuration Management policies • Create a new configuration item (CI) type • Modify a configuration item (CI) type • Delete a configuration item (CI) type • Validate a configuration creation change task • Create a new model • Create a new configuration item (CI) • Modify a configuration item • Plan CI relationship changes from a change record • Review a modification to a configuration item • Create a new configuration report • Conduct a configuration audit • Validate data sets • Add, update and delete primary data records • Check related configurations
Inactive	N/A

Related topics

Configuration Management overview
Service assets
Processes and Best Practices

1.11.3.1. Delete a configuration item (CI) type

Applies to User roles: System Administrator

When a Change Coordinator creates a task that requests an update to the Configuration Management System (CMS) data model, the Configuration Manager reviews the task to make sure that the change is required. If the Configuration Manager determines that the CMS data model must be updated to delete an existing configuration item (CI) type, it is the task of the System Administrator to delete the CI type in Service Manager.

To delete a CI type, follow these steps:

1. Find the task that requests deletion of an existing CI type in your To Do Queue, and then double-click the record to open it. If you don't see the task in your To Do Queue, click **Change Management** > **Search Tasks**. Use search or advanced search to find one or more records.
2. Review the task instructions regarding deletion of the existing CI type.
3. Click **Configuration Management** > **Resources** > **CI Types**.
4. Click **Search** to view a list of existing CI types.
5. Click the CI type you want to delete and view it in the Manage CI Types form.
6. Click **Delete**.
7. Click **Yes** to confirm the deletion.

   By default, Service Manager will also delete items associated with the CI type (View Format, Format Control, and so on).

8. Deselect the items that you don't want to delete, and click **Confirm** to delete the CI type.
9. After the CI type is deleted, return to the task record and update it with the following information:
   - CI data
   - Actual start date
   - Actual end date
10. Close the task. To do this, follow these steps:
    1. Click **Close**.
    2. Select **Closure Code** and then type a Closure Comment.
    3. Click **Finish**. Service Manager sets the task state to **Closed**.

11. Determine whether the Configuration Management policies require an update:
    - If an update to the Configuration Management policies is required, contact the CMS or Tools Administrator to request implementation.
    - If no update to the Configuration Management policies is required, no further action is required.

Related topics

Searching records

Create a new configuration item (CI) type
Modify a configuration item (CI) type
Update Configuration Management policies

1.11.3.2. Modify a configuration item (CI) type

Applies to User roles: Configuration Administrator, CMS and Tools Administrator

When a Change Coordinator creates a task that requests an update to the Configuration Management System (CMS) data model, the Configuration Manager reviews the task to make sure the change is required. If the Configuration Manager determines that the CMS data model must be updated to modify the definition of an existing CI type, it is the task of the Configuration Administrator or Tools Administrator to modify the CI type in Service Manager.

To modify a CI type, follow these steps:

1. Find the task that requests the modification of an existing CI type in your To Do Queue, and then double-click the record to open it. If you don't see the task in your To Do Queue, click **Change Management** > **Search Tasks**. Use search or advanced search to find one or more records.
2. Review the task instructions regarding modification of the existing CI type.
3. Click **Configuration Management** > **Resources** > **CI Types** to access CI type records.
4. Use search or advanced search to find one or more records.
5. Click the CI type you want to modify and view it in the Manage CI Types form.
6. Create or modify the definition of the CI type. For example, you may want to add or change the CI subtypes or attribute definitions.
7. Click **Save**.
8. After the new CI type is modified, return to the task record and update it with the following information:
    - CI data
    - Actual start date
    - Actual end date
9. Close the task. To do this, follow these steps:
    1. Click **Close**.
    2. Select a **Closure Code** and then type a comment in the **Closure Comment** field.
    3. Click **Finish**. Service Manager sets the task state to **Closed**.
10. Determine whether the Configuration Management policies require an update:
    - If an update to the Configuration Management policies is required, contact the CMS or Tools Administrator to request implementation.

- If no update to the Configuration Management policies is required, no further action is required.

Related topics

Searching records
Create a new configuration item (CI) type
Delete a configuration item (CI) type
Update Configuration Management policies

1.11.3.3. Create a new configuration item (CI) type

Applies to User roles: CMS and Tools Administrator

When a Change Coordinator creates a task that requests an update to the Configuration Management System (CMS) data model, the Configuration Manager reviews the task to verify that the change is required. If the Configuration Manager determines that the CMS data model must be updated to add a new configuration item (CI) type, the Tools Administrator creates the new CI type in Service Manager.

To create a new CI type, follow these steps:

1. Find the change task that requests the creation of a new CI type in your To Do Queue, and then double-click the record to open it. If you don't see the task in your To Do Queue, click **Change Management** > **Search Tasks**. Use search or advanced search to find one or more records.
2. Review the task instructions regarding the new CI type.
3. Create the new CI type. To do this, follow these steps:
    1. Click **Configuration Management** >**Administration** > **Add New CI Type**.
    2. After the wizard opens, follow the onscreen instructions.
4. After the CI type is created, return to the task record and update it with the following information:
    - CI data
    - Actual Start date
    - Actual End date
5. Close the task. To do this, follow these steps:
    1. Click **Close**.
    2. Select a **Closure Code** and then type a comment in the **Closure Comment** field.
    3. Click **Finish**. Service Manager sets the task state to **Closed**.
6. Determine whether the Configuration Management policies require an update:
    - If an update to the Configuration Management policies is required, update the policies or contact the Tools Administrator to request implementation.
    - If no update to the Configuration Management policies is required, no further action is required.

Related topics

Searching records
Modify a configuration item (CI) type

Delete a configuration item (CI) type
Update Configuration Management policies

1.11.3.4. Update Configuration Management policies

Applies to User roles: Configuration Administrator

CMS and Tools Administrator

>

User roles: Configuration Administrator, Tools Administrator

If the Configuration Manager determines that the Configuration Management policies must be updated or if an update to the Configuration Management System (CMS) data model requires a change to the policies, the Configuration Administrator or Tools Administrator implements the policy changes.

The policies include business rules and requirements for specific information to be maintained in the CMS. For example, the policies may affect the contract monitoring or determine how often a configuration audit is required. Other items that are covered by policies and business rules include:

- Naming conventions
- Labeling rules
- Asset capitalization rules
- Procedures for lost or stolen items

Configuration Management policies and requirements are translated into tool settings, such as required fields, the schedule for automated inventory and discovery, and reconciliation rules.

Implement any required updates to the Configuration Management policies in accordance with your company procedures.

1.11.3.5. Evaluate the need for a Configuration Management System (CMS) data model update

Applies to User roles: Configuration Manager

Individuals and organizational groups submit requests to create or to modify configurations as part of a variety of management processes, including User Interaction Management, Incident Management, Problem Management, and Release Management. You can also request the creationor modification of configurations based on the results of the periodic review of the Configuration Management plan. It is the Change Coordinator's role to request an update to the Configuration Management System (CMS) data model by creating a change task after there has been a successful test implementation of the change in the production environment.

The Configuration Manager evaluates any task that requests a change to the CMS data model before the manager releases the task for implementation. For example, the introduction of a new configuration item (CI) into the IT infrastructure would require a request for change and a review of that request prior to implementation of the change.

Be sure to verify that no existing CI type meets the needs of the change and that the proposed data model change doesn't conflict with other parts of the model.

To evaluate the need for a CMS data model update, follow these steps:

1. From your To Do Queue, click **Change Task** in the Queue list, and then click **All Open Tasks** in the View list.
2. Find the task requesting an update to the CMS data model, and double-click the record to open it. Use search or advanced search to find one or more records.
3. Review the task to verify whether an update to the CMS data model is required:
    - If an update is required, notify the System Administrator or CMS Administrator that the task is ready for implementation.
    - If no update is required, reject the task.
4. Determine whether the Configuration Management policies require an update.
    - If an update to the Configuration Management policies is required, contact the Configuration Administrator to request implementation.
    - If no update to the Configuration Management policies is required, no further action is required.

Related topics

Update Configuration Management policies
Create a new configuration item (CI) type
Modify a configuration item (CI) type
Delete a configuration item (CI) type

1.11.3.6. Create a new configuration item (CI)

Applies to User roles: Configuration Administrator

After you validate a configuration creation change task and confirm that configuration item (CI) types and models already exist in Service Manager for the new configuration, you can create the new CIs that make up the new configuration.

Note

- If your system is integrated with Universal CMDB, Universal CMDB sends CI records to Service Manager and attempts to match them to existing records. Your Service Manager system can be configured to automatically create a new CI record whenever Universal CMDB can't match the actual state of a CI to the values of an existing CI record. If this integration is active, you will see an Actual State section in the Service Manager CI record.
- You can't use this method to create Baseline CI Groups but you can use the Add New Baseline wizard instead. To access the wizard, click **Configuration Management** > **Resources** > **Baseline**.
- Instead of starting from scratch, you can also create a new configuration item from an existing one by using the **Duplicate Configuration Item** option.

To create a new CI, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Click **New**.

   Note

   Alternatively, you can enter information in some fields and then click **New**. For details about fields on a CI form, see Configuration items details.

3. Select the CI type and then click **Next**.
4. The Configuration Item Details form opens, and the CI Identifier field is automatically populated with a unique identification number (for example, CI11090).
5. If the **Allow User-Defined CI Identifier Values for New CIs** option in the Configuration Management Environment record is enabled, you can edit the system-generated CI Identifier value according to your needs (for example, change the CI Identifier to Color-Printer-A1234). For more information about this option, see Configuration Management environment record.

   Note

   When this option is disabled, although you are still allowed to enter a CI Identifier value before clicking **New**, a system-generated value overwrites your value when the CI is created.

6. In the Display Name field, enter a display name for the CI.

   Note

   CIs of the same CI type must use a unique display name, otherwise a validation error occurs later when you attempt to save the CI record. However, CIs of different CI types can have the same display name.

7. In the Part Number field, enter the part number of the model that's used for the new CI.
8. Enter other required or optional information to complete the CI record, and then click **Save**.

   Service Manager creates the CI record with a unique identifier.

9. If your Configuration Management policies require that CIs must be labeled, create and then attach a label to the CI. The label should include a unique identifier.
10. Repeat the steps above for all CIs that make up the configuration.
11. After you create the CIs, return to the change task and update it with the following information:
    - CI data
    - Actual start date
    - Actual end date

12. Close the task.
13. Notify the change task requester upon task completion.

Related topics

Searching records
Create a new model

1.11.3.7. Create a new model

Applies to User roles: Configuration Administrator

After you validate a configuration creation change task and determine that there is no existing model in Service Manager for the new configuration, you can create a new model and manufacturer. A model refers to the product catalog that defines the approved and certified list of components that can be deployed within the IT environment. Models contain information such as model name, manufacturer, and the ID of a component. You must create a model before you can create new configuration items (CIs) that are based on the model.

To create a new model, follow these steps:

1. Click **Request Fulfillment** > **Configuration** > **Product Catalog**.
2. Click **New**.
3. Provide the required information (a unique part number, a brief description, a model name, and a product catalog category) and other optional information.
4. Click **Save**.

You can now create new configuration items based on the new model.

Related topics

Searching records
Create a new configuration item (CI)

1.11.3.8. Validate a configuration creation change task

Applies to User roles: Configuration Administrator

A Service Manager configuration is a group of configuration items (CIs) that work together to deliver an IT service or a recognizable part of an IT service. A configuration can also refer to the parameter settings for one or more CIs. When a change task that requests configuration creation is assigned to you or your group, you must first verify that the information required to create the new configuration is complete and correct before you perform the task. You must also verify that the CI types and models exist in Service Manager before you can create new CIs.

Your Service Manager system may be configured to automatically create a change record whenever the actual state of a CI doesn't match the values in the Service Manager CI record. If this integration is active, you will see an Actual State section in the Service Manager CI record.

To validate a configuration creation change task, follow these steps:

1. From your To Do Queue, click **Change Task** in the Queue list and then click **All Open Tasks** in the View list.
2. Find the task that requests configuration creation, and double-click the record to open it. Use the search or advanced search to find one or more records.
3. Review the task to verify that all information needed to create the configuration is complete and correct:
    - If the task information is incomplete or incorrect, reject the task.
    - If the task information is complete and correct, determine the CI types needed to create the configuration and search to verify that they are available in Service Manager. To do this, follow these steps:
        1. Click **Configuration Management** > **Resources** > **CI Types**.
        2. Use search or advanced search to find one or more records.
        3. Find any existing CI types that are suitable as templates for the new CIs.
4. Verify whether CI types that match the CI creation task exist:
    - If there are no existing CI types that match the CI creation task, reject the task.
    - If there are existing CI types that can be used as templates, verify that the models from the manufacturer or supplier exist in Service Manager. To do this, follow these steps:
        1. Click **Request Fulfillment** > **Configuration** > **Product Catalog**.

2. Use search or advanced search to find one or more records.
3. Find the models from the manufacturer or supplier.

5. Verify that the models exist in Service Manager:
   - If there are no existing models, create a new model. For details, see Create a new model.
   - If there are existing models, create the new CIs that make up the configuration. For details, see Create a new configuration item (CI).

Related topics

UCMDB
Discovery Event Manager
Searching records
Create a new model
Create a new configuration item (CI)

1.11.3.9. View the audit history of a configuration item

In addition to recording activity updates, the Configuration Management module also provides field-level audit history for updates to a configuration item (CI). An audit log entry is created when a field value is updated.

Note

Audit log entries are created for a CI record only when:

- the CI type joindef is `joinnode` or `joinbizservice`
- updates are made to the fields that are defined in auditspecs table `joinnode` or `joinbizservice`.

To view the audit history of a configuration item, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Perform a search to locate the CI.
3. In the CI record, click **More** > **Audit History**.

   If the CI has a list of audit log entries, the list of entries is displayed. If only one entry, the entry is displayed.

   If the CI has no audit history, a message is displayed.
4. Click one entry from the list to open it.

   Each audit log entry provides the following information:

   - File: The dbdict that was audited.
   - Unq A: By default, the record ID is displayed.
   - Unq B: By default, this field is empty.
   - Unq C: By default, this field is empty.
   - Unq D: By default, this field is empty.
   - Date: The date when the audit log entry was created
   - Oper: The ID of the operator who modified the source record, causing (through some actions) the audit log entry to be created
   - Field: The name of the field that was modified.
   - Old: The old value/array of the field
   - New: The new value/array of the field

> **Note**
>
> To define what information to display in audit log entries, modify auditspecs table with filename `joinnode` or `joinbizservice` in database manager utility. For details, see Define an audit specifications entry

5. Click **Show Detail** to view more details.

Related topics

Configuration items details
Create a new configuration item (CI)

1.11.3.10. Plan CI relationship changes from a change record

Applies to User roles: Configuration Manager

Configuration Administrator

Configuration Auditor

>

As of version 9.31, Service Manager provides the ability to plan Configuration Item (CI) relationship changes from a change record of the Planned Change category. Two tables (dataModEventRel and dataModEventRelItem) were added to store the data modification events of relationship records and their downstream CIs. In addition, the same data modification event status change applies for both CI attributes and CI relationships. All changes are logged incrementally. For example, if a change plans to add one downstream CI to an existing relationship that already has two downstream CIs, only the new downstream CI is listed for tracking or approval.

To add a CI relationship to be changed in a Planned Change record, follow these steps:

1. From Change Management, open a change record that's not approved.
2. Click **More** > **Set Expected States**. Alternatively, open the **Associated CIs** section, and then click the **Add Attributes/Relationships** button.
3. Select one CI that you want to modify, and then click **Next**.

Note

If the change record has only one affected CI, this step is automatically skipped; if the change record has no affected CIs, the **Set Relationship** option is not selectable in the next step.

Caution

You can select only one CI in this step; otherwise the **Set Relationship** option will not be selectable in the next step. This is because you can add relationships for only one CI at one time.

4. Select **Set Relationship**, and then click **Next**.

   A new form opens. If the selected CI has existing relationships, two options are displayed: **Add New Relationship** and **Update/Delete Existing Relationship**; if the selected CI has no existing relationships, only the first option is displayed.

5. Select **Add New Relationship**, and then click **Next**. The Relationship Definition form opens.

6. Complete necessary fields on the **Relationship Detail** and **Outage Dependency** tabs, and then click **Finish**. See the following table for an example.

Field	Value
Upstream CI	CI1000800
Relationship Name	desk104connection101
Relationship Type	Physical
Relationship Subtype	Connects
Downstream CIs	adv-afr-desk-101
Outage Dependency	(selected)
This Configuration Item will be considered down if	1 or more of the supporting configuration items are down

   A message similar to the following is displayed:

   `Change Event for relationship "desk104connection101" linked to Change "C10002"` .

7. Go to the **Associated CIs** section of the change record, and locate the new relationship in the **CMDB relationships need to be changed for CIs in the list** table. The new relationship has a status of **Proposed**.
8. Click the link of the new relationship to view the relationship details.

9. If needed, change the **Status** from **Proposed** to **Canceled**, and then save your change. If the **Status** is set to **Canceled**, the relationship changes aren't processed.

Note

When the change record has been approved and moved to the implementation phase, the status of the new relationship (if not canceled) changes from **Proposed** to **Planned**, and the relationship changes then go through the rest of the change workflow. The change record can't be closed until the CI relationship changes have been implemented and validated.

Next, from the **Relationship Changes** sections of the upstream and downstream CI records you can view the new CI relationship, which has a link to the change record. In this example, the affected CIs are **adv-afr-desk-104** and *adv-afr-desk-101*, and the new relationship is *desk104connection101*.

1.11.3.11. Modify a configuration item

Applies to User roles: Configuration Administrator

You may be assigned a change task by the Change Coordinator that requests you to modify a configuration item (CI). Follow the procedure below to ensure that changes to configurations are properly controlled and records of any modifications are maintained.

To modify a CI, follow these steps:

1. Find the task that requests the modification of an existing CI in your To Do Queue, and then double-click the record to open it. If you don't see the task in your To Do Queue, click the **Change Tasks** queue to view the task.
2. Review the task to determine whether the information required to modify the CI is complete and accurate.
3. In the **Affected CI** list, click the CI that you have been requested to modify and then click **Find** to display the CI details.
4. Review the current information about the CI to make sure that the starting point for the modifications is as expected.
5. Verify that the proposed modifications are correct, applicable, and in accordance with Configuration Management policies. If the modification is not applicable, reject the change task.
6. Modify the necessary fields in the configuration record.
7. Click **Save & Exit**.
8. Notify the Change Coordinator that the CI modification is complete and that the task can be closed.

Related topics

Searching records

1.11.3.12. View component reliability

Applies to User roles: System Administrator

Reliability is a measure of how long a service or component can perform its function without interruption. The Component Reliability functionality in the Configuration Management module automatically calculates the reliability of a service or component, and presents the information in a monthly report.

To view the reliability of a service or component, follow these steps:

1. Click **Configuration Management** > **Reports** > **Component Reliability**. By default, the SLT report for last month opens.
2. Specify **CI Name**, **Year** and **Month** as needed. If you don't have the exact CI name, click the **Fill Field CI Name** icon to search for the CI name by specifying search criteria such as location, building and part number.

   The RAG (Red, Amber, and Green) status indicator on the right of the service or component shows at a glance whether the SLTs for the selected month / year have been achieved for each item. To supplement the SLT / RAG information, Mean Time Between Failures (MTBF) and Total Outages information is also displayed (where available).
3. Select a record link to see more details about the SLT information.
4. Click **Back** to exit the report.

Related topics

View the availability plan of a business service

1.11.3.13. View component availability

Applies to User roles: System Administrator

The Configuration Management module allows you to view outage information of a business or component, as well as the records affected by each outage.

To view component availability information, follow these steps:

1. Click **Configuration Management** > **Reports** > **Component Availability**.

   A list of services and components is displayed.

2. Specify **CI Name**, **Outage Start** and **Outage End** as needed.

   Tip

   If you don't have the exact CI name, click the **Fill Field CI Name** icon to search for the CI name by specifying search criteria such as location, building and part number.

3. Double-click a record from the list to view the outage details.

   A list of outages is displayed, with information about each outage start and end, as well as the outage duration.

4. Double-click each outage. All associated incident records are listed.

1.11.3.14. View the availability plan of a business service

Applies to User roles: Configuration Administrator

The availability requirements for any service are documented in the Service Level Agreement and agreed by the IT Service provider and the customer. An Availability Plan is a document which outlines the calculated forecast needs for a service to be able to meet those needs for the next business year. Its primary purpose is to define and document a plan to fulfill the forecasted requirements.

To view the availability plan of a business service, follow these steps:

1. Click **Configuration Management** > **Reports** > **Availability Plans**.

   A list of services which contain availability plans is displayed.

2. In the record list toolbar, specify **CI Name** and **Status** to narrow down the results, if needed.

   Tip

   If you don't have the exact CI name, click the **Fill Field CI Name** icon to search for the CI name by specifying search criteria such as location, building and part number.

3. Select a service from the list to open the service record.
4. Open the **Related Documents** section.
5. Click the availability plan document link to view the document.

Tip

To attach an availability plan to a service record, create a knowledge document of the Reference document type and Availability Plan subtype, and associate the availability plan document with the business service by using the Associated Service field. Once the availability plan document is published externally or internally, the document is then automatically linked to the service record under the Related Documents section.

Related topics

Create a knowledge document

1.11.3.15. Create a new configuration report

Applies to User roles: Configuration Administrator

>

Service Manager provides a number of standard configuration reports that you can view from **Configuration Management** > **CI Overview**. You may also want to define and generate your own reports. If you receive a stakeholder request for a specific report, you can prepare a customized configuration report to meet specific stakeholder requirements.

To create a configuration report, follow these steps:

1. Click **Reporting** > **Create New Report**.

   Alternatively, click **Configuration Management** > **CI Overview** > **Add Content** > **Create New Report**.

2. In the **Name** field, enter a descriptive name for the report.
3. In the **Source Data** field, select **Configuration Item - device**.
4. In the **Report Category** field, select **Configuration Item**.
5. Click **Query Builder**, and specify a query. For example, select **Application** for **CI Type** and select **Development Tools** for **CI Subtype**. Click **OK**.

   The Query box is automatically populated with a query expression based on your selections, for example, **type#"application" and subtype#"Development Tools"**.

6. On the **Chart properties** tab, configure the report properties as needed.
7. On the **Permission** tab, configure the owner and audience of this report.

   Note

   By default, you are the report owner. You can change the owner as needed.

8. On the **Advanced settings** tab, modify the default settings if needed.
9. Click **Add**.

10. Add this report to a new or an existing dashboard.

    When you open the dashboard, the report is displayed.

1.11.3.16. Review a modification to a configuration item

Applies to User roles: Configuration Auditor

When you review the modifications made to configuration item (CI) records, you should verify that any change in status is properly recorded. Often changes in status have implications to other stakeholders and departments in the organization. For example, when an asset's status changes from **In Stock** to **In Use**, this new status may have financial accounting/depreciation implications. It is important to correctly record the status of each configuration item throughout its entire life cycle.

To review a modification to a CI, follow these steps:

1. Click **Configuration Management** >**Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. Enter the CI Identifier or other information that will help you find the modified CI.
3. Select the configuration item that was modified.
4. Determine whether the information in the CI record must be reviewed or verified, based on documented Configuration Management policies and policies related to finance, procurement, Contract Management, and security procedures.
5. Verify if a life cycle event that must be reported to various stakeholders has occurred. Examples include:
    - Acceptance of a received item
    - Installation of a new asset
    - Item was lost or stolen
    - Item was retired

    If such an event has occurred, notify the stakeholders (for example, Finance, Procurement, or Contract Management).
6. Verify that all the information in the CI record is accurate and complete, based on documented Configuration Management policies and policies related to finance, procurement, Contract Management, and security procedures. You must also make sure that the modification is the result of an authorized change request.
    - If everything is correct and complete according to Configuration Management policies, open the **Audit** section and select **Audited** from the Audit Status list.
    - If the CI modification or details are incorrect and incomplete, record the exception by selecting **Minor discrepancy found** or **Critical discrepancy**

from the Audit Status list. If you maintain an exception report outside of Service Manager, you can also enter it there.

7. If there is a discrepancy or exception, create a new incident by clicking **Incident Management** > **Create New Incident**. Enter the affected CI (or copy it from the Configuration Item record screen) and other required information, and then click **Save**.

Related topics

Modify a configuration item (CI) type
Searching records

1.11.3.17. Conduct a configuration audit

Applies to User roles: Configuration Auditor

You should conduct configuration audits on a regular periodic basis, and also before or after major changes or releases. The objective is to ensure that information in the configuration management database is correct, and that all configuration items are accurately and completely recorded.

You can conduct an audit manually, or by using automated inventory and discovery tools. If you find discrepancies between the actual configuration and the details as recorded in Service Manager, you need to determine whether there is an authorized change to account for the difference, or whether further investigation or corrective action is necessary.

Note

The general procedures for conducting a configuration audit are described below. If you have the Service Manager UCMDB Integration, you can see the Actual State section, which contains detailed data about each of the CIs discovered. In addition, you can use the Service Manager Discovery Event Manager feature to set rules about handling discrepancies between the actual discovered and managed CIs.

To conduct a configuration audit, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Compare each configuration item listed in your inventory or discovered by your discovery tool to the data for the configuration item contained in the **Managed State** section and in other sections of the CI record.
4. If you discover an item that's not registered in Service Manager, follow these steps:

    a. Determine whether the item needs to be managed, based on its scope and your policies.

    b. If the item must be managed, determine its CI type and then create a new CI record in Service Manager.

c. Open the **CI Changes** section. Determine whether the mismatch resulted from an authorized change, or whether it was an unauthorized change that requires further investigation or corrective action.

d. If the discrepancy resulted from an unauthorized change, create a new incident.

5. If Service Manager lists a configuration item that can't be found, it may be lost or stolen. To check the configuration item, follow these steps:

   a. Open the **CI Changes** section. Determine whether the mismatch resulted from an authorized change (such as the authorized retirement and disposal of a piece of equipment that no longer works properly), or whether it was an unauthorized change that requires further investigation or corrective action.

   b. If the discrepancy resulted from an unauthorized change, create a new incident.

6. If the discovered item is registered in Service Manager but there is some discrepancy between the actual, registered, and managed data, follow these steps:

   a. Investigate the discrepancy to define the differences.

   b. Determine whether the managed data can be updated directly without opening a formal change.

   c. If the CI details can be updated directly, make the necessary changes in the appropriate fields in the CI record.

   d. Determine whether the discrepancy requires further investigation or corrective action.

   e. If further action is necessary, determine the appropriate investigative or corrective actions, and create an incident to execute those actions.

   f. If no further action is necessary, update the audit log with the audit status and last audit date.

Related topics

Discovery Event Manager
Discovery Event Manager managed fields
UCMDB
View or modify rules in Discovery Event Manager
Searching records

1.11.3.18. Check related configurations

Applies to User roles: System Administrator, Configuration Administrator

Note

In ITIL best practices, a Configuration Administrator manages master data. In Service Manager, however, only a System Administrator has rights to master data, because the master data is shared by all Service Manager applications. You may need to contact a System Administrator to implement master data changes or request permission to perform this function.

After you implement master data record changes, you must search for configuration items (CIs) that are related to the master data record to ensure that the updates don't conflict with the configuration administration. For example, after you update a master data record to retire a particular contact record, you should search for subscriptions or CIs that were assigned to the contact.

Caution

These steps are specific to finding related CIs, but master data is shared by all Service Manager applications. We recommend that you search for related records in all applications after updating a master data record.

To check related configurations:

1. Click **Configuration Management** > **Resources** > **Search CIs**. The Search Configuration Items form opens.
2. Specify the name of the master data record that was updated in the designated field, and then click **Search**.

   For example, if a master data record for a contact was updated to indicate that the contact is retired, select the contact in the Contact Name field.

   Tip: If you don't see a field for the master data record you want to search for, click **Advanced Filter** > **Add New Filter Criteria** to see additional fields.

   Service Manager returns a list of CI records that are related to the master data record.

3. If there are no related CIs, create a data reconciliation report summarizing the data modifications and any reconciliation errors in accordance with your company procedures.
4. If there are related CIs, determine the corrective actions and create a new incident to implement the fix. For example, if the master data record was retired, you might request that the related CIs be updated to replace the retired master data value.

Validate data sets

1.11.3.19. Validate data sets

Applies to User roles: System Administrator, Configuration Administrator

Note

In ITIL best practices, a Configuration Administrator manages primary data. In Service Manager, however, only a System Administrator has rights to primary data, because the primary data is shared by all Service Manager applications. You may need to contact a System Administrator to implement primary data changes or to request permission to perform this function.

If you receive a request from a trusted source to manually reconcile primary data in the Configuration Management system (CMS), you must first validate the data sets before implementing the changes.

Primary data is often provided by functions outside of Information Technology, such as Human Resource Management, Finance, and Facilities, and requests to change the data can come from different sources. For example, requests can be generated via Change Management as a Request for Change or via Request Management as the result of a new quote. Examples of primary data are company records, which include contact information and preferences about the companies you work with or vendor records, which contain manufacturer and retail vendor/supplier information.

Note

Service Manager can automatically update primary data if your system is integrated with other applications, such as Universal CMDB.

To validate data sets:

1. Confirm that the format and content of the data sets adhere to your company's defined specifications.
2. If the data sets meet the defined specifications, implement the changes to the primary data records in Service Manager.
3. If the data sets don't meet the defined specifications, contact the requestor for additional information.

Related topics

Check related configurations

1.11.3.19.1. Add, delete, or update a primary data record

Primary data are key data that are shared by all Service Manager applications, and are stored as records in support tables. primary data are often provided by functions outside of Information Technology, such as Human Resource Management, Finance, and Facilities.

As a System Administrator, you can create, update, and delete primary data records from the **System Administration** > **Base System Configuration** menu.

Note: In ITIL best practices, a Configuration Administrator manages primary data. In Service Manager, however, only a System Administrator has rights to primary data, because the primary data are shared by all Service Manager applications. You may need to contact a System Administrator to implement primary data changes or to request permission to perform this function.

primary data record name	Purpose
Companies	Contains contact information and preferences about the companies you work with.
Contacts	Contains contact information about users who contact the service desk to initiate a service desk interaction, incident, or change, or a person who uses components tracked in Configuration Management. This is a generally larger group than the Service Manager operators, but may include them.
Conversion Rates	Contains the conversion rates for international currencies.
Currencies	Contains the ISO codes and formats for international currencies. These records specify whether an individual currency uses European Union Currency (EUR) as its root.
Departments	Contains identifying information and service level agreements for company departments.
Fax	Contains device name and address information for your office fax configurations.
Holidays	Contains holiday dates by group (worldwide, unique to United States, etc.)

primary data record name	Purpose
Locations	Contains address and organization information about the locations in your company. You can define the site category with the following types: Critical Site, Major Site, Satellite Site, Home Site.
Models	Contains a detailed bill of materials for a part. This includes the hardware, software and services that comprise the item, as well as controls on the part and its components. The record contains information regarding what operators can do with the part, how components of this part are selected, how quote line items turn into orders, and whether to order or to consume from stock.
Standard File	Contains file configuration information, which defines file handling (import, export, etc.) for different operating systems.
Vendors	Contains manufacturer and retail vendor/supplier information. It also holds data on internal and external service providers. A vendor record for a particular manufacturer must exist before items from the manufacturer can be admitted to the Configuration Management and Request Management catalogs.

primary data record name	Purpose
Work Schedules	Defines the work hours for one or more operators. Service Manager can generate a complex 24×7 schedule that spans multiple time zones, includes all shift and break information, accommodates any regional shift to Daylight Savings time, and automatically accounts for local or national holidays.

Add a primary data record

Applies to User roles: System Administrator

Note: In ITIL best practices, a Configuration Administrator manages primary data. In Service Manager, however, only a System Administrator has rights to primary data, because the primary data is shared by all Service Manager applications. You may need to contact a System Administrator to implement primary data changes or to request permission to perform this function.

After you validate that the data sets for a new primary data record request meet your company's specifications, you can add the primary data record to Service Manager. For example, you may need to add a primary data record for a new company location or a new vendor or supplier.

To add a primary data record:

1. Click **System Administration** > **Base System Configuration**, and then double-click the primary data category you want to add a record to.

   For example, if you want to add a primary data record for a new company location, click **Locations**.

2. Specify the new primary data information and preferences.

3. Click **Add**.

   Service Manager adds the new primary data record.

Create a data reconciliation report summarizing the data modifications and any reconciliation errors in accordance with your company procedures.

Delete a primary data record

User roles: System Administrator, Configuration Administrator

Note

In ITIL best practices, a Configuration Administrator manages primary data. In Service Manager, however, only a System Administrator has rights to primary data, because the primary data is shared by all Service Manager applications. You may need to contact a System Administrator to implement primary data changes or to request permission to perform this function.

After you validate that the data sets for a primary data record delete request meet your company's specifications, you can delete a primary data record in Service Manager. For example, you may need to delete a location record because the site was closed.

Caution

When you delete a primary data record, it is possible that some configuration item records will still include the old primary data value. Before you delete primary data records, we recommend you conduct a mass update to replace the primary data value scheduled for deletion with the new approved value. This prevents obsolete data from remaining in Service Manager CI records.

To delete a primary data record:

1. Click **System Administration** > **Base System Configuration**
2. Double-click the primary data category for the record you want to delete.

   For example, if you need to delete a primary data record for a user who is no longer with the company, click **Contacts**.
3. Fill in optional search criteria, and then click **Search**.

   Service Manager returns a list of records matching your search request.
4. Double-click the primary data record you want to delete.
5. Click **Delete**.

> **Note**
>
> When you delete the contact record, the corresponding operator record will also be deleted.

6. Click **Yes** to confirm the deletion.

   Service Manager deletes the primary data record and notifies you that the corresponding operator record has also been deleted.

Related topics

Data maps
Exporting records
Importing records
Purging and archiving records
Unload script utility
Check related configurations

1.11.3.20. View scheduled maintenance for a configuration item

Applies to User roles: Configuration Manager

Configuration Administrator

Configuration Auditor

To view scheduled maintenance for a configuration item, follow these steps:

1. Click **Configuration Management** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select a record to display it in the configuration item form.
4. Click **More** or the **More Actions** icon from the Configuration Item form.
5. Click **Scheduled Maintenance** and then click **Maintenance Tasks** to view the tasks or **Maintenance History** to view the item's history.

1.11.3.21. Service life cycle management

Services are represented as Configuration Items (CIs) in Service Manager, which enables CI access to all related support and delivery processes. Life cycle processes include the following:

- Define a new service.
- Discover, define, and manage CI relationships.
- Add Service Levels (service-oriented SLA).
- Publish a service to the Service Catalog.
- Sign up customers and manage their subscriptions.
- Fulfill subscription requests through Change Management.
- Monitor and support by using Incident, Change, and Problem Management.
- Monitor SLAs and optimize service delivery.

Services are provided by IT to satisfy a range of business needs. Services can be delivered to individuals, departments, or an entire enterprise. Sample services can include email, billing system, workstation backup, or office automation. The service instance is the deployment of a service, modeled as a CI in the CMDB, and it can have related CIs, customers, incidents, and changes.

Once the service is deployed, the Service Catalog lists and describes all of the services that IT offers, providing customers with a view to browse and make requests. Catalog items can range from single-user workstations to department or enterprise application support.

Note

For more information on Service life cycle management, refer to the *Service life cycle management* white paper (document ID KM524933).

Related topics

Visual design of CI relationships
Publish services in Service Catalog
Requesting services
Service-oriented SLAs
Service support
Subscriptions

1.11.3.21.1. Visual design of CI relationships

The following are the visualization features in Service Manager.

Feature	Description
CI Visualization	A Configuration Manager user can define CI relationships and can visualize and edit them with a graphical CI diagram tool.
CI node grouping	Flexible CI aggregation gives you the option on any folder grouping of CIs to display the individual CIs.
Display of outage spreading in CI visualization	CI Visualization provides enhanced outage spreading modeling. It displays the Item Down decorator to indicate that a Configuration Item (CI) is down.
Out-of-box icons for decorators	Several CI Visualization decorators, including an SLA breached decorator, are available with the out-of-box application. You can optionally customize decorators or add new decorators.

Related topics

Service life cycle management
Publish services in Service Catalog
Requesting services
Service-oriented SLAs
Service support

1.11.3.21.2. Publish services in Service Catalog

The Service Catalog serves as the comprehensive archive of all services provided by IT. You can manage the publication of services by using the following features.

Feature	Description
Out-of-box Service Catalog	The Service Catalog includes one hundred (100) services and bundles that span major categories such as Personal Productivity, Business Services, Employee Life Cycle, and Technical IT.
Add an item	You can define the external view of a service.
Adding delivery objectives for a catalog item or bundle	You can enter service options such as delivery objectives that are dynamically exposed through the self-service shopping interface.
Defining a catalog item that results in the creation of a subscription	You can set the criteria for subscriptions.
Service Catalog connectors	You can select the subscription change connector.

Related topics

Service life cycle management
Visual design of CI relationships
Requesting services
Service-oriented SLAs
Service support

1.11.3.21.3. Requesting services

The Service Catalog self-service interface serves as a single point of entry to request any service provided by IT. Subscribers utilize the following features when using the request interface.

Feature	Description
Subscriptions	Service subscriptions track the relationship between IT customers and the services they use.
Choosing a subscription mode	Service Manager offers two modes of support for requesting service subscriptions: Service Catalog requesting and direct request with Change Management fulfillment.
Define a department requestor for subscriptions	You can define and manage the list of users that are allowed to request services for the department subscribed to the service.
Service Catalogs	Users can browse for services in the Service Catalog.
Request for a department	You can request a service for a department.
Request a service catalog item or bundle	You can request a service for an individual.

1.11.3.21.4. Service-oriented SLAs

Service SLAs define the level of ongoing service for items described and available for order in the Service Catalog.

Feature	Description
Adding Agreements to a catalog item or bundle	You can define service-oriented SLAs for items in the Service Catalog.
Request a service catalog item or bundle	Requesters can view the SLAs provided to support an item or service in the catalog.
Subscriptions and service level agreements	You can drill down from a catalog item or subscription to view the details of the SLA.

Related topics

Service life cycle management
Visual design of CI relationships
Publish services in Service Catalog
Requesting services
Service support

1.11.3.21.5. Service support

Service Manager enables service support with features that help you assess risk and plan changes according to service impacts.

Feature	Description
Affected Item field in Incident Management	Incident records include a link to the Contact's affected service. Based on the contact, a list of contact subscribed services are presented to the operator for more accurate collection of incident information.
Affected CI field in Service Desk	Interaction records include a link to the Contact's affected service. Based on the contact, a list of contact subscribed services are presented to the operator for more accurate collection of interaction information.
Viewing subscriptions on the contacts and department records	You can drill into the details for the subscription by double-clicking on it in the Windows client or clicking the link in the Web client.
View the subscribers for a subscription	Operators managing a CI have access to the list of all subscribers enabled to access the CI, both through individual and departmental subscriptions.
Notifying service subscribers of a planned outage	You can send an email notification of the planned outage to all the individuals and members of any departments that are subscribed to the service.

Related topics

Service life cycle management
Visual design of CI relationships
Publish services in Service Catalog
Requesting services
Service-oriented SLAs

1.11.3.21.6. Service life cycle status

The Service Status indicates at what point the service or Configuration Item (CI) is within its life cycle. Service status options are modeled on the Information Technology Infrastructure Library (ITIL) v3 life cycle statuses. The following are the available service status options:

- Requirements
- Analyzed
- Approved
- Chartered
- Designed
- Developed
- Build
- Test
- Release
- Operational
- Retired

Related topics

Service life cycle management
Visual design of CI relationships
Publish services in Service Catalog
Requesting services
Service-oriented SLAs
Service support

1.11.3.21.7. Subscriptions

Service subscriptions track the relationships between IT customers and the services they use. A service subscriber, either an individual user or an entire department, can request subscriptions to various services listed in Service Catalog. A subscriber's list of subscriptions may reference access to shared services and individually assigned CIs. Subscriptions can include SLAs, history, custom options, and pending change requests.

Service Manager offers two modes of support for requesting service subscriptions: requests by using Service Catalog and direct requests by using Change Management fulfillment.

1.11.3.21.8. Choosing a subscription mode

An administrator can choose between two subscription modes in the Configuration Management (CM) environment record. The selected mode determines whether a user subscribes to a Configuration Item (CI) by using Change Management or by using Service Catalog.

If the mode is set to Change Management, a Self-Service user can subscribe and cancel subscriptions from the subscription list view on the Self-Service page. If the mode is set to Service Catalog, a user can only subscribe through Service Catalog, although canceling a subscription can still be performed within the subscription record. The default mode is Service Catalog.

Choose a subscription mode

Applies to User roles: System Administrator

You can choose a subscription mode on the Configuration Management environment record to control whether a user subscribes to a configuration item (CI) by using Change Management or by using Service Catalog.

To choose a subscription mode, follow these steps:

1. Click **Configuration Management** > **Administration** > **Environment**. The Configuration Management Environment Profile record opens.
2. In the **Subscriptions** section, select **Service Catalog** or **Change Management** for the Subscription Request Mode.
3. Click **Save**.
4. Click **OK**. Service Manager updates the Configuration Management environment record.

Contents

- 1 Choose a subscription mode

1.11.3.21.9. Defining the change category to use when adding or canceling subscriptions

To define what change category you want to use when subscribing and unsubscribing to a specific subscription, set the change categories in the business service record (CI). If no values are specified in the business service record, the values in the Configuration Management environment record are used.

If you want to define the change categories for subscriptions at a global level, set the change category values in the Configuration Management environment record. If no values are specified in the CM environment record, the default value used is `Subscription`.

If you want to create a new change category for subscriptions, the best practice is to copy the `Subscription` category and give it a new name. After you create the new change category, you can change the CM environment record to specify the new category.

Caution

This is only relevant when subscription requests are handled directly by Change Management without the use of Service Catalog.

Define the change category to use for a specific subscription

Applies to User roles: System Administrator

If your system is designed to handle subscription requests through Change Management, you can define the change category that Service Manager uses when users subscribe or unsubscribe to a specific subscription by updating the configuration item (CI) record.

Caution

If your system is designed to handle subscription requests through Service Catalog, you need to use Service Catalog options to control how subscription requests are handled.

To define the change category to use for a specific subscription, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records with the CI type of **Business Service**.
3. To define the change category that Service Manager uses when a user subscribes to the subscription, select a change category from the Subscription Add list in the Change Categories area.
4. To define the change category that Service Manager uses when a user unsubscribes to the subscription, select a change category from the Subscription Remove list in the Change Categories area.
5. Click **Save**.
6. Click **OK**. Service Manager updates the CI record.

Define the change category to use globally for subscriptions

Applies to User roles: System Administrator

If your system is designed to handle subscription requests through Change Management, you can define the change category Service Manager uses for subscriptions at a global level by updating the Configuration Management environment record.

Caution

If your system is designed to handle subscription requests through Service Catalog, you will need to use Service Catalog options to control how subscription requests are handled.

To define the change category to use globally for subscriptions, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Environment Records** > **Configuration Management Environment**. The Configuration Management environment record opens.
2. To define the change category Service Manager uses when a user subscribes to a subscription, select a change category from the Subscription Add list in the Change Categories for Subscriptions area.
3. To define the change category Service Manager uses when a user unsubscribes to a subscription, select a change category from the Subscription Remove list in the Change Categories for Subscriptions area.
4. Click **Save**.
5. Click **OK**. Service Manager updates the Configuration Management environment record.

Contents

- 1 Define the change category to use for a specific subscription
- 2 Define the change category to use globally for subscriptions

1.11.3.21.10. Viewing subscriptions on the contacts and department records

When a power user views a Contacts record, there is a Subscription tab. The Subscription tab includes a Personal subscription table and a Department subscription table for the contact's department. The user can drill into the details for the subscription by double-clicking on it in the Windows client or clicking the link in the Web client.

When a power user views a Department record, there is a Subscription tab, which shows all of the subscriptions for the department. The user can drill into the details for the subscription by double-clicking on it in the Windows client or clicking the link in the Web client.

1.11.3.21.11. Notifying service subscribers of a planned outage

Whenever a change record is opened against a configuration item (CI) and the **Down Start** and **Down End** fields are populated, all individual users and members of any departmental group that are subscribed to the CI receive an email notification of the planned outage. If the **Down Start** and **Down End** values are changed, an email is sent notifying subscribers that the outage is rescheduled.

1.11.3.21.12. Subscriptions and service level agreements

Each subscription can be associated with one Service SLA. The subscription table stores the SLA ID for the Service SLA associated with the subscription. Service Manager supports multiple Service SLAs when an interaction, a change request, an incident, or a problem record is created.

A power user can search for and view the Service SLA title when viewing a subscription record. A self-service user can view the Service SLA in a read-only find field. When a self-service user clicks the **Find** button, a detail window opens that contains the Service SLA ID, title, and description.

Related topics

Service life cycle management
Subscriptions
Change Management and Service Level Agreements
Incident Management and service level agreements
Problem Management and Service Level Management
View my subscriptions
View subscriptions for the departments you manage
Administer subscriptions

1.11.3.21.13. View my subscriptions

Applies to User roles: Self-Service User

You can view your subscriptions for services you use from the Self-Service home page. Your list of subscriptions may reference access to both shared services and individually assigned configuration items (CIs). Subscriptions can include service level agreements (SLAs), history, custom options, and pending change requests.

To view your subscriptions, follow these steps:

1. Click **Subscriptions** > **For My Use** on the Self-Service home page. A list of all your individual and department service subscriptions opens.
2. Select a name to view the service description, status, history, and Service SLA.
3. In the **Service SLA** field, click the **Find Related Information** button to view a detail screen that contains the Agreement ID, Title, and Description.
4. Click **Cancel**, and then click **Back** to return to the list of service subscriptions.

Note

If it is an individual subscription, your name appears in the **Subscriber** field. If it is a department subscription, the name of the department appears in the **Subscriber** field.

Related topics

Service life cycle management
Business service groups
Defining the change category to use when adding or canceling subscriptions
Searching records
View subscriptions for the departments you manage
Add a subscription
Add a subscription for the departments you manage
Subscribe to a service you canceled previously
Cancel a subscription
Cancel a subscription for the departments you manage
Define which services are available for a subscription
Define a department requestor for subscriptions
View the subscribers for a subscription
Administer subscriptions
View related changes for a subscription

1.11.3.21.14. View subscriptions for the departments you manage

Applies to User roles: Self-Service User

You can view subscriptions for services used by departments you manage from the Self-Service home page. The list of subscriptions displays both shared services and individually assigned configuration items (CIs). Subscriptions can include service level agreements (SLAs), history, custom options, and pending change requests.

To view subscriptions for the departments you manage, follow these steps:

1. Click **Subscriptions** > **For Departments I Manage** on the Self-Service home page. A list of subscriptions for the department opens.
2. Select a name to view the service description, status, history, and Service SLA.
3. In the **Service SLA** field, click the **Find Related Information** button to view a detail screen that contains the Agreement ID, Title, and Description.
4. Click **Cancel**, and then click **Back** to return to the list of service subscriptions.

Note

If you manage more than one department, you can see the subscriptions for all the departments.

Related topics

Service life cycle management
Business service groups
Defining the change category to use when adding or canceling subscriptions
Searching records
View my subscriptions
Add a subscription
Add a subscription for the departments you manage
Subscribe to a service you canceled previously
Cancel a subscription
Cancel a subscription for the departments you manage
Define which services are available for a subscription
Define a department requestor for subscriptions
View the subscribers for a subscription

Administer subscriptions

View related changes for a subscription

1.11.3.21.15. Add a subscription

Applies to User roles: Self-Service

User

From the self-service home page, you can request subscriptions to various services listed in Service Catalog.

Note

This procedure assumes that your system manages subscriptions from Change Management. If you don't see the options described in this procedure, it means that your system manages subscriptions from Service Catalog.

To add a subscription, follow these steps:

1. From the self-service home page, click **Subscriptions** > **For My Use**. A list of all your subscriptions opens.
2. Click **Subscribe** to view to a list of available subscriptions.
3. Select a Service ID to view the details of a subscription.
4. Type a reason for subscribing to this service.
5. Click **Subscribe**. A list of all your subscriptions opens. The status of the service is now set to **Requested**.
6. Click the name of the subscription you added to view its status, history, and any comments you added.
7. Click **Back** to return to the list of your subscriptions.

1.11.3.21.16. Add a subscription for the departments you manage

Applies to User roles: Self-Service

User

On the self-service home page, you can request subscriptions to various services listed in Service Catalog for departments you manage.

Note

This procedure assumes that your system manages subscriptions from Change Management. If you don't see the options described in this procedure, it means that your system manages subscriptions from Service Catalog.

To add a subscription for the departments you manage, follow these steps:

1. Click **Subscriptions** > **For Departments I Manage** on the self-service home page. A list of your department subscriptions opens.
2. Click **Subscribe**.
3. If you are the department requestor for more than one department, select a department name from the list.
4. Select a Service ID from the list of available subscriptions to view the details of that subscription.
5. Type a reason for subscribing to this service.
6. Click **Subscribe**. A list of your department subscriptions opens. The status of the service is now set to **Requested**.
7. Click the name of the subscription you added to view its status, history, and any comments you added.
8. Click **Back** to return to the list of your subscriptions.

1.11.3.21.17. Subscribe to a service you canceled previously

Applies to User roles: Self-Service

User

You can subscribe to a previously canceled service from the Self-Service home page.

Note

This procedure assumes that your system manages subscriptions from Change Management. If you don't see the options described in this procedure, then your system manages subscriptions from Service Catalog.

To subscribe to a service you canceled previously, follow these steps:

1. Click **Subscriptions** > **For My Use** on the Self-Service home page. A list of your subscriptions opens.
2. Select a name to view the subscription detail for the canceled service.
3. Click **Subscribe**.
4. Type a reason for resubscribing to this service.
5. Click **OK**. You return to the list of all your subscriptions. The status of the subscription is now set to **Requested**.
6. Click the name of the subscription you added to view its status, history, and any comments you added.
7. Click **Back** to return to the list of all your subscriptions.

1.11.3.21.18. Cancel a subscription

Applies to User roles: Self-Service User

You can cancel a subscription to a service you no longer need from the Self-Service home page.

> **Note**
>
> This procedure assumes that your system manages subscriptions from Change Management. If you don't see the options described in this procedure, then your system manages subscriptions from Service Catalog.

To cancel a subscription, follow these steps:

1. Click **Subscriptions** > **For My Use** on the Self-Service home page. A list of your subscriptions opens.
2. Select a name to view the subscription details for the service you want to cancel.
3. Click **Unsubscribe**.
4. Type a reason for unsubscribing to this subscription.
5. Click **OK**. You return to the list of all your subscriptions. The status of the subscription is now set to **Cancelation Requested**.
6. Click the name of the subscription you canceled to view a detail of the status, history, and the comments you added.
7. Click **Back** to return to the list of your subscriptions.

Related topics

Service life cycle management
Business service groups
Defining the change category to use when adding or canceling subscriptions
Searching records
View my subscriptions
View subscriptions for the departments you manage
Add a subscription
Add a subscription for the departments you manage
Subscribe to a service you canceled previously
Cancel a subscription for the departments you manage
Define which services are available for a subscription

Define a department requestor for subscriptions
View the subscribers for a subscription
Administer subscriptions
View related changes for a subscription

1.11.3.21.19. Cancel a subscription for the departments you manage

Applies to User roles: Self-Service

User

You can cancel a subscription to a service used by the departments you manage from the Self-Service home page.

> **Note**
>
> This procedure assumes that your system manages subscriptions from Change Management. If you don't see the options described in this procedure, then your system manages subscriptions from Service Catalog.

To cancel a subscription for the departments you manage, follow these steps:

1. Click **Subscriptions** > **For Departments I Manage** on the Self-Service home page.
   A list of your department subscriptions opens.
2. Select a name to view the subscription details for the service you want to cancel.
3. Click **Unsubscribe**.
4. Type a reason for unsubscribing to this subscription.
5. Click **OK**. You return to the list of all your subscriptions. The status of the subscription is now set to **cancelation Requested**.
6. Click the name of the subscription you canceled to view its status, history, and any comments you added.
7. Click **Back** to return to the list of your subscriptions.

1.11.3.21.20. Define which services are available for subscription

Applies to User roles: System Administrator

In the Change Management subscription model, users can only subscribe to business services for which you have explicitly allowed subscriptions. If you are using the Service Catalog subscription model, you manage subscriptions by adding or removing business services from the catalog.

To define which services are available for a subscription, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. In the **Type** field list, select **Business Service**.
3. Select the Business Service definition record for the service to which you want to allow user subscriptions.
4. Check the **Allow Subscriptions** checkbox. Users can now subscribe to this service.
5. Click **Save**.

1.11.3.21.21. Define a department requestor for subscriptions

Applies to User roles: System Administrator

You can grant specific users the right to request subscriptions for their department. Only these users can add new subscriptions for the department, but all users in the department can view the subscribed records.

To define a department requestor for subscriptions, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Departments**.
2. Use search or advanced search to find one or more records.
3. Open the Department Data record for the department name that requires a requestor definition.
4. In the **Department Requestors** field, type the names of the users who are allowed to request services for the department.
5. Click **Save**.

1.11.3.21.22. View the subscribers for a subscription

Applies to User roles: System Administrator

When a power user views a Contacts record or a Department record, a Subscription tab is displayed. The Subscription tab displays all of the subscriptions for personal subscriptions and department subscriptions.

When the Subscription tab is available, you can define services and department requestors; add subscriptions and administer subscriptions; and view subscribers and related changes.

To view the subscribers for a subscription, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select the target record.
4. Open the **Subscribers** section. A list of subscribers with the types of subscriptions (individual or department) and their statuses is opened.
5. Click a **Subscriber** in the list to display the subscription details.

1.11.3.21.23. Administer subscriptions

Applies to User roles: Configuration Administrator, System Administrator

You can manage the subscriptions of other users to define what services are available to individual users and department requester. You can also view all the personal and department subscriptions to a particular record.

To administer subscriptions, follow these steps:

1. Click **Configuration Management** > **Resources** > **Subscriptions**.
2. Use search or advanced search to find one or more records.
3. Select the record with subscriptions you want to manage.
4. Each subscription contains the following information:
   - Subscription ID (can't edit)
   - CI
   - Status
   - Subscriber
   - Subscriber Type
   - History (can't edit)
   - Related Changes (view change record)
   - Service SLA
5. Add or modify the subscription fields as needed.
6. When you complete your changes, click **Next** to select another record. You are prompted to save your changes.
7. Click **Save** to save your changes and exit the record.

Caution

If you click **Back**, you exit the record without saving your changes.

1.11.3.21.24. View related changes for a subscription

Applies to User roles: Self-Service

User

You can view changes related to your individual or department subscriptions from the Self-Service home page.

To view related changes for a subscription, follow these steps:

1. Click **Subscriptions** > **For My Use** or **For Departments I Manage** on the Self-Service home page. A list of your individual or department subscriptions opens.
2. Select a name record to view the description detail.
3. Select a change number to view the change record.
4. A subscription that has one or more related changes contains the following information:
    - Change No.
    - Status
    - Approval Status
    - Change Coordinator
    - Phase
    - Subscription Action

1.11.3.22. Configuration item actual states

By default, Service Manager only stores and displays the expected or managed state of CIs. The information Service Manager displays in the Configuration Management form is essentially the definitive list of attributes that the CI should have. However, the actual state of the CI may differ from the expected state.

To view the actual state of the CI, you must first create an integration to an Universal CMDB server. The Universal CMDB server periodically discovers the actual state of CIs and records the actual state in the Configuration Management database. Service Manager accesses the actual state information by using a Web services connection. Service Manager sends the CI ID to the Universal CMDB server and receives a full list of the attributes for that CI. Service Manager displays the CI attributes in the Actual State section of the Configuration Management form.

If a Service Manager CI doesn't have a matching CI in the Universal CMDB server, then Service Manager doesn't display the Actual State section. For example, you may track office furnishing CIs in Service Manager that can't be discovered and tracked in the Universal CMDB.

View the actual state of a configuration item

Applies to User roles: Configuration Manager

Configuration Administrator

Configuration Auditor

This procedure requires that your Service Manager system has an active integration to an Universal CMDB server.

To view the actual state of a configuration item, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select the CI with the actual state you want to see.
4. From the CI detail form, open the **Actual State** section. Service Manager displays the actual state of the CI, as well as its UCMDB Id value.

Note

The **Actual State** section is only visible if your system has an active integration to an Universal CMDB server and the CI has a matching entry in the Configuration Management database. Service Manager doesn't display the **Actual State** section if there is no matching CI in the Universal CMDB server.

Contents

- 1 View the actual state of a configuration item

1.11.3.23. Set the expected state of a configuration item

This procedure provides a standard process to change the state of one or multiple configuration items in Service Manager. This procedure is also applicable if your Service Manager system has an active integration to an Universal CMDB server.

To set the expected state of one or multiple configuration items, follow these steps:

1. Sign in to Service Manager.
2. Go to **Change Management**, create a new change record or open an existing change record.
3. Make sure the change record has one or multiple affected configuration items specified. You will set the expected change states of some attributes of these configuration items.
4. Click **More** or the **More Actions** menu, and then select **Set Expected States**. The Set Expected Change State wizard opens.

   Note

   - The wizard can be started only when at least one configuration item has been specified in the **Affected CIs** field.
   - If the change record associates with multiple configuration items, the Select CI wizard opens first and you need to select the configuration items for which you want to set the expected state.

5. There are two options in the Set Expected Change State wizard: **Select From a Baseline** and **Set Attribute Name and Values**. The **Set Attribute Name and Values** option is selected for the following steps. For detailed information about how to set the baseline for a configuration item, see Associate a configuration item (CI) to a baseline.
6. Select **Set Attribute Name and Values** and then click **Next**.

> **Note**
>
> If the selected configuration items belong to various CI types, only attribute fields listed in the `device` table can be planned. If the selected configuration items have the same CI type, new attribute fields can be planned for all attributes of this CI type.

7. In the Attribute Names and Values page, you can add new fields, edit fields or remove fields.
   - To add a new field, follow these steps:
     1. Click **Add New Field**. The Baseline Attributes page opens.
     2. Select a field name from the **Field in** drop-down list, and then type or select the corresponding value in the **Value** field. Click **Next**.

        > **Tip**
        >
        > - The supported field types include simple data type (string, number, logical, and date), array field of simple data types and array of structure field. Array of structure field refers to a field which is an array. The element type within the array is structure, which contains simple data types. For detailed information about how to configure the value of array of structure field, see *Example: How to plan array of structure field* below.
        > - To plan the new attribute values, click the **Format Based Changes** button.

     3. The new field is displayed in the Attribute Names and Values page. Click **Finish**.
   - To edit an existing field, select the field you want to edit and then click **Edit Field**.
   - To remove a field, select the field you want to remove and then click **Remove Field**.

   All attribute changes are saved in the `dataModEvent` tables. You can open the change record and review the planned attribute change in the **Associated CIs** section and the **Status** is **Proposed**.

8. When the change is approved and implemented, the status of the planned attribute change is **Planned**.
9. Click the attribute change record. In the CMDB Data Change Event page, change the **Status** from **Planned** to **Implemented**.
10. If your Service Manager system has an active integration to an Universal CMDB server, Universal CMDB detects the change that's implemented as planned and the value is pushed to Service Manager later. If the new value is the same as planned, the **Status** is automatically changed to **Validated**.

    If your Service Manager system is not integrated to an Universal CMDB server, you can also manually check whether the actual new values are the same as planned or not and set the **Status** to **Validated**.
11. Close the change and the validated planned changes will be applied to the attribute values of the configuration item in Service Manager.

    The whole process of setting the expected state for the configuration item is complete.

Example: How to plan an array of structure field

When planning array of structure fields, you need to do it field by field with the structure and combine these fields together to perform validation and apply the values.

Take the **addIIPAddr** field in the `computer` table as an example. As a typical example of array of structure fields, this field consists of **addIIPAddress** and **addISubnet** sub fields. The following table lists the original values.

addIIPAddress	addISubnet
192.168.0.1	255.0.0.0
192.168.0.2	255.0.0.1
192.168.0.3	255.0.0.2

Change the values as listed in the following table.

addIIPAddress	addISubnet
192.168.0.4	255.0.0.3
192.168.0.5	255.0.0.4
192.168.0.6	255.0.0.5

You need to plan for **addIIAddress** and **addISubnet** separately, and then change the values as listed in the following table.

Field Name	Old Value	New Value
addIIAddress	{“192.168.0.1”, “192.168.0.2”, “192.168.0.3” }	{“192.168.0.4”, “192.168.0.5”, “192.168.0.6” }
addISubnet	{“255.0.0.0”, “255.0.0.1”,” 255.0.0.2”}	{“255.0.0.3”, “255.0.0.4”,” 255.0.0.5”}

Tip

To be more straightforward and much easier, you can click the **Format Based Changes** button to plan array of structure fields.

Data Modification Event

If a change modifies a CI's attributes through the **Set Expected States** wizard, the modification history is stored in the **dataModEvent** table and presented on the **Data Modification Event** page.

The following example demonstrates how the **dataModEvent** table works:

1. Create a standard change and select **Adobe Reader** in the **Affected CI** field.
2. In the **Change** record, click **More** > **Set Expected States** to plan the following attribute change: **Secuity Classification: 2. Restricted**.
3. Go to **Tailoring** > **Web Services** > **Data Modification Event**, and then click **Search**. A record is displayed, showing the planned CI attribute change.
4. Go to the CI record **Adobe Reader**, and update the **Security Classification** field to value "2".

5. Go to the **Associated CIs** tab in the change, click **Security Classification** and update the status to **Implemented** and then **Validated**.
6. Click **Close** in the change record to close the change. Note that the change can't be closed unless the CI attribute change is implemented and then validated.
7. Go to **Data Modificateion Event**, and then click **Search**. The record status is updated to **Processed**.

1.11.3.24. Configuration items

In Configuration Management, a company's assets are referred to as Configuration Items (CIs) and comprise items of value to the company, such as software, hardware, or furnishings. Configuration Management identifies, defines, and tracks organizational CIs by creating and managing records for those items. Service Manager CI records are stored in the device table and are therefore tracked through a device record in the Configuration Management application.

CIs can vary in complexity, size, and type. For example, objects or processes within the Service Desk infrastructure are recorded in the database as Configuration Items (CIs). A CI can include an entire system with all hardware, software, and documentation, a single software application, a software license, a hardware device, or office furnishings.

1.11.3.24.1. Configuration items details

Use the following information as a guide when you search for, add or modify a Configuration Item (CI).

Field	Description
CI Identifier	The system generated unique ID for the CI.
Display Name	Mandatory. The human readable name for the CI. **Note** This is the only field that identifies the CI in the system. The value of this field must be unique. If you enter a duplicate value, an error is displayed and the CI can't be saved.
CI Type	The type of CI (for example, Business Service, Computer or Running Software). **Note** When you search for a CI, the **Service Status** field is displayed once you select Business Service for CI type.
CI Subtype	The subtype of the CI (for example, Application Service).
Asset Tag	The asset tag number assigned to the CI by your organization.

Field	Description
Environment	The environment in which the CI currently exists (for example, Development, Test, Production, or Failover).
Status	Mandatory. The current status of the CI (for example, In Use, Installed, Retired/Consumed, or Returned to Vendor/Supplier).
Security classification	Specifies any security classification if any (for example, Restricted, Unrestricted, or Confidential).
SOX classification	Specifies the SOX classification (for example, Critical, Non-critical).
Owner	Identifies a specific person who owns the CI.
Export control classification	Specifies whether there are any Export controls on the CI.
Config admin group	Mandatory. The group responsible for configuring the CI. **Note** This field must be a valid assignment group value. Click **Fill Field Config admin group** to select a value from the list.

Field	Description
Support Groups	The assignment groups responsible for answering any support calls regarding the CI.
Support Remarks	A free text field used for any notes regarding the Support Groups (for example, "Contact via email first").
IT service continuity plan enabled checkbox	Indicates whether there is a continuity plan enabled for the CI.
Vital Business Function checkbox	Specifies whether this CI is a Vital Business Function.
Priority	The default priority incidents, changes, an so on will be given if this CI is selected in the record.
Default Impact	The default impact incidents, changes, an so on will be given if this CI is selected in the record.
User Base	Identifies the general users for the CI.
DML Location	The Definitive Media Library (DML) location. For example: f:\\shared\software\definitive\client
Outage Cost Per Hour	A numeric value that specifies the outage costs to the business per hour.
Currency	The currency in which the cost of the outage is specified.
System Down checkbox	Specifies whether the system is currently down.

Field	Description
Pending Change checkbox	Specifies whether the CI is awaiting a change.
Allow Subscribe checkbox	Specifies whether the CI could be subscribed.
Count Related Records	Calculates counts of related records except documents, such as open incidents, open problems, open known errors, open changes, and open requests.
Part Number	The part number of the CI.
Service Contract	Identifies the Service Contract associated with the CI.
Manufacturer	Identifies the manufacturer for the CI (for example, HP, Cisco). This is a read-only field, which is automatically populated with the manufacturer of the part when a Part Number is specified.
Model	The model number of the CI. This is a read-only field, which is automatically populated with the model of the part when a Part Number is specified.
Version	The version number of the CI.

Field	Description
Serial Number	The serial number of the CI. **Note** If the CI Type is Documentation, you can enter the ISBN number of the book in this field.
Title	A free text field for the title of the CI.
Description	A free text field to provide a brief description of the CI.

In addition, most CI types share the following sections for additional details about the CI, although the content may vary.

Section	Description
Baseline	The baseline version of the CI.

Section	Description
Managed State	This section contains subsections that show different information, depending on the CI Type. The following describes the information for three example CI types. *Business Service*: • Network: Specifies network information, such as the network name, IP address and default gateway. • Service: Specifies the business service CI information, such as the service lifecycle, administration URL/port, business importance level, disaster/recovery coverage, disaster/recovery tier and primary directory path. • Additional: Other information about the business service. Computer • Hardware: Specifies the hardware information related to the computer, such as the machine name, MAC address, Bios information, memory, and operating system • Network: Specifies network information, such as the network name, IP address and default gateway. • Additional: Other information about the manufacturer of the computer.

Section	Description
	• System Attribute: Specifies the CPU and file system information Running Software • Application/Running Software: Specifies information related to the application/running software such as software lifecycle, software status, the administration URL/port, business importance level, disaster/recovery coverage, disaster/recovery tier, primary directory path, data classification, license type, service hours and notification group. • Network: Specifies network information, such as the network name, IP address and default gateway. • Additional: Other information about the manufacturer of the application/running software.

Section	Description
Actual State	**Note** The Actual State section is visible only if your system has an active integration to a Universal CMDB server and the CI has a matching entry in the Configuration Management database. Service Manager doesn't display the Actual State section if there is no matching CI in the Universal CMDB server. Out-of-the-box, Actual State is provided for several CI Types. Computer • Shows the original discovery information, if any, for the computer. Running Software • Shows the original discovery information, if any, for the computer.

Section	Description
Service Catalog	**Note** This section is available only for CIs of the Business Service type. Shows the related service catalog items. If the current CI is specified as the parent service (in the Business Service field) of a bundle or item in Service Catalog, a link to the bundle or item is automatically created in this section.
CI Changes	Lists pending and historical changes to the CI.
Relationship Changes	Lists pending and historical changes to the CI's relationships.
Relationships	Lists the current relationships for the CI. **Note** Click **Add** to add upstream or downstream relationships between this CI and other CIs.

Section	Description
Relationship Graph	A graphical representation of the CI's relationships. The graphical viewer displays relationships and also the upstream or downstream CI affected where a specific CI is unavailable. This assists with incident and problem management as well as facilitating availability management techniques. For information about the CI icons and decorator icons used in relationship graphs, see Out-of-box icons for CI representation and Out-of-box icons for decorators.

Section	Description
Software	**Note** Available only when application software (an application) is installed on the CI (a computer system). This section consists of two subsections: Applications, and Drivers. Applications This subsection lists all software installation records created for the CI: application name, version, installation date, and license ID. This information is retrieved from the software installation records, which are accessible from **Configuration Management** > **Resources** > **Manage Software**. In each of the software installation records, the current CI is specified in the **Installed System** field. Drivers This subsection allows you to enter information about the drivers of the hardware components of the CI.
CI Owner	Shows the primary contact and support contacts for the CI.
Subscribers	Lists any subscribers to the CI.
Location	Shows the physical location information for the CI.

Section	Description
Vendor/Supplier	Shows information about the vendor/supplier for the CI.
Audit	Shows information about the audit policies currently running on the CI.
Metrics	Shows Service metrics related to the CI, such as the historical outage record.
Financial	Shows financial information for the CI, such as applicable service contracts, expense lines, or parts and labor incurred against the CI.
Related Documents-(<x>)	**Note** Available only for CI type Business Service. In this tab, you can only view published knowledge documents to which you have access from the business service CI record. Lists any knowledge documents associated with the CI. In these knowledge documents, the current CI is specified in the **Associated Service** field.
Attachments - <x> file(s) attached	Lists any attachments associated with the CI. For example, you can attach the book if the CI Type is Documentation.

Section	Description
Activities	Activity updates of the CI. Each update entry includes the following information: Date/Time, Type, Operator, and Description.

1.11.3.24.2. Service assets

A service asset is any resource or capability that could contribute to the delivery of a service. You can select an asset to be managed for operational, financial, or contractual purposes. Once you select an asset for management, it becomes a Configuration Item (CI). Assets can be one of the following types:

- Management
- Organization
- Process
- Knowledge
- People
- Information
- Applications
- Infrastructure
- Financial capital

1.11.3.24.3. Baselines

Baselines are an optional feature of Configuration Management that allow you to define a set of attributes that all instances of a configuration item (CI) should have. A baseline is a template CI that defines the expected or authorized attributes of a CI. Typically, a baseline only describes the attributes that you expect CIs to share in common and doesn't include attributes that you expect to vary. For example, a baseline describing PCs might require that all PC CIs are assigned the same model number and operating system version but not the same owner or serial number. In this example, the model number and the operating system would be authorized attributes of the baseline, while the owner and the serial number would be individually managed attributes.

Note

Baseline records replace baseline configuration item groups from previous versions of Service Manager. The upgrade process converts existing baseline configuration item groups to query groups.

Baseline records and CI records

Baseline records are separate from the CI records they manage. You must first create a baseline record before you can associate it with one or more CIs. All baseline records must have a name, a list of authorized attributes, and a state. Baseline records can optionally have a version number, which administrators can configure from the Configuration Management environment record. A baseline record's status determines whether you can add or edit attributes, and whether you can associate CIs to the baseline. After you authorize a baseline record, its attributes are locked and you can only associate or remove CIs from the baseline.

CI record baseline section

Each CI record has a baseline section that lists details about the baseline, if any, that's currently managing the CI. The baseline section lists the name of the managing baseline, its version, and a list of the attribute names and attribute values the baseline expects. If the CI has a value other than the baseline value, Service Manager displays a warning message that the Configuration Item is out of compliance with Baseline.

CI compliance with baseline

It is up to a Configuration Management manager to determine whether a CI that's out of compliance with its baseline is acceptable or requires a change. Keep in mind that both the CI record and the baseline record describe the expected or managed state of a CI. A baseline record is intended to describe the expected state across many similar items. A CI record describes the expected state of an individual item.

CI managed states

There may be cases where it is acceptable for an individual CI to have a different managed state than other CIs in the same baseline. For example, you might have a baseline requiring that all application servers have 8 GB of RAM. However, you may also want one of your application servers, the Web server, to have 16 GB of RAM. You may want to authorize this exception to the baseline rather than create a new baseline record to describe just one CI.

Contents

- 1 Baseline records and CI records
- 2 CI record baseline section
- 3 CI compliance with baseline
- 4 CI managed states

1.11.3.24.3.1. Baseline attributes

Each baseline record requires one or more baseline attributes. Baseline attributes are a list of fields that have a predefined expected value. Each baseline attribute lists both an attribute name and an attribute value. The attribute name is the field that contains the expected value. The attribute value is the value Service Manager expects to find in the CI record.

Note

To make selecting expected values easier, Service Manager provides lists or lookups to help you select the attribute values of some fields. If an attribute value provides such an option, you can either select an existing value or type in a new value.

CI compliance with baseline

If a CI field matches the expected value in the baseline, then that field is in compliance with the baseline. If a CI field doesn't match the expected value in the baseline, then the field is out of compliance with the baseline, and Service Manager marks the entire CI record as out of compliance with the baseline. For example, suppose you have a baseline record that manages PCs and that contains attributes for the operating system, operating system version, and manufacturer fields. If any one of these fields has a value other than the value listed in the baseline record's attribute fields, then the CI record for the PC is out of compliance with the baseline.

Changing baseline attributes

You can only change baseline attributes when the baseline record is in the analysis, design, or develop status. After a baseline record reaches the authorized state, the baseline attributes for that version can't be changed. To change attributes of an authorized baseline requires either creating a new baseline version or creating an entirely new baseline record. Baseline records in the retiring and retired status also require a new version or a new record to change baseline attributes.

Selecting baseline device types

When you create a baseline, you select the device type that manages the baseline. Each device type maps to a Service Manager join table that contains a list of attributes relevant to the CI. For example, if you select the device type computer, then you can select attributes relevant to computers such as OS manufacturer, BIOS manufacturer, or display manufacturer.

When selecting baseline attributes, you can select fields from either the device table or the associated device type join table. However, after you create a baseline record, you can't change the baseline's device type. You must create a new baseline record and select a different device type to view attributes from a different device type.

Contents

- 1 CI compliance with baseline
- 2 Changing baseline attributes
- 3 Selecting baseline device types

1.11.3.24.3.2. Baseline statuses

Each baseline record requires a status, which determines where the record is in its life cycle. It also requires that you indicate whether to allow the changing of baseline attributes and whether to allow associating CIs. Baseline records list the following status indicators in sequential order.

Status	Sequence	Description	Allows editing attributes?	Allows associating CIs?
Analysis	1	ITIL statuses that you can use to evaluate, plan, and implement your baseline record. Baselines in any of these statuses allow you to add, remove, or change attributes and attribute values, but prevent you from associating CIs because associated changes aren't yet authorized.	**Yes**	No

Status	Sequence	Description	Allows editing attributes?	Allows associating CIs?
Design	2	ITIL statuses that you can use to evaluate, plan, and implement your baseline record. Baselines in any of these statuses allow you to add, remove, or change attributes and attribute values, but prevent you from associating CIs because associated changes aren't yet authorized.	**Yes**	No

Status	Sequence	Description	Allows editing attributes?	Allows associating CIs?
Develop	3	ITIL statuses that you can use to evaluate, plan, and implement your baseline record. Baselines in any of these statuses allow you to add, remove, or change attributes and attribute values, but prevent you from associating CIs because associated changes aren't yet authorized.	**Yes**	No

Status	Sequence	Description	Allows editing attributes?	Allows associating CIs?
Authorized	4	This status indicates that the baseline is ready for use with your Configuration Management and Change Management processes. You can no longer edit attributes and attribute values without creating a new version of baseline. You can associate or disassociate CIs to an authorized baseline.	No	**Yes**

Status	Sequence	Description	Allows editing attributes?	Allows associating CIs?
Retiring	5	This status indicates that you are disassociating CIs from this baseline and associating them to another baseline or version. After disassociating all CIs with this baseline, it's ready for the next status.	No	No
Retired	6	This status indicates a past baseline. You shouldn't associate baselines in the retired status with any CIs. Associate all CIs with another version or baseline.	No	No

You can only change the baseline status from a lower sequence to a higher sequence. For example, you can change a baseline in the analysis status (sequence 1)

directly to the authorized status (sequence 4). However, you can't change a retired baseline (sequence 6) to the analysis state (sequence 1). If you want to create an editable copy of an existing baseline, you can create a new version of the baseline. The new version starts with all the same attribute names and values as the old baseline, but the status returns to the analysis state.

Authorize a baseline

Applies to User roles: System Administrator

To associate configuration items (CIs) to a baseline record, you must first authorize the baseline record. After you authorize a baseline record, you can no longer change its attributes, but you can create a new version.

To authorize a baseline, follow these steps:

1. Click **Configuration Management** > **Resources** > **Baseline**.
2. Use search or advanced search to find one or more records.
3. Select the baseline record you want to authorize from the list of search results. Service Manager displays the details of the selected baseline record.
4. Click **Authorized** in the **Status** field.

   Note

   You can only authorize baseline records that are currently in the **Analysis**, **Design**, or **Develop** states.

5. Click **Save**. Service Manager authorizes the baseline record, and you can now associate CIs to it.

Retire a baseline

Applies to User roles: System Administrator

You can retire a baseline record when you no longer want to associate configuration items (CIs) to it. After you retire a baseline record, you can no longer associate CIs to it, but you can create a new version.

Note

You can only retire a baseline record that doesn't have any associated CIs. You can always retire baselines records that are in the **Analysis**, **Design**, and **Develop** states because those states don't have associated CIs. You must disassociate CIs from a baseline before you can retire the baseline.

To retire a baseline, follow these steps:

1. Click **Configuration Management** > **Resources** > **Baseline**.
2. Type any search criteria that you want to use to find the baseline record.
3. Use search or advanced search to find one or more records.
4. Select the baseline record that you want to retire from the list of search results. Service Manager displays the details of the selected baseline record.
5. Click **Retiring** in the **Status** field.
6. Click **Save**. Service Manager marks the baseline record as **Retiring**, and you can now disassociate CIs from it.
7. Search for CIs that are associated to the baseline you want to retire. To do this, follow these steps:

    a. Click **Configuration Management** > **Resources** > **Search CIs**.

    b. Click the **Advanced Filter** tab.

    c. Click the **Add New Filter Criteria** button. Service Manager opens the Add Filter Criteria wizard.

    d. Select the following filtering criteria.

Field in Configuration Item	Comparison	Value
Baseline	Equals	*<Baseline name>*

    For *<Baseline name>*, select the baseline you want to retire.

    e. Click **Search**. Service Manager opens a list of CI records that are associated to the baseline you want to retire.

8. Disassociate the CIs from the baseline you want to retire. To do this, follow these steps:

    a. Click **Mass Update**. Service Manager opens a list of fields you can update.

    b. Select the following mass update values.

Caption	Value
Baseline	*<New baseline name>*
Baseline Version	*<New baseline version number>*

    For *<New baseline name>*, type the name of the new baseline you want to manage the CIs.

    For *<New baseline version number>*, type the version number of the new baseline you want to manage the CIs. Don't add this field if you're not using baseline versions.

    c. Click **Execute**. Service Manager updates the CI records to use the new baseline name and version.

9. Retire the baseline. To do this, follow these steps:

    a. Click **Configuration Management** > **Resources** > **Baseline**.

    b. Use search or advanced search to find one or more records.

    c. Select the baseline record you want to retire from the list of search results. Service Manager displays the details of the selected baseline record.

    d. Click **Retired** in the **Status** field.

    Note

    Service Manager displays an error message if you try to retire a baseline that still has associated CIs.

    e. Click **Save**. Service Manager marks the baseline record as retired.

1.11.3.24.3.3. Baseline versions

You can enable baseline versions to manage how many different versions are available for a given baseline name. Service Manager controls baseline versions from the Configuration Management environment record. By enabling the **Enforce Baseline Version Control** option, you can set the number of available levels (the number of decimal separators), the initial value of each level, and the number of increments available for each level. The following table describes how the baseline version control fields determine the maximum number of versions available for a baseline.

Field name	Default value	Possible values	Description
Available level	5 levels	1 to 5 levels. Service Manager only supports up to 5 levels.	The number of decimal places of the baseline version.
Initial value	1.0.0.0.0	0-9. The initial value should not be higher than the maximum increment.	The starting value of the version level.
Maximum increment	5	0-9. The maximum value should be greater than or equal to the initial value.	The maximum value of the version level.

You can create a maximum of ten thousand (10,000) baseline versions from 0.0.0.0.0 to 9.9.9.9.9. To minimize the number of baseline versions, you can delete one or more of the higher available levels. For example, to restrict the maximum number of baseline versions to one thousand (1,000), you can delete levels 4 and 5, but not levels 1 and 2.

A new baseline version must be of the same device type and attributes as the original version, but the new version is a separate record that starts in the analysis status. You can modify baseline attributes or attribute values as needed in the new baseline record, but the device type is fixed.

You can only create a new baseline version from the most recent version of the baseline record. For example, if you have a baseline consisting of three versions (1.0.0.0.0, 1.0.0.0.1, and 1.0.0.0.2), then you can only create a new baseline version from the 1.0.0.0.2 baseline record. The baseline record can be in any status.

Create a new baseline version

Applies to User roles: System Administrator

You can create a new baseline version to change the attributes managed by the baseline. You can only create a new version from the most recent version of a baseline record.

To create a new baseline version, follow these steps:

1. Click **Configuration Management** > **Resources** > **Baseline**.
2. Use search or advanced search to find one or more records.
3. Select the most recent version of the baseline record from the list of search results. Service Manager displays the details of the selected baseline record.
4. Click **Add Version**. Service Manager opens the Baseline Attributes wizard.
5. Follow the onscreen instructions to modify attributes or attribute values as needed. Service Manager creates a new baseline record in the **Analysis** state. The baseline record version increments to the version number you specified.

Contents

- 1 Create a new baseline version

1.11.3.24.3.4. Create a baseline

Applies to User roles: System Administrator

You can create a baseline record to manage the expected state of multiple related CIs. Typically, a baseline record contains attributes that CIs share in common.

To create a baseline record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Baseline**. The Search Baseline Records form opens.
2. Click **New**. The Add New Baseline wizard opens.
3. Follow the onscreen instructions to create the baseline, and then click **Save**. Service Manager creates a baseline record in the **Analysis** state.

1.11.3.24.3.5. Associate a configuration item (CI) to a baseline

Applies to User roles: Configuration Manager

Configuration Administrator

A System Administrator must first create and authorize a baseline record before you can associate a configuration item (CI) to it. A CI can only be associated with one baseline record at a time.

Tip

You can use Mass Update to associate multiple CIs to a baseline record.

To associate a CI to a baseline record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select the CI record that you want to associate to a baseline record from the list of search results. Service Manager displays the details of the selected CI record.
4. Open the **Baseline** section.
5. From the Baseline field, click the **Fill** button. Service Manager opens a list of available, authorized baselines or displays an error message if you don't have any authorized baselines.
6. Double-click the baseline record that you want to use to manage the CI. Service Manager displays the baseline name and attributes.
7. Click **Save**. Service Manager associates the CI to the baseline record and compares the baseline record attributes to the CI attributes. If any of the CI attributes differ from the baseline attributes, Service Manager displays the warning "Configuration Item is out of compliance with Baseline."

Associate multiple configuration items (CIs) to a baseline

Applies to User roles: Configuration Manager

Configuration Administrator

A System Administrator must first create and authorize a baseline record before you can associate a configuration item (CI) to it. A CI can only be associated with one baseline record at a time.

To associate multiple CIs to a baseline record, follow these steps:

1. Click **Configuration Management** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Click **Mass Update**. Service Manager displays a list of fields you can update.
4. Select the following mass update values.

Caption	Value
Baseline	*<New baseline name>*
Baseline Version	*<New baseline version number>*

   For *<New baseline name>*, type the name of the new baseline record that you want to manage the CIs.

   For *<New baseline version number>*, type the version number of the new baseline record that you want to manage the CIs. Do *not* add this field if you're not using baseline versions.

5. Click **Execute**. Service Manager updates the CI records to use the new baseline name and version.

Contents

- 1 Associate multiple configuration items (CIs) to a baseline

1.11.3.24.3.6. Change baseline attributes

Applies to User roles: System Administrator

You can change the attributes that make up a baseline record by selecting different fields from the device type join table. You can't change a baseline record's device type after you create it. If you want to select a different device type, you must create a new baseline record.

> **Note**
>
> You can only modify attributes of baseline records in the **Analysis**, **Design**, and **Develop** states.

To change baseline attributes, follow these steps:

1. Click **Configuration Management** > **Resources** > **Baseline**.
2. Use search or advanced search to find one or more records.
3. Select the baseline record that has attributes you want to change from the list of search results.
4. Click **Modify Attributes**. Service Manager opens the Baseline Attributes wizard.
5. Follow the onscreen instructions to complete the wizard. Service Manager updates the baseline record to use the new attributes you select.

   > **Note**
   >
   > Service Manager only displays fields from the device table and the associated device type.

1.11.3.24.3.7. Determine if a configuration item (CI) is in compliance with its baseline

Applies to User roles: Configuration Manager

Configuration Administrator

If you associate a configuration item (CI) to a baseline, Service Manager displays whether it is compliant with the baseline, and if not, Service Manager shows you which attributes are out of compliance.

To determine if a CI is in compliance with its baseline, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select the CI whose baseline compliance you want to see from the list of search results.
4. Open the **Baseline** section. Service Manager displays the baseline attributes managing the CI. If one or more baseline attributes is out of compliance, Service Manager displays the error message "Configuration Item is out of compliance with Baseline".

View all configuration items (CIs) out of compliance with a baseline

Applies to User roles: Configuration Manager

Configuration Administrator

In certain instances, you may want to perform a one-time advanced search in order to view all CIs that are out of compliance with a baseline. If you expect to perform this search frequently, consider creating a view instead.

To view all CIs out of compliance with a baseline, follow these steps:

1. Click **Configuration Management** > **Search CIs**.
2. Use search or advanced search to find one or more records.

3. Click the **Advanced Filter** tab, and then click **Add New Filter Criteria**. The Add Filter Criteria wizard opens.
4. Select the following filtering criteria.

Field in Configuration Item	Comparison	Value
Baseline Noncompliant	Equals	*True*
Baseline	Equals	*<Baseline name>*
Baseline Version	Equals	*<Baseline version number>*

   For *<Baseline name>*, select the name of the baseline you want to query for compliance. Omit this search criteria if you want to search all baselines.

   For *<Baseline version number>*, type the version number of the baseline you want to query. Omit this criteria if you're not using baseline versions.

5. Click **Search**. Service Manager opens a list of the CI records that are out of compliance with the specified baseline.

Contents

- 1 View all configuration items (CIs) out of compliance with a baseline

1.11.3.24.4. Configuration item groups

A business enterprise may have a large number of Configuration Items (CIs) to acquire, install, repair, or manage in a variety of activities. Because many of these CIs are based on the same model or share other defining characteristics, creating CI groups is a simple way to classify CIs, to simplify tracking, and to manage changes.

Making global changes to CI groups is easier than updating CIs one at a time. Change Management can automatically open tasks for all CIs in a group. Service Level Management enables you to attach a CI group to a service level agreement (SLA) instead of attaching each CI individually to an SLA. Thereafter, Service Level Management can generate response and outage information at a CI group level.

Group types

Groups can be static, such as all monitors purchased on the same day, or dynamic, such as all monitors on the third floor. Configuration items can be grouped in many ways. For example, you might create the following groups:

- All printers
- Laser printers
- Ink jet printers
- IBM® printers
- Xerox® printers
- Out-of-service printers

Related topics

Configuration Management overview

1.11.3.24.4.1. Types of Configuration Item groups

To search for Configuration Items (CIs), you can create a search query and the system returns a list of CIs that match your search criteria. You can use a list group to create a group of CIs. A wizard leads you through the process after you enter the applicable search criteria and the system generates a list of records for the search criteria.

You can save and edit this search to regenerate a list or group of CIs as one of the following types:

- **List group:** A static snapshot of all the CI records that appears when you save a list generated by your selected search criteria. Configuration Management saves enough information to reconstruct the list with the same components each time that you view the list. The list doesn't add new CI records or remove obsolete CI records.
- **Query group:** A dynamic list of CI records that appears when you save a query generated by your selected search criteria. Configuration Management saves the query to regenerate the list. Depending on the query parameters, the components can vary each time that you view the list. New CI records can appear; obsolete CI records can disappear from the generated list.

Create a configuration item group

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To create a configuration item group, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. In the **Type** field list, select **CI Group** and then click **Search**.
3. From the record list, click **More** or the **More Actions** icon and then choose **Create/Edit AdHoc CI Groups**.
4. In the Manage Configuration Item Groups wizard, select **Create Query group** or **Create List Group** and then click **Next**.
5. Enter a group name and then click **Next**.
6. Select or deselect **Display group record**.
7. Click **Finish**.
8. Click **Save & Exit**.

Edit a configuration item group

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To edit a configuration item group, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**. The Search Configuration Item Records form opens.
2. In the **Type** field list, select **CI Group**. Use search or advanced search to find one or more records.
3. Select a CI group.
4. Click **More** or the **More Actions** icon.
5. Select **Edit AdHoc CI Groups**.
6. Select either **Add Configuration Item to list** or **Remove Configuration Item from list** and then click **Next**.
7. Choose a CI to add or remove.
8. Click **Save**.
9. Click **OK**.

View the members of a configuration item group

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To view the members of a CI group, follow these steps:

1. Click **Configuration Management** > **Search CIs**.
2. Use search or advanced search to find one or more records. In the **Type** field list, select **CI Group**.
3. Select a CI group.
4. Click **Show Members**. Configuration Management generates a record list of current group members.
5. Select any record to view its details.
6. Click **Save & Exit**.

View associated configuration item groups

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To view associated configuration item groups, follow these steps:

1. Click **Configuration Management** > **Search CIs**.

2. Use search or advanced search to find one or more records. In the **Type** field list, select **CI Group**.
3. Select a CI group.
4. Click **More** or the **More Actions** icon and then choose **Show Associated Groups**.
5. Click **Save & Exit**.

1.11.3.24.4.2. Ad hoc groups

An ad hoc group is a query that produces a list of items that's not associated within a hierarchical or peer relationship within Configuration Management. Ad hoc groups can't be related to service level agreements or processes.

An ad hoc group consists of components that you can associate at random, or that have a special relationship. For example, AdHocHardware might be the name of a group of all the Configuration Items (CIs) temporarily assigned to a user to complete a special assignment. By collecting the CIs into an ad hoc group, you maintain a global strategy for addressing and managing the special assignment CIs within the group.

You can move the CIs in an ad hoc group to another group and they're not subject to change control.

Ad hoc group attributes

The following table lists the ITIL attributes of a Configuration Item (CI) in an ad hoc group.

Attribute	Description
CI name	The unique name for the selected CI by type.
Comment	Text comments, such as a description of how the selected CI version is different from the former version.
Unique ID	A unique number that identifies the CI. For example, a software copy number, a serial number, or an employee ID.
Relationships	Describes the relationship of the selected CI to other CIs that aren't parent or child CIs. For example, a description might be "uses the *xxxx* CI," "connects to the *xxxx* CI," "accesses the *xxxx* CI," and so on.
Type	Describes the CI type with more detailed category information, such as software bundle, hardware device, employee name, or vendor/supplier name.

1.11.3.24.4.3. Baseline configuration groups

Baseline configuration groups are obsolete in Service Manager and are replaced with baseline records. Service Manager converts any existing baseline configuration groups in your system to configuration item query groups.

1.11.3.24.4.4. Business service groups

A business service group collects information about all resources that support an enterprise business service. The group lists the Configuration Items (CIs), documents, resources, service level agreements (SLAs), and other components. A business service group can contain:

- One or more SLAs
- Owners, such as a Change Manager or a Problem Manager
- CIs that have hierarchical relationships with other CIs

The CIs in a business service group are usually subject to change control.

Business service group attributes

The following table lists the ITIL attributes of a Configuration Item (CI) in a business service group.

Attribute	Description
Category	CI classification, such as hardware, software, documentation, or employee.
Change numbers	Contains the identification numbers of all change records that affect the selected CI.
Child CI relationship	The unique identifiers for all child CIs of the selected CI.
CI name	The unique name for the selected CI by type.
Comment	Contains text, such as a description of how the selected CI version is different from the former version.
Unique ID	A unique number that identifies the selected CI. For example, a software copy number, a serial number, or an employee ID.
Incident numbers	The identification numbers of all incident records that affect the selected CI.
Location	The physical location of the CI. For example, software might be on a CD in a media library; a hardware CI would be in a room or office, on a specific floor, and in a specific building.
Owner responsible	Name or title of the person responsible for the CI.
Parent CI relationship	The unique identifier for the parent CI of the selected CI.

Attribute	Description
Problem numbers	The identification numbers of all problem records that affect the selected CI.
Relationships	Describes the relationship of the selected CI to other CIs that aren't parent or child CIs. For example, a description might be "uses the *xxxx* CI," "connects to the *xxxx* CI," "accesses the *xxxx* CI," and so on.
Responsibility date	The date that the owner took responsibility for the CI.
RFC numbers	The identification numbers of all change request records that affect the selected CI.
Status (current)	The current status of the CI, such as test, live, or archive.
Status (scheduled)	The next scheduled status of the CI with the date of (or name of the event that triggers) the status change.
Supply date	When the organization acquired the CI.
Type	Describes the selected CI type with more detailed category information, such as software bundle, hardware device, employee name, or vendor/supplier name.
Version number	The version number of the CI.
SLA numbers	The identification numbers of all Service Level Agreement records that affect the selected CI.

Add a business service CI record

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To add a business service CI record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Click **New**. The Add New Device wizard opens.
3. Select **Business Service** from the list, and then click **Next**. The New Configuration Item form opens.
4. Enter the information to complete the configuration item (CI) record, and then click **Add**.
   Service Manager creates the CI record and assigns it a unique identifier.
5. Click **OK**.

Update a business services CI record

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To update a business services CI record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. In the **Type** field list, select **Business Service**.
3. Select a record from the record list.
4. Type new values into the applicable fields in the record.
5. Click **Save & Exit**.

View related records for a CI record

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

To view related records for a CI record, follow these steps:

1. From the detail list toolbar of a CI record, click **More** or the More Actions icon.
2. Click **Related**.
3. Click the type of related record that you want to view.
   The menu list includes the following types of related records:

- Interactions
- Incidents
- Changes
- Problems
- Requests
- Documents
- Known Errors

4. Click **Open** to open a new record of the selected type or click **View Existing** to open an existing record of the selected type.

Note

- For **Requests** and **Documents**, only the **View Existing** option is available.
- The **Documents** option is available only for CIs of the Business Service type, because only a Business Service CI can be specified in the Associated Service field in a knowledge document.
- You can only view published knowledge documents to which you have access from the business service CI record.

Tip

You can also go to the **Related Documents** tab to view related knowledge documents associated with a business service CI. For more information, see Configuration items details.

To view the counts of each related record type except Documents, click **Count Related Records**. The count of related knowledge documents is displayed on the label of the **Related Documents** tab.

1.11.4. Configuration management administrator tasks

Access Configuration Management reports

View a service level agreement for a configuration item

Rename a configuration item

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1.11.4.1. Access Configuration Management reports

Applies to User roles: Configuration Administrator

The Reporting tool in Service Manager provides a number of out-of-the-box reports on the CI data in your system. You can view these reports through dashboards named Availability Overview (Global). You can also create your own dashboards to display other reports of your interest.

To access Configuration Management reports, follow these steps:

1. Click **Configuration Management** > **Reports**.
2. Click **Configuration Management** > **Reports** > **Availability Overview** to display the **Availability Overview (Global)** dashboard.

   Click **Configuration Management** > **Reports** > **CI Inventory** to display the **CI Inventory (Global)** dashboard.

   Click **Configuration Management** > **Reports** > **CI Overview** to display the **CI Overview (Global)** dashboard.

   You can also open one of these reports and then switch to other reports using the dashboard drop-down list on the toolbar.
3. Modify the dashboard that you opened as needed:

   • To add content to the dashboard, click **Add Content** on the toolbar. For details, see Add reports to a dashboard.

   • To remove a report from a dashboard, click **X** on the report window.

   • To change the position of a report, drag the report window and drop it to a new position.

   • To change the size of a report, drag the border or corner of the report window.
4. Add your own dashboards as needed. For details, see Create a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

Tip

You can click **Export** to export the reports on a dashboard to PDF format.

5. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default one.

1.11.4.2. View a service level agreement for a configuration item

Applies to User roles: System Administrator

To view a service level agreement for a configuration item, follow these steps:

1. Click **Service Level Management** > **SLA Administration** > **Availability SLOs**. The Availability Service Level Objective form opens.
2. Click the **Fill** button for the **Affected CI** field, and then select the CI.
3. Click **Search**.
4. If necessary, double-click a record to display it in the SLO Information form.

1.11.4.3. Rename a configuration item

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor

There are many cases in which you need to rename a configuration item (CI). As a best practice, you are recommended to run the Change Management process to modify CI names; alternatively, if Universal CMDB (UCMDB) is available, you can directly rename CIs in UCMDB.

Prior to version 9.41, renaming a CI in Service Manager did not trigger automatic updates to the related records, such as CI relationships or records that reference the CI as an affected service or CI.

Starting with version 9.41, when a CI is renamed, Service Manager automatically updates the CI information in all related records (incidents, interactions, changes, and so on) to maintain data integrity across modules.

The process of CI renaming depends on whether you are using the UCMDB integration or Service Manager for Configuration Management.

If you are using the UCMDB integration

If you are using the UCMDB integration for Configuration Management, use the following CI renaming process:

1. The requestor submits a request to rename a CI (for example, a laptop).
2. The Configuration Analyst creates a request for change in Service Manager for renaming the CI, and then go through the change management process.
3. The Configuration Manager changes the CI name in UCMDB.
4. The UCMDB integration pushes the renamed CI to Service Manager.

If you are using only Service Manager

If you are using only Service Manager for Configuration Management, use the following CI renaming process:

1. The requestor submits a request to rename a CI (for example, a laptop).
2. The Configuration Analyst creates a request for change in Service Manager for renaming the CI. The change request then goes through the Change Management process.
3. The Configuration Manager changes the CI display name in Service Manager.

Caution

Service Manager doesn't allow duplicate display names for CIs of the same CI type.

1.11.4.4. Delete multiple configuration item records

To delete all records with a specified value, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Enter the desired value. For this example, type **adv** in the **Display Name** field.
3. Click **Search**.

   Service Manager performs the search and displays a list of matching records, if any, in the `device.qbe` form.
4. If the query produces no matching records or only one matching record, Service Manager doesn't display a record list.

   > **Note**
   >
   > To display the Mass function buttons, you can use a query that returns more than one record.

5. From the toolbar, click **More** > **Mass Delete** to remove the records in the record list from the database.

   Service Manager confirms the deleted request with a prompt screen.
6. Click **No** to cancel the deletion or click **Yes** to confirm the deletion.

Service Manager deletes all records in the list.

> **Note**
>
> - Service Manager performs the Mass Delete function in the foreground, which means your terminal devotes to this function until it completes. If you are deleting a large quantity of records, this may take a while.
> - Upon completion, Service Manager returns to the search form and displays the *nnn* `records deleted from the device file` message, where *nnn* is the number of deleted records.

1.11.4.5. View the status of a configuration item

To view the status of a configuration item, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Enter the search criteria and then click **Search**.
3. Database Manager performs the search and displays the record list of matching records, if any, in the device.qbe form.
4. Select the configuration item.
5. View the value in the **Status** field on the CI record, for example, "In Use", "In Stock", or "Retired/Consumed".
6. Click **OK** to return to the search results list.

1.11.4.6. Delete a configuration item

You may be assigned a change task by the Change Coordinator that requests you to delete a configuration item (CI). Follow the procedure below to ensure those changes to configurations are properly controlled.

To delete a CI, follow these steps:

1. Find the task that requests the deletion of an existing CI in your To Do Queue, and double-click the record to open it. If you don't see the task in your To Do Queue, click the **Change Tasks** queue to view the task.
2. In the Affected CI list, click the CI that you have been requested to delete and then click **Find** to display the CI details.
3. Review the current information about the CI to determine if deleting the CI is in accordance with Configuration Management policies. For example, verify there are no open Service Desk, Incident, Problem, or Change records relating to the CI. If the modification is not applicable, reject the change task.
4. Click **Delete**.
5. Click **Yes** to confirm the deletion.
6. After the CI type is deleted, return to the task record.
7. Close the task. To do this, follow these steps:
    1. Click **Close Task**.
    2. Select the **Closure Code** and type a **Closure Comment**.
    3. Click **Save & Exit**. Service Manager sets the task state to **Closed**.

1.11.4.7. Add a relationship

Applies to User roles: Configuration Manager

Configuration Administrator

Configuration Auditor

CMS and Tools Administrator

To add a relationship for a CI record, follow these steps:

1. On the CI record, scroll down to the **Relationships** section.
2. To add a new downstream relationship, click **Add**. If you are creating a new CI, click **Save** before adding the relationship.
3. On the Configuration Item Relationship form, fill in the **Relationship Name**, **Relationship Type**, and the list of **Downstream CIs**.
4. (Optional) Fill in the **Outage Dependency** fields, and then click **OK** to return to the CI record.
5. When you finish creating or modifying the configuration item record, click **Save**.

   The upstream and downstream CIs are displayed with their display names.

1.11.4.8. Configuration Management environment record

Applies to User roles: System Administrator

The Configuration Management environment record defines options affecting basic Configuration Management functionality for all users. Service Manager provides a default Configuration Management environment record that you can modify for your system.

Configure the Configuration Management environment record

The Configuration Management environment record defines options that enable or disable Configuration Management application functionality for all Configuration Management users. The Configuration Management environment record provides default settings that you can customize based on your specific business needs.

To configure the Configuration Management environment settings, follow these steps:

1. Click **Configuration Management** > **Administration** > **Environment**.
2. Select new options or clear default options. Your changes redefine the Configuration Management environment for all users.

Option	Description
Allow Operator Access without Security Role Record?	If selected (set to true), indicates that a user can access the Configuration Management module even if no security role is defined for the user.
Max Number of Members in Group	The maximum number of members a Configuration Items group can have. This option applies to CI groups (Baseline or Ad Hoc).
Max Size For Operations	The maximum number of members a group can have for operations, including filling group members in Problem Management, creating tasks for each group member in Change Management, and so on. This option applies to CI groups (Baseline or Ad Hoc).

Option	Description
Enforce Baseline Group Version Control	Indicates whether to enable version control for baseline Configuration Items Group. This setting is used whenever a user creates a baseline group. • **Available level**: Levels for each version. This setting is used whenever a user creates a baseline group. • **Initial value**: The initial values for each level when a baseline Configuration Items Group was first created. This setting is used whenever a user creates a baseline group. • **Maximum increment**: The maximum increment allowed for each level. This setting is used whenever a user creates a baseline group.
Max Levels for Relationship Tree Search	The maximum level of downstream CIs to retrieve when you click the **Show Members** button from a CI record of the “Business Service” or “CI Group” type. For example, a business service email may have 3 child CIs, and each child CI may have 3 children. If this field is set to 1, the **Show Members** option retrieves only 3 children; if the level is set to 2, the **Show Members** option retrieves 3 children plus 9 grandchildren.

Option	Description
Max Levels for Related CI's for Service	The maximum level of downstream CIs of an affected business service to retrieve when you click the Fill button to select an affected CI. In an interaction/incident/problem/known error/change record, you select an affected business service, and then click the **Fill** button to select an affected CI. The Fill option retrieves all downstream CIs of the business service that you select. For example, a business service named Email may have 3 children, and each child CI may have 3 children. If this field is set to 1, the Fill option retrieves only the 3 children; if the field is set to 2, the Fill option retrieves 3 children plus 9 grandchildren.
Subscription Add	Indicates the Change category that will be used for the fulfillment of an item being subscribed to.
Subscription Remove	Indicates the Change category that will be used for the fulfillment of an item being unsubscribed to.

Option	Description
Subscription Request Mode	Use this option to select the way to subscribe to items: via Service Catalog or directly from the Subscription list. The Administrator is able to choose between two Subscription modes - Change Management or Service Catalog. If the value is left empty, it will be defaulted to Service Catalog. If the mode is set to Service Catalog, users can only subscribe to items through Service Catalog; if the mode is set to Change Management, a Self Service Ticketing user (SST user) can subscribe or cancel subscriptions to items directly from the Subscription list.
Allow User-Defined CI Identifier Values for New CIs	If selected (set to true), indicates that the user can manually specify a CI Identifier value or edit the system-generated CI Identifier value when creating a new CI. If the user doesn't specify a CI Identifier value, Service Manager automatically generates one as the initial value. Once the new CI is created (saved), the user-defined CI Identifier value is no longer editable. If not selected, when users create a new CI, Service Manager creates a CI Identifier value, which is not editable. This setting applies to all CI types.

Option	Description
Outage Cost Per Hour:	The outage cost to CI/Service per hour. The default value is 500.
Currency	The currency in which the outage cost is calculated. The default currency is US dollar.

3. When you finish making your changes, click **Save**.
4. Click **OK**.
5. Log out and log back in so that your changes take effect.

1.11.4.9. Contract Management overview

Contract Management enables you to perform the following actions:

- Create contracts from contract types or user-defined templates.
- Track contract and payment details.
- Define payments for contracts and generate payment schedules.
- Associate contracts with multiple assets.
- Maintain a table of standard contract terms and conditions.
- Create contract templates.
- Renew contracts automatically.

The contract life cycle

You can manage a contract throughout the contract life cycle. To do so, follow these steps:

1. Create the contract.
   The initial status of the contract is draft. You can delete a contract only while it is in **Draft** status.
2. The contract term begins on the start date entered in Service Manager.
   On the start date, the status of the contract changes to **Current**.
3. The contract term ends on the expiration date entered in the system.

At the end of the contract term, you must renew the contract, otherwise it expires.

Contract types

Contract Management is useful to companies that have assets under contract. To use Contract Management, a company must be licensed to use the Configuration Management application. Service Manager supports the following contract types.

Type	Definition
Lease	Define general leasing conditions.
Maintenance	Define a maintenance contract.
Software	Define a software license agreement.
Support	Define asset support.
Warranty	Track warranty data about your asset.

1.11.4.9.1. Create and update contracts

Contract Management enables you to create contracts from contract types or user-defined templates. You can also update the terms and conditions of a contract to meet your business needs.

When you create a new contract, you can:

- Select a contract type by using the Add New Contract wizard.
- Select a template by using the Add New Contract wizard.
- Bypass the wizard by filling in the General Contract Information form directly.

Update the terms and conditions of a contract

Applies to User roles: System Administrator

You can update the terms and conditions of a current contract from a repository of terms and conditions.

To update the terms and conditions for a contract, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use search or advanced search to find one or more records.
3. Double-click a record to display it in the contract information form.
4. Open the **Terms** section.
5. Insert the cursor in the **Term/Condition** table cell, and then click the **Fill** button in the tool tray. A list of terms and conditions is displayed.
6. Double-click the terms and conditions that you want to apply to the selected contract record.
7. In the **Financial** section, make sure the cost information is correct. Because you are updating the terms and conditions of the selected contract, make sure that the cost information, such as the currency expiration date, is current as well.
8. Click **Save**.

Create a new contract terms and conditions record

Applies to User roles: System Administrator

You can create contract terms and conditions records to use with your contract templates. If you add a new contract or update an existing contract, you can choose an applicable set of terms and conditions from the list of available records.

To create a new contract terms and conditions record, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Terms and Conditions**.
2. Select from the **Contract Type** list to choose an applicable contract type. You can choose from the following contract types:
   - **Lease**
   - **Maintenance**
   - **Software**
   - **Support**
   - **Warranty**

   Note

   If you want the new terms or conditions to apply to all contract types, leave the **Contract Type** field blank.
3. Type a brief description in the **Term/Condition** field.
4. Click **Add**.

Configuration Management generates a unique internal system ID for the new contract terms and conditions record. This value is displayed in the **Internal System ID** field.

Contents

- 1 Update the terms and conditions of a contract
- 2 Create a new contract terms and conditions record

1.11.4.9.2. Adding assets to a contract

You can add assets to only the following contract types:

- Lease
- Maintenance
- Support
- Warranty

Add configuration items to a contract

Applies to User roles: System Administrator

In Configuration Management, a company's assets are referred to as configuration items (CIs). CIs comprise items of value to the company, such as software, hardware, or furnishings. Configuration Management identifies, defines, and tracks organizational CIs by creating and managing records for those items. To associate an asset with a contract, you must first add the CI (asset) to the contract. If the terms and conditions of a contract change, you must update your list of assets.

To add configuration items to a contract, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use search or advanced search to find one or more records.
3. Select a contract record to associate one or more assets to that contract.
4. Do one of the following to launch the Select Assets to Add to Contract wizard:
   - (For a contract other than software) In the **Assets** section, click **Add Assets**.
   - (For a software contract) In the **Licenses** section, click **Add Licenses**.
5. Insert the cursor in the first blank line of the **Asset** field, and then click **Fill** .
6. Type or select optional search criteria, and click **Search** to generate a record list of assets.
7. Select the asset to associate with the contract, and then click **Next**

   Note

   If there are multiple assets for this contract, continue to select each of the assets. When you are finished, click **Next**. Configuration Management returns you to the original **Assets** or **Licenses** section and displays all the new assets that you specified.

8. To assign costs equally for all the assets in the contract, click **Generate %**.

9. To assign costs to an individual asset, double-click the asset link. Service Manager displays the Allocation Information form. Type the percentage of the cost allocated to this asset in the **% Cost Allocation** field.
10. Click **Save** and then **Back** to return to the Assets or Licenses section.

Note

- If you want assets to have unequal cost allocations, select them individually.
- You can assign costs manually for some assets, and then choose **Generate %** to assign the remaining assets equally.

1.11.4.9.3. Contract expiration

When a contract reaches its expiration date, the system changes the status to expired if the Auto-Renew flag is not on. After a contract is expired, you can no longer make any modifications to the contract, without first reopening the contract.

To make changes to an expired contract, you must reopen the contract by opening the **More Actions** menu and clicking **Reopen Contract**. Reopening the contract changes its status to draft and clears the **Start Date** and **Expiration Date** fields.

If you change your mind about reopening a contract, you can click **Back** to undo the reopen command.

If you select another contract from the record list, click **Previous** or **Next**. Service Manager automatically saves the reopened contract with a draft status.

You can cancel a contract any time before its expiration date. When you cancel a contract, the status of the contract changes to canceled. To cancel a contract, click **More** or the **More Actions** icon, and then **Cancel Contract**.

Contract renewal

Configuration Management enables you to set contracts to renew automatically by setting the Auto-Renew flag before the expiration date. At the time of renewal, the system updates the start and expiration dates to reflect the new renewal term of the contract.

When Service Manager renews a contract, the status changes to renewed.

Renew a contract

Applies to User roles: System Administrator

When you renew a contract, the system updates the start and expiration dates of that contract to reflect the renewal term of the contract.

To renew a contract, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use search or advanced search to find one or more records.
3. Select the contract you want to renew.

Note

If you want to renew an expired contract, you must reopen the contract first. To reopen the contract, follow these steps:

1. Click **More** or the **More Actions** icon.
2. Choose **Reopen Contract**.
3. Make sure that you update the cost information in the **Financial** section. Information, such as the currency expiration date, must be current.
4. When you are ready to renew the contract, proceed to the **Renewal** section

4. In the **Renewal** section, select **Auto-Renew?**. The **Renewal Period** field and **Notify in Advance?** checkbox appear.
5. Type the number of days before the contract expires in the **Renewal Period** field.
6. Select **Notify in Advance?** to receive advance notification before the next expiration date.
7. When you select notify in advance, type the number of days to notify before the contract expires in the **How Many Days?** field.
8. Click **Save**.

Cancel a contract

Applies to User roles: System Administrator

If you want to cancel a contract, you can do so until its expiration date. When you cancel a contract, it's not deleted from the system. The contract remains on file and the status changes to canceled.

Note

If you cancel a contract, you can't reinstate it. When you cancel a contract, you also cancel all outstanding payments against that contract.

To cancel a contract, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use search or advanced search to find one or more records.
3. Double-click the record that you want to cancel.
4. Click **More** or the **More Actions** icon.
5. Choose **Cancel Contract**. The contract remains on the list of contract records, but the status field changes to canceled.

Contents

- 1 Contract renewal
- 2 Renew a contract
- 3 Cancel a contract

1.11.4.9.4. Contract payments

Configuration Management enables you to track payment amounts made against a contract. The payment amount is dependent on weighted allocations assigned to Configuration Items (CIs). When payments post, an internal application adds the weighted amount allocation and compares the result to the total. This calculates the amount of the first payment.

Calculating contract payments

When you view a contract, the **Assets** section displays a line for each contract asset. The amount shown on each contract asset line depends on the weighted allocations that are assigned to the assets. When you record payments, there is an internal application that adds the weighted amount allocation, compares the result to the total, and calculates the first payment.

Example: Calculating contract payments

Assume that you have a contract covering three assets. An expense line is created for each asset in the **Assets** section of the Contract form. You make a payment of $127.50 (USD) for all three assets. Contract Management allocates approximately one-third of the payment to each asset's expense line.

Asset	Payment
First asset	42.59
Second asset	42.46
Third asset	42.46
Total	127.51

When Contract Management adds the allocated amounts, the total is 127.51, which is one cent over the total payment amount. Contract Management adjusts the first allocation amount to ensure the total allocated amount equals the actual payment amount. The final expense line amounts are as follows:

Asset	Payment
First asset	42.58
Second asset	42.46
Third asset	42.46
Total	127.50

Cancel a payment record

Applies to User roles: System Administrator

Contract Management integrates service contract information and tracking into Service Manager. Unlike service level agreements, which describe how services in a contract are to be rendered, service contracts are financial agreements that define the services to be provided and the financial implications of using those services. Contract Management tracks each service contract associated between a customer and service provider, and compiles individual expenses and costs for those services. When using the Configuration Management Payments option, you can submit, cancel, or delete a payment record.

Note

You can't cancel a payment if there is no outstanding balance.

To cancel a payment record, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Payments**.
2. Use search or advanced search to find one or more records.
3. Select the target payment record you want to cancel.
4. Click **Cancel Payment**. The **Status** changes from **Pending** to **Canceled**.
5. Click **Save**.

Delete a payment record

Applies to User roles: System Administrator

Contract Management integrates service contract information and tracking into Service Manager. Unlike service level agreements, which describe how services in a contract are to be rendered, service contracts are financial agreements that define the services to be provided and the financial implications of using those services. Contract Management tracks each service contract between a customer and service provider, and compiles individual expenses and costs for those services. When using the Configuration Management Payments option, you can submit, cancel, or delete a payment record.

To delete a payment record, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Payments**.
2. Use search or advanced search to find one or more records.
3. Select the target payment record that you want to delete.
4. Click **Delete**. The payment record is deleted from the list of payment records.

Generate a payment schedule for a contract

Applies to User roles: System Administrator

Contract Management tracks each service contract between a customer and service provider, and compiles the individual expenses and costs for those services. Configuration Management allows you to generate a payment schedule for a contract as long as there is an outstanding balance for that contract.

Note

The payment schedule you set up should fit within the start and end dates of the contract.

To generate a payment schedule for a contract that's not expired, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use search or advanced search to find one or more records.
3. Select the record that you want to update.
4. Open the **Financial** section.
5. Go to the **Payment Information** subsection.

6. Click **Generate Payment Schedule** to start the Generate Payment Schedule wizard.
7. Complete the following required fields:
    - **No. of Payments**: Enter the total number of payments to be made.
    - **Frequency**: Select Daily, Monthly, or Annually.
    - Do one of the following to choose the **Currency** value:
        - Select a currency from the list. For example, British Pound.
        - If the Currency box is empty, click **Fill** to choose an ISO currency code. For example, GBP for British Pound.
        - Click **Find** to generate a record list of currency records. Double-click any record to display the record in the Currency form.
    - **Currency EX Date**: Required only if the system currency and payment currency aren't the same.
8. Click **Next** to return to the Contract Information form. The pending payments are listed in the payment schedule table at the bottom of the form. For example, if you have scheduled five monthly payments of $100.00 each, these payments are listed with their due dates and a status of pending.

> **Note**
>
> Configuration Management verifies that the payment schedule fits within the start and end dates of the contract. If the amount and frequency of payments don't fit within those limits, Configuration Management displays an error message.

Schedule a payment for a contract

Applies to User roles: System Administrator

Contract Management tracks each service contract between a customer and service provider, and compiles individual expenses and costs for those services. Configuration Management allows you to schedule a payment for a contract as long as there is an outstanding balance for that contract.

To schedule a payment for a contract that's not expired, follow these steps:

1. If you use the **Manage Payments** option, follow these steps:
    1. Click **Configuration Management** > **Contracts** > **Manage Payments**.
    2. Click **New** to create a new payment record.

3. Use search or advanced search to find one or more records.
4. Select your target record.

2. If you use the **Contracts** option, follow these steps:
    1. Click **Configuration Management** > **Contracts** > **Contracts**.
    2. Use search or advanced search to find one or more records.
    3. Select your target record.
    4. Open the **Financial** section, and then go to the **Payment Information** subsection. In this subsection you can view actual costs, such as rent and renewal.
    5. Click **Schedule Single Payment**.
3. Type the payment amount you want to schedule in the **Amount** field.
4. Choose a value in the **Currency** field. For example, British Pound.
5. Select a date in the **Currency EX Date** field if the system currency and payment currency aren't the same.
6. Select the payment status in the **Status** field. The following are available status types:
    - **Pending**: The initial status when you first add a payment to a contract.
    - **Submitted**: Indicates that the payment processing is complete and can't be retrieved for modification or deletion.
    - **Canceled**: The payment is stopped. The payment appears in the list of payments, but it can't be submitted for payment.

      If you are scheduling a payment, you can select **Pending**.
7. Select a **Payment Code** to indicate the type of payment being scheduled. You can choose from the following payment options:
    - **One-time**: Single payment.
    - **Purchase**: Payment to purchase an asset or to apply toward acquiring a leased asset.
    - **Renewal**: Payment toward renewal costs.
    - **Buyout**: Pay off a lease contract.
    - **Other**: Payment of other costs, such as administrative expenses.
8. Type the date the payment is due in the **Date Due** text box. The format is *mm/dd/yy*. You can also select the due date and time from the Service Manager calendar. The due date must be a future date to ensure that the payment processes correctly.
9. Enter additional payment information in the available payment fields.
10. Click **Add**.

Submit a payment record

Applies to User roles: System Administrator

Contract Management integrates service contract information and tracking into Service Manager. Unlike service level agreements, which describe how services in a contract are to be rendered, service contracts are financial agreements that define the services to be provided and the financial implications of using those services. Contract Management tracks each service contract between a customer and service provider, and compiles individual expenses and costs for those services. When using the Configuration Management Payments option, you can submit, cancel, or delete a payment record.

Note

You can't submit a payment if there is no outstanding balance.

To submit a payment record, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Payments**.
2. Use search or advanced search to find one or more records.
3. Select the target payment record you want to submit.
4. Click **Submit Payment**. The **Status** changes from **Pending** to **Submitted**.

   Note

   Submitting a payment doesn't make an actual payment.

5. Click **Save**.

Contents

- 1 Calculating contract payments
- 2 Example: Calculating contract payments
- 3 Cancel a payment record
- 4 Delete a payment record
- 5 Generate a payment schedule for a contract
- 6 Schedule a payment for a contract

- 7 Submit a payment record

1.11.4.9.5. Contract Management administration

Contracts are agreements that describe contract terms, the method by which contract payments are calculated, and payment due dates. Administrative tasks include setting up contract profiles, calculating payments and payment dates, posting payments, and providing balance information for the duration of the contract.

Contract Management environment record

The Contract Management environment record defines options affecting basic Contract Management functionality for all users. Service Manager provides you with a default Contract Management environment record that you can modify for your system. The Contract Management environment record allows administrators to enable or disable access for all operators to Contract Management functionality with or without having a related Contract Management profile record. The Contract Management profile record specifies the activities or operations allowed for contracts such as adding, updating, canceling, or renewing contracts.

Configure the Contract Management environment

Applies to User roles: System Administrator

The Contract Management environment record defines options that enable or disable Contract Management application functionality for all Contract Management users. The Contract Management environment record provides default out-of-box settings. You can customize these settings to meet your specific business needs.

To configure the Contract Management environment settings, follow these steps:

1. Click **Configuration Management** > **Administration** > **Contract Management Environment**.
2. Select new options or clear default options. Your changes redefine the Contract Management environment for all users.
3. When you finish making your changes, click **Save**.
4. Click **OK**.

Contract Management profile records

After you modify the operator record to give a user Contract Management rights, you must add a Contract Management profile for that user. Service Manager provides a default series of profile records, which is based on user roles. The DEFAULT profile

sets permissions for users who aren't defined by any other profile record. Users can access Contract Management by using this profile only if you select the **Allows Access Without Profile Record** option in the Contract Management environment record.

Add a Contract Management profile

Applies to User roles: System Administrator

Contract Management profiles enable administrators to selectively assign access and security to users through the interaction of user roles and profiles. You can add Contract Management profiles to the out-of-box profiles.

To add a new contract security profile, follow these steps:

1. Click **Configuration Management** > **Administration** > **Contract Management Profile**.
2. Type the name of the new profile in the **Profile Name** field.
3. Select the applicable profile elements: Threading, contract, software counter, payment, asset allocation, terms and conditions, and general privileges options for the profile.
4. Click **Add**.

To add a new profile by using an existing profile , follow these steps:

1. Click **Configuration Management** > **Administration** > **Contract Management Profile**.
2. Use search or advanced search to find one or more records.
3. Select the profile from the list that most closely matches the profile you want to add.
4. Type the name of the new profile in the **Profile Name** field.
5. Add or change information on the form to select the applicable contract elements: Threading, contract, software counter, payment, asset allocation, terms and conditions, and general privileges options for the new profile.
6. Click **Add**.

Caution

Make sure that you do *not* click **Save** because doing so will replace the existing profile with the new profile you are attempting to add.

Creating a contract

You can create a variety of contract types using the forms provided in the Configuration Management application. Information is entered into the selected form and that information models the specifics for the contract. Some forms provide multiple sections that contain information fields. For example, if you choose *lease*, the Lease Contract Information form opens. This form contains several sections that apply specifically to leased configuration items (CIs).

Create a contract template

Applies to User roles: System Administrator

Contract Management tracks each service contract between a customer and service provider. Configuration Management enables you to create contracts from contract types or user-defined templates.

To create a contract template, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Use search or advanced search to find one or more records.
3. Select a contract to use as a template.
4. Click **More** or the **More Actions** icon.
5. Choose **Create Template**.
6. Type the name of the new template in the **Name** field.
7. Add or change information on the form to select the applicable contract, lease, financial, terms and conditions, and other information to ensure the contract template is standardized for all related contracts.
8. Click **Add**.

Caution

Make sure that you do *not* click **Save** because doing so will replace the existing contract with the new contract you are attempting to create.

Add a contract

Applies to User roles: System Administrator

Contract Management enables you to track service contracts between customers and service providers. You can add new contracts from contract types or user-defined templates.

To create a new contract, follow these steps:

1. Click **Configuration Management** > **Contracts** > **Contracts**.
2. Click **New** to start the Add New Contract wizard.
3. To create a contract by type, follow these steps:
    1. Choose **Create what type of contract?**, and then select a contract type from the list of records.
    2. Click **Next**. Configuration Management displays the contract form with the applicable fields to enter information about the contract. The form may have multiple sections with multiple information fields. For example, if you choose lease, the Lease Contract Information form opens. It has several sections that apply to leased assets.
4. To create a contract from a user-defined template, follow these steps:
    1. Choose **Create a contract from which template?**, and then choose a template from the list of records.

        Note

        If there are no contract templates available for selection, you need to create one or more first.

    2. Click **Next**. Configuration Management displays the contract template form in which to enter information about the new contract. The form may have multiple sections with multiple information fields. For example, if you choose the Software template, the Software Contract Information form opens.
5. Type a ID in the **Contract ID** field.
6. Choose a date in the **Start Date** and **Expiration Date** fields.
7. Complete the remaining fields, as required.
8. Provide as much information as possible in each of the sections.
9. Click **Add**. The contract record is added and the status is noted as a draft.

    Note

    You can delete a contract only while it is in draft status. When the contract term begins on the start date, the status of the contract changes to current.

Contract Management schedule records

Service Manager uses a background scheduler (called Contract) and the information in the schedule record (called Contract Status) to check for start date, expiration date,

and status to determine whether contract updates are required.

Currency conversion utility

By using the Contract Management currency conversion utility, you can automatically convert national currencies, depending on the exchange rates at the time you initiate the contract. Contract Management manages all currencies in compliance with European Union currency regulations. By default, it uses the fixed inter-European exchange rates. You can enter daily exchange rates into the system to ensure accurate rate conversions.

Access a currency definition record

Applies to User roles: System Administrator

Contract Management enables you to automatically convert between national currencies, depending on exchange rates at the time you initiate the contract. You can view a currency definition record to make sure it is current.

To access a currency definition record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Currencies**.
2. Use search or advanced search to find one or more records.
3. Click a record to view it in the Currency Information form.
4. Edit information about the currency as necessary.
5. Click **Save**.

Set daily currency exchange rates

Applies to User roles: System Administrator

Contract Management provides a currency conversion utility that automatically converts national currencies, depending on exchange rates at the time you initiate the contract. You can enter daily exchange rates into the system to ensure accurate rate conversions.

To set daily currency exchange rates, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Conversion Rates**.
2. Use search or advanced search to find one or more records.
3. Click a record to view it in the Conversion Rate Information form.
4. Update the **Date** and **Exchange Rate** fields with the daily exchange rate information.
5. Click **Save**.

Note

You can tailor Service Manager to automatically import the daily currency exchange rates. For more information, see Tailoring.

1.11.4.10. Enable Configuration Management access

Applies to User roles: System Administrator

Capability words provide a security mechanism to control access to Service Manager applications by enabling or disabling parts of the interface. You can add capability words to a user role or individual operator to control user access to Configuration Management.

To enable Configuration Management access, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Use search or advanced search to find one or more records.
3. Select the operator record that you want to grant access to Configuration Management.
4. Click the **Startup** tab.
5. Add the desired Configuration Management capability words to **Execute Capabilities**. Click **Find** or **Fill** to select an applicable capability word.

   For example: Select **ICMAdmin** to grant administrative access to Configuration Management or select **inventory management**to grant access to Configuration Management.
6. Click **Save**.

1.11.4.11. Change a Configuration Item device type

Applies to User roles: System Administrator

Each asset in the Configuration Management database is assigned a configuration item (CI) device type. The CI device type contains one record for each CI defined. When you define a new CI type, the `devtype` file adds a new record.

To change a Configuration Item device type, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Click a record to display it in the Configuration Item form.
4. To change the device type for a single device, click **More** or the **More Actions** icon on the Configuration Item form.
5. Click **Change Device Type**. Configuration Management starts the wizard.
6. Select a **New Device Type** from the list of records to choose a new device type and then click **Next**.
7. Select a **Subtype** from the list of records if you want to choose a subtype for the device and then click **Next**. Configuration Management opens a Change Device Type Status of Attribute Files form to confirm current and new attribute file names.

   Caution

   If any fields contain non-matching data in the old and the new attribute files, the data in those fields will be lost. Also be aware that any data in fields with matching field names, but with different data types, will become corrupt.

8. To finish, click **Next**.

The updated asset record displays the new device type and optional subtype information.

Note

To change the device type for a list of devices, click **More** or the **More Actions** icon for the record list and complete steps 5 to 8.

1.11.4.12. Using Mass Update with Configuration Management record lists

Mass Update enables you to select multiple records from a list of records in order to update the value in a field or several fields across all selected records. The system provides a template form that displays the fields for the selected records and allows a user to change the value of any of the fields displayed. The Mass Update template form doesn't display all fields in the records. For example, read-only fields in the data policy don't display. You can also perform a Complex Update on the selected records. A Complex Update uses rad expressions that contain variables and concatenated fields to populate other fields in the selected records.

Mass Update is available for device records (Manage CIs), software records (Manage Software), and CI Queue records. By default, the System Administrator security role includes the Mass Update capability.

Note

It is possible to use Mass Update to make changes across configuration item (CI) types and groups; however, it is important to be careful when you are using this feature. A best practice is to perform mass updates only within a specific device type, for example, computer or telecom.

Update multiple configuration item records

Applies to User roles: System Administrator

You can use mass update to update a value in one or more fields for multiple configuration item (CI) records. For example, if you want to update CI records that aren't compliant with baseline, select the Devices not compliant with the baseline view of the CI queue. The value you enter for a particular field becomes the value for that field for all the records you selected for mass update.

You can use complex update to enter expressions as instructions for actions performed on multiple CI records.

Caution

If you exit the Mass Update template form or Complex Update form without clicking **Execute**, all of your changes are cleared on the respective form.

To update multiple CI records by using mass update, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. Select the View list of CI records that you want to update.
3. Using the scroll bar and the **ctrl** key, select the individual records that you want to include in the mass update. Each record you select remains highlighted.
4. Click **Mass Update**. A list of fields available for update opens.
5. Update the fields included in the mass update. To do this ,follow these steps:
    1. Double-click the field you want to update, and then type the value for the field in the box or use the **Find** or **Fill** feature to display a list of potential values for the field.
    2. Click **Next**.
    3. Continue updating the fields that you want included in the mass update.
    4. Click **Next** after each update.
6. Click **Execute**.
7. If a CI record can't be included in the mass update, you can choose the following options:
    - **Retry**: Retry the mass update.
    - **Skip**: Skip the CI record to exclude it from the mass update and continue with the next record.
    - **End**: End the mass update and retry later.

To update multiple CI records by using complex update, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. Select the View list of CI records you want to update.
3. Using the scroll bar and the **ctrl** key, select the individual records that you want to include in the complex update. Each record you select remains highlighted.
4. Click **Mass Update**.
5. Click **Complex Update**.
6. Follow the instructions in the complex update form and type the expressions in the **Instructions to be executed ONCE at the beginning of mass add/update**

field and the **Instructions for action on EACH RECORD** field.
7. Click **Execute**.
8. If a CI record can't be included in the complex update, you can choose the following options:
   - **Bypass**: Bypass the invalid message and include the CI record in the complex update.
   - **Skip**: Skip the CI record to exclude it from the complex update and continue with the next record.
   - **End**: End the complex update and retry later.

Update multiple configuration item software records

Applies to User roles: System Administrator

You can use mass update to update one or several fields in a set of records. Mass update consists of three related functions: Mass Update, Simple Mass Update, and Complex Mass Update. In a Mass Update, a template displays a list of fields that can be updated. You enter a value for each field updated to complete the process. In a Simple Mass Update, the system displays a record form in which you can enter values for selected fields. The values are then applied to all the records selected for the Simple Mass Update. In a Complex Mass Update, the system displays a form in which you type expressions containing variables and concatenated fields to populate another field in the selected records.

Caution

If you exit the Mass Update template form or Complex Update form without clicking **Execute**, all of your changes are cleared on the respective form.

To update multiple CI software records by using mass update, follow these steps:

1. Click **Configuration Management** > **Resources** > **Manage Software**.
2. Use search or advanced search to find one or more records.
3. Using the scroll bar and the **ctrl** key, select the individual records that you want to include in the mass update. Each record you select remains highlighted.
4. Click **Mass Update**. A list of fields available for update opens.

5. Update the fields included in the mass update, follow these steps:
    1. Double-click the field you want to update, and then type the value for the field in the box or use the **Find** or **Fill** feature to display a list of potential values for the field.
    2. Click **Next**.
    3. Continue updating the fields that you want included in the mass update.
    4. Click **Next** after each update.
6. Click **Execute**.
7. If a CI record can't be included in the mass update, you can choose to:
    - **Retry**: Retry the mass update.
    - **Skip**: Skip the CI record to exclude it from the mass update and continue with the next record.
    - **End**: End the mass update and retry later.

To update multiple CI software records by using simple update, follow these steps:

1. Click **Configuration Management** > **Resources** > **Manage Software**.
2. Use search or advanced search to find one or more records.
3. Using the scroll bar and the **ctrl** key, select the individual records that you want to include in the simple update. Each record you select remains highlighted.
4. Click **Mass Update**.
5. Click **Simple Update**.
6. Type new values into the applicable fields on the form.
7. Click **Execute**.
8. If a CI record can't be included in the simple update, you can choose the following options:
    - **Retry**: Retry the simple mass update.
    - **Skip**: Skip the CI record to exclude it from the simple mass update and continue with the next record.
    - **Force**: Force the update on this record and continue with the next record.
    - **End**: End the simple mass update and retry later.

To update multiple CI software records by using complex update, follow these steps:

1. Click **Configuration Management** > **Resources** > **Manage Software**.
2. Use search or advanced search to find one or more records.
3. Using the scroll bar and the **ctrl** key, select the individual records that you want to include in the complex update. Each record you select remains highlighted.
4. Click **Mass Update**.
5. Click **Complex Update**.
6. Follow the instructions in the complex update form and type the expressions in the **Instructions to be executed ONCE at the beginning of mass add/update**

field and the **Instructions for action on EACH RECORD** field.

7. Click **Execute**.
8. If a CI record can't be included in the complex update, you can choose the following options:
    - **Bypass**: Bypass the invalid message and include the CI record in the complex update.
    - **Skip**: Skip the CI record to exclude it from the complex update and continue with the next record.
    - **End**: End the complex update and retry later.

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- 1 Update multiple configuration item records
- 2 Update multiple configuration item software records

1.11.4.13. Checklist: Software license tracking and compliance

You can use Configuration Management to manage the software used in your organization. You can perform the following software management tasks:

- Set up software inventories.
- Verify that the number of actual software installations is within the number of installations allowed by your licenses.
- Link each software license to a specified contract.
- Manage software suites.

1.11.4.13.1. Software license tracking and compliance files

Managing software includes managing the application, license, installations, and contracts. The following files track information about the software in your organization.

File	Description
model	Tracks different software models, including software license and software installation models.
device	Tracks data about each application and software license.
contractsoftware	Documents any financial and procurement information for licensing agreements. You can also attach multiple licenses to a software contract.
contractitem	Tracks the associations between contracts and assets.
pcsoftware	Contains one record for each software installation. The software compliance engine uses information in these records to count each installation during software compliance checks. The information also appears on Configuration Item (CI) records in the device table for CIs that have installed software.
softwarecounter	Verifies that the number of actual software installations conforms with the number of installations allowed by your licenses.

1.11.4.13.2. Software license and installation information

The model file describes different asset types in the model catalog. To effectively track software licenses, the model file contains two types of descriptions:

- Software licenses
- Software installations

The Software tab displays the information about licenses and installations. If the data in the LI (Line Item) Category field on the Catalog tab references a software license or a software installation, the Software tab is populated with a description of the license and installation information.

Create a software installation record

Applies to User roles: System Administrator

After you verify that there are existing software installation models in Service Manager, you can create a software installation record. If there are no software installation models in Service Manager, you must add a software installation model to the catalog.

A software installation is an asset that tracks where software is installed, who is using it, and the date it was installed. You must create separate records for each installation.

Note

If your system is integrated with Universal CMDB, it can be configured to automatically create a new CI record.

To create a software installation record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Manage Software**.
2. Click **Fill** to generate a list of applications.
3. Select the application for the software installation record you want to create. The Manage Software form opens.
4. Complete the remaining fields, as required to document the installation.
5. Click **Add**. Service Manager adds a pcsoftware record.

View software installation information

Applies to User roles: System Administrator

You can view software installation information, such as Version, Install Date, and License ID, from configuration item (CI) records or from software installation records.

To view software installation information from a CI record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select the CI record containing the software installation information you want to view.
4. Open the **Software** section to view a list of all installed software.
5. Double-click an installation record to view software information.

To view software installation information from a software installation record, follow these steps:

1. Click **Configuration Management** > **Resources** > **Manage Software**. The Manage Software form opens.
2. Click **Search** to generate a record list of applications, versions, and the associated installations.
3. Double-click the application record containing the software installation information you want to view, to display it in the Manage Software form.
4. View the software installation information in the **Application Information** and **Installed Computer System** sections.

1.11.4.13.3. Multiple license types

There are three types of multiple licenses:

- **Per named workstation** enables you to install a software application on a given number of identified workstations. For example, a license for an office automation suite that can be installed on 1,000 workstations.
- **Per named user** enables access to an application or database for a given number of users. For example, a license enabling access to a database for 500 named users.
- **Per concurrent accesses** enables a specified number of concurrent accesses to a database. For example, a license enabling 1,000 concurrent users access to the database.

For each multiple license, Service Manager has a Rights field, which specifies the number of:

- Workstations where you can install the software
- Named users
- Concurrent users

Software counters use these values to verify that your company has not exceeded the number of installations or concurrent users granted by a license.

1.11.4.13.4. Software licenses are assets

A software license is an asset record stored in the device file. It describes the number of authorized software uses within an organization. For example, you may purchase a 50-user license for Microsoft Office that supports 50 installations of Microsoft Office. Information about that license is stored as an asset record in the device file.

Create a software license configuration item (CI)

Applies to User roles: Configuration Administrator

After you verify that there are existing software license models in Service Manager, you can create a software license configuration item (CI). If there are no software license models in Service Manager, you must add a software license model to the catalog.

A software license is an asset that represents the number of authorized software users within an organization. The software license CI also contains information, such as whether the software can be used on more than one operating system, the current software version, and whether the software resides on the network or on the client. After the software license CI is created, you can create one or more installation records to track where the software is installed and who is using it.

To create a software license CI, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Click **New**. The Add New Device wizard opens.
3. Select **Software License** in the **CI type** field, and then click **Next**. The New Configuration Item form opens.
4. Enter the information about the software license to complete the CI record, and then click **Save & Exit**.

> **Note**
>
> The part number you select determines the software license CI model that's used for the new CI.

Service Manager creates the CI record and assigns a unique identifier.

Contents

- 1 Create a software license configuration item (CI)

1.11.4.13.5. What's a software counter?

Software counters enable you to verify that the number of software installations you actually have doesn't exceed the number authorized by your license. The number of software installations or access rights authorized by a license is indicated by the number of rights.

For certain licenses, the software installation corresponds to the consumption of a certain number of points. For example, a Microsoft Select license specifies that installing Microsoft Word uses 10 points. Therefore, 100 of these installations uses 1,000 points. When you use software installation counters, you can specify if the installation count should also count the number of points allocated to each installation.

Create a software counter

Applies to User roles: System Administrator

You can create a software counter in Service Manager to track software installations and to ensure that the number of actual software installations doesn't exceed the number authorized by your license.

Choosing a calculation method for the number of software installations enables you to verify that you have not exceeded the number of rights specified in the software license. For example, if an office software license credits you with 1,000 rights and each software installation consumes 10 points, the software installation counter enables you to verify that the software has not been installed more than allowed. In this case, 100 installations X 10 points = 1,000 rights.

To create a software counter, follow these steps:

1. Click **Configuration Management** > **Administration** > **Software Counters** The Software Counter Information form opens.
2. Type a name in the **Name** field for the software counter.
3. Select the **Licenses** tab to choose which models should be included in the software counter information.
    - Select **All License Models?** to count the licenses linked to all software models.

    - Type only the **License Models** that you want included in the software counter.
4. Select the **Installs** tab to specify which installations and parts of those installations you want included in the software counter.
    - Select **All Installation Models?** to count the licenses linked to the models included in all installations.
    - Type only the **Installation Models** that you want included in the software license count.
5. Choose one of the following calculation methods in the **Calculation Method** field:
    - **Each Installation**: Count all licenses in each installation.
    - **Each Different Machine**: Count licenses by each different machine within each installation.
    - **Each Different User**: Count licenses by each different user within each installation.
6. Select **Count Software Suite Components?** to count licenses linked to all components within a software suite.
7. Select **Count Removed or Unknown Installs?** to count licenses of those components that have been removed or unknowingly installed.
8. Select **Count Non-Authorized Installs?** to count licenses of those components that are included in non-authorized installs.
9. Click **Add**.

When you click **Add** to create the software counter, Configuration Management adds the following items:

- Results tab, which displays the following fields and buttons:

Field/Button	Description
No. of Rights	The total number of rights acquired for a particular software license.
Rights Counted	Locates the number of instances of the software that are in use.
Last Processed	Indicates the last time a compliance check was performed.
Compliance Check button	Immediately runs the compliance check.
Create Schedule Record button	Starts the **Schedule Software Compliance Check** wizard, which sets up the compliance check calendar, when the initial check should be performed, and if a recurring license check should occur.

- Message Log tab

Choosing a calculation method

Choosing the calculation method for the number of installations enables you to verify that you have not exceeded the number of rights specified in the software license. For example, an office software license credits you with one thousand (1,000) rights. Each software installation consumes ten points. The software installation counter enables you to verify that the software has not been installed more than one hundred (100) times (100 × 10 = 1,000). Service Manager provides three calculation methods:

- Count each installation regardless of the number of workstations or users.
- Count each workstation regardless of the number of installations on each workstation or users on the workstation.
- Count each user regardless of the number of workstations or installations.

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- 1 Create a software counter
- 2 Choosing a calculation method

1.11.4.13.6. Software license compliance

You can configure specifics for software license compliance manually or automatically by using the Configuration Management application. After you configure software license compliance, you can run compliance check wizards, add or modify terms, schedule compliance checks, and view results.

Compliance messages

The Message Log tab displays one of the following compliance check error messages:

- No license models have been selected for the software counter.
- No installation models have been selected for the software counter.
- There are no license models in the model table.
- A software license doesn't exist for the license model with Part No.=%S.
- Asset ID=%S is not a software license but is linked to license model with Part No.=%S.
- The software license with Asset ID=%S doesn't indicate any rights.
- There are no software install models in the model table.
- A software install doesn't exist for the software installation model with Part No.=%S.
- Installed system=%S doesn't exist for the software install with the License ID=%S and Application Name=%S.

Check software license compliance automatically

Applies to User roles: System Administrator

You can automatically check software license compliance in the Configuration Management application.

To check software license compliance automatically, follow these steps:

1. Click **Configuration Management** > **Administration** > **Software Counters**.
2. Use search or advanced search to find one or more records.
3. Select the software counter record for which you want to check compliance.
4. Click **Create Schedule Record** on the **Results** tab to launch the Schedule Software Compliance Check wizard.

5. Select a date for the first compliance check to occur.
6. If you want to schedule recurring compliance checks, select a date for the next compliance check to occur, and then click **Monthly**, **Quarterly**, **Semi-Annually**, or **Annually**.
7. Click **Finish** to complete the automatic schedule.

Configuration Management creates a schedule record with the same name as the software counter record.

Note

After Configuration Management completes a compliance check, it displays all results on the **Results** tab. Click **No. of Rights** on the **Results** tab to display a list of all the software licenses counted during the compliance check.

Check software license compliance manually

Applies to User roles: System Administrator

You can manually check software license compliance in the Configuration Management application.

To check software license compliance manually, follow these steps:

1. Click **Configuration Management** > **Administration** > **Software Counters**.
2. Use search or advanced search to find one or more records.
3. Select the software counter record for which you want to check compliance.
4. Click **Compliance Check** on the **Results** tab. Configuration Management returns a compliance message.
5. Click **OK**.

Note

After Configuration Management completes a compliance check, results are displayed in the Results tab. Click **No. of Rights** on the Results tab to display a list of all the software licenses counted during the compliance check.

Clear the compliance message log

Applies to User roles: System Administrator

Service Manager saves all compliance check error messages in the compliance check message log. You can clear the log to remove old messages.

To clear the compliance message log, follow these steps:

1. Click **Configuration Management** > **Administration** > **Software Counters**.
2. Use search or advanced search to find one or more records.
3. Select a software counter record to clear the compliance message log.
4. Click **More** or the **More Actions** icon.
5. Click **Clear Message Log**. Service Manager clears the message log.

Contents

- 1 Compliance messages
- 2 Check software license compliance automatically
- 3 Check software license compliance manually
- 4 Clear the compliance message log

1.11.4.13.7. Software license tracking and compliance task review

To effectively manage software licenses and compliance, perform the following sequence of tasks:

1. Add the software license and software installation information to the catalog.
2. Create an asset record for the software application.
3. When you receive the software license, add a Software License type asset record and an Installation type asset record to the Configuration Management database.
4. Create a software contract record.
5. Associate the software license asset record with the software contract record.
6. Create a support contract and associate the application asset to the contract.
7. Install the software on multiple workstations and create one new software installation record for each installation.
8. Create a software counter to verify that the number of actual installations conforms with the number of installation licenses.

1.11.4.14. CI Visualization

CI Visualization enables you to easily visualize relationships between Configuration Item (CI) components and the current state of each item in the configuration. An extensive library of graphics display the relationship in a variety of views. You can customize the views to fit your organization, including using your own graphics for display. You can view, add, or update relationships using the graphical interface. CI visualization enables you to manage resources and outages more effectively.

1.11.4.14.1. Print a visualization diagram

Applies to User roles: Configuration Manager, Configuration Administrator, Configuration Auditor, Change Manager, Change Coordinator, Incident Manager, Incident Analyst, Incident Coordinator

You can print a visualization diagram from the Configuration Management, Change Management, or Incident Management modules.

To print a visualization diagram, follow these steps:

1. Launch configuration item (CI) visualization from Configuration Management, Change Management, or Incident Management. The following steps take Configuration Management as an example:
    1. Click **Configuration Management** > **Resources** > **Search CIs**.
    2. Use search or advanced search to find one or more records. If you know the name of the CI or any other identifying information, type that information into the form.
    3. Open the **Relationship Graph** section. The visualization diagram opens and displays the CI and all the first-level relationships and CIs. You can expand a node to see additional levels of CIs and relationships.
    4. To expand the graph to the entire size of the form, click **More** or the More Actions icon, and then select **Visualize CI**.
2. Click the **Print Preview** icon.
3. For Windows client, complete the following steps. For web tier, skip this step.
    1. Select the magnification level of the diagram.
    2. Click **Options** to select the number of rows and columns to print.
    3. Select or clear the **Add Poster Coords** option to display or hide the coordinates.
    4. From the **Clip Area** menu, select **Graph** or **View** for the print type.
    5. Click the **Title** tab.
    6. In the **Text** field, type the text you want to appear as the title on the printed diagram. You can optionally select the title bar color, text color, and font size for the printed title. The default font size is 12.
    7. Click **OK**.
4. Print the visualization diagram. Service Manager prints the visualization diagram.

1.11.4.14.2. Out-of-box icons for CI representation

The following icons are available with the out-of-box application to represent Configuration Items (CIs). You can customize these icons or add your own.

Icon	Device type	Device subtype	Description
	Default	Default	Icon used to represent a generic CI. This icon appears by default if no other icon is associated with an item.
	Application		Icon used to represent a software application.
	Application	Back-up	Icon used to represent a backup software application.
	Application	Business	Icon used to represent a business software application.
	Application	Internet/Web	Icon used to represent a Web-based application.
	Business Service		Icon used to represent a business service.
	Business Service	Application Service	Icon used to represent a predefined business application.

Icon	Device type	Device subtype	Description
	Business Service	Infrastructure Service	Icon used to represent the components that define a business service.
	CI Group	Baseline	Icon used to represent a CI group.
	Computer		Icon used to represent a desktop computer system.
	Computer	Laptop	Icon used to represent a laptop computer.
	Computer	Server	Icon used to represent a network server.
	Computer	Terminal	Icon used to represent a dumb terminal..
	Computer	Tower	Icon used to represent a PC tower.
	Display Device	Monitor	Icon used to represent a computer monitor.
	Hand Held Devices		Icon used to represent a wireless device.

Icon	Device type	Device subtype	Description
	Hand Held Devices	PDA	Icon used to represent a handheld device, or Personal Digital Assistant (PDA).
	Hand Held Devices	GPS Device	Icon used to represent a Global Positioning System (GPS) device.
	Mainframe		Icon used to represent an installation of a centralized computer mainframe
	Network Components		Icon used to represent a grouping of connected network components.
	Network Components	Firewall	Icon used to represent dedicated gateway serving as a firewall. A firewall can be hardware, software, or both.

Icon	Device type	Device subtype	Description
	Network Components	Gateway	Icon used to represent an entrance point to a network (gateway).
	Network Components	Hub	Icon used to represent a network connection hub.
	Network Components	Modem	Icon used to represent a modem.
	Network Components	Network Interface Card	Icon used to represent a network card.
	Network Components	Router	Icon used to represent a network router.
	Network Components	Switch	Icon used to represent a router switch.
	Office Electronics		Icon used to represent a grouping of computer peripherals or other office equipment.
	Office Electronics	Camera	Icon used to represent a camera.

Icon	Device type	Device subtype	Description
	Office Electronics	Copy Machine	Icon used to represent a copier machine.
	Office Electronics	Fax Machine	Icon used to represent a connected installation of a facsimile (fax) machine.
	Office Electronics	Multifunction	Icon used to represent an all-in-one device, or multifunction printer (MFP) that comprises several pieces of equipment into one unit, such as a printer, copier, fax, and scanner.
	Office Electronics	Printer	Icon used to represent a printer.
	Software License		Icon used to represent software installation.
	Software License	Outlook	Icon used to represent an instance of Microsoft Outlook.

Icon	Device type	Device subtype	Description
	Software License	Utility Software License	Icon used to represent a software utility or tool.
	Storage		Icon used to represent a media storage device.
	Storage	CDRW	Icon used to represent a Rewritable CD.
	Storage	Direct Attached Storage (DAS)	Icon used to a represent a data storage device that's connected to computers directly.
	Storage	Network Attached Storage (NAS)	Icon used to represent a hard disk storage device that's set up with its own network address.
	Telecommunications		Icon used to represent a public telephone system with multiple connections.
	Telecommunications	Desk Phone	Icon used to represent an individual telephone instrument.

Icon	Device type	Device subtype	Description
	Telecommunications	PBX	Icon used to represent an installation of a Private Branch Exchange (PBX) phone system.

1.11.4.14.3. Out-of-box icons for decorators

The following decorators are available with the out-of-box application. You can optionally customize them, or add your own icons.

Icon	Name	Description
	Availability SLO Breach	Indicates when a CI has breached its availability SLO.
	Critical CI (background)	When used with an icon as a binding box, indicates that the item is a critical part of the relationship structure.
	Current Change	Indicates that a CI is currently down.
	Future Change	Indicates that a CI has a scheduled downtime greater than the current time.
	Group	When used with an icon as a binding box, indicates that the item is part of a group of multiple items of the same type.
	Incident	Indicates that an incident is related to the item.
	Item Down	Indicates that the item is not in service.
	Item Down (background)	When used with an icon as a binding box, indicates that the item is not in service.

Icon	Name	Description
	Point of Focus	When used with an icon as a binding box, indicates what item is currently the focal point of the graph.

1.11.4.14.4. Launch a configuration item (CI) Visualization directly from Configuration Management

Applies to User roles: Configuration Manager

Configuration Administrator

Configuration Auditor

You can easily view the relationships between configuration item (CI) components and the current state of each item in the configuration from Configuration Management by viewing the CI in a visualization diagram.

To launch CI Visualization directly from Configuration Management, follow these steps:

1. Click **Configuration Management** > **Resources** > **Search CIs**.
2. Use search or advanced search to find one or more records. If you know the name of the CI or any other identifying information, type that information into the form.
3. Select the CI with the visualization diagram you want to view.
4. Open the **Relationship Graph** section. The visualization diagram opens and displays the CI and all the first-level relationships and CIs. You can expand a node to see additional levels of CIs and relationships.
5. To expand the graph to the entire size of the form, click **More** or the More Actions icon, and select **Expand CI Visualization**.
6. Click **Back** to return to the CI record.

1.11.4.14.5. Launch a configuration item (CI) Visualization directly from Change Management

Applies to User roles: Change Manager

Change Coordinator

You can easily view the relationships between configuration item (CI) components and the current state of each item in the configuration from Change Management by viewing the CI in a visualization diagram.

To launch CI Visualization directly from Change Management, follow these steps:

1. Click **Change Management** > **Changes** > **Search Changes**.
2. Use search or advanced search to find one or more records. If you know the name of the change or any other identifying information, type that information into the form.
3. Select the change with the CI visualization diagram you want to view.
4. Click **More** or the More Actions icon and then select **Visualize CI**.
    - If there is only one associated CI, the visualization diagram opens and displays the CI and all the first-level relationships and CIs. You can expand a node to see additional levels of CIs and relationships.
    - If the Change type allows for multiple CIs and it has more than one associated CI, the Select CI window opens.
5. Select a CI from the list of associated CIs.
6. Click **Finish**. The visualization diagram opens for the selected CI. You can expand a node to see additional levels of CIs and relationships.
7. Click **Back** to return to the Change record.

1.11.4.14.6. Launch a configuration item (CI) Visualization directly from a change task

Applies to User roles: Change Manager

Change Coordinator

From the configuration item (CI) visualization diagram, you can easily view the relationships between CI components as well as the current state of each item in the configuration.

To launch CI Visualization directly from a change task, follow these steps:

1. Click **Change Management** > **Tasks** > **Search Tasks**.
2. Use search or advanced search to find one or more records. If you know the name of the task or any other identifying information, type that information into the form.
3. Select the task whose CI visualization diagram you want to view.
4. Click **More** or the More Actions icon, and then select **Visualize CI**.
    - If there is only one associated CI, the visualization diagram opens and displays the CI and all the first level relationships and CIs. You can expand a node to see additional levels of CIs and relationships.
    - If the Change type allows for multiple CIs and it has more than one associated CI, the Select CI window opens.
5. Select a CI from the list of associated CIs.
6. Click **Finish**. The visualization diagram opens for the selected CI. You can expand a node to see additional levels of CIs and relationships.
7. Click **Back** to return to the task record.

1.11.4.14.7. Launch a configuration item (CI) Visualization directly from Incident Management

User roles: Incident Manager, Incident Analyst, Incident Coordinator

You can easily view the relationships between configuration item (CI) components and the current state of each item in the configuration from Incident Management by viewing the CI in a visualization diagram.

To launch CI Visualization directly from Incident Management, follow these steps:

1. Click **Incident Management** > **Search Incidents**.
2. Use search or advanced search to find one or more records. If you know the name of the incident or any other identifying information, type that information into the form.
3. Select the incident whose CI visualization diagram you want to view.
4. Click **More** or the More Actions icon, and then select **Visualize CI**. The visualization diagram opens and displays the CI and all the first-level relationships and CIs. You can expand a node to see additional levels of CIs and relationships.
5. Click **Back** to return to the incident record.

1.11.4.14.8. View the details of a configuration item (CI) in CI Visualization

Applies to User roles: Configuration Manager

Configuration Administrator

Configuration Auditor

You can view the details of a CI by opening the detail record from CI Visualization.

To view the details of a configuration item in CI Visualization, follow these steps:

1. Click **Configuration Management** > **Search CIs**.
2. Use search or advanced search to find one or more records.
3. Select the CI with the details you want to view.
4. Open the **Relationship Graph** section. The CI Visualization diagram displays the relationships between the CI components and the current state of each item in the configuration.
5. Click on any node in the graph and select the **View CI Details** option. The detail record for that CI opens on a new tab.
6. Click **OK** to return to the CI Visualization diagram.

1.11.4.14.9. Configuring CI Visualization

You can configure and customize the CI visualization implementation by performing the following administrative actions.

- Specify the images that represent Configuration Items (CIs) in the visualization diagram.
- Define the criteria and images used to indicate CI status.
- Configure settings and display preferences.
- Create or modify labels for menu items used to control display of decorators.

Adding custom icons and decorators to the graph

You can add your own custom decorators and icons to the library to use with your application. The objects you create are saved as attachments in the database. For peak performance and to align with the look and feel of the out-of-box (OOB) application, use the following standards:

Object	Dimensions	Color palette	Format	Rules
Decorator	16×16	256	.gif	Ensure that the image name is not the same as any of the OOB image names. The image must be saved in the SYSATTACHMENTS table.
Decorator (background)	76×64 maximum	256	.gif	Ensure that the image name is not the same as any of the OOB image names. The image must be saved as transparent or it will cover the binding icon in the visualization diagram. The image must be saved in the SYSATTACHMENTS table.

Object	Dimensions	Color palette	Format	Rules
Icon	32×32 in 48×48 frame	256	.gif	Icons must be 32×32 images centered within a 48×48 frame. Ensure that the image name is not the same as any of the OOB image names. The image must be saved in the SYSATTACHMENTS table.
Line —	None	None	N/A	A solid line between icons indicates a physical relationship between the items, they are physically connected.

Object	Dimensions	Color palette	Format	Rules
Line —	None	None	N/A	A dotted line between icons indicates a logical relationship between the items; they are connected with software and dependent on each other.

Modify an out-of-box icon

Applies to User roles: System Administrator

You can modify the out-of-box icons that represent configuration items (CIs) in visualization diagrams from the Configuration Management module.

To modify an out-of-box icon, follow these steps:

1. Click **Configuration Management** > **Administration** > **Configure CI Visualization** > **CI Images**.
2. Use search or advanced search to find one or more records. Select the CI Type and CI Subtype, if applicable, from the CI Type menu.
3. Select the icon you want to modify from the record list. Service Manager opens the image associated with that CI Type.
4. To Add, Save As, or Remove the icon, right-click the image and select an action from the menu.
5. Click **Add or modify fields** to use the Add or Modify Tooltip Information wizard. Follow the instructions in the wizard to add, modify, or remove the fields that determine what text the system displays when a user hovers over the CI in a visualization diagram.
6. When you finish modifying the icon, click **Save**. Service Manager updates the CI Image record.

7. Click **OK**.
8. Log off and log back on to view your changes.

Add a custom icon

Applies to User roles: System Administrator

You can add a custom icon to represent a new configuration item (CI) in the visualization diagrams from the Configuration Management module.

To add a custom icon, follow these steps:

1. Add a new device type record in Configuration Management. For details, see Create a new configuration item (CI) type.
2. Create the image you want to use for the CI icon.
3. Click **Configuration Management** > **Administration** > **Configure CI Visualization** > **CI Images**.
4. Select the device type and optionally a subtype.
5. Click **Add**.
6. Start adding the image by doing one of the following:
   - For Windows clients: right-click the **Image** field, select **Add**, browse to the file, and then click **Open**.
   - For Web tier clients: click **Specify Image**, type the full path of the icon file in the **File location** text field or browse to the icon file, and then click **OK**.

     The icon opens on the CI Image record.
7. Click **Add or modify fields** to use the **Add or modify tooltip information** wizard. Follow the instructions in the wizard that allow you to add, modify, or remove the fields and text that are displayed when hovering over the CI in the visualization diagram.

   Note

   When displaying the tooltip for a multi-line text field, CI visualization inserts a line break for each element in the tooltip. Additionally, if the length of one element is greater than 55 characters, the line is truncated with an ellipsis mark (...).
8. Click **Save**.

   Service Manager maps the custom icon to the specified CI Type and saves it in the SYSATTACHMENTS table in the database.

Modify an out-of-box decorator

Applies to User roles: System Administrator

You can modify the out-of-box decorators that are shown in the visualization diagrams from the Configuration Management module.

To modify an out-of-box decorator, follow these steps:

1. Click **Configuration Management** > **Administration** > **Configure CI Visualization** > **Decorators**.
2. Use search or advanced search to find one or more records.

   > **Note**
   >
   > Make sure that the Disable Decorator option is disabled. If you enable this option, the decorator is not displayed in the visualization diagram.

3. Edit the icon if needed.
   - For Windows clients: To Add, Save As, or Remove the icon, right-click the image and select an action from the menu.
   - For web clients: Click **Remove** to remove the existing image, and then click **Specify Image** to replace the existing image with a new one.
4. In the **Category Label** menu, the decorator label type is selected with the out-of-box settings. You can modify this field with **Change**, **Critical**, or **Incident**.
5. Add or modify Conditions.
6. You can use the **Decorator Position** menu to select where the decorator appears relative to the base image. The default setting is the lower right corner.
7. Select a priority from the **Priority** menu or accept the default setting.
8. Modify the line color to show upstream and downstream relationships:
   - Modify the line color for upstream relationships by selecting the highlight color from the **Upstream Highlight Color** menu or accept the default settings.
   - Modify the line color for downstream relationships by selecting the highlight color from the **Downstream Highlight Color** menu or accept the default settings.
9. Edit the information that displays when users hover over the decorator image:
   - In the **Tooltip** section of the record, you can add or edit the fields that show data when hovering over the decorator image in the visualization diagram.

- In the **Group Label** field, you can type a group label to use when a user hovers the mouse pointer over the decorator image with a group label or accept the default settings.

10. When you finish modifying the decorator, click **Save** and click **OK**.
11. Log off and log back on to view your changes.

Add a custom decorator

Applies to User roles: System Administrator

You can add a custom decorator that's shown in the visualization diagrams from the Configuration Management module.

To add a custom decorator, follow these steps:

1. Create an icon to represent the new decorator by following these steps:
    1. Click **Configuration Management** > **Administration** > **Configure CI Visualization** > **Decorators**.
    2. Type a name in the **Name** field for the new decorator.
    3. Click **Specify Image**, and then browse for an image in the Specify Image Location window.
    4. Select an image and click **OK**.
    5. Select **Add** from the menu.
    6. Navigate to the directory where you saved the custom decorator and select the file name.
    7. Click **Open**. Service Manager displays the icon on the decorator record.

    Note

    Make sure that the Disable Decorator box is not checked. If you select the box, the decorator is not displayed in the visualization diagram.

2. In the **Category Label** menu, select **Change**, **Critical**, or **Incident** in the decorator label type.
3. Add or modify Conditions.
4. You can optionally use the **Decorator Position** menu to select where the decorator appears relative to the base image. The default setting is the lower right corner.
5. Select a priority from the **Priority** menu or accept the default setting.
6. Modify the line color to show upstream and downstream relationships:

1. Modify the line color for upstream relationships by selecting the color from the **Upstream Highlight Color** menu or accept the default settings.
2. Modify the line color for downstream relationships by selecting the color from the **Downstream Highlight Color** menu or accept the default settings.

7. Edit the information that displays when users hover over the decorator image:
    1. In the **Tooltip** section of the record, you can add or edit the fields that show data when hovering over the decorator image in the visualization diagram.
    2. In the **Group Label** field, you can type a label to use when a user hovers over the decorator image with a group label or accept the default settings.
8. Click **Save**.
9. Click **OK**.
10. Log off and log back on to view your changes.

1.11.4.14.10. CI node grouping

When you expand a node that contains multiple related Configuration Items (CIs) in a visualization diagram, it can slow system performance. To prevent performance issues, you can display CIs of the same type as a group by using node grouping.

You can select the option on any folder grouping of CIs to display the individual CIs. Activating this option opens a window that allows you to choose one or more CIs from a list. When you are finished selecting the CIs, the graph updates to show only the extracted CIs.

Group configuration items (CIs) of the same type

Applies to User roles: System Administrator

You can group configuration items (CIs) of the same type from the Configuration Management module. Node grouping prevents performance issues by displaying CIs of the same type as a group in visualization diagrams that contain multiple related CIs.

To group CIs of the same type, follow these steps:

1. Click **Configuration Management** > **Administration** > **Configure CI Visualization** > **Settings**. The CI Node Grouping form opens.
2. In the **When expanding a node, display a group when CIs of the same type exceed** field, type a threshold value, which controls the number of CIs of the same type that trigger the formation of a CI group.
3. Log off and log back on to view your changes. Service Manager visually represents CIs of the same type that exceed the new threshold as a grouped item.

Expand a node group to view individual configuration items (CIs)

User roles: Configuration Manager, Configuration Administrator, Configuration Auditor, Change Manager, Change Coordinator, Incident Manager, Incident Analyst, Incident Coordinator

You can view individual configuration items (CIs) in any folder grouping in a visualization diagram by expanding the node.

To expand a node group to view individual CIs, follow these steps:

1. Launch CI Visualization from the Configuration Management, Change Management, or Incident Management module.
2. Click one time on a node grouping or folder inside of the CI Visualization diagram, and then select **Expand Group**. A list of CIs opens.
3. Press and hold the **ctrl** button, and then click each item to select one or more CIs from the list.
4. Click **Next**. The graph is updated to show the extracted CIs.

1.11.4.14.11. Display of outage spreading in CI Visualization

CI visualization displays the Item Down decorator to indicate that there is an active outage of a configuration item (CI). The edge connecting two CIs is highlighted red if both CIs are down. You can customize the Item Down decorator to meet your organization's specific requirements.

1.11.4.15. Calculate CI descendant relationships

Service Manager tracks upstream and downstream relationships between CIs. A relationship between CIs means that there is some dependency between CIs. Any given CI typically has one upstream relationship and one or more downstream relationships. If an upstream CI has an interruption of service, Service Manager assumes that all CIs with a downstream relationship to the affected CI also have an interruption of service. For example, if a network router has an interruption of service, then all servers and PCs that connect to that router also have an interruption of service.

After you select an Affected Service and then select an Affected CI in an Incident, a Change, an Interaction, a Request, or a Problem record, the system browses the configuration relationship records in the cirelationship table and uses the BSGRelationsCalculation ScriptLibrary to retrieve all downstream CIs of the business service that you select.

Currently, there are two business service group relations (BSG relations) implementations available in Service Manager and you can use either of them to calculate CI descendant relationships.

- BSGRelationsCalculation1 ScriptLibrary.
  This implementation can quickly retrieve downstream CIs but consumes more system memory. If you have a large number of CIs, using this ScriptLibrary may bring an out-of-memory issue.
- BSGRelationsCalculation2 ScriptLibrary.
  This implementation retrieves downstream CIs slower but consumes less system memory. If you have many CI relationships, using this ScriptLibrary is a better choice.

If you are working with the BSGRelationsCalculation1 ScriptLibrary but it constructs huge redundant data in the cirelationbsg table and downgrades the system performance, follow these steps to switch to BSGRelationsCalculation2.

1. Open the BSGRelationsCalculation ScriptLibrary.
2. Update `var bsg = lib.BSGRelationsCalculation1;` to `var bsg = lib.BSGRelationsCalculation2;`, and then save your changes.
3. Create a ScriptLibrary and then add the following codes:

   ```
   lib.BSGRelationsCalculation2.run();
   ```

4. Click the **Execute** button to initialize the data of CI relationship to improve your system performance.

1.11.4.16. Create a remote desktop connection

A remote desktop connection is a useful tool to connect one Windows computer to another Windows computer. As a system administrator, you might want to troubleshoot a computer associated with a CI record. From Service Manager, you can locate the CI record that relates to a computer. If the computer is a networked Windows device, Service Manager enables you to connect to it.

Follow these steps to create a remote desktop connection from a CI:

1. Search for a CI with the CI Type of **Computer**.
2. Click **Remote Connection** on the toolbar.

   The system opens the **Service Manager Remote Desktop Connection** window. Connections appear as an IP address, a network name, or a display name.
3. Select an available connection by clicking **Connect by IP Address**, **Connect by Network Name**, or **Connect by Display Name**.

   If you are working with the Service Manager web client, the system automatically creates a *.rdp file and downloads it to your computer. You can double-click this *.rdp file to launch a remote desktop connection. If you are working with the Service Manager Windows client, the system directly launches a remote desktop connection.

1.11.5. Configuration Management security

Configuration Management security areas

The security areas for Configuration Management are Configuration Item and Configuration Management Configuration. These areas contain the default security rights and settings for the Configuration Management module. The security right settings will be inherited by any new roles created in an area when no other settings are specified for that security role.

These security areas are used to set permissions to operators to provide access to particular area of Configuration Management. The following table lists the areas and the relevant Configuration Management menu items the operators can access.

Area	System Navigator menu items for this area
Configuration Item	This area contains the default security rights and settings for Configuration Item. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.

Area	System Navigator menu items for this area
Configuration Management Configuration	This area contains the default security rights and settings for Configuration Management configuration. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the Role.

Area	System Navigator menu items for this area
	Note When you set the security rights for a security role in the Configuration Management Configuration area: • The **View** right is to view the settings defined in the **Administration** menu and the **Configuration** menu. • The **Update** right is to update the values of existing settings defined in the **Administration**menu and the **Configuration** menu. • The **New** and **Delete** rights are to create and delete a setting in the **Configuration** menu. • The **Admin** right is to add, edit, or delete the settings in the **Administration** > **Settings** menu.

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the

Configuration Item and Configuration Management Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Configuration Item	TRUE	FALSE	Never	Never	FALSE	FALSE
Configuration Management Configuration	FALSE	FALSE			FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Configuration Item and Configuration Management Configuration areas.

Configuration Management security roles and settings

The out-of-box security roles for the Configuration Management module include the following:

- Configuration Administrator
- Configuration Auditor

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Configuration Management area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Configuration Item and Configuration Management Configuration areas. This table only lists the new security roles that have different settings with the default rights.

Area Name	Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin
Configuration Item	Configuration Administrator	TRUE	TRUE	Always	Never	FALSE	FALSE	FALSE
	Configuration Auditor	TRUE	TRUE	Always	Never	FALSE	FALSE	FALSE
Configuration Management Configuration	Configuration Administrator	FALSE	FALSE	Never	Never	FALSE	FALSE	FALSE
	Configuration Auditor	FALSE	FALSE	Never	Never	FALSE	FALSE	FALSE

Configuration Management security options

Security options are available to users with the SysAdmin and ICMAdmin capability words. Security options are independent of the options selected in the Configuration Management security roles.

Field	Definition
Add	Add component records.
Update	Update component records.
Delete	Delete component records.
Print	Enables the print option for the user or group.
Fill	Enables the Fill function for the user or group.
Find	Enables the Find function for the user or group.
Mass Delete	Select a group of configuration records and delete them.
Change Devtype	Change the CI type of an asset.
Mass Update	Select a group of configuration records and modify fields in those records.
Open Incident	Open an incident record from within Incident Management.
Request Change	Open a change request in Change Management from within Configuration Management.
Query Window	Run structured queries in Configuration Management.
Export	Export component records to an external file.
Count Records	Count the records in a record list.

Field	Definition
Views	View alternate forms in Configuration Management and additional information, such as Baseline version queries.

The following options are available only when selected.

Field	Definition
Allow Inefficient Queries	Execute an incomplete or partially keyed query. This option is overridden if you select Skip Inefficient Query Warning.
Skip Inefficient Query Warning	Disable the message warning the user that a non-keyed query will be slow. If you choose this option, you will override Allow Inefficient Queries.

Contents

- 1 Configuration Management security areas
- 2 Configuration Management security roles and settings
- 3 Configuration Management security options

1.12. Knowledge Management help hub

Service Manager Knowledge Management supports Knowledge-Centered Support (KCS) standards and guidelines by providing a natural language search engine and a rich-text authoring tool that enables users to search, update, and author knowledge articles.

User roles

Describes the ITIL user roles as implemented by Service Manager.

KCS I

KCS II

KCS III

Key concepts

Describes the most important things that you need to know about how the Knowledge Management module functions.

Knowledge Management overview

Knowledge-Centered Support (KCS)

Knowledge Management workflow

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Managing document categories

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User workflows

Describes the ITIL workflows as implemented by Service Manager and the sub tasks that end users must perform in order to execute the workflows.

Contribute and approve knowledge document (ST 7.1)

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Administrator tasks

Describes the Administrator tasks in the Knowledge Management module.

Upgrade to the Process Designer KM workflow

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Additional resources

Processes and Best Practices

1.12.1. Knowledge Management overview

Knowledge Management supports Knowledge-Centered Support (KCS) standards and guidelines by providing a natural language search engine and rich-text authoring tool that enables your users to search for, update, and author knowledge articles. Knowledge Management has a strong interface with the Interaction, Incident, and Problem Management modules so that your users can search for and use knowledge from existing Incident or Problem records while attempting to resolve new Incident and Problem records. Users can also use this integration to create new knowledge.

You can use Knowledge Management to meet your business needs in the following ways:

- Knowledge Management integrates with Service Desk Interaction so that users responding to an Interaction record can search for similar interaction records with solutions. In addition to searching for knowledge, the out-of-box mapping allows users with applicable privileges to create solutions and author knowledge documents from resolved interaction records.
- Knowledge Management integrates with Incident Management so that users updating incident records can search for similar issues with solutions in the knowledgebases. Users can also create knowledge to contribute a new knowledge document from a resolved Incident record, so that data in the mapped fields are used in the knowledge document.
- Knowledge Management integrates with Problem Management so that users can perform the following tasks:
  - Search for similar Problem records, Incident records, Known Error records, or knowledge documents
  - Use the resolution from an existing resolved issue in a knowledgebase as a solution for the problem record
  - Use the information from a problem resolution to create a new knowledge document
- Publish documents, retire documents, and delete documents from the knowledgebases
- Track knowledge documents at various stages with statistics generated in the **kmusagehistory** table

1.12.1.1. Knowledge Management profiles

The Knowledge Management (KM) profiles control knowledge document access, creation, editing and administration. The administrative privilege allows users to manage document types, document views, and KM profiles.

Users have a KM security profile and an operator record which determine what privileges the users has. The knowledge document categories that a user can access are determined by the knowledge groups the user is assigned.

Within each security profile, there is a direct association between a defined (named) profile and the categories and sub-categories for which the selected capabilities apply. Each security profile maps to a document category or sub-category. A user with that profile has access to the documents specified by the profile-to-category mapping and all subcategories in that branch of the category tree.

Profiles can be modified to select only some of the available capabilities for a profile, effectively creating sub-profiles. For example, some users may not be allowed to publish knowledge documents externally as well as internally.

The out-of-box KM security profiles are briefly described in the following table.

KM profile	Privilege
DEFAULT	With this profile, you can search and view externally published knowledge documents for those documents in categories to which this profile has access. You can also submit feedback on these documents.
INTERNAL USER	With this profile, you can search and view externally published knowledge documents for those documents in categories to which this profile has access. You can also search and view internally approved knowledge documents for those documents in categories to which this profile has access. You can also submit feedback on these documents.
KCS I	With this profile, you can author and contribute knowledge documents.
KCS I Editor	With this profile, you can author knowledge documents and edit working copy documents in the workflow for those categories to which this profile has access.
KCS II	With this profile, you can author knowledge documents, contribute knowledge documents, edit published documents without placing them in the workflow (edit in place), and publish documents internally only. Also, you can view adaptive learning data when adaptive learning is enabled in the system.

KM profile	Privilege
KCS III	With this profile, you can author knowledge documents, contribute knowledge documents, edit published documents without placing them in the workflow (edit in place), and publish documents internally and externally. Also, you can view adaptive learning data when adaptive learning is enabled in the system.
KM ADMIN	With this profile, you can administer KM document security profiles, document types, document views, and document categories. You can also author knowledge documents, contribute knowledge documents, edit published documents without placing them in the workflow (edit in place), and publish documents internally and externally. Also, you can edit adaptive learning data when adaptive learning is enabled in the system.

KCS I profile

The KCS I profile in the out-of-box system allows you to perform the following tasks:

- Contribute new knowledge documents
- Suggest a document category or categories for approval
- Save draft versions of the knowledge document
- View unapproved working-copy documents

A user with a KCS I profile is sometimes known as a document submitter or contributor. A KCS I can view documents approved for internal and external use.

The KCS I capabilities apply to the knowledge document categories associated with the KCS I profile and knowledge group in which the user is a member. The three-way profile-to-group-to-category mapping is defined under the Permissions tab of the Category Details form.

The Knowledge Management administrator creates a KCS I contributor knowledge group and then associates the knowledge group with a KCS I profile and document category. The users in the knowledge group can contribute documents to that category and to the subcategories below that category.

For example, in the out-of-box sample data, the KM 1 operator has a **KCS I** profile and is a member of the **KCS I KM Group**. The profile-to-group-to-category mapping is KCS I (profile)-to-KCS I for KM Group (group)-to-Knowledge Management (category).

The category hierarchy in the out-of-box sample data is:

```
Service Manager
                                Knowledge Management

Knowledge-Centered Support (KCS)

.Document Maintenance and Lifecycle
```

The KM 1 operator, as a member of the KCS I for KM Group, can contribute knowledge documents to the Knowledge Management category and to the sub-categories below it. Those are Knowledge-Centered Support (KCS) and Document Maintenance and Lifecycle. The KM 1 operator can't contribute documents to the **Service Manager** category. Also, the KM 1 operator can't view documents in the **Service Manager** category.

KCS I editor profile

The KCS I editor profile in the out-of-box system allows you to perform the following tasks:

- Contribute new knowledge documents
- Suggest a document category or categories for published documents
- Save draft versions of knowledge documents
- Edit working copy knowledge documents that are assigned to the KCS I editor in the **Pending Documents** queue
- View unpublished working copy documents

A KCS I editor can view documents published for internal and external use.

The KCS I editor capabilities apply to the knowledge document categories associated with the KCS I editor profile and knowledge group in which the user is a member. The three-way profile-to-group-to-category mapping is defined under the **Permissions** tab of the **Category Details** form.

The Knowledge Management administrator creates a KCS I editor knowledge group and then associates the knowledge group with a KCS I editor profile and document category. The users in the knowledge group can contribute documents to that category and to the sub-categories below that category.

KCS II profile

The KCS II profile in the out-of-box system allows you to do all of tasks a KCS I profile can do and also the following tasks:

- Publish immediately internal knowledge documents
- Retire and unretire knowledge documents
- Delete retired knowledge documents
- View and delete feedback for a knowledge document
- Edit and review knowledge documents in the Pending Documents queue
- Assign documents for editing and reviewing
- Edit published documents in place without sending them to workflow
- View adaptive learning data when adaptive learning is enabled in the system

A user with a KCS II profile may also be assigned as a document owner or editor of a document in the document workflow.

The KCS II capabilities apply to the knowledge document categories associated with the KCS II profile and knowledge group in which the user is a member. The three-way profile-to-group-to-category mapping is defined under the **Permissions** tab of the **Category Details** form.

Similar to the KCS I profile, the Knowledge Management administrator creates a KCS II knowledge group and then associates the knowledge group with a KCS II profile and document category. The users in the knowledge group can contribute, review, and edit documents to that category and to the sub-categories below that category. With a KCS II profile, you can also publish documents for internal access for that category and the sub-categories below that category.

KCS III profile

The KCS III profile in the out-of-box system allows you to do all the tasks that a KCS I or KCS II profile can do. A KCS III profile also enables a user to publish documents immediately or any documents in the workflow intended for external audiences (publish external) and edit published documents without sending them to the workflow.

Like a KCS II user, a user with a KCS III profile may also be assigned as a document owner or editor of a document in the workflow. Also, like a KCS II user, a user with a

KCS III profile can view adaptive learning data when adaptive learning is enabled in the system.

KCS III capabilities apply to the knowledge document categories associated with the KCS III profile and knowledge group in which the user is a member. The three-way profile-to-group-to-category mapping is defined under the **Permissions** tab of the **Category Details** form.

Similar to the KCS I profile, the Knowledge Management administrator creates a KCS III knowledge group and then associates the knowledge group with a KCS III profile and document category. The users in the knowledge group can contribute, review, and edit assigned documents for that category and the sub-categories below that category. With a KCS III profile, you can also publish documents for external access for that category and the sub-categories below that category.

KM ADMIN profile

The KM ADMIN profile in the out-of-box system allows you to do all the tasks that a KCS I, KCS II, or KCS III profile can do. With the KM ADMIN profile, you can also perform these tasks without specific permissions in the document categories. A KM ADMIN user can perform the following tasks:

- Search, view, retire, unretire, and delete documents in any document category
- Author knowledge documents, contribute knowledge documents, edit published documents without placing them in workflow, and publish documents internally and externally
- Assign documents for editing and reviewing when adaptive learning is enabled in the system

Note

The KM ADMIN profile is unlike all other out-of-box Knowledge Management profiles, because the KM ADMIN profile is not assigned in the **Category Details** form.

1.12.1.1.1. Knowledge Administrator role

Knowledge Administrator is a security role specified in a user's operator record and allows a user to perform all the administrative tasks associated with maintaining the Knowledge Management system. A Knowledge Administrator can perform the following tasks:

- Manage profiles and environment variables
- Manage document categories and permissions on the categories
- Manage document types and views
- Manage shared content for the knowledge documents
- Maintain the knowledgebases
- Configure and customize the layout of search results

The Knowledge Administrator security role is assigned to a user in the operator record. Users in this role can maintain knowledge groups and Knowledge Management profiles, and update Knowledge Management profiles as user duties change.

As part of the integration of Knowledge Management with Incident Management, Service Desk Interactions, and Problem Management, the Knowledge Administrator can modify the default field mappings. These mappings can change which fields are searched, from which fields data is retrieved, and which fields can be used by users when creating knowledge from solutions and workarounds. The out-of-box tables for these field mappings are **incidents**, **kmdocument**, **knownerror**, **rootcause**, and **probsummary**.

1.12.1.1.2. Security profiles and Knowledge Management document display lists

Knowledge Management (KM) uses knowledge groups, document categories, and the KM security profiles to determine which documents to display in the lists of knowledge documents available from the following KM menus:

- Published Documents
- Draft Documents
- Retired Documents
- Pending Documents
- Documents for Review

Note

A user with **SysAdmin** or **KMAdmin** capabilities is able to view all the knowledge documents that appear in any of these lists.

Document list	View access
Published Documents	KCS II or III in the out-of-box (OOB) system or a user in a knowledge group associated with the document category and a profile with the publish internal capability can view all published documents. A KCS II or III user can only perform other actions such as create pending or retire documents if the user's privileges include the combination of profile, knowledge group and category privileges to allow these actions.
Draft Documents	Displays a list of knowledge documents with the status "draft" for the user who authored the draft document. All draft documents in the system display for users with SysAdmin or KMAdmin capabilities.
Retired Documents	Users with KCS II or III in the OOB system or for a user in a knowledge group associated with a category and a profile with the retire documents capability, KM displays all retired knowledge documents in the categories associated with a profile. Also, users can create working copies of these documents based on the capabilities provided by the combination of profile - knowledge group - category privileges.

Document list	View access
Pending Documents	Documents display in this list based on the user's access to the document category and a profile that allows viewing unapproved working copy documents. The documents displayed are currently in the approval workflow (Triage > Revise > Review > Publish phases). Also, users with the proper permissions, based on their profile, can move documents through the workflow, publish documents, revert documents, or retire documents in this list.
Documents for Review	Documents display in this list based on the user's access to the document category and a profile that allows viewing unapproved working copy documents. The documents displayed are currently in the Review phase of the approval workflow. Users with capabilities based on profile permissions are able to forward the Publish phase, or return it to the Revise phase for further revision.approve internal. Also, users with capabilities based on profile - knowledge group - category can approve, revert or retire documents in this list.

Add a KM security profile

User role: System Administrator, KMAdmin

Published documents are documents that are available for searching and viewing. Some documents may only be viewed by internal users, while others can be viewed by internal and external users. Published documents can be edited by users with the **Edit in place a published document** option selected for their KM security profile.

To add a KM security profile, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Profiles**.
2. Type the name of the new profile in **Profile Name**.
3. Select the applicable checkboxes on each of the security profile tabs. For example, go to the KCS II tab to select the **Edit in place a published document** option.
4. Click **Add**.
5. Click **Save**.
6. Click **OK**.

View a KM security profile

User profile: System Administrator, KMAdmin

Knowledge Management (KM) uses knowledge groups, document categories, and the KM security profiles to determine which documents to display in the lists of knowledge documents available from the KM menus. The KM security profile and operator record together determine what KM privileges are granted to users.

To view a KM security profile, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Profiles**.
2. Click **Search**.
3. Click a record to view the security profile.

Contents

- 1 Add a KM security profile
- 2 View a KM security profile

1.12.1.2. Adaptive learning

Adaptive learning is a feature of Knowledge Management where the system collects words or specific phrases used to search a knowledgebase for those documents that are considered useful or can be used as solutions for incidents, problems, or interactions.

Knowledge-Centered Support (KCS) users are encouraged to improve Knowledge Management service levels for customers, gain operational efficiencies, and increase an organization's value to their company. With a defined set of principles and practices, KCS II and KCS III users are able to view the adaptive learning list. Knowledge Management (KM) administrative (KM ADMIN) users can edit the list by adding, deleting, or modifying words in the list or modifying the count.

Note

System Administrators can turn on or off adaptive learning with a setting on the Knowledge Management Environment form.

Knowledge Management maintains a count which specifies the number of times search words or phrases resulted in finding a useful document (whether marked as useful or used as a solution). Each time a user performs a search using the same words or phrases, the count is incremented or a new entry is created when the search returns useful results. Also, each time an operator clicks the **Use Solution** button, the phrase used by the operator to search for the solution is saved or the count is incremented for the phrase for the adaptive learning record associated with the solution.

Note

For a working copy of a document that has no published version, the system collects the words and phrases for adaptive learning. For a working copy of a document that already has a published version, the system doesn't collect the words and phrases for adaptive learning.

When a user performs a search using advanced search and then marks a document as useful, the system creates or increments the associated adaptive learning record as follows:

- When the search results are filtered by "Any of these words," the system adds separate entries for each word in the search phrase.

- When the search results are filtered by "All of these words" or "This exact phrase," the system creates a single entry for all the words in the search phrase.

The useful count becomes a weighted value so that when another user performs a different search using the same words or phrase, documents containing the same words or phrase are returned higher on the hit list in the next set of search results.

Example: If words like "monitor" or "blue monitor" are used in a document and it resolves an incident and is used as a solution, the phrase blue monitor or the word monitor are appended to the adaptive learning terms for that document and the count is incremented. For additional searches, a document with one blue monitor in it will come up with a reasonable relevancy. You can also change the count of monitor to 15, for example, to make this document appear higher on the search results list than another document with the word monitor in it.

Searching with adaptive learning

When adaptive learning is enabled, data about the queries that returned documents that were used for solving issues is stored and integrated into the documents when they are indexed. This allows the scope of the search to grow beyond the actual content of the document itself. Adaptive learning is only available for searches of sclib-type knowledgebases.

Searching is enhanced by adaptive learning as follows:

- Adding additional words from the saved queries to the content of the document means that additional hits on the document occur when these words are supplied in a new query.
- The words and phrases collected by adaptive learning can be further weighted to increase the relevancy value of a particular document, which means some documents are found more frequently and appear higher on the hitlist.

Viewing adaptive learning data

There are several ways to view adaptive learning phrases and counts when adaptive learning is enabled.

Note

You must have a KCS II or KCS III profile to view adaptive learning data, and a KM ADMIN profile to view or edit adaptive learning data.

- Search using a word or phrase and then from the hit list select one of the documents to view. There is an **Adaptive Learning** button available from the document view if the document is an sclib-type document. Click this button to view a list of phrases and counts for the document.
- View a document from the list of published documents. The view includes an **Adaptive Learning** button. Click this button to view a list of phrases and counts for the document.
- The **Knowledge Management** > **Configuration** menu includes an **Adaptive Learning** selection. When you click **Adaptive Learning**, a form appears, from which you can search for adaptive learning data using knowledgebase, document ID, or phrase.
- For knowledge candidates such as Incident or Interaction records found when doing a search, the **Adaptive Learning** button is available for KM ADMIN users when viewing a specific document.
- The **Adaptive Learning** button appears for users with a KCS II, KCS III,or KM ADMIN profile when viewing or editing documents from the Pending Documents queue that are new working copy documents (that is, documents without a previously-published version).

Edit adaptive learning data

User role: KM Admin or System Administrator

Note

You must have a KM ADMIN profile to edit adaptive learning data.

When adaptive learning is enabled, data about the queries that returned documents that were used for solving issues is stored and integrated into the documents when they are indexed. If you want to edit the actual content of a document, you can edit data that will be used as adaptive learning data.

To edit adaptive learning data, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **Adaptive Learning**.
2. Type the document ID if you are looking for the adaptive learning data for a specific document.
3. Select the knowledgebase if you are looking for the adaptive learning data for a specific knowledgebase.
4. Type the phrase if you are looking for the adaptive learning data for a specific phrase.
5. Click **Search**.

6. Select a document from the list of documents.
7. Click the phrase or count for the document you want to update and then type the new value for the count.
8. Add the phrase to the next blank line.
9. Click **Delete Phrase** after you select the phrase you want to delete.
10. Click **Save**.
11. Edit the knowledge article.
12. Perform the **Publish External** or **Publish Internal** action to publish the knowledge article, so that the article will be freshly indexed and the changes of kmadaptivelearning will take effect in the search results.

You can also edit adaptive learning data by viewing a published document or by viewing a working copy of a document that also has a published version of the document.

Enable or disable adaptive learning

Applies to User roles: System Administrator

When adaptive learning is enabled, data about the queries that returned documents that were used for solving issues is stored and integrated into the documents when they are indexed.

To enable or disable adaptive learning, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Environment**.
2. Select the **Use Adaptive Learning to enhance search results** field to enable adaptive learning.

   Note

   Clear the **Use Adaptive Learning to enhance search results** field if you want to disable adaptive learning.

3. Save your changes.
4. Reindex all sclib-type knowledgebases.

   Note

   This step is needed whenever you enable or disable adaptive learning.

1.12.1.3. Searching the knowledgebases

When searching the knowledgebases, you can either perform a basic search where you search for a text string or an advanced search where you can provide several search parameters. When you perform an advanced search, you can specify the knowledgebases and the document categories to search. You can also specify a set of filtering parameters, such as exact phrase and creation date. After doing an initial search or advanced search, you can then search within the search results.

Each of the knowledgebases has different fields that are indexed for searching, so when you search a knowledgebase, provide search parameters that match the fields in the knowledgebase. For example, the knowledge articles have a title and author field. When you view an Incident record, the out-of-box system displays the Incident record number, description, and solution for closed incidents.

If your Service Manager uses Smart Analytics as the search engine, you can perform cross-module search from selectable sources inside and outside of Service Manager. For more information, see Use Smart Search as a general search tool.

You can use the Knowledgebase Maintenance feature to add additional knowledgebases for searching. These knowledgebases are created from a table in Service Manager(**sclib**) or by using web crawling to browse and index an external web site (**weblib**).

Knowledge documents

In the Knowledge Management (KM) system, a knowledge document typically consists of information about a particular topic. The amount of detail and length depends on the complexity of the topic and the audience for the document. Knowledge documents can include attachments that can be images, text files, PDF files, or Word files for example. In the out-of-box system, the types of knowledge documents are error messages, external documents, question/answer documents, problem/solution documents, and reference documents. All knowledge documents are assigned to at least one document category and may be placed in multiple categories. A typical document category might be technical documents or HR procedures.

- **Draft documents** - These are documents that are being created, but have not yet been submitted to be published..
- **Published documents** - These are documents that are available for searching and viewing. Some documents may only be viewed by internal users, while others can be viewed by internal and external users. Published documents can

be edited by users with the **Edit in place a published document** option selected for their KM security profile. Users can add feedback to published documents. Users with appropriate privileges can create a working copy of the document and then make revisions.

- **Working copy** - A document becomes a working copy when a published document goes into the workflow to be updated. When the working copy of the document is published, it replaces the original document. A working copy of a published document has an "R" appended to the document ID. A new document submitted to be published is also considered a working copy while it is in the workflow waiting to be published, but there is no "R" appended to the document ID.
- **Retired documents** - These are documents that become out-of-date and have been archived for some reasons. Retired documents aren't searched when users perform a knowledgebase search. KM ADMIN users can delete retired documents.

Knowledge documents typically have a life cycle that includes draft, triage, pending documents in the approval workflow, published, and retired.

Published knowledge documents

Published knowledge documents are documents that are available for searching and viewing. Some documents may only be viewed by internal users, while others can be viewed by internal and external users. Published documents can be edited by users with the **Edit in place** option selected for their Knowledge Management (KM) security profile. Users can add feedback to published documents. Users with appropriate privileges can create a working copy of the document and then make revisions.

The system displays a search screen instead of a list of published documents so that users can quickly find a published document or a group of published documents.

In the out-of-box system, a Knowledge-Centered Support (KCS) II user profile can publish internal documents and a KCS III user profile can publish internal and external documents.

Reverting a working copy document

Working copies of published documents can be reverted to the original version of the published document while the document is in the document workflow. There is a **Revert** button available for a working copy of a published document. If changes have been made to the working copy of the document, it can still be reverted. In the out-of-box system, users with a KCS II or KCS III profile can revert working copy documents

in those categories to which they have access. A KM ADMIN profile allows the user to revert any working copy document.

Retired knowledge documents

A retired document is a knowledge document that's no longer searched during a knowledgebase search. A retired document can be deleted or unretired.

Before you retire a document, you should update any documents that link to the document you are retiring. When you attempt to retire a document that other documents link to, the system will not allow you to retire it. The system alerts you to the situation and then allows you to send these linked documents to workflow for review. You can then update the links or remove the links in these documents because Knowledge Management will not allow you retire a document that links to other documents in the knowledgebase.

Note

The alert message includes the document IDs of the documents that need to be updated and placed in the Pending Documents queue. If you click OK, it then places those documents in the Pending Documents queue.

In the out-of-box system, users with KCS II or KCS III profiles can retire any of the published documents that are assigned to the document categories and subcategories to which they have access. A user with a KCS ADMIN profile can retire any draft or published document in any document category.

Using the Pending Documents queue

The Pending Documents queue manages knowledge documents that have been submitted to the document workflow. The standard workflow for documents is to determine what processing the knowledge document requires, assign the document for editing or review, assign the document for review, and publish the document for internal or external audiences. In some cases, documents can be published immediately without going through a formal workflow.

Users with a KCSII or KCS III profile are primarily responsible for coordinating the Knowledge Management documents in the workflow. They are able to view and act on documents in the categories or subcategories to which they have access. A user with a KM ADMIN profile or a System Administrator can access all the Knowledge Management documents for all categories and subcategories.

If the resolution of an Incident or Interaction record is used frequently, the KCS Coach or System Administrator may want to manually create a knowledge document by

locating the Incident or Interaction record and clicking the **Create Knowledge** link.

In the workflow, an editor can initiate the following changes:

- Make any required changes to the document

  > **Note**
  >
  > If you are editing an attachment for an existing knowledge document, you must send the document to the workflow. You can then save the attachment to your desktop, make the necessary updates, save the changes, and then re-upload the attachment to the knowledge document.

- Assign the document to a member of the document group for the document category or subcategory and move the document to the next phase in the workflow
- Publish the document immediately
- Use activity update
- Retire the document

For each document, a user will select a document owner and an assigned editor from a list of users with edit privileges.

> **Note**
>
> Documents don't need to progress through every phase and end in the final phase. A user with the correct permissions in the document categories may publish or retire a document before it reaches the final phase. Additionally, the owner or editor may decide to revert back to the original version of a published document.

Tips for searching the knowledgebases

These tips will help you to narrow your search for knowledge documents. While it is helpful to be specific to find the knowledge documents you need, it is also sometimes helpful to broaden your search by eliminating some of the search parameters.

Here are some suggestions for using the **Search for** text box.

- Enter words and phrases in plain text.
- To search for an exact phrase, surround it with a pair of double quotation marks ("").

- When a search returns a large number of hits, limit your search by searching within the search results (search within results).
- If you have performed an advanced search, terms entered in the advanced search screen are still active even if the advanced search screen is not visible. The advanced query terms are added to the simple search query unless you go to the advanced search screen and remove them.

Search engine relevance ranks

A simple search by the search engine ranks documents found according to relevance of the criteria you enter for the search. For example, based on word density, word count, word proximity, word stemming, and thesaurus expansion, the documents found are listed in order based on their relevance to your search criteria. The most relevant documents are displayed at the top of the list.

1.12.1.4. Knowledge Administrator role

Knowledge Administrator is a security role specified in a user's operator record and allows a user to perform all the administrative tasks associated with maintaining the Knowledge Management system. A Knowledge Administrator can perform the following tasks:

- Manage profiles and environment variables
- Manage document categories and permissions on the categories
- Manage document types and views
- Manage shared content for the knowledge documents
- Maintain the knowledgebases
- Configure and customize the layout of search results

The Knowledge Administrator security role is assigned to a user in the operator record. Users in this role can maintain knowledge groups and Knowledge Management profiles, updating Knowledge Management profiles as user duties change.

As part of the integration of Knowledge Management with Incident Management, Service Desk Interactions, and Problem Management, the Knowledge Administrator can modify the default field mappings. These mappings can change which fields are searched, from which fields data is retrieved, and which fields can be used by users when creating knowledge from solutions and workarounds. The out-of-box tables for these field mappings are **incidents**, **kmdocument**, **knownerror**, **rootcause**, and **probsummary**.

Related topics

Knowledge Management profiles
Knowledge groups

1.12.1.5. Knowledge-Centered Support (KCS)

Knowledge-Centered Support (KCS) is a knowledge management strategy for service and support organizations. It defines a set of principles and practices that enable organizations to improve service levels for customers, gain operational efficiencies, as well as increase the organization's value to their company.

KCS practices involve collaborating, sharing, using and improving knowledge. KCS practices encourage knowledge creation and improvement by all members of the support team and promote effective knowledge searches by maintaining up-to-date knowledgebases. The out-of-box Knowledge Management system enables all internal and external users to provide user feedback on the knowledge documents. This helps to identify useful articles and to mark other articles for revision, which contributes to the overall quality and value of the information in the knowledgebase.

The goal of KCS is to solve a problem once and use the solution often. KCS-based systems take advantage of the knowledge available in all phases of the service desk operations by making the knowledge captured by the customer support process available for reuse throughout the support organization.

Related topics

Knowledge Management overview

1.12.1.6. Knowledge Management statistics tracked in the kmusagehistory table

The Service Manager kmusagehistory table is intended for generating statistics to track knowledge documents at various stages.

The following document statistics are tracked in the kmusagehistory table.

Statistic	Description
Viewed	Tracks documents viewed as the result of a search. This indicates if a knowledge document can be found and used in its existing state. If a document is viewed from a URL linked directly to the kmdocument record, the viewed statistic in kmusagehistory is not incremented. You can change this default behavior by specifying a parameter with the URL request. When the parameter is specified, the usage history and view count will be logged and incremented.
Created	Tracks new knowledge documents that have been created.
Submitted for approval:	Tracks new and updated knowledge documents submitted for content verification and approval.
Approved internal	Tracks knowledge documents published internally.
Approved external	Tracks knowledge documents published externally.
Modify	Tracks modified knowledge documents.
Revert	Tracks knowledge documents reverted to their original state after modifications have already been made.

Statistic	Description
Retired	Tracks knowledge documents taken out of the knowledgebase and retired either because the knowledge documents reached their expiration date or the content was no longer applicable.
Deleted	Tracks deleted knowledge documents.
Unretired	Tracks knowledge documents put back into circulation and made available in the knowledgebase after having been earlier retired and pulled out of the documentation queue.
Used Solution	Tracks the number of times a knowledge document was used as a solution for Interaction, Incident, and Problem records.

Related topics

Knowledge Management workflow

1.12.1.7. Knowledge Management sample data

The following table provides an overview of the out-of-box sample data for Knowledge Management users. It specifies the knowledge groups, profiles, and document categories each user is associated with or has access to. These data demonstrate how knowledge groups, security profiles, and document categories interact to manage user access to knowledge documents.

User	KM Group	KM Profile	Document Categories
Amanda.Admin	HR KCS III	KCS III	HR
Harry.Incident, Clark.Change, and Rick.Problem	HR VIEWER	INTERNAL USER	HR
all operators with user role: incident manager, incident coordinator, problem manager, problem coordinator, change manager, change coordinator, and change approver	KCS III for Service Manager group	KCS III	Service Manager
all operators with userrole: incident analyst, problem analyst, change analyst	KCS II for Service Manager group	KCS II	Service Manager
all operators with userrole: service desk agent	KCS I for Service Manager Group	KCS I	Service Manager
all operators with userrole: self service	Service Manager External Viewers	DEFAULT	Service Manager
KM 3	KCS III for KM Group	KCS III	Service Manager > Knowledge Management

User	KM Group	KM Profile	Document Categories
KM 2	KCS II for KM Group	KCS II	Service Manager > Knowledge Management
KM 1	KCS I for KM Group	KCS I	Service Manager > Knowledge Management

The following table lists the out-of-box document categories.

Service Manager		
	Knowledge Managemen	
		Knowledge Centered Support (KCS)
		Document Maintenance and Lifecycle
	Support	
	Change Management	
HR		
	Benefits	
	Policy	

Related topics

Knowledge Management administration

1.12.2. Knowledge Management user roles

KCS I

The KCS I user role has the following responsibility:

- Contribute new knowledge articles

To execute this role, the KCS I has the following tasks available:

Note

For an overview of how these tasks fit in the Knowledge Management workflow, see Knowledge Management workflows and tasks.

ITIL Workflow	Task
Contribute and approve knowledge document (ST 7.1)	• Create a knowledge document • Create knowledge from an interaction, incident, problem, or known error • Preview and view a document • Contribute hot news • Edit a document in place • Submit a document for approval
Search, view and utilize knowledge document (ST 7.2)	• Search for published documents • Search for adaptive learning data • Search for a solution • Use a solution from a knowledgebase search • View adaptive learning data

KCS II

The KCS II user role has the following responsibilities:

- Contribute new knowledge articles

- Approve articles for internal audiences

To execute this role, the KCS II has the following tasks available:

Note

For an overview of how these tasks fit in the Knowledge Management workflow, see Knowledge Management workflows and tasks.

ITIL Workflow	Task
Contribute and approve knowledge document (ST 7.1)	• Create a knowledge document • Create knowledge from an interaction, incident, problem, or known error • Preview and view a document • Contribute hot news • Edit a document in place • Submit a document for approval • Create a working copy of a knowledge document • Add feedback to a knowledge document • Submit a document for revision • Submit a document for review • Accept a document
Search, view and utilize knowledge document (ST 7.2)	• Search for published documents • Search for adaptive learning data • Search for a solution • Use a solution from a knowledgebase search • View adaptive learning data
Maintain and continually improve knowledge (ST 7.3)	• Retire a knowledge document • Retire a knowledge document when other documents link to it • Unretire a knowledge document

KCS III

The KCS III user role has the following responsibilities:

- Contribute new knowledge articles
- Approve articles for internal audiences

- Approve articles for external audiences

To execute this role, the KCS III has the following tasks available:

Note

For an overview of how these tasks fit in the Knowledge Management workflow, see Knowledge Management workflows and tasks.

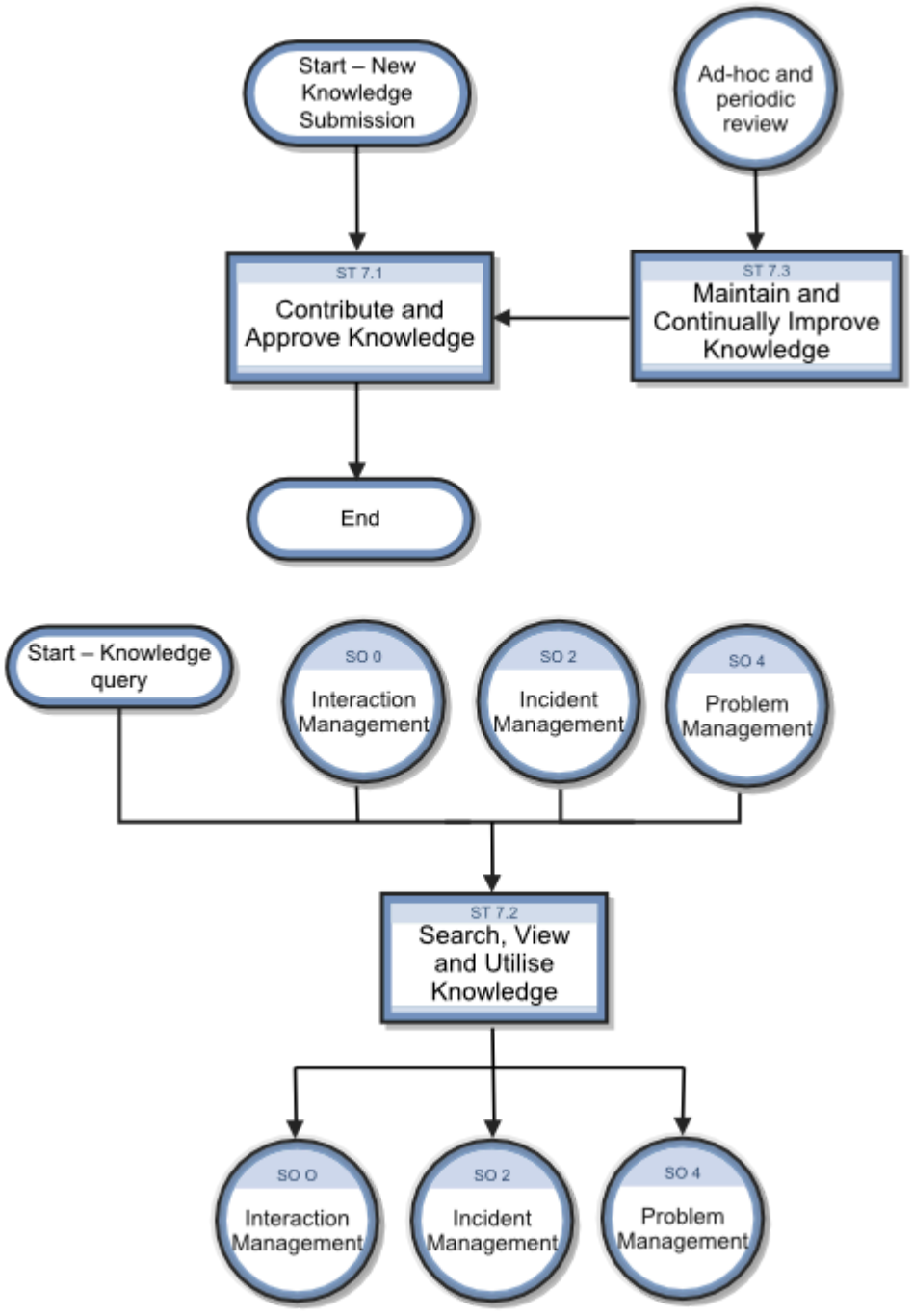
ITIL Workflow	Task
Contribute and approve knowledge document (ST 7.1)	• Create a knowledge document • Create knowledge from an interaction, incident, problem, or known error • Preview and view a document • Contribute hot news • Edit a document in place • Submit a document for approval • Create a working copy of a knowledge document • Add feedback to a knowledge document • Submit a document for revision • Submit a document for review • Accept a document
Search, view and utilize knowledge document (ST 7.2)	• Search for published documents • Search for adaptive learning data • Search for a solution • Use a solution from a knowledgebase search • View adaptive learning data
Maintain and continually improve knowledge (ST 7.3)	• Retire a knowledge document • Retire a knowledge document when other documents link to it • Unretire a knowledge document

1.12.3. Knowledege Management Process

The Knowledge Management process collects, organizes, structures and distributes knowledge for ongoing use. It is responsible for ensuring that an organization can archive and retrieve knowledge in an efficient manner.

The Knowledge Management process consists of the following workflows and tasks.

Knowledge Management Workflow

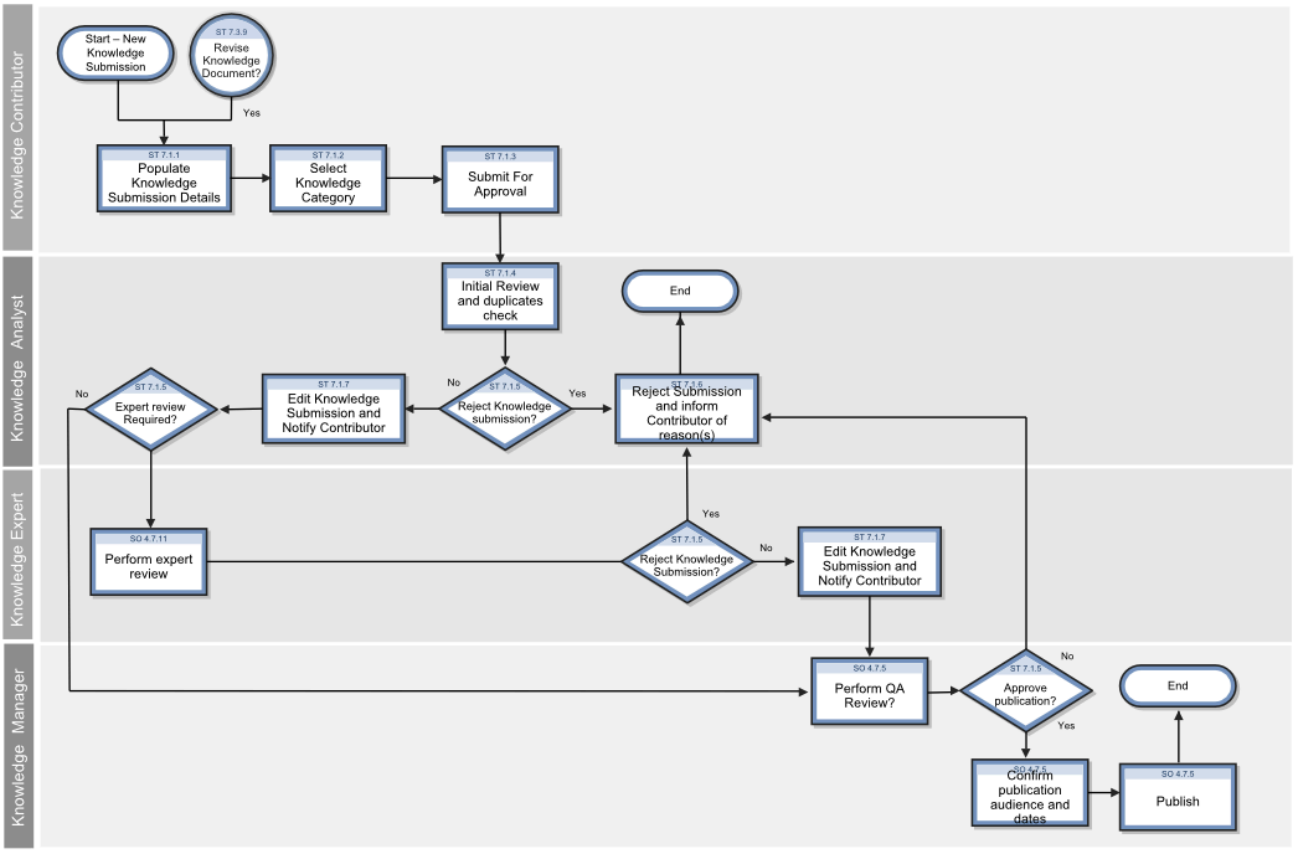


ITIL Workflow	Task
Contribute and approve knowledge document (ST 7.1)	• Create a knowledge document • Create knowledge from an interaction, incident, problem, or known error • Preview and view a document • Contribute hot news • Edit a document in place • Submit a document for approval • Create a working copy of a knowledge document • Add feedback to a knowledge document • Submit a document for revision • Submit a document for review • Accept a document
Search, view and utilize knowledge document (ST 7.2)	• Search for published documents • Search for adaptive learning data • Search for a solution • Use a solution from a knowledgebase search • View adaptive learning data
Maintain and continually improve knowledge (ST 7.3)	• Retire a knowledge document • Retire a knowledge document when other documents link to it • Unretire a knowledge document

1.12.3.1. Contribute and approve knowledge document (ST 7.1)

The Contribute and Approve Knowledge Document process starts with the submission of a KM document. You can contribute a knowledge document by authoring knowledge articles or using external documents that are uploaded into a knowledge base. You can use the rich-text editor to author documents, and you can add documents as attachments that can include images, text files, Word files, or PDFs.

- The process begins when a contributor creates a new submission or revising an existing knowledge document. The contributor selects a documentation type, enters basic details, selects a category type, and submits the draft for review by a Knowledge Analyst.
- The analyst can edit, accept, or reject the document. If necessary, the document can be passed to a Knowledge Expert, who can also edit, accept, or reject it. The Knowledge Expert is typically a subject-matter expert with more in-depth understanding of a given topic and the ability to determine the accuracy and currency of the KM document.
- The Knowledge Manager makes the final decision about whether to accept the document or return it for further revision. If the document meets all KM publication criteria, the Knowledge Manager accepts and publishes the document.



Related topics

Knowledge Management overview
Knowledge Management workflows and tasks
Processes and Best Practices

1.12.3.1.1. Create a knowledge document

User roles: KCS I, KCS II, KCS III, KM Admin, System Administrator

You contribute a knowledge document by authoring knowledge articles or using external documents that are uploaded into a knowledgebase. When you author a document, you will be asked to specify a document type. Knowledge documents can be saved as drafts before they are complete. You can also preview a document while you are creating it.

As part of the task to create a knowledge document, you need to assign it to a document category or several document categories, depending upon the topic of the knowledge document. After you have created the knowledge document, you need to either submit it for approval or publish it for either internal or external viewing. In the out-of-box system, a user with a KCS II profile can immediately publish a document for internal viewing and a user with a KCS III profile can immediately publish a document for an internal or external audience.

Using the rich-text editor

You can use the rich-text editor to author documents, and you can add documents as attachments that can include images, text files, Word files, or PDFs. The rich-text editor allows you to author documents in a WYSIWYG environment. It provides an embedded, rich-text editor in the Contribute New Knowledge function, and it provides most of the standard features available for authoring and editing knowledge documents. You can also use the editor to create new document views.

Its features include:

- Standard text formatting
- Font formatting
- Table creation
- Supports paste from Word documents
- Support links and anchors

Note

Tabbing is unavailable in the rich-text editor. Use the space bar to create indents. Using the tab key will exit you from the editor. For additional information regarding the editor, see the related topics links below.

The rich-text editor toolbar contains many of the common icons and editing functions available with most word processors. For example, with the rich-text editor you can cut, paste, and copy text. You can search for a text string within the text. You can select from a small group of font types and size to stylize the text in the document you are authoring. There is a table wizard you can use to add a table to your document.

In addition to the standard functions of a word processor, you can also use the rich-text editor to add links in your document to other documents that are either document attachments for the document you are authoring or links to other documents in the knowledgebase. There is also a Source button that displays the HTML code used in the document. This is particularly useful when you are creating particular document views for the document types.

Note

There is a **Preview** button on the rich-text editor toolbar that opens the document in another browser window. It is best to use the **Preview** button on the Contribute Knowledge form toolbar rather than the one on the rich-text editor toolbar to preview a document. The **Preview** button on the Contribute Knowledge form displays the document in the document view associated with the document you are authoring.

Attachments to knowledge documents

Knowledge document attachments consist of shared content and document- or record-type attachments. Shared content usually consists of company logos or other branding elements used in the knowledge documents stored in the knowledgebase. The document- or record-type attachments are files or records attached to a particular knowledge document or several documents. These attachments include summary documents with links to more detailed documents, a knowledge document with a link to a particular incident or known error record, error logs, screenshots, and policy or procedure documents to which many other documents refer. Generally, shared content has a one-to-many relationship with a document and linked attachments have a one-to-one relationship with a document.

When knowledge document attachments need updates, users must send those knowledge documents into the workflow. Editors can then view, edit, save, and re-upload updated attachments for knowledge documents.

Attachments may be indexed for searching on a per-knowledgebase basis. Consult your Knowledge Management administrator to determine if attachments are indexed

for the **Knowledge Library** knowledgebase.

To create a knowledge document, follow these steps:

1. Click **Knowledge Management** > **Contribute Knowledge**.

   A list of document types is displayed. The out-of-box document types are:

   - Error Message/Cause
   - External
   - Question/Answer
   - Problem/Solution
   - Reference

2. Select the document type you want from the list.

3. Complete the mandatory and optional fields on the form.

Field	Description
Title	A general or descriptive heading for this knowledge document. This is a mandatory field. **Note** If you install and enable the latest Smart Analytics for your Service Manager, the content in the **Title** field of this knowledge document is used as the search string for context-aware search. Depending on your Smart Search configurations, context-aware search can be either automatically triggered as you type in **Title**, or manually triggered by clicking the Context-aware search icon. You can also put the cursor in the **Title** field, and press **Shift + Enter** to trigger a context-aware search.
Summary	A brief statement that presents the main points of the Knowledge document in a concise form. If this field is empty, the search engine may provide a dynamically generated summary.

Field	Description
Expiration date	The date when this Knowledge document should be flagged as retired, for example, when the document may be out-of-date and require a review. A retired document can also be seen from the Knowledge KPI (Global) dashboard. After viewing such a report, a KCS II. KCS III, or an administrator can search for a retired document by id or title, review the document, and elect to send the document through a workflow for revision.
Legislation	A drop-down field that you can modify to include specific legislation and regional requirements. Out-of-box options are listed as HIPAA, ISO/IEC 20,000, and SOX.
Confirmed legal rights to publish this material	A checkbox that you must select to confirm that you have the legal rights to use the data contained within the document. If this option is not selected, when you attempt to publish the document, the system prompts you to confirm your legal rights to publish this document.

Field	Description
Tags	Enter one or more free-form tags (keywords or terms) of this document. The tags you enter are searchable and can be used as keywords to highlight important knowledge information. For more information, see Working with tags.
Document ID	This is a system-generated unique identification number for this document. **Note** This field is read-only.
Document type	This read-only field indicates the document type you selected.

Field	Description
Document subtype	**Note** Available only when you selected the Reference document type. You can optionally specify a document subtype as appropriate. By default, the following subtypes are available: ◦ **Agreement**: a document describing a formal understanding between two or more parties. An agreement is not legally binding unless it forms part of a Contract. ◦ **Service Design Package**: a document that provides the initial outline of a service agreement and denotes what agreements will be required, who the signatories shall be, and other process oriented details. ◦ **Service Quality Plan**: SQP, a long-term quality improvement plan ◦ **Service Improvement Plan**: SIP, a limited, short-term improvement plan ◦ **Business Case**: provides a business justification for a change proposed by a formal Change Proposal

Field	Description
	◦ **Project Information**: provides information about the project proposed by a formal Change Proposal ◦ **Availability Plan**: a plan to ensure that existing and future availability requirements for IT services can be provided cost effectively ◦ **Process Documentation**: a storage of process related documentation, including Standard Operating Procedures, Organizational Charts & Responsibilities, Escalation Procedures, Contact Lists, Work Instructions, Process Schedules, and so on.
Status	This read-only field indicates the current status of the document based on the pre-defined knowledge management workflow.
Locale	In the drop-down list, choose the locale to be used by the search engine when searching and indexing the document. By default, the language code matches the language you logged in as. Changing this field overrides the Knowledgebase default locale setting.

Field	Description
Reason for information	This text field can be used for describing the reason for this document or for additional copyright information.
Associated Services	Click the **Fill Field Associated Services** icon to select one or multiple business services that are associated with this document. Specifying an associated service will cause this knowledge document and the service record to be linked to each other in their **Related Records** tab and **Related Documents** respectively. Once this knowledge document is published, it is listed in the **Related Documents** section of the service records. After you select the services, the **Notify Service Subscribers on Publish** option is displayed. Select this option if you want to notify users who have subscribed to this service once the document is published. To view subscribers to the services, click the **View Subscribers** icon. A list of users and departments actively subscribed to the selected services is displayed. Click the subscription ID link of a department to see users in that department.

4. Assign the document to a category.

   All documents in the knowledgebase are assigned to a document category. Categories for knowledge documents are ordered into hierarchies.

Note

When you contribute a new document, a draft document doesn't require a document category assignment, but any document that's going to be submitted for approval or published immediately must be assigned to at least one document category or subcategory.

1. Click the **Categorization / Flags** tab.
2. Click **Add** to display a list of document categories.
3. Double-click the document category.
4. Double-click the subcategory or click **Skip** to end the category or subcategory selection.

   Note

   If you assign the document to the wrong category, you can remain in the edit mode, select the category you want to delete, click **Remove**, and then click **Add** to assign the document to a new category.

5. If needed, repeat the steps until you finish adding all the categories for the document.
6. If you want to create this document as hot news, select the **Hot News** checkbox. For more information, see Contribute hot news.
7. In the **Associated CIs** field, you can specify the CIs that are associated with this knowledge document if needed.

   Note

   Once the knowledge document is published, it is listed in the **Related Documents** section of each associated CI record.

   Note

   The **Associated CIs** field is available for the Reference document type only.

5. Use the rich-text editor to enter text for the document. You can use the built-in spell checker of a web browser to resolve spelling errors. If there is a typo, press Ctrl and then right-click to display the spell suggestions.

6. If needed, add a file attachment to the document.

    1. Click the **Attachments** tab. Administrators can define supported attachment file types.
    2. Add a file by doing one of the following:
        - For Web-tier clients: click **Add File**, type the full path of the file in the **File location** text field or browse to the file, and then click **OK**.
        - For Windows clients: right-click in the blank area of the attachments list, click **Add**, browse to the file, and then click **Open**.

        Note

        If you are editing an attachment for an existing knowledge document, you must send the document to the workflow. You can then save the attachment to your desktop, make the necessary updates, save the changes, and then re-upload the attachment to the knowledge document.

7. If needed, insert a link to the attached file.

    1. Click in the rich-text editor, place the cursor where you want to insert the link, and then click the **Insert a link toward an attached file** icon on the rich-text editor toolbar.
    2. Type the name of the attached file in the **Filter Text** field, or leave the field blank so that all the attachments are listed for selection.
    3. In the **Link Text** field, specify the **Alt Text** for the link.
    4. Click **Next** and then double-click the name of the attached file from the list of files.

8. If needed, insert a link to another knowledge document.

    1. Click in the rich-text editor field to position the location for the link.
    2. Click the **Insert a link toward another record** icon on the rich-text editor toolbar. The message "Document saved to allow attachment linking" appears.
    3. Click **Knowledge Document** in the Select Link Type wizard and edit the **Link Text** label.

4. Click **Next** and then type the document ID. The document must be a published document in the database, and you must have view access to at least one of the categories to which the document belongs.
5. Click **Finish** and then click **Preview** to view your document and the new link.
6. Click the link to verify that it links correctly.

   Note

   The link doesn't open until the document in which you are trying to add the link is either published internally or externally.

7. Click **Back**, and then click **Back** again.

9. If needed, insert an image to the document.

   Note

   Thel toolbar that allows you to insert an image is not visible if you select the document type as **External**.

   1. If your document doesn't yet have the image as an attached file, attach the image as described above.
   2. Click the **Insert an image** icon on the rich-text editor toolbar.
   3. Type the name of the image file in the **Filter Text** field, or leave the field blank so that all of the image files listed on the **Attachments** tab are included in the selection wizard.

      Note

      You can attach only .jpg and .gif format images.

   4. In the **Link Text** field, specify the **Alt Text** for the image.
   5. Update the default **Image Height** and **Image Width** as required.
   6. Click **Next**.
   7. Double-click the image file name in the list.
10. If needed, click the **Link** icon to insert a hyperlink to an external web site.
11. Click the **Preview** button to view the document. The following information is also displayed:

- Document Id
- Document Type
- Document subtype (available only for the Reference type)
- Summary
- Creation Date
- Expiration Date
- Author
- Associated Services (if selected)
- Tags
- Related Documents

12. Click **Back**.
13. If you want to save the document as a draft, click **Save Draft**.
14. Click **Save** or click the applicable button to submit the document for approval or to publish the document immediately if you are a KCS II or KCS III for the document category.

    Once the document is published, the following information is also displayed to users: Document Id, Document Type, Document subtype, Summary, Creation Date, Expiration Date, Author, Tags, Associated Services, and Related Documents (Document subtype is available only for the Reference type).

To delete a draft knowledge document you created:

1. Click **Knowledge Management** > **Draft Documents**. A list of draft documents is displayed.
2. Select the document you want to delete.
3. Click **Delete**.
4. Click **Yes** in the confirmation window.

1.12.3.1.2. Link a knowledge document to another

Applies to User Roles:KCS I, KCS II, KCS III, KM Admin, System Administrator

The Knowledge Management module allows you to link related knowledge documents to each other. You can link a published document to another published document or a new document. You can do so when editing a published document.

To link a published knowledge document to another knowledge document, follow these steps:

1. Navigate to **Knowledge Management** > **Published Documents**. The **Search Knowledge Documents** page is displayed.
2. Do either of the following:
    - In the **Document ID** field, type the document ID, and then click **Search**. The document that you want to link to another document is displayed.
    - If you don't know the exact document ID, you can also fill in any of the other fields on the search form such as **Title**, **Summary**, **Author** and **Associated Service**, and then click **Search**. A list of published documents that meet your search criteria is displayed. Click the document that you want from the list.
3. Click **Edit**.
4. Click the **Related Records** tab.
5. In the **Link Type** field, select **Related Knowledge**.
6. If you want to link the document to another published document, click **Link Existing Knowledge**. A form is displayed, allowing you to link the document to another document published internally or externally. Associate the documents in either way:
    - Directly enter the document ID, for example "KM0110", and then click **OK**.
    - If you don't know the exact document ID, click **Search**. In the search form that's displayed, enter your search criteria and then click **Search**. From the search results, select the document ID that you want. The selected document ID appears in the form. Click **OK**.

    On the **Related Records** tab of each document, a link to the other document is added.

7. If you want to link the document to a new document, click **Link New Knowledge**. Select a document type to create a new document. Fill in the document form as needed. For details, see Create a knowledge document. Then, on the **Related Records** tab of each of the two documents, a link to the other document is added.

   Once the new document is published, a link to the other document is also displayed under the **Related Documents** section of both of the two documents when users browse knowledge documents.

   To unlink a document from another document, select the document from the **Related Records** tab, and then click **Unlink Knowledge**. Service Manager removes the links from the **Related Records** tabs of both documents respectively.

1.12.3.1.3. Preview and view a document

You can preview a document while you are authoring a new document or working on a draft document. To preview a document, click the **Preview** button available in the document edit form.

You can also view a retired or published document. When you select the document from the list of retired or published documents, the document appears in the document view.

When you search, use the hyperlink in the search results list to view a document.

Related topics

Create a knowledge document

1.12.3.1.4. Submit a document for approval

User roles: KCS I

Once you have created a draft knowledge document, you submit it for review and approval.

To submit a document for approval, follow these steps:

1. Create a document and save it as a draft.
2. Open the draft document from **Knowledge Management** > **Draft Documents**.
3. Click **Submit**.

   The document is submitted for approval and its status changes to **triage**. The document will appear in the **Pending Documents** queue. Depending on the document quality, the approver takes further actions. For example:

   - Click **Add Feedback** to add feedback or additional information to the document.
   - Click **Submit for Revision** to send the document for further revision.
   - Click **Publish Internal** or **Publish External** to publish the document internally or externally.
   - Click **Retire** to retire the document.

1.12.3.1.5. Contribute hot news

User role: System Administrator, KMAdmin, KCS I, KCS II, KCS III

Hot news entries contain very recent information regarding server outages or other infrastructure failures. Hot news can also include solutions to a new incident or problem.

To contribute hot news, follow these steps:

1. Click **Knowledge Management** > **Contribute Knowledge**.
2. Double-click the document type from the list.
3. Type a title and summary for hot news.
4. Specify an expiration date for the hot news document. This is the date by which the document content should be reviewed for accuracy and timeliness.+
5. Select a document category for the hot news on the **Categorization/Flags** tab.
6. Check the **Hot News** checkbox on the **Categorization/Flags** tab.
7. Use the calendar to specify a **Valid from** and **Valid to** date range. These dates specify when the document is marked as hot news so that a hot news search displays the document. You can specify a date in the future for the document to become hot news and a date when the document is no longer hot news.
8. Use the rich-text editor to enter text for the hot news.
9. Click the **Preview** button to preview the hot news document.
10. Click the applicable button to submit the document for approval or to publish the document immediately.

Related topics

Create a knowledge document

1.12.3.1.6. Create a working copy of a knowledge document

User roles: KCSII, KCS III, KM Admin, System Administrator

A working copy of a knowledge document becomes a working copy when a published document goes into the document workflow to be updated. When the working copy of the document is published, it replaces the original document. A working copy of a published document has an "R" appended to the document ID. A new document submitted for review to be published is also considered a working copy while it is in the workflow waiting to be published, but there is no "R" appended to the document ID.

To create a working copy of a knowledge document by using Published Documents, follow these steps:

1. Click **Knowledge Management** > **Published Documents**.
2. Click **Search**.
3. Select the document from the document list.
4. Click the **Create Working Copy** button.
5. (Optional) Type your feedback in the add feedback box.
   Note: It is useful to add feedback because the KCS II or III user who creates the working copy might not be the one responsible for updating the content. By providing feedback, you can explain what needs to be updated or the reason for creating a working copy of the document.
6. Select Useful? or Not Useful? if required.
7. Click **OK** or **Skip Feedback**.
8. Click **Knowledge Management > Pending Documents**.
9. Select the knowledge document to be approved in the workflow. The title column in the list contains the document ID with an R suffix for working copy and the title of the document.
10. Update the working copy of the document in the embedded rich text editor, and click **Save**.
11. Move it to the next phase of the approval workflow.

    For example, click **Submit for Revision**. The status of the working copy changes to revise.

1.12.3.1.7. Create knowledge from an interaction, incident, problem, or known error

User roles: All users

To create knowledge from an Interaction, Incident, Problem, or Known Error record, follow these steps:

1. While working with the record, click **More** or the **More Actions** icon to select **Create Knowledge**.
2. If a document type selection form opens, select a document type for the document.
3. As required, view and update the fields the system has completed with the mapped information from the record you selected during your search. The data inserted in these fields depends on the document type and the defined field mappings.
4. Enter an expiration date for the document.
5. Select at least one category for the document.
6. Use the rich-text editor to enter or edit text for the document.
7. Click the **Preview** button to view the document.
8. Click the appropriate button to submit the document for approval or to approve the document immediately.

1.12.3.1.8. Notify service subscribers of the publication of a document

User roles: KCS I, KCS II, KCS III, KM Admin, System Administrator

The Knowledge Management module provides an option Notify Service Subscribers on Publish that enables the system to immediately notify users upon the publication of a document that's important to them. Each knowledge document can be associated to a service so that the system can immediately notify users who have subscribed to the associated service upon the publication of the document.

To enable Notify Service Subscribers on Publish for a knowledge document, follow these steps:

1. Create a new knowledge document or edit an existing knowledge record that you want the system to notify the associated service subscribers upon document publication.
2. In the **Associated Services** field, select a service that's associated with this document.

   The **Notify Service Subscribers on Publish** option is then displayed.

   Tip

   You can click the **View Subscribers** icon to the right of this option to view a list of subscriptions of this service. You can further click each subscription record to view contact information about the users who have subscribed to this service.
3. Select the **Notify Service Subscribers on Publish** option.
4. Submit the document for approval or directly publish it.

   Once the document is published, the system immediately sends a notification to all users who have subscribed to the associated service. For more information about service subscriptions, see Subscriptions.

Note

For a document with Notify Service Subscribers on Publish enabled, the system also sends a notification to the associated service subscribers if any updates to that document are published.

Related topics

Subscriptions
Create a knowledge document

1.12.3.1.9. Edit a document in place

User roles: KCS II, KCS III, System Administrator

Editing a document in place is an option that allows you to edit published documents and then re-publish the document immediately for either internal or external use without sending the document to workflow. This allows you to update documents quickly that for some reason need to be updated and made available immediately.

The edit in place option is available for published documents only. It doesn't apply to knowledge candidates, such as working copies of a document or draft documents.

Caution

Knowledge document attachments that are updated using the edit in place option will result in a broken attachment link and a broken knowledge document link. When a knowledge document attachment needs updates, then the document must be put into the workflow so that the editor can save the attachment to the editor's desktop, make the necessary updates, save the changes, and then re-upload the attachment to the knowledge document.

In the out-of-box system, the permission for edit in place is set for KCS II, KCS III and KM ADMIN Knowledge Management profiles. A user with a KM ADMIN profile manages assigning the edit in place permission to KM profiles.

You contribute a knowledge document by authoring knowledge articles or using external documents that are uploaded into a knowledgebase. You can use the rich-text editor to create or edit documents, and you can add documents as attachments that can include images, text files, Word files, or PDFs.

To edit a document in place, follow these steps:

1. Click **Knowledge Management** > **Published Documents**, or use Search to find the published document.
2. Click the document in the list that you want to edit.
3. Click **Edit**.
4. Make the changes to the document in the editable view of the document.
5. Add needed attachments by doing one of the following:

- For Web-tier clients: click **Add File**, type the full path of the file in the **File location** text field or browse to the file, and then click **OK**.
- For Windows clients: right-click in the blank area of the attachments list, click **Add**, browse to the file, and then click **Open**.

> **Note**
>
> If you are editing an attachment for an existing knowledge document, you need to save the attachment to your desktop, make the necessary updates, save the changes, and then re-upload the attachment to the knowledge document.

6. Continue creating or editing your document
7. (Optional) Specify an expiration date for the document.
8. Update the category for the document, if required.
9. Use the rich-text editor to enter text for the document.
10. Click the appropriate button to publish the document immediately for the document category or click **Cancel** to revert the document to the published version.

1.12.3.1.10. Add feedback to a knowledge document

User roles: All users

A default Knowledge Management security profile allows users to search and view externally published knowledge documents for those documents in categories to which the profile has access. Users can also submit feedback on these documents.

To add feedback by using Published Documents, follow these steps:

1. Click **Knowledge Management** > **Published Documents**.
2. Select the document from the document list.
3. Click **Add Feedback**.
4. Type your feedback in the text box.
5. Select **Useful** or **Not Useful**.
6. Click **OK**.

To add feedback by using the Pending Documents queue:

1. Click **Knowledge Management** > **Pending Documents**.
2. Select the document record in the queue.
3. Select the **Feedback** tab.
4. Click **Add**.
5. Type your feedback in the text box.
6. Select **Useful** or **Not Useful**.
7. Click **OK**.

1.12.3.1.11. Submit a document for revision

User roles: KCS II, KCS III

Once the contributor has submitted a document for approval, if needed you can send it for further revision.

To submit a document for revision, follow these steps:

1. Open the document from **Knowledge Management** > **Pending Documents**.
2. Click **Submit for Revision**.

   The document is submitted for further revision and its status changes to **revise**. You or somebody else can further revise the document and then submit it for review.

1.12.3.1.12. Submit a document for review

User roles: KCS II, KCS III

Once you have made revisions to a document in **revise** status, you can submit the document for review.

To submit a document for review, follow these steps:

1. From **Knowledge Management** > **Pending Documents**, open a document in **revise** status.
2. Click **Submit for Review**.

   The document is submitted for review and its status changes to **review**. The document will appear in the **Documents for Review** queue. The reviewer can then accept the document or if needed return it for revision.

1.12.3.1.13. Accept a document

User roles: KCS II, KCS III

Once you have completed your review of a document and believe all necessary revisions have been made to the document, you can accept the document.

To accept a document, follow these steps:

1. Open the document from **Knowledge Management** > **Documents for Review**.
2. Select the knowledge document that you want to accept.
3. Click **Accept Document**.

   The document's status changes to **conclude**.

   You can still return the document for revision or review by clicking the **Return for Revision** or **Return to Review** button, or you can then directly publish it internally or externally.
4. Make sure you have the legal rights to publish the document, and then select the **Confirmed legal rights to publish this material** checkbox.
5. To publish the document, click **Publish Internal** or **Publish External**.

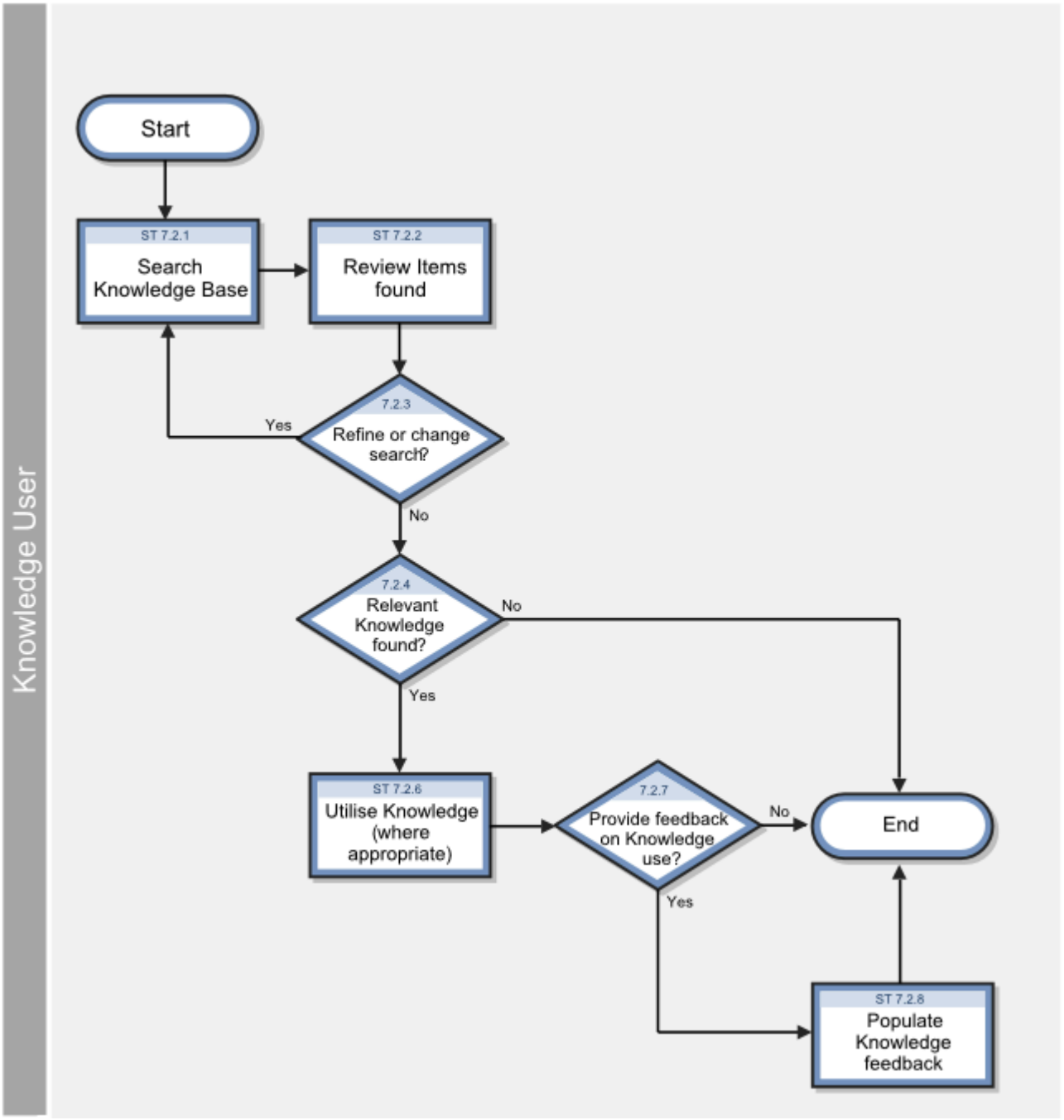
Related topics

Knowledge Management profiles
Knowledge groups
Managing document categories
Submit a document for approval
Submit a document for revision
Submit a document for review

1.12.3.2. Search, view and utilize knowledge document (ST 7.2)

The Knowledge Base is the central repository of the Knowledge Management system. Service Manager provides powerful search and retrieval functionality for accessing, retrieving, and displaying knowledge documents. Searches can be performed using filters, and even Boolean searches are allowed.

Each type of knowledgebases has different fields that are indexed for searching, so specific search parameters that match the fields in the knowledgebase must be provided. For example, the knowledge articles have a title and author field. When you view an Incident record, the out-of-box system displays the Incident number, description, and solution for closed incident records.



Related topics

Knowledge Management overview
Knowledge Management workflows and tasks
Processes and Best Practices

1.12.3.2.1. Search for published documents

User roles: All users

Published knowledge documents are documents that are available for searching and viewing. Some documents may only be viewed by internal users, while others can be viewed by internal and external users. Published documents can be edited by users with applicable privileges. Users can also provide feedback to published documents.

To search for published knowledge documents, follow these steps:

1. Click **Knowledge Management** > **Published Documents**.
2. Type as much information available in the search form. For example, you can type a portion of the title of the document or a phrase from the summary of the document in the appropriate field.
3. Click **Search**.
4. Select and click the document in the list to view the document.
5. Depending upon your profile, you can edit the document, view and update the adaptive learning data, or retire the document.

1.12.3.2.2. Search for adaptive learning data

User roles: KM Admin, System Administrator

When adaptive learning is enabled, data about the queries that returned documents that were used for solving issues is stored and integrated into the documents when they are indexed.

Note

You must have a KM ADMIN profile to use the Adaptive Learning search function on the Knowledge Management Administration menu.

To search for adaptive learning data, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **Adaptive Learning**.
2. Type the document ID if you are looking for the adaptive learning data for a specific document.
3. Select the knowledgebase if you are looking for the adaptive learning data for a specific knowledgebase.
4. Type the phrase if you are looking for the adaptive learning data for a specific phrase.
5. Click **Search**.

Note

You can provide the search parameters for adaptive learning in various combinations to narrow or enlarge your search for adaptive data.

1.12.3.2.3. Search for a solution

User roles: All users

While creating or updating an Interaction, Incident, or Problem record, you can search for a solution to assist in solving your problem.

To search for a solution, follow these steps:

1. While creating or updating a record, click **More** or the **More Actions** icon, and then select **Search Knowledge**.

   The search uses the mapped fields to display a search results list of possible similar records or other knowledge documents.

2. Click the hyperlink to view a particular document.
3. Use the results from your search to determine how to proceed. For example, you can view another record from the search results or use the data from the search to update the record you are working on.

1.12.3.2.4. Use a solution from a knowledgebase search

At any time while you create or update an Interaction or Problem record or update an Incident record, you can perform a knowledgebase search to find solutions. When you find a solution, you can add the solution to the resolution or workaround field of the record you are processing.

1. While you are processing a record, perform a knowledgebase search to find a solution.
2. In the knowledgebase search results, click the hyperlink to view a solution document.
3. When you find a document with the correct solution, click **Use Solution** to add the solution to the resolution or workaround field of the record. The **Use Solution** button only appears if the selected document contains a resolution or workaround.

   Note

   If the knowledge document is a working copy of an approved published document, the display option, **Show Approved Version**, appears, which enables you to view the published document and then select it as a solution.

4. After you select **Use Solution**, the system redisplays the record with the resolution or workaround field updated. You can continue processing the Interaction, Incident, or Problem record that prompted your knowledgebase search.

1.12.3.2.5. View adaptive learning data

User roles: KCSII, KCS III, KM Admin, System Administrator

When adaptive learning is enabled, data about the queries that returned documents that were used for solving issues is stored and integrated into the documents when they are indexed.

Note

You must have a KCS II or KCS III profile to view adaptive learning data, and a KM ADMIN profile to view or edit adaptive learning data.

To view adaptive learning data for published documents by using the **Published Documents** list, follow these steps:

1. Click **Knowledge Management** > **Published Documents**.
2. Select a document to view.
3. Click **Adaptive Learning**.
4. You can now view or edit the adaptive learning data for the document you selected, depending upon your profile.

1.12.3.2.6. Print a knowledge document when using the Windows client

User roles: All users

You can search knowledgebases to determine if your current issue matches any existing incidents or problems with known solutions or workarounds. When located, you can print out those documents.

To print a knowledge document using the Windows client, follow these steps:

1. Display the document you want to print in the Knowledge Management Document Viewer.
2. Click **File** > **Print** or **Print Preview**.
3. Click **Detail**.
4. Click **OK**.
5. Select a printer and then click **Print**.
6. If you selected **Print Preview**, you can print the document from the **Print Preview** window. Click the **Print** icon, select a printer, and then click **Print**.

1.12.3.2.7. Print a knowledge document when using the web client

User roles: All users

You can search knowledgebases to determine if your current issue matches any existing incidents or problems with known solutions or workarounds. When located, you can print out those documents.

To print a knowledge document using the web client, follow these steps:

1. Display the document you want to print in the Knowledge Management Document Viewer.
2. Click the **Print Page** icon at the top right corner.
3. Select **Detail View**.

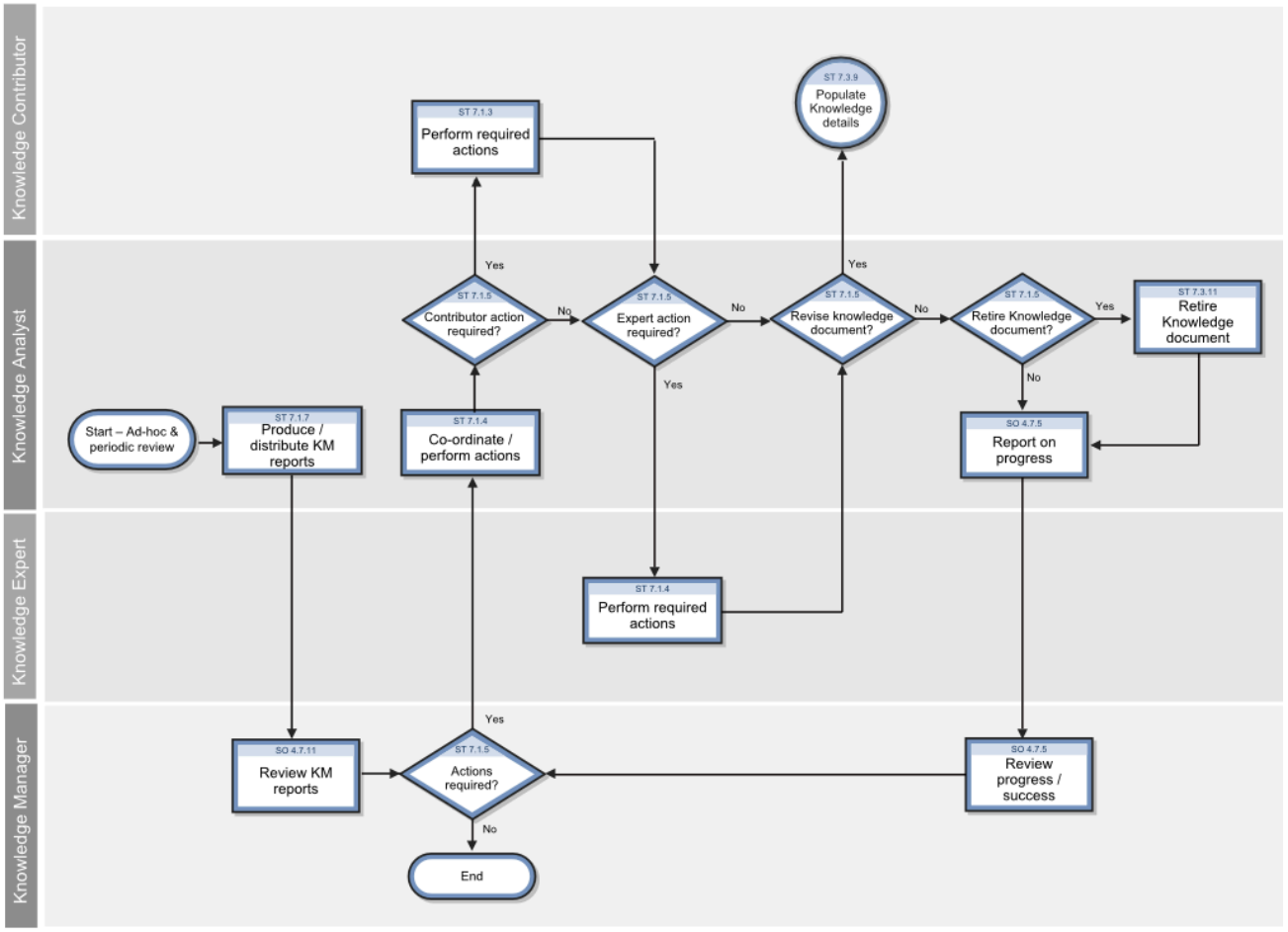
   The document opens in a page.
4. Right-click the document page, and select **Print**.
5. Select a printer, and then click **Print**.

1.12.3.3. Maintain and Continually Improve Knowledge (ST 7.3)

The Maintain and Continually Improve Knowledge process involves the following tasks:

- Records the details of new knowledge (documents)
- Updates existing knowledge (documents) where there is any inaccuracy or incompleteness
- Removes obsolete knowledge (documents)

The process is performed by the Knowledge Contributor, Knowledge Analyst, Knowledge Expert, or Knowledge Manager.



Retire a knowledge document

User roles: KCSII, KCS III, KM Admin, System Administrator

Before you can retire a document, the system verifies that there are no published documents that link to the document you want to retire. If the system finds documents with links to the document you want to retire, it alerts you to this situation and gives you the option of putting these documents into the document workflow to review and update.

To retire a knowledge document, follow these steps:

1. Before retiring a document, be sure to update any documents that link to the document you wish to retire.
2. Click **Knowledge Management** > **Published Documents**.
3. Click **Search**.
4. Select the document that you wish to retire from the list of documents.
5. Click **Retire**.

Retire a knowledge document when other documents link to it

User roles: KCSII, KCS III, KM Admin, System Administrator

Before you can retire a document, the system verifies that there are no published documents that link to the document you want to retire. If the system finds documents with links to the document you want to retire, it alerts you to this situation and gives you the option of putting these documents into the document workflow to review and update.

Note

The alert message includes the document IDs of the documents that need to be updated and placed in the **Pending Documents** queue. If you click **OK**, it then places those documents in the **Pending Documents** queue.

To retire a knowledge document when other documents link to it, follow these steps:

1. Review the **Pending Documents** queue and select the documents for editing. You can identify the documents by the document ID, which appears in the title column of the list of document changes in the **Pending Documents** queue. These documents also have a standard feedback message: "This document links to another document (KMnnn) that was attempted to be retired. Please review now." added by the system.

2. From the **Pending Documents** queue, you can edit each document to remove any links to documents you want to retire, and then publish the edited document.
3. Return to the **Published Documents** list and select the document you want to retire.
4. Click **Retire**.

Unretire a knowledge document

User roles: KCSII, KCS III, KM Admin, System Administrator

A user with a KM ADMIN profile can view and unretire a document that doesn't have a category associated with it. If the KM ADMIN unretires a retired document without a category, the document is put in the workflow as a draft document.

To unretire a knowledge document, follow these steps:

1. Click **Knowledge Management** > **Retired Documents**.
2. Select the document from the list of documents.
3. Click **Unretire**.

Contents

- 1 Retire a knowledge document
- 2 Retire a knowledge document when other documents link to it
- 3 Unretire a knowledge document

1.12.3.4. Access Knowledge Management reports

Applies to User roles: KM Admin

The Reporting tool in Service Manager provides a number of out-of-the-box reports on the knowledge data in your system. You can view these reports through dashboards named Knowledge Overview (Global), Knowledge Admin (Global) and Knowledge KPI (Global). You can also create your own dashboards to display other reports of your interest.

To access Knowledge Management reports, follow these steps:

1. Click **Knowledge Management** > **Knowledge Overview** to display the **Knowledge Overview (Global)** dashboard.

   Click **Knowledge Management** > **Knowledge Admin** to display the **Knowledge Admin (Global)** dashboard.

   Select **Knowledge KPI (Global)** from the dashboard drop-down list on the toolbar to display the **Knowledge KPI (Global)** dashboard.

2. Modify the dashboard as needed:

   • To add content to the dashboard, click **Add Content** on the toolbar.

   • To remove a report from a dashboard, click **X** on the report window.

   • To change the position of a report, drag the report window and drop it to a new position.

   • To change the size of a report, drag the border or corner of the report window.

3. Add your own dashboards as needed. For details, see Create a dashboard.

   Your custom dashboards are added to the dashboard list on the toolbar of the dashboard page.

Tip

You can click **Export** to export the reports on a dashboard to PDF format.

4. Click the **Open dashboard settings** icon on a dashboard to set its properties, or click the **Set as Default Dashboard** button to set it as your default one.

1.12.4. Knowledge Management administration

Knowledge Management administration provides Knowledge Management Administrators (KM Admin) with the capability to manage document categories, document groups, and user profiles so that users have access to the documents they need to view and maintain while at the same time limiting access to only those documents that are appropriate for particular groups of users and categories of documents. Knowledge Management also provides KM Admins with document types and document views to manage the amount and type of data displayed in knowledge documents. The out-of-box system provides the standard document types (error message/cause, external, question/answer, problem/solution, and reference).

Knowledge Management administration consists of the following:

- Environment
- Knowledge groups
- Profiles

Rights to the Knowledge Management administration functions are controlled by the Knowledge Administration security area. To perform these functions, users must belong to a security role that has rights in this area.

The primary function of Knowledge Management administration is to create a set of document categories, groups, types, and views that allow users to access all the documents they need to do their jobs and prevent users from accessing documents that contain sensitive information that they don't need to access.

Knowledge Management administrators can also use the security profiles to provide users different capabilities in the knowledge document workflow. You can use the security profile to set which users can edit, approve, and author knowledge documents.

Related topics

Managing document categories
Knowledge groups
Knowledge Management profiles
Managing document views
Knowledge Management user roles

1.12.4.1. Document types and document views

Document types and document views specify how Knowledge Management displays a knowledge document. Each document type can have several document views so that not all of the information in a document is accessible to all users. Each document view has one or more document groups associated with it. In this way, document views combined with document groups manage what information appears in a particular document view and what users have access to that view.

Document types are templates for documents in Knowledge Management. Knowledge Management includes several default document types. These include **Question/Answer**, **Problem-Solution**, **Reference**, **Error Message/Cause**, and **External**. These default document types provide the basic structure for new document types that you can create. If you later delete your customized document types, the default document types retain their basic features.

For example, you can use Forms Designer to create a document type called **Payroll Error** based on the out-of-box document type, **Error Message/Cause**. When there is an error in processing payroll, the staff uses this **Payroll Error** form to write up the problem.

- Change the **Error Message** tab to **Error** and the error message field to **Error**. Users describe the error in this field.
- Change the **Cause** tab to **Problem** and the cause field to **Problem**. Users describe the underlying problem that caused the error in this field.
- Change the **Workaround / Fix** tab to **Solution** and the fix field to **Solution**. Users provide a solution to the problem.
- Use the default document view for **Error Message/Cause** to create at least one document view for this document type.

Document views have two purposes. Document views provide users a way to customize how the information displays in each form and to limit users to see only information that their group permission allows them to see. This way, users with different permissions view the same document records, but only those users with certain permissions can view sensitive information in certain fields of the record.

You can manage the documents in your knowledgebase by effectively using document types and groups. You can add new views to existing document types and then control the access to these views by selecting only certain document groups.

You can also add new document types to the standard document types provided by the system.

The tasks for managing document types and views are:

- Add a new document type
- Add a view for an existing document type
- Add a document group to a view

Tip

You can add document types and document views for users by copying the out-of-box templates for document types and views.

To add document types, you must log in with a user profile that has either the System Administrator or Knowledge Administrator role.

Adding a rich-text field

If you add a new rich-text field to the kmdocument table, add it to the dbdict and to the document views that will display the new field.

In order for three of the buttons on the rich-text editor for this new field to work properly, also create three `displayoption` records for the kmdocument.open displayscreen and three for the kmdocument.view displayscreen.

- Insert Image
- Insert Link to Attachment
- Insert Link to Record

For more information, see the existing displayoption records in the kmdocument.open displayscreen. For example, 555, 556, and 557. Note that:

- the Condition is

  ```
  $L.preview="false"
  ```

- on the **Rad** tab and **Pre Rad Expressions** tab in all three displayoption records, the field name is set as follows:

  ```
  $kminsertlinkfieldname="error"
  ```

Also,if your new field name was **comments**, for example, then you would create the three displayoption records and set the following on the **Rad** tab and **Pre Rad Expressions** tab in the three new displayoption records:

```
$kminsertlinkfieldname="comments"
```

1.12.4.1.1. Managing document views

A document view determines how the document displays to a user. Document views are associated with document types. Each document type has a default document view, and each document type must have at least one document view associated with it.

Users with a KM ADM profile can add, delete, and modify document views. Use the document view editor to add a new view or modify an existing document view. When you add a new document view or modify an existing view, you should select the user profiles that you want to have access to the document view.

With good planning in combination with well-defined user profiles and document views, you can manage access to knowledge documents so that users are able to view the knowledge documents or portions of the knowledge documents they need to perform their job. At the same time you can maintain control over which users are able to access knowledge documents or sensitive information in a knowledge document.

Typically, you create a new document view by modifying an existing document view and then assigning it a new name when you save the new document view.

Rich text fields (RTFs) in document views

A rich text field is a field that has content that includes formatting properties such as font type and size. When you modify a document view, you can use the rich text editor to add rich text fields to the document view or add special sections in the document. For example, the default view contains a header section and a details section that's separated by a solid line. You can add labels to fields and use rich text formatting to distinguish the label from the contents of the field. When the system displays a document in a particular document view, it applies the formatting you specified for each rich text field. In addition, document views also include a marker (*SCKM*) that a document parser uses when rendering a document to replace the field name marked by *SCKM* with data from the file for the specified field.

Security level in document views

The security level of a document view is used to determine which document view the system displays when a user is a member of several knowledge groups. The system uses the document view that has the highest security level for the particular user.

A group permission is formed by the association of a document view with a knowledge group. Each document view has a security level and one or more groups associated with it. Zero (0) is the lowest security level. Each document view for a document type must have a unique security level. You should assign a higher number to the security level for those document views with higher security requirements. Assign security level values relative to the other security level values of the other document views for a document type.

Create a new document view

User role: System Administrator, KMAdmin

A document view determines how the document displays to a user. Document views are associated with document types. Each document type has a default document view, and each document type must have at least one document view associated with it.

Note

In some cases, before creating a new document view you will need to add a field or fields to the database and use Forms Designer to create new forms or to update existing forms. Before beginning these procedures, you should be familiar with the information in the Database Administration and Tailoring online help modules.

To create a new document view, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **Document Types**.
2. Click the document type for the document view you want to create.
3. Click a document view from the list of **Associated Document Views** that's similar to the document view you want to create.
4. Click **Modify**.
5. Click **Source** in the editor to view the HTML source.
6. Right-click and then click **Select All**.
7. Right-click and then select **Copy**.
8. Click **Cancel**.
9. Click **New** for the document views of the document type for which you want to create a new view.
10. Click the **Source** button for the editor.
11. Paste the old document view into the editor.
12. Click the **Source** button to see the view as it will display.

13. Type a name for your new view in View name.

    For example, if you added a field called **notes** to the **kmdocument** table in dbdict and added a **Notes** tab to the **Error Message Cause** form (kmdocument.errormsg), you can copy and paste the last two lines in the document view:

    ```
    <div class="formSectionHeading">Workaround/Fix</div>
    *SCKM*fix*SCKM*
    ```

    to make a new section Heading "Notes" that contains the new "Notes" field:

    ```
    <div class="formSectionHeading">Notes</div>
    *SCKM*notes*SCKM*
    ```

14. Modify the security level for this document view.
15. To insert a field in the view, position your cursor in the rich text editor window at the location where you want the field to appear on the form or select the field you are replacing and click the **Insert a Field Reference** icon.
16. Select a new field in the **Insert a Field Reference** window.
17. Click **OK**.
18. You can use the formatting features of the editing tool to format the appearance of the field label and its position in the form.
19. Repeat adding or modifying the fields on the document view as required.
20. Click the **Groups** tab and select the document groups that can use this document view.
21. Click **OK**.
22. Click **Save**.
23. Click **OK**.

Add a knowledge group to a document view

User role: System Administrator, KMAdmin

Knowledge groups enable you to collect users into groups that have access to work with the same set of documents. A knowledge group might be the user for a department, a group of document authors, or a special group of users within your organization, such as subject matter experts.

To add a knowledge group to a document view, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **Document Types**.
2. Click the document type for the document view.
3. Select the document view from the list of **Associated Document Views**.
4. Click **Modify**.
5. Click the **Groups** tab.
6. Click in the next blank line of the **Specify the knowledge user groups that can access this view** section.
7. Type the name of the knowledge group to add to this view.
8. Click **OK**.
9. Click **Save**.
10. Click **OK**.

Create labels for multilingual document views

Applies to User roles: System Administrator

In addition to creating new document views, administrators can configure the labels for the document view to display in languages other than English. The labels contain heading delimiters which the systems uses to determine the label title. A label delimiter is in the form: *SCMSG*123*SCMSG* where the number between the replacement tokens identifies the message string in knowledge management **scmessage** table. When the document displays, the system replaces the message number token (123) and the delimiters with the appropriate string from the s**cmessage** table.

A document view must be multilingual enabled before you can create multilingual labels. A document view is multilingual enabled when you place the delimiter *SCMSG*123*SCMSG* in the label of the document view form using the document view editor. The default view is multilingual capable for all document types.

To configure the document views to display in languages other than English, you need to add additional message strings in the **scmessage** table. In doing this you provide the class km message identifier (a unique message number), the actual message string you want to display and a language code. For each language, create a new record with the same message number but the language code will be different and the message text will be the same word(s) in the appropriate language. Repeat the process for each label that needs to be multilingual. Ensure each set that represents a single label value has a unique message identifier.

Once new records exist in the **scmessage** table, replace the tokens in the label heading delimiter with the message number created in the **scmessage** table for the

appropriate message text string.

> **Note**
>
> The default view for all out-of-box document types is multilingual capable.

This procedure is for a document view that's not multilingual enabled. Before you begin, make sure that you create a message number ID for each multilingual label on the document view form and that there are message numbers defined to display the labels in multiple languages.

To create labels for multilingual document views, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **Document Types**.
2. Select a document type record from the list.
3. Select the document view from the View List.
4. Click **Modify**.
5. Select the item to update and then select the label delimiter (*SCKM*Title*SCKM*) and replace it with the label delimiter (*SCMSG*123*SCMSG*) for the label you are making multilingual. The 123 should be the message number ID from the **scmessage** table for the unique message number ID for this label.
6. Continue updating the other labels with delimiters for multilingual labels.
7. Click **OK** to save your changes.

Contents

- 1 Rich text fields (RTFs) in document views
- 2 Security level in document views
- 3 Create a new document view
- 4 Add a knowledge group to a document view
- 5 Create labels for multilingual document views

1.12.4.1.2. Document type management

You can use the **Document Types** menu option to add new document types in addition to the ones provided by the out-of-box system. The process for adding new document types is complex and requires a thorough knowledge of Forms Designer. When you add a new document type, you must provide a document creation form and a document edit form for the new document type. Before you add a new document type, create the document creation form and the document edit form to use with the new document type.

You must also create at least one document view for the new document type. You can do this after you have created the new document type, but the document type will not be usable until you create at least one document view for the document type.

You also need to customize the **KMCustom** JavaScript if a new document type is added or a new field is added for a document type. Its content may resemble the following.

```
 /**
* This method is used to return a json object {'<document type'>: [<'field_1',
'feidl_2'>]} for links.
* Related type and fields should be appended to this json object if a new
document type is added or a new field is added for a document type.
**/
function getKmFieldsForLinks() {
   return {
      "probsol": ["problem", "solution"],
      "howto": ["question", "answer"],
      "errormsg": ["error", "cause", "fix"],
      "reference": ["reference", "SP.test"],
      "Test": ["question", "answer", "SP.test"],
      "Reference2": ["reference", "SP.test"]
   };
}
```

1.12.4.1.3. Managing shared content

In Knowledge Management, shared content can be a file that's shared by many document views. This type of shared content is referred to as system-managed shared content. Examples of this type of shared content are an image file used as a logo on the document view or a text file such as a copyright notice or license agreement.

The Manage Shared Content function helps you to manage your system-managed shared content by providing a search function, an add function, delete function, and a save function. Shared content can't be updated. If you need to modify a file being used as shared content, you should rename the file, make the changes and then add it as a new file. You can then delete the original file.

All shared content is assigned an ID by the system. You can assign it a name and description when you are adding it to the system. Each shared content item must have a unique ID, which means that you can only add one item of shared content for each ID.

Add new shared content

User roles: KM Admin, System Administrator

In Knowledge Management, shared content can be a file that's shared by many document views. This type of content is referred to as system-managed shared content. Examples of this type of shared content are an image file used as a logo on the document view or a text file, such as a copyright notice or license agreement.

To add new shared content, follow these steps:

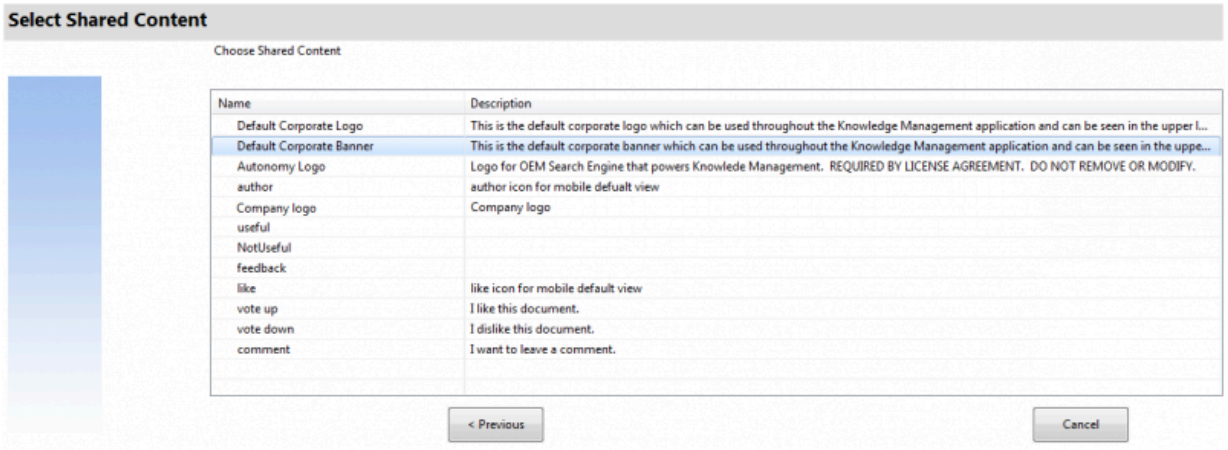
1. Click **Knowledge Management** > **Configuration** > **Shared Content**.
2. Click **New** to create the new shared content.
3. Type a **Name** and **Description** for the shared content.
4. Specify an attachment by doing one of the following:
    - For Web Tier clients: Click **Specify File**, type the full path of the file in the **File location** text field or browse to the file, and then click **OK**.
    - For Windows clients: Right-click in the **Select a single attachment below to add to system-managed shared content** field, click **Add**, browse to the file, and then click **Open**.
5. Click **Save**.
6. Click **OK**.

Example: Using the shared content

User roles: KM Admin, System Administrator

This section provides an example for using the shared content. Follow these steps:

1. Sign in to Service Manager by using the Windows client.
2. Click **Knowledge Management** > **Configuration** > **Document Types**.
3. Select a document type, and then select a view in the Associated Document Views list. For example, Default View.
4. Click **Modify**.
5. Click the **Insert an Image** button to open the Select Content Attributes page.
6. Click **Next**. The system opens the Select Content Attributes page where you can see a list of shared content.



7. Double-click a shared content to insert it in the Document View.
8. Click **OK**.

1.12.4.2. Knowledge groups

Knowledge groups enable you to collect users into groups that have access to work with the same set of documents. A knowledge group might be the user for a department, or a group of document authors, or a special group of users within your organization such as subject matter experts. All knowledge groups have a unique name, an assigned manager, and a group of operators.

You associate a knowledge group with at least one document category or subcategory to give the members of a knowledge group access to documents in the category or subcategory and all subcategories below it. Access privileges can range from search and view to publishing privileges depending upon what knowledge profile(s) you assign to the category. You assign a knowledge group access to a document category by associating a knowledge profile and its associated knowledge group with a document category.

You should assign a new user to the appropriate knowledge groups when you create an operator record for the user. You can use the **Find Related Information** icon on the **Login Profiles** tab of the Operator record form under Knowledge Groups to display a list of defined knowledge groups.

Default knowledge view group

All users are automatically members of the **Default Knowledge View Group** if the Knowledge Management environment file option, **Assign the Default Knowledge View Group to all operators**, is checked. This option is checked for the out of box system.

The **Default Knowledge View Group** is a convenient way to associate all the users in the system with a uniform set of permissions on a document category. For example, if you wanted all the externally published documents under the Human Resources category to be viewable by all users, you could associate the **Default Knowledge View Group** and the **DEFAULT** profile with that category.

In the out-of-box system, each document type has a default document view and this default document view has the **Default Knowledge View Group** assigned to it. This means that if a user has access to the category the document is in, that user can view the document with the default view if there are no other document views available to that user to view the document. However, because the default view has the lowest security level, if another view is available to the user then the higher security view is the view the system uses. By associating the **Default Knowledge View Group** with

each of the document types, there can never be a case where there is a document in that system that can't be viewed.

The **Default Knowledge View Group** has a unique ID in the system, so that if the name is changed, the capabilities of the group persists. This means that if you choose to rename this group, the capabilities of the group don't change. All of the users in the system can still view all of the documents in those categories to which this group has access.

Access knowledge group information

User roles: KM Admin or System Administrator

Knowledge groups enable you to collect users into groups that have access to work with the same set of documents. A knowledge group might be the user for a department, a group of document authors, or a special group of users within your organization, such as subject matter experts.

To view the information of a knowledge group, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Knowledge Groups**.
2. Click **Search**.
3. Click the name of the knowledge group.

Add a user to a knowledge group

User roles: KM Admin, System Administrator

Only a user with a KM ADMIN profile or the System Administrator can add users to knowledge groups. There are two methods for adding a user to a knowledge group.

- In the operator record, on the **Login Profiles** tab there is a **Knowledge Group** section for the system administrator to specify the knowledge groups for the user.
- The **Knowledge Management** > **Administration** > **Knowledge Groups** function allows a user with a KM ADMIN profile to add users to a knowledge group.

To add a user to a knowledge group in the operator record, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.
2. Click **Search**.
3. Select an operator.
4. Click the **Login Profiles** tab.
5. In the Knowledge Groups section use the Fill icon to display a list of the knowledge groups.

6. Select the appropriate knowledge groups.
7. Click **Save**.
8. Click **OK**.

To use the Knowledge Group function to add a user to a knowledge group, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Knowledge Groups**.
2. Click **Search** to display a list of knowledge groups.
3. Click the knowledge group.
4. Type the name of the user in the Operators section, or use the Find or Fill button to locate the user.
5. Click **Save**.
6. Click **OK**.
7. Repeat steps 2 to 6 until you have added the user to all of the appropriate knowledge groups.

Add a knowledge group

User role: System Administrator, KM Admin

Knowledge groups enable you to collect users into groups that have access to work with the same set of documents. A knowledge group might be the user for a department, a group of document authors, or a special group of users within your organization, such as subject matter experts.

To create a new knowledge group, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Knowledge Groups**.
2. Type the name of the new knowledge group in **Knowledge Group**.
3. Type the name of the manager of the knowledge group in **Manager Name**.
4. Type the names of the members of the knowledge group in the Operators area.
5. Click **Add**.

To copy and rename an existing knowledge group, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Knowledge Groups**
2. Click **Search**.
3. Select a knowledge group from the list.
4. Click **Copy/Rename**.
5. Type a new name for the knowledge group in **Knowledge Group**.
6. Select **Copy** or **Rename**.
7. Click **Finish**.

8. Type a new name of the manager of the knowledge group or keep the current manager.
9. Type the names of the members of the knowledge group or keep the current members in the Operators area.
10. Click **Save** and the **OK**.

Assign a knowledge group access to a document category

User role: System Administrator, KMAdmin

All documents in the knowledgebase are assigned to a document category. Categories for knowledge documents are ordered hierarchies. The document categories order documents into top-level categories and subcategories. You can assign a knowledge group access to a specific category.

To assign a knowledge group access to a document category, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **KM Document Categories**.
2. When you have a large number of top-level categories, use the category filter to display only those top-level categories you are interested in viewing. Type the name of the top-level category in the Category filter text box.
3. Use the **Category Search** tab to search a list of all occurrences of a category or subcategory name. You can type the complete name of the category or subcategory you are searching for, or you can type a portion of the name in the Search box. For example, if you are searching for all test categories, you might type tes to see **test**, **test1**, and **testall**.
4. Select the category or subcategory in the list and click **Details**.
5. Click the **Permissions** tab.
6. Click the **Add** button.
7. Select the user profile for the knowledge group to which you want to assign access to the category you selected.
8. Double-click the user profile you selected.
9. Select the knowledge group you want to assign to the document category.
10. Double-click the knowledge group you selected.
11. Click the **Permissions** tab to verify your selections.

Determine which knowledge groups can contribute documents to a category

User role: System Administrator, KMAdmin

To view which knowledge groups have access to a document category, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **KM Document Categories**.
2. When you have a large number of top-level categories, use the category filter to display only those top-level categories you are interested in viewing. Type the name of the top-level category in the Category filter text box.
3. Use the **Category Search** tab to search a list of all occurrences of a category or subcategory name. You can type the complete name of the category or subcategory you are searching for, or you can type a portion of the name in the Search text box. For example, if you were searching for all test categories you might type tes to see **test**, **test1**, and **testall**.
4. Select the category or subcategory in the list and click **Details**.
5. Click the **Permissions** tab.
6. View the profiles and knowledge groups for the category.

Assign rights for publishing a document for a document category

Applies to User roles: System Administrator

To assign rights for publishing a document for a document category, follow these steps:

1. Identify the document category or categories to which you want to assign the document.
2. Identify the knowledge management profile and document group that have publishing privileges for the category or categories.
3. Make sure that the users who will have the document publishing rights are members of the appropriate document group.

For example, to give a user the ability to publish documents for external users in **HR** category for the out-of-box system, you would make the user a member of the HR KCS III document group.

1.12.4.3. Managing document categories

All knowledge documents in the knowledgebase are assigned to a document category. Categories for knowledge documents are ordered into hierarchies. The document categories order documents into top-level categories and subcategories. Each subcategory has a parent category that's the category immediately above it in the ordered list of categories. A top-level category is also a parent category of the subcategory immediately below it in the ordered list of categories.

The following is an example of category and subcategory hierarchy:

Animal > Dog > Retriever or Category > subcategory > subcategory

- Animal is the top-level category for Dog and Retriever.
- Animal has no top-level category or parent category.
- Dog is the parent category for Retriever.

For a top-level category, the level in the hierarchy is always zero (0).

A category and subcategory has associated with it a set of permissions. These permissions specify the profiles and knowledge groups that have access to the category or subcategory. When a profile and knowledge group is given access to a category or subcategory that profile and knowledge group also has access to all of the subcategories below that point in the hierarchy.

If there are any published documents assigned to the category, they can't be deleted. You can't delete a document category before deleting a category. Assign any of the published documents to another document category. You can delete a document category if there are only retired documents assigned to the category. If you decide to unretire those documents, a KM Admin will provide a new document category for the document as part of the unretire process.

Deleting document categories

You can delete a document category, but you should verify first that there are no documents in the category or in the subcategories under the category. When you delete a category, the system will not allow you to delete the category if there are documents in the category or one of its subcategories. However, if the documents are retired, the system will warn you that there are documents present in the category or its subcategories, but it will still allow you to delete the category. The category being

deleted is removed from the retired documents record so that the document is no longer associated with the deleted category.

Documents can't be managed in the knowledge management system without an assigned category because the Knowledge Management uses document categories to control access to the knowledge documents. All published documents and pending documents must have a category associated with them. Draft and retired documents don't require an assigned category.

A user with a KM ADMIN profile can view and unretire a document that doesn't have a category associated with it. If the KM ADMIN unretires a retired document without a category, the document is put in the workflow as a draft document.

When you unretire a document that has at least one category associated with it, the system puts the document into the workflow as a pending document.

Add subcategories to a document category

User role: System Administrator, Knowledge Administrator

All documents in the knowledgebase are assigned to a document category. The document categories order documents into top-level categories and subcategories. Each subcategory has a parent category that's the category immediately above it in the ordered list of categories.

To add subcategories to a document category, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **KM Document Categories**.
2. Select a top-level category.
3. On the **Subcategories** tab, click **New**.
4. Type the name of the new subcategory.
5. Click **Add**.
6. You can continue adding subcategories by clicking **New Subcategory**, or you can begin to specify permissions for the subcategories.
7. Click **Save** to continue saving your changes.
8. Click **OK**.

Add a new top-level document category

User role: System Administrator, Knowledge Administrator

All documents in the knowledgebase are assigned to a document category. Categories for knowledge documents are ordered into hierarchies. The document

categories order documents into top-level categories and subcategories. A top-level category is a parent category of the subcategory immediately below it in the ordered list of categories.

To add a new top-level document category, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **KM Document Categories**.
2. Click **New**.
3. Type the name of the new top-level category.
4. Click **Add**.
5. Click **Save**.

Add permissions for a document category or subcategory

User role: System Administrator, Knowledge Administrator

A Knowledge Management document category or subcategory has associated with it a set of permissions. These permissions specify the profiles and knowledge groups that have access to the category or subcategory.

To add permissions, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **KM Document Categories**.
2. Double-click a top-level category or search for a subcategory to display the category for which you are adding permissions.
3. Click the **Permissions** tab on the **Category Details** form.
4. Click **Add**.
5. Select a knowledge group from the list, and then double-click the knowledge group you select.
6. Select a profile from the list, and then double-click the profile you select.
7. Repeat the steps to add additional permissions.
8. When you are finished adding permissions, click the **Permissions** tab to verify your selections.
9. Click **Save**.
10. Click **OK**.

Provide all users access to a document category

Applies to User roles: System Administrator

You can make all externally published documents associated with a single document category (such as Human Resources) to be viewable by all users.

To provide all users access to a document category, follow these steps:

1. Click **Knowledge Management** > **Configuration** > **KM Document Categories**.
2. Click the **Category Search** tab.
3. Type HR in the Search box.
4. Click the **Search** button.
5. Double-click **HR**.
6. Click the **Permissions** tab.
7. Click the **Add** button.
8. Select and then double-click **Default Knowledge View Group**.
9. Select and then double-click the **DEFAULT** profile. In the out-of-box system, the DEFAULT profile is set to search external access.
10. Click the **Permissions** tab to verify your changes.
11. Click **Save**.

Note

If you want all users to be automatically included in the Default Knowledge View Group, be sure to select the **Assign the Default Knowledge Group to all operators** checkbox in the **Knowledge Management Application Environment** form.

Contents

- 1 Deleting document categories
- 2 Add subcategories to a document category
- 3 Add a new top-level document category
- 4 Add permissions for a document category or subcategory
- 5 Provide all users access to a document category

1.12.4.4. Knowledge Management environment

The **Knowledge Management Application Environment** form displays the settings used to configure Knowledge Management.

You can use this form to view and update the environment settings, including assigning default knowledge to all operators who have permission to search knowledge and use adaptive learning to weight search results based on usage.

Note

Whenever you make changes to the Knowledge Management environment, all users must sign out of Service Manager and then sign back in for the changes to be visible and in effect.

Edit the Knowledge Management environment record

User role: System Administrator, KMAdmin

Before you can use Knowledge Management and the search engine, you must configure settings. The Knowledge Management Environment record allows you to set up connectivity and define options, such as enabling adaptive learning and setting the number of documents to be returned from a search. This record contains default settings. However, you can configure these settings to meet your business needs.

To configure the Knowledge Management application environment settings, follow these steps:

1. Click **Knowledge Management** > **Administration** > **Environment**.
2. Select new options or clear default options. Your changes redefine the Knowledge Management environment for all users.

   The **Assign the Default Knowledge View Group to all operators** option ensures that all operators who will have permission to search knowledge are able to view any documents in those document categories to which the default knowledge view group has access.

   The **Use Adaptive Learning to enhance search results** option enables Service Manager to weight search results based on usage. You can artificially weight

documents using Adaptive Learning by adding a phrase and adding a number of occurrences with the phrase to specify that the term or phrase occurs in the document a specified number of times. If the phrase is not literally present, this can have the effect of putting the phrase, or word, in the document. It also simulates the number of times the phrase is added to the document index, based on the quantity you choose to apply.

Note

- Clear the checkbox to disable Adaptive Learning.
- When you disable Adaptive Learning, you need to go to the "Manage Knowledgebases" screen and reindex all knowledgebases of type sclib.
- Adaptive Learning data is not deleted when Adaptive Learning is disabled.

3. Click **Save** and **OK** to save your changes.
4. Sign out of Service Manager and then sign in again for your changes to take effect.

Caution

When these fields aren't set correctly and a user attempts to access a knowledgebase, the **Manage Knowledgebase** form displays an error, stating that the search engine is incorrect or not found, and asks you to check the environment settings.

1.12.4.5. Knowledge Management workflow

Knowledge Management uses a multi-step workflow to manage the lifecycle of a knowledge document. In the workflow, the document goes through the following phases:

- Draft
- Triage
- Revise
- Review
- Publish
- Published Internal
- Published External
- Retired

When a user submits a document for approval, the document enters the workflow in the **Triage** phase. In a typical document workflow, the document is routed to the **Revise**, **Review**, and **Publish** phases before it is actually published internally or externally. In some cases, users with applicable access rights can publish a document immediately after creating it, without going through the formal **Revise** and **Review** phases.

Note

When you need to update a knowledge document attachment, you must save the attachment to your desktop, make the necessary updates, save the changes, and then re-upload the attachment to the knowledge document.

Knowledge Management uses knowledge groups and document categories to control access to knowledge documents along with document view and Knowledge Management profiles. Individual users are able to view, contribute, edit, and publish documents based on the knowledge groups in which they are members and the security mappings between the knowledge groups, knowledge profiles, and knowledge categories.

Upgrading to the Process Designer Knowledge Management workflow

As of Service Manager 9.30, Change Management no longer implements the workflow for Knowledge Management documents. Change Management records with Category “KM Document” close and convert to the applicable Knowledge Management (KM) Document records, as Process Designer now manages KM Documents. Additionally, you don't need a Change Management license to operate Knowledge Management.

KM Document records are created with an applicable status, based on the current phase of the Change Management records. The mapping is described in the following table.

cm3r (based on current.phase)	kmdocument (status)
KM 1 Triage	Triage
KM 2 Revise	Revise
KM3 Review read-only	Review
KM 4 Conclude	Conclude

Two fields are also mapped from a Change Management record to a Knowledge Management document, as described in the following table.

Change Management	Knowledge Management
assigned.to	assignedEditor
Coordinator	documentOwner

Another functionality is included in the upgrade script. Operators with the capability word “SysAdmin” or “KMAdmin” will have a security role in their operator record as System Administrator and Knowledge Administrator.

Note

When using the new Process Designer workflow, the following applies:

- All open change records of category ‘KM Document’ close with the completion code

‘1’ – Successful.

- The corresponding KM document status changes from working copy to the status based on the current phase of the change record.
- The upgrade to Process Designer for Knowledge Management is mandatory.

Contents

- 1 Upgrading to the Process Designer Knowledge Management workflow

1.12.4.6. Knowledge Management migration

Knowledge Management migration consists of several migration utilities, including a utility to import data from the core table in Service Management to the **kmdocument** table used by Knowledge Management.

The script **wsdl2java.bat** or **wsdl2java.sh** imports documents into the Knowledge Management tables in Service Manager. The script library file **KMSCCoreImport** imports data from the core table in Service Manager to the **kmdocument** table in the Knowledge Management application. This import migrates the IRExpert knowledge documents from previous versions of ServiceCenter.

To run the Knowledge Management migration utilities, you need a working knowledge of Service Manager. To modify these utilities you need a working knowledge of Service Manager applications and JavaScript.

Related topics

Import documents into Service Manager
XML file specification for Knowledge Management migration
KMSCCoreImport utility
Configure the runDocumentImport.bat file
Configure the runDocumentImport.sh file

1.12.4.6.1. Import documents into Service Manager

To import documents into Service Manager, perform the following steps:

1. Review the information described in General considerations for importing documents and ensure that all listed considerations are being adhered to.
2. Ensure that the XML source document file conforms to the requirements described in XML source document considerations for importing.

   Note

   The data in the Rich Text fields must be XHTML formatted.

3. Review the information described in Document import considerations for attachments and ensure that all listed considerations are being adhered to.
4. Update the wsdl2java.bat or wsdl2java.sh file.
5. Run the wsdl2java.bat or wsdl2java.sh file.
6. Update the runDocumentImport.bat or runDocumentImport.sh file.
7. Run the runDocumentImport.bat or runDocumentImport.sh file.
8. Review the knowledgecreator.log file.

Related topics

Knowledge Management migration
General considerations for importing documents
XML source document considerations for importing
Document import considerations for attachments

1.12.4.6.2. XML file specification for Knowledge Management migration

The KMMigrate script file processes the XML data file that you want to import into Service Manager. This script expects the data to be in the XML format specified.

There is no DTD for the XML data being imported because the elements can be any column name in any Service Manager table. This XML file specification is important if you edit the XML files to include any customized fields that you might have added and you want to import to the Knowledge Management application in Service Manager.

All required fields must be defined for a table element, or the imported records can't be inserted into the Service Manager database. The records can all be in the same file or in separate files. For example, the multiple usersAndContactsN.xml files can each hold 1000 users.

The root document element is `<migrationdoc>` and it must be the parent element of all table elements and record elements.

```
        <migrationdoc>
                ...
        </migrationdoc>
```

Each element tag within the `<migrationdoc>` tag must be the name of a Service Management table.

```
<migrationdoc>
        <contacts>
                <contact.name _type="Text"><!
[CDATA[owner,otto]]></contact.name>
                <user.id _type="Text"><![CDATA[1296]]></user.id>
                <full.name _type="Text"><![CDATA[owner,otto]]>
</full.name>
                <email _type="Text"><![CDATA[oowner@hp.com]]>
</email>
                <contact.phone _type="Text"><!
[CDATA[858.792.3416]]></contact.phone>
                <operator.id _type="Text"><![CDATA[oowner]]>
</operator.id>
         </contacts>
...
</migrationdoc>
```

Each element within the table element must be the name of a text, boolean, or decimal data field or the name of an Array or ArrayStructure in the record. A data field has an attribute of `_type="Text"`, `_type="Boolean"`, or `_type="Decimal"`.
The actual data value can be enclosed in a CDATA tag.

```
<email _type="Text"><![CDATA[oowner@hp.com]]></email>
<login.revoked_type="Boolean"><![CDATA[false]]></login.revoked>
<msglog.lvl _type="Decimal"><![CDATA[1]]></msglog.lvl>
```

The actual data value can also appear as plain text, if it has no special characters in it.

```
<email _type="Text">oowner@hp.com</email>
<login.revoked _type="Boolean">false</login.revoked>
<msglog.lvl _type="Decimal">1</msglog.lvl>
```

An element within a table that's an Array is imported into the kmgroup table as follows:

```
<operators _type="Array">
        <operators _type="Text"><![CDATA[jowens]]></operators>
        <operators _type="Text"><![CDATA[rtillman]]></operators>
        <operators _type="Text"><![CDATA[ssmith]]></operators>
</operators>
```

An element within a table that's an ArrayStructure is imported into the kmcategory table as follows:

```
<permission _type="ArrayStructure">
        <kmprofile _type="Text"><![CDATA[KMP005]]></kmprofile>
        <kmgroup _type="Text"><![CDATA[KMG02]]></kmgroup>
</permission>
```

Related topics

Knowledge Management migration

1.12.4.6.3. Sample import record migration file

This sample XML file shows one of each of the five export records. The ExportData utility exports four XML files:

- usersAndContactsN.xml
- groups.xml
- categories.xml
- doctypes.xml

The usersAndContactsN.xml file contains contact and operator records similar to the following.

```
<migrationdoc>
        <contacts>
                <contact.name _type="Text"><!
[CDATA[owner,otto]]>
                </contact.name>
                <user.id _type="Text"><![CDATA[1296]]></user.id>
                <full.name _type="Text"><![CDATA[owner, otto]]>
</full.name>
                <email _type="Text"><![CDATA[oowner@hp.com]]>
</email>
                <contact.phone _type="Text">
                <![CDATA[858792.3416]]></contact.phone>
                <operator.id _type="Text"><![CDATA[oowner]]>
</operator.id>
        </contacts>
        <operator>
                <application.name_type="Text">
                <![CDATA[menu.manager]]></application.name>
                <command _type="Boolean"><![CDATA[true]]>
</command>
                <company _type="Text"><![CDATA[PRGN]]></company>
                <contact.name _type="Text">
                <![CDATA[Tremain, Johnny]]></contact.name>
                <email _type="Text"><![CDATA[jt@hp.com]]>
</email>
                <ess.access.only _type="Boolean">
                <![CDATA[false]]></ess.access.only>
                <expire.password _type="Boolean">
                <![CDATA[false]]></expire.password>
                <full.name _type="Text"><![CDATA[Tremain,
Johnny]]>
                </full.name>
                <locked.reason _type="Decimal"><![CDATA[0]]>
                </locked.reason>
                <login.revoked _type="Boolean"><![CDATA[false]]>
                </login.revoked>
                <msglog.lvl _type="Decimal"><![CDATA[1]]>
</msglog.lvl>
                <name _type="Text"><![CDATA[johnny]]></name>
                <names _type="Array">
```

```
                 <names _type="Text"><![CDATA[name]]></names>
                 <names _type="Text"><![CDATA[string1]]></names>
                </names>
                <no.lock.user _type="Boolean"><![CDATA[true]]>
                </no.lock.user>
                <password _type="Text"><![CDATA[johnny]]>
</password>
                <profile.change _type="Array">
                 <profile.change _type="Text">
                 <![CDATA[KM COORDINATOR]]></profile.change>
                </profile.change>
                <values _type="Array">
                 <values _type="Text"><![CDATA[MAIN MENU]]>
</values>
                 <values _type="Text"><![CDATA[HOME]]></values>
                </values>
        </operator>
                        .
                        .
                        .
</migrationdoc>
```

The groups.xml file contains Knowledge Management group records as follows:

```
<migrationdoc>
        <kmgroup>
                <id _type="Text"><![CDATA[KMGM01311]]></id>
                <name _type="Text"><![CDATA[HR Owner Group]]>
</name>
                <operators _type="Array">
                <operators _type="Text"><![CDATA[jowens]]>
</operators>
                <operators _type="Text"><![CDATA[rtillman]]>
                </operators>
                <operators _type="Text"><![CDATA[ssmith]]>
</operators>
                </operators>
        </kmgroup>
                .
                .
                .
</migrationdoc>
```

The categories.xml file contains kmcategory records similar to the following record. Note that the id in this sample record is **KM42T** and ends with a "T". Each id in the kmcategory table must end with a unique character such as "T" because the search engine logic depends on having a category id with a unique delimiter to determine where one category id ends and the next category id begins.

```
<migrationdoc>
        <kmcategory>
                <level.in.hierarchy
_type="Decimal">0</level.in.hierarchy>
                <name _type="Text">HR Employment FAQs</name>
                <id _type="Text">KM42T</id>
                <permission _type="ArrayStructure">
                 <kmprofile _type="Text">KMP005</kmprofile>
                 <kmgroup _type="Text">KMG02</kmgroup>
                </permission>
                <permission _type="ArrayStructure">
                 <kmprofile _type="Text">KMP002</kmprofile>
                 <kmgroup _type="Text">KMG011</kmgroup>
                </permission>
                <permission _type="ArrayStructure">
                 <kmprofile _type="Text">KMP001</kmprofile>
                 <kmgroup _type="Text">KMG012</kmgroup>
                </permission>
        </kmcategory>
                .
                .
                .
</migrationdoc>
```

The **doctypes.xml** file contains kmdoctype records similar to the following:

```
<migrationdoc>
        <kmdoctype>
                <name _type="Text"><![CDATA[errormsg]]></name>
                <description _type="Text"><![CDATA[Description
of an error
                message</description>
                <display.name _type="Text"><![CDATA[Error
Message/
                Cause]]></display.name>
                <format.edit _type="Text"><!
[CDATA[kmdocument.errormsg]]>
                </format.edit>
                <format.new _type="Text"><!
[CDATA[kmdocument.errormsg]]>
                </format.new>
                <format.view _type="Array">
                        <format.view _type="ArrayStructure">
                         <view.data _type="Text"><![CDATA[<table
style=]>
                         </view.data>
                         <view.description _type="Text"><!
[CDATA[This is the
                         Default ]]></view.description>
                         <view.seclevel _type="Decimal"><!
[CDATA[60]]>
                         </view.seclevel>
                         <view.name _type="Text"><!
                         [CDATA[Default ErrMsgCauseView]]>
</view.name>
                         <view.roles _type="Array">
                                 <view.roles _type="Text"><!
[CDATA[KMGM010]]>
                                 </view.roles>
                         </view.roles>
                        </format.view>
                </format.view>
        </kmdoctype>
                .
                .
```

```
                .
</migrationdoc>
```

Related topics

Knowledge Management migration
XML file specification for Knowledge Management migration

1.12.4.6.4. General considerations for importing documents into Service Manager

The following are general considerations for importing documents into Service Manager:

- If the Service Manager license doesn't include SOAP API license, the sm.ini file needs to include the following parameter: *allowwsdlretrieval:1*.
- Set refresh interval in the Knowledge_Library to -1 to turn off incremental updates during a large import.
- We recommend using a <Legacyid> element. The value of this element should be unique across ALL documents. If <Legacyid> is specified, the import queries the kmdocument table for the legacyid. If an existing record in kmdocument contains a legacyid matching the value in the <Legacyid> element of the source xml, the kmdocument record is updated with the information in the source xml. If no existing record in kmdocument is found with a legacyid matching the value in the <Legacyid> element, the xml source is inserted as a new kmdocument.
- Elements in the source xml files must have corresponding fields in the kmdocument table in Service Manager (the kmdocument record in the dbdict table), and they must be included in the Web services API.
- Fields that are in a document xml file that aren't in the dbdict are omitted on import, but the documents are imported.
- Elements with data containing HTML markup or any special characters that are illegal in XML should be CDATA wrapped to prevent the XML parser from throwing an exception.

Related topics

Knowledge Management migration

1.12.4.6.5. XML source document considerations for importing

The XML source documents processed by the KMImport utility must meet the following requirements:

- Root element must be <rKmDocument>.
- The source document may contain elements only at one level under the root element. Elements containing child elements cause the import to fail.
- Source documents must contain a <Title> element.
- Source documents must contain a valid <DocType> element. Valid means the DocType value must match an existing DocType (name field) in Service Manager.
- Source documents must contain a <Categories> element with a semicolon delimited string of published (approved) categories. Categories to publish to are specified using XPATH format. For example a document to be published to the "Incident Management" subcategory of the "Service Manager" category would appear as follows:

`<Categories> Service Manager/Incident Management</Categories>`
Additional categories are appended and preceded by a semicolon;.
`<Categories> Service Manager/Incident Management ;anothercategory; </Categories>`

- Elements with data containing HTML markup or any special characters that are illegal in XML should be CDATA wrapped to prevent the XML parser from throwing an exception.
- XML source documents that contain meta data for an "external" DocType must also contain a <ExternalDocumentName> element that contains the file name of the external document used as the document body. This file must reside in the same directory as the XML file that references it.
- When using the `kmimport` utility to reimport an external document, the external document name must remain the same as the original. Also, any attachments must keep the same name.
- If you have tailored the **kmdocument** dbdict, be sure to not include your tailored Array fields, Structure fields, and Arrayed Structures fields in the source XML documents; otherwise the import will fail.

Related topics

Knowledge Management migration

1.12.4.6.6. Document import considerations for attachments

Before importing documents into Service Manager, consider the following regarding attachments.

- All attachments should be in an **attachments** folder under the XML source directory.
- Shared Content attachments should have the file name prefixed with "shared_" followed by a unique string used as the legacyid for attachments followed by an underscore and the file name.

`shared_<legacyId>_<filename>`
Example:
`shared_1234567890_Logo.jpg` .

- Attachment links (shared or owned) in the source XML should have the format `href=/answers/attachments/1234567890` where 1234567890 is the legacyid of the attachment. Embedded attachments should use `src=` instead of `href=` .
- Document owned attachments (that is, documents not shared) should have the file name prefixed with a unique string used as the legacyid for attachments followed by an underscore and the file name.

`<legacyId>_<filename>`
Example:
`1234567890_Logo.jpg`

Related topics

Knowledge Management migration
General considerations for importing documents

1.12.4.6.7. Configure the wsdl2java.bat file

Applies to User roles: System Administrator

To use the wsdl2java.bat file to import documents into Service Manager, you must first configure the file as follows:

1. Set JAVA_HOME to your jdk directory. For example, `JAVA_HOME=C:/jdk11.0.xxxx`.
2. Change SC_SERVER and SC_PORT to reflect your service target.

```
SET SC_SERVER=localhost
SET SC_PORT=13080
```

3. Set the CLASSPATH.

```
SET CLASS_PATH=./lib/axis-1.2.jar;./lib/saaj-1.2.jar;./lib
/commons-logging-1.0.3.jar;./lib/jaxrpc-1.1.jar;./lib
/mail-1.3.2.jar;./lib/activation-1.0.2.jar;./lib
/commons-discovery-0.2.jar;./lib/wsdl4j-1.5.1.jar
```

Related topics

Running the wsdl2java command file

1.12.4.6.8. Configure the wsdl2java.sh file

Applies to User roles: System Administrator

To use the wsdl2java.sh file to import documents into Service Manager, you must first configure the file as in the following example:

1. Set JAVA_HOME to your jdk directory. For example, `JAVA_HOME=/jdk11.0.xxxx`.
2. Change SC_SERVER and SC_PORT to reflect your service target.

```
SC_SERVER=localhost
SC_PORT=13080
```

3. Set the CLASSPATH

```
SET CLASS_PATH=./lib/axis-1.2.jar;./lib/saaj-1.2.jar;./lib
/commons-logging-1.0.3.jar;./lib/jaxrpc-1.1.jar;./lib
/mail-1.3.2.jar;./lib/activation-1.0.2.jar;./lib
/commons-discovery-0.2.jar;./lib/wsdl4j-1.5.1.jar
```

Related topics

Knowledge Management migration
Running the wsdl2java command file

1.12.4.6.9. Running the wsdl2java command file

Run the **wsdl2java** command file to build JAVA stubs, compile and jar the class, and clean up.

On Windows, the **wsdl2java.bat** file executes the following operations:

- Builds the JAVA stubs from the wsdl

```
%JAVA_HOME%/bin/java -cp %CLASS_PATH%
org.apache.axis.wsdl.WSDL2Java http://%SC_SERVER%:%SC_PORT%
/sc61server/ws/KnowledgeManagement.wsdl
```

- Compiles and jars the classes into **libcenterpws.jar** with the command, **md classes**

```
%JAVA_HOME%/bin/javac -d classes -classpath %CLASS_PATH% .
/com/peregrine/servicecenter/PWS/Common/*.java
%JAVA_HOME%/bin/javac -d classes -classpath %CLASS_PATH%;.
/; ./com/peregrine/servicecenter/PWS/*.java
%JAVA_HOME%/bin/jar cvf ./lib/servicecenterpws.jar -C
classes .
```

- Cleans up with the following commands

```
rd com /s /q
rd classes /s /q
```

On UNIX, the **wsdl2java.sh** file executes the following operations:

- Builds the JAVA stubs from the wsdl

```
$JAVA_HOME/bin/java -cp $CLASS_PATH
org.apache.axis.wsdl.WSDL2Java http://$$SC_SERVER:$SC_PORT
/sc61server/ws/KnowledgeManagement.wsdl
```

- Compiles and jars the classes into **lib/servicecenterpws.jar** with the command, **mkdr classes**

```
$JAVA_HOME/bin/javac -d classes -classpath $CLASS_PATH$.
/com/peregrine/servicecenter/PWS/Common/*.java
$JAVA_HOME/bin/javac -d classes -classpath $CLASS_PATH:.
/:./com/peregrine/servicecenter/PWS/*.java
$JAVA_HOME%/bin/jar cvf./lib/servicecenterpws.jar -C
classes.
```

- Cleans up with the commands

```
rd -R com
rd -R classes
```

Related topics

Configure the wsdl2java.bat file
Configure the wsdl2java.sh file

1.12.4.6.10. Configure the runDocumentImport.bat file

User role: System Administrator

> **Note**
>
> The user needs execute capability.

The runDocumentImport command file is run to import documents into Service Manager. Documents successfully imported remain in the migration root directory and rejected documents are placed in _rejects.

To configure the runDocumentImport.bat file, follow these steps:

1. Set JAVA_HOME to your jdk directory. For example, `SET JAVA_HOME=C:/jdk11.0.xxxx`.
2. Change SC_SERVER, SC_PORT, SC_USER, and SC_PASSWORD to reflect your service target.

   ```
   SET SC_SERVER=localhost
   SET SC_PORT=13080
   SET SC_USER=%1
   SET SC_PASSWORD=%2
   ```

   > **Note**
   >
   > Setting the user name and password to the batch file parameters means that you don't need to store sensitive information in the batch file. The user must have the KMAdmin or SysAdmin execute capability in order to import the documents.
   >
   > For example, if SC_USER is rmorgan and SC_PASSWORD is pa$$w0rd, you can call runDocumentImport.bat like this:
   > `C:\> runDocumentImport.bat rmorgan pa$$w0rd`
   > If no SC_USER or SC_PASSWORD is defined, the default user is falcon with a blank password.

3. Change MIGRATION_DOCS_DIR to specify location of the documents being imported.

```
SET MIGRATION_DOCS_DIR=C:/MigrationDocs
```

4. Change DOC_STATUS to default status of the imported documents.

```
SET DOC_STATUS=internal
```

5. Set the CLASSPATH.

```
SET CLASS_PATH=./lib/activation-1.0.2.jar;./lib
/axis-1.2.jar;./lib/commons-discovery-0.2.jar;./lib
/commons-logging-1.0.3.jar;./lib
/jaxrpc-1.1.jar;./lib/kmimport.jar;./lib/log4j-1.2.8.jar;.
/lib/mail-1.3.2.jar;./lib/saaj-1.2.jar;./lib
/wsdl4j-1.5.1.jar;./lib/servicecenterpws.jar;
```

Related topics

Running the runDocumentImport command file
Using the knowledgecreator.log file

1.12.4.6.11. Configure the runDocumentImport.sh file

User role: System Administrator

> **Note**
>
> The user needs execute capability.

The runDocumentImport command file is run to import documents into Service Manager. Documents successfully imported remain in the migration root directory and rejected documents are placed in _rejects.

To configure the runDocumentImport.sh file, follow these steps:

1. Set JAVA_HOME to your jdk directory. For example, `SET JAVA_HOME=C:/jdk11.0.xxxx`.
2. Change SC_SERVER, SC_PORT, SC_USER, and SC_PASSWORD to reflect your service target.

   ```
   SET SC_SERVER=localhost
   SET SC_PORT=13080
   SET SC_USER=$1
   SET SC_PASSWORD=$2
   ```

   > **Note**
   >
   > Setting the user name and password to the batch file parameters means that you don't need to store sensitive information in the batch file. The user must have the KMAdmin or SysAdmin execute capability in order to import the documents.
   >
   > For example, if SC_USER is rmorgan and SC_PASSWORD is pa$$w0rd, you can call runDocumentImport.sh like this: `$runDocumentImport.sh rmorgan pa$$w0rd`
   > If no SC_USER or SC_PASSWORD is defined, the default user is falcon with a blank password.

3. Change MIGRATION_DOCS_DIR to specify location of the documents being imported.

```
MIGRATION_DOCS_DIR=/MigrationDocs
```

4. Change DOC_STATUS to default status of the imported documents.

   ```
   DOC_STATUS=internal
   ```

5. Set the CLASSPATH.

   ```
   CLASS_PATH=./lib/activation-1.0.2.jar:./lib/axis-1.2.jar:.
   /lib/commons-discovery-0.2.jar:./lib
   /commons-logging-1.0.3.jar:./lib/jaxrpc-1.1.jar:./lib
   /kmimport.jar:./lib/log4j-1.2.8.jar:./lib/mail-1.3.2.jar:.
   /lib/saaj-1.2.jar:./lib/wsdl4j-1.5.1.jar:.
   /lib/servicecenterpws.jar:
   ```

Configure the wsdl2java.sh file

1.12.4.6.12. Running the runDocumentImport command file

You run the command file **runDocumentImport** to import the documents into Service Manager. Documents successfully imported remain in the migration root directory and rejected documents are placed in `_rejects`.

For Windows, **unDocumentImport.bat** executes the following statement to import the documents.

```
%JAVA_HOME%/bin/java -cp %CLASS_PATH% com.peregrine.webservice.knowledgemanagement.CreateKnowledge -host %SC_SERVER% -port %SC_PORT% -username %SC_USER% -password %SC_PASSWORD% -xmlSourceDirectory %MIGRATION_DOCS_DIR% -defaultStatus %DOC_STATUS%
```

For UNIX, **unDocumentImport.sh** executes the following statement to import the documents.

```
$JAVA_HOME/bin/java -cp $CLASS_PATH com.peregrine.webservice.knowledgemanagement.CreateKnowledge -host $SC_SERVER -port $SC_PORT -username $SC_USER -password $SC_PASSWORD -xmlSourceDirectory $MIGRATION_DOCS_DIR -defaultStatus $DOC_STATUS
```

Related topics

Import documents into Service Manager
Configure the runDocumentImport.bat file
Configure the runDocumentImport.sh file

1.12.4.6.13. Reconciling unresolved links

When two documents contain links to each other, one of the links will not be resolvable when the document is imported. This broken circular reference is caused by one of the link target documents not being imported at the time the source document was resolving its links. The document import utility imports the documents, and logs the ID of the document that contains an unresolved link. After importing a set of documents, open and execute the **KMReconcileLinks** script in the script library. This script processes the documents logged as having unresolved links, and attempts to reconcile them. Once the import is finished, the document with the unresolved links should be reconciled, as the **KMReconcileLinks** process will fix the broken links.

Related topics

Knowledge Management migration
Import documents into Service Manager
General considerations for importing documents

1.12.4.6.14. Using the knowledgecreator.log file

After running document import (either runDocumentImport.bat or runDocumentImport.sh), examine the knowledgecreator.log file. Correct any file errors noted, and then re-run the import, if desired.

The end of the file contains the following import statistics:

-Imported Documents.....: 4008
-Skipped Documents.....: 12
-Reject Documents.....: 4
-Imported Attachments.....: 1
-Skipped Attachments.....: 4

Skipped files include any non-xml files that are left in the import folder. For example, an error such as " org.xml.sax.SAXParseException: The content of elements must consist of well-formed character data or markup. " indicates that the XML probably contains special characters that could not be parsed.

You can edit and correct the files that were skipped and re-run the import.

If the meta data for any imported file contains a field that was not defined in the kmdocument data dictionary (dbdict), then this field's value was not imported. The log file lists any undefined fields. You can add the field to the kmdocument dbdict and re-run the import to populate the field's value.

Related topics

Knowledge Management migration
Running the runDocumentImport command file

1.12.4.6.15. KMSCCoreImport utility

The utility `KMSCCoreImport` imports data from the core table in Service Manager to the `kmdocument` table in the Knowledge Management application. This import migrates the IRExpert knowledge documents from previous versions of ServiceCenter. By importing data from the core table used by IRExpert, you make the data available to the Knowledge Management application.

The `KMSCCoreImport` utility is for a one-time transfer of all files from the core table into the Knowledge Management application. If you run this code more than once, it imports duplicate data into the Knowledge Management application.

You can edit `KMSCCoreImport` utility script to take into account any customizations of either the core table or the data contained in the core table. The areas in the `KMSCCoreImport` script where changes are allowed and expected are clearly marked. Also, those sections of the code that can't be changed are also commented accordingly.

Tips for using the KMSCCoreImport script

Before you use the KMSCCoreImport script, consider the following:

- Back up the database before you run the script against the database.
- Rename the script to create a copy and then add the copy to the ScriptLibrary before you begin modifying the script.
- Run the modified script against a copy of your database to verify that the script works correctly.
- Run the complete script in a test environment to verify the script and the resulting records.

Modifying the KMSCCoreImport script file

Before running the import script, you should modify the **KMSCCoreImport** script to support any customizations or modifications to the core table.

To modify the **KMSCCoreImport** script, find the section of the script file that begins with the comment `"this is the section of the customized code where additional code should be added ..."` then follow the instructions provided by these comments. Essentially, you need to map any fields added to the out-of-box core table and update the field names.

The following is a sample:

```
//????????????????????????????????????????????????????????????
??
//@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@@
//this is the section of the customized code where additional
code
//should be added if the user so desires...the following are a
few
//example lines of code
//NOTE: if a field name has a dot delimiter such as in entry.id,
in
//javascript it has to be changed to an underscore as in
// entry_id...you can see another //example of this in the
example
//code below
//newKMDoc.reference = corerecs.description + " " +
//corerecs.mycustom_field;
//newKMDoc.doctype = corerecs.mycustomdoctype;
//if the customizations to the core table include fields that do
//not directly translate to any fields in the default kmdocument
//table, and yet the user still wants to maintain that data,
//they should follow the instructions in the user guide for how
//to add a new field to the kmdocument table (WARNING: this is
//a very serious and involved task so be sure you know what you
//are doing before taking this step - it is advised that only
//admin users perform any sort //of changes like this). Once the
//new field has been added to the kmdocument table, you can use
//lines like the following example:
//newKMDoc.mynewcustomfield = corerecs.myothercustomizedfield;
//PLEASE NOTE however the caveat that adding new customized
fields
//to the kmdocument table impacts many different aspects of the
//program that MUST be considered such as the searching and
//document type/document view structure(s) that represent the
raw
//data in end user form...as well as the forms used to edit the
//documents once they have been imported...all of these areas
must
//also be considered before making any changes...
//@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
```

```
@@
If you want to create document categories that match the
document categories
used in IRExpert, see the section of the script that begins:
//in the default setup, the category and subcat fields of the
core
//table appear not to be used so we will just shove all core
//records into a single category within
//KM - Service Manager/KM/CoreImport
        var newcat =
        system.library.KMCategory.getCategoryIdString
        ("Service Manager/Knowledge Management/CoreImport",
true);
        newKMDoc.categories[0] = newcat;
        newKMDoc.generatedcats[0] =
        "Service Manager/Knowledge Management/CoreImport";
//if however, the cat and subcat fields are used and the user
//would like to use them for categorization, then comment out
the
//three lines above and remove the commenting before the next
//lines var newcat =
system.library.KMCategory.getCategoryIdString
        (corerecs.category+"/"+corerecs.subcategory, true);
//newKMDoc.categories[0] = newcat;
//newKMDoc.generatedcats[0] =
        corerecs.category+"/"+corerecs.subcategory;
//in the core table, the description and resolution fields are
//arrays of strings so we have to cycle through them and put
them
//together to get our question and answer fields' values
//respectively.
```

Running the KMSCCoreImport script

Run the **KMSCCoreImport** script from Service Manager. You can use either the command line (**sl**) or click **Tailoring > Script Library** to access and run the script.

To run the **KMSCCoreImport** script:

1. Locate the importCoreToKMNow function in the script. It is the single line of code at the bottom of the script that calls the single function in the script file.
2. If the import is based on the out-of-box core table data then the parameter value for this call should be true. This creates records of the Problem / Solution type.
   `importCoreToKMNow(true);`
3. If the import is based on customized core table data then the parameter value should be false.
   `importCoreToKMNow(false);`
4. With this parameter change, you can modify the document type to match a customized document type. In the script, find the **second** occurrence of the line: `newKMDoc.doctype = "howto"`.
   Change the doctype to match the name of your customized document type.
5. If you change the value of the parameter and update the doctype, save the file before continuing.
6. Click **Execute**. The script runs. You are notified when the process completes with the message, `"The import from the core table is complete"`.

1.12.4.6.16. KMDocumentExport utility

The KMDocumentExport utility is a knowledge document export tool to achieve the article migration from Service Manager to Service Anywhere. You can use this utility to export knowledge documents from Service Manager to a pre-defined directory in the following formats:

- A metadata.xml file - shows all meta information of an article including the title, status, category, create time, last modify time, attachment, and links between articles, etc.
- An articles folder - contains files that shows other information about an article except for the meta information.
- A resources folder - includes all embedded images and attachment files.

Running the KMDocumentExport script

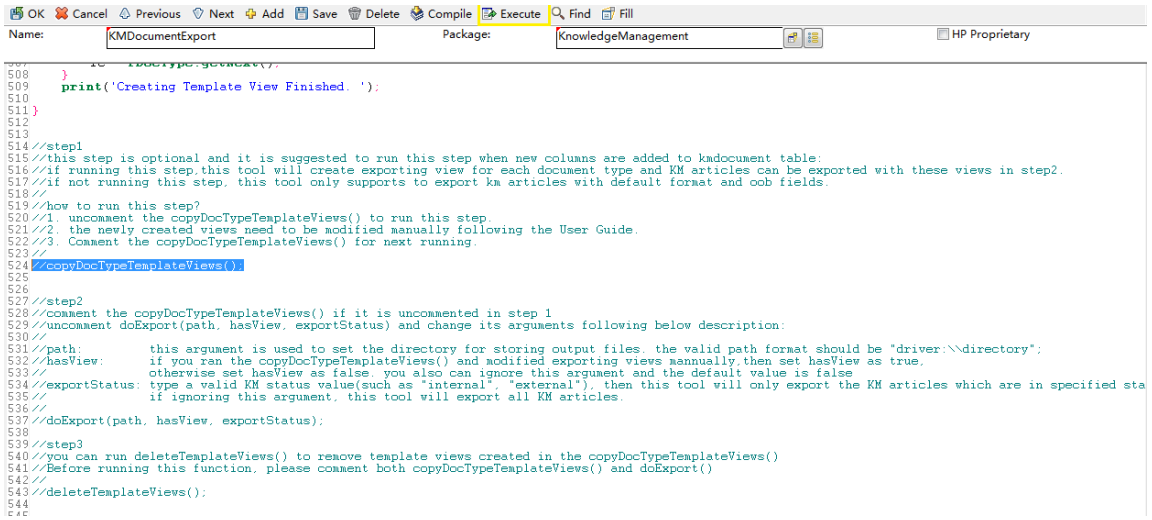
Follow these steps to run the **KMDocumentExport** script:

1. Sign in to Service Manager as a system administrator.
2. Type sl in the Service Manager command line field, and then press Enter. Alternatively, you can click **Tailoring** > **Script Library**.
3. Type KMDocumentExport in the Name field, and then click **Search**.
4. (Optional) Create the exporting view for each document type so that the KM articles can be exported with these views in the next step. Follow these steps:

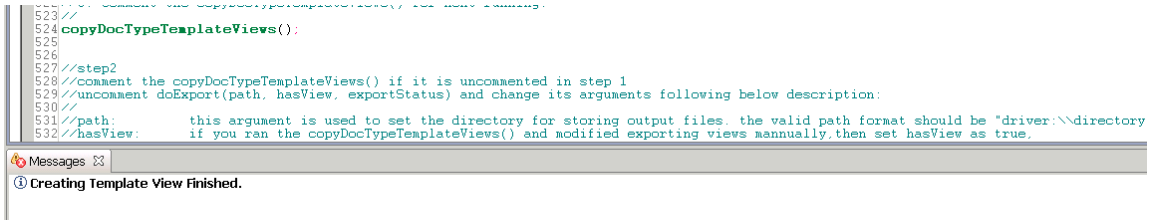
   Note

   The exporting view created in this step duplicates with the existing view that owns the highest security level. However, you are recommended to perform this step especially when you have added new columns to the kmdocument table. If you don't perform this step, the Migration tool only exports KM articles with the default format and the OOB fields.

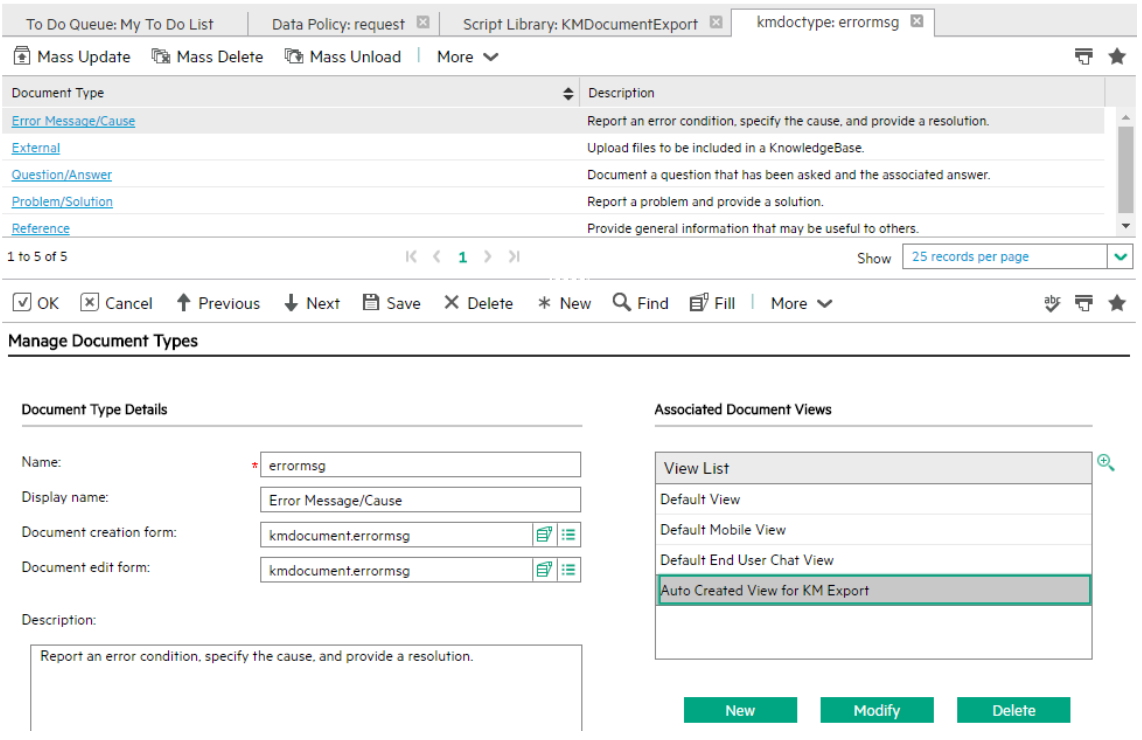
   1. Uncommon the `//copyDocTypeTemplateViews();` line, and then save your modifications.



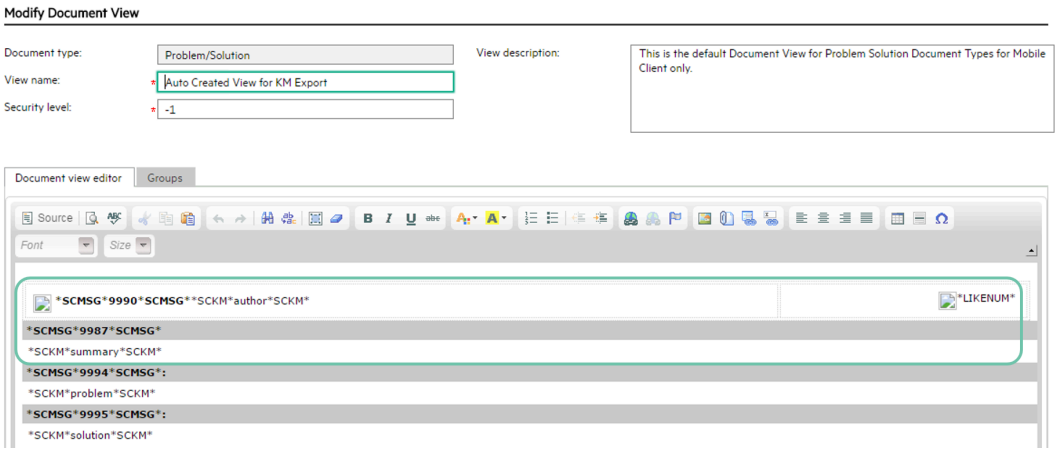
2. Click **Execute** to create the new Auto Created View for KM Export template view which is used to export the KM articles for each document type.
3. When the copy process is complete, the system displays the following message.



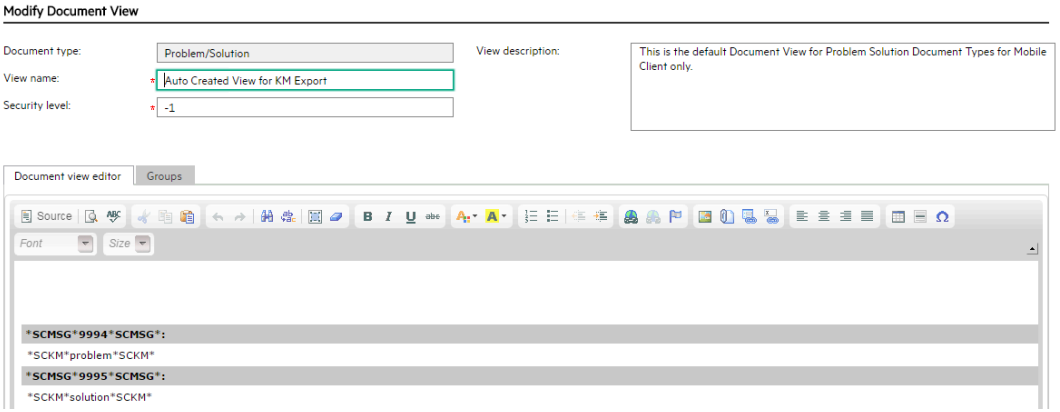
4. Click **Knowledge Management** > **Configuration** > **Document Types**, and update the newly created Auto Created View for KM Export view as follows:



1. Remove the highlighted sections (such as title, km id, doctype, summary, creation date, and so on). Keep the knowledge content left.



2. Click **OK** to return to the Manage Document Types page, and then click **Save** to save your modifications. The modified template view is displayed as illustrated follows.



5. Continue to follow *step i* and *step ii* to modify other document type views for KM exporting as necessary.

5. Comment out the `copyDocTypeTemplateViews();` line if it is uncommented in *step 4*.

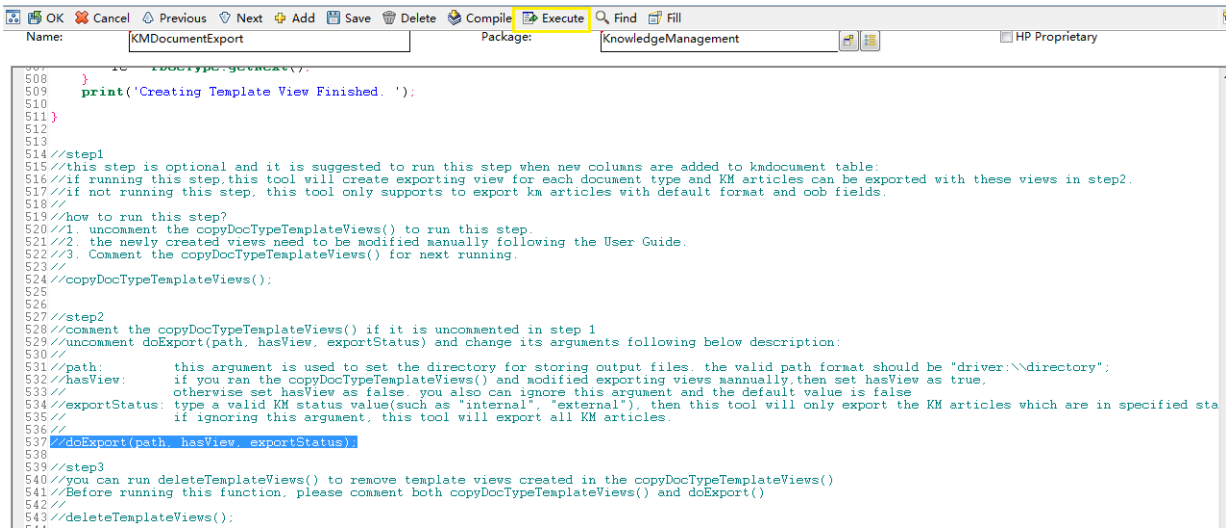
6. Uncomment the `//doExport(path, hasView, exportStatus);` line, and then change its arguments by following the descriptions below:

   - path: Set the directory for storing the output files in this argument. For example, `C:\\ KMDoumentExport` is for the Windows server, and `/home/KMDoumentExport` is for Linux.

> **Note**
>
> Make sure the export path is located in the Service Manager server.

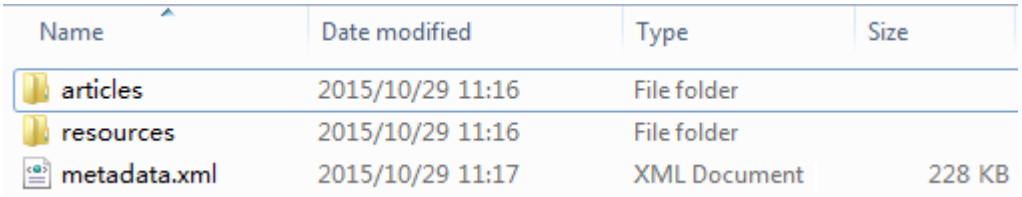
- hasView: Set this argument to `true` if you have run `copyDocTypeTemplateViews()` and modified the exporting views manually. Otherwise, set this argument to `false`. You can also ignore this argument and the default value is `false`.
- exportStatus: Set this argument to a valid KM status value (such as `draft`, `internal`, or `external`) so that the KMDocumentExport script only exports the specified KM articles. You can also ignore this argument and the tool exports all KM articles.

7. Save your modifications, and then click **Execute** to start the exporting process.



8. When the exporting process is completed, the system displays the following messages as illustrated below.

You can check the exported files in the export directory.

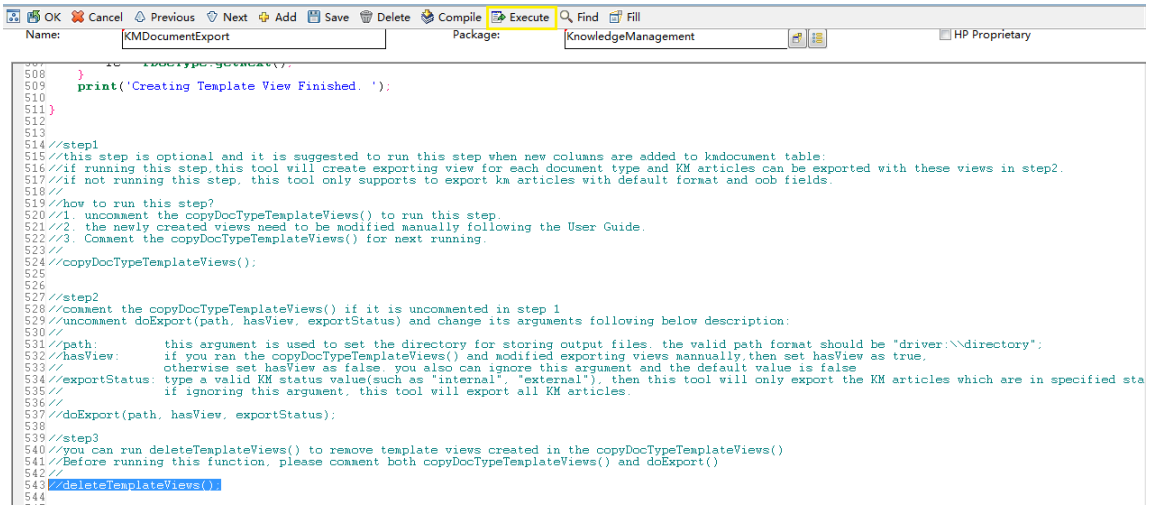


Note

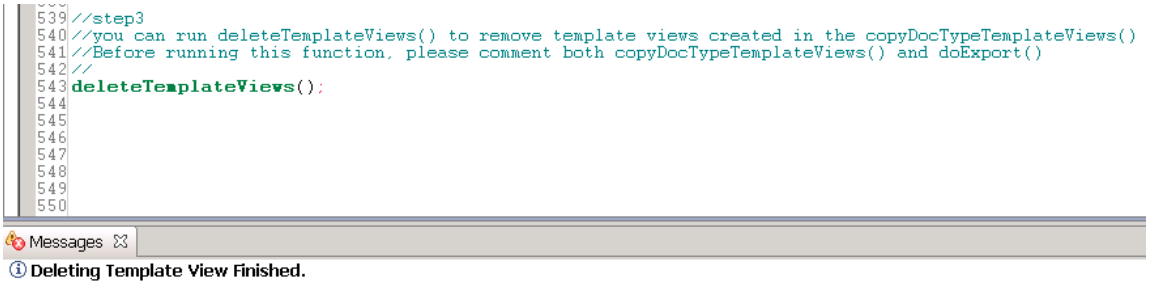
The metadata.xml file is overwritten with the latest results after each exporting. However, the files created by previous exporting in both the articles folder and the resources folder aren't removed during the next exporting process. To keep your exporting data contains the latest articles and resources only, you need to use a new export path and folder each time. If your KM article has no content or has no content in the exporting view, the exported article detail HTML file displays the `There is no content for this KM Article.` sentence only.

9. Follow these steps to remove the exporting views created in `copyDocTypeTemplateViews()` :

    1. Uncommon the `//deleteTemplateViews();` line, save your modification and then click **Execute** to remove the exporting views.



    2. When the removal process is completed, the system displays the following message as illustrated below:



1.12.4.7. Knowledge Management integration

Knowledge Management integrates with Service Desk, Incident Management, and Problem Management to help meet your business needs. When Knowledge Management is integrated with each of these Service Manager modules, users can do the following:

- Starting at Service Desk, users responding to an Interaction record can search for similar interaction records with solutions.
- Users updating Incident records can search for similar issues with solutions in the knowledgebases.
- When working on Problem records, users can perform the following tasks:
  - search for similar problems, incidents, known errors, or knowledge documents
  - use the resolution from an existing resolved issue in a knowledgebase as a solution for the problem
  - use the information from a problem resolution to create a new knowledge document

Knowledge Management integration with Service Desk Interaction

Knowledge Management integrates with Service Desk so that users responding to an Interaction record can search for similar interaction records with solutions. The knowledge search uses the description by the Service Desk user to search for the knowledgebases for similar interaction records with solutions. When searching for knowledge, Knowledge Management uses fields defined in the integration mapping for searches.

The search results display a list of relevant documents. The user can view each document and then has the option of using the text from one the documents as a resolution for the original Interaction record. If the user chooses to use a solution, the system automatically populates the resolution field in the Interaction record with data from the field mapped to the resolution field. The **Use Solution** option is available from knowledge documents and knowledge candidates with a workaround or resolution.

Note

If the knowledge document is a working copy of a published document, the **Use Solution** button doesn't appear because the system views it as a temporary document. However, you can usually find the published version in the hitlist and use that as a solution.

In addition to searching for knowledge, the out-of-box mapping allows those users with appropriate privileges to create solutions and author knowledge documents from resolved Interaction records. For example, if in your search you find a knowledge document that includes in the description a resolution for the call, you can select the **Create Knowledge** option to begin authoring a new knowledge document. The field mappings for the knowledgebase determine which fields in the knowledge document are populated with data from the record.

Knowledge Management integration with Incident Management

Knowledge Management integrates with Incident Management so that users updating incidents can search for similar issues with solutions in the knowledgebases. The knowledge search uses the incident description to search. When searching for knowledge, Knowledge Management uses what is mapped by the integration field mappings to search each knowledgebase. This mapping specifies the fields and tables to search in the knowledgebases to which the user has access. For example in the out-of-box system, a user with browse access in Service Desk and Incident Management can view all relevant Incident and Interaction records in a search and those knowledge documents in categories to which the user has access.

The search results display a list of relevant documents. The user can view each document and then has the option of displaying and then using the text in one of the documents as a resolution to the Incident record. If the user chooses to use a resolved record, the system automatically populates the solution in the record based on the record mapping being reviewed for the knowledgebase. For example, if in a search you find an Incident record that includes a resolution, you can use **Use Solution** to copy the data from the resolution or workaround field to insert it into the solution (resolution field) of the open Incident record.

You can use **Create Knowledge** to contribute a new knowledge document from a resolved Incident record so that the data in the mapped fields is used in the

knowledge document. The field mappings and the document type determine which fields in the knowledge document are populated with data from the resolved record.

The field mappings specify:

- The fields passed from the incident to the search
- The fields passed from a knowledgebase record to the incident
- The fields passed from an Incident record to create a knowledge document

Note

The **Use Solution** option is available from knowledge documents, Incident and Interaction knowledge candidates with resolutions, and Known Error and Problem knowledge candidates with workarounds.

Knowledge Management integration with Problem Management

The Knowledge Management integration with Problem Management enables users to perform the following tasks:

- Search for similar Problems, Incidents, Known Errors, or knowledge documents
- Use the resolution from an existing resolved issue in a knowledgebase as a solution for the problem
- Use the information from a problem resolution to create a new knowledge document

When searching for knowledge, Knowledge Management uses the fields mapped in the Integration Mapping function for searches. This mapping specifies the fields and tables to search in the knowledgebases to which a user has access. The user can view each document and then has the option of using one of the resolutions as a resolution to the problem. If the user chooses to use a solution or workaround, the system automatically populates the resolution field in the Problem record.

For example, if in your search you find a Problem record that includes a workaround for the Incident record you are working with, use **Use Solution** to copy the data from the mapped fields of the **rootcause** table to insert it into the mapped fields of the Incident record.

Users can also create knowledge to contribute a new knowledge document from a resolved record so that the data in the mapped fields is used in the knowledge

document. The field mappings and the document type determine which fields in the knowledge document are populated with data from the resolved record.

Note

The **Use Solution** option is available from knowledge documents, Incident and Interaction knowledge candidates with resolutions, and Known Error and Problem knowledge candidates with workarounds.

Knowledge Management record mapping

The Integration Mapping function enables you to map fields between two tables in Service Manager. This allows users to search, retrieve, and author documents more efficiently since data is shared between operations.

In the out-of-box system, there are record maps predefined for the **incidents**, **knownerror**, **probsummary**, and **rootcause** tables. The Integration Mapping function provides a form for mapping fields, searching, retrieval, and authoring.

The **Map Name** field is the name of the mapping. By default the out-of-box mappings are named according to the Service Manager file name they provide a mapping for. If a **Map Name** other than the Service Manager file name is used, the thread variable **$gamapname** must be set to the value of the **Map Name** from the Display Option that's used to call the mapping.

The **Source Field in Current Record** column specifies the field in Service Manager that's sent to the Knowledge Management **Target Search Field**. A literal string value can be specified in quotation marks. Any string of characters specified within quotation marks is copied literally (without the quotations) into the **Target Search Field**.

The **Target Search Field** column specifies the field in Knowledge Management that receives the data sent from the source field. If a field is specified as a target for more than one source field, the data from the source field is appended (in the order specified in the mapping) into the target field. A value of "QueryString" specified in this column indicates that the data from the source field is sent as part of the non-fielded search string (such as if a user typed into the "Enter your question" box in the Knowledge Management search interface). A value of "contextTag" is used to indicate that the source field value will be used to track all search activity in the **kmsearchhistory** and **kmusagehistory** tables.

Note

The viewed statistic in **kmusagehistory** is intended to track documents that were viewed as the result of a search. This indicates if a knowledge document can be found and used in its existing state. If a document is viewed from a URL linked directly to the **kmdocument** record, the viewed statistic in **kmusagehistory** is not incremented. You can change this default behavior by specifying a parameter with the URL request. When the parameter is specified, the usage history and view count will be logged and incremented.

Tip

The additional parameter to add at the end of the URL query is as follows: **&relatedSystem=URLQuery**

Field mappings between the knownerror and kmdocument tables

Note

These descriptions are based on the out-of-box field mappings between the **knownerror** table and **kmdocument** table. If these field mappings have been modified on your system, the data for the fields on your **Contribute New Knowledge** form may be different than what this topic describes.

You can create a knowledge document for a known error. When you do this, the out-of-box system provides the following information on the **Contribute New Knowledge** form for the known error. This information is only available for the knowledge document if it is present in the **knownerror** table record.

Field	Description
Title	This entry comes from the brief description in the **knownerror** record.
Error Message	This entry comes from the description field in the **knownerror** record.
Cause	This entry comes from the root cause description in the **knownerror** record.
Workaround	This entry comes from the workaround and resolution description in the **knownerror** record.

Contents

- 1 Knowledge Management integration with Service Desk Interaction
- 2 Knowledge Management integration with Incident Management
- 3 Knowledge Management integration with Problem Management
- 4 Knowledge Management record mapping
  - 4.1 Field mappings between the knownerror and kmdocument tables

1.13. Service Catalogs

A Service Catalog contains a comprehensive list of enterprise products and services available to internal and external customers, depending on their business role. Users submit service requests that follow the usual planning and approval workflows. Service Manager fulfills service requests by using an internal interface to route the requested data to the appropriate Service Manager application. The application that delivers the requested product or service opens any required fulfillment tasks.

Service Catalogs support the Information Technology Infrastructure Library (ITIL) goals for Service Management. They integrate seamlessly with Service Level Management to measure and report on service level objectives for service request fulfillments. Service Catalogs set expectations for both customers and service providers about the delivery, quality, and level of services. Any Service Manager Business Service owner can create or maintain a Service Catalog using a simplified Service Manager interface and wizards to guide them in adding or changing service offerings.

Service Catalogs can include Information-only items as well as items that your customers can purchase. Purchasable items include item picture, description, attachments, and purchasing options. Information-only items, such as instructions on how to order something from an outside service, include only the item picture description and attachments and don't include purchasing options.

1.13.1. Out-of-box Service Catalog

The Service Catalog includes hundreds of services and bundles that span major categories. The four main Service Catalog categories include:

- **Business and Department Services** – This category contains enterprise services for business units and departments, including Application Access, New Application Hosting, and Professional Consulting.
- **Employee Lifecycle Services** – This category contains life cycle services available to employees and managers, including New Employee requests, Personal Information Change, and Employment Status Change,
- **Personal Productivity Services** – This category contains common employee services such as telecommunications, collaboration support, accounts, network access, software, and hardware bundles.
- **Technical and Support Services** – This category contains common technical IT requests such as setup and support of Servers, Database, Applications, and Network.

You can use the categories, subcategories, and items provided, or you have the flexibility to modify, copy, and remove them to fit your needs.

1.13.2. Service Catalog capability words

Service Catalog controls access through Profile records and Operator records. This allows different people to see and order different items from the catalog. A catalog manager can also set capability words for catalog items, categories, or bundles on the Access tab of any definition record. When the system verifies that the user has at least one of the capability words from the list defined on the catalog item definition record, the user is able to see and order that item from the catalog. If there are no capability words set in the catalog item definition record, the item is available to all users.

Note that the Operator record overrides the Profile record. Enabling functional access with a capability word in the Profile record, but omitting it in the Operator record, means that access is disabled. Users must have an Operator record, but may (or may not) have a Profile record. The Execute Capabilities table in the Operator record specifies a user's permissions. Following are some of the capability words that enable user access to Service Catalog:

Capability word	Description
svcCatAdmin	Allows access to Service Catalog administration and is the hierarchical parent for all other capability words.
svcDeptRequester	Allows a department to request items from catalog.
svcEmployeeRequester	Allows an employee to request items from catalog.
svcCatManagerRequester	Allows a manager to request items from catalog.
svcCatTechRequester	Allows a technician to request items from catalog.
svcCatRequestOnBehalf	Allows an employee self-service (ESS) user to submit a service catalog request on behalf of another user.

1.13.3. Filter access to catalog items

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type the **Name** of a definition record.
3. Select the **Type** of record (category, bundle, or item) from the drop-down list and then click **Search**.
   The definition record you want to modify opens.
4. Click on the **Access** tab.
5. In the **Available to:** field, select a capability word from the list to control access to the catalog item.
   A user must have at least one of the capability words in the list in order to select the item from the catalog. If you don't select any capability words, the item is available to all users.
6. If you want to edit any filter criteria for this Service Catalog entry, click **Edit Access Filter** and follow the wizard instructions.
7. Click **Save**.
8. Click **OK**.

1.13.4. Shopping cart

When a user requests one or more services, products, or bundles, Service Manager uses a shopping cart to collect each service request that you make during that session. When the user completes their selections, Service Manager processes the contents of the shopping cart (one or more service requests) internally and submits each one to the appropriate application that fulfills the service request.

1.13.5. Service Catalog interface

The Service Catalog management Web interface is simple enough for non-technical business managers to build and manage their own catalog. A visual workflow makes it easy to see the status of a service request. The simple interface enables managers and approvers to add approvals or make assignments to modify the workflow.

The Service Catalog interface allows users to print an order from the catalog. Users with appropriate access privilege can also print record lists for Service Catalog requests, a detailed record for a request, catalog item definition record lists and detailed records a catalog item definition.

1.13.6. Self-service access

Employee Self-Service (ESS) enables you to connect with a Service Manager application to request a service, provide information, or track previous requests. You don't need any special training or software. If you can connect to the Service Manager server using a supported browser, you can open, modify, and cancel service requests. The **Standard ESS View** is an interface for self-service access and provides a simplified Service Desk interface for you to create, view, and update service requests. To learn more, see the related topics links.

Note: The ESS interface has been designed so that users can easily understand what they need to do; therefore, no Service Manager help icon is available. If users want to see the online help, their System Administrator can provide them with a URL that they can enter into their web browser.

Searching the Service Catalog

The Service Catalog enables users to search for services, items, and bundles in the language in which the user is logged on to the system. For example, if a user logs on in French, the user enters a description in French when searching and the items or bundles available display in French. Also, when the user selects a category to search within, the list of categories displays all categories available to the user in French.

If a manager or a user searches the catalog in a localized language and there is not a localized version of the item, the search will not find the item. However, if a user or manager uses a search filtered on a category, any non-localized items or bundles within the category display in English because the base language in Service Manager is English.

This also means that items created in English but not localized are found by a search, but these items don't display in the search results if a user is logged in to the system in a language other than English. This occurs because there is no localized display name for the item in the system. To prevent this situation, as a best practice, localize all data to the desired language before users start using the Service Catalog.

Search for catalog items

1. Click **Service Catalog** > **Order from Catalog**.
2. Specify the **Category**, if you know it.
3. Specify the item name, if you know it.
4. Specify the search options if needed:

- **Bundles and Items only**: Select this option to limit your search to catalog items whose Type is **Bundle** or **Item**.
- **Any of these words**: Select this option to limit your search to catalog items that contain any of the keywords you enter in the **Search for** box.
- **All of these words**: Select this option to limit your search to catalog items that contain all of the keywords you enter in the **Search for** box.

5. Click **Search**.
6. To view more items in the catalog, click **More results**.

What are Most Popular Requests?

The **Most Popular Requests** are listed on the **Order from Catalog** home page. This lists and tracks the most requested items in the catalog for the current calendar month. The results are based on all system activity and not individual requestor activity.

The Most Popular Requests panel displays as many items as your screen size allows.

Request a service catalog item or bundle

Applies to User roles: User

You can easily order products and services from Service Catalog without needing to contact the service desk. Items in the Service Catalog can include an item picture, description, attachments, and purchasing options. Some items in the Service Catalog are for information only, such as instructions on how to order items from an outside service. Information-only items only have an item picture, description, and attachments; they don't include purchasing options.

Note

You may also order one or more products and related service tasks by choosing a bundle.

To request an item or bundle from the Service Catalog, follow these steps:

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.
3. Click the item you want to request to view the Order Information for that item.

> **Note**
>
> Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.

4. In the Requested for field, select the Individual who will receive the good or service.
5. Select the quantity and any other item options, and then click **Add to Cart** to add the item you want to your cart.
6. From the Order from Catalog home page, click **View Cart/Checkout**.
7. Click **Submit Request**.
8. Type any required information justifying the request.
9. Click **Submit**.

Request a non-cart item

Applies to User Roles:

Self-Service user

When ordering services from the Service Catalog, you can order non-cart items. These items are for services, such as resetting a password, printer support, or desktop support. When you order a non-cart item, you don't need to place the service request in a cart and submit the cart. Instead, your request creates an interaction record and doesn't require any fulfillment.

To request a non-cart item from the Service Catalog:

1. Click **Non-cart Catalog Requests**.
2. In the **Service Catalog** pane, view the list of non-cart services available. Notice that each link is the detailed description of the item. For example, to select a Password Reset item, click the linked detailed item description **Help user reset password**.

   The non-cart service item opens.
3. Click **Submit** to order the item.
4. Enter the following information to describe what the request is for and when you need it:
   1. **Purpose**
   2. **Needed by** (select a date from the calendar)
   3. Urgency: **1-Critical**, **2-High**, **3-Average**, or **4-Low**

4. **This Request is for**
5. **Contact for this Request**
6. **Preferred method of contact**

5. Click **Add Attachment** to add additional information.
6. Click **Submit**.

   Service Manager displays a message stating that the request has been submitted and an interaction record has been opened.

   - To go back to the Service Catalog and browse the available services, click **Back to catalog**.
   - To cancel the request and exit, click **Cancel**.
7. Click **Continue** to exit.

Make a request for multiple recipients

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.
3. Click the item you want to request to view the Order Information for that item.

   Note: Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.
4. Change the Quantity to reflect the number of recipients you are requesting the item for.
5. Click **Split Items**.
6. From the Order from Catalog home page click **View Cart/Checkout**.
7. Click the name of the individual Item/service in your cart.
8. Select the recipient's name in the Requested for field.
9. Click **Save changes**.
   You are taken back to view the contents of your cart.
10. Repeat step 6 through step 9 for the other recipients.
11. Click **Submit request**.
12. Type any required information justifying the request.
13. Click **Submit**.

Save a shopping cart

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.

3. Click the item you want to request to view the Order Information for that item.

   Note: Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.

4. In the Requested for field, select the Individual who will receive the good or service.
5. Select the quantity and any other item options, and then click **Add to Cart** to add the item you want to your cart.
6. From the Order from Catalog home page click **Save your Cart for later**.
7. Type a cart description.
8. Click **Finish**.
9. Click **OK** to confirm the saved cart.

Save a shopping cart template

1. Click **Order from Catalog**.
2. Select a category, and then click **Search** to view available products and services.
3. Click the item you want to request to view the Order Information for that item.

   Note: Some catalog items may be Information-only. You will not be able to add Information-only items to your cart.

4. In the Requested for field, select the Individual who will receive the good or service.
5. Select the quantity and any other item options, and then click **Add to Cart** to add the item you want to your cart.
6. From the Order from Catalog home page click **Save your Cart as a Template**.
7. Type a cart description.
8. Click **Finish**.
9. Click **OK** to confirm the saved cart.

Note: The items in a shopping cart template can't be modified before you submit the request.

Submit a saved shopping cart

1. Click **Service Catalog** > **Saved Carts and Templates**.
2. Select the saved cart that you want to submit from the record list.
3. Click **Checkout**.
   The saved items are added to your active cart.
4. Click **Submit Request**.

5. Type the required information regarding this request.
6. Click **Submit**.
7. Click **OK** to confirm the request was submitted.

Submit a shopping cart template request

1. Click **Service Catalog** > **Saved Carts and Templates**.
2. Select the template cart that you want to submit from the record list.
3. Click **Checkout**.
   The items are added to your active cart.
4. Click **Submit Request**.
5. Type the required information regarding this request.
6. Click **Submit**.

Note: The items in a shopping cart template can't be modified before you submit the request.

Check the status of a request

1. Click **View Open Requests**.
2. Select an interaction that you created. The **Status** field displays the request status.

Modify a catalog request in progress

1. Click **View Open Requests**.
2. Select the interaction that you created.
3. Click **View/Edit Cart**.
   A list of the items currently in your cart opens.
4. Click **View Catalog**. The Service Catalog opens.
5. Add another item to your cart or remove an existing item.
6. Click **Return to Request**. The interaction you created opens.
7. Click **View/Edit Cart**.
   Your modified cart opens.

Cancel a request

1. Click **View Open Requests**.
2. Select the interaction that you created.
3. Click **Cancel Request**.
   The interaction is closed.

- 1 Searching the Service Catalog

- 1.1 Search for catalog items
- 2 What are Most Popular Requests?
- 3 Request a service catalog item or bundle
- 4 Request a non-cart item
- 5 Make a request for multiple recipients
- 6 Save a shopping cart
- 7 Save a shopping cart template
- 8 Submit a saved shopping cart
- 9 Submit a shopping cart template request
- 10 Check the status of a request
- 11 Modify a catalog request in progress
- 12 Cancel a request

1.13.7. Service Catalog approvals

When an employee self-service (ESS) requestor submits an order from Service Catalog, Service Manager automatically creates an interaction that, based on approval requirements, may have to be approved before it can be fulfilled.

The Service Catalog manager can define two types of approvals for an item or bundle: request level, or line item level.

Additionally, Service Manager provides a default request-level approval definition record, named "Service Catalog Approval," that applies to the entire Service Catalog. This default approval definition can be configured by Approval Roles and Approval Activities.

	Request-level approval	Line item level approval
When a user submits a request that requires approvals	Request-level approvals start.	Line item level approvals don't start until all request-level approvals are approved.
When an approval rule creates an approval record	Each approval record that's created is for the whole request. The **Applies To** field in the **Approvals** section in the interaction record displays "Request."	Each approval record that's created is specific to a cart item. The **Applies To** field in the **Approvals** section in the interaction record displays the name of the cart item.
When an approver views an approval record in the approval inbox	The approval record shows the interaction ID in the title, such as "Interaction SD10327."	The approval record shows the cart item ID in the title, such as "Cart Item 20."
If a request is denied and returned	The denial record indicates that the request was denied. Line item level approvals for this request will not start.	The denial record indicates that the request was denied on one or more specific items. (The submitter can know which items were rejected, make modifications, and then resubmit the request.)

1.13.7.1. Approve a request

Applies to User Roles: Service Catalog approver

1. Sign in to Service Manager as a Service Catalog approver, for example, as a requestor’s manager.
   Note: The manager is defined in a requestor's Contact Information.
2. You can approve requests through both the standard and the employee self-services web clients.
   - Through an `index.do` URL:
     Click **Approve Requests**. A list of all the interactions requiring approval opens.
     Note: It's possible to see the Approval Inbox as the default starting page, which is pre-configured in your operator record.
   - Through an `ess.do` URL:
     Click **Approve a Request**. A list of all the interactions requiring approval opens.
     Note: You can access the **Approve a Request** menu item only when **ESSSM-Approval** is selected for the Self-Service Menu field on the Self-Service tab of your operator record.
3. Click **View** to see the request details.
4. To approve the individual request, click **Approve**.
5. To approve multiple requests, select the checkboxes you want and then click **Approve All Selected**.

   The records are now approved and disappear from the view list.

1.13.7.2. Deny a request

Applies to User roles: Service Catalog approver

As a Service Catalog approver, you can deny a request.

To deny a Service Catalog request, follow these steps:

1. Sign in to Service Manager as a Service Catalog approver, for example, as a requestor's manager.
   Note: You can define the manager in a requestor's Contact Information.
2. You can approve requests through both the standard and the employee self-services web clients.
   - Through an `index.do` URL:
     Click **Approve Requests**. A list of all the interactions requiring approval opens.
     Note: It's possible to see the Approval Inbox as the default starting page. You can configure this setting in your operator record.
   - Through an `ess.do` URL:
     Click **Approve a Request**. A list of all the interactions requiring approval opens.
     Note: You can access the **Approve a Request** menu item only when you have selected **ESSSM-Approval** for the Self Service Menu field on the Self-Service tab of your operator record.
3. Click **View** to see the request details.
4. To deny the individual request, click **Deny**.
5. To deny multiple requests, select multiple checkboxes and then click **Deny All Selected**.
6. Type a denial reason and click **OK**. The records are now denied and disappear from the view list.

Note

By default, if a Service Catalog request gets denied and no one has updated this request after seven days, this Service Catalog request will be automatically closed with the "Denied Service Catalog Request" closure code.

1.13.7.3. Default Service Catalog approvals

You can define default approvals that apply globally to all Service Catalog items.

To set up default approvals, the Service Catalog Manager needs to configure the Approval Roles and the Approval Activities that define the approvers, approval conditions and approval levels. For example, different levels of approval roles are required to approve a request depending on the order's cost value. After that, approvers can perform several types of approval tasks.

Note: Settings in Approval Roles and Approval Activities are saved in an approval definition record that's named "Service Catalog Approval," which is a request-level approval.

Define approval roles

User roles: Service Catalog Manager, System Administrator

Approval Roles are a way to identify which users have the capability to approve Service Catalog requests. Service Manager has two OOB Service Catalog Approval Roles: "Use Group" and "1st Level Manager."

Step 1: Create an approval role

1. Sign in to Service Manager.
2. Click **Service catalog** > **Administration** > **Approval Roles**.
3. Click **Search**. Two OOB roles are displayed in the list: "1st Level Manager" and "Use Group."

Note

- 1st Level Manager is the manager specified in the user's contact record.
- Use Group is a group comprising several operators; it is a sample role that shows how you can specify a group as an approver.
- You can select **1st Level Manager** and view the JavaScript that defines the approver dynamically, based on the Contact Record of the requestor. Or, select **Use Group** and view the Expression that defines the approval role Use Group as SYSTEMS ADMIN.

4. Select **Use Group** from the list.
5. Modify the role information as follows:
    - Role Name: testRole
    - Description: testRole
    - Select the **Define Approver** option
    - Enter the following expression under the **Expressions** tab:

      `$L.approval.role.user=callback.contact in $L.file`
6. Click **Add**. A message that confirms the addition of the svcApprovalRole record is displayed.
7. Sign out of Service Manager.

Step 2: Create an approval definition record

1. Type db in the command line, enter ApprovalDef in the **Table** field, and then click the Search icon.
2. In the **Name** field, enter SVCTest.
3. In the Group/Oper column of the table, enter testRole.
4. Click **Add**.

Step 3: Add an approval activity to a Catalog Item

1. Click **Service catalog** > **Administration** > **Manage Items**, and then perform a search for "Digital Badge."
2. On the **Approvals** tab, add "SVCTest" to the **Request Level Approvals** list.
3. Click **Add/Edit Cart Item Approvals**, and then click **Edit Approval Activity**.
4. Select the **Role** option, enter testRole in the **Role** field, and click **Next**.
5. Click **Next**, click **Finish**, and then click **OK** to save the update of the Catalog Item.

Step 4: Verification

1. Click **Service Catalog** > **Order from Catalog**.
2. Click **Services** > **Browse Catalog** > **Personal Productivity Services** > **Authentication and Security** > **Digital Badge** > **Request** > **Next** > **Request Now**.
3. Complete all mandatory fields, and then click **Submit**.
4. Sign in to Service Manager, and search for the interaction that you just created. You should see two pending approvals on the **Current Approvals** tab. One is "General Approval" and the other is the new approval, in which the Approver is the contact of the interaction.

Define approval activities

Applies to User Roles: Service Catalog Manager

The Service Catalog Manager can configure the approval sequence and approval conditions when defining Approval Activities. In this example, the Service Catalog Manager defines required approvals based on the request cost. If the cost value of an order is more than 1000 US dollars, the 1st Level Manager is required to approve as the second level approver; if the order costs over 5000 US dollars, the Technical IT Approver is also involved as the third level approver.

1. Sign in to Service Manager as Service Catalog Manager.
2. Click **Administration** > **Approval Activities**.
3. Select **Add New Approval Activity**. The Add / Edit an Approval Activity window opens.
4. To define approval activities for 1st Level Manager:
    1. Enter a Description. For example, **First Level Manager**.
    2. Select **Role.**
       Note: The option User means you can only specify a particular operator while the option Role allows you to dynamically configure who can approve. For instance, you can set it up so that the manager defined in a user's contact record can approve.
    3. Use the Fill button to select **1st Level Manager** from the list.
    4. Select **2** for the Approval Sequence.
    5. Click **Next**. The Approval Condition window opens.
5. To set up an approval condition for 1st Level Manager:
    1. Click **Add Condition**.
    2. Provide your approval condition. For example, enter the following values:
        - Field in Interaction – svc.cost
        - Comparison - Greater Than
        - Value - 1000
    3. Click **Next**. The Approval Condition is added.
    4. Click **Next**. The Add/Edit Approval window opens, with the new Approval Activity on the list.
6. To define approval activities for Technical IT Approver:
    1. Select **Add New Approval Activity**.
    2. Enter a Description. For example, **Technical IT Approval**.
    3. Select **Role**.
    4. Use the Fill button to select **Technical IT Approver** from the list.
    5. Select **3** for the Approval Sequence.
    6. Click **Next**. The Approval Condition window opens.

7. To set up an approval condition for Technical IT Approver:
    1. Click **Add Condition**.
    2. Provide your approval condition. For example, enter the following values:
        - Field in Interaction – svc.cost
        - Comparison - Greater Than
        - Value - 5000
    3. Click **Next**. The Approval Condition is added.
    4. Click **Next**. The Add/Edit Approval window opens, with the new Approval Activity on the list.
8. Click **Finish**.

Note: When an employee self-service (ESS) requestor submits an order successfully from Service Catalog, a Service Desk interaction is created with each request. You can view all the required approvers in the Approvals section of the interaction.

1.13.8. Service Catalog management

The Service Catalog requires some management so that users have access to the approriate items and services and that requests are approved efficiently. Service Catalog managers are able to organize items and services into categories and bundles and use Service Catalog capability words and catalog items definitions to manage access.

Best practice recommendation is for all Catalog Management to be done in a base language. Localization can be done item by item (using the Localization wizard), or it can be done in mass via export/import of the svcDisplay table contents. This allows for users in a particular locale to view approriate catalog items in a search results. However, note that this alone should not be used as means for access restriction of catalog content according to region or country. Customers who desire such access restrictions should also follow the normal procedures for defining access control to each catalog item.

1.13.8.1. Service Category hierarchies

A hierarchy describes a collection of IT products or services in a Service Catalog. This collection is broad enough to break down into subordinate categories and subcategories. For example, Desktop Services, Employee Services, and Telecommunication Services are examples of hierarchies with categories and subcategories. The following example shows the hierarchy of Desktop Services that includes Software and Hardware categories.

Desktop Services (hierarchy)

Software (category)

Productivity Tools (subcategory)
Catalog item 1
Catalog item 2
Catalog item 3

Microsoft Windows (subcategory)
Catalog item 1
Catalog item 2
Catalog item 3

Hardware (category)

Service Catalog categories

A category describes similar IT products or services that users can select from an online catalog. Similar categories can become components of a product hierarchy. The Service Manager Service Catalog feature includes out-of-box hierarchies, categories, and subcategories. Hierarchies contain categories; categories contain subcategories. Administrators can create new categories and subcategories to meet their operational needs. For example, the hierarchy of Desktop Services includes Software and Hardware categories. The Software category contains the Productivity Tools and Microsoft Windows subcategories.

Desktop Services (hierarchy)

Software (category)

Productivity Tools (subcategory)
Catalog item 1
Catalog item 2
Catalog item 3

Microsoft Windows (subcategory)
Catalog item 1
Catalog item 2
Catalog item 3

Hardware (category)

Add a Service Catalog category

1. Click **Service Catalog** > **Manage Items**.
2. Click **Add New Category**.
   When you launch the New Service Catalog Category wizard, Service Manager prompts you for the required information.
3. Complete the following fields:

   Category Name (mandatory): The unique name of the Service Catalog entry.

   Display Name: The display name of the Service Catalog entry. If left empty, the Category Name is used.

   Description (mandatory): A brief description of the Service Catalog entry.
4. Click **Next**.
5. For a new category, click **Top Level**.
6. Click **Next**.
7. Specify whether the new category will contain subcategories or items/bundles.
8. Click **Next**.
9. Click **OK** to confirm the new category.

   Service Manager adds the new category.

Add a Service Catalog non-cart category

Applies to User roles: Service Catalog Manager

Before you can add non-cart items, you must create a non-cart category for them.

To add a non-cart category:

1. Click **Administration** > **Manage Items**.
2. Click **Add New Category**.
   When you launch the New Service Catalog Category wizard, Service Manager prompts you for the required information.
3. Complete the following fields:

   Category Name (mandatory): The unique name of the Service Catalog entry.

   Display Name: The display name of the Service Catalog entry. If left empty, the Category Name is used.

   Description (mandatory): A brief description of the Service Catalog entry.
4. Click **Next**.
5. For a new category, click **Top Level**.
6. Click **Next**.
7. Specify that the new category will contain items/bundles.
8. Click **Next**.
9. Click **OK** to confirm the new category.

   Service Manager adds the new category, and the new category record is displayed.
10. Select the **Non-cart category** checkbox.
11. On the **Access** tab, select one or more capability words to make this item available to certain users; or leave this field blank to make it available to all users.
12. Click **Save**.

    Now you can add non-cart items to this category. The items will be displayed on the **Items/Bundles in this Category** tab of this category record.

Edit a category

Applies to User roles: Service Catalog Manager

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Select an existing category from the **In Category** drop-down list.
3. Click **Search**.
4. Select a category from the record list.

5. Add or change information on the form. If necessary, press F1 (web client) or Ctrl+H (Windows client) to view help for each field.

   For example, add or change the following fields on the **Category Details** tab.

Field	Description
Name	The unique name of the bundle or item. This is a mandatory field.
Display Name	The display name of the bundle or item.
Owner Group	The assignment group to which the owner of this category belongs.
Owner	The contact who is the owner of the process related to this category.
Parent Categories	The hierarchical parents of the bundle or item. A parent must be a category.
Description	A short description of the bundle or item.
Current Phase	This read-only field indicates the current phase of the item in the service lifecycle.
Status	The current lifecycle status of the bundle or item. Available Status values vary depending on the current workflow phase of the bundle or item: ◦ Service Catalog phase: Requirements, Defined, Analyzed, Approved, and Chartered, Designed, Developed, Built, Tested, Released, Operational, and Retiring. ◦ Retired Service phase: Retired.

Field	Description
Non-cart category	A checkbox that indicates if the category is a non-cart category that includes non-cart items. Once you select the **Non-cart category** checkbox, the **Support category** checkbox appears.
Contains	Select either Categories or Items/bundles to indicate contents of the category.
Sort Order	The sort order of the values in the Status list.

6. On the **Access** tab, update the list of capability words that control access to the category, or leave it blank to make the category available to all users.
7. On the **Attachments** tab, upload supporting documents for the category.
8. On the **Subcategories** tab, add subcategories for the category.
9. On the **Image** tab, specify an image for the category or remove the existing image.
10. Click **Save**.
11. Click **OK**.

Delete a Service Catalog category

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Select an existing category from the **In Category** drop-down list.
3. Click **Search**.
4. Select a category from the record list.
5. Click **Delete**.
6. Click **OK**.

Service Catalog subcategories

Subcategories are a way to refine a category into smaller groups. For example, the hierarchy of Desktop Services includes Software and Hardware categories. The Software category contains the Productivity Tools and Microsoft Windows subcategories. You could add more subcategories as the need arises.

Desktop Services (hierarchy)

Software (category)

Productivity Tools (subcategory)
Catalog item 1
Catalog item 2
Catalog item 3

Microsoft Windows (subcategory)
Catalog item 1
Catalog item 2
Catalog item 3

Hardware (category)

Subcategories aren't localized and display in English.

Add a Service Catalog subcategory

1. Click **Service Catalog** > **Manage Items**.
2. Click **Add New Category**.
   When you launch the New Service Catalog Category wizard, Service Manager prompts you for the required information .
3. Type a **Category Name** and brief **Description**.
4. Click **Next**.
5. For a new subcategory, click **Subcategory of** and choose a parent category from the drop-down list.
6. Click **Next**.
7. Specify whether the new category will contain subcategories or items/bundles.
8. Click **Next**.
9. Click **OK** to confirm the new subcategory.

   Service Manager adds the new subcategory.

Contents

- 1 Service Catalog categories
  - 1.1 Add a Service Catalog category
  - 1.2 Add a Service Catalog non-cart category
  - 1.3 Edit a category

- 1.4 Delete a Service Catalog category
- 2 Service Catalog subcategories
  - 2.1 Add a Service Catalog subcategory

1.13.8.2. Service Catalog items and bundles

Items are standard catalog offerings with either a product or a service task. Catalog items can be Information-only. Information-only items can be viewed in the catalog but can't be ordered.

Bundles are standard catalog offerings with both product and service components that a user can select with a single service request. Each bundle contains one or more products and related service tasks. You can't put Information-only items in a bundle.

You can create new items and bundles or edit the existing ones.

1.13.8.2.1. Add a Service Catalog item

Applies to User roles: Service Catalog Manager

You can add two types of Service Catalog items: service item, and support item. Follow these steps if you want to add a service item.

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Click **Add New Service Item**.

   Note

   To add a support item, click **Add New Support Item** instead.

3. Complete the following fields. Fields that are marked with a red asterisk are mandatory fields.

Field	Description
Item Name	The unique name of the item. This must be unique for each item, as it is the unique identifier of the item in the catalog.
Display Name	The display name of the item. The display name you assign to the item doesn't have to be unique within the catalog.
Business Service	The business service that's supported by this item.
Description	A short description of the item.
Detailed Description	A detailed description of the item. This may include instructions on how to request the item. **Note** You can format the text in this field. Refer to the HTML Editor help for more information.
Currency	The currency in which the cost of the item is specified.
Internal Costs	The costs that the internal team needs to provide the item.

Field	Description
Charges	A numeric value that represents the amount of money that should be charged to the end user who requests the item.
Recurring Charges	A numeric recurring charge for the item.
Per	Specifies how often the recurring cost payment is due. The drop-down list provides options including Hour, Day, Week, Month, or Year.
Information-only item checkbox	Select this option, if desired. Some items in the Service Catalog are for information only, such as instructions on how to order items from an outside service. Information-only items only have an item picture, description, and attachments; they don't include purchasing options. **Note** If you selected Information-only, you will not be prompted to specify the Internal Costs, Charges, Recurring Charges, Currency, or Per information.

Field	Description
Restrict request to single item checkbox	This option limits the quantity of this item that a user can order to one. If you select this option, the quantity field on the order form will be inactive.

4. Click **Next**.
5. Select a connector and parent category.
6. Click **Next**.
7. Depending on the connector you choose, you need to select either the Request Category, Request Model, Change Category, or Change Model.
8. Follow the screen prompts to complete the information required.
9. Click **Finish**, and then click **OK** to confirm the new item.
10. In the Status field, update the status from **Requirements** to **Operational**. This procedure moves the workflow phase from **Planned** to **Published**. The new item is active and visible now in Service Catalog.
11. Click **Save** and **OK**.

1.13.8.2.2. Add a Service Catalog non-cart item

Applies to User roles: Service Catalog Manager

Non-cart items can be added to define a separate catalog of support services. For example, you can add items such as:

- Password Reset Service
- Printer Support
- Desktop Support

Each non-cart item is requested with a single-click; therefore, it doesn't require the user to place the service request in a cart and submit the cart. Instead, these are one-click items, meaning the request creates an interaction record and doesn't require any fulfillment.

Note

Before you can add a non-cart item, you must create a non-cart category for it.

To add a non-cart item to the Service Catalog:

1. Click **Administration** > **Manage Items** .

   The Search Catalog Item Definitions form opens.

2. Select **Active** to make this new non-cart item available in the Service Catalog.

3. Select **Non-cart item** to distinguish this catalog item as not requiring the fulfillment process.

4. Specify the following information to create the new non-cart catalog item.

   a. Type the **Name** of the new catalog item. This must be unique, as it is the unique identifier of the item in the catalog.

   b. Type the **Display Name**.

> **Note**
>
> The display name you assign to the item doesn't have to be unique within the catalog.

c. In the Type field, select **Item**.

d. In the **Available to** field, select one or more capability words to make this item available to certain users; or leave this field blank to make it available to all users.

e. Select a **Language**.

f. Leave the **In category** field blank.

Important: You must leave this field blank at this point, as non-cart categories aren't available from the list. If you choose a category at this point, the classification of non-cart item is automatically changed and the item becomes a cart item.

g. In the **Interface Type** field, select **Open a Service Desk Interaction**. This is how the user's request is going to be fulfilled.

h. In the **Description** field, type the description of how this request should be fulfilled.

5. Click **Add New Service Item**.

   You will continue to add this new item in the New Service Item Wizard form.

6. Type the **Detailed Description**.

> **Note**
>
> You can format the text in the Detailed Description field. Refer to the HTML Editor help for more information.

7. Click **Next**.

   The New Service Catalog Category wizard opens. The Connector is set to **Open a Service Desk Interaction**.

8. In the **In Category** field, select the non-cart category you have created for this item.

9. Click **Next**.

   The Service Catalog Interaction wizard opens.

10. In the **Initial Impact Assessment** field, select an impact level for interaction records requesting for this item.
11. In the **Assignment** field, select an assignment group.
12. Click **Next**.

    Service Manager adds the catalog item.

13. Click **OK** to confirm the new item.

    The new catalog item record is displayed.

1.13.8.2.3. Add a Service Catalog bundle

Applies to User roles: Service Catalog Manager

Note

If a bundle has a connector specified, an interaction is created when it is ordered. Once the interaction is approved, a request (for example, RF10002) is created and related to the interaction. The request contains all the items in the bundle.

If a bundle has no connector specified, an interaction is created when it is ordered. Once the interaction is approved, a related record is created for each item in the bundle based on the item's connector setting on its Connector Details tab.

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Click **Add New Bundle**.
3. Complete the following fields:

Field	Description
Bundle Name	The unique name of the bundle. This must be unique for each bundle, as it is the unique identifier of the bundle in the catalog.
Display Name	The display name of the bundle.
Description	A short description of the bundle.
Detailed Description	A detailed description of the bundle. This may include instructions on how to request the bundle. **Note** You can format the text in the Detailed Description field. Refer to the HTML Editor help for more information.
Restrict request to single bundle checkbox	If desired, select this option to limit the quantity of this bundle that a user can order to one. If you select this option, the quantity field on the order form will display "1" and will not be editable.

4. Click **Next**.
5. Choose a parent category for the bundle to belong to from the drop-down list, and then click **Next**.
6. Choose whether it is a New Item or an Existing Item.

    • If you select **New Item**, the New Service Item Wizard opens. You must complete the steps in the New Service Item Wizard before you can finish

adding a new bundle.
- If you select **Existing Item**, continue to follow the steps in this topic.

7. Specify the information for the existing item or bundle that you want to add to the new bundle.

> **Note**
>
> You can't add items tagged Information-only to a bundle.

   - Choose an item or a bundle to add from the drop-down list. The drop-down list will not include information-only items.
   - Select the quantity of the item or bundle to add.
   - Select whether the item or bundle is Mandatory, Default, or Optional.

8. Click **Next**.
9. Select the number of items that should be included in the bundle, and then click **Next**.
10. Continue to add additional items and bundles to this new bundle until you are satisfied with its contents.
11. When the bundle is complete, select **No** if you don't want to add additional items to this bundle now.
12. Click **Next**.
13. (Optional) Perform the following actions to add a Create New Request connector for the bundle:

    a. On the **Connector Details** tab, click **Change Connector** to open the Service Catalog Connector page.

> **Note**
>
> You can update the connector by clicking the **Add/Edit Information** button or remove the connector by clicking the **Clear Connector** button on this tab after you add a new connector successfully.

    b. Select **Create New Request** from the **Connector** drop-down list.

    c. Click **Next**.

d. Select a Request Category, Request SubCategory, and Request Model for the bundle.

e. Click **Next**.

f. Configure the **Urgency**, **Impact**, and **Assignment** fields.

g. Click **Finish**.

14. Click **OK** to confirm the new bundle.

1.13.8.2.4. Add user selections in a Service Catalog item

You can add user selections for a Service Catalog item so that users can provide additional information in their requests. For example, after you add a new Service Catalog item that has a Currency value, you can add user selection options to it for cost adjustment purposes.

To add user selections, follow these steps:

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type or select search criteria for the item you want to modify and click **Search**.

   Click the item in the search list to select it, if necessary. The catalog item definition record opens.
3. On the **User Selections** tab, click **Definition**, and then click **Add User Selection**. The Dynamic Field Description form opens.
4. Complete the following fields:
   - In the **Name** field, type a field name for the user selection. For example, **City**.

     Note

     The field name must be alphanumeric, contain no spaces and start with a character.
   - In the **Label** field, type a display name for the user selection. For example, **City**.
   - In the **Help Description** field, type instructional text about how to use the user selections you will create. The text will be displayed as field help information on the user selection subform being created. Currently, the SMA Service Portal supports help description. In the web client, when you preview the user selection or order from catalog, you can press F1 to display the help description as well.
   - In the **Display Type** field, select a display type.

5. Click **Next**. The next wizard screen varies depending on your selection for display type. Here are some highlights.

   Text

   Select a type in the **Type** drop-down list.

   - If you select **Date/Time**, you can only enter a static date/time value, for example: `07/31/2008 00:00:00`. The value you entered will be parsed according to the Date Format defined in your operator record. Dynamic date/time values (for example, `tod()+'10 00:00:00'`) are unsupported.
   - If you select **String**, don't select the field in array type when you define the field validation rule as the field in array type is unsupported in user selections.

   In the **Mandatory Condition**, **Visible Condition**, and **Read-Only Condition** fields, enter a RAD expression that specifies under which condition the user selection is mandatory, visible, or read-only.

   Tip

   Be sure to use the correct syntax for RAD expressions. The following is an example: `affected.item in $L.file>"222323"`. To learn the syntax of RAD expressions, see System language.

   For example, enter the following conditions:

   - Mandatory condition: `false` (which indicates the user selection is not mandatory when users order the item)

     Note

     If you set the Visible condition to `false` or set the Read-Only condition to `true`, the system always validates the Mandatory condition as `false` regardless of your input to the **Mandatory Condition** field.

   - Visible condition: `true` (which indicates that the user selection is always visible in Service Manager)

Note

The behavior of the Visible condition has the following limitations:

- If the Visible condition depends on a MultiText type field, the DVD selection statement in dynamic forms is not triggered in time to apply the condition.

  For example, you have a MultiText field "test1" and a Text field "test2" that has the Visible Condition set to `[test1]="ok"`. When you enter ok into the "test1" field and then shift the focus, the "test2" field is not displayed until you save the form.

- The DVD select statement in dynamic forms can't interact with fields outside the dynamic form. Therefore if a field outside the dynamic form (such as the status of an Interaction) is changed, the change is not recognized by the widget inside the dynamic form. Additionally, any changes to the dynamic form don't trigger the DVD outside the dynamic form.

- Read-only condition: `true` (which indicates that the user selection is always read-only in Service Manager)

MultiText

Enter RAD expressions in the **Mandatory Condition**, **Visible Condition**, and **Read-Only Condition** fields. See the description above for more information.

Checkbox

Enter the **Additional Cost** field as necessary, and then enter **Visible Condition**, and **Read-Only Condition** fields. See the description above for more information.

Pick List

Complete the following fields:

- **Value**: the actual value of the field selection. For example, **California**, and **New York**.
- **Label**: the display value of the field selection. For example, **CA**, and **NY**.

- **Cost Adjustment**: the cost adjustment value (numeric). For example, **10**, and **20**.

  > **Note**
  >
  > Do not include the currency symbol (for example, $) in the **Cost Adjustment** value. Otherwise, the cost calculation of the item will be incorrect.

Select **Drop-Down List** or **Radio Buttons** to display the user selection options. Next, enter RAD expressions in the **Mandatory Condition**, **Visible Condition**, and **Read-Only Condition** fields. See the description above for more information.

6. Click **Next**. The next wizard screen varies depending on your selection for display type. Here are some highlights.

   Text

   If you select the **String** type, select a validation rule.

   - **Record in Table:** Select a table in the **Table Name** drop-down list, and then select a field in the **Field Name** drop-down list. Enter a query in the **Query** field. Enter a number in the **Multiple Selections** field to display multiple rows of text. Select a value in the **Group by Fields** drop-down list to group the records.
   - **Regular Expression:** In the **Regular Expression** field, type the regular expressions you want to use. In the **Regular Expression Error Message** field, type an error message. If the user's input in a portal fails the regular expressions validation, the system displays this error message on the user selection subform as you defined.
   - **JavaScript:** In the **Script Name** field, select the associated JavaScript created in the ScriptLibrary. In the **Parameter** field below, define the input parameters if required. For more information about the rules when defining JavaScript as the validation rule, see the *JavaScript coding guidelines to validate user selections* section below.

   If you select the **Number** or **Date/Time** type, enter values in the **Greater than** and **Less than** fields.

   MultiText, Checkbox, and Pick List

Continue with the next step.

7. Click **Next**. The **Default Value Configuration** form opens. You can either configure the default values for this user selection or click **Next** to skip this step.

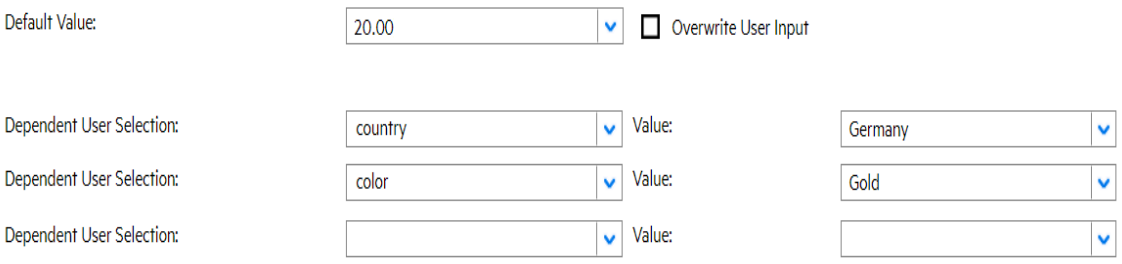
   When adding a default value:

   - If the default value has no dependency on other user selections, enter a default value or select a value from the **Default value** drop-down list. See the following screenshot as an example:

     In this example, the user selection is a Checkbox and the default value has no dependency on other user selections. Service Manager doesn't display the dependent fields and values. You can select a default value from the drop-down list and then directly click **Next**.

   - If the default value's judgement conditions are based on the settings of other user selections, click **Add**. On the **Add a new default value** form, enter a default value or select a value from the **Default value** drop-down list, and then set up dependencies between the value and other user selections. Note that the value of a user selection only depends on the value selections of Pick List and Checkbox on top of the current user selection in the Service Catalog item.

     You can select up to three user selections together with their values. The conditions are in coordination relations and a default value must meet all conditions. See the following screenshot as an example:

     In the example, the user selection is a Pick List and the default value depends on prior user selections including Country and Color, both of which are Pick List as well. The default value is 20.00 when the Country is Germany and the Color is Gold.

Caution

If the user selection is a Text field and you have configured validation rules in *step 6*, make sure the default value you entered is valid and follows these validation rules. See the following examples.

- If you select the **Record in Table** rule, set the Table Name to `device`, and set the Field Name to `Display Name`, all display names in the device table are valid default values. For example, `Adobe Reader`.
- If you select the **Regular Expression** rule and then set the expression to `[a-zA-Z]+.[a-zA-Z]@advantage.com`, `sm.user@advantage.com` is one of the valid default values.
- If you select the **JavaScript** rule, do not configure a default value for the user selection because the JavaScript output is dynamic. Skip this step.

Note

After selecting the **Overwrite User Input** checkbox, even if a user manually modifies the default value of the user selection when placing an order, Service Manager overwrites the user's modification with the default value as long as the conditions are met. This user selection becomes read-only.

You can add more default values as necessary. If a condition matches multiple default values, Service Manager assigns default values according to their sequences. The highest-ranking value in the sequence will be given the highest priority. You can click **Move Up** or **Move Down** to arrange the ranking of the default values. To remove a default value, select this value and then click **Remove**.

8. Click **Next**. The user selection is added.
9. Click **Save** to save the item.
10. If desired, repeat the steps above to add more user selections.

    All the user selections you added are now displayed on the **User Selections** > **Definition** tab of the item.

11. You can change how your list will be displayed to users if desired.

- Select one or more items in the list and click **Remove User Selection** to remove them from the list.
- Select one or more items in the list and click **Move Up** or **Move Down** to move them up or down on the list.

> **Note**
>
> You can't remove a user selection if it is depended by other user selections. In addition, as mentioned in *step 7* above, the value of a user selection only depends on the value selections of Pick List and Checkbox on top of the current user selection in a Service Catalog item. If you have configured default values for user selections, you may not move a user selection freely because the move up and move down actions can't break this rule.

12. Click the **Preview** tab to see a preview of the updated user selection subform.

> **Note**
>
> This subform will appear on the order form when a user orders this item from the Service Catalog. If the user selects one or more options on the subform, Service Manager automatically adjusts the total cost of the order.

13. Click **Save** to update the catalog record.

JavaScript coding guidelines to validate user selections

When you define your own JavaScript validation rule for a string field in user selections, follow the guidelines below:

- Put your JavaScript under the **ExternalLookup** package.
- Use the following attributes of **context** variable object to define the JavaScript:
  - start: Starting from where the returned value list will be displayed in Service Portal. Currently, this attribute is hardcoded to 1.
  - pageSize: Size per page that will be displayed in Service Portal. Currently, this attribute is hardcoded to 50.
  - searchText: What is used for searching inside the return value list from Service Portal. For example, `var search=context.searchText;`

    - paramList: Array for input parameter list. For example, `var region=context.paramList[0];`
- Include the following attributes in the Return object of the JavaScript:
    - total: Total count of all records.

      Note: If you are working with the SMA Service Portal and the total count is greater than 50, be sure to return the total count. Otherwise, the Portal UI will only get the first 50 records. If you are working with the SMA Service Portal and the total count is less than 50, it is optional to include this attribute in the Return object.
    - fields: Columns displayed for each record in Service Portal. If one field should be displayed as a display field, set the attribute "display" to true. For example, `fields: [{name: 'id', label: 'ID'}, {name: 'value', label: 'Full Name', display: true}]`. According to this example, the system displays two columns: ID and Full Name; In the comfill of the user selection in Service Manager, Full Name will be displayed; In the request tracking page, the value of user selection will be displayed as Full Name.
    - data: data array containing values.

Limitations:

1. There are 30 seconds timeouts in the SMA Service Portal user interface. The SMA Service Portal displays an error message if the JavaScript execution time is longer than 30 seconds.
2. In Service Manager, a Service Catalog administrator cannot define the default value for the user selections defined with JavaScript validation (via “Preview” panel).
3. Neither the SMA Service Portal nor the ESS portal recognizes or calculates the additional price defined on the return value of the JavaScript.

1.13.8.2.5. Add multiple selections for Service Catalog item user options

Applies to User Roles: Service Catalog Manager

You can add multiple selections for Service Catalog item user options so that users can select multiple values in an Item Option field when ordering a catalog item.

In addition, you can save the existing User Selections in a catalog item as a template, and then apply the template to another or multiple catalog items.

Add multiple user selections

To add multiple user selections for a Service Catalog item, follow these steps:

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type or select search criteria for the item you want to modify, and then click **Search**.

   Click the item in the search list to select it, if necessary. The catalog item definition record opens.

   > **Note**
   >
   > You must select an item of the "item" type. You can't select a bundle or category.

3. On the Definition tab in the User Selections page, click **Add User Selection** to open the Dynamic Filed Description form.
4. Populate the fields in the form as follows:
   - In the **Name** field, type a field name for the user selection. For example, City.

     > **Note**
     >
     > The field name must be alphanumeric, contain no spaces, and start with a character.

   - In the **Label** field, type a display name for the user selection. For example, City.

- In the **Display Type** drop-down list, select **Text**.

5. Click **Next**.
6. Populate the fields in the form as follows:
    - In the **Type** drop-down list, select **String**.
    - Specify the **Mandatory Condition** and **Visible Condition** fields based on your business need.
7. Click **Next**.
8. In the **Validation Rule** option box, select **Record in Table**.
9. Complete the following fields:
    - In the **Table Name** field, type or select the name of a lookup table.
    - In the **Field Name** field, type or select a field of the lookup table as a lookup field.

      Note

      Do not select the field in array type as it's not supported in user options.

    - In the **Multiple Selections** field, type an integer equal to or greater than 2 to allow users to select multiple values that match the selected table and fields.

Note

For The value specified here defines the minimum number of rows that are displayed on the section of item options in the fulfillment record.

- If the actual number of the selected items is less than the value specified here, the remaining rows still appear as blank. For example, if you set the value to “5” and the user selects two items from user options, you can see two rows with items plus three empty rows on the **Options** tab in the fulfillment record.
- If the actual number of the selected items is more than or equal to the value specified here, the rows increase automatically to show all the selected items with an additional empty row. For example, if you set the value to “5” and the user selects six items from user options, you can see six rows with items plus one empty row on the **Options** tab in the fulfillment record.

    - Complete the other fields as needed.

10. Click **Next**.

Now you have successfully created multiple user selections for one service catalog item.

To edit the User Selection, double-click it. The User Selection modification wizard opens. You can click **Next** to modify the User Selection, and click **Finish** to save it.

Define a User Selection template

To define a User Selection template, follow these steps:

1. On the Definition tab in the User Selections page, select the desired user selection, and then click **Save As Template** to open the User Selection Templates Information wizard.
2. Populate the fields in the form as follows:
    - In the **Template Name** field, type a name for your template.
    - In the **Template Description** field, type description of your template.
3. Click **Finish**.

Note

- To modify the saved template, navigate to **Service Catalog** > **User Selection Templates**, click **Search**, and then select the template to modify it.
- To add User Selections in the selected template, click **Add User Selection**.
- To remove User Selections in the selected template, click **Remove User Selection**.

Apply a User Selection template

You can apply the created template to one single catalog item, or apply it to multiple catalog items.

To apply the template to a single catalog item, follow these steps:

1. Click **Service Catalog** >**Administration** > **Manage Items**, and then open the catalog item by performing a search.
2. On the Definition tab in the User Selections page, click **Apply template** to display a list of templates.
3. Select the template from the list to apply. The User Selections from the template are automatically added to the catalog item.

To apply the template to multiple catalog items, follow these steps:

1. Click **Service Catalog** >**Administration** > **Manage Items**.
2. Perform a search by using appropriate search criteria to display a list of desired catalog items.
3. Select all items or some items, and then click **Mass Apply User Selection Template** from the list toolbar.
4. Select the template from the list. The User Selections from the template are automatically added to the selected catalog items.

Note

- The system displays a dialog box and prompts you to confirm when there are duplicated names between the template and the current user selections. If the names are different, the system imports the template directly.
- If there are are duplicated names between the template and the current user selections but you still click **Yes**, the template importing process will overwrite your current user selections. Before importing a template, you must check and make sure that the type, dynamic field validations, dynamic field description of pick lists, and dependencies of the imported user selections match those ones in the template. After importing a template, you also need to check and make sure that the sequence of the user selections matches those ones in the template. Otherwise, these user selections may work incorrectly.

1.13.8.2.6. Edit a bundle or item

Applies to User roles: Service Catalog Manager

To edit a bundle or item, follow these steps:

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. In the **Type** field, select **Bundle** or **Item** from the drop-down list.
3. Optionally, enter additional search criteria.
4. Click **Search**.
5. Select a record from the record list.
6. Add or change information on the form.

   For example, add or change the following fields on the **Bundle Details** or **Item Details** tab.

Field	Description
Name	The unique name of the bundle or item. This is a mandatory field.
Display Name	The display name of the bundle or item.
Owner Group	The assignment group to which the owner of this bundle or item belongs.
Owner	The contact who is the owner of the process related to this bundle or item.
Business Service	The parent service that's supported by this bundle or item.
Parent Categories	The hierarchical parents of the bundle or item. A parent must be a category.
Review Frequency	The frequency at which the bundle or item needs to be reviewed: Weekly, Monthly, Quarterly, or Yearly.
Next Review Date	The next review date of the bundle or item.
Description	A short description of the bundle or item.
Detailed Description	A detailed description of the bundle or item. This may include instructions on how to request the bundle or item.

Field	Description
Current Phase	This read-only field indicates the current phase of the item in the service lifecycle: planning phase, published phase or retired phase.
Status	The current lifecycle status of the bundle or item. Available Status values vary depending on the current workflow phase of the bundle or item: ◦ Planning phase: Requirements, Defined, Analyzed, Approved, and Chartered, Designed, Developed, Built, Tested, Released, and Operational. ◦ Published phase: Operational, Retiring, and Retired ◦ Retired phase: Requirements, Analyzed, Approved, and Retired.
Information-only item checkbox	Indicates if the bundle or item is for information only.
Non-cart item checkbox	Indicates if the item is a non-cart item. it doesn't require the user to place the service request in a cart and submit the cart.
Restrict request to single bundle checkbox Restrict request to single item checkbox	Indicates if only one bundle or item is allowed in each request.

Field	Description
Currency	The currency in which the cost of the bundle or item is specified.
Internal Costs	The costs that the internal team needs to provide the item.
Charges	A numeric value that represents the amount of money that should be charged to the end user who requests the item. You can manually enter a numeric value or click the **Cost Calculation Wizard** link to select the method that should be used to calculate the actual cost of this bundle or item. **Note** The **Cost Calculation Wizard** link is available only in bundles.
Recurring Charges	A text box in which you can enter a numeric recurring cost for the bundle or item.
Per	Indicates how often the recurring cost payment is due: Hour, Day, Week, Month, or Year.
Sort Order	The sort order of categories and items in the catalog. Note that this field is not used in the SMA Service Portal.

7. On the **Additional Details** tab, type the following information:

- Fulfillment Summary : a description of the fulfillment steps (the associated tasks, groups to which they are assigned, and so on) of the bundle or item.
- Value Proposition: a value proposition of the service.

8. On the **Connector Details** tab, click one of the following links, and then follow the wizard to update the connector configuration.
   - **Change Connector**: Add a connector or change the existing connector.
   - **Add/Edit Information**: Edit the settings of the existing connector.
   - **Clear Connector**: Remove the existing connector.
9. (For a bundle only) On the **Components** tab, click **Add Items to Bundle** or **Remove Items from Bundle** to update the list of items in the bundle.
10. On the **Access** tab, update the list of capability words that control access to the bundle or item, or leave it blank to make the bundle or item available to all users.
11. On the **Approval** tab, follow the on-screen instructions to update the list of additional approvals that are required for this specific bundle or item.
12. On the **Agreements** tab, add or remove SLAs to restrict access to this bundle or item only to users with the specified SLAs. For details, see **Add a Service SLA to an item or bundle**.
13. On the **Delivery Targets** tab, add or remove delivery targets for the bundle or item.
14. On the **Attachments** tab, upload supporting documents for the bundle or item. The attachments can be any types of supporting documents, including SLA (Service Level Agreement), SQP (Service Quality Plan), SIP (Service Improvement Plan), SDP (Service Design Package) and so on.

    Note

    The supported attachment file types and sizes are customizable.

15. (For an item only) On the **User Selections** tab, configure user selections for the item. For details, see Add user selections in a Service Catalog item.
16. (For an item only) On the **Activities** tab, select an update type and provide a description of the update.
17. (For an item only) On the **Related Records** tab, link or unlink an existing record that's related to the item, such as a change record and a knowledge record.
18. On the **Image** tab, specify an image for the bundle or remove the existing image.
19. Click **Save**.
20. Click **OK**.

1.13.8.2.7. Accessing user selections

After a fulfillment record is generated, the options selected by the user are saved to the "userOption" table together with the fulfillment record ID and the interaction ID. Compared to saving user options in xml format to fulfillment records, you can easily access the user selection information from this table for purposes such as reporting.

Related topics

Service Catalogs
Service Catalog items and bundles
Add an item
Add a Service Catalog bundle

1.13.8.2.8. Displaying currency in the Service Catalog

Catalog items prices are stored in the currency in which the item was added to the catalog. This allows the definition of catalogs in a baseline currency. A catalog administrator views the catalog item in the currency in which it was added, which is not necessarily the item's display currency. When editing an item, the catalog manager views the catalog items in the currency in which the items were added.

There is a display currency field defined in the operator record for each operator. The value is selected from a drop-down list of the active currencies in the system. The value of display currency defaults to the root currency of the system as defined in the company record. In addition, the currency information record for a currency defines how to display the currency and any associated currency symbol. There is a currency information record in the system for each of the available currency codes. You can use this record to define how you want a currency to display.

The currency of items in a bundle defaults to the main bundle currency. This means that a bundle can't have items with different base currencies. For example, a line item of $20 US dollars within a bundle with a currency of Franks will be viewed by the bundle as Franks, and the $20 item then displays as 20 Franks. The system doesn't convert currencies within bundles.

Warning: You should not configure a bundle with any items (bundles or line items) of a different currency as they will not be converted to the proper currency value. The sub bundles and line items will simply have the face value taken and have the bundle's currency applied to it.

When a users orders from the catalog, the item price displays in the display currency set by the user's operator record. A currency conversion table is used for this display. Conversions are rounded to two decimal places. The conversion rates don't display. Approvers see the cart item price displayed in the display currency setting for the approver's operator record.

The user selection items always displays in the original currency that the requester (operator) used to order the item. For example, if an approver is viewing the request, but the approver uses a different currency than the requester, the item cost displays in the currency of the requester.

New or modified catalog items use the conversion rate available in the conversion table at the time item is added to the cart. The item is not approved at this point in the process. If there is no rate for that date in the conversion table, then the system uses a fuzzy logic to select a rate before the date. Catalog managers are responsible for the maintenance of the conversion table and should update their conversion tables only as often as they need for accuracy. The cart item prices are stored in the root currency of Service Manager as converted from the catalog currency using the rate specified in the conversion table.

1.13.8.2.9. Format a currency for display

Applies to User Roles:

System Administrator

You must have the **SysAdmin** capability word to use this procedure.

To format a currency for display:

1. Click **System Administration** > **Base System Configuration** > **Currencies**.
2. Click **Search**.
3. Select the currency to format.
4. Specify the symbol to use when the currency is displayed.
5. Use the drop-down list to specify whether the currency symbol displays before or after the numeric display.
6. Specify the symbol for decimal placement (. or ,).
7. Specify the number of digits after the decimal.
8. Specify the symbol used to separate groups of numbers in the currency display (. or ,).
9. Click **Save**.

1.13.8.3. Defining a catalog item that results in the creation of a subscription

The Service Catalog manager can define whether a request for a Catalog Item results in the creation of a subscription linked to the goods and services delivered. When the **Create Subscription** checkbox is selected on the Catalog Item Definition record, a subscription record is created after the request is approved.

Note

The **Create Subscription** checkbox is unavailable for Support Catalog Items.

The Service Catalog manager can flag a Catalog Item as available for individual and/or department subscription. When browsing the catalog to request a subscription, the Department Requester can select either individual or department based on what's selected in the Catalog Item definition record.

A collection of items within a bundle can also be made available for subscription. It is important to note that a bundle can only contain items that are defined as available for Individual subscription. If a bundle includes an item that's defined exclusively as Department, the user receives an error message when they attempt to select the bundle from the catalog listing.

A user is not be able to request items that they are already subscribed to or that are in a pending status.

Define which catalog items are available for subscription

1. Verify that the subscription mode is set to *Service Catalog* in the Configuration Management environment record.
2. Click **Service Catalog** > **Administration** > **Manage Items**.
3. Type `Item` in the **Type** field and click **Search**.
4. Select a Catalog Item from the record list and click on the **Connector Details** tab.

5. Select the **Create Subscription** checkbox.

   When the checkbox is selected, two additional fields open.

   > **Note**
   >
   > The **Create Subscription** checkbox is unavailable for Support Catalog Items.

6. In the **Subscription Item** field, fill or find the name of the item that will post to the Subscription record and Change record that's created during the fulfillment process.
7. In the **Subscription Availability** field, select either **Individual** or **Department**. This indicates whether an individual or department subscription can be made against the Catalog Item.

   > **Note**
   >
   > If you select Department, the user will not be able to specify a quantity.

8. Click **Save**.
9. Click **OK**.
   Users can now subscribe to this catalog item.

Contents

- 1 Define which catalog items are available for subscription

1.13.8.4. Adding delivery objectives for a catalog item or bundle

Each catalog item or bundle definition record contains a Delivery Targets tab where a catalog manager can add or remove multiple objectives. The following information is available on the Delivery Targets tab:

Column name	Description
Name	A character field that contains the title of a delivery objective.
Deliver Within	A numerical field that contains the length of time that it is expected for the delivery of the catalog item or bundle.
Calendar	A character field that contains a custom work schedule.
Limit Access to Users with these SLAs	An array field that allows the service provider to offer delivery objectives only to requesters with a listed SLA. For example, consider a delivery objective of 3 Business Days that provides one service level with associated limitations: **Name** **Limit Access to ...** 3 Business Days Sales Gold, Marketing Gold, Executive Gold As defined above, the three business day delivery objective is only available as a choice to requesters with other listed Gold SLAs. Everyone else requesting the service is given only one (and thus defaulted) choice: Silver.

Add a delivery objective to a catalog item or bundle

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type or select any search criteria for the item or bundle you want to modify and click **Search**.

   The catalog item or bundle definition record opens.

3. On the Delivery Targets tab, click **Add**.

   This launches a wizard that prompts you to select an objective from a list.

4. Click **Save**.

5. Click **OK**.

   Service Manager updates the item or bundle definition record.

Remove a delivery objective from a catalog item or bundle

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type or select any search criteria for the item or bundle you want to modify and click **Search**.
   The catalog item or bundle definition record opens.
3. On the Delivery Targets tab, select the objective that you want to delete and click **Remove**.
   The delivery objective is deleted from the list.
4. Click **Save**.
5. Click **OK**.

The item or bundle definition record is now updated.

Contents

- 1 Add a delivery objective to a catalog item or bundle
- 2 Remove a delivery objective from a catalog item or bundle

1.13.8.5. Adding Service SLAs to a catalog item or bundle

Each catalog item or bundle definition record contains a Agreements tab where a catalog manager can add or remove multiple Agreements. Note that access to the SLA record is based on the user's SLA privileges. The following information is available on the Agreements tab:

Column name	Description
Name	A character field that contains the title for the Service SLA.
Limit Access to Users with these Customer SLAs	An array field that allows the service provider to offer a Service SLA only to requesters with a listed Customer SLA. For example, consider an Email service that provides two service levels with associated limitations: **Service SLA** **Limit Access to ...** Email Gold Sales Gold, Marketing Gold, Executive Gold Email Silver none As defined above, the Email Gold SLA is only available as a choice to requesters with other listed Gold SLAs. Everyone else requesting the service is given only one (and thus defaulted) choice: Email Silver.

Add a Service SLA to a catalog item or bundle

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type or select any search criteria for the item or bundle you want to modify and click **Search**. The catalog item or bundle definition record opens.

3. On the Agreements tab, click **Add**.

   This launches a wizard that prompts you to select an SLA. You can click the **Find** button to view the details of an SLA before selecting it.

4. After selecting the SLA, the wizard prompts you to create the Limit Access to Users with these Customer SLAs list.
5. Click the **Fill** button and select the Customer SLAs that you want to associate with the Service SLA.
6. Click **Finish**.
7. Click **Save**.
8. Click **OK**. Service Manager updates the item or bundle definition record.

Remove a Service SLA from a catalog item or bundle

1. Click **Service Catalog** > **Administration** > **Manage Items**.

2. Type or select any search criteria for the item or bundle you want to modify and click **Search**.

   The catalog item or bundle definition record opens.

3. On the Agreements tab, select the SLA that you want to delete and then click **Remove**.

   The SLA is deleted from the Service SLA list.

4. Click **Save**.

5. Click **OK**.

   Service Manager updates the item or bundle definition record.

Edit a Service SLA for a catalog item or bundle

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Type or select any search criteria for the item or bundle you want to modify and click **Search**.
   The catalog item or bundle definition record opens.
3. On the Agreements tab, double-click on the Name of the SLA that you want to modify.
   This launches a wizard that allows you to change the Service SLA and access list information.
4. Edit the Service SLA and access list information as necessary.

5. Click **Save**.
6. Click **OK**.
   The item or bundle definition record is now updated.

1.13.8.6. Localizing Service Catalog items, bundles, and categories

A Service Catalog manager can localize a catalog item, category, and bundle. The Service Catalog localization wizard enables a catalog manager to view and localize the name, descriptions (short and long), and user options in any of the languages Service Manager supports for catalog items and bundles. The localization wizard also enables a Service Catalog manager to localize the name and descriptions (short and long) for a Service Catalog category.

To localize a Service Catalog item, bundle, or category the Service Catalog manager should first log on to the system in English and then use the localization wizard to localize items, bundles, and categories.

If you have a large number of Service Catalog records to localize or you want you use an external service to localize your data, use the import/export functions to create text files and then import the translated files into Service Manager. You can also export data to Excel using the functions available in Service Manager.

The parameter, `language:utf8`, in the Service Manager initialization file (**sm.ini)** is required to Import and Export catalog items. The best practice is to localize all data to the desired language before a user starts using the Service Catalog.

Viewing localized catalog items

In the Service Catalog, catalog managers are able to search for and view a list of catalog items in any of the languages supported by the system. From this list of records, managers can export the data for localization by using the Export/Unload functionality. Managers can also use this view to update any of the details of the record. This view is also useful for identifying those item records that need to be localized and verifying that the localized records will display correctly to end users.

View localized Service Catalog items

1. Click **Service Catalog** > **Localized Items**.
2. Select the language you want to use to search.
3. If you are searching for a particular item, provide the additional search parameters to limit your search.
4. Click **Search**.
5. Click the catalog record to view the record details for the item.

6. You can use the detail record to update any of the details on the record currently being displayed.

Localize a Service Catalog item

1. Click **Service Catalog** > **Manage Items**.
2. Click **Search** or use the Search Catalog Items Definition form to search for a specific item record.
3. Select an item record.
4. From the detail list toolbar of the selected record, click **More** or the More Actions icon, and then click **Localize this record**.
5. Select the language to use for localization.
6. Click **Next**.
7. Specify the localization data for the catalog item.

Field	Description
Display Name	The is the localized name of the item.
Short Description	This is the localized description from the catalog definition that displays to the user.
Long Description	This is the localized detailed description from the catalog definition that displays to the user.
Form Instructions	These are any form instructions defined by the catalog item definition.
Options	This field has XML that represents the user options defined for the item in the catalog definition. For example, here is XML for an item with a single checkbox option: `<form><checkbox id="XYZ" label="This is a label"/> </form>.` The label text enclosed in quotes (" ") should be localized.

8. Click **Finish**.
9. Continue to localize by selecting another item record or exit by clicking **OK** and then **Back**.

Note

A user selection in a Service Catalog item may have different renderings under different conditions. It may be visible in one language but invisible in another. To achieve this, you can set different Dynamic Field Validations (such as **Mandatory Condition** and **Visible Condition**) to this user selection in a certain language and then localize this Service Catalog item to other languages. For example, to make a user option visible in English and French only, set the **Visible Condition** to `($G.my.language isin {"en","fr"})` in English before localization.

If you update Dynamic Field Validations of a user option in any language after you localize a Service Catalog item, your update applies to all available languages of that user option as well.

Localize a Service Catalog bundle

1. Click **Service Catalog** > **Manage Items**.
2. Click **Search** or use the Search Catalog Items Definition form to search for a specific bundle record.
3. Select a bundle record.
4. From the detail list toolbar of the selected record, click **More** or the More Actions icon, and then click **Localize this record**.
5. Select the language to use for localization.
6. Click **Next**.
7. Specify the localization data for the bundle.

Field	Description
Display Name	This is the localized name of the bundle.
Short Description	This is the localized description from the catalog definition that displays to the user.
Long Description	This is the localized detailed description from the catalog definition that displays to the user.
Form Instructions	These are any form instructions defined by the catalog bundle definition.
Options	This field has XML that represents the user options defined for the item in the catalog definition. For example, here is XML for an item with a single checkbox option: `<form><checkbox id="XYZ" label="This is a label"/> </form>.` The text enclosed in quotes (" ") should be localized.

8. Click **Finish**.
9. Continue to localize by selecting another bundle record or exit by clicking **OK** and then **Back**.

Localize a Service Catalog category

1. Click **Service Catalog** > **Manage Items**.
2. Click **Search** or use the Search Catalog Items Definition form to search for a specific category record.
3. Select a category record.

4. From the detail list toolbar of the selected record, click **More** or the More Actions icon, and then click **Localize this record**.
5. Select the language to use for localization.
6. Click **Next**.
7. Specify the localization data for the bundle.

Field	Description
Display Name	The is the localized name of the category.
Short Description	This is the localized description from the catalog definition that displays to the user.
Long Description	This is the localized detailed description from the catalog definition that displays to the user.

8. Click **Finish**.
9. Continue to localize by selecting another category record or exit by clicking **OK** and then **Back**.

Using export and import for localization

When you have a large number of items to localize in the Service Catalog, you can use the export and import capabilities of Service Manager to create a text file or Excel file. You can then edit the file or provide it to a translation service for localizing. For additional information about exporting and importing files see the related topics.

When using export and import consider the following:

- When you generate the list of records to export, be sure to include key fields (for example, the name field and syslanguage) and all of those fields you intend to localize (description, display name, detailed description).
- The language parameter is the language code defined by the language identifier in the language table. You need to have SysAdmin access to view the language table in the database.
- Select characters as delimiters that aren't found in the data you are exporting. For example use ^^ as delimiters instead of tabs, commas, or spaces.
- Use a UTF-8 compliant text editor such as Textpad for a text file.

- When you import the translated file, use the same delimiter you used when exporting.
- Define the same fields in the correct order in the import wizard to match the exported fields and field order.

1.13.8.7. Service Catalog connectors

Service Catalog connectors are the fulfillment mechanisms for creating new records in the database. The following table lists the out-of-box connectors for Service Catalog.

Connector	Description
Open a Change	Adds a Change record.
Open a Record Using Web Service	Adds a record in an external system using web services.
Open a Standard Request in Asset Manager	Creates a record in Asset Manager using a web service.
Open a Subscription Change	Adds a Change record when a user requests a subscription to a published service.
Open an Incident	Adds an Incident record.
Open New Request	Adds a Request fulfillment record.

Connector Detail tab

The Connector Detail tab contains information about each of the Service Catalog connectors.

Field	Description
Name	Contains the name of the service request.
Record Type	Contains the type of record to be created for this service request.
Link Record	Contains a link record that describes the fields that relate information from one type of record to the other.
Process	Contains a process for this service request. This process routes the service to the appropriate Service Manager application.
Wizard	Contains the name of the wizard that runs when you add or edit the connector information of a Service Catalog item that uses this connector.
Description	Contains a description of the process for this service request.
XML Fields	Contains a field name that's used in the process to create the appropriate record in the appropriate Service Manager table.
Field Name	Contains the actual database field name that maps to the XML field in the record.
Validity Check	Contains an expression to validate the value of the field
Target Files	Contains the name of the file where this target field matches with.

Field	Description
Target Fields	Contains the name of the field in the target file where the XML field matches with.
XML Caption	Contains a field caption for the field that will be used in the process to create the appropriate record in the appropriate Service Manager table.

Expressions tab

The Expressions tab contains information about the JavaScript expressions contained within each of the Service Catalog connectors. It contains an initialization expression that needs to be processed before adding the record.

This is an example of an initialization expression for the Create New Request connector.

```
if (filename($L.file)="request") then (category in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "category"))
if (filename($L.file)="request") then (subcategory in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "subcategory"))
if (filename($L.file)="request") then (bill.to.dept in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "department"))
if (filename($L.file)="request") then (company in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "company"))
if (filename($L.file)="request") then (model in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "requestModel"))
if (filename($L.file)="request") then (modelname in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "requestModelName"))
if (filename($L.file)="request") then (severity in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "severity"))
if (filename($L.file)="request") then (impact in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "impact"))
if (filename($L.file)="request") then (assigned.group in
$L.file=jscall("svcCatInterface.getXMLFieldValue",
interface.info in $L.item, "assigngroup"))
if (bundle in $L.cartItem=true and bundle.options in
$L.cartItem~=NULL and bundle.options in $L.cartItem~="<bundle>
</bundle>") then (svc.options in $L.file=bundle.options in
$L.cartItem)
if ((bundle in $L.cartItem=false or bundle in $L.cartItem=NULL)
and options in $L.cartItem~=NULL and options in $L.cartItem~="
</form>") then (svc.options in $L.file=options in $L.cartItem)
```

Modify an existing Service Catalog connector

1. Click **Service Catalog** > **Tailoring** > **Catalog Connectors**.
2. To bring up a record list of all the out-of-box connectors click **Search**.
3. Select a connector. For example, click **Open a Change**.

4. Add **XML Fields** for any customized fields you have in your forms.
5. Change the **Link Record** to move any additional fields from the interactions record to the change record.
6. Change **Process** to include anything that needs to happen to create the record.
7. Change the wizard messages to include any new fields.
8. Add, delete, or modify any of the RAD expressions on the Expression tab.
9. Click **Save**.

The connector is modified.

Contents

- 1 Connector Detail tab
- 2 Expressions tab
- 3 Modify an existing Service Catalog connector

1.13.8.8. Service Catalog status mapping

Service Catalog status mapping enables a system administrator to set the appropriate status on Service Catalog cart item (svcCartItem) records. The status of a cart item can then be consumed by the Service Level Management module to define process targets based on the delivery targets that are defined in catalog items.

Catalog item definition

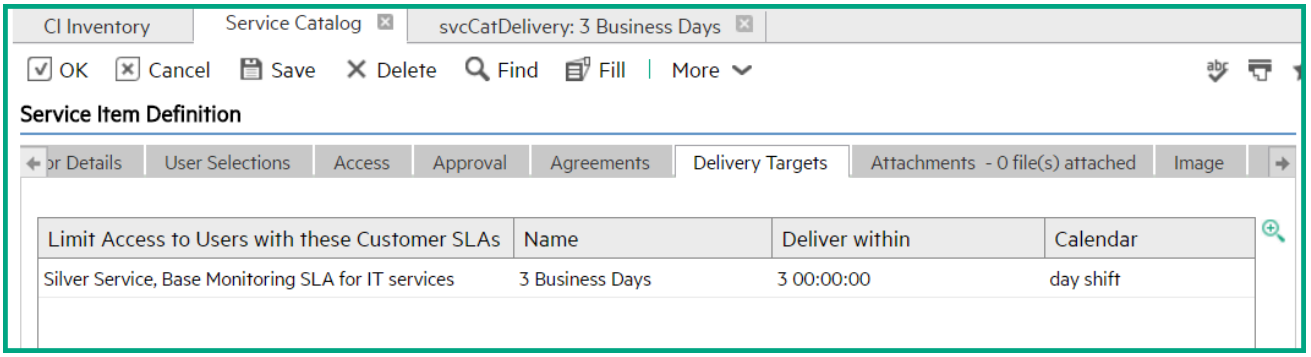
Tip

To access a catalog item or bundle definition, click **Service Catalog** > **Administration** > **Manage Items**.

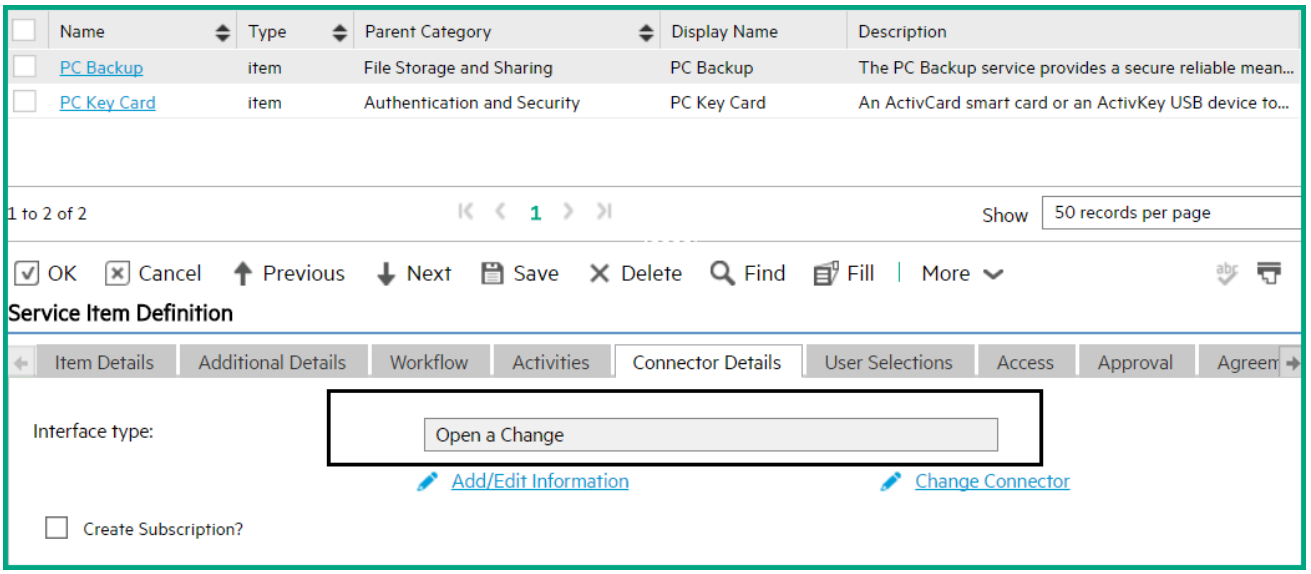
A catalog item record may have a list of delivery objectives defined on the **Delivery Targets** tab. Each delivery objective describes a period of time that's expected for the delivery of the catalog item or bundle. The following figure shows that a catalog item named PC Backup is expected to be delivered within three business days.

Tip

For more information about delivery targets, see Adding delivery objectives for a catalog item or bundle.



Also, the **Connector Details** tab defines a connector for the catalog item. A connector defines which record will be created once the user's request is approved. The following figure shows that the PC Backup catalog item uses the **Open a Change** connector. When a user orders this catalog item, an Interaction is created first, and, once the Interaction record is approved, a Change record is automatically created and associated to the Interaction record.



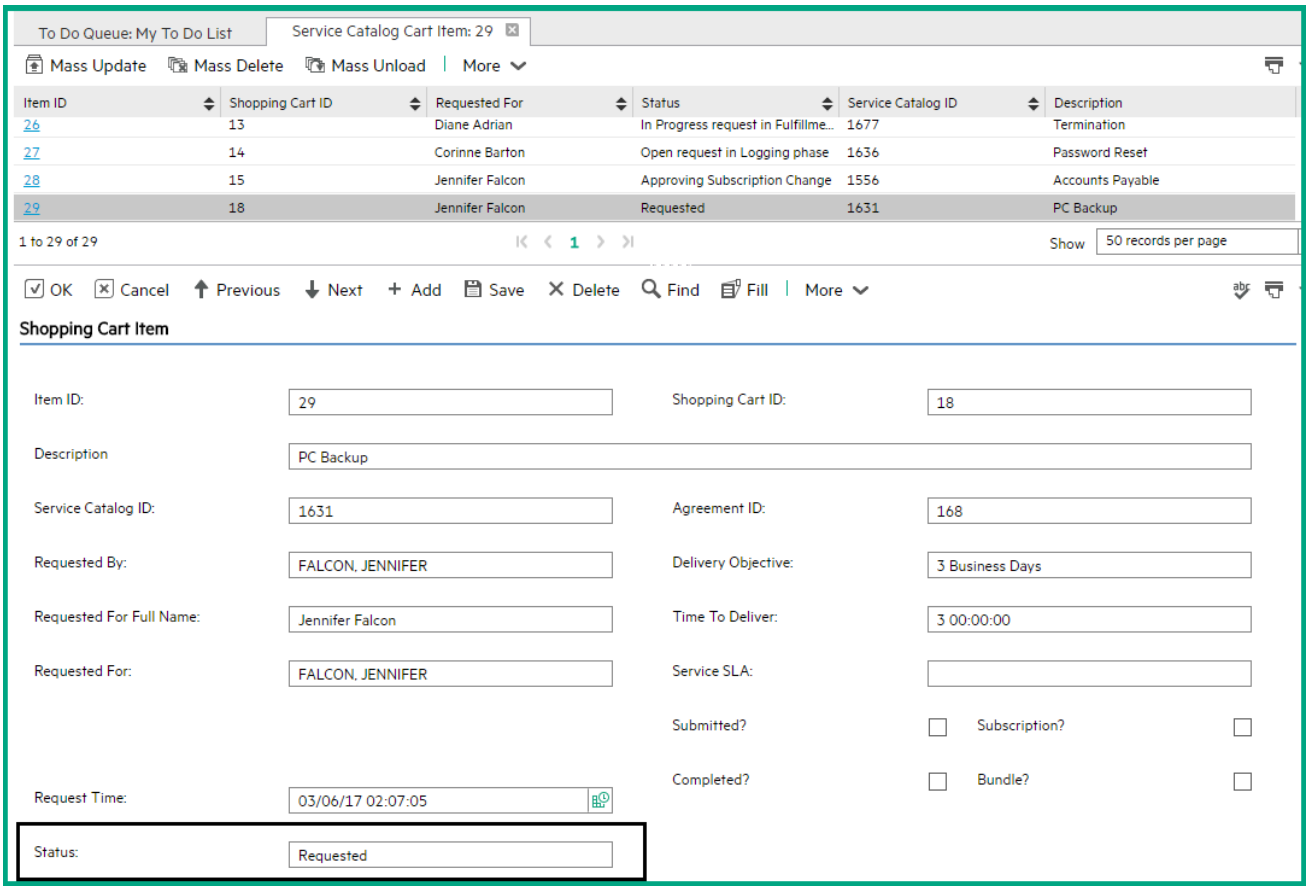
Service Catalog cart item records

Tip

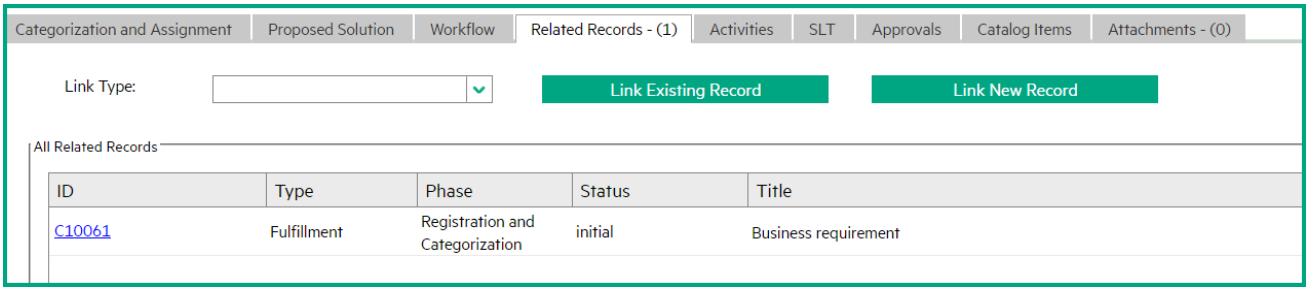
Cart item records are stored in the svcCartItem table. To access them, enter db in the command line, press Enter to open the Database Manager, and then perform a search by entering svcCartItem in the Table field.

When a user has submitted a request for this catalog item by using the **Order from Catalog** feature, a cart item record is created.

The following figure shows an example of the PC Backup catalog item, with an initial status of **Requested**.



Meanwhile, an Interaction record is created. Once the Interaction is approved, a fulfillment record is automatically created and linked to the Interaction record. See the following figure for an example.

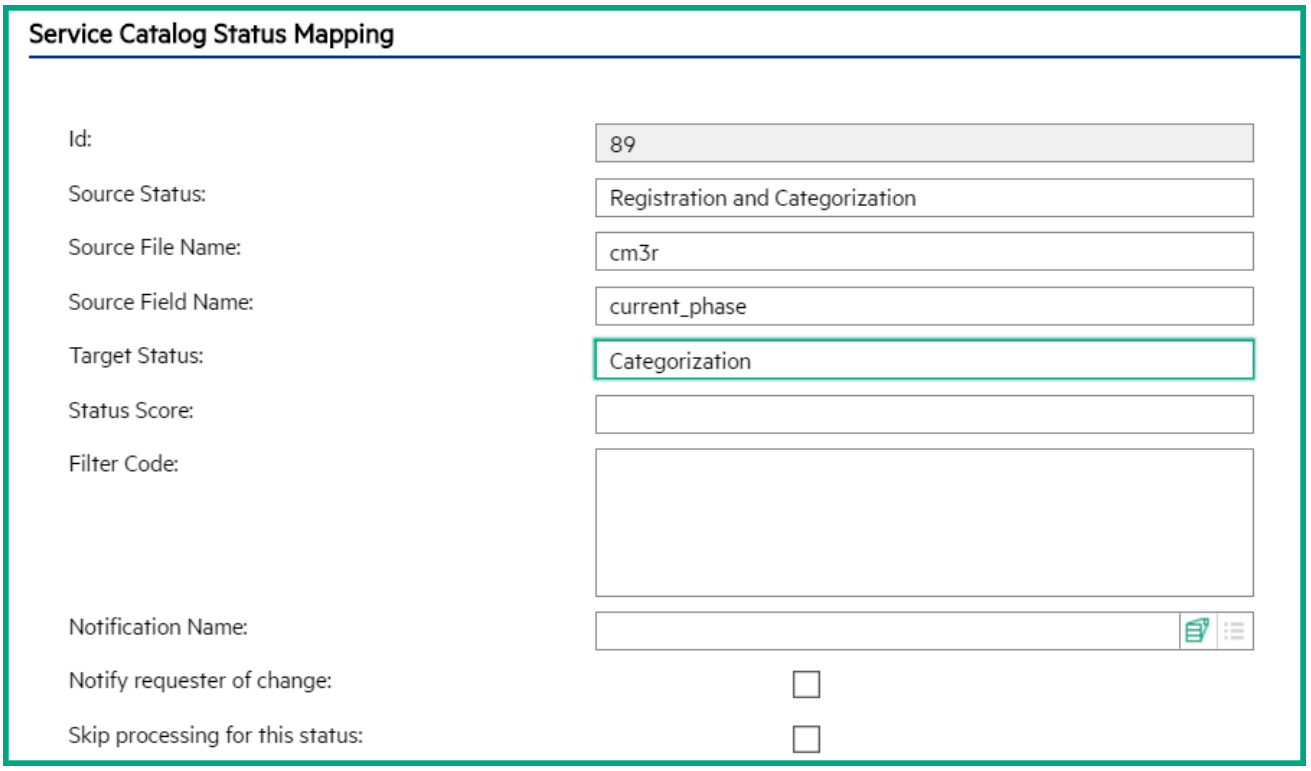


The fulfillment record goes through the phases of a workflow, which you specify in the catalog item connector. You can use the Service Catalog status mapping functionality to define status mapping between the status values of the cart item record and the phases of the fulfillment record so that you can use the cart item status values for SLA (delivery targets) monitoring.

The following figure shows an example Service Catalog status mapping record. This record maps the Registration and Categorization phase of the Standard Change workflow to the Categorization status of cart item records. When the Change record moves to this phase, the corresponding cart item automatically changes to the Categorization status.

Tip

To access a Service Catalog status mapping record, click **Service Catalog** > **Tailoring** > **Add Field Mapping**.



You can create a mapping record for the rest of the phases in the corresponding fulfillment record workflow, and then configure Service Catalog SLA monitoring (including suspending) based on the target status values that you have specified.

Configure Service Catalog status mapping for delivery target monitoring

Service Catalog status mapping enables you to set the appropriate status on Service Catalog cart item (svcCartItem) records. When an order for a catalog item is being fulfilled, the cart item status is automatically synchronized based on status mapping that you specify.

You can add these cart item status values to the slamodulecontrol table so that you can create a process target based on these status values to monitor the delivery target of a catalog item during its fulfillment process.

To achieve this, complete the following tasks.

Task 1: Create Service Catalog status mapping records for the catalog item

In this task, you need to create status mapping records for different phases of the fulfillment record that's created after a user orders the catalog item. The type of fulfillment record varies depending on the connector that's defined in the catalog item. For example, if the **Open a Change** connector is defined, the fulfillment record is a Change record.

To complete this task, follow these steps:

1. Add a Service Catalog status mapping record for a phase of the fulfillment record.
    1. Click **Service Catalog** > **Tailoring** > **Add Field Mapping**.
    2. Provide the following information.

Field	Description
Id	This is a system-generated identification number for the record.
Source Status	The status of the source field that will be mapped to the status of the request. This field is referenced as SOURCEVALUE in the transformation code. Example: Registration and Categorization
Source File Name	The source file where the request's status will be mapped from. Example: cm3r **Note** The possible values for this field include cm3r, request, and ocmq. Don't enter other values except these three.

Field	Description
Source Field Name	The field name that will be mapped to the status of the request. This field name can be referenced in the transformation code. Example: current_phase
Target Status	The status text that the request maps to. This field is referenced as TARGETVALUE in the transformation code. Example: Categorization
Status Score	The order in which to process this field. When mapping more than one field to the status of a request, you must specify the order in which to process the fields. Mapping more than one field makes it possible to build phrases based on the status of various fields in the source record. Example: 1

Field	Description
Filter Code	JavaScript code to transform the target status of fields based on the source status of this or other mapped fields in this source file. The source status and target status of this and other fields in the same source file can be referenced with the strings "@[field name].SOURCEVALUE@" and "@[field name].TARGETVALUE@". Using these strings and JavaScript, it is possible to transform the target values. Example: if (@current_phase.SOURCEVALUE@=="Registration and Categorization") {@status.TARGETVALUE@="";@approval_status.TARGETVALUE@="";}
Notification Name	The name of the notification to be sent when the request reaches this status.
Notify requester of change	Indicates whether to send a notification when the request reaches this status.
Skip processing for this status	Indicates whether to skip processing if the source record has the specified status.

3. Click **Add** to add the new record to the svcCatStatusMap table.

2. Repeat the previous step to add a status mapping record for each of the remaining workflow phases that you want to use for delivery target monitoring.

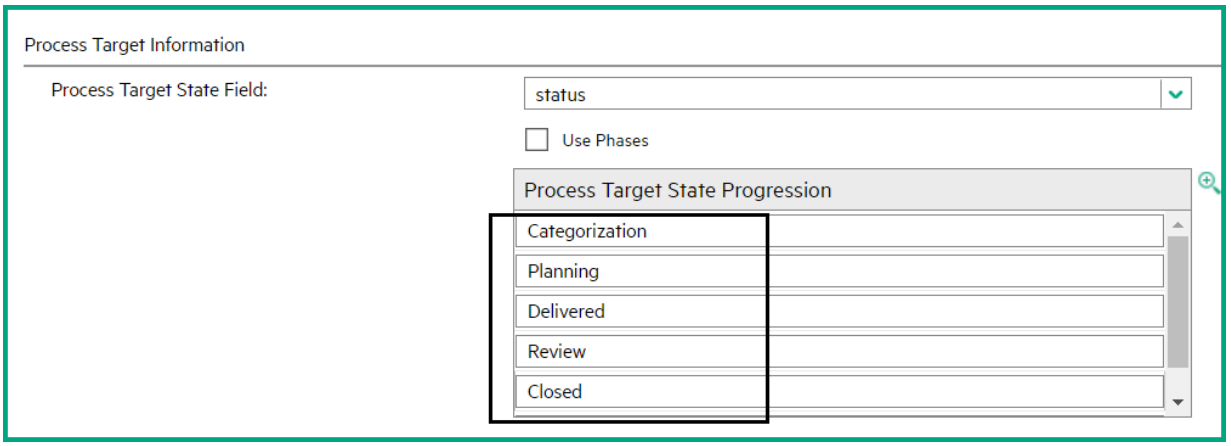
Task 2: Add the new target status in the slamodulecontrol table

The purpose of this task is to make the target status values that you define in the status mapping records available in Service Level Agreements so that you can use them to create process targets for the Service Catalog area.

To complete this task, follow these steps:

1. Enter db in the Service Manager command line, and press Enter.
2. In the Table field, enter slamodulecontrol. Click **Search**.
3. In the Table Name field, enter svcCartItem. Click **Search**.
4. Go to the Process Target Information section, and make sure the **Process Target State Field** value is **status**.
5. Add the new target status that you specified previously to the **Process Target State Progression** array. For example, enter Categorization in the array field.

   The following figure shows an example.



Task 3: Add a process target to the relevant Service Level Agreement (SLA)

In this task, you need to go to the relevant Service Level Agreement (SLA) that's defined in the catalog item definition, and add a process target for the catalog item. If the catalog item has no SLAs defined, use the default SLA that's defined in Service Level Management.

Tip

You can define suspending statuses for the process target. For example, if the cart item reaches a "Pending customer information" status, you may want to suspend the delivery target calculation.

Important

To use the Service Catalog status mapping functionality for SLA monitoring for catalog items, you need to create a process target for the Service Catalog area. The process target must use the delivery target defined in the catalog item definition.

To complete this task, follow these steps:

1. Go to **Service Catalog** > **Administration** > **Manage Items**, and open the catalog item definition by performing a search. For example, open the **PC Backup** record.
2. On the **Agreements** tab, find the service level agreements defined for the item.

   Important

   If the catalog item has multiple agreements defined, you need to add a process target for each of the agreements, as described in the following steps.

3. Click **Service Level Management** > **Agreements**, and open an agreement record of the catalog item.

   For example, open the **Base Monitoring SLA for IT services** agreement.

4. Click the **Process Targets** tab, and then click **Add SLT(s)**.
5. In the Service Area field, select **Service Catalog**. Click **Next**.
6. Enter a name for the SLT (for example, Cart Item SLA monitoring) and enter an optional description. Click **Next**.
7. Select a schedule and time zone. Click **Next**.
8. Provide information as follows:

   1. Select a catalog item or leave the Catalog Item field blank to use this process target for multiple items.

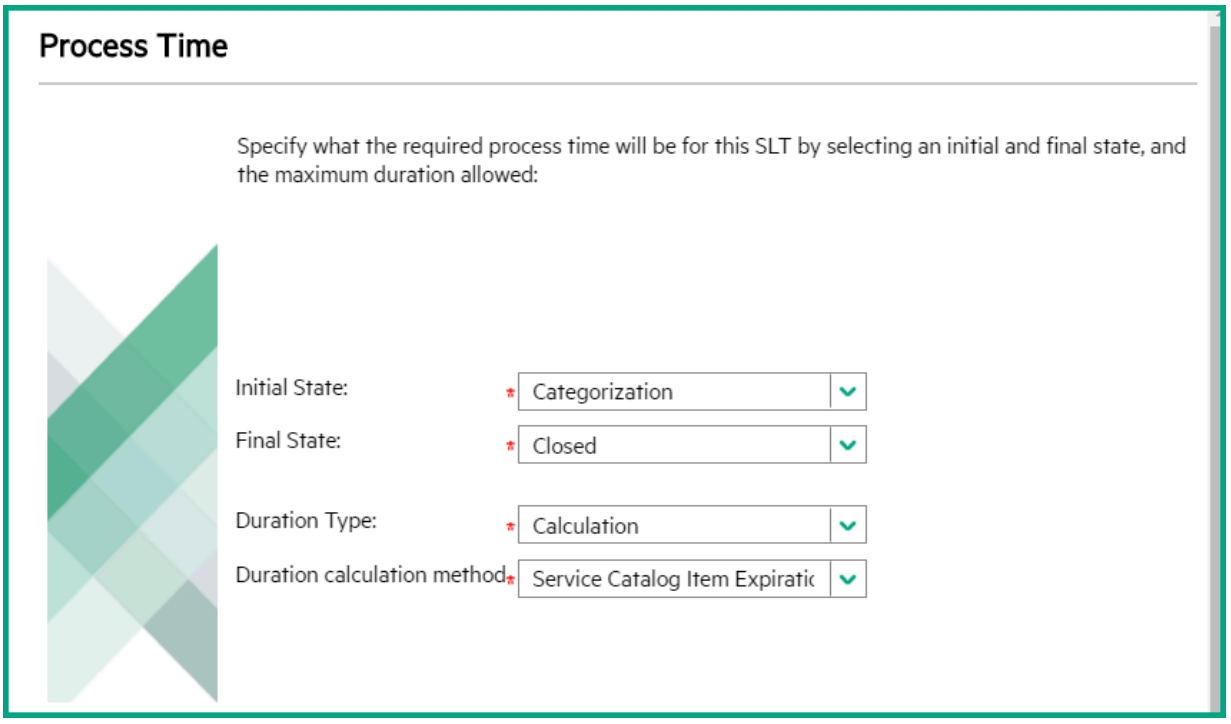
2. Select a service recipient or leave the **Requested For** field blank to use this process target for multiple recipients.
3. In the **Delivery** field, select the delivery target defined in the catalog item definition. For example, select **3 Business Days** for PC Backup.
4. In the **Quantity** field, specify a quantity condition if necessary.
5. Click **Next**.

9. On the Process Time page, specify the following information:

> **Note**
>
> On this page, the target status values that you specified previously when creating the Service Catalog status mapping record are displayed in the **Initial State** and **Final State** drop-down lists.

- Initial State: Select the initial state to be used for SLA monitoring.
- Final State: Select the final state to be used for SLA monitoring.
- Duration Type: Select **Calculation**.
- Duration calculation method: Select **Service Catalog Item Expiration**.

The following figure shows an example for PC Backup.



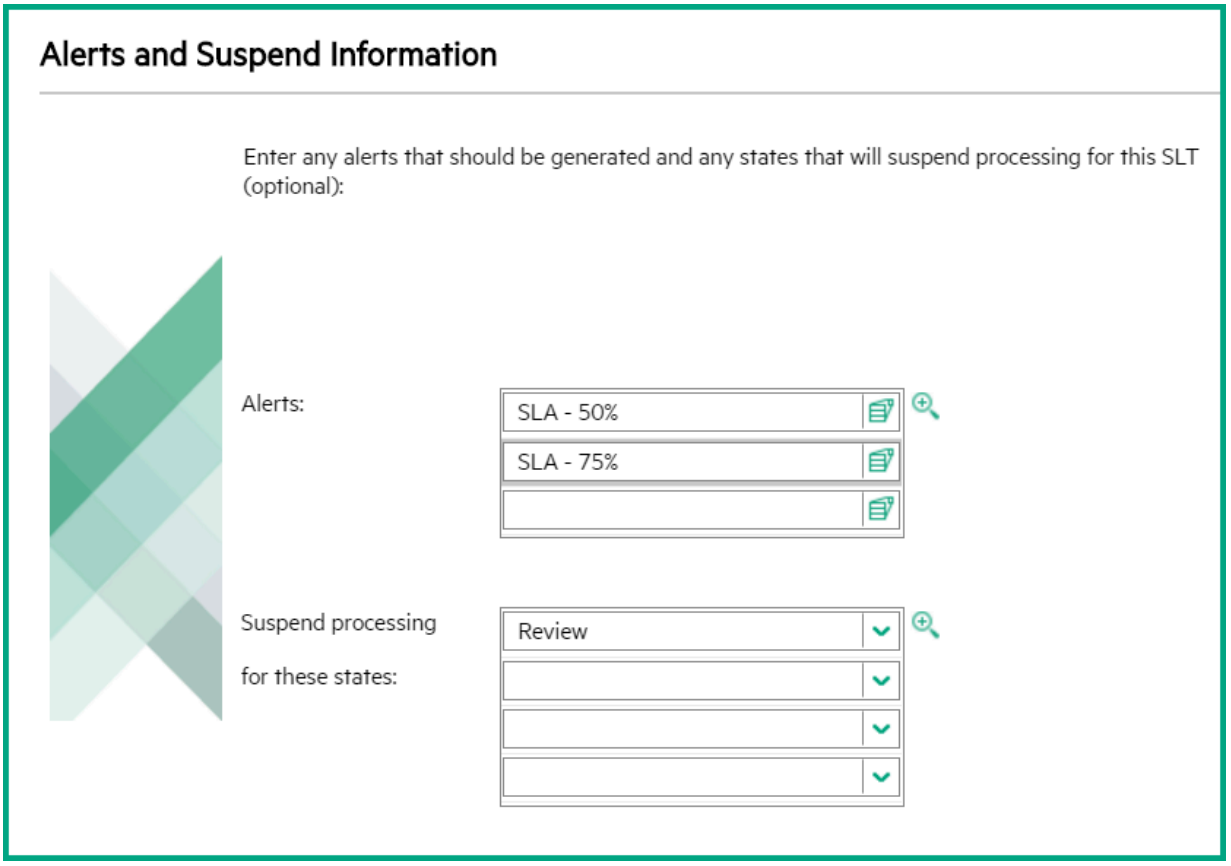
Click **Next**.

10. On the Alerts and Suspend Information page, provide the following information:

- If needed, select one or more alerts; otherwise, leave the Alerts field blank.

- If appropriate, specify one or more states when processing for the Service Level Target (SLT) should be suspended; otherwise, leave this field blank.

The following figure shows an example for PC Backup.



11. Click **Finish** to add the Process Target, and then click **OK**.
12. If the catalog item has multiple agreements defined, repeat these steps to add a process target for the rest of the agreements.

1.13.8.9. Customizing Service Catalog search result

By default, you can submit a Service Catalog request from **Service Catalog** > **Order from Catalog** and then search for the desired cart items. Or, submit a Support Catalog request from **Service Catalog** > **Non-cart Catalog Requests** and then search for the services that aren't in the cart. These two search results are under different formats in an out-of-the-box Service Manager system and you can't merge them. However, you can customize the Service Catalog search result to include both cart items and services that aren't in the cart for a better user experience.

Follow these steps:

Applies to User roles: System Administrator

1. Sign in to Service Manager as System Administrator.
2. Type `ds` in the command line, and then press **Enter**.
3. Type `svcCat.display.catalog` in the Screen ID field, and then click **Search**.
4. Update the initialization expressions as follows:
    1. Locate line 9. Replace both `$L.valueCat=$G.svcCat.noncart.category.values` and `$L.valueCat=$G.svcCat.category.values` with `$L.valueCat=$G.svcCat.category.values+$G.svcCat.noncart.category.values`.
    2. Locate line 10. Replace both `$L.displayCat=$G.svcCat.noncart.category.display.values` and `$L.displayCat=$G.svcCat.category.display.values` with `$L.displayCat=$G.svcCat.noncart.category.display.values+$G.svcCat.category.display.values`.
5. Click **Save** and **OK**.
6. Type `sl` in the command line, and then press **Enter**.
7. Type `svcCatCategory` in the Name field, and then click **Search**.
8. Update the scripts as follows:
    1. Locate `function getAllItemsbyType`, and then comment the following codes:

```
if (nonCart)
fSql+=" and non.cart=true ";
else
fSql+=" and (non.cart=NULL or non.cart=false)";
```

2. Locate `function buildRADInitFieldsQuery`, and then comment the following codes:

```
if ("noncart"==vars["$L.type"]){
catalogSql = catalogSql + " and non.cart=true";
} else {
catalogSql = catalogSql + " and (non.cart=NULL or
non.cart=false)";
}
```

3. Locate `function buildRADSetSearchQuery`, and then comment the following codes:

```
if (type == "noncart")
{
subSql=" and non.cart=true ";
} else {
subSql=" and (non.cart=NULL or non.cart=false)";
}
```

9. Click **Save** and **OK**.

Note

After these configurations, you can click **Service Catalog** > **Order from Catalog** to order both cart items and services that aren't in the cart. However, the search results from **Non-cart Catalog Requests** still include services that aren't in the cart only.

1.13.8.10. Importing an existing Request Management catalog into the Service Catalog

Applies to User roles: System Administrator

1. Sign in to Service Manager as System Administrator.
2. Click **Service Catalog** > **Administration** > **Request Management Conversion**. The Request Management to Service Catalog Utility wizard opens.
3. Click **Next** to convert the existing Request Management catalog structure to the new Service Catalog structure.
4. Go to **Service Catalog** > **Administration** > **Manage Items** > **Search**.
5. Confirm that the items converted in the Service Catalog are correct.

1.13.8.11. Importing user options for an item through Web Services

After importing Service Catalog items using web services, any options for that item can also be made available when it is ordered. At the time of import and when the **optionList** and **optionOptions** fields are populated in the *svcCatalog* record, a JavaScript function is called via a *svcCatalog extaccess* expression that parses these two fields, generates an XML string, and populates the **Options** field for the catalog item.

These are the two new character fields in the *svcCatalog extaccess* record that are required to import user options.

- **optionList** - provides a placeholder for options.
- **optionOptions** - provides a placeholder for sub-options related to the options of *select* type.

The fields for **optionList** need to be comma-delimited character fields that are listed in the following order. If any field is blank, then you must use a comma (",") as a placeholder for that field.

`option type, option id, option label, option attribute, option attribute value`

The fields for **optionOptions** need to be comma-delimited character fields in the following order. The option ID must match the option ID used in the **optionList** field above.

`option id, options id, options label`

Data examples

The following is an example of cart item options that need to be imported for a record.

`"text,comment,Comment"`

`"text,delivery,Delivery Instructions,multiline,true"`

`"checkbox,printer,Printer Included"`

`"select,color,Color,style,combo"`

`"select,size,Size,style,radio"`

The data for **optionList** uses the following format:

```
optionList="text,comment,Comment,,,text,delivery,Delivery Instructions,multiline,true,checkbox,printer,Printer Included,,,select,color,Color,style,combo,select,size,Size,style,radio,"
```

The data for **optionOptions** or sub-options of the *select* type have the following format:

```
optionOptions="color,green,Green,color,red,Red,color,blue,Blue,color,yellow,Yellow,size,small,Small,size,medium,Medium [+$100.00],size,large,Large [+$200.00],"
```

Results in the Options field

The following is the XML string that populates the **Options** field in the *svcCatalog* record.

```
<form>
<text id="comment" label="Comment"/>
<text id="delivery" label="Delivery Instructions"
multiline="true"/>
<checkbox id="printer" label="Printer Included"/>
<select id="color" label="Color" style="combo">
<option label="Green">green</option>
<option label="Red">red</option>
<option label="Blue">blue</option>
<option label="Yellow">yellow</option>
</select>
<select id="size" label="Size" style="radio">
<option label="Small">small</option>
<option label="Medium [+$100.00]">medium</option>
<option label="Large [+$200.00]">large</option>
</select>
</form>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12. Importing connector details for an item through Web Services

After importing Service Catalog items using web services, any connector information for that item can also be imported. At the time of import and when the **connectorInfo** and **connectorSubInfo** fields are populated in the *svcCatalog* record, a JavaScript function is called via *svcCatalog extaccess* expression that parses these two fields, generates an XML string and populates the **interface.info** field for the catalog item.

The following two fields in the *svcCatalog* record are required to import connector information for an item. The first one is a character field and the second one is an array of characters.

- **connectorInfo** - provides a placeholder for basic connector information.
- **connectorSubInfo** - provides a placeholder for information on repeating fields related to one or more basic connector fields.

The fields for **connectorInfo** need to be comma delimited character fields. If any field is blank, then you must use a comma (", ") as a placeholder for that field. The number of fields and the order of them can be different depending on the type of connector being imported. The following is a list of possible fields that can be imported via **connectorInfo**:

```
Interface type, category, subcategory, initial impact, severity, reason, assignment, assigned department, coordinator, type, release type, part number, parts, assets
```

The fields for **connectorSubInfo** are passed as an array of characters. Each element of the array needs to be comma delimited character fields. The first field in each element must match a corresponding field in **connectorInfo** field above. The following is a list of possible fields for **connectorSubInfo**:

```
parts,partNo,vendorNo,quantity,option,group
```

```
assets,logical.name
```

The following is a list of all the out-of-box connectors and the fields they use that can be imported for a record. To see data examples for each out-of-box connector, see the related topics.

It is important to note that only the **Open a Change of RFC category** and **Create New Request** connectors contain sub-information and use the **connectorSubInfo** field.

- "Open a Subscription Change,Subscription,initial.impact,severity"
- "Open a Change,Application,initial.impact,severity"
- "Open a Change,CI Group,initial.impact,severity"
- "Open a Change,HW server,initial.impact,severity"
- "Open a Change,Hardware,initial.impact,severity"
- "Open a Change,Security,initial.impact,severity"
- "Open a Change,Subscription,initial.impact,severity"
- "Open a Change,RFC - Advanced,subcategory,initial.impact,severity,assign.dept"
- "Open a Change,Release Management,coordinator,reason,initial.impact,severity,release.type"
- "Open a Service Desk Interaction,impact,assignment"
- "Open a Record Using Web Services,partNumber"
- "Open a Standard Request in Asset Manager,partNumber"
- "Open a Change,RFC,coordinator,reason,initial.impact,severity,assets"
- "Open New Request,category,parts,parent_group,dependent_group,dependency_group"

Related topics

Importing user options for an item through Web Services
Example: Importing details for Open a Change of Application category
Example: Importing details for Open a Change of CI Group category
Example: Importing details for Open a Change of Hardware category
Example: Importing details for Open a Change of HW Server category
Example: Importing details for Open a Change of Release Management category
Example: Importing details for Open a Change of RFC category
Example: Importing details for Open a Change of Security category
Example: Importing details for Open a Change of Subscription category
Example: Importing details for Open a Record Using Web Services
Example: Importing details for Open a Service Desk Interaction
Example: Importing details for Open a Subscription Change
Example: Importing details for creating new requests

1.13.8.12.1. Example: Importing details for Open a Change of Application category

The following is an example of the format required to import the connector details for Open a Change of Application category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,Application,4 - User,3 - Average,"`

Result

```
<info>
<category>Application</category>
<initial.impact>4 - User</initial.impact>
<severity>3 - Average</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.2. Example: Importing details for Open a Change of CI Group category

The following is an example of the format required to import the connector details for Open a Change of CI Group category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,CI Group,4 - User,3 - Average,"`

Result

```
<info>
<category>CI Group</category>
<initial.impact>4 - User</initial.impact>
<severity>3 - Average</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.3. Example: Importing details for Open a Change of Hardware category

The following is an example of the format required to import the connector details for Open a Change of Hardware category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,Hardware,4 - User,3 - Average,"`

Result

```
<info>
<category>Hardware</category>
<initial.impact>4 - User</initial.impact>
<severity>3 - Average</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.4. Example: Importing details for Open a Change of HW Server category

The following is an example of the format required to import the connector details for Open a Change of HW Server category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,HW server,4 - User,3 - Average,"`

Result

```
<info>
<category>HW server</category>
<initial.impact>4 - User</initial.impact>
<severity>3 - Average</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.5. Example: Importing details for Open a Change of Release Management category

The following is an example of the format required to import the connector details for Open a Change of Release Management category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,Release Management,CM 1,Incident/Problem Resolution,2 - Site/Dept,1 - Critical,Hardware Release,"`

Result

```
<info>
<category>Release Management</category>
<coordinator>CM 1</coordinator>
<reason> Incident/Problem Resolution</reason>
<initial.impact>2 - Site/Dept</initial.impact>
<severity>1 - Critical</severity>
<release.type> Hardware Release </release.type>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.6. Example: Importing details for Open a Change of RFC - Advanced category

The following is an example of the format required to import the connector details for Open a Change of RFC - Advanced category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,RFC - Advanced,facilities,2 - Site/Dept,3 - Average,CAB,"`

Result

```
<info>
<category>RFC - Advanced</category>
<subcategory>facilities</subcategory>
<initial.impact>2 - Site/Dept</initial.impact>
<severity>3 - Average</severity>
<assign.dept>CAB</assign.dept>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.7. Example: Importing details for Open a Change of RFC category

The following is an example of the format required to import the connector details for Open a Change of RFC category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,RFC,CM 1,Incident/Problem Resolution,1 - Enterprise,1 - Critical,assets,"`

connectorSubInfo data

```
['assets,DEFAULTpc012,',
'assets,DEFAULTpc015,',
'assets,DEFAULTserver002,']
```

Result

```
<info>
<category>RFC</category>
<coordinator>CM 1</coordinator>
<reason>Incident/Problem Resolution</reason>
<initial.impact>1 - Enterprise</initial.impact>
<severity>1 - Critical</severity>
<assets>
<logical.name>DEFAULTpc012</logical.name>
<logical.name>DEFAULTpc015</logical.name>
<logical.name>DEFAULTserver002</logical.name>
</assets>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.8. Example: Importing details for Open a Change of Security category

The following is an example of the format required to import the connector details for Open a Change of Security category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,Security,4 - User,3 - Average,"`

Result

```
<info>
<category>Security</category>
<initial.impact>4 - User</initial.impact>
<severity>3 - Average</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.9. Example: Importing details for Open a Change of Subscription category

The following is an example of the format required to import the connector details for Open a Change of Subscription category. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Change,Subscription,4 - User,3 - Average,"`

Result

```
<info>
<category>Subscription</category>
<initial.impact>4 - User</initial.impact>
<severity>3 - Average</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.10. Example: Importing details for Open a Record Using Web Services

The following is an example of the format required to import the connector details for Open a Record Using Web Services. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Record Using Web Services,12345,"`

Result

```
<info>
<partNumber>12345</partNumber>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.11. Example: Importing details for Open a Service Desk Interaction

The following is an example of the format required to import the connector details for an Open a Service Desk Interaction. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

```
"Open a Service Desk Interaction,1 - Enterprise,AUTO,"
```

Result

```
<info>
<impact>1 - Enterprise</impact>
<assignment>AUTO</assignment>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.12. Example: Importing details for Open a Standard Request in Asset Manager

The following is an example of the format required to import the connector details for Open a Standard Request in Asset Manager. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open a Standard Request in Asset Manager, DEMO-REQ1,"`

Result

```
<info>
<partNumber>DEMO-REQ1</partNumber>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.13. Example: Importing details for Open a Subscription Change

The following is an example of the format required to import the connector details for Open a Subscription Change. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

```
"Open a Subscription Change,Subscription,2 - Site/Dept,2 - High,"
```

Result

```
<info>
<category>Subscription</category>
<initial.impact>2 - Site/Dept</initial.impact>
<severity>2 - High</severity>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.12.14. Example: Importing details for Open New Request

The following is an example of the format required to import the connector details for creating a new request. The result is an XML string that populates the **interface.info** field in the *svcCatalog* record.

connectorInfo data

`"Open NEW Request,customer,parts,parent_group,dependent_group,dependency_group,"`

connectorSubInfo data

```
['parts,266,10,1,required,speaker,',
'parts,211,10,1,required,cpu,',
'parts,281,51,1,optional,upgrade,',
'parts,261,4,1,optional,monitor,',
'parent_group,speaker,',
'parent_group,monitor,',
'dependent_group,cpu,',
'dependent_group,upgrade,',
'dependency_group,instock,',
'dependency_group,closed,']
```

Result

```
<info>
<category>customer</category>
<parts>
<part>
<partno>266</partno>
<vendorno>10</vendorno>
<quantity>1</quantity>
<option>required</option>
<group>speaker</group>
</part>
<part>
<partno>211</partno>
<vendorno>10</vendorno>
<quantity>1</quantity>
<option>required</option>
<group>cpu</group>
</part>
<partno>281</partno>
<vendorno>51</vendorno>
<quantity>1</quantity>
<option>optional</option>
<group>upgrade</group>
</part>
<part>
<partno>261</partno>
<vendorno>4</vendorno>
<quantity>1</quantity>
<option>optional</option>
<group>monitor</group>
</part>
</parts>
<parent_group>
<item>speaker</item>
<item>monitor</item>
</parent_group>
<dependent_group>
<item>cpu</item>
<item>upgrade</item>
</dependent_group>
<dependency_group>
```

```
<item>instock</item>
<item>closed</item>
</dependency_group>
</info>
```

Related topics

Importing connector details for an item through Web Services

1.13.8.13. Using the Service Catalog WSDL for external fulfillment of requests

Service Manager allows you to connect to and consume external Web Services. When you connect to an external Web Service, Service Manager retrieves the Web Service Description Language (WSDL) file for the service and converts it into a set of JavaScript functions. Service Manager stores the JavaScript functions in its local ScriptLibrary. You can then write custom JavaScript to call these Web Services JavaScript functions and send and receive SOAP messages to the remote Web Service.

For example, you can query external Web services to enable fulfillment of Service Catalog requests from external applications such as Asset Manager. The Service Catalog Web Services and Asset Manager Out-of-Box (OOB) connector is already setup to consume external web services through Service Manager's standard support. See the related topics for an example of the Service Catalog WSDL.

Example: Service Catalog fulfillment WSDL

The following is the Service Catalog fulfillment Web Service Description Language (WSDL) that conforms to Web Services standard practices. This procurement services WSDL is used to connect to external applications such as Asset Manager.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<definitions xmlns="http://schemas.xmlsoap.org/wsdl/" xmlns:http="http://schemas.xmlsoap.org/wsdl/http/"
xmlns:mime="http://schemas.xmlsoap.org/wsdl/mime/" xmlns:ns="http://schemas.hp.com/ServiceCatalog/1"
 xmlns:soap="http://schemas.xmlsoap.org/wsdl/soap/" xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" targetNamespace="
http://schemas.hp.com/ServiceCatalog/1" xsi:schemaLocation="http://schemas.xmlsoap.org/wsdl/
http://schemas.xmlsoap.org/wsdl/">
   <types>
      <xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" attributeFormDefault="unqualified"
elementFormDefault="qualified" targetNamespace="http://schemas.hp.com/ServiceCatalog/1" xmlns="
http://schemas.hp.com/ServiceCatalog/1" >

            <xs:complexType name="ArrayType">
               <xs:attribute name="type" type="xs:string" use="required" fixed="Array"/>
            </xs:complexType>

            <xs:complexType name="MessageType">
               <xs:simpleContent>
                  <xs:extension base="xs:string">
                     <xs:attribute name="severity" type="xs:int" use="optional"/>
                     <xs:attribute name="module" type="xs:string" use="optional"/>
                  </xs:extension>
               </xs:simpleContent>
            </xs:complexType>

            <xs:complexType name="MessagesType">
               <xs:sequence>
                  <xs:element name="message" type="MessageType" maxOccurs="unbounded"/>
               </xs:sequence>
            </xs:complexType>

                     <xs:complexType name="OptionsType">
                           <xs:sequence>
                                 <xs:element name="option">
                                       <xs:complexType>
                                             <xs:attribute name="name" type="xs:string"/>
                                             <xs:attribute name="value" type="xs:string"/>
                                       </xs:complexType>
                                 </xs:element>
                           </xs:sequence>
                     </xs:complexType>

            <xs:simpleType name="StatusType">
               <xs:restriction base="xs:string">
                  <xs:enumeration value="SUCCESS"/>
                  <xs:enumeration value="FAILURE"/>
               </xs:restriction>
            </xs:simpleType>

         <xs:complexType name="CreateFulfillmentRequest">
            <xs:sequence>
               <xs:element  name="RequesterRequestID" nillable="true" type="xs:string"/>
               <xs:element  name="RequestedBy" nillable="true" type="xs:string"/>
               <xs:element  name="CallBackContact" nillable="true" type="xs:string"/>
               <xs:element  name="NotifyBy" nillable="true" type="xs:string"/>
               <xs:element  name="Description" nillable="true" type="xs:string"/>
               <xs:element  name="NeededBy" nillable="true" type="xs:dateTime" minOccurs="0"/>

               <xs:element  name="Severity" nillable="true" type="xs:string" />
               <xs:element  name="CartItemID" nillable="true" type="xs:int"/>
               <xs:element  name="Completed" nillable="true" type="xs:boolean"/>
```

```
<xs:element  name="Submitted" nillable="true" type="xs:boolean"/>
```

```
            <xs:element  name="Status" nillable="true" type="xs:string"/>
            <xs:element  name="SLAAgreementID" nillable="true" type="xs:int"/>
            <xs:element  name="Cost" nillable="true" type="xs:decimal"/>
            <xs:element  name="Duration" nillable="true" type="xs:duration"/>
            <xs:element  name="Quantity" nillable="true" type="xs:int"/>
            <xs:element  name="SLADeadline" nillable="true" type="xs:dateTime" minOccurs="0"/>
            <xs:element  name="UnitCost" nillable="true" type="xs:decimal"/>
            <xs:element  name="Currency" nillable="true" type="xs:string"/>
            <xs:element  name="RequestTime" nillable="true" type="xs:dateTime" minOccurs="0"/>
            <xs:element  name="StartTime" nillable="true" type="xs:dateTime" minOccurs="0"/>
            <xs:element  name="ServiceCatalogID" nillable="true" type="xs:int"/>
            <xs:element  name="ItemDescription" nillable="true" type="xs:string"/>
            <xs:element  name="PartNumber" nillable="true" type="xs:string"/>
            <xs:element  name="RequestedFor" nillable="true" type="xs:string"/>
            <xs:element  name="options" minOccurs="0" maxOccurs="unbounded" type="OptionsType"/>
         </xs:sequence>
      </xs:complexType>

      <xs:complexType name="CreateFulfillmentResponse">
         <xs:sequence>
            <xs:element  name="ProviderRequestID" nillable="true" type="xs:string"/>
            <xs:element  name="RequesterRequestID" type="xs:string" />
            <xs:element  name="messages" type="MessagesType"/>
         </xs:sequence>
         <xs:attribute name="status" type="StatusType" use="required"/>
         <xs:attribute name="message" type="xs:string" use="required"/>
         <xs:attribute name="returnCode" type="xs:decimal" use="optional"/>
      </xs:complexType>
   </xs:schema>
</types>

<message name="CreateFulfillmentRequest">
   <part name="CreateFulfillmentRequest" type="ns:CreateFulfillmentRequest"/>
</message>
<message name="CreateFulfillmentResponse">
   <part name="CreateFulfillmentResponse" type="ns:CreateFulfillmentResponse"/>
</message>

<portType name="ServiceCatalog">
   <operation name="CreateFulfillment">
      <documentation/>
      <input message="ns:CreateFulfillmentRequest"/>
      <output message="ns:CreateFulfillmentResponse"/>
   </operation>
</portType>
<binding name="ServiceCatalog" type="ns:ServiceCatalog">
   <soap:binding style="document" transport="http://schemas.xmlsoap.org/soap/http"/>
   <operation name="CreateFulfillment">
      <soap:operation soapAction="CreateFulfillment" style="document"/>
      <input>
         <soap:body use="literal"/>
      </input>
      <output>
         <soap:body use="literal"/>
      </output>
   </operation>
</binding>
<service name="ServiceCatalog">
   <port name="ServiceCatalog" binding="ns:ServiceCatalog">
      <soap:address location="http://yourservicelocationURL"/>
   </port>
```

```
    </service>
</definitions>
```

Field mappings between Service Catalog and Asset Manager

These are the Asset Manager fields that are populated from the procurement WSDL fields when a Service Catalog user requests an item that's mapped to the **Open a Standard Request in Asset Manager** connector. After a user requests the item and the request is approved, the information is routed to Asset Manager. All of the Asset Manager fields are found within the *amRequest* table.

Service Catalog field	Asset Manager field
RequestedBy	Requester.BarCode
RequestedFor	User.BarCode
Description	Description and Purpose (truncates the string to the first 150 characters)
NeededBy	dtService
Severity	Priority
Quantity	ReqLines.fQty
PartNumber	The PartNumber is not stored in an AC field. This number should match the ID of a Standard Request instance in AC.

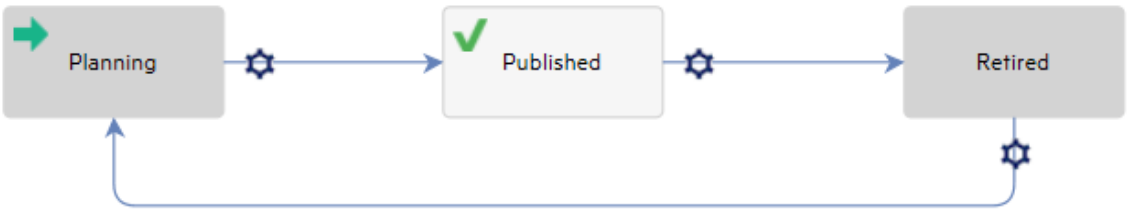
Contents

- 1 Example: Service Catalog fulfillment WSDL
- 2 Field mappings between Service Catalog and Asset Manager

1.13.8.14. Service Catalog Management process

The Service Catalog Management process aligns to ITIL 2011 Edition workflows and process integrations. In Service Manager, the out-of-box Service Catalog workflow conforms to the principles of ITIL 2011 Edition. You can adapt this default workflow to meet your specific process and business needs.

To access the default workflow definition, go to **Service Catalog** > **Tailoring** > **Catalog Workflow**, and then open the workflow **Service Catalog Item**. The following figure illustrates this workflow.



This workflow is composed of three phases by default:

- Planning: In this phase, the service portfolio item is still under development and is inactive (the read-only **Active** flag is automatically set to false).

  To move the phase from Planning to Published, you need to update the item's status from **Requirements** to **Operational**. Hence, this item becomes active and visible in Service Catalog.

- Published: In this phase, the service portfolio item is already taken from the pipeline and put into the service catalog and becomes active (the read-only **Active** flag is automatically set to true).
- Retired: In this phase, the service is retired and therefore not available for use. A retired service is inactive (the read-only **Active** flag is automatically set to false).

To view the current phase of a category, bundle, or item, open the record from the Service Catalog and then select the **Workflow** tab.

Related topics

Service Catalogs
Service Catalog items and bundles

Add an item
Add a Service Catalog bundle

1.13.8.15. Service Catalog security areas and roles

Service Manager provides the following out-of-box security areas and security roles to control access to the Service Catalog module.

Security areas

Service Manager provides two out-of-box security areas for the Service Catalog module. To access these security areas and view the Service Catalog access rights of different security roles, go to **System Administration** > **Security** > **Areas**.

Area	Description
Service Catalog	This area contains the default security rights and settings for categories, bundles, and items. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the role.
Service Catalog Configuration	This area contains the default security rights and settings for Service Catalog administration. The rights will be copied to new roles created for this area. However, the settings will only be inherited if there are no settings specified on the role.

The following table describes their default rights.

Area	View	New	Update	Delete/Close	Modify Template
Service Catalog	True	False	Never	Never	False
Service Catalog Configuration	True	True	Always	Always	True

Security role

Service Manager provides one out-of-box security role, Service Catalog Administrator, for the Service Catalog module. To access this security role and its rights and settings, go to **System Administration** > **Security** > **Roles**. This role has full administration rights to the Service Catalog and Service Catalog Configuration security areas. The following table lists the rights of this role to different security areas.

Area	Rights
Calendar Administration	Partial Rights
Change	Partial Rights
Change Management Configuration	No Rights
Change Tasks	Partial Rights
Common Configuration	Partial Rights
Incident	Partial Rights
Incident Management Configuration	No Rights
Incident Tasks	Partial Rights
Knowledge	No Rights
Knowledge Administration	No Rights
OO Integration for Tasks	Partial Rights
Problem	Partial Rights
Problem Management Configuration	No Rights
Problem Tasks	Partial Rights
Report	No Rights
Report Configuration	No Rights
Report Schedule	No Rights
Request	Partial Rights
Request Management Configuration	No Rights
Request Tasks	Partial Rights

Area	Rights
Security	No Rights
Service Catalog	Admin Rights
Service Catalog Configuration	Admin Rights
Service Desk	Partial Rights
Service Desk Configuration	No Rights
Service Level Management	Partial Rights
Service Level Management Configuration	No Rights
Survey	No Rights
Survey Configuration	No Rights
Tailoring	No Rights
Timeperiod	Partial Rights
Timeperiod Configuration	No Rights

Contents

- 1 Security areas
- 2 Security role

1.13.9. Service Catalog Access

The Service Catalog—along with all other Service Manager modules—is available through a web interface. All catalog management features are accessible through this interface.

Restricting Service Catalog Access

Service Manager security utilities allow administrators to define role-based views of the Service Catalog. This allows administrators to easily enable a Service view or a Technical view so the Catalog is specific to the end user and his/her role. Role-based views can include full access for designing the catalog, catalog hierarchy, orderable items, service levels, and request fulfillment. Users will have a view of the Service Catalog for ordering only, and access to catalog items can also be limited based on the user's job role, position in the organization, business unit, and so on. Finally, other views into the Service Catalog can be defined to meet an organization's unique requirements.

Service Catalog Self Help

The Service Catalog provides an easy-to-use web interface for Business Users (end users) that allows them to browse and order available services and follow up their requests' status.

Service Catalog Access Control

Service Catalog Management involves configuration management, which is controlled by security profiles. These user profiles allow authorized users to create, modify, and close CI records. The system provides options for Never, Always, and When Assigned. For example, one user may be able to create and close CIs but only modify items assigned to them. Access to functionality within each module is governed by Service Manager security utilities, which define access based on role and user.

Service Manager also provides a security control to segment data between multiple customers. "Mandanten" controls identify which customer data a particular user or group can access, update, or both.

1.13.10. Catalog Design

Catalog Managers can design the catalog using native tools within Service Manager. The catalog structure is flexible and can be modified to accommodate any content structure. Here are some examples of the flexibility of the framework:

- Any Service Manager business service owner can create or maintain a catalog using a simplified Service Manager interface and wizards to guide them in adding or changing service offerings.
- Catalog items can include any type of service, from password reset to new application access, from mail box capacity upgrade to new service request (for example, a request for application hosting service not yet provided).
- Each business service can be associated with one or more categories thus reflecting the multiple facets of some of the more complex business services.
- The categories themselves are organized in a hierarchical manner, which is not only useful to the Catalog Manager, but also to the business users when they need to find a catalog item.
- The Service Catalog module allows Catalog Managers to create their own hierarchy of categories to more closely match their organization’s nomenclature.

Service Catalog Attachments

The Service Catalog provides a way to attach documentation to the catalog items. This allows Catalog Managers to include additional information about a catalog item, such as functional specifications of a requested workstation, or other instructions related to the service requested.

Bundle Design

Services are represented as CIs in Service Manager, which enables their access to all related support and delivery processes. Out-of-box lifecycle processes include:

- Defining a new service
- Discovering, defining, and managing CI relationships
- Adding service levels (service-oriented SLA)
- Publishing a service to the Service Manager Service Catalog
- Signing up customers and managing their subscriptions
- Fulfilling subscription requests through Service Manager Change Management
- Monitoring and supporting through Incident, Change, and Problem Management
- Monitoring SLAs and optimizing service delivery

Services are provided by IT to satisfy a range of business needs. Services can be delivered to individuals, departments, or an entire enterprise as defined by service subscriptions. Sample services can include email, a billing system, workstation backup or office automation. The service instance is the deployment of a service, modeled as a CI in the CMDB, and it can have related CIs, customers, incidents, and changes.

Once the service is deployed, the Service Catalog lists and describes all of the services that IT offers, providing customers with a view to browse and make requests. Catalog items can range from single user workstations to department or enterprise application support.

The Service Catalog module provides the ability to create bundles of services in customer- and business-relevant packages. Out-of-box it includes dozens of predefined bundles that span major categories. Bundles are exposed in the service catalog as other services. They are part of the same service category tree as simple catalog items. Bundles can be described with the same expressiveness as regular catalog items, with multiple descriptions, an image, attached documents, service level options, access rights, and other information. A wizard allows the Catalog Managers to add bundles in a very simple way, letting them add components from existing catalog items, or from brand new ones created specifically for the bundle. A bundle can be composed of simple catalog items, and can also contain other bundles. Some of the components can be mandatory, others can be optional.

Service Catalog Connectors Overview

The Service Catalog module allows the Catalog Manager define how a service request must be routed for fulfillment.

Request processing is automated using other appropriate Service Manager modules. For example, the Service Manager Request Management module may be used to automate a “new employee setup” request, because this type of request has goods and services components that require defined ordering and sequencing processes that can be facilitated by the Request Management module.

Other types of service requests can be fulfilled in the Service Desk, Change Management, or other modules.

A catalog administrator defines the modules required to fulfill each service catalog item. Fulfillment can include creating sequenced or parallel tasks to fulfill the request, routing tasks to appropriate groups or staff, gathering required approvals, generating notifications and alerts, measuring response time based on applied SLAs, and so on.

The service catalog uses an open connector architecture, where new fulfillment connectors can be added to integrate with third-party or legacy tools.

Service Descriptions

The Service Manager Service Catalog provides a user interface that allows the Catalog Manager to create and publish a comprehensive list of services available to the business user.

Within each catalog item, the Service Catalog collects information for a service, including such details as service descriptions, categorization, costs, access rights, service request approval rules, delivery and support service levels, and custom options. The out-of-box definitions can be used as templates to start defining the specific services supported in an enterprise. In addition, new services can easily be defined through a web-based, wizard-guided process.

The service offering description can be made as detailed as desired, from a simple text description, to a more detailed rich-text description, to attaching documents to the catalog item.

The Service Catalog lets the catalog manager define the service's base price, which can be a onetime cost, a recurring cost, or both.

The Catalog Manager can also:

- Expose service options, which, when selected, can affect the price of subscribing to the service (for example, it is possible to describe access to optional modules of an application, or an optional data plan to a cell phone service).
- Expose optional service levels.
- Restrict access to business users based on their role.

Finally, the management user interface also allows Catalog Managers to duplicate existing services, thereby facilitating the creation of new catalog items.

After a service is defined, its details can be exposed to end users browsing or requesting services on the corporate intranet. For example, an email service may list available options, costs, and service levels.

Service Level

The Service Catalog user interface allows the Catalog Manager to describe the different service level available to the business users. The Service Catalog also has the ability to limit each business user's access to specific service levels.

When ordering a service, users can select their service SLAs (if applicable) on the order screen.

Catalog Templates

The Service Catalog is hierarchical. Companies can use the categories, subcategories, and items that are provided. Companies also have the flexibility to modify, copy, and remove any of these to fit their needs. Any Service Manager business service owner can create or maintain a catalog using a simplified Service Manager interface and wizards to guide them in adding or changing service offerings.

Types of catalog entries within categories and subcategories can include the following:

- A pure service request—for example: office move
- A pure goods request—for example: business cards
- A single good with a service delivery component—for example: memory upgrade
- Bundle of goods and services—for example: new employee setup
- Project or Request for Services—for example: application-hosting service

It is very easy to create service templates from existing catalog items, and conversely to create services from services templates.

1.13.11. Service Catalog Configuration Management: Service States

In the Service Catalog, which represents the business-user view of the catalog, a catalog item can be simply made active or inactive. In the Configuration Management database, where business services are represented as CIs, Service Manager allows Catalog Managers to designate the state of the service. Service state codes can be created to meet an organization’s requirements.

1.13.12. Service Catalog Subscriptions

The Service Manager Service Catalog can capture and display information on services that can be obtained by individual users or departments on a usage or subscription basis. In particular, business users are able to see all their subscriptions to business services—those subscribed directly or subscribed through a department. In addition, managers can see all the subscriptions to business services subscribed through the department under their control.

1.13.13. Service Catalog Integration

The Service Catalog is fully integrated with other Service Manager modules, including Service Desk, Change Management, Configuration Management, and Request Management.

Service Manager can automatically generate records to fulfill Service Catalog requests in these other modules. As the request is being fulfilled, status information is passed back to the service request so that the business user can be informed of the progress.

Service Catalog Integration: Request Fulfillment

Service requests are created and approved in the Service Catalog module. The end user, through the Employee Self-Service web interface, can browse the service catalog and submit service requests. The Service Catalog allows users with approval permission to review and approve service requests through the same Employee Self-Service web interface:

- The IT technicians can monitor the service request processing in more depth. They have access to the resulting service request, which is a Service Desk interaction of a specific type, or can monitor the progress of the request through the fulfillment module.
- End users can track the status of their requests through the Service Catalog end-user interface. End users can also review the details of their requests. They can make changes to a request and resubmit it. If a request is held up in the approval process, the requester can see the problem and act on it. Empowering the users helps minimize frustration and prevents more calls to the help desk.
- Service managers can track the status of service delivery through fulfillment records created in Service Manager.

The Service Catalog will automatically synchronize the state of the service request with the state of the underlying fulfillment processes, even when, for some complex requests, the fulfillment is carried out in parallel in different modules.

Service Catalog Integration: CMDB

The Service Catalog documents services at a general level so that they can be requested by business users. Catalog items can be linked to services managed in the Configuration Management database. This is how, when a request is fulfilled, a new subscription can be added to the service CI.

The technical service catalog itself is implemented in the Configuration Management database. The technical services and their components are modeled as CIs in the Service Manager Configuration Management module (the Configuration Management Database (CMDB)). At the top level, each service is also modeled as a CI with links to its related systems and components definition in the CMDB.

Note: Service Manager also defines business services as a CIs.

Service Catalog Integration: Business Agreements

Several business agreements can be managed in the Service Catalog. You can define operational SLAs that apply to the catalog item (service). Or, you can define agreed-upon delivery objectives and the delivery objective options in the catalog item.

Service Catalog Integration: Vendor Performance

Performance is measured against Service Level Objectives (SLOs), which are defined within SLAs and Operational Level Agreements [OLA], and assigned to relevant records. SLOs can be created for Service Catalog requests, as well as the records created in other Service Manager modules to fulfill these requests. Fulfillment records can be assigned to vendors/suppliers, and their performance can be measured against applied SLOs.

Service Catalog Integration: Service Level Management

There are several ways to measure the request fulfillment using Service Level Agreements (SLAs).

Monitoring the Service Request

All catalog requests are opened and associated with the requester's default SLA. This SLA is defined in the requester's department or at the company record level, or at the system level.

Note: There is a default SLA defined in the Service Manager Service Level Management module.

In the SLA, you can define Response Time Service Level Objectives against catalog requests. These objectives will typically define the agreed-upon time for the request to move from one state to another. In particular, administrators can define a time agreement from request approval to request fulfillment. It is possible to have multiple response time objectives that will each only apply to certain kind of service requests. Service Manager offers flexibility in that area. In particular, it is possible to link the

response time to selected delivery objectives. Delivery objectives are delivery options that can be defined as part of the Service Catalog module, which helps define the target delivery date.

Each catalog item or bundle definition record contains a Delivery Targets tab where a catalog manager can add or remove multiple objectives and list dependencies for these objectives.

Delivery Targets can be limited to specific group of users, based on their default SLA. For example, consider a delivery objective of three business days that provides one service level with associated limitations:

Delivery Target	Limit Access to ...
3 Business Days	Sales Gold, Marketing Gold, Executive Gold
6 Business Days	All other users

As defined above, the three-business-day delivery objective is only available to specific requesters with listed Gold SLAs. Everyone else requesting the service is given only one (and thus defaulted) choice: six business days.

Monitoring the Fulfillment Process

It is also possible to monitor directly the fulfillment processes that are spawned from the service requests. The same default user SLA that was used for the service request will be used for the fulfillment request. Any response-time service level objective that applies will help measure the fulfillment process.

1.13.14. Service Catalog Ordering: Service Request Forms

The Service Catalog module provides the ability to add dynamic fields to a catalog item, as well as specific instructions to fill in those values. The application then uses this information to generate a custom request form. The dynamic fields are described by their name (not user visible), a label, a widget type and a data type.

The Catalog Manager has the ability to preview the dynamic fields the way the users would see them. The Service Catalog takes care of all the validation logic based on the simple definitions.

1.13.15. Service Catalog Audit Trail

Service Manager doesn't audit the Service Catalog items but has a utility by which the system can create a copy of the catalog item record every time it is updated. In this way all changes to the items are fully documented.

Service Manager date/time and user stamps to each record when the record is created or saved. Service Manager uses this to provide an audit trail capability that identifies the updates to the CIs/services. Each time the CI record is updated (whether manually or automatically), updates document the attributes that were updated.

Service Manager also provides an auditing feature that records modifications to fields within the Service Manager database. Field modifications are detected by comparing the fields in the original version of a record to the updated version. When modifications are detected, an Audit Log entry is recorded for each changed field showing the name of the modified field, the old and new version of the data, the current date/time, and the current operator’s user ID.

Related topics

Audit processing

1.13.16. Service Catalog: Search

The Service Catalog interface has built-in search capabilities. The end user can search the entire catalog or can restrict the service to a specific category. In either case, Service Manager allows the end user to see search results based on items available to that user.

The search engine performs a word search in the service catalog item description field. The search technology is flexible enough to find a match if the search terms share a common root. For example, a search for “security” will return items whose description uses the words “security,” “secure,” or “securely.”

This leniency speeds up the search throughout the catalog, especially for business users that aren't exposed daily to the catalog, or may not be familiar with its exact wording of the item being sought.

Note

- IR search must be enabled for Service Catalog search. To enable IR search, set the **ir_disable** parameter to 0.
- To enable Service Catalog search, the **description** and **displayName** fields in the svcDisplay DBDICT are set as the IR key by default.

1.14. Service Level Management

Service Manager Service Level Management uses Process Designer to manage Service Level Agreements, Operational Level Agreements, and Underpinning Contracts (UCs). Service Level Management also creates and maintains Service Level Targets to monitor and measure the availability and response times for services.

Key concepts

Describes the most important things that you need to know about how the Service Level Management module functions.

Structure of a service level agreement

Defining Service Level Targets

Service Reviews

Service Level Target selection process

Workflows

Describes the workflows implemented by Service Manager and the sub-tasks that end users must perform in order to execute the workflows.

Default agreement workflow

User Roles

Service Level Management uses the Administrator or System Administrator user roles for most tasks.

Tasks

Describes the Administrator tasks in the Service Level Management module.

Create a new Service Level Agreement

Create an Operational Level Agreement

Create an Underpinning Contract

Add Service Level Targets

Create an SQP or SIP

Define availability Service Level Targets

Define response Service Level Targets

Assign a default Service Level Agreement

Assign a Service Level Agreement to a company

Assign a Service Level Agreement to a department

Additional resources

Processes and Best Practices

1.14.1. Service Level Management overview

Service management best practices describe the goals for Service Level Management (SLM) as follows:

The goal for SLM is to maintain and improve IT Service quality, through a constant cycle of agreeing, monitoring, and reporting upon IT Service achievements and instigation of actions to eradicate poor service – in line with business or cost justification. Through these methods, a better relationship between IT and its customers can be developed.

Service Manager supports these goals by providing a service management best practices compliant application framework with a built-in workflow that incorporates service management best practices. The primary Service Level Management goal is to ensure the delivery of services within agreed Service Level Targets (SLTs), including the following:

- Availability of the Configuration Item, which includes planned and unplanned outages
- Response time, which tracks the amount of time it takes for the incident, service desk interaction, or change request to advance to the next state (For example, the amount of time required for an incident record status to change from Open to Work in Progress.)

Configuring Service Level Targets and the various types of service agreements are the tools that Service Level Management uses to accomplish these goals.

Related topics

Availability
Service Level Management drivers
Service agreements overview
Working with service agreements
Service Contracts overview
Working with service contracts

1.14.1.1. Availability

Availability describes the amount of time a Configuration Item (CI) is able to perform its role or function in the infrastructure. The usual metric for reporting availability is a percentage of time that the CI is in service during agreed hours. Service Level Management gathers availability data from change requests, change tasks, and incidents.

Related topics

Service Level Management overview

1.14.1.2. Service Level Management drivers

The impetus for a robust Service Level Management system is the improvement in the quality of service, reduced number of incidents, and increased customer satisfaction. Over time, you can quantify the financial benefits in reduced incidents, outages, and time invested in system failures and downtime. The time and money recovered can be invested in improved customer relationships, more sophisticated monitoring, better training, and improved business efficiency.

The following Service Manager applications provide input to the Service Level Management process:

- Change Management (changes and change tasks)
- Incident Management (incidents and incident tasks)
- Problem Management (problems and problem tasks)
- Request Fulfillment (requests and request tasks)
- Service Desk (interactions)
- Items from the Service Catalog

The output of Service Level Management is data that drives incident queue prioritization, generates immediate response for breaches, and monitors the ongoing status.

Related topics

Service Level Management overview
Service agreements overview
Working with service agreements
Service Contracts overview
Working with service contracts

1.14.1.3. Service agreements overview

Service agreements come in three types. A Service Level Agreement (SLA) describes the agreed level of service between service providers and customers. It defines service goals and responsibilities for Configuration Items (CIs). An SLA is an external document between the IT department and its customer. An Operational Level Agreement (OLA) is an internal document describing the level of service among the departments within an organization. An Underpinning Contract (UC) is an external document describing the level of service between an organization and a third-party vendor/supplier. While all agreements are equal, UCs and OLAs are created to support the implementation of an SLA.

These service agreements define the following aspects of service management:

- Availability, which describes the availability of a CI within a specified time frame.
- Response, which describes the performance targets.

Service agreements are monitored automatically to continuously recalculate availability and response metrics. Service Manager applications provide availability and response metrics to Service Level Management. For example, Service Level Management summarizes outage information from incidents and change requests to determine the performance of service guarantees for affected CIs.

Service Manager integrates service agreement support into Change Management (changes and tasks), Incident Management (incidents and tasks), Problem Management (problems and tasks), Request Fulfillment (requests and tasks) and Service Desk. Administrators can use these metrics to help prioritize incident resolution, schedule tasks, and escalate incidents when the service agreement is in jeopardy.

1.14.1.3.1. Why are Service Level Agreements important?

Service agreements manage performance tracking information and provide system feedback on service agreements among departments within a company. You can use this information to quantify the level of service you receive within your organization or with outside vendors and suppliers, and to determine if resources are available when you need them. It is important to know when an outage occurs so that the provider responds as agreed.

Service Level Management collects performance information automatically to track service targets. If you can detect the breach of a service target, it is easier to protect yourself from future occurrences and limit the economic impact.

1.14.1.3.2. Working with service agreements

There are several tasks that relate to service agreements:

- Accessing Service Level Agreements from other applications
- Assigning Service Level Targets to an application record
- Assigning a default SLA or OLA to a service desk interaction, incident, problem, request, or change
- Assigning an SLA to a contact or company
- Creating and updating an SLA
- Creating reports
- Creating dashboards
- Creating Operational Level Agreements
- Creating Underpinning Contracts
- Linking service contracts
- Monitoring availability and response performance
- Recalculating performance
- Tracking outages
- Working with clocks

Related topics

Service Level Management overview
Service Level Management drivers
Service agreements overview
Operational Level Agreements
Service Level Agreement components
Service agreement selection process
Outages
Service Level Agreement performance and reporting
Service Contracts overview
Working with service contracts
Create a new Service Level Agreement
Edit a Service Level Agreement
Access service agreements from other applications
View service agreements from Configuration Management

1.14.1.3.2.1. Operational Level Agreements

Service guarantees exist between IT and other departments in the organization to track hardware or software asset availability and response performance. For example, the IT department might guarantee that a department server will be available 98% of the time and that 99% of the time IT will respond to an outage involving that device within one hour, or that a change will be complete by a specified time.

An Operational Level Agreement (OLA) is a service agreement that describes these guarantees, tracks compliance, and calculates the potential economic impact of outages within the organization. Most organizations use OLAs to:

- Focus on discrete performance metrics, such as hardware availability.
- Add metrics for service desk performance, technician response time, and customer satisfaction.
- Assess the economic impact of outages on the enterprise.
- Publish satisfaction statistics to the internal user community.

The OLA produces metrics through links to the contact records of all responsible personnel, including problem owners. An administrator must ensure that the contact records of all affected personnel have valid Service Manager IDs.

The out-of-box examples show how to define OLAs for internal services, such as Change Management or Problem Management.

Related topics

Working with service agreements
Service Level Agreement components
Service agreement selection process
Outages
Service Level Agreement performance and reporting
Create a new Service Level Agreement
Edit a Service Level Agreement
Access service agreements from other applications
View service agreements from Configuration Management

1.14.1.3.2.2. Service Review

To ensure optimum service provisioning, it is imperative that you periodically review all service agreements and take appropriate corrective actions if needed. To achieve this, every agreement has fields to specify the Review Date of the agreement, the Next Service Review Date, and the frequency with which the service review should take place.

Generally, actual Service Level Agreements should be reviewed at least once per year with the signatories of the agreement. This is to ensure that the agreement is still valid, business requirements have not changed, or organization capabilities have note been extended or increased.

Additionally, service reviews, which are distinct from the agreement review, should occur more frequently. Service reviews are often held quarterly, and use quantifiable data to ensure that all agreements are fulfilled, whether between your organization and your customer (e.g. Service Level Agreements), within the various departments of your organization (e.g. Operational Level Agreements) or between your organization and a third party (e.g. Underpinning Contracts).

Service reviews produce the following two types of output:

- Service improvement plans (SIPs)
- Service quality plans (SQPs)

Service improvement plans are usually the result of regularly monthly meetings and are limited in scope. They usually describe short term fixes. In contrast, service quality plans are usually long term strategies to provide better service over months or years. For example, an organization may find that their employees require more comprehensive technical training, the results of which become apparent much later.

Note

SQPs and SIPs are stored in Service Manager as knowledge documents of the corresponding subcategory.

1.14.1.3.2.3. Service Level Agreement components

A Service Level Agreement (SLA) has few required fields. You must specify this information:

- Type that identifies an SLA as either a Service or Customer SLA
- Customer name
- Title (name) of the SLA
- Date when the SLA begins
- Date when the SLA expires

The optional information describes Service Level Targets, a linked contract (if applicable), and related agreements.

An SLA produces metrics through a link to a department or company record. An administrator must ensure that department records specify which SLA governs the department, and that company records specify which SLA governs the company.

Details tab

The **Details** tab is a repository for additional notes, a record of verbal agreements, or comments that relate to the Service Level Agreement.

Linked service contracts

You can create a connection between a Service Level Agreement (SLA) and a service contract. The SLA tracks response and availability objectives. The service contract tracks which services, such as service desk interactions or incidents, are consumed and which services remain.

Service contract information in an SLA is informational; however, it is important during the SLA selection process to populate information in an application record.

Link a service contract to a Service Level Agreement

Applies to User roles: System Administrator, Administrator

Service contracts describe the type of service available to a customer. You create a connection between a Service Level Agreement (SLA) and a service contract to track which services are used as part of the SLA.

To link a service contract to an SLA:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record.
4. Click the **Service Contract** list and choose the service contract to link to the Service Level Agreement.
5. Click **Save**.
6. Click **OK**.

1.14.1.3.2.4. Service Level Agreement alerts

The Service Level Agreement (SLA) include the following five out-of-box alerts:

- SLA - 25%
- SLA - 50%
- SLA - 75%
- SLA - BREACH
- SLA - Objective 30 minutes from expiration

These alerts trigger standard notifications when SLA deadlines approach a pre-defined threshold. Administrators can change these out-of-box settings or create new ones that apply your business rules to SLA alerts.

Alert levels

You can set alerts at these five alert levels:

- You can add or edit SLA alerts for a Service Level Target (SLT). Select the Service Criteria tab or Process Criteria tab of the SLT record and go to the Escalations section. Double-click an existing alert to edit the alert, or insert the cursor in an empty row to add an existing alert to the individual SLT.
- You can specify alerts in an SLM Integration record in the **slamodulecontrol** table. Click **Service Level Management** > **Administration** > **Configure Application**. Click **Search** to display a list of existing SLM Integration records.
- You can define alerts for all Service Targets in the SLA Control record. Double-click an existing alert to edit the alert, or click in an empty row to add an existing alert to all Service SLTs.

Standard alerts and Service Level Agreements

Standard alerts have components that enable you to reference Service Level Agreements (SLAs) if you create new alerts or modify out-of-box alerts.

Expression	References
$L.slo.expiration	Service Level Target (SLT) expiration date and time
$L.slo.name	SLT name
$L.slo.id	SLT unique identifier
$L.slo.condition	SLT condition
$L.slo.record.active	Active condition of the record
$L.slo.allowed	Amount of time until a breach occurs

1.14.1.3.2.5. Service Level Agreement performance and reporting

You can use Service Manager dashboards to gather service information, or use any standard report package to customize your reports. When using reporting, you can see achievements by assignment group or by customer; overall Service Level Agreement (SLA) achievement levels; or breaches by category, group, or priority.

Use the data that you gather to plan service improvements. For example, knowing which SLAs are breached more often than others enables you to identify the point of failure. If you can identify a pattern of response failure, such as time of day or day of the week, you can add coverage at critical points of failure, or implement other preventive measures that interrupt the failure pattern.

Tracking availability failures through reports can also show performance patterns, such as equipment failures, peak overload periods, or under-performing external vendors/suppliers.

Process metrics

Each service desk interaction, change request, incident, or problem record displays metric information when you enable the Service Level Agreement (SLA) component to work with these applications. Thereafter, each new service desk interaction, change request, incident, or problem record has an SLT section that identifies the related SLA and shows the Service Level Target (SLT) performance metrics.

The process information includes:

- Status
  - Achieved
  - Running
  - Breached
  - Suspended
  - Inactive
- SLT Name
- Agreement Name
- Agreement ID
- From
- To
- Expiration
- Total Elapsed Time

Response time data

Service Level Targets (SLTs) measure response time guarantees in the case of an outage. Service Level Management gathers the following information about response times from SLT records and stores it in the Service Level Agreement (SLA):

- Application name
- Initial state
- Final state

When you view the SLA, Service Level Management displays a complete list of all process and service SLTs and their current status. Service Level Management gathers response data from service desk interactions, change requests, change tasks, incidents, problems, problem tasks, requests and request tasks.

Calculation of Process Target metrics

Based on predefined Process Targets, a record must be handled within the specified period of time. Otherwise, the target is breached. A duration is usually used to measure record handling performance. You can measure the record handling performance of a service provider, an internal group, or a supplier, and calculations can be done either without considering the groups that are assigned to handle a record, or by considering groups that are assigned to handle the record.

Metric calculation without considering assignment groups

When Process Target metrics are calculated without considering assignment groups, no specific group is involved in the calculation. You can specify one of the following methods for the system to calculate the metrics. If a record that's measured by a Process Target is not processed from its initial state to its final state within a predefined duration, the system treats the target as being breached.

- Using a specified duration

  With this method, you specify a duration for a Process Target when you define the target.

  To enable the system to use this calculation method, select **Interval** in the **Duration Type** field and specify a duration in the **Duration** box when you define a Process Target.

  The duration must be specified in this format: *dd hh:mm:ss*. In this duration type, one day is counted as 24 hours. For example, if the working hour lasts 8 hours based on the current schedule, and you specify the interval as 1 05:00:00, the

duration (24+5=29 working hours) is treated as 3 business days and 5 business hours.

The duration takes effect once a record that's associated with the Process Target enters its initial state.

- Using the duration specified in the record

  With this method, you specify a duration for a Process Target in the record that's associated with the target. For example, you can add a new field in the incident table (probsummary), and then use the specified value in that field as the duration for the Process Target.

  To enable the system to use this calculation method, select **Duration in Record** in the **Duration Type** field and specify the field that contains the predefined duration in the **Field in Record** box when you define a Process Target.

  Similar to using a specified duration, the duration must be specified in the format of *dd hh:mm:ss*, and one day is counted as 24 hours in this duration type. The duration takes effect once a record that's associated with the Process Target enters its initial state.

- Using a calculated duration

  With this method, the duration of a Process Target is calculated based on different conditions.

  To enable the system to use this method, select **Calculation** in the **Duration Type** field when you define a Process Target. You can further choose one of the following conditions.

  - End of business day

    You can usually use this condition to specify that a record must be processed to its final state before the end of a business day.

    You can optionally specify a buffer period for record handling. For example, if the business day ends at 5:00 pm and the buffer period is 3 hours, records that are submitted before 2:00 pm must be handled before the end of the current business day whereas records that are submitted after 2:00 pm can be handled before the end of the next business day.

    To enable this condition, select **End of Business Day** in the **Duration Calculation Method** field. If you want to specify a buffer period, type it in the **End of Day Window** box, with the format of *dd hh:mm:ss*. If no buffer

period is set, the actual end time of a business day that's defined in the schedule is used.

- End of business week

  You can usually use this condition to specify that a record must be processed to its final state before the end of a business week.

  You can optionally specify a buffer period respectively for the current business day and for the current business week for record handling. For example, if the business week ends on Friday and you specify that the "End of Week" day as Thursday so as to have one day buffer, records that are submitted on or before Thursday must be handled before the end of the current business week whereas records that are submitted after Thursday and before the end of the current business week can be handled in the next business week. When buffer periods are specified for both the business day and the business week, the "end of business week" calculation is combined with the "end of business day" calculation.

  To enable this condition, select **End of Business Week**in the **Duration Calculation Method** field. If you want to specify a buffer period for the business day, type it in the **End of Day Window** box, and if you want to specify a buffer period for the business week, select a day in the **End of Week** drop-down list for the system to calculate the buffer.

- End of business day plus interval (Work Schedule based Expiration)

  You can optionally specify an additional period for record handling. This condition can extend the record handling duration by several business days and hours.

  The system starts to count the additional period based on the setting of the buffer time. For example, if the business day ends at 5:00 pm and the buffer period for the business day is 3 hours, the system starts to count the additional period of the Process Target from the beginning of the next business day for records that are submitted before 2:00 pm whereas it starts to count the period from the beginning of the third business day for records that are submitted after 2:00 pm.

  In this calculation, the length of the actual business day might vary based on the definition of the work schedule, and only business hours are counted by the system, which means that public holidays, vacations, record suspending time, and other non-working time are excluded. The non-working time is added back after regular calculation. For example, if a

record handling process is delayed by a public holiday (1 business day), which counts for 3 business hours based on the work schedule, the 3 business hours are added back after the regular calculation.

Consider the following scenario:

- The current work schedule is from 9:00 am to 3:00 pm on Mondays, 10:00 am to 1:00 pm on Tuesdays and Wednesdays, and 9:00 am to 2:00 pm on Thursdays and Fridays.
- The end of business day buffer is 3 hours.
- The additional period for record handling is 2 4:00:00.
- The record is submitted at 11:00 am on Monday.

In this scenario, the system starts to count the additional period from 10:00 am on Tuesday, and Tuesday and Wednesday are counted as 2 business days. The remaining 4 hours of the period are counted from 9:00 am on Thursday. After this calculation, the target expiration time is 1:00 pm on Thursday.

If non-working time is involved, the time is added after the regular calculation above. For example, if Wednesday is a public holiday in the scenario, the system adds the 3 hours (10:00 am to 1:00 pm on Wednesday) back after the regular calculation. Then the new target expiration time is 11:00 am on Friday.

To enable this condition, select **Work Schedule based Expiration** in the **Duration Calculation Method** field. Optionally, you can specify a buffer period in the **End of business day buffer** box, and specify the additional period for record processing in the **Additional interval (business hour)** box. If no buffer period is set, the actual end time of a business day that's defined in the schedule is used. If no buffer period is set, the actual end time of a business day that's defined in the schedule is used.

- Service Catalog Item Expiration

  With this condition, you specify that the system must use the duration defined in a Service Catalog item as the duration of the Process Target. For this option to take effect, Delivery Targets must be defined for the corresponding Service Catalog item.

  To enable this condition, select **Service Catalog Item Expiration** in the **Duration Calculation Method** field when you define the Process Target. For more information about how to make the Service Catalog Item

Expiration calculation method take effect, see Example: How to make Service Catalog Item Expiration work.

Metric calculation for assignment groups

When you define a Process Target for an Operation Level Agreement (OLA) or a Underpinning Contract (UC), you might want to better understand the performance of the groups that are assigned to handle the record. In this case, you can specify to which assignment group the duration of a target applies. Based on your specified option, the system either stops or continues its counting for the Process Target when the record is reassigned among different groups.

Note

- This calculation only applies to Process Targets for OLAs or UCs.
- If multiple groups are involved, they must agree on the same OLA. Otherwise, the targets associated with different OLAs are measured separately.

As vendor/supplier is managed as an external group in Service Level Management, the measurement methods for internal groups and external groups are the same except that internal groups agree on an OLA while external groups agree on a UC. Therefore, the following sections don't differentiate between internal groups and external groups.

Metric calculation for all assignment groups

When this method is used, the system counts the duration of a Process Target together for all the groups that are assigned to handle the record. The counting doesn't stop when the record is reassigned among different groups.

Consider the following scenario:

- The work schedule is 24×7, and the default time zone of the system server is used.
- The duration of a Process Target is 4 hours.
- The record is assigned to group 1 at 9:00 am.
- The record is reassigned to group 2 at 11:00 am.

In this scenario, as the record has already been processed by group 1 for 2 hours, only 2 hours are left for group 2, and the target becomes breached from 1:00 pm.

The breaching time is different if group 1 and group 2 have different schedules. For example, if the work schedule of group 1 is from 9:00 am to 5:00 pm and that of

group 2 is from 1:00 pm to 9:00 pm, group 2 is not working when the the record is reassigned to it at 11:00 am, and thus the record is suspended from 11:00 am to 1:00 pm. The counting resumes when group 2 starts to work at 1:00 pm, and the target becomes breached from 3:00 pm.

To enable the system to use this method, select **Total time for all assignment groups** from the **Duration scope** drop-down list.

Metric calculation for each assignment group

When this method is used, the system counts the duration of a Process Target for each individual group that's assigned to handle the record. The counting pauses when the record is reassigned to another group and it resumes when the record is assigned back.

Consider the following scenario:

- The work schedule is 24×7, and the default time zone of the system server is used.
- The duration of a Process Target is 4 hours.
- The record is assigned to group 1 at 9:00 am.
- The record is reassigned to group 2 at 11:00 am.
- The record is reassigned back to group 1 at 2:00 pm.

In this scenario, group 1 uses 2 hours to process the record, and when the record is assigned back, it has 2 hours left before the target becomes breached, which means the breaching time is 4:00 pm for group 1. Group 2 uses 2 hours to process the record, and 2 hours of the target duration is left, which means the target is not breached for group 2 when it processes the record.

To enable the system to use this method, select **Total time for each assignment group** from the **Duration scope** drop-down list.

Metric calculation for the current assignment group

When this method is used, the system counts the duration of a Process Target for the group that's currently assigned to handle the record. The counting starts when the record is assigned and it restarts each time when the record is reassigned, including the situation in which the record is reassigned back to the original group.

Consider the following scenario:

- The work schedule is 24×7, and the default time zone of the system server is used.
- The duration of a Process Target is 4 hours.

- The record is assigned to group 1 at 9:00 am.
- The record is reassigned to group 2 at 12:00 pm (noon).
- The record is reassigned back to group 1 at 2:00 pm.

In this scenario, group 1 uses 3 hours to process the record initially. When the record is reassigned back, the system restarts its duration counting for group 1, so the target won't be breached by group 1 until 6:00 pm. As to group 2, it uses 2 hours to process the record, so the target is not breached by that group.

To enable the system to use this method, select **Time for the current assignment group** from the **Duration scope** drop-down list.

Monitoring the peformance of an assignment group

When you monitor the peformance of assignment groups, you might be interested in the following two types of targets:

- Response time targets: These targets reflect how soon a group can start processing a record that's assigned to it. Usually it is a period in which the status of a record changes from "open" to "work-in-progress".
- Resolution time targets: These targets reflect how soon a group can resolve a reported issue. Usually it is a period in which the status of a record changes from "open" to "resolved".

As a general practice, you can usually select **Time for the current assignment group** in the **Duration Scope** field when you measure the response time targets, because there should be enough time for a group each time when a record it assigned to it.

For resolution time targets, you can usually select **Total time for each assignment group** in the **Duration Scope** field, because the total time for each group to resolve an issue is measured individually, and the performances of different groups are comparable.

Service metrics

Service Level Management enables you to view Service Level Agreement (SLA) performance from an overall perspective and target individual devices and response types. Service Level Management collects different types of metrics:

- Overall performance of all SLAs by month and year.
- Availability of a resource for a specified time:
  - Availability of all SLAs by month and year
  - Availability of one SLA by year
  - Availability of components named in one SLA by month and year

- Availability of a single device by year
- Outage history of a single device by month and year

Each SLA displays the current status of its named Service Targets. You can view the status, required availability, current availability, and the next expiration date and time. Service Level Management gathers service metrics from change requests, tasks, and incidents.

Availability data

Service Level Targets (SLTs) measure Configuration Item (CI) availability, such as a server or an application. Service Level Management gathers the following information about availability from SLT records and stores it in the Service Level Agreement (SLA):

- Availability (uptime) objectives
- Affected CI
- Current performance
- Expiration date and time for the objective

When you view the SLA, Service Level Management displays a complete list of all availability and response SLTs and their current status. Service Level Management gathers availability data from change requests, tasks, and incidents.

Research response and availability results

The slaresponse table contains summary records with response time information collected from sloresponse records. Includes the number of affected and breached records, mean, median, and standard deviation data.

The slamonthly table contains one record with a summary of monthly outage information for each Configuration Item (CI). You can trace the input data for each response or availability record whenever you view a record in the sloresponse or slamonthly table.

The slamonthlyag table contains one record with a summary of monthly outage information for each Service Level Agreement.

Recalculate Service Level Agreement data

Service Level Management enables you to view Service Level Agreement (SLA) performance from an overall perspective, and targets individual devices and response types. The Regen SLM Partials function recalculates Service Level Management response and availability data for a specified month and year.

Service Level Agreement dashboards

You can display statistical metrics using a dashboard. For example, you can show the current status of incidents that contain Service Level Target (SLT) statistics:

- SLTs that meet expectations
- Breached SLTs
- SLTs pending expiration

You can filter the criteria to show categories, assignment groups, total loss of service, historical achievement, or any other relevant statistical information.

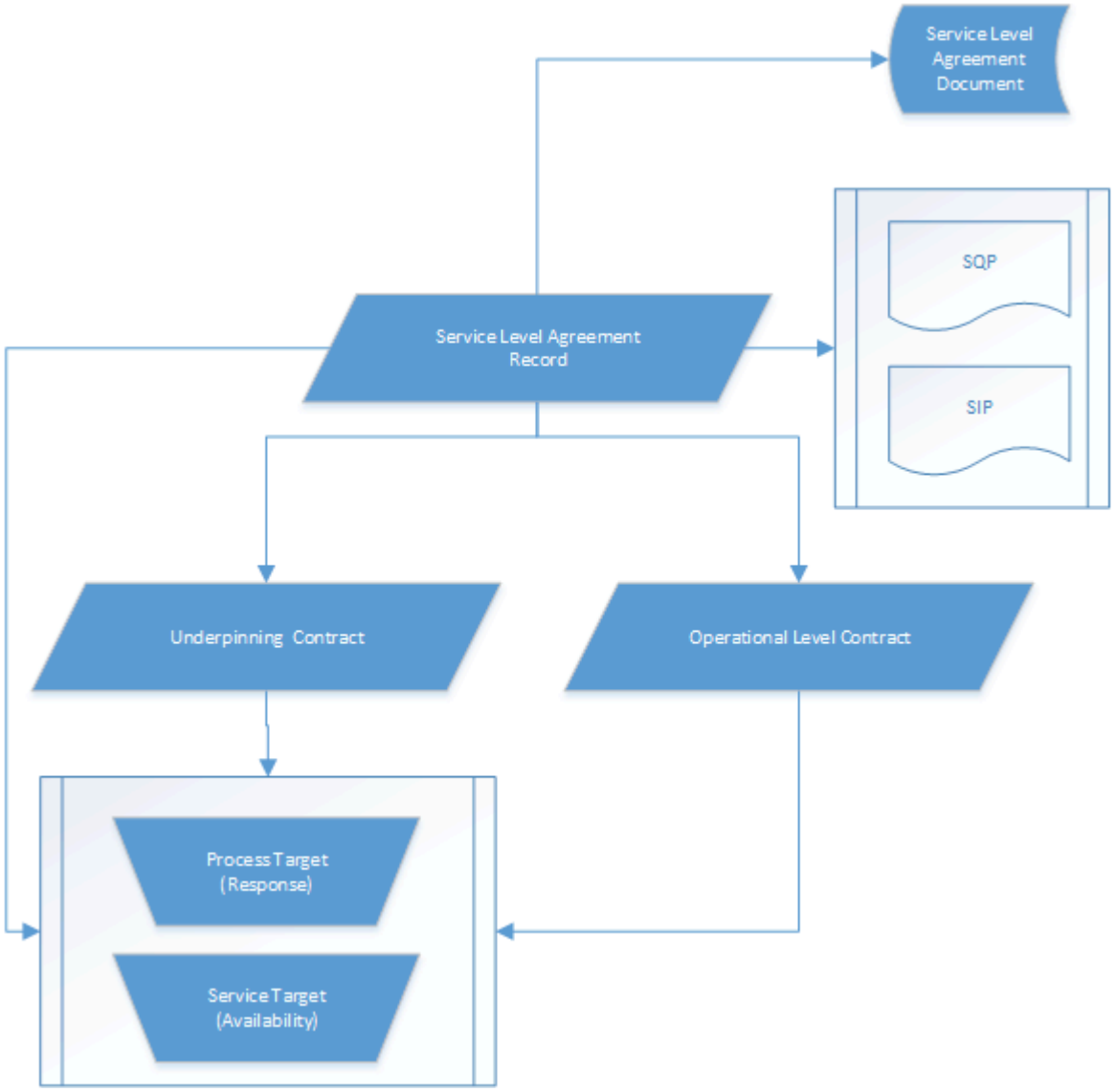
For customized reports, use any reporting package to design the report and specify which data appears in the report.

Service Level Agreement reports

Service Level Management produces metrics that you can view within summary records. For example, a Process Service Level Target (SLT) record displays a monthly performance summary. You can also create simple dashboards to display data visually. For customized reports, use any standard reporting package.

1.14.1.3.3. Structure of a Service Level Agreement

The following diagram illustrates the relationship of the various components of Service Level Agreements (SLAs) in Service Manager:



In this diagram, the centerpiece is the Service Level Agreement record. The Service Level Agreement contains several components:

- The official Service Level Agreement document, which is usually a scanned and signed document held in the record as an attachment.
- Underpinning Agreements, which are the supporting agreements of the following types:
  - Underpinning Contract: An agreement between your organization and another party outside of your organization.

    - Operational Level Agreement: An agreement with another department of your organization (for example, an agreement between Level I and Level II Support).
- Specific Service Level Targets to measure response and availability.
- Service agreement start and end dates.
- The date of the last review, the date of the next review, and the frequency with which that review should occur.
- For Service Level Requirements (SLRs, which become Service Level Agreements after the agreed stage) of the “Service” type, an additional tab for Subscriptions appears. This allows you to see what individuals have subscribed to the SLA.

Note

OLAs and UCs follow the same structure, except that those agreements don't have underpinning agreements.

1.14.1.3.4. Service Level Targets

Service Level Targets (SLTs) are the goals that you set for response time and availability. Some SLTs define the expectations between customers and the IT organization. For example, the email server must be available 99.9% of all business hours. Other targets fall under the domain of Operational Level Agreements (OLAs) that define expectations among various internal support teams, such as Level I and Level II support.

Process targets set goals for service desk interactions, incidents, change requests, problems, and user requests. They define the amount of time required to move the record from one state to another. For example, the status of an incident must change from *Open* to *Work in progress* within 3 hours.

Service targets define the percentage of availability during a month, or the maximum amount of time for a single outage. For example, an email server must be available 99.999% of the time during normal work hours.

Service Level Target catalog

The first step to implement Service Level Management is to create a catalog of pre-defined response and availability service goals called Service Level Targets (SLTs). The catalog contains SLT templates that you can apply to any Service Level Agreement (SLA).

SLT catalog records always contain the SLT name and SLT type (process or service), and may include the following information:

- Configuration Item
- Service and status
- Schedule
- Availability
- Escalations
- Text description

The process to define SLT templates is separate from the process to define a service agreement. Service Level Management stores all SLT templates in the slocatalog table. When you create or edit a service agreement, you can add SLTs from the template catalog, customize them, or you can define new ones.

Contents

- 1 Service Level Target catalog

1.14.1.3.5. Application integration

Service Level Management integrates with Change Management (requests and tasks), Incident Management (incidents and tasks), Problem Management (problems and tasks), Request Fulfillment and Service Desk. This integration enables Service Level Management to collect metric information automatically and monitor the performance of associated service agreements.

Service Manager has an out-of-box configuration record in the **slamodulecontrol** table for each application. You can modify these records instead of creating new records. To modify the records, click **Service Level Management** > **Administration** > **Configure Application** in the system navigator, and then click **Search** in the SLM Module Control form.

You can disable Service Level Management by clearing the **Enable SLM in this Application** option.

Linking application records to service agreements

When you enable an application to work with Service Level Management and you create new service desk interactions, incidents, problems, requests, service catalog items, or change records, you can link the application record to a service agreement in the following ways:

- Customer SLA

  Service Manager automatically assigns a department or company SLA as the Customer SLA to the application record if it exists. This would be an SLA defined for the contact's department or company.

- Service SLA

  The SLA for the contact's subscription to a Configuration Item (CI). This is true only if the application record references a Business Service, the contact has a subscription to the service, and the subscription references an SLA.

- Default SLA

  If there is no SLA but there is a default SLA specified in the SLA control record, Service Manager automatically assigns the default SLA to the application record.

Calculating response time

Service Manager calculates response time with the **sla.state.change** application. To measure response time for an application event, Service Manager uses the out-of-box (or customized) **slamodulecontrol** record for the external application. In addition, the administrator must ensure that the application record has fields that are referenced by the **sla.state.change** application, such as:

- sla.breach
- next.breach
- logical.name.vj.duration
- logical.name.vj.sloavail

For example, to enable process metrics for a module, an administrator must add the **sla.breach**, **next.breach**, **logical.name.vj.duration**, and **logical.name.vj.sloavail** fields to the record table before SLM can be configured for the module. Most of the major tables (probsummary, request, etc) that can use SLM already contain these fields out of box.

Example: Enabling Service Level Agreement alerts

The following example shows you how to generate response Service Level Targets (SLTs) for Change Management requests and to generate alerts for SLTs. At the end of this example, you will be able to complete the following tasks:

- Create a Service Level Agreement (SLA)
- Add a Service Level Target (SLT)

Note

To enable Service Level Agreement alerts for Change Management, you must perform the entire sequence of example tasks. You must be a System Administrator, or have equivalent system privileges, to perform tasks 1 through 7.

Task 1: Ensure the background processes are running

1. Click **Menu Navigation** > **System Status**.
2. If the processes aren't running, click **Start Scheduler**.
3. In the **Name** field, double-click **startup** to start all the background processes.

Task 2: Enable the SLM application

1. Click **Service Level Management** > **Administration** > **Edit Control Record**.
2. Select **Enable SLM Application**.
3. Click **Save**.
4. Log off Service Manager, and then log back on for the changes to take effect.

Task 3: Configure Change Management

1. Click **Service Level Management** > **Administration** > **Configure Application**.
2. In the **Table Name** field, type cm3r, and then click **Search**.

   The SLA module control form for cm3r opens. Ensure the **Enable SLM in this application** checkbox is selected.
3. Select **Run in Foreground**.
4. Click **Save**.
5. Click **OK**.

Task 4: Create an SLA

1. Click **Service Level Management** > **Agreements** > **New Agreement**.
2. Click **Service Level Requirement**.
3. In the **Type** field, select the contract type **Customer**.
4. In the **Customer** field, select a customer name from the list of records.
5. In the **Title** field, type SLA for change requests.
6. In the **Effective From** field, choose today's date in the calendar options for the SLA start date.
7. In the **Expiration Date** field, choose a date one year from today in the calendar options for the SLA end date.
8. Add an optional **Description** for the new SLA.
9. Click **Save**.

   You receive a message that states the service level agreement has been added. Service Level Management assigns an Agreement ID number to the record. Make note of the new agreement ID.
10. Click **Review** to move the record into the Review phase.
11. Click the **Agreed By** tab and specify the people who agreed to the SLA.
12. Specify appropriate values for the **Next Agreement Review Date**, **Service Review Frequency**, and **Next Service Review Date** fields.
13. Click **Agree** to move the record into the Agreed phase.

The category of the SLA is moved from Service Level Requirement to Service Level Agreement.

14. Click **Save** or **Save&Exit**.
15. Log off Service Manager, and then log back on to see that the new SLA is available in the selection lists.

Task 5: Add an SLT

Note

You can add a service level target immediately after you add a service level agreement, or you can add it later.

1. Click **Service Level Management** > **Agreements** > **Search Agreement**.
2. Type the **Agreement ID**, and then click **Search**.

   The SLA record opens.
3. In the **Process Targets** tab, click **Add SLT(s)**.
4. In the **Service Area** field, select **Change Requests**, and then click **Next**.
5. Fill in the following fields to add the SLT. Make sure you click **Next**, as required.

Field	Value
SLT Name	Response SLT for change requests
Description	Add an (optional) description.
Schedule	24×7
Priority	1-Critical
Category	Default
Initial State	Registration and Categorization
Final State	Closure
Duration Type	Interval
Duration	The format should be DDD HH:MM where DDD is the number of days, HH is the number of hours, and MM is the number of minutes. Type 00 01:00
Alerts	Click **Fill**, and then select **SLA-50%** from the list of records.

6. Click **Finish**.

   You receive a message that states a record has been added to Process Target. Make sure the record is displayed in the **Process Targets** tab.

7. Click **OK**.
8. Log out, and then log in again for your changes to take effect.

Task 6: Select the SLA definition for an ESS user

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. In the **Contact Name** field, type the user's name, and then click **Search**.

   The user record opens.

3. Place your cursor in the **Dept Name** field, and then click **Find**.

The department data form opens.

4. In the SLA field, select an SLA definition. For this example, select **SLA for change requests**.
5. Click **Save**.
6. Click **OK**.

Task 7: Service Desk Agent creates a change from a user interaction

As a Service Desk Agent, you can escalate an interaction to a change record.

To create a change request from a user interaction, follow these steps:

1. Click **Service Desk** > **Interaction Queue** or select **Interaction** from your To Do queue.
2. From the record list, double-click the Open-Idle interaction that you want to escalate to a change.
3. Click **Fill** for the Category field.
4. Double-click **Request for Change** and select the subcategory.
5. Click **Escalate**. The Escalation Details wizard appears.
6. Click the list fields to select the Location, Assignment, and Requested End Date for the change.

> **Note**
>
> A Change Coordinator is automatically assigned based on your selections.

7. Click **Finish**. A notification message with the change number displays above the interaction detail fields and the Service Desk application notifies the original end user of the actions taken.
8. Click **OK**.

Task 8: Change Coordinator goes to the change queue to review the upcoming alerts

After the Service Desk Agent escalates an interaction to a change, the change request moves to the Change Review phase. The Change Coordinator can then review the change request.

The Change Coordinator reviews the SLA and the Process Targets associated with the change. For this example, the duration was 1 hour, and the alert was 50%. The alert is

30 minutes from the time displayed on the Open Time in the **History** section.

1. Click **Change Management** > **Changes** > **Change Queue**.
2. From your To Do queue, select **Change** from the Queue list and **My Open Changes** from the View list.
3. Expand **Phase: Change Review** from the record list, and then click the change request to review it. Determine if it is complete and verify that the change is assigned to the correct support group. If more information is needed or the support group is incorrect, send the request back to the Service Desk Agent.
4. Open the **SLT** section. The new SLA definition is displayed.
5. Select the **Upcoming Alerts** subsection in the **SLT** section. the listed alert should be 30 minutes from the time displayed in Open Time, in the **History** section.

   The duration was one hour and the alert was 50%.

Access Service Level Agreements from Configuration Management

User roles: Coordinator [Configuration Management], System Administrator

You can access Service Level Agreements (SLAs) from Configuration Management in order to view an SLA and the details of its associated Process Targets. If you have System Administrator rights, you can also modify or add SLAs.

To access SLAs from Configuration Management, follow these steps:

1. Click **Configuration Management** > **Resources** > **Agreement Information**.
2. Add optional search criteria, and then click **Search** to generate a record list.
3. Click a record to view it in the Agreement form.

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    - 3.8 Task 8: Change Coordinator goes to the change queue to review the upcoming alerts
- 4 Access Service Level Agreements from Configuration Management

1.14.1.3.6. Defining Service Level Targets

You can define two types of Service Level Targets (SLTs) in Service Manager. They are process targets (response) and service targets (availability) .

You can add targets in two ways. You can select an existing SLT Catalog template with a pre-defined target, or you can define a customized target that meets the requirements of the individual Service Level Agreement (SLA).

When you create a new SLA, you can access the SLT catalog and select one or more existing template targets that support the SLA. If the existing SLT templates don't meet your requirements, then you can create a customized SLT.

Service Level Target duration types

The duration of a Service Level Target (SLT) specifies a span of time in which the target must be met. You can assign the following duration types to an SLT:

- **Interval**: Directly specifies the span of time in days or hours. The number of days and hours must be specified in this format: *ddd hh:mm:ss*.

  Note

  In this duration type, one day is counted as 24 hours.

- **Calculation**: Specifies the ways of calculating the duration. Only business hours are counted in these calculations. Service Level Management has the following calculated durations.
  - End of Business Day: Specifies when to start counting the time for an SLT in a business day. You can optionally specify a buffer before the end of a business day, so that records submitted within the buffer time are handled later.
  - End of Business Week: Specifies when to start counting the time for an SLT in a business week. You must specify when the business week ends and optionally specify a buffer before the end of a business day, so that records submitted after a certain day in the week or within the buffer time are handled later.

- Work Schedule based Expiration: On top of the "End of Business Day" calculation, this option offers an additional period of time for record handling.
- Service Catalog Item Expiration: Specifies that the duration defined in the Service Catalog item request is used.

> **Note**
>
> This option takes effect only when Delivery Targets are defined for the corresponding Service Catalog item request.

- **Duration in Record**: Specifies that the system uses the duration defined in the record as the duration of the SLT. When this option is selected, you need to enter a valid field name for the record in the **Field in Record**) box.

> **Note**
>
> In addition to fixed intervals, this option supports flexible intervals that are calculated from record fields. For example, calculations can be executed using a rule set from the workflow.

Service Targets

Service Targets monitor the availability of a Business Service or Configuration Item (CI), such as a service, application, hardware, software, or other infrastructure component. SLTs report any change to the availability of the CI in change requests, changes, tasks, and incidents.

Service Targets require the following information. You can use the SLT wizard or type required information directly into the form.

Field	Content	Comments
Service Level Target ID	Numeric value	Automatically assigned by Service Level Management.
Agreement ID	SLA name	Name of an existing SLA. Required.
Name	Service Target name	Descriptive name for the objective. for example, 95% Availability. Required.
Owner	SLT owner	Optional.
Affected CI	The Configuration Item (CI) related to this objective	Required.
Required Uptime (%)	0.0 – 100.0	Express as a decimal value. Do not use the percentage sign (%). **Required Uptime(%)** or **Max Outage Duration** is required.
Max Outage Duration	Number of days, hours, minutes, and seconds for a deadline (*ddd hh:mm:ss*)	The acceptable amount of time for an outage. **Required Uptime(%)** or **Max Outage Duration** is required.
Schedule	A pre-defined work schedule	Limits availability monitoring to the hours specified by the work schedule. Optional.

Field	Content	Comments
Time Zone	Time zone name	Select the time zone for the work schedule, the CI, or other related component. Optional.
Alerts	Alerts related to the remaining time on the Service Target	Select the out-of-box alerts for 50%, 75%, and 30 minutes until the SLT expires, or create your own. Optional.

Note

- You can define global alerts for all SLAs, instead of individual SLTs, in the SLA Control record.
- The **Description** describes additional information about customer expectations or the internal requirements of an Operational Level Agreement.
- The **History** tab for an availability SLT record shows the target and actual availability of the CI on a monthly basis.

Process Targets

A Process Target is a rule that describes the maximum amount of time allowable for a response to a business event. For example, if you report a priority 1 incident during business hours, work should be assigned within 15 minutes and a resolution should occur within an hour. A similar rule would be if you make a self-service request, it should be resolved or linked to an incident record within an hour.

Process Target rules contain the following information. You can use the SLT wizard or type required information directly into the form.

Field	Content	Comments
Service Level Target ID	Numeric value	Automatically assigned by Service Level Management.
Agreement ID	SLA name	Name of an existing SLA. Required.
Name	Process Target name	Descriptive name for the target. For example, One hour response. Required.

Field	Content	Comments
Condition	A status or condition value	Service Level Management uses the condition to determine if the SLT applies to the application record. If you use the SLT wizard, choosing filter criteria generates the condition in this field. For example, if the only filtering criteria selected is Severity 1, the SLT applies only to Severity 1 incidents. This expression is the value in the condition field: severity.code in $L.file="1" This expression must be written by using RAD. The criteria list changes depending on the application for service desk interactions, change requests, change tasks, incidents, problems, problem tasks, requests or request tasks. Optional.
Owner	Process Target owner	Name of the Process Target owner

Field	Content	Comments
Service Level Category	Category of the Process Target	The category can be **Service Level Agreement**, **Operational Level Agreement**, or **Underpinning Contract**.
Service Area	Application related to the SLT	For example, if you want to apply the response SLT to Change Management, choose Change Requests or Change Tasks. Required.
Initial State	A status or phase value (depending on module configuration)	The initial state when Service Level Management begins measuring the response time. The list changes depending on the application and whether phases are used, such as Request for Change phases. Required.
Final State	A status or phase value (depending on module configuration)	The final state when Service Level Management stops measuring the response time. The list changes depending on the application and whether phases are used, such as Request for Change phases. Required.

Field	Content	Comments
Duration Type	"Interval" or "Duration in Record" directly specifies the duration for your SLT.	The format is *ddd hh:mm:ss*. The system automatically completes the duration if you don't specify it in the complete form. For example, you can use **3:00** for 3 hours or **3** for 3 days. Required.
	"Calculation" calculates the duration for your SLT based on different conditions.	For example, choose **End of Business Day**, **End of Business Week**, **Work Schedule based Expiration**, or **Service Catalog Item Expiration**. Required.
Schedule	Shift information	Choose a pre-defined work schedule. If you omit a pre-defined shift, Service Level Management defaults to a 24 × 7 schedule. Optional.

Field	Content	Comments
Time Zone	Choose: Server time zone Technician time zone Customer time zone CI time zone Specific value Assignment Group	The process metric depends on the time zone selected. Optional. **Note** The **Assignment Group** option is only available for the OLA or UC service level category.
Alerts	Alerts related to the remaining time on the SLT	Select the out-of-box alerts for 25%, 50%, 75%, BREACH, or 30 minutes until the SLT expires, or create your own. Optional.
Suspend Processing	Suspension states	Click **Fill** on a blank row to view a list of states. The list changes depending on the application and whether phases are used, such as Request for Change phases. Optional.

Field	Content	Comments
Duration scope	Specifies to which assignment group the calculation applies.	The method that's used to calculate the expiration time of an Operational Level Agreement (OLA) or Underpinning Contract (UC) process target. This calculation is based on assignment groups.

Note

- You can define global alerts for all SLAs, instead of individual SLTs, in the SLA Control record.
- The **Description** provides additional information about customer expectations or the internal requirements of an Operational Level Agreement.
- The **History** tab for a response SLT record shows the percentage of time that the objective was met and number of breaches on a monthly basis.

Define scheduling information for Process Targets

System administrators can customize the values for work schedule, holiday group, and time zone in a Process Target record. By default, the Process Targets record uses the holiday group defined in the work schedule when calculating an interval. An administrator can select a value for the holiday group independently of the work schedule. When the administrator selects a value for the holiday group, it overrides the holiday group value in the work schedule.

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- 1 Service Level Target duration types
- 2 Service Targets
- 3 Process Targets
- 4 Define scheduling information for Process Targets

1.14.1.4. Outages

An outage is a temporary loss of service of a Configuration Item. It can be the result of planned maintenance or operational failure.

Outage classes

Outages fall into different classifications:

- Planned (and approved) outages
- Degraded service
  - Partial loss of service
  - Slow response time
  - Intermittent interruptions
- Total loss of service

Availability criteria should specify what the maximum outage threshold is on a monthly basis. For example, you can choose a metric based on the required percentage of up time, or the maximum time allowed for an outage, when you define an availability Service Level Target.

Planned outages

A planned outage is usually the result of a routine maintenance schedule, upgrade action, or faulty equipment replacement. Incidents, change requests, and change tasks can generate outage records. The change request and task records show scheduled downtime. Incident, change request, and change task records have a checkbox to specify if the Configuration Item is out of service.

Note

Some planned outages may fall within pre-defined service windows when the outage doesn't count against availability metrics.

Calculating outage costs

When an outage occurs and the associated CI has an SLT, the outage costs to the business are calculated and stored in the outagedetail records. You can configure the default Outage Cost per Hour in the Configuration Management Environment record. The default Outage Cost per Hour is only applied if you don't specify Outage Cost per Hour in the CI form.

Automatic outage posting

When you enable Service Level Management to integrate SLA information with other applications, you can configure automatic outage posting to ensure that Service Level Management automatically records when an outage begins and when it ends for the affected Configuration Item.

When you configure an application integration record in the **slamodulecontrol** table, complete the **Service Target Information** section to capture outage data. To configure application integration, click **Service Level Management** > **Administration** > **Configure Application** and then search for the appropriate record.

If you omit automatic outage posting, Service Manager displays the outage start and end times and requests confirmation when you close the application record.

Outage spreading

Although a Service Level Target describes a single Configuration Item (CI), the availability of one CI may also affect the availability of other CIs. For example, if a web server has an outage, the corporate web site also experiences an outage, even though it has no defects. Outage spreading captures data about the related CIs affected by an outage.

An slamodulecontrol application integration record has two settings to configure outage spreading. Select **Spread Outages** to enable outage spreading. When you choose this setting, Service Level Management displays the **Extend outage spreading to more than one level?** checkbox.

Outage posting in other applications

Service Level Management tracks Configuration Item outages related to incidents, changes, and change tasks. There are two ways to post the outage information:

- If you select **Auto Post Outage Information** in the slamodulecontrol record for an application, Service Level Management automatically uses the time when you open the incident as the outage start and the time when you close the incident as the outage end time.
- If you don't select **Auto Post Outage Information** in the slamodulecontrol record for the probsummary file, Service Level Management displays an Outage Confirmation dialog box when you close the record. The open and close time for the incident is the default start and end time for the outage. You can change the outage start time or the outage end time before you close the record.

Example: Configure and verify planned outages

A planned outage is usually the result of a routine maintenance schedule, upgrade action, or faulty equipment replacement. Incidents, change requests, and change tasks can generate outage records. The change request and task records show scheduled downtime. Incident, change request, and change task records have a checkbox to specify if the Configuration Item is out of service. The following example shows an interaction record that's escalated to an incident record. Once the Service Level Agreement application is enabled and configured to spread outages, the System Down field is checked for the down Configuration Items (CIs). The incident record contains the outage details with outage start and end times.

- For information on ensuring the background processes are running, see the related topics.
- For information on enabling the Service Level Management application, see the related topics.

At the end of this example, you will be able to:

- Configure the Service Level Management application (Task 1)
- Set the outage dependency counters in Configuration Management (Task 2)
- Create an interaction record and escalate it to an incident record (Tasks 3 and 4)
- Verify planned outages, including showing that the Configuration Items (CIs) are down (Tasks 5 and 6)

Task 1: Configure the Service Level Management application

1. Sign in as a System Administrator.
2. Click **Service Level Management** > **Administration** > **Configure Application**.

   The SLM Integration Information record opens.
3. In the Table Name field, type **probsummary**, and then click **Search**.

   The SLM module control form for probsummary opens.
4. Go to the **Service Target Information** section of the form.
5. Select the following options:
   - **Spread Outages**
   - **Extend outage spreading to more than one level?**
6. Click **Save**.
7. Click **OK**.

Task 2: Set the outage dependency counters in Configuration Item (CI) relationships

Set the outage dependency counter for:

- **adv-nam-switch**
- **adv-nam-router**
- **adv-nam-modem**

1. Click **Configuration Management** > **Resources** > **Configuration Item Relationships**.
2. Click **Search**.
3. Select the **adv-nam-switch** CI relationship to set the outage dependency.
    1. Go to the **Outage Dependency** section of the form.
    2. Select the **Outage Dependency** option.
    3. Set the counter to **3** for **This Configuration Item will be considered down if ___ or more of the supporting configuration items are down**.
    4. Click **Save**.
    5. Click **OK**.
4. Click **Search**.
5. Select the **adv-nam-router** CI relationship to set the outage dependency.
    1. Go to the **Outage Dependency** section of the form.
    2. Select the **Outage Dependency** option.
    3. Set the counter to **1** for **This Configuration Item will be considered down if ___ or more of the supporting configuration items are down**.
    4. Click **Save**.
    5. Click **OK**.
6. Click **Search**.
7. Select the **adv-nam-modem** CI relationship to set the outage dependency.
    1. Go to the **Outage Dependency** section of the form.
    2. Select the **Outage Dependency** option.
    3. Set the counter to **1** for **This Configuration Item will be considered down if ___ or more of the supporting configuration items are down**.
    4. Click **Save**.
    5. Click **OK**.
8. Verify that the System Down field is unchecked in the adv-nam-modem CI record.
    1. Click **Configuration Management** > **Resources** > **CI Queue**.
    2. Click **Search**.
    3. Type **adv-nam-modem** in the Display Name field of the search form, and then click **Search**.
    4. Verify the **System Down** option is unchecked.
    5. Click **Save**.
    6. Click **OK**.

9. Sign out of Service Manager, and then sign in to Service Manager again for the changes to take effect.

Task 3: Create an interaction and incident for CI adv-nam-switch-hr

1. Sign in as a self-service user and submit an interaction request.
2. Sign in as a Service Desk Agent and bring up the interaction request submitted by the self-service user.
3. Click **Service Desk** > **Interaction Queue**.
4. On the **To Do** queue, select the following:
   - **Interactions** in the Queue field
   - **Self-Service Interactions** in the View field
5. Fill in the following information:

Field	Value
Assignment Group	Network
Service	Intranet/Internet (North America)
Manually enter CI	adv-nam-switch-hr
Category	Incident
Area	Hardware
Subarea	Hardware failure
Impact	1

6. Click **Escalate**.
7. On the Escalation Details screen, do the following:
   1. Select **advantage/Africa** in the Location field.
   2. Fill in the rest of the required fields.
   3. Click **Submit**.

      An Incident record is opened.

> **Note**
>
> If there is already an incident record related to this CI, click **Create New Incident**.

8. Sign out of Service Manager.

Task 4: Create a separate interaction and incident for CI adv-nam-switch-mar.

1. Sign in as a self-service user to submit an interaction request.
2. Sign in as a Service Desk Agent and bring up the interaction request submitted by the self-service user.
3. Click **Service Desk** > **Interaction Queue**.
4. On the **To Do** queue, select the following:
   - **Interactions** in the Queue field
   - **Self-Service Interactions** in the View field
5. Fill in the following information:

Field	Value
Assignment Group	Network
Service	Intranet/Internet (North America)
Manually enter CI	adv-nam-switch-mar
Category	Incident
Area	Hardware
Subarea	Hardware failure
Impact	1

6. Click **Escalate**.
7. On the Escalation Details screen, do the following:
   1. Select **advantage/Africa** in the Location field.
   2. Fill in the rest of the required fields.
   3. Click **Submit**.

An Incident record is opened.

> **Note**
>
> If there is already an incident record related to this CI, click **Create New Incident**.

8. Sign out of Service Manager.

Task 5: Verify "System Down" is checked for CIs selected on incidents opened earlier.

1. Sign in as a System Administrator.
2. Click **Configuration Management** > **Resources** > **Search CIs**.
3. Type **adv-nam-modem** in the Display Name field, and then click **Search**.
4. Select the CIs selected on the records opened previously, as well as the CIs "adv-nam-switch," "adv-nam-router," and "adv-nam-modem". These CIs appear in red.
5. Verify that each of the CIs has the "System Down" checkbox selected.
6. Click **Cancel** and **Back** until you get back to the main menu.

Task 6: See outage details with confirmation for the outage start and end times

1. Sign in as a System Administrator.
2. Click **Incident Management** > **Search Incidents**.
3. Type an incident record ID number in the Incident ID field, and then click **Search**.

   The incident record opens.
4. Click **Fill** in the Closure Code field to select an applicable closure code.
5. Type a solution in the Solution field.
6. Click **Close Incident**.
7. In the Confirm Outage Information form, update or accept the specified outage start and end times, and then click **OK**.

   The status of the incident record is updated to Closed.
8. Click **OK**.

1.14.1.5. Service Level Management data

Service Level Management contains Service Agreements and Service Contract functionality. Each of these functions requires many supporting tables to manage information about:

- SLAs
- Service Level Targets
- Service Contracts
- Parts detail
- Labor detail
- Currency

Service agreement tables and applications

Service agreements use several tables to store and manage data.

To view a table, use the Database Manager utility (**Tailoring** > **Database Manager**) or the Database dictionary (**Tailoring** > **Database Dictionary**).

Table name	Description	Comments
outage	Contains one record for each planned or unplanned outage. These records may be merged together, linked to outageevent records, or they can be split or deleted when Service Level Management recalculates an outage event.	
outagedetail	Contains one record for each outage and includes downtime information.	
outageevent	Contains one record for each planned or unplanned outage.	
sla	Contains one record for each service agreement (SLA, OLA, or UC).	

Table name	Description	Comments
slaactive	Contains one slaactive record for each enabled application record that has an associated service agreement. Slaactive records contain response time and Service Level Target information. Each record includes the current state and time, and an array of past state changes and times.	
slacontrol	Contains only one record with settings that enable the Service Level Management application, check or override service hours, specify the default SLM, and identify standard availability alerts.	There are no default values; you must edit the record to enable any feature or setting.
slamodulecontrol	Contains one out-of-box record for each Service Manager application that can integrate with Service Level Management.	Records in the table don't automatically enable application integration. You must select or clear the **Enable SLMs in this application** checkbox in each record to enable or disable SLM integration with each Service Manager application.

Table name	Description	Comments
slamonthly	Contains a summary of monthly outage information for affected Configuration Items.	
slamonthlyag	Contains one record for each SLM with a summary of monthly response information.	
slaprofile	Contains one record for each user profile.	
slaresponse	Contains records with response time information collected from sloresponse records. Includes the number of affected and breached records, mean, median, and standard deviation data.	
slo	Contains one record for each response time Service Level Target (SLT). Each record contains an agreement.id field that identifies the parent service agreement.	
sloavail	Contains one record for each Service Target. Each record contains the agreement.id field that identifies the parent service agreement.	

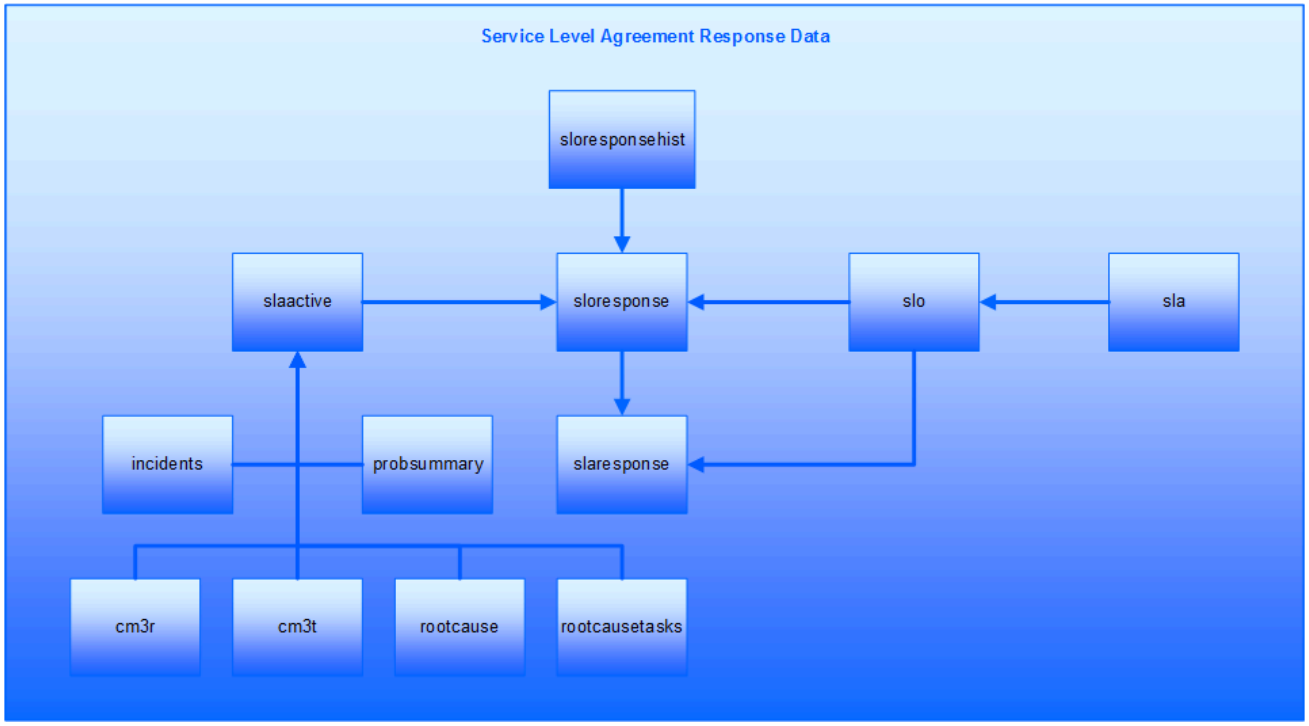
Table name	Description	Comments
slocatalog	Contains one record for each Process Target or Service Target.	
sloresponse	Contains one record for each enabled application record that has an associated SLT. Sloresponse records describe response time metrics, including a breach flag for missed targets.	One application record can have multiple associated sloresponse records that display as a virtual join in the application record.
sloresponsehist	Contains records that include historical information for service level targets, such as the total elapsed time per target itself or the total elapsed time an assignment group spends on a record.	You can use the information in this table to better understand and analyze the elapsed time of the service level targets that are associated with a record.

Service agreement response data

Service Level Management captures Service agreement (Service Level Agreement, Operational Level Agreement, Underpinning Contract) information from application records, such as problems or service desk interaction records. The responses are summarized in the slaresponse table.

- Integrated applications:
  - The slaactive table contains one record for each application record that has an associated service agreeement.
- Service agreement and SLT information:
  - The sla table contains one record for each service agreement.
  - The slo table contains one record for each Process Target. Each record contains an agreement.id field that identifies the parent service agreement.
- Summary information:

- The sloresponse table contains one record for each enabled application record that has an associated SLT. Sloresponse records describe response time metrics, including a breach flag for missed objectives.
- The slaresponse table contains summary records with response time information collected from sloresponse records. Includes the number of affected and breached records, mean, median, and standard deviation data.



Service agreement outage relationships

Service Level Management captures outage information and Service Level Target (SLT) information, which produces an outage summary for each Configuration Item (CI).

- Outage information:

- - The outageevent table contains one record for each outage.
  - The outage table contains one record for each outage. When Service Manager recalculates an outage event, it may merge, split, or delete these records, or link them to the outageevent records.
- SLT information:

- - The sla table contains one record for each Service Level Agreement (SLA).
  - The sloavail table contains one record for each availability SLT. Each sloavail record contains an agreement.id field that identifies the parent SLA.

- Summary information:
- - The outagedetail table contains one record for each outage and includes downtime information.
  - The slamonthly table contains a summary of monthly outage information for affected Configuration Items.

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Service Contract tables

Service Contracts use several tables to store and manage data.

To view a table, use the Database Manager utility (**Tailoring** > **Database Manager**) or the Database dictionary (**Tailoring** > **Database Dictionary**).

Table name	Description	Comments
cmlabor	Contains detail data about the labor performed.	
cmparts	Contains detail data about the parts used.	
contract	Contains contract data.	Contains software, lease, support, warranty, and maintenance contracts.
contractcategory	Defines the type of contract.	
contractdecide		
contractexpense	Contains expense information.	
contractitem	Defines expense allocation by item.	
contractlease	Contains lease contract data.	
contractmaintenance	Contains maintenance contract data.	
contractsoftware	Contains software contract data.	
contractsupport	Contains support contract data.	
contracttemplate	Defines contract templates for different types of contracts.	
contractterms	Defines payment terms.	

Table name	Description	Comments
contractwarranty	Contains warranty contract data.	
curconvert	Contains one record with the exchange rate from one monetary unit to another.	For example, one record that describes the exchange rate of one USD to one other monetary unit. In this case, for 185 monetary units, there would be 186 records.
currency	Contains one record for each monetary unit by international currency code.	For example, EUR (Euro) or USD (US dollar).
expline	Contains expense line data.	
payment	Contains payment data.	

Expense line records

An expense line record is an itemized accounting of expenses incurred by the provider while servicing a contract. Expense line records are generated by Service Level Management as services are rendered. Service Level Management automatically calculates the money spent for each part or service in the currency of the contract.

Currency conversion records

Service Level Management has a currency conversion utility for Service Contracts that automatically converts national currencies, depending on exchange rates at the time the contract is granted. You can enter daily exchange rates into the curconvert table to ensure accurate rate conversions. Service Level Management is compliant with European Union currency regulations, and has out-of-box records pre-defined with the fixed inter-European exchange rates.

Currency definition records

The currency table contains currency definition records that define currency codes for most of the international currencies. Each record specifies whether the monetary unit has the European Union Currency (Euro) as its root.

Labor Performed Detail records

The Labor Performed Detail records describe the details of where and how labor was performed on components for service records and billing purposes.

Parts Detail records

Parts Detail records define the details of used parts, including device type, part number, model number, and the location of the asset.

Contents

- 1 Service agreement tables and applications
- 2 Service agreement response data
- 3 Service agreement outage relationships
- 4 Service Contract tables
  - 4.1 Expense line records
  - 4.2 Currency conversion records
  - 4.3 Currency definition records
  - 4.4 Labor Performed Detail records
  - 4.5 Parts Detail records

1.14.1.6. Service Contracts overview

Service Contracts describe the type of service available to a customer and usually describe the hours, available services, and costs for special services. For example, a Service Contract might specify 24×7 Service Desk availability, or limit requests for service to 8am–5pm, Monday–Friday. Weekend requests might have a billable hourly rate, or there could be other service hour combinations specified in the Service Contract.

Service Level Agreements measure how often these contractual obligations were met or breached by the Service Desk.

Related topics

Service Level Management overview
Service Level Management drivers
Service agreements overview
Working with service agreements
Contract overruns
Working with service contracts

1.14.1.6.1. Service Contract administration

Service Contracts are financial agreements that describe which services are available, how services in a contract are to be rendered, and the financial impact when you use available services.

Administrative tasks include creating new contracts with the Contract wizard, assigning service level agreements to a department or company, editing the Service Contract Control record, defining response level objectives, and integrating a service level agreement with applications.

Service Contract control record

The Service Contract control record contains options that enable you to use service contracts, specify currency denomination and basic unit, and enable other aspects of contract management.

Tip

If necessary, press **Ctrl** + **H** in the Windows client or press **F1** in the Web client to view help from each field in the control record.

Service Contract entitlement records

When a customer contacts a client company and requests service, it is important to determine whether the customer is entitled to additional service. If the contract provides for five service desk interactions, and the customer has five interactions completed, service may be denied.

Because the precise nature of language and rules varies from contract to contract, the process of determining which contract applies at a given time is complex and difficult to generalize. The entitlement determination process identifies the contract and the Service Level Agreement that applies to the current situation. Service Manager uses entitlement records to verify that:

- The referenced contract permits service at this time.
- The contact has not exceeded the allocated number of service desk interactions.
- The contact has not exceeded the contracted number of incidents.

Service Manager checks all of this information automatically from the incident, service desk interaction, or change request if the Service Contract application is

enabled. The Service Contract entitlement information appears in each contract on the Details and Rules tabs.

Checklist: Service Contract Management

Before you create individual Service Contracts, you must ensure that related information is also available by configuring each of the following records.

Tip

If necessary, press Ctrl+H in the Windows client or F1 in the Web client to view help for each field.

Record	Access path
Control record	Click **Service Level Management** > **Service Contracts** > **Edit Control Record**
Currency exchange rate	Click **System Administration** > **Base System Configuration** > **Conversion Rates**
Currency definition	Click **System Administration** > **Base System Configuration** > **Currencies**
Labor performed detail	Click **Service Level Management** > **Supporting Data** > **Labor Performed Detail**
Part usage detail	Click **Service Level Management** > **Supporting Data** > **Parts Usage Detail**

1.14.1.6.2. Working with service contracts

The Contract Management application integrates information and tracking into the Service Desk. Unlike Service Level Agreements, which describe how services in a contract are to be rendered, service contracts are financial agreements that define the services to be provided and the financial implications of using those services.

1.14.1.6.2.1. Service contracts

Service contracts are the principal records that determine which services are used and which services are available. Service Level Management does the following:

- Stores contracts in a structured format for automated analysis, or as the original contract document.
- Uses the Contract wizard to determine the contract thresholds quickly.
- Charges the customer for costs incurred while working with incidents, handling service desk interactions, or implementing changes to a specific service contract.
- Links discrete incidents and interactions to service contracts to provide up-to-date information about the state of each contract, including its budgeted allocations, and the actual number of interactions and incidents applied against each contract.
- Associates service contracts with time and materials expended through Service Desk, Incident Management, and Change Management to compute the real cost of handling each incident and service desk interaction, as well as to calculate the cost of managing each service contract.

1.14.1.6.2.2. Service contract tasks

Service Contract management includes:

- Creating, updating, and viewing contracts.
- Managing service desk interaction and incident allocations.

1.14.1.6.2.3. Contract thresholds

Contract thresholds are limits on the number of service desk interactions or incidents that can be opened, and the cost of those services offered under the contract. Contract Management can determine when an interaction or incident exceeds these budgeted thresholds.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks

1.14.1.6.2.4. Contract overruns

Service Contracts options enable you to view contracts that have reached certain limits. You can query Contract Management for a list of contracts that meet certain search parameters. The following table describes the types of contract overruns.

Option	Description
Expired contracts	Displays all contracts in the system with an expired time limit for service.
Overspent contracts	Displays all contracts in the system with an exceeded budget.
Exceeded incidents	Displays all contracts in the system with an exceeded Incident allocation limit.
Exceeded service desk interactions	Displays all contracts in the system with an exceeded call allocation limit.

Related topics

Service Level Management
Service Contracts overview

1.14.1.6.2.5. Entitlement checking

When a customer contacts a client company and requests service, it is important to determine whether that customer is entitled to additional service. If the contract provides five service requests, and the customer has already reached that limit, service may be denied. Because the precise nature of language and rules varies from one contract to another, determining what the contract covers is complex.

Entitlement checking identifies the contract and a Service Level Agreement (SLA) that apply to the current situation. Service Level Management automatically verifies whether the requested service is within the specified service hours and the service budget.

1.14.1.6.2.6. Expense line records

An expense line record is an itemized accounting of expenses incurred by the provider while servicing a contract. Expense line records are generated by Service Level Management as services are rendered. Service Level Management automatically calculates the money spent for each part or service in the currency of the contract.

Itemizing costs

Contract Management enables you to itemize the cost of fixing a particular incident. Detailed cost data helps the user make informed decisions by answering such questions as:

- What type of incidents are the most expensive to fix?
- What percentage of your costs are parts?
- What percentage of your costs are labor?

View an expense line record

1. Click **Service Level Management** > **Service Contracts** > **Expense Lines**.
2. Click **Search** to generate a record list.
3. Select a record to view in the Expense Lines Information form.

Display a cost table and expense line record

1. Click **Incident Management** > **Search Incidents**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record to view it in the Incident form.
4. Click the **Cost** tab and check the information for the cost related to the incident.

Note: An administrator must select the **Enable Service Contracts Application** option in the Service Contracts Control record to provide access to Service Contract information or the Contract wizard.

1.14.1.6.2.7. Handle time costs

The time spent processing a request for service costs a provider money. A 10-minute phone conversation with the customer represents an expense that should be charged back to the customer. A large number of brief service desk interactions consumes a significant amount of a provider's resources. The time spent in service desk interactions is *handle time*.

Service Manager integrates Contract Management with Service Desk. When the service desk operator processes a service desk interaction, Service Desk automatically records the elapsed time for the interaction. In Service Desk, accounting for handle time is when a service desk interaction begins, and then terminates when the service desk operator ends the interaction. Handle time is multiplied by the technician's billing rate and recorded as an Expense Line record against the relevant service contract.

1.14.1.6.2.8. Cost assessment

Each time a client company provides a service to a customer, there is real cost incurred from three sources:

- The cost of interacting with the customer. This is the time spent by the service desk operator handling the customer's service request, multiplied by the operator's hourly rate.
- The actual cost of labor associated with fixing an incident. This is the time spent by one or more technicians working on an incident, multiplied by their hourly rate.
- The cost of any parts used in the repair process.

1.14.1.6.2.9. Labor costs

As an Incident record or change request is managed, more than one person might work on the issue. A technician can work on the issue more than once over a period of days or weeks. Contract Management integrates with Incident Management, Change Management, and Request Fulfillment to enable technicians to record the hours they spent working on a change request, Incident record, or request record.

As technicians record their labor in the Incident record, line item, or change request, Service Manager automatically translates this information into Expense line records that link to Incident Management, Request Fulfillment, and Change Management.

Labor Performed Detail records

The Labor Performed Detail records describe the details of where and how labor was performed on components for service records and billing purposes.

Contents

- 1 Labor Performed Detail records

1.14.1.6.2.10. Parts costs

Service Manager integrates Contract Management with Incident Management, Request Management, and Change Management to enable technicians to record any parts they use to resolve an incident. As these parts are recorded, Service Manager:

- Creates Expense Line records against the relevant contract.
- Adjusts the on-hand quantity of parts used.
- Automatically tracks the number of parts in stock and, through Request Management, places replacement part orders when the quantities in stock dip below a user-defined threshold.

Parts Detail records

Parts Detail records define the details of used parts, including device type, part number, model number, and the location of the asset.

Contents

- 1 Parts Detail records

1.14.1.6.2.11. Currency records

The currency table contains currency definition records that define currency codes for most of the international currencies. Each record specifies whether the monetary unit has the European Union Currency (Euro) as its root. The curconvert table contains currency exchange rate records.

1.14.1.7. Service Level Management administration

Service agreements guarantee service between providers and customers. They define availability and response service goals, such as availability of certain Configuration Items (CIs) and response time expectations.

Administrative tasks include configuring the SLA Control record and the application integration records, and defining a catalog of Service and Process Targets.

1.14.1.7.1. SLM control record

The Service Level Management control record configures the following settings:

- Enable the SLM application
- Verify that a new service desk interaction occurs during the supported service hours
- Specify if the service desk operator can accept a new interaction outside of the supported service hours
- Choose a default Service Level Agreement (SLA) to apply when no SLA is assigned to the contact or to the contact's company
- Define alerts that occur if no other application-specific alerts occur first

Tip

If necessary, press Ctrl+H in the Windows client or F1 in the Web client to view help from each field in the SLM control record.

Service Level Management control record options

The following table describes Service Level Management (SLM) control record options.

Option	Description
Enable SLM Application	Integrate Service Level Management with the external application. You must log out and log back in for changes to this record to take effect.
Check Service Hours	Prevent users from opening Incident records outside of the service hours defined in the service agreement. A dialog box advises the operator that service is unavailable.
Allow Override of Service Hours Violation	Permit new Service Desk interactions outside of the hours specified in the service agreement.
Log Process Target History	Save the historical information for Process Targets, such as the total elapsed time per target itself or the total elapsed time an assignment group spends on a record. You can use the log information to better understand and analyze the elapsed time of the targets that are associated with a record.
Default SLA	Choose the default SLA to apply when no SLA is specified for assignment groups.
Default OLA	Choose the default Operational Level Agreement (OLA) to apply when no OLA is specified for assignment groups.
Effective Condition	Specifies the condition under which the service agreements are operating (e.g. the"agreed" phase).

Option	Description
Group Table Name	Specifies the default group list for an OLA.
Standard Alerts	List the alerts that occur when the application doesn't generate specific alerts.

Contents

- 1 Service Level Management control record options

1.14.1.7.2. Service Level Agreement links

Service Level Agreements (SLAs) must have links to other system records to ensure they gather the right information at the right time. If users have permission to create a new SLA, they must also have permission to update related records.

Record	Comments
Department record	If the new SLA collects service or process metrics for a department, the SLA field in the department record must identify the new SLA.
Company record	If the new SLA collects service or process metrics for a company, the SLA field in the company record must identify the new SLA.
Contact record	If the service agreement is an Operational Level Agreement, and it collects process metrics related to a problem, the ID field in the contact record for the problem owner must identify that owner.

Linking a service agreement to other records

Depending on the objective, you might link a customer Service Level Agreement (SLA) to a department, company, or contacts record to ensure that the system gathers the necessary information to track SLM metrics.

Contents

- 1 Linking a service agreement to other records

1.14.1.7.3. Service Level Agreement selection process

Service Level Management assigns a service agreement to a new service desk interaction, change, incident, request or problem by using the following selection process:

- The value in the slamodulecontrol record **Customer Field** is the search argument used to match a contact specified in an SLA candidate.
- The first candidate SLA is the one assigned to the contact's department. The departmental SLA sets expectations for the response time and service entitlements when department personnel request a service.
- If no department SLA exists, the next candidate SLA is the one assigned to the contact's company.
- If no company SLA exists, the next candidate is the default SLA specified by the SLA Control record.
- At any point in the selection process, if the current date and time doesn't fall within the limits of an SLA candidate, the next SLA candidate is considered. It is a good idea to ensure that the default SLA doesn't contain highly restrictive date and time limitations and describes general support for all services.

When Service Level Management selects the appropriate SLA, it also populates the new service desk interaction, change, incident, or problem record with the name of any service contract linked to the SLA.

Note: You can override the automatic selection process if you click the list in the SLA field and select a different one.

Example

A new manager (the contact) who requests email setup could expect a response within four business hours, based on whether a response SLA requires that services for managers have a higher priority than services for an intern. If there is no departmental policy that prioritizes this service, the next SLA selection level might be a company-wide response policy that doesn't prioritize by job description but instead by key services.

The final selection level would be the default SLA that describes general support for all services.

Assigning Service Level Targets to an application record

When you create a new interaction, change request, request, incident, or problem record, you can choose a Customer SLA for the contact or a Customer SLA for the contact and one or more applicable Service SLAs for the contact's subscriptions to a service. Service SLAs only apply if the application record references a Business Service, the contact has a subscription to the service, and the subscription references an SLA. The following describes the system's process for adding SLAs to an application record.

- If one SLA is associated with the application record based on the contact, the Customer SLA is added to the record.
- If the contact has an Individual Subscription for the CI, the Service SLA from that subscription is added to the application record.
- If the contact has a Department Subscription for the CI, the Service SLA from that subscription is added to the application record.
- If the contact has neither, then no Service SLA is added to the application record.

The SLAs should contain all Service Level Targets (SLTs) that define the business rules for all process and service metrics. You can choose as many SLTs as necessary to describe your process and service commitment. If necessary, you can add more SLTs that meet your criteria.

When you view the new record, the SLA section lists the SLTs that apply to the application record.

See the related topics to view the definitions for Customer SLA and Service SLA.

Service Level Target selection process

An administrator assigns one or more Service Level Targets (SLTs) to a new Service Level Agreement (SLA). When Service Level Management evaluates whether the process or service targets objectives are met, it uses the following logical processes.

Process Targets

Service Level Management examines the condition specified in the SLT filter criteria. This value is a Boolean expression that can include any valid field values from the record. For example, you can create a condition that includes these values:

- Affected Configuration Item
- Category

- Priority
- Risk level
- Company
- Department
- Location

If there is more than one Process Target, Service Level Management uses the strictest SLT as the measurement for responsiveness.

Service Targets

Service Level Management examines the schedule and the required uptime or maximum outage duration specified in the SLT. If there is more than one Service Target, Service Level Management uses the strictest SLT as the criteria.

Contents

- 1 Example
- 2 Assigning Service Level Targets to an application record
- 3 Service Level Target selection process
  - 3.1 Process Targets
  - 3.2 Service Targets

1.14.1.7.4. Service Level Agreement application integration records

Service Manager provides an out-of-box record for each application in the slamodulecontrol table that you can modify to enable SLM integration with other applications and associated settings. To edit the records, click **Service Level Management** > **Administration** > **Configure Application** in the system navigator.

The name of the primary application table is the name of the application integration record.

Choose this record	To enable this application
cm3r	Change Management requests
cm3t	Change Management tasks
probsummary	Incident Management
imTask	Incident Management tasks
incidents	Service Desk
request	Request Fulfillment
requestTask	Request Fulfillment tasks
rootcause	Problem Management problems
rootcausetask	Problem Management tasks
svcCartItem	Service Catalog

Ensure that you select **Enable SLM in this application** to integrate service agreements with the individual application.

The following table describes Service Level Management application integration control record options.

Option	Description
Table Name	The name of the application table that Agreements are configured for.
Enable SLM in this application	Select this checkbox to enable Service Level Management (SLM) processing for the indicated table.
Run in Foreground	Select this checkbox to process Agreement calculations in the foreground for the indicated table. Clear this checkbox and the Agreement calculations will be scheduled to run as background processes. The Agreement information will not be immediately visible to the technicians.
Process Targets	Select this checkbox to enable Process Target processing for the indicated table.
Service Targets	Select this checkbox to enable Service Target processing for the indicated table.
Record ID Field	The field name in the indicated table that's used to store the unique ID of the ticket. For example, values in the Record ID Field contains "Number" if the unique ID is stored in the Number field in the probsummary table.
Agreement ID Field	The field name in the indicated table that's used to store the Agreement ID of the ticket.

Option	Description
Start Time Field	The field name in the indicated table that's used to store the date and time value corresponding to the time when the ticket becomes active (open).
End Time Field	The field name in the indicated table that's used to store the date and time value corresponding to the time when the ticket becomes inactive (closed).
Customer Field	The field name in the indicated table that's used to store the customer name of the ticket. Service Manager uses this field when the following behaviors occur: • The customer's time zone is necessary for Agreement calculations. • The Agreement selection process is trying to determine which Agreement needs to be used for the ticket being opened.
Technician Field	The field name in the indicated table that's used to store the name of the technician who is responsible for working on the ticket. Service Manager uses this field when the technician's time zone is necessary for Agreement calculations.

Option	Description
Active Condition	Determines whether the record is active or not. This condition only refers to the valid field names in the current record and must use the $L.file variable. For example, the "flag in $L.file=true" condition is valid when it refers to the fields in Incident Management.
Record Group Field Name	The field name in the indicated table that's used to store the name of the ticket group.
Process Target State Field	The field name in the indicated table that's used to store the ticket's current state or status. Service Manager uses this field when processing Process Targets.
Use Phases	Select this checkbox to use phases to measure service levels for the ticket's Process Targets. Clear this checkbox and Service Manager uses the Response State Progression list. When Service Manager uses phases for the application that's configured for Agreements, a ticket is processed through the series of phases defined in the Category phase table. The name of the Category phase table is specified in the Object record for the indicated table.

Option	Description
Process Target State Progression	The set of ticket states that are used to measure the process service levels for the ticket's Process Targets. The states must be valid states for the indicated table ans must exist in valid succession. For example, the state of "closed" must exist in the list after the state of "open". For Incident Management, the valid states are defined in the pmstatus table.
Standard Alerts	The Agreement alert(s) to be generated when processing Process Targets for tickets in the indicated table. The definitions for the alert(s) must exist in the AlertDef table. Service Manager generates these alerts in addition to those as already defined for the Process targets of the ticket's Agreement.

Option	Description
Use Legacy Unordered Suspend Process	Select this checkbox to use the legacy unordered suspend state for the record's Response Service Level Targets (SLTs). Clear this checkbox to use the ordered suspend state. When Service Manager uses the legacy unordered suspend state, if a record moves to a suspend state, the running SLT will be suspended directly without checking the suspend state order in the response states list or phase lists. Otherwise, when Service Manager uses the ordered suspend state, the SLT state will be calculated by comparing the suspend state order with the initial and the final state order in the SLT definition. Example: Assume that an Incident has the following SLT definitions: The initial state is Open, the final state is Working Progress, and the suspend processing for these states are Pending Vendor/Supplier and Pending Change. The corresponding response state progression list in the slamodulecontrol definition for the Incident is as follows: • Open • Pending Vendor/Supplier • Working Progress • Pending Change • Resolved • Closed

Option	Description

	You create an incident and set the SLT state to Running. In this situation, the following behaviors occur:
	• When this checkbox is selected, the incident is moved to Pending Vendor/Supplier or to Pending Change (which is in the suspend list), and the SLT state will be set to suspended directly without comparing their orders.
	• When this checkbox is cleared:
	  ◦ If this incident is moved to Pending Vendor/Supplier, which is prior to Working Progress (the final state of SLT), this running SLT will be set to Suspended.
	  ◦ If this incident is moved to Pending Change, which is after Working Progress, the Working Progress state is automatically finished and passed and this SLT will be set to Achieved.

CI Fields for Outage Processing	The CI field names that are referenced when processing outages.
CI Fields for Subscription Processing	The CI field names that are referenced when selecting the record's Subscription Agreement(s).

Option	Description
Outage Condition	Determines whether to process outages for the tickets that associate to the specified table. This condition only refers to the valid field names in the current record and must use the $L.file variable. For example, the "nullsub(ci.down in $L.file, false)=true" condition is valid when refers to the fields in Change Management.
Outage Start Field	The field name in the indicated table that's used to store the start date or time of the outage for the CI(s) in the ticket.
Outage End Field	The field name in the indicated table that's used to store the end date or time of the outage for the CI(s) in the ticket.
Auto Post Outage Information	Select this checkbox to automatically post outage information to the outage table for the CIs in the ticket being closed. Clear this checkbox and Service Manager will prompt the operator who closes the ticket to verify the start and end time of the outage before posting the outage information.

Option	Description
Spread Outages	Select this checkbox to generate outages for the child CIs of the ticket's primary CIs. Service Manager uses the information in the deviceparent table to determine which CIs are affected.
Extend outage spreading to more than one level	Select this checkbox to generate outages for any child CIs of the ticket's primary CIs that exist after the first level in the parent-child hierarchy. Service Manager uses the information in the deviceparent table to determine which CIs are affected. **Note** This option is available only when the **Spread Outages** option is selected.

1.14.1.7.5. Prioritizing incidents, problems, requests and changes

When you enable Service Level Management to work with other applications, the service agreement information added to the incident, problem, requests or change record helps you prioritize tasks by Expiration date and time.

To use SLM information, create a record list of pending incidents, problems, requests, changes, or associated tasks. Sort the record list using the Breached or Expiration date as the sort order. Administrators can escalate or reassign work depending on the amount of time left before the SLA is breached or reassign priority 1 to those records with breached SLAs.

When you view incident, problem, change, or task records, the SLA section shows all pending Service Level Target expiration dates, alerts, and other relevant information.

1.14.1.7.6. Time duration data processing

Service Manager records the system data provided by the SLM module license and configuration. Service Manager extracts the system data against status, phase and assignment group changes, and then calculates the duration values according to your metrics. You can then configure calculation or time duration reports to display the time duration data based on the calculation results.

Note

If you modify calculation metrics for duration reports , the modified configurations only take effect on newly created tickets.

The time duration data can be processed for further analysis. Currently, Service Manager supports the following tables and fields.

Module	Monitor field changes
incidents	State, Phase, Assignment
probsummary	State, Phase, Assignment
imTask	State, Assignment
cm3r	State, Assignment
cm3t	State, Assignment
rootcause	State, Phase, Assignment
rootcausetask	State, Phase, Assignment
request	State, Phase, Assignment
requestTask	State, Phase, Assignment
svcCartItem	State, Phase

The raw time stamp of the field status changes can be extracted from the following tables. To configure the time duration settings, see Configure time duration settings.

Field	Table
State	slaactive **Note** The information in this table requires that SLM is enabled in this application. The default field is the **Process target state field** set from **Service Level Management** > **Administration** > **Configure Application**. If SLM is disabled, no data is available for this field and the time duration process is not working.
Phase	workflowhistory (for PD only)

Field	Table
Assignment	assignmentlog **Note** The information in this table requires that SLM is enabled in this application. The default field is the **Record Group Field Name** set from **Service Level Management** > **Administration** > **Configure Application**. If SLM is disabled, no data is available for this field and the time duration process is not working.

After the data is extracted and processed, Service Manager saves the field change metrics in the following tables.

Note

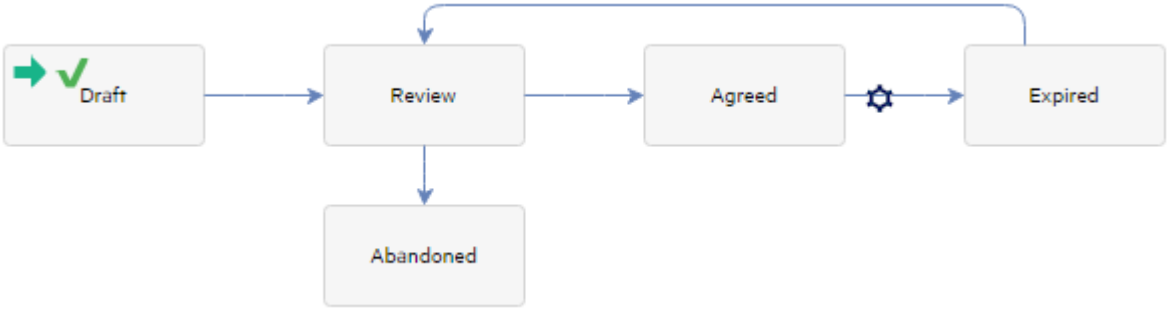
Make sure that the background schedule of **timedurationdata** process is started to process the raw time stamp information.

- timeDuration4probsummary
- timeDuration4cm3r
- timeDuration4request
- timeDuration4incidents
- timeDuration4imTask
- timeDuration4cm3t
- timeDuration4requestTask
- timeDuration4rootcause
- timeDuration4rootcausetask
- timeDuration4svcCartItem

You can apply the data in these tables to create calculation or time duration reports. For details about how to create a report by using the time duration metrics, see Create a calculation report by using duration metrics.

1.14.2. Service Level Management agreement workflow and tasks

Process Designer is used to manage service agreements. You can use Process Designer to create a workflow, rules, and transitions to control the behavior of the various types of service agreements.



The following table illustrates the default service agreement workflow phases and shows what can be done in each in phase.

Workflow Phase	Common Tasks
Draft	Create a new Service Level Agreement Create an Operational Level Agreement Create an Underpinning Contract Add Service Level Targets Attach a file to a service agreement
Review	Review a service agreement Create an SQP or SIP Edit a Service Level Agreement Edit Service Level Targets View service agreement metrics
Agreed	
Expired	
Abandoned	

Other Activities	Tasks
Defining Service Level Targets	Add Service Level Targets Delete Service Level Targets Edit Service Level Targets Define Service Targets Define Process Targets Use the Service Level Target wizard

1.14.2.1. Create a Service Level Agreement

To create a Service Level Agreement (SLA), follow these steps:

1. Click **Service Level Management** > **Agreements** > **New Agreement**.
2. Click **Service Level Requirement**.
3. In the **Type** field, select the contract type that identifies it as either a Service or Customer SLA.
4. In the **Customer** field, select from the list of records to choose a customer name.
5. In the **Title** field, type a descriptive name for the SLA.
6. In the **Effective from** field, choose a date in the calendar options for the SLA start date.
7. In the **Expiration Date** field, choose a date in the calendar options for the SLA end date.
8. In the **Next Review Date** field, choose a date for the first review of the SLA.
9. In the **Service review Frequency** field, choose the time period for the review.
10. Add an optional **Description** for the new SLA.
11. Click **Save**. The agreement is now in the "Draft" phase and an Agreement ID number added to the record.
12. Click the **Process Targets** or **Service Targets** tabs. You can add, edit, or remove Service Level Targets on these tabs.
13. The following steps are optional:
    1. Click the **Agreed By** tab, and specify the appropriate signatories to the agreement.
    2. Click the **Attachments** tab, and add upload any formal documents between the two parties signing the contract.
    3. Click the **Underpinning Agreements** tab and review the underpinning contracts and operational level agreements.

       Note

       All operational level agreements and underpinning contracts are listed in the **Underpinning Agreements** tab.
14. Click **Save**.

After you create a new SLA, ensure that you link the new SLA to a department, company, or contacts record.

Note

When you create a new SLA, you must log out and log in again to view the new SLA on the drop-down selection lists.

1.14.2.2. Create an Operational Level Agreement

To create an operational level agreement (OLA), follow these steps:

1. Click **Service Level Management** > **Agreements** > **New Agreement**.
2. Click **Operational Level Agreement**.
3. In the **Title** field, type a descriptive name for the OLA.
4. In the **Effective from** field, choose a date in the calendar options for the OLA start date.
5. In the **Expiration Date** field, choose a date in the calendar options for the OLA end date.
6. In the **Next Review Date** field, choose a date for the first review of the OLA.
7. In the **Service review Frequency** field, choose the time period for the review.
8. Add an optional **Description** for the new OLA.
9. Click the **Assignment Group** tab. Specify the appropriate group for this OLA.
10. Click **Save**. Note that an Agreement ID number was added to the record.
11. Click the **Process Targets** or **Service Targets** tabs. You can add, edit, or remove Service Level Targets on these tabs.
12. The following steps are optional:
    1. Click the **Agreed By** tab, and specify the appropriate signatories to the agreement.
    2. Click the **Attachments** tab, and add upload any formal documents between the two parties signing the contract.
13. Click **Save**.

Note: After you create a new OLA, you must log out and log in again to view the new OLA on the drop-down selection lists.

1.14.2.3. Create an Underpinning Contract

To create an underpinning contract (UC), follow these steps:

1. Click **Service Level Management** > **Agreements** > **New Agreement**.
2. Click **Underpinning Contract**.
3. In the **Title** field, type a descriptive name for the UC.
4. In the **Effective from** field, choose a date in the calendar options for the UC start date.
5. In the **Expiration Date** field, choose a date in the calendar options for the UC end date.
6. In the **Next Review Date** field, choose a date for the first review of the UC.
7. In the **Service review Frequency** field, choose the time period for the review.
8. Add an optional **Description** for the new UC.
9. Click the **Assignment Group** tab. Specify the appropriate group for this UC.
10. Click **Save**. Note that an Agreement ID number was added to the record.
11. Click the **Process Targets** or **Service Targets** tabs. You can add, edit, or remove Service Level Targets on these tabs.
12. The following steps are optional:
    1. Click the **Agreed By** tab, and specify the appropriate signatories to the agreement.
    2. Click the **Attachments** tab, and add upload any formal documents between the two parties signing the contract.
13. Click **Save**.

1.14.2.4. Edit a service agreement

Follow these steps:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Click **Search** to generate a record list.
3. Select the agreement that you want to edit.
4. Add or change information on the form. If necessary, press Ctrl+H to view help for each field.
5. Click **Save**.

Related topics

Working with service agreements
Operational Level Agreements
Service Level Agreement components
Service agreement selection process
Outages Service Level Agreement performance and reporting
Create a new Service Level Agreement
Access service agreements from other applications
View service agreements from Configuration Management

1.14.2.5. Review a service agreement

In the Review phase of a service agreement, follow these steps:

1. Navigate to the agreement you want to review.
2. Verify that following fields or tabs are filled in appropriately:
    - **Effective From**
    - **Expiration Date**
    - **Next Agreement Review Date**
    - **Service Review Frequency**
    - **Next Service Review Date**
    - **Agreed by** (on the Agreed By tab)
    - **Assignment Groups** (Operational Level Agreements only)
    - **External Assignment Groups** (Underpinning Contracts only)
3. Verify that following optional fields are filled in appropriately:
    - **Service Contract**
    - **Service Hours**
    - **Owner**
    - **Owner Group**
    - **Next Service Review Date**
4. If this is a new agreement, verify that appropriate SLTs exist for the agreement under the **Process Targets** and **Service Targets** tabs.
5. If this is an existing agreement that's back in the review phase, navigate to the **Improvement Plans** tab and then click to view any service improvement plans (SIP) or service quality plans (SQP) that may exist. If no improvement plan exists, you may need to create one, depending on your organization's service review process.

1.14.2.6. Attach a file to a Service Level Agreement

Applies to User Roles: System Administrator, administrator (SLA)

The Attachments tab stores links to the documents associated with the service agreement. For example, you can attach text or graphic files that contain signatures, invoices, related contracts, or any other reference materials.

To attach a file to a service agreement:

1. Click **Service Level Management** > **Agreements** > **Search agreements**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record.
4. Do the following on the **Attachments** tab:
   - For Web tier clients: click **Add File**, browse to the file, and then click **OK**.
   - For Windows clients: right-click in the attachment space, and then click **Add**. Browse to the file, and click **Open**.

   Service Manager attaches the file to the service agreement.
5. Click **Save**.

1.14.2.7. Create or add an SQP or SIP

To create a service quality plan (SQP) or service improvement plan (SIP) to a Service Level Agreement, follow these steps:

1. Navigate to the SLA for which you want to create or add an SQP or SIP.
2. Navigate to the **Improvement Plans** tab and then click the **Create Improvement Plan** button.
3. Enter a title in the **Title** field.
4. Add a summary, tags, and expiration date as appropriate.
5. In the **Document Subtype** drop down, select **Service Improvement Plan** or **Service Quality Plan** as appropriate.
6. To save your plan and associate the plan with the SLA, click **Save Draft**.

Note

After completing step 6, the plan follows the normal Knowledge Management document workflow and you should follow those procedures to publish or index the KM document as needed.

1.14.2.8. Define Service Targets

Applies to User roles: System Administrator, Administrator

You define a Service Target to monitor the availability of a Configuration Item (CI), such as a service, application, hardware, software, or other infrastructure component.

To define a Service Target, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Service Targets**
2. Fill in all the required fields for the new service level target.
3. Click **Add**.
4. Click **OK**.

Define Service Targets in the SLT Catalog

You can also use the wizard in the SLT Catalog to create a Service Target.

To define an Service Target by using the SLT Catalog, follow these steps:

1. Click **Service Level Management** > **Agreements** > **SLT Catalog**.
2. Click **More** or the More Actions icon, and then select **Create SLT using Wizard**.
3. In the **SLT Type** field, select **Service**, and then click **Next**.
4. Continue providing the applicable information and clicking **Next** for each wizard form until you have created a Service Target.

   You receive a message that states the new service level target catalog record is added.

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- 1 Define Service Targets in the SLT Catalog

1.14.2.9. Define Process Targets

Applies to User roles: System Administrator, Administrator

You define a Process Target to specify the maximum allowable amount of time for a response to a business event.

To define a Process Target, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Process Targets**
2. Fill in all the required fields for the new service level target.
3. Click **Add**.
4. Click **OK**.

Define Service Level Targets in the SLT Catalog

You can also use the wizard in the SLT Catalog to create a Process Target.

To define a Process Target by using the SLT Catalog wizard, follow these steps:

1. Click **Service Level Management** > **Agreements** > **SLT Catalog**.
2. Click **More**, and then select **Create SLT using Wizard**.
3. In the **SLT Type** field, select **Process**, and then click **Next**..
4. Continue providing the applicable information and clicking **Next** for each wizard form until you have created a Process Target.

   You receive a message that states the new service level target catalog record is added.

Contents

- 1 Define Service Level Targets in the SLT Catalog

1.14.2.10. Add Service Level Targets

Applies to User roles: System Administrator, Administrator

You can create a new Service Level Target (SLT) or select an existing SLT from the SLT catalog. SLTs are for either availability or response time.

To add a Service Level Target to a service agreement:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**
2. Add optional search criteria and then click **Search**.
3. Select a target record.
4. In the **Process Targets** tab or **Service Targets** tab, do the following to add an SLT using the SLT wizard. This adds a new SLT. You can also select an SLT from the catalog, in which case you will not need to complete steps a through d.
    1. Click **Add SLT** Service Level Management starts the SLT wizard.
    2. In the **SLT Name** field, type the name of the new service level target.
    3. In the **Description** field, type the description of the new service level target.
    4. Click **Next** to advance through the wizard. Depending on the type of service level target, the SLT wizard collects information about:
        - Schedule
        - Affected CI
        - Availability criteria
        - Filter criteria
        - Initial and final states
        - Duration type
        - Alerts
5. Click **Save**.
6. Click **OK**.

To add a SLT directly, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Process Target** or **Service Target**.
2. Add information to the Process or Service Target form.
3. Click **Add**.

1.14.2.11. Edit Service Level Targets

Applies to User roles: System Administrator, Administrator

To edit a Service Level Target (SLT) from a service agreement, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Click **Search**.
3. Click the **Process** or **Service** tab.

   There are two methods to edit an existing SLT. Method One is for an SLT that was created with the SLT wizard. If you choose Method One but the SLT was created directly, Service Level Management automatically switches to Method Two.

   Method One (the SLT Wizard)
   - Click **Edit SLT**.
   - When Service Level Management displays a list of SLTs, double-click an SLT to launch the SLT Wizard.
   - Click **Next** through the wizard to change settings in the SLT.

   Method Two (Manual editing)
   - Double-click an SLT from the **Process Targets** or **Service Targets** table.
   - The Service Level Targets form displays the SLT information. Add or change information on the form. If necessary, press Ctrl+H in the Windows client or F1 in the Web client for each field.
     Method Two works for all SLTs.

4. Click **Save**.
5. Click **OK**.

To edit SLT records directly, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Process Targets** or **Service Targets**.
2. Click **Search**.
3. Select an existing SLT.

4. The Service Level Targets form displays the SLT information. Add or change information on the form. If necessary, press Ctrl+H in the Windows client or F1 in the Web client for each field.
5. Click **Save**.
6. Click **OK**.

1.14.2.12. Delete Service Level Targets

Applies to User roles: System Administrator, Administrator

Process or Service Targets can be deleted if for some reason they are no longer needed; however, once one is deleted its associated history is lost.

To remove a Service Level Target from a service agreement, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record.
4. In the **Process** tab or **Service** tab, do the following:
    1. Click **Remove SLT**.
    2. Double-click an SLT from the record list. Service Level Management deletes the SLT and returns to the original SLA.
5. Click **Save**.
6. Click **OK**.

To delete an SLT directly:

1. Click **Service Level Management** > **Administration** > **Process** or **Service**.
2. Add optional search criteria and then click **Search**.
3. Select a target record.

    The Service Level Targets form displays the SLT information.
4. Once you make sure that these are the service level targets that you want to delete, click **Delete**.
5. To confirm the deletion, click **Yes**.
6. Click **OK**.

1.14.2.13. Use the Service Level Target wizard

Applies to User roles: System Administrator, Administrator

You can use the Service Level Target (SLT) wizard to build a catalog of pre-defined SLTs that you can add to new or existing service agreements.

To add a new SLT by using the wizard, follow these steps:

1. Click **Service Level Management** > **Agreements** > **SLT Catalog**.
2. Click **More** or the More Actions icon, and then select the **Create SLT using Wizard** option.
3. In the **SLT Type** field, select **Process** or **Service**, and then click **Next**.
4. Continue providing the information the wizard requests, and then click **Next** until you have provided all of the information needed to create an SLT.
5. Click **Finish**.

To add new SLTs to an existing service agreement, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Click **Search**.
3. Select an existing service agreement.
4. Select the **Process Targets** tab or **Service Targets** tab.
5. Click **Add SLT**.
6. Type a name in the **SLT Name** field.
7. Type a **Description** of the service level objective and then click **Next**.
8. Use the **Fill** function to choose an **Affected CI**.

   Note

   Click **Search** to add optional search criteria and search through a specific list of CI records, or use the **Fill** function to choose from the list of CI records.

9. Click **Next** to provide other required information.
10. Click **Finish**.

1.14.2.14. Access Service Level Agreements from other applications

Applies to User Roles: Users who have the right to view an interaction, incident, change request, change request task, request, or problem record

To access service agreements from other applications:

1. Follow the steps to search for a service desk interaction, change request, change request task, incident, request, problem, or problem task record.
2. Select your target record.
3. Open the **SLT** section to display the name of all the service agreements and targets.
4. Click **Process Targets**, **Uptime Objectives**, and **Max Duration Objectives** tabs to view related information.

Notes:

- If you create a new service agreement, you must log out and log in again to view the new service agreement in the selection lists.
- You can't view service agreement information in related application records unless the `SLM Control` record has **Enable Application** selected, or there is an `slamodulecontrol` record for the affected application that has **Enable SLM** selected.

Related topics

- Working with service agreements
- Operational Level Agreements
- Service Level Agreement components
- Service agreement selection process
- Outages
- Service Level Agreement performance and reporting
- Create a new Service Level Agreement
- Edit a Service Level Agreement
- View service agreements from Configuration Management

1.14.2.15. View service agreements from Configuration Management

Follow these steps:

1. Click **Configuration Management** > **Resources** > **SLM Information**.
2. Click **Search**.
3. Select a service agreement.
4. If you make changes, click **Save**.
5. Click **OK**.

Related topics

Working with service agreements
Operational Level Agreements
Service Level Agreement components
Service agreement selection process
Outages
Service Level Agreement performance and reporting
Create a new Service Level Agreement
Edit a Service Level Agreement
Access service agreements from other applications

1.14.3. Service Level Management administration tasks

Service Level Management administration covers these areas of responsibility:

Service Level Administration Activities	Common Tasks
Service Level Management data	Recalculate availability and response time Research overall results Research process metrics Research service metrics Check Service Level Target metrics on a dashboard
Service Contract administration	View an expired service contract View a service contract Link a service contract to a Service Level Agreement Create a new service contract Delete a service contract Edit a service contract Edit the Service Contract control record Configure Process Targets for assignment groups
Service Level Management administration	Edit the Service Level Management control record Assigning Service Level Targets to an application record

Service Level Administration Activities	Common Tasks
Outages	Change outage start and end times Disable outage spreading Enable outage spreading Enable automatic outage posting Disable automatic outage posting

1.14.3.1. Create a new service contract

1. Click **Service Level Management** > **Service Contracts** > **Service Contracts**.
2. Type the name of the new service contract in the **Reference Name** field.
3. Complete the required information in the Service Contract form, adding applicable information in the following tabs:
   - General Information
   - Details
   - Rules
   - Named Users
   - Comments
   - Additional Services
4. Click **Add**.
5. Click **OK**.

To use an existing contract to create a new contract:

1. Click **Service Level Management** > **Service Contracts** > **Service Contracts**.
2. Add optional search criteria, and then click **Search**.
3. Select the target record.
4. Clear the **Reference Name** field and type the new name of the new service contract.
5. Add or change information on the form to select the applicable information in the following tabs:
   - General Information
   - Details
   - Rules
   - Named Users
   - Comments
   - Additional Services
6. Click **Add**.

   Caution: Make sure that you don't click Save because doing so will replace the existing contract with the new contract you are attempting to add.
7. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts

Service contracts
Service contract tasks
Delete a service contract
Edit a service contract
View a service contract
View an exceeded interaction allocation service contract
View an exceeded incident allocation service contract
View an expired service contract
View an overspent service contract

1.14.3.2. Delete a service contract

Follow these steps:

1. Click **Service Level Management** > **Service Contracts** > **Service Contracts**.
2. Add optional search criteria, and then click **Search** to generate a record list.
3. Select a target record to view in the Service Contract form.
4. Make sure this is the record you want to delete, and then click **Delete**.
5. To confirm the record deletion, click **Yes**.
6. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Edit a service contract
View a service contract
View an exceeded interaction allocation service contract
View an exceeded incident allocation service contract
View an expired service contract
View an overspent service contract

1.14.3.3. Edit a service contract

Follow these steps:

1. Click **Service Level Management** > **Service Contracts** > **Service Contracts**.
2. Click **Search** to generate a record list.
3. Select a contract record to view in the Service Contract form.
4. Add or change information on the form. If necessary, press Ctrl+H to view help for each field.
5. Click **Save**.
6. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Delete a service contract
View a service contract
View an exceeded interaction allocation service contract
View an exceeded incident allocation service contract
View an expired service contract
View an overspent service contract

1.14.3.4. Edit the Service Contract Control record

Applies to User roles: System Administrator

The Service Contract control record specifies whether or not to enable service contract management in the applications and also specifies the default values used for calculating the statistics the system maintains for managed service contracts.

To edit a Service Contract control record, follow these steps:

1. Click **Service Level Management** > **Service Contracts** > **Edit Control Record**.
2. Select or clear any Contract Management options.

Option	Description
Enable Service Contracts Application	Enables Service Contract Management in Incident Management, Request Management, and Service Desk.
Default Rate	Specifies the default labor rate for the application.
Currency Code	Specifies the currency code for all contracts and conversions.
Problem Parts	Calculates the cost of parts from the Parts & Labor tab in an incident record.
Calculate Incident Labor	Calculates the cost of labor from the Parts & Labor tab in an incident record.
Calculate Service Labor	Calculates the cost of labor from the time.spent field in an incident record.
Calculate Change Management Parts	Calculates the cost of parts from the Parts & Labor tab in a change request.
Calculate Change Management Labor	Calculates the cost of labor from the Parts & Labor tab in a change request.
Calculate Request Management Labor	Calculates the cost of labor from the Parts & Labor tab in a change request.
Calculate Request Management Parts	Calculates the cost of parts from the Parts & Labor tab in a change request.

3. Click **Save**.
4. Click **OK**.
5. Restart the Service Manager server for your changes to take effect.

1.14.3.5. View a service contract

1. Click **Service Level Management** > **Service Contracts** > **Service Contracts**.
2. Click **Search** to generate a record list.
3. Select a contract record to view in the Service Contract form.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Delete a service contract
Edit a service contract
View an exceeded interaction allocation service contract
View an exceeded incident allocation service contract
View an expired service contract
View an overspent service contract

1.14.3.6. View an exceeded interaction allocation service contract

1. Click **Service Level Management** > **Service Contracts** > **Exceeded Call Allocation**.
2. Click **Search** to generate a record list.
3. Select a contract record.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Delete a service contract
Edit a service contract
View a service contract
View an exceeded incident allocation service contract
View an expired service contract
View an overspent service contract

1.14.3.7. View an exceeded incident allocation service contract

1. Click **Service Level Management** > **Service Contracts** > **Exceeded Incidents**.
2. Click **Search** to generate a record list.
3. Select a contract record to view in the Service Contract form.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Delete a service contract
Edit a service contract
View a service contract
View an exceeded interaction allocation service contract
View an expired service contract
View an overspent service contract

1.14.3.8. View an expired service contract

1. Click **Service Level Management** > **Service Contracts** > **Expired Contracts**.
2. Click **Search** to generate a record list.
3. Select a contract record to view in the Service Contract form.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Delete a service contract
Edit a service contract
View a service contract
View an exceeded interaction allocation service contract
View an exceeded incident allocation service contract
View an overspent service contract

1.14.3.9. View an overspent service contract

1. Click **Service Level Management** > **Service Contracts** > **Overspent Contracts**.
2. Click **Search** to generate a record list.
3. Select a contract record to view in the Service Contract form.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Create a new service contract
Delete a service contract
Edit a service contract
View a service contract
View an exceeded interaction allocation service contract
View an exceeded incident allocation service contract
View an expired service contract

1.14.3.10. Create a Labor Performed Detail record

Follow these steps:

1. Click **Service Level Management** > **Configuration** > **Labor Performed Detail**.
2. Type or select data for each field.

   Note: If you want to edit an existing record, click **Search** to locate the target record.
3. Click **Add**. The **ID** field displays a new record number.
4. Click **Save**.
5. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Labor costs
View a Labor Performed Detail record

1.14.3.11. Create a Parts Detail record

Follow these steps:

1. Click **Service Level Management** > **Configuration** > **Part Usage Detail**.
2. Type or select data for each field.

   Note: If you want to edit an existing record, click **Search** to locate the target record.
3. Click **Add**.

   The **ID** field displays a new record number.
4. Click **Save**.
5. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Labor costs
Parts costs
View a Parts Detail record

1.14.3.12. View a Labor Performed Detail record

1. Click **Service Level Management** > **Configuration** > **Labor Performed Detail**.
2. Click **Search** to generate a record list.
3. Click the record you want to view in the Labor Performed Detail form.
4. Edit the record as required.
5. Click **Save**.
6. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Labor costs
Create a Labor Performed Detail record

1.14.3.13. View a Parts Detail record

1. Click **Service Level Management** > **Configuration** > **Part Usage Detail**.
2. Click **Search** to generate a record list.
3. Click the record you want to view in the Parts Detail form.
4. Edit the record as required.
5. Click **Save**.
6. Click **OK**.

Related topics

Service Contracts overview
Working with service contracts
Service contracts
Service contract tasks
Labor costs
Parts costs
Create a Parts Detail record

1.14.3.14. Enable automatic outage posting

1. Click **Service Level Management** > **Administration** > **Configure Application**.
2. Click **Search**.
3. Select an existing record.
4. Service Level Management populates the Service Target Information section to generate outage data correctly. You can change the out-of-box field names to match your field names.
    - **Configuration Item** fields identify the item to be tracked.
    - **Outage Condition** is the signal to gather service metrics. For example, when a Configuration Item is unavailable, the condition is `nullsub(ci.down in $L.file, false)=true`. However, the condition for other tables may vary.
    - **Outage Start** and **Outage End** fields specify when the outage begins and ends for manual or automatic outage posting.
5. Select the **Auto Post Outage Information** checkbox.
6. Click **Save**.
7. Click **OK**.

1.14.3.15. Disable automatic outage posting

1. Click **Service Level Management** > **Administration** > **Configure Application**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record.
4. In the Service Target Information section, clear the **Auto Post Outage Information** checkbox.

   This disables the option to automatically post outage information.
5. Click **Save**.
6. Click **OK**.

1.14.3.16. Change outage start and end times

User roles: System Administrator, Incident Analyst, Incident Coordinator, Incident Manager, Service Desk Agent, Service Desk Manager

The outage times track the amount of time a Configuration Item (CI) is unavailable. You can't update outage time for a closed incident record. Whenever you update the outage times for an incident, you must also update the Activities section of the incident record.

To change an outage start and end times:

1. Cick **Incident Management** > **Search Incidents**
2. Add optional search criteria and then click **Search**.
3. Select a target record with outage information.
4. In the Incident Details section, set new date and time values in the **Outage Start** and **Outage End** fields.
5. Click **Save & Exit**.

1.14.3.17. Enable outage spreading

1. Click **Service Level Management** > **Administration** > **Configure Application**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record.
4. In the Service Target Information section, select **Spread Outages**.
5. In the Service Target Information section, you can also select **Extend outage spreading to more than one level?**
6. Click **Save**.
7. Click **OK**.

1.14.3.18. Disable outage spreading

1. Click **Service Level Management** > **Administration** > **Configure Application**.
2. Add optional search criteria, and then click **Search**.
3. Select a target record.
4. In the Service Target Information section, clear the **Spread Outages** checkbox.

   This disables the spread outages option.
5. Click **Save**.
6. Click **OK**.

1.14.3.19. Edit the Service Level Management control record

Applies to User roles: System Administrator, Administrator

The Service Level Management (SLM) control record specifies the basic configuration options for the service agreements in Service Manager.

To edit the SLM control record options, follow these steps:

1. Click **Service Level Management** > **Administration** > **Edit Control Record**.
2. Select or clear any service agreement options.

Option	Description
Enable SLM Application	Integrate Service Level Management with Incident Management, Problem Management, Request Management, and Service Desk. You must log out and log back in for changes to this record to take effect.
Check Service Hours	Prevent users from opening Incident records outside of the service hours defined in the service agreement. A dialog box advises the operator that service is unavailable.
Allow Override of Service Hours Violation	Permit new service desk interactions outside of the hours specified in the service agreement.
Default SLA	Choose the default SLA to apply when no SLA is specified for the contact’s department or company.
Default OLA	Choose the default Operational Level Agreement (OLA) to apply when no OLA is specified for the contact’s department or company.
Effective Condition	Specifies the condition under which the service agreements are operating (e.g. the"agreed" phase).
Group Table Name	Specifies the default group list for an OLA and UC.

Option	Description
Schedule Time Interval	Specify in how many minutes the corresponding sla schedule record will expire after a record (for example, a Change or Incident record) is created. You can select one of the three options: ◦ 2 minutes ◦ 5 minutes ◦ 10 minutes By default, no option is selected and 2 minutes is used.
Standard Alerts	List the alerts that occur when the application doesn't generate specific alerts.

3. Click **Save**.
4. Click **OK**.

1.14.3.20. Link an SLA to a contact record

Applies to User roles: System Administrator

To link an SLA to a contact record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Contacts**.
2. Click **Search**.
3. Select the contact record for the problem owner.
4. Click **Fill** to choose the **Service ManagerID** for the problem owner.
5. Click **Save**.
6. Click **OK**.

1.14.3.21. Link an SLA to a company record

Applies to User roles: System Administrator

To link an SLA to a company record, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Companies**.
2. Click **Search**.
3. Select the company record to update.
4. Click the drop-down list to choose the **SLA** for the company.
5. Click **Save**.
6. Click **OK**.

1.14.3.22. Recalculate availability and response time

1. Click **Service Level Management** > **Administration** > **Regen SLM Partials**.
2. Choose the **Month**.
3. Choose the **Year**.
4. Click **Proceed**.

When you view metrics in a service agreements, Service Level Management recalculates the availability and response data for the month and year that you selected.

1.14.3.23. View service agreement metrics

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Click **Search**.
3. Select a service agreement.
4. Select the Process tab or Service tab to view current metrics for all Service Level Targets.

1.14.3.24. Configure Process Targets for assignment groups

Before you can check the record handling performance of a group that's assigned to handle a record, you must configure the associated Process Target so that the system applies the duration of a target to specific group(s).

Note

This configuration only applies to the Operation Level Agreement (OLA) or Underpinning Contract (UC) service level categories.

To configure a Process Target for assignment groups, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Process Targets**.
2. If you want to define a new target, go to the next step.
   If you want to edit an exiting target, specify your search criteria in appropriate fields, click **Search**, and then click an appropriate **SLT ID** .
3. Select appropriate assignment groups from the **Duration scope** drop-down list.
4. (Optional) If the duration type is **Interval** or **Duration in Record**, you can select Assignment Group as the applicable schedule and time zone by clicking **Assignment Group**.

The schedule and time zone that's defined for the assignment group is used. If no schedule or time zone is defined for the assignment group, the system uses 24×7 as the default schedule and the time zone of the system server as the default time zone.

1.14.3.25. Research overall results

1. Click **Service Level Management** > **Metrics** > **Overall Metrics**.
2. Add optional search criteria, and then click **Search** to generate a list of all SLM monthly aggregate records.
3. Select a target record.
4. To view service metrics:
    1. Click **Service Target** to display the parent Outage Detail record.
    2. Click **View Outage Events** to display the parent Outage Event record.
    3. Click **View Record** to view the parent request for a change record.
5. To view process Metrics:
    1. Click **View Process Target Details** to display the parent SLT process record.
    2. Click **View Details** to display the parent service agreement Time Activity record.

1.14.3.26. Research service metrics

1. Click **Service Level Management** > **Metrics** > **Service Metrics**.
2. Add optional search criteria, and then click **Search** to generate a list of all SLM monthly aggregate records.
3. Select a target record.
4. Click **View Details** to display the parent Outage Detail record.
5. Click **View Outage Events** to display the parent Outage Event record.
6. Click **View Record** to view the parent request for a change record.

1.14.3.27. Research process metrics

1. Click **Service Level Management** > **Metrics** > **Process Metrics**.
2. Add optional search criteria, and then click **Search** to generate a list of all SLM response records.
3. Select a target record.
4. Click **View Details** to display the parent process target record.
5. Click **View Details** to display the parent service agreement process Time Activity record.

1.14.3.28. Check Service Level Target metrics on a dashboard

You can check the overall status of the Service Targets or Process Targets to better understand the performance of your business. The Service Level Agreement (SLA) dashboards provide comprehensive information about the targets.

To check the Service Level Target metrics on the SLA dashboards, follow these steps:

1. Click **Service Level Management** > **SLA Overview**.
2. Select a category of information that you want to check from the drop-down list on the toolbar.
3. (Optional) To save the current page as the default view of the dashboard, click the **Save as home page** icon .
4. (Optional) To get the latest data, click **Refresh** .
5. (Optional) To export the current page as a PDF document, click **Export** .
6. (Optional) To create a new dashboard, click **Create New Dashboard** and then follow the instructions.
7. (Optional) To add new content into the dashboard, click **Add Content** and then follow the instructions. For example, you can add new content to the dashboard to monitor the SLA, OLA, or UC status.

Note

A new table "sloresponsehist" is added, and the table is mainly a copy of the standard "sloresponse" dbdict with necessary amendments for the working time of assignment groups. You can use the table for report generation.

1.14.3.29. Assign a Service Level Agreement to a department

Applies to User Roles:

System Administrator

Service Level Management assigns an SLA to a new service desk interaction, change, incident, or problem according to a prescribed selection process. As part of this process, Service Level Management uses the SLA specified for the contact's department if there is one available.

To assign a default SLA to a department:

1. Click **System Administration** > **Base System Configuration** > **Departments**.
2. Add optional search criteria and then click **Search**.
3. Select the department record to update.
4. In the **SLA** field, select the SLA to assign to the department.
5. Click **Save**.
6. Click **OK**.

1.14.3.30. Assign a Service Level Agreement to a company

Applies to User Roles:

System Administrator

Service Level Management assigns an SLA to a new service desk interaction, change, incident, or problem according to a prescribed selection process. As part of this process, Service Level Management uses the default SLA specified for the contact's company if no department Service Level Agreement (SLA) exists for the contact.

To assign a default SLA to a company:

1. Click **System Administration** > **Base System Configuration** > **Companies**.
2. Add optional search criteria and then click **Search**.
3. Select the company record to update.
4. In the **Default SLA for Company** field, select from the list to assign the SLA to the company.
5. Click **Save**.
6. Click **OK**.

1.14.3.31. Assign a default Service Level Agreement

Applies to User Roles:

administrator (SLA profile), System Administrator

Service Level Management assigns a default Service Level Agreement (SLA) when there is no SLA specified for the contact's department or company.

1. Click **Service Level Management** > **Administration** > **Edit Control Record**.
2. In the **Default SLA** field, choose the default service level agreement to apply when no agreement is specified for a contact's department or company.
3. Select or clear any other Service Level Agreement options.
4. Click **Save**.
5. Click **OK**.

1.14.3.32. Assign an Operational Level Agreement or an Underpinning Contract to an assignment group

Applies to User Roles:

System Administrator

An Operational Level Agreement (OLA) or an Underpinning Contract (UC) can be assigned to an assignment group to measure its performance. You can either make the assignment when you create a new agreement or do that when you edit an existing agreement.

To assign an OLA/UC when you create a new agreement, follow these steps:

1. Click **Service Level Management** > **Agreements** > **New Agreement**.
2. In the Select Agreement Category form, click **Operational Level Agreement** or **Underpinning Contract** as necessary.
3. In the New Agreement form, click the **Assignment Groups** tab for an OLA or the **External Assignment Group** tab for a UC.
4. Click the Fill Field button to add an assignment group for the OLA or UC.
5. Click **Save** or **Save & Exit** to save your changes.

To add an assignment group for an existing OLA or UC, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Agreement Queue**.
2. In the Agreement Queue: All Agreements form, click an appropriate agreement ID.
3. Click the **Assignment Group** tab for an OLA or the **External Assignment Group** tab for a UC.
4. Click the Fill Field button to add an assignment group for the OLA or UC.
5. Click **Save** or **Save & Exit** to save your changes.

1.14.3.33. Configure time duration settings

To configure the time duration settings, follow these steps:

1. Click **Service Level Management** > **Administration** > **Time Duration Settings**.
2. Click **Search** to open a list of the time duration control tables.
3. Click a specific table you want to modify.

   When you open the Time Duration Settings form for a specific table, the time duration property fields are automatically populated.
4. Specify the settings on the form, as described in the following table.

Attribute	Description	Default value
Table Name	Specifies the name of the table from which the data is extracted for time duration processing.	N/A
Enable	If this option is selected, the data in this specific object table is enabled for time duration processing. **Note:** This option only works when **Enable processing for time duration reporting data** in **Report Settings** is selected. Otherwise, all tables are disabled for time duration data processing.	Enable
Calculation From	Specifies the exact start date and time for the data to be processed.	N/A
Last Calculation Time	Shows the last time when the background scheduler runs to process the data. This field is read-only.	N/A

Attribute	Description	Default value
Enable Fields	Shows the available labels and their corresponding field names in the table. Select the checkbox to assign labels for time duration calculation. The available labels include State, Phase, and Assignment Group. **Note:** The field names for the State and Assignment Group labels dynamically change according to the system configuration, while the field name for the Phase label is static.	N/A
Metrics > Field	Specifies the field applied for calculation.	N/A
Metrics > KPI	Sets the metrics number for the selected field. Service Manager supports up to five rules for one field.	N/A
Metrics > Schedule Type	Specifies the schedule type applied for calculation. The available types include specific values and the assignment group.	N/A

Attribute	Description	Default value
Metrics > Schedule Value	Specifies the exact schedule value applied for calculation.	N/A
Metrics > Time Zone Type	Specifies the time zone type applied for calculation. **Note:** Only if you select specific value as the time zone type, you need to specify a time zone value.	N/A
Metrics > Time Zone Value	Specifies the time zone value applied for calculation. **Note:** This field only takes effect when you select specific value as the time zone type and you need to specify a value here. The other time zone types already specify a time zone value by their own settings. In this case, any value you specify here is not taking effect.	N/A

5. Click **Save**.

Customize the time duration data processing schedules and threads

Currently, there is an out-of-box schedule set for processing the time duration data. However, you can customize the schedule to add more threads to process the data for a table. Different schedules divide the time duration data by the time stamp that you specify. In this case, each schedule processes the data during the specified period and each schedule process is assigned with a thread to implement the data processing.

For each table containing the time duration data to be processed, it has a main schedule which you need to specify a start time for the data to be calculated. This schedule handles the data starting from the time stamp you specified to the current time. To set the time stamp for the main schedule, follow these steps:

1. Click **Service Level Management** > **Administration** > **Time Duration Settings**.
2. Click **Search** to open a list of the time duration control tables.
3. Click a specific table you want to modify.
4. Specifies the exact start date for the data to be processed in the **Calculation From** field.
5. Click **Save**.

1. Click **Service Level Management** > **Administration** > **Time Duration Settings**.
2. Click **Search** to open a list of the time duration control tables.
3. Click a specific table you want to modify.
4. Specifies the exact start date for the data to be processed in the **Calculation From** field.
5. Click **Save**.

To add more new schedules for and specify the time stamp for a table, follow these steps:

1. Type **schedule** in the Service Manager command-line box and click **Enter**.
2. Click **Search** to open a list of the schedules.
3. Click the **Process Time Duration Reporting Data** schedule record.
4. Change the **Name**, **Class**, and **Scheduled Class** values. For example, you can modify the **Name** value to **Process Time Duration Reporting Data1**, and modify the **Class** and **Scheduled Class** values to **timedurationdata1**.
5. Modify the content in the Javascript tab to the following script:

   `lib.timeDurationEtl.doWorkOnFile(<log_level>, '<table name>', <start date>, <end date>);`

   For example: `lib.timeDurationEtl.doWorkOnFile(2, 'probsummary', 10/01/14, 12/31/14);`

> **Note**
>
> Service Manager defines four log_level values as: 1 DEBUG, 2 INFO, 3 WARN, and 4 ERROR.

6. Click **Add** to add this record.
7. Repeat steps 4 to 6 to add more schedules as it requires and use a different number for each new record. For example, add the **Name** value of **Process Time Duration Reporting Data2** with the **Class** and **Scheduled Class** values as **timedurationdata2**.

The **Class** value of the new schedules will be needed for adding the background processes later.

To assign a background process for the new schedules, follow these steps:

1. Type **info** in the Service Manager command-line box and click **Enter**.
2. Click **Search** to open a list of the background process records.
3. Click the **timedurationdata.processor** record to open the **Background Processor Initialization Registry** schedule form for this record.
4. Add a new record under the main record. Add the **Class** value you have specified to the **Name** and **Class** fields and copy the other required values from the previous record.
5. Click **Save** to save the record and the system adds a new blank record for you to add the next one, if there is any.
6. Repeat steps 4 and 5 to assign all the new schedules you have added with a thread to process the data.
7. Click **Save**.

After you have added the schedules and the threads, start a new Service Manager process by running the following command line, or by adding the command line to the sm.cfg file:

`sm scheduler.start <name of the background process> -log:../logs/timeduration.log`

In this example: `sm scheduler.start timedurationdata.processor -log:../logs/timeduration.log`

1.14.3.34. Visualize Service Level Management

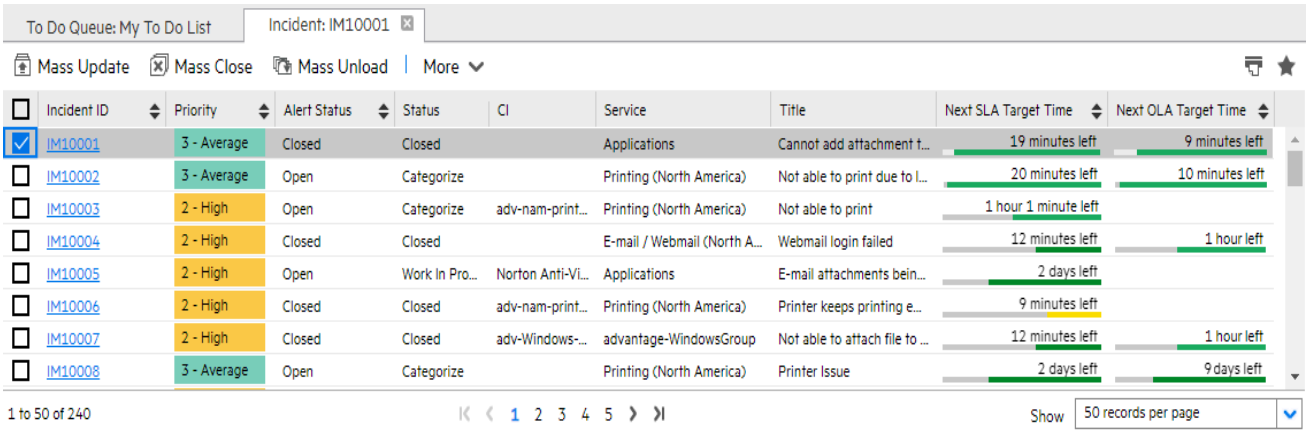
Service Level Management visualization in the record list

As of version 9.70, both the Incident and Interaction queues provide the **Next SLA Target Time** and **Next OLA Target Time** columns that display the operation status of the Service Level Targets (SLTs). You can click the column headers to sort the records by their natural remaining time.

In the columns, the progress bars visualize the natural time and the color indicators highlight the status of the remaining time for the latest expiring SLT. The default color indicator definitions are as follows:

- Green: The remaining time is more than 50% of the SLT.
- Yellow: The remaining time is between 25% and 50% of the SLT.
- Orange: The remaining time is less than 25% of the SLT.
- Red: The SLT is going to breach.

See the following screenshot as an example.



To configure the color indicator settings, go to **Tailoring** > **Color Indicator Setting** and then search for **sla**. See Configuring color indicator for more information about color indicator settings.

Tip

To display the **Next SLA Target Time** and **Next OLA Target Time** columns in the record list for other Service Manager modules, you need to manually add the next.sla.target.time and next.ola.target.time fields to the corresponding tables in the **database dictionary**. For example, to display these two columns in the Change queue, add two fields to the cm3r table as follows:

Name	Type	Level	Index	SQL Name	SQL Type	SQL Table
next.sla.target.time	character	1	114	NEXT_SLA_TARGET_TIME	NVARCHAR(60)	m1
next.ola.target.time	character	2	115	NEXT_OLA_TARGET_TIME	NVARCHAR(60)	m1

Note that the Index values and the SQL Table values may vary depending on your actual implementation.

Service Level Management visualization on the SLT View tab

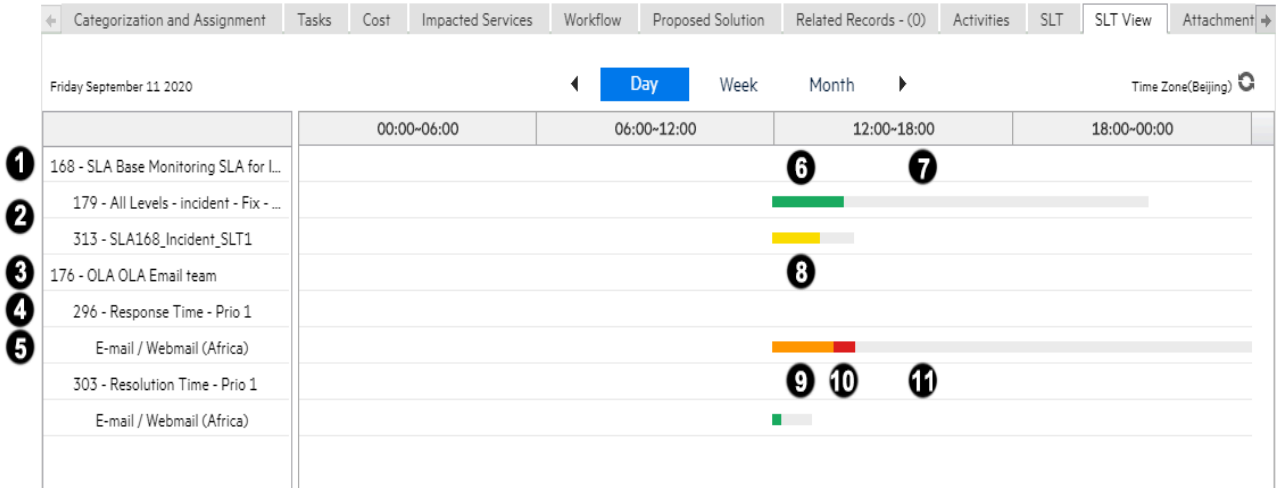
As of version 9.70, the out-of-the-box Incident and Interaction forms have the **SLT View** tab that displays the operation status for all related SLA, OLA, and UC of a record. SLAs, OLAs, SLTs, and the related assignment groups are displayed in hierarchies on the left side. On the right side, the progress bars consist of various color indicators and visualize the actual working time of the process targets. You can view the definition of all indicators and duration of a color indicator by tooltips.

In the progress bar, the default color indicator definitions are as follows:

- Green: The elapsed time is less than 50% of the next SLT breach time.
- Yellow: The elapsed time is between 50% and 75% of the next SLT breach time.
- Orange: The elapsed time is between 75% and 100% of the next SLT breach time.
- Red: The time after the SLT breach.

- Light grey: The remaining time to the next SLT breach, or the remaining time after the SLT breach.
- Dark grey: The suspended time.

See the following screenshot as an example.

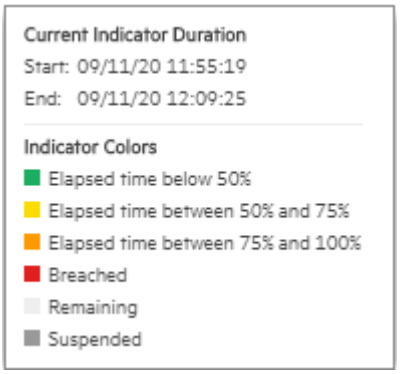
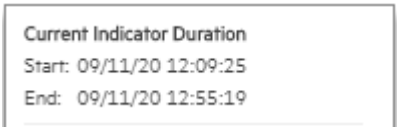


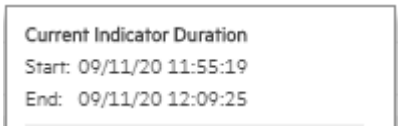
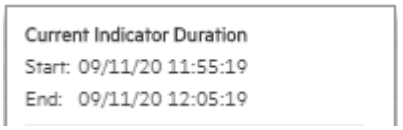
Note

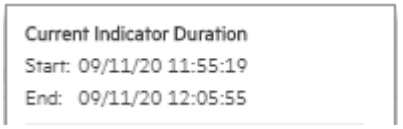
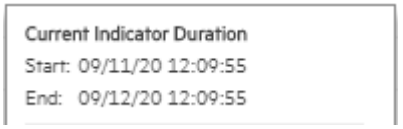
The system doesn't automatically refresh the progress bars' status on the SLT View tab. You need to perform a manual refresh to view the latest status of the progress bars.

See the following table for a detailed description of the sample screenshot.

Item	Description
1	The SLA that relates to the current record. In the hierarchy, the SLA level doesn't have progress bars by design.
2	The SLT that relates to the SLA. In the hierarchy, the SLTs under SLAs have progress bars by design.
3	The OLA that relates to the current record. In the hierarchy, the OLA level doesn't have progress bars by design.
4	The SLT that relates to the OLA. In the hierarchy, the SLTs with the duration scopes of either **Total time for each assignment group** or **Time for the current assignment group** do not have progress bars by design. Only the SLTs with the duration scope of **Total time for all assignment groups** have progress bars.
5	The assignment group that works on the SLT. In the hierarchy, the assignment groups under SLTs have progress bars by design.

Item	Description
6	A green color indicator that indicates the elapsed time is less than 50% of the SLT breach time. When you hover your mouse over this indicator, the tooltip may resemble the following:  The Start time is the SLT start time, and the End time is the actual time when you refresh the SLT View tab and display this tooltip.
7	A light grey indicator that indicates the remaining time to the SLT breach. The time information on the tooltip may resemble the following:  The Start time is the actual time when you refresh the SLT View tab and display this tooltip, and the End time is the SLT breach time.

Item	Description
8	A yellow color indicator that indicates the elapsed time is between 50% and 75% of the SLT breach time. The time information on the tooltip may resemble the following:  The Start time is the SLT start time, and the End time is the actual time when you refresh the SLT View tab and display this tooltip.
9	An orange color indicator that indicates the elapsed time is between 75% and 100% of the SLT breach time. The time information on the tooltip may resemble the following:  In the sample screenshot, the current SLT is breached and the orange color indicator displays the entire duration as you have defined for the SLT. The Start time is the SLT start time, and the End time is the SLT breach time.

Item	Description
10	A red color indicator that indicates the SLT has breached. The time information on the tooltip may resemble the following:  The Start time is the SLT breach time, and the End time is the actual time when you refresh the SLT View tab and display this tooltip.
11	A light grey indicator that indicates the time after the SLT breach. This indicator appears here because the ticket's status has not reached the Final State as defined in the SLT when the SLT breaches. The time information on the tooltip may resemble the following:  The Start time is the actual time when you refresh the SLT View tab and display this tooltip, and the End time is 24 hours later. In the sample screenshot, this light grey indicator disappears once you update the ticket's status to the Final State as defined in the SLT.

1.14.3.34.1. Example: SLM visualization in the record list

The following example illustrates how the SLM visualization works in the record list when an SLA and an OLA work on an Incident. You can view the indicators' color changes as time elapses.

Prepare SLA and OLA

This example uses the following OLAs, SLAs, and Process Targets (SLTs) and most of them are available out-of-the-box. Create similar data if you're not working with the out-of-the-box data and don't have the examples.

Item	Values
OLA 176	• Title: OLA Email Team • Assignment Groups: E-mail / Webmail (Africa), E-mail / Webmail (Asia) • Related process targets: 296, 303
Process Target 296	• Name: Response Time - Prio 1 • Schedule: 24×7 • Initial State: Open • Final State: Work In Progress • Duration Type: Interval • Duration: 00:10:00 • Duration scope: Time for the current assignment group
Process Target 303	• Name: Resolution Time - Prio 1 • Schedule: 24×7 • Initial State: Open • Final State: Resolved • Duration Type: Interval • Duration: 01:00:00 • Duration scope: Total time for each assignment group
SLA 168	• Title: Base Monitoring SLA for IT Services • Related process targets: 179, 313

Item	Values
Process Target 179	• Name: All Levels - incident - Fix - All Priorities • Schedule: Mon to Fri from 8am to 4pm • Initial State: Open • Final State: Resolved • Duration Type: Interval • Duration: 20:00:00
Process Target 313	• Name: SLA168_Incident_SLT1 • Schedule: 24×7 • Initial State: Open • Final State: Assign • Duration Type: Interval • Duration: 00:20:00

Create an Incident record

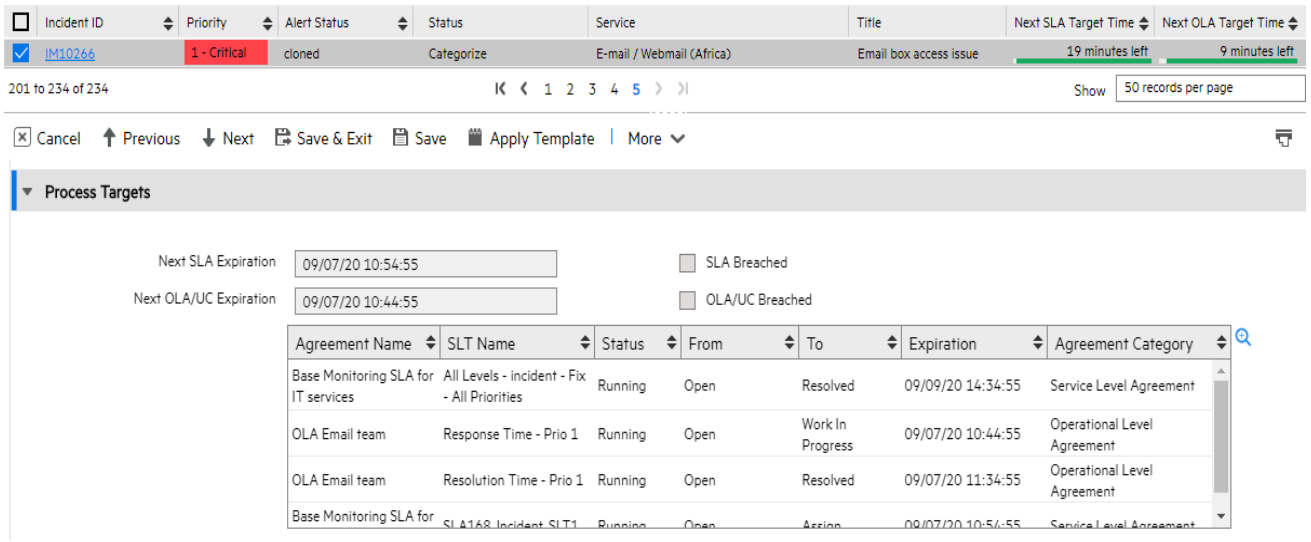
Create an Incident record that both the SLA and OLA can apply. This example uses the Incident record IM10266.

View the indicators' status change

Green

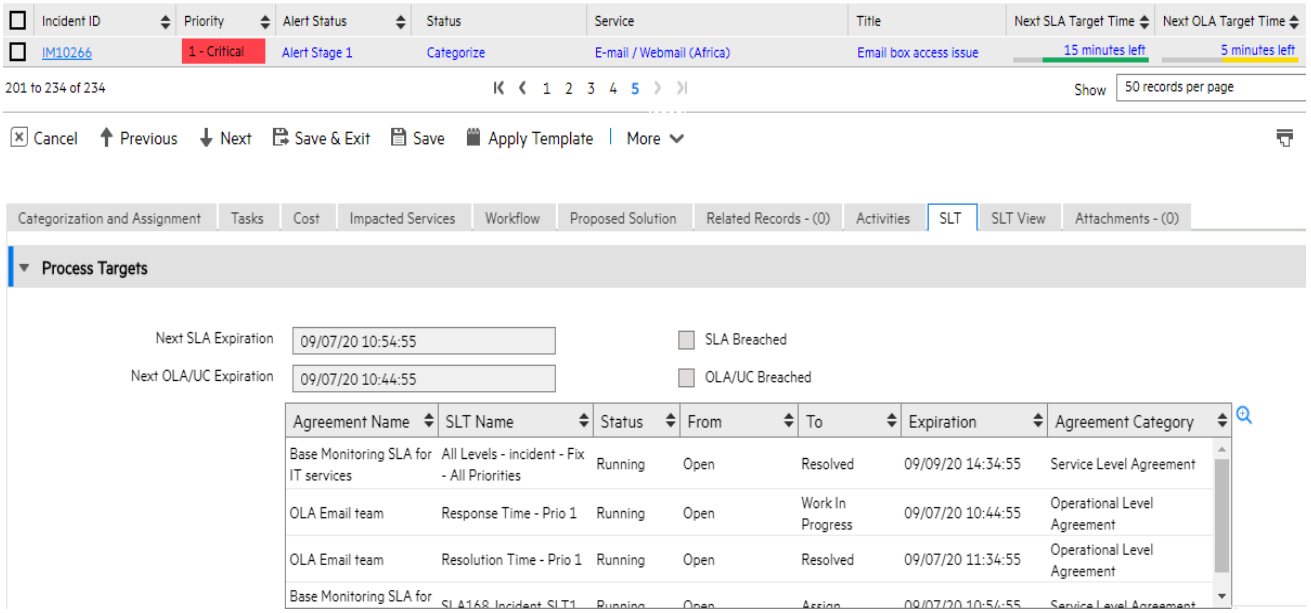
Shortly after you create IM10266, open this record and then click the **SLT** tab to view the status of the SLTs. All of them are **Running**.

In the record list, the **Next SLA Target Time** column displays the status of the next SLA target (313) and **Next OLA Target Time** column displays the status of the next OLA target (296). Both indicators are green because the remaining time is more than 50% of the next SLT.



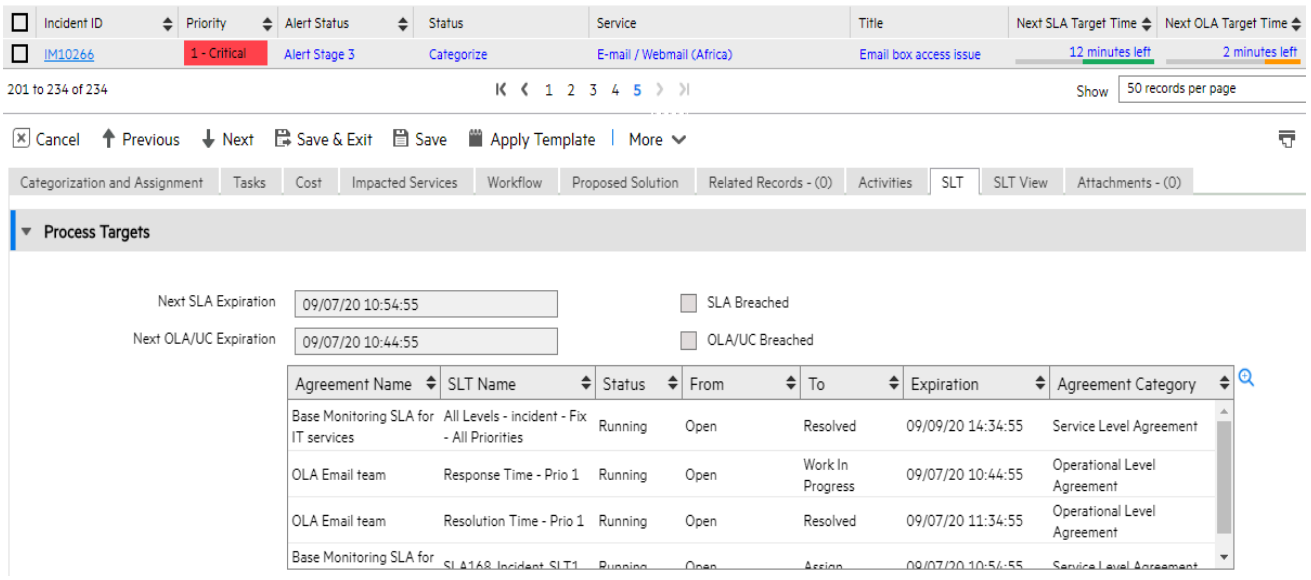
Yellow

Refresh the record list after a short while. Now the color indicator in the **Next OLA Target Time** column turns yellow because the remaining time is between 25% and 50% of the next SLT (296). The color indicator in the **Next SLA Target Time** column is still green because the duration of the next SLT (313) is 20 minutes.



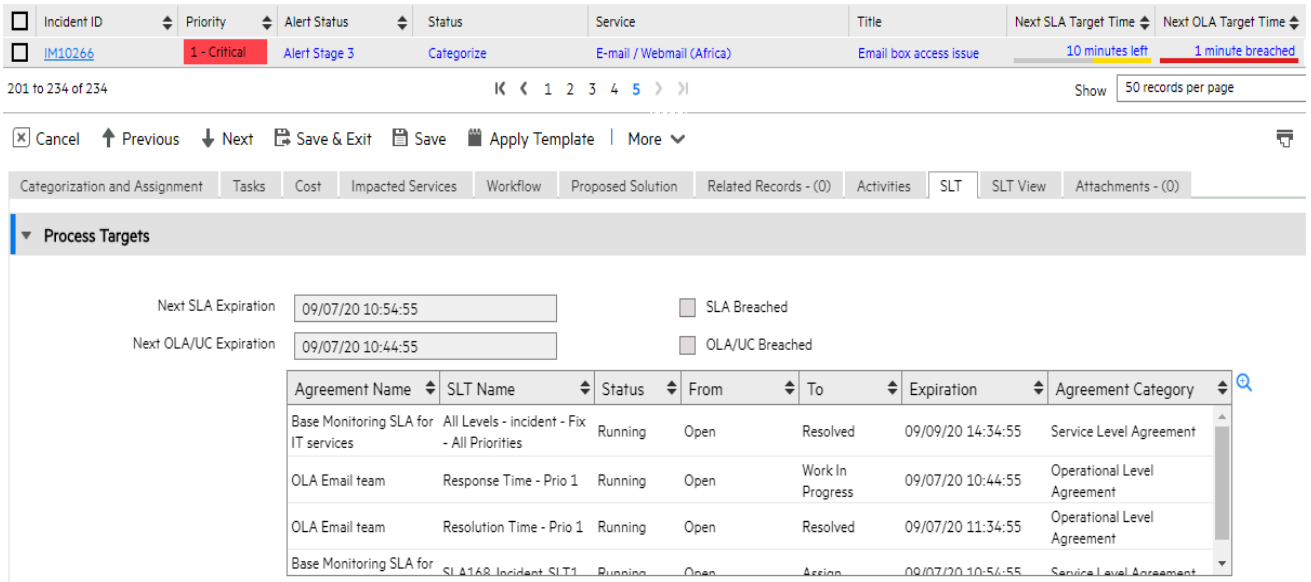
Orange

Refresh the record list after a short while. Now the color indicator in the **Next OLA Target Time** column turns orange because the remaining time is less than 25% of the next SLT (296).



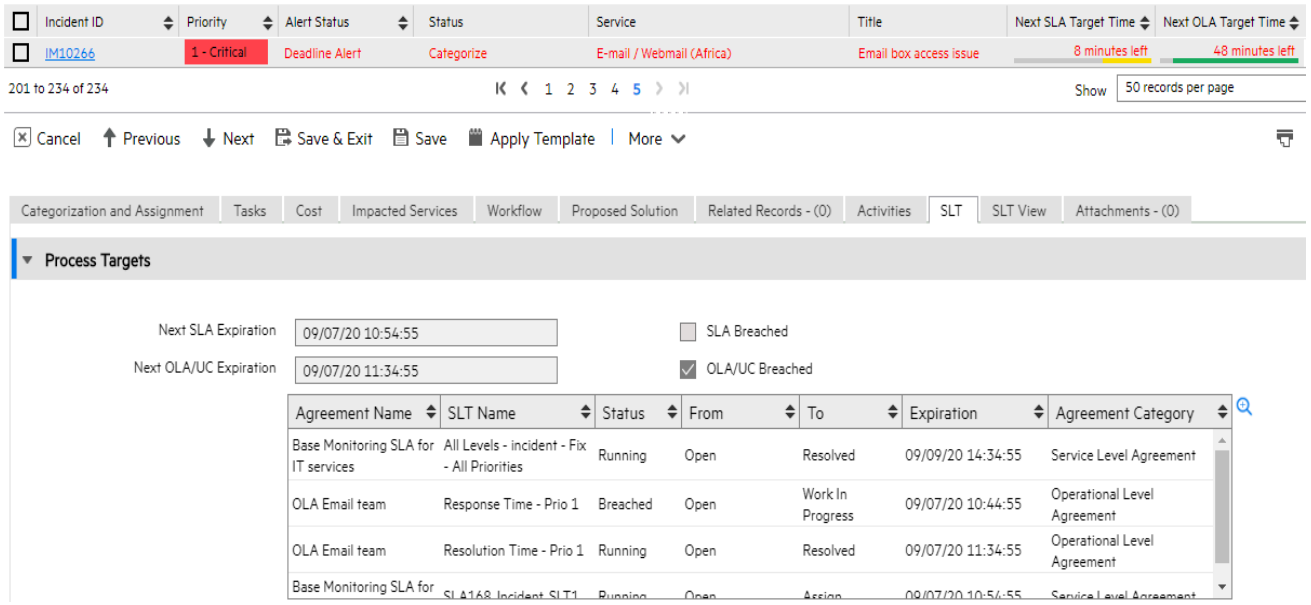
Red

Refresh the record list after a short while. Now the color indicator in the **Next OLA Target Time** column turns red because the SLT (296) has been breached for 1 minute. The color indicator in the **Next SLA Target Time** column turns yellow because the remaining time is between 25% and 50% of the next SLT (313).

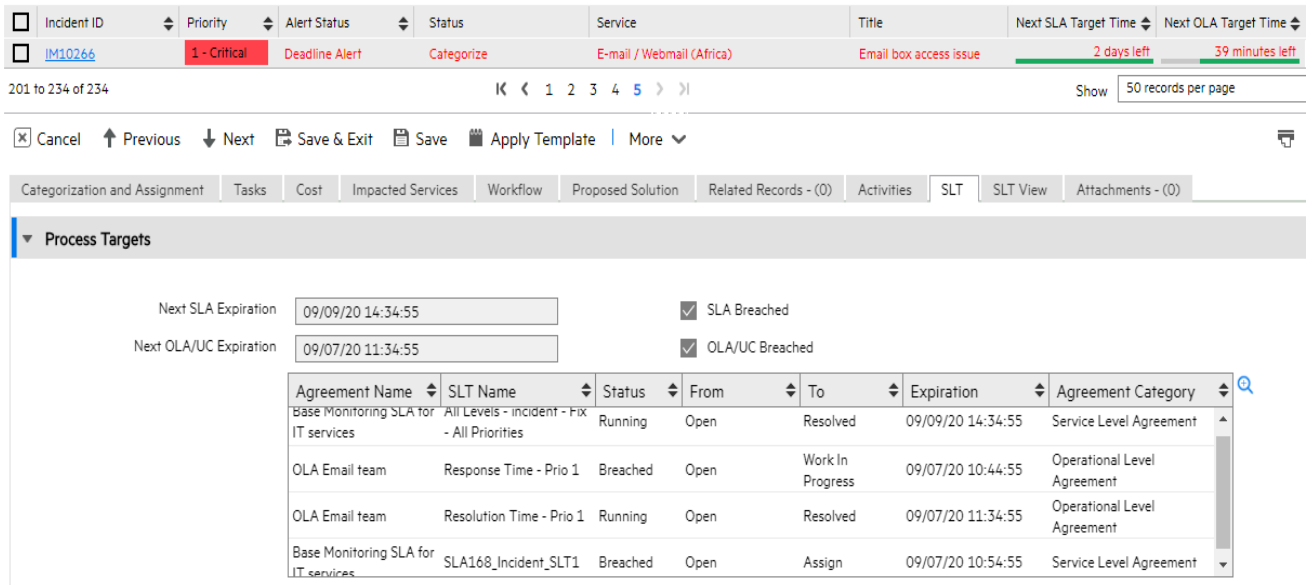


Green again

After SLT (296) breaches, the color indicator in the **Next OLA Target Time** column is green again and displays the status of the next SLT (303).

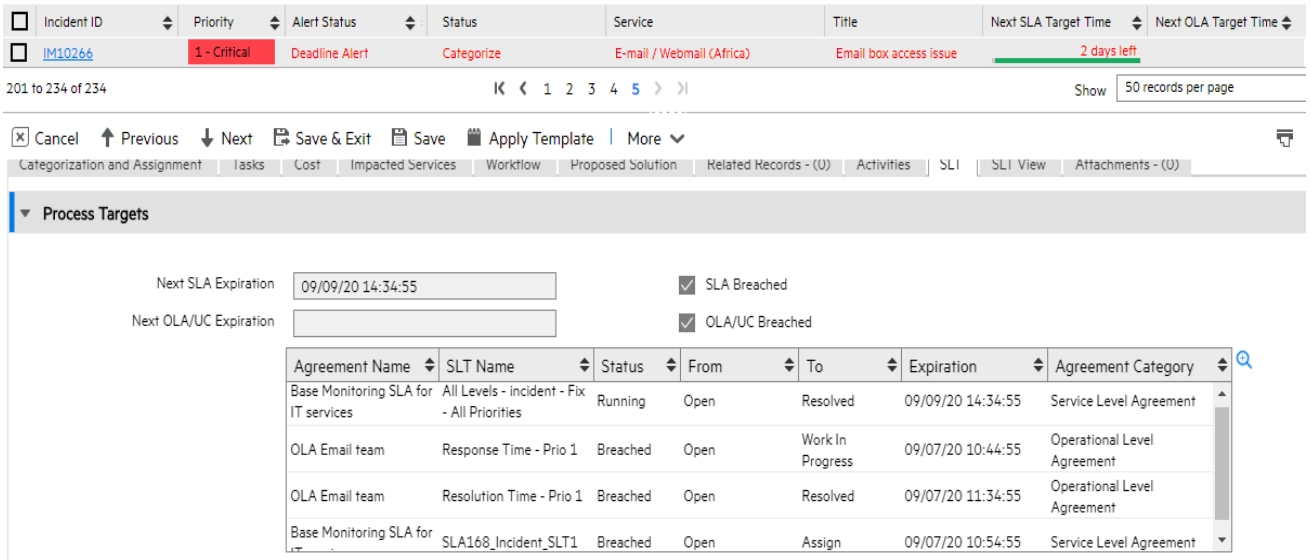


After SLT (313) breaches, the status of the next SLT (179) is displayed in the **Next SLA Target Time** column.



Blank

Refresh the record list after some time. Now SLT (303) is also breached and the system displays the **Next OLA Target Time** column as blank.


Incident ID
Priority
Alert Status
Status
Service
Title
Next SLA Target Time
Next OLA Target Time
IM10266
1 - Critical
Deadline Alert
Categorize
E-mail / Webmail (Africa)
Email box access issue
2 days left
201 to 234 of 234
1 2 3 4 5
Show
50 records per page
Cancel
Previous
Next
Save & Exit
Save
Apply Template
More
Categorization and Assignment
Tasks
Cost
Impacted Services
Workflow
Proposed Solution
Related Records - (0)
Activities
SLT
SLT View
Attachments - (0)
Process Targets
Next SLA Expiration
09/09/20 14:34:55
SLA Breached
Next OLA/UC Expiration
OLA/UC Breached
Agreement Name
SLT Name
Status
From
To
Expiration
Agreement Category
Base Monitoring SLA for IT services
All Levels - incident - Fix - All Priorities
Running
Open
Resolved
09/09/20 14:34:55
Service Level Agreement
OLA Email team
Response Time - Prio 1
Breached
Open
Work In Progress
09/07/20 10:44:55
Operational Level Agreement
OLA Email team
Resolution Time - Prio 1
Breached
Open
Resolved
09/07/20 11:34:55
Operational Level Agreement
Base Monitoring SLA for
SLA168_Incident_SLT1
Breached
Open
Assign
09/07/20 10:54:55
Service Level Agreement

1.14.3.34.2. Example: SLM visualization on SLT View tab with a suspended ticket

The following example illustrates how the SLM visualization works on the SLT View tab when an SLA works on an Interaction with suspended status. You can view the indicators' color changes as time elapses. When this Interaction's status becomes **Suspended**, the indicator's color can reflect this update as well.

Prepare SLA

This example uses the following SLA and Process Targets (SLTs) and most of them are available out-of-the-box. Create similar data if you're not working with the out-of-the-box data and don't have the sample data.

Item	Values
SLA 172	• Title: Platinum Service • Related process targets: 307, 311
Process Target 307	• Name: Platinum - Initial Response • Schedule: 24×7 • Initial State: Open • Final State: Dispatched • Duration Type: Interval • Duration:20:00:00 • Suspend processing: Pending Vendor • for these states: Suspended
Process Target 311	• Name: Platinum - Time to Fix • Schedule: 24×7 • Initial State: Open • Final State: Closed • Duration Type: Interval • Duration: 20:00:00

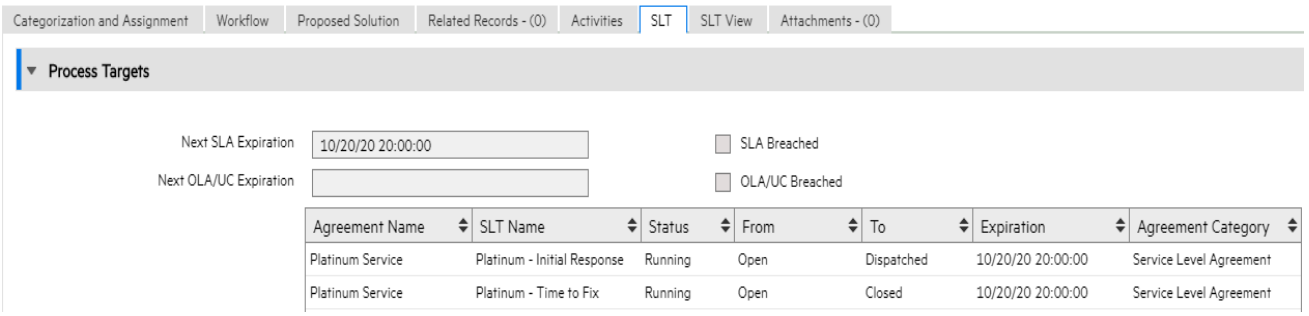
Create a Service Desk Interaction record

Create a Service Desk Interaction record that both the SLA can apply. This example uses the Interaction record SD10222.

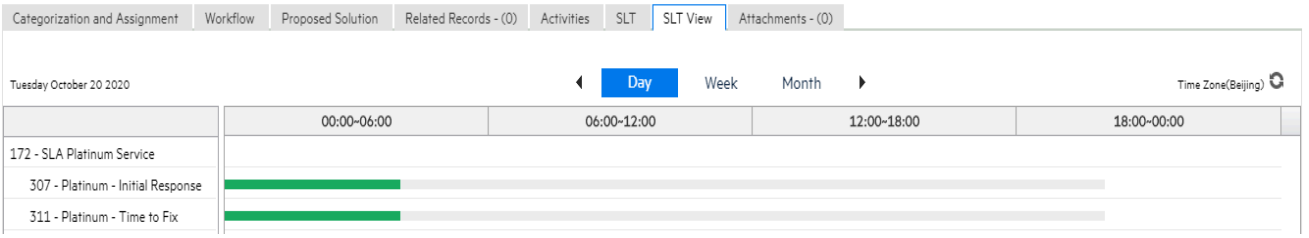
View the indicators' status change

Green

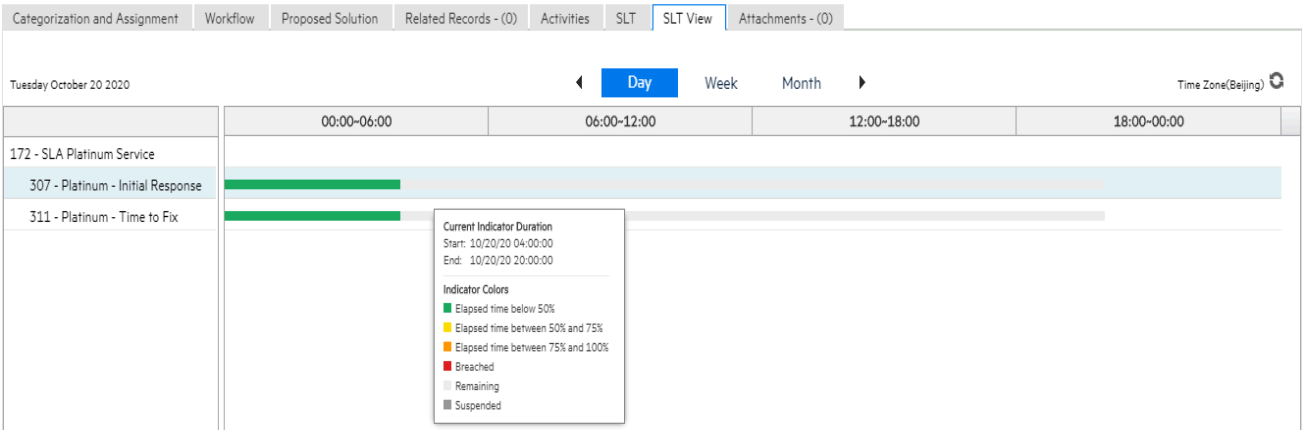
Refresh the record list and open SD10222 four hours later. Click the **SLT** tab to view that the SLTs' status is Running.



Next, click the **SLT View** tab to view the status of the SLTs.



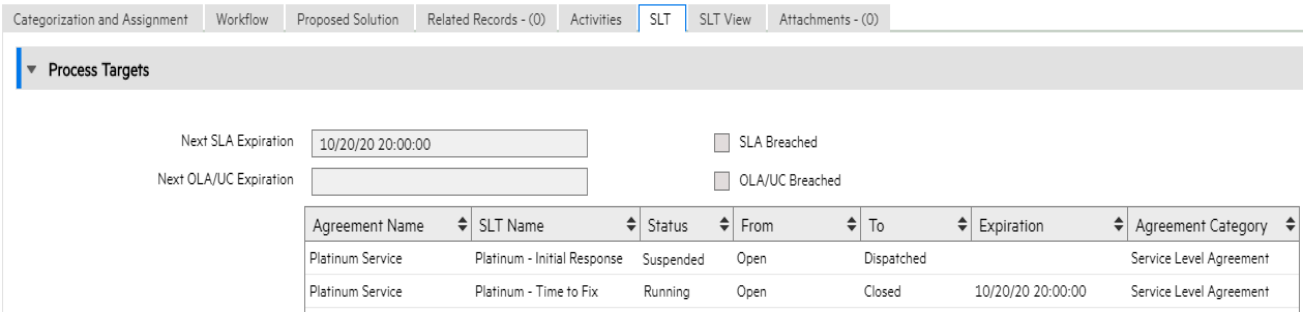
Hover your mouse over the remaining time area (the light grey indicator) of SLT 307, the time information on the tooltip may resemble the following. The Start time is the actual time when this tooltip is displayed, and the End time is the SLT breach time.



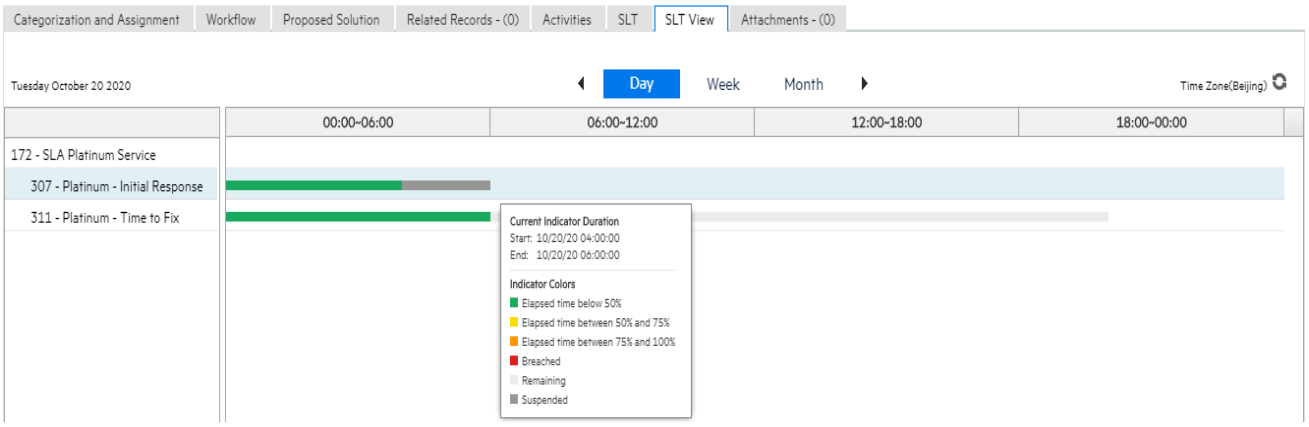
Update the ticket's status to **Suspended**, and then set the suspend time to 2 hours. Save your changes.

Dark grey

Refresh the record list and check the ticket's status after 2 hours. Click the **SLT** tab to view the SLTs' status.

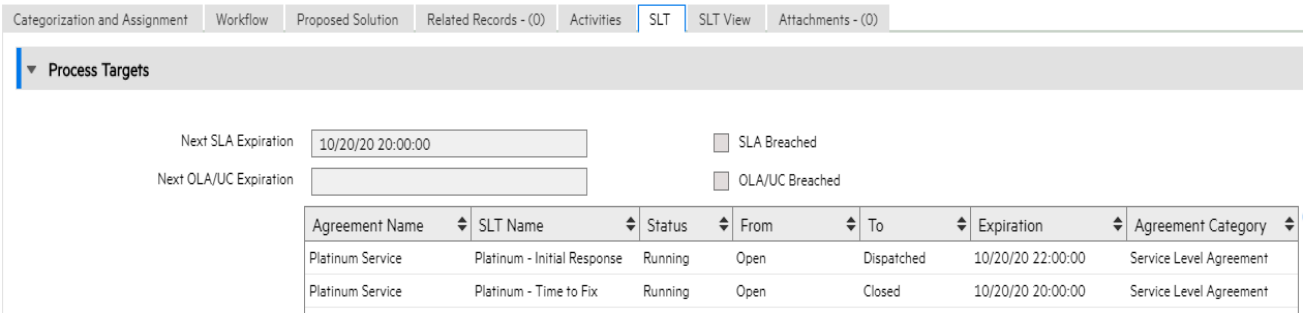


Next, click the **SLT View** tab to view the status of the SLTs. Hover your mouse over the dark grey indicator of SLT 307, the time information on the tooltip may resemble the following. The Start time is the start time when this ticket is suspended, and the End time is the actual time when this tooltip is displayed.



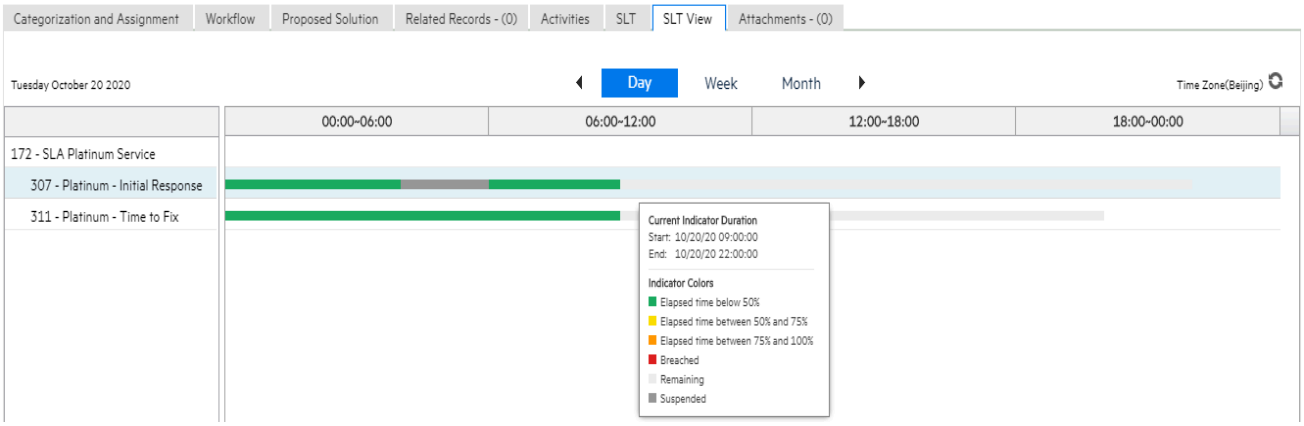
Green again

Refresh the record list and check the ticket's status after 3 hours. Now the ticket is unsuspended. Click the **SLT** tab to view the SLTs' status.



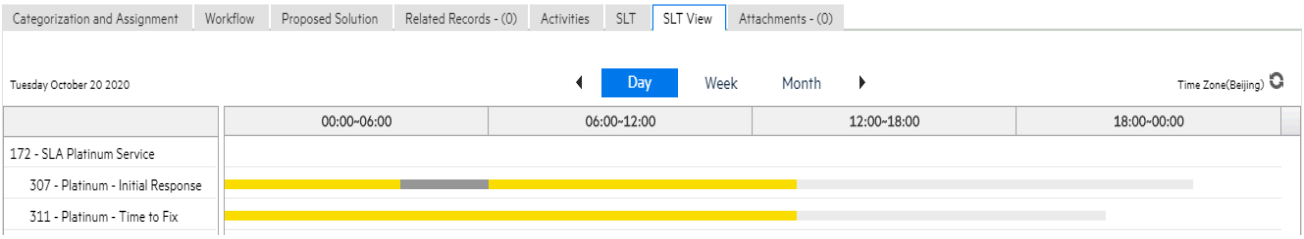
Next, click the **SLT View** tab to view the status of the SLTs. Hover your mouse over the remaining time area (the light grey indicator) of SLT 307, the time information on the tooltip may resemble the following. The Start time is the actual time when this

tooltip is displayed, and the End time is the SLT breach time. Note that the SLT breach time equals to the original breach time plus the suspended time.



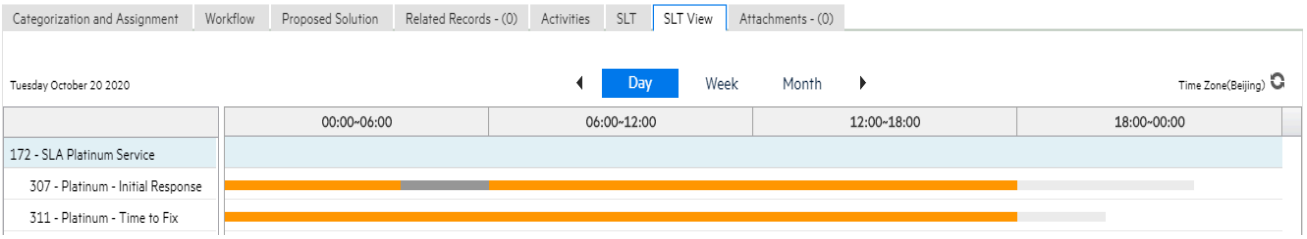
Yellow

Refresh the record list and check the ticket's status after 4 hours. Click the **SLT View** tab to view the status of the SLTs.



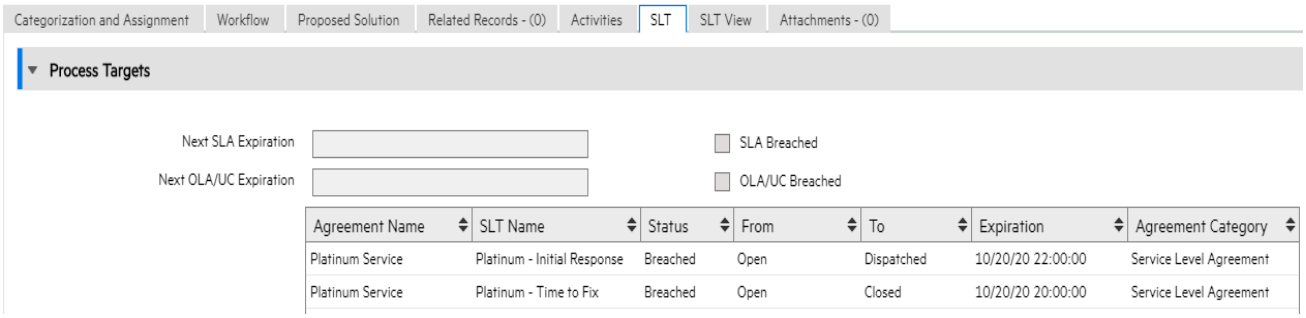
Orange

Refresh the record list and check the ticket's status after 6 hours. Click the **SLT View** tab to view the status of the SLTs.

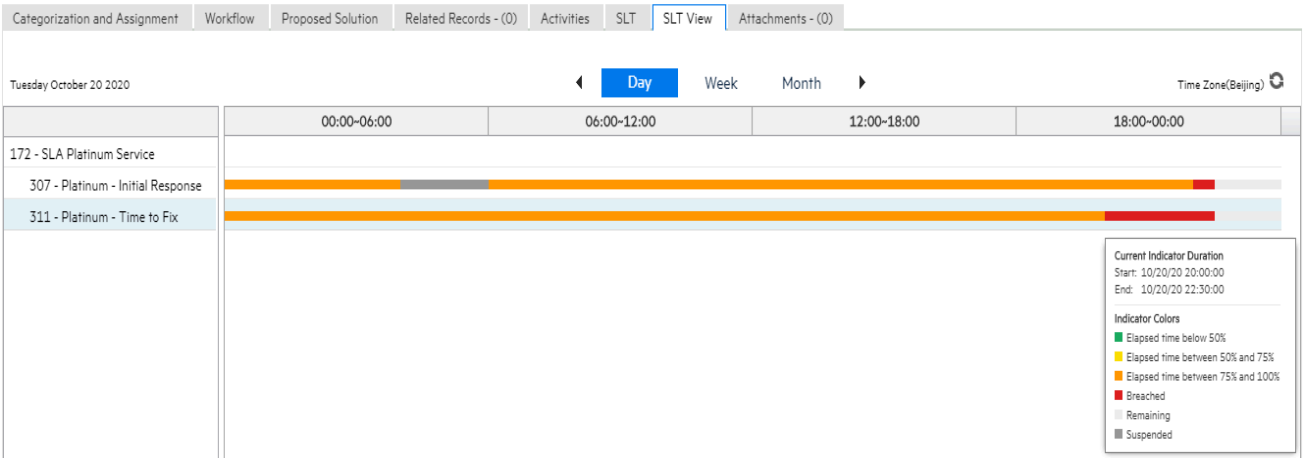


Red

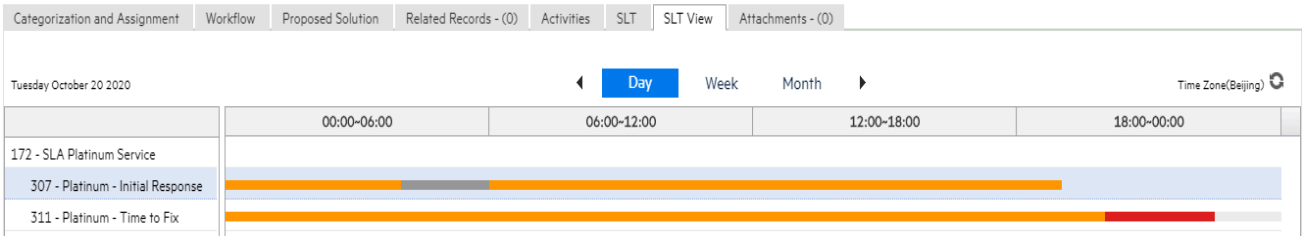
Refresh the record list and check the ticket's status after 4.5 hours. Click the **SLT** tab to view that both SLTs' status is Breached.



Next, click the **SLT View** tab to view the status of the SLTs. Hover your mouse over the remaining time area (the light grey indicator) of SLT 307, the time information on the tooltip may resemble the following. The Start time is the SLT breach time, and the End time is the actual time you refresh the SLT View tab and display this tooltip.



Note that if you escalate the Service Desk Interaction record and its status become Dispatched (the Final State as defined in SLT 307), the light grey indicator for SLT 307 disappears, whereas the light grey indicator for SLT 311 sustains because the ticket has not reached the Final State as defined in SLT 311 yet. Suppose you escalate the Service Desk Interaction record at 19:00 and then check the status of the SLTs at 22:30, the screenshot may resemble the following:



1.14.3.34.3. Example: SLM visualization on SLT View tab with altered assignment groups

The following example illustrates how the SLM visualization works on the SLT View tab when an OLA works on an Incident with altered assignment groups. You can view the indicators' color changes as time elapses. When you change this Incident's assignment group setting, the progress bars can reflect this update as well.

Prepare OLA

This example uses the following OLAs and Process Targets (SLTs) and most of them are available out-of-the-box. Create similar data if you're not working with the out-of-the-box data and don't have the examples.

Item	Values
OLA 176	• Title: OLA Email Team • Assignment Groups: E-mail / Webmail (Africa), E-mail / Webmail (Asia) • Related process targets: 296, 303, 314
Process Target 296	• Name: Response Time - Prio 1 • Schedule: 24×7 • Initial State: Open • Final State: Work In Progress • Duration Type: Interval • Duration: 12:00:00 • Duration scope: Time for the current assignment group
Process Target 303	• Name: Resolution Time - Prio 1 • Schedule: 24×7 • Initial State: Open • Final State: Resolved • Duration Type: Interval • Duration: 12:00:00 • Duration scope: Total time for each assignment group
Process Target 314	• Name: Fix Time - Prio 1 • Schedule: 24×7 • Initial State: Open • Final State: Work In Progress • Duration Type: Interval • Duration: 12:00:00 • Duration scope: Total time for all assignment groups

Create an Incident record

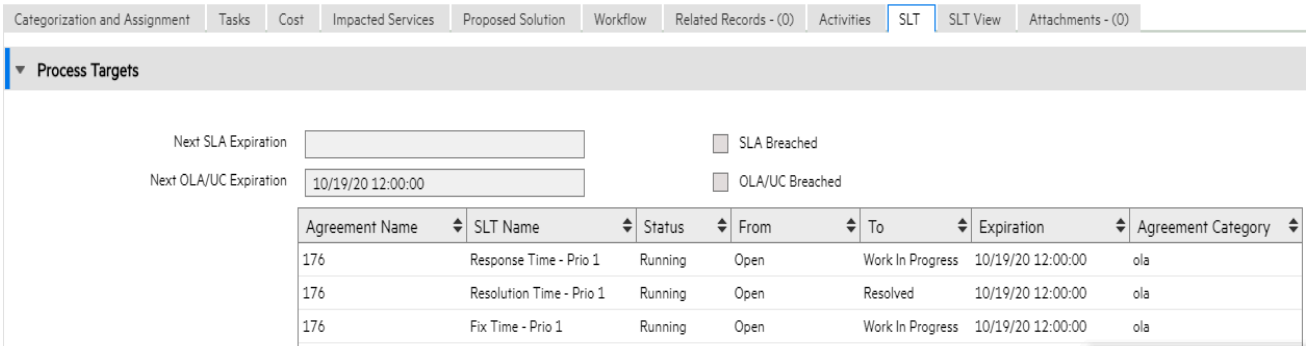
Create an Incident record that the OLA can apply. This example uses the Incident record IM10421 with the following specific values and you need to fill other fields as required.

- Assignment Group: E-mail / Webmail (Africa)
- Assignee: Incident.Coordinator

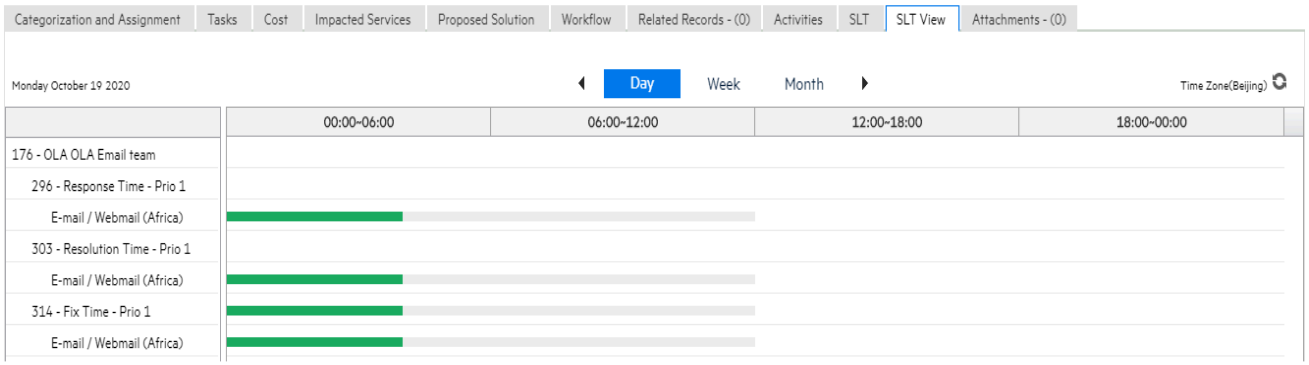
View the indicators' status change

Green

Refresh the record list and open IM10421 four hours later. Click the **SLT** tab to view the status of the SLTs.

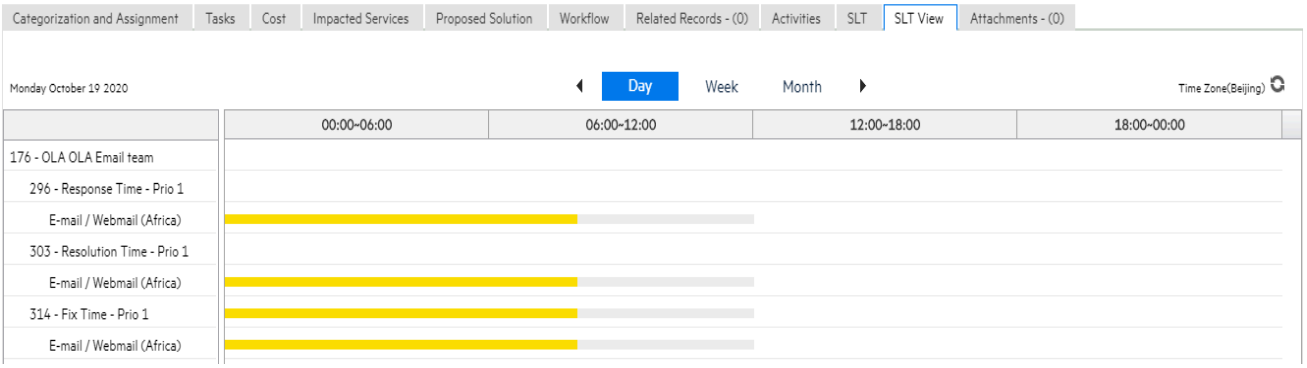


Next, click the **SLT View** tab to view the status of the SLTs. All related OLA, SLT, and assignment groups that are working on the OLA are displayed hierarchically. The color indicator for SLT 296, 303, and 314 are green. Note that SLT 314 has a progress bar because the duration scope of this SLT is **Total time for all assignment groups**.



Yellow

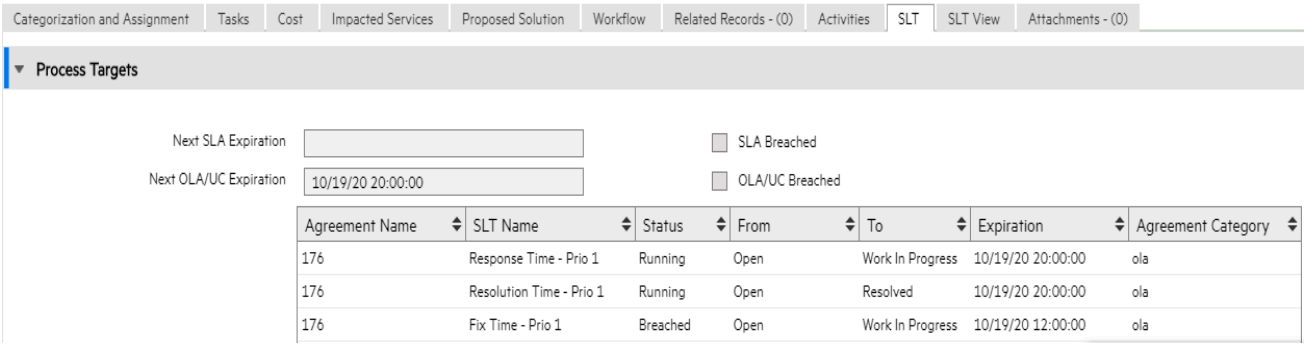
Refresh the record list and check the ticket's status after 4 hours.



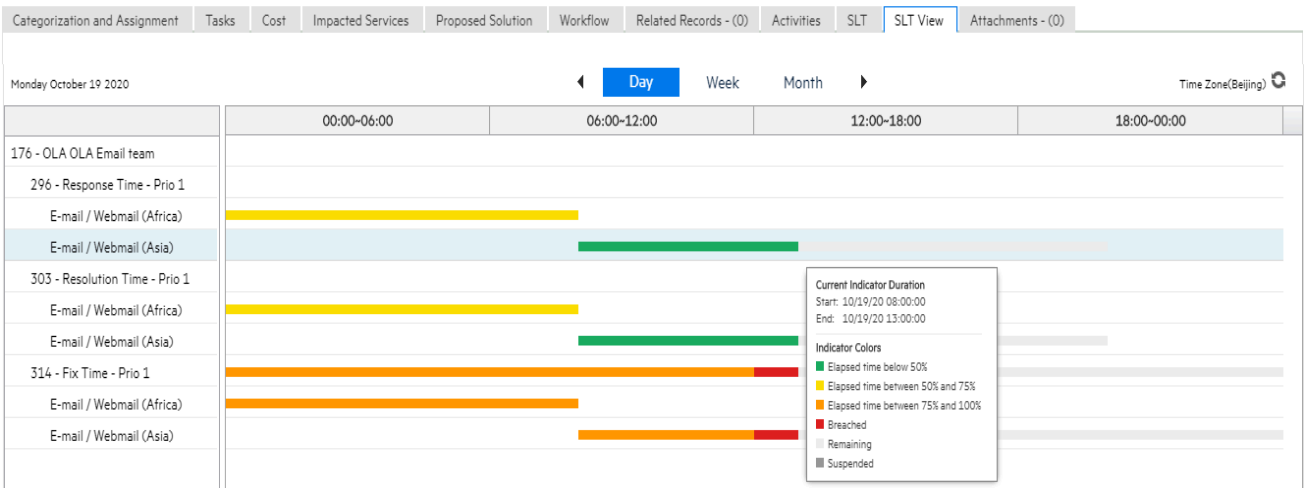
Update the ticket's assignment group to **E-mail / Webmail (Asia)**, and then save your changes.

Green again

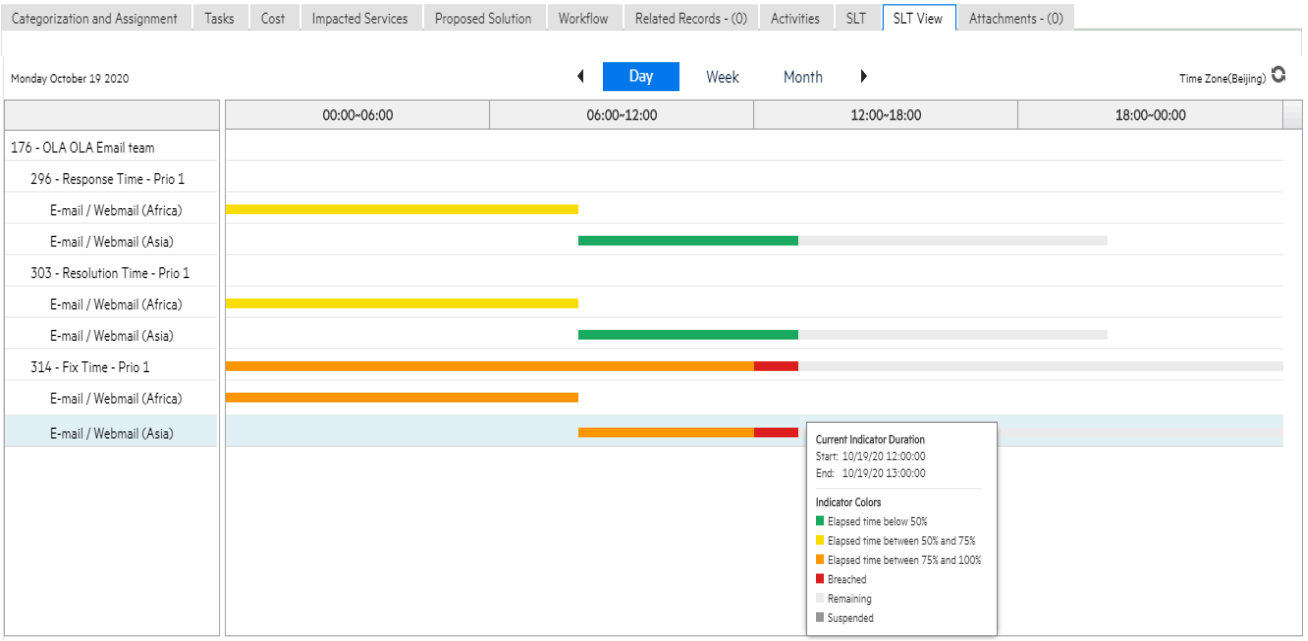
Refresh the record list and check the ticket's status after 4 hours. Click the **SLT** tab to view the status of the SLTs.



Next, click the **SLT View** tab to view the status of the SLTs. Now the new assignment group E-mail / Webmail (Asia) appears on this tab. Hover your mouse over the green indicator for the **E-mail / Webmail (Africa)** assignment group under SLT 296, the time information on the tooltip may resemble the following. The Start time is when you update the ticket's assignment group, and the End time is the actual time you refresh the SLT View tab and display this tooltip.

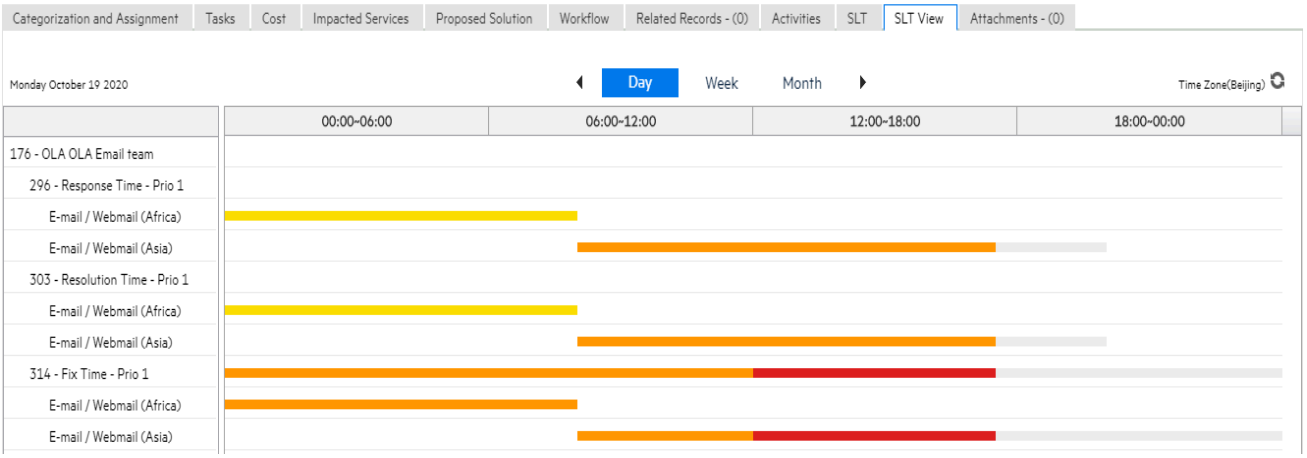


Hover your mouse over the red indicator for the **E-mail / Webmail (Africa)** assignment group under SLT 314, the time information on the tooltip may resemble the following. The Start time is the SLT breach time, and the End time is the actual time you refresh the SLT View tab and display this tooltip.



Orange

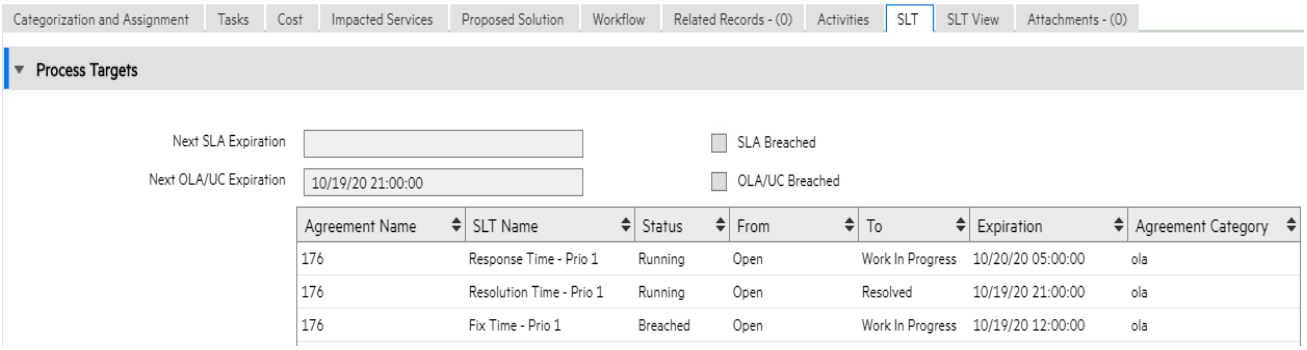
Refresh the record list and check the ticket's status after 5 hours. Click the **SLT View** tab to view the status of the SLTs.



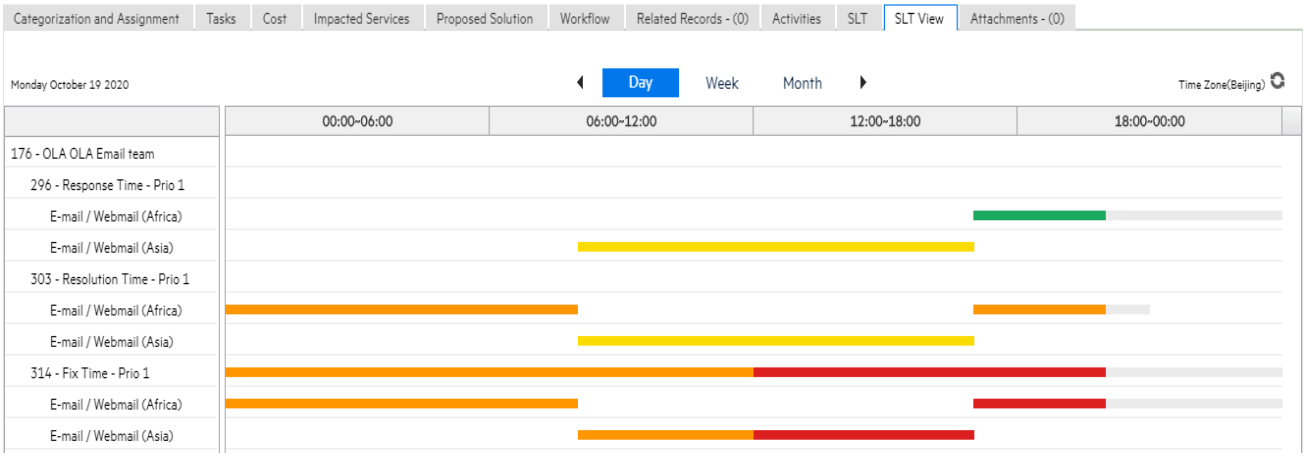
Update the ticket's assignment group to **E-mail / Webmail (Africa)** again, and then save your changes.

Green again

Refresh the record list and check the ticket's status after 3 hours. Click the **SLT** tab to view the status of the SLTs. The system recalculates the expiration time for both SLT 296 and SLT 303 due to assignment group changes.

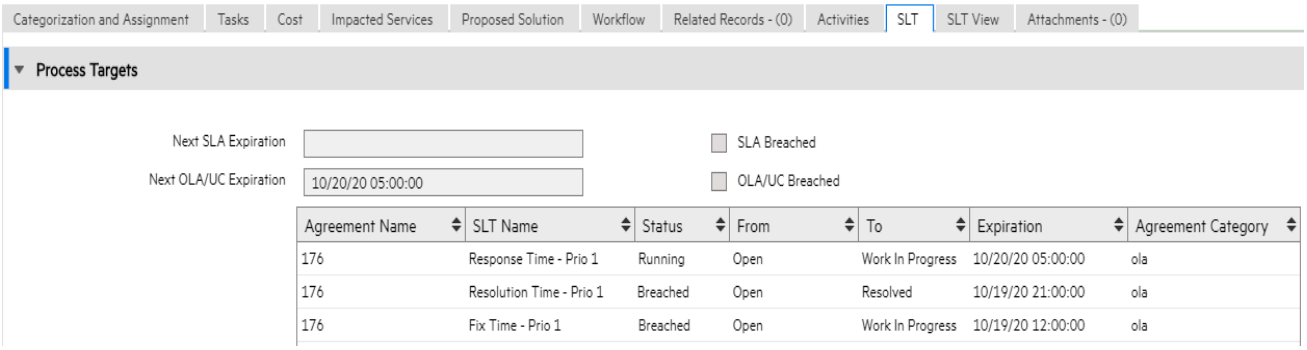


Next, click the **SLT View** tab to view the status of the SLTs. Note that the progress bar for the **E-mail / Webmail (Africa)** assignment group under SLT 303 now has a gap due to the group changes.

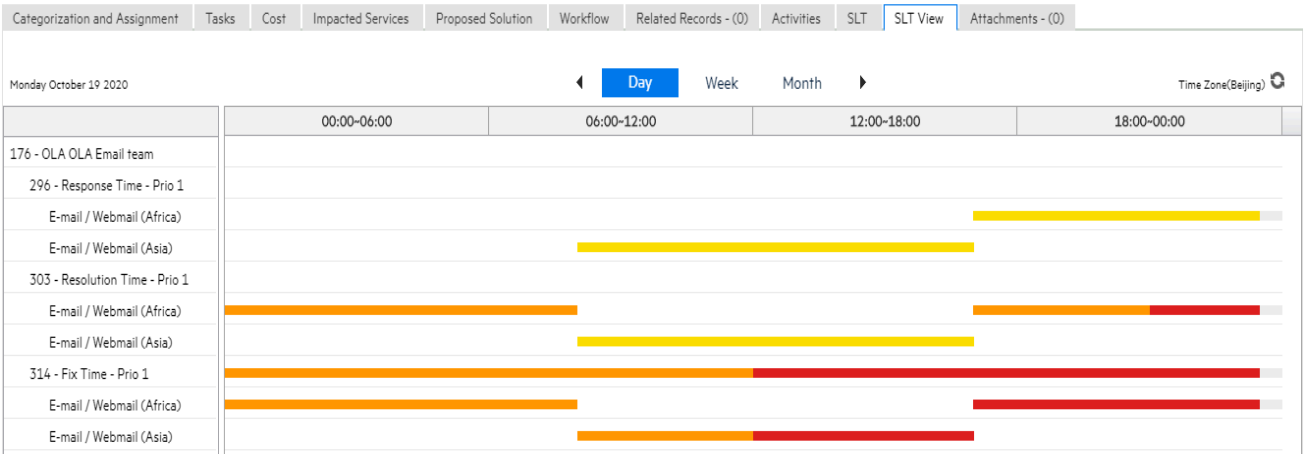


Yellow again

Refresh the record list and check the ticket's status after 3.5 hours.



Next, click the **SLT View** tab to view the status of the SLTs.


Categorization and Assignment
Tasks
Cost
Impacted Services
Proposed Solution
Workflow
Related Records - (0)
Activities
SLT
SLT View
Attachments - (0)
Monday October 19 2020
Day
Week
Month
Time Zone(Beijing)
00:00-06:00
06:00-12:00
12:00-18:00
18:00-00:00
176 - OLA OLA Email team
296 - Response Time - Prio 1
E-mail / Webmail (Africa)
E-mail / Webmail (Asia)
303 - Resolution Time - Prio 1
E-mail / Webmail (Africa)
E-mail / Webmail (Asia)
314 - Fix Time - Prio 1
E-mail / Webmail (Africa)
E-mail / Webmail (Asia)

1.14.3.35. Example: How to make Service Catalog Item Expiration work

To make the Service Catalog Item Expiration calculation method work, you need to add or modify a set of configurations. By doing this, after a user has ordered the catalog item and the Interaction record created is approved, a record is related to the Interaction, and the Service catalog item request expiration time is successfully attached to the Interaction and to its related records.

The following tasks describe example steps to configure Service Manager to achieve this goal.

Note

As an example, the following tasks use the "3 Days" delivery target and the default SLA named "Base Monitoring SLA for IT services" in the out-of-box Service Manager system.

Task 1: Allow request users to access the delivery target

In this task, you will allow request users to access a delivery target that you want to use for the SLTs.

To do this, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Click **Service Catalog** > **Administration** > **Delivery Targets**.
3. Click **Search**, and then select the delivery target to be used. In this example, select **3 Days**.
4. Set request users valid SLAs in the **Limited access to users with these Customer SLAs** field, or clear the value so that the target becomes valid for all users.
5. Click **Save** and **OK**.

Task 2: Associate updated Delivery Target to a Catalog Item

In this example, you will associate the delivery target that you updated previously to a catalog item so that users can select the delivery target when ordering the catalog

item by using the **Order from catalog** menu option.

To do this, follow these steps:

1. Click **Service Catalog** > **Administration** > **Manage Items**.
2. Click **Search**, and then select the Catalog Item to which you want to associate the updated Delivery Target. In this example, select **PC Key Card**.
3. Click the **Delivery Targets** tab, and then click **Add**.
4. Select **3 Days** as the Delivery Target.
5. Click **Save** and **OK**.

Task 3: Locate the default SLA

To do this, follow these steps:

1. Click **Service Level Management** > **Administration** > **Edit Control Record**.
2. Make a note of the value in the **Default SLA** field. In the out-of-box system, the value is Base Monitoring SLA for IT services.

Task 4: Add a Service Catalog SLT to the default SLA

To do this, follow these steps:

1. Click **Service Level Management** > **Agreements** > **Search Agreements**.
2. Open the default SLA record by performing a search.

   In this example, it is Base Monitoring SLA for IT services and the Agreement ID of this agreement is 168.
3. Click the **Process Target** tab of the default SLA, and then click **Add SLT(s)**.
4. In the Create Process Target window, select **Service Catalog** in the **Service Area** drop-down list. Click **Next**.
5. Type a name (for example, SLT For Service Catalog) in the **SLT Name** field, and then click **Next**.
6. Set **Schedule** to **24×7**, and then select a time zone. Click **Next**.
7. Select **3 Days** in the **Delivery** drop-down list so that this SLT will work for all Service Catalog requests for a catalog item to be delivered within 3 days. Click **Next**.
8. Provide a value for the following fields:

- Initial State: Select a value (for example, **Requested**)
- Final State: Select a value, for example, **Release verification**
- Duration Type: select **Interval**
- Duration: enter **3 00:00:00** (The Duration value must exactly match the delivery target that you selected in previous steps. In this example, 3 days)

Click **Next**.

9. Leave the **Alerts** field and the **Suspend processing** field empty. Click **Finish**.
10. Click **Save**.

Note

- Make sure the **Duration** value in *step 8* is the one that you want to monitor the Delivery Targets. This value must be the same value that you selected in the **Delivery** drop-down list in *step 7*.
- The newly added SLT will work for all Service Catalog Item requests with a **Deliver within** value of **3 Days**.
- To monitor other Delivery Target requests, you can add more SLTs of this kind.

Task 5: Add an Interaction SLT to the default SLA

In this task, you will add an Interaction SLT to the default SLA. This SLT will work for Interactions created for relevant Service Catalog item requests.

To do this, follow these steps:

1. In the Base Monitoring SLA for IT services SLA record, click the **Process Target** tab, and then click **Add SLT(s)**.
2. In the Create Process Target window, select **Service Desk Interactions** in the **Service Area** drop-down list. Click **Next**.
3. Type a name (for example, Interaction SLT for Service Catalog) in the **SLT Name** field, and then click **Next**.
4. Set **Schedule** to **24×7**, and then select a time zone. Click **Next**.
5. Skip the filters so that the SLT is always triggered. Click **Next**.
6. Provide a value for the following fields:
   - Initial State: Select a value (for example, **Open**)
   - Final State: Select a value (for example, **Closed**)

- Duration Type: select **Calculation**
- Duration: select **Service Catalog Item Expiration**

Click **Next**.

7. Leave the **Alerts** field and the **Suspend processing** field empty. Click **Finish**.
8. Click **Save**.

Task 6: Add a Request SLT to the default SLA

In this task, you will add a Request SLT to the default SLA. This SLT will work for Request records that are related to the Interactions created from a Service Catalog item request.

To do this, follow these steps:

1. In the Base Monitoring SLA for IT services SLA record, click the **Process Target** tab, and then click **Add SLT(s)**.
2. In the Create Process Target window, select **Request** in the **Service Area** drop-down list. Click **Next**.
3. Type a name (for example, Request SLT for Service Catalog) in the **SLT Name** field, and then click **Next**.
4. Set **Schedule** to **24×7**, and then select a time zone. Click **Next**.
5. Provide a value for the following fields:

   - Initial State: Select a value (for example, **Open**)
   - Final State: Select a value (for example, **Closed**)
   - Duration Type: select **Calculation**
   - Duration: select **Service Catalog Item Expiration**

   Click **Next**.

6. Leave the **Alerts** field and the **Suspend processing** field empty. Click **Finish**.
7. Click **Save**.

Task 7: Add a Request Task SLT to the default SLA

To do this, follow these steps:

1. In the Base Monitoring SLA for IT services SLA record, click the **Process Target** tab, and then click **Add SLT(s)**.
2. In the Create Process Target window, select **Request Tasks** in the **Service Area** drop-down list. Click **Next**.

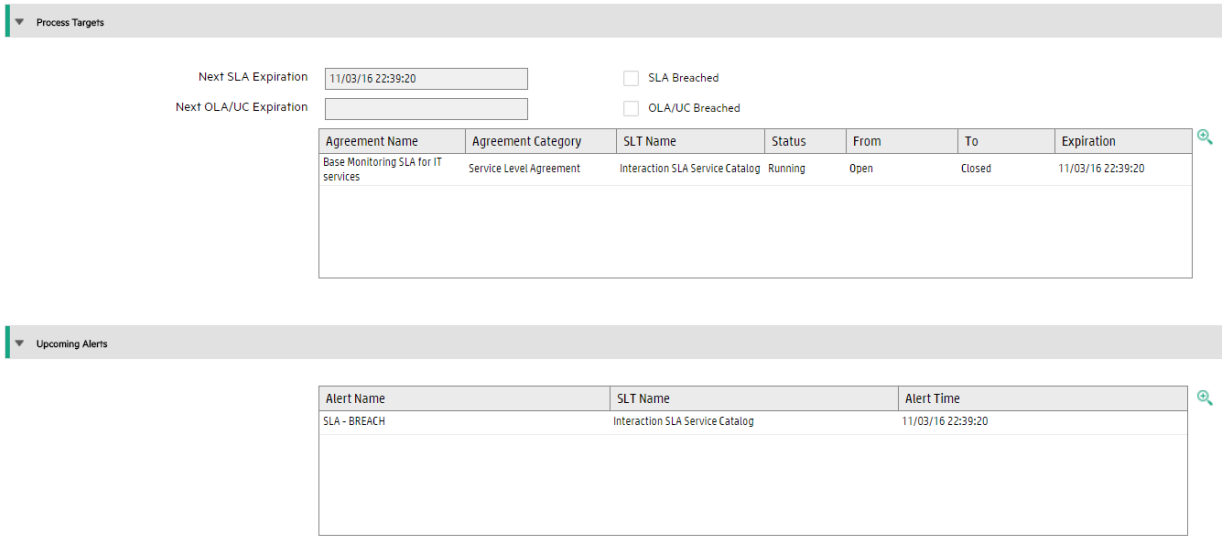
3. Type Request Task SLA Service Catalog in the **SLT Name** field, and then click **Next**.
4. Set **Schedule** to **24×7**, and then select a time zone. Click **Next**.
5. Provide a value for the following fields:
   - Initial State: Select a value (for example, **Open**)
   - Final State: Select a value (for example, **Closed**)
   - Duration Type: select **Calculation**
   - Duration: select **Service Catalog Item Expiration**

   Click **Next**.
6. Leave the **Alerts** field and the **Suspend processing** field empty. Click **Finish**.
7. Click **Save**.

Task 8: Verify the Service Level Targets are working

You can use this sample task to verify that the SLTs are working. The following steps use the Interaction SLT as an example.

1. Click **Service Catalog** > **Order from Catalog**.
2. Search for **PC Key Card**, and click the link.
3. Select **3 Days** from the **Deliver within** drop-down list, and then click **Add to cart**.
4. Click **View Cart/Checkout**, and then click **Submit Request**.
5. Fill in the mandatory **Purpose** field, select an urgency type, and then click **Submit**. The system automatically creates an Interaction.
6. Sign out and then sign in to the system as a Service Desk Approver, and approve the newly created Interaction.
7. Sign out and then sign in to the system as a System Administrator again.
8. Click **Service Desk** > **Search Interactions**, and then search for the approved Interaction.
9. Click the **SLT** tab to verify the SLA calculation of Service Catalog Item Expiration works for this Interaction. See the following screenshot for an example:


Process Targets
Next SLA Expiration
11/03/16 22:39:20
SLA Breached
Next OLA/UC Expiration
OLA/UC Breached
Agreement Name
Agreement Category
SLT Name
Status
From
To
Expiration
Base Monitoring SLA for IT services
Service Level Agreement
Interaction SLA Service Catalog
Running
Open
Closed
11/03/16 22:39:20
Upcoming Alerts
Alert Name
SLT Name
Alert Time
SLA - BREACH
Interaction SLA Service Catalog
11/03/16 22:39:20

1.14.4. Service Level Management security

Service Level Management security areas

The security areas for Service Level Management are Service Level Management and Service Level Management Configuration. These areas contain the default security rights and settings for the Service Level Management module. The security right settings will be inherited by any new roles created in an area when no other settings are specified for that security role.

These security areas are used to set permissions to operators to provide access to particular area of Service Level Management. The following table lists the areas and the relevant Service Level Management menu items the operators can access.

Area
Service Level Management
Service Level Management Configuration

Default rights

The default rights defined in areas will be inherited when you create new security roles. The following table shows the out-of-box default rights defined in the Service Level Management or Service Level Management Configuration areas.

Area Name	View	New	Update	Delete/ Close	Expert	Admin
Service Level Management	TRUE	FALSE	Never	Never	FALSE	FALSE
Service Level Management Configuration	FALSE	FALSE	Never	Never	FALSE	FALSE

Default settings

The default settings defined in areas will be inherited when you create new security roles. In an out-of-box system, none of the default settings is checked or set in the Service Level Management or Service Level Management Configuration areas.

Service Level Management security roles and settings

The out-of-box security roles for the Incident module include the following:

- SLM Manager

Mapping between previous security profiles and current PD security roles

The following table lists the mapping relationship between previous Service Level Management security profiles and current PD security roles in the Service Level Management module.

Security Profile	Security Role/Area
DEFAULT	DEFAULT/Service Level Management
administrator	administrator/Service Level Management
sysadmin	sysadmin/Service Level Management

Field mapping between security profiles and PD security rights/settings

Note

In the following two tables, AND and OR relate to whether a mapped security role has the corresponding right. For example, if any of the Service Level Agreement, Response Objectives, or Availability Objective has the "New" security profile setting, then the Process Designer security rights and settings will also be "New."

For Service Level Agreements, Response Objectives and Availability Objectives

Security profile settings		**Process Designer security rights and settings**
New	OR	New
Delete	OR	Delete/Close
Update	OR	Update
View	OR	View
Allowed Statuses	AND	Allowed Statuses

SLT Catalog Records

Security profile settings		**Process Designer security rights and settings**
New Delete Update View	OR	Manage SLT Catalog Records

Settings:

Security profile settings	Process Designer security rights and settings
View Monthly Aggregate Results	View Monthly Aggregate Results
View Monthly Response Time Results	View Monthly Process Time Results
View Response Detail Records	View Process Detail Records
View SLA Active Records	View SLM Active Records
View Monthly Availability Results	View Monthly Service Results
View Outage Detail Records	View Outage Detail Records
View Outage Event Records	View Outage Event Records

Out-of-box role rights

Based on the mapping rules, the rights and settings in previous security profiles are mapped to the rights and settings in the Service Level Management area specified in the corresponding security roles. See the table below for the out-of-box security rights in the Service Level Management and Service Level Management Configuration areas. This table only lists the new security roles that have different settings with the default rights.

Area Name	Role Name	View	New	Update	Delete/Close	Modify Template	Expert	Admin
Service Level Management	SLM Manager	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
	System Administrator	TRUE	TRUE	Always	Always	FALSE	TRUE	TRUE
Service Level Management Configuration	SLM Manager	TRUE	TRUE	Always	Always	TRUE	TRUE	TRUE
	System Administrator	TRUE	TRUE	Always	Always	FALSE	TRUE	TRUE

Contents

- 1 Service Level Management security areas
- 2 Service Level Management security roles and settings

1.15. Smart Analytics

Built on Service Manager (SM) and using an OEM-licensed version of IDOL, SM Smart Analytics heralds the debut of the "Big Data" edition of Service Manager. This powerful SM-IDOL integration drives automation further into ITSM processes by mining unstructured data and by extracting information from different types of data. Smart Analytics in the current release focuses on the following:

- Improving the processes of Help Desk management
- Reducing the time and effort expended on interaction submittals by end users and IT professionals
- Accelerating the process of problem management
- Improving search experience across a variety of internal and external content

Smart Analytics enables your Service Manager to become a more intelligent and efficient system by extracting and understanding your content. In this release, Smart Analytics includes the following features:

Smart Ticket

With Smart Ticket, you can quickly submit a Service Desk record by just entering a description or attaching a picture. Smart Analytics intelligently populates other fields, such as category or affected services, by extracting and analyzing the content that you entered in the record.

Hot Topic Analytics

Hot Topic Analytics intelligently displays an interactive diagram that indicates the hot topics among recent interactions, incidents, problems, or surveys. Hot Topic Analytics enables you to easily discover common issues, identify escalation candidates, and create new records for escalation based on the selected candidates.

Smart Search

Smart Search enables you to search across a variety of content, including Service Manager records (such as Incidents and Changes), SharePoint documents, static web pages, and KM documents. You can integrate multiple knowledge libraries by configuring different search connectors so that Service Manager users can search all the information that they can access.

Note

In Service Manager, Smart Email and Virtual Agent provide intelligent solutions using these Smart Analytics features.

Related topics

Install Smart Analytics
Configure Smart Analytics
Administer Smart Analytics
Smart Analytics best practices
Smart Analytics tailoring
Smart Analytics APIs

1.15.1. Use Smart Search as a general search tool

Smart Search supports cross-module search from selectable sources inside and outside of Service Manager. It can also perform fuzzy search, and use advanced filters. Smart Search used as a general search tool is supported for self-service users, mobility users, and Service Request Catalog users.

Search capabilities

Input - search keywords

Scenario	Sample or explanation
None of these words	NOT email
Any of these words	email OR access
All of these words	Email AND ACCESS
Location between these words	email BEFORE access email AFTER access email NEAR*N* access **Note:** This means that the term **access** is within *N* words from the term **email**. For example, **email NEAR1 access** returns the results in which the term **email** and the term **access** are adjacent.
This exact phrase	"email access"
Wildcard	*email*, email?
Punctuation	\\, \?, *
Free-text search/exact match search	Email/status = "true"

Output - Search accuracy

Scenario	Sample or explanation
Configurable weight to determine the ranking of search result.	Relevance scores for matches in certain fields over matches in other less important fields
Adaptive learning if use as solution	
Configurable Predefined condition for suggested solution based on context-aware search	

Output - search list

Summary of search result plus the Customizable hitlist fields will be showed in search main page

Search Utility

Scenario	Sample or explanation
Fuzzy Search	Search Emal, return result for correct typing: Email
Support language-specific word stemming	Search break, return broken and breaks as well
Attachment search	Separately search attachment
Global filter for all libraries	Last modify time, status, assignment, affected services
Facet oriented search per library	Configurable, all of matched field can be treated as facet to narrow down the search result
Store last search condition for next usage per user	
Check all libraries in one click	
Clear all conditions in one click	
Show count per each value for search result.	
Language detection	Auto detect language according to the search key words
Allow index duplicated fields for different search purpose, "Multivalue=true"	

Use Smart Search

To use Smart Search, follow these steps:

1. Click the **Smart Search** icon on the top-right corner of the Service Manager UI to display the search box.

2. Enter the desired search terms in the search box, and then press Enter. For example, type IM10005 in the Smart Search box, and then press Enter.

   The Smart Search window opens and a list of documents that match your search terms is displayed. If no result is displayed, type a new search string for your search or use Advanced Search to specify additional search criteria.

   Note

   - You need to set up and verify the server connectivity for multiple servers and connectors. For detailed information to configure and monitor the connectors and servers, see Configure connectors.
   - Automatic Smart Search is triggered only if you set **Enable instant search** in the Smart Search Tailoring page to true. For more information, see Customize context-aware search.

3. Optionally, configure the following additional search conditions to narrow the Smart Search results, and then click **Search**.

Field	Description
Search within results	This option is only activated when there are results returned. Click this option button to add it as a filter to the filter panel on the right. You can click the remove button on the filter to remove it. For example, type Service in the search box and then click **Enter**. After the results are returned, you click this option and a **Previous Search: Service** filter is added to the search results panel. Then, type **Server** in the search box and then click **Enter**. The results that are returned are the same as those returned if you search for **Service AND Server**.
General filter	Click the **Filter** button to cascade the general filter to filter the content by the last modified time. The option you select is added to the filter panel on the right. You can click **Clear All** to remove all the filters.
Library	Specifies the knowledgebases for your search repository.
All	Click the checkbox to select all the knowledgebases as your search repository.

Field	Description
<Knowledgebase name>	Click the checkbox to include this knowledgebase to your search repository. **Note** Your library options in Smart Search are remembered for your next login.

4. Click a document title or identifier to open it.

In the search results, if you click the button to vote up or vote down an article, this will affect the ranking of the search result next time. This is not configurable. Also, if the document is re-used for a solution, the ranking for this document is also affected. For detailed information, see Use adaptive learning in Smart Search.

1.15.2. Use Smart Search to find a solution

At any time while you create or update an Interaction or Incident record or update a Problem record, you can perform a smart search to find solutions. When you find a solution, you can add the solution to the resolution or workaround field of the record you are processing.

Note

To remove the legacy search icon, you need to edit the form to remove the legacy icon manually.

To use Smart Search to find a solution, follow these steps:

1. By default, while you are submitting a record, context-aware search is automatically triggered as you type in the **Description** field. If you have configured your Smart Analytics to manually trigger context-aware search, click the **Similar Document** icon to open the quick Smart Search window on the right panel. Smart Search uses the content in the **Description** field of this record as the search string.

   Tip

   You can also put the cursor in the **Description** field, and press **Shift** + **Enter** to trigger a context-aware search.

2. In the search results, click the hyperlink to view a solution document.
3. When there's a document with the correct solution or workaround, the **Use Solution** button appears, and you can click **Use Solution** to add the solution to the **Resolution** or **Workaround** field of the record. For KM articles, the button appears regardless of whether there's a solution or workaround available.
4. After you select **Use Solution**, the system displays the record with the **Resolution** or **Workaround** field updated. You can continue processing the Interaction, Incident, or Problem record that prompted your quick Smart Search.

1.15.3. Use adaptive learning in Smart Search

Smart Search adaptive learning is enabled by default to collect words or specific phrases used to search a knowledgebase for those documents that are considered useful or can be performed with actions.

Smart Search maintains a count which specifies the number of times search words or phrases resulted in finding a useful document (whether marked as useful or used as a solution). Each time a user performs a search using the same words or phrases, the count is incremented or a new entry is created when the search returns useful results. Also, each time an operator clicks the **Use Solution** button, the phrase used by the operator to search for the solution is saved or the count is incremented for the phrase for the adaptive learning record associated with the solution.

Note

For a working copy of a document that has no published version, the system collects the words and phrases for adaptive learning. For a working copy of a document that already has a published version, the system doesn't collect the words and phrases for adaptive learning.

Example: If **monitor** or **blue monitor** is used in a document that resolves an incident and is used as a solution or performed with other actions, the phrase **blue monitor** or the word **monitor** is appended to the adaptive learning terms for that document and the count is incremented. For additional searches, a document with one **blue monitor** in it will come up with a reasonable relevancy. You can also change the count of **monitor** to 15, for example, to make this document appear higher on the search results list than another document with the word **monitor** in it.

At the same time when an operator clicks the **Use Solution** button, the system creates or increments the associated AuTNRANK value of this record. When an operator performs a search by using advanced search and then clicks the **Vote** button to mark a document as useful, the system also creates or increments the associated AuTNRANK value of this record as follows:

- When the search results are filtered by "Any of these words," the system adds separate entries for each word in the search phrase.
- When the search results are filtered by "All of these words" or "This exact phrase," the system creates a single entry for all the words in the search phrase.

The AuTNRANK value becomes a weighted value so that when another user performs a different search using the same words or phrase, documents containing the same words or phrase are returned higher on the hit list in the next set of search results.

Note

If you perform a full re-index of a knowledgebase, all the AuTNRANK values are removed from the library.

1.15.4. Create a Smart Ticket in ESS

User Role: Self-service Users

If you have installed and enabled Smart Analytics, a new menu, **Submit a Smart Request**, is automatically added to leverage the power of the Smart Ticket feature. Clicking it opens a new, simplified request form that only requires you to add attachments or comments to submit a request, which simplifies the process of submitting the ESS support requests.

To submit a self-service request by using Smart Ticket in self-service user view (ess.do), follow these steps:

1. Sign in to Service Manager from the ESS portal.
2. Click **Submit a Smart Request**.
3. Click **Add Files** to attach an image file. For example, a screenshot of the error message.

   > **Note**
   >
   > For image files containing multiple languages, Service Manager provides a language list according to the language priorities. The following restrictions apply to the language list:
   >
   > - Arabic can't be combined with any other alphabets or languages.
   > - Hebrew can't be combined with either Japanese, Simplified Chinese, or Traditional Chinese.

4. Type comments for your request.
5. Click **Submit**.

   An Interaction is now created. The fields defined in Smart Ticket configuration are automatically filled by SM Smart Analytics.

1.15.5. Create a Smart Ticket in Mobility client

User Role: Self-service Users

To submit a self-service request in the Mobility client, follow these steps:

1. Sign in to Service Manager on your mobile device.
2. Click **Submit a Smart Request** on the main menu.
3. Add an attachment. For example, a screenshot of the error message.
4. Type comments for your request.
5. Click **Submit**.

   An Interaction is now created. The fields defined in Smart Ticket configuration are automatically filled by SM Smart Analytics.

1.15.6. Create a Smart Ticket in power user view

User Role: Service Desk Agent

To submit a request on behalf of a user in the power user view (`index.do`), follow these steps:

1. Sign in to Service Manager.
2. Click **Service Desk** > **Create Smart Interaction**.
3. Fill in the name of the contact.
4. Type a description of the issue.
5. Click **Smart Classification**.

   The fields such as **Category** and **Affected Service** are intelligently populated with the most likely values based on the analysis by SM Smart Analytics. Meanwhile, SM Smart Analytics also suggests some other possible values for you to choose from.

   > **Note**
   >
   > If you're not satisfied with the values suggested by SM Smart Analytics, you can click the **Fill Field** icon to manually choose a value for each field.

6. Click **Continue**. Service Manager displays the full interaction form and populates the corresponding fields with the values that you specified in the previous step.
7. Complete the interaction with additional information if needed, and then proceed with your record.

1.15.7. Use Hot Topic Analytics to create an incident, problem, change, or interaction

User Role: Incident Manager, Problem Manager, Change Manager

You can identify Incident, Problem, or Change candidates at ease based on the hot areas automatically suggested by Hot Topic Analytics.

> **Note**
>
> You must use the web client instead of the Windows client to view the dynamic topic map in Hot Topic Analytics.

To use Hot Topic Analytics to find Incident, Problem, or Change candidates, follow these steps:

1. Sign in to Service Manager from the web client.
2. Do one of the followings to access Hot Topic Analytics for your module:
   - Click **Service Desk** > **Hot Topic Analytics**. On the Hot Topic Analytics for Interactions screen, you can analyze the interactions that are categorized for the suggested hot topics, and then create new incidents based on your selected interactions.

     > **Note**
     >
     > If Process Designer and the streamlined interaction solution are enabled for the Service Desk module, it's advisable that you remove the **Create Incident** action from the Hot Topic Analytics for Interactions screen to accommodate the streamlined interaction solution. For information on tailoring the actions, see Customize actions in hot topic map.

   - Click **Incident Management** > **Hot Topic Analytics**. On the Hot Topic Analytics for Incidents screen, you can analyze the incidents that are categorized for the suggested hot topics, and then create new problems or changes based on your selected incidents.

- Click **Problem Management** > **Hot Topic Analytics**. On the Hot Topic Analytics for Problems screen, you can analyze the problems that are categorized for the suggested hot topics, and then create new changes based on your selected problems.
- Click **Survey Management** > **Hot Topic Analytics**. On the Hot Topic Analytics for Surveys screen, you can analyze the surveys that are categorized for the suggested hot topics, and then create new interactions or list surveys based on your selected surveys.

3. View the hot topics suggested by Hot Topic Analytics.
   - The size of a topic indicates the heat of the topic. The background color of a topic serves to identify the topic.
   - You can click a hot topic to drill down to the subtopics.

     Note

     In the out-of-the-box system, for the Hot Topic Analytics for incidents, the last level of the topic map is the affected service.

   - The records that belong to a topic are displayed in the list on the right.
4. If you want to run a custom analysis, enter your keywords, and then click **Find Hot Topics**.
5. If you want to further refine the results, click **Advanced Filter**, specify your filters, and then click **Find Hot Topics** again. Service Manager refreshes the graphic with the filtered results.

   Tip

   If you want to build complex queries based on your specific needs, you can click **Edit Query** to use Query Editor to define your queries. When you define your queries, be sure to use only the fields that are specified in the **Properties Fields** section of the Hot Topic Analytics configuration. Otherwise, your queries willn't take effect because the fields that you use aren't included in the analysis of Hot Topic Analytics.

6. Review the filtered records that are listed to the right of the graphic to identify the candidates for escalation. You can click the record ID to view the record detail.

7. After you identify the candidates, select the checkboxes before the record IDs, and then click the corresponding button (such as **Create Problem**). Service Manager creates a form for the new record and populates some fields based on the selected candidates.

   > **Note**
   >
   > For the Hot Topic Analytics for Incidents, you need to select **Create Problem** or **Create Change/Article** from the **Analytics Action** drop-down list so that Service Manager can display the **Create Problem** or **Create Change** button when you select the candidates.

8. Update the form as needed, and then click **Save** to create the new record.

   When the new record gets created, the selected candidates are listed in the **Related Records** section of the new record.

1.15.8. Use Hot Topic Analytics to create a KM article

User Role: Service Desk agents, Incident Manager, Problem Manager

You can create KM articles for interactions, incidents, or problems based on the hot topics automatically suggested by Hot Topic Analytics.

> **Note**
>
> You must use the web client instead of the Windows client to view the dynamic topic map in Hot Topic Analytics.

To use Hot Topic Analytics to create a KM article, follow these steps:

1. Sign in to Service Manager from the web client.
2. Do one of the following to access Hot Topic Analytics:
   - Click **Service Desk** > **Hot Topic Analytics**.
   - Click **Incident Management** > **Hot Topic Analytics**.
   - Click **Problem Management** > **Hot Topic Analytics**.

   The Hot Topic Analytics for the specific module opens.
3. View the hot topics suggested by Hot Topic Analytics.
   - The size of a topic indicates the heat of the topic. The background color of a topic serves to identify the topic.
   - You can click a hot topic to drill down to the subtopics.

     > **Note**
     >
     > In the out-of-the-box system, for the Hot Topic Analytics for incidents, the last level of the topic map is the affected service.

   - The records that belong to a topic are displayed in the list on the right.
4. If you want to run a custom analysis, enter your keywords, and then click **Find Hot Topics**.

5. If you want to further refine the results, click **Advanced Filters**, specify your filters, and then click **Find Hot Topics** again. The graphic refreshes with the filtered results.
6. Review the categorized records as listed to the right of the graphic to identify the candidates for creating the new KM article. You can click the record ID to view the record details.
7. After you identify the candidates, select the checkboxes before the record IDs, and then click the **Create Article** button.

   Note

   For the Hot Topic Analytics for Incidents, make sure that you select **Create Change/Article** in the **Analytics Action** drop-down list so that the system displays the **Create Article** button on top of the record list.

8. Select a document type. Service Manager displays the New Knowledge Document form with the information in your selected records copied.
9. Enter information in the mandatory fields (such as **Title**) and other optional fields if necessary.
10. Click **Submit**. Service Manager creates the new KM record.

1.15.9. Use Hot Topic Analytics to set incident parent

User Role: Incident Analyst

By using Hot Topic Analytics, you can identify similar incidents at ease, and then group them by setting the parent incident.

To use Hot Topic Analytics to set the incident parent, follow these steps:

1. Sign in to Service Manager from the web client.

   > **Note**
   >
   > You must use the web client instead of the Windows client to view the dynamic topic map in Hot Topic Analytics.

2. Click **Incident Management** > **Hot Topic Analytics**. Service Manager displays the Hot Topic Analytics for Incidents screen.
3. From the **Analytics Action** drop-down list, select **Set Parent**. Service Manager refreshes the hot topic map.

   > **Note**
   >
   > In the out-of-the-box system, the hot topic analysis includes only the incidents that are neither the parent or child incidents. The administrator can set particular queries for different actions in **Analytics Action** from the **Advanced** tab in the Hot Topic Analytics configuration. For more information, see Configure Hot Topic Analytics.

4. View the hot topics suggested by Hot Topic Analytics.
   - The size of a topic indicates the heat of the topic. The background color of a topic serves to identify the topic.
   - You can click a hot topic to drill down to the subtopics.
   - The records that belong to a topic are displayed in the list on the right. You can click the record ID to view the record detail.
5. If you want to run a custom analysis, enter your keywords, and then click **Find Hot Topics**.

6. If you want to further refine the results, click **Advanced Filters**, specify your filters, and then click **Find Hot Topics** again. Service Manager refreshes the graphic with the filtered results.

Tip

If you want to build complex queries based on your specific needs, you can click **Edit Query** to use Query Editor to define your queries. When you define your queries, be sure to use only the fields that are specified in the **Properties Fields** section of the Hot Topic Analytics configuration. Otherwise, your queries won't take effect because the fields that you use aren't included in the analysis of Hot Topic Analytics.

7. After you identify a group of incidents for which you want to set the parent/child relationship, select the checkboxes before the record IDs, and then click the **Set Parent** button. Service Manager displays the Set Parent screen.
8. From the **Link to Parent Incident** drop-down list, select the parent incident. The system will suggest the parent incident (the status of the incident isn't closed) based on the similarity among incidents.
9. From the Child Incidents list, select at least one child incident.
10. Click the **Proceed** button to set the Parent/Child relationship for the selected incidents.

Note

If you want to create a new incident as the parent instead of selecting one from the list, you can click the **New Parent Incident** button to create a new parent incident.

1.15.10. Access Hot Topic Analytics from SM reports

User Role: Service Desk Agent, Incident Manager, Problem Manager

When viewing SM reports for interactions, incidents, problems, and surveys, you can click a section in the report to drill down to the detailed record list, and then access Hot Topic Analytics for further analysis. Your selected filter condition in the report is automatically used in Hot Topic Analytics if the fields are indexed by Smart Analytics.

Note

The fields and queries that are used to generate reports may not be supported by the Hot Topic Analytics configuration. In this case, Hot Topic Analytics doesn't take effect and can't provide hot topic suggestions.

To access Hot Topic Analytics from SM reports, follow these steps:

1. Sign in to Service Manager from the web client.

   You must use the web client instead of the Windows client to view the dynamic topic map in Hot Topic Analytics.

2. Do one of the following to open the report dashboard for the Service Desk, Incident Management, Problem Management, or Survey Management module:

   • Click **Service Desk** > **Service Desk Overview**. The Service Desk Overview dashboard is displayed.

   • Click **Incident Management** > **Incident Overview**. The Incident Overview dashboard is displayed.

   • Click **Problem Management** > **Problem Overview**. The Problem Overview dashboard is displayed.

   • Click **Survey Management** > **Survey Overview**. The Survey dashboard is displayed.

3. Click a section in a report where you can drill down the report and view the detailed record list. For example, if you click the section indicating "High" priority from the "Incident Backlog by Priority" report, the incidents with high priority are listed in the record list. The record list is displayed.

4. From the record list, click the **Hot Topic Analytics** button or **More** > **Hot Topic Analytics**. The Hot Topic Analytics is displayed and the hot areas are suggested based on the specific filter that's used to drill down the report. For example, high-priority incidents.

   Note

   To use Hot Topic Analytics for Surveys, you need to create a KM knowledgebase library for survey. For more information, see Manage Smart Search Knowledgebases.

5. Review the hot topics suggested by Hot Topic Analytics, and then decide your further actions.

1.15.11. Add stop phrases for Hot Topic Analytics

Hot Topic Analytics provides many out-of-the-box stop words and the authorized Smart Analytics users can add some stop phrases to an analysis.

Follow these steps to add a word to the stop phrase list:

1. Open a Hot Topic Analytics form.
2. Click the **Add Stop Phrase** button. The system opens the **Hot Topic Analytics - Add stop phrase** form that contains all words in the current hot topic form.
3. Select the words to be added to the stop phrase list.
4. Click **Save** to add the selected words to the stop phrase list.

1.16. Service Manager Reports

The Service Manager Reports feature provides reports and dashboards to enable faster analysis and improved time to resolution. Service Manager reports organize data into various chart formats, and dashboards display one or more reports. These reports display relationships among categories of data. For example, one report might display the number of incidents per customer. Another report might display the number of incidents by priority. Viewing these reports together as a dashboard enables you to make better business decisions, such as assigning resources to close incidents.

Service Manager Reports aims to provide a light-weight reporting feature for active operational data and you can use the reports to retrieve, represent and visualize at most 100,000 active records out of millions. To define analytic reports against the entire dataset, use third-party business intelligence tools.

Reports provide a dynamic view of records, data, and relationships. Dashboards collect similar reports into an overview of data relationships. For example, separate reports can illustrate how many change requests, open incidents, and requests are pending. A dashboard shows you the balance among change requests as well as open incidents, and pending requests. Viewing these reports in a dashboard can help you make decisions about resource allocation and prioritization.

Security roles

Describes Service Manager Reports security roles as implemented by Service Manager.

Report Viewer

Report Designer

Report Manager

Report Administrator

Key concepts

Describes the most important things that you need to know about how Service Manager Reports module functions.

My reports

My dashboards

Report schedule and distribution

Report types and settings

Administrator tasks

Describes the Administrator tasks in the Service Manager Reports module.

Configure report settings

Enable Replicated Database

Configure time duration settings

Additional resources

Processes and Best Practices

1.16.1. Set home page preferences

Service Manager enables you to configure your personalized and role based homepage. If an administrator has configured one of the pre-configured **Dashboard** and **To Do Queue** home pages for you, it will be the first screen that's displayed when you log in to the Service Manager web client.

To set your preferences, click **Miscellaneous** > **My Preferences**, and then specify your settings.

This Preferences dialog enables you to customize the following items.

Preference	Description	Default setting
Home Page	Specifies the home page as **Dashboard** or the **To Do Queue**, which you see when logging in to Service Manager.	To Do Queue
Queue	Specifies your **To Do Queue**.	To Do
View	Specifies your view in **To Do Queue**. If you change and save the view in **To Do Queue**, the view here changes automatically.	N/A
Language	Specifies the primary language.	English
Time Zone	Specifies the time zone information.	N/A
Dashboard	Specifies your dashboard.	N/A
Date Format	Specifies the date format information.	`mm/dd/yy`
First Day of Week	Specifies the first day of week.	N/A

Preference	Description	Default setting
Preferred Notebook Style	Specifies the notebook style as tab or group.	N/A **Note:** The notebook style of **Tabs** overrides your configuration in Forms Designer. If you select the **Tabs** style for notebooks but you have enabled group rendering for the notebook control in Forms Designer, Service Manager still displays notebooks as tabs.

1.16.2. Service Manager Reports user roles

The Service Manager Reports module has the following user roles, each of which has different rights for reports.

Report Viewer

	View	Update/Delete	Create	Set Audience	Set Owner
As an owner	N/A	N/A	N/A	N/A	N/A
As an audience	Yes	No	N/A	No	No

Report Designer

	View	Update/Delete	Create	Set Audience	Set Owner
As an owner	Yes	Yes	Yes	No	No
As an audience	Yes	Yes You need to have the update privilege when the owner shares the report with you.	N/A	No	No

Report Manager

	View	Update/Delete	Create	Set Audience	Set Owner
As an owner	Yes	Yes	Yes	Yes You can only set the audience to all the audience type except everyone.	No
As an audience	Yes	Yes You need to have the update privilege when the owner shares the report with you.	N/A	No	No
Not as an audience	No	No	N/A	No	No

Report Administrator

	View	Update/Delete	Create	Set Audience	Set Owner
As an owner	Yes	Yes	Yes	Yes	Yes
As an audience	Yes	Yes	N/A	Yes	Yes
Not as an audience	Yes	Yes	N/A	Yes	Yes

Note

Only the report administrator can share a report or dashboard with everyone. If a report administrator shares a report or dashboard with everyone, then only the owner or report administrator can update this report or dashboard.

1.16.3. My reports

The reports provide detailed reporting and analysis information that help maximize the abilities of your organization. Data monitoring and tracking, using Service Manager data from your environment, provide valuable insight to daily operations. Service Manager enables you to build graphic reports dynamically from lists of records for a quick view of data and relationships.

With this feature	You can
Set permission for a report	Configure reports globally or individually by setting permission for a report.
Create different types of reports	• Create chart reports that total how many records have the same value in a group by field. • Create list reports that list each record according to the value in a field. • Create pivot table reports that list records in two-dimension view and you can view the content dynamically by dragging and dropping the columns.
Query	Build or write different query conditions to draw the report. If a query is cached, multiple users can view the report results from cache.
Legends	• Identify the data source for each report element by using color-coded legends. • Customize the legend color by setting the color indicator.
View the data details	• Click a component to open a record list of the data that generate the report. • Hover the mouse over a report component to view the data count for that component.

1.16.3.1. Create a report

You can create a new report from the following access points:

- **Create New Report** button on the **Add Content** page
- **Create New Report** button on the search report page
- **More** option on the list detail page
- **More** option on the To Do queue

To create a report from SM Reports menu, follow these steps:

1. In the System Navigator pane, select **Reporting** > **Create New Report**.
2. In the **New Report** window, complete the necessary Report properties.
3. (Optional) If you have already created a default dashboard, click **Add to Default Dashboard** below the **Report ID** to add this report to your default dashboard.

To create a report from a list, follow these steps:

1. Use search or advanced search to open a list of records.
2. Click **More** or the More Actions icon, and select **Create New Report**.

   > **Note**
   >
   > This option is visible when you have the `report.from.list` capability word. However, to successfully create a new report from list, you should be assigned with the correct roles.

3. In the **New Report** window, complete the necessary Report properties.
4. (Optional) If you have already created a default dashboard, click **Add to Default Dashboard** below the **Report ID** to add this report to your default dashboard.

Report properties

> **Note**
>
> You must complete all fields marked with a red asterisk *.

The following fields and tabs are available both for defining a new report in the **New report** page and for editing an existing report in the **Search Report** result page.

Fields and tabs	Description
Name (*)	Type a word or phrase that's a unique identifier for this report. It should be a value that makes it easy to understand the purpose of the report.
Description	Enter a description of the report. It is helpful for other users to understand the purpose and objectives of this report. The description is displayed under the report name when you add the reports to the dashboard.
Source Data (*)	Select the file for which you want to generate this report.
Report type (*)	Select the report type from the drop down list. For detailed information about report types and the corresponding report properties, see Report types and settings.
Report category	Select the report category from the drop down list. You can also type a new category in the field. This category only works as a group tag to categorize your own reports. Service Manager will not save this category as a real category.
Report ID	The report ID is automatically assigned to the report and displayed in the **Report ID** field after you save your report. The **Add to default dashboard** button is enabled at the same time for you to add this report to the default dashboard.

Fields and tabs	Description
Disable Report	If this option is checked, this report is disabled and Service Manager sends a notification to the owner of the report. For detailed information about when to disable a report, see Monitor report running status .
Use Replicated Database	If this option is checked, Service Manager generates the report data from Replicated Database. You can retain the default configuration from global settings, or you can change the settings for an individual report. Make sure that the Replicated Database is already set up and enabled before this option takes effect. If Replicated Database is not set up, an error occurs when Service Manager generates this report. If Replicated Database is disabled in sm.ini file, this option is disabled. For details to enable Replicated Database, see Enable Replicated Database.

Fields and tabs	Description
List/Chart/Pivot table properties	Different report types have different properties fields. Standard reports provide basic reports to display single-source data. Calculation reports provide comprehensive reports that can display both the source and calculated data. Duration reports provide the duration field reports. All the chart types support calculation and duration reports. List and Pivot table don't support calculation and duration reports. For detailed information about standard report types and the corresponding report properties, see Report types and settings. For details about report calculation and time duration properties, see Report calculation chart properties and Report duration chart properties.
Permission	The default owner is the current operator. You can change the owner and set the audience of this report. **Note** You need to have the privilege to change the permission. Only the Report Manager can change the report owner and the audience of their owned reports.

Fields and tabs	Description
Advanced settings	Different report types have different settings fields. For detailed information about report types and the corresponding settings, see Report types and settings.

Note

When you save a report, you save the query definition that retrieves data from the database to generate a real-time report.

1.16.3.1.1. Report calculation chart properties

Service Manager provides calculation functionality to reorganize the standard report data. You can configure the properties to display in the same report both the raw data from the system and the new data calculated from the raw data. The configured report can display the data of up to ten standard reports together with several calculation results based on the standard reports.

Note

- It may take some time if you configure to calculate the raw duration for historical data.
- The content in a shared standard report varies depending on the report viewer’s privilege. However, the content in a shared calculation or duration report only displays the data information based on the report owner's privilege. The reports which are calculated from the back end apply the same language as which the report owner defines in the **My Preference** settings.
- To process the report data, make sure that all the background schedule processes for reporting are started.

Here are the calculation setting preferences for your reference.

Note

Select **Calculation** from the drop-down box next to the **Type of chart** field, and then you can see the calculation properties.

Fields	Description
Type of chart	Shows the current chart type. You can change the chart type by selecting the chart type from the drop-down list. **Note** Report calculation capability doesn't support the List or Pivot table types of report.
Run in Frontend	If this option is selected, the current calculation report can be run immediately. When clicking the **Refresh** button on the Preview tab of a saved report, the system generates a preview of the current report at once. The related report running task will be removed from the background queue if it was run in backend before. We don't recommend running a calculation report frequently because this report type consumes considerable database resources. **Note** After you select **Run in Frontend**, both the **Refresh Interval** drop-down list and the **Run Now** button become unavailable.

Fields	Description
Refresh Interval	Specifies the interval at which a new report is generated according to your report configurations.
Run Now	Click this button to send the current calculation report to the processing queue. This also enables the report to get the latest source data from the system. **Note** This button is only available for a saved report.

Fields	Description
Aggregator	Shows the data aggregation method. For different chart types, the report calculation supports different aggregation types, including: sum, count, average, max, and min. **Note** When you select the count aggregation type, the **Time Duration** option is invisible. When you select the sum, average, max, or min aggregation types, you can select the **Time Duration** option to enable the **Duration Unit** drop-down list. Service Manager provides second, minute, hour, day, week, month, and year for **Duration Unit** field.
Trend field	Shows the trend field for the report. It should be a date time type of field. You can select multiple level fields by clicking the **Multi Level Field** icon in the calculation report. **Note** This field is only available for some type of charts.

Fields	Description
Stacked field	Shows the field that the records are stacked by in the report. You can select multiple level fields by clicking the **Multi Level Field** icon. **Note** This field is only available for some type of charts.

Fields	Description
Group By Field	Selects the field in the report for calculation. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator. Service Manager provides year, quarter, month, week, and date for "datetime type" field. **Note** You can specify up to ten standard reports. If you select the checkbox for one of the calculation fields, the standard report for this field is displayed together in the final report. However, the label is the field name by default. If you want to specify a customized label, you can add this field name directly to the calculation formula and then specify the label.
Metrics Field	Selects the aggregation value used for calculation from the drop-down list.

Fields	Description
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display the desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.
Test Query	Validates the query. A new message pops up to display the diagnostic information for this query string.

Fields	Description
Calculation Formula	Specifies the calculation formula you defined for this report. **Label:** Adds a label for this calculation formula, and the label is displayed as the legend in the report. **Formula:** Service Manager supports +, -, *, /, (), and numbers in the formula. For example, you can define a formula as (A1+A2)/A3*100. **Drilled:** Sets the value to **True** so that you can drill down to specific record lists in a Calculation type of chart. **Note** If you have defined the formula as A1+A2 or A1-A2 and then set the **Drilled** value to **True**, the record list expands according to A1. That is, the record list always expands according to the first item in the calculation formula. If you want to use the data in a standard report, or a calculated result to compare with a constant, you can specify the constant as one variable in this section. In this case, in the final report, the constant is displayed as a straight line and you can view the contrast of your data and this base line.

Fields	Description
X-Axis Label	Specifies the label for the X-axis in the report. **Note** This label field is unavailable for Pie charts.
Y-Axis Label	Specifies the label for the Y-axis in the report. **Note** This label field is unavailable for Pie and Number charts.

1.16.3.1.2. Sample report for calculation

Here is a sample calculation type report for your reference. This report displays the number of incidents opened, incidents closed, and the difference between these two numbers by quarter.

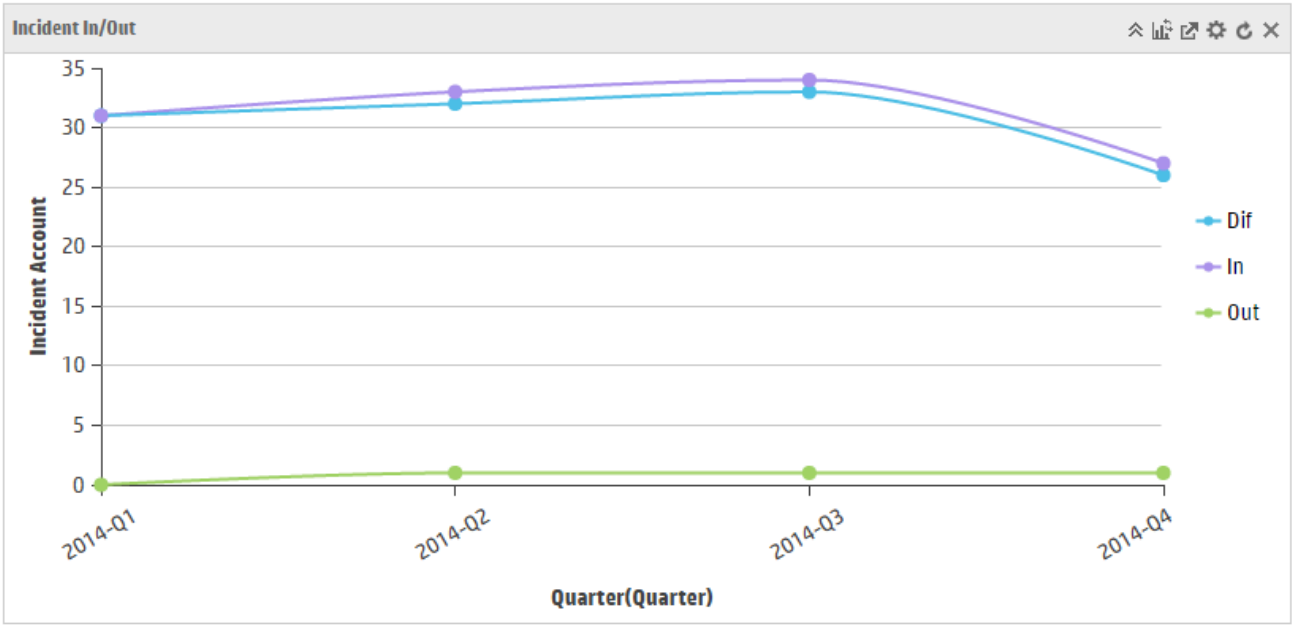
Note

Select **Calculation** from the drop-down box next to the **Type of chart** field, and then you can see the calculation properties.

Fields	Value	Description
Type of chart	Line	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Refresh Interval	Recurs every 1 day	Specifies the refresh interval to generate a new report according to your report configurations. **Note**: This is for report data refresh only. Users need to manually refresh the report content or the dashboard to get the latest report data.
Aggregator	Count	Shows the data aggregation method. In this example, the report data is the count number of Incidents.
Group By Field - A1	Open Time: Quarter	Specifies the first field for calculation. This field shows that the number of open Incidents in each quarter during the specified time period.

Fields	Value	Description
Group By Field - A2	Close Time: Quarter	Specifies the second field for calculation. This field shows that the number of closed Incidents in each quarter during the specified time period.
Query - A1	open.time>startofyear(startofyear() - 1) and open.time< startofyear()	Shows the query content for the current report. This query means that the report collects all the open incidents from the start of the last year to the start of this year.
Query - A2	close.time>startofyear(startofyear() - 1) and close.time<startofyear()	Shows the query content for the current report. This query means that the report collects all the closed incidents from the start of the last year to the start of this year.

Fields	Value	Description
Calculation Formula	A1-A2 (Label: Dif)	Specifies the calculation formula you defined for this report. The formula in this example calculates that number difference between the open Incidents and the closed incidents. The Dif label is assigned for the calculated result and shown as the legend in the report.
	A1 (Label: In)	This formula means that the report also displays all the open incidents and assign the In label for this data. The label is shown as the legend in the report.
	A2 (Label: Out)	This formula means that the report also displays all the closed incidents and assign the Out label for this data. The label is shown as the legend in the report.
X-Axis Label	Quarter	Specifies the label for the X-axis in the report.
Y-Axis Label	Incident Count	Specifies the label for the Y-axis in the report.


Incident In/Out
35
30
25
20
15
10
5
0
Incident Account
2014-Q1
2014-Q2
2014-Q3
2014-Q4
Quarter(Quarter)
Dif
In
Out

1.16.3.1.3. Report duration chart properties

Service Manager provides duration functionality to count and display the duration information.

Note

- The content in a shared standard report varies depending on the report viewer’s privilege. However, the content in a shared calculation or duration report only displays the data information based on the report owner's privilege. The reports which are calculated from the back end apply the same language as which the report owner defines in the **My Preference** settings.
- To process the report data, make sure that all the background schedule processes for reporting are started.

Here are the duration setting preferences for your reference.

Note

Select **Duration** from the drop-down box next to the **Type of chart** field, and then you can see the duration properties.

Fields	Description
Type of chart	Shows the current chart type. You can change the chart type by selecting the chart type from the drop-down list. **Note** Report duration capability doesn't support the List or Pivot table types of report.
Run in Frontend	If this option is selected, the current duration report can be run immediately. When clicking the **Refresh** button on the Preview tab of a saved report, the system generates a preview of the current report at once. The related report running task will be removed from the background queue if it was run in backend before. We don't recommend to run a duration report frequently because this report type consumes considerable database resources. **Note** After you select **Run in Frontend**, both the **Refresh Interval** drop-down list and the **Run Now** button become unavailable.
Refresh Interval	Specifies the interval at which a new report is generated according to your report configurations.

Fields	Description
Run Now	Click this button to send the current calculation report to the processing queue. This also enables the report to get the latest source data from the system. **Note** This button is only available for a saved report.
Aggregator	Shows the data aggregation method. For different chart types, the report calculation supports different aggregation types, including: sum, average, max, and min.
Group By Field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator. Service Manager provides year, quarter, month, week, and date for **datetime** type field.
Duration Start Field	Shows the duration start field used for calculation in the report.
Duration End Field	Shows the duration end field used for calculation in the report.

Fields	Description
Duration Field	Shows the duration field displayed in the report. This option is only available when the **Time Duration** option is selected. In this situation, you just need to select the duration field directly and save the system from the duration calculation.
Duration Unit	Shows the duration unit displayed in the report. Service Manager provides second, minute, hour, day, week, month, and year for **Duration Unit** field. **Note** For different time duration units, Service Manager sets different unit calculation precision, as shown below. **Duration unit:** year/month/week/day/hour/minute **Calculation precision:** minute **Duration unit:** second **Calculation precision:** second
Duration Label	Shows the duration label for the report. The label is displayed as the legend in the report.

Fields	Description
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.
Test Query	Validates the query. A new message pops up to display the diagnostic information for this query string.

1.16.3.1.4. Sample report for duration

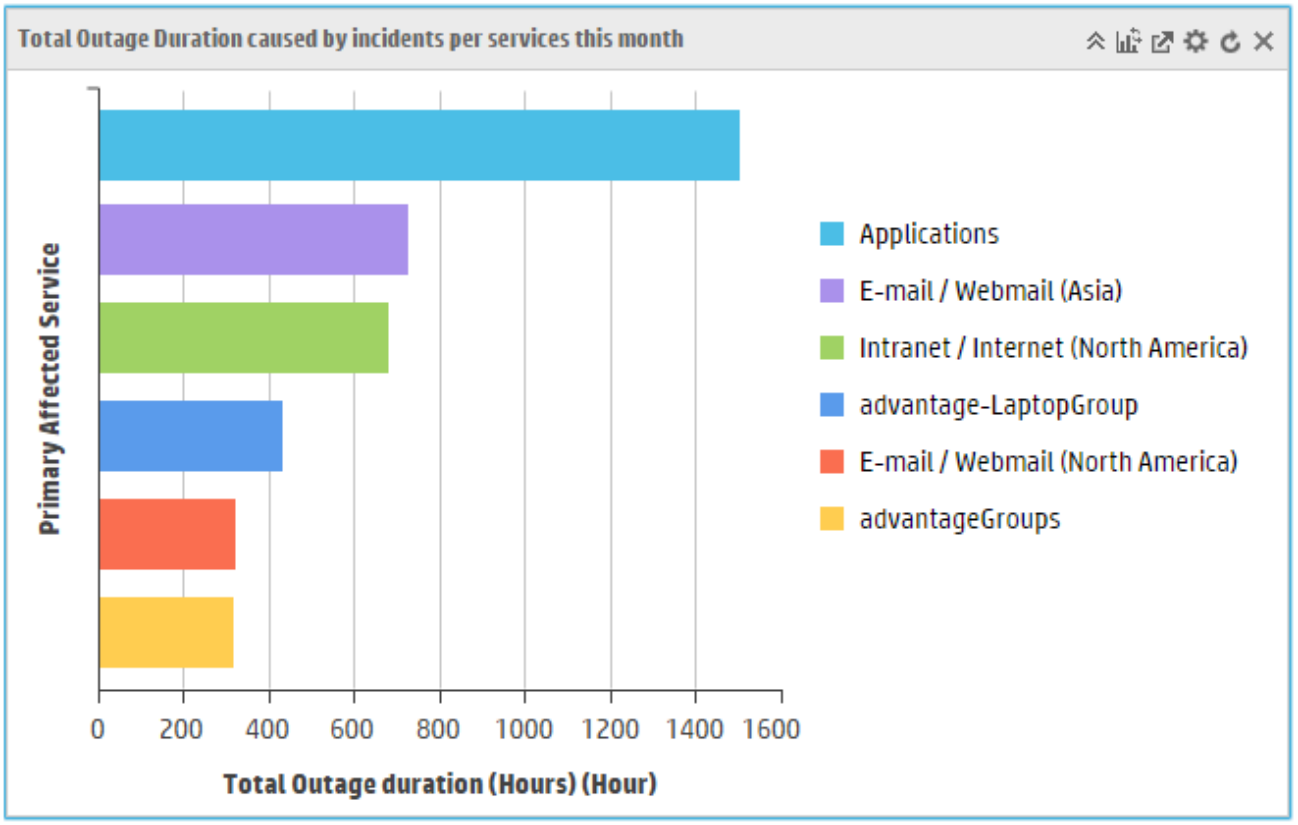
Here is a sample duration type report for your reference. This report displays a summary of the outage duration of incidents, sorted by service.

Note

Select **Duration** from the drop-down box next to the **Type of chart** field, and then you can see the duration properties.

Fields	Value	Description
Type of chart	Horizontal bar	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Refresh Interval	Recurs every 1 day	Specifies the refresh interval to generate a new report according to your report configurations. **Note** This is for report data refresh only. Users need to manually refresh the report content or the dashboard to get the latest report data.
Aggregator	Sum	Shows the data aggregation method. In this example, the report data is the time summary of the duration.
Duration Start Field	Outage Start	Shows the duration start field used for calculation in the report.

Fields	Value	Description
Duration End Field	Outage End	Shows the duration end field used for calculation in the report.
Duration Unit	Hour	Shows the duration unit displayed in the report.
Duration Label	Total Outage duration	Shows the duration label for the report. The label is displayed as the legend in the report.
Query	open.time>startofmonth( )	Shows the query content for the current report. This query means that the open time of the Incidents to be collected should be earlier than the start of the this month.


Total Outage Duration caused by incidents per services this month
Primary Affected Service
0
200
400
600
800
1000
1200
1400
1600
Total Outage duration (Hours) (Hour)
Applications
E-mail / Webmail (Asia)
Intranet / Internet (North America)
advantage-LaptopGroup
E-mail / Webmail (North America)
advantageGroups

1.16.3.1.5. Create a ticket aging report

You may want to create a Service Manager report to display the ticket aging information for incidents. The following example guides you to complete the related tasks:

1. Create a Service Manager report to display the Incident tickets that are in the Open state for more than 30 minutes.
2. Add the newly created report to a new or an existent dashboard.

Follow these steps:

5) Click **Save**.

1. From the System Navigator pane, click **Reporting** > **Create New Report**.
2. On the **Create New Report** form, complete the necessary Report properties.

   For example, type a report name such as **Incident ticket aging report**, and then select **List** in the Report Type drop-down list.
3. Configure the list properties, permission, advanced settings in the related tabs as required. Or you can just use the default values.
4. Update the Query field with the following statement to search or filter out the Incident tickets that are submmited for more than 30 minutes and are still in the Open State.

   `open.time<tod()-'00:30:00' and problem.status="Open"`
5. Follow the instructions as described in My dashboards to add the newly created report to a new or an existent dashboard.

1.16.3.1.6. Create a calculation report by using duration metrics

Service Manager extracts and processes the time duration data, and then saves the field change metrics in the intermediate data tables. You can use calculation reports to further process the time duration metrics. For example, the metrics provide the duration for which an incident is processed among each assignment group, then you can use a calculation report to calculate the average handling time among different assignment groups.

Note

You must have SLM module license to configure the time duration settings and get the field change metrics tables for calculation. Service Manager extracts the system data against status, phase, and assignment group changes, and then calculates the duration values according to your metrics. For more information, see Time duration data processing.

Here is a sample report that uses the calculation report to process the time duration metrics for your reference.

Note

Select **Calculation** from the drop-down box next to the **Type of chart** field, and then you can see the calculation properties.

Fields	Value
Name	Average Incident handling time per assignment group
Description	This report enables the user to review the average Incident handling time per assignment group by using the processed time duration data.
Source data	Incident Time Duration - timeDuration4probsummary

Fields	Value	Description
Type of chart	Line	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Refresh Interval	Recurs every 1 day	Specifies the refresh interval to generate a new report according to your report configurations.
Aggregator	Average	Shows the data aggregation method. In this example, the report data is the average time duration of the Incident handling time.
Time Duration	Enabled	When you select the sum, average, max, or min aggregation types, you can select the **Time Duration** option to enable the **Duration Unit** drop-down list.
Duration Unit	Hour	Service Manager provides second, minute, hour, day, week, month, and year for the **Duration Unit** field.
Group by Field	to Value	Shows the field that the records are grouped by in the report.

Fields	Value	Description
Trend Field - A1	Change Time: Week	Specifies the first field for calculation. This field shows that the assignment group change time of the incident in each week during the specified time period.
Metrics Field	Sum Metrics 1	Specifies the aggregation value used for calculation. This value means that the data used for calculation is from the Sum Metrics 1 value in the `timeDuration4probsummary` table.
Query	field.name="assignment" and is.sum=true	Shows the query content for the current report. This query means that the report collects the time duration information for the assignment field.

Fields	Value	Description
Calculation Formula	Average handle time: A1	Specifies the calculation formula you defined for this report. The formula in this example means that the final report only displays the average time duration data based on the assignment group change.
X-Axis Label	Re-assign time	Specifies the label for the x-axis in the report.
Y-Axis Label	Average handle time (Hours)	Specifies the label for the y-axis in the report.

1.16.3.2. Update a report in the dashboard

Service Manager provides the UI elements on a single report that enables you to easily update the report in this dashboard. The following table describes the UI elements in the report.

UI Element	Description
	Collapses the report. This UI element is displayed when a report is expanded.
	Expands the report. This UI element is displayed when a report is collapsed.
	Shows the chart switch options. It displays only the options available for the current report data type. You can switch and save the chart types by directly clicking the icons without opening the report definition page.
	Exports the report as PDF or Excel files. A dialogue box pops up and you can change the name or type of the exported file and select the orientation as portrait or landscape. You can export all types of reports as PDF files, and only three types of reports, including Pivot table, title list, and standard list, as Excel files.
	Opens the report definition page, which enables you to modify the report properties.
	Refreshes the report.
	Removes this report from the dashboard.

1.16.3.3. Export reports from the dashboard

You can export a single report in a dashboard, or the whole dashboard, as a PDF.

Tip

You can also configure scheduled reports to be automatically exported. For more information about how to do this, see Create a report schedule.

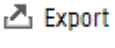
Note

The copyright information in the exported reports is defined as a Service Manager Message with `Class="Reporting"` and `Message Number="copyright"`. You can customize this message as required.

To export a single report:

1. Click the export icon ( ) on the top-right of the report.
2. In the dialog box that appears, enter the name of the exported file (the default value is the report name).
3. Select the file type. You can export all types of reports as PDF files. Additionally, you can export Pivot tables, title lists, and standard lists as Excel files.
4. Select the page orientation (PDF files only).
5. Click **Export**.

To export a dashboard (that is, all the reports on a dashboard):

1. Click the export icon (

) in the dashboard menu.
2. In the dialog box that appears, enter the name of the exported file (the default value is the dashboard name).
3. Select the page orientation.
4. Click **Export**.

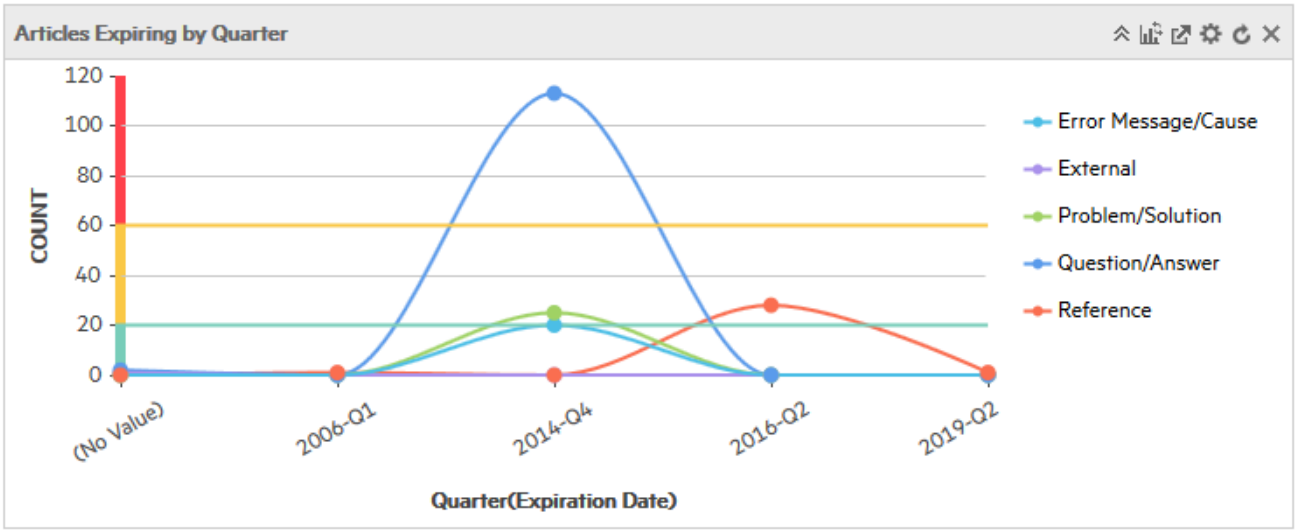
Note

You can only export dashboards as PDF files.

If you have configured any colored ranges for reports, these are included in the exported file.

1.16.3.4. Add colored ranges to a report

You can add colored ranges to charts in reports to highlight thresholds and ranges. When you define colored ranges for a graphical chart, the section of the relevant axis falling within each range is displayed in the defined color, and threshold markers are displayed on the report to mark the limits of the ranges. When you define colored ranges for a number chart, the color of the number changes according to the range in which it falls.



You can configure colored ranges on the following types of report:

- Chart- Vertical bar
- Chart- Stacked vertical bar
- Chart- Horizontal bar
- Chart- Stacked horizontal bar
- Chart- Line
- Chart- Area
- Chart- Number

To configure colored ranges for a chart, follow these steps:

1. Navigate to the **Advanced settings** tab of the report to which you want to add colored ranges. To do this, view the report on a dashboard, click the **Settings** (⚙) icon on the top-right of the report, and then click the **Advanced settings** tab. Or, click **Reporting** > **Search Report**, search for the desired report, and then click the **Advanced settings** tab.
2. In the **From** column of the **Colored Ranges** table, enter the lower threshold value of each color range.
3. In the **To** column, enter the upper threshold value of each color range.

4. In the **Color** column, select the color for each range. You can use the same color for multiple ranges. Available colors are defined in the Color Indicator Settings table.
5. Click **Save**. To preview the colored ranges, click the **Preview** tab.

Note

- The **From** value of the first range and the **To** value of the last range can be left blank, to indicate infinite values.
- Ranges must be defined in ascending order.
- Ranges must be contiguous. That is, the **From** value of each range must equal the **To** value of the preceding range.

1.16.3.5. Build the query strings

You can use query filters to define the report content logic when you create a report. This topic provides specifications to explain how to configure the strings for the query parameter when you create a report. The query display field is read-only, and you can use the query builder to define the condition for the report. Service Manager also supports cross-table queries.

Note

If you want to use query filters in a duration or calculation report for a user who is not the report owner, you must run the report in the frontend (to do this, select the **Run in Frontend** option). Reports run in the background will always use the report owner's information to populate the query parameters.

Note that reports run in the frontend typically take longer to execute, as the data is not prepared and cached in the backend.

Service Manager provides the following types of query parameters.

Query type	Sample	Procedure
Simple query	Open incident backlog group by priority	Click **Query Builder** to fill in the requirements for the query results. You can edit the filter fields and check the checkbox to specify the required conditions. Click **OK** to generate the query strings directly in the query display field.
Cross-table query	Open incident group by affected CI	Click **Query Builder** and select **Advanced Filter** to specify the query conditions. For more information about how to use advanced filter, see Advanced filter and other related topics. **Note:** All relationships between tables are stored in relatedObjectMap .

Query type		Sample	Procedure
Advanced Query	Support variables	All incidents assigned to my group	Select one assignment group. The query will be generated automatically, and then you can change the query to use the variables. `assignment isin $lo.pm.assignments`
	Support JSCall	incident for falcon	`function getDefaultComp() {return "falcon";}` `assignee.name =jscall("test.getDefaultComp")`
	Support RTE function	All incident assigned to me	`assignee.name = operator()`
	Support new method	startofyear, startofmonth, and startofweek	`startofweek()` returns the first day of the current week `startofweek(startofweek() - 1)` returns the first day of the last week

After you enter a query string, you can click **Test Query** to validate the query. A new message pops up, displaying the diagnostic information for this query string.

- When you enter 1=1 condition, Service Manager displays the following message:

  Current search criteria is equal to "(1=1)", full table query would be needed.

- When the field is CLOB/BLOB, Service Manager displays the following message:

  Partial File Scan might be needed since field "action" (on file probsummary) is mapped to BLOB/CLOB/TEXT/IMAGE [Description field in cm3r]

- When the field is not indexed, Service Manager displays the following message:

  There is no index defined in dbdict for field "sysmodtime" (on file kmfeedback). [sysmodtime in kmfeedback]

1.16.3.6. Report types and settings

Service Manager provides 12 report types, including eight types of charts, two types of lists, one pivot table, and one Hot Topic Map. The reports properties settings vary according to the type of report.

You can review the examples of all the different report types and their setting preferences. These are the chart properties for the standard type of charts only. For the calculation and duration chart properties, see Report calculation chart properties and Report duration chart properties.

Chart types:

- Chart > Pie
- Chart > Horizontal bar
- Chart > Vertical bar
- Chart > Stacked Horizontal bar
- Chart > Stacked Vertical bar
- Chart > Number
- Chart > Line
- Chart > Area

List types:

- List > Standard list
- List > Title list

Pivot Table
Hot Topic Map

1.16.3.6.1. Chart > Pie

Here are the setting preferences and examples for your reference.

Chart Properties tab

Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Group by field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator.
Metrics	Shows the data aggregation method. Pie supports two aggregation types: sum and count. **Note**: When you select the count aggregation type, the aggregation value field is invisible. When you select the sum aggregation type, you can select the aggregation value from the drop-down list.

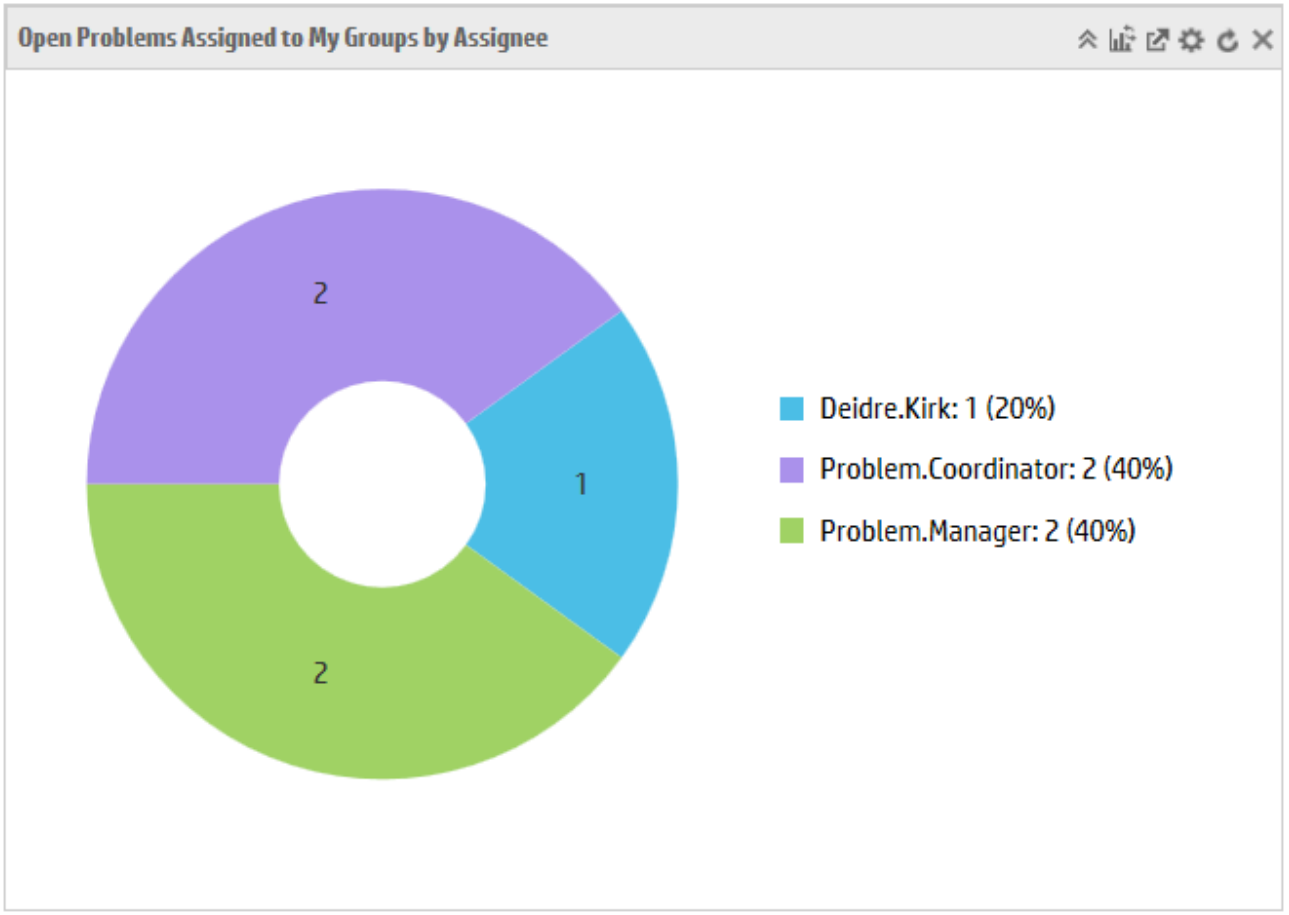
Fields	Description
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.
Show Legend	Specifies the legend location in the report. Service Manager displays no legend when you select **None**.
Show Records	Specifies the permitted maximum number of records displayed in a report, ordered by the legend or the top N.
Donut	Specifies the donut space in the pie chart. Service Manager displays a complete pie when you select **None**.
Support Top N	If you select this option, the report displays the top number records as defined in the **Show Records** field.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description



1.16.3.6.2. Chart > Horizontal bar

Here are the horizontal bar setting preferences and examples for your reference.

Chart Properties tab

Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Group by field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator. And if the two group by fields are both date and time fields, they should be the same one.
Stacked field	Shows the field that the records are stacked by in the report. You can select multiple level fields by clicking the **Multi Level Field** icon.

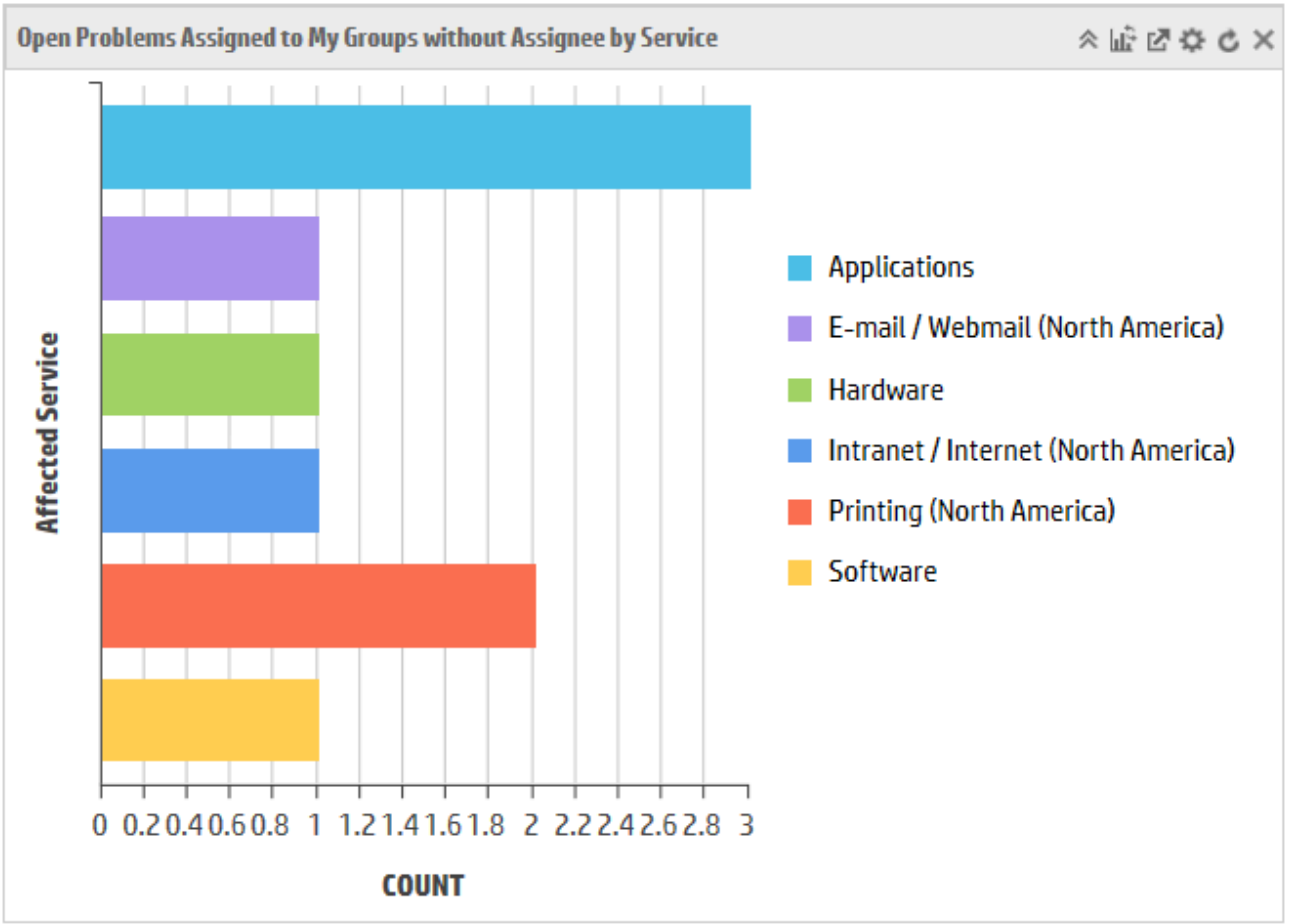
Fields	Description
Metrics	Shows the data aggregation method. Horizontal bar supports five aggregation types: sum, count, average, max, and min. **Note:** When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.
Show Legend	Specifies the legend location in the report. Service Manager displays no legend when you select **None**.
Show Records	Specifies the permitted maximum number of records displayed in a report ordered by the legend or the top N.
Show Data Labels	If you select this checkbox, the report displays the count labels for data items. When you export this report, the count labels are exported as well.
Support Top N	If you select this option, the report displays the top number records as defined in the **Show Records** field.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Axis Bounds (Minimum/Maximum)	Specifies the minimum and the maximum values on the value axis.
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.3. Chart > Vertical bar

Here are the vertical bar setting preferences and examples for your reference.

Chart Properties tab

Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Group by field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator. And if the two group by fields are both date and time fields, they should be the same one.
Stacked field	Shows the field that the records are stacked by in the report. You can select multiple level fields by clicking the **Multi Level Field** icon.

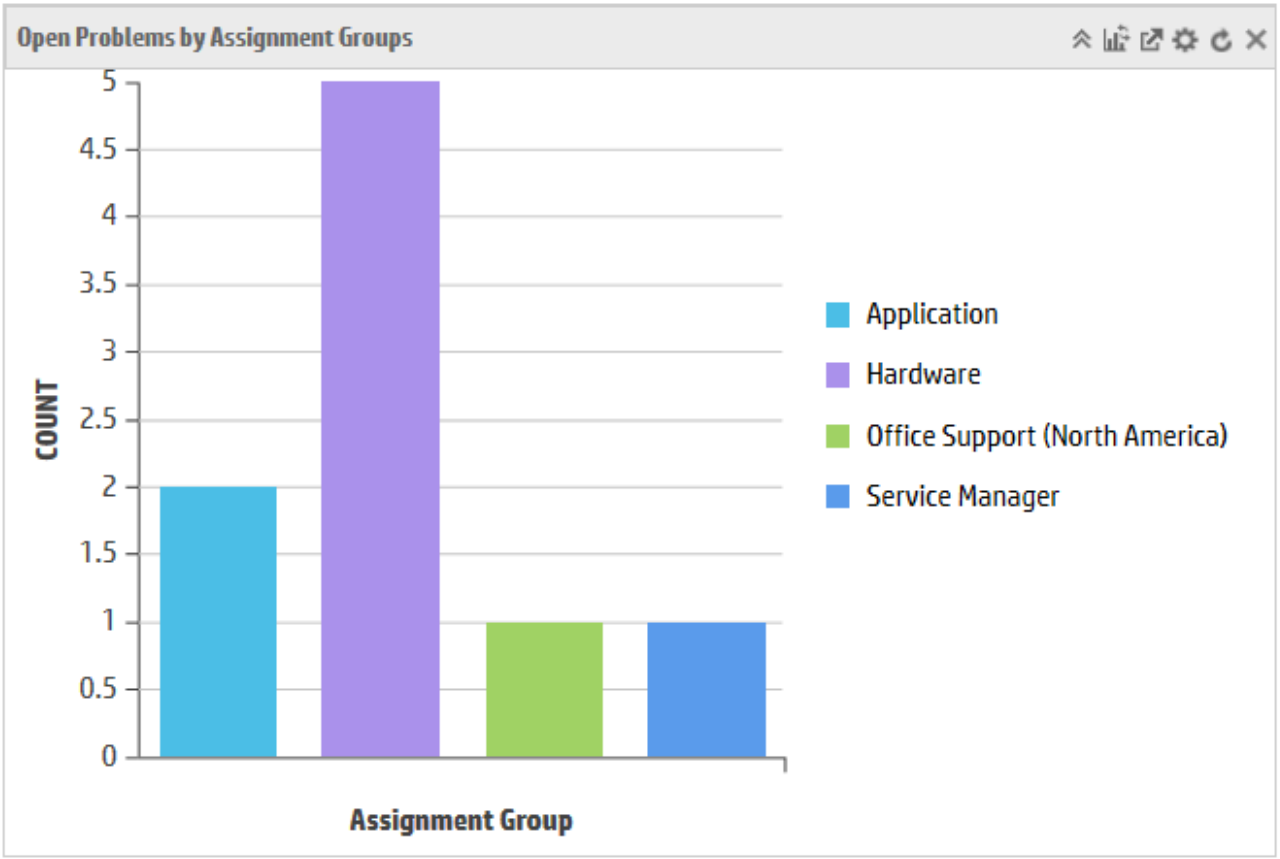
Fields	Description
Metrics	Shows the data aggregation method. Vertical bar supports five aggregation types: sum, count, average, max, and min. Note When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display the desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.
Show Legend	Specifies the legend location in the report. Service Manager displays no legend when you select **None**.
Show Records	Specifies the permitted maximum number of record displayed in a report ordered by the legend or the top N.
Show Data Labels	If you select this checkbox, the report displays the count labels for data items. When you export this report, the count labels are exported as well.
Support Top N	If you select this option, the report displays the top number records as defined in the **Show Records** field.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Axis Bounds (Minimum/Maximum)	Specifies the minimum and the maximum values on the value axis.
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.4. Chart > Stacked Horizontal bar

Here are the stacked horizontal bar setting preferences and examples for your reference.

Chart Properties tab

Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Group by field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator. And if the two group by fields are both date and time fields, they should be the same one.
Stacked field	Shows the field that the records are stacked by in the report. You can select multiple level fields by clicking the **Multi Level Field** icon.

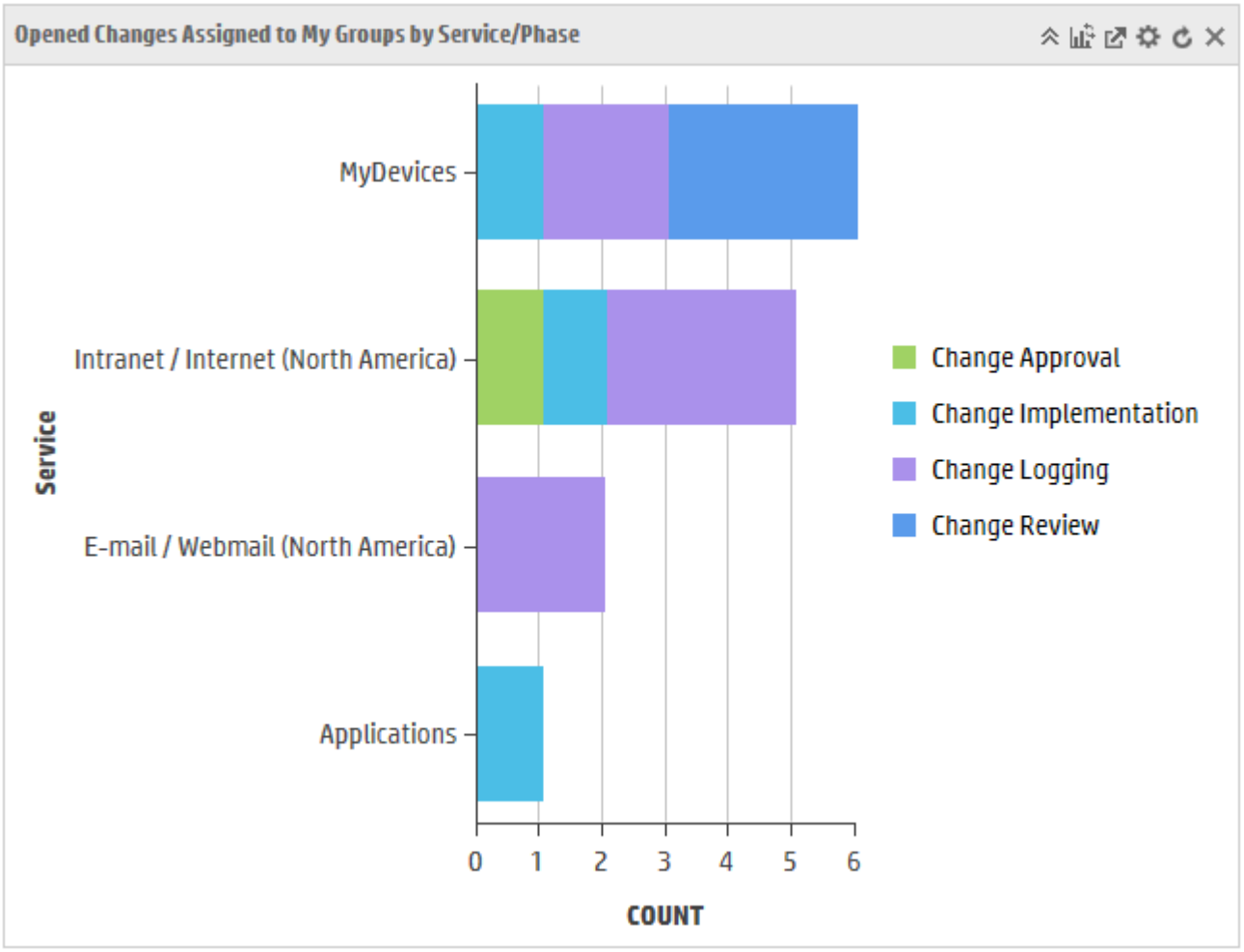
Fields	Description
Metrics	Shows the data aggregation method. Stacked horizontal bar supports five aggregation types: sum, count, average, max, and min. Note When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report is calculating the record percentage. Service Manager provides up to five decimal places.
Show Legend	Specifies the legend location in the report. Service Manager displays no legend when you select **None**.
Show Records	Specifies the permitted maximum number of record displayed in a report.
Show Data Labels	If you select this checkbox, the report displays the count labels for data items. When you export this report, the count labels are exported as well.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Axis Bounds (Minimum/Maximum)	Specifies the minimum and the maximum values on the value axis.
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.5. Chart > Stacked Vertical bar

Here are the stacked vertical bar setting preferences and examples for your reference.

Chart Properties tab

Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Group by field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon. If you choose a field of date or time, you need to choose a calculator. And if the two group by fields are both date and time field, they should be the same one.
Stacked field	Shows the field that the records are stacked by in the report. You can select multiple level fields by clicking the **Multi Level Field** icon.

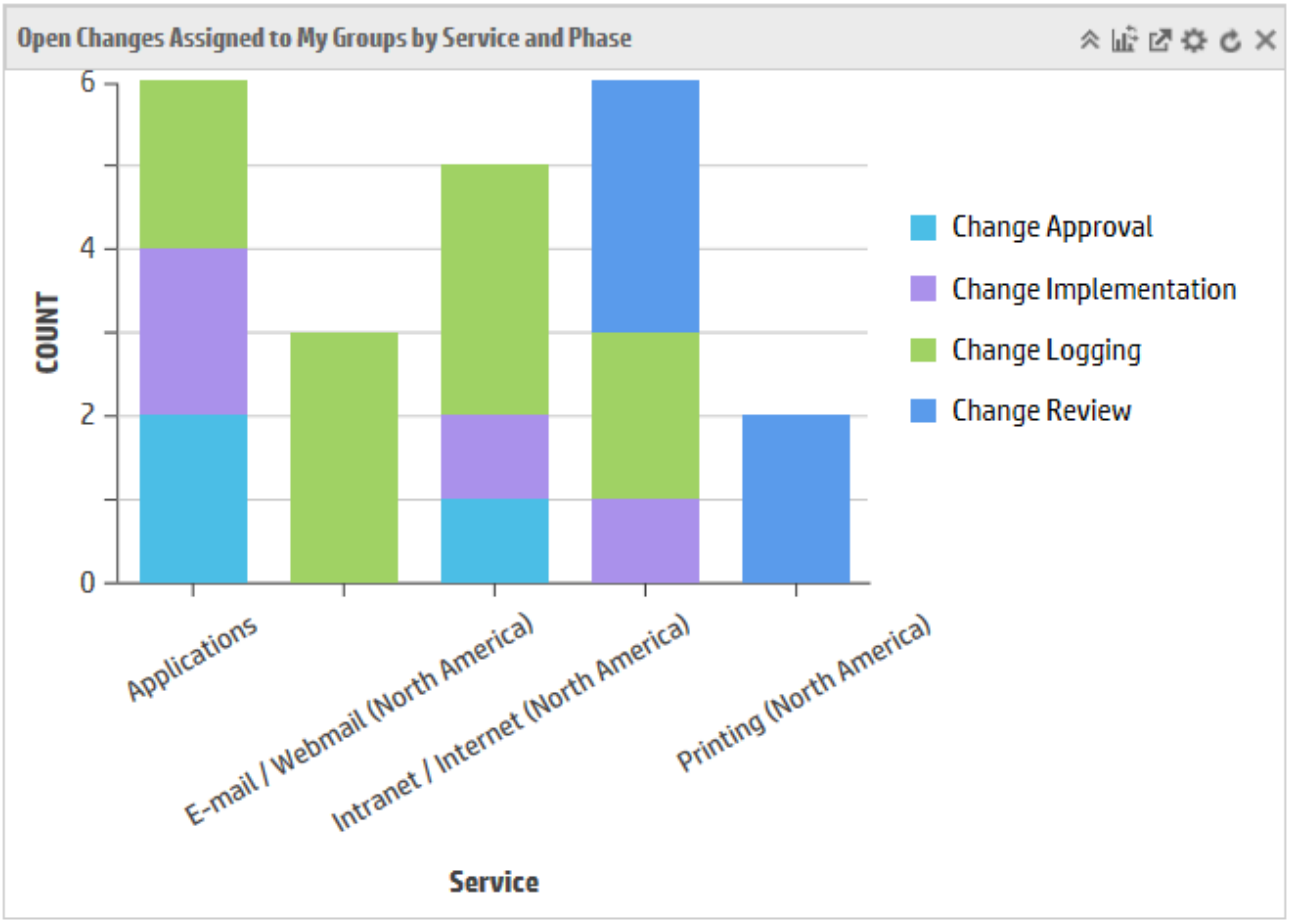
Fields	Description
Metrics	Shows the data aggregation method. Stacked vertical bar supports five aggregation types: sum, count, average, max, and min. Note When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.
Show Legend	Specifies the legend location in the report. Service Manager displays no legend when you select **None**.
Show Records	Specifies the permitted maximum number of records displayed in a report.
Show Data Labels	If you select this checkbox, the report displays the count labels for data items. When you export this report, the count labels are exported as well.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Axis Bounds (Minimum/Maximum)	Specifies the minimum and the maximum values on the value axis.
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.6. Chart > Number

Here are the number setting preferences and examples for your reference.

Chart Properties tab

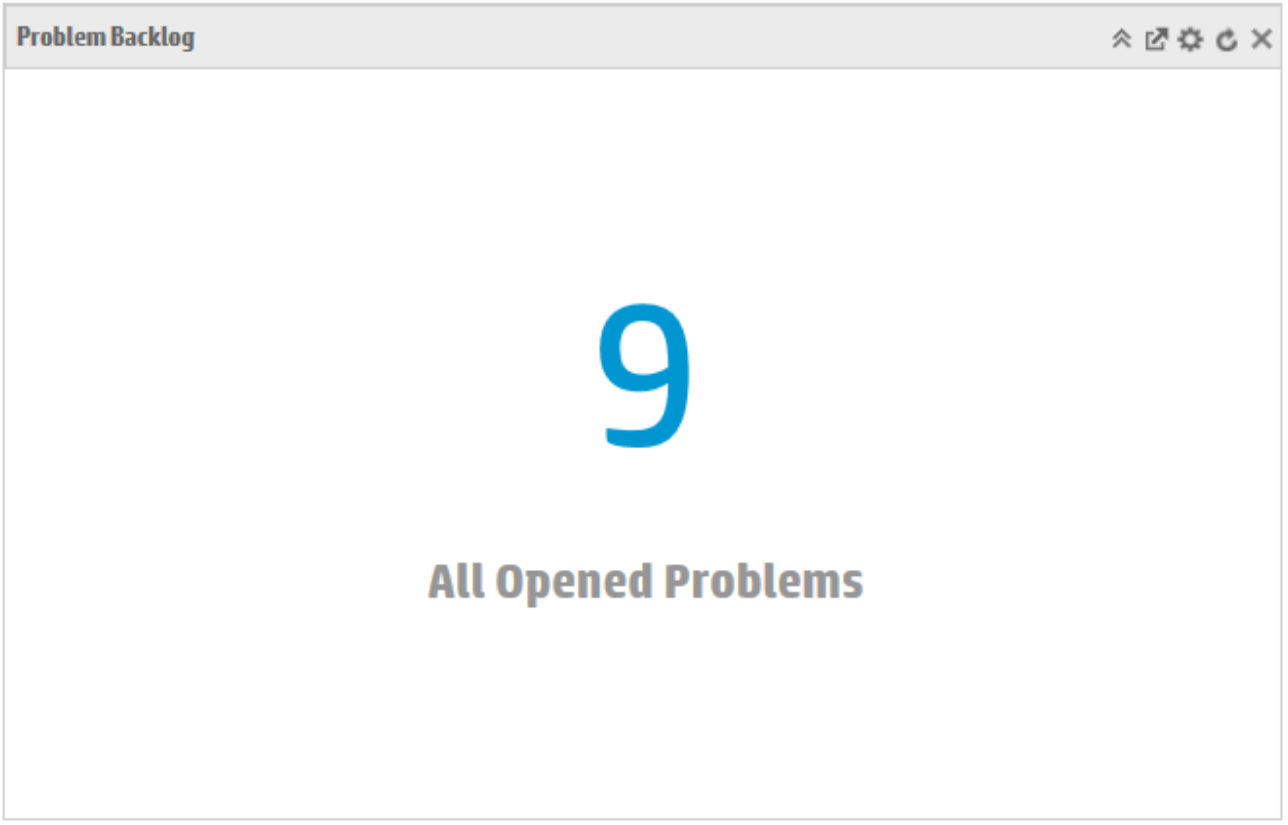
Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Label	Specifies the label content to show under the number.
Metrics	Shows the data aggregation method. Number supports five aggregation types: sum, count, average, max, and min. Note When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display the desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.
Default Color	Specifies the color of the number if you set no colored range or if the value falls outside all ranges.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.7. Chart > Line

Here are the line setting preferences and example for your reference.

Chart Properties tab

Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Trend field	Shows the trend field for the report. It should be a date time type of field.
Group by field	Shows the field that the records are grouped by in the report. For example, one report might show the number of Incidents by customer; another report might show the number of Incidents by priority. You can select multiple level fields by clicking the **Multi Level Field** icon.
Metrics	Shows the data aggregation method. Line supports five aggregation types: sum, count, average, max, and min. **Note:** When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.

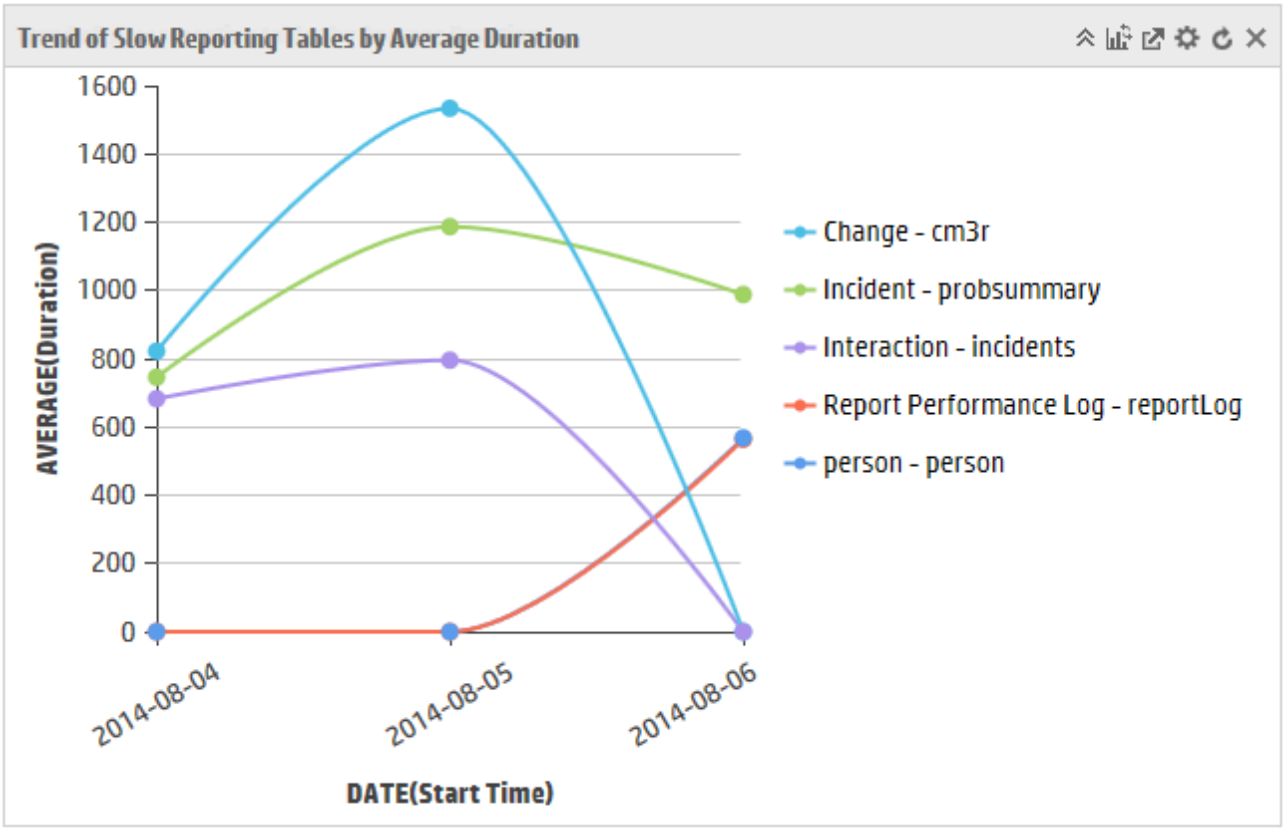
Fields	Description
Query	Shows the query content for the current report. You can click the **Query Builder**to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.
Show Records	Specifies the permitted maximum number of records displayed in a report.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Axis Bounds (Minimum/Maximum)	Specifies the minimum and the maximum values on the value axis.
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.8. Chart > Area

Here are the area setting preferences and example for your reference.

Chart Properties tab

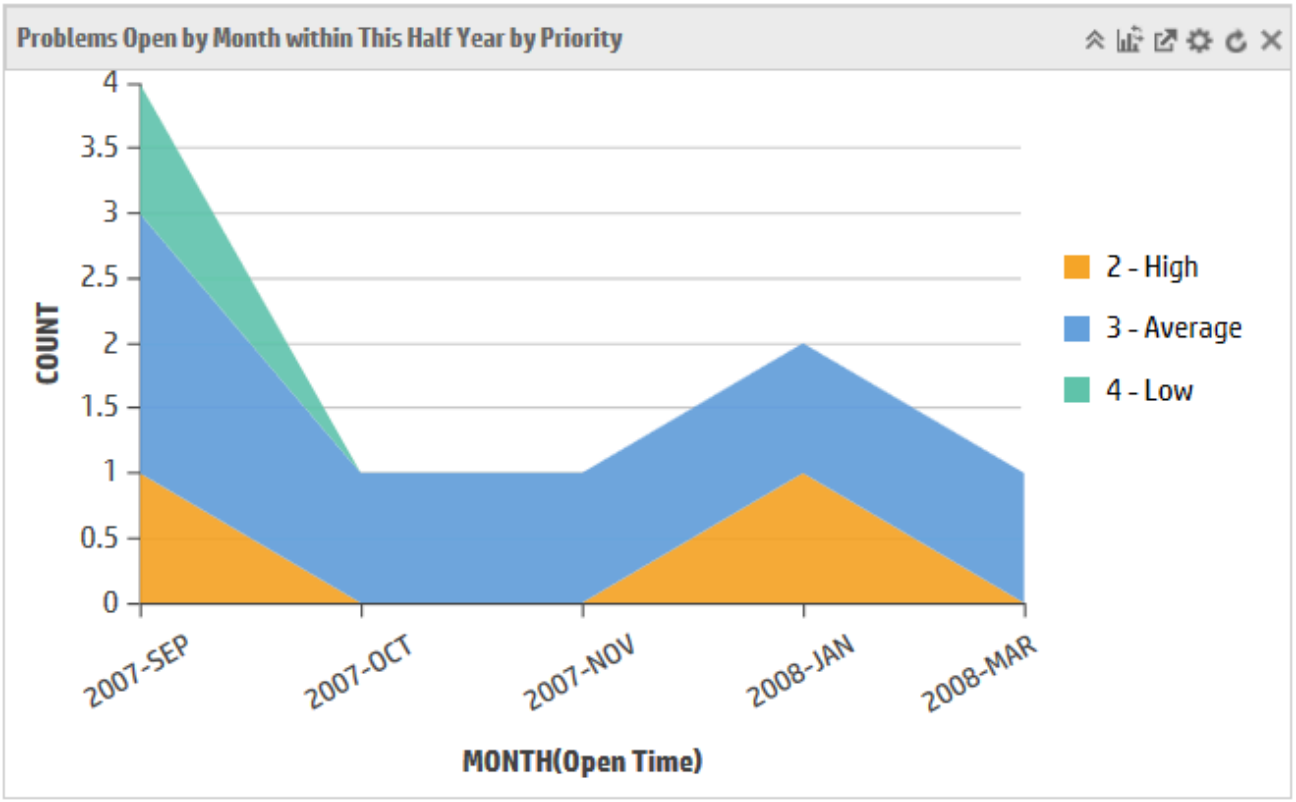
Fields	Description
Type of chart	Shows the current chart type. You can change it by selecting the chart type from the drop-down list.
Chart by	Shows the query type for the report. The default value is **Aggregate query**.
Trend field	Shows the trend field for the report. It should be a date or time type of field.
Stacked field	Shows the field that the records are stacked by in the report. You can select multiple level fields by clicking the **Multi Level Field** icon.
Metrics	Shows the data aggregation method. Area supports five aggregation types: sum, count, average, max, and min. Note When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

Fields	Description
Decimal Places	Specifies the number of decimal places at which the report is calculating the record percentage. Service Manager provides up to five decimal places.
Show Records	Specifies the permitted maximum number of records displayed in a report.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description
Axis Bounds (Minimum/Maximum)	Specifies the minimum and the maximum values on the value axis.
Colored Ranges	Specifies any colored ranges that are displayed on the report. For more information about setting colored ranges, see Add colored ranges to a report.



1.16.3.6.9. List > Standard list

Here are the standard list setting preferences and examples for your reference.

List Properties tab

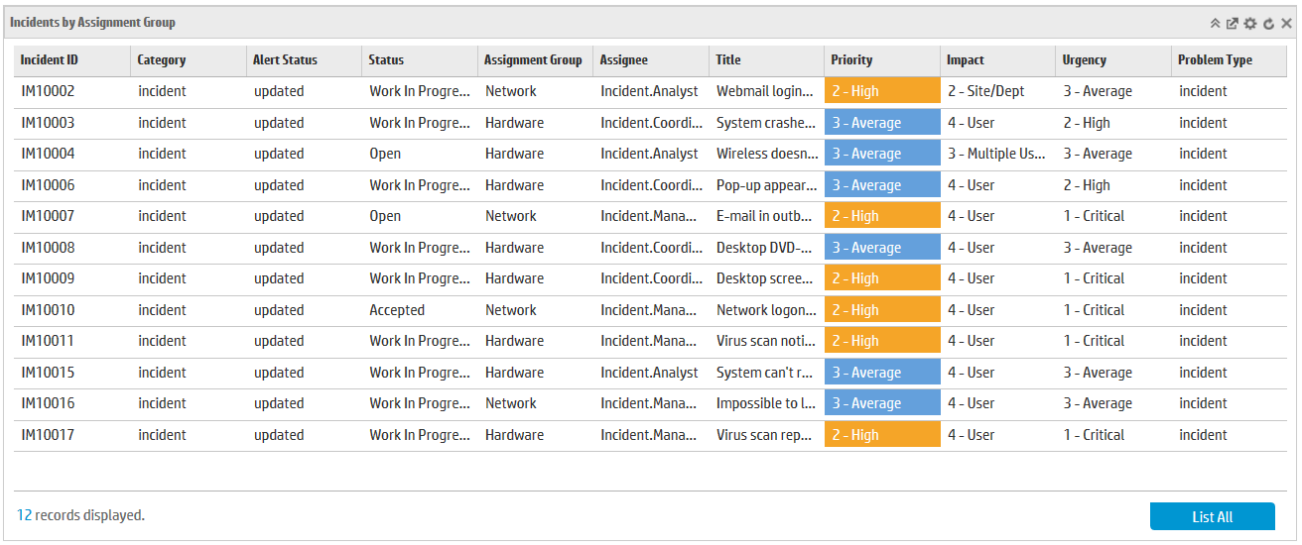
Fields	Description
View Fields	Click the button to open the **Choose View Fields** wizard and select the desired fields that the report displays.
Sort by	Click the button to open the **Sort By** wizard and select the fields to sort by.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.
Use normalized query	Select the checkbox to use the normalized query to search and display the result.

Advanced settings tab

Fields	Description
Show Records	Specifies the permitted maximum number of records displayed in a report.
Show as	Shows the current list type. You can change it by selecting the list type from the drop-down list.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description



1.16.3.6.10. List > Title list

Here are the title list setting preferences and examples for your reference.

List Properties tab

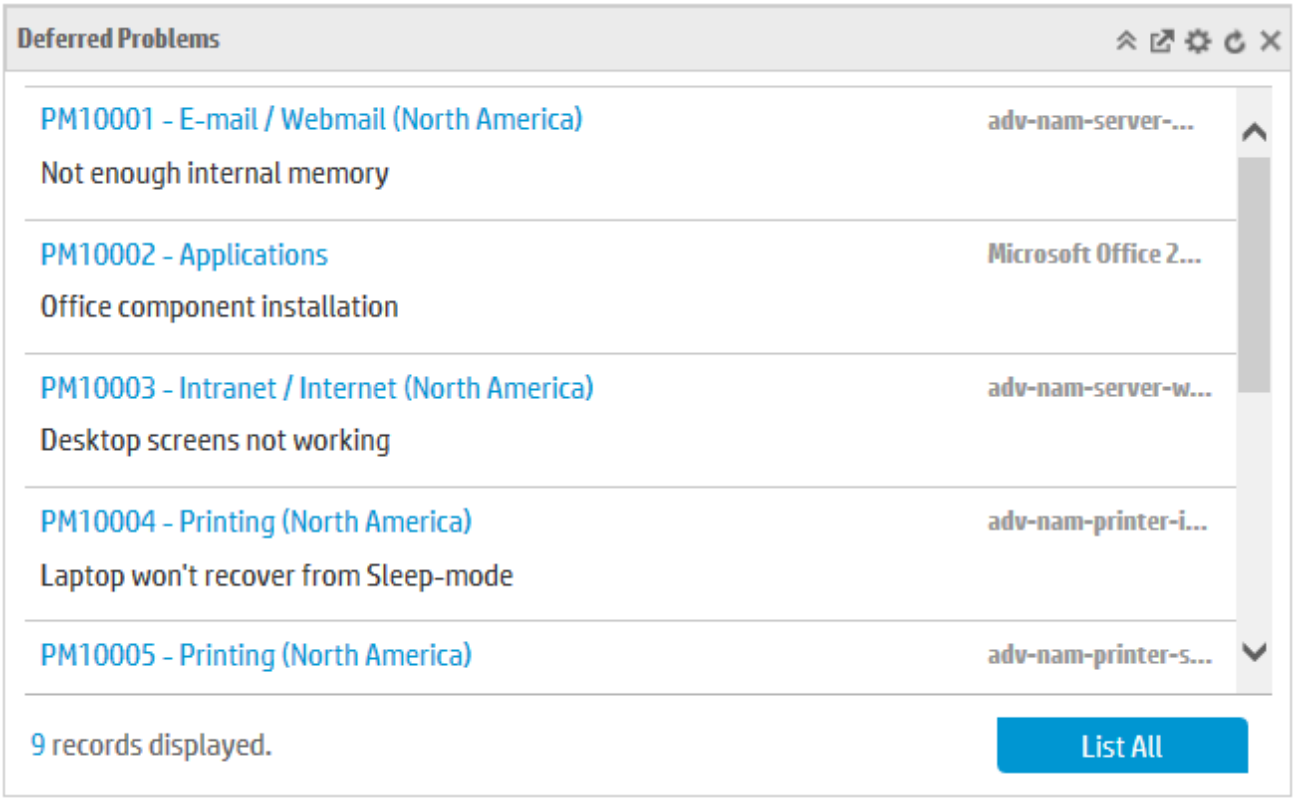
Fields	Description
View Fields	Click the button to open the **Choose View Fileds** wizard and select the desired fields that the report displays.
Sort by	Click the button to open the **Sort By** wizard and select the fields to sort by.
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.
Use normalized query	Select the checkbox to use the normalized query to search and display the result.

Advanced settings tab

Fields	Description
Show Records	Specifies the permitted maximum number of records displayed in a report.
Show as	Shows the current list type. You can change it by selecting the list type from the drop-down list.
Header Field	Selects the header information of each title list item.
Flag Field	Selects the flag information of each title list item on the right of the report. Color indicators can control this field.
Summary Field	Selects the summary information of each title list item under the header field.
Footer Field	Selects the footer information of each title list item under the summary field.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description



1.16.3.6.11. Pivot Table

Here are the pivot table setting preferences and examples for your reference. Service Manager supports five pivot display types: **Table**, **Table BarChart**, **HeatMap**, **Row HeatMap**, and **Col HeatMap**. You can click the row, the column, or the candidate field to filter the data in the pivot.

Pivot Table Properties tab

Fields	Description
Aggregator	Shows the data aggregation method. The pivot table supports five aggregation types: sum, count, average, max, and min. Note When you select the count aggregation type, the aggregation value field is invisible. When you select the sum, average, max, or min aggregation types, you can select the aggregation value from the drop-down list.
Row Fields	Click to select the row field and the calculator type from the drop-down list. The report displays the row field as an active button. You can drag the button and drop it directly to the pivot table to redesign the report.
Column Fields	Click to select the column field and the calculator type from the drop-down list. The report displays the column field as an active button. You can drag the button and drop it directly to the pivot table to redesign the report.
Candidate Fields	Click to select the candidate field and the calculator type from the drop-down list. The report displays the candidate field as an inactive button. You can drag the button and drop it directly to the pivot table to redesign the report. Once you have added this field to the pivot table, the button becomes active.

Fields	Description
Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display the desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.

Advanced settings tab

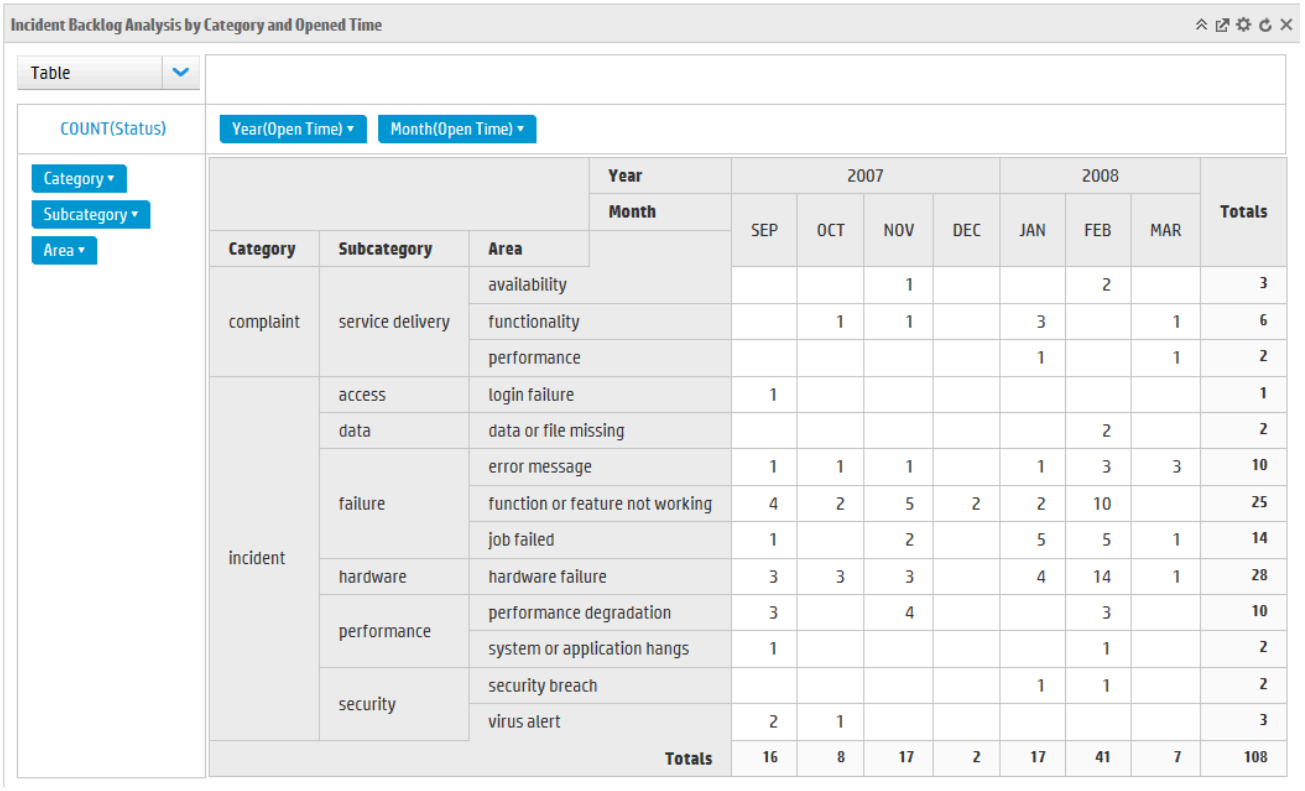
Fields	Description
Show Records	Specifies the permitted maximum number of records displayed in a report.
Decimal Places	Specifies the number of decimal places at which the report calculates the record percentage. Service Manager provides up to five decimal places.

Fields	Description
Drill Down Process	Specifies whether clicking a record in the report will drill down to the record stored in the `ToDo` table or to the record details page. • **Empty (default):** drill down to the record search results page. • **To-Do Drill Down:** drill down to the To Do Queue. Configure the settings as follows so that the **To-Do Drill Down** option can take effect: **Source Data:** `To Do - Todo` • **Change Drill Down:** drill down to the record stored in the `ToDo` table. Configure the settings as follows so that the **Change Drill Down** option can take effect: **Source Data:** `Change - cm3r` **Report type:** List **Show as:** Title List **Header Field:** Change ID • **Kmusagestats Drill Down:** drill down to the knowledge document record. Configure the settings as follows so that the **Kmusagestats Drill Down** option can take effect: **Source Data:** `kmusagestats - kmusagestats` **Report type:** List **Show as:** Standard List

Fields	Description

Note

You can only save the pivot definition change of the row, the column, and the candidate.



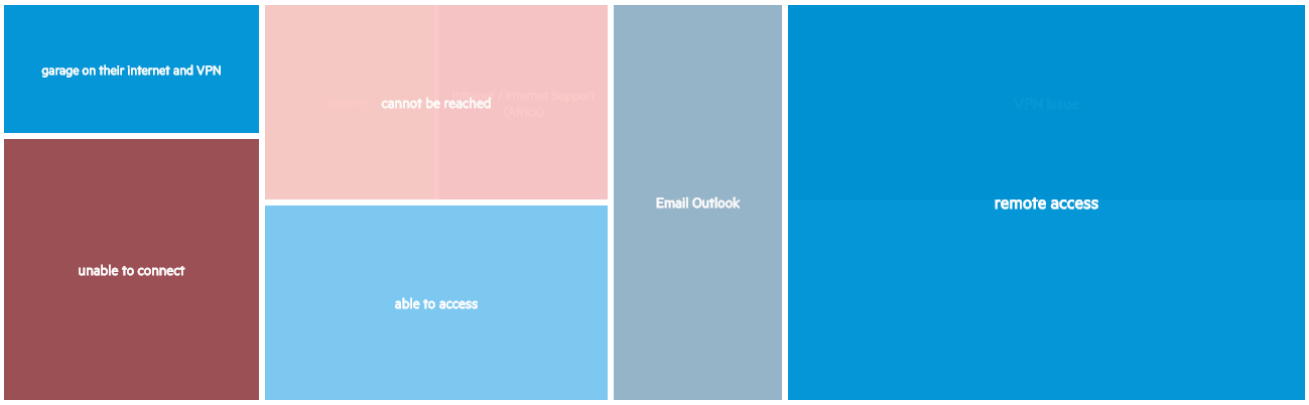
			Year	2007				2008			Totals
			Month	SEP	OCT	NOV	DEC	JAN	FEB	MAR	
Category	Subcategory	Area									
complaint	service delivery	availability				1			2		3
		functionality			1	1		3		1	6
		performance						1		1	2
incident	access	login failure		1							1
	data	data or file missing							2		2
	failure	error message		1	1	1		1	3	3	10
		function or feature not working		4	2	5	2	2	10		25
		job failed		1		2		5	5	1	14
	hardware	hardware failure		3	3	3		4	14	1	28
	performance	performance degradation		3		4			3		10
		system or application hangs		1					1		2
	security	security breach						1	1		2
		virus alert		2	1						3
			Totals	16	8	17	2	17	41	7	108

1.16.3.6.12. Hot Topic Map

After you have set up Smart Analytics in your Service Manager system, you can create Hot Topic Map reports. Here are the setting preferences and examples for your reference.

Hot Topic Map Properties tab

Fields	Description
Key Words	Type the keywords to filter the hot topics.
From	Specify a starting date to filter the data that are created after the specified date.
To	Specify an ending date to filter the data that are created before the specified date.
Analytics Actions	Filter data and available actions in hot topic map. The available conditions are: **Set Parent**, **Create Problem**, and **Create Change/Article**. **Note:** Service Manager displays this drop-down list only when you select **Incident - probsummary** in the Source Data field. To enable Hot Topic Analytics for more modules, see Enable Hot Topic Analytics for other modules. Also, to add more Analytics action queries in the Hot Topic Analytics for Incidents and other modules, see the steps described in Add more "Analytics Action" queries in the Hot Topic Analytics for Incidents as an example.
Customized Query	Shows the query content for the current report. You can click the **Query Builder** to define the query to display desired information. You can also write a new query directly. For more information about how to edit a query, see Build the query strings and other related topics.


garage on their internet and VPN
cannot be reached
Email Outlook
remote access
unable to connect
able to access

1.16.4. My dashboards

Service Manager displays all reports as objects in the Dashboard view. You can group reports to create a dashboard that offers a quick view of an entire system's performance data.

Dashboards provide dynamic reports in the web client which enable you to perform the following actions:

- Re-size and relocate a report

  You can resize a report by dragging the vertical border on the right, the horizontal border on the bottom, or the bottom right corner. You can relocate a report by drag-and-drop. In addition, press Ctrl and then drag-and-drop to swap the locations of two reports.

  Note: Do not drag-and-drop a report out of the dashboard. Otherwise, the report's borders may have overlap issues.

  The Dashboard view uses the responsive grid layout that helps you position and resize reports precisely. The grid is virtually consists of twelve aequilate columns and unlimited rows. The height of each row occupies fourty pixels. The minimum dimension of a report is three columns in width and four rows in height. When you position or resize a report, the system automatically aligns this report to the rows and columns. A report will rise if there is empty space above it. You can add up to four reports to each row in the Dashboard view.

- Remove a report

  To remove a report, click **Remove content from dashboard**. After you remove a report, the report in the same colum underneath moves up.

- Hover the mouse pointer to view data counts

  Pause the mouse pointer on the area of interest in a chart. For example, you can pause the mouse pointer on a pie chart to display a popup window that indicates the number of records that a slice of the pie represents.

- Refresh the dashboard to get updated reports based on database changes

- Drill down to a subset of the records, in order to display those records in greater detail

Create a dashboard

A dashboard enables you to view multiple reports about data in your system.

To create a dashboard, follow these steps:

1. In the System Navigator pane, select **Reporting** > **Create New Dashboard**.

   The **New Dashboard** button is also available on the **Search Dashboard** page and any dashboard view pages.

2. In the **New Dashboard** window, complete the necessary Dashboard properties.
3. Click **Save**. A unique dashboard ID is automatically assigned to this dashboard and displayed in the **Dashboard ID** field.
4. (Optional) Click **Dashboard Content** below the dashboard ID to open the dashboard and manage the dashboard contents. For more information about how to add reports to a dashboard, see **Add reports to a dashboard**.

Dashboard properties

Note

You must complete all fields marked with a red asterisk *.

The following fields and tabs are available both for defining a new report in the **New Dashboard** page and for editing an existing report in the **Search Dashboard** result page.

Fields and tabs	Description
Name (*)	Type a word or phrase that is a unique identifier for this dashboard. It should be a value that makes it easy to understand the purpose of the dashboard.
Description	Enter a description for the dashboard. It is helpful for other users to understand the purpose and objectives of this dashboard. The description is displayed in the dashboard properties in the right panel of the Reports page when that report is displayed.
Dashboard ID	The dashboard ID is automatically assigned to the dashboard and displayed in the **Dashboard ID** field after you save your dashboard. The **Dashboard Content** button is enabled at the same time for you to manage contents in the dashboard, such as adding or removing reports, dragging and dropping reports, or re-sizing reports. For more information about how to add reports to a dashboard, see **Add reports to a dashboard**.

Fields and tabs	Description
Permission	The default owner is the current operator. You can change the owner and set the audience of this dashboard. **Note** You need to have the privilege to change the permission. Report Manager can share the dashboard to others. Only the Report administrator can change the dashboard owner.

Add reports to a dashboard

A dashboard enables you to view multiple reports about data in your system.

Note

If the database changes, the dashboard contents change.

1. Open the dashboard you want to modify.
2. Click **Add Content**. A floating window is displayed to show the categories and the reports.

   There are several ways to filter the displayed reports. You can select different report categories or chart types, and sort the results by name or update time. You can also search a report by report ID, or report name, which is always case insensitive.

> **Note**
>
> The system doesn't display Hot Topic Map in the chart type drop-down list. However, you can search for a Hot Topic Map report by its report name or report ID, and then add this report to the dashboard.

Alternatively, if there is already a default dashboard, you can create a new report and add it directly to the dashboard. For more information, see Create a report.

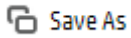
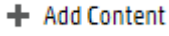
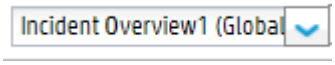
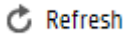
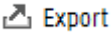
3. (Optional) Clear the **Add reports to the top of the dashboard** checkbox so that the system displays the newly added reports on the bottom of the dashboard.
4. Click the plus ("+") button beside a report to add this report to the dashboard. The button changes to a check mark after this report is added to the dashboard. By default, you can add up to sixteen reports to a dashboard. You can change the default value of **Max reports per dashboard** from **Reporting** > **Administration** > **Report Settings**.
5. Click **Close** when you finished adding the reports.

Service Manager saves the modified dashboard automatically.

Update a dashboard

Service Manager provides you with UI elements with which you can easily update the dashboard view. The following table describes the UI elements in a dashboard.

For the options on individual report, see Update a report in a dashboard.

UI Element	Description
	Creates a new dashboard.
	Saves the current dashboard as a new dashboard.
	Adds reports to the dashboard. It displays a report list and you can find, select, and add any report to the dashboard directly.
	Selects the dashboard view from the drop-down list. This list content is not automatically refreshed when you click the drop down button. Therefore, you need to click the refresh button before you can get the latest list of the dashboards you can view.
	Sets the dashboard view you have selected as the home page.
	Refreshes the dashboard. The dashboard reads the data from the database.
	Exports the dashboard as a PDF file. You can change the name of the exported file and select the orientation as portrait or landscape.
	Opens the dashboard definition page, which enables you to modify the dashboard properties.
	Prints the current dashboard.
	Add the current dashboard to Favorites.

Note

If your Service Manager Reports role can update a dashboard, Service Manager automatically saves the following layout changes: resizing, relocating or deleting a report, and rearranging fields in a pivot table.

Enable automatic refresh for a dashboard

Only when your report administrator has enabled the Auto Refresh functionality for dashboards in the front end, you have the option to enable automatic refresh for a specific dashboard so that the dashboard owner and audience can view up-to-date data in the dashboard.

Note

Unlike the **Refresh** button on the dashboard form, the Auto Refresh functionality refreshes only the data being displayed in each report of the dashboard. If the dashboard content is modified by someone else (that is, the dashboard has a report added or removed), Auto Refresh doesn't display this modification to you until you log off and log in again.

Note

You must have the Edit Dashboard permission to perform this task.

To enable automatic refresh for a dashboard, follow these steps:

1. Go to **Reporting** > **Search Dashboard** to open the dashboard.
2. Select the **Support Auto Refresh** checkbox on the right side.

   Note

   If your report administrator has disabled automatic refresh for all dashboards, this checkbox is hidden.

   The **Refresh Interval** field is then displayed.
3. In the **Refresh Interval** field, select an interval from the list. For example, select **Recurs every 2 minutes**.

Tip

Follow these steps to add more intervals to the list:

1. Click **Tailoring** > **Tailoring Tools** > **Global Lists**.
2. Search for the **Reporting Dashboard Refresh Interval** global list, and then append more values to the Value List field, the Display List field, and the SM message List field, respectively.
3. Save your changes.
4. Add the corresponding new messages to the scmessage table.

4. Click **Save**.

   Now, automatic refresh is enabled for the dashboard owner. To enable automatic refresh for the dashboard audience, your system administrator must select the **Enable dashboard auto refresh for audience** option. For more information, see Configure report settings.

1.16.5. Report schedule and distribution

Report schedules assist report managers or administrators to send the reports or the dashboards information on scheduled intervals to the notified users or user groups.

A report schedule specifies who is to be notified, when, and the notification method. An administrator must make sure that the report or dashboard has been created in Service Manager before Service Manager can create a report schedule record.

Scheduled reports or dashboards enable users to view the updated content without logging in to Service Manager. Service Manager Reports provide two ways to distribute the schedule reports or dashboards: sending email notifications by attaching the generated PDF files, or using the file server to save the generated files.

For example, if you want to embed the scheduled reports to your applications, you can follow these steps:

1. Generate scheduled reports and specify a folder for these reports.
2. Set up an Apache instance to access the folder and provide static URLs for the reports.
3. Embed the Apache URLs in the products.

In this way, users can view the updated content from your applications timely and directly.

Create a report schedule

Applies to User roles: Report Manager

To create a new report schedule, follow these steps:

1. Click **Reporting** > **Report Schedule** > **Create New Schedule**.
2. Type the name of the new report schedule in the **Name** field.
3. Select the **Owner** for this report schedule. The owner is the operator by default. You can change this value only if you are a Report Administrator.
4. Type the description of the new report schedule in the **Description** field.
5. Select the dashboards or reports for which you want to schedule notifications. By default, you can add up to five reports and dashboards to a scheduled task. You can change the default value of **Max reports/dashboards per scheduled task** from **Reporting** > **Administration** > **Report Settings**.

6. Selects the **Export to** method to send the notification. There are two methods available.
    - **Email (PDF)**: Email notification by sending the PDF files

      After you select **Email (PDF)**, the **Use Notification** checkbox becomes available. You can select this checkbox and then select a notification definition from the Combo Box to send emails by following the notification rules.

      To add a new notification definition record, click the **View Detailed Information** button in the Combo Box and then follow the instructions as documented in Add a notification definition record. Note that you must select **email** in the Notify Method column.

Note

- When you select the **Use Notification** checkbox, the **Email** tab is disabled and hidden.
- When sending an email using a customized HTML template for the exported reports, be sure to add the following codes to the Expressions tab of the HTML template. Otherwise, the system doesn't attach the generated report to the email.

```
// please always keep "##schedule task
id:**##" in email_title
// It will not be displayed in email and
will be discarded in
eventout trigger
email_title += "##Schedule Task ID:"+
$RECORD.id+"##"
```

- When sending an email using a format for the exported reports, be sure to add the following codes to the Text field of the message record.

```
##Schedule Task ID:%S##
```

For example, the value in the Text field is `format test` in a format. To support email notification using this format, you must add `##Schedule Task ID:%S##` to the Text field. The new value for the Text field is `format test ##Schedule Task ID:%S##`.

- **File Server**: Save the generated files to a file server.

  After you select **File Server**, the **File Type to Save** checkboxes become available. You can select either **PDF** or **HTML** file type to save the notification.

Note

- If you selected a report of the List type, the number of records that can be exported from this report is limited by the **Max number of records from a list report that can be exported in the back end** setting. See Configure report settings.
- If you have configured any colored ranges for reports, these are included in the exported file.

7. Selects the page direction of the exported file. Two directions are available: **Portrait** and **Landscape**.
8. On the **Rule** tab, set the rules for the schedule.

Field	Description
On an arbitrary schedule	If this option is selected, the regular schedule definition is not displayed and you need to provide a specific time for the schedule.
Range of recurrence	Specifies the effective period of this report schedule.
Recurrence	Specifies the recurrence of this report schedule.
Start time	Specifies the start time of this report schedule.

9. On the **Email** tab, set the properties for the Email notification.

Field	Description
Subject	The name of the Email notification that's sent to the recipients when the action occurs.
Email To/ Groups	The user or user group who receive the email notification.

10. The **Workflow** tab shows the status of the current report schedule. There are three types of report schedule phases: **Planning**, **Scheduled**, and **Retired**.

    When a report schedule is in the Scheduled phase, the **Next Event** and **Recent Histories (30 days)** tabs become available. The **Next Event** tab shows the next event for this report schedule. The **Recent Histories (30 days)** tab lists the tasks which have been run during the last 30 days and shows the task detail message.

11. The **Activities** tab shows the activities related to this report schedule. The activities show information about who has created or updated the report schedule, and at what time.

12. Click **Add**. A unique report ID is automatically assigned to this report and displayed in the **Schedule ID** field.

To update an existing report schedule, follow these steps:

1. Click **Reporting** > **Report Schedule** > **Search Schedule**.

    Note

    Report Administrator can search all report schedules while others can only search their own schedules.

2. Use search or advanced search to find one or more record schedules. A list of report schedules opens.

3. Select a report schedule from the list.

4. Update the required information.

    The following list describes the phases in the schedule lifecycle and their corresponding system actions:

    - Planning → Scheduled: the task will be generated and the schedule can't be updated
    - Scheduled → Planning: the scheduled task will be removed and the schedule is editable again
    - Scheduled → Retired: the scheduled task will be removed and an email notification will be sent to the schedule owner
    - Planning → Retired: an email notification will be sent to the schedule owner
    - Retired → Planning: the schedule can be updated and saved

5. Click **Save**. A message appears that confirms that the record schedule is updated.

Report background schedule process

You can start a background schedule process by running the **sm report.export** command. Make sure you have already specified the path to save the exported reports and dashboards by adding the *dashboard_export_path* parameter to the sm.ini file.

> **Note**
>
> If you set too many threads for one process, you have to add the JVMOption parameter to expand the memory capacity. For example, you can start the process by running the following command:
>
> **sm -JVMOption0:-Xms1024m -JVMOption1:-Xmx1024m report.export**

To check the running status of the background schedule process, click **Miscellaneous** > **System Status**.

There are six types of threads that will run:

- report.export.misc
  - Cleans the report performance log
  - Regenerates the schedule task in 30 days
  - Moves the task from **reportscheduletask** to **reportschedulehistory**
- report.export.distribute

  Distributes the reports or the dashboards by email or to the file server.

  > **Note**
  >
  > The *clean:1* parameter excludes a message from email notifications about exported reports or dashboards. See Startup parameter: emailout. By default, this message is included.
- report.export.prepare

  Prepares the data to generate the reports and dashboards.
- report.export

  This schedule thread will synchronize the status of the other three types of schedule threads. Restart the schedule thread if it stops responding. Make sure you set the correct maximum counts of running threads. You can change the

default value of **Max report-exporting threads per process** and **Max report-preparing threads per process** from **Reporting** > **Administration** > **Report Settings**.

- report.calccharts

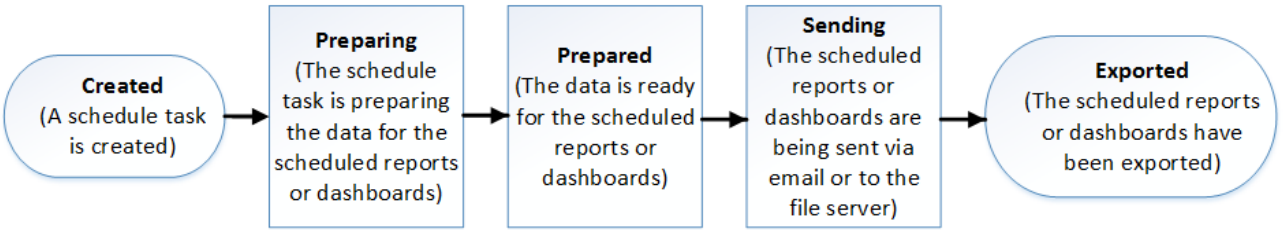
  Scans the system every one minute to check if there is any new report or updated report for calculation.

- timedurationdata

  Extracts the raw time stamp information from the source tables. Service Manager will process the data and save the change metrics in the intermediate tables, for example, in the timeDuration4probsummary table.

Report schedule task lifecycle

Service Manager creates the schedule tasks for a report schedule. A schedule record can generate several schedule tasks. Each schedule task includes lifecycle states relating to the process from task created, to preparing report data, and to exporting report data, as shown in the following figure.



The state transition from **Created** to **Preparing** can be interrupted when the administrator:

- Disables reporting
- Stops sending reports or dashboards via email, if the scheduled reports are sent only via email

The state transition from **Prepared** to **Sending** can be interrupted when the administrator:

- Stops sending reports or dashboards via email, if the scheduled reports are sent only via email.
- Does not specify a location for the exported reports, if the scheduled reports are only exported to the file server

Note

- Service Manager can't interrupt the state transition from **Preparing** to **Prepared** or the state transition from **Sending** to **Exported**.
- If you disable reporting, it will prevent Service Manager from generating any more schedule tasks.

Apply a template to complete a report schedule

Templates enable users to quickly complete a report schedule by automatically populating fields with the applicable information.

To apply a template to complete a report schedule, follow these steps:

1. Open a report schedule.
2. Click **Apply Template**. The **Select Report Schedule Template** wizard opens.
3. Double-click the template that you want to apply to the selected report schedule. You are returned to the selected report schedule, and the fields are automatically filled in with the values set in the template.
4. Complete the other fields, as needed.
5. Click **Save & Exit**.

Note: Users authorized to use this template can also modify the template to meet their needs.

1.16.6. Report administration

Report administration covers these areas of responsibility:

- Report categories
- Report running status
- Report settings
- Time duration settings

1.16.6.1. Report category

Report category enables you to collect reports into different groups. The Service Manager Reports feature requires a category value when it creates a new report, or when it searches for a report to be added to the dashboard.

Service Manager defines the following out-of-box report categories.

Name	Description	Applies To (Table Name)
Alert	Alert	AlertLog Alert
Approval	Approval	Approval ApprovalLog ApprovalDelegation

Name	Description	Applies To (Table Name)
Assignment Groups	Assignment Groups	assignment ocmgroups cm3groups subcategory producttype cm3rcategory cm3rsubcat cm3tcategory sdCategory sdSubcategory sdArea pbmCategory pbmSubcategory pbmArea pbmTaskCat imCategory imSubcategory imArea
Change	Change Management	cm3r cm3t activitycm3r activitycm3t

Name	Description	Applies To (Table Name)
Configuration Item	Configuration Item	device Subscription cirelationship
Contracts and Payments	Contracts and Payments	contract payment
Data Management	Data Management	contacts dept operator company city location
Incident	Incident Management	probsummary imTask activityincidenttasks activity

Name	Description	Applies To (Table Name)
KM	Knowledge Management	kmdocument kmfeedback kmsearchhistory kmstatus kmupdateinfo kmusagestats
Problem	Problem Management	rootcause knownerror rootcausetask activityproblem activityproblemtasks knownerrortask
Report Monitor	Report Monitor	reportLog
Request	Request Management	request requestTask acitivityRequest acitivityRequestTask

Name	Description	Applies To (Table Name)
SLA	Service Level Management	sla slo slaresponse sloresponse slocatalog sloavail slamonthlyag slaactive slamonthly outagedetail outageevent
Service Catalog	Service Catalog	svcCatalog userOption svcDisplay svcCart svcCartItem
Service Desk	Service Desk	incidents activityservicemgt
Survey	Survey	SurveyInternal SurveyDefinition

Name	Description	Applies To (Table Name)
Templates and Models	Templates and Models	Template changeModel requestModel productCatalog
ToDo	ToDo	Todo
Work Schedule	Work Schedule	caldutyhours calholidays calholtable timeperiodDefinition timeperiodOccurrence timeperiodCategory timeperiodConflict

Add a report category

Report Administrator can create, update, and delete a report category. Report Designer, Report Manager, and Report Administrator can categorize a report, which enables users to find a report easily when adding content.

To create a new report category, follow these steps:

1. Click **Reporting** > **Administration** > **Report Categories**.
2. Type the name of the new report category in the **Name** field.
3. Type the description of the new report category in the **Description** field.
4. Select the table names to be grouped in this category from the drop-down list in the **Applies To** field.
5. Click **New**.
6. Click **Save**. A message appears that confirms that the report category record is added.

To update an existing report category, follow these steps:

1. Click **Reporting** > **Administration** > **Report Categories**.
2. Use search or advanced search to find one or more records. A list of report categories opens.
3. Select a report category from the list.
4. Update the required information.
5. Click **Save**. A message appears that confirms that the report category record is updated.

Note

If you type a new report category when creating a report, this report category is only local and you can see this category only when you are adding a report to a dashboard.

1.16.6.2. Configure report settings

Administrators can configure some global settings for the Service Manager Report module.

To configure the report settings, follow these steps:

1. Click **Reporting** > **Administration** > **Report Settings**.

   The Report Settings form opens, in which Report Module settings and report property fields are automatically populated.

2. Specify the settings on the form, as described in the following table.

Attribute	Description	Default value
Enable dashboard auto refresh in the front end	This option shows or hides the **Support Auto Refresh** checkbox in each dashboard. ◦ If this option is selected, the checkbox is displayed. Users can then enable automatic refresh for a specific dashboard and then select a refresh interval. ◦ If this option is not selected, the checkbox is hidden in each dashboard and automatic refresh is disabled for all dashboards.	Disabled

Attribute	Description	Default value
Enable dashboard auto refresh for audience	This option enables a dashboard that has auto refresh enabled to be automatically refreshed for the audience of the dashboard. If this option is selected, when the report data contained in the dashboard has changed, the dashboard is automatically refreshed for the dashboard audience based on the specified refresh interval.	Disabled

Attribute	Description	Default value
	Note For performance considerations, this option is disabled by default and the dashboard is automatically refreshed only for the dashboard owner.	
Enable processing for time duration reporting data	If this option is selected, the Service Manager time duration reporting data is processed by the system and ready for use. Users can select tables containing the processed data to generate reports. If this option is not selected, all tables for time duration reporting are disabled.	Enabled

Attribute	Description	Default value
Enable reporting	If this option is selected, the Service Manager Reports feature is enabled and users can access the Service Manager Reports module from the navigation pane. If this option is not selected, the Service Manager Reports module is invisible from the navigation pane and only the administrator can enable it by selecting this option.	Enabled
Export HTML reports to file server by default	If this option is selected, Service Manager generates HTML files while exporting the scheduled reports or dashboards to the file server.	Enabled
Export PDF reports to file server by default	If this option is selected, Service Manager generates PDF files while exporting the scheduled reports or dashboards to the file server.	Disabled

Attribute	Description	Default value
Export reports to file server by default	If this option is selected, Service Manager sends the exported reports or dashboards to the file server by default.	Disabled
Export reports via email by default	If this option is selected, Service Manager sends the exported reports or dashboards via email by default.	Enabled
Stop exporting reports via email	If this option is selected, the system stops sending any reports or dashboards via email. If the scheduled report is sent via email only, selecting this option will terminate all scheduled tasks.	Disabled

Attribute	Description	Default value
Use production database when replicated database is disabled	If this option is selected, Service Manager disables Replicated Database and uses Production Database to generate the reports. When Replicated Database is disabled, or if there is any problem with Replicated Database, selecting this option can switch all the reports to Production Database. **Note** Switching all the reports to Production Database can increase the number of queries to Production Database and thus have an impact on system performance.	Disabled

Attribute	Description	Default value
Use replicated database by default when users create report	If this option is selected, the user is creating a new report by using the Replicated Database by default, which will enable Service Manager to generate the report data from Replicated Database. Make sure that a Replicated Database is already set up and enabled before this option takes effect. For more information about how to enable Replicated Database, see Enable Replicated Database.	Disabled
Default max records displayed in a report	Specifies the default maximum number of records displayed in a report.	12
Default decimal places	Specifies the default decimal places in reports.	2
Default donut value for pie chart (0-90)	Specifies the default donut value for pie charts.	35

Attribute	Description	Default value
Max number of active dashboards that can automatically refresh	Specifies, at the system level, the maximum number of active dashboards that can automatically refresh. **Note** When two or more users are viewing the same dashboard concurrently, it is regarded as one dashboard.	5
Max number of records from a list report that can be exported in the back end	Specifies the maximum number of records that a report schedule can export from a report of the List type.	200

Attribute	Description	Default value
Discard scheduled tasks expired more than (days)	Specifies the number of days before expired tasks are discarded. The default value for this is one, which means if a scheduled task has expired for more than one day, then the task is discarded.	1
Max background calculation threads per process	Specifies the maximum number of background threads per process to calculate formulas defined for the report generation.	1
Max records to process each time	Specifies the maximum number of records to process each time. The records to be processed are used for time duration.	200

Attribute	Description	Default value
Max report-exporting threads per process	Specifies the maximum number of threads per process to execute export operations. The default value for this is 1 and you need to add more threads when many reports or dashboards are scheduled. When Service Manager performs too many scheduled tasks, it requires more threads to perform different tasks at the same time. Allowing too many threads per process could also have an impact on system performance.	1

Attribute	Description	Default value
Max report-preparing threads per process	Specifies the maximum number of threads per process to prepare reports or dashboards data. The default value for this is 1 and you need to add more threads when many reports or dashboards are scheduled. When Service Manager performs too many scheduled tasks, it requires more threads to perform different tasks at the same time. Allowing too many threads per process could also have an impact on system performance.	1
Max reports per dashboard	Specifies the maximum number of reports allowed in a single dashboard.	16
Max reports/dashboards per scheduled task	Specifies the maximum number of reports and dashboards allowed in each scheduled task. This number controls the attachments in an email.	5
Report performance log clean up interval (days)	Specifies the interval to clean up the report performance log.	365

Attribute	Description	Default value
Time threshold for report generation in production database (milliseconds)	Specifies the time threshold for report generation in Production Database in milliseconds. Service Manager monitors the time spent on report generation and provides the information in the report performance log. If the actual time spent on generating a report exceeds this time threshold, this report is displayed as a slow report on the **Report Running Status** dashboard. For more information about how to handle the slow reports, see Monitor report running status .	500

Attribute	Description	Default value
Time threshold for report generation in Replicated Database (milliseconds)	Specifies the time threshold for report generation in the Replicated Database in milliseconds. Service Manager monitors the time spent on report generation and provides the information in the report performance log. If the actual time spent on generating a report exceeds this time threshold, this report is displayed as a slow report on the **Report Running Status** dashboard. For more information about how to handle the slow reports, see Monitor report running status .	500

Attribute	Description	Default value
Location for exported reports	Specify a file server directory in which report users can access the generated reports. When a user creates a report schedule and saves it, a message is sent to the user (who may or may not be the report schedule owner), indicating that the generated reports can be found in this directory. **Note** If this location is left empty, the user will not receive this message. For example, you can: ◦ Specify this attribute as C:\Reports, which is same with the parameter specified in Startup parameter: dashboard_export_path.	None

Attribute	Description	Default value
	◦ Specify this attribute as \\<*host name of another file server*>\Reports In this case, the generated reports are stored in the Reports folder on another file server for user access.	

3. Click **Save**.

1.16.6.3. Monitor report running status

The **Report Running Status** dashboard helps you to identify whether Service Manager is working well when the system is generating reports. This prevents the system from returning errors if the status is showing that generating some reports has a significant impact on the system performance.

Service Manager monitors performance impact of reports and dashboards. Service Manager takes some time to generate reports. If this time exceeds the time threshold that the administrator sets, the report performance log collects the information and displays the slow reports information on the **Report Running Status** dashboard. If Service Manager spends too much time loading one of the slow reports, the loading of the dashboard that includes this report could be very slow.

You can click **Disable this Report** to disable the slow reports displayed on the Report Running Status dashboard directly. Or, you can click **Open Report** to diagnose the reason, and then decide whether to modify the report or to disable the report. For those reports that are diagnosed as the cause of the problem, the administrator can take the following measures to eliminate the impact of this particular report.

- Enable cache. For details, see Enable query cache.
- Enable Replicated Database. For details, see Enable Replicated Database.
- Enable indexing for the table query

If the problem still exists, then you can disable the report. Service Manager sends a notification to the owner of the report.

1.16.6.4. Enable Replicated Database

Generating the reports from the server will consume additional system resources (memory and CPU). Service Manager provides the database replication to preserve Production Database performance.

Before you enable Replicated Database for Service Manager Reports, prepare a database with the same database type and version as the Production Database. For example, if your Service Manager uses an SQL Server database, the replicated database should be an SQL Server database of the same version. Perform database synchronization periodically by using the standard database synchronization mechanism.

To enable Replicated Database, follow these steps:

1. Open the `sm.ini` file in the `<Service Manager installation path>\Server\RUN` folder.
2. Add the `Replicated Database` parameter and value to the file. For example:

   ```
   [your database type_replicate]
   sqldb:sm980db
   sqllogin:username/password
   dashboardonreplicatedb
   ```

   In this example:

   Change `your database type` to one of the following values: `oracle`, `sqlserver`, and `postgresql`.

   The `sqllogin` parameter is mandatory.

   The three parameter lines of `Replicated Database` should be in one block in the `sm.ini` file.
3. Save the `sm.ini` file.

1.16.6.5. Enable query cache

For system performance consideration, Service Manager enables the query cache by default. When the query cache is enabled, Service Manager creates a cache table in the Production Database to save query results and the query SQL. Both the queries against the Production Database and Replicated Database are all saved in the cache table

Note

When the query result is expired or you refresh the report or dashboard, Service Manager will retrieve the data by executing the query from the Production Database or Replicated Database again.

To enable the query cache, follow these steps:

1. Open the sm.ini file. By default, the file is located on the Service Manager server in the *<Service Manager installation directory>*\Server\RUN folder.
2. Enter the query cache parameters and values in the file. Service Manager enables query cache by default and the default values for the parameters are shown as follows:

   ```
   dashboardquerycache_enable:1
   dashboardquerycache_dbtime:100
   dashboardquerycache_expire:10
   ```

3. Save the sm.ini file.

1.16.6.6. Define the colors used for colored ranges

The colors that are available for selection when users create colored ranges are customizable. By default, four colors are available (gray, green, yellow, and red). However, you can add more colors, edit existing colors, and remove colors.

Add a new color

To add a new color to the drop-down list that's displayed when users create colored ranges in reports, follow these steps:

1. Type db in the command box, and then press Enter.
2. Enter scmessage in the Table field and then click the Search icon.
3. Select **message.edit** in the **Format Name** column.
4. In the **Class** field, type Global.
5. In the **Message Number** field, enter chart.*<color name>*. For example, type chart.black.
6. In the **Text** field, type the name of the color as you want it to be displayed on the UI. For example, type Black.
7. Click **Add** to add this record to the scmessage table.
8. Type gl in the command box, and then press Enter.
9. Enter Chart Threshold Color Indicator in the **List Name** field, and then click **Search** to open the Chart Threshold Color Indicator global list.
10. Add new number to the comma-delimited list in the **Value List** field. For example, update `{"1","2","3","4"}` to `{"1","2","3","4","5"}`.
11. In the **Display List** field, enter the value that you entered in the **Message Number** field of the scmessage table in *step 5*. For example, type chart.black.
12. Click **Save**, click **Rebuild Global List** in the **More** menu, and then click **Save** again. The color is then added to the list in the SM message List field.
13. Type db in the command box, and then press Enter.
14. Enter colorindicatorsetting in the Table field and then click the Search icon.
15. Select **colorindicatorsetting** in the **Format Name** column.
16. Enter Inbox in the **File** field, and then click **Search**.
17. Click **Add** to open the Color Indicator Setting Wizard.
18. In the **Value** field, enter the name of the color that you want to add.
19. In the **Foreground Color** field, specify the foreground color. To do this, enter the hex code of the color, or you click the text field, and then select a color from the color palette.

20. In the **Background Color** field, specify the background color (this is the color that's displayed in reports). To do this, enter the hex code of the color, or you click the text field, and then select a color from the color palette.
21. Click **OK** to add the color to the table, and then click **Save** to save your changes.
22. Log out and then log back in to Service Manager for your changes to take effect in reports.

Modify an existing color

To modify a color in the drop-down list that's displayed when users create colored ranges in reports, follow these steps:

1. Type db in the command box, and then press Enter.
2. Enter colorindicatorsetting in the Table field and then click the Search icon.
3. Select **colorindicatorsetting** in the **Format Name** column.
4. Enter Inbox in the **File** field, and then click **Search**.
5. Click **Edit** beside the color that you want to modify to open the Color Indicator Setting Wizard.
6. Modify the colors in the **Foreground Color** and **Background Color** fields as desired. To do this, enter the hex code of the color, or you click the text field, and then select a color from the color palette.
7. Click **OK**, and then click **Save** to save your changes.
8. Log out and then log back in to Service Manager for your changes to take effect in reports.

Remove an existing color

Caution

As a general best practice, we recommend that you don't remove colors that are currently used in reports.

To remove a color from the drop-down list that's displayed when users create colored ranges in reports, follow these steps:

1. Type gl in the command box, and then press Enter.
2. Enter Chart Threshold Color Indicator in the **List Name** field, and then click **Search** to open the Chart Threshold Color Indicator global list.
3. From the following fields, delete the values that correspond to the color you want to remove:
   - **Value List**
   - **Display List**

    - **Message Number**
4. Click **Save**, click **Rebuild Global List** in the **More** menu, and then click **Save** again. The color is then removed from the list in the SM message List field.
5. Type db in the command box, and then press Enter.
6. Enter colorindicatorsetting in the Table field and then click the Search icon.
7. Select **colorindicatorsetting** in the **Format Name** column.
8. Enter Inbox in the **File** field, and then click **Search**.
9. Click **Delete** beside the color that you want to remove from the drop-down list.
10. Click **Save** to save your changes.
11. Log out and then log back in to Service Manager for your changes to take effect in reports.

1.16.7. Report descriptions and usage

Some of the out-of-box (OOB) reports are listed here to provide users with realistic examples of business reports requirements. All reports listed here can be executed on an out-of-box Service Manager system. The reports are intended to be used as a starting point for the development of more detailed, customer-specific reports.

Incident Management reports

Report name	Report description	Chart type
Open Incidents Assigned to Me - Breached within 3 Days	This report provides the number of all open incidents that are assigned to the user and will be breached within three days.	Number
Incidents Assigned to Me by Priority	This report enables the user to review the number of open incidents that are assigned to the user, sorted by priority.	Pie
Open Incidents Assigned to My Groups by Assignee and Status	This report enables the user to review the number of open incidents that are assigned to the user's groups, sorted by assignee and then by status.	Stacked horizontal bar
Open Incidents without Assignee Assigned to My Groups	This report enables the user to review the number of open incidents without assignee information that are assigned to the user's groups, sorted by priority.	Pie
Top 5 Affected Services with Open Incidents assigned to My Groups	This report enables the user to review the top five affected services that have open incidents assigned to the user's groups.	Vertical bar

Report name	Report description	Chart type
Open Incidents Assigned to My Groups - Breached within 3 Days	This report enables the user to review all open incidents that are assigned to the user's groups and will be breached within three days.	Title list
High Priority Incidents Open longer than 7 Days	This report enables the user to review all the high priority incidents that were opened more than seven days ago.	Title list
Incidents Opened within This Half Year by Month	This report enables the user to review all incidents opened in each month within this half of the year.	Line
Incidents Closed within This Half Year by Month	This report enables the user to review all incidents closed in each month within this half of the year.	Line
Incident Backlog	This report provides the number of all incidents that aren't resolved yet.	Number
Open incidents Group by category and Area	This report enables the user to review all open incidents, sorted by category and then by area.	Stacked vertical bar

Report name	Report description	Chart type
Incidents Distribution on Affected Services - Opened within This Half Year	This report enables the user to review the distribution over the affected services of incidents opened within this half of the year.	Horizontal bar
Closed Incidents meeting SLA Target By Closed Month	This report enables the user to review all the closed incidents that meet SLA target, sorted by their closed month.	Area
Open Incidents Breached Today	This report provides the number of all the open incidents that will be breached today.	Number
Open Incidents Breached Already by Status and Priority	This report enables the user to review the number of all open incidents that have been breached already, sorted by status and then by priority.	Stacked vertical bar
Major Open Incidents by Service and Assignment Group	This report enables the user to review the number of all major open incidents, sorted by the affected services and then by the assignment group.	Horizontal bar

Report name	Report description	Chart type
Incident Backlog Analysis by Category and Opened Time	This report enables the user to review incident backlog analysis, sorted by category and then by the opened time.	Pivot Table
Incident Backlog by Priority	This report enables the user to review incident backlog, sorted by priority.	Pie
Closed Incidents Assigned to My Groups	This report enables the user to review the list of closed incidents assigned to the user's groups.	Standard list
Closed Incidents Assigned to My Groups by Priority	This report enables the user to review closed incidents assigned to the user's groups, sorted by priority.	Pie
Closed Incidents Assigned to Me by Priority	This report enables the user to review closed incidents assigned to the user, sorted by priority.	Pie

Change Management reports

Report name	Report description	Chart type
Open Changes Assigned to My Groups by Service and Phase	This report enables the user to review the number of open changes that are assigned to the user's groups, sorted by service and then by phase.	Stacked vertical bar
Open Changes Assigned to Me by Scheduled Implementation Start Date	This report enables the user to review the number of open changes assigned to the user, sorted by the scheduled implementation start date.	Vertical bar
Open Changes Assigned to My Groups without Assignee by Priority	This report enables the user to review the number of open changes assigned to the user's groups without the assignee information, sorted by priority.	Pie
Open Changes Assigned to My Groups by Phase and Priority	This report enables the user to review the number of open changes assigned to the user's groups, sorted by phase and then by priority.	Stacked vertical bar
Changes Closed Today by Me	This report provides the number of changes that are closed today and closed by the user.	Number

Report name	Report description	Chart type
Changes Assigned to Me and Opened Today	This report provides the number of changes that are assigned to the user and opened today.	Number
Changes Closed by Month within This Half Year	This report enables the user to review the trend of changes closed in each month within this half of the year.	Line
Changes Opened by Month within This Half Year	This report enables the user to review the trend of changes opened in each month within this half of the year.	Line
Number of Successful Changes by Closed Month	This report enables the user to review the trend of the number of successful changes, sorted by their closed month.	Line
Emergency Changes by Opened Month	This report enables the user to review the trend of emergency changes, sorted by their opened month.	Line
Failed Changes within This Half Year	This report enables the user to review the list of all failed changes within this half of the year.	Title list

Report name	Report description	Chart type
Overdue Changes	This report enables the user to review the list of all change requests that have attained an overdue escalation value.	Title list
Changes Open longer than 1 Week	This report provides the number of all change requests that have stayed open for longer than one week.	Number
Closed Changes Analysis	This report enables the user to review the pivot table of closed changes analysis, sorted by the completion code, the affected services, the open date, and the closed date.	Pivot table
Changes Opened within This Half Year by Month and Category	This report enables the user to review the changes opened within this half of the year, sorted by month and then by category.	Stacked vertical bar
Parallel Changes to Involved CIs within the Last 7 Days	This report enables the user to review parallel changes to the involved CIs within the last seven days.	Vertical bar

Report name	Report description	Chart type
Open Changes by Assignment Group	This report enables the user to review the number of open changes, sorted by the assignment group.	Pie
Changes Opened within This Half Year by Category and Status	This report enables the user to review the changes opened within this half of the year, sorted by category and then by status.	Stacked vertical bar
Changes Awaiting Approval	This report provides the number of all the changes that are awaiting approval.	Number
Changes Scheduled for This Week	This report provides an overview of changes that are scheduled for this week.	Title list
Critical Open Changes	This report enables the user to review all open change requests that have critical priority.	Title list
Number of Successful Changes by Closed Month	This report enables the user to review the trend of successful changes, sorted by their closed month.	Line
Changes Scheduled for the Next 30 Days	This report provides an overview of changes that are scheduled next 30 days.	Title list

Report name	Report description	Chart type
Changes Opened Within This Half Year by Category and Status	This report enables the user to review the number of changes that are opened this half year, sorted by category and then by status.	Stacked vertical bar
Overall change risk distribution for open changes	This report enables the user to review the risk distribution for all open changes.	Pie
Risk of successful vs. unsuccessful changes	This report enables the user to review the risk level of successful changes compared with unsuccessful ones, to see which kind of changes are more successful.	Vertical bar
Average risk trend of normal changes	This report enables the user to review the risk score of Normal Changes. It also defines the goal, to identify which changes are of higher risk than the planned goal.	Line

Report name	Report description	Chart type
Average risk trend of standard changes	This report enables the user to review the risk score of Standard Changes. Since standard changes are by nature of low risks, this report also defines the goal to identify which changes are of higher risk than the planned goal.	Line
Open changes with more than one risk assessment	This report enables the user to review the change requests where the risk assessment was executed more than once.	List
Changes with collisions	This report enables the user to review the lists of changes which have a collision.	List
Risk result for open changes per service	This report enables the user to review the risk level of all open changes group by the affected service.	Pivot Table

Problem Management reports

Report title	Report description	Chart type
Open Problems Assigned to Me by Priority	This report enables the user to review the number of open problems assigned to the user, sorted by priority.	Pie
Open Problems Assigned to My Groups by Assignee	This report enables the user to review the number of open problems assigned to the user's groups, sorted by the assignee.	Pie
Open Problem Tickets Owned by Me by Subcategory	This report enables the user to review the number of open problem tickets owned by the user, sorted by subcategory.	Horizontal bar
Total Count of Open Problem Tickets Owned by Me	This report provides the number of open problem tickets owned by the user.	Number
Open Problems Assigned to My Groups without Assignee	This report enables the user to review the number of open problems assigned to the user's groups without the assignee information.	Number

Report title	Report description	Chart type
Open Problems Assigned to My Groups without Assignee by Service	This report enables the user to review the number of open problems assigned to the user's groups without the assignee information, sorted by the affected service.	Horizontal bar
Open Problems by Subcategory	This report enables the user to review the number of open problems, sorted by subcategory.	Vertical bar
Open Problems by Service	This report enables the user to review the number of open problems, sorted by service.	Vertical bar
Problems Open by Month within This Half Year by Priority	This report enables the user to review the trend of problems open in each month within this half of the year, sorted by priority.	Area
Problems Closed by Month Within This Half Year by Priority	This report enables the user to review the trend of problems closed in each month within this half of year, sorted by priority.	Line

Report title	Report description	Chart type
Open Problems by Priority	This report enables the user to review the number of open problems, sorted by priority.	Pie
Open Problems by Assignment Groups	This report enables the user to review the number of open problems, sorted by the assignment group.	Vertical bar
Problem Backlog	This report provides the number of all open problems.	Number
Deferred Problems	This report enables the user to review all deferred problems with the problem ID, the title, and the service information.	Title list

Service Desk reports

Report name	Report description	Chart type
All Interactions Pending My Approval	This report provides the total number of all interactions pending on the user's approval.	Number
Open Interactions Assigned to Me by Priority	This report enables the user to review all open interactions assigned to the user, sorted by priority.	Pie
Open Interactions Assigned to Me - Breached within 1 Day	This report enables the user to review all open interactions that are assigned to the user that will be breached within one day.	Title list
Open Interactions Assigned to Me by Category	This report enables the user to review the interaction distribution chart for all open interactions assigned to the user, sorted by category.	Pie
Interaction Backlog	This report provides the number of all open interactions.	Number
Interaction Backlog Analysis by Category and Subcategory	This report enables the user to review the number of all open interactions, sorted by category and then by subcategory.	Stacked vertical bar

Report name	Report description	Chart type
Fulfilled and Aborted Interactions Within This Half Year	This report enables the user to review the number of all fulfilled and aborted interactions within this half of the year.	Vertical bar
Open Interactions within This Half Year by Status	This report enables the user to review the number of all interactions open within this half of the year, sorted by status.	Pie
Open Interactions by Priority	This report enables the user to review the number of all open interactions, sorted by priority.	Pie
Breached Interactions by Category and Subcategory	This report enables the user to review the number of all breached interactions, sorted by category and then by subcategory.	Title list
Interactions Breached within 1 Day	This report enables the user to review all interactions that will be breached within one day.	Title list
Interactions Awaiting Approval	This report enables the user to review all interactions that await for approval.	Title list

Report name	Report description	Chart type
Open Interactions Assigned to My Groups	This report enables the user to review the list of all open interactions assigned to the user's groups.	Standard list
Closed Interactions Assigned to My Groups	This report enables the user to review the list of all closed interactions assigned to the user's groups.	Standard list
Open Interactions Assigned to My Groups by Priority	This report enables the user to review all open interactions assigned to the user's groups, sorted by priority.	Pie
Closed Interactions Assigned to My Groups by Priority	This report enables the user to review all closed interactions assigned to the user's groups, sorted by priority.	Pie
Closed Interactions by Assignment Group	This report enables the user to review all closed interactions, sorted by the assignment group.	Vertical bar

Request Fulfillment reports

Report name	Report description	Chart type
Requests Awaiting Approval	This report enables the user to review the requests that are waiting for approval.	Number
Open Requests by Phase	This report enables the user to review the number of opened requests, sorted by phase.	Pie
Request Backlog	This report enables the user to review the number of all open requests.	Number
Requests Created by Month within This Half Year	This report enables the user to review the trend of requests created in each month within this half of the year.	Line
Total Cost of Open Requests by Month	This report enables the user to review the trend of the total cost of open requests in each month.	Vertical bar
Unassigned Requests by Priority	This report enables the user to review unassigned requests, sorted by priority.	Pie
Total Cost of Open Requests Last 6 Months per Location	This report enables the user to review the trend of the total cost of open requests in last 6 months, sorted by location.	Vertical bar

Report name	Report description	Chart type
Total Cost of Open Requests Last 6 Months per Department	This report enables the user to review the trend of the total cost of open requests in last 6 months, sorted by department.	Vertical bar
Average Cost of a Request	This report enables the user to review the trend of the average cost of a request in each month.	Line
Percentage of Open Requests	This report enables the user to review the percentage of open requests in each month.	Line
Percentage of Closed Requests	This report enables the user to review the percentage of closed requests in each month.	Line

Configuration Item reports

Report Title	Report Description	Report Type
Total Count of CIs	This report enables the user to review all the CIs related to other CIs.	Number
CIs by Location	This report enables the user to review all the CIs, sorted by the location.	Pie
CIs by Service Provider	This report enables the user to review all the CIs, sorted by the service provider.	Pie
Count of CIs related to Services	This report enables the user to review the number of CIs that are related to one or more IT services as a percentage of the total number of registered CIs that can be related to IT services.	Pie
CI Changes within the Last 7 Days	This report enables the user to review CI changes within the last seven days.	Vertical bar
CIs without Owner	This report provides the number of CIs without an owner.	Number
CIs Related to Change	This report enables the user to review all CIs that are currently related to an open change.	Vertical bar

Report Title	Report Description	Report Type
Configuration Item Summary by Type and Status	This report enables the user to view all the configuration items within their organization, sorted by type and then by status.	Vertical bar

Knowledge Management reports

Report Title	Report description	Report type
Number of Articles Awaiting Review	This report enables the user to review the number of the articles that are waiting for review.	Number
Number of Articles Retired	This report enables the user to review the number of the articles that get retired.	Number
Number of Externally Published Documents	This report enables the user to review the number of the externally published documents.	Number
Number of Internally Published Documents	This report enables the user to review the number of the internally published documents.	Number
Number of Articles Created by Month within This Half Year	This report enables the user to review the number of the articles created in each month within this half of the year.	Stacked vertical bar
Document Usage by Article	This report enables the user to review the document usage statistics, sorted by each article.	Horizontal bar
Document Usage	This report enables the user to review the document usage list.	Standard list

License reports

Report title	Report description	Report type
High Water Mark of Module License Usage by Module and by Day	This report enables the user to review the high water mark of module license usage, sorted by module and then by day.	Vertical bar
License Statistics within the Last 3 Days	This report enables the user to review the license statistics within the last three days.	Vertical bar
License Usage Details	This report enables the user to review the license usage details.	Standard list

SLA reports

Report title	Report description	Report type
Response SLO Metrics This Year_Base Monitoring SLA for IT Services	This report enables the user to review the target response metrics for the ten lowest performing Service Level Agreements (SLAs) in the specified month.	Stacked horizontal bar
Availability-Summary_Base Monitoring SLA for IT Aervices_Duration Met %	This report enables the user to review the target response and availability metrics for a single Service Level Agreement (SLA) within a given month.	Vertical bar
Availability-Summary_Base Monitoring SLA for IT Services_Total Outages	This report enables the user to review the target response and availability metrics for a single Service Level Agreement (SLA) within a given month.	Vertical bar
Availability-Summary_Base Monitoring SLA for IT Services_Availability %	This report enables the user to review the target response and availability metrics for a single Service Level Agreement (SLA) within a given month.	Vertical bar

Time duration and calculation reports

Report title	Report description	Report type
Incident In/Out	This report enables the user to review open and closed incidents by quarters in one year. It calculates the numerical difference of the open and closed Incidents in each quarter of the past year.	Line
Average Incident handling time for last year by month	This report enables the user to review the average Incident handling time in each month during the past year.	Line
Average incidents handling time per assignment group	This report enables the user to review the average Incident handling time per assignment group by using the processed time duration data.	Line
Incident handling time per assignment group for breached tickets	This report enables the user to review SLA breaches, which may be caused by OLA breaches. The Incident handling time data in this report is stacked by the assignment group and shows how much time a groupis spending on that record.	Stacked Horizontal bar

Report title	Report description	Report type
Total outage duration caused by Incidents per Service this month	This report enables the user to review the total outage duration in hours caused by Incidents, sorted by the affected service during this month.	Horizontal bar
Success change rate by Service	This report enables the user to review the success change rate, sorted by Service.	Vertical bar
Weekly success change rate	This report enables the user to review the weekly success change rate.	Line
Standard change rate this week	This report enables the user to review the standard change rate during this week.	Number
Weekly automated change rate trend	This report enables the user to review the weekly automated change rate trend.	Line
Weekly standard change rate trend	This report enables the user to review the weekly standard change rate trend.	Line

End User Chat reports

Report title	Report description	Report type
All Pending Chats	This report enables the user to review all pending chats.	List
All Accepted Chats	This report enables the user to review all accepted chats.	List
All Closed Chats	This report enables the user to review all closed chats.	List
All Chat First Resolution Rate within Last 30 Days	This report enables the user to review all chat first resolution rate within last thirty days.	Vertical bar
My Average 1st Response Time Within Last 7 Days	This report enables the user to review his/her first response time within last seven days.	Line
My Team Average 1st Response Time Within Last 7 Days	This report enables the user to review the first response time of his/her team within last seven days.	Vertical bar
My Accepted Chat Today	This report enables the user to review the number of chats accepted by him/her today.	Number
My Average Chat Duration within Last 7 Days	This report enables the user to review his/her average chat duration within last seven days.	Line

Report title	Report description	Report type
My Team Average Chat Duration within Last 7 Days	This report enables the user to review the average chat duration of his/her team within last seven days.	Vertical bar
My Team Chat Abandonment Rate within Last 7 Days	This report enables the user to review the chat abandonment rate of his/her team within last seven days.	Line
My Chat Average Resolution Time within Last 7 days	This report enables the user to review his/her average chat resolution time within last seven days.	Vertical bar
My Team Chat Average Resolution Time within Last 7 Days	This report enables the user to review the average chat resolution time of his/her team within last seven days.	Vertical bar
My Team First Resolution Rate of Chat Tickets within Last 30 Days	This report enables the user to review his/her team first resolution rate of chat tickets within last thirty days.	Stacked vertical bar
Percentage of Interactions Initiated from Chat	This report enables the user to review the percentage of Interactions initiated from End Use Chat.	Number

Report title	Report description	Report type
Percentage of Chat Solved by Virtual Agent	This report enables the user to review the percentage of chat solved by Virtual Agents (Sages).	Number

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- 2 Change Management reports
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1.17. Service Manager Collaboration

Service Manager Collaboration is an instant messaging solution combining IT Collaboration and End User Chat. IT Collaboration enables Service Manager IT operators to collaborate in real-time (or anytime) when handling an Interaction, Incident, Incident Task, Request, Request Task, Problem, Problem Task, Change, or Change Task. End User Chat builds a communication channel between the Service Manager Service Desk agents and the Employee Self-Service (ESS) portal users, or the SMA Service Portal users so as to increase IT efficiency, improve customer satisfaction and reduce IT support costs.

Note

Service Manager Collaboration is available on the Service Manager web client only.

1.17.1. Basic concepts in Service Manager Collaboration conversations

This section describes the basic concepts in Service Manager Collaboration conversations.

Conversation roles

Service Manager Collaboration defines the following conversation roles.

- Participants:

  A participant is defined as a user who has posted at least one message to a conversation, or a user who has been added to a conversation.

  Any participant can add other IT Collaboration users to a conversation, as well as remove themselves and other participants. A maximum of 200 participants can participate in a conversation.

- IT Collaboration users:

  All valid Service Manager IT operators are defined as IT Collaboration users.

- End users:

  The Employee Self-Service (ESS) users who access Service Manager web client by using a web browser. The end users can collaborate with Service Manager Service Desk agents in End User Chat.

Conversations

Conversations are a series of messages that are saved in the Service Manager database. By default, the title of an IT Collaboration conversation consists of the record ID and the record title, whereas the title of an End User Chat displays the end user's name only. For example, *IM10001 Printer malfunction*. To access a conversation, you need to locate the corresponding Service Manager record, and then click the **Start Conversation** button.

Notifications

With Service Manager Collaboration notifications, you are always up-to-date with real-time incoming messages and new conversation invitations. The Collaboration notifications are rendered as notification badges (for example,), which are

displayed in different places, depending on whether the conversation window is minimized or not.

When the conversation window is minimized, the following behavior occurs:

- A notification badge appears on the Collaboration notification button showing the sum of the message number of all unread messages in all updated conversations.

When the conversation window is not minimized and there are multiple conversation tabs open, the following behaviors occur:

- A notification badge appears on the Collaboration notification button showing the sum of incoming messages in all conversations that are minimized. The incoming message numbers on all active/inactive conversation tabs in the conversation window aren't included in the sum.
- New messages and new conversation invitations are displayed in the Conversation area of the currently active conversation immediately, without showing a red notification badge on the conversation tab.
- Notification badges appear on the inactive conversation tabs showing the incoming message number of updated conversations, respectively.

Service Manager Collaboration recounts the numbers on the notification badges every time when you log in.

Note

You will no longer receive messages or notifications from a conversation after clicking **Quit Conversation**. See Quit a conversation for more information.

Note

Suppose an IT agent starts multiple sessions in Service Manager by logging on to the web tier with the same user account in different web browsers. When the notifications are read in a session, the unread number displayed on the red badge doesn't decrease in other sessions.

HTML5 desktop notification

As of version 9.50, Service Manager web client also supports HTML5 desktop notification for incoming Collaboration conversation messages and new conversation

invitations in the following scenarios:

- When the browser is open and the Service Manager web client is not the currently active webpage, the notification appears on the lower-right corner. Click this notification to return to the Service Manager web client webpage.
- When the browser is minimized, the browser window icon flashes in the taskbar. Click the browser window icon to return to the Service Manager web client webpage.

Note

Before using the HTML5 desktop notification feature, pay attention to your browser settings. If you are working with Google Chrome and Mozilla Firefox, you must configure the browser settings and allow the Service Manager web client URL to show notifications.

Note

You can't get desktop notifications if the web browser is not active but is not minimized.

1.17.2. UI navigation

This section introduces the UI of Service Manager Collaboration.

Collaboration buttons in Service Manager UI

Conversation window

Participant window

Chat request panel (End User Chat only)

Chat button (End User Chat only)

Chat window (End User Chat only)

1.17.2.1. Collaboration buttons in Service Manager UI

The Collaboration buttons embedded in Service Manager UI are illustrated in the following screenshot:

Callouts	Button	Description
1	Notifications	Click to display or hide the notification list for new conversation invitations and incoming messages. This icon is gray when there is no open conversation. See Check updated conversations for more information.
2	Notification badge	Displays the number of new conversation invitations and incoming messages in the updated conversations. If the count exceeds 99, the number is displayed as N. See Notifications for more information.

Callouts	Button	Description
3	Chat Request button	Click to display or hide the chat request panel for End User Chat.

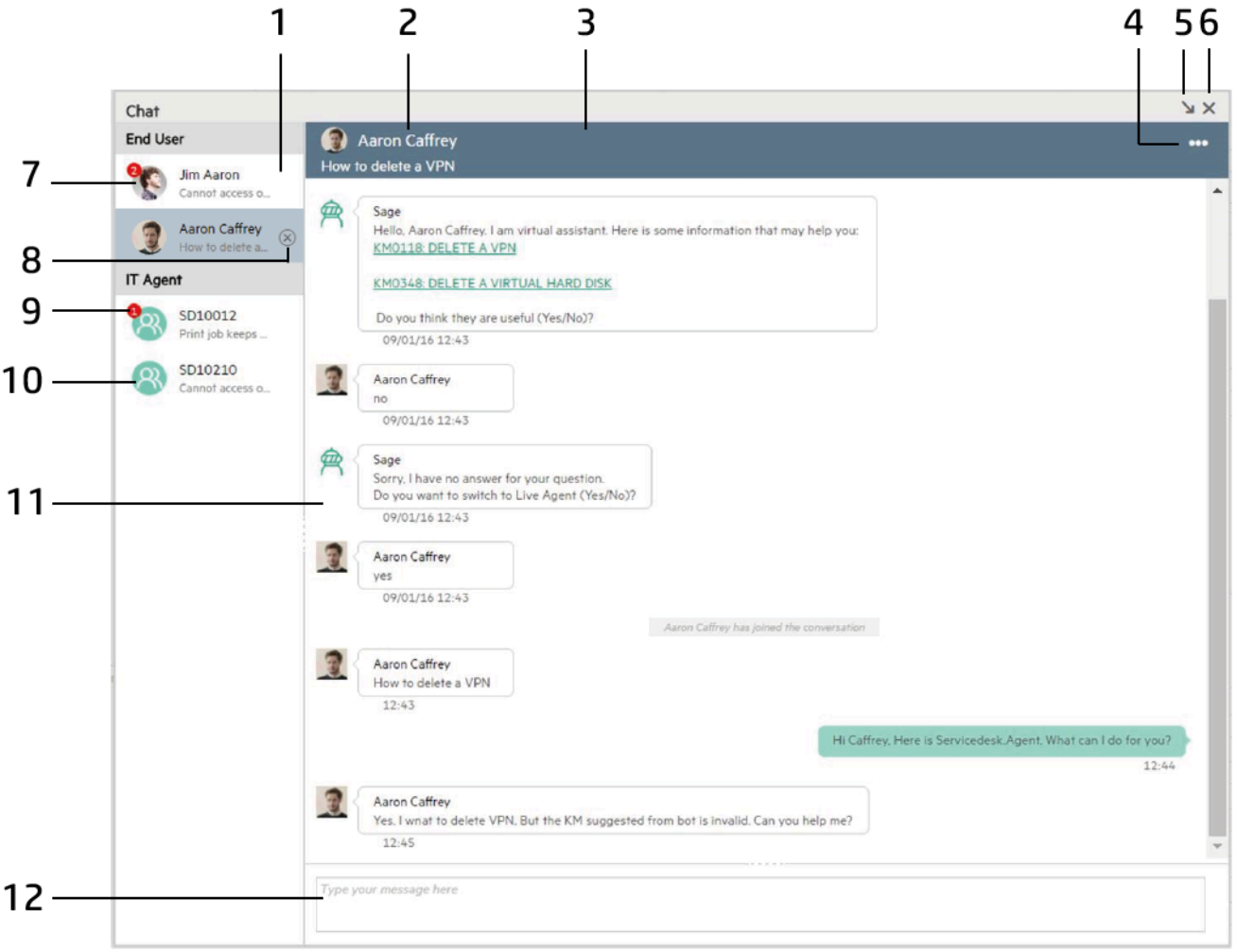
Callouts	Button	Description
		Note • This button is availabl e only when Service Manage r Collabo ration is enabled , and the IT agent must belong to an assign ment group that support s End User Chat. To update the settings of an assign ment group to support End Use Chat,

Callouts	Button	Description
		see Add a Service Desk assignment group. • The IT agent who is removed from an assignment group will not be able to see the pending requests for the group from the next login to Service Manager. • The IT agent who is added to an assignment group will be able to

Callouts	Button	Description
		see the pending requests for the group from the next login to Service Manager.
4	Chat Request number badge	Displays the total number of pending end user chat requests. If the count of new chat requests exceeds 99, the number is displayed as 99+.
5	User Information	Click to display or hide the User Basic Information Card. See Display user information for more information.

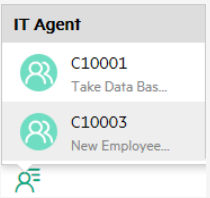
1.17.2.2. Conversation window

The Service Manager Collaboration conversation window is illustrated in the following screenshot:



Callouts	Component	Description
1	Conversation tab	Click to switch between multiple conversations that are currently open. The active conversation appears in dark gray and inactive conversations appear in light gray. The corresponding record ID is also displayed on this tab.

Callouts	Component	Description
		Tip To achieve a better user experience, you may want to hide the conversation tabs. To do so, place your cursor on the left edge of the conversation window and then drag rightwards. When the conversation tabs are hidden, an icon appears on the bottom-left corner of the conversation window. To display the content of a hidden conversation tab, click the icon and then click the correspondin g tab. Refer to the following

Callouts	Component	Description
		screenshot as an example:  To restore the hidden conversation tabs, place your cursor on the left edge of the conversation window and then drag leftwards.

Callouts	Component	Description
		Note • Service Manager Collaboration displays the last received message on the Conversation tab of the inactive conversations. Long messages are displayed as an ellipsis on the Conversation tab. • If you have opened more than 10 conversation tabs in IT Collabo

Callouts	Component	Description
		ration and End User Chat altogether, the system displays the last 10 active conversation tabs only. You are still able to receive notifications from other conversations.

Callouts	Component	Description
2	Record ID and title	In IT Collaboration: ID and title of the record that the conversation belongs to. Click the record ID to open the record. **Note** The record can only be opened by the users with the corresponding authorization to view these types of records. In End User Chat: The end user's full name and title of the Service Desk chat request record. The record ID is not clickable.
3	Conversation window header	Header of the conversation window.

Callouts	Component	Description
4	More	In IT Collaboration: Click to display the following options: • **Add Participant**: Click to open the Participant window so as to add one or more participants to the conversation or remove participants from the conversation. See Invite participants and Remove participants for more information. • **Quit Conversation**: Click to leave a conversation. You will no longer receive messages or notifications from this conversation. See Quit a conversation for more information. In End User Chat: Click to display the following options: • **Add Participant**: Click to open the Participant window so as to add one or

Callouts	Component	Description
		more participants to the conversation or remove participants from the conversation. See Invite participants and Remove participants for more information. • **Create Interaction**: Click to create an additional Interaction from the chat so that the Service Desk chat ticket can be related to multiple Interactions. See Create an additional Interaction from chat for more information. • **View Related Interactions**: Click to view all Interactions that are related to the current chat request. • **Close Chat Request**: Click to close the chat request. You will no longer receive messages or notifications from this conversation. See Close a chat

Callouts	Component	Description
		request for more information. • **Quit Conversation**: Click to leave a conversation. See Quit a conversation for more information. **Note** The option is available only after you have closed a chat request.

Callouts	Component	Description
		Note If there is only one conversation in the conversation window, the conversation window will be closed after you quit this conversation.

Callouts	Component	Description
5	Pin	Click this button to scale down and pin the conversation window to the right side of the Service Manager UI. After the conversation window is pinned, this button changes to ↖ . You can click this button again to unpin the conversation window. **Tip** When you pin the conversation window, the conversation tabs are hidden. Refer to the *Tip* in Conversation tab for more information.
6	Minimize conversation window	Click to minimize the conversation window. When a conversation window is minimized, the incoming messages and new conversation invitations are displayed as notifications.

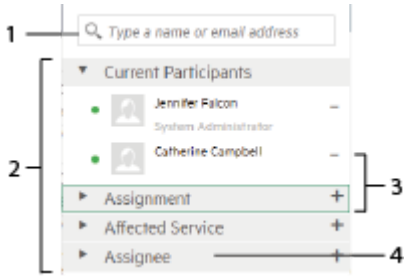
Callouts	Component	Description
7	End users' photo	The end user's photo in the current End User Chat conversation. When hovering your mouse over the photo thumbnails, the participants name list is shown.

Callouts	Component	Description
8	Close conversation tab	Click to close the conversation tab. When a conversation tab is closed, the incoming messages and new conversation invitations are displayed as notifications. **Note** • This button is visible for mouse-over only. • If there is only one conversation in the conversation window, click this button to minimize the conversation window.

Callouts	Component	Description
9	Notification badge	Displays the number of new conversation invitations and incoming messages in the updated conversations. If the count exceeds 99, the number is displayed as **N...** . See Basic concepts in Service Manager Collaboration conversations for more information.
10	Mock-up of participants' photos	Mock-up of the participants' photos in the current IT Collaboration conversation. When hovering your mouse over the photo thumbnails, the participants name list is shown.
11	Conversation area	Displays all the conversation activities.
12	Text input area	Type your input in this area.

1.17.2.3. Participant window

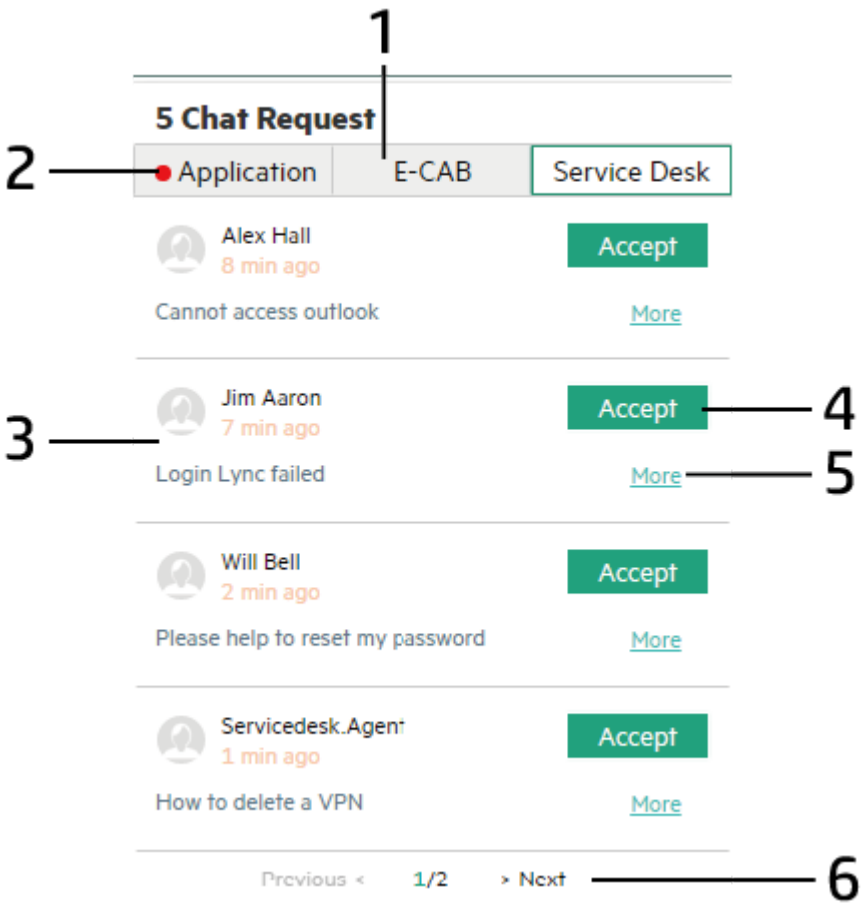
The Participant window is illustrated in the following screenshot:



Callouts	Component	Description
1	Search participant	Type a user's email address or a user name in this field to automatically display the search result. See Invite participants for more information.
2	Suggested participant list	Displays the suggested participants. Service Manager Collaboration retrieves participants based on the Suggested User Fields configuration that's associated with the record type. See Invite participants and IT Collaboration administrator tasks for more information.
3	Add/Remove button	Click to add/remove the selected participants. See Invite participants and Remove participants for more information.
4	Suggested participant group title	Displays the title of the suggested participants group. To customize this title, see step 9 in Add supported record types.

1.17.2.4. Chat request panel (End User Chat only)

The chat request panel is illustrated in the following screenshot:



Callouts	Component	Description
1	Assignment group tab	Displays the pending chat request list of a chat assignment group that a Service Manager Service Desk agent belongs to.
2	Chat request notification	Symbolizes that there are new chat requests for the inactive assignment group.
3	Chat request	Displays the brief information of a pending chat request, including the full name of chat request submitter, the brief description of the chat request and how long the submitter has been waiting. The incoming chat requests are displayed in chronological order in this list.

Callouts	Component	Description
4	Accept chat request	Click to accept a chat request and open the conversation window. The accepted chat request disappears from the Chat Request list.

Callouts	Component	Description
		Note The Accept button is disabled due to the following possible causes: • The chat request is accepted by another IT agent. The agents who failed to accept this chat request receives a system message this request has already been accepted. • The end user

Callouts	Component	Description
		canceled the chat request. To remove the accepted chat request from the Chat Request list, close the chat request panel and open it again.
5	More	Click or hover your mouse to display the preview of the related chat request. For more information about how to customize the preview information, see Customize the chat request preview information.
6	Previous/Next	Click to browse through the chat requests in the current assignment group.

1.17.2.5. Chat button (End User Chat only)

The ESS portal provide a chat button for the end users to start a chat. Various statuses of the chat button are illustrated in the following screenshots:

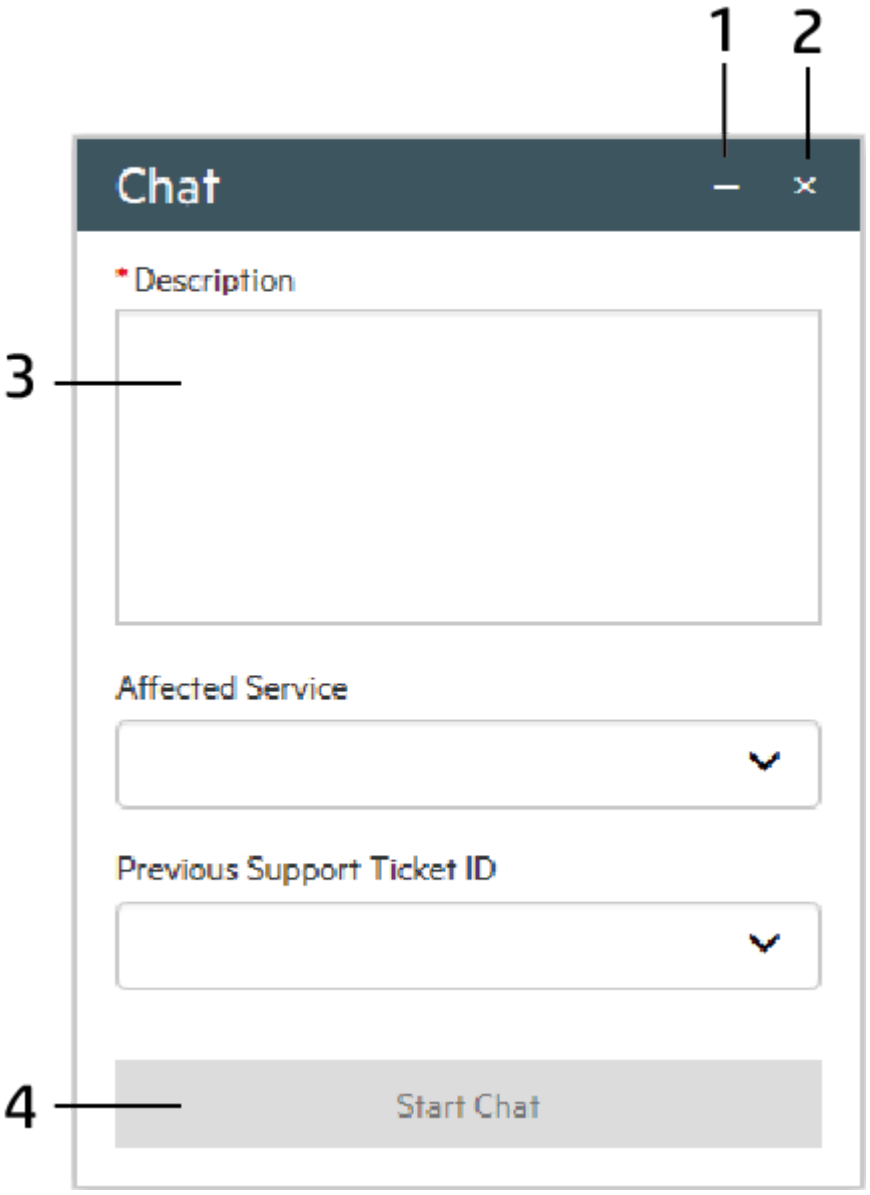
Status	Description
	Displays a swirling ellipsis when the system is loading to launch End User Chat.
	Displays the initial status of the chat button. You can click this button to start a chat. See Start/End a live chat for more information.
	When the End User Chat window is minimized, displays a flashing ellipsis if you are waiting for a Service Desk agent or a virtual agent (Sage) to join the chat.
The Service Desk agent's photo / the virtual agent's photo	When the End User Chat window is minimized, displays the photo of the Service Desk agent or the virtual agent who has accepted the chat request.

1.17.2.6. Chat window (End User Chat only)

The default End User Chat window is illustrated in the following screenshot:

Note

Depending on the customization in Service Manager, this window may consist of additional fields, checkboxes, and drop-down menus.



Callouts	Component	Description
1	Minimize button	Click to minimize the chat window. To restore the chat window, click the chat button again.
2	Close button	Click to leave the chat.
3	Text input area	Type your question in this area.
4	Start chat	Click to start chat after you have typed your question in the text input area. When you are waiting for a Service Desk agent or a virtual agent (Sage) to join the chat, this button changes to **Cancel Chat** and enables you to cancel your chat request.

1.17.3. IT Collaboration

As of version 9.41, IT Collaboration supersedes the previous Enterprise Collaboration (EC) based instant messaging solution. As an instant messaging tool embedded in Service Manager, IT Collaboration enables Service Manager IT operators to collaborate in real time (or anytime) when handling an Interaction, Incident, Incident Task, Request, Request Task, Problem, Problem Task, Known Error, Change, or Change Task.

Note

The system supports these record types by default, including Interaction, Incident, Incident Task, Request, Request Task, Problem, Problem Task, Known Error, Change, or Change Task.

1.17.3.1. IT Collaboration audiences

This section is intended for the following audiences:

- System Administrators (for Service Manager Collaboration deployment and initial configuration)
- Incident Analysts (and others involved in Incident Management, such as Incident Coordinators)
- Change Analysts (and others involved in Change Management, such as Change Coordinators)
- Problem Analysts (and others involved in Problem Management, such as Problem Coordinators)
- Request Analysts (and others involved in Request Fulfillment, such as Request Coordinators)
- Service Desk Agents
- Service/Configuration Item Owners

1.17.3.2. IT Collaboration administrator tasks

This section describes the IT Collaboration administration tasks.

Add supported record types

By default, Service Manager Collaboration supports the following types of records in this release:

- Interaction
- Request and Request Task
- Incident and Incident Task
- Change and Change Task
- Problem, Known Error, and Problem Task

In addition to these record types, you can add a new record type that's supported by Service Manager Collaboration. After certain configurations, the **Start Conversation** button becomes available on the detailed view of the new record type and you can start a conversation.

Note

Service Manager Collaboration doesn't support the Knowledge Management module.

To add a new record type that's supported by Service Manager Collaboration, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Support List** to open the Service Manager Collaboration Configuration form.
3. Type the name of the reference table of a new record type in the File Name field. For example, type `probsummary` for Incidents.
4. Select the **Service Manager Collaboration Enabled** checkbox to enable Service Manager Collaboration for the new record type.
5. Type the key that represents the record ID in the ID Field. This is the record ID that's displayed on the Conversation window header. For example, `number`.
6. Type the key that represents the record description in the Display Field. This is the record description that's displayed on the Conversation window header. For example, `brief.description`.

7. Customize the suggested participant list for the new record type in the Suggested User Fields section.

    1. Type the key that represents the field as a suggested group in the Source Field column, and then type its associated file name in the Target File column. For example, to add the Opened By field to the suggested participant list for Interactions, type `opened.by` in the Source Field column, and then type `operator` in the Target File column.
    2. Refer to the existing configurations to add more suggested groups that are listed in the suggested participant list.

    Note

    Only character fields and array character fields are supported.

8. Click **Add** and **OK**.

9. (Optional) Customize the suggested participant group title.

    Every suggested participant group has a default group title. However, you can customize the existing group titles or add new group titles.

    1. Click **Tailoring** > **Notifcations** > **Messages**.

    2. To edit an existing suggested participant group title, follow these steps:

        1. Type `smcconfig` in the Class field.
        2. Type the reference table name, a semicolon (;), and the key that represents the field as a suggested group in the Message Number field. You must delimit these two values by a semicolon with no spaces between them. For example, to edit the title of the Contact field for Incidents, type `probsummary;contact.name`.
        3. Click **Search**.
        4. Update the suggested participant group title in the Text field.
        5. Click **Save** and **OK**.

    3. To add a new suggested participant group title, follow these steps:

        1. Type the language code in the Language Code field. For example, en.
        2. Type `smcconfig` in the Class field.
        3. Type the reference table name, a semicolon (;), and the key that represents the field as a suggested group in the Message Number field. You must delimit these two values by a semicolon with no spaces between them. For example, type `probsummary;vendor`.

4. Type the suggested participant group title in the Text field.
5. Click **Add**.

10. Sign out and then sign in to Service Manager again.

Now you can start a Service Manager Collaboration conversion by clicking the **Start Conversation** button in the detailed view of the new record type.

1.17.3.3. IT Collaboration user tasks

This section describes the most typical actions you will perform to collaborate with other IT operators.

Display user information

Edit user profile

Change the presence status

Send an email message

Start a conversation

Participate in multiple conversations

Invite participants

Remove participants

Quit a conversation

Check updated conversations

View conversation history

For information about how to install and configure IT Collaboration, see Install Service Manager Collaboration.

1.17.3.3.1. Display user information

After you have logged on to Service Manager, click **User Information** on the top-right corner of the Service Manager UI to show your User Basic Information Card. The User Basic Information Card displays your photo, presence status, name, title, email address, and telephone number, and also provides a logout button to log out of Service Manager.

Note

The system retrieves the user information from the Service Manager `contacts` table. If your system uses LDAP authentication only, the contacts table is not synchronized automatically from LDAP. Therefore, some user information is incorrectly displayed on the User Basic Information Card. In this case, you need to manually synchronize the `contacts` table. To enable Collaboration with LDAP, comment the follow line in sm.ini and then restart the Service Manager server.

`#ldapauthenticateonly:1`

1.17.3.3.2. Edit user profile

You can change your photo that's displayed on the User Basic Information Card and update other profile information as well.

To change your photo that's displayed on the User Basic Information Card, follow these steps:

1. Click your current photo on the User Basic Information Card.
2. Browse for another photo that's saved on your computer.

   Note

   The supported image types are *.png, *. jpg, *.bmp, and *.jpeg.

3. Select your new photo and then click **Open**. Alternatively, double-click the new photo.

   Your photo that's displayed on the User Basic Information Card is updated automatically.

To update other profile information such as name, title, email address, and telephone number, you must log on to Service Manager as a system administrator and then update the users' profile in the `contacts` table.

Note

If your system uses LDAP authentication only, the contacts table is not synchronized automatically from LDAP. Therefore, some user information is incorrectly displayed on the User Basic Information Card and you can't change your photo.

1.17.3.3.3. Change the presence status

The statuses provided by Service Manager Collaboration are: **Online** 🟢 , **Busy** 🔴 , **Away** 🟡 , and **Offline** ⚪ . These presence statuses are pre-defined and you can't create a custom status in Service Manager Collaboration. Your presence status is set to **Online** 🟢 once you have logged on to Service Manager. To change your presence status, click the status menu below your photo and then click another status.

1.17.3.3.4. Send an email message

To send an email message to an IT operator, follow these steps:

1. Click the IT operator's photo in a conversation.
2. When the User Basic Information Card is displayed, click the email address. Your default email program (for example, Microsoft Outlook) opens a blank email message addressed to the IT operator.

1.17.3.3.5. Start a conversation

Conversations can only be started in the detailed view of the records supported by IT Collaboration, including Interaction, Incident, Incident Task, Request, Request Task, Problem, Known Error, Problem Task, Change, and Change Task. By default, the **Start Conversation** button floats on the upper-right corner of the detailed view of these records after you enable IT Collaboration. You can drag this button to any place on the form. However, this button will restore to its default position once the detail view is refreshed.

Tip

To display the **Start Conversation** button on the detailed view of other types of records that's supported by Service Manager Collaboration, refer to Add supported record types.

To start a conversation, follow these steps:

1. Open a record in Service Manager.
2. Click **Start Conversation** to open the conversation window, and then click **OK** in the system confirmation dialog. A conversation starts with the record's ID and title displayed on the header of the conversation window.

   Important

   The **Start Conversation** button is displayed only when both the ID field and the description field are defined as text fields on a form.

   Note

   The **Start Conversation** button has a different icon if the record has an existing conversation.

   Tip

   You can also press **Alt** + **Ctrl** + **C** to open or close the conversation window.

3. Refer to Invite participants to add one or more participants to the conversation. On the contrary, refer to Remove participants to remove participants from the conversation as necessary.

4. Type your message in the text input area at bottom of the conversation window. IT Collaboration conversation supports plain text only.

   Tip

   Press **Shift** + **Enter** to start a new line.

5. Press **Enter** to post the message. This message appears in the conversation window with your photo and the current time.

   Posting a message makes the current IT operator a participant automatically.

   Tip

   Click the message sender's photo to view his/her User Basic Information Card.

   Tip

   A **Resend Message** button appears beside your message when your message is not posted successfully. Click the **Resend Message** button to post your message again.

Only the activity of starting a conversation is stored in the journal update of a record if this record has an activity log. To view the conversation history, you can slide up the scroll bar in a conversation window. See View conversation history for more information.

1.17.3.3.6. Participate in multiple conversations

You can participate in multiple conversations at the same time, and click the conversation tabs to switch from one to another.

To participate in multiple conversations, follow these steps:

1. Initiate a conversation as instructed in Start a conversation.
2. Open another record in Service Manager.
3. Click **Start Conversation**. The new conversation starts in another tab in the conversation window.

1.17.3.3.7. Invite participants

To simultaneously consult with other experts in an IT Collaboration conversation, follow these steps to invite participants:

1. Open a conversation directly or open a record that has a conversation.
2. Click **More** > **Add Participant** on the header of the conversation window to open the Participants window. IT Collaboration retrieves participants based on the Suggested User Fields configuration in Service Manager.

   > **Tip**
   >
   > For more information about how to customize the Suggested User Fields, see step 7 in Add supported record types.

3. Add a new participant by performing either of the following tasks:
   - Select a user name or a group from the suggested participant list, and then click the **Add** button.
   - Enter a user's email address or a user name in the Search Participant field. Select the participant from the search result, and then click the **Add** button.

     > **Tip**
     >
     > - You can view participants' presence statuses that are shown beside the user names in the suggested participant list, the participant list, and the participant search result list.
     > - To close the Participants window, press **ESC** or click anywhere outside of the participants list.

   After you invite a participant to the current conversation, the system broadcasts a message to all the other participants to identify whom is invited. Meanwhile, the invited participant receives a system message that shows the members in the current conversation.

1.17.3.3.8. Remove participants

Participants can be removed from conversations if they no longer need to contribute. Participants can remove themselves or the other participants. When being removed, participants will not receive any incoming message notifications through IT Collaboration.

To remove participants, follow these steps:

1. Click **More** > **Add Participant** to display the participants list.
2. Select the person in the participant list, and then click the **Remove** button.
3. The system prompts you to confirm the operation, click **OK**.

1.17.3.3.9. Quit a conversation

To quit a conversation, click **More** > **Quit Conversation** on the header of the conversation window, and then click **Yes** in the system prompt.

You will no longer receive messages or notifications from this conversation. After you quit the current conversation, the system broadcasts a message to all the other participants to identify that you have left the conversation.

The conversation is automatically saved in the Service Manager database. Quitting the conversation doesn't close the record in the main workspace.

Note

Leaving the conversation doesn't delete the conversation. Conversations can't be deleted.

1.17.3.3.10. Check updated conversations

To view the updated conversations, follow these steps:

1. Click **Notifications** to open the notifications panel.

   > **Tip**
   >
   > You can also press **Alt** + **Ctrl** + **V** to open or close the notifications panel. When there is no notification, this shortcut opens a conversation window.

2. Click a notification to view its context in the corresponding conversation. Or, press the **Up/Down Arrow** keys to highlight a notification in the panel, and then press **Enter**.

   - If you don't minimize the conversation window and there are multiple conversation tabs open:

     Collaboration pushes the conversation that relates to your selected notification to the last tab, and displays the notification context in the Conversation area.

   - If you minimize the conversation window:

     Service Manager Collaborations first launches the conversation window, and then pushes the conversation that relates to your selected notification to the last tab.

   > **Note**
   >
   > When you open an updated conversation, the record that the conversation belongs to doesn't automatically open. To open the corresponding record, click the record ID on the conversation window header.

1.17.3.3.11. View conversation history

IT Collaboration conversations are saved in the Service Manager database. You can slide up the scroll bar in a conversation window to view the conversation history.

Note

If you previously slide up the scroll bar to display conversation history and there are new messages appear in the conversation area in real-time, a **New Message** pop-up is displayed on the bottom of the conversation area. You can return to the initial view (with only several latest messages) by clicking the **New Message** pop-up.

1.17.4. End User Chat

Being an indispensable part of Service Manager Collaboration, End User Chat is a self-service solution which operates seamlessly with Service Manager Service Desk. By clicking the chat button in either the Employee Self-Service (ESS) portal or the SMA Service Portal, end users can communicate with either a virtual agent or a Service Desk IT agent in real-time to quickly address the service requests and support requests as they arise. Service Manager Process Designer Assignment rule settings streamline the End User Chat process and automatically route the chat requests to a right Service Desk assignment group. The End User Chat virtual agent simulates the way a human being responds to a question by suggesting Knowledge Management document links and Service Catalog item links or automating common cases. Therefore, the end users can receive prompt, concise answers when they need them the most.

Note

- To use End User Chat, you must install and configure Service Manager IT Collaboration first. For the related concepts and detailed instructions, see Install Service Manager Collaboration.
- The chat service fails to start a chat in Service Portal if there are no valid assignment groups configured in Service Manager. For detailed instructions, see Configure chat assignment settings.

1.17.4.1. Service Desk chat module

As of version 9.50, Service Manager introduces a minimum Service Desk chat module to handle end user chat requests. Service Manager automatically creates a Service Desk chat request record as long as an end user starts a chat. When a chat is closed or a chat request is canceled, the corresponding chat request closes asynchronously by the Process Designer rule engine.

Service Desk chat request form details

The following table identifies and describes some of the features on the Service Desk chat request forms.

Service Desk chat request form details

Label	Description
Chat ID	The system-generated unique ID for this Service Desk chat request.
Phase	This is a system-generated field. These phases are available out-of-box: • Logged • Accepted • Closed
Assignment Group	The group assigned to accept the Service Desk chat request. The Service Desk assignment group is available out-of-box.
Contact	The name of the service recipient from whom the end user chat was initiated. This field ensures that the correct person will be notified about updates to the interaction.
Assignee	The name of the person who accepted the Service Desk chat request. This person is a member of the assigned support group. Assignees may belong to one or multiple assignment groups, based on the needs of your company.
Chat Bot	The name of the virtual agent which accepted the Service Desk chat request.

Label	Description
Affected Service	The Affected service is the user's individual and department subscriptions that can be defined in Service Manager. This field is populated with data from the end user input. These business services are available out-of-box: • Applications • E-mail/Webmail • Handheld, PDA & Telephony • Intranet/Internet • My Devices (The My Devices service represents all personal devices that the user would use.) • Printing • Other - See Description
Chat ended by end user	Displays a check mark in the checkbox if the chat was terminated by the end user.
End Reason	Displays an end reason if the chat was terminated by the end user. These end reasons are available out-of-box: • Problem Resolved • Problem Unresolved
Description	A detailed description of the end user's problem. This field is prepopulated with input from the end user.

Label	Description
Closure Code	Displays a closure code if the chat was closed by the Service Desk agent. These closure codes are available out-of-box: • Problem Resolved • Problem Unresolved • No Response from End User
Workflow tab	Displays a figure of Service Desk chat workflow. It also indicates the current phase which the chat request is in, and traces the phase transition history. These phases are available out-of-box: • Logged: This is the phase that's initiated when an end user initiates a chat request. • Accepted: This is the phase that's transited to when a Service Desk agent or a virtual agent accepts a chat request. • Closed: This is the final phase when a chat is closed or a chat request is canceled.
Related Records tab	Displays the related Interaction record.

Label	Description
History tab	Displays the related history of the Service Desk chat request. These fields are available out-of-box: • Open Time • Opened by • Transfer to Live time • Accept Time • Accepted by • Close Time • Closed by
Activity tab	Displays the activities of the conversation being opened and the OO flow execution results when an end user triggered an OO flow during the chat. See Set up End User Chat automation for more information.

1.17.4.2. End User Chat administrator tasks

This section describes the IT Collaboration administration tasks.

Configure end user profile

Configure chat assignment settings

Set live chat assignment End Lead Time

Customize the End User Chat window

Customize the chat request preview information

Set up End User Chat virtual agent

Set up End User Chat automation

Set up surveys for the Service Desk chat module

1.17.4.2.1. Configure end user profile

All end users must also be users of Service Manager. The end users' profile are defined in various fields in the `contacts` table in Service Manager, including company (if multi-tenancy is enabled), department, location, and language preference, and on service if the service field is mandatory. The Service Manager system administrator must configure these profiles properly so that each end user chat request can be properly routed. Meanwhile, an end user must have New, Update, and Close rights on Service Desk security area for End User Chat.

Note

For virtual agent and assignment group routing, the system uses following language settings instead of the user's profile language:

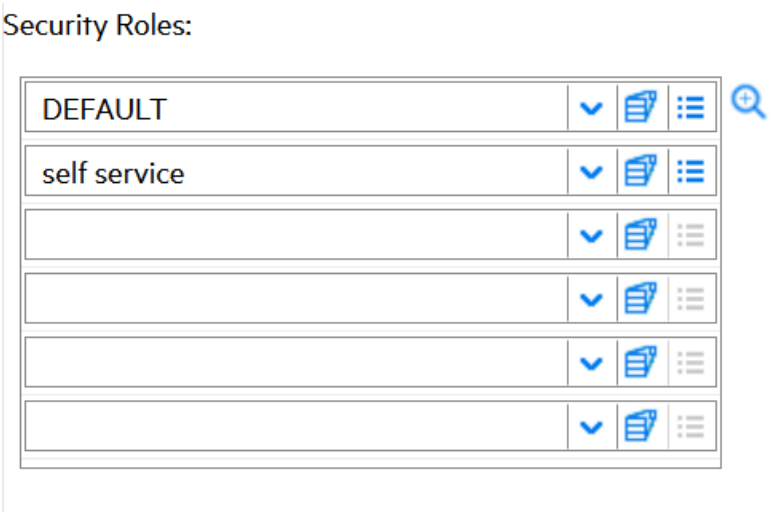
- The language that a user selected when signing in to the ESS portal.

Follow these steps to configure the user profiles in the contacts table:

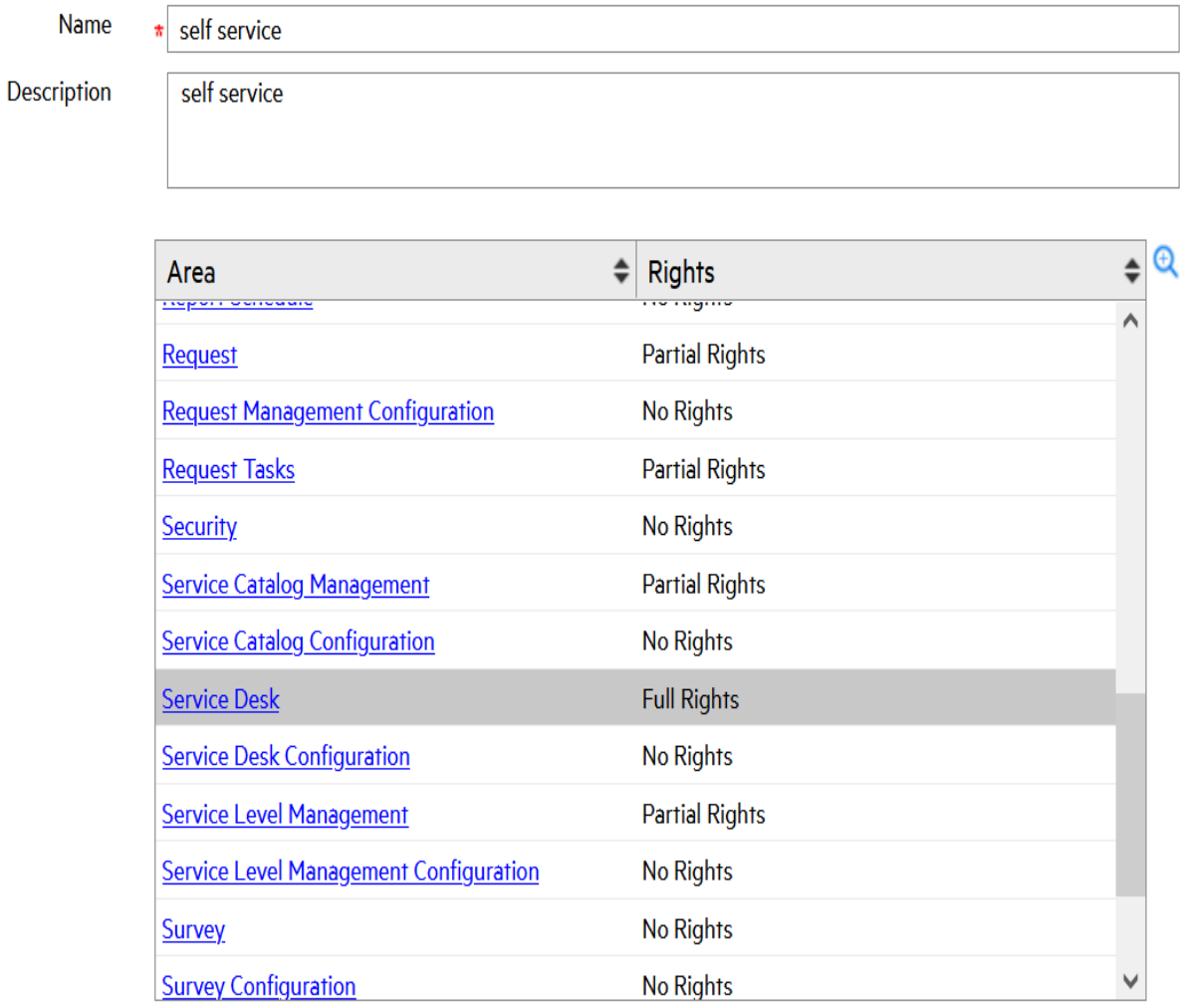
1. Sign in to Service Manager as a system administrator.
2. Type `db` in the Service Manager command line field, and then press Enter.
3. Type `contacts` in the Table field, and then click **Search**.
4. Click **contacts** in the Format Name list.
5. Type the end users' name in the Contact Name field, and then click **Search**. For example, CAFFREY, AARON.
6. Update the users' profile as necessary. This user must have New, Update, and Close rights on Service Desk security area for End User Chat.

    a. Click the **General** tab, and then click the **Application Profiles** tab.

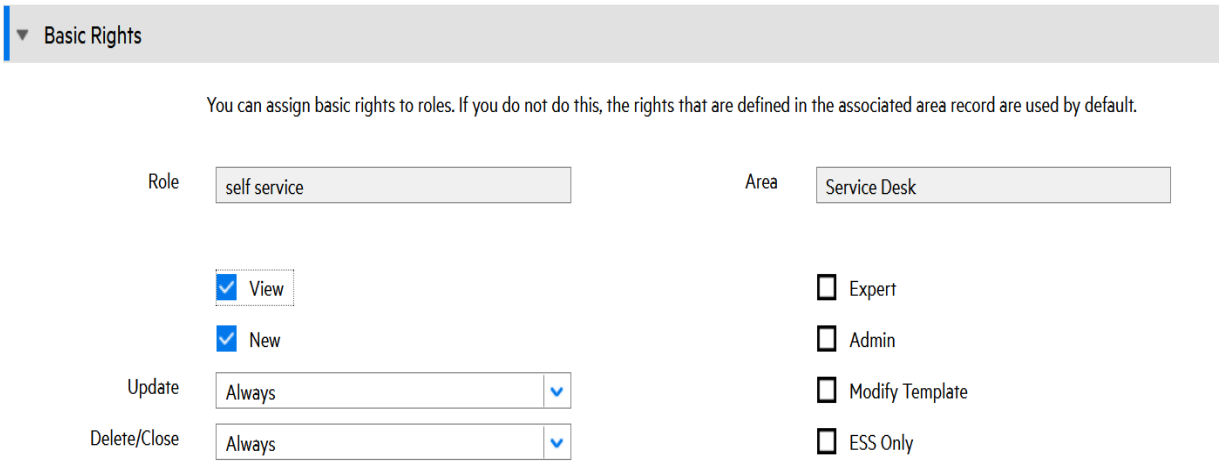
    b. Click the **View Detailed Information** button of a security role. For example:



c. Click the **Service Desk** area.



d. Check and update the rights. Select the **New** checkbox, and then select **Always** in the **Delete/Close** drop-down list.



7. Click **Save** and **OK**.

1.17.4.2.2. Configure chat assignment settings

When an end user initializes a chat request, the chat request will be automatically accepted by a virtual agent or be assigned to a specific chat assignment group based on the Service Manager Process Designer Assignment rule set configurations. Consider the following scenarios:

- If your system administrator has set up virtual agent, and then defined a rule to find a virtual agent based on user's requirement, Service Manager detects the end user's language selection when signing in to the ESS portal and then assigns this chat request to a virtual agent which has the same language setting. See Set up End User Chat virtual agent for more information.
- If the virtual agent is disabled or not found, Service Manager assigns this chat request to an available chat assignment group according to the assignment rule configuration and chat assignment group configuration. These configurations enable an assignment group's members to respond to End User Chat requests and to provide instant support.

Note

If an end user starts a chat with the previous support ticket ID, Service Manager automatically assigns this chat request to an available chat assignment group.

Configure Chat Assignment Group

To support basic chat routing and provide a most effective resolution to an end user's issue, a Service Desk manager needs to create a new assignment group or update an existent assignment group as a chat assignment group. A service desk manager also needs to set chat related settings for a chat assignment group, including support departments, locations, languages, and companies.

To update the settings of an assignment group to support End Use Chat, see Add a Service Desk assignment group.

Configure Chat Assignment Rules

To assign each End User Chat request to a virtual agent or to an assignment group, a system administrator needs to save a copy of the out-of-box Service Manager

Process Designer Assignment rule set `sdc.assignment.routing`, and then perform either of the following changes on the copied rule.

Rules	Tasks	Reference
Live chat Assignment rule	• To update the out-of-box chat assignment rule, select the rule and then click **Edit Rule/Group**. • To create a new chat assignment rule to suit the business needs, click **Add Rule**.	See Add an Assignment rule for detailed information and instructions. **Important** In the **Assignment Rule** pane, you must use `Helpdesk Group = true` in Additional Query to filter the Help Desk groups for live chat assignment.
Virtual agent Assignment rule	• To update the out-of-box chat assignment rule, select the rule and then click **Edit Rule/Group**. • To create a new chat assignment rule to suit the business needs, click **Add Rule**.	Write your own JavaScript.

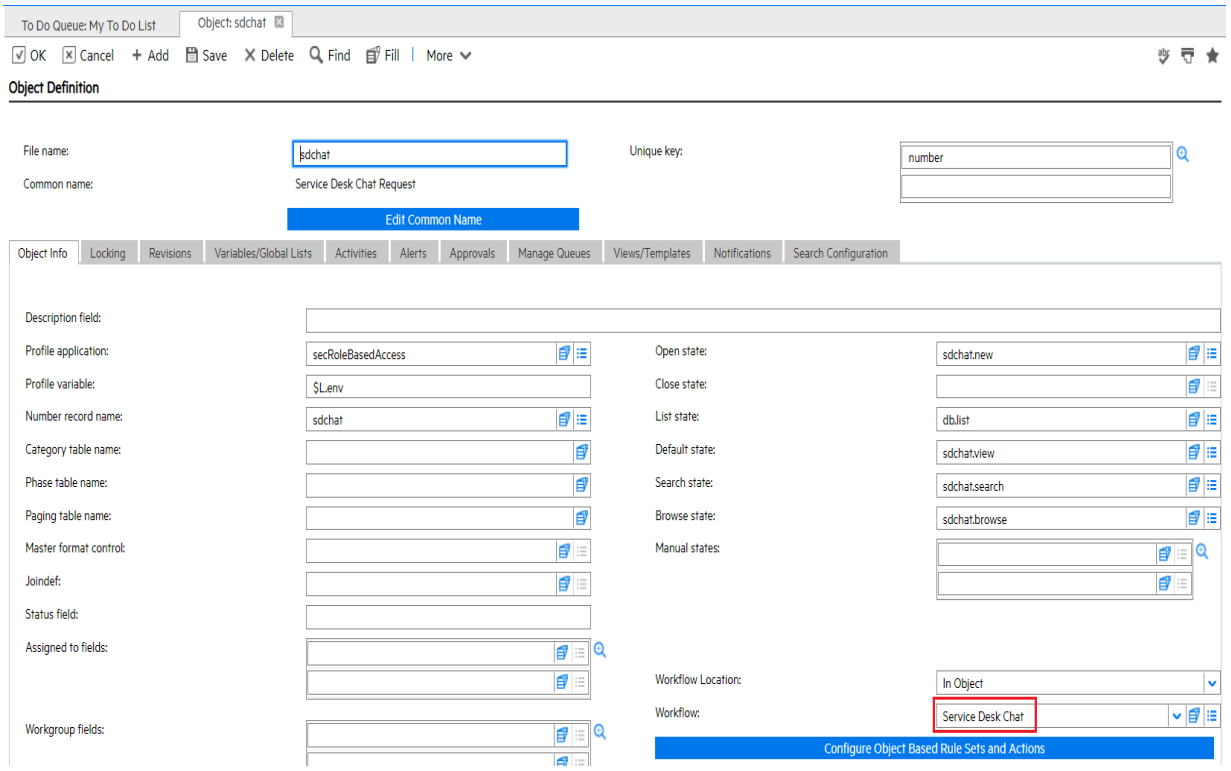
Note

If there are multiple assignment groups available for an End User Chat request, the system assigns only one group to this chat based on the sequence of the records stored in database.

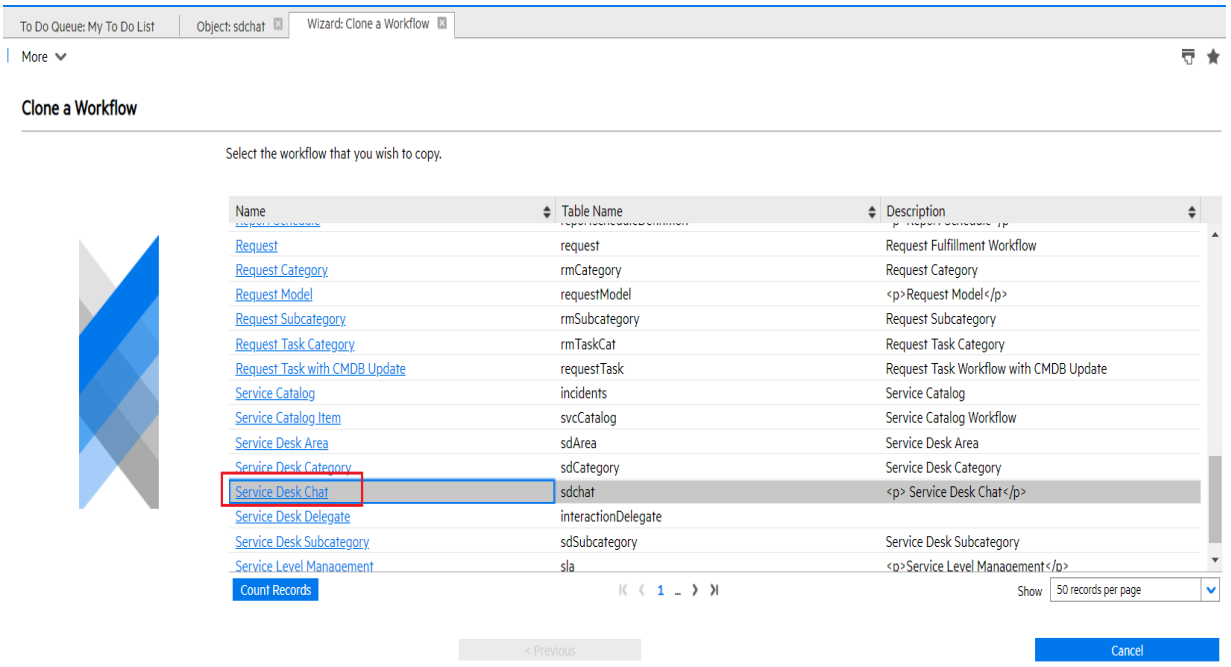
Service-based chat assignment example

In this example, we modify an out-of-the-box workflow to demonstrate the service based chat assignment.

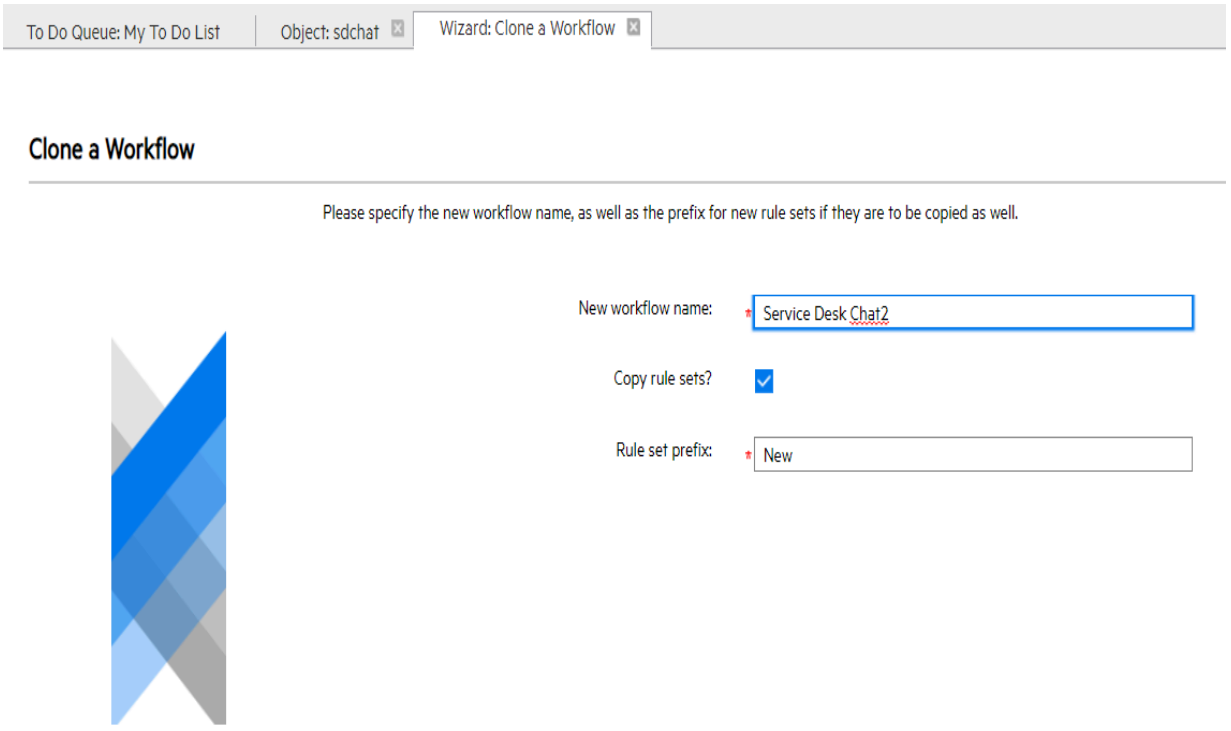
1. Go to **Tailoring** > **Document Engine** > **Objects** and find the workflow used in the **sdchat** Object. It is **Service Desk Chat** here.



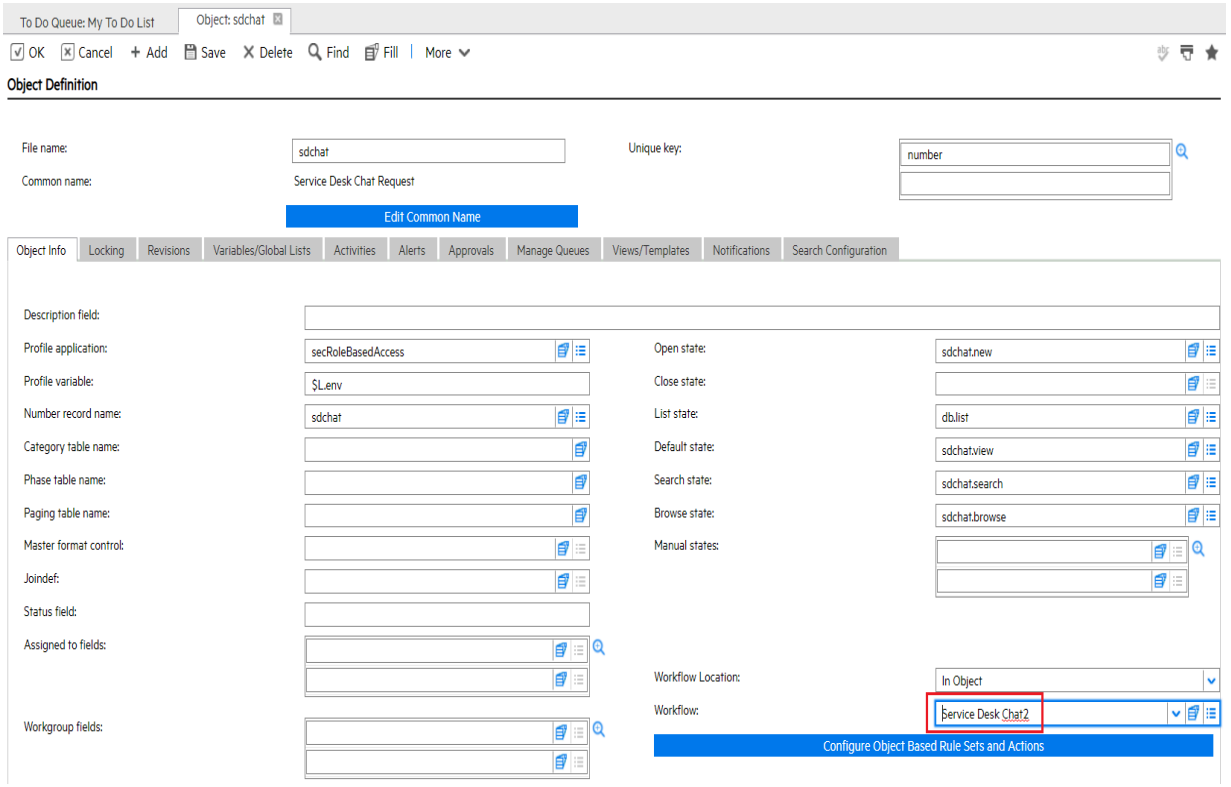
2. Go to **Tailoring** > **Process Designer** > **Copy Existing Workflow**. Click the **Service Desk Chat** link.



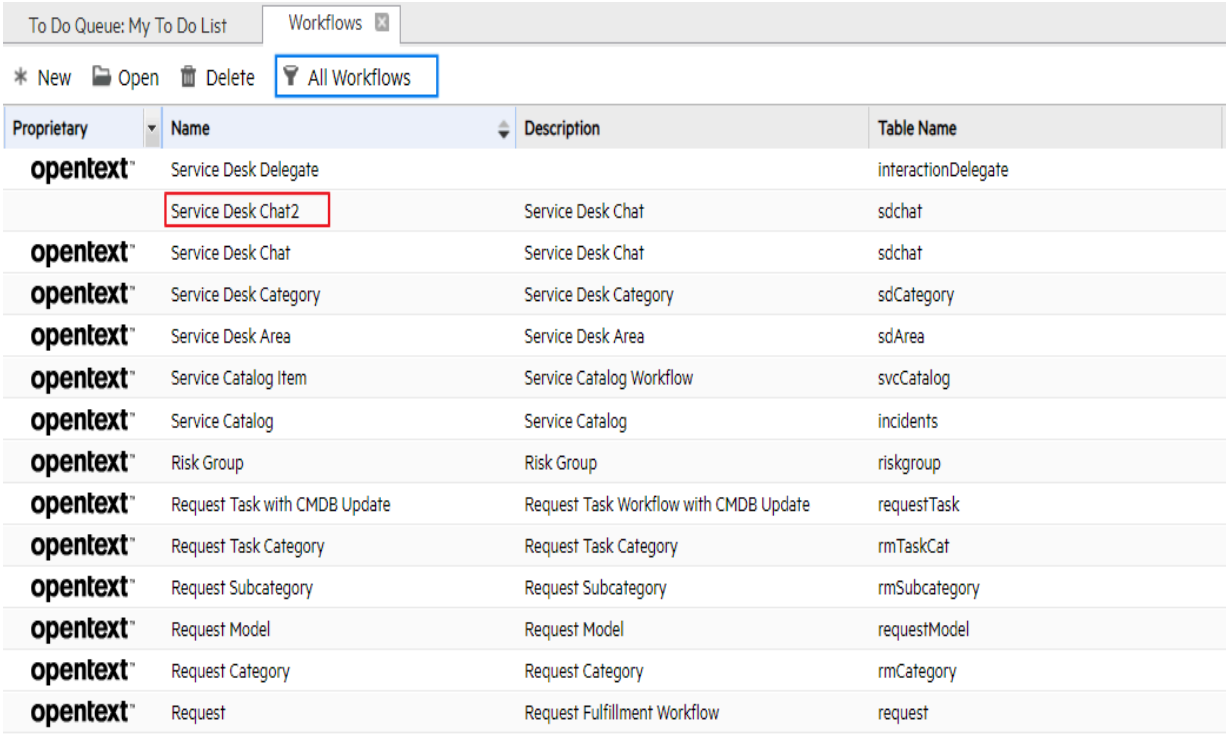
3. Type a new name to the workflow, select the **Copy rule sets?** checkbox and then provide a prefix (such as New). Click **OK**.



4. Go back to the sdchat object opened in step 1, and then select the newly cloned workflow in the Workflow drop-down list. Save the change.

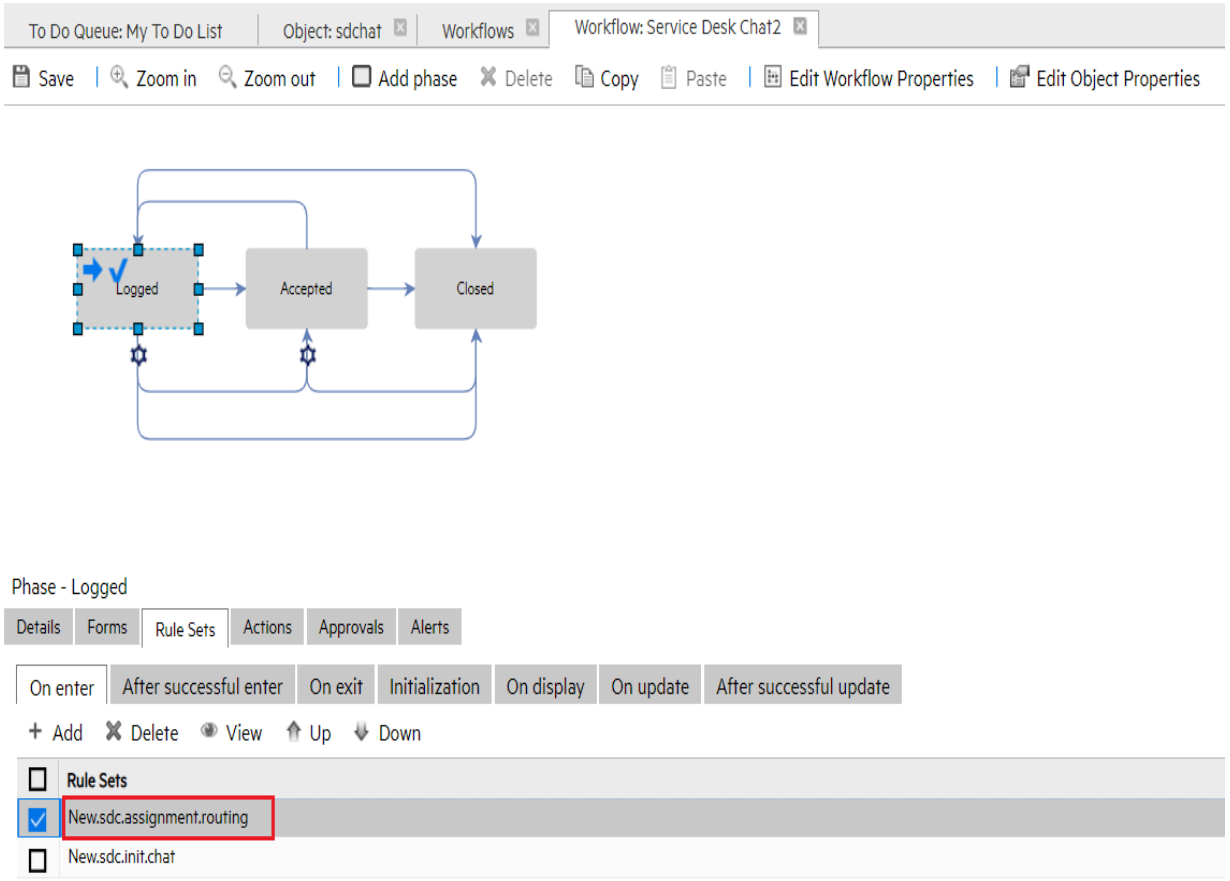


5. Go to **Tailoring** > **Process Designer** > **Workflows**, and then double-click to open the newly cloned workflow **Service Desk Chat2**.

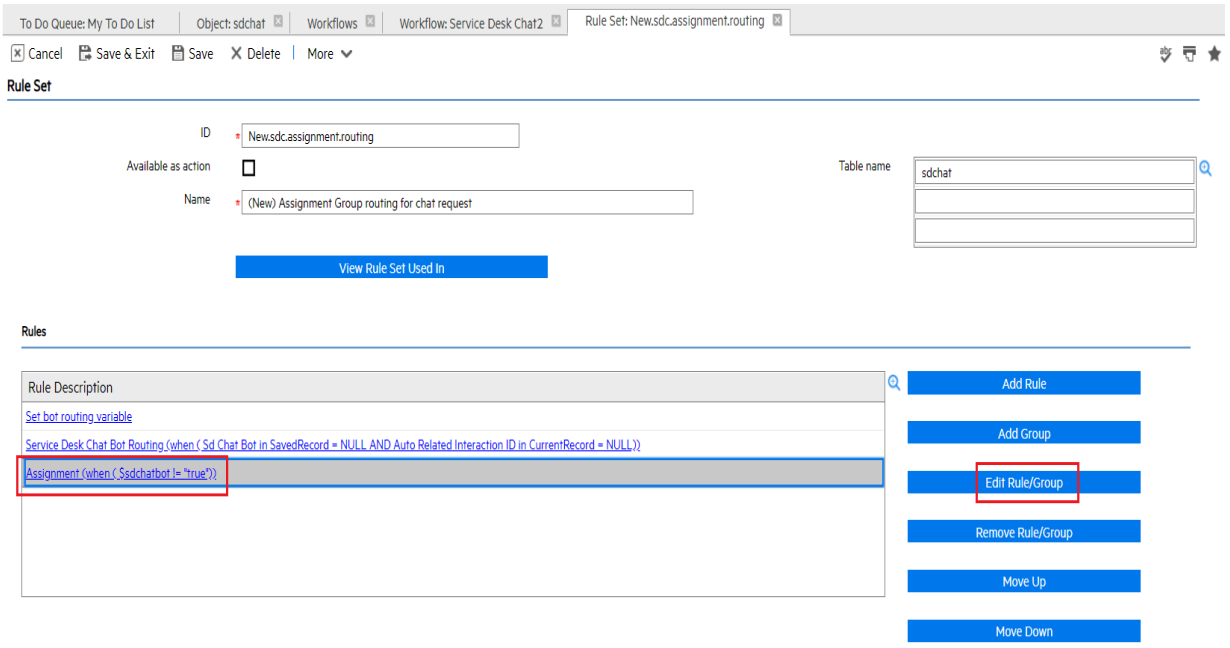


Proprietary	Name	Description	Table Name
opentext	Service Desk Delegate		interactionDelegate
	Service Desk Chat2	Service Desk Chat	sdchat
opentext	Service Desk Chat	Service Desk Chat	sdchat
opentext	Service Desk Category	Service Desk Category	sdCategory
opentext	Service Desk Area	Service Desk Area	sdArea
opentext	Service Catalog Item	Service Catalog Workflow	svcCatalog
opentext	Service Catalog	Service Catalog	incidents
opentext	Risk Group	Risk Group	riskgroup
opentext	Request Task with CMDB Update	Request Task Workflow with CMDB Update	requestTask
opentext	Request Task Category	Request Task Category	rmTaskCat
opentext	Request Subcategory	Request Subcategory	rmSubcategory
opentext	Request Model	Request Model	requestModel
opentext	Request Category	Request Category	rmCategory
opentext	Request	Request Fulfillment Workflow	request

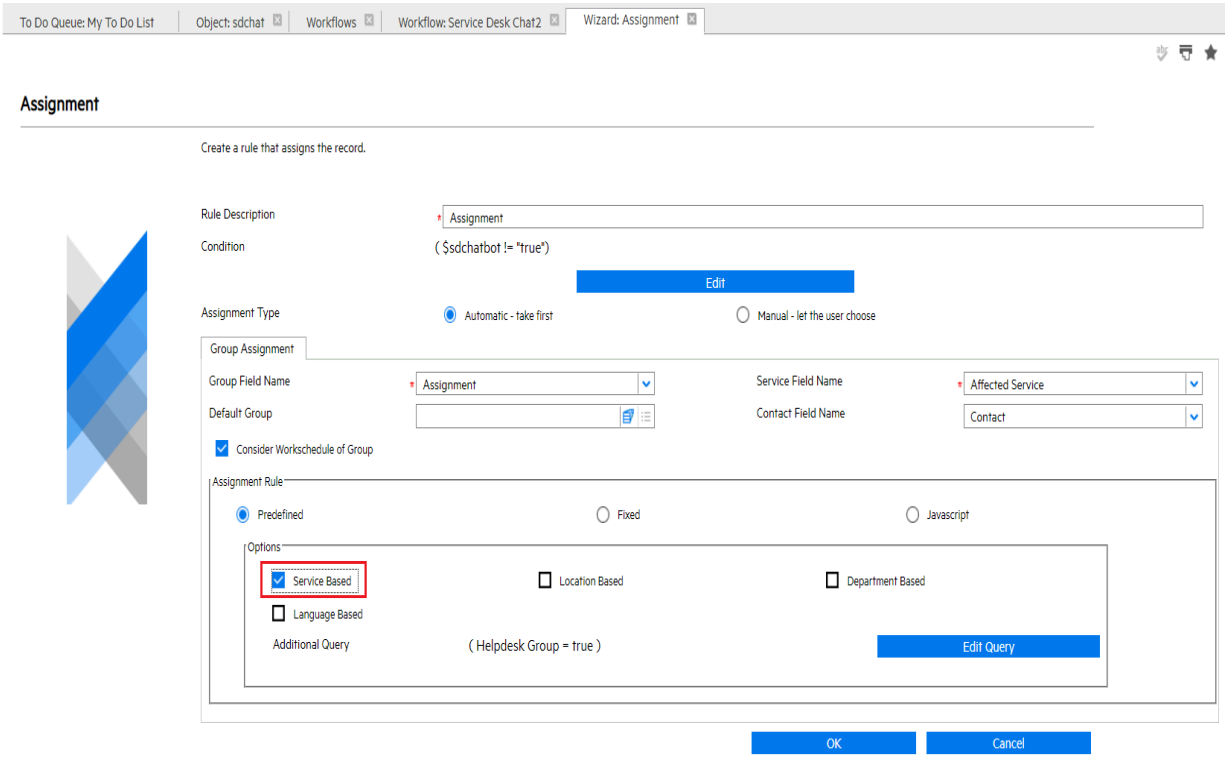
6. Select the **Logged** phase, go to the **Rule Sets** tab and then double-click to open the **New.sdc.assignment.routing** rule set.



7. Select the third rule set and then click the **Edit Rule/Group** button.

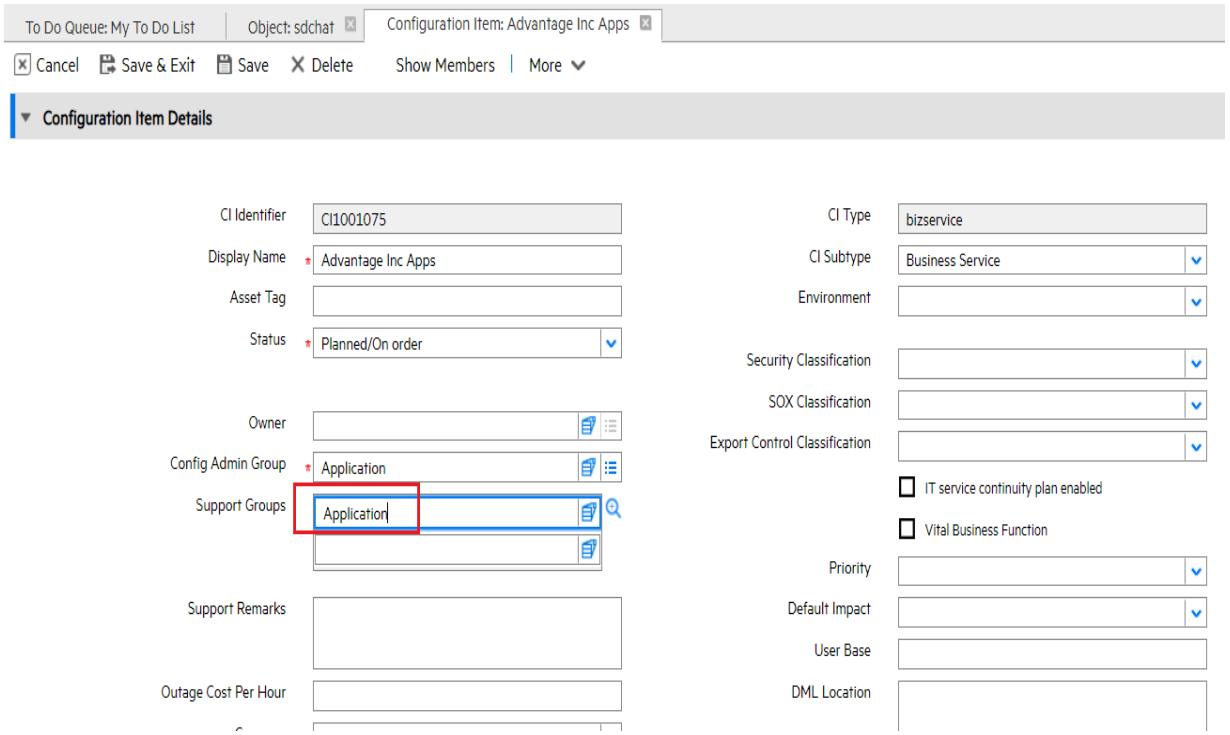


8. Select the **Service Based** checkbox, and then click **OK**.



9. Click **Save & Exit** in the Rule Set page.

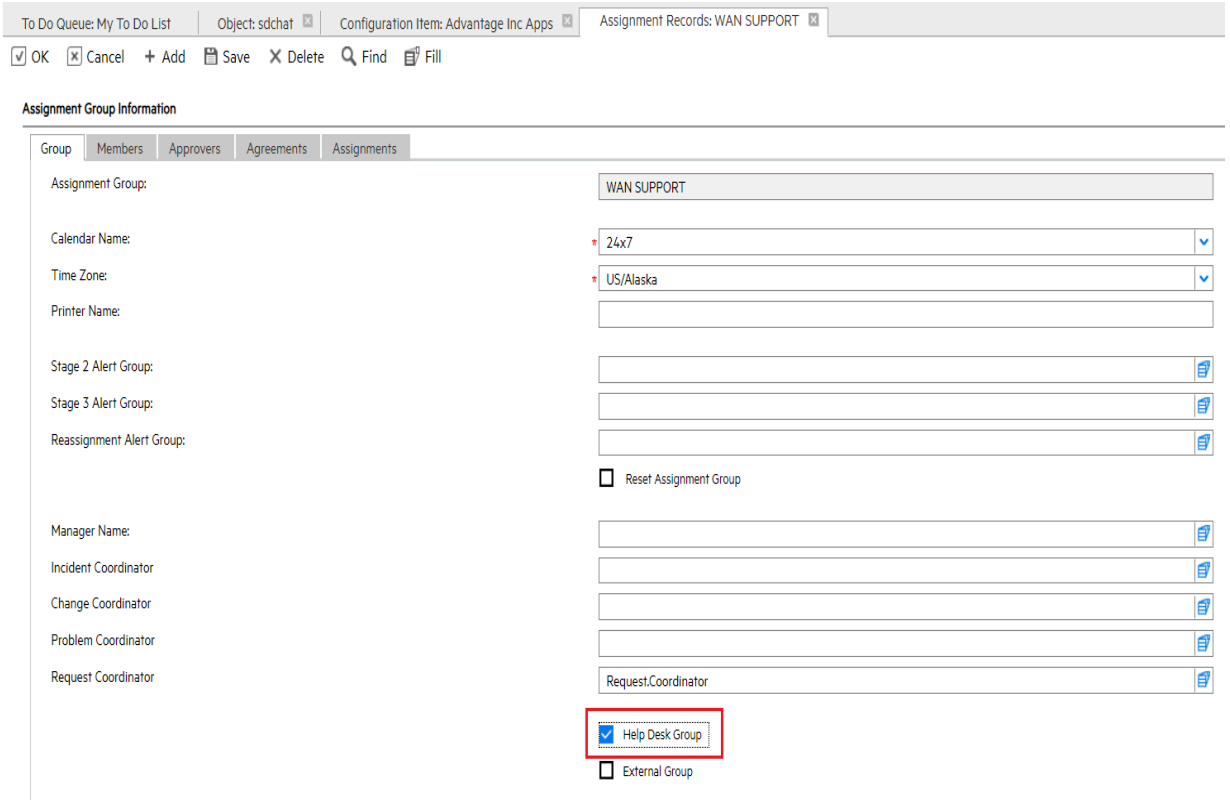
10. Go to **Configuration Management** > **Resources** > **Search Cis**. Search for a CI such as **Advantage Inc Apps**, and then set the value of Support Group to **Application**. Save your changes.



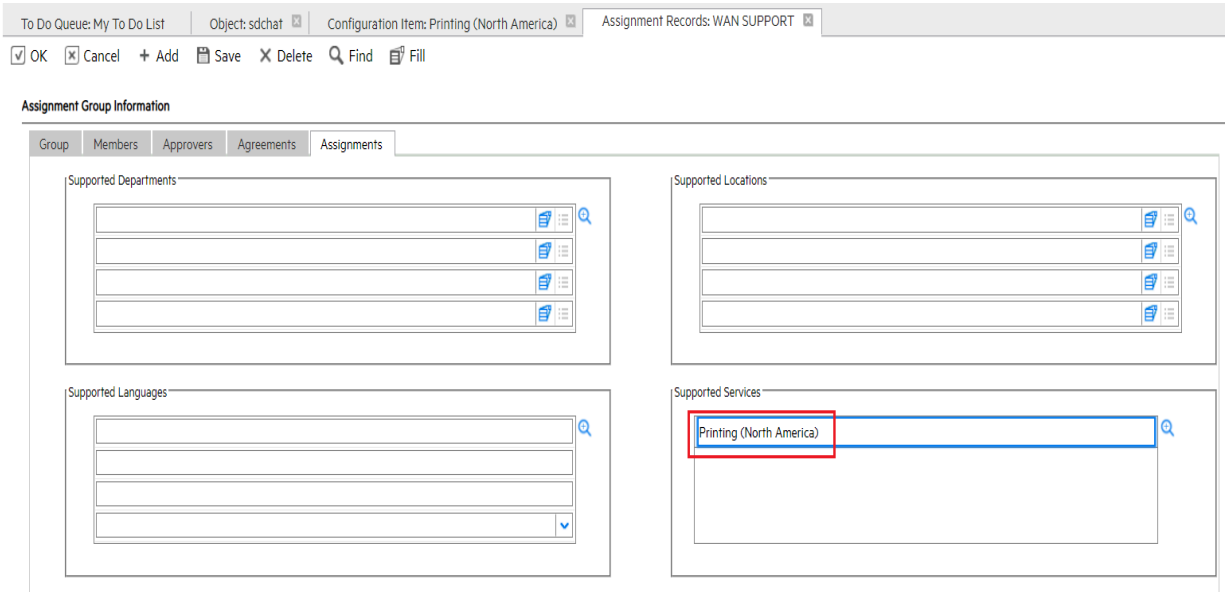
11. Repeat step 10 to update the CIs in the following table.

CI or Supported Services	Assignment Group
Advantage Inc Apps	Application
Exchange	FACILITIES
Education	Service Desk
Printing (North America)	WAN SUPPORT

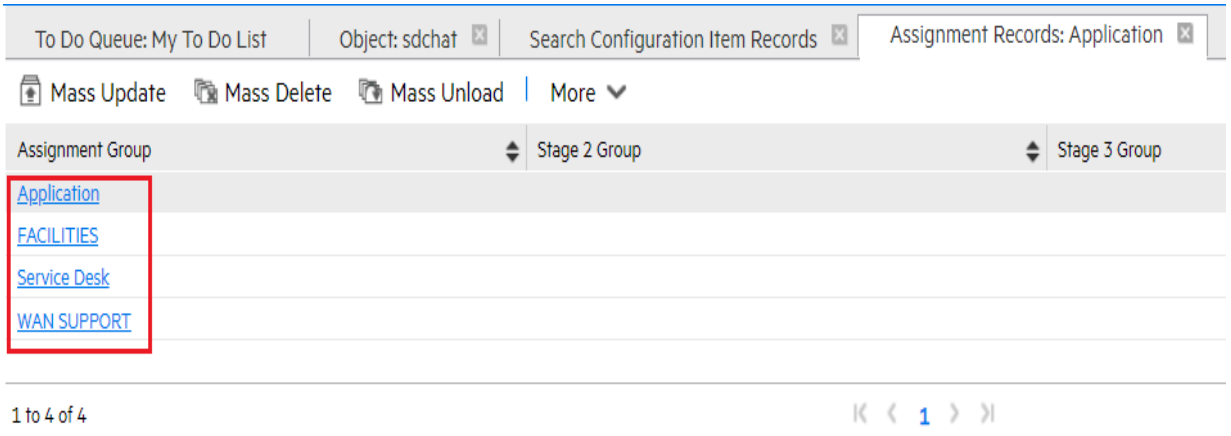
12. Go to **Service Desk** > **Configuration** > **Assignment Groups**. Search for and update the above assignment groups one-by-one. You need to fill in the mandatory fields and then select the **Help Desk Group** checkbox to grant the assignment group help desk capability.



13. For each assignment group, check that the Supported Services field is configured based on the table in step 11.

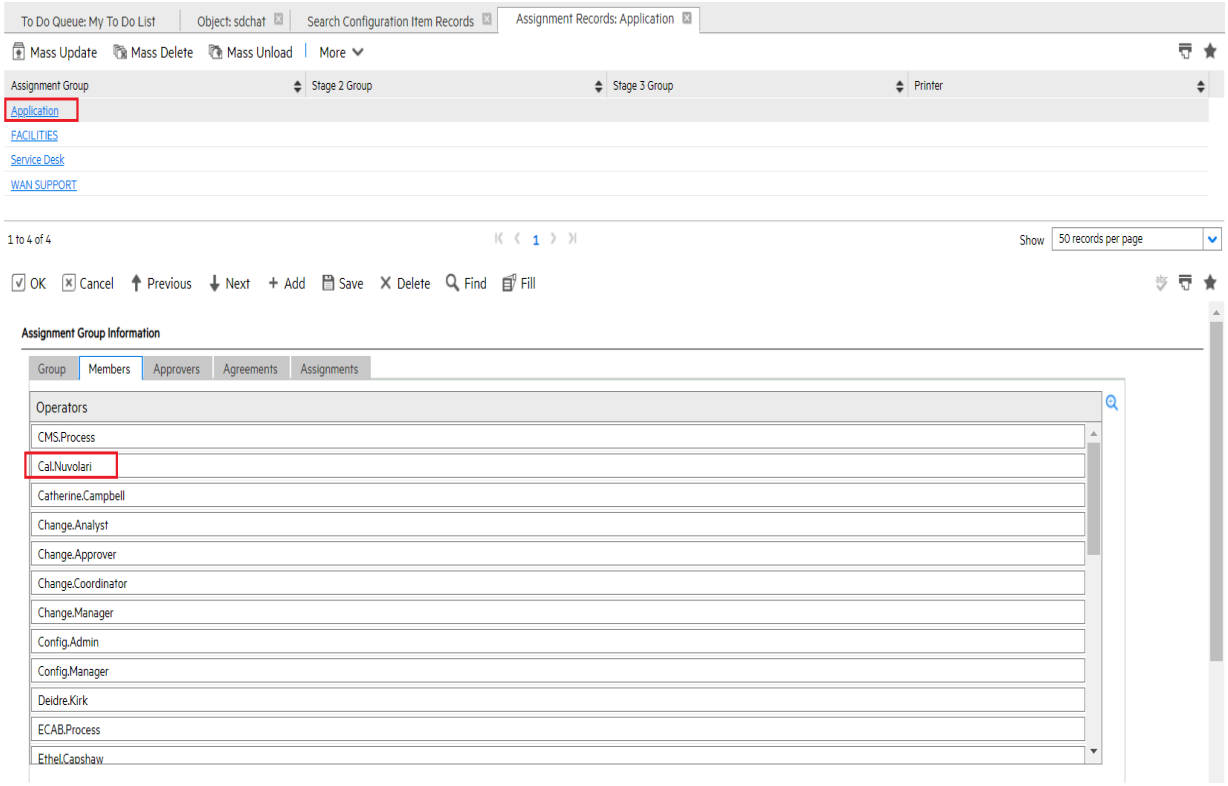


14. Go to **Service Desk** > **Configuration** > **Assignment Groups**. Select the **Help Desk Group** checkbox and then click **Search**. There are four Help Desk Groups now.

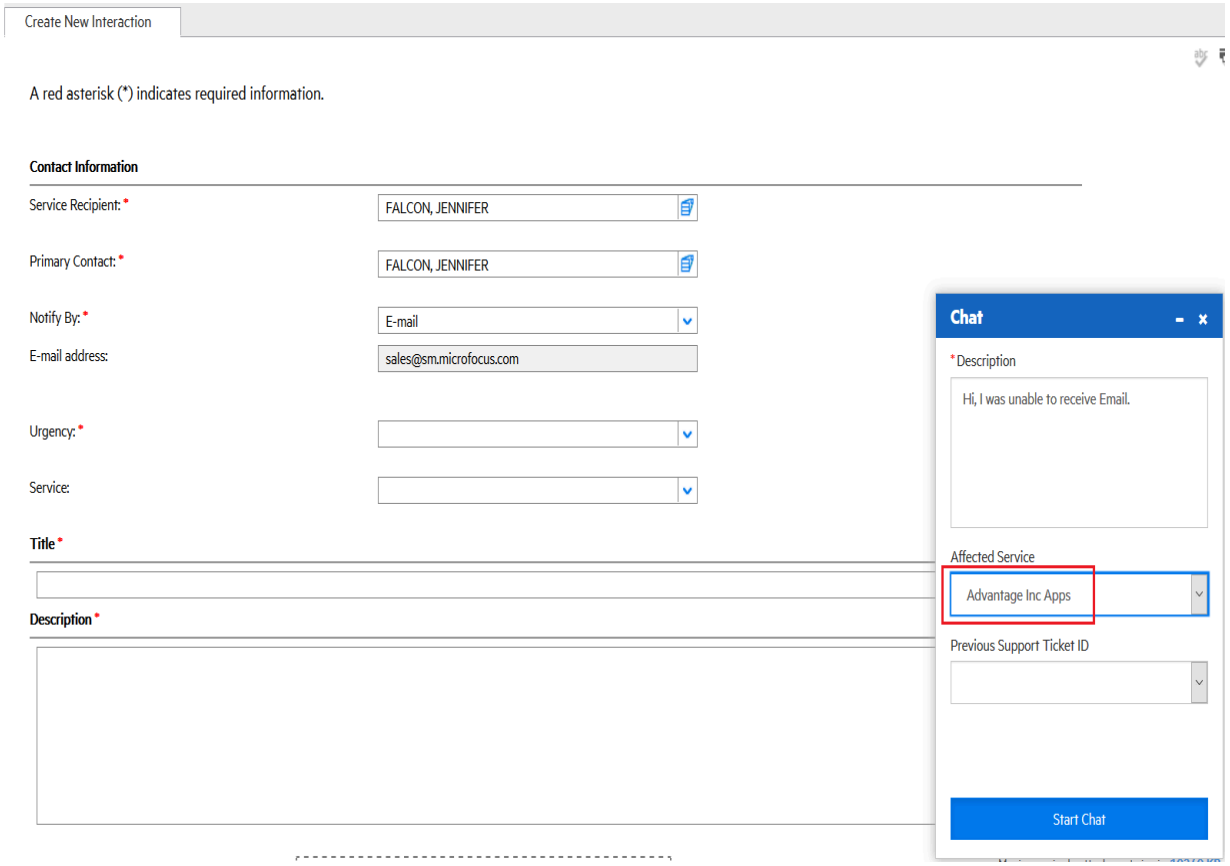


15. Check the members of each assignment group. Make sure that the operators in the following table exist in the corresponding group.

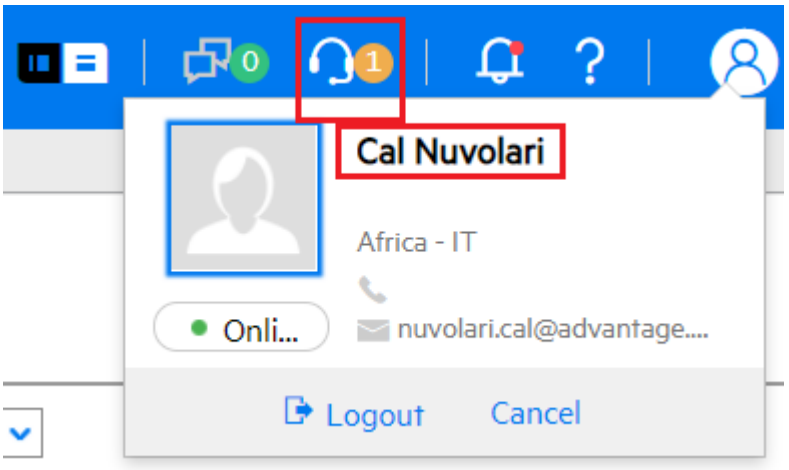
CI or Supported Services	Assignment Group	Operator
Advantage Inc Apps	Application	Cal.Nuvolari
Exchange	FACILITIES	Belinda.Morris
Education	Service Desk	Jaco.Staple
Printing (North America)	WAN SUPPORT	Adrian.Baxt, Belinda.Morris, Jaco.Staple



16. Sign in to Service Manager with the above four operator accounts on different computers or using different browsers.
17. Sign in to the ESS web client (ess.do) as an ESS user. Start a chat with the **Affected Service** field set to **Advantage Inc Apps**.



Only the operator Cal.Nuvolari can receive the chat request.



18. Repeat the test for each Affected Service in the above table to verify the service-based routing.

	Cal.Nuvolari	Belinda.Morris	Jaco.Staple	Adrian.Baxt
Advantage Inc Apps	√	X	X	X
Exchange	X	√	X	X
Education	X	X	√	X
Printing (North America)	X	√	√	√

1.17.4.2.3. Set live chat assignment End Lead Time

The Chat Assignment End Lead Time setting enables you to set a general time period for all the Service Desk chat assignment groups to stop responding to new chat requests prior to the service end hour. For example, if the working hour of a chat assignment group is 9:00 to 18:00 and the value of the End Lead Time is set to 00:30:00, the group stops accepting new chat requests at 17:30. Consequently, when live chat is unavailable, the system displays a message in the Service Portal chat window indicating that chat is unavailable.

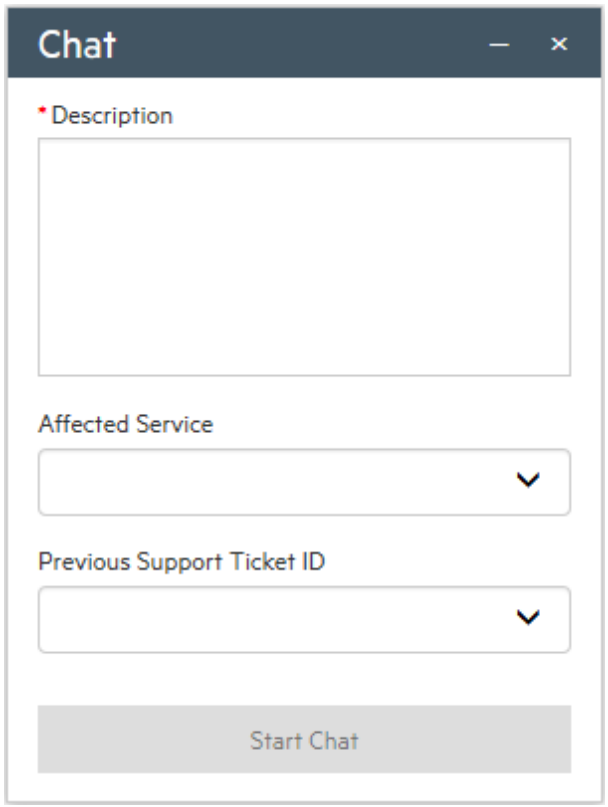
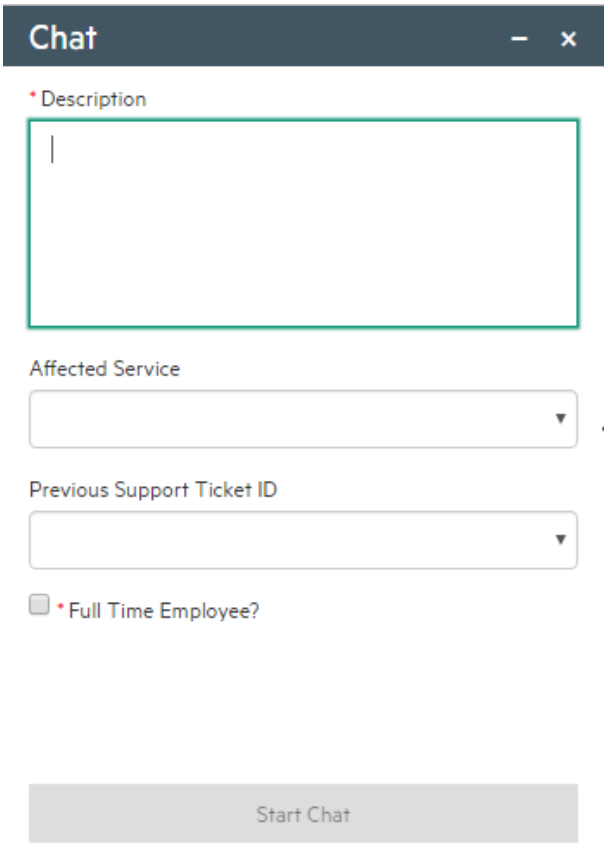
To configure the Chat Assignment End Lead Time setting:

1. From the System Navigator, click **Service Desk** > **Administration** > **Settings**.
2. In the Chat Assignment End Lead Time field, set the time period in the `hh:mm:ss` format. For example, 00:15:00.

1.17.4.2.4. Customize the End User Chat window

As a Service Desk manager, you can use the Service Desk Pre-chat Configuration form to customize the fields that are displayed on the End User Chat window. These customized fields, together with a list of custom fields types, enable you to gather additional information from end users before the chat begins.

See the following sample screenshots as a comparison:

Default End User Chat window	End User Chat window after configuration
	

Update the out-of-box End User Chat window configuration

One company can only have one End User Chat window configuration record. If you don't enable Multiple Company mode in your Service Manager system, you can update the out-of-box End User Chat window configuration record only.

Follow these steps to update the out-of-box End User Chat window configuration:

1. Sign in to Service Manager as a Service Desk manager.
2. Click **Service Desk** > **Administration** > **Service Desk Pre-chat Configuration** to open the Service Desk Prechat Configuration form.
3. Click **Search** to display the out-of-box Service Desk Prechat Configuration form, on which the default fields are Description, Affected Service, and Previous Support Ticket ID.
4. Follow these steps to add a new field to the End User Chat window:
    1. Click **Add New Custom Field**.
    2. Fill in the fields as necessary. See the following table for details.

Label	Description
ID	Specify the ID for the custom field. This ID uniquely identifies the field so that it can be referenced programmatically.
Label	Specify a label for the custom field. This label is displayed on the End User Chat window.
Mapped Field	Select a mapped field from the Service Manager sdchat table. To append more selections to this drop-down menu, see the *Append a Customized Mapped Field* section.
Max Length	The maximum text input length of the custom field. This number is populated automatically.
Mandatory	Select this checkbox to make the custom field mandatory.

Label	Description

Display Type	Specify the display type of the custom field. The format of the field on the End User Chat window depends on the type you select. Refer to the following explanations:
	▪ Select **Text** to display the custom field as a single-line text field.
	▪ Select **Check Box** to display the custom field as a checkbox.
	▪ Select **Multitext** to display the custom field as a multiple-line text field.
	▪ Select **Pick List** to display the custom field as a drop-down menu. The system prompts you for the following additional information:
	  ▪ Select **Global List**, and then click the **Fill Field** button to select a Service Manager global list definition.
	  ▪ Select **Options**, and then provide entries to display in the drop-down menu manually. For example, type 1 in the Value field, and then type Emergency in the Display Value field, this entry is display as 1 - Emergency in the

Label	Description
	customized drop-down menu. ▪ Select **JavaScript**, and then set the *valueList* JavaScript array and the *displayList* JavaScript array. For example: `var user=system.functions.operator(); var list=lib.EndUserChat.getSubscriptions(user); valueList=list['valueList']; displayList=list['displayList'];`

3. Click **Finish**. Your customized field is displayed in the Prechat Form section on the Service Desk Prechat Configuration form.

5. (Optional) Select a field in the Prechat Form section, and then click **Move Field Up** or **Move Field Down** to adjust the sequence of the fields.

6. (Optional) To remove a customized field, select the field and then click **Remove Custom Field**.

> **Note**
>
> You can't remove the default Description field.

7. (Optional) To update the localized labels, click the ID of a customized field in the Prechat Form section and then click **Edit localized labels**.

For each language available, you can localize the label text. After you localize the text, click **Save** and then **OK** to return to the Service Desk Prechat Configuration form.

8. Click **Save**.

Now your customized field is displayed on the End User Chat window. End users need to sign in to the Service Portal or the ESS portal again to see the newly customized fields that are displayed on the End User Chat window.

Add a new End User Chat window configuration

If your Service Manager system has enabled Multiple Company mode, you can add a new End User Chat window configuration record for each company.

Follow these steps to add a new End User Chat window configuration:

1. Sign in to Service Manager as a Service Desk manager.
2. Click **Service Desk** > **Administration** > **Service Desk Pre-chat Configuration** to open the Service Desk Prechat Configuration form.
3. Type your description in the Brief Description field.
4. Select a company in the Company drop-down menu.
5. Click **Add**. The system automatically generates a new configuration record number in the Number field.
6. Follow *step 4* to *step 7* as described in the *Update the Out-Of-Box Service Portal Chat Window Configuration* section.

Note

If you are working with an out-of-the-box End User Chat window configuration record and your Service Manager system has enabled Multiple Company mode, make sure you have selected your company in the Company drop-down menu in **Service Desk Pre-chat Configuration**.

Append a customized mapped field

Follow these steps to append a customized mapped field:

1. Sign in to Service Manager as a system administrator.
2. Click **System Administration** > **Tailoring** > **Database Dictionary**.
3. Type sdchat in the File Name field, and then click **Search**.
4. Click **sdchat** in the list, and then click **New Field/Key**.

5. Type a name for your customized field in the Name field, and then select **character** in the Type drop-down list.
6. Click **Add**.
7. Sign out from Service Manager and then sign in again.

Now your customized mapped field appears in the Mapped Field drop-down menu.

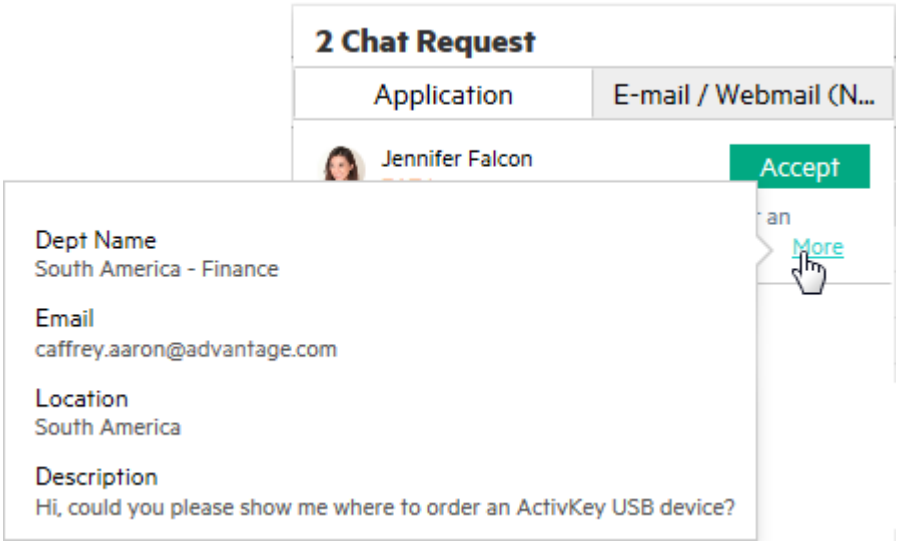
View the customized field on a Service Manager End User Chat request record

Follow these steps to view the customized field on a Service Manager End User Chat request record:

1. Sign in to Service Manager as a Service Desk manager.
2. Select **Service Desk Chat Request** in Service Manager To-Do queue.
3. Click **More** > **Customize Current View**.
4. Click **Fields...**, and then select your customized field in the Service Desk Chat Request drop-down list.
5. Click **Add to List**.
6. Click **Next** and **Finish**.

1.17.4.2.5. Customize the chat request preview information

When you hover your mouse over the **More** button of a chat request in the chat request panel, the system displays some preview information of the chat request. By default, this information includes the department, email, location, and the end user's issue description. See the following screenshot as an example:



As a Service Manager system administrator, you can use the Script Library to update the JavaScript and define the preview information of a chat request. To customize the chat request preview information in the chat request panel, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Click **Tailoring** > **Script Library**.
3. Open the `EndUserChatCustomization` JavaScript in the Script Library, and then locate the `function overrideChatRequestInfo(chatReqId,chatReqTableName)` script. The out-of-the-box script is as follows:

```
function overrideChatRequestInfo(chatReqId,chatReqTableName)
{
/*
//Below is an example
var chat=new SCFile(chatReqTableName,SCFILE_READONLY);
var sql="number=\""+chatReqId+"\"";
if(chat.doSelect(sql)===RC_SUCCESS){
var
contact=lib.EndUserChat.getContactByName(chat['contact']);
var rtn=[];
rtn.push({"label":lib.EndUserChat.getFieldLocalizedCaption("
company", "contacts"),"value":contact['company']});
rtn.push({"label":lib.EndUserChat.getFieldLocalizedCaption("
dept.name", "contacts"),"value":contact['dept.name']});
rtn.push({"label":lib.EndUserChat.getFieldLocalizedCaption("
email", "contacts"),"value":contact['email']});
rtn.push({"label":lib.EndUserChat.getFieldLocalizedCaption("
location", "contacts"),"value":contact['location']});
rtn.push({"label":lib.EndUserChat.getFieldLocalizedCaption("
description",
chatReqTableName),"value":chat['description']});
return rtn;
}
else return null;
*/
return null;
}
```

4. Update the script to suit your business needs. See the following sample script for detailed descriptions for this function:

```
/**
* This function is left to customers to customize the chat
request info which is displayed when accepting a chat
request, system will display
* OOB default configuration (Department, Email, Location,
and Description of the Service Portal user's issue) if this
function returns null
* @param chatReqId (String)
*          The chat request ticket id, it will be passed in
automatically
* @param chatReqTableName (String)
*          The chat request dbdict name, it will be passed
in automatically
* @return [
*             {"label":"Company","value":"advantage"},
*             {"label":"Dept Name","value":"North America -
HR & Administration"},
*
{"label":"Email","value":"user@microfocus.com"},
*             {"label":"Location","value":"BLDG1"},
*             {"label":"Description","value":"Can not
connect to email server"}
*          ]
*/
```

Note

The system displays the default configuration (Department, Email, Location, and Description of the Service Portal user's issue) when the function returns `null`.

5. Click **Save** and **OK**.

1.17.4.2.6. Set up End User Chat virtual agent

The End User Chat virtual agent simulates the way a human being responds to a question and helps to reduce IT support costs by providing intelligent answers to the end users all the time. When an end user asks a question, the animated virtual agent joins the chat and then retrieves solutions from a variety of resources, including Service Manager Service Catalog items, Knowledge Management (KM) documents, SharePoint documents, static web pages, and file systems. At any time during the working hours, the end users can escalate their issues and choose to chat with a live Service Manager Service Desk agent.

Note

When a virtual agent accepts a chat request, Service Manager automatically creates a Service Desk chat request record but doesn't open a new Interaction record.

Enable virtual agent

Follow these steps to enable virtual agent:

1. Sign in to Service Manager as a system administrator.
2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Configuration**.
3. Select the **End User Chat with chatbot enabled** checkbox.

After you enabled virtual agent, the system firstly routes a Service Portal or ESS portal chat request to a virtual agent according to the assignment rule settings for virtual agent. During the chat with a virtual agent, the Service Portal user or ESS portal user can escalate their issues at any time to a live Service Desk agent.

Configure language setting for virtual agent

As a system administrator, you need to either updating the out-of-box language settings or add language settings for a new virtual agent. Consequently, Service Manager detects the end user's language selection when signing in to the ESS portal and then assigns this chat request to a virtual agent which has the same language setting.

Note

In this release, virtual agent provides support for English, German, French, Spanish, Italian, and Dutch only. If the end user's input language is other than these six languages, Service Manager assigns the chat request to Service Manager Service Desk assignment groups based on the assignment rule configuration.

Caution

Be careful with the language settings because the AIML files are related to the languages.

Follow these steps to update the out-of-box language setting for an existing virtual agent:

1. Sign in to Service Manager as a system administrator.
2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Service Desk ChatBot**.
3. Click **Search** to display a list of existing virtual agents.
4. Select a virtual agent, and then update the language settings on the Service Desk Chat Bot definition form.

   Tip

   You can also rename the virtual agent in this step.

5. Click **Save**.

Follow these steps to configure the language setting for a new virtual agent:

1. Sign in to Service Manager as a system administrator.
2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Service Desk ChatBot**.
3. Type the name for the virtual agent in the Name field.
4. Select a language from the Language drop-down menu.
5. Click **Add** and **Save**.

Configure portrait for virtual agent

Follow these steps to update the out-of-box portrait for a virtual agent:

1. Sign in to Service Manager as a system administrator.
2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Service Desk ChatBot**.
3. Click **Search** to display a list of existing virtual agents.
4. Select a virtual agent, and then click **Specify Image** to open the Specify Image Location window.
5. Click **Choose File** to browse for another photo that's saved on your computer.

   Note

   The supported image types are *.png, *. jpg, *.bmp, and *.jpeg.

6. Select your new photo and then click **Open**. Alternatively, double-click the new photo.

   The system displays the new photo in the Portrait section automatically. To remove this photo, click **Remove**.
7. Click **Save**.
8. Restart the chat service to make the new portrait effective immediately.

   Note

   If you don't want to restart the chat service, the new portrait will be effective after 30 minutes.

Configure IDOL search for virtual agent

Follow these steps to configure the IDOL search settings for virtual agent:

1. Sign in to Service Manager as a system administrator.
2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Service Desk ChatBot**.
3. Click **Search** to display a list of existing virtual agents.
4. Select a virtual agent, and then configure the following settings on the IDOL Search Configuration tab:
   - Max Return Results: Define the maximum number of Knowledge Management articles returned from IDOL search.
   - Knowledgebase Libraries: Select the knowledgebase libraries to narrow the IDOL search. These libraries may include Catalog library, Knowledge

library, and external libraries such as a SharePoint repository, a file system, or a website if the corresponding connectors are set up in Smart Analytics.

5. Click **Save**.

Use AIML with virtual agent

Service Manager End User Chat virtual agent uses Artificial Intelligence Markup Language (AIML) to receive end user's questions, search for answers, and then return solutions. You can define a series of AIML configuration records in Service Manager to represent a variety of responses that a virtual agent may provide to an end user. These responses and their corresponding questions can be published to an AIML file, which is a simple text file saved in the chat service folder.

Note

The system automatically adds an AIML record when you add a new virtual agent.

Follow these steps to update the out-of-box AIML configuration for a virtual agent:

1. Sign in to Service Manager as a system administrator.
2. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Service Desk Chat Bot**.
3. Click **Search** to display a list of existing virtual agents.
4. Select a virtual agent, and then click **Edit AIML** to display a list of AIML configuration forms that are used by this virtual agent.
5. Select an AIML configuration record, and then fill in the fields as necessary. See the following table for details.

Label	Description
Category ID	The system-generated unique ID for this AIML category. AIML objects are composed of knowledge units called categories.
Bot ID	The system-generated unique ID for this virtual agent.
Input Pattern Text	Each AIML category contains a single pattern element that represent the user's question. Type a string of characters intended to match one or more end user inputs. You can use the asterisk (*) wildcard in this field. **Note** This field is greyed out if you select **Any Word** in the Input Matching Type list.

Label	Description
Input Matching Type	Select a matching type from the drop-down list. Depending on your selections, the input pattern syntax may match an infinite number of user inputs. Refer to the following explanations: ◦ Select **Starts With The Text**, the system appends a caret (^) to the pattern text when publishing the configuration to an AIML file. ◦ Select **Ends With The Text**, the system adds a caret (^) before the pattern text when publishing the configuration to an AIML file. ◦ Select **Contains The Text**, the system adds a caret (^) before the pattern text and then appends a caret (^) to the pattern text when publishing the configuration to an AIML file. ◦ Select **Exactly Matching The Text**, the system uses the pattern text directly when publishing the configuration to an AIML file. ◦ Select **Any Word**, the system generates an asterisk (*) for the pattern text when publishing the configuration to an AIML file.

Label	Description
That Pattern Text	Type a string of characters intended to specify the virtual agent's previous utterance. **Note** This field is greyed out if you select **Any Word** in the That Matching Type list.
That Matching Type	See the explanations in **Input Matching Type** above.
Topic Pattern Text	Type a string of characters intended to specify the name of an AIML topic group.
Template	Each AIML category contains a single template element that represent the virtual agents response. Type your AIML template, which may consist of literal texts, variables, conditional responses, random responses, customized tags, and so on. See the **Example: an AIML template with customized tags** section below for more information.

6. Click **Save** to save your updates to the existent AIML configuration record. Otherwise, click **Add** to save your updates as a new AIML configuration record.
7. Click **Back** to return to the Service Desk Chatbot definition form.
8. Click **Publish AIML** to publish the configuration to an AIML file, which is stored in the *<chat service installation directory>*\bots*<virtual agent ID>*\aiml directory.

Example: an AIML template with customized tags

The following script is an excerpt from an out-of-box AIML configuration template, which consists of customized tags that are specific for Service Manager:

```
<SMJS>
        <JSCALL>
                lib.EndUserChatBotIDOLSearch.IDOLSearch
        </JSCALL>
        <JSPARAMS>
                <JSPARAM name="smLoginUser"><get
name="smLoginUser"/></JSPARAM>
                <JSPARAM name="smBotId"><get name="smBotId"/>
</JSPARAM>
                <JSPARAM name="problemDesc"><get
name="problemDesc"/></JSPARAM>
        </JSPARAMS>
</SMJS>
```

Refer to the following table for detailed descriptions about the tags:

Tag	Description
<SMJS>	Indicates that the system invokes a Service Manager JavaScript and returns the executed value of this Javascript to a end user as the real response.

Tag	Description
<JSCALL>	Indicates the Service Manager JavaScript to be invoked. The format is `lib.<Service Manager ScriptLibrary Name>.<function name>`. The sample JavaScript is `lib.EndUserChatBotIDOLSearch.IDOLSearch`, which consists of the following codes:

Tag	Description

	<pre>var predicatePrefix = "BOTCUSTPREDICATE";
	var IDOLDSearchPredicate = "SMIDOLSEARCH";
	
	function IDOLSearch() {
	    var endUserLoginName = arguments[0]['smLoginUser'];
	    var botId = arguments[0]['smBotId'];
	    var content = arguments[0]['problemDesc'];
	    var smSdChatId = arguments[0]['smSdChatId'];
	    ... ...
	        var lng = funcs.lng(results);
	
	        if (lng > 0) {
	            returnResults['predicates'] = {};
	            var i;
	            for (i = 0; i < lng; i++) {
	                if (lib.IDOLGlobalSearchOKM.isOkmDocument(results[i].query, results[i].database, results[i].reference)) {
	                    returnResults['predicates'][predicatePrefix + "_" + IDOLDSearchPredicate + (i + 1)]
	
	                    = genereateChatLinkForOKM(results[i]);
	                } else {
	                    returnRe</pre>

Tag	Description

	<pre>sults['predicates'] [predicatePrefix + "_" + IDOLDSearchPredicate + (i + 1)]
	
	                =
	genereateChatLink(results[i]);
	                }
	            }
	            return returnResults;
	        } else {
	            return {};
	        }
	    }
	
	    return {};
	}</pre>
	In the `return` value:
	• `predicates`: This key word is to notify a virtual agent to set customized predicates that can be used in the following AIML categories as the template response by `<get name="predicate name"/>`. See the **Service Manager out-of-box predicates** section below for more information.
	• Predicate with the `BOTCUSTPREDICATE` prefix: This is a Service Manager reserved predicate for IDOL

Tag	Description
	search result. You can define your own predicates. • Predicate value with the `BOTCUSTPREDICATE` prefix: This is Service Manager "key word". The system uses the content after `BOTCUSTPREDICATE` as the template text response and filters `BOTCUSTPREDICATE` out of the response. **Note** After you "Configure IDOL search for virtual agent", you must update the AIML configuration by setting related predicates amount along with the search results so that the virtual agent's IDOL search can work successfully.
<JSPARAMS>	Indicates the parameters that JavaScript required.
<JSPARAM>	Specifies the parameter name and value.

Service Manager out-of-box predicates

Service Manager provides the following out-of-box predicates that can be used directly in AIML templates.

Predicate	Description
problemDesc	Problem description.
smLoginUser	End User's ID for Service Manager login.
smLoginUserFullName	End User's full name for Service Manager login.
smBotId	Virtual agent ID.
smSdChatId	Service Desk chat request record ID.
MULTIPLE_SENTENCES_INPUT	In AIML standard, a user input sentence is split to multiple sub-sentences if the sentence includes periods (.), question marks (?), or exclamation marks (!). The system performs a match for each sub-sentence and combines all the matched responses together as the final output to the end user during the conversation. However, this is not expected when a Service Manager virtual agent is solving an end user's issue. You can use the MULTIPLE_SENTENCES_INPUT predicate to keep the integrity of an end user's issue description and perform an IDOL search for the complete sentence.

Predicate	Description
ATE_MULTIPLE_SENTENCES_INPUT	ATE_MULTIPLE_SENTENCES_INPUT is a chat predicate key to set a flag and indicate whether a virtual agent needs to continue handling the subsequent sentences or not. When you set this predicate to `true`, a virtual agent regards the current template as the final output to the end user. All previously matched responses are abandoned and all subsequent sentences in the same request are ignored.

In addition, you need to define and assign values to the following predicates in either a JavaScript or in an AIML template.

Locations	Predicate	Description
Define in Javascript	BOTCUSTPREDICATE_SMIDOLSEARCH1/ BOTCUSTPREDICATE_SMIDOLSEARCH2/ BOTCUSTPREDICATE_SMIDOLSEARCH3...	Solutions from IDOL search.
Define in AIML template	smChatClosureCode	Closure codes when a virtual agent closes a Service Desk chat request ticket.

Out-of-box commands for virtual agents

Service Manager virtual agents use the following out-of-box commands in AIML templates to execute particular actions. However, virtual agents neither regards these commands as the template responses nor output these commands to an end user.

Command	Action
BOTCMD_GO_TO_LIVE_CHAT	Transfer to Live Chat.
BOTCMD_VIRTUAL_AGENT_END_CHAT	Close a Service Desk chat request ticket.

1.17.4.2.7. Set up End User Chat automation

End User Chat automation enables Service Manager system administrators and Knowledge Management (KM) users to link Operations Orchestration (OO) flows to a KM document. Consequently, when a Service Desk agent or a virtual agent shares this KM document to an end user, the user can run the linked OO flow and get the returned result in the End User Chat window. The Service Desk agent is also able to check the OO flow execution result from the Activity tab of the related chat request record.

Follow these steps to set up End User Chat automation:

1. Sign in to Service Manager as a system administrator.
2. Set up SM-OO integration and enable the SM-OO integration instance by following the instructions as described in Operations Orchestration integration setup.
3. Click **Knowledge Management** > **Configuration** > **Operations Orchestration Flows**, and then click **Search**. Service Manager displays a list of OO flows that were retrieved from the integrated OO end point.
4. Search for an existent KM article, or click **Knowledge Management** > **Contribute Knowledge** to contribute a particular type of knowledge. For example, Error Message/Cause.
5. Click the **OO Flow Links** tab on your contributed knowledge form, and then select an OO flow to be displayed in End User Chat in the Select Path drop-down list.
6. Click **Add Link** to open the OO Flow Link Detail form, and then verify the details. The General Parameters section may consist of some parameters that are used to collect the end user's input according to this configuration.

   Tip

   If you set the **Required?** field of a parameter to true, the system displays this parameter as a mandatory field in the End User Chat window.

7. Click **Add**, and then click **Save** and **OK** to return to the contributed knowledge form.

8. Click **Submit** to submit this KM article. The system activates the **Show/hide in chat** button.
9. Select the OO flow to be displayed in End User Chat, and then click **Show/hide in chat** to open the End User Chat OO Flow Run Configuration form.
10. Update the fields as necessary. See the following table for details.

Label	Description
Number	The system-generated unique ID for this KM article. This field is read-only.
OO Flow ID	The system-generated unique ID for this OO flow. This field is read-only.

Label	Description
OO Run Mode	Select a run mode for the OO flow: ◦ Select **Sync**, Service Manager invokes OO central to run this workflow with user input parameters. When the OO flow is completed, the system returns a success or failure message to the end user as customized in the **Message Success** field and the **Message Failure** field, respectively. ◦ Select **Async**, Service Manager generates a run schedule record to perform this task later, and then makes the following behaviors: 1. Returns a success message to the end user as customized in the **Message Success** field. This indicates that the user's request to launch an OO flow has been accepted. 2. When the OO flow is completed successfully, sends the end user an email notification which embeds the message as customized in the **Email notification message template** field. 3. When the OO flow failed, sends the end user an

Label	Description
	email notification which embeds the message as customized in the **Message Failure** field.
Message Success Message Failure	In Sync mode: Both messages reflect the actual OO flow completion status. Edit the OO flow completion status messages in the **Message Success** field and the **Message Failure** field, respectively. In Async mode: The message in the Message Success field indicates that the user's request to launch an OO flow has been accepted, whereas the message in the Message Failure field reflect the actual failed status upon an OO flow completion. For each language available, you can localize the message texts by clicking **Edit localized message**. After you localize the text, click **Save** and then **OK** to return to the End User Chat OO Flow Run Configuration form.

Label	Description
Email notification message template	Edit the email notification message when an OO flow is successfully completed in the Async mode. For each language available, you can localize the message texts by clicking **Edit localized message**. After you localize the text, click **Save** and then **OK** to return to the End User Chat OO Flow Run Configuration form.

11. Click **Add**.
12. The Run Parameters section displays the input parameters of this OO flow. When an end user opens this KM document in the End User Chat window, the system generates a small form, prompts this user to fill in these parameters, and then passes the parameters to the OO flow. You can edit the imported OO flow parameters by performing the following tasks:
    - To add an OO flow parameter, click **Add Parameter** and then provide inputs on the Service Desk Automation Parameter Configuration form.
        1. Type an ID for the OO parameter, and then provide a label.
        2. Select the **Secure** checkbox so that an end user's input for this parameter becomes confidential in the End User Chat window.
        3. Select the **Required** checkbox so that an end user must provide input for this parameter in the End User Chat window.
        4. Click **Finish**.

> **Note**
>
> After you added a new OO flow parameter, the system displays the following message:
>
> Invoke parameters in OO Flow links is different with parameters in Chat OO run configuration. Please synchronize it manually.
>
> In this case, click the OO flow link on the OO Flow Links tab and then add the new OO flow parameter to the **General Parameters** section accordingly.

- To remove an OO flow parameter, select this parameter and then click **Remove Parameter**.
- To edit an OO flow parameter, click this parameter and then update the Service Desk Automation Parameter Configuration form.
- Click **Move Field Up** and **Move Field Down** to adjust the sequence of the parameters.

13. Click **Save** and **Back** to return to the contributed knowledge form. Verify that the value in the **chat run active?** column of the selected OO flow is **true**. That is, the current OO flow is the default configuration for this KM document in End User Chat.
14. (Optional) Continue to follow *step 5* to *step 8* and add more OO flow links to this KM document.

> **Note**
>
> - Every KM document can have only one active OO flow configuration in End User Chat.
> - If a KM document links to multiple OO flows, to activate another OO flow as the default configuration, select that flow and then click **Show/hide in chat**. The previously active OO flow is automatically disabled with its **chat run active?** column shows **false**.

15. (Optional) To remove a linked OO flow from a KM document, select this flow and then click the **Remove Link** button. The corresponding chat OO flow run configuration data is removed as well.
16. Click **Save** to save your changes.

Related topic

Operations Orchestration integration setup

1.17.4.2.8. Set up surveys for the Service Desk chat module

You can define a Process Designer rule set for the Service Desk chat module and add it to the Closure phase of the **Service Desk Chat** workflow to trigger a rule-based survey. When a Service Desk chat record enters the Closure phase, Service Manager automatically sends a survey email to the end user.

> **Note**
>
> For rule-based surveys, the system selects a survey template as described in Automatic selection of survey templates. You can use this rule set for reference.

To use a rule set for rule-based surveys from the Service Desk chat module, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Go to **Tailoring** > **Process Designer** > **Rule Sets**.
3. In the **ID** field, type `send.survey`.
4. Click **Search**.
5. Click **More** > **Clone** to copy the rule set to a new one.
6. Rename the rule set and save it. For example, save it with the following attributes:
   - ID: `send.survey.sdc`
   - Name: `send survey for Service Desk chat request`
7. Select the rule description, and then click **Edit Rule/Group**.
8. (Optional) Click **Edit** under the **Condition** field, and then specify a condition that applies to rule-based surveys sent from the module. For example, set the Rapid Application Development (RAD) expression to `true` means always run this rule.
9. Update the JavaScript as follows:

```
vars["$survey.template"]=lib.SurveyTemplateHelpers.getDefaul
tTemplate4SingleSurvey(record, true);
vars["$survey.users"]=[];
vars["$survey.users"].push(record["contact"]);
lib.SurveyTemplateHelpers.sendSurvey(record,
vars["$survey.users"], "sdchat");
```

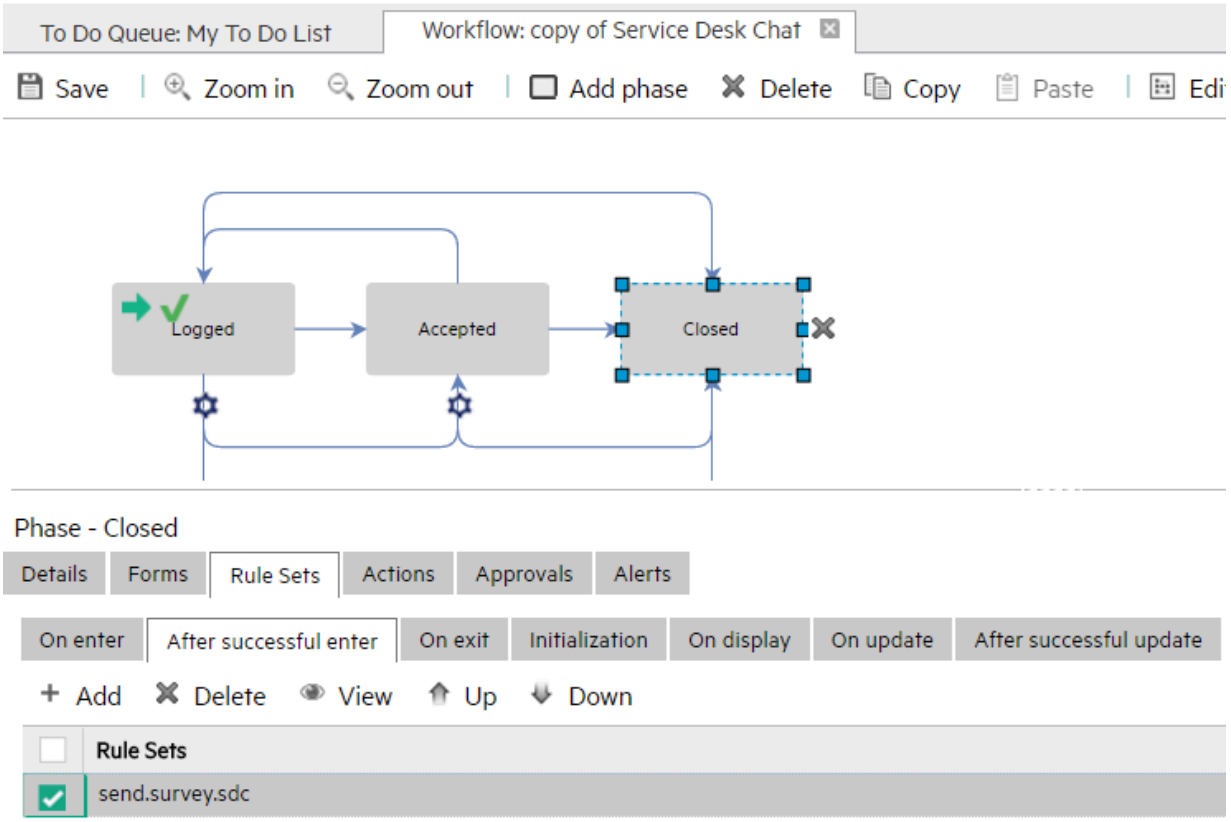
Note

The JavaScript needs to define recipients and a survey template for the survey.

10. Click **OK** to save your changes to the rule set, and then click **Save & Exit**.
11. Go to **Tailoring** > **Process Designer** > **Copy Existing Workflow**, and then click **Service Desk Chat** in the list.
12. Rename the copied workflow and save it. For example, save it with the following attributes:
    - New workflow name: `copy of Service Desk Chat`
    - Select the **Copy rule sets?** checkbox.
    - Rule set prefix: `copy of`
13. Click **OK** to save your changes.
14. Click **Tailoring** > **Document Engine** > **Objects**.
15. In the **File name** field, type `sdchat`.
16. Click **Search**.
17. Change the value in the **Workflow** field to **copy of Service Desk Chat**.

18. Click **OK** to save your changes.
19. Go to **Tailoring** > **Process Designer** > **Workflow**, and then open **copy of Service Desk Chat**.
20. Select the **Closed** phase.

21. Go to **Rule Sets** tab > **After successful enter** tab, and then click **Add** to add the **send.survey.sdc** set to the selected phase. See the following screenshot for an example.



22. Click **Save**.

1.17.4.3. Service Desk agent tasks

This section describes the Service Desk agent tasks in End User Chat.

Accept a chat request

Create an additional Interaction from chat

(Optional) Provide support using Smart Search

Participate in multiple conversations

Search for a Service Desk chat request record

Invite participants

Remove participants

Close a chat request

Quit a conversation

View reports for End User Chat

1.17.4.3.1. Accept a chat request

As a Service Manager Service Desk agent, you may belong to one or more assignment groups. You are notified when a chat request is assigned to one of your assignment groups, and then you can accept the chat.

Follow these steps to provide live support to the end users:

1. Sign in to Service Manager as a Service Desk agent.
2. Click the **Chat Request** button to open the Chat Request panel. The incoming chat requests that are assigned to the Service Desk agent's assignment groups are displayed in various tabs.
3. Select a chat request, and then click **Accept** to open the conversation window. A conversation starts with the end user's photo and question displayed on the header of the conversation window. Meanwhile, the following behaviors occur:

   • If an end user starts the chat without providing the previous support ticket ID, Service Manager automatically creates a Service Desk chat request record and opens a new Interaction record with the contact and location information prefilled. You can update the rest of this Interaction record at any time during the conversation.

   Tip

   You can configure which fields can be mapped from a Service Desk chat request record to an Interaction record by updating the `screlate.sdchat.incidents.bg` link in Service Manager.

   • If an end user starts the chat with the previous support ticket ID, Service Manager automatically creates a Service Desk chat request record and relates the previous Interaction record to the Service Desk chat request record. The system opens the previous Interaction record as soon as the conversation is established between the Service Desk Agent and the end user.

Tip

In the Interaction record, click the **Related Record** tab to view the associated Service Desk chat request. Contrariwise, click the **Related Record** tab in a Service Desk chat request to view the associated Interaction record. Each chat with the user on a given ticket will create a separate chat record and all chat records are listed on the **Related Record** tab of the Interaction.

4. Type your message in the text input area at bottom of the conversation window.

   End User Chat supports plain text only. You can press Shift + Enter to start a new line.

5. Press Enter to post the message. This time-stamped message appears in the conversation window with your photo.

   A **Resend Message** button appears beside your message when your message is not posted successfully. Click the **Resend Message** button to post your message again.

Note

If you lost connection to the chat server due to certain reasons such as network issues, the pending chat requests and the conversation window show the most recent information as soon as you are reconnected to the chat server again.

1.17.4.3.2. Create an additional Interaction from chat

Besides the new Service Desk chat record that's automatically created when an end user starts a chat, Service Manager also opens a new associated Interaction record when a Service Desk agent accepts a chat request. Depending on the chat outcome, you can create an additional Interaction from the chat so that the Service Desk chat ticket can be related to multiple Interactions.

To create an additional Interaction from chat, click **More** > **Create Interaction** from the conversation window. Service Manager automatically creates a new Interaction record with some prefilled information, including Contact, Location, Title, Description, and so on. You can update the rest of this Interaction record at any time during the conversation.

Tip

You can configure which fields can be mapped from a Service Desk chat request record to an Interaction record by updating the `screlate.sdchat.incidents.bg` link in Service Manager.

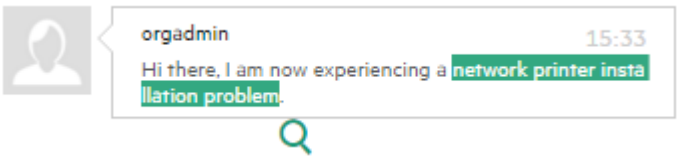
1.17.4.3.3. (Optional) Provide support using Smart Search

If your Service Manager system administrator has set up Smart Analytics, you can use Smart Search to search across a variety of resources, including Service Manager Service Catalog items, Knowledge Management (KM) documents, SharePoint documents, static web pages, and file systems while providing support to end users.

Follow these steps to provide support using Smart Search:

1. Click the **Search** button on the top-right corner of the Service Manager UI. The Smart Search window opens.
2. Type the key words or phrases you want to search for. Service Manager displays a list of items matching your search request when you are typing. If no result is displayed, type a new search string for your search.

   Alternatively, you can simply highlight the keywords in the end user's chat message and then click the automatically appeared magnifier icon to start a Smart Search. Refer to the following screenshot as an example:

3. Click **Share** to send the link of a search result item to the end user.

   Note

   The system displays a notification in the conversation window when the following scenarios occur:

   - The end user is not authorized to access the shared link.
   - The end user did not receive the shared link.

1.17.4.3.4. Participate in multiple conversations

While you are chatting with an end user, you may notice that the Chat Request number badge is growing. You can participate in multiple conversations at the same time, and click the conversation tabs to switch from one to another.

Follow these steps to participate in multiple conversations:

1. Click the **Chat Request** button to open the Chat Request panel again.
2. Browse through the chat assignment group tabs, select another chat request, and then click **Accept**.

The new conversation starts in another tab under the End User category in the conversation window.

1.17.4.3.5. Search for a Service Desk chat request record

To view all Service Desk chat request records, select **Service Desk Chat Request** in Service Manager To-Do queue. Alternatively, you can follow these steps to search for a specific chat request:

1. Sign in to Service Manager as a Service Desk agent.
2. From the System Navigator, click **Service Desk** > **Search Chat Requests**.
3. Type the ID of the chat request record in the Chat ID field, and then click **Search**.

1.17.4.3.6. Invite participants

To simultaneously consult with other experts in a End User Chat, follow these steps to invite participants:

1. Open a conversation directly or open a record that has a conversation.
2. Click **More** > **Add Participant** on the header of the conversation window to open the Participants window. End User Chat retrieves participants based on the Suggested User Fields configuration in Service Manager.
3. Add a new participant by performing either of the following tasks:
    - Select a user name or a group from the suggested participant list, and then click the **Add** button.
    - Enter a user's email address or a user name in the Search Participant field. Select the participant from the search result, and then click the **Add** button.

      Tip

      - You can view participants' presence statuses that are shown beside the user names in the suggested participant list, the participant list, and the participant search result list.
      - To close the Participants window, press **ESC** or click anywhere outside of the participants list.

    After you invite a participant to the current conversation, the system broadcasts a message to all the other participants to identify whom is invited. Meanwhile, the invited participant receives a system message that shows the members in the current conversation.

1.17.4.3.7. Remove participants

Participants can be removed from conversations if they no longer need to contribute. Participants can remove themselves or the other participants. When being removed, participants will not receive any incoming message notifications.

To remove participants, follow these steps:

1. Click **More** > **Add Participant** to display the participants list.
2. Select the person in the participant list, and then click the **Remove** button.
3. The system prompts you to confirm the operation, click **OK**.

1.17.4.3.8. Close a chat request

Follow these steps to end a chat:

1. Click **More** > **Close Chat Request** on the header of the conversation window.
2. Select a closure code from the **Completion Code** drop-down list.
3. Click **OK** in the system prompt.

You will no longer receive messages or notifications from this conversation. After you have ended the current chat, the system broadcasts a message to the end user and identifies that you have ended the chat. Neither the end user nor you can send messages, but you can still view the chat content.

When a chat is closed, the corresponding chat request closes asynchronously by the Process Designer rule engine. The chat is automatically saved in the Service Manager database.

Note

End a chat doesn't delete the chat. Chats can't be deleted.

1.17.4.3.9. Quit a conversation

To quit a conversation, click **More** > **Quit Conversation** on the header of the conversation window, and then click **Yes** in the system prompt.

Note

The **Quit Conversation** option is available only after you have closed a chat request.

To view the End User Chat conversation history after you quit a conversation, you need to access the corresponding Service Desk chat request record and then click the **Start Conversation** button to open the conversation window. The system displays the conversation history and you can slide up the scroll bar to view the content.

1.17.4.3.10. View reports for End User Chat

Service Manager provides a series of out-of-box End User Chat reports. Follow these steps to view these report:

1. From the System Navigator, click **Miscellaneous** > **Dashboard**.
2. Select **End User Chat Overview (Global)** from the drop-down list.
3. (Optional) To add more End User Chat reports to the dashboard, click **Add Content** and then select more reports from the Service Desk category.

For more information about the End User Chat reports, see Report descriptions and usage. For more information about creating custom reports, see Create a report.

Note

The automatic refresh functionality for the End User Chat Overview dashboard is disabled by default. You need to click the **Refresh** button to retrieve the latest reporting data. To enable the automatic refresh functionality, See Enable automatic refresh for a dashboard.

1.17.4.4. End user tasks

This section describes the end user tasks in End User Chat.

Chat with a Service Desk agent

If virtual agent is disabled, you can obtain live support from a Service Desk agent by using End User Chat:

1. Click the chat button to open a chat window.
2. Fill in all mandatory fields, and then click **Start Chat**. The chat window automatically displays the number of people who are already in the queue. If no agent is available or it is out of the chat service hour, click **Submit a Request** in the chat window to submit an Interaction.

   Tip

   If you have submitted a ticket before the chat, you can specify the ticket from the **Previous Support Ticket ID** drop-down list.

3. When a Service Desk agent joins the chat, you can type messages to state your issue and chat with the Service Desk agent. The Service Desk agent may guide you through troubleshooting, suggest possible resolutions, or create an Incident record in Service Manager to handle your question.

Note

If you refresh the webpage or re-login to the Service Portal or ESS portal, the chat window retains its status as before.

You can close the chat after the chat is completed. If further work is needed to resolve your issue, the agent might notify you by e-mail when this work is completed, as appropriate.

To end a chat:

1. Click the close button on the chat window.

2. The system prompts you to select a reason to end the chat. Select a reason in the drop-down menu, and then click **End the Chat**.

You will no longer receive messages from this chat. After you end the current chat, the system broadcasts a message to the Service Desk agent to identify that you have ended the chat. Your closure reason is also displayed to the Service Desk agent.

Chat with a virtual agent

When a Service Manager system administrator has properly configured and enabled virtual agent, you can firstly obtain support from a virtual agent by using End User Chat. However, if the answers from a virtual agent is not satisfactory, you are still able to chat with a live Service Desk agent.

Follow these steps:

1. Click the chat button to open a chat window.
2. Fill in all mandatory fields, and then click **Start Chat**. A virtual agent will join the chat and send you greetings.

   Tip

   If you have submitted a ticket before the chat, you can specify the ticket from the **Previous Support Ticket ID** drop-down list. In this case, the chat will be directly routed to a Service Desk agent.

3. Type your questions or issues, the virtual agent performs Smart Search based on your access rights and provides you several links that are relevant to your question. These links include Knowledge Management document links, Service Catalog item links, other knowledge resource links such as SharePoint pages, documents on a file system, or web pages from external libraries. Click the links to view the content in the Service Portal or ESS portal window.

   Note

   You can only view a KM document in the chat window if this KM document has an OO flow linked. In this case, the system may prompt you to fill in some parameters for the OO flow. Click **Submit** to launch the OO flow and you will get the returned result in the End User Chat window.

Tip

You can type #menu to display an option menu. The out-of-box menu displays the following options:

```
Please type the number before an option to continue
1 - List three most frequently asked questions
2 - List three hot news
3 - Update my account information
```

4. If the virtual agent addresses your question correctly, type Yes. The virtual agent will inquire of another question. Type Yes to describe another problem, or type No to close the chat.

   Otherwise, If you can't find an appropriate answer, type No and the virtual agent inquires whether to transfer you to a live chat. Type Yes to switch to live support from a Service Desk agent, or type No to close the chat.

Tip

At anytime during the chat, you can click **Live Chat with Agent** on the chat window to switch to live support from a Service Desk agent.

Tip

Click **Submit a Request** in the chat window to submit a support ticket if your chat request is timeout or no live chat assignment group is currently available.

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- 1 Chat with a Service Desk agent
- 2 Chat with a virtual agent

1.18. Service Manager Mobile Applications

The field engineers are typically tasked to work with business users at a customer site, and they spend most of their time away from the office to resolve tickets. When Change Approvers are away from the office, they need a way to quickly approve or deny changes, so that pending work orders can be assigned. Service Manager Mobile Applications connect your company's people and information by providing your team access to the Change Management, Incident Management, Service Desk, and Request applications through the use of smartphones. When these IT operators are away from the office, they can receive and view work assignments and perform the tasks below by using the Mobile Applications:

- Approve and deny change requests.
- Approve and deny interaction.
- Reassign incidents/changes to the proper support group.
- Search the knowledge base.
- Open a support ticket.
- Continue working on-site with customers.

These quick responses improve business metrics, as service level objectives are met.

This section provides details about how to work with Service Manager Mobile Applications. Topics in this section include:

- Using Service Manager Mobile Applications in power user view
- Using Service Manager Mobile Applications in self-service user view

1.18.1. Using Service Manager Mobile Applications in power user view

The Service Manager Mobile Applications power user view is intended for Service Desk technicians, managers, and provides various views for the following tasks:

- Understand the views within Service Manager Mobile Applications
- Manage user interactions
- Manage incidents
- Manage changes
- Manage requests

1.18.1.1. Understand the views within Service Manager Mobile Applications

After you logged in to Service Manager Mobile Applications, the default view is displayed depending on your user role. Service Manager Mobile Applications provides the following default views to access individual and group assignments and enables you to address any critical and pending issues that need immediate attention. If you are a Change Approver, you will also see change requests waiting for your approval.

Service Manager Mobile Applications Default Modules and Views

Module	View
Incident	• Assigned to Me • Assigned to My Groups
Change	• Awaiting My Approval • Assigned to My Groups
Interaction	• All My Interaction • Awaiting Approval
Approval	• My Pending Approvals • My Pending Delegated Approvals As of version 9.40, Mobile Applications provides additional approvals for time periods and requests (for Service Manager Codeless).
Request	• Open Requests Assigned to Me • Open Requests Assigned to My Groups

List view

Being the default home page after you log on to Service ManagerMobile Applications, the List view enables you to navigate through lists of Incidents and Change requests. In the List view, you can perform the following tasks:

- Switch between different modules (Incident, Change, Interaction, Approval, and Request).
- Switch between different list views (For example, Assigned to Me, Assigned to My Groups, and so on).
- Scroll vertically through a list of records.
- Search for specific incidents, change requests and interactions by record ID.

Common tasks in the List view

Refer to the following content for some common user tasks in the List view.

- To switch between different modules:

  Tap the Module icon at the top and select other categories from the list. The default categories includes **Incident**, **Change** and so on.

- To switch between different list views:

  Refer to the Incident module as an example. Tap **Assigned to My Groups** at the top and scroll down to select **Assigned to Me** and so on from the list.

  Tip

  For more information about how to add views on Mobile Applications client, see Add a view for Mobile Applications.

- To access the Detail view of a record:

  When navigating through the records list, tap a record to access its Detail view.

- To search for specific incidents and change requests by record ID:

  1. Tap the Magnifier icon at the top. A Search field opens.
  2. In the Search field, type the `IM` or `C` prefix with ID of the incident record or change request. For example: `IM10002` or `C10003`.
  3. Tap **Enter**. The specific incident record or change request opens in the Detail view.

Note

- When using the search function, make sure the prefix of the incident record or change request aligns with those in your Service Manager environment. For example, `C` is the prefix for a change request. For more information about how to configure the prefix of an incident record or a change request, see Configure the prefix of a record.
- Service ManagerMobile Applications supports wildcards in the Search field. If you type `IM1000*` and then tap **Enter**, the multiple search results will be displayed as a list of Incident records.

- **Tap the Ellipsis icon at the top** and select corresponding option in the list opened to:
  - View or clear system messages for the current user
  - Log out of the mobile session
- **Tap the floating action bar** and select corresponding option in the list opened to:
  - Refresh the current page
  - Sort the records in the List View

Detail view

The Detail record view enables you to view or update the details of a record. In the Detail view, you can perform the following tasks:

- Browse the summary information and the updates of a record.
- Update the editable fields of a record (for example, Description, Status, Affected Service/CI).
- Reassign a record (update the Assignee and Assignment Group).
- Use the click-to-dial functionality and the click-to-email functionality. For example, if a field engineer is going to stop by a customer and wants to notify this customer, this field engineer can drill into the contact information of a record and use the single-click method to call or email the customer.

Common tasks in the Detail view

Refer to the following content for some common user tasks in the Detail view.

- To access the List view of a record:

When navigating through detailed records, you can tap **Back** to go back to the List view.

- **Tap the tabs** at the top to display the detailed information of a record (for example, Categorization and Assignment, Related Records, Attachments, and so on).
- **Tap the Fill icon** to update the editable fields of a record (for example, Status, Affected Service/CI, Assignee/Assignment Group).
- **Tap the bulb icon** in the contact information of a record, and then tap the telephone number to use the click-to-dial functionality. The dial dialog box will open.
- **Tap the bulb icon** in the contact information of a record, and then tap the email address to use the click-to-email functionality. The email application embedded in your smartphone opens.
- **Tap the Ellipsis icon at the top** and select corresponding option in the list opened to:
  - View or clear system messages for the current user.
  - Log out of the mobile session.
- **Tap the floating action bar** to display the options that you can perform on the current record.

1.18.1.2. Manage user interactions

After a user contact with the Service Desk is logged as an interaction, User Interaction Management provides the Service Desk Agents the ability to perform the following tasks from their smartphones:

- Browse and review interactions.
- Approve, deny, or close an interaction.
- Update interactions. For example, a field engineer can update the details and add an activity (or journal entry) update to an interaction record.
- Approve or deny a Service Catalog request.

Update an open interaction

User role: Service Desk Agent

To update an open Service Desk interaction, follow these steps:

1. Tap the Service Desk interaction record you want to update. The Detail view of this record opens.
2. On the **Interaction Details** tab, update the interaction details as needed.
3. Tap the **Activities** tab.
4. In the **New Update Type** field, scroll up or down to select a type, and then tap **Done**.
5. In the **New Update** field, add the details for the new activity (or journal entry).
6. Tap **Save** from the action bar to save the record.

The details of the new activity entry will become available for customer viewing on your customer support portal.

Approve or deny an interaction

User role: Service Desk Agent

When an order is submitted from Service Desk, Service Manager automatically creates an interaction that, based on approval requirements, may have to be approved before its fulfillment or to be denied.

To process an interaction, follow these steps:

1. Tap a Service Desk interaction record from the **Awaiting Approvals** view. The Detail view of this record opens.

2. Review the Service Desk interaction and determine how you will process the approval interaction.
3. To approve an Service Desk interaction, tap **Approve** from the Ellipsis icon options.

   The record is now approved and disappear from the view list.
4. To deny an Service Desk interaction, follow these steps:

   1. Tap **Deny** from the action bar.
   2. Type the reason you are denying the request, and then tap **Save**.

   The record is now denied and disappears from the view list.

Close an interaction

User role: Service Desk Agent

You can close an existing Service Desk interaction if the user accepts the proposed solution. If you or the user disagree with the proposed solution, you need to resubmit the related incident for further investigation.

To close an existing Service Desk interaction, follow these steps:

1. Tap the Service Desk interaction record you want to close. The Detail view of this record opens.
2. Tap the Fill icon in the **Contact** field to determine which user will be notified of the solution.
3. If necessary, update the **Notify By** field to determine how the user will be notified.
4. Tap on the Fill icon in the **Closure Code** field to select a closure code.
5. Type the solution in the **Solution** field.
6. Tap **Close Interaction** from the action bar. The status of this interaction record changes to Closed.

Approve or deny a Service Catalog request

User role: Service Catalog approver

When an order is submitted from Service Catalog, Service Manager automatically creates an interaction that, based on approval requirements, may have to be approved before its fulfillment or to be denied.

To process a Service Catalog request, follow these steps:

1. Tap a request record from the My Pending Approvals view. The Detail view of this record opens.
2. Review the request and determine how you will process the approval request.
3. To approve a Service Catalog request, tap **Approve** from the action bar.

   The record is now approved and disappear from the view list.

   > **Note**
   >
   > You can also approve delegated approvals from the **My Pending Delegated Approvals** view.

4. To deny a Service Catalog request, follow these steps:
   1. Tap **Deny** from the action bar.
   2. Type the reason you are denying the request, and then tap **Save**.

The record is now denied and disappear from the view list.

1.18.1.3. Manage incidents

When incidents are escalated from Service Desk interactions, opened by support staff, or reported by event monitoring tools, Incident Management provides the Incident Management staff the ability to perform the following tasks from their smartphones:

- Browse and review incidents.
- Assign or reassign an incident.
- Investigate incidents.
- Update incidents. For example, a field engineer can add an activity (or journal entry) update to an incident record and set the customer-visible flag to make the update available for customer viewing on the customer support portal.
- Resolve or close a ticket. When a field engineer attempts to resolve or close an incident, Service Manager determines the business logic and displays Resolve or Close accordingly.

Set the customer-visible flag for an incident's activity

User role: IT Operator

When you want to publish the activity (or journal entry) details of an incident for customer viewing, you can set the customer-visible flag in a new activity entry.

To set the customer-visible flag in a new activity (or journal entry) entry, follow these steps:

1. Tap the incident record you want to resolve. The Detail view of this record opens.
2. Tap the **Activities** tab.
3. In the **New Update Type** field, scroll up or down to select a type, and then tap **Done**.
4. Tap to highlight the checkmark in the **Visible to Customer** field.
5. In the New Update field, add the details for the new activity (or journal entry).
6. Tap **Save** from the action bar to save the record.

The details of the new activity entry will become available for customer viewing on your customer support portal.

Reassign an incident

User role: Incident Analyst, Incident Coordinator, Incident Manager

At times you will need to reassign incident records when an Incident Analyst is unavailable.

To resolve an incident, follow these steps:

1. Tap the incident record you want to resolve. The Detail view of this record opens.
2. Tap the Fill icon in the **Assignee** field, and then select an operator from the list of names to reassign the incident to that person.
3. Tap **Save** from the action bar to save the record.

Resolve an incident

User role: Incident Analyst, Incident Coordinator, Incident Manager

To resolve an incident, follow these steps:

1. Tap the incident record you want to resolve. The Detail view of this record opens.
2. Tap the arrow icon in the **Status** field, and then tap **Resolved** to close the incident record.
3. Tap the Fill icon in the **Closure Code** field to select a closure code.
4. Type the solution in the Solution field.
5. Tap **Close Incident** from the action bar. The status of this incident record changes to Closed.

1.18.1.4. Manage changes

When a change request is logged, the Change Analyst assesses the change request, implements a plan for delivering the change, and then notifies the Change Coordinator as to the impact of the change. The change request is then submitted for Change Approver, or Change Advisory Board (CAB)'s approval. Service Manager Mobile Applications provide the Change Management staff the ability to perform the following tasks from their smartphones:

- Review change requests.
- Approve or deny changes.
- Coordinate change implementation.
- Handle emergency change requests.
- Assign or reassign change requests.
- Add activity (or journal entry) entries.
- Review change requests.

An example of a change approver's possible actions for a change record include approving or denying change requests. To deny a pending change request, a Change Approver may do the following:

- Drill down into a single record.
- Deny the ticket.
- In the Update field, type detailed notes about the denied request.

Users can also drill down to the details of a field within a record. For example, if a Change Management staff wants to add an activity (or journal entry) update to a change request, he/she nay do the following:

- Search for the change record.
- Drill down into the record's activities.
- Add the necessary activity update.
- Make any other necessary changes.

Set the customer-visible flag for a change's activity

User role: All users

When you want to publish the activity (or journal entry) details of a change for customer viewing, you can set the customer-visible flag in a new activity entry.

To set the customer-visible flag in a new activity (or journal entry) entry, follow these steps:

1. Tap the change record you want to resolve. The Detail view of this record opens.
2. Tap the **Activities** tab.
3. In the **New Update Type** field, scroll up or down to select a type, and then tap **Done**.
4. Tap to highlight the check mark for the **Visible to Customer** field.
5. In the **New Update** field, add the details for the new activity (or journal entry).
6. Tap **Save** from the action bar to save the record.

The details of the new activity entry will become available for customer viewing on your customer support portal.

Approve, deny or retract a change

User role: Change Approver

You can approve or deny a change that's pending approval only if you are a member of the necessary approval group and you are assigned the appropriate Change Management user profile. You can also retract a change that has been previously approved or denied, if you are unwilling to commit resources or know of technical incidents that affect the request.

To process a change approval request, follow these steps:

1. Tap a change record from the **Awaiting My Approval** view. The Detail view of this record opens.
2. Review the change information and determine how you will process the approval request.
3. To approve a change, tap **Approve** from the action bar.

   Service Manager changes the Approval Status to approved, and the Change Manager updates the change and passes it to the Change Coordinator for implementation.
4. To deny a change, tap **Deny** from the action bar. Type the reason you are denying the change, and then tap **Save**.

   Service Manager changes the Approval Status to denied, and no further approvals are possible until the denial is retracted.
5. To retract a change, tap **Retract** from the action bar. Type the reason you are retracting the change, and then tap **Save**.

Service Manager changes the Approval Status to pending, and the change request requires a new approval cycle to progress.

1.18.1.5. Manage requests

When requests are logged, Request Fulfillment provides the Request Fulfillment staffs the ability to perform the following tasks from their smartphones:

- Browse and review requests.
- Update requests.
- Approve, deny or close requests.

Note

If you have upgraded your Service Manager system from version 9.4x or 9.5x to 9.70, you need to manually enable both the **All Open Requests Assigned to Me** view and the **All Open Requests Assigned to My group** view for the Mobile Applications client. For detailed instructions, see Add a view for Mobile Applications.

Update a request

User role: Request Coordinator, Request Analyst

To update a request record, follow these steps:

1. Tap the request record you want to update. The Detail view of this record opens.
2. In the **Request Details** tab, update the request details as needed.
3. Tap the **Activities** tab.
4. In the **New Update Type** field, scroll up or down to select a type, and then tap **Done**.
5. In the **New Update** field, add the details for the new activity (or journal entry).
6. Tap **Save** from the action bar to save the record.
7. If needed, click the appropriate button to move the request record the next or previous phase.

Approve or deny a request

User role: Request Approver

To process a request, follow these steps:

1. Tap a request record from the **Awaiting Approvals** view. The Detail view of this record opens for your review.
2. To approve a request, tap **Approve** from the action bar.

The record is now approved and disappears from the view list.

3. To deny a request, tap **Deny** from the action bar. Type the reason you are denying the request, and then tap **Save**.

   The record is now denied and disappears from the view list.

Close a request

User role: Request Coordinator

After request fulfillment and review, you can close the request record. Follow these steps:

1. Tap the request you want to close. The Detail view of this record opens.
2. Tap **Close** from the action bar.
3. Enter or change the subcategory for the request.
4. Tap on the Fill icon in the **Closure Code** field to select a closure code.
5. Type the solution in the **Solution** field.
6. Enter the closure comments.
7. Click **Finish**. The request record is now successfully closed.

1.18.2. Using Service Manager Mobile Applications in self-service user view

The Service Manager Mobile Applications self-service user view is intended for end-users as an entry point to Service Desk and provides a simplified Service Desk interface for users to perform the following tasks:

- Search the knowledge base
- Submit a self-service request
- Submit a smart request
- View opened and closed tickets
- View, approve, or deny pending approval requests

1.18.2.1. Search the knowledge base

User role: IT Operator

To search the knowledge base for your questions, follow these steps:

1. Type your issue, and then tap the magnifier icon to search the knowledge base.

   > **Note**
   >
   > The maximum length of the query key words string is limited to 1024 characters. Any query string longer than 1024 characters will be truncated to 1024 characters.

2. Tap the result record and view the knowledge article. You can tap **Cancel** on the bottom to return to the search results.
3. If you like an knowledge article, tap .
4. To add feedback to the knowledge article, tap **Add Feedback**.

1.18.2.2. Perform smart search

After you have purchased a Smart Analytics module license and enabled Smart Analytics in Service Manager, Service Manager Mobile Applications supports Smart Search to perform advanced cross-module search actions in Service Manager resources (Knowledge Base articles and attachments) and external resources (SharePoint resources and static web pages). Meanwhile, Smart Search can perform a spell check to return meaningful results. For example, if you search for "offica," the search result will return records that include the term "office" if there is no exact result for "offica."

You need to set up and verify the server connectivity for multiple servers and connectors before you can use the Smart Search. For detailed information to configure and monitor the connectors and servers, refer to the related topics from *Service Manager Help Center*.

For detailed steps about how to perform Smart Search, see Search the knowledge base.

1.18.2.3. Submit a self-service request

User role: IT Operator

If you can't find the knowledge article to address your issue, you can also submit a self-service request.

To submit a self-service request, follow these steps:

1. Tap **Submit a Request** on the bottom of the screen. You can also tap the menu icon , and then tap **Submit a Request** in the menu.
2. Type the required information for a support ticket, including title, description and urgency.
3. Tap to attach a file as necessary. The supported file types including images (*.jpg, *.png, *.tiff and *.bmp) and PDF.

   Note

   Attachment is unsupported on BlackBerry 6.x, 7.0 and 7.1 devices.

4. Tap **Submit** to submit the request.

1.18.2.4. Submit a smart request

User role: IT Operator

If you have installed and enabled Service Manager Smart Analytics, **Submit a Smart Request** is automatically added to leverage the power of the Smart Ticket feature. You can tap **Submit a Smart Request** to open a new, simplified request form that only requires “description” or “attachment” to submit a request, which simplifies the process of submitting the ESS support requests.

To submit a smart request, follow these steps:

1. Tap **Submit a Request** on the bottom of the screen. You can also tap the menu icon , and then tap **Submit a Smart Request** in the menu.
2. Tap to attach an image file. For example, a screenshot of the error message.
3. (Optional) Type the comment for your request.
4. Tap **Submit** to submit the request.

Service Manager Smart Analytics will analyze your attached image file, fill in the necessary information automatically, and then generate the interaction directly.

Note

Submit a Smart Request is visible to self-service users only when Service Manager Smart Analytics is enabled. However, this option is visible to the administrator no matter Smart Analytics is enabled or not.

1.18.2.5. View opened and closed tickets

User role: IT Operator

To view your opened and closed tickets, follow these steps:

1. Tap **My Closed Request** or **My Opened Request** to view your submitted tickets.
2. Tap the record to view the content of a ticket.
3. Scroll down to the bottom of the screen to see a list of the related records.
4. You can update, close, or resubmit the ticket by tapping the buttons on the bottom of the screen.

1.18.2.6. View, approve, or deny pending approval requests

Applies to user role: IT Operator

Users with the self-service Approval menu is allowed to view, approve, or deny pending approval requests.

To view, approve, or deny pending approval requests, follow these steps:

1. Tap the menu icon and then tap **Pending Approval** to view your pending approvals.
2. Tap one record and then tap **View**, **Approve** or **Deny**.
3. If there are more than 10 pending approvals, tap the **More** button to view the records on the next page.

1.19. Survey

Service Manager not only allows you to set up a scheduled survey through survey integration but also provides a Service Manager Survey tool. You can use Service Manager Survey to implement email surveys in Service Manager independently without integration with third-party survey tools. With this tool, you can either manually send a survey from a record, or set up scheduled surveys.

Related topics

Service Manager Survey
Survey Integration

1.19.1. Service Manager Survey

The Service Manager Survey (SM Survey) feature is based on the HTML Email solution (JavaMail) and doesn't require an external survey tool to complete the survey process. You can manually send individual surveys directly form a record to selected users, or configure the system to automatically send scheduled surveys or rule-based surveys to specified users through an out-of box SM Survey connector (that is, the "internal connector"):

- Manual surveys: are individual surveys that users manually send from a record by using the **More** > **Send Survey** option.
- Scheduled surveys: are surveys that the system automatically sends to specified recipients based on predefined scheduling rules.
- Rule-based surveys: are surveys that a rule set in a Process Designer workflow automatically triggers and sends to specified users.

Note

SM Survey doesn't support Multiple Company mode. Sending surveys based on each company's records (Incident, Interaction, and so on) is supported; however, designing company specific survey questionnaires is unsupported.

Supported user portals

SM Survey supports Employee Self-Service (ESS) Portal and SMA Service Portal.

Survey process

When ESS Portal users receive a survey link from an email, they can directly open the link in a web browser and then log in to the portal to complete the survey. They can also check the surveys opened for them or closed by them in the portal.

Once the surveys are completed and submitted, the survey senders can check the survey results from the TO-DO queue or from survey reports on the Survey dashboard.

Support of multiple survey templates

SM Survey sends a survey based on a predefined survey template. You can easily create multiple survey templates and apply them to different surveys to suit your various business needs.

SM Survey provides an out-of-box survey template named DEFAULT, which is designed to achieve the following goals:

- Collect user satisfaction ratings on the following areas: the level of service provided, the speed of response that the user received, and the level of communication rating from 1 to 5 regarding the service.
- Collect additional comments from the user (if any)
- Allow the user to select if a complaint or compliment needs to be registered. If the user selects to do so, the system automatically generates an interaction record of a Complaint or Compliment category with the information the user provided, and sends an email to the user with information about the interaction record. The user can click the record link to check the record details and status.

Survey reports

In addition to individual survey results, administrators can also check the statistics reports that the system automatically generates based on the existing survey results. All this can help your organization to improve your services and user satisfaction. For more information, see View SM survey reports.

Note

To view survey reports, users must have the View permission for the Report area.

Hot Topic Analytics

If your system has Smart Analytics enabled, Hot Topic Analytics is enabled for SM Survey. For more information, see Use Hot Topic Analytics with SM Survey.

Note

To use Hot Topic Analytics for SM Survey, users must have the idol.assistant capability word defined in their operator record.

Related topics

SM Survey setup

1.19.1.1. SM Survey setup

SM surveys fall into three categories:

- Manual surveys: surveys that the user manually sends from the **Send Survey** option in a record. For more information, see Send a survey from a record.
- Scheduled surveys: surveys that the user creates and then the system automatically sends out based on a scheduler. For more information, see *Create an SM survey for a module*.
- Rule-based surveys: surveys that are automatically triggered by a rule set specified in a Process Designer workflow. For more information, see Use a rule set to send surveys.

You need to configure these types of SM surveys differently.

Common configurations

All types of SM surveys require the following configurations:

- The HTML Email solution must be set up. For details, see *Set up HTML Email for SM surveys*.
- The survey portal URL must be specified. For details, see *Specify the survey portal URL*.
- The **survey service agent** scheduler must be started. For more information, see *Start the survey service agent scheduler*.

Once these configurations are ready, you can send manual SM surveys from a record by selecting the **Send Survey** option. However, rule-based and scheduled SM surveys require additional configurations.

Set up HTML Email for SM surveys

User roles: System Administrator

For the SM Survey feature to work, you need to set up HTML Email (JavaMail) by configuring the required emailout parameters. For more information about the emailout parameters, see Emailout parameters in the sm.ini file.

To set up HTML Email, follow these steps:

1. Navigate to the RUN folder of the Service Manager Server.
2. Open the sm.cfg file in a text editor, and add the emailout parameter in the file:

sm -emailout

3. Open the sm.ini file in the RUN folder, and add the following emailout parameters as well as the **queryhashcode** parameter:

```
smtphost:smtp3.mycompany.com
smtpport:25
mailFrom:sender@mycompany.com
queryhashcode:<hostname>:<port>
```

   Where:

   - **smtphost** is the name of the SMTP server host for client requests
   - **smtpport** is the communications port that the SMTP server uses
   - **mailFrom** is any email address that you want to appear as the sender in the survey emails. The **mailFrom** parameter must use an email address format, but doesn't have to be a real email address.
   - ***<hostname>*** is the Service Manager Server host name and must be identical to the value specified in the **serverHost** parameter in the Service Manager web tier configuration file (web.xml)
   - <**port**> is the communications port of the the Service Manager Server. For example: 13080.

   Tip: Instead of configuring the parameters in the two configuration files, you can also configure all of these emailout parameters in the sm.cfg file:

   sm -emailout -smtphost:smtp3.mycompany.com -smtpport:25 -mailFrom:sender@mycompany.com-queryhashcode:<hostname>:13080

4. Save the files and restart the Service Manager server.
5. Sign in to Service Manager, and do the following:
   1. Navigate to **System Status**.
   2. Make sure that the **SMTPMail** background process is running. If it's not displayed in the list, click **Start Scheduler** and double-click the **emailout.startup** scheduler.

Next, you need to specify your survey portal URL. For details, see **Specify the survey portal URL**.

Specify the survey portal URL

User roles: System Administrator

SM surveys are sent to recipients through an email with a survey link, which is generated based on the survey portal URL. For this reason, you need to specify the URL for your survey portal.

To specify the survey portal URL, follow these steps:

1. Click **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
2. Click the **Active Integrations** tab.
3. Specify the portal URL for your portal, as described in the following table.

Portal	Steps

ESS	**Note:** For ESS surveys, you must configure both the standard web client URL and the ESS web client URL in the System Information Record. Both URLs are used as predefined parameters for SM Survey.
	1. In the **WebServer URL** field, type the fully qualified index.do URL to your web tier. For example:
	`http://myserver.mydomain.com:8080/SM/index.do`
	2. In the **ESS URL** field, type the fully qualified ess.do URL to your web tier. For example:
	`http://myserver.mydomain.com:8080/SM/ess.do`

4. Save the System Information Record.
5. Sign out and then sign in to Service Manager again.

Tip: If you have already set up HTML Email, now users can send surveys from a record by using the **Send Survey** option from the **More** menu. For details, see Send a survey from a record.

To enable the system to automatically send scheduled surveys or rule-based surveys, additional configurations are needed:

Enable a survey integration instance

Create an SM survey for a module

Start the survey service agent scheduler

User roles: System Administrator

Scheduled surveys are running based on the **survey service agent** scheduler.

For this reason, when setting up SM Survey, make sure this scheduler is started. To do this, follow these steps:

1. Go to **System Status**, and make sure that the **survey service agent** scheduler is started.
2. If not, click **Start Scheduler**, and then double-click this scheduler to start it.

Additional configurations for rule-based SM surveys

You need to create a rule set and add it to the workflow of a desired module (for example, the Incident module). For details, see Use a rule set to send surveys.

Additional configurations for scheduled SM surveys

Scheduled surveys are based on the survey integration framework, and therefore require the following additional configurations:

- A survey integration instance is set up and enabled in Integration Manager. For details, see *Enable a survey integration instance*.
- A survey definition for each supported module is created based on the SM Survey connector (that is, the Internal Connector). For details, see *Create an SM survey for a module*. You can create an SM survey for each of the following modules: Change, Configuration Item, Interaction, Incident, Request, and Problem.

Note: These are the minimum configurations required for SM Survey, based on the assumption that the out-of-box survey template and HTML Email template will be used. If users want to design their own survey questionnaires or tailor the out-of-box HTML Email template, further configurations are needed. For details, see the following topics:

Add fields to the SurveyInternal dbdict record

Create a survey template

Tailor the SM Survey HTML template

Enable a survey integration instance

User roles: System Administrator

Scheduled SM surveys are based on the survey integration framework. The major difference between the scheduled SM Survey feature and the survey integration is that the former uses an SM Survey internal connector while the latter uses a connector for an external survey tool.

They both require a survey integration instance to be set up and enabled in Integration Manager and the **survey service agent** scheduler to be started in System Status. For detailed steps, see Create a survey integration instance.

Create an SM survey for a module

User roles: Survey Administrator, Survey Manager, Survey Analytics

SM Survey supports the following modules: Change, Configuration Item, Interaction, Incident, Request, and Problem. To enable the system to automatically send scheduled surveys based on a specific module, you need to create a scheduled survey for that module.

To create a scheduled survey for a module, follow these steps:

1. Click **Survey Management** > **Scheduled Surveys** > **Create New Survey** in the system navigator to display the **Create a new survey** wizard.
2. Select **Internal Connector** from the connector list.
3. Select the module on which the survey will be based from the **Object** drop-down list. For example, select **Interaction**.
4. Select an assignment group in the **Owner Group** field. For example, select **COORDINATOR**.
5. In the **Name** field, enter a descriptive title for the survey.
6. Click **Finish** to display the survey definition record.
7. (Optional) Enter a description for the survey definition as needed.

8. In the **Owner** field, specify an owner for the survey.
9. In the **Survey Template** field, select an appropriate template.

   Before you make a selection, the **Survey Template** field is automatically populated with a survey template name based on certain rules. For more information, see Automatic selection of survey templates. When you make a selection, only published survey templates whose **Module** field is set to either this module or an empty value are displayed in the **Survey Template** list.

   Note: The Condition setting in the selected survey template is ignored for scheduled surveys. The condition is evaluated only for manual surveys and rule-based surveys.

   Note: If you have not yet created a survey template of your own, you can use the out-of-box survey template named DEFAULT. If needed, you can edit it or create your own ones. For details about how to update an existing template or create a new survey template, see Edit an existing survey template and Create a survey template.
10. Click **Preview** next to the Survey Template field, to see if the survey template meets your needs.
11. Configure the scheduling rules as described below.
    1. Select a **Send Interval** value. For example, select **Daily**.
    2. In the **Active From** and **To** fields, specify a future time period during which the survey definition will be active.

    For more information, see Configure the survey scheduling rules.
12. In the **User Selection Rules** section, select one or more rules. For example, select **Randomly select 10% users as survey recipients**.

    For more information, see Filter the survey recipients.
13. Apply data filters to the survey.
    1. Specify a predefined filter. For example, select **Closed interactions since last processing**.
    2. Define additional filters if needed. For details, see Create an additional filter.
14. In the **Recipients** section, select a recipient from the **Available Recipients** box, and then click the right arrow to move the recipient to the **Selected Recipients** box.

Note: To remove a recipient from the **Selected Recipients** box, select the recipient, and then click the left arrow.

Note: An available recipient here is actually a mapping field in the corresponding DBDICT record, which is defined in the Internal Connector and can be either the group or the person type. To view or update the out-of-box recipient setting, follow these steps:

1. Go to **Survey Management** > **Administration** > **Survey Connectors**, and open the **Internal Connector** by performing a search.
2. Browse to the **Recipients and Parameters** section, and add new parameters as needed. For example, in the out-of-box system, for Interactions, the recipient **Contact** is mapped to the **callback.contact** field in the **incidents** DBDICT record, which is the **Contact** field in the interaction detail form. If you need to add a new recipient for a module, see Create a new recipient.

15. In the **Email Settings** section, do either of the following:
    - Manually configure your email template.
    - Click **Select HTML Template** to select an existing one. Service Manager provides an out-of-box HTML template as **Internal Survey Template**. In the **Test Record** field, enter an existing record ID (for example, SD10002), and then click **Preview** to make sure the template satisfies your needs. If it doesn't, you can customize it. For details, see Tailor the SM Survey HTML template.
16. Click **Back** to return to the survey definition.
17. Click **Add Translation** if you want to add a localized version of the email template. For details, see Create a localized email.
18. (Optional) In the **Parameters** section, click the **Optional** tab to add your own parameter mappings.

    Parameters are predefined values that provide surveys in Service Manager with the correct metadata. You can create additional parameters that are mapped to specific fields in the module.
19. Click **Save** to create the survey definition.
20. Click **Schedule** to start the survey.

    Surveys will be automatically sent to the specified recipients at the defined time interval.

1.19.1.2. SM Survey upgrade

If you are upgrading from a Service Manager version earlier than 9.50, the SM Survey functionality needs an upgrade. The upgrade process is simple for the Employee Self-Service (ESS) user portal.

After your SM application upgrade, do the following to upgrade SM Survey:

1. Specify the ESS portal as the survey portal. For detailed steps, see SM Survey setup.
2. Create a survey template with your original questionnaire.

   SM Survey no longer requires you to tailor any forms for ESS surveys. You only need to recreate your original questionnaire by creating a survey template. Additionally, you have the option to create multiple survey questionnaires because SM Survey supports multiple survey templates. For detailed steps, see Create a survey template.

Related topics

SM Survey setup

1.19.1.3. Automatic selection of survey templates

In the out-of-box system, only one survey template (named DEFAULT) is provided. Because SM Survey supports multiple templates, there may be multiple survey templates in your system. Each survey must be associated to a survey template through the **Survey Template** field.

When you create a scheduled survey or send a manual survey, before you select a survey template, the system prepopulates the **Survey Template** field based on certain rules; when a survey is automatically triggered by a rule set in a workflow, the system selects a survey template based on certain rules as well. These rules are described in the following section.

Scheduled surveys

When you create a scheduled survey, the system prepopulates the **Survey Template** field with a published template according to the following fields (listed from the highest to the lowest priority):

- **Module**: The system first looks for templates whose Module field is set to the same module defined in the survey. If no such templates are found, the system looks for templates whose **Module** field is empty.
- **Template ID**: If multiple templates are found, the system selects the one with the smallest template ID.

Note

The Condition field is ignored when the system selects a survey template for a scheduled survey.

Manual surveys

When you send a manual survey, the system prepopulates the **Survey Template** field with a published template according to the following fields (listed from the highest to the lowest priority):

- **Module**: The system first looks for templates whose **Module** field is set to the same module defined in the survey. If no such templates are found, the system continues to look for templates whose **Module** field is empty.

- **Condition**: If multiple survey templates are found, the system continues to evaluate the **Condition** setting defined in each template.

  > **Note**
  >
  > For templates that contain an empty Module value, this step is not needed and skipped.

- **Template ID**: If multiple survey templates whose condition evaluates to true are found, the system selects the one with the smallest template ID.

Rule-based surveys

Rule-based surveys are surveys that are automatically triggered by a rule set in a Process Designer workflow. For more information, see Use a rule set to send surveys.

When sending a rule-based survey, the system selects a survey template as follows:

1. Locate a survey template based on the same rules for manual surveys.
2. If no applicable templates are found, the system selects the survey template that's defined in the **Default survey template name** field in the Survey Settings form. For more information, see Configure the global survey settings.

1.19.1.4. SM Survey administrator tasks

The scheduled SM Survey feature shares the same framework as the survey integration. Most of the administrator tasks are the same as those for the survey integration. See the following topics in the Survey Integration section.

- Configure the global survey settings
- Create a new parameter
- Create a new recipient
- Create a predefined filter

The following administrator tasks are only for Service Manager surveys.

Add fields to the SurveyInternal dbdict record

User roles: System Administrator

SM surveys are based on survey templates. Each template contains a set of questions, which must be mapped to a field in the ServeyInternal dbdict record. Before users can create their own survey templates, you must add the required mapping fields to the ServeyInternal dbdict record.

To add fields to the dbdict record, follow these steps:

1. Sign in to the Service Manager Windows client as a system administrator.
2. Enter dbdict in the command line, and then press Enter.
3. Enter SurveyInternal in the **File Name** field, and then click **Search**.
4. On the **Fields** tab, select the **descriptor** row and then click **New Field/Key**.
5. Add a field to the dbdict.

   Caution

   The following field types are supported: character, number, logical, and array. For the array type, only the character subtype is supported.

6. Repeat the steps to add more fields as needed.
7. When you have finished adding all of the required fields, click **OK**.
8. Click **SM Alters** to save your changes to the dbdict record.

Note

These fields will be available to select from the **Field Name** field when users add a question to a survey template. For more information, see **Create a survey template**.

Tailor the SM Survey HTML template

User roles: System Administrator

When you manually send a survey from a record, an HTML Email template named **Internal Survey Template** is applied by default to generate the default subject and message body. You can tailor the content in the template or use your own template instead of this default one.

To tailor the default SM Survey template, follow these steps:

1. Click **Tailoring** > **Notifications** > **HTML Templates**.
2. Enter **Internal Survey Template** in the Name field, and then click **Search**.
3. Update the survey template content based on your requirements.
4. Click **Save** to save the updated template.

To replace the default SM Survey template, follow these steps:

1. Click **Tailoring** > **Notifications** > **Notifications**.
2. Enter **Send Survey** in the Name field and click **Search**.
3. In the Arguments column corresponding to the HTMLTemplate class, replace the string "Internal Survey Template" in the jscall function with your new template name. You need to make this update in both the **Message** and the **Email/Mail Subject Line** tabs.
4. Click **Save** to save the updated notification definition.

Manage the self-service survey menu

User roles: System Administrator

The SM Survey menu is added to out-of-box self-service menus, which allows an end user to view and complete an SM survey. You can update an operator record to grant survey access by assigning the ESSSM-Survey self-service menu.

There are multiple self-service menus available, you can choose to assign only the SM Survey menu to an operator. To do this, follow these steps:

1. Click **System Administration** > **Ongoing Maintenance** > **Operators**.

2. Type or select the optional search criteria.
3. Click **Search**.
4. Select an existing operator record.
5. Click the **Self Service** tab. Make sure that the operator record has these settings:
   - The **Self Service Access Only** checkbox must be selected.
   - Select or type **ESSSM-Survey** in the **Self Service Menu** field. In this way, the SM Survey menu is displayed when this user logs on using a self-service URL.
6. Click **Save**.

View the SM Survey workflows

User roles: System Administrator

SM Survey incorporates Process Designer technology. Scheduled SM surveys and SM survey templates are managed by a Process Designer workflow.

To view the workflows, follow these steps:

1. Click **Survey Management** > **Administration** > **Survey Workflows**.
2. Double-click the **Survey** workflow. The survey workflow is also displayed in the **Workflow** section of the survey definition record.

   Note

   This workflow applies for both SM Survey and Survey Integration.

   Out-of-box, the Survey workflow comprises the following three phases.

Phase	Description
Planning	The survey manager configures survey settings such as data filters, recipients, and parameters. Survey invitations aren't sent during this phase.
Scheduled	The survey runs for the configured period. The majority of settings are no longer editable during this phase.
Retired	The survey is no longer active.

3. Double-click the **Survey Template** workflow.

   Out-of-box, the Survey Template workflow comprises the following three phases.

Phase	Description
Draft	The survey template is editable in this phase. The survey template is not yet available for user selection when users create surveys.
Published	The survey template is published to users. Users can select the template when creating applicable surveys.
Retired	The survey template is no longer active.

Specify a maximum recommended number of recipients

User roles: Survey Administrator

For individual surveys that users manually send from a record, you can specify a maximum number of recipients that's recommended. If a user selects a number of recipients that exceeds this value, when the user clicks the **Send** button, the system

displays a warning to the user. The user has the option to either reduce the number of recipients or ignore the warning and continue.

To specify this setting, follow these steps:

1. Go to **Survey Management** > **Administration** > **Survey Settings**.
2. In the **Maximum recommended number of recipients per manual survey** field, enter an appropriate number. For example, 50.

   By default, this field has an empty value, and the feature is disabled. When a user clicks **Send** to send a manual survey, the system doesn't check how many recipients the user has selected and displays no warning to the user.

   Note

   Be aware that this setting takes effect only for manual surveys.

Use a rule set to send surveys

You can use a Process Designer rule set to trigger a survey from a module, which is therefore referred to as a rule-based survey. For example, you can define a rule set for the Interaction module and add it to the Closure phase of the **ServiceDesk** workflow. You can configure this rule set so that a survey is automatically sent to the Contact person of an Interaction record once the record enters the Closure phase.

Note

For rule-based surveys, the system selects a survey template as described in Automatic selection of survey templates. You can use this rule set for reference.

To use a rule set for rule-based surveys for a module (for example, Incident), follow these steps:

1. Go to **Tailoring** > **Process Designer** > **Rule Sets**.
2. In the **ID** field, enter send.survey.

   Note

   This sample rule set is intended for the Interaction module. You can use it as a reference.

3. Click **Search**.
4. Click **More** > **Clone** to copy the rule set to a new one.
5. Rename the rule set, and save it. For example, save it with the following attributes:
   - ID: send.survey.sd
   - Name: Send survey for Interaction
6. Select the rule description, and then click **Edit Rule/Group**.
7. (Optional) Click **Edit** under the **Condition** field, and then specify a condition that applies to rule-based surveys sent from the module.

   Note

   Be aware that the Condition setting in a survey template applies not only to manual surveys, but also to rule-based surveys.

8. Update the JavaScript as needed.

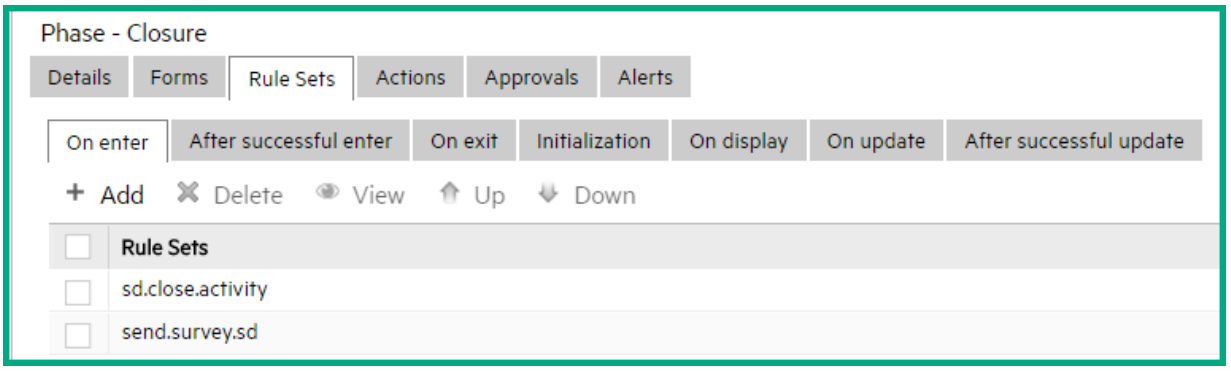
   Note

   The JavaScript needs to define recipients and a survey template for the survey.

   Tip

   For an Interaction workflow, you can use the sample JavaScript code without any modifications.

9. Click **OK** to save your changes to the rule set.
10. Go to **Tailoring** > **Process Designer** > **Workflows** to open your workflow for the module. In this example, open the **ServiceDesk** workflow.
11. Select a phase. In this example, select the **Closure** phase.
12. Select the **Rule Sets** tab, and then click **Add** to add the new rule set to the selected phase. See the following figure for an example.



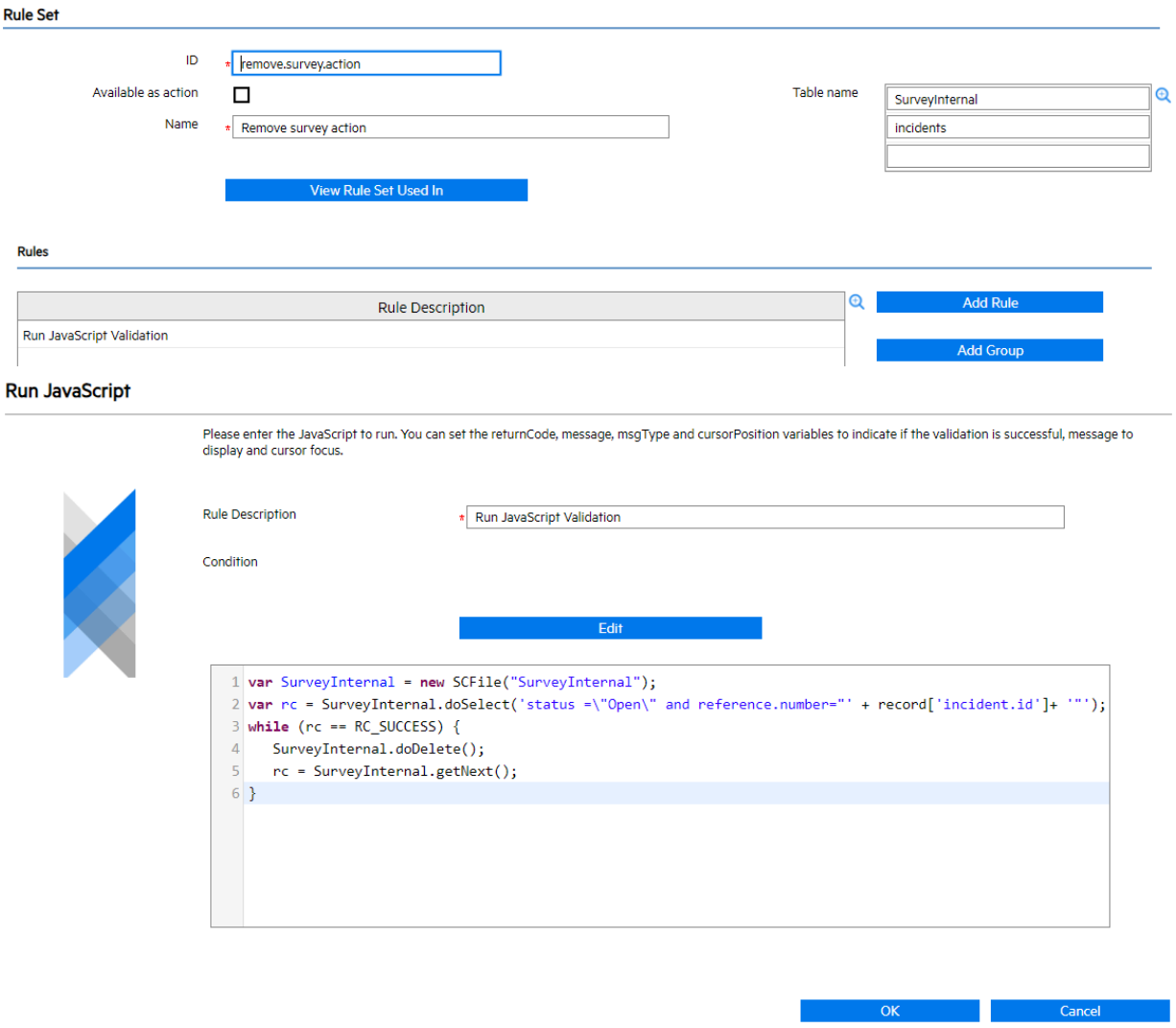
Use a rule set to remove open surveys

You can use a Process Designer rule set to remove surveys from a module.

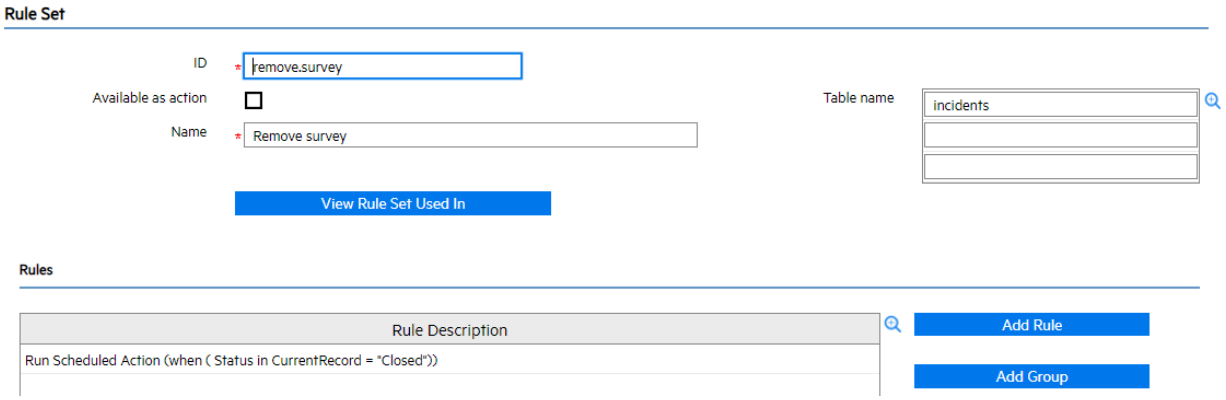
For example, you can define rule sets for the Interaction module and add it to the Closure phase of the service desk workflow. Once a interaction has been closed, system can automatically remove related open surveys.

In the below example, you can define a duration between Interaction close date/time and Survey deletion date/time. Later when an open survey ticket has exceeded the duration time, it will be removed automatically.

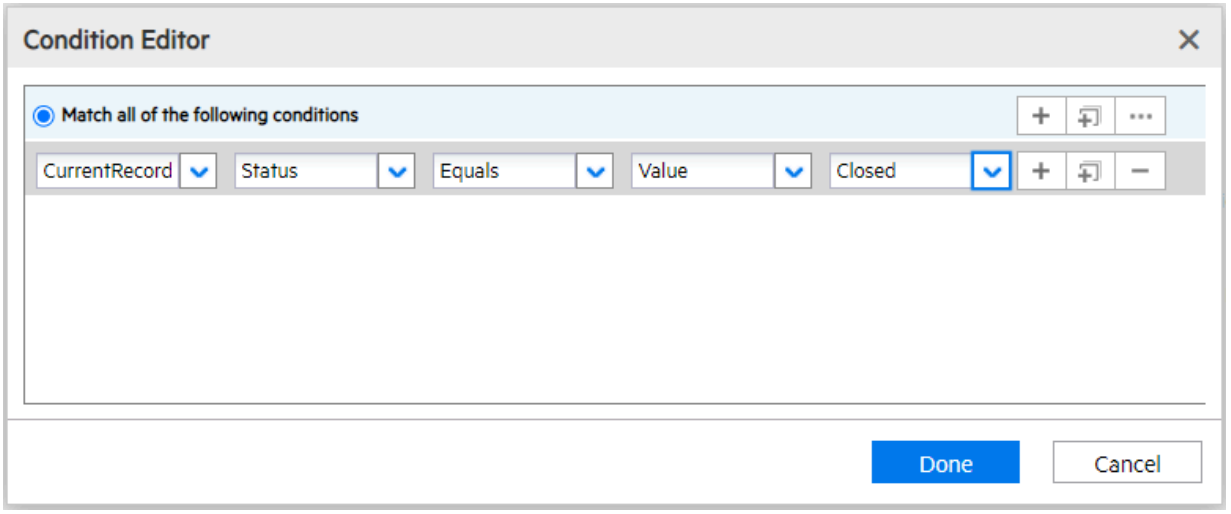
1. Define a rule set **remove.survey.action** with the **Run JavaScript** rules. You should define the rule set on the table **SurveyInternal** and **incidents**. See the following figures as an example.



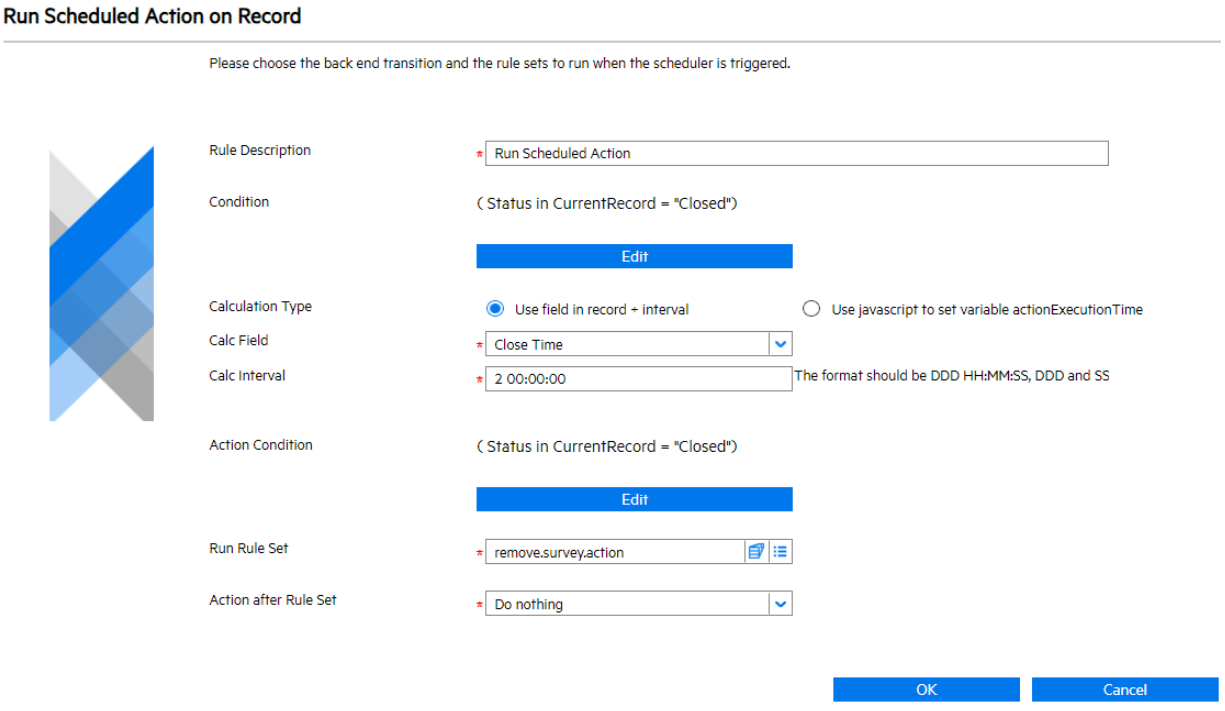
2. Define another rule set **remove.survey** with the **Run Scheduled Action** rule to use the rule set **remove.survey.action**. You should define the rule set on the table **incidents**. See the following figure as an example.



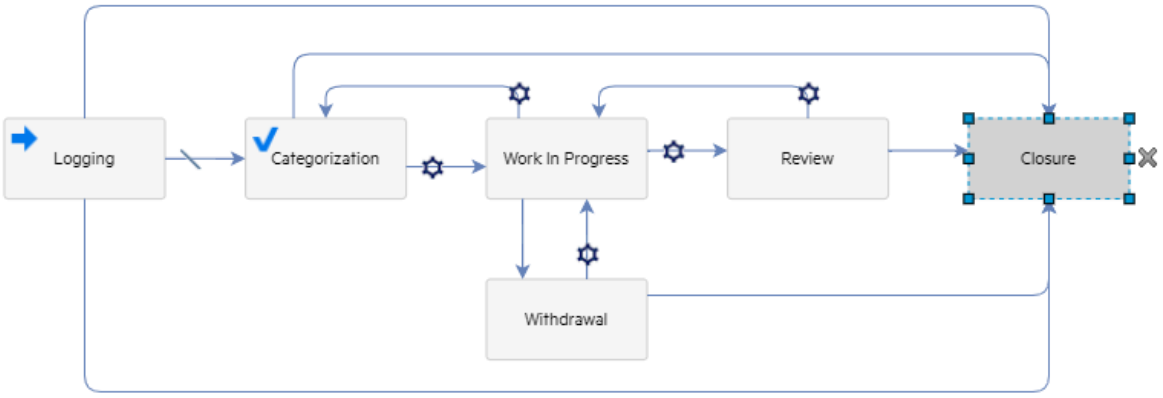
In the **Run Scheduled Action on Record** page, edit the condition to take the closed incidents into account.



Select **Use field in record + interval** for the **Calculation Type** field.
Select **Close Time** for the **Calc Field** field.
Set **2 00:00:00** for the **Calc Interval** field, which means the scheduled action is executed after two days.
Add **Action Condition** to the rule: **(Status in CurrentRecord = "Closed")**.
Search rule set **remove.survey.action** for the **Run Rule Set** field.
Select **Do nothing** for the **Action after Rule Set** field.



3. Configure the rule set **remove.survey** into the **After successfully enter** event of the service desk workflow, which means when the status of the interaction record is changed to **Closed**, the rule set is triggered. See the following figure for an example.



Phase - Closure

Details | Forms | Rule Sets | Actions | Approvals | Alerts

On enter | After successful enter | On exit | Initialization | On display | On update | After successful update

+ Add | Delete | View | Up | Down

Rule Sets
sd.solvedirectly
remove.survey

1.19.1.5. SM Survey user tasks

Once SM Survey is set up, users can perform the following tasks.

Create a survey template

Localize survey template

Publish a survey template

Edit an existing survey template

Retire a survey template

Send a survey from a record

Use Hot Topic Analytics with SM Survey

View SM survey reports

Respond to an SM survey

View my SM survey results

Check my SM surveys

1.19.1.5.1. Create a survey template

User roles: Survey Administrator, Survey Manager

Service Manager Survey enables you to send surveys based on different survey templates. The support of multiple survey templates enables users to choose appropriate questionnaires that suit their different needs.

Each survey template defines a questionnaire, and is managed by a workflow that consists of three phases: **Draft**, **Published**, and **Retired**. Users can create surveys based on published templates.

You can create a template for a specific module or for all supported modules. Additionally, you can specify a condition for a template. For example, you can use a condition to make a template available only for a specific assignment group.

Tip

In the out-of-box system, a default survey template named **DEFAULT** is provided. You can create a new template from scratch or from an existing template.

Create a survey template from scratch

To create a new survey template from scratch, follow these steps:

1. Click **Survey Management** > **Survey Templates** > **Create New Template** in the system navigator.
2. In the **Name** field, enter a unique name for the template.

   Note

   This is a mandatory field.

3. Click **Save**.

   Caution

   The template name must be unique. If you enter a duplicate name, an error will occur when you click **Save**.

The template is automatically populated with a template ID and set to the **draft** phase.

4. (Optional) In the **Description** field, enter descriptive text for the template.
5. (Optional) In the **Module** field, select a module to which you want to apply the current template. For example, if you select **Interaction**, this template is available from the Survey Template list only when users create a survey for the Interaction module. For more information, see Create an SM survey for a module.

   Note

   If you leave the **Module** field empty, the template will be available to use for all supported modules: Change, Configuration Item, Interaction, Incident, Request, and Problem.

6. Specify a condition that applies to the template.

   Note

   The condition setting in a survey template applies only for manual surveys and rule-based surveys; it doesn't apply for scheduled surveys. For more information about how the condition setting is used for template selection, see Automatic selection of survey templates.

   Tip

   If you want to specify a condition to trigger a rule-based survey from data records, use a Process Designer rule set instead. For more information, see Use a rule set to send surveys.

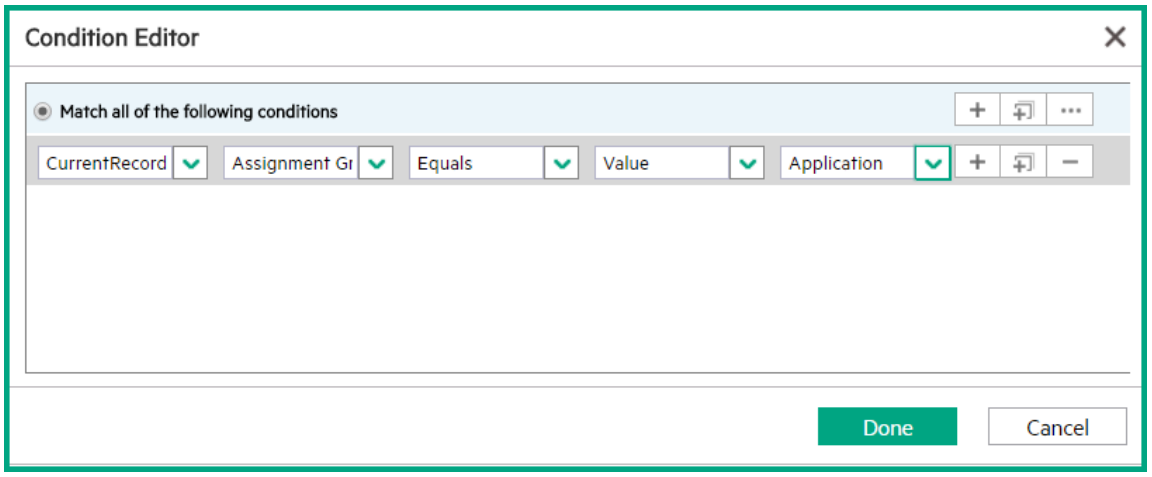
   1. Click **Edit** under the Condition field to open the Condition Editor.

      Note

      The **Condition** field is editable only after you selected a module in the previous step.

   2. Specify a condition.

For example, you can specify that the survey template can be used only for records that are assigned to a certain assignment group, as shown in the following figure.



> **Note**
>
> For SM Survey, all data types, operators, and value types available in the Condition Editor are supported.

3. Click **Done**.

   The **Condition** field is populated with the setting you specified. For example:

   ```
   ( Assignment Group in CurrentRecord = "Application")
   ```

7. Create a new question and link it to the template.

   1. On the **Questions** tab, click **Add Question**.
   2. In the **Field Name** field, select a field from the list.

      > **Note**
      >
      > This list displays applicable fields from the SurveyInternal table. Supported field types include: character, number, logical, and array (only the character subtype is supported). If one of these fields is already added to the current template, it's not available to select from the list.

      The Field Type field is automatically populated.

3. Click **Select a Question Label** to select an existing question label, or click **Add a Question Label** to specify a new label.
4. Select an answer type. The following table describes the answer types supported for different field types.

Answer Type	Explanation	Supported Field Types	Global List Available?	Allow Customized Answer?
Text	Enables users to enter a text answer in a single row.	number, character, array	No	No
Multiple Text	Enables users to enter a text answer in multiple rows.	array	No	No
Checkbox	Enables users to choose an answer for a logical (True or False) data type field.	logical	No	No

Answer Type	Explanation	Supported Field Types	Global List Available?	Allow Customized Answer?
Multiple Selection	Enables users to choose multiple answers. Multiple Selection entries are displayed as a drop-down list.	array	Yes	Yes
Dropdown List	Enables users to select one value from a set of mutually exclusive values provided in the drop-down list format.	number, character, array	Yes	Yes

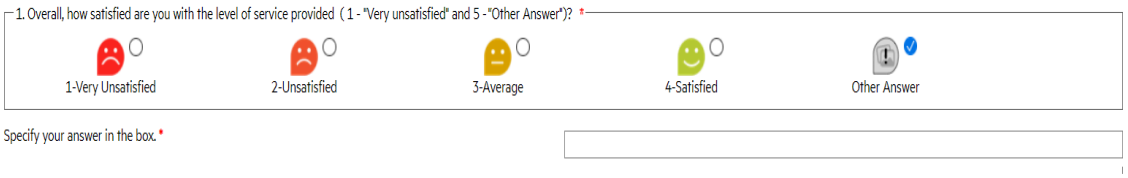
Answer Type	Explanation	Supported Field Types	Global List Available?	Allow Customized Answer?
Radio Buttons	Enables users to select one value from a set of mutually exclusive values provided in the radio button format.	number, character, array	Yes	Yes

5. If the question provides multiple options for users to choose from, the **Global List** field is displayed. Select the appropriate global list.

> **Note**
>
> If an appropriate global list doesn't already exist, you need to create it first.

6. (Optional) In the **Image List** field, choose an image group for the current question. This option applies to the **Radio Buttons** answer type only.

   You can use this option to add icons or emojis to the answers in Service Manager Surveys so that the questionnaires in both the SMA Service Portal and the Employee Self-Service (ESS) Portal have a more vivid display. See the following questionnaire as an example.

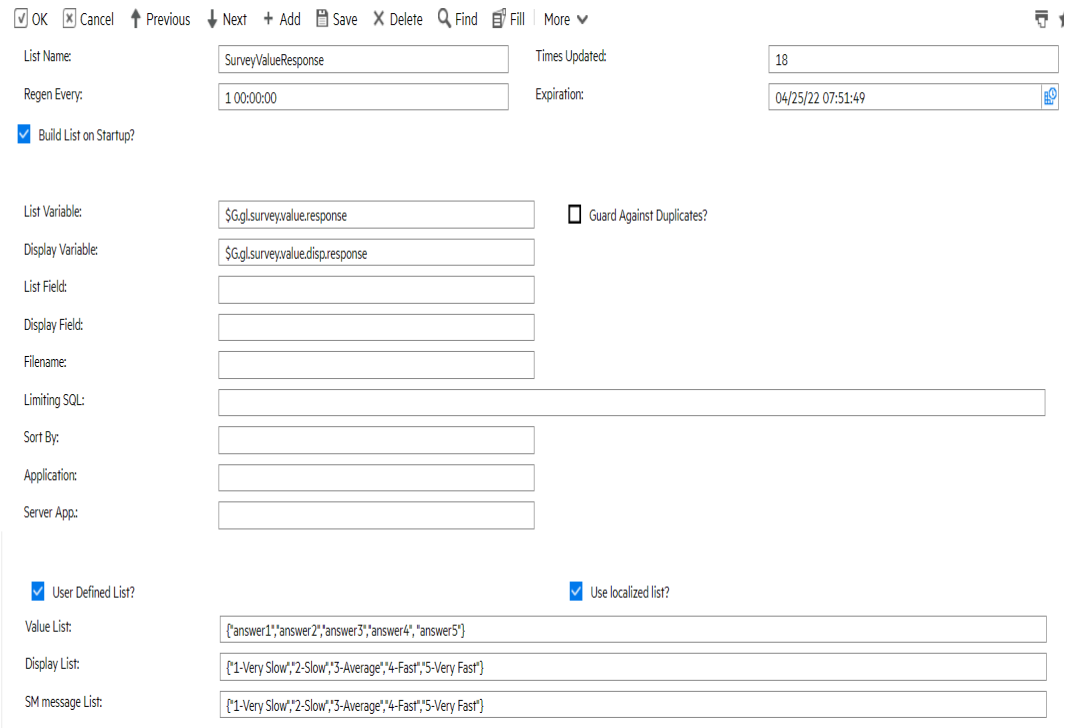


Service Manager provides a series of out-of-the-box emojis and a default image group named **Default Satisfaction** for your choice. For tailoring purposes, you may want to add more icon images and image groups so that your questionnaire has a customized display. Refer to the following steps for some guidelines.

1. Upload more icon images. Click **Survey Management** > **Survey Templates** > **Survey Image**. Enter a name for the image, and then click **Add**. Click **Specify Image...** to choose an image file, and then click **OK**. For example, add five new image files named image1, image2, image3, image4, and image5.

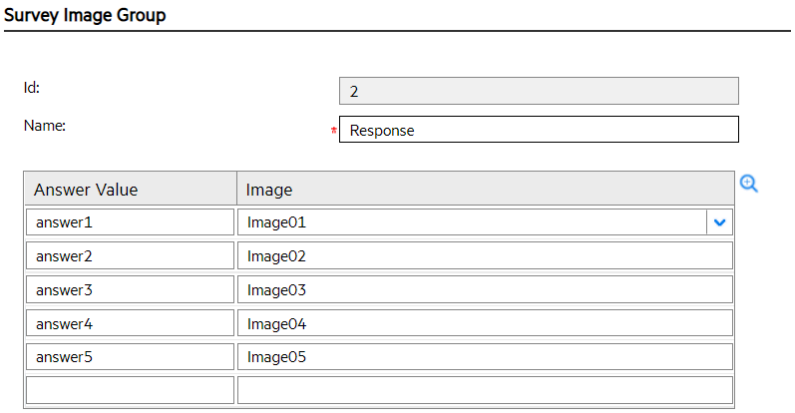
   Note: If you have used an icon image in a survey and the survey is closed later, when you remove or update this icon image, your changes apply to the closed survey as well.

2. Refer to the out-of-the-box global list **SurveyValue** as an example and then create your own one. Value for both the **List Variable** and the **Display Variable** fields must start with `$G.gl.` . For example, `$G.gl.survey.value.response` . In your global list, take notes of the values for the **Value List** field. For example, {"answer1","answer2","answer3","answer4", "answer5"}.



3. Add an image group. Click **Survey Management** > **Survey Templates** > **Survey Image Group**. Enter a name for the group and then click **Add**.

4. In the **Answer Value** column, enter the values as included in the **Value List** field in your global list. Each value starts with a new row. Next, select an image for each answer. For example:



5. Save your changes.

7. (Optional) In the **Layout** field, choose a layout for the answers. This option applies to the **Radio Buttons** answer type only.

8. Select **Mandatory** if this question is mandatory.

> **Caution**
>
> Be careful about making a question of the Checkbox type mandatory. If you make it mandatory, it can only have a value of True.

9. Select **Allow Customized Answer** if choices are extensive or if the user may need to enter a choice manually. Note that if you use this option, the system refreshes the survey form every time your select an answer to a question.

   When adding a question, if you have selected an image group for your answers and then select this checkbox, the system displays a default emoji for the **Other Answer** option in your survey. In the out-of-the-box emojis, the **Other Answer** emoji is designed for the customized answer and you don't need to specify an emoji for it again.

   Note: Don't remove the **Other Answer** emoji because the system may fail to find an emoji for the customized answer.

10. When you are satisfied with the result, click **Finish**.

11. Click **Save**. The question is added to the survey template.

> **Note**
>
> The question is not linked to the template until you click **Save**.

8. If needed, repeat the previous step to add more questions to the template.
9. Link one or more existing questions to the template if appropriate.
    1. Click **Link Existing Question**.
    2. In the search form that's displayed, enter your search criteria and then click **Search**.

    > **Tip**
    >
    > The **Question Label** field supports wildcard searches.

    3. Select the checkboxes of the questions that you want to link, and then click **Select**. Only questions that aren't yet linked to the current survey template are returned in the search results.

    > **Note**
    >
    > If you link the **Do you want to register an interaction?** question to the survey template, the system creates an interaction after the user selects the positive answer in the survey.

    The selected questions are added to the list of questions.

    4. Click **Save**.
10. If needed, remove one or more existing questions from the template.
    1. Select the checkboxes of the questions that you want to remove.
    2. Click **Remove Question**.
11. If needed, edit one or more existing questions in the template.
    1. On the **Questions** tab, click the question ID to open the question.
    2. Edit the question as needed.
    3. Click **Finish**.

> **Note**
>
> You can edit a question only when it's not linked to another template. If it is already linked to another template, when you click **Finish**, a warning is displayed. If you click **Yes** and then **Finish**, the updated question is automatically saved as a new question and added to the current template. If you don't want to create a new question, click **No** to cancel.

12. If needed, adjust the order of the questions.
    1. To do this, select the questions that you want to move.
    2. Click **Move Question Up** or **Move Question Down**.
13. (Optional) In the **Help Description** field, type some introductory texts to the survey. These texts will be displayed on top of the questions list when you access the survey from the Service Manager ESS portal.
14. Click **Save** to save your changes.
15. Preview the template.
    1. Make sure you have clicked **Save** to save the survey template.
    2. Click **Preview** on the task bar to see if the template satisfies your needs.
    3. Click **Back**.
    4. If the template doesn't satisfy your needs, continue to update the template.
16. When you are satisfied with the template, click **Publish** to publish it.

> **Note**
>
> Only published templates are available for selection when users create a survey.

17. If you created new global lists, restart the Service Manager Server.

Copy an existing survey template

To create a new survey template from an existing one, follow these steps:

1. Click **Survey Management** > **Survey Templates** > **Search Templates**.
2. Enter your search criteria and then click **Search** to open an existing template.
3. Click **More**, and then select **Copy**.
4. Click **Save** to create the new template.

Service Manager automatically copies all questions from the source template to the new template.

5. Update the template name, and edit the list of questions as needed.
6. If you think the template is ready for use, click **Publish** to publish the template.

1.19.1.5.2. Localize survey template

User roles: System Administrator

Suppose you have enabled multiple login languages in the Service Manager system. After creating a survey template, a system administrator may need to create localized versions of the template. When sending surveys, Service Manager sends an email according to each recipient's login language. When each recipient opens the survey link in the email, the survey gets displayed in the recipient's login language.

Tip

As a best practice, once a creator creates new survey questions and these questions need localization, the creator can have the new question labels localized into different languages and then send the localized text to a system administrator, who can then get the new questions localized in the system.

To localize a survey template into a language, follow these steps:

1. Sign in to Service Manager with that language.
2. In the command line, enter `db` and press Enter.
3. Enter `scmessage` in the **Table** field, and then click **Search**.
4. Click **message.edit** from the list.
5. Enter `srcsurvey` in the **Class** field, and then click **Search**.

   Service Manager displays a list of messages. This list includes only messages that are created for your login language.

   Note

   When you add and save a question to a survey template, Service Manager automatically creates a record with the new question label in the **scmessage** table for each language that has **Active for Survey Management** enabled in the **language** table.

6. Select a record from the list, and update the **Text** field with a translated version of the question label.

7. Click **Save**.
8. Repeat the steps for the rest of the questions that you want to localize.

Or, you can localize a survey template in the following way:

1. Sign in to Service Manager with that language.
2. Click **Survey Management** > **Survey Templates** > **Search Templates**.
3. (Optional) Enter your search criteria.
4. Click **Search** to directly open a survey template or open one from a list of survey templates.

   Note

   Service Manager displays the template or list of templates according to your login language.

5. On the **Questions** tab, click a question to open it.
6. Click the **View Detailed Information** icon for the question label field.

   A message record created for the current language opens.

7. Update the **Text** field of this record with a translated version of the question label.
8. (Optional) If you have added introductory texts to the survey in the **Help Description** field, click the **Localize** button. The system opens a wizard where you can select a language from the list and then type the localized help description.

   Tip

   System administrators can configure which languages appear in the list by using the Languages table. To include a language, select that language in the Languages table, and then select the **Active for Survey Management** option.

9. Select a language from the drop-down list, and then type the localized content.
10. Repeat the steps for the rest of the questions in the template.

Related topics

SM Survey setup
Create a survey template

1.19.1.5.3. Publish a survey template

Applies to User roles: Survey Administrator, Survey Manager, Survey Analytics

When you create a new survey template and save it, the template is in the Draft phase and therefore not available for selection when users create surveys. If the template is ready for use, you must publish it to make it available to users.

Note

A survey template must have at least one question in it before it can be published.

To publish a survey template, follow these steps:

1. Click **Survey Management** > **Survey Templates** > **Search Surveys** in the system navigator.
2. In the search form, enter your search criteria.

   You can search survey templates by Template ID, Name, Module, or Phase. You can also use advanced search.
3. Click **Search** to directly open the survey template or open it from a list of survey templates.
4. Click **Publish**. The template is moved to the Published phase.

   If the template has no questions linked to it, an error message is displayed.

Related topics

SM Survey setup
Create a survey template
Edit an existing survey template
Retire a survey template

1.19.1.5.4. Edit an existing survey template

Applies to User roles: Survey Administrator, Survey Manager

Survey templates are managed by a workflow of three phases: Draft, Published, and Retired. You can edit a survey template only when it is in the Draft phase. If a survey template is in the Published or Retired phase, you have to move it back to the Draft phase before you can edit it.

Tip

You can always edit or retire a survey template, no matter whether there are open surveys associated to it. When a survey is created, SM Survey takes a snapshot of the survey questions and stores the snapshot in the survey record in the database. After the creation of the survey, no matter whether the associated survey template is updated or retired and whether the survey questions are updated, the survey is always rendered based on the snapshot.

Note

In some cases, a survey may not have a snapshot stored in the database, for example, a survey that's upgraded from a version earlier than Service Manager 9.50. Considering this scenario, SM Survey renders surveys based on one of the following items, which are listed from the highest to lowest priority: snapshot, template ID, and the default survey template defined in the **Survey Settings** form (**Survey Management** > **Administration** > **Survey Settings**).

To edit an existing survey template, follow these steps:

1. In the system navigator, click **Survey Management** > **Administration** > **Search Templates**.
2. In the search form, enter your search criteria.

   You can search survey templates by Template ID, Name, Module, or Phase. You can also use advanced search.
3. Click **Search** to directly open the survey template or open it from a list of survey templates.

4. If the template is in the Published phase, click **Revise** to move it to the Draft phase.
5. If the template is in the Retired phase, click **Unretire** move it to the Draft phase.
6. If the template is in the Draft phase, edit it directly. For example, you can add or remove questions or change the module. For details, see Create a survey template.
7. Click **Save**.

1.19.1.5.5. Retire a survey template

User roles: Survey Administrator, Survey Manager, Survey Analytics

When a survey template no longer suits your needs, you may want to retire it. Retried templates aren't available for selection when users create surveys.

To retire a survey template, follow these steps:

1. Click **Survey Management** > **Survey Templates** > **Search Surveys** in the system navigator.
2. In the search form, enter your search criteria.

   You can search survey templates by Template ID, Name, Module, or Phase. You can also use advanced search.
3. Click **Search** to directly open the survey template or open it from a list of survey templates.
4. Click **Retire**. The template is moved to the Retired phase.

Note

If you want to unretire it at a later time, click **Unretire**.

1.19.1.5.6. Send a survey from a record

User roles: Service Desk Agent

Once your administrator has set up HTML Email (JavaMail) and configured the survey portal and survey portal URL, you can manually send a survey from a record at any state in one of the following modules:

- Interaction
- Request
- Incident
- Problem
- Change
- Configuration Item

To manually send an SM survey from a record, follow these steps:

1. Open a record. For example, open an interaction record.
2. Click **More** > **Send Survey** to display the **Manual Customer Survey** interface.
3. In the **Survey Template** field, click Fill to select the appropriate survey template.

   Tip

   SM Survey provides an out-of-box sample survey template with sample questions. In a testing environment, you can use this template for testing purposes.

4. Click Fill to select one or more recipients from the list.

   Note

   If you manually enter recipients, auto-complete is supported for recipient names. If you enter an invalid recipient name, the recipient is skipped when you send the survey.

5. Click **Preview** to preview the survey before sending, and then click **Back**.
6. Click **Send**.

   A confirmation dialog opens.

7. Click **Yes** to send the survey to the selected recipients.

The specified recipients receive an email notification, which contains a survey link and instructions on how to respond to this survey.

1.19.1.5.7. Use Hot Topic Analytics with SM Survey

Applies to User roles: Survey Administrator, Survey Manager, Survey Analyst

If Smart Analytics is enabled in the system, you can use Hot Topic Analytics with SM Survey.

Access Hot Topic Analytics from survey reports

For details, see Access Hot Topic Analytics from SM reports.

Use Hot Topic Analytics to create an interaction

For details, see Use Hot Topic Analytics to create an incident, problem, change, or interaction.

Related topics

Service Manager Survey

1.19.1.5.8. View SM survey reports

Applies to User Roles: Survey Administrator, Survey Manager, Survey Analyst

The out-of-box system provides a survey dashboard, which contains reports that the system generates based on your survey data. These reports can help your organization improve your services and user satisfaction.

To view SM survey reports, follow these steps:

1. Navigate to **Survey Management** > **Survey Overview**.
2. View the reports on the dashboard.

Related topics

Service Manager Survey

1.19.1.5.9. Respond to an SM survey

User roles: ESS users

> **Tip**
>
> The following steps are only for ESS users.

If you receive an email notification inviting you to respond to an SM survey, follow the instructions to view and respond to the survey.

To view and respond to an SM survey from the ESS portal, follow these steps:

1. Open the survey link in a web browser.
2. Sign in to Service Manager. The survey details page opens.

   > **Tip**
   >
   > Instead of following the survey link in an email notification, you can directly click **Survey** > **View Open Surveys** and then click the survey number to respond to the survey.

3. Select your answers to the survey questions.
4. (Optional) Enter additional comments.
5. (Optional) Click **Register a compliment** or **Register a complaint**.

   If you select either option, the system will automatically create an interaction record that has a category of Compliment or Complaint and send you another email with the record details after you submit the survey. By default, the **No interaction registration** option is selected.
6. Click **Submit Survey**.

   The survey now appears in your list of closed surveys (**Survey** > **View Closed Surveys**).

Related topics

Service Manager Survey
Send a survey from a record
Check my SM surveys

1.19.1.5.10. View my SM survey results

User roles: Service Desk Agent

From the TO-DO queue, you can view the results of the SM surveys you have sent out.

To view your SM survey results, follow these steps:

1. From the TO-DO queue, select **Internal Survey**.
2. Select the **All internal surveys created by me** view. The surveys are grouped into closed surveys and open surveys.
3. Open the closed surveys to view the results.

Related topics

Service Manager Survey
Send a survey from a record
Respond to an SM survey

1.19.1.5.11. Check my SM surveys

Applies to User Roles: Employee Self-Service users

You can view your surveys, including those that you have and haven't completed.

To check your surveys, follow these steps:

1. Sign in to the Service Manager web client Employee Self-Service User view (`ess.do`).
2. Click **Survey** from the system navigator, and then select **View Open Surveys** or **View Closed Surveys**.

Related topics

Service Manager Survey
Send a survey from a record
Respond to an SM survey

1.19.1.6. SM Survey security areas and roles

The Survey module uses the same set of security areas and security roles for SM Survey and Survey Integration.

For details, see Survey Integration reference material.

1.19.2. Survey Integration

Service Manager Survey Integration enables you to effectively configure and manage the surveys in your organization.

Survey Integration includes three integration methods (connectors) that enable you to integrate Service Manager with third-party survey tools.

1.19.2.1. Create a survey integration instance

Before you can create and configure surveys in Service Manager Survey Integration, you must create a survey integration instance. Survey Integration only requires one survey integration instance, even if you have multiple surveys. Once you have created a survey integration instance, you can't create a second one.

To create a survey integration instance, follow these steps:

1. In the System Navigator, click **Tailoring** > **Integration Manager** > **Add**.
2. Select **SMSurvey** from the **Integration Template** drop-down list, and then click **Next**.
3. Populate the **Interval Time** and **Max Retry Times** fields.
4. Click **Next** repeatedly until the **Finish** button is displayed.
5. Click **Finish**.
6. Select the **SMSurvey** integration instance, and click **Enable** to enable the integration.
7. Go to **System Status**, and make sure that the **survey service agent** scheduler is started. If not, click **Start Scheduler**, and then double-click survey service agent scheduler to start it.

1.19.2.2. Survey administrator tasks

This section introduces administrator tasks for survey integration.

Configure the global survey settings

The System Administrator can configure a number of settings that apply to every scheduled survey in SM Survey and Survey Integration.

To view and configure these settings, follow these steps:

1. Click **Survey Management** > **Administration** > **Survey Settings**.
2. Configure the following attributes.

Attribute	Description	Notes
Use HTML emails	When this option is selected, the HTML editor is displayed in the survey definition record. If the option is not selected, a plain text editor is displayed instead.	
Use HTML templates	When this option is selected, HTML templates are available to help you create survey emails.	
Default language	The language in which the survey email is sent when no localized version is available in the recipient's language.	This attribute is mandatory.

Attribute	Description	Notes
Maximum recommended number of recipients per manual survey	This is the maximum number of recipients that's recommended for each survey manually sent from a record. When the number of recipients that the user has specified exceeds this number, a warning message is displayed when the user clicks **Send**. The user still has the option to ignore the warning and send the survey. **Note** If you leave this field empty, no warning about the number of recipients is displayed to the user. See also Specify a maximum recommended number of recipients.	For SM Survey only.

Attribute	Description	Notes
Additional Filter Change form	The form that's used to filter additional records against the cm3r table. You can create your own forms that have the filter fields you require.	This attribute is mandatory.
Additional Filter Configuration Item form	The form that's used to filter additional records against the device table. You can create your own forms that have the filter fields you require.	This attribute is mandatory.
Additional Filter Incident form	The form that's used to filter additional records against the probsummary table. You can create your own forms that have the filter fields you require.	This attribute is mandatory.
Additional Filter Interaction form	The form that's used to filter additional records against the incidents table. You can create your own forms that have the filter fields you require.	This attribute is mandatory.

Attribute	Description	Notes
Additional Filter Problem form	The form that's used to filter additional records against the rootcause table. You can create your own forms that have the filter fields you require.	This attribute is mandatory.
Additional Filter Request form	The form that's used to filter additional records against the request table. You can create your own forms that have the filter fields you require.	This attribute is mandatory.
Default survey template name	This is the survey template that the system applies to an automatic survey triggered by a rule set in a workflow when no applicable survey templates are found. In the out-of-box system, this value is **DEFAULT**. For more information, see Automatic selection of survey templates and Use a rule set to send surveys.	For SM Survey only.

Attribute	Description	Notes
HTML format for email body	Specifies the form that contains the HTML email body.	This attribute is mandatory.
Link for copy (default: SurveyDefinition.copy)	Enables you to specify a link to decide which fields are copied from the source to the target record.	
Plain text format for email body	Specifies the form that contains the plain text email body.	This attribute is mandatory.

3. Click **Save**.
4. Log off and log back in to Service Manager so that your configuration takes effect.

Create a new survey connector

In addition to the out-of-box survey connectors, you can create new survey connectors in Service Manager Survey Integration. To do this, follow these steps:

1. In the system navigator, click **Survey Management** > **Administration** > **Survey Connectors** > **New** to display a blank connector form. The **ID**, **Last Updated by**, and **Last Update time** fields are populated automatically.
2. Click to clear the **Active** checkbox if you don't want the new survey connector to be displayed in the list of available survey connectors when you create a new survey.
3. Select a connector type from the drop-down list in the **Type** field.
4. Type a name for the connector in the **Name** field.
5. Type a description of the connector in the **Description** field.
6. Click **Save** to save the connector. The **Parameters and Recipients** section of the form is displayed.

Search for a survey connector

To search for an existing survey connector in Service Manager, follow these steps:

1. In the system navigator, click **Survey Management** > **Administration** > **Survey Connectors**.
2. Use the search form to specify search criteria. For example, in the **Type** drop-down list, select **Send emails with a survey link**.
3. Click **Search** to display the list of records that match your search criteria.

Note

If Service Manager generates a message that states no records were found, you can broaden the search by eliminating some search parameters.

Create a new parameter

Parameters are predefined values that provide surveys in Service Manager with the correct metadata. Survey Administrators can create additional parameters that are available to survey users when they configure a survey connector. Parameters are associated with specific Service Manager modules.

To create a new parameter, follow these steps:

1. Open a survey connector by creating a new connector or by searching for an existing connector.
2. Browse to the **Parameters and Recipients** section, and then expand the Service Manager module to which you want to add a parameter.
3. Expand **Parameters**, and then click **Add Parameter** to display the Survey Parameter Mapping form. The **ID**, **Connector ID**, **Last update time**, **Last updated by**, and **Object** fields and the **Active** check-box are populated automatically.
4. Type a user-friendly description of the parameter in the **Description** field.
5. Type the internal name for the parameter in the **Parameter Name (Internal)** field.
6. Select the order in which the parameter is processed in the **Order ID** field (this step applies only to API-based connectors).
7. Select a mapping type from the **Mapping Type** drop-down list.
8. Type a constant value, variable, field name, or RAD expression in the **Mapping Value** field, and then click **Save**.

Create a new recipient

Survey Administrators can create predefined recipients that are available to survey users when they assign recipients to a survey in Service Manager. Recipients are associated with specific Service Manager modules.

To create a new recipient, follow these steps:

1. Open a survey connector by creating a new connector or by searching for an existing connector.
2. Browse to the **Parameters and Recipients** section, and then expand the Service Manager module to which you want to add recipients.
3. Expand **Recipients**, and then click **Add Recipient** to display the Survey Recipients Configuration form. The **ID**, **Connector ID**, **Last update time**, **Last updated by**, and **Object** fields and the **Active** check-box are populated automatically.
4. Type a user-friendly description of the recipient in the **Description** field.
5. Type the internal name for the recipient in the **Parameter Name (internal)** field.
6. Select a mapping type from the **Mapping Type** drop-down list.
7. Select a recipient type from the **Recipient Type** drop-down list.

   Note

   If you select **Group** as the recipient type, you must also select values from the **Group** and **Subgroup** drop-down lists.

8. Type a constant value, variable, field name, or RAD expression call in the **Recipient(s)** field, and then click **Save**.

Create a predefined filter

Data filters enable you to configure which records in the database are selected to be recipients of Service Manager survey requests. Survey Administrators can create predefined filters, which are available for survey users to choose from, by creating views in Service Manager.

To create a view to use as a predefined filter, follow these steps:

1. In the system navigator, click **System Administration** > **Base System Configuration** > **Miscellaneous** > **Views/Favorites** > **New**.
2. Select an object from the **Area** drop-down list, and then click **Next**.
3. Type a name in the **Name** field, select a view type from the **View type** drop-down list, and then click **Next**.

4. Select the **Selected Groups** option, click to select the **Survey** group, and then click **Next**.
5. Configure the parameters for the view, and then click **Finish**.

To ensure the survey solution can filter for records that are exclusively related to the survey data, the "@FieldName@" directive is added to the filter queries. When the query is run, the directive is replaced by the value in the specified field. For example, the following query filters for records that were closed since the last survey run:

```
close.time>@lastProcessingTime@
```

Note

- Ensure that you configure the predefined filter so that the system can process all the records within the configured send interval.
- The filter results must be ordered correctly to ensure that the required users are surveyed. This is because Survey selects records in the order in which they appear in the database after a query is run.

1.19.2.3. Survey user tasks

This section introduces the Survey Integration user tasks.

Create a scheduled survey from menu

Search for a scheduled survey

Configure the survey header

Add an activity to the survey activities log

Configure the survey scheduling rules

Filter the survey recipients

Apply a data filter to the survey

Create an additional filter

Configure the survey recipients

Configure the survey email settings

Create a localized email

Configure the survey parameters

View the survey workflow

View the survey history

1.19.2.3.1. Create a scheduled survey from menu

To create a new survey in Service Manager Survey Integration, follow these steps:

1. Click **Survey Management** > **Scheduled Surveys** > **Create New Survey** in the system navigator to display the Create a New Survey wizard.
2. If there is more than one survey connector, select an external survey connector from the list. The **Connector type** field is populated automatically, based on your choice of connector.
3. Select the Service Manager module on which the survey will be based from the **Object** drop-down list.
4. Enter values in the **Owner Group** and **Name** fields.
5. Click **Finish** to display the survey definition record.
6. Populate all the mandatory fields in the survey definition record to configure the survey.
7. Click **Save** to create the survey record.

1.19.2.3.2. Search for a scheduled survey

To search for an existing survey definition record in Service Manager Survey Integration framework, follow these steps:

1. In the system navigator, click **Survey Management** > **Scheduled Surveys** > **Search Schedules**.
2. Use the search form to specify search criteria. For example, in the **Current Phase** field, select **retired**.
3. Click **Advanced Filter** to create custom filters. To create new search criteria, follow these steps:
    1. Click **Add New Filter Criteria**.
    2. Populate all the fields as required, and then click **Next** to add the criterion to the **Conditions** list.
    3. Click **Add New Compound** to add a compound rule to a filter that has more than one criterion.
4. Click **Search** to display the list of records that match your search criteria.

> **Note**
>
> If Service Manager generates a message that states no records were found, you can broaden the search by eliminating some search parameters.

Related topics

Create a new survey
Configure the survey parameters
Configure the survey recipients

1.19.2.3.3. Configure the survey header

The **Header** section of the survey management form in Service Manager Survey Integration contains the survey metadata. The content of the section is dependent on whether the survey uses a URL-based or API-based connector.

Surveys that use a URL-based connector

To configure the survey header of a survey that uses a URL-based connector, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. The **Active** check-box and the following fields are populated automatically:
    - **ID**
    - **Object**
    - **Current Phase**
    - **Connector Type**
    - **Owner Group**
    - **Name**
3. Type a value in the **External Survey ID** field.
4. Type values in the **Owner** and **Description** fields, and then click **Save**.

Surveys that use an API-based connector

To configure the survey header of a survey that uses an API-based connector, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. The **Active** and **Keep Default Connection** check-boxes and the following fields are populated automatically:
    - **ID**
    - **Object**
    - **Current Phase**
    - Connector Type
    - **Owner Group**
    - **Name**
3. Type a value in the **External Survey ID** field.
4. Type values in the **Owner** and **Description** fields.
5. To create a custom connection to the survey solution, follow these steps:

1. Clear the **Keep Default Connection** check-box to display the **Connection Parameters** sub-section.
2. Type the URL, user name, and password that are required to connect to the survey solution.

6. Click **Save**.

1.19.2.3.4. Add an activity to the survey activities log

In Service Manager Survey Integration, the **Activities** section of the survey definition record displays the major activities that are associated with the survey. You are able to add an activity to the record in the **Activities** section as well. To add an activity, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Activities** section.
3. Select the type of update that you want to add from the **New Update Type** drop-down list.
4. Type a description of the update in the **New Update** field, and then click **Save**.

Note

If too many activities are recorded, you can use the **Activity Type** and **User** drop-down lists to filter the activity logs.

Related topics

Create a new survey
Search for a survey

1.19.2.3.5. Configure the survey scheduling rules

The **Scheduling Rules** section of the survey definition record in Service Manager Survey Integration enables you to configure the duration of the survey and how often survey requests are sent out.

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Scheduling Rules** section.
3. Select the interval at which survey requests are sent from the **Send Interval** drop-down list.

   > **Note**
   >
   > Survey Administrators can configure the options that appear in the Sending Interval drop-down list by modifying the SurveySendInterval types in the Code table. The value is in seconds.

4. Enter the survey start date in the **Active From** field.

   > **Note**
   >
   > This date can't be in the past.

5. Enter the survey end date in the **To** field, and then click **Save**.

   > **Note**
   >
   > This date must be later than the date in the **Active From** field.

To configure a survey that runs one time only, follow these steps:

1. In the **Scheduling Rules** section of a survey definition record, set the **Send Internal** field to **Every hour**.
2. Set the **Active From** field to the start time that you require.
3. Set the **To** field to a time that's 10 minutes later than the time that you entered in the **Active from** field.

Note

If the time difference between the **Active From** and the **To** fields is less than 5 minutes, the survey may not run.

Related topics

Create a new survey
Search for a survey

1.19.2.3.6. Filter the survey recipients

The **User Selection Rules** section of the survey definition record in Service Manager Survey Integration enables you to configure how often users are sent survey requests and the percentage of users that are sent a survey request. To filter the survey recipients, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **User Selection Rules** section.
3. Select one or more of the following options:
    - Do not survey a user more than once in a survey cycle

      When this option is selected, users who appear in the survey database more than once aren't sent multiple survey requests.
    - **Do not send a user a new survey for <***time***>**

      When this option is selected, you can configure a period of time in which users aren't sent new survey requests.
    - **Randomly select <***percentage***> % of users as survey recipients**

      When this option is selected, you can configure the percentage of users that are sent a survey request.
4. Click **Save**.

Related topics

Create a new survey
Search for a survey

1.19.2.3.7. Apply a data filter to the survey

The **Data Filter** section of the survey definition record in Service Manager Survey Integration enables you to configure which records in the database are selected to be recipients of survey requests. The section comprises of predefined filters (that are managed by the Survey Administrator) and additional filters (that are user-generated). The two types of filter are connected with an AND operator.

To apply a data filter, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Data Filter** section.
3. Select a predefined filter.
4. Create an additional filter, and then click **Save**.

1.19.2.3.8. Create an additional filter

Data filters enable you to configure which records in the database are selected to be recipients of in Service Manager Survey Integration survey requests. Additional data filters can be created by users to supplement the predefined data filters.

To create an additional data filter, follow these steps:

1. Open a survey definition record, and then browse down to the **Data Filter** section.
2. Click **Build Additional Filter** to display the Query Generator form.
3. Populate the appropriate fields with values to create the required filter.
4. Click **Advanced Filter** to create custom filters. To create new search criteria, follow these steps:
    1. Click **Add New Filter Criteria**.
    2. Populate all the fields as required, and then click **Next** to add the criterion to the **Conditions** list.
    3. Click **Add New Compound** to add a compound rule to a filter that has more than one criterion.
5. Click **OK** to display the additional data filter in the **Data Filter** section of the survey management form.

    Tip

    You can click **Test Filter** to see how many records are returned.

6. Click **Save**.

To ensure the survey solution can filter for records that are exclusively related to data the survey, the "@FieldName@" directive is added to the filter queries. When the query is run, the directive is replaced by the value in the specified field. For example, the following query filters for records that were closed since the last survey run:

```
close.time>@lastProcessingTime@
```

Related topics

Apply a data filter to a survey

1.19.2.3.9. Configure the survey recipients

The **Recipients** section of the survey definition record in Service Manager Survey Integration enables you to configure the recipients of survey requests. Recipients are defined in the survey connector, and are based on the specific Service Manager module to which the survey applies.

To configure the survey recipients, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Recipients** section.
3. Select a recipient from the **Available Recipients** box, and then click the right arrow to move the recipient to the **Selected Recipients** box.

   > **Note**
   >
   > To remove a recipient from the **Selected Recipients** box, select the recipient, and then click the left arrow.

4. Click **Save**.

1.19.2.3.10. Configure the survey email settings

The **Email Settings** section of the survey definition record in Service Manager Survey Integration enables you to configure the email survey requests that are sent for surveys that use URL-based connectors.

To configure the email settings, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Email Settings** section.
3. Type the URL of the external survey in the **External Survey URL** field.
4. Type an email subject in the **Email Subject** field.
5. Type the email body text in the **Email Body** field. Or, click **Select HTML Template** to use an out-of-box template.

   Note

   HTML templates are only available if this option is selected in the global survey settings.

6. To preview the email, type a value in the **Test Record** field, and then click **Preview**.

Note

You can reference parameter names in the **External Survey URL**, **Email Subject**, and **Email Body** fields. For example, you can type "Survey for record {interactionId}" as the subject, where "{interactionId}" is a parameter for the interaction ID that triggered the survey request.

1.19.2.3.11. Create a localized email

The Survey Integration email functionality in Service Manager Survey Integration enables you to create alternate language versions of the survey request email. Survey recipients receive a localized email if their default language matches the language of the localized email. Otherwise, the user receives the default language version of the email that's configured in the global survey settings, regardless of the user's default language.

To create a localized email, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Email Settings** section.
3. Configure the default language survey request email, and then click **Save**.
4. In the **Translations** sub-section, click **Add Translation** to display the Survey Email Localization form. The **ID**, **Last updated by**, and **Last update time** fields are populated automatically.
5. Select a language from the **Language** drop-down list.

   Note

   - The default language that's configured in the global survey settings doesn't appear in this list.
   - System administrators can configure which languages appear in the list by using the Languages table. To include a language, select that language in the Languages table, and then select the **Active for Survey Management** option.

6. Type a subject in the **Email Subject** field.
7. Type the email body text in the **Email Body** field.
8. Click **Save**, and then click **Cancel** to return to the survey definition record.

Related topics

Create a new survey
Search for a survey
Configure the email settings

1.19.2.3.12. Configure the survey parameters

The **Parameters** section of the survey definition record in Service Manager Survey Integration enables you to configure the survey parameters that aren't inherited from the connector. Parameters are defined in the survey connector, and are based on the specific Service Manager module to which the survey applies.

To configure the survey parameters, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to the **Parameters** section.
3. Mandatory parameters are predefined by the connector. To create additional parameters, click **Optional**.
4. Type a name in the **Parameter Name** field, select a field from the **Field Name** drop-down list, type a description of the parameter in the **Description** field, and then click **Save**.

1.19.2.3.13. View the survey workflow

Service Manager Survey Integration incorporates Process Designer technology. To view the Survey Integration workflow, click **Survey Management** > **Administration** > **Survey Workflows**, and then double-click the **Survey** workflow. The survey workflow is also displayed in the **Workflow** section of the survey definition record.

To view specific information about a transition or phase in the survey workflow, click the transition or phase.

Out-of-box Survey Integration workflows comprise the following three phases.

Phase	Description
Planning	The survey manager configures survey settings such as data filters, recipients, and parameters. Survey invitations aren't sent during this phase.
Scheduled	The survey runs for the configured period. The majority of settings are no longer editable during this phase.
Retired	The survey is no longer active.

Note

When the date that's configured in the **Active To** field of the survey settings passes, the survey automatically transitions from the Scheduled phase to the Retired phase.

1.19.2.3.14. View the survey history

The **History** section of the survey definition record in Service Manager Survey Integration displays basic information about the life of a survey. The fields in this section can't be edited.

To view the survey history, follow these steps:

1. Open a survey definition record by creating a new survey or by searching for an existing survey.
2. Scroll down to and expand the **History** section to display the following information:
    - Who created the survey
    - When the survey was created
    - Who last updated the survey
    - When the survey was last updated
    - Who retired the survey
    - When the survey was retired
3. Click the **Show Send History** button to display the following information:
    - When the survey was sent
    - The status of the uploaded and sent data
    - Detailed information about the uploaded and sent data
4. Click the **Show Send Log** button to display the following information:
    - When the survey was sent
    - The recipient of the survey request email
    - The record that triggered the survey

Related topics

Create a new survey
Search for a survey

1.19.2.4. Survey reference material

This section introduces the Survey reference material.

Survey connectors

The survey connector is the integration method that SM Survey and Survey Integration use to send survey data to survey recipients. The survey connector defines some elements of the survey, such as the survey parameters, survey recipients, and connection parameters.

Out-of-box deployments of Survey contain the following types of connector. For scheduled SM surveys, users can create surveys based on the out-of-box internal connector, and there is no need to create any new connectors.

ID	Name	Description
001	API Connector - Market Tools	An API-based connector that uploads data directly to the Market Tools survey solution
002	URL Connector	An URL-based connector. When you use this connector to manage a survey, survey recipients are emailed a URL that points to the survey
003	API Connector - Medallia	An API-based connector that uploads data directly to the Medallia survey solution
004	Internal Connector	The internal connector for scheduled SM surveys using Internal URL

To view the available survey connectors, follow these steps:

1. In the system navigator, click **Survey Management** > **Administration** > **Survey Connectors** to display the Survey Connector form.
2. Enter search criteria into any of the fields, and then click **Search**.

Security areas

The Survey module includes three security areas that ensure correct security in the Survey module.

The following table displays the mapping between the security areas and the security roles in the Survey module.

Area	Applies to Interactions in	Roles
Survey	Survey definitions	Survey Administrator Survey Analyst Survey Manager
Survey configuration	Survey connectors Survey templates	Survey Administrator
Survey configuration	Survey templates	Survey Manager
Tailoring	Survey settings Survey workflow	Survey Administrator

Note: Survey templates are supported only for SM Survey (not for Survey Integration).

Security roles

The Survey module uses the following three security roles.

Note

Be aware that survey templates are supported only for SM Survey (not supported for Survey Integration).

Survey Administrator

Area	Rights
Survey	Full rights for the area
Survey configuration	Full rights for the area
Tailoring	Full rights for the area

Survey Manager

Area	Rights
Survey	Expert rights for the area (can always update all survey definition records, and retire or unretire surveys)
Survey configuration	Expert rights for the area (can always update all survey templates, and retire or unretire survey templates)
Tailoring	No rights

Survey Analyst

Area	Rights
Survey	Partial rights for the area (can only update and retire or unretire assigned surveys)
Survey configuration	No rights
Tailoring	No rights

The following rights can be configured for Survey security roles.

Right	Description
View	Can view survey definition records
New	Can create new survey definition records
Update	Determines when a user can update a survey. This right has the following options: • Never • Always • When assigned • When assigned to workgroup
Delete / Close	Determines when a user can retire a survey. This right has the following options: • Never • Always • When assigned • When assigned to workgroup
Expert	No special functionality related
Admin	Has administrative rights (by default, full rights to the Survey module)
Modify Template	Can tailor template records in this area
ESS Only	When selected, indicates that the specified rights take effect only when the role is logged in to the Employee Self-Service (ESS) web client.

Note

The **When assigned** and **When assigned to workgroup** values in the **Update** and **Delete/Close** fields aren't supported in the Survey Configuration security area.

The following settings can be configured for Survey security roles.

Setting	Description
Can copy record	Can copy survey definition records
Can create global inbox	Can create global time period inboxes
Initial Inbox	Initial inbox for time period module
Manage format	Sc manage format
Can create personal inbox	Can create personal inboxes for survey definition records
Un-retire	Can unretire (reopen) survey definition records
New Thread List → Edit	Determines whether a new thread is opened when a user moves from a List to an Edit screen
New Thread Search → List	Determines whether a new thread is opened when a user moves from a Search screen to a List
New Thread View → Edit	Determines whether a new thread is opened when a user moves from an Inbox to an Edit screen
New Thread View → Search	Determines whether a new thread is opened when a user moves from an inbox to a Search screen

Parameter mapping

The following mapping types are available when you create a new parameter in Service Manager Survey Integration framework.

Predefined mapping type	Description	Example syntax
Value	A constant value	• John.doe@*xxxx*.com (email) • jdoe (operator name) • DOE, JOHN (contact name)
Variable	Any variable that's available when the survey runs	$G.serviceDesk
Field	A valid field name in the selected Service manager module	Incident.id
RAD expression	Any valid RAD expression that returns the required value	jscall(<lib.function>, contact.name in $L.file)
Predefined	The following predefined options are available: • Recipient Email • Recipient Name • Language • Web URL • ESS URL • External URL	

Recipient mapping

The following mapping types are available when you create a new recipient in Service Manager Survey framework.

Predefined mapping type	Description	Example Syntax
Value	A constant value. This value must be a valid email address, contact name, or operator name. Multiple values can be entered and must be separated by a semi-colon (;).	• John.doe@*xxxx*.com (email) • jdoe (operator name) • DOE, JOHN (contact name)
Variable	Any variable that's available when the survey runs. If the recipient type is set to **Person**, this value must be a valid email address, contact name, or operator name. If the recipient type is set to **Group**, this variable must contain a valid group name.	• $G.serviceDesk (person) • $lo.pm.assignment (group)
Field	A valid field name in the selected Service manager module. If the recipient type is set to **Person**, this value must be a valid email address, contact name, or operator name. If the recipient type is set to **Group**, this variable must contain a valid group name.	• contact.name (person) • assignment (group)

Predefined mapping type	Description	Example Syntax
RAD expression	Any valid RAD expression that returns an email address, a contact name, an operator name, or an array of values. Multiple values can be entered and must be separated by a semi-colon (;).	jscall(<lib.function>, contact.name in $L.file)

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